

**VOLUME I: FOUNDATIONS – EXERCISE EDITION**

# The Quantum Theory of Fields

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ABSTRACT: Weinberg's QFT in the repository's modernized notation. This independent exercise edition adds editorial solutions and credited supplementary problems; the added material is not part of Weinberg's original text.

*To Louise*



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## Preface to Volume I

Why another book on quantum field theory? Today the student of quantum field theory can choose from among a score of excellent books, several of them quite up-to-date. Another book will be worth while only if it offers something new in content or perspective.

As to content, although this book contains a good amount of new material, I suppose the most distinctive thing about it is its generality; I have tried throughout to discuss matters in a context that is as general as possible. This is in part because quantum field theory has found applications far removed from the scene of its old successes, quantum electrodynamics, but even more because I think that this generality will help to keep the important points from being submerged in the technicalities of specific theories. Of course, specific examples are frequently used to illustrate general points, examples that are chosen from contemporary particle physics or nuclear physics as well as from quantum electrodynamics.

It is, however, the perspective of this book, rather than its content, that provided my chief motivation in writing it. I aim to present quantum field theory in a manner that will give the reader the clearest possible idea of *why* this theory takes the form it does, and why in this form it does such a good job of describing the real world.

The traditional approach, since the first papers of Heisenberg and Pauli on general quantum field theory, has been to take the existence of fields for granted, relying for justification on our experience with electromagnetism, and ‘quantize’ them — that is, apply to various simple field theories the rules of canonical quantization or path integration. Some of this traditional approach will be found here in the historical introduction presented in Chapter 1. This is certainly a way of getting rapidly into the subject, but it seems to me that it leaves the reflective reader with too many unanswered questions. Why should we believe in the rules of canonical quantization or path integration? Why should we adopt the simple field equations and Lagrangians that are found in the literature? For that matter, why have fields at all? It does not seem satisfactory to me to appeal to experience; after all, our purpose in theoretical physics is not just to describe the world as we find it, but to explain — in terms of a few fundamental principles — why the world is the way it is.

The point of view of this book is that quantum field theory is the way it is because (aside from theories like string theory that have an infinite number of particle types) it is the only way to reconcile the principles of quantum mechanics (including the cluster decomposition property) with those of special relativity. This is a point of view I have held for many years, but it is also one that has become newly appropriate. We have learned in recent years to think of our successful quantum field theories, including quantum electrodynamics, as ‘effective field theories,’ low-energy approximations to a deeper theory that may not even be a field theory, but something different like a string theory. On this basis, the reason that quantum field theories describe physics at accessible energies is that *any* relativistic quantum theory will look at sufficiently low energy like a quantum field theory. It is therefore important to understand the rationale for quantum field theory in terms of the principles of relativity and quantum mechanics. Also, we think differently now about some of the problems of quantum field theories, such as non-renormalizability and ‘triviality,’ that used to bother us when we thought of these theories as truly fundamental,



and the discussions here will reflect these changes. This is intended to be a book on quantum field theory for the era of effective field theories.

The most immediate and certain consequences of relativity and quantum mechanics are the properties of particle states, so here particles come first — they are introduced in Chapter 2 as ingredients in the representation of the Poincaré group in the Hilbert space of quantum mechanics. Chapter 3 provides a framework for addressing the fundamental dynamical question: given a state that in the distant past looks like a certain collection of free particles, what will it look like in the future? Knowing the generator of time-translations, the Hamiltonian, we can answer this question through the perturbative expansion for the array of transition amplitudes known as the  $S$ -matrix. In Chapter 4 the principle of cluster decomposition is invoked to describe how the generator of time-translations, the Hamiltonian, is to be constructed from creation and annihilation operators. Then in Chapter 5 we return to Lorentz invariance, and show that it requires these creation and annihilation operators to be grouped together in causal quantum fields. As a spin-off, we deduce the CPT theorem and the connection between spin and statistics. The formalism is used in Chapter 6 to derive the Feynman rules for calculating the  $S$ -matrix.

It is not until Chapter 7 that we come to Lagrangians and the canonical formalism. The rationale here for introducing them is not that they have proved useful elsewhere in physics (never a very satisfying explanation) but rather that this formalism makes it easy to choose interaction Hamiltonians for which the  $S$ -matrix satisfies various assumed symmetries. In particular, the Lorentz invariance of the Lagrangian density ensures the existence of a set of ten operators that satisfy the algebra of the Poincaré group and, as we show in Chapter 3, this is the key condition that we need to prove the Lorentz invariance of the  $S$ -matrix. Quantum electrodynamics finally appears in Chapter 8. Path integration is introduced in Chapter 9, and used to justify some of the hand-waving in Chapter 8 regarding the Feynman rules for quantum electrodynamics. This is a somewhat later introduction of path integrals than is fashionable these days, but it seems to me that although path integration is by far the best way of rapidly deriving Feynman rules from a given Lagrangian, it rather obscures the quantum mechanical reasons underlying these calculations.

Volume I concludes with a series of chapters, 10–14, that provide an introduction to the calculation of radiative corrections, involving loop graphs, in general field theories. Here too the arrangement is a bit unusual; we start with a chapter on non-perturbative methods, in part because the results we obtain help us to understand the necessity for field and mass renormalization, without regard to whether the theory contains infinities or not. Chapter 11 presents the classic one-loop calculations of quantum chromodynamics, both as an opportunity to explain useful calculational techniques (Feynman parameters, Wick rotation, dimensional and Pauli–Villars regularization), and also as a concrete example of renormalization in action. The experience gained in Chapter 11 is extended to all orders and general theories in Chapter 12, which also describes the modern view of non-renormalizability that is appropriate to effective field theories. Chapter 13 is a digression on the special problems raised by massless particles of low energy or parallel momenta. The Dirac equation for an electron in an external electromagnetic field, which historically appeared almost at the very start of relativistic quantum mechanics, is not seen here until Chapter 14, on bound

state problems, because this equation should not be viewed (as Dirac did) as a relativistic version of the Schrödinger equation, but rather as an approximation to a true relativistic quantum theory, the quantum field theory of photons and electrons. This chapter ends with a treatment of the Lamb shift, bringing the confrontation of theory and experiment up to date.

The reader may feel that some of the topics treated here, especially in Chapter 3, could more properly have been left to textbooks on nuclear or elementary particle physics. So they might, but in my experience these topics are usually either not covered or covered poorly, using specific dynamical models rather than the general principles of symmetry and quantum mechanics. I have met string theorists who have never heard of the relation between time-reversal invariance and final-state phase shifts, and nuclear theorists who do not understand why resonances are governed by the Breit–Wigner formula. So in the early chapters I have tried to err on the side of inclusion rather than exclusion.

Volume II will deal with the advances that have revived quantum field theory in recent years: non-Abelian gauge theories, the renormalization group, broken symmetries, anomalies, instantons, and so on.

I have tried to give citations both to the classic papers in the quantum theory of fields and to useful references on topics that are mentioned but not presented in detail in this book. I did not always know who was responsible for material presented here, and the mere absence of a citation should not be taken as a claim that the material presented here is original. But some of it is. I hope that I have improved on the original literature or standard textbook treatments in several places, as for instance in the proof that symmetry operators are either unitary or antiunitary; the discussion of superselection rules; the analysis of particle degeneracy associated with unconventional representations of inversions; the use of the cluster decomposition principle; the derivation of the reduction formula; the derivation of the external field approximation; and even the calculation of the Lamb shift.

I have also supplied problems for each chapter except the first. Some of these problems aim simply at providing exercise in the use of techniques described in the chapter; others are intended to suggest extensions of the results of the chapter to a wider class of theories.

In teaching quantum field theory, I have found that each of the two volumes of this book provides enough material for a one-year course for graduate students. I intended that this book should be accessible to students who are familiar with non-relativistic quantum mechanics and classical electrodynamics. I assume a basic knowledge of complex analysis and matrix algebra, but topics in group theory and topology are explained where they are introduced.

This is not a book for the student who wants immediately to begin calculating Feynman graphs in the standard model of weak, electromagnetic, and strong interactions. Nor is this a book for those who seek a higher level of mathematical rigor. Indeed, there are parts of this book whose lack of rigor will bring tears to the eyes of the mathematically inclined reader. Rather, I hope it will suit the physicists and physics students who want to understand *why* quantum field theory is the way it is, so that they will be ready for whatever new developments in physics may take us beyond our present understandings.

\* \* \*

Much of the material in this book I learned from my interactions over the years with numerous other physicists, far too many to name here. But I must acknowledge my special intellectual debt to Sidney Coleman, and to my colleagues at the University of Texas: Arno Bohm, Luis Boya, Phil Candelas, Bryce DeWitt, Cecile DeWitt-Morette, Jacques Distler, Willy Fischler, Josh Feinberg, Joaquim Gomis, Vadim Kaplunovsky, Joe Polchinski, and Paul Shapiro. I owe thanks for help in the preparation of the historical introduction to Gerry Holton, Arthur Miller, and Sam Schweber. Thanks are also due to Alyce Wilson, who prepared the illustrations and typed the L<sup>A</sup>T<sub>E</sub>X input files until I learned how to do it, and to Terry Riley for finding countless books and articles. I am grateful to Maureen Storey and Alison Woollatt of Cambridge University Press for helping to ready this book for publication, and especially to my editor, Rufus Neal, for his friendly good advice.

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## Notation

*Typesetter's note:* see NOTATION.md for the specific notation upgrades I implemented. Some include switching to Dirac notation, switching  $a(\mathbf{p}) \rightarrow a_{\mathbf{p}}$ , and getting rid of the terrible-to-look-at `mathscr` letters, e.g.  $\mathscr{J} \rightarrow \mathcal{J}$ .

Latin indices  $i, j, k, \dots$  generally run over the three spatial coordinates 1, 2, 3. Greek indices  $\mu, \nu, \dots$  run over spacetime coordinates in the order 0, 1, 2, 3, with  $x^0$  the time coordinate. Repeated indices are summed unless otherwise indicated.

The spacetime metric is mostly plus,

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1),$$

so an on-shell four-momentum satisfies  $p^2 = -m^2$ , and spacelike separations satisfy  $(x - x')^2 \geq 0$ . The d'Alembertian is

$$\square = \partial_\mu \partial^\mu = \nabla^2 - \partial_t^2.$$

The Levi-Civita tensor is totally antisymmetric, with  $\epsilon^{0123} = +1$ .

Spatial three-vectors are bold, and a hat denotes a unit vector:  $\hat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$ . A dot denotes a time derivative.

Dirac matrices obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \quad \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3.$$

With this metric convention,  $\beta = i\gamma^0$  and the Dirac adjoint is  $\bar{u} = u^\dagger \beta$ . The step function  $\theta(s)$  equals 1 for  $s > 0$  and 0 for  $s < 0$ .

Complex conjugation, transpose, and Hermitian conjugation are written  $A^*$ ,  $A^\top$ , and  $A^\dagger$ , respectively. The identity is  $\mathbf{1}$ . The abbreviations +H.c. and +c.c. indicate addition of the Hermitian conjugate and complex conjugate of the preceding terms.

States use Dirac notation:  $|p, \sigma\rangle$  for one-particle states,  $|\Omega\rangle$  for the vacuum, and  $\langle\phi|\psi\rangle$  for inner products. Asymptotic states are labeled  $|\alpha\rangle_{\text{in}}$  and  $|\beta\rangle_{\text{out}}$ . Positive- and negative-frequency field pieces are written  $\psi^{(+)}$  and  $\psi^{(-)}$ , avoiding the original overloading of + and − for both field frequencies and asymptotic states.

The interaction Hamiltonian is denoted

$$H_I(t) = \int d^3x \mathcal{H}_I(x),$$

in place of the original bare  $V$  notation. Other operators are generally left unhatted when the context is unambiguous.

Except in Chapter 1,  $\hbar = c = 1$ . Throughout,  $-e$  is the rationalized charge of the electron, so  $\alpha = e^2/(4\pi) \simeq 1/137$ . Parentheses at the end of quoted numerical data give the uncertainty in the final digits.

## 1 Historical Introduction

Our immersion in the present state of physics makes it hard for us to understand the difficulties of physicists even a few years ago, or to profit from their experience. At the same time, a knowledge of our history is a mixed blessing—it can stand in the way of the logical reconstruction of physical theory that seems to be continually necessary.

I have tried in this book to present the quantum theory of fields in a logical manner, emphasizing the deductive trail that ascends from the physical principles of special relativity and quantum mechanics. This approach necessarily draws me away from the order in which the subject in fact developed. To take one example, it is historically correct that quantum field theory grew in part out of a study of relativistic wave equations, including the Maxwell, Klein–Gordon, and Dirac equations. For this reason it is natural that courses and treatises on quantum field theory introduce these wave equations early, and give them great weight. Nevertheless, it has long seemed to me that a much better starting point is Wigner’s definition of particles as representations of the Poincaré group, even though this work was not published until 1939 and did not have a great impact for many years after. In this book we start with particles and get to the wave equations later.

This is not to say that particles are necessarily more fundamental than fields. For many years after 1950 it was generally assumed that the laws of nature take the form of a quantum theory of fields. I start with particles in this book, not because they are more fundamental, but because what we know about particles is more certain, more directly derivable from the principles of quantum mechanics and relativity. If it turned out that some physical system could not be described by a quantum field theory, it would be a sensation; if it turned out that the system did not obey the rules of quantum mechanics and relativity, it would be a cataclysm.

In fact, lately there has been a reaction against looking at quantum field theory as fundamental. The underlying theory might not be a theory of fields or particles, but perhaps of something quite different, like strings. From this point of view, quantum electrodynamics and the other quantum field theories of which we are so proud are mere ‘effective field theories,’ low-energy approximations to a more fundamental theory. The reason that our field theories work so well is not that they are fundamental truths, but that any relativistic quantum theory will look like a field theory when applied to particles at sufficiently low energy. On this basis, if we want to know why quantum field theories are the way they are, we have to start with particles.

But we do not want to pay the price of altogether forgetting our past. This chapter will therefore present the history of quantum field theory from earliest times to 1949, when it finally assumed its modern form. In the remainder of the book I will try to keep history from intruding on physics.

One problem that I found in writing this chapter is that the history of quantum field theory is from the beginning inextricably entangled with the history of quantum mechanics itself. Thus, the reader who is familiar with the history of quantum mechanics may find some material here that he or she already knows, especially in the first section, where I discuss the early attempts to put together quantum mechanics with special relativity. In

this case I can only suggest that the reader should skip on to the less familiar parts.

On the other hand, readers who have no prior familiarity with quantum field theory may find parts of this chapter too brief to be altogether clear. I urge such readers not to worry. This chapter is not intended as a self-contained introduction to quantum field theory, and is not needed as a basis for the rest of the book. Some readers may even prefer to start with the next chapter, and come back to the history later. However, for many readers the history of quantum field theory should serve as a good introduction to quantum field theory itself.

I should add that this chapter is not intended as an original work of historical scholarship. I have based it on books and articles by real historians, plus some historical reminiscences and original physics articles that I have read. Most of these are listed in the bibliography given at the end of this chapter, and in the list of references. The reader who wants to go more deeply into historical matters is urged to consult these listed works.

A word about notation. In order to keep some of the flavor of past times, in this chapter I will show explicit factors of  $\hbar$  and  $c$  (and even  $h$ ), but in order to facilitate comparison with modern physics literature, I will use the more modern rationalized electrostatic units for charge, so that the fine structure constant  $\alpha \simeq 1/137$  is  $e^2/4\pi\hbar c$ . In subsequent chapters I will mostly use the ‘natural’ system of units, simply setting  $\hbar = c = 1$ .

## 1.1 Relativistic Wave Mechanics

Wave mechanics started out as relativistic wave mechanics. Indeed, as we shall see, the founders of wave mechanics, Louis de Broglie and Erwin Schrödinger, took a good deal of their inspiration from special relativity. It was only later that it became generally clear that relativistic wave mechanics, in the sense of a relativistic quantum theory of a fixed number of particles, is an impossibility. Thus, despite its many successes, relativistic wave mechanics was ultimately to give way to quantum field theory. Nevertheless, relativistic wave mechanics survived as an important element in the formal apparatus of quantum field theory, and it posed a challenge to field theory, to reproduce its successes.

The possibility that material particles can like photons be described in terms of waves was first suggested [1] in 1923 by Louis de Broglie. Apart from the analogy with radiation, the chief clue was Lorentz invariance: if particles are described by a wave whose phase at position  $\mathbf{x}$  and time  $t$  is of the form  $2\pi(\boldsymbol{\kappa} \cdot \mathbf{x} - \nu t)$ , and if this phase is to be Lorentz invariant, then the vector  $\boldsymbol{\kappa}$  and the frequency  $\nu$  must transform like  $\mathbf{x}$  and  $t$ , and hence like  $\mathbf{p}$  and  $E$ . In order for this to be possible  $\boldsymbol{\kappa}$  and  $\nu$  must have the same velocity dependence as  $\mathbf{p}$  and  $E$ , and therefore must be proportional to them, with the same constant of proportionality. For photons, one had the Einstein relation  $E = h\nu$ , so it was natural to assume that, for material particles,

$$\boldsymbol{\kappa} = \mathbf{p}/h, \quad \nu = E/h, \quad (1.1.1)$$

just as for photons. The group velocity  $\partial\nu/\partial\boldsymbol{\kappa}$  of the wave then turns out to equal the particle velocity, so wave packets just keep up with the particle they represent.

By assuming that any closed orbit contains an integral number of particle wavelengths  $\lambda = 1/|\boldsymbol{\kappa}|$ , de Broglie was able to derive the old quantization conditions of Niels Bohr

and Arnold Sommerfeld, which though quite mysterious had worked well in accounting for atomic spectra. Also, both de Broglie and Walter Elsasser [2] suggested that de Broglie's wave theory could be tested by looking for interference effects in the scattering of electrons from crystals; such effects were established a few years later by Clinton Joseph Davisson and Lester H. Germer [3]. However, it was still unclear how the de Broglie relations (1.1.1) should be modified for non-free particles, as for instance for an electron in a general Coulomb field.

Wave mechanics was by-passed in the next step in the history of quantum mechanics, the development of matrix mechanics [4] by Werner Heisenberg, Max Born, Pascual Jordan and Wolfgang Pauli in the years 1925–1926. At least part of the inspiration for matrix mechanics was the insistence that the theory should involve only observables, such as the energy levels, or emission and absorption rates. Heisenberg's 1925 paper opens with the manifesto: 'The present paper seeks to establish a basis for theoretical quantum mechanics founded exclusively upon relationships between quantities that in principle are observable.' This sort of positivism was to reemerge at various times in the history of quantum field theory, as for instance in the introduction of the S-matrix by John Wheeler and Heisenberg (see Chapter 3) and in the revival of dispersion theory in the 1950s (see Chapter 10), though modern quantum field theory is very far from this ideal. It would take us too far from our subject to describe matrix mechanics in any detail here.

As everyone knows, wave mechanics was revived by Erwin Schrödinger. In his 1926 series of papers [5], the familiar non-relativistic wave equation is suggested first, and then used to rederive the results of matrix mechanics. Only later, in the sixth section of the fourth paper, is a relativistic wave equation offered. According to Dirac [6], the history is actually quite different: Schrödinger first derived the relativistic equation, then became discouraged because it gave the wrong fine structure for hydrogen, and then some months later realized that the non-relativistic approximation to his relativistic equation was of value even if the relativistic equation itself was incorrect! By the time that Schrödinger came to publish his relativistic wave equation, it had already been independently rediscovered by Oskar Klein [7] and Walter Gordon [8], and for this reason it is usually called the 'Klein–Gordon equation.'

Schrödinger's relativistic wave equation was derived by noting first that, for a 'Lorentz electron' of mass  $m$  and charge  $e$  in an external vector potential  $\mathbf{A}$  and Coulomb potential  $\phi$ , the Hamiltonian  $H$  and momentum  $\mathbf{p}$  are related by<sup>1</sup>

$$0 = (H + e\phi)^2 - c^2(\mathbf{p} + e\mathbf{A}/c)^2 - m^2c^4. \quad (1.1.2)$$

For a free particle described by a plane wave  $\exp\{2\pi i(\boldsymbol{\kappa} \cdot \mathbf{x} - \nu t)\}$ , the de Broglie relations (1.1.1) can be obtained by the identifications

$$\mathbf{p} = h\boldsymbol{\kappa} \longrightarrow -i\hbar\nabla, \quad E = h\nu \longrightarrow i\hbar\frac{\partial}{\partial t}, \quad (1.1.3)$$

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<sup>1</sup>This is Lorentz invariant, because the quantities  $\mathbf{A}$  and  $\phi$  have the same Lorentz transformation property as  $c\mathbf{p}$  and  $E$ . Schrödinger actually wrote  $H$  and  $\mathbf{p}$  in terms of partial derivatives of an action function, but this makes no difference to our present discussion.

where  $\hbar$  is the convenient symbol (introduced later by Dirac) for  $h/2\pi$ . By an admittedly formal analogy, Schrödinger guessed that an electron in the external fields  $\mathbf{A}, \phi$  would be described by a wave function  $\psi(\mathbf{x}, t)$  satisfying the equation obtained by making the same replacements in (1.1.2):

$$0 = \left[ \left( i\hbar \frac{\partial}{\partial t} + e\phi \right)^2 - c^2 \left( -i\hbar \nabla + \frac{e\mathbf{A}}{c} \right)^2 - m^2 c^4 \right] \psi(\mathbf{x}, t). \quad (1.1.4)$$

In particular, for the stationary states of hydrogen we have  $\mathbf{A} = 0$  and  $\phi = e/4\pi r$ , and  $\psi$  has the time-dependence  $\exp(-iEt/\hbar)$ , so (1.1.4) becomes

$$0 = \left[ \left( E + \frac{e^2}{4\pi r} \right)^2 - c^2 \hbar^2 \nabla^2 - m^2 c^4 \right] \psi(\mathbf{x}). \quad (1.1.5)$$

Solutions satisfying reasonable boundary conditions can be found for the energy values [9]

$$E = mc^2 \left[ 1 - \frac{\alpha^2}{2n^2} - \frac{\alpha^4}{2n^4} \left( \frac{n}{\ell + \frac{1}{2}} - \frac{3}{4} \right) + \dots \right], \quad (1.1.6)$$

where  $\alpha = e^2/4\pi\hbar c$  is the ‘fine structure constant,’ roughly  $1/137$ ;  $n$  is a positive-definite integer, and  $\ell$ , the orbital angular momentum in units of  $\hbar$ , is an integer with  $0 \leq \ell \leq n-1$ . The  $\alpha^2$  term gave good agreement with the gross features of the hydrogen spectrum (the Lyman, Balmer, etc. series) and, according to Dirac [6], it was this agreement that led Schrödinger eventually to develop his non-relativistic wave equation. On the other hand, the  $\alpha^4$  term gave a fine structure in disagreement with existing accurate measurements of Friedrich Paschen [10].

It is instructive here to compare Schrödinger’s result with that of Arnold Sommerfeld [10c], obtained using the rules of the old quantum theory:

$$E = mc^2 \left[ 1 - \frac{\alpha^2}{2n^2} - \frac{\alpha^4}{2n^4} \left( \frac{n}{k} - \frac{3}{4} \right) + \dots \right]. \quad (1.1.7)$$

where  $m$  is the electron mass. Here  $k$  is an integer between 1 and  $n$ , which in Sommerfeld’s theory is given in terms of the orbital angular momentum  $\ell\hbar$  as  $k = \ell + 1$ . This gave a fine structure splitting in agreement with experiment: for instance, for  $n = 2$  Eq. (1.1.7) gives two levels ( $k = 1$  and  $k = 2$ ), split by the observed amount  $\alpha^4 mc^2/32$ , or  $4.53 \cdot 10^{-5}$  eV. In contrast, Schrödinger’s result (1.1.6) gives an  $n = 2$  fine structure splitting  $\alpha^4 mc^2/12$ , considerably larger than observed.

Schrödinger correctly recognized that the source of this discrepancy was his neglect of the spin of the electron. The splitting of atomic energy levels by non-inverse-square electric fields in alkali atoms and by weak external magnetic fields (the so-called anomalous Zeeman effect) had revealed a multiplicity of states larger than could be accounted for by the Bohr–Sommerfeld theory; this led George Uhlenbeck and Samuel Goudsmit [11] in 1925 to suggest that the electron has an intrinsic angular momentum  $\hbar/2$ . Also, the magnitude of the Zeeman splitting [12] allowed them to estimate further that the electron has a magnetic moment

$$\mu = \frac{e\hbar}{2mc}. \quad (1.1.8)$$



It was clear that the electron's spin would be coupled to its orbital angular momentum, so that Schrödinger's relativistic equation should not be expected to give the correct fine structure splitting.

Indeed, by 1927 several authors [13] had been able to show that the spin-orbit coupling was able to account for the discrepancy between Schrödinger's result (1.1.6) and experiment. There are really two effects here: one is a direct coupling between the magnetic moment (1.1.8) and the magnetic field felt by the electron as it moves through the electrostatic field of the atom; the other is the relativistic 'Thomas precession' caused (even in the absence of a magnetic moment) by the circular motion of the spinning electron [14]. Together, these two effects were found to lift the level with total angular momentum  $j = \ell + \frac{1}{2}$  to the energy (1.1.7) given by Sommerfeld for  $k = \ell + 1 = j + \frac{1}{2}$ , while the level with  $j = \ell - \frac{1}{2}$  was lowered to the value given by Sommerfeld for  $k = \ell = j + \frac{1}{2}$ . Thus the energy was found to depend only on  $n$  and  $j$ , but not separately on  $\ell$ :

$$E = mc^2 \left[ 1 - \frac{\alpha^2}{2n^2} - \frac{\alpha^4}{2n^4} \left( \frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) + \cdots \right]. \quad (1.1.9)$$

By accident Sommerfeld's theory had given the correct magnitude of the splitting in hydrogen ( $j + \frac{1}{2}$  like  $k$  runs over integer values from 1 to  $n$ ) though it was wrong as to the assignment of orbital angular momentum values  $\ell$  to these various levels. In addition, the multiplicity of the fine structure levels in hydrogen was now predicted to be 2 for  $j = \frac{1}{2}$  and  $2(2j + 1)$  for  $j > \frac{1}{2}$  (corresponding to  $\ell$  values  $j \pm \frac{1}{2}$ ), in agreement with experiment.

Despite these successes, there still was not a thorough relativistic theory which incorporated the electron's spin from the beginning. Such a theory was discovered in 1928 by Paul Dirac. However, he did not set out simply to make a relativistic theory of the spinning electron; instead, he approached the problem by posing a question that would today seem very strange. At the beginning of his 1928 paper [15], he asks 'why Nature should have chosen this particular model for the electron, instead of being satisfied with the point charge.' To us today, this question is like asking why bacteria have only one cell: having spin  $\hbar/2$  is just one of the properties that define a particle as an electron, rather than one of the many other types of particles with various spins that are known today. However, in 1928 it was possible to believe that all matter consisted of electrons, and perhaps something similar with positive charge in the atomic nucleus. Thus, in the spirit of the times in which it was asked, Dirac's question can be restated: 'Why do the fundamental constituents of matter have to have spin  $\hbar/2$ ?'

For Dirac, the key to this question was the requirement that probabilities must be positive. It was known [16] that the probability density for the non-relativistic Schrödinger equation is  $|\psi|^2$ , and that this satisfies a continuity equation of the form

$$\frac{\partial}{\partial t} |\psi|^2 - \frac{i\hbar}{2m} \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*) = 0$$

so the space-integral of  $|\psi|^2$  is time-independent. On the other hand, the only probability density  $\rho$  and current  $\mathbf{J}$ , which can be formed from solutions of the relativistic Schrödinger

equation and which satisfy a conservation law,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0, \quad (1.1.10)$$

are of the form

$$\rho = N \operatorname{Im} \psi^* \left( \frac{\partial}{\partial t} - \frac{ie\phi}{\hbar} \right) \psi, \quad (1.1.11)$$

$$\mathbf{J} = Nc^2 \operatorname{Im} \psi^* \left( \nabla + \frac{ie\mathbf{A}}{\hbar c} \right) \psi, \quad (1.1.12)$$

with  $N$  an arbitrary constant. It is not possible to identify  $\rho$  as the probability density, because (with or without an external potential  $\phi$ )  $\rho$  does not have definite sign. To quote Dirac's reminiscences [17] about this problem

I remember once when I was in Copenhagen, that Bohr asked me what I was working on and I told him I was trying to get a satisfactory relativistic theory of the electron, and Bohr said 'But Klein and Gordon have already done that!' That answer first rather disturbed me. Bohr seemed quite satisfied by Klein's solution, but I was not because of the negative probabilities that it led to. I just kept on with it, worrying about getting a theory which would have only positive probabilities.

According to George Gamow [18], Dirac found the answer to this problem on an evening in 1928 while staring into a fireplace at St John's College, Cambridge. He realized that the reason that the Klein–Gordon (or relativistic Schrödinger) equation can give negative probabilities is that the  $\rho$  in the conservation equation (1.1.10) involves a time-derivative of the wave function. This in turn happens because the wave function satisfies a differential equation of second order in the time. The problem therefore was to replace this wave equation with another one of first order in time derivatives, like the non-relativistic Schrödinger equation.

Suppose the electron wave function is a multi-component quantity  $\psi_n(\mathbf{x})$ , which satisfies a wave equation of the form,

$$i\hbar \frac{\partial \psi}{\partial t} = \mathcal{H}\psi, \quad (1.1.13)$$

where  $\mathcal{H}$  is some matrix function of space derivatives. In order to have a chance at a Lorentz-invariant theory, we must suppose that because the equation is linear in time-derivatives, it is also linear in space-derivatives, so that  $\mathcal{H}$  takes the form:

$$\mathcal{H} = -i\hbar c \boldsymbol{\alpha} \cdot \nabla + \beta mc^2, \quad (1.1.14)$$

where  $\alpha_1, \alpha_2, \alpha_3$ , and  $\beta$  are constant matrices. From (1.1.13) we can derive the second-order equation

$$\begin{aligned} -\hbar^2 \frac{\partial^2 \psi}{\partial t^2} = \mathcal{H}^2 \psi = & -\hbar^2 c^2 \alpha_i \alpha_j \frac{\partial^2 \psi}{\partial x_i \partial x_j} \\ & - i\hbar mc^3 (\alpha_i \beta + \beta \alpha_i) \frac{\partial \psi}{\partial x_i} + m^2 c^4 \beta^2 \psi. \end{aligned}$$

(The summation convention is in force here;  $i$  and  $j$  run over the values 1, 2, 3, or  $x, y, z$ .) But this must agree with the free-field form of the relativistic Schrödinger equation (1.1.4), which just expresses the relativistic relation between momentum and energy. Therefore, the matrices  $\alpha$  and  $\beta$  must satisfy the relations

$$\alpha_i \alpha_j + \alpha_j \alpha_i = 2\delta_{ij} \mathbf{1}, \quad (1.1.15)$$

$$\alpha_i \beta + \beta \alpha_i = 0, \quad (1.1.16)$$

$$\beta^2 = \mathbf{1}, \quad (1.1.17)$$

where  $\delta_{ij}$  is the Kronecker delta (unity for  $i = j$ ; zero for  $i \neq j$ ) and  $\mathbf{1}$  is the unit matrix. Dirac found a set of  $4 \times 4$  matrices which satisfy these relations

$$\begin{aligned} \alpha_1 &= \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, & \alpha_2 &= \begin{bmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{bmatrix}, \\ \alpha_3 &= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, & \beta &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}. \end{aligned} \quad (1.1.18)$$

To show that this formalism is Lorentz-invariant, Dirac multiplied Eq. (1.1.13) on the left with  $\beta$ , so that it could be put in the form

$$\left[ \hbar c \gamma^\mu \frac{\partial}{\partial x^\mu} + mc^2 \right] \psi = 0, \quad (1.1.19)$$

where

$$\gamma^i \equiv -i\beta\alpha_i, \quad \gamma^0 \equiv -i\beta. \quad (1.1.20)$$

(The Greek indices  $\mu, \nu$ , etc. will now run over the values 0, 1, 2, 3, with  $x^0 = ct$ . Dirac used  $x_4 = ict$ , and correspondingly  $\gamma_4 = \beta$ .) The matrices  $\gamma^\mu$  satisfy the anticommutation relations

$$\frac{1}{2}(\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) = \eta^{\mu\nu} \equiv \begin{cases} +1, & \mu = \nu = 1, 2, 3, \\ -1, & \mu = \nu = 0, \\ 0, & \mu \neq \nu. \end{cases} \quad (1.1.21)$$

Dirac noted that these anticommutation relations are Lorentz-invariant, in the sense that they are also satisfied by the matrices  $\Lambda^\mu{}_\nu \gamma^\nu$ , where  $\Lambda$  is any Lorentz transformation. He concluded from this that  $\Lambda^\mu{}_\nu \gamma^\nu$  must be related to  $\gamma^\mu$  by a similarity transformation:

$$\Lambda^\mu{}_\nu \gamma^\nu = S^{-1}(\Lambda) \gamma^\mu S(\Lambda).$$

It follows that the wave equation is invariant if, under a Lorentz transformation  $x^\mu \rightarrow \Lambda^\mu{}_\nu x^\nu$ , the wave function undergoes the matrix transformation  $\psi \rightarrow S(\Lambda)\psi$ . (These matters are discussed more fully, from a rather different point of view, in Chapter 5.)

To study the behavior of electrons in an arbitrary external electromagnetic field, Dirac followed the ‘usual procedure’ of making the replacements

$$i\hbar\frac{\partial}{\partial t} \longrightarrow i\hbar\frac{\partial}{\partial t} + e\phi, \quad -i\hbar\nabla \longrightarrow -i\hbar\nabla + \frac{e}{c}\mathbf{A} \quad (1.1.22)$$

as in Eq. (1.1.4). The wave equation (1.1.13) then takes the form

$$\left(i\hbar\frac{\partial}{\partial t} + e\phi\right)\psi = (-i\hbar c\nabla + e\mathbf{A}) \cdot \boldsymbol{\alpha}\psi + mc^2\beta\psi. \quad (1.1.23)$$

Dirac used this equation to show that in a central field, the conservation of angular momentum takes the form

$$[\mathcal{H}, -i\hbar\mathbf{r} \cdot \nabla + \hbar\boldsymbol{\sigma}/2] = 0, \quad (1.1.24)$$

where  $\mathcal{H}$  is the matrix differential operator (1.1.14) and  $\boldsymbol{\sigma}$  is the  $4 \times 4$  version of the spin matrix introduced earlier by Pauli [19]

$$\boldsymbol{\sigma} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \boldsymbol{\alpha}. \quad (1.1.25)$$

Since each component of  $\boldsymbol{\sigma}$  has eigenvalues  $\pm 1$ , the presence of the extra term in (1.1.24) shows that the electron has intrinsic angular momentum  $\hbar/2$ .

Dirac also iterated Eq. (1.1.23), obtaining a second-order equation, which turned out to have just the same form as the Klein–Gordon equation (1.1.4) except for the presence on the right-hand-side of two additional terms

$$[-e\hbar c\boldsymbol{\sigma} \cdot \mathbf{B} - ie\hbar c\boldsymbol{\alpha} \cdot \mathbf{E}]\psi. \quad (1.1.26)$$

For a slowly moving electron, the first term dominates, and represents a magnetic moment in agreement with the value (1.1.8) found by Goudsmit and Uhlenbeck [11]. As Dirac recognized, this magnetic moment, together with the relativistic nature of the theory, guaranteed that this theory would give a fine structure splitting in agreement (to order  $\alpha^4 mc^2$ ) with that found by Heisenberg, Jordan, and Charles G. Darwin [13]. A little later, an ‘exact’ formula for the hydrogen energy levels in Dirac’s theory was derived by Darwin [20] and Gordon [21]

$$E = mc^2 \left( 1 + \frac{\alpha^2}{\left\{ n - j - \frac{1}{2} + \left[ (j + \frac{1}{2})^2 - \alpha^2 \right]^{1/2} \right\}^2} \right)^{-1/2}. \quad (1.1.27)$$

The first three terms of a power series expansion in  $\alpha^2$  agree with the approximate result (1.1.9).

This theory achieved Dirac’s primary aim: a relativistic formalism with positive probabilities. From (1.1.13) we can derive a continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0 \quad (1.1.28)$$

with

$$\rho = |\psi|^2, \quad \mathbf{J} = c\psi^\dagger \boldsymbol{\alpha} \psi, \quad (1.1.29)$$

so that the positive quantity  $|\psi|^2$  can be interpreted as a probability density, with constant total probability  $\int |\psi|^2 d^3x$ . However, there was another difficulty which Dirac was not immediately able to resolve.

For a given momentum  $\mathbf{p}$ , the wave equation (1.1.13) has four solutions of the plane wave form

$$\psi \propto \exp \left[ \frac{i}{\hbar} (\mathbf{p} \cdot \mathbf{x} - Et) \right]. \quad (1.1.30)$$

Two solutions with  $E = +\sqrt{\mathbf{p}^2 c^2 + m^2 c^4}$  correspond to the two spin states of an electron with  $J_z = \pm \hbar/2$ . The other two solutions have  $E = -\sqrt{\mathbf{p}^2 c^2 + m^2 c^4}$ , and no obvious physical interpretation. As Dirac pointed out, this problem arises also for the relativistic Schrödinger equation: for each  $\mathbf{p}$ , there are two solutions of the form (1.1.30), one with positive  $E$  and one with negative  $E$ .

Of course, even in classical physics, the relativistic relation  $E^2 = \mathbf{p}^2 c^2 + m^2 c^4$  has two solutions,  $E = \pm \sqrt{\mathbf{p}^2 c^2 + m^2 c^4}$ . However, in classical physics we can simply assume that the only physical particles are those with positive  $E$ . Since the positive solutions have  $E > mc^2$  and the negative ones have  $E < -mc^2$ , there is a finite gap between them, and no continuous process can take a particle from positive to negative energy.

The problem of negative energies is much more troublesome in relativistic quantum mechanics. As Dirac pointed out in his 1928 paper [15], the interaction of electrons with radiation can produce transitions in which a positive-energy electron falls into a negative-energy state, with the energy carried off by two or more photons. Why then is matter stable?

In 1930 Dirac offered a remarkable solution [22]. Dirac's proposal was based on the exclusion principle, so a few words about the history of this principle are in order here.

The periodic table of the elements and the systematics of X-ray spectroscopy had together by 1924 revealed a pattern in the population of atomic energy levels by electrons [23]. The maximum number  $N_n$  of electrons in a shell characterized by principal quantum number  $n$  is given by twice the number of orbital states with that  $n$

$$N_n = 2 \sum_{\ell=0}^{n-1} (2\ell + 1) = 2n^2 = 2, 8, 18, \dots \quad (1.1.31)$$

Wolfgang Pauli [24] in 1925 suggested that this pattern could be understood if  $N_n$  is the total number of possible states in the  $n$ th shell, and if in addition there is some mysterious 'exclusion principle' which forbids more than one electron from occupying the same state. He explained the puzzling factor 2 in (1.1.31) as due to a 'peculiar, classically non-describable duplexity' of the electron states, and as we have seen this was understood a little later as due to the spin of the electron [11]. The exclusion principle answered a question that had remained obscure in the old atomic theory of Bohr and Sommerfeld: why do not all the electrons in heavy atoms fall down into the shell of lowest energy? Subsequently Pauli's exclusion principle was formalized by a number of authors [25] as the requirement that

the wave function of a multi-electron system is antisymmetric in the coordinates, orbital and spin, of all the electrons. This principle was incorporated into statistical mechanics by Enrico Fermi [26] and Dirac [27], and for this reason particles obeying the exclusion principle are generally called ‘fermions,’ just as particles like photons for which the wave function is symmetric and which obey the statistics of Bose and Einstein are called ‘bosons.’ The exclusion principle has played a fundamental role in the theory of metals, white dwarf and neutron stars, etc., as well as in chemistry and atomic physics, but a discussion of these matters would take us too far afield here.

Dirac’s proposal was that the positive energy electrons cannot fall down into negative energy states because ‘all the states of negative energy are occupied except perhaps a few of small velocity.’ The few vacant states, or ‘holes,’ in the sea of negative energy electrons behave like particles with opposite quantum numbers: positive energy and positive charge. The only particle with positive charge that was known at that time was the proton, and as Dirac later recalled [27a], ‘the whole climate of opinion at that time was against new particles’ so Dirac identified his holes as protons; in fact, the title of his 1930 article [22] was ‘A Theory of Electrons and Protons.’

The hole theory faced a number of immediate difficulties. One obvious problem was raised by the infinite charge density of the ubiquitous negative-energy electrons: where is their electric field? Dirac proposed to reinterpret the charge density appearing in Maxwell’s equations as ‘the departure from the normal state of electrification of the world.’ Another problem has to do with the huge dissimilarity between the observed masses and interactions of the electrons and protons. Dirac hoped that Coulomb interactions between electrons would somehow account for these differences but Hermann Weyl [28] showed that the hole theory was in fact entirely symmetric between negative and positive charge. Finally, Dirac [22] predicted the existence of an electron–proton annihilation process in which a positive-energy electron meets a hole in the sea of negative-energy electrons and falls down into the unoccupied level, emitting a pair of gamma ray photons. By itself this would not have created difficulties for the hole theory; it was even hoped by some that this would provide an explanation, then lacking, of the energy source of the stars. However, it was soon pointed out [29] by Julius Robert Oppenheimer and Igor Tamm that electron–proton annihilation in atoms would take place at much too fast a rate to be consistent with the observed stability of ordinary matter. For these reasons, by 1931 Dirac had changed his mind, and decided that the holes would have to appear not as protons but as a new sort of positively charged particle, of the same mass as the electron [29a].

The second and third of these problems were eliminated by the discovery of the positron by Carl D. Anderson [30], who apparently did not know of this prediction by Dirac. On August 2, 1932, a peculiar cosmic ray track was observed in a Wilson cloud chamber subjected to a 15 kG magnetic field. The track was observed to curve in a direction that would be expected for a positively charged particle, and yet its range was at least ten times greater than the expected range of a proton! Both the range and the specific ionization of the track were consistent with the hypothesis that this was a new particle which differs from the electron only in the sign of its charge, as would be expected for one of Dirac’s holes. (This discovery had been made earlier by P.M.S. Blackett, but not immediately published

by him. Anderson quotes press reports of evidence for light positive particles in cosmic ray tracks, obtained by Blackett and Giuseppe Occhialini.) Thus it appeared that Dirac was wrong only in his original identification of the hole with the proton.

The discovery of the more-or-less predicted positron, together with the earlier successes of the Dirac equation in accounting for the magnetic moment of the electron and the fine structure of hydrogen, gave Dirac's theory a prestige that it has held for over six decades. However, although there seems little doubt that Dirac's theory will survive in some form in any future physical theory, there are serious reasons for being dissatisfied with its original rationale:

(i) Dirac's analysis of the problem of negative probabilities in Schrödinger's relativistic wave equation would seem to rule out the existence of any particle of zero spin. Yet even in the 1920s particles of zero spin were known—for instance, the hydrogen atom in its ground state, and the helium nucleus. Of course, it could be argued that hydrogen atoms and alpha particles are not elementary, and therefore do not need to be described by a relativistic wave equation, but it was not (and still is not) clear how the idea of elementarity is incorporated in the formalism of relativistic quantum mechanics. Today we know of a large number of spin zero particles— $\pi$  mesons,  $K$  mesons, and so on—that are no less elementary than the proton and neutron. We also know of spin one particles—the  $W^\pm$  and  $Z^0$ —which seem as elementary as the electron or any other particle. Further, apart from effects of the strong interactions, we would today calculate the fine structure of 'mesonic atoms,' consisting of a spinless negative  $\pi$  or  $K$  meson bound to an atomic nucleus, from the stationary solutions of the relativistic Klein–Gordon–Schrödinger equation! Thus, it is difficult to agree that there is anything fundamentally wrong with the relativistic equation for zero spin that *forced* the development of the Dirac equation—the problem simply is that the electron happens to have spin  $\hbar/2$ , not zero.

(ii) As far as we now know, for *every* kind of particle there is an 'antiparticle' with the same mass and opposite charge. (Some purely neutral particles, such as the photon, are their own antiparticles.) But how can we interpret the antiparticles of charged *bosons*, such as the  $\pi^\pm$  mesons or  $W^\pm$  particles, as holes in a sea of negative energy states? For particles quantized according to the rules of Bose–Einstein statistics, there is no exclusion principle, and hence nothing to keep positive-energy particles from falling down into the negative-energy states, occupied or not. And if the hole theory does not work for bosonic antiparticles, why should we believe it for fermions? I asked Dirac in 1972 how he then felt about this point; he told me that he did not regard bosons like the pion or  $W^\pm$  as 'important.' In a lecture [27a] a few years later, Dirac referred to the fact that for bosons 'we no longer have the picture of a vacuum with negative energy states filled up', and remarked that in this case 'the whole theory becomes more complicated.' The next section will show how the development of quantum field theory made the interpretation of antiparticles as holes unnecessary, even though unfortunately it lingers on in many textbooks. To quote Julian Schwinger [30a], 'The picture of an infinite sea of negative energy electrons is now best regarded as a historical curiosity, and forgotten.'

(iii) One of the great successes of the Dirac theory was its correct prediction of the mag-

netic moment of the electron. This was particularly striking, as the magnetic moment (1.1.8) is twice as large as would be expected for the orbital motion of a charged point particle with angular momentum  $\hbar/2$ ; this factor of 2 had remained mysterious until Dirac's theory. However, there is really nothing in Dirac's line of argument that leads unequivocally to this particular value for the magnetic moment. At the point where we brought electric and magnetic fields into the wave equation (1.1.23), we could just as well have added a 'Pauli term' [31]

$$\kappa c \beta [\gamma^\mu, \gamma^\nu] \psi F_{\mu\nu} \quad (1.1.32)$$

with arbitrary coefficient  $\kappa$ . (Here  $F_{\mu\nu}$  is the usual electromagnetic field strength tensor, with  $F^{12} = B_3$ ,  $F^{01} = E_1$ , etc.) This term could be obtained by first adding a term to the free-field equations proportional to  $[\gamma^\mu, \gamma^\nu](\partial^2/\partial x^\mu \partial x^\nu)\psi$ , which of course equals zero, and then making the substitutions (1.1.22) as before. A more modern approach would be simply to remark that the term (1.1.32) is consistent with all accepted invariance principles, including Lorentz invariance and gauge invariance, and so there is no reason why such a term should not be included in the field equations. (See Section 12.3.) This term would give an additional contribution proportional to  $\kappa$  to the magnetic moment of the electron, so apart from the possible demand for a purely formal simplicity, there was no reason to expect any particular value for the magnetic moment of the electron in Dirac's theory.

As we shall see in this book, these problems were all eventually to be solved (or at least clarified) through the development of quantum field theory.

## 1.2 The Birth of Quantum Field Theory

The photon is the only particle that was known as a field before it was detected as a particle. Thus it is natural that the formalism of quantum field theory should have been developed in the first instance in connection with radiation and only later applied to other particles and fields.

In 1926, in one of the central papers on matrix mechanics, Born, Heisenberg, and Jordan [32] applied their new methods to the free radiation field. For simplicity, they ignored the polarization of electromagnetic waves and worked in one space dimension, with coordinate  $x$  running from 0 to  $L$ ; the radiation field  $u(x, t)$  if constrained to vanish at these endpoints thus has the same behavior as the displacement of a string with ends fixed at  $x = 0$  and  $x = L$ . By analogy with either the case of a string or the full electromagnetic field, the Hamiltonian was taken to have the form

$$H = \frac{1}{2} \int_0^L \left\{ \left( \frac{\partial u}{\partial t} \right)^2 + c^2 \left( \frac{\partial u}{\partial x} \right)^2 \right\} dx. \quad (1.2.1)$$

In order to reduce this expression to a sum of squares, the field  $u$  was expressed as a sum of Fourier components with  $u = 0$  at both  $x = 0$  and  $x = L$ :

$$u(x, t) = \sum_{k=1}^{\infty} q_k(t) \sin \left( \frac{\omega_k x}{c} \right), \quad (1.2.2)$$

$$\omega_k = k\pi c/L, \quad (1.2.3)$$



so that

$$H = \frac{L}{4} \sum_{k=1}^{\infty} \{ \dot{q}_k^2(t) + \omega_k^2 q_k^2(t) \}. \quad (1.2.4)$$

Thus the string or field behaves like sum of independent harmonic oscillators with angular frequencies  $\omega_k$ , as had been anticipated 20 years earlier by Paul Ehrenfest [32a].

In particular, the ‘momentum’  $p_k(t)$  canonically conjugate to  $q_k(t)$  is determined, as in particle mechanics, by the condition that if  $H$  is expressed as a function of the  $ps$  and  $qs$ , then

$$\dot{q}_k(t) = \frac{\partial}{\partial p_k(t)} H(p(t), q(t)).$$

This yields a ‘momentum’

$$p_k(t) = \frac{L}{2} \dot{q}_k(t), \quad (1.2.5)$$

so the canonical commutation relations may be written

$$[\dot{q}_k(t), q_j(t)] = \frac{2}{L} [p_k(t), q_j(t)] = \frac{-2i\hbar}{L} \delta_{kj}, \quad (1.2.6)$$

$$[q_k(t), q_j(t)] = 0. \quad (1.2.7)$$

Also, the time-dependence of  $q_k(t)$  is governed by the Hamiltonian equation of motion

$$\ddot{q}_k(t) = \frac{2}{L} \dot{p}_k(t) = -\frac{2}{L} \frac{\partial H}{\partial q_k(t)} = -\omega_k^2 q_k(t). \quad (1.2.8)$$

The form of the matrices defined by Eqs. (1.2.6)–(1.2.8) was already known to Born, Heisenberg, and Jordan through previous work on the harmonic oscillator. The  $q$ -matrix is given by

$$q_k(t) = \sqrt{\frac{\hbar}{L\omega_k}} \left[ a_k \exp(-i\omega_k t) + a_k^\dagger \exp(+i\omega_k t) \right] \quad (1.2.9)$$

with  $a_k$  a time-independent matrix and  $a_k^\dagger$  its Hermitian adjoint, satisfying the commutation relations

$$[a_k, a_j^\dagger] = \delta_{kj}, \quad (1.2.10)$$

$$[a_k, a_j] = 0. \quad (1.2.11)$$

The rows and columns of these matrices are labelled with a set of positive integers  $n_1, n_2, \dots$ , one for each normal mode. The matrix elements are

$$(a_k)_{n'_1, n'_2, \dots; n_1, n_2, \dots} = \sqrt{n_k} \delta_{n'_k, n_k-1} \prod_{j \neq k} \delta_{n'_j, n_j}, \quad (1.2.12)$$

$$(a_k^\dagger)_{n'_1, n'_2, \dots; n_1, n_2, \dots} = \sqrt{n_k + 1} \delta_{n'_k, n_k+1} \prod_{j \neq k} \delta_{n'_j, n_j}. \quad (1.2.13)$$

For a single normal mode, these matrices may be written explicitly as

$$a = \begin{bmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & 0 & 0 & \sqrt{3} & \cdots \\ 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \quad a^\dagger = \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots \\ \sqrt{1} & 0 & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

It is straightforward to check that (1.2.12) and (1.2.13) do satisfy the commutation relations (1.2.10) and (1.2.11).

The physical interpretation of a column vector with integer components  $n_1, n_2, \dots$  is that it represents a state with  $n_k$  quanta in each normal mode  $k$ . The matrix  $a_k$  or  $a_k^\dagger$  acting on such a column vector will respectively lower or raise  $n_k$  by one unit, leaving all  $n_\ell$  with  $\ell \neq k$  unchanged; they may therefore be interpreted as operators which annihilate or create one quantum in the  $k$ th normal mode. In particular, the vector with all  $n_k$  equal to zero represents the vacuum; it is annihilated by any  $a_k$ .

This interpretation is further borne out by inspection of the Hamiltonian. Using (1.2.9) and (1.2.10) in (1.2.4) gives

$$H = \sum_k \hbar\omega_k \left( a_k^\dagger a_k + \frac{1}{2} \right). \quad (1.2.14)$$

The Hamiltonian is then diagonal in the  $n$ -representation

$$(H)_{n'_1, n'_2, \dots, n_1, n_2, \dots} = \sum_k \hbar\omega_k \left( n_k + \frac{1}{2} \right) \prod_j \delta_{n'_j, n_j}. \quad (1.2.15)$$

We see that the energy of the state is just the sum of energies  $\hbar\omega_k$  for each quantum present in the state, plus an infinite zero-point energy  $E_0 = \frac{1}{2} \sum_k \hbar\omega_k$ . Applied to the radiation field, this formalism justified the Bose method of counting radiation states according to the numbers  $n_k$  of quanta in each normal mode.

Born, Heisenberg, and Jordan used this formalism to derive an expression for the r.m.s. energy fluctuations in black-body radiation. (For this purpose they actually only used the commutation relations (1.2.6)–(1.2.7).) However, this approach was soon applied to a more urgent problem, the calculation of the rates for spontaneous emission of radiation.

In order to appreciate the difficulties here, it is necessary to go back in time a bit. In one of the first papers on matrix mechanics, Born and Jordan [33] had assumed in effect that an atom, in dropping from a state  $\beta$  to a lower state  $\alpha$ , would emit radiation just like a classical charged oscillator with displacement

$$\mathbf{r}(t) = \mathbf{r}_{\beta\alpha} \exp(-2\pi i\nu t) + \mathbf{r}_{\beta\alpha}^* \exp(2\pi i\nu t), \quad (1.2.16)$$

where

$$h\nu = E_\beta - E_\alpha \quad (1.2.17)$$

and  $\mathbf{r}_{\beta\alpha}$  is the  $\beta, \alpha$  element of the matrix associated with the electron position. The energy  $E$  of such an oscillator is

$$E = \frac{1}{2}m \{ \dot{\mathbf{r}}^2 + (2\pi\nu)^2 \mathbf{r}^2 \} = 8\pi^2 m \nu^2 |\mathbf{r}_{\beta\alpha}|^2. \quad (1.2.18)$$

A straightforward classical calculation then gives the radiated power, and dividing by the energy  $h\nu$  per photon gives the rate of photon emission

$$A(\beta \rightarrow \alpha) = \frac{16\pi^3 e^2 \nu^3}{3hc^3} |\mathbf{r}_{\beta\alpha}|^2. \quad (1.2.19)$$

However, it was not at all clear why the formulas for emission of radiation by a classical dipole should be taken over in this manner in dealing with spontaneous emission.

A little later a more convincing though even less direct derivation was given by Dirac [34]. By considering the behavior of quantized atomic states in an oscillating classical electromagnetic field with energy density per frequency interval  $u$  at frequency (1.2.17), he was able to derive formulas for the rates  $uB(\alpha \rightarrow \beta)$  and  $uB(\beta \rightarrow \alpha)$  for absorption or induced emission:

$$B(\alpha \rightarrow \beta) = B(\beta \rightarrow \alpha) \simeq \frac{2\pi^2 e^2}{3h^2} |\mathbf{r}_{\beta\alpha}|^2. \quad (1.2.20)$$

(Note that the expression on the right is symmetric between states  $\alpha$  and  $\beta$ , because  $\mathbf{r}_{\alpha\beta}$  is just  $\mathbf{r}_{\beta\alpha}^*$ .) Einstein [34a] had already shown in 1917 that the possibility of thermal equilibrium between atoms and black-body radiation imposes a relation between the rate  $A(\beta \rightarrow \alpha)$  of spontaneous emission and the rates  $uB$  for induced emission or absorption:

$$A(\beta \rightarrow \alpha) = \left( \frac{8\pi h \nu^3}{c^3} \right) B(\beta \rightarrow \alpha). \quad (1.2.21)$$

Using (1.2.20) in this relation immediately yields the Born–Jordan result (1.2.19) for the rate of spontaneous emission. Nevertheless, it still seemed unsatisfactory that thermodynamic arguments should be needed to derive formulas for processes involving a single atom.

Finally, in 1927 Dirac [35] was able to give a thoroughly quantum mechanical treatment of spontaneous emission. The vector potential  $\mathbf{A}(\mathbf{x}, t)$  was expanded in normal modes, as in Eq. (1.2.2), and the coefficients were shown to satisfy commutation relations like (1.2.6). In consequence, each state of the free radiation field was specified by a set of integers  $n_k$ , one for each normal mode, and the matrix elements of the electromagnetic interaction  $e\dot{\mathbf{r}} \cdot \mathbf{A}$  took the form of a sum over normal modes, with matrix coefficients proportional to the matrices  $a_k$  and  $a_k^\dagger$  defined in Eqs. (1.2.10)–(1.2.13). The crucial result here is the factor  $\sqrt{n_k + 1}$  in Eq. (1.2.13); the probability for a transition in which the number of photons in a normal mode  $k$  rises from  $n_k$  to  $n_k + 1$  is proportional to the square of this factor, or  $n_k + 1$ . But in a radiation field with  $n_k$  photons in a normal mode  $k$ , the energy density  $u$  per frequency interval is

$$u(\nu_k) = \left( \frac{8\pi \nu_k^2}{c^3} \right) n_k \cdot h\nu_k,$$

so the rate for emission of radiation in normal mode  $k$  is proportional to

$$n_k + 1 = \frac{c^3 u(\nu_k)}{8\pi h \nu_k^3} + 1.$$

The first term is interpreted as the contribution of induced emission, and the second term as the contribution of spontaneous emission. Hence, without any appeal to thermodynamics, Dirac could conclude that the ratio of the rates  $uB$  for induced emission and  $A$  for spontaneous emission is given by the Einstein relation, Eq. (1.2.21). Using his earlier result (1.2.20) for  $B$ , Dirac was thus able to rederive the Born–Jordan formula [33] (1.2.19) for spontaneous emission rate  $A$ . A little later, similar methods were used by Dirac to give a quantum mechanical treatment of the scattering of radiation and the lifetime of excited atomic states [36], and by Victor Weisskopf and Eugene Wigner to make a detailed study of spectral line shapes [36a]. Dirac in his work was separating the electromagnetic potential into a radiation field  $\mathbf{A}$  and a static Coulomb potential  $A^0$ , in a manner which did not preserve the manifest Lorentz and gauge invariance of classical electrodynamics. These matters were put on a firmer foundation a little later by Enrico Fermi [36b]. Many physicists in the 1930s learned their quantum electrodynamics from Fermi’s 1932 review.

The use of canonical commutation relations for  $q$  and  $p$  or  $a$  and  $a^\dagger$  also raised a question as to the Lorentz invariance of the quantized theory. Jordan and Pauli [37] in 1928 were able to show that the commutators of fields at different spacetime points were in fact Lorentz-invariant. (These commutators are calculated in Chapter 5.) Somewhat later, Bohr and Leon Rosenfeld [38] used a number of ingenious thought experiments to show that these commutation relations express limitations on our ability to measure fields at spacetime points separated by time-like intervals.

It was not long after the successful quantization of the electromagnetic field that these techniques were applied to other fields. At first this was regarded as a ‘second quantization’; the fields to be quantized were the wave functions used in one-particle quantum mechanics, such as the Dirac wave function of the electron. The first step in this direction seems to have been taken in 1927 by Jordan [39]. In 1928 an essential element was supplied by Jordan and Wigner [40]. They recognized that the Pauli exclusion principle prevents the occupation number  $n_k$  of electrons in any normal mode  $k$  (counting spin as well as position variables) from taking any values other than 0 or 1. The electron field therefore cannot be expanded as a superposition of operators satisfying the commutation relations (1.2.10), (1.2.11), because these relations require  $n_k$  to take all integer values from 0 to  $\infty$ . Instead, they proposed that the electron field should be expanded in a sum of operators  $a_k, a_k^\dagger$  satisfying the *anticommutation* relations

$$a_k a_j^\dagger + a_j^\dagger a_k = \delta_{jk}, \tag{1.2.22}$$

$$a_k a_j + a_j a_k = 0. \tag{1.2.23}$$

The relations can be satisfied by matrices labelled by a set of integers  $n_1, n_2, \dots$ , one for

each normal mode, each integer taking just the values zero and one:

$$(a_k)_{n'_1, n'_2, \dots, n_1, n_2, \dots} = \begin{cases} 1 & n'_k = 0, \ n_k = 1, \ n'_j = n_j \text{ for } j \neq k, \\ 0 & \text{otherwise,} \end{cases} \quad (1.2.24)$$

$$(a_k^\dagger)_{n'_1, n'_2, \dots, n_1, n_2, \dots} = \begin{cases} 1 & n'_k = 1, \ n_k = 0, \ n'_j = n_j \text{ for } j \neq k, \\ 0 & \text{otherwise.} \end{cases} \quad (1.2.25)$$

For instance, for a single normal mode we have just two rows and two columns, corresponding to the values unity and zero of  $n'$  and  $n$ ; the  $a$  and  $a^\dagger$  matrices take the form

$$a = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad a^\dagger = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The reader may check that (1.2.24) and (1.2.25) do satisfy the anticommutation relations (1.2.22) and (1.2.23).

The interpretation of a column vector characterized by integers  $n_1, n_2, \dots$  is that it represents a state with  $n_k$  quanta in each normal mode  $k$ , just as for bosons. The difference is, of course, that since each  $n_k$  takes only the values 0 and 1, there can be at most one quantum in each normal mode, as required by the Pauli exclusion principle. Again,  $a_k$  destroys a quantum in normal mode  $k$  if there is one there already, and otherwise gives zero; also,  $a_k^\dagger$  creates a quantum in normal mode  $k$  unless there is one there already, in which case it gives zero. Much later it was shown by Fierz and Pauli [40a] that the choice between commutation and anticommutation relations is dictated solely by the particle's spin: commutators must be used for particles with integer spin like the photon, and anticommutators for particles with half-integer spin like the electron. (This will be shown in a different way in Chapter 5.)

The theory of general quantum fields was first laid out in 1929, in a pair of comprehensive articles by Heisenberg and Pauli [41]. The starting point of their work was the application of the canonical formalism to the fields themselves, rather than to the coefficients of the normal modes appearing in the fields. Heisenberg and Pauli took the Lagrangian  $L$  as the space-integral of a local function of fields and spacetime derivatives of fields; the field equations were then determined from the principle that the action  $\int L dt$  should be stationary when the fields are varied; and the commutation relations were determined from the assumption that the variational derivative of the Lagrangian with respect to any field's time-derivative behaves like a 'momentum' conjugate to that field (except that commutation relations become anticommutation relations for fermion fields). They also went on to apply this general formalism to the electromagnetic and Dirac fields, and explored the various invariance and conservation laws, including the conservation of charge, momentum, and energy, and Lorentz and gauge invariance.

The Heisenberg–Pauli formalism is essentially the same as that described in our Chapter 7, and so for the present we can limit ourselves to a single example which will turn out to be useful later in this section. For a free complex scalar field  $\phi(x)$  the Lagrangian is taken

as

$$L = \int d^3x \left[ \dot{\phi}^\dagger \dot{\phi} - c^2 (\nabla \phi)^\dagger \cdot (\nabla \phi) - \left( \frac{mc^2}{\hbar} \right)^2 \phi^\dagger \phi \right]. \quad (1.2.26)$$

If we subject  $\phi(x)$  to an infinitesimal variation  $\delta\phi(x)$ , the Lagrangian is changed by the amount

$$\begin{aligned} \delta L = \int d^3x \left[ \dot{\phi}^\dagger \delta \dot{\phi} + \dot{\phi} \delta \dot{\phi}^\dagger - c^2 \nabla \phi^\dagger \cdot \nabla \delta \phi - c^2 \nabla \phi \cdot \nabla \delta \phi^\dagger \right. \\ \left. - \left( \frac{mc^2}{\hbar} \right)^2 \phi^\dagger \delta \phi - \left( \frac{mc^2}{\hbar} \right)^2 \phi \delta \phi^\dagger \right]. \end{aligned} \quad (1.2.27)$$

It is assumed in using the principle of stationary action that the variation in the fields should vanish on the boundaries of the spacetime region of integration. Thus, in computing the change in the action  $\int L dt$ , we can immediately integrate by parts, and write

$$\delta \int L dt = c^2 \int d^4x \left[ \delta \phi^\dagger \left( \square - \left( \frac{mc}{\hbar} \right)^2 \right) \phi + \delta \phi \left( \square - \left( \frac{mc}{\hbar} \right)^2 \right) \phi^\dagger \right].$$

But this must vanish for any  $\delta\phi$  and  $\delta\phi^\dagger$ , so  $\phi$  must satisfy the familiar relativistic wave equation

$$\left[ \square - \left( \frac{mc}{\hbar} \right)^2 \right] \phi = 0 \quad (1.2.28)$$

and its adjoint. The ‘momenta’ canonically conjugate to the fields  $\phi$  and  $\phi^\dagger$  are given by the variational derivatives of  $L$  with respect to  $\dot{\phi}$  and  $\dot{\phi}^\dagger$ , which we can read off from (1.2.27) as

$$\pi \equiv \frac{\delta L}{\delta \dot{\phi}} = \dot{\phi}^\dagger, \quad (1.2.29)$$

$$\pi^\dagger \equiv \frac{\delta L}{\delta \dot{\phi}^\dagger} = \dot{\phi}. \quad (1.2.30)$$

These field variables satisfy the usual canonical commutation relations, with a delta function in place of a Kronecker delta

$$[\pi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = [\pi^\dagger(\mathbf{x}, t), \phi^\dagger(\mathbf{y}, t)] = -i\hbar \delta^3(\mathbf{x} - \mathbf{y}), \quad (1.2.31)$$

$$[\pi(\mathbf{x}, t), \phi^\dagger(\mathbf{y}, t)] = [\pi^\dagger(\mathbf{x}, t), \phi(\mathbf{y}, t)] = 0, \quad (1.2.32)$$

$$[\pi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = [\pi^\dagger(\mathbf{x}, t), \pi^\dagger(\mathbf{y}, t)] = [\pi(\mathbf{x}, t), \pi^\dagger(\mathbf{y}, t)] = 0, \quad (1.2.33)$$

$$[\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = [\phi^\dagger(\mathbf{x}, t), \phi^\dagger(\mathbf{y}, t)] = [\phi(\mathbf{x}, t), \phi^\dagger(\mathbf{y}, t)] = 0. \quad (1.2.34)$$

The Hamiltonian here is given (just as in particle mechanics) by the ‘sum’ of all canonical momenta times the time-derivatives of the corresponding fields, minus the Lagrangian:

$$H = \int d^3x \left[ \pi \dot{\phi} + \pi^\dagger \dot{\phi}^\dagger \right] - L \quad (1.2.35)$$

or, using (1.2.26), (1.2.29), and (1.2.30):

$$H = \int d^3x \left[ \pi^\dagger \pi + c^2 (\nabla \phi)^\dagger \cdot (\nabla \phi) + \left( \frac{m^2 c^4}{\hbar^2} \right) \phi^\dagger \phi \right]. \quad (1.2.36)$$

After the papers by Heisenberg and Pauli one element was still missing before quantum field theory could reach its final pre-war form: a solution to the problem of the negative-energy states. We saw in the last section that in 1930, at just about the time of the Heisenberg–Pauli papers, Dirac had proposed that the negative-energy states of the electron were all filled, but with only the holes in the negative-energy sea observable, rather than the negative-energy electrons themselves. After Dirac’s idea was seemingly confirmed by the discovery of the positron in 1932, his ‘hole theory’ was used to calculate a number of processes to the lowest order of perturbation theory, including electron–positron pair production and scattering.

At the same time, a great deal of work was put into the development of a formalism whose Lorentz invariance would be explicit. The most influential effort was the ‘many-time’ formalism of Dirac, Vladimir Fock, and Boris Podolsky [42], in which the state vector was represented by a wave function depending on the spacetime and spin coordinates of all electrons, positive-energy and negative-energy. In this formalism, the total number of electrons of either positive or negative energy is conserved; for instance, production of an electron–positron pair is described as the excitation of a negative-energy electron to a positive-energy state, and the annihilation of an electron and positron is described as the corresponding deexcitation. This many-time formalism had the advantage of manifest Lorentz invariance, but it had a number of disadvantages; In particular, there was a profound difference between the treatment of the photon, described in terms of a quantized electromagnetic field, and that of the electron and positron. Not all physicists felt this to be a disadvantage; the electron field unlike the electromagnetic field did not have a classical limit, so there were doubts about its physical significance. Also, Dirac [42a] conceived of fields as the means by which we observe particles, so that he did not expect particles and fields to be described in the same terms. Though I do not know whether it bothered anyone at the time, there was a more practical disadvantage of the many-time formalism: it would have been difficult to use it to describe a process like nuclear beta decay, in which an electron and antineutrino are created without an accompanying positron or neutrino. The successful calculation by Fermi [43] of the electron energy distribution in beta decay deserves to be counted as one of the early triumphs of quantum field theory.

The essential idea that was needed to demonstrate the equivalence of the Dirac hole theory with a quantum field theory of the electron was provided by Fock [43a] and by Wendell Furry and Oppenheimer [44] in 1933–4. To appreciate this idea from a more modern standpoint, suppose we try to construct an electron field in analogy with the electromagnetic field or the Born–Heisenberg–Jordan field (1.2.2). Since electrons carry a charge, we would not like to mix annihilation and creation operators, so we might try to write the field as

$$\psi(x) = \sum_k u_k(\mathbf{x}) e^{-i\omega_k t} a_k, \quad (1.2.37)$$

where  $u_k(\mathbf{x})e^{-i\omega_k t}$  are a complete set of orthonormal plane-wave solutions of the Dirac equation (1.1.13) (with  $k$  now labelling the three-momentum, spin, and sign of the energy):

$$\mathcal{H}u_k = \hbar\omega_k u_k, \quad (1.2.38)$$

$$\mathcal{H} \equiv -i\hbar c \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + \beta mc^2, \quad (1.2.39)$$

$$\int u_k^\dagger u_\ell d^3x = \delta_{k\ell}, \quad (1.2.40)$$

and  $a_k$  are the corresponding annihilation operators, satisfying the Jordan–Wigner anti-commutation relations (1.2.22)–(1.2.23). According to the ideas of ‘second quantization’ or the canonical quantization procedure of Heisenberg and Pauli [41], the Hamiltonian is formed by calculating the ‘expectation value’ of  $\mathcal{H}$  with a ‘wave function’ replaced by the quantized field (1.2.37)

$$H = \int d^3x \psi^\dagger \mathcal{H} \psi = \sum_k \hbar\omega_k a_k^\dagger a_k. \quad (1.2.41)$$

The trouble is, of course, that this is not a positive operator—half the  $\omega_k$  are negative while the operators  $a_k^\dagger a_k$  take only the positive eigenvalues 1 and 0. (See Eqs. (1.2.24) and (1.2.25).) In order to cure this disease, Furry and Oppenheimer picked up Dirac’s idea [42] that the positron is the absence of a negative-energy electron; the anticommutation relations are symmetric between creation and annihilation operators, so they defined the positron creation and annihilation operators as the corresponding annihilation and creation operators for negative-energy electrons

$$b_k^\dagger \equiv a_k, \quad b_k \equiv a_k^\dagger \quad (\text{for } \omega_k < 0) \quad (1.2.42)$$

where the label  $k$  on  $b$  denotes a positive-energy positron mode with momenta and spin opposite to those of the electron mode  $k$ . The Dirac field (1.2.37) may then be written

$$\psi(x) = \sum_k^{(+)} a_k u_k(x) + \sum_k^{(-)} b_k^\dagger u_k(x), \quad (1.2.43)$$

where  $(+)$  and  $(-)$  indicate sums over normal modes  $k$  with  $\omega_k > 0$  and  $\omega_k < 0$ , respectively, and  $u_k(x) \equiv u_k(\mathbf{x})e^{-i\omega_k t}$ . Similarly, using the anticommutation relations for the  $b$ s, we can rewrite the energy operator (1.2.41) as

$$H = \sum_k^{(+)} \hbar\omega_k a_k^\dagger a_k + \sum_k^{(-)} \hbar|\omega_k| b_k^\dagger b_k + E_0, \quad (1.2.44)$$

where  $E_0$  is the infinite c-number

$$E_0 = - \sum_k^{(-)} \hbar|\omega_k|. \quad (1.2.45)$$

In order for this redefinition to be more than a mere formality, it is necessary also to specify that the physical vacuum is a state  $|\Omega\rangle$  containing no positive-energy electrons or positrons:

$$a_k |\Omega\rangle = 0 \quad (\omega_k > 0), \quad (1.2.46)$$

$$b_k |\Omega\rangle = 0 \quad (\omega_k < 0). \quad (1.2.47)$$



Hence (1.2.44) gives the energy of the vacuum as just  $E_0$ . If we measure all energies relative to the vacuum energy  $E_0$ , then the physical energy operator is  $H - E_0$ ; and Eq. (1.2.44) shows that this is a positive operator.

The problem of negative-energy states for a charged spin zero particle was also resolved in 1934, by Pauli and Weisskopf [45], in a paper written in part to challenge Dirac's picture of filled negative-energy states. Here the creation and annihilation operators satisfy commutation rather than anticommutation relations, so it is not possible to interchange the roles of these operators freely, as was the case for fermions. Instead we must return to the Heisenberg–Pauli canonical formalism [41] to decide which coefficients of the various normal modes are creation or annihilation operators.

Pauli and Weisskopf expanded the free charged scalar field in plane waves in a cube of spatial volume  $V = L^3$ :

$$\phi(\mathbf{x}, t) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} q(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}} \quad (1.2.48)$$

with the wave numbers restricted by the periodicity condition, that the quantities  $k_j L/2\pi$  for  $j = 1, 2, 3$  should be a set of three positive or negative integers. Similarly the canonically conjugate variable (1.2.29) was expanded as

$$\pi(\mathbf{x}, t) \equiv \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} p(\mathbf{k}, t) e^{-i\mathbf{k} \cdot \mathbf{x}}. \quad (1.2.49)$$

The minus sign is put into the exponent here so that (1.2.29) now becomes:

$$p(\mathbf{k}, t) = \dot{q}^\dagger(\mathbf{k}, t). \quad (1.2.50)$$

The Fourier inversion formula gives

$$q(\mathbf{k}, t) = \frac{1}{\sqrt{V}} \int d^3x \phi(\mathbf{x}, t) e^{-i\mathbf{k} \cdot \mathbf{x}}, \quad (1.2.51)$$

$$p(\mathbf{k}, t) = \frac{1}{\sqrt{V}} \int d^3x \pi(\mathbf{x}, t) e^{+i\mathbf{k} \cdot \mathbf{x}}, \quad (1.2.52)$$

and therefore the canonical commutation relations (1.2.31)–(1.2.34) yield for the  $qs$  and  $ps$ :

$$[p(\mathbf{k}, t), q(\mathbf{l}, t)] = \frac{-i\hbar}{V} \int d^3x e^{i\mathbf{k} \cdot \mathbf{x}} e^{-i\mathbf{l} \cdot \mathbf{x}} = -i\hbar \delta_{\mathbf{k}\mathbf{l}} \quad (1.2.53)$$

$$\begin{aligned} [p(\mathbf{k}, t), q^\dagger(\mathbf{l}, t)] &= [p(\mathbf{k}, t), p(\mathbf{l}, t)] = [p(\mathbf{k}, t), p^\dagger(\mathbf{l}, t)] \\ &= [q(\mathbf{k}, t), q(\mathbf{l}, t)] = [q(\mathbf{k}, t), q^\dagger(\mathbf{l}, t)] = 0 \end{aligned} \quad (1.2.54)$$

together with other relations that may be derived from these by taking their Hermitian adjoints. By inserting (1.2.48) and (1.2.49) in the formula (1.2.36) for the Hamiltonian, we can also write this operator in terms of  $ps$  and  $qs$ :

$$H = \sum_{\mathbf{k}} \left[ p^\dagger(\mathbf{k}, t) p(\mathbf{k}, t) + \omega_k^2 q^\dagger(\mathbf{k}, t) q(\mathbf{k}, t) \right], \quad (1.2.55)$$

where

$$\omega_k^2 = c^2 \mathbf{k}^2 + \left( \frac{mc^2}{\hbar} \right)^2. \quad (1.2.56)$$

The time-derivatives of the  $p$ s are then given by the Hamiltonian equation

$$\dot{p}(\mathbf{k}, t) = -\frac{\partial H}{\partial q(\mathbf{k}, t)} = -\omega_k^2 q^\dagger(\mathbf{k}, t) \quad (1.2.57)$$

(and its adjoint), a result which in the light of Eq. (1.2.50) is just equivalent to the Klein–Gordon–Schrödinger wave equation (1.2.28).

We see that, just as in the case of the 1926 model of Born, Heisenberg, and Jordan [4], the free field behaves like an infinite number of coupled harmonic oscillators. Pauli and Weisskopf could construct  $p$  and  $q$  operators which satisfy the commutation relations (1.2.53)–(1.2.54) and the ‘equations of motion’ (1.2.50) and (1.2.57), by introducing annihilation and creation operators  $a, b, a^\dagger, b^\dagger$  of *two* different kinds, corresponding to particles and antiparticles:

$$q(\mathbf{k}, t) = i\sqrt{\frac{\hbar}{2\omega_k}} \left[ a_{\mathbf{k}} \exp(-i\omega_k t) - b_{\mathbf{k}}^\dagger \exp(i\omega_k t) \right], \quad (1.2.58)$$

$$p(\mathbf{k}, t) = \sqrt{\frac{\hbar\omega_k}{2}} \left[ b_{\mathbf{k}} \exp(-i\omega_k t) + a_{\mathbf{k}}^\dagger \exp(+i\omega_k t) \right] \quad (1.2.59)$$

where

$$[a_{\mathbf{k}}, a_{\mathbf{l}}^\dagger] = [b_{\mathbf{k}}, b_{\mathbf{l}}^\dagger] = \delta_{\mathbf{k}\mathbf{l}}, \quad (1.2.60)$$

$$[a_{\mathbf{k}}, a_{\mathbf{l}}] = [b_{\mathbf{k}}, b_{\mathbf{l}}] = 0, \quad (1.2.61)$$

$$\begin{aligned} [a_{\mathbf{k}}, b_{\mathbf{l}}] &= [a_{\mathbf{k}}, b_{\mathbf{l}}^\dagger] = [a_{\mathbf{k}}^\dagger, b_{\mathbf{l}}] \\ &= [a_{\mathbf{k}}^\dagger, b_{\mathbf{l}}^\dagger] = 0. \end{aligned} \quad (1.2.62)$$

It is straightforward to check that these operators do satisfy the desired relations (1.2.53), (1.2.54), (1.2.50), and (1.2.57). The field (1.2.48) may be written

$$\begin{aligned} \phi(\mathbf{x}, t) = \frac{i}{\sqrt{V}} \sum_{\mathbf{k}} \sqrt{\frac{\hbar}{2\omega_k}} & \left[ a_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega_k t) \right. \\ & \left. - b_{-\mathbf{k}}^\dagger \exp(-i\mathbf{k} \cdot \mathbf{x} + i\omega_k t) \right] \end{aligned} \quad (1.2.63)$$

and the Hamiltonian (1.2.55) takes the form

$$H = \sum_{\mathbf{k}} \frac{1}{2} \hbar \omega_k \left[ b_{\mathbf{k}}^\dagger b_{\mathbf{k}} + b_{\mathbf{k}} b_{\mathbf{k}}^\dagger + a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + a_{\mathbf{k}} a_{\mathbf{k}}^\dagger \right]$$

or, using (1.2.60)–(1.2.62)

$$H = \sum_{\mathbf{k}} \hbar \omega_k \left[ b_{\mathbf{k}}^\dagger b_{\mathbf{k}} + a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \right] + E_0, \quad (1.2.64)$$

where  $E_0$  is the infinite c-number

$$E_0 = \sum_{\mathbf{k}} \hbar \omega_{\mathbf{k}}. \quad (1.2.65)$$

The existence of two different kinds of operators  $a$  and  $b$ , which appear in precisely the same way in the Hamiltonian, shows that this is a theory with two kinds of particles with the same mass. As emphasized by Pauli and Weisskopf, these two varieties can be identified as particles and the corresponding antiparticles, and if charged have opposite charges. Thus, as we stressed above, bosons of spin zero as well as fermions of spin 1/2 can have distinct antiparticles, which for bosons cannot be identified as holes in a sea of negative energy particles.

We now can tell whether  $a$  and  $b$  or  $a^\dagger$  and  $b^\dagger$  are the annihilation operators by taking the expectation values of commutation relations in the vacuum state  $|\Omega\rangle$ . For instance, if  $a_k^\dagger$  were an annihilation operator it would give zero when applied to the vacuum state, so the vacuum expectation value of (1.2.60) would give

$$-||a_{\mathbf{k}}|\Omega\rangle||^2 = \langle\Omega|[a_{\mathbf{k}}, a_{\mathbf{k}}^\dagger]|\Omega\rangle = +1 \quad (1.2.66)$$

in conflict with the requirement that the left-hand side must be negative-definite. In this way we can conclude that it is  $a_k$  and  $b_k$  that are the annihilation operators, and therefore

$$a_{\mathbf{k}}|\Omega\rangle = b_{\mathbf{k}}|\Omega\rangle = 0. \quad (1.2.67)$$

This is consistent with all commutation relations. Thus, the canonical formalism forces the coefficient of the  $e^{+i\omega_{\mathbf{k}}t}$  in the field (1.2.58) to be a creation operator, as it also is in the Furry–Oppenheimer formalism [44] for spin 1/2.

Equations (1.2.64) and (1.2.67) now tell us that  $E_0$  is the energy of the vacuum state. If we measure all energies relative to  $E_0$ , then the physical energy operator is  $H - E_0$ , and (1.2.64) shows that this again is positive.

What about the problem that served Dirac as a starting point, the problem of negative probabilities? As Dirac had recognized, the only probability density  $\rho$ , which can be formed from solutions of the Klein–Gordon–Schrödinger free scalar wave equation (1.2.28), and which satisfies a conservation law of the form (1.1.10), must be proportional to the quantity

$$\rho = 2 \operatorname{Im} \left[ \phi^\dagger \frac{\partial \phi}{\partial t} \right] \quad (1.2.68)$$

and therefore is not necessarily a positive quantity. Similarly, in the ‘second-quantized’ theory, where  $\phi$  is given by Eq. (1.2.63),  $\rho$  is not a positive operator. Since  $\phi^\dagger(x)$  does not commute with  $\phi(x)$  here, we can write (1.2.68) in various forms, which differ by infinite c-numbers; it proves convenient to write it as

$$\rho = \frac{i}{\hbar} \left[ \frac{\partial \phi}{\partial t} \phi^\dagger - \frac{\partial \phi^\dagger}{\partial t} \phi \right]. \quad (1.2.69)$$

The space-integral of this operator is then easily calculated to be

$$N \equiv \int \rho d^3x = \sum_{\mathbf{k}} \left( a_{\mathbf{k}}^\dagger a_{\mathbf{k}} - b_{\mathbf{k}}^\dagger b_{\mathbf{k}} \right) \quad (1.2.70)$$

and clearly has eigenvalues of either sign.

However, in a sense this problem appears in quantum field theory for spin 1/2 as well as spin zero. The density operator  $\psi^\dagger\psi$  of Dirac is indeed a positive operator, but in order to construct a physical density we ought to subtract the contribution of the filled electron states. In particular, using the plane-wave decomposition (1.2.43), we may write the total number operator as

$$N \equiv \int d^3x \psi^\dagger\psi = \sum_{\mathbf{k}}^{(+)} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \sum_{\mathbf{k}}^{(-)} b_{\mathbf{k}} b_{\mathbf{k}}^\dagger.$$

The anticommutation relations for the  $b$ s allow us to rewrite this as

$$N - N_0 = \sum_{\mathbf{k}}^{(+)} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} - \sum_{\mathbf{k}}^{(-)} b_{\mathbf{k}}^\dagger b_{\mathbf{k}}, \quad (1.2.71)$$

where  $N_0$  is the infinite constant

$$N_0 = \sum_{\mathbf{k}}^{(-)} 1. \quad (1.2.72)$$

According to Eqs. (1.2.46) and (1.2.47),  $N_0$  is the number of particles in the vacuum, so Furry and Oppenheimer reasoned that the physical number operator is  $N - N_0$ , and this now has both negative and positive eigenvalues, just as for spin zero.

The solution to this problem provided by quantum field theory is that neither the  $\psi$  of Furry and Oppenheimer nor the  $\phi$  of Pauli and Weisskopf are probability amplitudes, which would have to define conserved positive probability densities. Instead, the physical Hilbert space is spanned by states defined as containing definite numbers of particles and/or antiparticles in each mode. If  $|\Phi_n\rangle$  are a complete orthonormal set of such states, then a measurement of particle numbers in an arbitrary state  $|\Psi\rangle$  will yield a probability for finding the system in state  $|\Phi_n\rangle$ , given by

$$P_n = |\langle\Phi_n|\Psi\rangle|^2, \quad (1.2.73)$$

where  $\langle\Phi_n|\Psi\rangle$  is the usual Hilbert space scalar product. Hence, no question as to the possibility of negative probabilities will arise for any spin. The wave fields  $\phi$ ,  $\psi$ , etc., are not probability amplitudes at all, but *operators* which create or destroy particles in the various normal modes. It would be a good thing if the misleading expression ‘second quantization’ were permanently retired.

In particular, the operators  $N$  and  $N - N_0$  of Eqs. (1.2.70) and (1.2.71) are not to be interpreted as total probabilities, but as number operators: specifically, the number of particles *minus* the number of antiparticles. For charged particles, the conservation of charge forces the charge operators to be proportional to these number operators, so the minus signs in (1.2.70) and (1.2.71) allow us immediately to conclude that particles and antiparticles have opposite charge. In this field-theoretic formalism, interactions contribute terms to the Hamiltonian which are of third, fourth, or higher order in field variables, and the rates of various processes are given by using these interaction operators in a time-dependent perturbation theory. The conceptual framework described in the above brief remarks will serve as the basis for much of the work in this book.

Despite its apparent advantages, quantum field theory did not immediately supplant hole theory; rather, the two points of view coexisted for a while, and various combinations of field-theoretic and hole-theoretic ideas were used in calculations of physical reaction rates. This period saw a number of calculations of cross sections to lowest order in powers of  $e^2$  for various processes, such as  $e^- + \gamma \rightarrow e^- + \gamma$  in 1929 by Klein and Nishina [46];  $e^+ + e^- \rightarrow 2\gamma$  in 1930 by Dirac [47];  $e^- + e^- \rightarrow e^- + e^-$  in 1932 by Møller [48];  $e^- + Z \rightarrow e^- + \gamma + Z$  and  $\gamma + Z \rightarrow e^+ + e^- + Z$  (where  $Z$  denotes the Coulomb field of a heavy atom) in 1934 by Bethe and Heitler [49]; and  $e^+ + e^- \rightarrow e^+ + e^-$  in 1936 by Bhabha [50]. (Rules for the calculation of such processes are given in Chapter 8, and worked out in detail there for the case of electron–photon scattering.) These lowest-order calculations gave finite results, in reasonable agreement with the experimental data.

Nevertheless, a general feeling of dissatisfaction with quantum field theory (whether or not in the form of hole theory) persisted throughout the 1930s. One of the reasons for this was the apparent failure of quantum electrodynamics to account for the penetrating power of the charged particles in cosmic ray showers, noted in 1936 by Oppenheimer and Franklin Carlson [50a]. Another cause of dissatisfaction that turned out to be related to the first was the steady discovery of new kinds of particles and interactions. We have already mentioned the electron, photon, positron, neutrino, and, of course, the nucleus of hydrogen, the proton. Throughout the 1920s it was generally believed that heavier nuclei are composed of protons and electrons, but it was hard to see how a light particle like the electron could be confined in the nucleus. Another severe difficulty with this picture was pointed out in 1931 by Ehrenfest and Oppenheimer [51]: the nucleus of ordinary nitrogen,  $N^{14}$ , in order to have atomic number 7 and atomic weight 14, would have to be composed of 14 protons and 7 electrons, and would therefore have to be a fermion, in conflict with the result of molecular spectroscopy [52] that  $N^{14}$  is a boson. This problem (and others) were solved in 1932 with the discovery of the neutron [53], and by Heisenberg’s subsequent suggestion [54] that nuclei are composed of protons and neutrons, not protons and electrons. It was clear that a strong non-electromagnetic force of short range would have to operate between neutrons and protons to hold nuclei together.

After the success of the Fermi theory of beta decay, several authors [54a] speculated that nuclear forces might be explained in this theory as due to the exchange of electrons and neutrinos. A few years later, in 1935, Hideki Yukawa proposed a quite different quantum field theory of the nuclear force [55]. In an essentially classical calculation, he found that the interaction of a scalar field with nucleons (protons or neutrons) would produce a nucleon–nucleon potential, with a dependence on the nucleon separation  $r$  given by

$$V(r) \propto \frac{1}{r} \exp(-\lambda r) \tag{1.2.74}$$

instead of the  $1/r$  Coulomb potential produced by electric fields. The quantity  $\lambda$  was introduced as a parameter in Yukawa’s scalar field equation, and when this equation was quantized, Yukawa found that it described particles of mass  $\hbar\lambda/c$ . The observed range of the strong interactions within nuclei led Yukawa to estimate that  $\hbar\lambda/c$  is of the order of 200 electron masses. In 1937 such ‘mesons’ were discovered in cloud chamber experiments [56]

by Seth Neddermeyer and Anderson and by Jabez Curry Street and Edward Carl Stevenson, and it was generally believed that these were the hypothesized particles of Yukawa.

The discovery of mesons revealed that the charged particles in cosmic ray showers are not all electrons, and thus cleared up the problem with these showers that had bothered Oppenheimer and Carlson. At the same time, however, it created new difficulties. Lothar Nordheim [56a] pointed out in 1939 that the same strong interactions by which the mesons are copiously produced at high altitudes (and which are required in Yukawa’s theory) should have led to the mesons’ absorption in the atmosphere, a result contradicted by their copious appearance at lower altitudes. In 1947 it was shown in an experiment by Marcello Conversi, Ettore Pancini, and Oreste Piccioni [57] that the mesons which predominate in cosmic rays at low altitude actually interact weakly with nucleons, and therefore could not be identified with Yukawa’s particle. This puzzle was cleared up by a theoretical suggestion [58], and its subsequent experimental confirmation [59] by Cesare Lattes, Occhialini, and Cecil Powell—there are two kinds of mesons with slightly different masses: the heavier (now called the  $\pi$  meson or pion) has strong interactions and plays the role in nuclear force envisaged by Yukawa; the lighter (now called the  $\mu$  meson, or muon) has only weak and electromagnetic interactions, and predominates in cosmic rays at sea level, being produced by the decay of  $\pi$  mesons. In the same year, 1947, entirely new kinds of particles (now known as  $K$  mesons and hyperons) were found in cosmic rays by George Rochester and Clifford Butler [60]. From 1947 until the present particles have continued to be discovered in a bewildering variety, but to pursue this story would take us outside the bounds of our present survey. These discoveries showed clearly that any conceptual framework which was limited to photons, electrons, and positrons would be far too narrow to be taken seriously as a fundamental theory. But an even more important obstacle was presented by a purely theoretical problem—the problem of infinities.

### 1.3 The Problem of Infinities

Quantum field theory deals with fields  $\psi(x)$  that destroy and create particles at a spacetime point  $x$ . Earlier experience with classical electron theory provided a warning that a point electron will have infinite electromagnetic self-mass; this mass is  $e^2/6\pi ac^2$  for a surface distribution of charge with radius  $a$ , and therefore blows up for  $a \rightarrow 0$ . Disappointingly this problem appeared with even greater severity in the early days of quantum field theory, and although greatly ameliorated by subsequent improvements in the theory, it remains with us to the present day.

The problem of infinities in quantum field theory was apparently first noted in the 1929–30 papers of Heisenberg and Pauli [41]. Soon after, the presence of infinities was confirmed in calculations of the electromagnetic self-energy of a bound electron by Oppenheimer [61], and of a free electron by Ivar Waller [62]. They used ordinary second-order perturbation theory, with an intermediate state consisting of an electron and a photon: for instance, the shift of the energy  $E_n$  of an electron in the  $n$ th energy level of hydrogen is given by

$$\Delta E_n = \sum_{m,\lambda} \int d^3k \frac{|\langle m; \mathbf{k}, \lambda | H' | n \rangle|^2}{E_n - E_m - |\mathbf{k}|c}. \quad (1.3.1)$$

where the sums and integral are over all intermediate electron states  $m$ , photon helicities  $\lambda$ , and photon momenta  $\mathbf{k}$ , and  $H'$  is the term in the Hamiltonian representing the interaction of radiation and electrons. This calculation gave a self-energy that is formally infinite; further, if this infinity is removed by discarding all intermediate states with photon wave numbers greater than  $1/a$ , then the self-energy behaves like  $1/a^2$  as  $a \rightarrow 0$ . Infinities of this sort are often called ultraviolet divergences, because they arise from intermediate states containing particles of very short wavelength.

These calculations treated the electron according to the rules of the original Dirac theory, without filled negative-electron states. A few years later Weisskopf repeated the calculation of the electron self-mass in the new hole theory, with all negative-energy states full. In this case another term appears in second-order perturbation theory, which in a non-hole-theory language can be described as arising from processes in which the electron in its final state first appears out of the vacuum together with a photon and a positron which then annihilate along with the initial electron. Initially Weisskopf found a  $1/a^2$  dependence on the photon wave-number cutoff  $1/a$ . The same calculation was being carried out (at the suggestion of Bohr) at that time by Carlson and Furry. After seeing Weisskopf's results, Furry realized that while Weisskopf had included an electrostatic term that he and Carlson had neglected, Weisskopf had made a new mistake in the calculation of the magnetic self-energy. After hearing from Furry and correcting his own error, Weisskopf found that the  $1/a^2$  terms in the total mass shift cancelled! However, despite this cancellation, an infinity remained: with a wave-number cutoff  $1/a$ , the self-mass was found to be [63]

$$m_{\text{em}} = \frac{3\alpha}{2\pi} m \ln \left( \frac{\hbar}{mca} \right). \quad (1.3.2)$$

The weakening of the cut-off dependence, to  $\ln a$  as compared with the classical  $1/a$  or the early quantum  $1/a^2$ , was mildly encouraging at the time and turned out to be of great importance later, in the development of renormalization theory.

An infinity of quite a different kind was encountered in 1933, apparently first by Dirac [64]. He considered the effect of an external static nearly uniform charge density  $\varepsilon(\mathbf{x})$  on the vacuum, i.e., on the negative-energy electrons in the filled energy levels of hole theory. The Coulomb interaction between  $\varepsilon(\mathbf{x})$  and the charge density of the negative-energy electrons produces a “vacuum polarization,” with induced charge density

$$\delta\varepsilon = A\varepsilon + B \left( \frac{\hbar}{mc} \right)^2 \nabla^2 \varepsilon + \dots. \quad (1.3.3)$$

The constant  $B$  is finite, and of order  $\alpha$ . On the other hand,  $A$  is logarithmically divergent, of order  $\alpha \ln a$ , where  $1/a$  is the wave-number cutoff.

Infinities also seemed to occur in a related problem, the scattering of light by light. Hans Euler, Bernard Kockel, and Heisenberg [65] showed in 1935–6 that these infinities could be eliminated by using a more-or-less arbitrary prescription suggested earlier by Dirac [66] and Heisenberg [67]. They calculated an effective Lagrangian density for the non-linear electrodynamic effects produced by virtual electron–positron pairs:

$$\mathcal{L} = \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) + \frac{e^4 \hbar}{360\pi^2 m^4 c^7} \left[ (\mathbf{E}^2 - \mathbf{B}^2)^2 + 7 (\mathbf{E} \cdot \mathbf{B})^2 \right] + \dots, \quad (1.3.4)$$



valid for frequencies  $\nu \ll m_e c^2/\hbar$ . Soon after, Nicholas Kemmer and Weisskopf [68] presented arguments that in this case the infinities are spurious, and that Eq. (1.3.4) can be derived without any subtraction prescription.

One bright spot in the struggle with infinities was the successful treatment of *infrared* divergences, those that arise from the low-energy rather than the high-energy part of the range of integration. In 1937 it was shown by Felix Bloch and Arne Nordsieck [68a] that these infinities cancel provided one includes processes in which arbitrary numbers of low-energy photons are produced. This will be discussed in modern terms in Chapter 13.

Yet another infinity turned up in a calculation by Sidney Michael Dancoff [69] in 1939 of the radiative corrections to the scattering of electrons by the static Coulomb field of an atom. The calculation contained a mistake (one of the terms was omitted), but this was not realized until later [69a].

Throughout the 1930s, these various infinities were seen not merely as failures of specific calculations. Rather, they seemed to indicate a gap in the understanding of relativistic quantum field theory on the most fundamental level, an opinion reinforced by the problems with cosmic rays mentioned in the previous section.

One of the symptoms of this uneasy pessimism was the continued exploration throughout the 1930s and 1940s of alternative formalisms. As Julian Schwinger [69b] later recalled, “The preoccupation of the majority of involved physicists was not with analyzing and carefully applying the known relativistic theory of coupled electron and electromagnetic fields but with changing it.” Thus in 1938 Heisenberg [70] proposed the existence of a fundamental length  $L$ , analogous to the fundamental action  $\hbar$  and fundamental velocity  $c$ . Field theory was supposed to work only for distances larger than  $L$ , so that all divergent integrals would effectively be cut off at distances  $L$ , or momenta  $\hbar/L$ . Several specific proposals [70a] were made for giving field theory a non-local structure. Some theorists began to suspect that the formalism of state-vectors and quantum fields should be replaced by one based solely on observable quantities, such as the  $S$ -matrix introduced by John Archibald Wheeler [71] in 1937 and Heisenberg [72] in 1943, whose elements are the amplitudes for various scattering processes. As we shall see, the concept of the  $S$ -matrix has now become a vital part of modern quantum field theory, and for some theorists a pure  $S$ -matrix theory became an ideal, especially as a possible solution to the problems of the strong interactions [73]. In yet another direction, Wheeler and Richard Feynman [74] in 1945 attempted to eliminate the electromagnetic field, deriving electromagnetic interactions in terms of an interaction at a distance. They were able to show that a pure retarded (or pure advanced) potential could be obtained by taking into account the interaction not only between source and test charges, but also between these charges and all the other charges in the universe. Perhaps the most radical modification of quantum mechanics suggested during this period was the introduction by Dirac [75] of states of negative probability, as a means of cancelling infinities in sums over states. This idea, of an “indefinite metric” in Hilbert space, has also flourished in quantum field theory, though not in the form originally suggested.

A more conservative idea for dealing with the infinities was also in the air during the 1930s. Perhaps these infinities could all be absorbed into a redefinition, a “renormalization” of the parameters of the theory. For instance, it was already known that in any Lorentz-



invariant classical theory the electromagnetic self-energy and self-momentum of an electron *must* take the form of corrections to the mass of the electron; hence the infinities in these quantities can be cancelled by a negative infinity in the “bare” non-electromagnetic mass of the electron, leaving a finite measurable “renormalized” mass. Also, Eq. (1.3.3) shows that the vacuum polarization changes the charge of the electron, from  $e = \int d^3x \varepsilon$ , to

$$e_{\text{TOTAL}} = \int d^3x (\varepsilon + \delta\varepsilon) = (1 + A)e. \quad (1.3.5)$$

Vacuum polarization gives finite results in lowest order if observables like scattering cross-sections are expressed in terms of  $e_{\text{TOTAL}}$  rather than  $e$ . The question was, whether all infinities in quantum field theory could be dealt with in this way. In 1936 Weisskopf [76] suggested that this is the case, and verified that known infinities could be eliminated by renormalization of physical parameters in a variety of sample calculations. However, it was impossible with the calculational techniques then available to show that infinities could always be eliminated in this way, and Dancoff’s calculation [69] seemed to show that they could not.

Another effect of the appearance of infinities was a tendency to believe that any effect which turned out to be infinite in quantum field theory was actually not there at all. In particular, the 1928 Dirac theory had predicted complete degeneracy of the  $2s_{1/2}$ – $2p_{1/2}$  levels of hydrogen to all orders in  $\alpha$ ; any attempt at a quantum electromagnetic calculation of the splitting of these two levels ran into the problem of the infinite self-energy of a bound electron; therefore the existence of such a splitting was generally not taken seriously. Later Bethe [80] recalled that “This shift comes out infinite in all existing theories, and has therefore always been ignored.” This attitude persisted even in the late 1930s, when spectroscopic experiments [77] began to indicate the presence of a  $2s_{1/2}$ – $2p_{1/2}$  splitting of order 1000 MHz. One notable exception was Edwin Albrecht Uehling [78], who realized that the vacuum polarization effect mentioned earlier would produce a  $2s_{1/2}$ – $2p_{1/2}$  splitting; unfortunately, as we shall see in Chapter 14, this contribution to the splitting is much smaller than 1000 MHz, and of the wrong sign.

The gloom surrounding quantum field theory began to lift soon after World War II. On June 1–4, 1947, the Conference on the Foundations of Quantum Mechanics at Shelter Island, NY brought theoretical physicists who had been working on the problems of quantum field theory through the 1930s together with a younger generation of theorists who had started scientific work during the war, and — of crucial importance — a few experimental physicists. The discussion leaders were Hans Kramers, Oppenheimer, and Weisskopf. One of the experimentalists (or rather theorist turned experimentalist), Willis Lamb, described a decisive measurement [79] of the  $2s_{1/2}$ – $2p_{1/2}$  shift in hydrogen. A beam of hydrogen atoms from an oven, many in  $2s$  and  $2p$  states, was aimed at a detector sensitive only to atoms in excited states. The atoms in  $2p$  states can decay very rapidly to the  $1s$  ground state by one-photon (Lyman  $\alpha$ ) emission, while the  $2s$  states decay only very slowly by two-photon emission, so in effect the detector was measuring the number of atoms in the metastable  $2s$  state. The beam was passed through a magnetic field, which added a known Zeeman splitting to any  $2s_{1/2}$ – $2p_{1/2}$  splitting naturally present. The beam was also exposed to a

microwave-frequency electromagnetic field, with a fixed frequency  $\nu \sim 10$  GHz. At a certain magnetic field strength the detector signal was observed to be quenched, indicating that the microwave field was producing resonant transitions from the metastable  $2s$  state to the  $2p$  state and thence by a rapid Lyman  $\alpha$  emission to the ground state. The total (Zeeman plus intrinsic)  $2s$ – $2p$  splitting at this value of the magnetic field strength would have to be just  $h\nu$ , from which the intrinsic splitting could be inferred. A preliminary value of 1000 MHz was announced, in agreement with the earlier spectroscopic measurements [77]. The impact of this discovery can be summarized in a saying that was current in Copenhagen when I was a graduate student there in 1954: “Just because something is infinite does not mean it is zero!”

The discovery of the Lamb shift aroused intense interest among the theorists at Shelter Island, many of whom had already been working on improved formalisms for calculation in quantum electrodynamics. Kramers described his work on mass renormalization in the classical electrodynamics of an extended electron [79a], which showed that the difficulties associated with the divergence of the self-energy in the limit of zero radius do not appear explicitly if the theory is reexpressed so that the mass parameter in the formalism is identified with the experimental electron mass. Schwinger and Weisskopf (who had already heard rumors of Lamb’s result, and discussed the matter on the trip to Shelter Island) suggested that since the inclusion of intermediate states involving positrons was known to reduce the divergence in energy level shifts from  $1/a^2$  to  $\ln a$ , perhaps the *differences* of the shifts in atomic energy levels might turn out to be finite when these intermediate states were taken into account. (In fact, in 1946, before he learned of Lamb’s experiment, Weisskopf had already assigned this problem to a graduate student, Bruce French.) Almost immediately after the conference, during a train ride to Schenectady, Hans Bethe [80] carried out a non-relativistic calculation, still without including the effects of intermediate states containing positrons, but using a simple cutoff at virtual photon momenta of order  $m_e c^2$  to eliminate infinities. He obtained the encouraging approximate value of 1040 MHz. Fully relativistic calculations using the renormalization idea to eliminate infinities were soon thereafter carried out by a number of other authors [81], with excellent agreement with experiment.

Another exciting experimental result was reported at Shelter Island by Isidor I. Rabi. Measurements in his laboratory of the hyperfine structure of hydrogen and deuterium had suggested [82] that the magnetic moment of the electron is larger than the Dirac value  $e\hbar/2mc$  by a factor of about 1.0013, and subsequent measurements of the gyromagnetic ratios in sodium and gallium had given a precise value [83]

$$\mu = \frac{e\hbar}{2mc} [1.00118 \pm 0.00003].$$

Learning of these results, Gregory Breit suggested [83a] that they arose from an order  $\alpha$  radiative correction to the electron magnetic moment. At Shelter Island, both Breit and Schwinger described their efforts to calculate this correction. Shortly after the conference Schwinger completed a successful calculation of the anomalous magnetic moment of the electron [84]

$$\mu = \frac{e\hbar}{2mc} \left[ 1 + \frac{\alpha}{2\pi} \right] = \frac{e\hbar}{2mc} [1.001162]$$

in excellent agreement with observation. This, together with Bethe’s calculation of the Lamb shift, at last convinced physicists of the reality of radiative corrections.

The mathematical methods used in this period presented a bewildering variety of concepts and formalisms. One approach developed by Schwinger [85] was based on operator methods and the action principle, and was presented by him at a conference at Pocono Manor in 1948, the successor to the Shelter Island Conference. Another Lorentz-invariant operator formalism had been developed earlier by Sin-Itiro Tomonaga [86] and his co-workers in Japan, but their work was not at first known in the West. Tomonaga had grappled with infinities in Yukawa’s meson theory in the 1930s. In 1947 he and his group were still out of the loop of scientific communication; they learned about Lamb’s experiment from an article in *Newsweek*.

An apparently quite different approach was invented by Feynman [87], and described briefly by him at the Pocono Conference. Instead of introducing quantum field operators, Feynman represented the  $S$ -matrix as a functional integral of  $\exp(iW)$ , where  $W$  is the action integral for a set of Dirac particles interacting with a *classical* electromagnetic field, integrated over all Dirac particle trajectories satisfying certain initial and final conditions for  $t \rightarrow \pm\infty$ . One result of great practical importance that came out of Feynman’s work was a set of graphical rules for calculating  $S$ -matrix elements to any desired order of perturbation theory. Unlike the old perturbation theory of the 1920s and 1930s, these Feynman rules automatically lumped together particle creation and antiparticle annihilation processes, and thereby gave results that were Lorentz-invariant at every stage. We have already seen in Weisskopf’s early calculation [63] of the electron self-energy, that it is only in such calculations, including particles and antiparticles on the same footing, that the nature of the infinities becomes transparent.

Finally, in a pair of papers in 1949, Freeman Dyson [88] showed that the operator formalisms of Schwinger and Tomonaga would yield the same graphical rules that had been found by Feynman. Dyson also carried out an analysis of the infinities in general Feynman diagrams, and outlined a proof that these are always precisely the sort which could be removed by renormalization. One of the most striking results that could be inferred from Dyson’s analysis was a criterion for deciding which quantum field theories are “renormalizable,” in the sense that all infinities can be absorbed into a redefinition of a *finite* number of coupling constants and masses. In particular, an interaction like the Pauli term (1.1.32), which would have changed the predicted magnetic moment of the electron, would spoil the renormalizability of quantum electrodynamics. With the publication of Dyson’s papers, there was at last a general and systematic formalism that physicists could easily learn to use, and that would provide a common language for the subsequent applications of quantum field theory to the problems of physics.

I cannot leave the infinities without taking up a puzzling aspect of this story. Oppenheimer [61] in 1930 had already noticed that most of the ultraviolet divergence in the self-energy of a bound electron cancels when one takes the difference between the shifts of two atomic energy levels, and Weisskopf [63] in 1934 had found that most of the divergence in the self-energy of a free electron cancels when one includes intermediate states containing positrons. It would have been natural even in 1934 to guess that including positron

intermediate states *and* subtracting the energy shifts of pairs of atomic states would eliminate the ultraviolet divergence in their relative energy shift.<sup>2</sup> There was even experimental evidence [77] for a  $2s_{1/2}$ – $2p_{1/2}$  energy difference of order 1000 MHz. So why did no one before 1947 attempt a numerical estimate of this energy difference?

Strictly speaking, there was one such attempt [88a] in 1939, but it focused on the wrong part of the problem, the charge radius of the *proton*, which has only a tiny effect on hydrogen energy levels. The calculation gave a result in rough agreement with the early experiments [77]. This was a mistake, as shown in 1939 by Lamb [88b].

A fully relativistic calculation of the Lamb shift including positrons in intermediate states could have been attempted during the 1930s, using the old non-relativistic perturbation theory. As long as one keeps all terms up to a given order, old-fashioned non-relativistic perturbation theory gives the same results as the manifestly relativistic formalisms of Feynman, Schwinger, and Tomonaga. In fact, after Bethe’s work, the first precise calculations [81] of the Lamb shift in the USA by French and Weisskopf and Norman Kroll and Lamb were done in just this way, though Tomonaga’s group [81] in Japan was already using covariant methods to solve this and other problems.

The one missing element was confidence in renormalization as a means of dealing with infinities. As we have seen, renormalization was widely discussed in the late 1930s. But it had become accepted wisdom in the 1930s, and a point of view especially urged by Oppenheimer [89], that quantum electrodynamics could not be taken seriously at energies of more than about 100 MeV, and that the solution to its problems could be found only in really adventurous new ideas.

Several things happened at Shelter Island to change this expectation. One was news that the problems concerning cosmic rays discussed in the previous section were beginning to be resolved; Robert Marshak presented the hypothesis [58] that there were two types of “meson” with similar masses; the muons that had actually been observed, and the pions responsible for nuclear forces. More important was the fact that now there were reliable experimental values for the Lamb shift and the anomalous magnetic moment that forced physicists to think carefully about radiative corrections. Probably equally important was the fact that the conference brought together theorists who had in their own individual ways been thinking about renormalization as a solution to the problem of infinities. When the revolution came in the late 1940s, it was made by physicists who though mostly young were playing a conservative role, turning away from the search by their predecessors for a radical solution.

## Weinberg Exercises

*Weinberg’s original Chapter 1 contains no end-of-chapter exercises.*

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<sup>2</sup>In fact, this guess would have been wrong. As discussed in Section 14.3, radiative corrections to the electron mass affect atomic energy levels not only through a shift in the electron rest energy, which is the same in all atomic energy levels, but also through a change in the electron kinetic energy, that varies from one level to another.

## Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

*There are no original Weinberg exercises to solve in this chapter.*

## Supplementary Exercises

*No supplementary exercises are assigned for this historical chapter.*

## Solutions to Supplementary Exercises

*There are no supplementary exercises to solve in this chapter.*

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## 2 Relativistic Quantum Mechanics

The point of view of this book is that quantum field theory is the way it is because (with certain qualifications) this is the only way to reconcile quantum mechanics with special relativity. Therefore our first task is to study how symmetries like Lorentz invariance appear in a quantum setting.

### 2.1 Quantum Mechanics

First, some good news: quantum field theory is based on the same quantum mechanics that was invented by Schrödinger, Heisenberg, Pauli, Born, and others in 1925–26, and has been used ever since in atomic, molecular, nuclear, and condensed matter physics. The reader is assumed to be already familiar with quantum mechanics; this section provides only the briefest of summaries of quantum mechanics, in the generalized version of Dirac [1].

(i) Physical states are represented by rays in Hilbert space. A Hilbert space is a kind of complex vector space; that is, if  $|\Phi\rangle$  and  $|\Psi\rangle$  are vectors in the space (often called ‘state-vectors’) then so is  $\xi|\Phi\rangle + \eta|\Psi\rangle$ , for arbitrary complex numbers  $\xi, \eta$ . It has a norm<sup>3</sup>: for any pair of vectors there is a complex number  $\langle\Phi|\Psi\rangle$ , such that

$$\langle\Phi|\Psi\rangle = \langle\Psi|\Phi\rangle^*, \quad (2.1.1)$$

$$\langle\Phi | \xi_1\Psi_1 + \xi_2\Psi_2\rangle = \xi_1\langle\Phi|\Psi_1\rangle + \xi_2\langle\Phi|\Psi_2\rangle, \quad (2.1.2)$$

$$\langle\eta_1\Phi_1 + \eta_2\Phi_2 | \Psi\rangle = \eta_1^*\langle\Phi_1|\Psi\rangle + \eta_2^*\langle\Phi_2|\Psi\rangle. \quad (2.1.3)$$

The norm  $\langle\Psi|\Psi\rangle$  also satisfies a positivity condition:  $\langle\Psi|\Psi\rangle \geq 0$ , and vanishes if and only if  $|\Psi\rangle = 0$ . (There are also certain technical assumptions that allow us to take limits of vectors within Hilbert space.) A ray is a set of normalized vectors (i.e.,  $\langle\Psi|\Psi\rangle = 1$ ) with  $|\Psi\rangle$  and  $|\Psi'\rangle$  belonging to the same ray if  $|\Psi'\rangle = \xi|\Psi\rangle$ , where  $\xi$  is an arbitrary complex number with  $|\xi| = 1$ .

(ii) Observables are represented by Hermitian operators. These are mappings  $|\Psi\rangle \rightarrow A|\Psi\rangle$  of Hilbert space into itself, linear in the sense that

$$A(\xi|\Psi\rangle + \eta|\Phi\rangle) = \xi A|\Psi\rangle + \eta A|\Phi\rangle \quad (2.1.4)$$

and satisfying the reality condition  $A^\dagger = A$ , where for any linear operator  $A$  the adjoint  $A^\dagger$  is defined by

$$\langle\Phi|A^\dagger|\Psi\rangle \equiv \langle A\Phi|\Psi\rangle = (\langle\Psi|A|\Phi\rangle)^*. \quad (2.1.5)$$

(There are also technical assumptions about the continuity of  $A|\Psi\rangle$  as a function of  $|\Psi\rangle$ .) A state represented by a ray  $\mathcal{R}$  has a definite value  $\alpha$  for the observable represented by an operator  $A$  if vectors  $|\Psi\rangle$  belonging to this ray are eigenvectors of  $A$  with eigenvalue  $\alpha$ :

$$A|\Psi\rangle = \alpha|\Psi\rangle \quad \text{for } |\Psi\rangle \text{ in } \mathcal{R}. \quad (2.1.6)$$

An elementary theorem tells us that for  $A$  Hermitian,  $\alpha$  is real, and eigenvectors with different  $\alpha$ s are orthogonal.

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<sup>3</sup>We shall often use the Dirac bra-ket notation: instead of  $\langle\Psi_1|\Psi_2\rangle$ , we may write  $\langle 1|2\rangle$ .

(iii) If a system is in a state represented by a ray  $\mathcal{R}$ , and an experiment is done to test whether it is in any one of the different states represented by mutually orthogonal rays  $\mathcal{R}_1, \mathcal{R}_2, \dots$  (for instance, by measuring one or more observables) then the probability of finding it in the state represented by  $\mathcal{R}_n$  is

$$P(\mathcal{R} \rightarrow \mathcal{R}_n) = |\langle \Psi | \Psi_n \rangle|^2, \quad (2.1.7)$$

where  $|\Psi\rangle$  and  $|\Psi_n\rangle$  are any vectors belonging to rays  $\mathcal{R}$  and  $\mathcal{R}_n$ , respectively. (A pair of rays is said to be orthogonal if the state-vectors from the two rays have vanishing scalar products.) Another elementary theorem gives a total probability unity:

$$\sum_n P(\mathcal{R} \rightarrow \mathcal{R}_n) = 1 \quad (2.1.8)$$

if the state-vectors  $|\Psi_n\rangle$  form a complete set.

## 2.2 Symmetries

A symmetry transformation is a change in our point of view that does not change the results of possible experiments. If an observer  $O$  sees a system in a state represented by a ray  $\mathcal{R}$  or  $\mathcal{R}_1$  or  $\mathcal{R}_2, \dots$ , then an equivalent observer  $O'$  who looks at the *same* system will observe it in a different state, represented by a ray  $\mathcal{R}'$  or  $\mathcal{R}'_1$  or  $\mathcal{R}'_2, \dots$ , respectively, but the two observers must find the same probabilities

$$P(\mathcal{R} \rightarrow \mathcal{R}_n) = P(\mathcal{R}' \rightarrow \mathcal{R}'_n). \quad (2.2.1)$$

(This is only a necessary condition for a ray transformation to be a symmetry; further conditions are discussed in the next chapter.) An important theorem proved by Wigner [2] in the early 1930s tells us that for any such transformation  $\mathcal{R} \rightarrow \mathcal{R}'$  of rays we may define an operator  $U$  on Hilbert space, such that if  $|\Psi\rangle$  is in ray  $\mathcal{R}$  then  $U|\Psi\rangle$  is in the ray  $\mathcal{R}'$ , with  $U$  either *unitary* and *linear*

$$\langle U\Phi | U\Psi \rangle = \langle \Phi | \Psi \rangle, \quad (2.2.2)$$

$$U(\xi|\Phi\rangle + \eta|\Psi\rangle) = \xi U|\Phi\rangle + \eta U|\Psi\rangle \quad (2.2.3)$$

or else *antiunitary* and *antilinear*

$$\langle U\Phi | U\Psi \rangle = \langle \Phi | \Psi \rangle^*, \quad (2.2.4)$$

$$U(\xi|\Phi\rangle + \eta|\Psi\rangle) = \xi^* U|\Phi\rangle + \eta^* U|\Psi\rangle. \quad (2.2.5)$$

Wigner's proof omits some steps. A more complete proof is given at the end of this chapter in Appendix A.

As already mentioned, the adjoint of a linear operator  $L$  is defined by

$$\langle \Phi | L^\dagger | \Psi \rangle \equiv \langle L\Phi | \Psi \rangle. \quad (2.2.6)$$

This condition cannot be satisfied for an antilinear operator, because in this case the right-hand side of Eq. (2.2.6) would be linear in  $|\Phi\rangle$ , while the left-hand side is antilinear in  $|\Phi\rangle$ . Instead, the adjoint of an antilinear operator  $A$  is defined by

$$\langle \Phi | A^\dagger | \Psi \rangle \equiv \langle A\Phi | \Psi \rangle^* = \langle \Psi | A | \Phi \rangle. \quad (2.2.7)$$



With this definition, the conditions for unitarity or antiunitarity both take the form

$$U^\dagger = U^{-1}. \quad (2.2.8)$$

There is always a trivial symmetry transformation  $\mathcal{R} \rightarrow \mathcal{R}$ , represented by the identity operator  $U = \mathbf{1}$ . This operator is, of course, unitary and linear. Continuity then demands that any symmetry (like a rotation or translation or Lorentz transformation) that can be made trivial by a continuous change of some parameters (like angles or distances or velocities) must be represented by a linear unitary operator  $U$  rather than one that is antilinear and antiunitary. (Symmetries represented by antiunitary antilinear operators are less prominent in physics; they all involve a reversal in the direction of time's flow. See Section 2.6.)

In particular, a symmetry transformation that is infinitesimally close to being trivial can be represented by a linear unitary operator that is infinitesimally close to the identity:

$$U = \mathbf{1} + i\epsilon t \quad (2.2.9)$$

with  $\epsilon$  a real infinitesimal. For this to be unitary and linear,  $t$  must be Hermitian and linear, so it is a candidate for an observable. Indeed, most (and perhaps all) of the observables of physics, such as angular momentum or momentum, arise in this way from symmetry transformations.

The set of symmetry transformations has certain properties that define it as a *group*. If  $T_1$  is a transformation that takes rays  $\mathcal{R}_n$  into  $\mathcal{R}'_n$ , and  $T_2$  is another transformation that takes  $\mathcal{R}'_n$  into  $\mathcal{R}''_n$ , then the result of performing both transformations is another symmetry transformation, which we write  $T_2 T_1$ , that takes  $\mathcal{R}_n$  into  $\mathcal{R}''_n$ . Also, a symmetry transformation  $T$  which takes rays  $\mathcal{R}_n$  into  $\mathcal{R}'_n$  has an inverse, written  $T^{-1}$ , which takes  $\mathcal{R}'_n$  into  $\mathcal{R}_n$ , and there is an identity transformation,  $T = \mathbf{1}$ , which leaves rays unchanged.

The unitary or antiunitary operators  $U(T)$  corresponding to these symmetry transformations have properties that mirror this group structure, but with a complication due to the fact that, unlike the symmetry transformations themselves, the operators  $U(T)$  act on vectors in the Hilbert space, rather than on rays. If  $T_1$  takes  $\mathcal{R}_n$  into  $\mathcal{R}'_n$ , then acting on a vector  $|\Psi_n\rangle$  in the ray  $\mathcal{R}_n$ ,  $U(T_1)$  must yield a vector  $U(T_1)|\Psi_n\rangle$  in the ray  $\mathcal{R}'_n$ , and if  $T_2$  takes this ray into  $\mathcal{R}''_n$ , then acting on  $U(T_1)|\Psi_n\rangle$  it must yield a vector  $U(T_2)U(T_1)|\Psi_n\rangle$  in the ray  $\mathcal{R}''_n$ . But  $U(T_2 T_1)|\Psi_n\rangle$  is also in this ray, so these vectors can differ only by a phase  $\phi_n(T_2, T_1)$

$$U(T_2)U(T_1)|\Psi_n\rangle = e^{i\phi_n(T_2, T_1)}U(T_2 T_1)|\Psi_n\rangle. \quad (2.2.10)$$

Furthermore, with one significant exception, the linearity (or antilinearity) of  $U(T)$  tells us that these phases are independent of the state  $|\Psi_n\rangle$ . Here is the proof. Consider any two different vectors  $|\Psi_A\rangle, |\Psi_B\rangle$ , which are not proportional to each other. Then, applying Eq. (2.2.10) to the state  $|\Psi_{AB}\rangle = |\Psi_A\rangle + |\Psi_B\rangle$ , we have

$$\begin{aligned} e^{i\phi_{AB}}U(T_2 T_1)(|\Psi_A\rangle + |\Psi_B\rangle) &= U(T_2)U(T_1)(|\Psi_A\rangle + |\Psi_B\rangle) \\ &= U(T_2)U(T_1)|\Psi_A\rangle + U(T_2)U(T_1)|\Psi_B\rangle \\ &= e^{i\phi_A}U(T_2 T_1)|\Psi_A\rangle + e^{i\phi_B}U(T_2 T_1)|\Psi_B\rangle. \end{aligned} \quad (2.2.11)$$



Any unitary or antiunitary operator has an inverse (its adjoint) which is also unitary or antiunitary. Multiplying (2.2.11) on the left with  $U^{-1}(T_2T_1)$ , we have then

$$e^{\pm i\phi_{AB}}(|\Psi_A\rangle + |\Psi_B\rangle) = e^{\pm i\phi_A}|\Psi_A\rangle + e^{\pm i\phi_B}|\Psi_B\rangle, \quad (2.2.12)$$

the upper and lower signs referring to  $U(T_2T_1)$  unitary or antiunitary, respectively. Since  $|\Psi_A\rangle$  and  $|\Psi_B\rangle$  are linearly independent, this is only possible if

$$e^{i\phi_{AB}} = e^{i\phi_A} = e^{i\phi_B}. \quad (2.2.13)$$

So as promised, the phase in Eq. (2.2.10) is independent of the state-vector  $|\Psi_n\rangle$ , and therefore this can be written as an operator relation

$$U(T_2)U(T_1) = e^{i\phi(T_2, T_1)}U(T_2T_1). \quad (2.2.14)$$

For  $\phi = 0$ , this would say that the  $U(T)$  furnish a representation of the group of symmetry transformations. For general phases  $\phi(T_2, T_1)$ , we have what is called a projective representation, or a representation ‘up to a phase’. The structure of the Lie group cannot by itself tell us whether physical state-vectors furnish an ordinary or a projective representation, but as we shall see, it can tell us whether the group has any intrinsically projective representations at all.

The exception to the argument that led to Eq. (2.2.14) is that it may not be possible to prepare the system in a state represented by  $|\Psi_A\rangle + |\Psi_B\rangle$ . For instance, it is widely believed to be impossible to prepare a system in a superposition of two states whose total angular momenta are integers and half-integers, respectively. In such cases, we say that there is a ‘superselection rule’ between different classes of states [3], and the phases  $\phi(T_2, T_1)$  may depend on which of these classes of states the operators  $U(T_2)U(T_1)$  and  $U(T_2T_1)$  act upon. We will have more to say about these phases and projective representations in Section 2.7. As we shall see there, any symmetry group with projective representations can always be enlarged (without otherwise changing its physical implications) in such a way that its representations can all be defined as non-projective, with  $\phi = 0$ . Until Section 2.7, we will just assume that this has been done, and take  $\phi = 0$  in Eq. (2.2.14).

There is a kind of group, known as a *connected Lie group*, of special importance in physics. These are groups of transformations  $T(\theta)$  that are described by a finite set of real continuous parameters, say  $\theta^a$ , with each element of the group connected to the identity by a path within the group. The group multiplication law then takes the form

$$T(\bar{\theta})T(\theta) = T(f(\bar{\theta}, \theta)) \quad (2.2.15)$$

with  $f^a(\bar{\theta}, \theta)$  a function of the  $\bar{\theta}$ s and  $\theta$ s. Taking  $\theta^a = 0$  as the coordinates of the identity, we must have

$$f^a(\theta, 0) = f^a(0, \theta) = \theta^a. \quad (2.2.16)$$

As already mentioned, the transformations of such continuous groups must be represented on the physical Hilbert space by unitary (rather than antiunitary) operators  $U(T(\theta))$ . For

a Lie group, these operators can be represented in at least a finite neighborhood of the identity by a power series

$$U(T(\theta)) = \mathbf{1} + i\theta^a t_a + \frac{1}{2}\theta^b \theta^c t_{bc} + \dots, \quad (2.2.17)$$

where  $t_a$ ,  $t_{bc} = t_{cb}$ , etc. are Hermitian operators independent of the  $\theta$ s. Suppose that the  $U(T(\theta))$  form an ordinary (i.e., not projective) representation of this group of transformations, i.e.,

$$U(T(\bar{\theta}))U(T(\theta)) = U(T(f(\bar{\theta}, \theta))). \quad (2.2.18)$$

Let us see what this condition looks like when expanded in powers of  $\theta^a$  and  $\bar{\theta}^a$ . According to Eq. (2.2.16), the expansion of  $f^a(\bar{\theta}, \theta)$  to second order must take the form

$$f^a(\bar{\theta}, \theta) = \theta^a + \bar{\theta}^a + f^a_{bc} \bar{\theta}^b \theta^c + \dots \quad (2.2.19)$$

with real coefficients  $f^a_{bc}$ . (The presence of any terms of order  $\theta^2$  or  $\bar{\theta}^2$  would violate Eq. (2.2.16).) Then Eq. (2.2.18) reads

$$\begin{aligned} & \left[ \mathbf{1} + i\bar{\theta}^a t_a + \frac{1}{2}\bar{\theta}^b \bar{\theta}^c t_{bc} + \dots \right] \left[ \mathbf{1} + i\theta^a t_a + \frac{1}{2}\theta^b \theta^c t_{bc} + \dots \right] \\ &= \mathbf{1} + i(\theta^a + \bar{\theta}^a + f^a_{bc} \bar{\theta}^b \theta^c + \dots) t_a \\ & \quad + \frac{1}{2}(\theta^b + \bar{\theta}^b + \dots)(\theta^c + \bar{\theta}^c + \dots) t_{bc} + \dots \end{aligned} \quad (2.2.20)$$

The terms of order 1,  $\theta$ ,  $\bar{\theta}$ ,  $\theta^2$ , and  $\bar{\theta}^2$  automatically match on both sides of Eq. (2.2.20), but from the  $\bar{\theta}\theta$  terms we obtain a non-trivial condition

$$t_{bc} = -t_b t_c - i f^a_{bc} t_a. \quad (2.2.21)$$

This shows that if we are given the structure of the group, i.e., the function  $f(\theta, \bar{\theta})$ , and hence its quadratic coefficient  $f^a_{bc}$ , we can calculate the second-order terms in  $U(T(\theta))$  from the generators  $t_a$  appearing in the first-order terms. However, there is a consistency condition: the operator  $t_{bc}$  must be *symmetric* in  $b$  and  $c$  (because it is the second derivative of  $U(T(\theta))$  with respect to  $\theta^b$  and  $\theta^c$ ) so Eq. (2.2.21) requires that

$$[t_b, t_c] = i C^a_{bc} t_a, \quad (2.2.22)$$

where  $C^a_{bc}$  are a set of real constants known as *structure constants*

$$C^a_{bc} \equiv -f^a_{bc} + f^a_{cb}. \quad (2.2.23)$$

Such a set of commutation relations is known as a *Lie algebra*. In Section 2.7 we will prove in effect that the commutation relation (2.2.22) is the single condition needed to ensure that this process can be continued: the complete power series for  $U(T(\theta))$  may be calculated from an infinite sequence of relations like Eq. (2.2.21), provided we know the first-order terms, the generators  $t_a$ . This does not necessarily mean that the operators  $U(T(\theta))$  are uniquely determined for all  $\theta^a$  if we know the  $t_a$ , but it does mean that the  $U(T(\theta))$  are uniquely

determined in at least a finite neighborhood of the coordinates  $\theta^a = 0$  of the identity, in such a way that Eq. (2.2.15) is satisfied if  $\theta$ ,  $\bar{\theta}$ , and  $f(\theta, \bar{\theta})$  are in this neighborhood. The extension to all  $\theta^a$  is discussed in Section 2.7.

There is a special case of some importance, that we will encounter again and again. Suppose that the function  $f(\theta, \bar{\theta})$  (perhaps just for some subset of the coordinates  $\theta^a$ ) is simply additive

$$f^a(\theta, \bar{\theta}) = \theta^a + \bar{\theta}^a. \quad (2.2.24)$$

This is the case for instance for translations in spacetime, or for rotations about any one fixed axis (though *not* for both together). Then the coefficients  $f^a_{bc}$  in Eq. (2.2.19) vanish, and so do the structure constants (2.2.23). The generators then all commute

$$[t_b, t_c] = 0. \quad (2.2.25)$$

Such a group is called *Abelian*. In this case, it is easy to calculate  $U(T(\theta))$  for all  $\theta^a$ . From Eqs. (2.2.18) and (2.2.24), we have for any integer  $N$

$$U(T(\theta)) = \left[ U \left( T \left( \frac{\theta}{N} \right) \right) \right]^N.$$

Letting  $N \rightarrow \infty$ , and keeping only the first-order term in  $U(T(\theta/N))$ , we have then

$$U(T(\theta)) = \lim_{N \rightarrow \infty} \left[ \mathbf{1} + \frac{i}{N} \theta^a t_a \right]^N$$

and hence

$$U(T(\theta)) = \exp(it_a \theta^a). \quad (2.2.26)$$

### 2.3 Quantum Lorentz Transformations

Einstein's principle of relativity states the equivalence of certain 'inertial' frames of reference. It is distinguished from the Galilean principle of relativity, obeyed by Newtonian mechanics, by the transformation connecting coordinate systems in different inertial frames. If  $x^\mu$  are the coordinates in one inertial frame (with  $x^0 = t$  a time coordinate and  $x^1, x^2, x^3$  Cartesian space coordinates, the speed of light being set equal to unity) then in any other inertial frame, the coordinates  $x'^\mu$  must satisfy

$$\eta_{\mu\nu} dx'^\mu dx'^\nu = \eta_{\mu\nu} dx^\mu dx^\nu \quad (2.3.1)$$

or equivalently

$$\eta_{\mu\nu} \frac{\partial x'^\mu}{\partial x^\rho} \frac{\partial x'^\nu}{\partial x^\sigma} = \eta_{\rho\sigma}. \quad (2.3.2)$$

Here  $\eta_{\mu\nu}$  is the diagonal matrix, with elements

$$\eta_{00} = -1, \quad \eta_{11} = \eta_{22} = \eta_{33} = +1, \quad (2.3.3)$$

and the summation convention is in force: we sum over any index like  $\mu$  and  $\nu$  in Eq. (2.3.2), which appears twice in the same term, once upstairs and once downstairs. These transformations have the special property that the speed of light is the same (in our units, equal to

unity) in all inertial frames;<sup>4</sup> a light wave travelling at unit speed satisfies  $|d\mathbf{x}/dt| = 1$ , or in other words  $\eta_{\mu\nu}dx^\mu dx^\nu = d\mathbf{x}^2 - dt^2 = 0$ , from which it follows that also  $\eta_{\mu\nu}dx'^\mu dx'^\nu = 0$ , and hence  $|d\mathbf{x}'/dt'| = 1$ .

Any coordinate transformation  $x^\mu \rightarrow x'^\mu$  that satisfies Eq. (2.3.2) is *linear* [3a]

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu \quad (2.3.4)$$

with  $a^\mu$  arbitrary constants, and  $\Lambda^\mu{}_\nu$  a constant matrix satisfying the conditions

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}. \quad (2.3.5)$$

For some purposes, it is useful to write the Lorentz transformation condition in a different way. The matrix  $\eta_{\mu\nu}$  has an inverse, written  $\eta^{\mu\nu}$ , which happens to have the same components: it is diagonal, with  $\eta^{00} = -1$ ,  $\eta^{11} = \eta^{22} = \eta^{33} = +1$ . Multiplying Eq. (2.3.5) with  $\eta^{\sigma\tau}\Lambda^\kappa{}_\tau$  and inserting parentheses judiciously, we have

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho(\Lambda^\nu{}_\sigma\Lambda^\kappa{}_\tau\eta^{\sigma\tau}) = \Lambda^\kappa{}_\rho = \eta_{\mu\nu}\eta^{\nu\kappa}\Lambda^\mu{}_\rho.$$

Multiplying with the inverse of the matrix  $\eta_{\mu\nu}\Lambda^\mu{}_\rho$  then gives

$$\Lambda^\nu{}_\sigma\Lambda^\kappa{}_\tau\eta^{\sigma\tau} = \eta^{\nu\kappa}. \quad (2.3.6)$$

These transformations form a group. If we first perform a Lorentz transformation (2.3.4), and then a second Lorentz transformation  $x'^\mu \rightarrow x''^\mu$ , with

$$x''^\mu = \bar{\Lambda}^\mu{}_\rho x'^\rho + \bar{a}^\mu = \bar{\Lambda}^\mu{}_\rho(\Lambda^\rho{}_\nu x^\nu + a^\rho) + \bar{a}^\mu$$

then the effect is the same as the Lorentz transformation  $x^\mu \rightarrow x''^\mu$ , with

$$x''^\mu = (\bar{\Lambda}^\mu{}_\rho\Lambda^\rho{}_\nu)x^\nu + (\bar{\Lambda}^\mu{}_\rho a^\rho + \bar{a}^\mu). \quad (2.3.7)$$

(Note that if  $\Lambda^\mu{}_\nu$  and  $\bar{\Lambda}^\mu{}_\nu$  both satisfy Eq. (2.3.5), then so does  $\bar{\Lambda}^\mu{}_\rho\Lambda^\rho{}_\nu$ , so this is a Lorentz transformation. The bar is used here just to distinguish one Lorentz transformation from the other.) The transformations  $T(\Lambda, a)$  induced on physical states therefore satisfy the composition rule

$$T(\bar{\Lambda}, \bar{a})T(\Lambda, a) = T(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a}). \quad (2.3.8)$$

Taking the determinant of Eq. (2.3.5) gives

$$(\text{Det } \Lambda)^2 = 1 \quad (2.3.9)$$

so  $\Lambda^\mu{}_\nu$  has an inverse,  $(\Lambda^{-1})^\nu{}_\mu$ , which we see from Eq. (2.3.5) takes the form

$$(\Lambda^{-1})^\nu{}_\mu = \Lambda_\mu{}^\nu \equiv \eta_{\mu\rho}\eta^{\nu\sigma}\Lambda^\rho{}_\sigma. \quad (2.3.10)$$

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<sup>4</sup>There is a larger class of coordinate transformations, known as *conformal transformations*, for which  $\eta_{\mu\nu}dx'^\mu dx'^\nu$  is proportional though generally not equal to  $\eta_{\mu\nu}dx^\mu dx^\nu$ , and which therefore also leave the speed of light invariant. Conformal invariance in two dimensions has proved enormously important in string theory and statistical mechanics, but the physical relevance of these conformal transformations in four spacetime dimensions is not yet clear.

The inverse of the transformation  $T(\Lambda, a)$  is seen from Eq. (2.3.8) to be  $T(\Lambda^{-1}, -\Lambda^{-1}a)$ , and, of course, the identity transformation is  $T(\mathbf{1}, 0)$ .

In accordance with the discussion in the previous section, the transformations  $T(\Lambda, a)$  induce a unitary linear transformation on vectors in the physical Hilbert space

$$|\Psi\rangle \longrightarrow U(\Lambda, a)|\Psi\rangle.$$

The operators  $U$  satisfy a composition rule

$$U(\bar{\Lambda}, \bar{a})U(\Lambda, a) = U(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a}). \quad (2.3.11)$$

(As already mentioned, to avoid the appearance of a phase factor on the right-hand side of Eq. (2.3.11), it is, in general, necessary to enlarge the Lorentz group. The appropriate enlargement is described in Section 2.7.)

The whole group of transformations  $T(\Lambda, a)$  is properly known as the *Poincaré group*. It has a number of important subgroups. First, those transformations with  $a^\mu = 0$  obviously form a subgroup, with

$$T(\bar{\Lambda}, 0)T(\Lambda, 0) = T(\bar{\Lambda}\Lambda, 0), \quad (2.3.12)$$

known as the *homogeneous Lorentz group*. Also, we note from Eq. (2.3.9) that either  $\text{Det } \Lambda = +1$  or  $\text{Det } \Lambda = -1$ ; those transformations with  $\text{Det } \Lambda = +1$  obviously form a subgroup of either the homogeneous Lorentz group or the Poincaré group. Further, from the 00-components of Eqs. (2.3.5) and (2.3.6), we have

$$(\Lambda^0_0)^2 = 1 + \Lambda^i_0 \Lambda^i_0 = 1 + \Lambda^0_i \Lambda^0_i, \quad (2.3.13)$$

with  $i$  summed over the values 1, 2, and 3. We see that either  $\Lambda^0_0 \geq +1$  or  $\Lambda^0_0 \leq -1$ . Those transformations with  $\Lambda^0_0 \geq +1$  form a subgroup. Note that if  $\Lambda^\mu_\nu$  and  $\bar{\Lambda}^\mu_\nu$  are two such  $\Lambda$ s, then

$$(\bar{\Lambda}\Lambda)^0_0 = \bar{\Lambda}^0_0 \Lambda^0_0 + \bar{\Lambda}^0_1 \Lambda^1_0 + \bar{\Lambda}^0_2 \Lambda^2_0 + \bar{\Lambda}^0_3 \Lambda^3_0.$$

But Eq. (2.3.13) shows that the three-vector  $(\Lambda^1_0, \Lambda^2_0, \Lambda^3_0)$  has length  $\sqrt{(\Lambda^0_0)^2 - 1}$ , and similarly the three-vector

$(\bar{\Lambda}^0_1, \bar{\Lambda}^0_2, \bar{\Lambda}^0_3)$  has length  $\sqrt{(\bar{\Lambda}^0_0)^2 - 1}$ , so the scalar product of these two three-vectors is bounded by

$$|\bar{\Lambda}^0_1 \Lambda^1_0 + \bar{\Lambda}^0_2 \Lambda^2_0 + \bar{\Lambda}^0_3 \Lambda^3_0| \leq \sqrt{(\Lambda^0_0)^2 - 1} \sqrt{(\bar{\Lambda}^0_0)^2 - 1}, \quad (2.3.14)$$

and so

$$(\bar{\Lambda}\Lambda)^0_0 \geq \bar{\Lambda}^0_0 \Lambda^0_0 - \sqrt{(\Lambda^0_0)^2 - 1} \sqrt{(\bar{\Lambda}^0_0)^2 - 1} \geq 1.$$

The subgroup of Lorentz transformations with  $\text{Det } \Lambda = +1$  and  $\Lambda^0_0 \geq +1$  is known as the *proper orthochronous Lorentz group*. Since it is not possible by a continuous change of parameters to jump from  $\text{Det } \Lambda = +1$  to  $\text{Det } \Lambda = -1$ , or from  $\Lambda^0_0 \geq +1$  to  $\Lambda^0_0 \leq -1$ , any Lorentz transformation that can be obtained from the identity by a continuous change of parameters must have  $\text{Det } \Lambda$  and  $\Lambda^0_0$  of the same sign as for the identity, and hence must belong to the proper orthochronous Lorentz group.

Any Lorentz transformation is either proper and orthochronous, or may be written as the product of an element of the proper orthochronous Lorentz group with one of the discrete transformations  $\mathcal{P}$  or  $\mathcal{T}$  or  $\mathcal{PT}$ , where  $\mathcal{P}$  is the space inversion, whose non-zero elements are

$$\mathcal{P}^0_0 = 1, \quad \mathcal{P}^1_1 = \mathcal{P}^2_2 = \mathcal{P}^3_3 = -1, \quad (2.3.15)$$

and  $\mathcal{T}$  is the time-reversal matrix, whose non-zero elements are

$$\mathcal{T}^0_0 = -1, \quad \mathcal{T}^1_1 = \mathcal{T}^2_2 = \mathcal{T}^3_3 = 1. \quad (2.3.16)$$

Thus the study of the whole Lorentz group reduces to the study of its proper orthochronous subgroup, plus space inversion and time-reversal. We will consider space inversion and time-reversal separately in Section 2.6. Until then, we will deal only with the homogeneous or inhomogeneous proper orthochronous Lorentz group.

## 2.4 The Poincaré Algebra

As we saw in Section 2.2, much of the information about any Lie symmetry group is contained in properties of the group elements near the identity. For the Poincaré group, the identity is the transformation  $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu$ ,  $a^\mu = 0$ , so we want to study those transformations with

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad a^\mu = \epsilon^\mu, \quad (2.4.1)$$

both  $\omega^\mu{}_\nu$  and  $\epsilon^\mu$  being taken infinitesimal. The Lorentz condition (2.3.5) reads here

$$\begin{aligned} \eta_{\rho\sigma} &= \eta_{\mu\nu}(\delta^\mu{}_\rho + \omega^\mu{}_\rho)(\delta^\nu{}_\sigma + \omega^\nu{}_\sigma) \\ &= \eta_{\rho\sigma} + \omega_{\sigma\rho} + \omega_{\rho\sigma} + O(\omega^2). \end{aligned}$$

We are here using the convention, to be used throughout this book, that indices may be lowered or raised by contraction with  $\eta_{\mu\nu}$  or  $\eta^{\mu\nu}$

$$\omega_{\sigma\rho} = \eta_{\mu\sigma}\omega^\mu{}_\rho, \quad \omega^\mu{}_\rho = \eta^{\mu\sigma}\omega_{\sigma\rho}.$$

Keeping only the terms of first order in  $\omega$  in the Lorentz condition (2.3.5), we see that this condition now reduces to the antisymmetry of  $\omega_{\mu\nu}$

$$\omega_{\mu\nu} = -\omega_{\nu\mu}. \quad (2.4.2)$$

An antisymmetric second-rank tensor in four dimensions has  $(4 \cdot 3)/2 = 6$  independent components, so including the four components of  $\epsilon^\mu$ , an inhomogeneous Lorentz transformation is described by  $6 + 4 = 10$  parameters.

Since  $U(\mathbf{1}, 0)$  carries any ray into itself, it must be proportional to the unit operator,<sup>5</sup> and by a choice of phase may be made equal to it. For an infinitesimal Lorentz transformation (2.4.1),  $U(\mathbf{1} + \omega, \epsilon)$  must then equal  $\mathbf{1}$  plus terms linear in  $\omega_{\rho\sigma}$  and  $\epsilon_\rho$ . We write

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<sup>5</sup>In the absence of superselection rules, the possibility that the proportionality factor may depend on the state on which  $U(\mathbf{1}, 0)$  acts can be ruled out by the same reasoning that we used in Section 2.2 to rule out the possibility that the phases in projective representations of symmetry groups may depend on the states on which the symmetries act. Where superselection rules apply, it may be necessary to redefine  $U(\mathbf{1}, 0)$  by phase factors that depend on the sector on which it acts.

this as

$$U(\mathbf{1} + \omega, \epsilon) = \mathbf{1} + \frac{i}{2} \omega_{\rho\sigma} J^{\rho\sigma} - i \epsilon_{\rho} P^{\rho} + \dots \quad (2.4.3)$$

Here  $J^{\rho\sigma}$  and  $P^{\rho}$  are  $\omega$ - and  $\epsilon$ -independent operators, and the dots denote terms of higher order in  $\omega$  and/or  $\epsilon$ . In order for  $U(\mathbf{1} + \omega, \epsilon)$  to be unitary, the operators  $J^{\rho\sigma}$  and  $P^{\rho}$  must be Hermitian

$$J^{\rho\sigma\dagger} = J^{\rho\sigma}, \quad P^{\rho\dagger} = P^{\rho}. \quad (2.4.4)$$

Since  $\omega_{\rho\sigma}$  is antisymmetric, we can take its coefficient  $J^{\rho\sigma}$  to be antisymmetric also

$$J^{\rho\sigma} = -J^{\sigma\rho}. \quad (2.4.5)$$

As we shall see,  $P^1$ ,  $P^2$ , and  $P^3$  are the components of the momentum operators,  $J^{23}$ ,  $J^{31}$ , and  $J^{12}$  are the components of the angular momentum vector, and  $P^0$  is the energy operator, or Hamiltonian.<sup>6</sup>

We now examine the Lorentz transformation properties of  $J^{\rho\sigma}$  and  $P^{\rho}$ . We consider the product

$$U(\Lambda, a) U(\mathbf{1} + \omega, \epsilon) U^{-1}(\Lambda, a),$$

where  $\Lambda^{\mu}_{\nu}$  and  $a^{\mu}$  are here the parameters of a new transformation, unrelated to  $\omega$  and  $\epsilon$ . According to Eq. (2.3.11), the product  $U(\Lambda^{-1}, -\Lambda^{-1}a)U(\Lambda, a)$  equals  $U(\mathbf{1}, 0)$ , so  $U(\Lambda^{-1}, -\Lambda^{-1}a)$  is the inverse of  $U(\Lambda, a)$ . It follows then from (2.3.11) that

$$\begin{aligned} U(\Lambda, a) U(\mathbf{1} + \omega, \epsilon) U^{-1}(\Lambda, a) \\ = U(\Lambda(\mathbf{1} + \omega)\Lambda^{-1}, \Lambda\epsilon - \Lambda\omega\Lambda^{-1}a). \end{aligned} \quad (2.4.6)$$

To first order in  $\omega$  and  $\epsilon$ , we have then

$$\begin{aligned} U(\Lambda, a) \left[ \frac{1}{2} \omega_{\rho\sigma} J^{\rho\sigma} - \epsilon_{\rho} P^{\rho} \right] U^{-1}(\Lambda, a) \\ = \frac{1}{2} (\Lambda\omega\Lambda^{-1})_{\mu\nu} J^{\mu\nu} - (\Lambda\epsilon - \Lambda\omega\Lambda^{-1}a)_{\mu} P^{\mu}. \end{aligned} \quad (2.4.7)$$

Equating coefficients of  $\omega_{\rho\sigma}$  and  $\epsilon_{\rho}$  on both sides of this equation (and using (2.3.10)), we find

$$U(\Lambda, a) J^{\rho\sigma} U^{-1}(\Lambda, a) = \Lambda_{\mu}^{\rho} \Lambda_{\nu}^{\sigma} (J^{\mu\nu} - a^{\mu} P^{\nu} + a^{\nu} P^{\mu}), \quad (2.4.8)$$

$$U(\Lambda, a) P^{\rho} U^{-1}(\Lambda, a) = \Lambda_{\mu}^{\rho} P^{\mu}. \quad (2.4.9)$$

For homogeneous Lorentz transformations (with  $a^{\mu} = 0$ ), these transformation rules simply say that  $J^{\mu\nu}$  is a tensor and  $P^{\mu}$  is a vector. For pure translations (with  $\Lambda^{\mu}_{\nu} = \delta^{\mu}_{\nu}$ ), they tell us that  $P^{\rho}$  is translation-invariant, but  $J^{\rho\sigma}$  is not. In particular, the change of the space-space components of  $J^{\rho\sigma}$  under a spatial translation is just the usual change of the angular momentum under a change of the origin relative to which the angular momentum is calculated.

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<sup>6</sup>We will see that this identification of the angular-momentum generators is forced on us by the commutation relations of the  $J^{\mu\nu}$ . On the other hand, the commutation relations do not allow us to distinguish between  $P^{\mu}$  and  $-P^{\mu}$ , so the sign for the  $\epsilon_{\rho} P^{\rho}$  term in (2.4.3) is a matter of convention. The consistency of the choice in (2.4.3) with the usual definition of the Hamiltonian  $P^0$  is shown in Section 3.1.

Next, let's apply rules (2.4.8), (2.4.9) to a transformation that is itself infinitesimal, i.e.,  $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega'^\mu{}_\nu$  and  $a^\mu = \epsilon'^\mu$ , with infinitesimals  $\omega'^\mu{}_\nu$  and  $\epsilon'^\mu$  unrelated to the previous  $\omega$  and  $\epsilon$ . Using Eq. (2.4.3), and keeping only terms of first order in  $\omega'^\mu{}_\nu$  and  $\epsilon'^\mu$ , Eqs. (2.4.8) and (2.4.9) now become

$$i \left[ \frac{1}{2} \omega'_{\mu\nu} J^{\mu\nu} - \epsilon'_\mu P^\mu, J^{\rho\sigma} \right] = \omega'^\rho{}_\mu J^{\mu\sigma} + \omega'^\sigma{}_\nu J^{\rho\nu} - \epsilon'^\rho P^\sigma + \epsilon'^\sigma P^\rho, \quad (2.4.10)$$

$$i \left[ \frac{1}{2} \omega'_{\mu\nu} J^{\mu\nu} - \epsilon'_\mu P^\mu, P^\rho \right] = \omega'^\rho{}_\mu P^\mu. \quad (2.4.11)$$

Equating coefficients of  $\omega'_{\mu\nu}$  and  $\epsilon'_\mu$  on both sides of these equations, we find the commutation rules

$$i[J^{\mu\nu}, J^{\rho\sigma}] = \eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}, \quad (2.4.12)$$

$$i[P^\mu, J^{\rho\sigma}] = \eta^{\mu\rho} P^\sigma - \eta^{\mu\sigma} P^\rho, \quad (2.4.13)$$

$$[P^\mu, P^\rho] = 0. \quad (2.4.14)$$

This is the Lie algebra of the Poincaré group.

In quantum mechanics a special role is played by those operators that are *conserved*, i.e., that commute with the energy operator  $H = P^0$ . Inspection of Eqs. (2.4.13) and (2.4.14) shows that these are the momentum three-vector

$$\mathbf{P} = \{P^1, P^2, P^3\} \quad (2.4.15)$$

and the angular-momentum three-vector

$$\mathbf{J} = \{J^{23}, J^{31}, J^{12}\} \quad (2.4.16)$$

and, of course, the energy  $P^0$  itself. The remaining generators form what is called the 'boost' three-vector

$$\mathbf{K} = \{J^{10}, J^{20}, J^{30}\}. \quad (2.4.17)$$

These are *not* conserved, which is why we do not use the eigenvalues of  $\mathbf{K}$  to label physical states. In a three-dimensional notation, the commutation relations (2.4.12), (2.4.13), (2.4.14) may be written

$$[J_i, J_j] = i\epsilon_{ijk} J_k, \quad (2.4.18)$$

$$[J_i, K_j] = i\epsilon_{ijk} K_k, \quad (2.4.19)$$

$$[K_i, K_j] = -i\epsilon_{ijk} J_k, \quad (2.4.20)$$

$$[J_i, P_j] = i\epsilon_{ijk} P_k, \quad (2.4.21)$$

$$[K_i, P_j] = iH\delta_{ij}, \quad (2.4.22)$$

$$[J_i, H] = [P_i, H] = [H, H] = 0, \quad (2.4.23)$$

$$[K_i, H] = iP_i, \quad (2.4.24)$$

where  $i, j, k$ , etc. run over the values 1, 2, and 3, and  $\epsilon_{ijk}$  is the totally antisymmetric quantity with  $\epsilon_{123} = +1$ . The commutation relation (2.4.18) will be recognized as that of the angular-momentum operator.



The pure translations  $T(\mathbf{1}, a)$  form a subgroup of the Poincaré group with a group multiplication rule given by (2.3.7) as

$$T(\mathbf{1}, \bar{a})T(\mathbf{1}, a) = T(\mathbf{1}, \bar{a} + a). \quad (2.4.25)$$

This is additive in the same sense as (2.2.24), so by using (2.4.3) and repeating the same arguments that led to (2.2.26), we find that finite translations are represented on the physical Hilbert space by

$$U(\mathbf{1}, a) = \exp(-iP^\mu a_\mu). \quad (2.4.26)$$

In exactly the same way, we can show that a rotation  $R_\theta$  by an angle  $|\theta|$  around the direction of  $\theta$  is represented on the physical Hilbert space by

$$U(R_\theta, 0) = \exp(i\mathbf{J} \cdot \theta). \quad (2.4.27)$$

It is interesting to compare the Poincaré algebra with the Lie algebra of the symmetry group of Newtonian mechanics, the Galilean group. We could derive this algebra by starting with the transformation rules of the Galilean group and then following the same procedure that was used here to derive the Poincaré algebra. However, since we already have Eqs. (2.4.18)–(2.4.24), it is easier to obtain the Galilean algebra as a low-velocity limit of the Poincaré algebra, by what is known as an *Inönü–Wigner contraction* [4, 5]. For a system of particles of typical mass  $m$  and typical velocity  $v$ , the momentum and the angular-momentum operators are expected to be of order  $J \sim 1$ ,  $P \sim mv$ . On the other hand, the energy operator is  $H = M + W$  with a total mass  $M$  and non-mass energy  $W$  (kinetic plus potential) of order  $M \sim m$ ,  $W \sim mv^2$ . Inspection of Eqs. (2.4.18)–(2.4.24) shows that these commutation relations have a limit for  $v \ll 1$  of the form

$$\begin{aligned} [J_i, J_j] &= i\epsilon_{ijk}J_k, & [J_i, K_j] &= i\epsilon_{ijk}K_k, & [K_i, K_j] &= 0, \\ [J_i, P_j] &= i\epsilon_{ijk}P_k, & [K_i, P_j] &= iM\delta_{ij}, \\ [J_i, W] &= [P_i, W] = 0, & [K_i, W] &= iP_i, \\ [J_i, M] &= [P_i, M] = [K_i, M] = [W, M] = 0, \end{aligned}$$

with  $K$  of order  $1/v$ . Note that the product of a translation  $\mathbf{x} \rightarrow \mathbf{x} + \mathbf{a}$  and a ‘boost’  $\mathbf{x} \rightarrow \mathbf{x} + \mathbf{v}t$  should be the transformation  $\mathbf{x} \rightarrow \mathbf{x} + \mathbf{v}t + \mathbf{a}$ , but this is not true for the action of these operators on Hilbert space:

$$\exp(i\mathbf{K} \cdot \mathbf{v}) \exp(-i\mathbf{P} \cdot \mathbf{a}) = \exp(iM\mathbf{a} \cdot \mathbf{v}/2) \exp(i(\mathbf{K} \cdot \mathbf{v} - \mathbf{P} \cdot \mathbf{a})).$$

The appearance of the phase factor  $\exp(iM\mathbf{a} \cdot \mathbf{v}/2)$  shows that this is a projective representation, with a superselection rule forbidding the superposition of states of different mass. In this respect, the mathematics of the Poincaré group is simpler than that of the Galilean group. However, there is nothing to prevent us from formally enlarging the Galilean group, by adding one more generator to its Lie algebra, which commutes with all the other generators, and whose eigenvalues are the masses of the various states. In this case physical states provide an ordinary rather than a projective representation of the expanded symmetry group. The difference appears to be a mere matter of notation, except that with this reinterpretation of the Galilean group there is no need for a mass superselection rule.

## 2.5 One-Particle States

We now consider the classification of one-particle states according to their transformation under the Poincaré group.

The components of the energy-momentum four-vector all commute with each other, so it is natural to express physical state-vectors in terms of eigenvectors of the four-momentum. Introducing a label  $\sigma$  to denote all other degrees of freedom, we thus consider state-vectors  $|p, \sigma\rangle$  with

$$P^\mu |p, \sigma\rangle = p^\mu |p, \sigma\rangle. \quad (2.5.1)$$

For general states, consisting for instance of several unbound particles, the label  $\sigma$  would have to be allowed to include continuous as well as discrete labels. We take as part of the definition of a *one-particle* state, that the label  $\sigma$  is purely discrete, and will limit ourselves here to that case. (However, a specific bound state of two or more particles, such as the lowest state of the hydrogen atom, is to be considered as a one-particle state. It is not an *elementary* particle, but the distinction between composite and elementary particles is of no relevance here.)

Eqs. (2.5.1) and (2.4.26) tell us how the states  $|p, \sigma\rangle$  transform under translations:

$$U(\mathbf{1}, a)|p, \sigma\rangle = e^{-ip \cdot a}|p, \sigma\rangle.$$

We must now consider how these states transform under homogeneous Lorentz transformations.

Using (2.4.9), we see that the effect of operating on  $|p, \sigma\rangle$  with a quantum homogeneous Lorentz transformation  $U(\Lambda, 0) \equiv U(\Lambda)$  is to produce an eigenvector of the four-momentum with eigenvalue  $\Lambda p$

$$\begin{aligned} P^\mu U(\Lambda)|p, \sigma\rangle &= U(\Lambda) [U^{-1}(\Lambda) P^\mu U(\Lambda)] |p, \sigma\rangle \\ &= U(\Lambda) (\Lambda^{-1\mu}{}_\rho P^\rho) |p, \sigma\rangle \\ &= \Lambda^\mu{}_\rho p^\rho U(\Lambda) |p, \sigma\rangle. \end{aligned} \quad (2.5.2)$$

Hence  $U(\Lambda)|p, \sigma\rangle$  must be a linear combination of the state-vectors  $|\Lambda p, \sigma'\rangle$ :

$$U(\Lambda)|p, \sigma\rangle = \sum_{\sigma'} C_{\sigma'\sigma}(\Lambda, p) |\Lambda p, \sigma'\rangle. \quad (2.5.3)$$

In general, it may be possible by using suitable linear combinations of the  $|p, \sigma\rangle$  to choose the  $\sigma$  labels in such a way that the matrix  $C_{\sigma'\sigma}(\Lambda, p)$  is block-diagonal; in other words, so that the  $|p, \sigma\rangle$  with  $\sigma$  within any one block by themselves furnish a representation of the Poincaré group. It is natural to identify the states of a specific particle type with the components of a representation of the Poincaré group which is irreducible, in the sense that it cannot be further decomposed in this way.<sup>7</sup> Our task now is to work out the structure

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<sup>7</sup>Of course, different particle species may correspond to representations that are isomorphic, i.e., that have matrices  $C_{\sigma'\sigma}(\Lambda, p)$  that are either identical, or identical up to a similarity transformation. In some cases it may be convenient to define particle types as irreducible representations of larger groups that contain the inhomogeneous proper orthochronous Lorentz group as a subgroup; for instance, as we shall see, for massless particles whose interactions respect the symmetry of space inversion it is customary to treat all the components of an irreducible representation of the Poincaré group including space inversion as a single particle type.

of the coefficients  $C_{\sigma'\sigma}(\Lambda, p)$  in irreducible representations of the Poincaré group.

For this purpose, note that the only functions of  $p^\mu$  that are left invariant by all proper orthochronous Lorentz transformations  $\Lambda^\mu{}_\nu$ , are the invariant square  $p^2 \equiv \eta_{\mu\nu} p^\mu p^\nu$ , and for  $p^2 \leq 0$ , also the sign of  $p^0$ . Hence, for each value of  $p^2$ , and (for  $p^2 \leq 0$ ) each sign of  $p^0$ , we can choose a ‘standard’ four-momentum, say  $k^\mu$ , and express any  $p^\mu$  of this class as

$$p^\mu = L^\mu{}_\nu(p) k^\nu, \quad (2.5.4)$$

where  $L^\mu{}_\nu$  is some standard Lorentz transformation that depends on  $p^\mu$ , and also implicitly on our choice of the standard  $k^\mu$ . We can then *define* the states  $|p, \sigma\rangle$  of momentum  $p$  by

$$|p, \sigma\rangle \equiv N(p) U(L(p)) |k, \sigma\rangle, \quad (2.5.5)$$

where  $N(p)$  is a numerical normalization factor, to be chosen later. Up to this point, we have said nothing about how the  $\sigma$  labels are related for different momenta; Eq. (2.5.5) now fills that gap.

Operating on (2.5.5) with an arbitrary homogeneous Lorentz transformation  $U(\Lambda)$ , we now find

$$\begin{aligned} U(\Lambda) |p, \sigma\rangle &= N(p) U(\Lambda L(p)) |k, \sigma\rangle \\ &= N(p) U(L(\Lambda p)) U(L^{-1}(\Lambda p) \Lambda L(p)) |k, \sigma\rangle. \end{aligned} \quad (2.5.6)$$

The point of this last step is that the Lorentz transformation  $L^{-1}(\Lambda p) \Lambda L(p)$  takes  $k$  to  $L(p)k = p$ , and then to  $\Lambda p$ , and then back to  $k$ , so it belongs to the subgroup of the homogeneous Lorentz group consisting of Lorentz transformations  $W^\mu{}_\nu$  that leave  $k^\mu$  invariant:

$$W^\mu{}_\nu k^\nu = k^\mu. \quad (2.5.7)$$

This subgroup is called the *little group* [5]. For any  $W$  satisfying Eq. (2.5.7), we have

$$U(W) |k, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}(W) |k, \sigma'\rangle. \quad (2.5.8)$$

The coefficients  $D(W)$  furnish a representation of the little group; that is, for any elements  $\bar{W}, W$  we have

$$\begin{aligned} \sum_{\sigma'} D_{\sigma'\sigma}(\bar{W}W) |k, \sigma'\rangle &= U(\bar{W}W) |k, \sigma\rangle = U(\bar{W}) U(W) |k, \sigma\rangle \\ &= U(\bar{W}) \sum_{\sigma''} D_{\sigma''\sigma}(W) |k, \sigma''\rangle \\ &= \sum_{\sigma'\sigma''} D_{\sigma'\sigma''}(\bar{W}) D_{\sigma''\sigma}(W) |k, \sigma'\rangle \end{aligned}$$

and so

$$D_{\sigma'\sigma}(\bar{W}W) = \sum_{\sigma''} D_{\sigma'\sigma''}(\bar{W}) D_{\sigma''\sigma}(W). \quad (2.5.9)$$

In particular, we may apply Eq. (2.5.8) to the little-group transformation

$$W(\Lambda, p) \equiv L^{-1}(\Lambda p) \Lambda L(p) \quad (2.5.10)$$

and then Eq. (2.5.6) takes the form

$$U(\Lambda)|p, \sigma\rangle = N(p) \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) U(L(\Lambda p))|k, \sigma'\rangle,$$

or, recalling the definition (2.5.5):

$$U(\Lambda)|p, \sigma\rangle = \left( \frac{N(p)}{N(\Lambda p)} \right) \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle. \quad (2.5.11)$$

Apart from the question of normalization, the problem of determining the coefficients  $C_{\sigma'\sigma}$  in the transformation rule (2.5.3) has been reduced to the problem of finding the representations of the little group. This approach, of deriving representations of a group like the Poincaré group from the representations of a little group, is called the method of induced representations [6].

Table 2.1 gives a convenient choice of the standard momentum  $k^\mu$  and the corresponding little group for the various classes of four-momenta.

Of these six classes of four-momenta, only (a), (c), and (f) have any known interpretations in terms of physical states. Not much needs to be said here about case (f)— $p^\mu = 0$ ; it describes the vacuum, which is simply left invariant by  $U(\Lambda)$ . In what follows we will consider only cases (a) and (c), which cover particles of mass  $M > 0$  and mass zero, respectively.

This is a good place to pause, and say something about the normalization of these states. By the usual orthonormalization procedure of quantum mechanics, we may choose the states with standard momentum  $k^\mu$  to be orthonormal, in the sense that

$$\langle k', \sigma' | k, \sigma \rangle = \delta^3(\mathbf{k}' - \mathbf{k}) \delta_{\sigma'\sigma}. \quad (2.5.12)$$

(The delta function appears here because  $|k, \sigma\rangle$  and  $|k', \sigma'\rangle$  are eigenstates of a Hermitian operator with eigenvalues  $\mathbf{k}$  and  $\mathbf{k}'$ , respectively.) This has the immediate consequence that the representation of the little group in Eqs. (2.5.8) and (2.5.11) must be unitary<sup>8</sup>

$$D^\dagger(W) = D^{-1}(W). \quad (2.5.13)$$

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<sup>8</sup>The little groups  $SO(2, 1)$  and  $SO(3, 1)$  for  $p^2 > 0$  and  $p^\mu = 0$  have no non-trivial finite-dimensional unitary representations, so if there were any states with a given momentum  $p^\mu$  with  $p^2 > 0$  or  $p^\mu = 0$  that transform non-trivially under the little group, there would have to be an infinite number of them.

Table 2.1. Standard momenta and the corresponding little group for various classes of four-momenta. Here  $\kappa$  is an arbitrary positive energy, say 1 eV. The little groups are mostly pretty obvious:  $SO(3)$  is the ordinary rotation group in three dimensions (excluding space inversions), because rotations are the only proper orthochronous Lorentz transformations that leave at rest a particle with zero momentum, while  $SO(2, 1)$  and  $SO(3, 1)$  are the Lorentz groups in  $(2 + 1)$ - and  $(3 + 1)$ -dimensions, respectively. The group  $ISO(2)$  is the group of Euclidean geometry, consisting of rotations and translations in two dimensions. Its appearance as the little group for  $p^2 = 0$  is explained below.

|                               | Standard $k^\mu$          | Little Group |
|-------------------------------|---------------------------|--------------|
| (a) $p^2 = -M^2 < 0, p^0 > 0$ | $(M, 0, 0, 0)$            | $SO(3)$      |
| (b) $p^2 = -M^2 < 0, p^0 < 0$ | $(-M, 0, 0, 0)$           | $SO(3)$      |
| (c) $p^2 = 0, p^0 > 0$        | $(\kappa, 0, 0, \kappa)$  | $ISO(2)$     |
| (d) $p^2 = 0, p^0 < 0$        | $(-\kappa, 0, 0, \kappa)$ | $ISO(2)$     |
| (e) $p^2 = N^2 > 0$           | $(0, 0, 0, N)$            | $SO(2, 1)$   |
| (f) $p^\mu = 0$               | $(0, 0, 0, 0)$            | $SO(3, 1)$   |

Now, what about scalar products for arbitrary momenta? Using the unitarity of the operator  $U(\Lambda)$  in Eqs. (2.5.5) and (2.5.11), we find for the scalar product:

$$\begin{aligned}\langle p', \sigma' | p, \sigma \rangle &= N(p) \langle U^{-1}(L(p)) p', \sigma' | k, \sigma \rangle \\ &= N(p) N^*(p') D_{\sigma\sigma'}(W(L^{-1}(p), p'))^* \delta^3(\mathbf{k}' - \mathbf{k}),\end{aligned}$$

where  $k' \equiv L^{-1}(p)p'$ . Since also  $k = L^{-1}(p)p$ , the delta function  $\delta^3(\mathbf{k} - \mathbf{k}')$  is proportional to  $\delta^3(\mathbf{p} - \mathbf{p}')$ . For  $p' = p$ , the little-group transformation here is trivial,  $W(L^{-1}(p), p) = \mathbf{1}$ , and so the scalar product is

$$\langle p', \sigma' | p, \sigma \rangle = |N(p)|^2 \delta_{\sigma'\sigma} \delta^3(\mathbf{k}' - \mathbf{k}). \quad (2.5.14)$$

It remains to work out the proportionality factor relating  $\delta^3(\mathbf{k} - \mathbf{k}')$  and  $\delta^3(\mathbf{p} - \mathbf{p}')$ . Note that the Lorentz-invariant integral of an arbitrary scalar function  $f(p)$  over four-momenta with  $-p^2 = M^2 \geq 0$  and  $p^0 > 0$  (i.e., cases (a) or (c)) may be written

$$\begin{aligned}\int d^4p \delta(p^2 + M^2) \theta(p^0) f(p) &= \int d^3\mathbf{p} dp^0 \delta((p^0)^2 - \mathbf{p}^2 - M^2) \theta(p^0) f(\mathbf{p}, p^0) \\ &= \int d^3\mathbf{p} \frac{f(\mathbf{p}, \sqrt{\mathbf{p}^2 + M^2})}{2\sqrt{\mathbf{p}^2 + M^2}}.\end{aligned}$$

( $\theta(p^0)$  is the step function:  $\theta(x) = 1$  for  $x \geq 0$ ,  $\theta(x) = 0$  for  $x < 0$ .) We see that when integrating on the ‘mass shell’  $p^2 + M^2 = 0$ , the invariant volume element is

$$d^3\mathbf{p} / \sqrt{\mathbf{p}^2 + M^2}. \quad (2.5.15)$$

The delta function is defined by

$$\begin{aligned} F(\mathbf{p}) &= \int F(\mathbf{p}') \delta^3(\mathbf{p} - \mathbf{p}') d^3\mathbf{p}' \\ &= \int F(\mathbf{p}') \left[ \sqrt{\mathbf{p}'^2 + M^2} \delta^3(\mathbf{p}' - \mathbf{p}) \right] \frac{d^3\mathbf{p}'}{\sqrt{\mathbf{p}'^2 + M^2}}, \end{aligned}$$

so we see that the invariant delta function is

$$\sqrt{\mathbf{p}'^2 + M^2} \delta^3(\mathbf{p}' - \mathbf{p}) = p^0 \delta^3(\mathbf{p}' - \mathbf{p}). \quad (2.5.16)$$

Since  $p'$  and  $p$  are related to  $k'$  and  $k$  respectively by a Lorentz transformation,  $L(p)$ , we have then

$$p^0 \delta^3(\mathbf{p}' - \mathbf{p}) = k^0 \delta^3(\mathbf{k}' - \mathbf{k})$$

and therefore

$$\langle p', \sigma' | p, \sigma \rangle = |N(p)|^2 \delta_{\sigma'\sigma} \left( \frac{p^0}{k^0} \right) \delta^3(\mathbf{p}' - \mathbf{p}). \quad (2.5.17)$$

The normalization factor  $N(p)$  is sometimes chosen to be just  $N(p) = 1$ , but then we would need to keep track of the  $p^0/k^0$  factor in scalar products. Instead, I will here adopt the more usual convention that

$$N(p) = \sqrt{k^0/p^0} \quad (2.5.18)$$

for which

$$\langle p', \sigma' | p, \sigma \rangle = \delta_{\sigma'\sigma} \delta^3(\mathbf{p}' - \mathbf{p}). \quad (2.5.19)$$

We now consider the two cases of physical interest: particles of mass  $M > 0$ , and particles of zero mass.

### Mass Positive-Definite

The little group here is the three-dimensional rotation group. Its unitary representations can be broken up into a direct sum of irreducible unitary representations [7]  $D_{\sigma'\sigma}^{(j)}(R)$  of dimensionality  $2j+1$ , with  $j = 0, \frac{1}{2}, 1, \dots$ . These can be built up from the standard matrices for infinitesimal rotations  $R_{ik} = \delta_{ik} + \Theta_{ik}$ , with  $\Theta_{ik} = -\Theta_{ki}$  infinitesimal:

$$D_{\sigma'\sigma}^{(j)}(\mathbf{1} + \Theta) = \delta_{\sigma'\sigma} + \frac{i}{2} \Theta_{ik} (J_{ik}^{(j)})_{\sigma'\sigma}, \quad (2.5.20)$$

$$\begin{aligned} (J_{23}^{(j)} \pm iJ_{31}^{(j)})_{\sigma'\sigma} &= (J_1^{(j)} \pm iJ_2^{(j)})_{\sigma'\sigma} \\ &= \delta_{\sigma', \sigma \pm 1} \sqrt{(j \mp \sigma)(j \pm \sigma + 1)}, \end{aligned} \quad (2.5.21)$$

$$(J_{12}^{(j)})_{\sigma'\sigma} = (J_3^{(j)})_{\sigma'\sigma} = \sigma \delta_{\sigma'\sigma}, \quad (2.5.22)$$

with  $\sigma$  running over the values  $j, j-1, \dots, -j$ . For a particle of mass  $M > 0$  and spin  $j$ , Eq. (2.5.11) now becomes

$$U(\Lambda)|p, \sigma\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}} \sum_{\sigma'} D_{\sigma'\sigma}^{(j)}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle, \quad (2.5.23)$$

with the little-group element  $W(\Lambda, p)$  (the Wigner rotation [5]) given by Eq. (2.5.10):

$$W(\Lambda, p) = L^{-1}(\Lambda p) \Lambda L(p).$$

To calculate this rotation, we need to choose a ‘standard boost’  $L(p)$  which carries the four-momentum from  $k^\mu = (M, \mathbf{0})$  to  $p^\mu$ . This is conveniently chosen as

$$\begin{aligned} L^i_k(p) &= \delta_{ik} + (\gamma - 1) \hat{p}_i \hat{p}_k, \\ L^i_0(p) &= L^0_i(p) = \hat{p}_i \sqrt{\gamma^2 - 1}, \\ L^0_0(p) &= \gamma, \end{aligned} \tag{2.5.24}$$

where

$$\hat{p}_i \equiv p_i / |\mathbf{p}|, \quad \gamma \equiv \sqrt{\mathbf{p}^2 + M^2} / M.$$

It is very important that when  $\Lambda^\mu_\nu$  is an arbitrary three-dimensional rotation  $\mathcal{R}$ , the Wigner rotation  $W(\Lambda, p)$  is the same as  $\mathcal{R}$  for all  $p$ . To see this, note that the boost (2.5.24) may be expressed as

$$L(p) = R(\hat{\mathbf{p}}) B(|\mathbf{p}|) R^{-1}(\hat{\mathbf{p}}),$$

where  $R(\hat{\mathbf{p}})$  is a rotation (to be defined in a standard way below, in Eq. (2.5.47)) that takes the three-axis into the direction of  $\mathbf{p}$ , and

$$B(|\mathbf{p}|) = \begin{pmatrix} \gamma & 0 & 0 & \sqrt{\gamma^2 - 1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sqrt{\gamma^2 - 1} & 0 & 0 & \gamma \end{pmatrix}.$$

Then for an arbitrary rotation  $\mathcal{R}$

$$W(\mathcal{R}, p) = R(\mathcal{R}\hat{\mathbf{p}}) B^{-1}(|\mathbf{p}|) R^{-1}(\mathcal{R}\hat{\mathbf{p}}) \mathcal{R} R(\hat{\mathbf{p}}) B(|\mathbf{p}|) R^{-1}(\hat{\mathbf{p}}).$$

But the rotation  $R^{-1}(\mathcal{R}\hat{\mathbf{p}}) \mathcal{R} R(\hat{\mathbf{p}})$  takes the three-axis into the direction  $\hat{\mathbf{p}}$ , and then into the direction  $\mathcal{R}\hat{\mathbf{p}}$ , and then back to the three-axis, so it must be just a rotation by some angle  $\theta$  around the three-axis

$$R^{-1}(\mathcal{R}\hat{\mathbf{p}}) \mathcal{R} R(\hat{\mathbf{p}}) = R(\theta) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & \sin \theta & 0 \\ 0 & -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Since  $R(\theta)$  commutes with  $B(|\mathbf{p}|)$ , this now gives

$$\begin{aligned} W(\mathcal{R}, p) &= R(\mathcal{R}\hat{\mathbf{p}}) B^{-1}(|\mathbf{p}|) R(\theta) B(|\mathbf{p}|) R^{-1}(\hat{\mathbf{p}}) \\ &= R(\mathcal{R}\hat{\mathbf{p}}) R(\theta) R^{-1}(\hat{\mathbf{p}}) \end{aligned}$$

and hence

$$W(\mathcal{R}, p) = \mathcal{R}$$

as was to be shown. Thus states of a moving massive particle (and, by extension, multi-particle states) have the same transformation under rotations as in non-relativistic quantum mechanics. This is another piece of good news—the whole apparatus of spherical harmonics, Clebsch–Gordan coefficients, etc. can be carried over wholesale from non-relativistic to relativistic quantum mechanics.

### Mass Zero

First, we have to work out the structure of the little group. Consider an arbitrary little-group element  $W^\mu{}_\nu$ , with  $W^\mu{}_\nu k^\nu = k^\mu$ , where  $k^\mu$  is the standard four-momentum for this case,  $k^\mu = (1, 0, 0, 1)$ . Acting on a time-like four-vector  $t^\mu = (1, 0, 0, 0)$ , such a Lorentz transformation must yield a four-vector  $Wt$  whose length and scalar product with  $Wk = k$  are the same as those of  $t$ :

$$(Wt)^\mu (Wt)_\mu = t^\mu t_\mu = -1, \quad (Wt)^\mu k_\mu = t^\mu k_\mu = -1.$$

Any four-vector that satisfies the second condition may be written

$$(Wt)^\mu = (1 + \zeta, \alpha, \beta, \zeta)$$

and the first condition then yields the relation

$$\zeta = (\alpha^2 + \beta^2)/2. \quad (2.5.25)$$

It follows that the effect of  $W^\mu{}_\nu$  on  $t^\nu$  is the same as that of the Lorentz transformation

$$S^\mu{}_\nu(\alpha, \beta) = \begin{pmatrix} 1 + \zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1 - \zeta \end{pmatrix}. \quad (2.5.26)$$

This does not mean that  $W$  equals  $S(\alpha, \beta)$ , but it does mean that  $S^{-1}(\alpha, \beta)W$  is a Lorentz transformation that leaves the time-like four-vector  $(1, 0, 0, 0)$  invariant, and is therefore a pure rotation. Also,  $S^\mu{}_\nu$  like  $W^\mu{}_\nu$  leaves the light-like four-vector  $(1, 0, 0, 1)$  invariant, so  $S^{-1}(\alpha, \beta)W$  must be a rotation by some angle  $\theta$  around the three-axis

$$S^{-1}(\alpha, \beta)W = R(\theta), \quad (2.5.27)$$

where

$$R^\mu{}_\nu(\theta) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & \sin \theta & 0 \\ 0 & -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The most general element of the little group is therefore of the form

$$W(\theta, \alpha, \beta) = S(\alpha, \beta)R(\theta). \quad (2.5.28)$$

What group is this? We note that the transformations with  $\theta = 0$  or with  $\alpha = \beta = 0$  form subgroups:

$$S(\bar{\alpha}, \bar{\beta})S(\alpha, \beta) = S(\bar{\alpha} + \alpha, \bar{\beta} + \beta) \quad (2.5.29)$$

$$R(\bar{\theta})R(\theta) = R(\bar{\theta} + \theta). \quad (2.5.30)$$



These subgroups are *Abelian*—that is, their elements all commute with each other. Furthermore, the subgroup with  $\theta = 0$  is *invariant*, in the sense that its elements are transformed into other elements of the same subgroup by any member of the group

$$R(\theta)S(\alpha, \beta)R^{-1}(\theta) = S(\alpha \cos \theta + \beta \sin \theta, -\alpha \sin \theta + \beta \cos \theta). \quad (2.5.31)$$

From Eqs. (2.5.29)–(2.5.31) we can work out the product of any group elements. The reader will recognize these multiplication rules as those of the group  $ISO(2)$ , consisting of translations (by a vector  $(\alpha, \beta)$ ) and rotations (by an angle  $\theta$ ) in two dimensions.

Groups that do *not* have invariant Abelian subgroups have certain simple properties, and for this reason are called *semi-simple*. As we have seen, the little group  $ISO(2)$  like the Poincaré group is *not* semi-simple, and this leads to interesting complications. First, let's take a look at the Lie algebra of  $ISO(2)$ . For  $\theta, \alpha, \beta$  infinitesimal, the general group element is

$$W(\theta, \alpha, \beta)^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad \omega_{\mu\nu} = \begin{pmatrix} 0 & -\alpha & -\beta & 0 \\ \alpha & 0 & \theta & -\alpha \\ \beta & -\theta & 0 & -\beta \\ 0 & \alpha & \beta & 0 \end{pmatrix}.$$

From (2.4.3), we see then that the corresponding Hilbert space operator is

$$U(W(\theta, \alpha, \beta)) = \mathbf{1} + i\alpha A + i\beta B + i\theta J_3, \quad (2.5.32)$$

where  $A$  and  $B$  are the Hermitian operators

$$A = -J^{13} + J^{10} = J_2 + K_1, \quad (2.5.33)$$

$$B = -J^{23} + J^{20} = -J_1 + K_2, \quad (2.5.34)$$

and, as before,  $J_3 = J_{12}$ . Either from (2.4.18)–(2.4.20), or directly from Eqs. (2.5.29)–(2.5.31), we see that these generators have the commutators

$$[J_3, A] = +iB, \quad (2.5.35)$$

$$[J_3, B] = -iA, \quad (2.5.36)$$

$$[A, B] = 0. \quad (2.5.37)$$

Since  $A$  and  $B$  are commuting Hermitian operators they (like the momentum generators of the Poincaré group) can be simultaneously diagonalized by states  $|k, a, b\rangle$

$$A|k, a, b\rangle = a|k, a, b\rangle, \quad B|k, a, b\rangle = b|k, a, b\rangle.$$

The problem is that if we find one such set of non-zero eigenvalues of  $A, B$ , then we find a whole continuum. From Eq. (2.5.31), we have

$$U[R(\theta)]A U^{-1}[R(\theta)] = A \cos \theta - B \sin \theta,$$

$$U[R(\theta)]B U^{-1}[R(\theta)] = A \sin \theta + B \cos \theta,$$

and so, for arbitrary  $\theta$ ,

$$\begin{aligned} A|k, a, b\rangle^\theta &= (a \cos \theta - b \sin \theta)|k, a, b\rangle^\theta, \\ B|k, a, b\rangle^\theta &= (a \sin \theta + b \cos \theta)|k, a, b\rangle^\theta, \end{aligned}$$

where

$$|k, a, b\rangle^\theta \equiv U^{-1}(R(\theta))|k, a, b\rangle.$$

Massless particles are not observed to have any continuous degree of freedom like  $\theta$ ; to avoid such a continuum of states, we must require that physical states (now called  $|k, \sigma\rangle$ ) are eigenvectors of  $A$  and  $B$  with  $a = b = 0$ :

$$A|k, \sigma\rangle = B|k, \sigma\rangle = 0. \quad (2.5.38)$$

These states are then distinguished by the eigenvalue of the remaining generator

$$J_3|k, \sigma\rangle = \sigma|k, \sigma\rangle. \quad (2.5.39)$$

Since the momentum  $\mathbf{k}$  is in the three-direction,  $\sigma$  gives the component of angular momentum in the direction of motion, or *helicity*.

We are now in a position to calculate the Lorentz transformation properties of general massless particle states. First note that by use of the general arguments of Section 2.2, Eq. (2.5.32) generalizes for finite  $\alpha$  and  $\beta$  to

$$U(S(\alpha, \beta)) = \exp(i\alpha A + i\beta B) \quad (2.5.40)$$

and for finite  $\theta$  to

$$U(R(\theta)) = \exp(iJ_3\theta). \quad (2.5.41)$$

An arbitrary element  $W$  of the little group can be put in the form (2.5.28), so that

$$U(W)|k, \sigma\rangle = \exp(i\alpha A + i\beta B) \exp(i\theta J_3)|k, \sigma\rangle = \exp(i\theta\sigma)|k, \sigma\rangle$$

and therefore Eq. (2.5.8) gives

$$D_{\sigma'\sigma}(W) = \exp(i\theta\sigma)\delta_{\sigma'\sigma},$$

where  $\theta$  is the angle defined by expressing  $W$  as in Eq. (2.5.28). The Lorentz transformation rule for a massless particle of arbitrary helicity is now given by Eqs. (2.5.11) and (2.5.18) as

$$U(\Lambda)|p, \sigma\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}} \exp(i\sigma\theta(\Lambda, p))|\Lambda p, \sigma\rangle \quad (2.5.42)$$

with  $\theta(\Lambda, p)$  defined by

$$W(\Lambda, p) \equiv L^{-1}(\Lambda p)\Lambda L(p) \equiv S(\alpha(\Lambda, p), \beta(\Lambda, p))R(\theta(\Lambda, p)). \quad (2.5.43)$$

We shall see in Section 5.9 that electromagnetic gauge invariance arises from the part of the little group parameterized by  $\alpha$  and  $\beta$ .

At this point we have not yet encountered any reason that would forbid the helicity  $\sigma$  of a massless particle from being an arbitrary real number. As we shall see in Section 2.7, there are topological considerations that restrict the allowed values of  $\sigma$  to integers and half-integers, just as for massive particles.

To calculate the little-group element (2.5.43) for a given  $\Lambda$  and  $p$ , (and also to enable us to calculate the effect of space or time inversion on these states in the next section) we need to fix a convention for the standard Lorentz transformation that takes us from  $k^\mu = (\kappa, 0, 0, \kappa)$  to  $p^\mu$ . This may conveniently be chosen to have the form

$$L(p) = R(\hat{\mathbf{p}})B(|\mathbf{p}|/\kappa) \quad (2.5.44)$$

where  $B(u)$  is a pure boost along the three-direction:

$$B(u) \equiv \begin{pmatrix} (u^2 + 1)/2u & 0 & 0 & (u^2 - 1)/2u \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (u^2 - 1)/2u & 0 & 0 & (u^2 + 1)/2u \end{pmatrix} \quad (2.5.45)$$

and  $R(\hat{\mathbf{p}})$  is a pure rotation that carries the three-axis into the direction of the unit vector  $\hat{\mathbf{p}}$ . For instance, suppose we take  $\hat{\mathbf{p}}$  to have polar and azimuthal angles  $\theta$  and  $\phi$ :

$$\hat{\mathbf{p}} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta). \quad (2.5.46)$$

Then we can take  $R(\hat{\mathbf{p}})$  as a rotation by angle  $\theta$  around the two-axis, which takes  $(0, 0, 1)$  into  $(\sin \theta, 0, \cos \theta)$ , followed by a rotation by angle  $\phi$  around the three-axis:

$$U(R(\hat{\mathbf{p}})) = \exp(i\phi J_3) \exp(i\theta J_2), \quad (2.5.47)$$

where  $0 \leq \theta \leq \pi$ ,  $0 \leq \phi < 2\pi$ . (We give  $U(R(\hat{\mathbf{p}}))$  rather than  $R(\hat{\mathbf{p}})$ , together with a specification of the range of  $\phi$  and  $\theta$ , because shifting  $\theta$  or  $\phi$  by  $2\pi$  would give the same rotation  $R(\hat{\mathbf{p}})$ , but a different sign for  $U(R(\hat{\mathbf{p}}))$  when acting on half-integer spin states.) Since (2.5.47) is a rotation, and does take the three-axis into the direction (2.5.46), any other choice of such an  $R(\hat{\mathbf{p}})$  would differ from this one by at most an initial rotation around the three-axis, corresponding to a mere redefinition of the phase of the one-particle states.

Note that the helicity is Lorentz-invariant; a massless particle of a given helicity  $\sigma$  looks the same (aside from its momentum) in all inertial frames. Indeed, we would be justified in thinking of massless particles of each different helicity as different species of particles. However, as we shall see in the next section, particles of opposite helicity are related by the symmetry of space inversion. Thus, because electromagnetic and gravitational forces obey space inversion symmetry, the massless particles of helicity  $\pm 1$  associated with electromagnetic phenomena are both called *photons*, and the massless particles of helicity  $\pm 2$  that are believed to be associated with gravitation are both called *gravitons*. On the other hand, the supposedly massless particles of helicity  $\pm 1/2$  that are emitted in nuclear beta decay have no interactions (apart from gravitation) that respect the symmetry of space inversion, so these particles are given different names: *neutrinos* for helicity  $+1/2$ , and *antineutrinos* for helicity  $-1/2$ .

Even though the helicity of a massless particle is Lorentz-invariant, the state itself is not. In particular, because of the helicity-dependent phase factor  $\exp(i\sigma\theta)$  in Eq. (2.5.42), a state formed as a linear superposition of one-particle states with opposite helicities will be changed by a Lorentz transformation into a different superposition. For instance, a general one-photon state of four-momenta may be written

$$|p; \alpha\rangle = \alpha_+ |p, +1\rangle + \alpha_- |p, -1\rangle,$$

where

$$|\alpha_+|^2 + |\alpha_-|^2 = 1.$$

The generic case is one of *elliptic polarization*, with  $|\alpha_\pm|$  both non-zero and unequal. *Circular polarization* is the limiting case where either  $\alpha_+$  or  $\alpha_-$  vanishes, and *linear polarization* is the opposite extreme, with  $|\alpha_+| = |\alpha_-|$ . The overall phase of  $\alpha_+$  and  $\alpha_-$  has no physical significance, and for linear polarization may be adjusted so that  $\alpha_- = \alpha_+^*$ , but the relative phase is still important. Indeed, for linear polarizations with  $\alpha_- = \alpha_+^*$ , the phase of  $\alpha_+$  may be identified as the angle between the plane of polarization and some fixed reference direction perpendicular to  $\mathbf{p}$ . Eq. (2.5.42) shows that under a Lorentz transformation  $\Lambda^\mu{}_\nu$ , this angle rotates by an amount  $\theta(\Lambda, p)$ . Plane polarized gravitons can be defined in a similar way, and here Eq. (2.5.42) has the consequence that a Lorentz transformation  $\Lambda$  rotates the plane of polarization by an angle  $2\theta(\Lambda, p)$ .

## 2.6 Space Inversion and Time-Reversal

We saw in Section 2.3 that any homogeneous Lorentz transformation is either proper and orthochronous (i.e.,  $\text{Det } \Lambda = +1$  and  $\Lambda^0{}_0 \geq +1$ ) or else equal to a proper orthochronous transformation times either  $\mathcal{P}$  or  $\mathcal{T}$  or  $\mathcal{PT}$ , where  $\mathcal{P}$  and  $\mathcal{T}$  are the space inversion and time-reversal transformations

$$\mathcal{P}^\mu{}_\nu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \mathcal{T}^\mu{}_\nu = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

It used to be thought self-evident that the fundamental multiplication rule of the Poincaré group

$$U(\bar{\Lambda}, \bar{a})U(\Lambda, a) = U(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a})$$

would be valid even if  $\Lambda$  and/or  $\bar{\Lambda}$  involved factors of  $\mathcal{P}$  or  $\mathcal{T}$  or  $\mathcal{PT}$ . In particular, it was believed that there are operators corresponding to  $\mathcal{P}$  and  $\mathcal{T}$  themselves:

$$P \equiv U(\mathcal{P}, 0), \quad T \equiv U(\mathcal{T}, 0)$$

such that

$$PU(\Lambda, a)P^{-1} = U(\mathcal{P}\Lambda\mathcal{P}^{-1}, \mathcal{P}a), \tag{2.6.1}$$

$$TU(\Lambda, a)T^{-1} = U(\mathcal{T}\Lambda\mathcal{T}^{-1}, \mathcal{T}a) \tag{2.6.2}$$

for any proper orthochronous Lorentz transformation  $\Lambda^\mu{}_\nu$ , and translation  $a^\mu$ . These transformation rules incorporate most of what is meant when we say that  $P$  or  $T$  are ‘conserved’.

In 1956–57 it became understood [8] that this is true for  $P$  only in the approximation in which one ignores the effects of weak interactions, such as those that produce nuclear beta decay. Time-reversal survived for a while, but in 1964 there appeared indirect evidence [9] that these properties of  $T$  are also only approximately satisfied. (See Section 3.3.) In what follows, we will make believe that operators  $P$  and  $T$  satisfying Eqs. (2.6.1) and (2.6.2) actually exist, but it should be kept in mind that this is only an approximation.

Let us apply Eqs. (2.6.1) and (2.6.2) in the case of an infinitesimal transformation, i.e.,

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad a^\mu = \epsilon^\mu$$

with  $\omega_{\mu\nu} = -\omega_{\nu\mu}$  and  $\epsilon_\mu$  both infinitesimal. Using (2.4.3), and equating coefficients of  $\omega_{\rho\sigma}$  and  $\epsilon_\rho$  in Eqs. (2.6.1) and (2.6.2), we obtain the  $P$  and  $T$  transformation properties of the Poincaré generators

$$PiJ^{\rho\sigma}P^{-1} = iP_\mu{}^\rho \mathcal{P}_\nu{}^\sigma J^{\mu\nu}, \quad (2.6.3)$$

$$PiP^\rho P^{-1} = iP_\mu{}^\rho P^\mu, \quad (2.6.4)$$

$$TiJ^{\rho\sigma}T^{-1} = iT_\mu{}^\rho \mathcal{T}_\nu{}^\sigma J^{\mu\nu}, \quad (2.6.5)$$

$$TiP^\rho T^{-1} = iT_\mu{}^\rho P^\mu. \quad (2.6.6)$$

This is much like Eqs. (2.4.8) and (2.4.9), except that we have not cancelled factors of  $i$  on both sides of these equations, because at this point we have not yet decided whether  $P$  and  $T$  are linear and unitary or antilinear and antiunitary.

The decision is an easy one. Setting  $\rho = 0$  in Eq. (2.6.4) gives

$$PiHP^{-1} = iH,$$

where  $H \equiv P^0$  is the energy operator. If  $P$  were antiunitary and antilinear then it would anticommute with  $i$ , so  $PHP^{-1} = -H$ . But then for any state  $|\Psi\rangle$  of energy  $E > 0$ , there would have to be another state  $P^{-1}|\Psi\rangle$  of energy  $-E < 0$ . There are no states of negative energy (energy less than that of the vacuum), so we are forced to choose the other alternative:  *$P$  is linear and unitary, and commutes rather than anticommutes with  $H$ .*

On the other hand, setting  $\rho = 0$  in Eq. (2.6.6) yields

$$TiHT^{-1} = -iH.$$

If we supposed that  $T$  is linear and unitary then we could simply cancel the  $i$ s, and find  $THT^{-1} = -H$ , with the again disastrous conclusion that for any state  $|\Psi\rangle$  of energy  $E$  there is another state  $T^{-1}|\Psi\rangle$  of energy  $-E$ . To avoid this, we are forced here to conclude that  *$T$  is antilinear and antiunitary.*

Now that we have decided that  $P$  is linear and  $T$  is antilinear, we can conveniently rewrite Eqs. (2.6.3)–(2.6.6) in terms of the generators (2.4.15)–(2.4.17) in a three-dimensional notation

$$P\mathbf{J}P^{-1} = +\mathbf{J}, \quad (2.6.7)$$

$$P\mathbf{K}P^{-1} = -\mathbf{K}, \quad (2.6.8)$$

$$P\mathbf{P}P^{-1} = -\mathbf{P}, \quad (2.6.9)$$

$$T\mathbf{J}T^{-1} = -\mathbf{J}, \quad (2.6.10)$$

$$T\mathbf{K}T^{-1} = +\mathbf{K}, \quad (2.6.11)$$

$$T\mathbf{P}T^{-1} = -\mathbf{P}, \quad (2.6.12)$$

and, as shown before,

$$PHP^{-1} = THT^{-1} = H. \quad (2.6.13)$$

It is physically sensible that  $P$  should preserve the sign of  $\mathbf{J}$ , because at least the orbital part is a vector product  $\mathbf{r} \times \mathbf{p}$  of two vectors, both of which change sign under an inversion of the spatial coordinate system. On the other hand,  $T$  reverses  $\mathbf{J}$ , because after time-reversal an observer will see all bodies spinning in the opposite direction. Note by the way that Eq. (2.6.10) is consistent with the angular-momentum commutation relations  $\mathbf{J} \times \mathbf{J} = i\mathbf{J}$ , because  $T$  reverses not only  $\mathbf{J}$ , but also  $i$ . The reader can easily check that Eqs. (2.6.7)–(2.6.13) are consistent with all the commutation relations (2.4.18)–(2.4.24).

Let us now consider what  $P$  and  $T$  do to one-particle states:

$P : M > 0$

The one-particle states  $|k, \sigma\rangle$  are defined as eigenvectors of  $\mathbf{P}$ ,  $H$ , and  $J_3$  with eigenvalues 0,  $M$ , and  $\sigma$ , respectively. From Eqs. (2.6.7), (2.6.9), and (2.6.13), we see that the same must be true of the state  $P|k, \sigma\rangle$ , and therefore (barring degeneracies) these states can only differ by a phase

$$P|k, \sigma\rangle = \eta_\sigma |k, \sigma\rangle$$

with a phase factor ( $|\eta_\sigma| = 1$ ) that may or may not depend on the spin  $\sigma$ . To see that  $\eta_\sigma$  is  $\sigma$ -independent, we note from (2.5.8), (2.5.20), and (2.5.21) that

$$(J_1 \pm iJ_2)|k, \sigma\rangle = \sqrt{(j \mp \sigma)(j \pm \sigma + 1)}|k, \sigma \pm 1\rangle, \quad (2.6.14)$$

where  $j$  is the particle's spin. Operating on both sides with  $P$ , we find

$$\eta_\sigma = \eta_{\sigma \pm 1}$$

and so  $\eta_\sigma$  is actually independent of  $\sigma$ . We therefore write

$$P|k, \sigma\rangle = \eta |k, \sigma\rangle \quad (2.6.15)$$

with  $\eta$  a phase, known as the *intrinsic parity*, that depends only on the species of particle on which  $P$  acts.

To get to finite momentum states, we must apply the unitary operator  $U(L(p))$  corresponding to the ‘boost’ (2.5.24):

$$|p, \sigma\rangle = \sqrt{M/p^0} U(L(p))|k, \sigma\rangle.$$

We note that

$$\mathcal{P}L(p)\mathcal{P}^{-1} = L(\mathcal{P}p), \quad \mathcal{P}p = (p^0, -\mathbf{p}),$$

so using Eqs. (2.6.1) and (2.6.15), we have

$$P|p, \sigma\rangle = \sqrt{M/p^0} U(L(\mathcal{P}p))\eta|k, \sigma\rangle$$

or in other words

$$P|p, \sigma\rangle = \eta|\mathcal{P}p, \sigma\rangle. \quad (2.6.16)$$

$T : M > 0$

From Eqs. (2.6.10), (2.6.12), and (2.6.13), we see that the effect of  $T$  on the zero-momentum one-particle state  $|k, \sigma\rangle$  is to yield a state with

$$\begin{aligned} \mathbf{P}(T|k, \sigma) &= 0, \\ H(T|k, \sigma) &= M(T|k, \sigma), \\ J_3(T|k, \sigma) &= -\sigma(T|k, \sigma), \end{aligned}$$

and so

$$T|k, \sigma\rangle = \zeta_\sigma|k, -\sigma\rangle,$$

where  $\zeta_\sigma$  is a phase factor. Applying the operator  $T$  to (2.6.14), and recalling that  $T$  anticommutes with  $\mathbf{J}$  and  $i$ , we find

$$(-J_1 \pm iJ_2)\zeta_\sigma|k, -\sigma\rangle = \sqrt{(j \mp \sigma)(j \pm \sigma + 1)} \zeta_{\sigma \pm 1}|k, -\sigma \mp 1\rangle.$$

Using Eq. (2.6.14) again on the left, we see that the square-root factors cancel, and so

$$-\zeta_\sigma = \zeta_{\sigma \pm 1}.$$

We write the solution as  $\zeta_\sigma = \zeta(-1)^{j-\sigma}$ , with  $\zeta$  another phase that depends only on the species of particle:

$$T|k, \sigma\rangle = \zeta(-1)^{j-\sigma}|k, -\sigma\rangle. \quad (2.6.17)$$

However, unlike the ‘intrinsic parity’  $\eta$ , the time-reversal phase  $\zeta$  has no physical significance. This is because we can redefine the one-particle states by a change of phase

$$|k, \sigma\rangle \longrightarrow |k, \sigma\rangle' = \zeta^{1/2}|k, \sigma\rangle$$

in such a way that the phase  $\zeta$  is eliminated from the transformation rule

$$T|k, \sigma\rangle' = \zeta^{*1/2}T|k, \sigma\rangle = \zeta^{*1/2}\zeta(-1)^{j-\sigma}|k, -\sigma\rangle = (-1)^{j-\sigma}|k, -\sigma\rangle'.$$

In what follows we will keep the arbitrary phase  $\zeta$  in Eq. (2.6.17), just to keep open our options in choosing the phase of the one-particle states, but it should be kept in mind that this phase is of no real importance.

To deal with states of finite momentum, we again apply the ‘boost’ (2.5.24). Note that

$$\mathcal{T}L(p)\mathcal{T}^{-1} = L(\mathcal{P}p), \quad \mathcal{P}p = (p^0, -\mathbf{p}).$$

(That is, changing the sign of each element of  $L^\mu{}_\nu$  with an odd number of *time*-indices is the same as changing the signs of elements with an odd number of *space*-indices.) Using Eqs. (2.6.2) and (2.5.5), we have then

$$T|p, \sigma\rangle = \zeta(-1)^{j-\sigma} |\mathcal{P}p, -\sigma\rangle. \quad (2.6.18)$$

$P : M = 0$

Acting on a state  $|k, \sigma\rangle$  that is defined as an eigenvector of  $P^\mu$  with eigenvalue  $k^\mu = (\kappa, 0, 0, \kappa)$  and an eigenvector of  $J_3$  with eigenvalue  $\sigma$ , the parity operator  $P$  yields a state with four-momentum  $(\mathcal{P}k)^\mu = (\kappa, 0, 0, -\kappa)$  and  $J_3$  equal to  $\sigma$ . Thus it takes a state of helicity (the component of spin along the direction of motion)  $\sigma$  into one of helicity  $-\sigma$ . As mentioned earlier, this shows that the existence of a space-inversion symmetry requires that any species of massless particle with non-zero helicity must be accompanied with another of opposite helicity. Because  $P$  does not leave the standard momentum invariant, it is convenient to consider instead the operator  $U(R_2^{-1})P$ , where  $R_2$  is a rotation that also takes  $k$  to  $\mathcal{P}k$ , conveniently chosen as a rotation by  $-180^\circ$  around the two-axis

$$U(R_2) = \exp(-i\pi J_2). \quad (2.6.19)$$

Since  $U(R_2^{-1})$  reverses the sign of  $J_3$ , we have

$$U(R_2^{-1})P|k, \sigma\rangle = \eta_\sigma |k, -\sigma\rangle \quad (2.6.20)$$

with  $\eta_\sigma$  a phase factor. Now,  $R_2^{-1}\mathcal{P}$  commutes with the Lorentz ‘boost’ (2.5.45), and  $\mathcal{P}$  commutes with the rotation  $R(\hat{\mathbf{p}})$  which takes the three-direction into the direction of  $\mathbf{p}$ , so by operating on (2.5.5) with  $P$ , we find for a general four-momentum  $p^\mu$

$$\begin{aligned} P|p, \sigma\rangle &= \sqrt{\frac{\kappa}{p^0}} U\left(R(\hat{\mathbf{p}})R_2B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right) U(R_2^{-1})P|k, \sigma\rangle \\ &= \sqrt{\frac{\kappa}{p^0}} \eta_\sigma U\left(R(\hat{\mathbf{p}})R_2B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right) |k, -\sigma\rangle. \end{aligned}$$

Note that  $R(\hat{\mathbf{p}})R_2$  is a rotation that takes the three-axis into the direction of  $-\hat{\mathbf{p}}$ , but  $U(R(\hat{\mathbf{p}})R_2)$  is not quite equal to  $U(R(-\hat{\mathbf{p}}))$ . According to (2.5.47),

$$U(R(-\hat{\mathbf{p}})) = \exp(i(\phi \pm \pi)J_3) \exp(i(\pi - \theta)J_2)$$

with azimuthal angle chosen as  $\phi + \pi$  or  $\phi - \pi$  according to whether  $0 \leq \phi < \pi$  or  $\pi \leq \phi < 2\pi$ , so that it remains in the range of 0 to  $2\pi$ . Then

$$\begin{aligned} U^{-1}(R(-\hat{\mathbf{p}}))U(R(\hat{\mathbf{p}})R_2) &= \exp(-i(\pi - \theta)J_2) \\ &\quad \cdot \exp(-i(\phi \pm \pi)J_3) \exp(i\phi J_3) \exp(i\theta J_2) \exp(-i\pi J_2) \\ &= \exp(-i(\pi - \theta)J_2) \exp(\mp i\pi J_3) \exp(-i(\pi - \theta)J_2). \end{aligned}$$

But a rotation of  $\pm 180^\circ$  around the three-axis reverses the sign of  $J_2$ , so

$$U(R(\hat{\mathbf{p}})R_2) = U(R(-\hat{\mathbf{p}})) \exp(\pm i\pi J_3). \quad (2.6.21)$$



Also,  $R(-\hat{\mathbf{p}})B(|\mathbf{p}|/\kappa)$  is just the standard boost  $L(\mathcal{P}p)$  in the direction  $\mathcal{P}p = (p^0, -\mathbf{p})$ . We have then finally

$$P|p, \sigma\rangle = \eta_\sigma \exp(\mp i\pi\sigma)|\mathcal{P}p, -\sigma\rangle \quad (2.6.22)$$

with the phase  $-\pi\sigma$  or  $+\pi\sigma$  according to whether the two-component of  $\mathbf{p}$  is positive or negative, respectively. This peculiar change of sign in the operation of parity for massless particles of half-integer spin is due to the convention adopted in Eq. (2.5.47) for the rotation used to define massless particle states of arbitrary momentum. Because the rotation group is not simply connected, some discontinuity of this sort is unavoidable.

$T : M = 0$

Acting on the state  $|k, \sigma\rangle$ , which has values  $k^\mu = (\kappa, 0, 0, \kappa)$  and  $\sigma$  for  $P^\mu$  and  $J_3$ , the time-reversal operator  $T$  yields a state which has values  $(\mathcal{P}k)^\mu = (\kappa, 0, 0, -\kappa)$  and  $-\sigma$  for  $P^\mu$  and  $J_3$ . Thus  $T$  does not change the helicity  $\mathbf{J} \cdot \hat{\mathbf{k}}$ , and by itself has nothing to say about whether massless particles of one helicity  $\sigma$  are accompanied with others of helicity  $-\sigma$ . Because  $T$  like  $P$  does not leave the standard four-momentum  $k$  invariant, it is convenient to consider the generator  $U(R_2^{-1})T$ , where  $R_2$  is the rotation (2.6.19), which also takes  $k$  into  $\mathcal{P}k$ . This commutes with  $J_3$ , so

$$U(R_2^{-1})T|k, \sigma\rangle = \zeta_\sigma|k, \sigma\rangle \quad (2.6.23)$$

with  $\zeta_\sigma$  another phase. Since  $R_2^{-1}\mathcal{T}$  commutes with the boost (2.5.45), and  $\mathcal{T}$  commutes with the rotation  $R(\hat{\mathbf{p}})$ , operating with  $T$  on the state (2.5.5) gives

$$T|p, \sigma\rangle = \sqrt{\frac{\kappa}{p^0}} U \left[ R(\hat{\mathbf{p}})R_2B \left( \frac{|\mathbf{p}|}{\kappa} \right) \right] \zeta_\sigma|k, \sigma\rangle. \quad (2.6.24)$$

Using Eq. (2.6.21), this yields finally

$$T|p, \sigma\rangle = \zeta_\sigma \exp(\pm i\pi\sigma)|\mathcal{P}p, \sigma\rangle. \quad (2.6.25)$$

Again, the top or bottom sign applies according to whether the two-component of  $\mathbf{p}$  is positive or negative, respectively.

\* \* \*

It is interesting that the square  $T^2$  of the time-reversal operator has a very simple action on both massive and massless one-particle states. Using Eq. (2.6.18), and recalling that  $T$  is antiunitary, we see that for massive one-particle states:

$$\begin{aligned} T^2|p, \sigma\rangle &= T\zeta(-1)^{j-\sigma}|\mathcal{P}p, -\sigma\rangle \\ &= \zeta^*(-1)^{j-\sigma}\zeta(-1)^{j+\sigma}|p, \sigma\rangle \end{aligned}$$

or in other words

$$T^2|p, \sigma\rangle = (-1)^{2j}|p, \sigma\rangle. \quad (2.6.26)$$

We get the same result for massless particles. If the two-component of  $\mathbf{p}$  is positive then the two-component of  $\mathcal{P}\mathbf{p}$  is negative, and vice-versa, so Eq. (2.6.25) gives

$$\begin{aligned} T^2|p, \sigma\rangle &= T\zeta_\sigma \exp(\pm i\pi\sigma)|\mathcal{P}p, \sigma\rangle \\ &= \zeta_\sigma^* \exp(\mp i\pi\sigma)\zeta_\sigma \exp(\mp i\pi\sigma)|p, \sigma\rangle \\ &= \exp(\mp 2\pi i\sigma)|p, \sigma\rangle. \end{aligned}$$

As long as  $\sigma$  is an integer or half-integer, this can be written

$$T^2|p, \sigma\rangle = (-1)^{2|\sigma|}|p, \sigma\rangle. \quad (2.6.27)$$

By the ‘spin’ of a massless particle, we usually mean the absolute value of the helicity, so Eq. (2.6.27) is the same as Eq. (2.6.26).

This result has an interesting consequence. When  $T^2$  acts on any state  $|\Psi\rangle$  of a system of non-interacting particles, either massive or massless, it yields a factor  $(-1)^{2j}$  or  $(-1)^{2|\sigma|}$  for each particle. Hence if the state contains an odd number of particles of half-integer spin or helicity (plus any number of particles of integer spin or helicity), we get an overall change of sign

$$T^2|\Psi\rangle = -|\Psi\rangle. \quad (2.6.28)$$

If we now ‘turn on’ various interactions, this result will be preserved, provided these interactions respect invariance under time-reversal, even if they do not respect rotational invariance. (For instance, these arguments will apply even if our system is subjected to arbitrary static gravitational and electric fields.) Now, suppose that  $|\Psi\rangle$  is an eigenstate of the Hamiltonian. Since  $T$  commutes with the Hamiltonian,  $T|\Psi\rangle$  will also be an eigenstate of the Hamiltonian. Is it the same state? If so, then  $T|\Psi\rangle$  can differ from  $|\Psi\rangle$  only by a phase

$$T|\Psi\rangle = \zeta|\Psi\rangle,$$

but then

$$T^2|\Psi\rangle = T(\zeta|\Psi\rangle) = \zeta^*T|\Psi\rangle = |\zeta|^2|\Psi\rangle = |\Psi\rangle,$$

in contradiction with Eq. (2.6.28). We see that any energy eigenstate  $|\Psi\rangle$  satisfying Eq. (2.6.28) must be degenerate, with another eigenstate of the same energy. This is known as a ‘Kramers degeneracy’ [10]. Of course, this conclusion is trivial if the system is in a rotationally invariant environment, because the total angular-momentum  $j$  of any state of this system would have to be a half-integer, and there would therefore be  $2j + 1 = 2, 4, \dots$  degenerate states. The surprising result is that at least a two-fold degeneracy persists even if rotational invariance is perturbed by external fields, such as electrostatic fields, as long as these fields are invariant under  $T$ . In particular, if any particle had an electric or gravitational dipole moment then the degeneracy among its  $2j + 1$  spin states would be entirely removed in a static electric or gravitational field, so such dipole moments are forbidden by time-reversal invariance.

For the sake of completeness, it should be mentioned that  $P$  and  $T$  can have more complicated effects on multiplets of particles with the same mass. This possibility will be considered in Appendix C of this chapter. No physically relevant examples are known.

## 2.7 Projective Representations<sup>9</sup>

We now return to the possibility mentioned in Section 2.2, that a group of symmetries may be represented projectively on physical states; that is, the elements  $T$ ,  $\bar{T}$ , etc. of the symmetry group may be represented on the physical Hilbert space by unitary operators  $U(T)$ ,  $U(\bar{T})$ , etc., which satisfy the composition rule

$$U(T)U(\bar{T}) = \exp(i\phi(T, \bar{T}))U(T\bar{T}) \quad (2.7.1)$$

with  $\phi$  a real phase. (A bar is used here just to distinguish one symmetry operator from another.) The basic requirement that any phase  $\phi$  in Eq. (2.7.1) would have to satisfy is the associativity condition

$$U(T_3)(U(T_2)U(T_1)) = (U(T_3)U(T_2))U(T_1),$$

which imposes on  $\phi$  the corresponding condition

$$\phi(T_2, T_1) + \phi(T_3, T_2T_1) = \phi(T_3, T_2) + \phi(T_3T_2, T_1). \quad (2.7.2)$$

Of course, any phase of the form

$$\phi(T, \bar{T}) = \alpha(T\bar{T}) - \alpha(T) - \alpha(\bar{T}) \quad (2.7.3)$$

will automatically satisfy Eq. (2.7.2), but a projective representation with such a phase can be replaced with an ordinary representation by replacing  $U(T)$  with

$$\tilde{U}(T) \equiv U(T) \exp(i\alpha(T))$$

for which

$$\tilde{U}(T)\tilde{U}(\bar{T}) = \tilde{U}(T\bar{T}).$$

Any set of functions  $\phi(T, \bar{T})$  that satisfy Eq. (2.7.2), and that differ only by functions  $\Delta\phi(T, \bar{T})$  of the form (2.7.3), is called a ‘two-cocycle’. A trivial cocycle is one that contains the function  $\phi = 0$ , and hence consists of functions of the form (2.7.3), which can be eliminated by a redefinition of  $U(T)$ . We are interested here in whether a symmetry group allows any non-trivial two-cocycles; that is, whether it may have a representation on the physical Hilbert space that is *intrinsically* projective, in the sense that the phase  $\phi(T, \bar{T})$  *cannot* be eliminated in this way.

In order to answer this question, it is useful first to consider the effect of a phase  $\phi$  in Eq. (2.7.1) on the commutation relations of the generators of infinitesimal transformations. When either  $\bar{T}$  or  $T$  is the identity, the phase  $\phi$  must clearly vanish

$$\phi(T, \mathbf{1}) = \phi(\mathbf{1}, \bar{T}) = 0. \quad (2.7.4)$$

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<sup>9</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

When both  $T$  and  $\bar{T}$  are *near* the identity the phase must be small. Using coordinates  $\theta^a$  to parameterize group elements (as in Section 2.2), with  $T(0) = \mathbf{1}$ , Eq. (2.7.4) tells us that the expansion of  $\phi[T(\theta), T(\bar{\theta})]$  around  $\theta = \bar{\theta} = 0$  must start with terms of order  $\theta\bar{\theta}$ :

$$\phi(T(\theta), T(\bar{\theta})) = f_{ab}\theta^a\bar{\theta}^b + \dots, \quad (2.7.5)$$

where  $f_{ab}$  are real numerical constants. Inserting this expansion in the power series expansion of Eq. (2.7.1), and repeating the steps that led to (2.2.22), we now have

$$[t_b, t_c] = iC_{bc}^a t_a + iC_{bc} \mathbf{1}, \quad (2.7.6)$$

where  $C_{bc}$  is the antisymmetric coefficient

$$C_{bc} = -f_{bc} + f_{cb}. \quad (2.7.7)$$

The appearance of terms on the right-hand side of the commutation relation proportional to the unit element (so-called *central charges*) is the counterpart for the Lie algebra of the presence of phases in a projective representation of a group.

The constants  $C_{bc}$  as well as  $C_{bc}^a$  are subject to an important constraint, which follows from the Jacobi identity. Taking the commutator of (2.7.6) with  $t_d$ , and adding the same expressions with  $b, c, d$  replaced with  $c, d, b$  and  $d, b, c$ , the sum of the three double-commutators on the left-hand side vanishes identically, and so

$$C_{bc}^a C_{ad}^e + C_{cd}^a C_{ab}^e + C_{db}^a C_{ac}^e = 0 \quad (2.7.8)$$

and also

$$C_{bc}^a C_{ad} + C_{cd}^a C_{ab} + C_{db}^a C_{ac} = 0. \quad (2.7.9)$$

Eq. (2.7.9) always has one obvious class of non-zero solutions for  $C_{ab}$ :

$$C_{ab} = C_{ab}^e \phi_e, \quad (2.7.10)$$

where  $\phi_e$  is an arbitrary set of real constants. For these solutions, we can eliminate the central charges from Eq. (2.7.6) by a redefinition of the generators

$$t_a \longrightarrow \tilde{t}_a \equiv t_a + \phi_a. \quad (2.7.11)$$

The new generators then satisfy the commutation relations without central charges

$$[\tilde{t}_b, \tilde{t}_c] = iC_{bc}^a \tilde{t}_a. \quad (2.7.12)$$

A given Lie algebra may or may not allow solutions of Eq. (2.7.9) other than Eq. (2.7.10).

We can now state the key theorem that governs the occurrence of intrinsically projective representations. The phase of any representation  $U(T)$  of a given group can be chosen so that  $\phi = 0$  in Eq. (2.7.1), if two conditions are met:

- (a) The generators of the group in this representation can be redefined (as in Eq. (2.7.11)), so as to eliminate all central charges from the Lie algebra.

- (b) The group is simply connected, i.e., any two group elements may be connected by a path lying within the group, and any two such paths may be continuously transformed into one another. (An equivalent statement is that any loop that starts and ends at the same group element may be shrunk continuously to a point.)

This theorem is proved in Appendix B of this chapter, which also offers comments about the case of groups that are not simply connected. It shows that there are just two (not exclusive) ways that intrinsically projective representations may arise: either algebraically, because the group is represented projectively even near the identity, or topologically, because the group is not simply connected, and hence a path from  $\mathbf{1}$  to  $T$  and then from  $T$  to  $\bar{T}$  may not be continuously deformable into some other path from  $\mathbf{1}$  to  $T\bar{T}$ . In the latter case, the phase  $\phi$  in Eq. (2.7.1) depends on the particular choice of standard paths, leading from the origin to the various group elements, that are used to define the corresponding  $U$ -operators.

Let's now consider each of these possibilities in turn for the special case of the Poincaré group.

### (A) Algebra

With central charges, the commutation relations of the generators of the Poincaré group would read

$$i[J^{\mu\nu}, J^{\rho\sigma}] = \eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\sigma\mu} J^{\rho\nu} + \eta^{\sigma\nu} J^{\rho\mu} + C^{\rho\sigma, \mu\nu}, \quad (2.7.13)$$

$$i[P^\mu, J^{\rho\sigma}] = \eta^{\mu\rho} P^\sigma - \eta^{\mu\sigma} P^\rho + C^{\rho\sigma, \mu}, \quad (2.7.14)$$

$$i[J^{\mu\nu}, P^\rho] = \eta^{\nu\rho} P^\mu - \eta^{\mu\rho} P^\nu + C^{\rho, \mu\nu}, \quad (2.7.15)$$

$$i[P^\mu, P^\rho] = C^{\rho, \mu} \quad (2.7.16)$$

in place of Eqs. (2.4.12)–(2.4.14). We see that the  $C$ s also satisfy the antisymmetry conditions

$$C^{\rho\sigma, \mu\nu} = -C^{\mu\nu, \rho\sigma}, \quad (2.7.17)$$

$$C^{\rho\sigma, \mu} = -C^{\mu, \rho\sigma}, \quad (2.7.18)$$

$$C^{\rho, \mu} = -C^{\mu, \rho}. \quad (2.7.19)$$

We will now show that all these constants have additional algebraic properties that allow them to be eliminated by shifting the definitions of  $J^{\mu\nu}$  and  $P^\mu$  by constant terms. (This corresponds to redefining the phase of the operators  $U(\Lambda, a)$ .) To derive these properties, we apply the Jacobi identities

$$[J^{\mu\nu}, [P^\rho, P^\sigma]] + [P^\sigma, [J^{\mu\nu}, P^\rho]] + [P^\rho, [P^\sigma, J^{\mu\nu}]] = 0, \quad (2.7.20)$$

$$[J^{\lambda\eta}, [J^{\mu\nu}, P^\rho]] + [P^\rho, [J^{\lambda\eta}, J^{\mu\nu}]] + [J^{\mu\nu}, [P^\rho, J^{\lambda\eta}]] = 0, \quad (2.7.21)$$

$$[J^{\lambda\eta}, [J^{\mu\nu}, J^{\rho\sigma}]] + [J^{\rho\sigma}, [J^{\lambda\eta}, J^{\mu\nu}]] + [J^{\mu\nu}, [J^{\rho\sigma}, J^{\lambda\eta}]] = 0. \quad (2.7.22)$$

(The Jacobi identity involving three  $P$ s is automatically satisfied, and hence yields no further information.) Using Eqs. (2.7.13)–(2.7.16) in Eqs. (2.7.20)–(2.7.22), we obtain algebraic conditions on the  $C$ s

$$0 = \eta^{\nu\rho} C^{\mu,\sigma} - \eta^{\mu\rho} C^{\nu,\sigma} - \eta^{\nu\sigma} C^{\mu,\rho} + \eta^{\mu\sigma} C^{\nu,\rho}, \quad (2.7.23)$$

$$0 = \eta^{\nu\rho} C^{\mu,\lambda\eta} - \eta^{\mu\rho} C^{\nu,\lambda\eta} - \eta^{\mu\lambda} C^{\rho,\eta\nu} + \eta^{\mu\eta} C^{\rho,\lambda\nu} \\ + \eta^{\lambda\nu} C^{\rho,\mu\eta} - \eta^{\eta\nu} C^{\rho,\mu\lambda} + \eta^{\rho\lambda} C^{\eta,\mu\nu} - \eta^{\rho\eta} C^{\lambda,\mu\nu}, \quad (2.7.24)$$

$$0 = \eta^{\nu\rho} C^{\mu\sigma,\lambda\eta} - \eta^{\mu\rho} C^{\nu\sigma,\lambda\eta} - \eta^{\sigma\mu} C^{\rho\nu,\lambda\eta} + \eta^{\sigma\nu} C^{\rho\mu,\lambda\eta} \\ + \eta^{\eta\mu} C^{\lambda\nu,\rho\sigma} - \eta^{\lambda\mu} C^{\eta\nu,\rho\sigma} - \eta^{\nu\lambda} C^{\mu\eta,\rho\sigma} + \eta^{\nu\eta} C^{\mu\lambda,\rho\sigma} \\ + \eta^{\sigma\lambda} C^{\rho\eta,\mu\nu} - \eta^{\rho\lambda} C^{\sigma\eta,\mu\nu} - \eta^{\sigma\eta} C^{\rho\lambda,\mu\nu} + \eta^{\rho\eta} C^{\sigma\lambda,\mu\nu}. \quad (2.7.25)$$

Contracting Eq. (2.7.23) with  $\eta_{\nu\rho}$  gives

$$C^{\mu,\sigma} = 0. \quad (2.7.26)$$

On the other hand, the constants  $C^{\mu,\lambda\eta}$  and  $C^{\rho\sigma,\mu\nu}$  are not necessarily zero, but their algebraic structure is simple enough so that they can be eliminated by shifting the definitions of  $P^\mu$  and  $J^{\mu\nu}$ , respectively. Contracting Eq. (2.7.24) with  $\eta_{\nu\rho}$  gives

$$C^{\mu,\lambda\eta} = \eta^{\eta\mu} C^\lambda - \eta^{\mu\lambda} C^\eta, \quad (2.7.27)$$

$$C^\lambda \equiv \frac{1}{3} \eta_{\rho\nu} C^{\rho,\lambda\nu}. \quad (2.7.28)$$

Also, contracting (2.7.25) with  $\eta_{\nu\rho}$  gives

$$C^{\mu\sigma,\lambda\eta} = \eta^{\eta\mu} C^{\lambda\sigma} - \eta^{\lambda\mu} C^{\eta\sigma} + \eta^{\sigma\lambda} C^{\eta\mu} - \eta^{\eta\sigma} C^{\lambda\mu}, \quad (2.7.29)$$

$$C^{\lambda\sigma} \equiv \frac{1}{2} \eta_{\nu\rho} C^{\lambda\nu,\sigma\rho}. \quad (2.7.30)$$

(These expressions automatically satisfy Eqs. (2.7.24) and (2.7.25), so there is no further information to be gained from the Jacobi identities.) We now see that if the  $C$ s are not zero, they can be eliminated by defining new generators

$$\tilde{P}^\mu \equiv P^\mu + C^\mu, \quad (2.7.31)$$

$$\tilde{J}^{\mu\sigma} \equiv J^{\mu\sigma} + C^{\mu\sigma}, \quad (2.7.32)$$

and the commutation relations are then what they would be for an ordinary representation

$$i[\tilde{J}^{\mu\nu}, \tilde{J}^{\rho\sigma}] = \eta^{\nu\rho} \tilde{J}^{\mu\sigma} - \eta^{\mu\rho} \tilde{J}^{\nu\sigma} - \eta^{\sigma\mu} \tilde{J}^{\rho\nu} + \eta^{\sigma\nu} \tilde{J}^{\rho\mu}, \quad (2.7.33)$$

$$i[\tilde{J}^{\mu\nu}, \tilde{P}^\rho] = \eta^{\nu\rho} \tilde{P}^\mu - \eta^{\mu\rho} \tilde{P}^\nu, \quad (2.7.34)$$

$$i[\tilde{P}^\mu, \tilde{P}^\rho] = 0. \quad (2.7.35)$$

The commutation relations will always be taken in the form Eqs. (2.7.33)–(2.7.35), but with tildas dropped.

Incidentally, the fact that there is no central charge in the algebra of the  $J^{\mu\nu}$  could have been immediately inferred from the fact that this algebra is of the type known as ‘semi-simple’. (Semi-simple Lie algebras are those that have no ‘invariant Abelian’ subalgebra, consisting of generators that commute with each other and whose commutators with any other generators also belong to the subalgebra.) There is a general theorem [11] that any central charges in semi-simple Lie algebras may always be removed by a redefinition of generators, as in Eq. (2.7.32). On the other hand, the full Poincaré algebra spanned by  $J^{\mu\nu}$  and  $P^\mu$  is not semi-simple (the  $P^\mu$  form an invariant Abelian subalgebra), and we needed a special argument to show that its central charges can also be eliminated in this way. Indeed, the non-semi-simple Galilean algebra discussed in Section 2.4 does allow a central charge, the mass  $M$ .

We see that the Poincaré group satisfies the first of the two conditions needed to rule out intrinsically projective representations. How about the second?

### (B) Topology

To explore the topology of the Poincaré group, it is very convenient to represent homogeneous Lorentz transformations by  $2 \times 2$  complex matrices. Any real four-vector  $V^\mu$  can be used to construct an Hermitian  $2 \times 2$  matrix

$$v \equiv V^\mu \sigma_\mu = \begin{pmatrix} V^0 + V^3 & V^1 - iV^2 \\ V^1 + iV^2 & V^0 - V^3 \end{pmatrix}, \quad (2.7.36)$$

where  $\sigma_\mu$  are the usual Pauli matrices with  $\sigma_0 = \mathbf{1}$ . Conversely any  $2 \times 2$  Hermitian matrix can be put in this form, and therefore defines a real four-vector  $V^\mu$ .

The property of Hermiticity will be preserved under the transformation

$$v \longrightarrow \lambda v \lambda^\dagger \quad (2.7.37)$$

with  $\lambda$  an arbitrary complex  $2 \times 2$  matrix. Furthermore, the covariant square of the four-vector is

$$V_\mu V^\mu = (V^1)^2 + (V^2)^2 + (V^3)^2 - (V^0)^2 = -\text{Det } v \quad (2.7.38)$$

and this determinant is preserved by the transformation (2.7.37) provided that

$$|\text{Det } \lambda| = 1. \quad (2.7.39)$$

Each complex  $2 \times 2$  matrix  $\lambda$  satisfying Eq. (2.7.39) thus defines a real linear transformation of  $V^\mu$  that leaves Eq. (2.7.38) invariant, i.e., a homogeneous Lorentz transformation  $\Lambda(\lambda)$ :

$$\lambda V^\mu \sigma_\mu \lambda^\dagger = (\Lambda^\mu{}_\nu(\lambda) V^\nu) \sigma_\mu. \quad (2.7.40)$$

Furthermore, for two such matrices  $\lambda$  and  $\bar{\lambda}$ , we have

$$\begin{aligned} (\lambda \bar{\lambda}) V^\mu \sigma_\mu (\lambda \bar{\lambda})^\dagger &= \lambda (\bar{\lambda} V^\mu \sigma_\mu \bar{\lambda}^\dagger) \lambda^\dagger \\ &= \lambda \Lambda^\mu{}_\nu(\bar{\lambda}) V^\nu \sigma_\mu \lambda^\dagger \\ &= \Lambda^\mu{}_\rho(\lambda) \Lambda^\rho{}_\nu(\bar{\lambda}) V^\nu \sigma_\mu \end{aligned}$$

and so

$$\Lambda(\lambda\bar{\lambda}) = \Lambda(\lambda)\Lambda(\bar{\lambda}). \quad (2.7.41)$$

However, two  $\lambda$ s that differ only by an overall phase have the same effect on  $v$  in Eq. (2.7.37), and so correspond to the same Lorentz transformation. It is therefore convenient to adjust the phase of the  $\lambda$ s so that

$$\text{Det } \lambda = 1, \quad (2.7.42)$$

which is consistent with Eq. (2.7.41). The  $2 \times 2$  complex matrices with unit determinant form a group, known as  $SL(2, \mathbb{C})$ . (The  $SL$  stands for ‘special linear’, with ‘special’ denoting a unit determinant, while  $C$  stands for ‘complex’.) The group elements depend on  $4 - 1 = 3$  complex parameters, or 6 real parameters, the same number as the Lorentz group. However,  $SL(2, \mathbb{C})$  is not the same as the Lorentz group; if  $\lambda$  is a matrix in  $SL(2, \mathbb{C})$ , then so is  $-\lambda$ , and both  $\lambda$  and  $-\lambda$  produce the same Lorentz transformation in Eq. (2.7.37). Indeed, it is easy to see that the matrix

$$\lambda(\theta) = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}$$

produces a Lorentz transformation  $\Lambda(\lambda(\theta))$  which is just a rotation by an angle  $\theta$  around the three-axis, and hence  $\lambda = -\mathbf{1}$  produces a rotation by an angle  $2\pi$ . The Lorentz group is not the same as  $SL(2, \mathbb{C})$ , but rather<sup>10</sup>  $SL(2, \mathbb{C})/\mathbb{Z}_2$ , which is the group of complex  $2 \times 2$  matrices with unit determinant, and with  $\lambda$  identified with  $-\lambda$ .

Now, what is the topology of the Lorentz group? By the polar decomposition theorem [12], any complex non-singular matrix  $\lambda$  may be written in the form

$$\lambda = ue^h,$$

where  $u$  is unitary and  $h$  is Hermitian

$$u^\dagger u = \mathbf{1}, \quad h^\dagger = h.$$

Since  $\text{Det } u$  is a phase factor, and  $\text{Det } \exp h = \exp \text{Tr } h$  is real and positive, the condition (2.7.42) requires both

$$\text{Det } u = 1, \quad \text{Tr } h = 0.$$

(The factor  $u$  simply provides the rotation subgroup of the Lorentz group; if  $u$  is unitary then  $\text{Tr}(uvu^\dagger) = \text{Tr } v$ , so  $V^0 = \frac{1}{2} \text{Tr } v$  is left invariant by  $\Lambda(u)$ .) Furthermore, this decomposition is unique, so  $SL(2, \mathbb{C})$  is topologically just the direct product (i.e., the set of pairs of points) of the space of all  $u$ s and the space of all  $h$ s. Any Hermitian traceless  $2 \times 2$  matrix  $h$  can be expressed as

$$h = \begin{pmatrix} c & a - ib \\ a + ib & -c \end{pmatrix}$$

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<sup>10</sup>The group  $\mathbb{Z}_2$  consists of just elements  $+1$  and  $-1$ . In general, when we write  $G/H$ , with  $H$  an invariant subgroup of  $G$ , we mean the group  $G$  with elements  $g$  and  $gh$  identified if  $g \in G$  and  $h \in H$ . The subgroup  $\mathbb{Z}_2$  is trivially invariant because its elements commute with all elements of  $SL(2, \mathbb{C})$ .



with  $a, b, c$  real but otherwise unconstrained, so the space of all  $h$ s is topologically the same as ordinary three-dimensional flat space,  $\mathbb{R}^3$ . On the other hand, any unitary  $2 \times 2$  matrix with unit determinant can be expressed as

$$u = \begin{pmatrix} d + ie & f + ig \\ -f + ig & d - ie \end{pmatrix}$$

with  $d, e, f, g$  subject to the single non-linear constraint

$$d^2 + e^2 + f^2 + g^2 = 1,$$

so the space  $SU(2)$  of all  $u$ s is topologically the same as  $S^3$ , the three-dimensional surface of a spherical ball in flat four-dimensional space. Thus  $SL(2, \mathbb{C})$  is topologically the same as the direct product  $\mathbb{R}^3 \times S^3$ . This is simply connected: any curve connecting two points of  $\mathbb{R}^3$  or  $S^3$  can be deformed into any other, and the same is true of the direct product. (All spheres  $S_n$  except the circle  $S_1$  are simply connected.) However, we are interested in  $SL(2, \mathbb{C})/\mathbb{Z}_2$ , not  $SL(2, \mathbb{C})$ . Identifying  $\lambda$  with  $-\lambda$  is the same as identifying the unitary factors  $u$  and  $-u$  (because  $e^h$  is always positive), so the Lorentz group has the topology of  $\mathbb{R}^3 \times S^3/\mathbb{Z}_2$ , where  $S^3/\mathbb{Z}_2$  is the three-dimensional spherical surface with opposite points of the sphere identified. This is *not* simply connected; for instance, a path on  $S^3$  from  $u$  to  $u'$  cannot be continuously deformed into a path on  $S^3$  from  $u$  to  $-u'$ , even though these two paths link the same points of  $S^3/\mathbb{Z}_2$ . In fact,  $S^3/\mathbb{Z}_2$  is *doubly* connected; the paths between any two points fall into two classes, depending on whether or not they involve an inversion  $u \rightarrow -u$ , and any path in one class can be deformed into another path of that class. An equivalent statement is that a double loop, that goes twice over the same path from any element back to itself, may be continuously contracted to a point. (As discussed in Appendix B, this is summarized mathematically in the statement that the fundamental group, or first homotopy group, of  $S^3/\mathbb{Z}_2$  is  $\mathbb{Z}_2$ .) Similarly, the Poincaré group has the same topology as  $\mathbb{R}^4 \times \mathbb{R}^3 \times S^3/\mathbb{Z}_2$ , and is therefore also doubly connected.

Because the Lorentz group (homogeneous or inhomogeneous) is not simply connected, it does have intrinsically projective representations. However, because the double loop that goes twice from  $\mathbf{1}$  to  $\Lambda$  to  $\Lambda\bar{\Lambda}$  and then back to  $\mathbf{1}$  can be contracted to a point, we must have

$$[U(\Lambda)U(\bar{\Lambda})U^{-1}(\Lambda\bar{\Lambda})]^2 = \mathbf{1}$$

and hence the phase  $e^{i\phi(\Lambda, \bar{\Lambda})}$  is just a sign

$$U(\Lambda)U(\bar{\Lambda}) = \pm U(\Lambda\bar{\Lambda}). \quad (2.7.43)$$

Likewise, for the Poincaré group

$$U(\Lambda, a)U(\bar{\Lambda}, \bar{a}) = \pm U(\Lambda\bar{\Lambda}, \Lambda\bar{a} + a). \quad (2.7.44)$$

These ‘representations up to a sign’ are familiar; they are just the states of integer spin, for which the signs in Eqs. (2.7.43) and (2.7.44) are always  $+1$ , and the states of half-integer spin, for which these signs are  $+1$  or  $-1$  according to whether the path from  $\mathbf{1}$  to

$\Lambda$  to  $\Lambda\bar{\Lambda}$  and then back to  $\mathbf{1}$  is or is not contractible to a point. This difference arises because a rotation of  $2\pi$  around the three-axis acting on a state with angular-momentum three-component  $\sigma$  produces a phase  $e^{2\pi i\sigma}$ , and thus has no effect on a state of integer spin and produces a sign change when acting on a state of half-integer spin. (These two cases correspond to the two irreducible representations of the first homotopy group,  $\mathbb{Z}_2$ .) Thus Eq. (2.7.43) or Eq. (2.7.44) imposes a superselection rule: we must not mix states of integer and half-integer spin.

For finite mass, the limitation to integer or half-integer spin was previously derived by purely algebraic means from the well-known representations of the generators of the little group, which here are just the angular-momentum matrices  $J^{(j)}$  with  $j$  integer or half-integer. On the other hand, for zero mass the action of the little group on physical one-particle states is just a rotation around the momentum, and here there is no *algebraic* reason for a limitation to integer or half-integer helicity. There is, however, a *topological* reason: a rotation by an angle  $4\pi$  around the momentum can be continuously deformed into no rotation at all, so the factor  $\exp(4\pi i\sigma)$  must be unity, and hence  $\sigma$  must be an integer or half-integer.

Instead of working with projective representations and imposing a superselection rule, we can just as well expand the Lorentz group, taking it as  $SL(2, \mathbb{C})$  itself, instead of  $SL(2, \mathbb{C})/\mathbb{Z}_2$  as before. Ordinary rotation invariance forbids transitions between states of integer and half-integer total spin, so the only difference is that now the group is simply-connected, and it therefore has only ordinary representations, not projective representations, so that we cannot infer a superselection rule. This does not mean that we actually can prepare physical systems in linear combinations of states of integer and half-integer spin, but only that the observed Lorentz invariance of nature cannot be used to show that such superpositions are impossible.

Similar remarks apply to any symmetry group. If its Lie algebra involves central charges, then we can always expand the algebra to include generators that commute with anything, and whose eigenvalues are the central charges, just as we did when we added a mass operator to the Lie algebra of the Galilean group at the end of Section 2.4. The expanded Lie algebra is then, of course, free of central charges, so the part of the group near the identity has only ordinary representations, and does not require any superselection rule. Likewise, even though a Lie group  $G$  may not be simply connected, it can always be expressed as  $C/H$ , where  $C$  is a simply connected group known as the ‘universal covering group’ of  $G$ , and  $H$  is an invariant subgroup<sup>11</sup> of  $C$ . In general, we may just as well take the symmetry group as  $C$  instead of  $G$ , because there is no difference in their consequences, except that  $G$  implies a superselection rule, while  $C$  does not. In short, the issue of superselection rules is a bit of a red herring; *it may or it may not be possible to prepare physical systems in arbitrary superpositions of states, but one cannot settle the question by reference to symmetry principles, because whatever one thinks the symmetry group of nature may be,*

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<sup>11</sup>The first homotopy group of  $C/H$  is  $H$ . We have seen that the covering group of the homogeneous Lorentz group is  $SL(2, \mathbb{C})$ , and the covering group of the three-dimensional rotation group is  $SU(2)$ . This connection with  $SL$  and  $SU$  groups is special to the case of three, four, or six dimensions; for general dimensions  $d$  the covering group of  $SO(d)$  is given a special name, ‘Spin( $d$ )’.

there is always another group whose consequences are identical except for the absence of superselection rules.

## Appendix A The Symmetry Representation Theorem

This appendix presents the proof of the fundamental theorem of Wigner [2] that any symmetry transformation can be represented on the Hilbert space of physical states by an operator that is either linear and unitary or antilinear and antiunitary. For our present purposes, the property of symmetry transformations on which we chiefly rely is that they are ray transformations  $T$  that preserve transition probabilities, in the sense that if  $|\Psi_1\rangle$  and  $|\Psi_2\rangle$  are state-vectors belonging to rays  $\mathcal{R}_1$  and  $\mathcal{R}_2$  then any state-vectors  $|\Psi'_1\rangle$  and  $|\Psi'_2\rangle$  belonging to the transformed rays  $T\mathcal{R}_1$  and  $T\mathcal{R}_2$  satisfy

$$|\langle\Psi'_1|\Psi'_2\rangle|^2 = |\langle\Psi_1|\Psi_2\rangle|^2. \quad (2.A.1)$$

We also require that a symmetry transformation should have an inverse that preserves transition probabilities in the same sense.

To start, consider some complete orthonormal set of state-vectors  $|\Psi_k\rangle$  belonging to rays  $\mathcal{R}_k$ , with

$$\langle\Psi_k|\Psi_l\rangle = \delta_{kl}, \quad (2.A.2)$$

and let  $|\Psi'_k\rangle$  be some arbitrary choice of state-vectors belonging to the transformed rays  $T\mathcal{R}_k$ . From Eq. (2.A.1), we have

$$|\langle\Psi'_k|\Psi'_l\rangle|^2 = |\langle\Psi_k|\Psi_l\rangle|^2 = \delta_{kl}.$$

But  $\langle\Psi'_k|\Psi'_k\rangle$  is automatically real and positive, so this requires that it should have the value unity, and therefore

$$\langle\Psi'_k|\Psi'_k\rangle = \delta_{kl}. \quad (2.A.3)$$

It is easy to see that these transformed states  $|\Psi'_k\rangle$  also form a complete set, for if there were any non-zero state-vector  $|\Psi'\rangle$  that was orthogonal to all of the  $|\Psi'_k\rangle$ , then the inverse transform of the ray to which  $|\Psi'\rangle$  belongs would consist of non-zero state-vectors  $|\Psi''\rangle$  for which, for all  $k$ :

$$|\langle\Psi_k|\Psi''\rangle|^2 = |\langle\Psi'_k|\Psi'\rangle|^2 = 0,$$

which is impossible since the  $|\Psi_k\rangle$  were assumed to form a complete set.

We must now establish a phase convention for the states  $|\Psi'_k\rangle$ . For this purpose, we single out one of the  $|\Psi_k\rangle$ , say  $|\Psi_1\rangle$ , and consider the state-vectors

$$|Y_k\rangle = \frac{1}{\sqrt{2}}[|\Psi_1\rangle + |\Psi_k\rangle] \quad (2.A.4)$$

belonging to some ray  $\mathcal{S}_k$ , with  $k \neq 1$ . Any state-vector  $|Y'_k\rangle$  belonging to the transformed ray  $T\mathcal{S}_k$  may be expanded in the state-vectors  $|\Psi'_l\rangle$ ,

$$|Y'_k\rangle = \sum_l c_{kl}|\Psi'_l\rangle.$$

From Eq. (2.A.1) we have

$$|c_{kk}| = |c_{k1}| = \frac{1}{\sqrt{2}}$$

and for  $l \neq k$  and  $l \neq 1$ :

$$c_{kl} = 0.$$

For any given  $k$ , by an appropriate choice of phase of the two state-vectors  $|Y'_k\rangle$  and  $|\Psi'_k\rangle$ , we can clearly adjust the phases of the two non-zero coefficients  $c_{kk}$  and  $c_{k1}$  so that both coefficients are just  $1/\sqrt{2}$ . From now on, the state-vectors  $|Y'_k\rangle$  and  $|\Psi'_k\rangle$  chosen in this way will be denoted  $U|Y_k\rangle$  and  $U|\Psi_k\rangle$ . As we have seen,

$$U \frac{1}{\sqrt{2}} [|\Psi_k\rangle + |\Psi_1\rangle] = U|Y_k\rangle = \frac{1}{\sqrt{2}} [U|\Psi_k\rangle + U|\Psi_1\rangle]. \quad (2.A.5)$$

However, it still remains to define  $U|\Psi\rangle$  for general state-vectors  $|\Psi\rangle$ .

Now consider an arbitrary state-vector  $|\Psi\rangle$  belonging to an arbitrary ray  $\mathcal{R}$ , and expand it in the  $|\Psi_k\rangle$ :

$$|\Psi\rangle = \sum_k C_k |\Psi_k\rangle. \quad (2.A.6)$$

Any state  $|\Psi'\rangle$  that belongs to the transformed ray  $T\mathcal{R}$  may similarly be expanded in the complete orthonormal set  $U|\Psi_k\rangle$ :

$$|\Psi'\rangle = \sum_k C'_k U|\Psi_k\rangle. \quad (2.A.7)$$

The equality of  $|\langle\Psi_k|\Psi\rangle|^2$  and  $|\langle U\Psi_k|\Psi'\rangle|^2$  tells us that for all  $k$  (including  $k = 1$ ):

$$|C_k|^2 = |C'_k|^2, \quad (2.A.8)$$

while the equality of  $|\langle Y_k|\Psi\rangle|^2$  and  $|\langle UY_k|\Psi'\rangle|^2$  tells us that for all  $k \neq 1$ :

$$|C_k + C_1|^2 = |C'_k + C'_1|^2. \quad (2.A.9)$$

The ratio of Eqs. (2.A.9) and (2.A.8) yields the formula

$$\text{Re}(C_k/C_1) = \text{Re}(C'_k/C'_1) \quad (2.A.10)$$

which with Eq. (2.A.8) also requires

$$\text{Im}(C_k/C_1) = \pm \text{Im}(C'_k/C'_1) \quad (2.A.11)$$

and therefore either

$$C_k/C_1 = C'_k/C'_1, \quad (2.A.12)$$

or else

$$C_k/C_1 = (C'_k/C'_1)^*. \quad (2.A.13)$$

Furthermore, we can show that the same choice must be made for each  $k$ . (This step in the proof was omitted by Wigner.) To see this, suppose that for some  $k$ , we have

$C_k/C_1 = C'_k/C'_1$ , while for some  $l \neq k$ , we have instead  $C_l/C_1 = (C'_l/C'_1)^*$ . Suppose also that both ratios are complex, so that these are really different cases. (This incidentally requires that  $k \neq 1$  and  $l \neq 1$ , as well as  $k \neq l$ .) We will show that this is impossible.

Define a state-vector  $|\Phi\rangle = \frac{1}{\sqrt{3}}[|\Psi_1\rangle + |\Psi_k\rangle + |\Psi_l\rangle]$ . Since all the ratios of the coefficients in this state-vector are real, we must get the same ratios in any state-vector  $|\Phi'\rangle$  belonging to the transformed ray:

$$|\Phi'\rangle = \frac{\alpha}{\sqrt{3}}[U|\Psi_1\rangle + U|\Psi_k\rangle + U|\Psi_l\rangle],$$

where  $\alpha$  is a phase factor with  $|\alpha| = 1$ . But then the equality of the transition probabilities  $|\langle\Phi|\Psi\rangle|$  and  $|\langle\Phi'|\Psi'\rangle|$  requires that

$$\left|1 + \frac{C'_k}{C'_1} + \frac{C'_l}{C'_1}\right|^2 = \left|1 + \frac{C_k}{C_1} + \frac{C_l}{C_1}\right|^2$$

and hence

$$\left|1 + \frac{C_k}{C_1} + \frac{C_l^*}{C_1^*}\right|^2 = \left|1 + \frac{C_k}{C_1} + \frac{C_l}{C_1}\right|^2.$$

This is only possible if

$$\operatorname{Re}\left(\frac{C_k C_l^*}{C_1 C_1^*}\right) = \operatorname{Re}\left(\frac{C_k C_l}{C_1 C_1}\right)$$

or, in other words, if

$$\operatorname{Im}\left(\frac{C_k}{C_1}\right) \operatorname{Im}\left(\frac{C_l}{C_1}\right) = 0.$$

Hence either  $C_k/C_1$  or  $C_l/C_1$  must be real for any pair  $k, l$ , in contradiction with our assumptions. We see then that for a given symmetry transformation  $T$  applied to a given state-vector  $\sum_k C_k |\Psi_k\rangle$ , we must have either Eq. (2.A.12) for all  $k$ , or else Eq. (2.A.13) for all  $k$ .

Wigner ruled out the second possibility, Eq. (2.A.13), because as he showed any symmetry transformation for which this possibility is realized would have to involve a reversal in the time coordinate, and in the proof he presented he was considering only symmetries like rotations that do not affect the direction of time. Here we are treating symmetries involving time-reversal on the same basis as all other symmetries, so we will have to consider that, for each symmetry  $T$  and state-vector  $\sum_k C_k |\Psi_k\rangle$ , either Eq. (2.A.12) or Eq. (2.A.13) may apply. Depending on which of these alternatives is realized, we will now define  $U|\Psi\rangle$  to be the particular one of the state-vectors  $|\Psi'\rangle$  belonging to the ray  $T\mathcal{R}$  with phase chosen so that either  $C_1 = C'_1$  or  $C_1 = C'_1^*$ , respectively. Then either

$$U\left(\sum_k C_k |\Psi_k\rangle\right) = \sum_k C_k U|\Psi_k\rangle \quad (2.A.14)$$

or else

$$U\left(\sum_k C_k |\Psi_k\rangle\right) = \sum_k C_k^* U|\Psi_k\rangle. \quad (2.A.15)$$

It remains to be proved that for a given symmetry transformation, we must make the same choice between Eqs. (2.A.14) and (2.A.15) for arbitrary values of the coefficients  $C_k$ . Suppose that Eq. (2.A.14) applies for a state-vector  $\sum_k A_k |\Psi_k\rangle$  while Eq. (2.A.15) applies for a state-vector  $\sum_k B_k |\Psi_k\rangle$ . Then the invariance of transition probabilities requires that

$$\left| \sum_k B_k^* A_k \right|^2 = \left| \sum_k B_k A_k \right|^2$$

or equivalently

$$\sum_{kl} \text{Im}(A_k^* A_l) \text{Im}(B_k^* B_l) = 0. \quad (2.A.16)$$

We cannot rule out the possibility that Eq. (2.A.16) may be satisfied for a pair of state-vectors  $\sum_k A_k |\Psi_k\rangle$  and  $\sum_k B_k |\Psi_k\rangle$  belonging to different rays. However, for any pair of such state-vectors, with neither  $A_k$  nor  $B_k$  all of the same phase (so that Eqs. (2.A.14) and (2.A.15) are not the same), we can always find a third state-vector  $\sum_k C_k |\Psi_k\rangle$  for which<sup>12</sup>

$$\sum_{kl} \text{Im}(C_k^* C_l) \text{Im}(A_k^* A_l) \neq 0 \quad (2.A.17)$$

and also

$$\sum_{kl} \text{Im}(C_k^* C_l) \text{Im}(B_k^* B_l) \neq 0. \quad (2.A.18)$$

As we have seen, it follows from Eq. (2.A.17) that the same choice between Eqs. (2.A.14) and (2.A.15) must be made for  $\sum_k A_k |\Psi_k\rangle$  and  $\sum_k C_k |\Psi_k\rangle$ , and it follows from Eq. (2.A.18) that the same choice between Eqs. (2.A.14) and (2.A.15) must be made for  $\sum_k B_k |\Psi_k\rangle$  and  $\sum_k C_k |\Psi_k\rangle$ , so the same choice between Eqs. (2.A.14) and (2.A.15) must also be made for the two state-vectors  $\sum_k A_k |\Psi_k\rangle$  and  $\sum_k B_k |\Psi_k\rangle$  with which we started. We have thus shown that for a given symmetry transformation  $T$  either all state-vectors satisfy Eq. (2.A.14) or else they all satisfy Eq. (2.A.15).

It is now easy to show that as we have defined it, the quantum mechanical operator  $U$  is either linear and unitary or antilinear and antiunitary. First, suppose that Eq. (2.A.14) is satisfied for all state-vectors  $\sum_k C_k |\Psi_k\rangle$ . Any two state-vectors  $|\Psi\rangle$  and  $|\Phi\rangle$  may be expanded as

$$|\Psi\rangle = \sum_k A_k |\Psi_k\rangle, \quad |\Phi\rangle = \sum_k B_k |\Psi_k\rangle$$

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<sup>12</sup>If for some pair  $k, l$  both  $A_k^* A_l$  and  $B_k^* B_l$  are complex, then choose all  $C$ 's to vanish except for  $C_k$  and  $C_l$ , and choose these two coefficients to have different phases. If  $A_k^* A_l$  is complex but  $B_k^* B_l$  is real for some pair  $k, l$ , then there must be some other pair  $m, n$  (possibly with either  $m$  or  $n$  but not both equal to  $k$  or  $l$ ) for which  $B_m^* B_n$  is complex. If also  $A_m^* A_n$  is complex, then choose all  $C$ 's to vanish except for  $C_m$  and  $C_n$ , and choose these two coefficients to have different phase. If  $A_m^* A_n$  is real, then choose all  $C$ 's to vanish except for  $C_k$ ,  $C_l$ ,  $C_m$ , and  $C_n$ , and choose these four coefficients all to have different phases. The case where  $B_k^* B_l$  is complex but  $A_k^* A_l$  is real is handled in just the same way.

and so, using Eq. (2.A.14),

$$\begin{aligned} U(\alpha|\Psi\rangle + \beta|\Phi\rangle) &= U \sum_k (\alpha A_k + \beta B_k) |\Psi_k\rangle = \sum_k (\alpha A_k + \beta B_k) U |\Psi_k\rangle \\ &= \alpha \sum_k A_k U |\Psi_k\rangle + \beta \sum_k B_k U |\Psi_k\rangle. \end{aligned}$$

Using Eq. (2.A.14) again, this gives

$$U(\alpha|\Psi\rangle + \beta|\Phi\rangle) = \alpha U|\Psi\rangle + \beta U|\Phi\rangle, \quad (2.A.19)$$

so  $U$  is *linear*. Also, using Eqs. (2.A.2) and (2.A.3), the scalar product of the transformed states is

$$\langle U\Psi | U\Phi \rangle = \sum_{kl} A_k^* B_l \langle U\Psi_k | U\Psi_l \rangle = \sum_k A_k^* B_k,$$

and hence

$$\langle U\Psi | U\Phi \rangle = \langle \Psi | \Phi \rangle, \quad (2.A.20)$$

so  $U$  is *unitary*.

The case of a symmetry that satisfies Eq. (2.A.15) for all state-vectors may be dealt with in much the same way. The reader can probably supply the arguments without help, but since antilinear operators may be unfamiliar, we shall give the details here anyway. Suppose that Eq. (2.A.15) is satisfied for all state-vectors  $\sum_k C_k |\Psi_k\rangle$ . Any two state-vectors  $|\Psi\rangle$  and  $|\Phi\rangle$  may be expanded as before, and so:

$$\begin{aligned} U(\alpha|\Psi\rangle + \beta|\Phi\rangle) &= U \sum_k (\alpha A_k + \beta B_k) |\Psi_k\rangle \\ &= \sum_k (\alpha^* A_k^* + \beta^* B_k^*) U |\Psi_k\rangle \\ &= \alpha^* \sum_k A_k^* U |\Psi_k\rangle + \beta^* \sum_k B_k^* U |\Psi_k\rangle. \end{aligned}$$

Using Eq. (2.A.15) again, this gives

$$U(\alpha|\Psi\rangle + \beta|\Phi\rangle) = \alpha^* U|\Psi\rangle + \beta^* U|\Phi\rangle, \quad (2.A.21)$$

so  $U$  is *antilinear*. Also, using Eqs. (2.A.2) and (2.A.3), the scalar product of the transformed states is

$$\langle U\Psi | U\Phi \rangle = \sum_{kl} A_k B_l^* \langle U\Psi_k | U\Psi_l \rangle = \sum_k A_k B_k^*,$$

and hence

$$\langle U\Psi | U\Phi \rangle = \langle \Psi | \Phi \rangle^*, \quad (2.A.22)$$

so  $U$  is *antiunitary*.

## Appendix B Group Operators and Homotopy Classes

In this appendix we shall prove the theorem stated in Section 2.7, that the phases of the operators  $U(T)$  for finite symmetry transformations  $T$  may be chosen so that these operators form a representation of the symmetry group, rather than a projective representation, provided (a) the generators of the group can be defined so that there are no central charges in the Lie algebra, and (b) the group is simply connected. We shall also comment on the projective representations encountered for groups that are not simply connected, and their relation to the homotopy classes of the group.

To prove this theorem, let us recall the method by which we construct the operators corresponding to symmetry transformations. As described in Section 2.2, we introduce a set of real variables  $\theta^a$  to parameterize these transformations, in such a way that the transformations satisfy the composition rule (2.2.15):

$$T(\bar{\theta})T(\theta) = T(f(\bar{\theta}, \theta)).$$

We want to construct operators  $U(T(\theta)) \equiv U[\theta]$  that satisfy the corresponding condition<sup>13</sup>

$$U[\bar{\theta}]U[\theta] = U[f(\bar{\theta}, \theta)]. \quad (2.B.1)$$

To do this, we lay down arbitrary ‘standard’ paths  $\Theta_\theta^a(s)$  in group parameter space, running from the origin to each point  $\theta$ , with  $\Theta_\theta^a(0) = 0$  and  $\Theta_\theta^a(1) = \theta^a$ , and define  $U_\theta(s)$  along each such path by the differential equation

$$\frac{d}{ds}U_\theta(s) = it_a U_\theta(s) h^a_b(\Theta_\theta(s)) \frac{d\Theta_\theta^b(s)}{ds} \quad (2.B.2)$$

with the initial condition

$$U_\theta(0) = \mathbf{1}, \quad (2.B.3)$$

where

$$[h^{-1}]^a_b(\theta) \equiv \left[ \frac{\partial f^a(\bar{\theta}, \theta)}{\partial \bar{\theta}^b} \right]_{\bar{\theta}=0}. \quad (2.B.4)$$

We are eventually going to identify the operators  $U[\theta]$  with  $U_\theta(1)$ , but first we must establish some of the properties of  $U_\theta(s)$ .

In order to check the composition rule, consider two points  $\theta_1$  and  $\theta_2$ , and define a path  $\gamma$  that runs from 0 to  $\theta_1$  and thence to  $f(\theta_2, \theta_1)$ :

$$\Theta_\gamma^a(s) \equiv \begin{cases} \Theta_{\theta_1}^a(2s), & 0 \leq s \leq \frac{1}{2}, \\ f^a(\Theta_{\theta_2}(2s-1), \theta_1), & \frac{1}{2} \leq s \leq 1. \end{cases} \quad (2.B.5)$$

At the end of the first segment, we are at  $U_\gamma(\frac{1}{2}) = U_{\theta_1}(1)$ . To evaluate  $U_\gamma(s)$  along the second segment, we need the derivative of  $f^a(\Theta_{\theta_2}(2s-1), \theta_1)$ . For this purpose, we use the fundamental associativity condition:

$$f^a(f(\theta_3, \theta_2), \theta_1) = f^a(\theta_3, f(\theta_2, \theta_1)). \quad (2.B.6)$$

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<sup>13</sup>Square brackets are used here to distinguish  $U$  operators constructed as functions of the group parameters from those expressed as functions of the group transformations themselves.



Matching the coefficients of  $\theta_3^c$  in the limit  $\theta_3 \rightarrow 0$  yields the result:

$$\frac{\partial f^a(\theta_2, \theta_1)}{\partial \theta_2^b} h^b{}_c(f(\theta_2, \theta_1)) = h^a{}_c(\theta_2). \quad (2.B.7)$$

Along the second segment, the differential equation (2.B.2) for  $U_\gamma(s)$  is thus the same as the differential equation for  $U_{\theta_2}(2s-1)$ . They satisfy different initial conditions, but  $U_\gamma(s)U_{\theta_1}^{-1}(1)$  also satisfies the same differential equation as  $U_{\theta_2}(2s-1)$ , and in addition the same initial condition: at  $s = \frac{1}{2}$ , both are unity. We therefore conclude that for  $\frac{1}{2} \leq s \leq 1$ ,

$$U_\gamma(s)U_{\theta_1}^{-1}(1) = U_{\theta_2}(2s-1)$$

and in particular

$$U_\gamma(1) = U_{\theta_2}(1)U_{\theta_1}(1). \quad (2.B.8)$$

However, this does *not* say that  $U_\theta(1)$  satisfies the desired composition rule (2.B.1), because although the path  $\Theta_\gamma(s)$  runs from  $\theta^a = 0$  to  $\theta^a = f^a(\theta_2, \theta_1)$ , in general it will not be the same as whatever ‘standard’ path  $\Theta_{f(\theta_2, \theta_1)}$  we have chosen to run directly from  $\theta^a = 0$  to  $\theta^a = f^a(\theta_2, \theta_1)$ . We need to show that  $U_\theta(1)$  is independent of the path from 0 to  $\theta$  in order to be able to identify  $U[\theta]$  as  $U_\theta(1)$ .

For this purpose, consider the variation  $\delta U$  of  $U_\theta(s)$  produced by a variation  $\delta\Theta(s)$  in the path from 0 to  $\theta$ . Taking the variation of Eq. (2.B.2) gives the differential equation

$$\begin{aligned} \frac{d}{ds}\delta U &= it_a \delta U h^a{}_b(\Theta) \frac{d\Theta^b}{ds} + it_a U h^a{}_{b,c}(\Theta) \delta\Theta^c \frac{d\Theta^b}{ds} \\ &\quad + it_a U h^a{}_b(\Theta) \frac{d\delta\Theta^b}{ds}, \end{aligned}$$

where  $h^a{}_{b,c} \equiv \partial h^a{}_b / \partial \theta^c$ . Using the Lie commutation relations (2.2.22) (without central charges) and rearranging a bit, this gives

$$\begin{aligned} \frac{d}{ds}(U^{-1}\delta U) &= \frac{d}{ds}(iU^{-1}t_a U h^a{}_b \delta\Theta^b) \\ &\quad + iU^{-1}t_a U \delta\Theta^b \frac{d\Theta^c}{ds} \left( h^a{}_{c,b} - h^a{}_{b,c} + C^a{}_{ed} h^e{}_b h^d{}_c \right). \end{aligned} \quad (2.B.9)$$

However, by taking the limit  $\theta_3, \theta_2 \rightarrow 0$  in the associativity condition (2.B.6), we find for all  $\theta$ :

$$h(\theta)^a{}_{b,c} = -f^a{}_{de} h(\theta)^d{}_b h(\theta)^e{}_c, \quad (2.B.10)$$

where  $f^a{}_{de}$  is the coefficient defined by (2.2.19). Antisymmetrizing in  $b$  and  $c$  shows that the last term in Eq. (2.B.9) vanishes

$$h^a{}_{c,b} - h^a{}_{b,c} + C^a{}_{ed} h^e{}_b h^d{}_c = 0. \quad (2.B.11)$$

Eq. (2.B.9) thus tells us that the quantity

$$U^{-1}\delta U - iU^{-1}t_a U h^a{}_b \delta\Theta^b$$

is constant along the path  $\theta(s)$ . It follows that  $U_\theta(1)$  is stationary under any infinitesimal variation of the path that leaves the endpoints  $\Theta(0) = 0$  and  $\Theta(1) = \theta$  (and  $U_\theta(0) = \mathbf{1}$ ) fixed. But assumption (b) tells us that any path from  $\Theta(0) = 0$  to  $\Theta(1) = \theta$  can be continuously deformed into any other, so we may now regard  $U_\theta(1)$  as a path-independent function of  $\theta$  alone:

$$U_\theta(1) \equiv U[\theta]. \quad (2.B.12)$$

In particular, since the path  $\gamma$  leads from 0 to  $f(\theta_2, \theta_1)$ , we have

$$U_\gamma(1) = U[f(\theta_2, \theta_1)] \quad (2.B.13)$$

so that Eq. (2.B.8) shows that  $U[\theta]$  satisfies the group multiplication law (2.B.1), as was to be proved.

Now that we have constructed a non-projective representation  $U[\theta]$ , it remains to prove that any projective representation  $\tilde{U}[\theta]$  of the same group with the same representation generators  $t_a$  can only differ from  $U[\theta]$  by a phase:

$$\tilde{U}[\theta] = e^{i\alpha(\theta)} U[\theta]$$

so that the phase  $\phi$  in the multiplication law for  $\tilde{U}[\theta]$ :

$$\tilde{U}[\theta'] \tilde{U}[\theta] = e^{i\phi(\theta', \theta)} \tilde{U}[f(\theta', \theta)]$$

can be removed by a simple change of phase of  $\tilde{U}[\theta]$ . To see this, consider the operator

$$U[\theta]^{-1} U[\theta']^{-1} \tilde{U}[\theta'] \tilde{U}[\theta] = U[f(\theta', \theta)]^{-1} \tilde{U}[f(\theta', \theta)] e^{i\phi(\theta', \theta)}.$$

Because  $U[\theta]$  and  $\tilde{U}[\theta]$  have the same generators, the derivative of the left-hand side with respect to  $\theta'^a$  vanishes at  $\theta' = 0$ , and so

$$0 = \frac{\partial}{\partial \theta'^b} \left\{ U[\theta]^{-1} \tilde{U}[\theta] \right\} + i\phi_b(\theta) U[\theta]^{-1} \tilde{U}[\theta],$$

where

$$\phi_b(\theta) \equiv h^a{}_b(\theta) \left[ \frac{\partial}{\partial \theta'^a} \phi(\theta', \theta) \right]_{\theta'=0}.$$

Differentiating this result with respect to  $\theta^c$  and antisymmetrizing in  $b$  and  $c$  gives immediately

$$0 = \frac{\partial \phi_b(\theta)}{\partial \theta^c} - \frac{\partial \phi_c(\theta)}{\partial \theta^b}.$$

A familiar theorem [13] tells us that in a simply connected space, this requires that  $\phi_b$  is just a gradient of some function  $\beta$ :

$$\phi_b(\theta) = \frac{\partial \beta(\theta)}{\partial \theta^b}.$$

Thus the quantity  $U[\theta]^{-1} \tilde{U}[\theta] e^{i\beta(\theta)}$  is actually constant in  $\theta$ . Setting it equal to its value at  $\theta = 0$ , we see that  $\tilde{U}$  is just proportional to  $U$ :

$$\tilde{U}[\theta] = U[\theta] \exp(-i\beta(\theta) + i\beta(0))$$

as claimed above.

\* \* \*

The above analysis provides some information about the nature of the phase factors that can appear in the group multiplication law when the Lie algebra is free of central charges but the group is not simply connected. Suppose that the path  $\gamma$  from zero to  $\theta$  to  $f(\bar{\theta}, \theta)$  cannot be deformed into the standard path we have chosen to go from zero to  $f(\bar{\theta}, \theta)$ , or in other words, that the loop from zero to  $\theta$  to  $f(\bar{\theta}, \theta)$  and then back to zero is not continuously deformable to a point. Then  $U[f(\theta_2, \theta_1)]^{-1}U[\theta_2]U[\theta_1]$  can be a phase factor  $\exp(i\phi(\theta_2, \theta_1)) \neq 1$ , but  $\phi$  will be the same for all other loops into which this can be continuously deformed. The set consisting of all loops that start and end at the origin and that can be continuously deformed into a given loop is known as the *homotopy class* [14] of that loop; we have thus seen that  $\phi(\theta_2, \theta_1)$  depends only on the homotopy class of the loop from zero to  $\theta$  to  $f(\bar{\theta}, \theta)$  and then back to zero. The set of homotopy classes forms a group; the ‘product’ of the homotopy class for loops  $\mathcal{L}_1$  and  $\mathcal{L}_2$  is the homotopy class of the loop formed by going around  $\mathcal{L}_1$  and then  $\mathcal{L}_2$ ; the ‘inverse’ of the homotopy class of the loop  $\mathcal{L}$  is the homotopy class of the loop obtained by going around  $\mathcal{L}$  in the opposite direction; and the ‘identity’ is the homotopy class of loops that can be deformed into a point at the origin. This group is known as the *first homotopy group* or *fundamental group* of the space in question. It is easy to see that the phase factors form a representation of this group: if going around loop  $\mathcal{L}$  gives a phase factor  $e^{i\phi}$ , and going around loop  $\bar{\mathcal{L}}$  gives a phase factor  $e^{i\bar{\phi}}$ , then going around both loops gives a phase factor  $e^{i\phi}e^{i\bar{\phi}}$ . Hence we can catalog all the possible types of projective representations of a given group  $\mathcal{G}$  (with no central charges) if we know the one-dimensional representations of the first homotopy group of the parameter space of  $\mathcal{G}$ . Homotopy groups will be discussed in greater detail in Volume II.

## Appendix C Inversions and Degenerate Multiplets

It is usually assumed that the inversions  $T$  and  $P$  take one-particle states into other one-particle states of the same species, perhaps with phase factors that depend on the particle species. In Section 2.6 we noted in passing that inversions might act in a more complicated way than this on degenerate multiplets of one-particle states, a possibility that seems to have been first suggested by Wigner [15] in 1964. This appendix will explore generalized versions of the inversion operators, in which finite matrices appear in place of the inversion phases, but without making some of Wigner’s limiting assumptions.

Let us start with time-reversal. Wigner limited the possible action of the inversion operators by assuming that their squares are proportional to the unit operator. Because  $T$  is antiunitary, it is easy to see that the corresponding proportionality factor for  $T^2$  can only be  $\pm 1$ , perhaps with different signs for subspaces separated by superselection rules. When the sign for  $T^2$  on the space of states with even or odd values of  $2j$  is opposite to the sign  $(-1)^{2j}$  found in Section 2.6, the physical states involved must furnish representations of the operator  $T$  that are more complicated than that assumed so far. But if we are willing to admit this possibility, there does not seem to be any good reason to impose Wigner’s condition that  $T^2$  is proportional to unity. It is not convincing to appeal to the structure of the extended Poincaré group; the only useful definition of any of the inversion operators

is one that makes the operator exactly or approximately conserved, and this may not be the definition that makes  $T^2$  proportional to the unit operator.

To explore more general possibilities for time-reversal, let us assume that on a massive one-particle state it has the action

$$T|\mathbf{p}, \sigma, n\rangle = (-1)^{j-\sigma} \sum_m \Theta_{mn} |-\mathbf{p}, -\sigma, m\rangle, \quad (2.C.1)$$

where  $\mathbf{p}$ ,  $j$ , and  $\sigma$  are the particle's momentum, spin, and spin  $z$ -component, and  $n, m$  are indices labelling members of a degenerate multiplet of particle species. (The appearance of the factor  $(-1)^{j-\sigma}$  and the reversal of  $\mathbf{p}$  and  $\sigma$  are deduced in the same way as in Section 2.6.) The matrix  $\Theta_{mn}$  is unknown, except that because  $T$  is antiunitary,  $\Theta$  must be unitary.

Now let us see how we can simplify this transformation by an appropriate choice of basis for the one-particle states. Defining new states by the unitary transformation  $|\mathbf{p}, \sigma, n\rangle' = \sum_m \mathcal{U}_{mn} |\mathbf{p}, \sigma, m\rangle$ , we find the same transformation (2.C.1), with the matrix  $\Theta_{mn}$  changed to

$$\Theta' = \mathcal{U}^{-1} \Theta \mathcal{U}^*. \quad (2.C.2)$$

We cannot in general make  $\Theta'$  diagonal by such a choice of basis of the one-particle states, as we could if  $T$  were unitary. But we can instead make it block-diagonal, with the blocks either  $1 \times 1$  phases, or  $2 \times 2$  matrices of the form

$$\begin{pmatrix} 0 & e^{i\phi/2} \\ e^{-i\phi/2} & 0 \end{pmatrix}, \quad (2.C.3)$$

where the  $\phi$  are various real phases.

(Here is the proof. First, note that Eq. (2.C.2) gives

$$\Theta' \Theta'^* = \mathcal{U}^{-1} \Theta \Theta^* \mathcal{U}.$$

This is a unitary transformation, so it can be chosen to diagonalize the unitary matrix  $\Theta \Theta^*$ . Assuming this to have been done, and dropping primes, we have

$$\Theta = D \Theta^T, \quad (2.C.4)$$

where  $D$  is a unitary diagonal matrix, say with phases  $e^{i\phi_n}$  along the main diagonal. One immediate consequence is that the diagonal component  $\Theta_{nn}$  vanishes unless  $e^{i\phi_n} = 1$ . Furthermore, if  $e^{i\phi_n} = 1$  but  $e^{i\phi_m} \neq 1$ , then Eq. (2.C.4) tells us that  $\Theta_{nm} = \Theta_{mn} = 0$ . By listing first all rows and columns for which  $e^{i\phi_n} = 1$ , the matrix  $\Theta$  is put in the form

$$\Theta = \begin{pmatrix} \mathcal{A} & 0 \\ 0 & \mathcal{B} \end{pmatrix}, \quad (2.C.5)$$

where  $\mathcal{A}$  is symmetric as well as unitary, and the diagonal elements of  $\mathcal{B}$  all vanish. Because  $\mathcal{A}$  is symmetric, it can be expressed as the exponential of a symmetric anti-Hermitian matrix, so it can be diagonalized by a transformation (2.C.2) acting only on  $\mathcal{A}$ , with the corresponding submatrix of  $\mathcal{U}$  real and hence orthogonal. It is therefore only necessary to

consider the submatrix  $\mathcal{B}$  that connects the rows and columns for which  $e^{i\phi_n} \neq 0$ . For  $n \neq m$ , Eq. (2.C.4) gives  $\Theta_{nm} = e^{i\phi_n}\Theta_{mn}$  and  $\Theta_{mn} = e^{i\phi_m}\Theta_{nm}$ , so  $\Theta_{nm} = e^{i\phi_n}e^{i\phi_m}\Theta_{nm}$  and also  $\Theta_{mn} = e^{i\phi_n}e^{i\phi_m}\Theta_{mn}$ . Hence  $\Theta_{nm} = \Theta_{mn} = 0$  unless  $e^{i\phi_n}e^{i\phi_m} = 1$ . If we list first all rows and columns of  $\mathcal{B}$  with a given phase  $e^{i\phi_1} \neq 1$ , and then all rows and columns with the opposite phase, and then all rows and columns with some other phase  $e^{i\phi_2} \neq 1$  not equal to  $e^{\pm i\phi_1}$ , and then all rows and columns with opposite phase, and so on, the matrix  $\mathcal{B}$  becomes of block diagonal form

$$\mathcal{B} = \begin{pmatrix} \mathcal{B}_1 & 0 & \cdots \\ 0 & \mathcal{B}_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}, \quad (2.C.6)$$

where

$$\mathcal{B}_i = \begin{pmatrix} 0 & e^{i\phi_i/2}\mathcal{C}_i \\ e^{-i\phi_i/2}\mathcal{C}_i^\dagger & 0 \end{pmatrix}. \quad (2.C.7)$$

Furthermore, the unitarity of  $\Theta$  and hence of  $\mathcal{B}$  requires that  $\mathcal{C}_i\mathcal{C}_i^\dagger = \mathcal{C}_i^\dagger\mathcal{C}_i = \mathbf{1}$ , and hence  $\mathcal{C}_i$  is square and unitary. By applying a transformation (2.C.2) with  $\mathcal{U}$  block-diagonal in the same sense as  $\Theta$ , and with the matrix in the  $i$ th block of form

$$\begin{pmatrix} V_i & 0 \\ 0 & W_i \end{pmatrix}$$

with  $V_i$  and  $W_i$  unitary, the submatrices  $\mathcal{C}_i$  are subjected to the transformations  $\mathcal{C}_i \rightarrow V_i^{-1}\mathcal{C}_iW_i^*$ , so we can clearly choose this transformation to make  $\mathcal{C}_i = \mathbf{1}$ . This establishes a correspondence between pairs of individual rows and columns within each block with phases  $e^{i\phi_i}$  and  $e^{-i\phi_i}$ . To put the matrix  $\mathcal{B}$  into block-diagonal form with  $2 \times 2$  blocks of form (2.C.3), it is now only necessary to rearrange the rows and columns so that within the  $i$ th block we list rows and columns with phase  $e^{i\phi_i}$  alternating with the corresponding rows and columns with phase  $e^{-i\phi_i}$ .

It is important to note that where  $e^{i\phi} \neq 1$ , it is not possible to choose states to diagonalize the time-reversal transformation. If we have a pair of states  $|\mathbf{p}, \sigma, \pm\rangle$  on which  $T$  acts with a matrix (2.C.3), then

$$T|\mathbf{p}, \sigma, \pm\rangle = e^{\pm i\phi/2}(-1)^{j-\sigma}|\mathbf{p}, -\sigma, \mp\rangle. \quad (2.C.8)$$

Then on an arbitrary linear combination of these states, time-reversal gives

$$T(c_+|\mathbf{p}, \sigma, +\rangle + c_-|\mathbf{p}, \sigma, -\rangle) = (-1)^{j-\sigma}(e^{i\phi/2}c_+^*|\mathbf{p}, -\sigma, -\rangle + e^{-i\phi/2}c_-^*|\mathbf{p}, -\sigma, +\rangle).$$

For  $c_+|\mathbf{p}, \sigma, +\rangle + c_-|\mathbf{p}, \sigma, -\rangle$  to be transformed under  $T$  by a phase  $\lambda$ , it is necessary that

$$e^{i\phi/2}c_+^* = \lambda c_-, \quad e^{-i\phi/2}c_-^* = \lambda c_+.$$

But combining these equations gives  $e^{\pm i\phi/2}c_\pm^* = |\lambda|^2 c_\pm^* e^{\mp i\phi/2}$ , which is impossible unless either  $c_+ = c_- = 0$  or  $e^{i\phi}$  is unity. Thus for  $e^{i\phi} \neq 1$ , time-reversal invariance imposes a two-fold degeneracy on these states, beyond that associated with their spin.

Of course, if there is an additional ‘internal’ symmetry operator  $S$  which subjects these states to the transformation

$$S|\mathbf{p}, \sigma, \pm\rangle = e^{\pm i\phi/2} |-\mathbf{p}, \sigma, \mp\rangle,$$

then we can redefine the time-reversal operator as  $T' \equiv S^{-1}T$ , and this operator would not mix the states  $|\mathbf{p}, \sigma, \pm\rangle$  with one another. It is only in the case where no such internal symmetry exists that we can attribute the doubling of particle states to time-reversal itself.

Let’s come back now to the question of the square of  $T$ . Repeating the transformation (2.C.8) gives

$$T^2|\mathbf{p}, \sigma, \pm\rangle = (-1)^{2j} e^{\mp i\phi} |\mathbf{p}, \sigma, \pm\rangle. \quad (2.C.9)$$

If we were to assume with Wigner that  $T^2$  is proportional to the unit operator, then we would have to have  $e^{i\phi} = e^{-i\phi}$ , and since the phase is then real it would have to be  $+1$  or  $-1$ . The choice  $e^{i\phi} = -1$  would still require a two-fold degeneracy of one-particle states beyond that associated with their spin, and under Wigner’s assumptions *all* particles would show this doubling. But there is no reason not to take a general phase  $\phi$  in Eq. (2.C.8), one that may vanish for some particles and not for others. Thus the fact that observed particles do not show the extra two-fold degeneracy does not rule out the possibility that others might.

We may also consider the possibility of more complicated representations of the parity operator  $P$ , with

$$P|\mathbf{p}, \sigma, n\rangle = \sum_m \Pi_{mn} |-\mathbf{p}, \sigma, m\rangle \quad (2.C.10)$$

with a unitary but otherwise unconstrained matrix  $\Pi$ . Unlike the case of time-reversal, here we may always diagonalize this matrix by a choice of basis for the states. But this choice of basis may not be the one in which time-reversal acts simply, so, in principle,  $P$  and  $T$  together can impose additional degeneracies that would not be required by  $P$  or  $T$  alone.

As discussed in Chapter 5, any quantum field theory is expected to respect a symmetry known as CPT, which acts on one-particle states as

$$CPT|\mathbf{p}, \sigma, n\rangle = (-1)^{j-\sigma} |\mathbf{p}, -\sigma, n^c\rangle, \quad (2.C.11)$$

where  $n^c$  denotes the antiparticle (or ‘charge-conjugate’) of particle  $n$ . No phases or matrices are allowed in this transformation (though of course we could always introduce such phases or matrices by combining CPT with good internal symmetries.) It follows that

$$(CPT)^2|\mathbf{p}, \sigma, n\rangle = (-1)^{2j} |\mathbf{p}, -\sigma, n\rangle, \quad (2.C.12)$$

so the possibility suggested by Wigner of a sign  $-(-1)^{2j}$  in the action of  $(CPT)^2$  does not arise in quantum field theory.

To the extent that  $T$  is a good symmetry of some class of phenomena, so is the inversion  $CP \equiv (CPT)T^{-1}$ . For the states that transform under  $T$  in the conventional way

$$T|\mathbf{p}, \sigma, n\rangle \propto |-\mathbf{p}, -\sigma, n\rangle, \quad (2.C.13)$$

the  $CP$  operator also acts conventionally

$$CP|\mathbf{p}, \sigma, n\rangle \propto |-\mathbf{p}, \sigma, n^c\rangle. \quad (2.C.14)$$

The operator  $C \equiv CPP^{-1}$  then just interchanges particles and antiparticles

$$C|\mathbf{p}, \sigma, n\rangle \propto |\mathbf{p}, \sigma, n^c\rangle. \quad (2.C.15)$$

On the other hand, where  $T$  has the unconventional representation (2.C.8), Eq. (2.C.11) gives

$$CP|\mathbf{p}, \sigma, \pm\rangle = e^{\mp i\phi/2} |-\mathbf{p}, \sigma, \mp^c\rangle \quad (2.C.16)$$

In particular, it is possible that the degeneracy indicated by the label  $\pm$  may be the same as the particle–antiparticle degeneracy, so that the antiparticle (as defined by CPT) of the state  $|\Psi_{\pm}\rangle$  is  $|\Psi_{\mp}\rangle$ . In this case, CP would have the unconventional property of *not* interchanging particles and antiparticles. As far as these particles are concerned, CP and T would be what are usually called P and CT. But this is not merely a matter of definition; on other particles CP and T would still have their usual effect.

No examples are known of particles that furnish unconventional representations of inversions, so these possibilities will not be pursued further here. From now on, the inversions will be assumed to have the conventional action assumed in Section 2.6.

### Weinberg Exercises

- W.2.1.** Suppose that observer  $\mathcal{O}$  sees a  $W$ -boson (spin one and mass  $m \neq 0$ ) with momentum  $\mathbf{p}$  in the  $y$ -direction and spin  $z$ -component  $\sigma$ . A second observer  $\mathcal{O}'$  moves relative to the first with velocity  $\mathbf{v}$  in the  $z$ -direction. How does  $\mathcal{O}'$  describe the  $W$  state?
- W.2.2.** Suppose that observer  $\mathcal{O}$  sees a photon with momentum  $\mathbf{p}$  in the  $y$ -direction and polarization vector in the  $z$ -direction. A second observer  $\mathcal{O}'$  moves relative to the first with velocity  $\mathbf{v}$  in the  $z$ -direction. How does  $\mathcal{O}'$  describe the same photon?
- W.2.3.** Derive the commutation relations for the generators of the Galilean group directly from the group multiplication law (without using our results for the Lorentz group). Include the most general set of central charges that cannot be eliminated by redefinition of the group generators.
- W.2.4.** Show that the operators  $P_{\mu}P^{\mu}$  and  $W_{\mu}W^{\mu}$  commute with all Lorentz transformation operators  $U(\Lambda, a)$ , where  $W_{\mu} \equiv \epsilon_{\mu\nu\rho\lambda} J^{\nu\rho} P^{\lambda}$ .
- W.2.5.** Consider physics in two space and one time dimensions, assuming invariance under a ‘Lorentz’ group  $SO(2, 1)$ . How would you describe the spin states of a single massive particle? How do they behave under Lorentz transformations? What about the inversions P and T?
- W.2.6.** As in Problem 5, consider physics in two space and one time dimensions, assuming invariance under a ‘Lorentz’ group  $O(2, 1)$ . How would you describe the spin states of a single massless particle? How do they behave under Lorentz transformations? What about the inversions P and T?

## Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

**W.2.1.** Write

$$p^\mu = (E, 0, p, 0), \quad E = \sqrt{p^2 + m^2}, \quad \sinh \xi = \frac{p}{m}.$$

If  $\mathcal{O}'$  moves with velocity  $v\hat{\mathbf{z}}$ , its coordinates are obtained with the passive boost  $B_z(-\eta)$ , where  $\tanh \eta = v$ . Thus

$$p'^\mu = B_z(-\eta)p^\mu = (\gamma_v E, 0, p, -\gamma_v v E), \quad \gamma_v = (1 - v^2)^{-1/2}.$$

The transformed state is not, in general, an eigenstate of the old  $z$ -component of spin. With rotation-free standard boosts  $L(p)$ , its spin is acted on by the Wigner rotation

$$R_W = L^{-1}(p')B_z(-\eta)L(p).$$

For the two perpendicular boosts here,  $R_W$  is a rotation about the  $x$ -axis. Choosing its positive sense by the displayed definition,

$$\tan \frac{\Omega}{2} = \tanh \frac{\eta}{2} \tanh \frac{\xi}{2}, \quad \tan \Omega = \frac{\sinh \eta \sinh \xi}{\cosh \eta + \cosh \xi}.$$

(Changing from the passive to the active convention reverses the sign of  $\Omega$ .) Equation (2.5.23) therefore gives

$$U(B_z(-\eta))|p\hat{\mathbf{y}}, \sigma\rangle = \sqrt{\frac{p'^0}{E}} \sum_{\sigma'=-1}^1 D_{\sigma'\sigma}^{(1)}(R_x(\Omega)) |\mathbf{p}', \sigma'\rangle.$$

The square-root is absent if covariantly normalized momentum states are used. Hence  $\mathcal{O}'$  sees the momentum above and a coherent superposition of the three  $z$ -spin values, rather than the original single value  $\sigma$ .

**W.2.2.** Take the original four-momentum and polarization to be

$$p^\mu = (\omega, 0, \omega, 0), \quad \epsilon^\mu = (0, 0, 0, 1).$$

The same passive boost as in the preceding solution gives

$$p'^\mu = (\gamma_v \omega, 0, \omega, -\gamma_v v \omega), \quad \epsilon'^\mu = (-\gamma_v v, 0, 0, \gamma_v).$$

They obey  $p' \cdot \epsilon' = 0$  and  $\epsilon'^2 = 1$ . The time component of  $\epsilon'$  is not a new physical polarization; it is removable by a gauge transformation. Adding  $(v/\omega)p'^\mu$  produces the temporal-gauge representative

$$\tilde{\epsilon}^\mu = \epsilon'^\mu + \frac{v}{\omega} p'^\mu = (0, 0, v, \gamma_v^{-1}).$$

Indeed,

$$\hat{\mathbf{p}}' = (0, \gamma_v^{-1}, -v), \quad \tilde{\boldsymbol{\epsilon}} = (0, v, \gamma_v^{-1}),$$

so  $\hat{\mathbf{p}}' \cdot \tilde{\boldsymbol{\epsilon}} = 0$  and  $\tilde{\boldsymbol{\epsilon}}^2 = 1$ . Thus  $\mathcal{O}'$  sees a photon moving obliquely in the  $yz$ -plane, with a transverse linear polarization rotated in that plane. No longitudinal photon state has been created.



**W.2.3.** Parameterize a Galilean transformation by  $(R, \mathbf{v}, \mathbf{a}, b)$ , acting on spacetime as

$$t' = t + b, \quad \mathbf{x}' = R\mathbf{x} + \mathbf{v}t + \mathbf{a}.$$

Composition gives a semidirect product: rotations act on  $\mathbf{v}, \mathbf{a}$ , while a boost followed by a time translation also shifts the spatial translation. Differentiating the multiplication law at the identity gives, before central terms,

$$\begin{aligned} [J_i, J_j] &= i\epsilon_{ijk}J_k, & [J_i, K_j] &= i\epsilon_{ijk}K_k, & [J_i, P_j] &= i\epsilon_{ijk}P_k, \\ [K_i, H] &= iP_i, & [K_i, K_j] &= [P_i, P_j] = [P_i, H] = [J_i, H] = 0. \end{aligned}$$

A ray representation may multiply the group law by a two-cocycle. Rotational covariance and the Jacobi identities eliminate every nontrivial central tensor except a scalar in the boost–translation bracket. After constant shifts of generators remove coboundary terms, the most general algebra in three spatial dimensions is therefore

$$[K_i, P_j] = i\delta_{ij}M, \quad [M, H] = [M, P_i] = [M, K_i] = [M, J_i] = 0,$$

together with the preceding commutators. A possible constant in  $H$ , or constant vector terms in  $\mathbf{P}$  or  $\mathbf{K}$ , are removed by redefining those generators. Terms such as  $[K_i, K_j] = i\epsilon_{ijk}C_k$  would require a central vector  $\mathbf{C}$ ; the Jacobi identity with  $J_i$  forces it to vanish in three dimensions. Thus the sole non-removable central charge is  $M$ , identified in an irreducible representation with the total mass.

**W.2.4.** Equation (2.4.9) says that translations leave  $P^\mu$  unchanged and a Lorentz transformation rotates it as a four-vector. Consequently

$$U(\Lambda, a)P_\mu P^\mu U^{-1}(\Lambda, a) = \eta_{\rho\sigma}P^\rho P^\sigma = P_\mu P^\mu.$$

For the Pauli–Lubanski operator, use Eqs. (2.4.8) and (2.4.9). Under a translation,

$$J^{\nu\rho} \mapsto J^{\nu\rho} - a^\nu P^\rho + a^\rho P^\nu.$$

The extra contribution to  $W_\mu$  contains the contraction of  $\epsilon_{\mu\nu\rho\lambda}$  with the symmetric product  $P^\rho P^\lambda$ , and hence vanishes. Under a proper Lorentz transformation the invariance of the Levi–Civita tensor gives

$$U(\Lambda, a)W_\mu U^{-1}(\Lambda, a) = \Lambda_\mu{}^\nu W_\nu.$$

Therefore

$$U(\Lambda, a)W_\mu W^\mu U^{-1}(\Lambda, a) = W_\mu W^\mu.$$

Both operators commute with every  $U(\Lambda, a)$ , as required. For an improper Lorentz transformation  $W_\mu$  is a pseudovector, but its square is still invariant.

**W.2.5.** Choose the standard massive momentum  $k^\mu = (m, 0, 0)$ . Its connected little group in  $2+1$  dimensions is  $SO(2)$ , generated by the single spatial rotation  $J$ . Every complex unitary irreducible representation of  $SO(2)$  is one-dimensional:

$$D_s(R(\theta)) = e^{is\theta}.$$

For the group  $SO(2, 1)$  itself, single-valuedness requires  $s \in \mathbb{Z}$ . For its double cover one obtains integer or half-integer  $s$ , while for the universal cover  $s$  may be any real number; the latter possibility is the origin of anyonic spin.

With standard boosts  $L(p)$ , a one-particle state is denoted simply  $|p, s\rangle$ , with no  $2s+1$  spin degeneracy, and

$$U(\Lambda)|p, s\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}} e^{is\theta(\Lambda, p)} |\Lambda p, s\rangle, \quad L^{-1}(\Lambda p)\Lambda L(p) = R(\theta).$$

A spatial reflection reverses the orientation of the plane and conjugates  $J$  to  $-J$ ; time reversal is antiunitary and also reverses  $J$ . Thus either  $P$  or  $T$  carries the representation  $s$  into  $-s$ . If either inversion is a symmetry, a particle with  $s \neq 0$  must have a degenerate partner of spin  $-s$  (unless the two are already the same representation). The combined operation  $PT$  preserves  $s$ .

**W.2.6.** For a standard null momentum  $k^\mu = (\kappa, 0, \kappa)$ , the connected little group in  $2+1$  dimensions is the one-parameter group of null rotations, isomorphic to  $(\mathbb{R}, +)$ . Its unitary irreducible representations are characters

$$D_\rho(S(\alpha)) = e^{i\rho\alpha}, \quad \rho \in \mathbb{R}.$$

Accordingly, the most general irreducible massless one-particle state is  $|p, \rho\rangle$ , and transforms as

$$U(\Lambda)|p, \rho\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}} e^{i\rho\alpha(\Lambda, p)} |\Lambda p, \rho\rangle.$$

There is no transverse rotation group  $SO(d-2)$  in  $d=3$ , so there is no ordinary helicity label. The particle representations normally used in local field theory take the null-rotation subgroup trivially,  $\rho=0$ ; they then have one state at each momentum.

An orientation- or time-reversing element conjugates the null-rotation generator to its negative. Hence  $P$  or  $T$  maps  $\rho \mapsto -\rho$ . A full  $O(2, 1)$  representation with  $\rho \neq 0$  must contain the pair  $(\rho, -\rho)$ . For the usual  $\rho=0$  representation, each inversion acts within the same one-particle species, by a phase for the unitary  $P$  and by an antiunitary phase convention for  $T$ .

## Supplementary Exercises

### S.2.1. Invariant On-Shell Measures and Lorentz-Normalized Wave Packets

(Adapted from MIT OpenCourseWare, 8.323, Spring 2023, Problem Set 1, Problem 2.)

- (a) Let  $\Lambda$  be a Lorentz transformation. Prove directly from the defining property of the Dirac delta that

$$\delta^4(\Lambda(p-q)) = \delta^4(p-q).$$

State where the absolute value of the determinant enters the argument.

- (b) For  $m \geq 0$ , prove that

$$d^4p \delta(p^2 + m^2) \theta(p^0) = \frac{d^3\mathbf{p}}{2E_{\mathbf{p}}}, \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2},$$

after the  $p^0$  integration, and explain precisely which component of the Lorentz group preserves the positive-energy sheet.

- (c) Let  $p$  and  $q$  lie on the same positive-energy mass shell. Show that  $E_{\mathbf{p}} \delta^3(\mathbf{p} - \mathbf{q})$  is Lorentz invariant. Deduce the covariant momentum-state normalization and compare it with the  $\delta^3$ -normalization used in Section 2.5.
- (d) A scalar one-particle wave packet is represented by a function  $\psi(\mathbf{p})$  on the positive mass shell. Find an invariant inner product, determine the scalar transformation law for  $\psi$ , and verify directly that the norm is frame independent.
- (e) For a boost of velocity  $u\hat{\mathbf{z}}$ , calculate the Jacobian  $\partial(p'_x, p'_y, p'_z)/\partial(p_x, p_y, p_z)$  on the mass shell. Use it to prove  $d^3\mathbf{p}'/E_{\mathbf{p}'} = d^3\mathbf{p}/E_{\mathbf{p}}$  without invoking a four-dimensional delta function.

### S.2.2. The Poincaré Algebra from Infinitesimal Generators

(Adapted from M. Gaberdiel, *ETH Zürich QFT I, HS 2011, Sheet 4, Exercise 2.*)

- (a) Acting on scalar functions of spacetime, take

$$P^\mu = -i\partial^\mu, \quad J^{\mu\nu} = -i(x^\mu\partial^\nu - x^\nu\partial^\mu).$$

Derive all three covariant Poincaré commutators. Check the signs by comparing with Eqs. (2.4.12) and (2.4.14).

- (b) Starting from the covariant Poincaré algebra, define  $J_i = \frac{1}{2}\epsilon_{ijk}J^{jk}$  and  $K_i = J^{i0}$ . Derive the commutators among  $\mathbf{J}, \mathbf{K}, \mathbf{P}, H$ , including  $[K_i, K_j]$ ,  $[K_i, P_j]$ , and  $[K_i, H]$ .

### S.2.3. The Poincaré Group Commutator and Its Lie Algebra

(Adapted from University of Cambridge, *Mathematical Tripos Part III, 2022, Paper 302, Symmetries, Fields and Particles, Question 2, printed p. 3.*)

A Poincaré transformation

$$x^\mu \longmapsto x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu$$

is written as  $(\Lambda, a)$ , where  $\Lambda^\mu{}_\nu$  is a Lorentz transformation.

- (i) Compute  $(L, b)(\Lambda, a)$ .
- (ii) Determine  $(\Lambda, a)^{-1}$  in terms of  $\Lambda$  and  $a$ .
- (iii) Defining

$$(\Lambda_c, a_c) := [(L, b), (\Lambda, a)],$$

where the brackets denote the group commutator, find  $\Lambda_c$  and  $a_c$  in terms of  $L, \Lambda, b, a$ .

- (iv) Define infinitesimal Lorentz transformations by

$$\Lambda := I + \omega_\Lambda + O(\omega_\Lambda^2), \quad L := I + \omega_L + O(\omega_L^2).$$

Find  $a_c$  to first order in  $\omega_L$  and first order in  $\omega_\Lambda$ .

- (v) Put  $\Lambda_c := I + \omega_c + O(\omega_c^2)$ , neglect terms of order  $\omega_L^2$  and  $\omega_\Lambda^2$ , and find  $\omega_c$  in terms of  $\omega_L$  and  $\omega_\Lambda$ .
- (vi) Write the defining relation for  $\Lambda^\mu{}_\nu$  in terms of the Minkowski metric  $\eta_{\rho\sigma}$ , and hence derive a symmetry relation for  $\omega_\Lambda$  under interchange of its indices.
- (vii) Introduce a unitary operator corresponding to a Poincaré transformation,  $U[(\Lambda, a)]$ , imposing

$$U[(\Lambda_2, a_2)]U[(\Lambda_1, a_1)] = U[(\Lambda_2, a_2)(\Lambda_1, a_1)].$$

Expanding in infinitesimal  $\omega$  and  $a$ , use the source normalization

$$U[(\Lambda(\omega), a)] := I + \frac{1}{2}\omega^{\mu\nu}M_{\mu\nu} + a^\mu P_\mu + \cdots, \quad M_{\mu\nu} = -M_{\nu\mu}.$$

Retaining constant, linear, and mixed bilinear terms in  $\omega_\Lambda, \omega_L, a, b$ , but neglecting pure squares and higher orders, show that

$$U[(\Lambda_c(\omega_c), a_c)] = I + \left[ \frac{1}{2}\omega_L^{\mu\nu}M_{\mu\nu} + b^\mu P_\mu, \frac{1}{2}\omega_\Lambda^{\rho\sigma}M_{\rho\sigma} + a^\rho P_\rho \right] + \cdots.$$

- (viii) Hence derive the Poincaré algebra: determine  $[P_\mu, P_\nu]$ ,  $[M_{\mu\sigma}, P_\rho]$ , and  $[M_{\mu\nu}, M_{\rho\sigma}]$ .

#### S.2.4. Pauli–Lubanski Spin and the Massive Casimir

(Adapted from S. Gieseke, C. Borschensky, and C. B. Duncan, KIT, Exercise Sheet 5, 2021, Problem 1.)

- (a) With

$$P^\mu = -i\partial^\mu, \quad J^{\mu\nu} = -i(x^\mu\partial^\nu - x^\nu\partial^\mu),$$

derive  $[J^{\mu\nu}, P^\rho]$ . Then define

$$\mathcal{W}_\mu = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}J^{\nu\rho}P^\sigma.$$

Prove  $\mathcal{W}_\mu P^\mu = 0$  and  $[\mathcal{W}_\mu, P^\nu] = 0$ . Finally derive the supplied Pauli–Lubanski commutator, translated to the conventions of this volume,

$$[\mathcal{W}_\mu, \mathcal{W}_\nu] = -i\epsilon_{\mu\nu\rho\sigma}\mathcal{W}^\rho P^\sigma.$$

- (b) For a massive irreducible representation, evaluate  $\mathcal{W}^\mu$  in the rest frame  $p^\mu = (m, \mathbf{0})$ . Show that  $\mathcal{W}^2 = m^2 j(j+1)$  in the mostly-plus metric and explain why the answer is frame independent.

- (c) Establish the algebraic identity

$$\mathcal{W}^2 = -\frac{1}{2}J_{\mu\nu}J^{\mu\nu}P^2 + J_{\mu\rho}J_{\nu}^{\rho}P^{\mu}P^{\nu}.$$

Use it, together with the Poincaré algebra, to show directly that  $P^2$  commutes with every Poincaré generator and that  $[\mathcal{W}^2, P^{\mu}] = 0$ . As in the source problem, do not perform the tedious direct expansion of  $[\mathcal{W}^2, J^{\mu\nu}]$ ; obtain that commutator in the next part.

- (d) Show directly that  $\mathcal{W}^{\mu}$  is unchanged by translations and transforms as a pseudovector under the full Lorentz group. Reconcile this finite-transformation argument with the preceding algebra and conclude that  $[\mathcal{W}^2, J^{\mu\nu}] = 0$ .

### S.2.5. Massive Little-Group Representations and Normalization

(Inspired by P. Woit, *Quantum Field Theory for Mathematicians*, Chapter 11, 2026.)

- (a) List the eigenvalues of  $P^2$  and  $\mathcal{W}^2$  for massive particles of spin  $0, \frac{1}{2}, 1$ . Then show that for a finite-helicity massless representation  $\mathcal{W}^{\mu} = \lambda P^{\mu}$ , so both Casimirs vanish for every helicity. What extra datum distinguishes the massless representations?
- (b) Find every proper orthochronous Lorentz transformation that fixes  $k^{\mu} = (m, 0, 0, 0)$ . Identify the little group and use its unitary irreducible representations to recover the  $2j + 1$  spin states of a massive particle.
- (c) Let  $|p, \sigma\rangle = N(p)U(L(p))|k, \sigma\rangle$  and normalize momentum states with  $\delta^3(\mathbf{p} - \mathbf{p}')$ . Determine  $N(p)$  for massive and massless standard momenta, and rederive the square-root factor in the Lorentz transformation law.

### S.2.6. Finite-Helicity Representations of the Null Little Group

(Inspired by J. Preskill, *Physics 236c notes*, Chapter 1, 1990.)

- (a) For  $k^{\mu} = (\kappa, 0, 0, \kappa)$ , verify that

$$A = J_2 + K_1, \quad B = -J_1 + K_2, \quad J_3$$

annihilate  $k^{\mu}$  in the vector representation. Derive their commutators and identify the resulting Lie algebra.

- (b) Suppose the commuting null-translation generators  $A, B$  of the massless little group have a simultaneous eigenstate with  $a^2 + b^2 = \rho^2 > 0$ . Show that rotations generate a continuous circle of orthogonal eigenstates. Explain why a finite set of helicity states therefore requires  $A = B = 0$ .
- (c) Assuming the null translations act trivially, derive the Lorentz transformation law of a massless state  $|p, \lambda\rangle$  in terms of the rotation angle of  $L^{-1}(\Lambda p)\Lambda L(p)$ . Check the result for a rotation through  $\theta$  about  $\mathbf{p}$ .

### S.2.7. Finite-Dimensional $SU(2)$ Representations and Charge Conjugation

(Adapted from *University of Cambridge, Mathematical Tripos Part III, 2023, Paper 302, Symmetries, Fields and Particles, Question 2, printed p. 3.*)

Consider a finite-dimensional irreducible representation  $d$  of the  $SU(2)$  Lie algebra. Denote its Hermitian generators by  $J_1, J_2, J_3$ , with

$$[J_i, J_j] = i\epsilon_{ijk}J_k.$$

Let  $I$  be the maximum weight of  $d$ , and let  $v_m$  be an eigenvector of  $d(J_3)$ , so that

$$d(J_3)v_m = mv_m.$$

- (a) Construct the  $J_3$ -eigenvector basis for the vector space on which  $d$  acts, showing that its dimension is  $2I + 1$ . *Hint:* use  $J_{\pm} = J_1 \pm iJ_2$ .
- (b) Write  $\|v_m\| = N_m$ , so that  $v_m^\dagger v_m = N_m^2$ , and define normalized eigenvectors by

$$|I, m\rangle := \frac{1}{N_m}v_m.$$

Show that

$$|I, m\rangle = A(I, m)(d(J_-))^{I-m}|I, I\rangle,$$

and determine the numerical factor  $A(I, m)$ . *Hint:* note that  $J_-^\dagger = J_+$ .

- (c) An operator  $C$  maps the  $SU(2)$  generators according to

$$CJ_1C^{-1} = -J_1, \quad CJ_2C^{-1} = +J_2, \quad CJ_3C^{-1} = -J_3.$$

Show that the set  $\{J'_i\}$ , where  $J'_i := CJ_iC^{-1}$ , also generates an  $SU(2)$  algebra.

- (d) Because the up and down quarks have very light masses, the strong force has an approximate  $SU(2)$  symmetry called isospin symmetry; for this problem, assume it is exact. The three pion states form an  $I = 1$  triplet

$$\{|\pi^-\rangle, |\pi^0\rangle, |\pi^+\rangle\},$$

corresponding respectively to  $m = -1, 0, 1$ . Given that  $C|\pi^0\rangle = |\pi^0\rangle$ , how are  $C|\pi^+\rangle$  and  $C|\pi^-\rangle$  related to the original states?

### S.2.8. A Wigner Rotation from Non-Collinear Boosts

(Inspired by *J. Preskill, Physics 236c notes, Chapter 1, 1990.*)

- (a) A rest momentum is first boosted by rapidity  $\xi$  along  $y$ , then by rapidity  $\eta$  along  $z$ . Show that the little-group element obtained after removing the final rotation-free boost is a rotation about  $x$ , with

$$\tan \frac{\Omega}{2} = \tanh \frac{\eta}{2} \tanh \frac{\xi}{2}.$$

Discuss the reversal of the angle under an active/passive change of convention.

- (b) For a spin-one particle initially in a  $J_z$  eigenstate, compute the probabilities for all three  $J_z$  outcomes after a rotation  $R_x(\Omega)$ . Give separate answers for initial  $\sigma = 0$  and  $\sigma = +1$ .

### S.2.9. The Lorentz Group through $SL(2, \mathbb{C})$

(Inspired by V. Kaplunovsky, Finite Multiplets of the Spin(3,1) Group, 2016, Exercises 3–5 / V. Kaplunovsky, Finite Multiplets of the Spin(3,1) Group, 2016, Exercise 1.)

- (a) Associate the Hermitian matrix  $X = x^0 \mathbf{1} + x^i \sigma_i$  with a four-vector. For  $A \in SL(2, \mathbb{C})$ , show that  $X \mapsto AXA^\dagger$  induces a proper orthochronous Lorentz transformation. Determine the kernel of the map.
- (b) Define

$$\mathbf{A} = \frac{1}{2}(\mathbf{J} + i\mathbf{K}), \quad \mathbf{B} = \frac{1}{2}(\mathbf{J} - i\mathbf{K}).$$

Show separately that  $\mathbf{A}$  and  $\mathbf{B}$  obey angular-momentum commutators, and show that the two sets commute. Explain the labels  $(j_A, j_B)$  for finite-dimensional Lorentz representations and why nontrivial finite-dimensional representations cannot be unitary.

### S.2.10. Bispinors and the Sign of a $2\pi$ Rotation

(Inspired by V. Kaplunovsky, Finite Multiplets of the Spin(3,1) Group, 2016, Exercises 4 and 7 / V. Kaplunovsky, Finite Multiplets of the Spin(3,1) Group, 2016, Exercises 3–5.)

- (a) Show that  $V_{\alpha\dot{\alpha}} = V^\mu(\sigma_\mu)_{\alpha\dot{\alpha}}$  transforms in the  $(\frac{1}{2}, \frac{1}{2})$  representation. Recover the Lorentz-vector transformation and account for all four complex components.
- (b) Follow the path  $A(\theta) = \exp(-i\theta\sigma_3/2)$  in  $SL(2, \mathbb{C})$  from  $\theta = 0$  to  $2\pi$  and  $4\pi$ . Determine its image in the Lorentz group and its action on a spin- $j$  state. Explain the origin of projective Lorentz representations.

### S.2.11. The Bargmann Cocycle and Mass Superselection

(Inspired by E. Annigoni and V. Moretti, arXiv:1205.1372, Section 2, 2012 / E. Annigoni and V. Moretti, arXiv:1205.1372, Sections 1–2, 2012.)

- (a) For a Schrödinger wavefunction of mass  $m$ , use

$$(U(\mathbf{v})\psi)(\mathbf{x}, t) = e^{im(\mathbf{v}\cdot\mathbf{x} - \mathbf{v}^2 t/2)} \psi(\mathbf{x} - \mathbf{v}t, t)$$

to compare a spatial translation with a boost. Extract the central commutator  $[K_i, P_j]$ .

- (b) Evaluate the operator corresponding to a translation by  $\mathbf{a}$ , a boost by  $\mathbf{v}$ , and their inverses in the group-commutator order. Although the net spacetime transformation is the identity, show that the state acquires a mass-dependent phase.

- (c) Apply the Bargmann loop to a coherent superposition of two different masses. Derive the usual mass-superselection conclusion. Then explain how treating mass as a central operator of an enlarged symmetry group changes the interpretation.

### S.2.12. The Heisenberg Group and Its Schrödinger Representation

(Adapted from *University of Cambridge, Mathematical Tripos Part III, 2024, Paper 302, Symmetries, Fields and Particles, Question 1, printed p. 2.*)

- (a) For each set and binary operation below, determine whether it forms a group. If it does, verify the group axioms; if it does not, give an axiom that fails.
- (i) The integers under addition,  $(\mathbb{Z}, +)$ .
  - (ii) The nonzero integers under multiplication,  $(\mathbb{Z}^*, \times)$ .
  - (iii) The set  $\mathcal{L}$  of linear functions under composition, where

$$\mathcal{L} = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f(x) = ux + b; \ b \in \mathbb{R}, \ u \in \mathbb{R}, \ u \neq 0\}.$$

- (b) Consider the matrix Lie group

$$G = \left\{ \begin{pmatrix} 1 & q & s \\ 0 & 1 & r \\ 0 & 0 & 1 \end{pmatrix} \mid q, r, s \in \mathbb{R} \right\},$$

with matrix multiplication as the group operation.

- (i) Find the centre  $Z(G)$ .
  - (ii) Find the Lie algebra  $L(G)$ . Choose a basis and determine the corresponding structure constants.
  - (iii) Show that  $\exp : L(G) \rightarrow G$  is both one-to-one and onto. Explain whether  $G$  is connected.
  - (iv) Explain whether  $L(G)$  is simple, semisimple, or neither.
- (c) Consider the inner-product space of square-integrable complex functions, written in the source with the normalization convention

$$V = \{\psi : \mathbb{R} \rightarrow \mathbb{C} \mid (\psi, \psi) = 1\}, \quad (\phi, \psi) = \int_{-\infty}^{\infty} dx \phi(x)^* \psi(x).$$

For checking linearity, regard  $V$  as the underlying space  $L^2(\mathbb{R})$ ; the displayed condition selects its normalized wavefunctions. For the same group  $G$ , define  $D : G \rightarrow GL(V)$  by

$$g = \begin{pmatrix} 1 & q & s \\ 0 & 1 & r \\ 0 & 0 & 1 \end{pmatrix}, \quad (D(g)\psi)(x) = e^{-is} e^{irx} \psi(x - q).$$

Show that  $D$  is a unitary representation of  $G$ . If  $\psi(x)$  is a time-independent quantum wavefunction in one spatial dimension, interpret this transformation physically.



### S.2.13. The Poincaré Group in 1 + 2 Dimensions

(Adapted from University of Cambridge, Mathematical Tripos Part III, 2025, Paper 302, Symmetries, Fields and Particles, Question 2, printed p. 3.)

This problem uses the source metric  $\eta_s = \text{diag}(+1, -1, -1)$ . In Minkowski spacetime with one time and two space dimensions, Poincaré transformations act as

$$x^\mu \mapsto \Lambda^\mu{}_\nu x^\nu + a^\mu, \quad \mu, \nu \in \{0, 1, 2\}.$$

- State the condition that a real  $3 \times 3$  matrix  $\Lambda$  must satisfy to belong to the Lorentz group in 1 + 2 dimensions. Show what this condition implies for  $\det \Lambda$  and  $\Lambda^0{}_0$ .
- Recall that  $SL(2, \mathbb{R})$  is the group of real  $2 \times 2$  matrices with determinant one. Show that there is a group homomorphism from  $SL(2, \mathbb{R})$  to the proper orthochronous Lorentz group in 1 + 2 dimensions, and find an explicit expression for the map. Is it an isomorphism?

*Hint.* Consider the space  $\mathcal{M}$  of real  $2 \times 2$  matrices spanned by

$$\tau_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \tau_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \tau_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Construct a map  $\varphi$  from 1 + 2-dimensional Minkowski space to  $\mathcal{M}$ ,  $\varphi(x) = X$ , such that  $\det X = x^2$ .

- Write the Poincaré group in 1 + 2 dimensions as a semidirect product, and define the subgroups known as little groups. A full derivation is not expected, but explain why the semidirect-product nature of the group is important.
- Determine the unitary irreducible representations of the Poincaré group suitable for describing states of a single massive particle.
- Determine the unitary irreducible representations of the Poincaré group suitable for describing states of a single massless particle.

### S.2.14. Parity Symmetry of Scalar Field Theory

(Adapted from J. McGreevy, Physics 215A QFT, Fall 2023, Assignment 9, Problem 8, printed pp. 4–5.)

Under parity,  $\mathbf{x} \mapsto \mathbf{x}' = -\mathbf{x}$ , and a real Klein–Gordon field transforms as

$$P\phi(t, \mathbf{x})P^{-1} = \eta_P \phi(t, -\mathbf{x}), \quad \eta_P = \pm 1,$$

where  $P$  is unitary,  $\eta_P$  is the intrinsic parity of  $\phi$ , and  $P|0\rangle = |0\rangle$ . For completeness, use

$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2,$$

and the mode conventions

$$\phi(t, \mathbf{x}) = \int \frac{d^3\mathbf{k}}{(2\pi)^3\sqrt{2E_{\mathbf{k}}}} \left( a_{\mathbf{k}} e^{-iE_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger e^{iE_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right), \quad [a_{\mathbf{k}}, a_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{q}).$$

- (a) Show that parity preserves the free Lagrangian, though not the Lagrangian density at a fixed point, for both values of  $\eta_P$ .
- (b) Show that an arbitrary  $n$ -particle state transforms as

$$P|\mathbf{k}_1, \dots, \mathbf{k}_n\rangle = \eta_P^n |-\mathbf{k}_1, \dots, -\mathbf{k}_n\rangle.$$

- (c) Here is an explicit realization of the parity operator. Let

$$P_1 = \exp\left[-\frac{i\pi}{2} \sum_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}}\right], \quad P_2 = \exp\left[\frac{i\eta_P \pi}{2} \sum_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}\right].$$

Show that

$$P_1 a_{\mathbf{k}} P_1^{-1} = i a_{\mathbf{k}}, \quad P_2 a_{\mathbf{k}} P_2^{-1} = -i \eta_P a_{-\mathbf{k}}.$$

*Hint.* Use the Campbell–Baker–Hausdorff formula

$$e^{i\alpha A} B e^{-i\alpha A} = \sum_{n=0}^{\infty} \frac{(i\alpha)^n}{n!} B_n, \quad B_0 = B, \quad B_n = [A, B_{n-1}] \quad (n \geq 1).$$

Show that  $P = P_1 P_2$  is unitary and satisfies the parity transformation above. (The source cross-reference reads “(24)” here, while the displayed transformation is its Eq. (14).)

- (d) Now take a complex scalar field. Using the preceding results and the particle current

$$j^\mu = i \left( \Phi^\dagger \partial^\mu \Phi - (\partial^\mu \Phi^\dagger) \Phi \right),$$

find the action of parity on  $j^\mu$ , namely  $j^\mu \mapsto P j^\mu P^{-1}$ . Extend the action of  $P$  from the real field to the complex field as needed.

## Solutions to Supplementary Exercises

### S.2.1. Invariant On-Shell Measures and Lorentz-Normalized Wave Packets

- (a) For any nonsingular real matrix  $A$ ,  $\delta^4(Ax) = \delta^4(x)/|\det A|$ , as follows by testing both sides against a smooth compactly supported function and changing variables. The Lorentz condition gives

$$(\det \Lambda)^2 = 1,$$

so  $|\det \Lambda| = 1$ , including for improper transformations. Taking  $x = p - q$  proves the stated invariance of the four-dimensional delta.

- (b) For a proper Lorentz transformation,  $|\det \Lambda| = 1$ , so  $d^4 p$  is invariant;  $p^2$  is a scalar, so is  $\delta(p^2 + m^2)$ . On the positive-energy shell,

$$\delta(p^2 + m^2) = \delta(-(p^0)^2 + \mathbf{p}^2 + m^2) = \frac{\delta(p^0 - E_{\mathbf{p}}) + \delta(p^0 + E_{\mathbf{p}})}{2E_{\mathbf{p}}}.$$

Multiplication by  $\theta(p^0)$  selects the first root, and hence

$$\int d^4p \delta(p^2 + m^2) \theta(p^0) F(p) = \int \frac{d^3\mathbf{p}}{2E_{\mathbf{p}}} F(E_{\mathbf{p}}, \mathbf{p}).$$

A proper orthochronous transformation maps every future-directed causal vector to another such vector. It therefore preserves the sign of  $p^0$ . Time reversal interchanges the two energy sheets, so the positive-sheet measure alone is not invariant under that disconnected component.

- (c) The invariant on-shell measure is  $d\mu(p) = d^3\mathbf{p}/E_{\mathbf{p}}$ , apart from an irrelevant constant. Its delta kernel  $\Delta(p, q)$  is defined by

$$f(p) = \int d\mu(q) \Delta(p, q) f(q).$$

Comparison with the ordinary three-dimensional delta function gives

$$\Delta(p, q) = E_{\mathbf{q}} \delta^3(\mathbf{q} - \mathbf{p}) = E_{\mathbf{p}} \delta^3(\mathbf{p} - \mathbf{q}).$$

Because both  $d\mu$  and the defining identity are invariant,  $\Delta$  is invariant.

The common covariant convention is

$$\langle p', \sigma' | p, \sigma \rangle_{\text{cov}} = 2E_{\mathbf{p}} (2\pi)^3 \delta_{\sigma'\sigma} \delta^3(\mathbf{p} - \mathbf{p}').$$

The convention of Eq. (2.5.19) omits the factor  $2E_{\mathbf{p}}(2\pi)^3$ . The compensating factor  $\sqrt{(\Lambda p)^0/p^0}$  then appears in the state transformation law.

- (d) Use

$$\langle \phi | \psi \rangle = \int \frac{d^3\mathbf{p}}{2E_{\mathbf{p}}} \phi(\mathbf{p})^* \psi(\mathbf{p}).$$

For a scalar representation,

$$(U(\Lambda)\psi)(\mathbf{p}) = \psi(\Lambda^{-1}\mathbf{p}),$$

where the energy component of  $p$  is fixed by the positive mass shell. Then, with  $q = \Lambda^{-1}p$ ,

$$\|U(\Lambda)\psi\|^2 = \int \frac{d^3\mathbf{p}}{2E_{\mathbf{p}}} |\psi(\Lambda^{-1}\mathbf{p})|^2 = \int \frac{d^3\mathbf{q}}{2E_{\mathbf{q}}} |\psi(\mathbf{q})|^2.$$

The last equality is precisely the invariance established in S.2.1. Thus the scalar pullback is unitary without an additional Jacobian factor.

- (e) For the passive boost

$$E' = \gamma(E - up_z), \quad p'_z = \gamma(p_z - uE), \quad p'_x = p_x, \quad p'_y = p_y.$$

Since  $\partial E / \partial p_z = p_z / E$ ,

$$\frac{\partial p'_z}{\partial p_z} = \gamma \left( 1 - u \frac{p_z}{E} \right) = \frac{\gamma(E - up_z)}{E} = \frac{E'}{E}.$$

The transverse coordinates are unchanged and their mixed derivatives do not alter the triangular determinant, so

$$\det \frac{\partial \mathbf{p}'}{\partial \mathbf{p}} = \frac{E'}{E}.$$

Consequently  $d^3 \mathbf{p}' = (E'/E)d^3 \mathbf{p}$ , which immediately yields  $d^3 \mathbf{p}'/E' = d^3 \mathbf{p}/E$ .

### S.2.2. The Poincaré Algebra from Infinitesimal Generators

(a) The elementary relations

$$[\partial^\mu, x^\nu] = \eta^{\mu\nu}, \quad [\partial^\mu, \partial^\nu] = [x^\mu, x^\nu] = 0$$

give  $[P^\mu, P^\nu] = 0$ . Applying the product rule once gives

$$[J^{\mu\nu}, P^\rho] = i(\eta^{\mu\rho} P^\nu - \eta^{\nu\rho} P^\mu),$$

which is equivalent to Eq. (2.4.13). Applying it twice gives

$$\begin{aligned} [J^{\mu\nu}, J^{\rho\sigma}] = & i(\eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\rho} J^{\mu\sigma} \\ & + \eta^{\nu\sigma} J^{\mu\rho} - \eta^{\mu\sigma} J^{\nu\rho}), \end{aligned}$$

equivalent to Eq. (2.4.12). No field equation was used: these relations express the action of infinitesimal spacetime isometries on scalar functions.

(b) Substituting  $J_i = \frac{1}{2}\epsilon_{ijk}J^{jk}$ ,  $K_i = J^{i0}$ , and  $H = P^0$  into the covariant algebra yields

$$\begin{aligned} [J_i, J_j] &= i\epsilon_{ijk}J_k, & [J_i, K_j] &= i\epsilon_{ijk}K_k, & [K_i, K_j] &= -i\epsilon_{ijk}J_k, \\ [J_i, P_j] &= i\epsilon_{ijk}P_k, & [J_i, H] &= 0, & [K_i, P_j] &= i\delta_{ij}H, & [K_i, H] &= iP_i, \\ [P_i, P_j] &= [P_i, H] = 0. \end{aligned}$$

The minus sign in  $[K_i, K_j]$  is the algebraic signature of the noncompact Lorentz boosts. The relations involving  $H$  also show why a boost does not preserve an energy eigenspace.

### S.2.3. The Poincaré Group Commutator and Its Lie Algebra

We use the group-commutator convention  $[g, h] = ghg^{-1}h^{-1}$ .

(i) Applying  $(\Lambda, a)$  first and  $(L, b)$  second gives

$$x \longmapsto L(\Lambda x + a) + b = L\Lambda x + La + b,$$

and hence

$$(L, b)(\Lambda, a) = (L\Lambda, La + b).$$

This is the semidirect-product law: the Lorentz factor acts on the translation before the translations are added.

(ii) Requiring the product with  $(\Lambda, a)$  to equal  $(I, 0)$  gives

$$(\Lambda, a)^{-1} = (\Lambda^{-1}, -\Lambda^{-1}a).$$

(iii) Successive use of the composition law gives

$$\begin{aligned} (\Lambda_c, a_c) &= (L, b)(\Lambda, a)(L^{-1}, -L^{-1}b)(\Lambda^{-1}, -\Lambda^{-1}a), \\ \Lambda_c &= L\Lambda L^{-1}\Lambda^{-1}, \\ a_c &= La + b - L\Lambda L^{-1}b - L\Lambda L^{-1}\Lambda^{-1}a. \end{aligned}$$

The result also makes clear that translations fail to commute with Lorentz transformations even though the translation subgroup itself is abelian.

(iv) To the required orders,

$$\begin{aligned} L\Lambda L^{-1} &= I + \omega_\Lambda + [\omega_L, \omega_\Lambda] + \cdots, \\ L\Lambda L^{-1}\Lambda^{-1} &= I + [\omega_L, \omega_\Lambda] + \cdots. \end{aligned}$$

Substitution in the exact answer from part (iii) yields

$$a_c = \omega_L a - \omega_\Lambda b - [\omega_L, \omega_\Lambda](a + b) + \cdots.$$

When all four infinitesimal parameters  $\omega_L, \omega_\Lambda, a, b$  are counted as first order, as in part (vii), the last displayed commutator term is cubic. The bilinear truncation is therefore

$$a_c = \omega_L a - \omega_\Lambda b + O(3).$$

(v) The Lorentz component found above immediately gives

$$\omega_c = [\omega_L, \omega_\Lambda] = \omega_L \omega_\Lambda - \omega_\Lambda \omega_L.$$

(vi) A Lorentz matrix obeys

$$\Lambda^\top \eta \Lambda = \eta, \quad \eta_{\rho\sigma} \Lambda^\rho{}_\mu \Lambda^\sigma{}_\nu = \eta_{\mu\nu}.$$

At first order,

$$\omega_\Lambda^\top \eta + \eta \omega_\Lambda = 0.$$

Thus, after lowering the first index,

$$(\omega_\Lambda)_{\mu\nu} := \eta_{\mu\rho} (\omega_\Lambda)^\rho{}_\nu = -(\omega_\Lambda)_{\nu\mu}.$$

(vii) Define

$$A = \frac{1}{2} \omega_L^{\mu\nu} M_{\mu\nu} + b^\mu P_\mu, \quad B = \frac{1}{2} \omega_\Lambda^{\rho\sigma} M_{\rho\sigma} + a^\rho P_\rho.$$

The representation property turns the group commutator into

$$U[(L, b)]U[(\Lambda, a)]U[(L, b)]^{-1}U[(\Lambda, a)]^{-1}.$$

Multiplying the four infinitesimal expansions, or using the Baker–Campbell–Hausdorff formula, gives

$$(I + A)(I + B)(I - A)(I - B) = I + [A, B] + O(3).$$

On the other hand, parts (iv)–(v) give

$$U[(\Lambda_c, a_c)] = I + \frac{1}{2}[\omega_L, \omega_\Lambda]^{\mu\nu} M_{\mu\nu} + (\omega_L a - \omega_\Lambda b)^\mu P_\mu + O(3),$$

which is precisely the asserted identity.

(viii) Equating independently the coefficients of  $a^\mu b^\nu$ ,  $\omega_L^{\mu\nu} a^\rho$ , and  $\omega_L^{\mu\nu} \omega_\Lambda^{\rho\sigma}$  gives, in the anti-Hermitian generator normalization supplied in the source,

$$[P_\mu, P_\nu] = 0,$$

$$[M_{\mu\nu}, P_\rho] = \eta_{\nu\rho} P_\mu - \eta_{\mu\rho} P_\nu,$$

and

$$\begin{aligned} [M_{\mu\nu}, M_{\rho\sigma}] &= \eta_{\nu\rho} M_{\mu\sigma} - \eta_{\mu\rho} M_{\nu\sigma} \\ &\quad + \eta_{\mu\sigma} M_{\nu\rho} - \eta_{\nu\sigma} M_{\mu\rho}. \end{aligned}$$

The absence of an explicit  $i$  follows from the source expansion  $U = I + \frac{1}{2}\omega M + aP + \dots$ : unitarity makes these generators anti-Hermitian. To return to the Hermitian convention used elsewhere in this volume, set

$$\widehat{M}_{\mu\nu} = -iM_{\mu\nu}, \quad \widehat{P}_\mu = -iP_\mu.$$

Then

$$[\widehat{M}_{\mu\nu}, \widehat{P}_\rho] = i(\eta_{\mu\rho} \widehat{P}_\nu - \eta_{\nu\rho} \widehat{P}_\mu),$$

and

$$\begin{aligned} [\widehat{M}_{\mu\nu}, \widehat{M}_{\rho\sigma}] &= i(\eta_{\mu\rho} \widehat{M}_{\nu\sigma} - \eta_{\nu\rho} \widehat{M}_{\mu\sigma} \\ &\quad + \eta_{\nu\sigma} \widehat{M}_{\mu\rho} - \eta_{\mu\sigma} \widehat{M}_{\nu\rho}), \end{aligned}$$

in agreement with the Chapter 2 convention.

#### S.2.4. Pauli–Lubanski Spin and the Massive Casimir

(a) The differential operators give

$$[J^{\mu\nu}, P^\rho] = i(\eta^{\mu\rho} P^\nu - \eta^{\nu\rho} P^\mu).$$

Antisymmetry alone gives

$$\mathcal{W}_\mu P^\mu = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} J^{\nu\rho} P^\sigma P^\mu = 0,$$

because the momentum operators commute, making  $P^\sigma P^\mu$  symmetric. Next,

$$[\mathcal{W}_\mu, P^\lambda] = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} [J^{\nu\rho}, P^\lambda] P^\sigma.$$

Inserting the Poincaré commutator produces two terms, each containing a symmetric product of momenta contracted with an antisymmetric pair of indices. They cancel, so  $[\mathcal{W}_\mu, P^\lambda] = 0$ .

For the remaining commutator, expand both Pauli–Lubanski operators, move momenta through one another, and use  $[J, J]$  plus the result just proved. Terms with two momenta cancel in antisymmetric pairs, leaving

$$[\mathcal{W}_\mu, \mathcal{W}_\nu] = -i \epsilon_{\mu\nu\rho\sigma} \mathcal{W}^\rho P^\sigma.$$

(b) At  $p^\mu = (m, \mathbf{0})$ , the antisymmetry of  $\epsilon_{\mu\nu\rho\sigma}$  gives

$$\mathcal{W}^0 = 0, \quad \mathbf{W} = \pm m \mathbf{J},$$

where the overall sign depends on the placement of indices in the Levi–Civita convention and is irrelevant to the square. Thus, on a spin- $j$  rest state,

$$\mathcal{W}^2 = m^2 \mathbf{J}^2 = m^2 j(j+1).$$

The next part proves directly that  $\mathcal{W}^2$  is a Poincaré Casimir, so its eigenvalue is the same in every inertial frame and throughout an irreducible massive multiplet.

(c) Contracting  $\epsilon_{\mu\nu\rho\sigma} \epsilon^{\mu\alpha\beta\gamma}$  and expanding the generalized Kronecker delta gives

$$\mathcal{W}^2 = -\frac{1}{2} J_{\mu\nu} J^{\mu\nu} P^2 + J_{\mu\rho} J_\nu{}^\rho P^\mu P^\nu.$$

The translation commutators of  $P^2$  vanish immediately. In  $[J^{\alpha\beta}, P^2]$ , the two terms from the Leibniz rule cancel after interchanging the contracted momentum indices. Thus  $P^2$  is a Casimir.

The result of the first part gives

$$[\mathcal{W}^2, P^\lambda] = \mathcal{W}^\mu [\mathcal{W}_\mu, P^\lambda] + [\mathcal{W}^\mu, P^\lambda] \mathcal{W}_\mu = 0.$$

The remaining Lorentz commutator is most cleanly obtained from the finite transformation law in the next part, avoiding the direct expansion explicitly waived in the source problem.

(d) Under a translation,

$$J^{\nu\rho} \mapsto J^{\nu\rho} - a^\nu P^\rho + a^\rho P^\nu.$$

The induced change in  $\mathcal{W}_\mu$  vanishes because it is proportional to  $\epsilon_{\mu\nu\rho\sigma} P^\rho P^\sigma$ . For a Lorentz transformation, substituting the tensor law for  $J$ , the vector law for  $P$ , and the determinant identity for the Levi–Civita tensor gives

$$\mathcal{W}'_\mu = (\det \Lambda) \Lambda_\mu{}^\nu \mathcal{W}_\nu.$$

It is a vector for proper transformations and changes the additional sign appropriate to a pseudovector for improper ones. In either case  $\mathcal{W}'^2 = \mathcal{W}^2$ . Its translation invariance and Lorentz scalar character are precisely the finite group statements whose infinitesimal versions are  $[\mathcal{W}^2, P^\mu] = [\mathcal{W}^2, J^{\mu\nu}] = 0$ , completing the Casimir result.

### S.2.5. Massive Little-Group Representations and Normalization

- (a) For a massive positive-energy representation,

$$P^2 = -m^2, \quad \mathcal{W}^2 = m^2 j(j+1).$$

Thus  $j = 0, \frac{1}{2}, 1$  give respectively

$$\mathcal{W}^2 = 0, \quad \mathcal{W}^2 = \frac{3}{4}m^2, \quad \mathcal{W}^2 = 2m^2.$$

For a finite-helicity massless representation the null translations of the little group vanish. The remaining relation is  $\mathcal{W}^\mu = \lambda P^\mu$ , and hence

$$P^2 = 0, \quad \mathcal{W}^2 = \lambda^2 P^2 = 0$$

for every  $\lambda$ . The missing datum is the one-dimensional representation of the rotational part of the little group—the helicity  $\lambda$ , including its sign.

- (b) If  $\Lambda k = k$  for  $k = (m, \mathbf{0})$ , then  $\Lambda^\mu_0 = \delta^\mu_0$ . The Lorentz condition forces  $\Lambda^0_i = 0$  and leaves a spatial matrix  $R$  satisfying  $R^\top R = \mathbf{1}$ . Properness gives  $\det R = 1$ , so

$$\Lambda = \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix}, \quad R \in SO(3).$$

The unitary irreducible representations of its double cover  $SU(2)$  are labeled by  $j = 0, \frac{1}{2}, 1, \dots$  and have dimension  $2j + 1$ . Inducing each such little-group representation over the positive mass shell gives the states  $|p, \sigma\rangle$ ,  $\sigma = -j, \dots, j$ .

- (c) Unitarity first gives

$$\langle p', \sigma' | p, \sigma \rangle = |N(p)|^2 \delta_{\sigma'\sigma} \frac{p^0}{k^0} \delta^3(\mathbf{p}' - \mathbf{p}).$$

Thus  $\delta^3$ -normalization requires

$$N(p) = \sqrt{\frac{k^0}{p^0}},$$

up to a momentum-independent phase. For a massive standard momentum  $k^0 = m$ ; for a massless standard momentum  $k^0 = \kappa$ , whose arbitrary fixed value cancels from physical transformation laws.



Applying  $U(\Lambda)$ , inserting  $\mathbf{1} = U(L(\Lambda p))U(L(\Lambda p))^{-1}$ , and using the ratio of the two normalization factors gives

$$\frac{N(p)}{N(\Lambda p)} = \sqrt{\frac{(\Lambda p)^0}{p^0}}.$$

This is the square-root multiplying the little-group matrix in Eq. (2.5.23).

### S.2.6. Finite-Helicity Representations of the Null Little Group

- (a) Acting with the vector-representation matrices on  $k = (\kappa, 0, 0, \kappa)$  shows  $(J_2 + K_1)k = (-J_1 + K_2)k = J_3k = 0$ . The Lorentz algebra gives

$$[J_3, A] = iB, \quad [J_3, B] = -iA, \quad [A, B] = 0.$$

Thus  $A, B$  are two commuting translations rotated into one another by  $J_3$ . This is the Lie algebra  $\mathfrak{iso}(2) = \mathbb{R}^2 \rtimes \mathfrak{so}(2)$ , the algebra of the two-dimensional Euclidean group. It is the connected little group of a null momentum in  $3 + 1$  dimensions.

- (b) Choose a simultaneous eigenstate  $|a, b\rangle$ . A rotation through  $\theta$  about the momentum sends its eigenvalue pair to

$$(a_\theta, b_\theta) = (a \cos \theta + b \sin \theta, -a \sin \theta + b \cos \theta).$$

If  $\rho^2 = a^2 + b^2 > 0$ , these pairs sweep a full circle. Distinct eigenvalues of commuting Hermitian operators label orthogonal generalized eigenstates, so the representation contains a continuous infinity of polarizations. A finite-dimensional polarization space is possible only for  $\rho = 0$ , which means  $A = B = 0$ . The remaining  $SO(2)$  character is  $e^{i\lambda\theta}$  and supplies the helicity.

- (c) Let

$$W(\Lambda, p) = L^{-1}(\Lambda p)\Lambda L(p).$$

When null translations act trivially, only the rotation angle  $\theta(\Lambda, p)$  of this little-group element remains. With the  $\delta^3$ -normalized states of Chapter 2,

$$U(\Lambda)|p, \lambda\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}} e^{i\lambda\theta(\Lambda, p)} |\Lambda p, \lambda\rangle.$$

For a spatial rotation by  $\theta$  about  $\mathbf{p}$ , the standard boost may be chosen so that  $W$  is the same rotation. Momentum is unchanged and the state acquires  $e^{i\lambda\theta}$ , confirming that  $\lambda$  is the angular momentum along the direction of motion.

### S.2.7. Finite-Dimensional $SU(2)$ Representations and Charge Conjugation

For readability we suppress the representation symbol  $d$  on the generators.

(a) The algebra implies

$$[J_3, J_{\pm}] = \pm J_{\pm}, \quad [J_+, J_-] = 2J_3.$$

Thus  $J_+v_m$  and  $J_-v_m$ , when nonzero, have weights  $m+1$  and  $m-1$ . Choose a highest-weight vector  $v_I$  with  $J_+v_I = 0$ . The quadratic Casimir

$$J^2 = J_3^2 + \frac{1}{2}(J_+J_- + J_-J_+)$$

commutes with all three generators. Acting on the highest-weight vector shows that its eigenvalue is  $I(I+1)$ . Positivity of norms then gives

$$\begin{aligned} \|J_-|I, m\rangle\|^2 &= (I+m)(I-m+1), \\ \|J_+|I, m\rangle\|^2 &= (I-m)(I+m+1). \end{aligned}$$

The descending ladder therefore terminates at  $m = -I$ , so  $2I \in \mathbb{Z}_{\geq 0}$ . Irreducibility says that this ladder exhausts the representation:

$$m = I, I-1, \dots, -I, \quad \dim d = 2I+1.$$

(b) With phases chosen so that the ladder coefficients are positive,

$$J_-|I, m\rangle = \sqrt{(I+m)(I-m+1)}|I, m-1\rangle.$$

Put  $n = I - m$ . Repeated lowering from the highest-weight state gives

$$(J_-)^n|I, I\rangle = \left[ \prod_{k=0}^{n-1} (2I-k)(k+1) \right]^{1/2} |I, I-n\rangle.$$

The product is

$$\prod_{k=0}^{n-1} (2I-k)(k+1) = \frac{(2I)!n!}{(2I-n)!} = \frac{(2I)!(I-m)!}{(I+m)!}.$$

Consequently

$$A(I, m) = \sqrt{\frac{(I+m)!}{(2I)!(I-m)!}}.$$

(c) The transformed generators are

$$J'_1 = -J_1, \quad J'_2 = J_2, \quad J'_3 = -J_3.$$

Directly,

$$[J'_1, J'_2] = iJ'_3, \quad [J'_2, J'_3] = iJ'_1, \quad [J'_3, J'_1] = iJ'_2.$$

Equivalently, conjugation preserves every commutator:  $[J'_i, J'_j] = C[J_i, J_j]C^{-1} = i\epsilon_{ijk}J'_k$ . It is also useful below that

$$CJ_+C^{-1} = -J_-, \quad CJ_-C^{-1} = -J_+.$$

(d) Fix the relative phases of the triplet by

$$J_+|\pi^0\rangle = \sqrt{2}|\pi^+\rangle, \quad J_-|\pi^0\rangle = \sqrt{2}|\pi^-\rangle.$$

Using  $C|\pi^0\rangle = |\pi^0\rangle$ , conjugating the first relation gives

$$\sqrt{2}C|\pi^+\rangle = CJ_+C^{-1}|\pi^0\rangle = -J_-|\pi^0\rangle = -\sqrt{2}|\pi^-\rangle.$$

The second relation similarly gives

$$\boxed{C|\pi^+\rangle = -|\pi^-\rangle, \quad C|\pi^-\rangle = -|\pi^+\rangle}.$$

The common minus sign depends on the stated ladder-phase convention; independent rephasings of the charged basis vectors can move it between the ladder relations and the action of  $C$ , without changing the fact that charge conjugation exchanges the two states.

### S.2.8. A Wigner Rotation from Non-Collinear Boosts

(a) Represent the two boosts by their rapidity matrices and factor

$$B_z(\eta)B_y(\xi) = L(p_{\text{final}})R_x(\Omega),$$

where  $L(p_{\text{final}})$  is the unique rotation-free boost taking the rest momentum to the final momentum. Comparing the off-diagonal entries (equivalently, using the polar decomposition of the corresponding  $SL(2, \mathbb{C})$  matrices) gives

$$\tan \frac{\Omega}{2} = \frac{\sinh(\eta/2) \sinh(\xi/2)}{\cosh(\eta/2) \cosh(\xi/2)} = \tanh \frac{\eta}{2} \tanh \frac{\xi}{2}.$$

The axis is parallel to the oriented cross product of the second and first boost directions, here the  $x$ -axis up to convention. Replacing an active boost of states by the passive boost of coordinates inverts one boost and therefore reverses  $\Omega$ , but leaves every mixing probability unchanged.

(b) The absolute squares of the spin-one rotation matrix elements are independent of whether the transverse rotation axis is chosen as  $x$  or  $y$ . For an initial  $\sigma = 0$ ,

$$P(0 \rightarrow 0) = \cos^2 \Omega, \quad P(0 \rightarrow +1) = P(0 \rightarrow -1) = \frac{1}{2} \sin^2 \Omega.$$

For an initial  $\sigma = +1$ ,

$$\begin{aligned} P(+1 \rightarrow +1) &= \cos^4 \frac{\Omega}{2}, \\ P(+1 \rightarrow 0) &= \frac{1}{2} \sin^2 \Omega, \\ P(+1 \rightarrow -1) &= \sin^4 \frac{\Omega}{2}. \end{aligned}$$

Each set sums to one. The phases of the amplitudes depend on the choice of  $x$ -versus  $y$ -axis and on active/passive conventions, but these probabilities do not.

### S.2.9. The Lorentz Group through $SL(2, \mathbb{C})$

- (a) The determinant is

$$\det X = (x^0)^2 - \mathbf{x}^2 = -x_\mu x^\mu.$$

For  $X' = AXA^\dagger$ ,

$$\det X' = \det A \det X \det A^\dagger = \det X,$$

so the induced real linear map preserves the Minkowski norm. Hermitian  $X$  remains Hermitian, so real four-vectors map to real four-vectors. Continuity from  $A = \mathbf{1}$  places the image in the proper orthochronous component.

If  $AXA^\dagger = X$  for every Hermitian  $X$ , then  $A$  commutes with the full matrix algebra and is proportional to  $\mathbf{1}$ . The determinant-one condition and the action on  $X = \mathbf{1}$  leave  $A = \pm \mathbf{1}$ . Thus

$$SL(2, \mathbb{C})/\{\pm \mathbf{1}\} \simeq SO^+(3, 1).$$

- (b) Using  $[J_i, J_j] = i\epsilon_{ijk}J_k$ ,  $[J_i, K_j] = i\epsilon_{ijk}K_k$ , and  $[K_i, K_j] = -i\epsilon_{ijk}J_k$ , one finds

$$[A_i, A_j] = i\epsilon_{ijk}A_k, \quad [B_i, B_j] = i\epsilon_{ijk}B_k, \quad [A_i, B_j] = 0.$$

After complexification, finite-dimensional irreducible representations are therefore tensor products labeled by  $(j_A, j_B)$ , of dimension  $(2j_A + 1)(2j_B + 1)$ .

The Lorentz group is noncompact. A faithful nontrivial finite-dimensional unitary representation would have compact image in a unitary group, which cannot represent the unbounded boost subgroup faithfully. Correspondingly,  $\mathbf{A}$  and  $\mathbf{B}$  are not separately Hermitian in these finite multiplets. Unitary particle representations instead act on infinite-dimensional momentum-space Hilbert spaces.

### S.2.10. Bispinors and the Sign of a $2\pi$ Rotation

- (a) Write  $V = V^\mu \sigma_\mu$ . Under  $A \in SL(2, \mathbb{C})$ ,

$$V_{\alpha\dot{\alpha}} \mapsto A_\alpha{}^\beta (A^\dagger)^{\dot{\beta}}{}_{\dot{\alpha}} V_{\beta\dot{\beta}},$$

which is the tensor product of the undotted  $(\frac{1}{2}, 0)$  and dotted  $(0, \frac{1}{2})$  representations:  $(\frac{1}{2}, \frac{1}{2})$ . Expanding the resulting matrix uniquely in the basis  $\{\sigma_\mu\}$  defines  $V'^\mu = \Lambda^\mu{}_\nu V^\nu$ , and S.2.9(a) proves that  $\Lambda$  is Lorentz.

A general bispinor has  $2 \times 2 = 4$  complex components, matching a complex four-vector. Imposing Hermiticity on  $V$  reduces these to four real components and gives an ordinary real Lorentz vector.

- (b) At  $\theta = 2\pi$ ,

$$A(2\pi) = e^{-i\pi\sigma_3} = -\mathbf{1},$$

whereas at  $4\pi$ ,  $A(4\pi) = +\mathbf{1}$ . Both  $\pm\mathbf{1}$  induce the same Lorentz transformation, so the image path at  $2\pi$  is already the identity rotation although the lifted path in  $SL(2, \mathbb{C})$  has not closed.

In the spin- $j$  representation,

$$U(R(2\pi)) = (-1)^{2j}\mathbf{1}, \quad U(R(4\pi)) = \mathbf{1}.$$

Thus half-integer spin gives a representation of the Lorentz group only up to a sign, or equivalently an ordinary representation of its double cover. This is the topological source of the familiar projective representation.

### S.2.11. The Bargmann Cocycle and Mass Superselection

- (a) Let  $T(\mathbf{a})\psi(\mathbf{x}, t) = \psi(\mathbf{x} - \mathbf{a}, t)$ . Using the stated boost law in the two possible orders gives

$$U(\mathbf{v})T(\mathbf{a}) = e^{im\mathbf{v}\cdot\mathbf{a}}T(\mathbf{a})U(\mathbf{v}).$$

The extra phase cannot be seen in the spacetime action of the Galilei group; it is a projective multiplier. Expanding both sides to first order in  $\mathbf{v}$  and  $\mathbf{a}$ , with the generator conventions of Chapter 2, yields

$$[K_i, P_j] = i\delta_{ij}M, \quad M|\psi\rangle = m|\psi\rangle.$$

The mass is central because the phase is a scalar and commutes with every spacetime transformation.

- (b) The relation in S.2.11(a) immediately gives

$$U(\mathbf{v})T(\mathbf{a})U(\mathbf{v})^{-1}T(\mathbf{a})^{-1} = e^{im\mathbf{v}\cdot\mathbf{a}}\mathbf{1}.$$

The associated sequence of coordinate transformations has zero net boost and zero net translation, so it is the identity in the unextended Galilei group. Quantum mechanically it is a nontrivial element of the center. Infinitesimally, the exponent is just the central boost–translation commutator.

- (c) For  $|\Psi\rangle = c_1|\psi_{m_1}\rangle + c_2|\psi_{m_2}\rangle$ , the loop gives

$$|\Psi\rangle \mapsto c_1 e^{im_1\mathbf{v}\cdot\mathbf{a}}|\psi_{m_1}\rangle + c_2 e^{im_2\mathbf{v}\cdot\mathbf{a}}|\psi_{m_2}\rangle.$$

Unless  $m_1 = m_2$ , this changes the relative phase even though the underlying spacetime transformation is the identity. Requiring the identity to preserve every ray therefore forbids coherent superpositions of different masses: Bargmann’s mass-superselection rule.

In the centrally extended Galilei group, however, the loop is not declared to be the identity; it is  $e^{i\alpha M}$ . If  $M$  is treated as an operator with an additional conjugate degree of freedom, different mass sectors are ordinary eigenspaces of the center. The algebra alone then does not forbid their superposition; a superselection rule requires an additional physical restriction on observables or accessibility of that central degree of freedom.

### S.2.12. The Heisenberg Group and Its Schrödinger Representation

- (a) (i) The integers are closed under addition, addition is associative, 0 is the identity, and the inverse of  $n$  is  $-n$ . Thus  $(\mathbb{Z}, +)$  is a group.
- (ii) Multiplication is associative and closed on  $\mathbb{Z}^*$ , with identity 1, but an element such as 2 has no multiplicative inverse in  $\mathbb{Z}^*$ . Therefore  $(\mathbb{Z}^*, \times)$  is not a group.
- (iii) Write  $f_{u,b}(x) = ux + b$ . Then

$$f_{u,b} \circ f_{u',b'} = f_{uu', ub'+b},$$

which remains in  $\mathcal{L}$ . Composition is associative, the identity is  $f_{1,0}$ , and

$$f_{u,b}^{-1} = f_{u^{-1}, -b/u}.$$

Hence  $(\mathcal{L}, \circ)$  is the affine group of the line.

- (b) Denote the displayed matrix by  $g(q, r, s)$ . Direct matrix multiplication gives

$$g(q, r, s)g(q', r', s') = g(q + q', r + r', s + s' + qr'),$$

and therefore

$$g(q, r, s)^{-1} = g(-q, -r, -s + qr).$$

- (i) Two elements commute precisely when  $qr' = q'r$ . For this to hold for every  $q', r'$ , one must have  $q = r = 0$ , while  $s$  is unrestricted. Thus

$$Z(G) = \{g(0, 0, s) \mid s \in \mathbb{R}\}.$$

- (ii) The Lie algebra consists of strictly upper-triangular matrices

$$X(q, r, s) = \begin{pmatrix} 0 & q & s \\ 0 & 0 & r \\ 0 & 0 & 0 \end{pmatrix} = qQ + rR + sS,$$

with

$$Q = E_{12}, \quad R = E_{23}, \quad S = E_{13}.$$

The only nonzero basis brackets are

$$[Q, R] = S, \quad [R, Q] = -S.$$

Hence  $f_{QR}^S = 1$ ,  $f_{RQ}^S = -1$ , and all other structure constants vanish.

- (iii) Since  $X^2 = qrS$  and  $X^3 = 0$ ,

$$e^{X(q,r,s)} = I + X + \frac{1}{2}X^2 = g\left(q, r, s + \frac{1}{2}qr\right).$$

Every  $g(q, r, s_0)$  therefore has the unique logarithm

$$X\left(q, r, s_0 - \frac{1}{2}qr\right).$$

Thus the exponential map is one-to-one and onto. It identifies the underlying manifold of  $G$  continuously with  $\mathbb{R}^3$ , so  $G$  is connected (indeed, simply connected).

- (iv) The one-dimensional space  $\mathbb{R}S$  is a nonzero central ideal, so  $L(G)$  is not simple. Moreover,

$$[L(G), L(G)] = \mathbb{R}S, \quad [\mathbb{R}S, \mathbb{R}S] = 0.$$

The algebra is nilpotent, its Killing form is degenerate, and it is not a direct sum of simple ideals. It is therefore neither simple nor semisimple.

- (c) The condition  $(\psi, \psi) = 1$  in the source describes the unit sphere rather than a vector space; the representation acts linearly on the full Hilbert space  $L^2(\mathbb{R})$  and preserves that unit sphere. Let  $g = g(q, r, s)$  and  $g' = g(q', r', s')$ . Then

$$\begin{aligned} (D(g)D(g')\psi)(x) &= e^{-i(s+s')}e^{irx}e^{ir'(x-q)}\psi(x-q-q') \\ &= e^{-i(s+s'+qr')}e^{i(r+r')x}\psi(x-q-q') \\ &= (D(gg')\psi)(x). \end{aligned}$$

Thus  $D$  respects the exact matrix-group multiplication law. It also preserves the inner product:

$$\begin{aligned} (D(g)\phi, D(g)\psi) &= \int_{-\infty}^{\infty} dx \phi(x-q)^*\psi(x-q) \\ &= \int_{-\infty}^{\infty} dy \phi(y)^*\psi(y) = (\phi, \psi). \end{aligned}$$

Since  $D(g^{-1}) = D(g)^{-1}$ , this is a unitary representation.

With  $\hat{p} = -i d/dx$  and  $\hat{x}$  multiplication by  $x$ , the operator can be written

$$D(g) = e^{-is}e^{ir\hat{x}}e^{-iq\hat{p}}.$$

Thus  $q$  translates position,  $r$  translates momentum, and  $s$  is an overall phase. The extra term  $qr'$  in the group law is the central phase forced by the canonical noncommutativity of position and momentum; this is the Schrödinger representation of the Heisenberg group.

### S.2.13. The Poincaré Group in 1 + 2 Dimensions

Throughout this solution, scalar products use the source metric

$$\eta_s = \text{diag}(+1, -1, -1).$$

(a) The Lorentz condition is

$$\Lambda^\top \eta_s \Lambda = \eta_s.$$

Taking determinants gives

$$(\det \Lambda)^2 = 1, \quad \det \Lambda = \pm 1.$$

The norm of the zeroth column gives

$$(\Lambda^0_0)^2 - (\Lambda^1_0)^2 - (\Lambda^2_0)^2 = 1,$$

and therefore  $|\Lambda^0_0| \geq 1$ . The proper component has  $\det \Lambda = +1$ , and the orthochronous component containing the identity has  $\Lambda^0_0 \geq 1$ .

(b) Map a Minkowski vector to the real symmetric matrix

$$X = \varphi(x) = x^0 \tau_0 + x^1 \tau_1 + x^2 \tau_2 = \begin{pmatrix} x^0 + x^1 & x^2 \\ x^2 & x^0 - x^1 \end{pmatrix}.$$

Its determinant is exactly the Minkowski norm:

$$\det X = (x^0)^2 - (x^1)^2 - (x^2)^2 = x^2.$$

For  $A \in SL(2, \mathbb{R})$ , define

$$X' = AXA^\top = \varphi(x').$$

The result is again real and symmetric, and

$$\det X' = \det A \det X \det A^\top = \det X.$$

Thus  $x' = \Lambda(A)x$  for a Lorentz transformation. Moreover,

$$\varphi(\Lambda(AB)x) = ABX(AB)^\top = A(BXB^\top)A^\top,$$

so  $\Lambda(AB) = \Lambda(A)\Lambda(B)$ .

For an explicit formula, write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc = 1.$$

Then

$$\Lambda(A) = \begin{pmatrix} \frac{1}{2}(a^2 + b^2 + c^2 + d^2) & \frac{1}{2}(a^2 - b^2 + c^2 - d^2) & ab + cd \\ \frac{1}{2}(a^2 + b^2 - c^2 - d^2) & \frac{1}{2}(a^2 - b^2 - c^2 + d^2) & ab - cd \\ ac + bd & ac - bd & ad + bc \end{pmatrix}.$$

Continuity places the image in  $SO^+(2, 1)$ . The differential at the identity is an isomorphism of the three-dimensional Lie algebras, so the image is all of the connected group. The kernel is

$$\ker \Lambda = \{I, -I\}.$$

Hence

$$SO^+(2, 1) \simeq SL(2, \mathbb{R})/\{\pm I\},$$

and the map is a two-to-one covering homomorphism, not an isomorphism.



(c) For the full Lorentz group,

$$\text{ISO}(2, 1) = \mathbb{R}^{1,2} \rtimes O(2, 1),$$

while its identity component is

$$\text{ISO}^+(2, 1) = \mathbb{R}^{1,2} \rtimes SO^+(2, 1).$$

The group law is

$$(a, \Lambda)(b, L) = (a + \Lambda b, \Lambda L).$$

Thus translations form a normal subgroup, but Lorentz transformations act non-trivially on them; this is a semidirect rather than a direct product. In momentum space that action organizes momenta into Lorentz orbits. For a representative momentum  $k$ , its little group is

$$G_k = \{\Lambda \in SO^+(2, 1) \mid \Lambda k = k\}.$$

Wigner's construction combines a translation character  $e^{-ia \cdot p}$  on a Lorentz orbit with a unitary irreducible representation of  $G_k$ .

(d) A future-directed massive orbit has  $p^2 = m^2$ ,  $p^0 > 0$ , and representative

$$k = (m, 0, 0).$$

Its connected little group is the spatial rotation group  $SO(2)$ . Every irreducible unitary representation is a character

$$D_s(R(\theta)) = e^{is\theta}.$$

For the group  $SO(2)$  itself, single-valuedness requires  $s \in \mathbb{Z}$ ; the double cover permits  $s \in \frac{1}{2}\mathbb{Z}$ ; and the universal cover has  $\theta \in \mathbb{R}$ , allowing arbitrary real  $s$ . The last possibility is the group-theoretic origin of anyonic spin in  $2+1$  dimensions. Choose a standard Lorentz transformation  $L(p)k = p$ . The induced massive representation may be written

$$U(a, \Lambda)|p, s\rangle = e^{-ia \cdot \Lambda p} e^{is\Omega(\Lambda, p)} |\Lambda p, s\rangle,$$

up to the normalization factor dictated by the choice of momentum-state normalization, where

$$L(\Lambda p)^{-1} \Lambda L(p) = R(\Omega(\Lambda, p)).$$

Thus a massive irreducible representation is labeled by  $m > 0$  and the allowed spin  $s$ .

(e) Take the future-directed null representative  $k = (\kappa, \kappa, 0)$ . In the  $SL(2, \mathbb{R})$  description it is

$$\varphi(k) = \begin{pmatrix} 2\kappa & 0 \\ 0 & 0 \end{pmatrix}.$$

Its connected stabilizer is represented by

$$A(u) = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}, \quad u \in \mathbb{R}.$$

Using the explicit map from part (b), the corresponding Lorentz matrix is

$$N(u) = \begin{pmatrix} 1 + \frac{1}{2}u^2 & -\frac{1}{2}u^2 & u \\ \frac{1}{2}u^2 & 1 - \frac{1}{2}u^2 & u \\ u & -u & 1 \end{pmatrix}, \quad N(u)k = k,$$

and  $N(u)N(v) = N(u+v)$ . Hence the connected massless little group is  $(\mathbb{R}, +)$ . Its irreducible unitary representations are the one-dimensional characters

$$D_\rho(N(u)) = e^{i\rho u}, \quad \rho \in \mathbb{R}.$$

Together with the null momentum orbit and translation character, each  $\rho$  induces a massless Poincaré representation. The conventional single-polarization massless particle described by a finite-component local field uses the trivial character  $\rho = 0$ ; nonzero  $\rho$  gives the nontrivial null-rotation family.

#### S.2.14. Parity Symmetry of Scalar Field Theory

(a) Differentiating the parity transformation gives

$$\begin{aligned} P \partial_0 \phi(t, \mathbf{x}) P^{-1} &= \eta_P \partial_0 \phi(t, -\mathbf{x}), \\ P \partial_i \phi(t, \mathbf{x}) P^{-1} &= -\eta_P \partial_i \phi(t, -\mathbf{x}). \end{aligned}$$

The two signs from each quadratic term cancel, as does  $\eta_P^2 = 1$ , so

$$P \mathcal{L}_0(t, \mathbf{x}) P^{-1} = \mathcal{L}_0(t, -\mathbf{x}).$$

The density at a fixed point is generally not unchanged. The Lagrangian is, because a change of variables in the spatial integral gives

$$P L_0(t) P^{-1} = \int d^3 \mathbf{x} \mathcal{L}_0(t, -\mathbf{x}) = \int d^3 \mathbf{y} \mathcal{L}_0(t, \mathbf{y}) = L_0(t).$$

This works for either intrinsic parity.

(b) Comparing the transformed mode expansion with  $\eta_P \phi(t, -\mathbf{x})$  gives

$$P a_{\mathbf{k}} P^{-1} = \eta_P a_{-\mathbf{k}}, \quad P a_{\mathbf{k}}^\dagger P^{-1} = \eta_P a_{-\mathbf{k}}^\dagger.$$

For

$$|\mathbf{k}_1, \dots, \mathbf{k}_n\rangle = a_{\mathbf{k}_1}^\dagger \cdots a_{\mathbf{k}_n}^\dagger |0\rangle,$$

vacuum invariance then gives

$$P |\mathbf{k}_1, \dots, \mathbf{k}_n\rangle = \eta_P^n |-\mathbf{k}_1, \dots, -\mathbf{k}_n\rangle.$$

(c) Let

$$N = \sum_{\mathbf{q}} a_{\mathbf{q}}^{\dagger} a_{\mathbf{q}}, \quad B = \sum_{\mathbf{q}} a_{\mathbf{q}}^{\dagger} a_{-\mathbf{q}}.$$

The canonical algebra gives

$$[N, a_{\mathbf{k}}] = -a_{\mathbf{k}},$$

so the supplied Campbell–Baker–Hausdorff series sums to

$$P_1 a_{\mathbf{k}} P_1^{-1} = e^{i\pi/2} a_{\mathbf{k}} = i a_{\mathbf{k}}.$$

For the second operator,

$$[B, a_{\mathbf{k}}] = -a_{-\mathbf{k}}, \quad [B, [B, a_{\mathbf{k}}]] = a_{\mathbf{k}},$$

and the nested commutators alternate. Therefore

$$e^{i\alpha B} a_{\mathbf{k}} e^{-i\alpha B} = \cos \alpha a_{\mathbf{k}} - i \sin \alpha a_{-\mathbf{k}}.$$

Putting  $\alpha = \eta_P \pi/2$  gives

$$P_2 a_{\mathbf{k}} P_2^{-1} = -i \eta_P a_{-\mathbf{k}}.$$

Both  $N$  and  $B$  are Hermitian: for  $B$ , take the adjoint and relabel  $\mathbf{q} \mapsto -\mathbf{q}$ . Hence  $P_1$ ,  $P_2$ , and their product are unitary. Finally,

$$P a_{\mathbf{k}} P^{-1} = P_1 (-i \eta_P a_{-\mathbf{k}}) P_1^{-1} = \eta_P a_{-\mathbf{k}}.$$

The adjoint relation follows similarly, and substitution in the field expansion proves

$$P \phi(t, \mathbf{x}) P^{-1} = \eta_P \phi(t, -\mathbf{x}).$$

(d) Extend parity to a complex field by

$$P \Phi(t, \mathbf{x}) P^{-1} = \eta_P \Phi(t, -\mathbf{x}), \quad P \Phi^{\dagger}(t, \mathbf{x}) P^{-1} = \eta_P^* \Phi^{\dagger}(t, -\mathbf{x}).$$

In oscillator language this is

$$P a_{\mathbf{k}} P^{-1} = \eta_P a_{-\mathbf{k}}, \quad P b_{\mathbf{k}} P^{-1} = \eta_P^* b_{-\mathbf{k}},$$

for particle and antiparticle annihilators. The phases cancel in the current. With

$$\mathcal{P}^{\mu}_{\nu} = \text{diag}(1, -1, -1, -1),$$

the result is the vector transformation law

$$\boxed{P j^{\mu}(t, \mathbf{x}) P^{-1} = \mathcal{P}^{\mu}_{\nu} j^{\nu}(t, -\mathbf{x})}.$$

Equivalently,

$$P j^0(t, \mathbf{x}) P^{-1} = j^0(t, -\mathbf{x}), \quad P \mathbf{j}(t, \mathbf{x}) P^{-1} = -\mathbf{j}(t, -\mathbf{x}).$$

Thus charge density is parity even and the spatial particle current is parity odd, as required for a Lorentz vector.

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### 3 Scattering Theory

The general principles of relativistic quantum mechanics described in the previous chapter have so far been applied here only to states of a single stable particle. Such one-particle states by themselves are not very exciting — it is only when two or more particles interact with each other that anything interesting can happen. But experiments do not generally follow the detailed course of events in particle interactions. Rather, the paradigmatic experiment (at least in nuclear or elementary particle physics) is one in which several particles approach each other from a macroscopically large distance, and interact in a microscopically small region, after which the products of the interaction travel out again to a macroscopically large distance. The physical states before and after the collision consist of particles that are so far apart that they are effectively non-interacting, so they can be described as direct products of the one-particle states discussed in the previous chapter. In such an experiment, all that is measured is the probability distribution, or ‘cross-sections’, for transitions between the initial and final states of distant and effectively non-interacting particles. This chapter will outline the formalism [1] used for calculating these probabilities and cross-sections.

#### 3.1 ‘In’ and ‘Out’ States

A state consisting of several non-interacting particles may be regarded as one that transforms under the Poincaré group as a direct product of one-particle states. To label the one-particle states we use their four-momenta  $p^\mu$ , spin  $z$ -component (or, for massless particles, helicity)  $\sigma$ , and, since we now may be dealing with more than one species of particle, an additional discrete label  $n$  for the particle type, which includes a specification of its mass, spin, charge, etc. The general transformation rule is

$$\begin{aligned}
 U(\Lambda, a)|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle &= \exp[-ia_\mu(p_1^\mu + p_2^\mu + \dots)] \sqrt{\frac{(\Lambda p_1)^0 (\Lambda p_2)^0 \dots}{p_1^0 p_2^0 \dots}} \\
 &\cdot \sum_{\sigma'_1 \sigma'_2 \dots} D_{\sigma'_1 \sigma_1}^{(j_1)}(W(\Lambda, p_1)) D_{\sigma'_2 \sigma_2}^{(j_2)}(W(\Lambda, p_2)) \dots \\
 &\cdot |\Lambda p_1, \sigma'_1, n_1; \Lambda p_2, \sigma'_2, n_2; \dots\rangle.
 \end{aligned} \tag{3.1.1}$$

where  $W(\Lambda, p)$  is the Wigner rotation (2.5.10), and  $D_{\sigma'\sigma}^{(j)}(W)$  are the conventional  $(2j+1)$ -dimensional unitary matrices representing the three-dimensional rotation group. (This is for massive particles; for any massless particle, the matrix  $D_{\sigma'\sigma}^{(j)}(W(\Lambda, p))$  is replaced with  $\delta_{\sigma'\sigma} \exp(i\sigma\theta(\Lambda, p))$ , where  $\theta$  is the angle defined by Eq. (2.5.43).) The states are normalized as in Eq. (2.5.19):

$$\begin{aligned}
 &\langle p'_1, \sigma'_1, n'_1; p'_2, \sigma'_2, n'_2; \dots | p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots \rangle \\
 &= \delta^3(\mathbf{p}'_1 - \mathbf{p}_1) \delta_{\sigma'_1 \sigma_1} \delta_{n'_1 n_1} \delta^3(\mathbf{p}'_2 - \mathbf{p}_2) \delta_{\sigma'_2 \sigma_2} \delta_{n'_2 n_2} \dots \\
 &\quad \pm \text{permutations}.
 \end{aligned} \tag{3.1.2}$$

with the term ‘ $\pm$  permutations’ included to take account of the possibility that it is some permutation of the particle types  $n'_1, n'_2, \dots$  that are of the same species of the particle

types  $n_1, n_2, \dots$  (As discussed more fully in Chapter 4, its sign is  $-1$  if this permutation includes an odd permutation of half-integer spin particles, and otherwise  $+1$ . This will not be important in the work of the present chapter.)

We often use an abbreviated notation, letting one Greek letter, say  $\alpha$ , stand for the whole collection  $p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots$ . In this notation, Eq. (3.1.2) is written simply

$$\langle \alpha' | \alpha \rangle = \delta(\alpha' - \alpha), \quad (3.1.3)$$

with  $\delta(\alpha' - \alpha)$  standing for the sum of products of delta functions and Kronecker deltas appearing on the right-hand side of Eq. (3.1.2). Also, in summing over states, we write

$$\int d\alpha \dots \equiv \sum_{n_1 \sigma_1 n_2 \sigma_2 \dots} \int d^3 p_1 d^3 p_2 \dots. \quad (3.1.4)$$

In particular, the completeness relation for states normalized as in Eq. (3.1.3) reads

$$|\Psi\rangle = \int d\alpha |\alpha\rangle \langle \alpha | \Psi \rangle. \quad (3.1.5)$$

The transformation rule (3.1.1) is only possible for particles that for one reason or another are not interacting. Setting  $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu$  and  $a^\mu = (\tau, 0, 0, 0)$ , for which  $U(\Lambda, a) = \exp(iH\tau)$ , Eq. (3.1.1) requires among other things that  $|\alpha\rangle$  be an energy eigenstate,

$$H|\alpha\rangle = E_\alpha |\alpha\rangle, \quad (3.1.6)$$

with an energy equal to the sum of the one-particle energies,

$$E_\alpha = p_1^0 + p_2^0 + \dots, \quad (3.1.7)$$

and with no interaction terms, terms that would involve more than one particle at a time.

On the other hand, the transformation rule (3.1.1) does apply in scattering processes at times  $t \rightarrow \mp\infty$ . As explained at the beginning of this chapter, in the typical scattering experiment we start with particles at time  $t \rightarrow -\infty$  so far apart that they are not yet interacting, and end with particles at  $t \rightarrow +\infty$  so far apart that they have ceased interacting. We therefore have not one but two sets of states that transform as in Eq. (3.1.1): the ‘in’ and ‘out’ states<sup>14</sup>  $|\alpha\rangle_{\text{in}}$  and  $|\alpha\rangle_{\text{out}}$  will be found to contain the particles described by the label  $\alpha$  if observations are made at  $t \rightarrow -\infty$  or  $t \rightarrow +\infty$ , respectively.

Note how this definition is framed. To maintain manifest Lorentz invariance, in the formalism we are using here, state-vectors do not change with time — a state-vector  $|\Psi\rangle$  describes the whole spacetime history of a system of particles. (This is known as the *Heisenberg picture*, in distinction with the Schrödinger picture, where the operators are constant and the states change with time.) Thus we do *not* say that  $|\alpha\rangle_{\text{in}}$  and  $|\alpha\rangle_{\text{out}}$  are the limits at  $t \rightarrow -\infty$  and  $t \rightarrow +\infty$  of a time-dependent state-vector  $|\Psi(t)\rangle$ .

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<sup>14</sup>The source uses the labels ‘+’ and ‘−’ for ‘in’ and ‘out’ states, respectively. This may seem backward, but the convention has become traditional; it arises from the signs in Eq. (3.1.16). This edition writes the labels explicitly as ‘in’ and ‘out’.

However, implicit in the definition of the states is a choice of the inertial frame from which the observer views the system; different observers see *equivalent* state-vectors, but not the *same* state-vector. In particular, suppose that a standard observer  $O$  sets his or her clock so that  $t = 0$  is at some time during the collision process, while some other observer  $O'$  at rest with respect to the first uses a clock set so that  $t' = 0$  is at a time  $t = \tau$ ; that is, the two observers' time coordinates are related by  $t' = t - \tau$ . Then if  $O$  sees the system to be in a state  $|\Psi\rangle$ ,  $O'$  will see the system in a state  $U(\mathbf{1}, -\tau)|\Psi\rangle = \exp(-iH\tau)|\Psi\rangle$ . Thus the appearance of the state long before or long after the collision (in whatever basis is used by  $O$ ) is found by applying a time-translation operator  $\exp(-iH\tau)$  with  $\tau \rightarrow -\infty$  or  $\tau \rightarrow +\infty$ , respectively. Of course, if the state is really an energy eigenstate, then it cannot be localized in time — the operator  $\exp(-iH\tau)$  yields an inconsequential phase factor  $\exp(-iE_\alpha\tau)$ . Therefore, we must consider wave-packets, superpositions  $\int d\alpha g(\alpha)|\alpha\rangle$  of states, with an amplitude  $g(\alpha)$  that is non-zero and smoothly varying over some finite range  $\Delta E$  of energies. The 'in' and 'out' states are defined so that the superposition

$$\exp(-iH\tau) \int d\alpha g(\alpha)|\alpha\rangle_{\text{in/out}} = \int d\alpha e^{-iE_\alpha\tau} g(\alpha)|\alpha\rangle_{\text{in/out}}$$

has the appearance of a corresponding superposition of free-particle states for  $\tau \ll -1/\Delta E$  or  $\tau \gg +1/\Delta E$ , respectively.

To make this concrete, suppose we can divide the time-translation generator  $H$  into two terms, a free-particle Hamiltonian  $H_0$  and an interaction  $H_{\text{int}}$ ,

$$H = H_0 + H_{\text{int}}, \quad (3.1.8)$$

in such a way that  $H_0$  has eigenstates  $|\alpha\rangle$  that have the same appearance as the eigenstates  $|\alpha\rangle_{\text{in}}$  and  $|\alpha\rangle_{\text{out}}$  of the complete Hamiltonian,

$$H_0|\alpha\rangle = E_\alpha|\alpha\rangle, \quad (3.1.9)$$

$$\langle\alpha'|\alpha\rangle = \delta(\alpha' - \alpha). \quad (3.1.10)$$

Note that  $H_0$  is assumed here to have the same spectrum as the full Hamiltonian  $H$ . This requires that the masses appearing in  $H_0$  be the physical masses that are actually measured, which are not necessarily the same as the 'bare' mass terms appearing in  $H$ ; the difference if there is any must be included in the interaction  $H_{\text{int}}$ , not  $H_0$ . Also, any relevant bound states in the spectrum of  $H$  should be introduced into  $H_0$  as if they were elementary particles.<sup>15</sup>

The 'in' and 'out' states can now be defined as eigenstates of  $H$ , not  $H_0$ ,

$$H|\alpha\rangle_{\text{in/out}} = E_\alpha|\alpha\rangle_{\text{in/out}}, \quad (3.1.11)$$

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<sup>15</sup>Alternatively, in non-relativistic problems we can include the binding potential in  $H_0$ . In the application of this method to 'rearrangement collisions,' where some bound states appear in the initial state but not the final state, or vice-versa, one must use a different split of  $H$  into  $H_0$  and  $H_{\text{int}}$  in the initial and final states.



which satisfy the condition

$$\begin{aligned}\int d\alpha e^{-iE_\alpha\tau} g(\alpha) |\alpha\rangle_{\text{in}} &\longrightarrow \int d\alpha e^{-iE_\alpha\tau} g(\alpha) |\alpha\rangle & (\tau \rightarrow -\infty), \\ \int d\alpha e^{-iE_\alpha\tau} g(\alpha) |\alpha\rangle_{\text{out}} &\longrightarrow \int d\alpha e^{-iE_\alpha\tau} g(\alpha) |\alpha\rangle & (\tau \rightarrow +\infty).\end{aligned}\quad (3.1.12)$$

Eq. (3.1.12) can be rewritten as the requirement that:

$$\exp(-iH\tau) \int d\alpha g(\alpha) |\alpha\rangle_{\text{in/out}} \longrightarrow \exp(-iH_0\tau) \int d\alpha g(\alpha) |\alpha\rangle$$

for  $\tau \rightarrow -\infty$  or  $\tau \rightarrow +\infty$ , respectively. This is sometimes rewritten as a formula for the ‘in’ and ‘out’ states:

$$\begin{aligned}|\alpha\rangle_{\text{in}} &= \Omega(-\infty) |\alpha\rangle, \\ |\alpha\rangle_{\text{out}} &= \Omega(+\infty) |\alpha\rangle,\end{aligned}\quad (3.1.13)$$

where

$$\Omega(\tau) \equiv \exp(+iH\tau) \exp(-iH_0\tau). \quad (3.1.14)$$

However, it should be kept in mind that  $\Omega(\mp\infty)$  in Eq. (3.1.13) gives meaningful results only when acting on a smooth superposition of energy eigenstates.

One immediate consequence of the definition (3.1.12) is that the ‘in’ and ‘out’ states are normalized just like the free-particle states. To see this, note that since the left-hand side of Eq. (3.1.12) is obtained by letting the unitary operator  $\exp(-iH\tau)$  act on a time-independent state, its norm is independent of time, and therefore equals the norm of its limit for  $\tau \rightarrow \mp\infty$ , i.e., the norm of the right-hand side of Eq. (3.1.12):

$$\begin{aligned}\int d\alpha d\beta \exp(-i(E_\alpha - E_\beta)\tau) g(\alpha) g^*(\beta)_{\text{in/out}} \langle \beta | \alpha \rangle_{\text{in/out}} \\ = \int d\alpha d\beta \exp(-i(E_\alpha - E_\beta)\tau) g(\alpha) g^*(\beta) \langle \beta | \alpha \rangle.\end{aligned}$$

Since this is supposed to be true for all smooth functions  $g(\alpha)$ , the scalar products must be equal:

$$_{\text{in/out}} \langle \beta | \alpha \rangle_{\text{in/out}} = \langle \beta | \alpha \rangle = \delta(\beta - \alpha). \quad (3.1.15)$$

It is useful for some purposes to have an explicit though formal solution of the energy eigenvalue equation (3.1.11) satisfying the conditions (3.1.12). For this purpose, write Eq. (3.1.11) as

$$(E_\alpha - H_0) |\alpha\rangle_{\text{in/out}} = H_{\text{int}} |\alpha\rangle_{\text{in/out}}.$$

The operator  $E_\alpha - H_0$  is not invertible; it annihilates not only the free-particle state  $|\alpha\rangle$ , but also the continuum of other free-particle states  $|\beta\rangle$  of the same energy. Since the ‘in’ and ‘out’ states become just  $|\alpha\rangle$  for  $H_{\text{int}} \rightarrow 0$ , we tentatively write the formal solutions as  $|\alpha\rangle$  plus a term proportional to  $H_{\text{int}}$ :

$$\begin{aligned}|\alpha\rangle_{\text{in}} &= |\alpha\rangle + (E_\alpha - H_0 + i\epsilon)^{-1} H_{\text{int}} |\alpha\rangle_{\text{in}}, \\ |\alpha\rangle_{\text{out}} &= |\alpha\rangle + (E_\alpha - H_0 - i\epsilon)^{-1} H_{\text{int}} |\alpha\rangle_{\text{out}},\end{aligned}\quad (3.1.16)$$

or, expanding in a complete set of free-particle states,

$$\begin{aligned} |\alpha\rangle_{\text{in}} &= |\alpha\rangle + \int d\beta \frac{T_{\beta\alpha}^{\text{in}} |\beta\rangle}{E_\alpha - E_\beta + i\epsilon}, \\ |\alpha\rangle_{\text{out}} &= |\alpha\rangle + \int d\beta \frac{T_{\beta\alpha}^{\text{out}} |\beta\rangle}{E_\alpha - E_\beta - i\epsilon}, \end{aligned} \quad (3.1.17)$$

$$T_{\beta\alpha}^{\text{in/out}} \equiv \langle \beta | H_{\text{int}} | \alpha \rangle_{\text{in/out}}, \quad (3.1.18)$$

with  $\epsilon$  a positive infinitesimal quantity, inserted to give meaning to the reciprocal of  $E_\alpha - H_0$ . These are known as the *Lippmann–Schwinger equations* [1a]. We shall use Eq. (3.1.17) at the end of the next section to give a slightly less unrigorous proof of the orthonormality of the ‘in’ and ‘out’ states.

It remains to be shown that Eq. (3.1.17), with a  $+i\epsilon$  or  $-i\epsilon$  in the denominator, satisfies the condition (3.1.12) for an ‘in’ or an ‘out’ state, respectively. For this purpose, consider the superpositions

$$|\Psi_g(t)\rangle_{\text{in/out}} \equiv \int d\alpha e^{-iE_\alpha t} g(\alpha) |\alpha\rangle_{\text{in/out}}, \quad (3.1.19)$$

$$|\Phi_g(t)\rangle \equiv \int d\alpha e^{-iE_\alpha t} g(\alpha) |\alpha\rangle. \quad (3.1.20)$$

We want to show that  $|\Psi_g(t)\rangle_{\text{in}}$  and  $|\Psi_g(t)\rangle_{\text{out}}$  approach  $|\Phi_g(t)\rangle$  for  $t \rightarrow -\infty$  and  $t \rightarrow +\infty$ , respectively. Using Eq. (3.1.17) in Eq. (3.1.19) gives

$$|\Psi_g(t)\rangle_{\text{in/out}} = |\Phi_g(t)\rangle + \int d\alpha \int d\beta \frac{e^{-iE_\alpha t} g(\alpha) T_{\beta\alpha}^{\text{in/out}} |\beta\rangle}{E_\alpha - E_\beta \pm i\epsilon}, \quad (3.1.21)$$

where the upper and lower signs refer to ‘in’ and ‘out’, respectively. Let us recklessly interchange the order of integration, and consider first the integrals

$$\mathcal{I}_\beta^{\text{in/out}} \equiv \int d\alpha \frac{e^{-iE_\alpha t} g(\alpha) T_{\beta\alpha}^{\text{in/out}}}{E_\alpha - E_\beta \pm i\epsilon}.$$

For  $t \rightarrow -\infty$ , we can close the contour of integration for the energy variable  $E_\alpha$  in the upper half-plane with a large semi-circle, with the contribution from this semi-circle killed by the factor  $\exp(-iE_\alpha t)$ , which is exponentially small for  $t \rightarrow -\infty$  and  $\text{Im } E_\alpha > 0$ . The integral is then given by a sum over the singularities of the integrand in the upper half-plane. The functions  $g(\alpha)$  and  $T_{\beta\alpha}^{\text{in/out}}$  may, in general, be expected to have some singularities at values of  $E_\alpha$  with finite positive imaginary parts, but just as for the large semi-circle, their contribution is exponentially damped for  $t \rightarrow -\infty$ . (Specifically,  $-t$  must be much greater than both the time-uncertainty in the wave-packet  $g(\alpha)$  and the duration of the collision, which respectively govern the location of the singularities of  $g(\alpha)$  and  $T_{\beta\alpha}^{\text{in/out}}$  in the complex  $E_\alpha$  plane.) This leaves the singularity in  $(E_\alpha - E_\beta \pm i\epsilon)^{-1}$ , which is in the upper half-plane for  $\mathcal{I}_\beta^{\text{out}}$  but not  $\mathcal{I}_\beta^{\text{in}}$ . We conclude then that  $\mathcal{I}_\beta^{\text{in}}$  vanishes for  $t \rightarrow -\infty$ . In the same way, for  $t \rightarrow +\infty$  we must close the contour of integration in the lower half-plane, and so  $\mathcal{I}_\beta^{\text{out}}$  vanishes in this limit. We conclude that  $|\Psi_g(t)\rangle_{\text{in/out}}$  approaches  $|\Phi_g(t)\rangle$  for  $t \rightarrow \mp\infty$ , in agreement with the defining condition (3.1.12).

\* \* \*

For future use, we note a convenient representation of the factor  $(E_\alpha - E_\beta \pm i\epsilon)^{-1}$  in Eq. (3.1.17). In general, we can write

$$(E \pm i\epsilon)^{-1} = \frac{\Pi_\epsilon}{E} \mp i\pi\delta_\epsilon(E), \quad (3.1.22)$$

where

$$\frac{\Pi_\epsilon}{E} \equiv \frac{E}{E^2 + \epsilon^2}, \quad (3.1.23)$$

$$\delta_\epsilon(E) \equiv \frac{\epsilon}{\pi(E^2 + \epsilon^2)}. \quad (3.1.24)$$

The function (3.1.23) is just  $1/E$  for  $|E| \gg \epsilon$ , and vanishes for  $E \rightarrow 0$ , so for  $\epsilon \rightarrow 0$  it behaves just like the ‘principal value function’  $\Pi/E$ , which allows us to give meaning to integrals of  $1/E$  times any smooth function of  $E$ , by excluding an infinitesimal interval around  $E = 0$ . The function (3.1.24) is of order  $\epsilon$  for  $|E| \gg \epsilon$ , and gives unity when integrated over all  $E$ , so in the limit  $\epsilon \rightarrow 0$  it behaves just like the familiar delta function  $\delta(E)$ . With this understanding, we can drop the label  $\epsilon$  in Eq. (3.1.22), and write simply

$$(E \pm i\epsilon)^{-1} = \frac{\Pi}{E} \mp i\pi\delta(E). \quad (3.1.25)$$

### 3.2 The $S$ -Matrix

An experimentalist generally prepares a state to have a definite particle content at  $t \rightarrow -\infty$ , and then measures what this state looks like at  $t \rightarrow +\infty$ . If the state is prepared to have a particle content  $\alpha$  for  $t \rightarrow -\infty$ , then it is the ‘in’ state  $|\alpha\rangle_{\text{in}}$ , and if it is found to have the particle content  $\beta$  at  $t \rightarrow +\infty$ , then it is the ‘out’ state  $|\beta\rangle_{\text{out}}$ . The probability amplitude for the transition  $\alpha \rightarrow \beta$  is thus the scalar product

$$S_{\beta\alpha} = {}_{\text{out}}\langle\beta|\alpha\rangle_{\text{in}}. \quad (3.2.1)$$

This array of complex amplitudes is known as the  $S$ -matrix [2]. If there were no interactions then ‘in’ and ‘out’ states would be the same, and then  $S_{\beta\alpha}$  would be just  $\delta(\alpha - \beta)$ . The rate for a reaction  $\alpha \rightarrow \beta$  is thus proportional to  $|S_{\beta\alpha} - \delta(\alpha - \beta)|^2$ . We shall see in detail in Section 3.4 what  $S_{\beta\alpha}$  has to do with measured rates and cross-sections.

Perhaps it should be stressed that ‘in’ and ‘out’ states do not inhabit two different Hilbert spaces. They differ only in how they are labelled: by their appearance either at  $t \rightarrow -\infty$  or  $t \rightarrow +\infty$ . Any ‘in’ state can be expanded as a sum of ‘out’ states, with expansion coefficients given by the  $S$ -matrix (3.2.1).

Since  $S_{\beta\alpha}$  is the matrix connecting two sets of orthonormal states, it must be unitary. To see this in greater detail, apply the completeness relation (3.1.5) to the ‘out’ states, and write

$$\int d\beta S_{\beta\gamma}^* S_{\beta\alpha} = \int d\beta {}_{\text{in}}\langle\gamma|\beta\rangle_{\text{out}} {}_{\text{out}}\langle\beta|\alpha\rangle_{\text{in}} = {}_{\text{in}}\langle\gamma|\alpha\rangle_{\text{in}}.$$

Using (3.1.15), this gives

$$\int d\beta S_{\beta\gamma}^* S_{\beta\alpha} = \delta(\gamma - \alpha), \quad (3.2.2)$$

or in brief,  $S^\dagger S = \mathbf{1}$ . In the same way, completeness for the ‘in’ states gives<sup>16</sup>

$$\int d\beta S_{\gamma\beta} S_{\alpha\beta}^* = \delta(\gamma - \alpha), \quad (3.2.3)$$

or, in other words,  $SS^\dagger = \mathbf{1}$ .

It is often convenient instead of dealing with the  $S$ -matrix to work with an operator  $S$ , defined to have matrix elements between *free-particle* states equal to the corresponding elements of the  $S$ -matrix:

$$\langle \beta | S | \alpha \rangle \equiv S_{\beta\alpha}. \quad (3.2.4)$$

The explicit though highly formal expression (3.1.13) for the ‘in’ and ‘out’ states yields a formula for the  $S$ -operator:

$$S = \Omega(+\infty)^\dagger \Omega(-\infty) = U(+\infty, -\infty), \quad (3.2.5)$$

where

$$U(\tau, \tau_0) \equiv \Omega(\tau)^\dagger \Omega(\tau_0) = \exp(iH_0\tau) \exp(-iH(\tau - \tau_0)) \exp(-iH_0\tau_0). \quad (3.2.6)$$

This will be used in the next section to examine the Lorentz invariance of the  $S$ -matrix, and in Sec. 3.5 to derive a formula for the  $S$ -matrix in time-dependent perturbation theory.

The methods of the previous section can be used to derive a useful alternative formula for the  $S$ -matrix. Let’s return to Eq. (3.1.21) for the ‘in’ state  $|\Psi_g\rangle_{\text{in}}$ , but this time take  $t \rightarrow +\infty$ . We must now close the contour of integration for  $E_\alpha$  in the *lower* half- $E_\alpha$ -plane, and although as before the singularities in  $T_{\beta\alpha}^{\text{in}}$  and  $g(\alpha)$  make no contribution for  $t \rightarrow +\infty$ , we now do pick up a contribution from the singular factor  $(E_\alpha - E_\beta + i\epsilon)^{-1}$ . The contour runs from  $E_\alpha = -\infty$  to  $E_\alpha = +\infty$ , and then back to  $E_\alpha = -\infty$  on a large semi-circle in the lower half-plane, so it circles the singularity in a *clockwise* direction. By the method of residues, this contribution to the integral over  $E_\alpha$  is given by the value of the integrand at  $E_\alpha = E_\beta - i\epsilon$ , times a factor  $-2i\pi$ . That is, in the limit  $\epsilon \rightarrow 0+$ , for  $t \rightarrow +\infty$  the integral over  $\alpha$  in (3.1.21) has the asymptotic behavior

$$\mathcal{I}_\beta^{\text{in}} \longrightarrow -2i\pi e^{-iE_\beta t} \int d\alpha \delta(E_\alpha - E_\beta) g(\alpha) T_{\beta\alpha}^{\text{in}},$$

and hence, for  $t \rightarrow +\infty$ ,

$$|\Psi_g(t)\rangle_{\text{in}} \longrightarrow \int d\beta e^{-iE_\beta t} |\beta\rangle \left[ g(\beta) - 2i\pi \int d\alpha \delta(E_\alpha - E_\beta) g(\alpha) T_{\beta\alpha}^{\text{in}} \right].$$

But expanding (3.1.19) for  $|\Psi_g\rangle_{\text{in}}$  in a complete set of ‘out’ states gives

$$|\Psi_g(t)\rangle_{\text{in}} = \int d\alpha e^{-iE_\alpha t} g(\alpha) \int d\beta |\beta\rangle_{\text{out}} S_{\beta\alpha}.$$

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<sup>16</sup>An alternative proof is given at the end of this section. Note that for infinite ‘matrices,’ the unitarity conditions  $S^\dagger S = \mathbf{1}$  and  $SS^\dagger = \mathbf{1}$  are not equivalent.

Since  $S_{\beta\alpha}$  contains a factor  $\delta(E_\beta - E_\alpha)$ , this may be rewritten

$$|\Psi_g(t)\rangle_{\text{in}} = \int d\beta |\beta\rangle_{\text{out}} e^{-iE_\beta t} \int d\alpha g(\alpha) S_{\beta\alpha},$$

and, using the defining property (3.1.12) for ‘out’ states, this has the asymptotic behavior for  $t \rightarrow +\infty$

$$|\Psi_g(t)\rangle_{\text{in}} \longrightarrow \int d\beta |\beta\rangle e^{-iE_\beta t} \int d\alpha g(\alpha) S_{\beta\alpha}.$$

Comparing this with our previous result, we find

$$\int d\alpha g(\alpha) S_{\beta\alpha} = g(\beta) - 2i\pi \int d\alpha \delta(E_\alpha - E_\beta) g(\alpha) T_{\beta\alpha}^{\text{in}},$$

or in other words

$$S_{\beta\alpha} = \delta(\beta - \alpha) - 2i\pi \delta(E_\alpha - E_\beta) T_{\beta\alpha}^{\text{in}}. \quad (3.2.7)$$

This suggests a simple approximation for the  $S$ -matrix: for a weak interaction  $H_{\text{int}}$ , we can neglect the difference between ‘in’ and free-particle states in (3.1.18), in which case Eq. (3.2.7) gives

$$S_{\beta\alpha} \simeq \delta(\beta - \alpha) - 2i\pi \delta(E_\alpha - E_\beta) \langle \beta | H_{\text{int}} | \alpha \rangle. \quad (3.2.8)$$

This is known as the *Born approximation* [3]. Higher-order terms are discussed in Section 3.5.

\* \* \*

We can use the Lippmann–Schwinger equations (3.1.16) for the ‘in’ and ‘out’ states to give a proof [4] of the orthonormality of these states and the unitarity of the  $S$ -matrix, as well as Eq. (3.2.7), without having to deal with limits at  $t \rightarrow \mp\infty$ . First, by using (3.1.16) on either the left- or right-hand side of the matrix element  ${}_{\text{in/out}}\langle \beta | H_{\text{int}} | \alpha \rangle_{\text{in/out}}$  and equating the results, we find that

$$\begin{aligned} {}_{\text{in/out}}\langle \beta | H_{\text{int}} | \alpha \rangle + {}_{\text{in/out}}\langle \beta | H_{\text{int}} (E_\alpha - H_0 \pm i\epsilon)^{-1} H_{\text{int}} | \alpha \rangle_{\text{in/out}} \\ = \langle \beta | H_{\text{int}} | \alpha \rangle_{\text{in/out}} + {}_{\text{in/out}}\langle \beta | H_{\text{int}} (E_\beta - H_0 \mp i\epsilon)^{-1} H_{\text{int}} | \alpha \rangle_{\text{in/out}}. \end{aligned}$$

Summing over a complete set  $|\gamma\rangle$  of intermediate states, this gives the equation:

$$\begin{aligned} T_{\alpha\beta}^{\text{in/out}*} - T_{\beta\alpha}^{\text{in/out}} = - \int d\gamma T_{\gamma\beta}^{\text{in/out}*} T_{\gamma\alpha}^{\text{in/out}} \\ \cdot ([E_\alpha - E_\gamma \pm i\epsilon]^{-1} - [E_\beta - E_\gamma \mp i\epsilon]^{-1}). \end{aligned} \quad (3.2.9)$$

To prove the orthonormality of the ‘in’ and ‘out’ states, divide Eq. (3.2.9) by  $E_\alpha - E_\beta \pm 2i\epsilon$ . This gives

$$\left( \frac{T_{\alpha\beta}^{\text{in/out}}}{E_\beta - E_\alpha \mp 2i\epsilon} \right)^* + \frac{T_{\beta\alpha}^{\text{in/out}}}{E_\alpha - E_\beta \pm 2i\epsilon} = - \int d\gamma \left( \frac{T_{\gamma\beta}^{\text{in/out}}}{E_\beta - E_\alpha \mp i\epsilon} \right)^* \frac{T_{\gamma\alpha}^{\text{in/out}}}{E_\alpha - E_\beta \pm i\epsilon}.$$

The  $2\epsilon$ s in the denominators on the left-hand side can be replaced with  $\epsilon$ s, since the only important thing is that these are positive infinitesimals. We see then that  $\delta(\beta - \alpha) + T_{\beta\alpha}^{\text{in/out}}/(E_\beta - E_\alpha \pm i\epsilon)$  is unitary. With (3.1.17), this is just the statement that the in and out states form two orthonormal sets of state-vectors. The unitarity of the  $S$ -matrix can be proved in a similar fashion by multiplying (3.2.9) with  $\delta(E_\beta - E_\alpha)$  instead of  $(E_\alpha - E_\beta \pm 2i\epsilon)^{-1}$ .

### 3.3 Symmetries of the $S$ -Matrix

In this section we will consider both what is meant by the invariance of the  $S$ -matrix under various symmetries, and what are the conditions on the Hamiltonian that will ensure such invariance properties.

#### Lorentz Invariance

For any proper orthochronous Lorentz transformation  $x \rightarrow \Lambda x + a$ , we may define a unitary operator  $U(\Lambda, a)$  by specifying that it acts as in Eq. (3.1.1) on either the ‘in’ or the ‘out’ states. When we say that a theory is Lorentz-invariant, we mean that the same operator  $U(\Lambda, a)$  acts as in (3.1.1) on both ‘in’ and ‘out’ states. Since the operator  $U(\Lambda, a)$  is unitary, we may write

$$S_{\beta\alpha} = {}_{\text{out}}\langle\beta|\alpha\rangle_{\text{in}} = {}_{\text{out}}\langle U(\Lambda, a)\beta | U(\Lambda, a)\alpha\rangle_{\text{in}},$$

so using (3.1.1), we obtain the Lorentz invariance (actually, covariance) property of the  $S$ -matrix: for arbitrary Lorentz transformations  $\Lambda^\mu{}_\nu$  and translations  $a^\mu$ ,

$$\begin{aligned} & S_{p'_1\sigma'_1n'_1;p'_2\sigma'_2n'_2;\dots,p_1\sigma_1n_1;p_2\sigma_2n_2;\dots} \\ &= \exp(ia_\mu(p_1^\mu + p_2^\mu + \dots - p_1'^\mu - p_2'^\mu - \dots)) \\ &\quad \cdot \sqrt{\frac{(\Lambda p_1)^0(\Lambda p_2)^0 \dots (\Lambda p'_1)^0(\Lambda p'_2)^0 \dots}{p_1^0 p_2^0 \dots p_1'^0 p_2'^0 \dots}} \\ &\quad \cdot \sum_{\bar{\sigma}_1 \bar{\sigma}_2 \dots} D_{\bar{\sigma}_1 \sigma_1}^{(j_1)}(W(\Lambda, p_1)) D_{\bar{\sigma}_2 \sigma_2}^{(j_2)}(W(\Lambda, p_2)) \dots \\ &\quad \cdot \sum_{\bar{\sigma}'_1 \bar{\sigma}'_2 \dots} D_{\bar{\sigma}'_1 \sigma'_1}^{(j'_1)*}(W(\Lambda, p'_1)) D_{\bar{\sigma}'_2 \sigma'_2}^{(j'_2)*}(W(\Lambda, p'_2)) \dots \\ &\quad \cdot S_{\Lambda p'_1 \bar{\sigma}'_1 n'_1; \Lambda p'_2 \bar{\sigma}'_2 n'_2; \dots, \Lambda p_1 \bar{\sigma}_1 n_1; \Lambda p_2 \bar{\sigma}_2 n_2; \dots}. \end{aligned} \tag{3.3.1}$$

(Primes are used to distinguish final from initial particles; bars are used to distinguish summation variables.) In particular, since the left-hand side is independent of  $a^\mu$ , so must be the right-hand side, and so the  $S$ -matrix vanishes unless the four-momentum is conserved. We can therefore write the part of the  $S$ -matrix that represents actual interactions among the particles in the form:

$$S_{\beta\alpha} - \delta(\beta - \alpha) = -2\pi i M_{\beta\alpha} \delta^4(p_\beta - p_\alpha). \tag{3.3.2}$$

(However, as we will see in the next chapter, the amplitude  $M_{\beta\alpha}$  itself contains terms that involve further delta function factors.)

Eq. (3.3.1) should be regarded as a definition of what we mean by the Lorentz invariance of the  $S$ -matrix, rather than a theorem, because it is only for certain special choices of Hamiltonian that there exists a unitary operator that acts as in (3.1.1) on both ‘in’ and ‘out’ states. We need to formulate conditions on the Hamiltonian that would ensure the Lorentz invariance of the  $S$ -matrix. For this purpose, it will be convenient to work with the operator  $S$  defined by Eq. (3.2.4):

$$S_{\beta\alpha} = \langle \beta | S | \alpha \rangle_0.$$

As we have defined the free-particle states in Chapter 2, they furnish a representation of the Poincaré group, so we can always define a unitary operator  $U_0(\Lambda, a)$  that induces the transformation (3.1.1) on these states:

$$\begin{aligned} & U_0(\Lambda, a) |p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle_0 \\ &= \exp(-ia_\mu(p_1^\mu + p_2^\mu + \dots)) \sqrt{\frac{(\Lambda p_1)^0 (\Lambda p_2)^0 \dots}{p_1^0 p_2^0 \dots}} \\ &\quad \cdot \sum_{\bar{\sigma}_1 \bar{\sigma}_2 \dots} D_{\bar{\sigma}_1 \sigma_1}^{(j_1)}(W(\Lambda, p_1)) D_{\bar{\sigma}_2 \sigma_2}^{(j_2)}(W(\Lambda, p_2)) \dots |\Lambda p_1, \bar{\sigma}_1, n_1; \Lambda p_2, \bar{\sigma}_2, n_2; \dots\rangle_0. \end{aligned}$$

Eq. (3.3.1) will thus hold if this unitary operator commutes with the  $S$ -operator:

$$U_0(\Lambda, a)^{-1} S U_0(\Lambda, a) = S.$$

This condition can also be expressed in terms of infinitesimal Lorentz transformations. Just as in Section 2.4, there will exist a set of Hermitian operators, a momentum  $\mathbf{P}_0$ , an angular momentum  $\mathbf{J}_0$ , and a boost generator  $\mathbf{K}_0$ , that together with  $H_0$  generate the infinitesimal version of inhomogeneous Lorentz transformations when acting on free-particle states. Eq. (3.3.1) is equivalent to the statement that the  $S$ -matrix is unaffected by such transformations, or in other words, that the  $S$ -operator commutes with these generators:

$$[H_0, S] = [\mathbf{P}_0, S] = [\mathbf{J}_0, S] = [\mathbf{K}_0, S] = 0. \quad (3.3.3)$$

Because the operators  $H_0$ ,  $\mathbf{P}_0$ ,  $\mathbf{J}_0$ , and  $\mathbf{K}_0$  generate infinitesimal inhomogeneous Lorentz transformations on the free states, they automatically satisfy the commutation relations (2.4.18)–(2.4.24):

$$[J_0^i, J_0^j] = i\epsilon_{ijk} J_0^k, \quad (3.3.4)$$

$$[J_0^i, K_0^j] = i\epsilon_{ijk} K_0^k, \quad (3.3.5)$$

$$[K_0^i, K_0^j] = -i\epsilon_{ijk} J_0^k, \quad (3.3.6)$$

$$[J_0^i, P_0^j] = i\epsilon_{ijk} P_0^k, \quad (3.3.7)$$

$$[K_0^i, P_0^j] = iH_0 \delta_{ij}, \quad (3.3.8)$$

$$[J_0^i, H_0] = [P_0^i, H_0] = [P_0^i, P_0^j] = 0, \quad (3.3.9)$$

$$[K_0^i, H_0] = iP_0^i. \quad (3.3.10)$$

where  $i, j, k$ , etc. run over the values 1, 2, and 3, and  $\epsilon_{ijk}$  is the totally antisymmetric quantity with  $\epsilon_{123} = +1$ .

In the same way, we may define a set of ‘exact generators,’ operators  $\mathbf{P}$ ,  $\mathbf{J}$ ,  $\mathbf{K}$  that together with the full Hamiltonian  $H$  generate the transformations (3.1.1) on, say, the ‘in’ states. (As already mentioned, what is not obvious is that the same operators generate the same transformations on the ‘out’ states.) The group structure tells us that these exact generators satisfy the same commutation relations:

$$[J^i, J^j] = i\epsilon_{ijk}J^k, \quad (3.3.11)$$

$$[J^i, K^j] = i\epsilon_{ijk}K^k, \quad (3.3.12)$$

$$[K^i, K^j] = -i\epsilon_{ijk}J^k, \quad (3.3.13)$$

$$[J^i, P^j] = i\epsilon_{ijk}P^k, \quad (3.3.14)$$

$$[K^i, P^j] = iH\delta_{ij}, \quad (3.3.15)$$

$$[J^i, H] = [P^i, H] = [P^i, P^j] = 0, \quad (3.3.16)$$

$$[K^i, H] = iP^i. \quad (3.3.17)$$

In virtually all known field theories, the effect of interactions is to add an interaction term  $H_{\text{int}}$  to the Hamiltonian, while leaving the momentum and angular momentum unchanged:

$$H = H_0 + H_{\text{int}}, \quad \mathbf{P} = \mathbf{P}_0, \quad \mathbf{J} = \mathbf{J}_0. \quad (3.3.18)$$

(The only known exceptions are theories with topologically twisted fields, such as those with magnetic monopoles, where the angular momentum of states depends on the interactions.) Eq. (3.3.18) implies that the commutation relations (3.3.11), (3.3.14), and (3.3.16) are satisfied provided that the interaction commutes with the free-particle momentum and angular-momentum operators

$$[H_{\text{int}}, \mathbf{P}_0] = [H_{\text{int}}, \mathbf{J}_0] = 0. \quad (3.3.19)$$

It is easy to see from the Lippmann–Schwinger equation (3.1.16) or equivalently from (3.1.13) that the operators that generate translations and rotations when acting on the ‘in’ (and ‘out’) states are indeed simply  $\mathbf{P}_0$  and  $\mathbf{J}_0$ . Also we easily see that  $\mathbf{P}_0$  and  $\mathbf{J}_0$  commute with the operator  $U(t, t_0)$  defined by Eq. (3.2.6), and hence with the  $S$ -operator  $U(\infty, -\infty)$ . Further, we already know that the  $S$ -operator commutes with  $H_0$ , because there are energy-conservation delta functions in both terms in (3.2.7). This leaves just the boost generator  $\mathbf{K}_0$  which we need to show commutes with the  $S$ -operator.

On the other hand, it is *not* possible to set the boost generator  $\mathbf{K}$  equal to its free-particle counterpart  $\mathbf{K}_0$ , because then Eqs. (3.3.15) and (3.3.8) would give  $H = H_0$ , which is certainly not true in the presence of interactions. Thus when we add an interaction  $H_{\text{int}}$  to  $H_0$ , we must also add a correction  $\mathbf{W}$  to the boost generator:

$$\mathbf{K} = \mathbf{K}_0 + \mathbf{W}. \quad (3.3.20)$$

Of the remaining commutation relations, let us concentrate on Eq. (3.3.17), which may now be put in the form

$$[\mathbf{K}_0, H_{\text{int}}] = -[\mathbf{W}, H]. \quad (3.3.21)$$



By itself, the condition (3.3.21) is empty, because for any  $H_{\text{int}}$  we could always define  $\mathbf{W}$  by giving its matrix elements between  $H$ -eigenstates  $|\alpha\rangle$  and  $|\beta\rangle$  as  $-\langle\beta|[\mathbf{K}_0, H_{\text{int}}]|\alpha\rangle/(E_\beta - E_\alpha)$ . Recall that the crucial point in the Lorentz invariance of a theory is not that there should exist a set of exact generators satisfying Eqs. (3.3.11)–(3.3.17), but rather that these operators should act the same way on ‘in’ and ‘out’ states; merely finding an operator  $\mathbf{K}$  that satisfies Eq. (3.3.21) is not enough. Eq. (3.3.21) does become significant if we add the requirement that matrix elements of  $\mathbf{W}$  should be smooth functions of the energies, and in particular should not have singularities of the form  $(E_\beta - E_\alpha)^{-1}$ . We shall now show that Eq. (3.3.21), together with an appropriate smoothness condition on  $\mathbf{W}$ , does imply the remaining Lorentz invariance condition  $[\mathbf{K}_0, S] = 0$ .

To prove this, let us consider the commutator of  $\mathbf{K}_0$  with the operator  $U(t, t_0)$  for finite  $t$  and  $t_0$ . Using Eq. (3.3.10) and the fact that  $\mathbf{P}_0$  commutes with  $H_0$  yields:

$$[\mathbf{K}_0, \exp(iH_0 t)] = -t\mathbf{P}_0 \exp(iH_0 t)$$

while Eq. (3.3.21) (which is equivalent to Eq. (3.3.17)) yields

$$[\mathbf{K}, \exp(iH t)] = -t\mathbf{P} \exp(iH t) = -t\mathbf{P}_0 \exp(iH t).$$

The momentum operators then cancel in the commutator of  $\mathbf{K}_0$  with  $U$ , and we find:

$$[\mathbf{K}_0, U(\tau, \tau_0)] = -\mathbf{W}(\tau)U(\tau, \tau_0) + U(\tau, \tau_0)\mathbf{W}(\tau_0), \quad (3.3.22)$$

where

$$\mathbf{W}(t) = \exp(iH_0 t)\mathbf{W} \exp(-iH_0 t). \quad (3.3.23)$$

If the matrix elements of  $\mathbf{W}$  between  $H_0$ -eigenstates are sufficiently smooth functions of energy, then matrix elements of  $\mathbf{W}(t)$  between smooth superpositions of energy eigenstates vanish for  $t \rightarrow \pm\infty$ , so Eq. (3.3.22) gives in effect:

$$0 = [\mathbf{K}_0, U(\infty, -\infty)] = [\mathbf{K}_0, S], \quad (3.3.24)$$

as was to be shown. This is the essential result: Eq. (3.3.21) together with the smoothness condition on matrix elements of  $\mathbf{W}$  that ensures that  $\mathbf{W}(t)$  vanishes for  $t \rightarrow \pm\infty$  provides a sufficient condition for the Lorentz invariance of the  $S$ -matrix. This smoothness condition is a natural one, because it is much like the condition on matrix elements of  $H_{\text{int}}$  that is needed to make  $H_I(t)$  vanish for  $t \rightarrow \pm\infty$ , as required in order to justify the very idea of an  $S$ -matrix.

We can also use Eq. (3.3.22) with  $\tau = 0$  and  $\tau_0 = \mp\infty$  to show that

$$\mathbf{K}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{K}_0, \quad (3.3.25)$$

where  $\Omega(\mp\infty)$  is according to (3.1.13) the operator that converts a free-particle state  $|\alpha\rangle_0$  into the corresponding ‘in’ or ‘out’ state. Also, it follows trivially from Eqs. (3.3.18) and (3.3.19) that the same is true for the momentum and angular momentum:

$$\mathbf{P}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{P}_0, \quad (3.3.26)$$

$$\mathbf{J}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{J}_0. \quad (3.3.27)$$

Finally, since all free and asymptotic states are eigenstates of  $H_0$  and  $H$  respectively with the same eigenvalue  $E_\alpha$ , we have

$$H\Omega(\mp\infty) = \Omega(\mp\infty)H_0. \quad (3.3.28)$$

Eqs. (3.3.25)–(3.3.28) show that with our assumptions, ‘in’ and ‘out’ states do transform under inhomogeneous Lorentz transformations just like the free-particle states. Also, since these are similarity transformations, we now see that the exact generators  $\mathbf{K}$ ,  $\mathbf{P}$ ,  $\mathbf{J}$ , and  $H$  satisfy the same commutation relations as  $\mathbf{K}_0$ ,  $\mathbf{P}_0$ ,  $\mathbf{J}_0$ , and  $H_0$ . This is why it turned out to be unnecessary in proving the Lorentz invariance of the  $S$ -matrix to use the other commutation relations (3.3.12), (3.3.13), and (3.3.15) that involve  $\mathbf{K}$ .

### Internal Symmetries

There are various symmetries, like the symmetry in nuclear physics under interchange of neutrons and protons, or the ‘charge-conjugation’ symmetry between particles and antiparticles, that have nothing directly to do with Lorentz invariance, and further appear the same in all inertial frames. Such a symmetry transformation  $T$  acts on the Hilbert space of physical states as a unitary operator  $U(T)$ , that induces linear transformations on the indices labelling particle species

$$\begin{aligned} & U(T)|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle_{\text{in/out}} \\ &= \sum_{\bar{n}_1 \bar{n}_2 \dots} \mathcal{D}_{\bar{n}_1 n_1}(T) \mathcal{D}_{\bar{n}_2 n_2}(T) \dots |p_1, \sigma_1, \bar{n}_1; p_2, \sigma_2, \bar{n}_2; \dots\rangle_{\text{in/out}}. \end{aligned} \quad (3.3.29)$$

In accordance with the general discussion in Chapter 2, the  $U(T)$  must satisfy the group multiplication rule

$$U(\bar{T})U(T) = U(\bar{T}T), \quad (3.3.30)$$

where  $\bar{T}T$  is the transformation obtained by first performing the transformation  $T$ , then some other transformation  $\bar{T}$ . Acting on Eq. (3.3.29) with  $U(\bar{T})$ , we see that the matrices  $\mathcal{D}$  satisfy the same rule

$$\mathcal{D}(\bar{T})\mathcal{D}(T) = \mathcal{D}(\bar{T}T). \quad (3.3.31)$$

Also, taking the scalar product of the states obtained by acting with  $U(T)$  on two different ‘in’ states or two different ‘out’ states, and using the normalization condition (3.1.2), we see that  $\mathcal{D}(T)$  must be unitary

$$\mathcal{D}^\dagger(T) = \mathcal{D}^{-1}(T). \quad (3.3.32)$$

Finally, taking the scalar product of the states obtained by acting with  $U(T)$  on one ‘out’ state and one ‘in’ state shows that  $\mathcal{D}$  commutes with the  $S$ -matrix, in the sense that

$$\begin{aligned} & \sum_{\bar{N}_1 \bar{N}_2 \dots} \sum_{\bar{N}'_1 \bar{N}'_2 \dots} \mathcal{D}_{\bar{N}'_1 n'_1}^*(T) \mathcal{D}_{\bar{N}'_2 n'_2}^*(T) \dots \mathcal{D}_{\bar{N}_1 n_1}(T) \mathcal{D}_{\bar{N}_2 n_2}(T) \dots \\ & \quad \cdot S_{p'_1 \sigma'_1 \bar{N}'_1; p'_2 \sigma'_2 \bar{N}'_2; \dots, p_1 \sigma_1 \bar{N}_1; p_2 \sigma_2 \bar{N}_2; \dots} \\ &= S_{p'_1 \sigma'_1 n'_1; p'_2 \sigma'_2 n'_2; \dots, p_1 \sigma_1 n_1; p_2 \sigma_2 n_2; \dots} \end{aligned} \quad (3.3.33)$$

Again, this is a definition of what we mean by a theory being invariant under the internal symmetry  $T$ , because to derive Eq. (3.3.33) we still need to show that the *same* unitary operator  $U(T)$  will induce the transformation (3.3.29) on both ‘in’ and ‘out’ states. This will be the case if there is an ‘unperturbed’ transformation operator  $U_0(T)$  that induces these transformations on free-particle states,

$$\begin{aligned} U_0(T)|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle_0 \\ = \sum_{\bar{N}_1 \bar{N}_2 \dots} \mathcal{D}_{\bar{N}_1 n_1}(T) \mathcal{D}_{\bar{N}_2 n_2}(T) \cdots |p_1, \sigma_1, \bar{N}_1; p_2, \sigma_2, \bar{N}_2; \dots\rangle_0 \end{aligned} \quad (3.3.34)$$

and that commutes with both the free-particle and interaction parts of the Hamiltonian

$$U_0^{-1}(T) H_0 U_0(T) = H_0, \quad (3.3.35)$$

$$U_0^{-1}(T) H_{\text{int}} U_0(T) = H_{\text{int}}. \quad (3.3.36)$$

From either the Lippmann–Schwinger equation (3.1.17) or from (3.1.13), we see that the operator  $U_0(T)$  will induce the transformations (3.3.29) on ‘in’ and ‘out’ states as well as free-particle states, so that we can derive Eq. (3.3.29) taking  $U(T)$  as  $U_0(T)$ .

A special case of great physical importance is that of a one-parameter Lie group, where  $T$  is a function of a single parameter  $\theta$ , with

$$T(\bar{\theta})T(\theta) = T(\bar{\theta} + \theta). \quad (3.3.37)$$

As shown in Section 2.2, in this case the corresponding Hilbert-space operators must take the form

$$U(T(\theta)) = \exp(iQ\theta) \quad (3.3.38)$$

with  $Q$  a Hermitian operator. Likewise the matrices  $\mathcal{D}(T)$  take the form

$$\mathcal{D}_{n'n}(T(\theta)) = \delta_{n'n} \exp(iq_n \theta), \quad (3.3.39)$$

where  $q_n$  are a set of real species-dependent numbers. Here Eq. (3.3.33) simply tells us that the  $q$ s are conserved:  $S_{\beta\alpha}$  vanishes unless

$$q_{n'_1} + q_{n'_2} + \cdots = q_{n_1} + q_{n_2} + \cdots. \quad (3.3.40)$$

The classic example of such a conservation law is that of conservation of electric charge. Also, all known processes conserve baryon number (the number of baryons, such as protons, neutrons, and hyperons, minus the number of their antiparticles) and lepton number (the number of leptons, such as electrons, muons,  $\tau$  particles, and neutrinos, minus the number of their antiparticles) but as we shall see in Volume II, these conservation laws are believed to be only very good approximations. There are other conservation laws of this type that are definitely only approximate, such as the conservation of the quantity known as strangeness, which was introduced to explain the relatively long life of a class of particles discovered by Rochester and Butler [5] in cosmic rays in 1947. For instance, the mesons now called

$K^+$  and  $K^0$ <sup>17</sup> are assigned strangeness +1 and the hyperons  $\Lambda^0$ ,  $\Sigma^+$ ,  $\Sigma^0$ ,  $\Sigma^-$  are assigned strangeness -1, while the more familiar protons, neutrons, and  $\pi$  mesons (or pions) are taken to have strangeness zero. The conservation of strangeness in strong interactions explains why strange particles are always produced in association with one another, as in reactions like  $\pi^+ + n \rightarrow K^+ + \Lambda^0$ , while the relatively slow decays of strange particles into non-strange ones such as  $\Lambda^0 \rightarrow p + \pi^-$  and  $K^+ \rightarrow \pi^+ + \pi^0$  show that the interactions that do not conserve strangeness are very weak.

The classic example of a ‘non-Abelian’ symmetry whose generators do not commute with one another is isotopic spin symmetry, which was suggested [6] in 1937 on the basis of an experiment [7] that showed the existence of a strong proton–proton force similar to that between protons and neutrons. Mathematically, the group is  $SU(2)$ , like the covering group of the group of three-dimensional rotations; its generators are denoted  $t_i$ , with  $i = 1, 2, 3$ , and satisfy a commutation relation like (2.4.18):

$$[t_i, t_j] = i\epsilon_{ijk}t_k.$$

To the extent that isotopic spin symmetry is respected, it requires particles to form degenerate multiplets labelled with an integer or half-integer  $T$  and with  $2T + 1$  components distinguished by their  $t_3$  values, just like the degenerate spin multiplets required by rotational invariance. These include the nucleons  $p$  and  $n$  with  $T = \frac{1}{2}$  and  $t_3 = \frac{1}{2}, -\frac{1}{2}$ ; the pions  $\pi^+$ ,  $\pi^0$ , and  $\pi^-$  with  $T = 1$  and  $t_3 = +1, 0, -1$ ; and the  $\Lambda^0$  hyperon with  $T = 0$  and  $t_3 = 0$ . These examples illustrate the relation between electric charge  $Q$ , the third component of isotopic spin  $t_3$ , the baryon number  $B$ , and the strangeness  $S$ :

$$Q = t_3 + (B + S)/2.$$

This relation was originally inferred from observed selection rules, but it was interpreted [8] by Gell-Mann and Ne’eman in 1960 to be a consequence of the embedding of both the isospin  $\mathbf{T}$  and the ‘hypercharge’  $Y \equiv B + S$  in the Lie algebra of a larger but more badly broken non-Abelian internal symmetry, based on the non-Abelian group  $SU(3)$ . As we will see in Volume II, today both isospin and  $SU(3)$  symmetry are understood as incidental consequences of the small masses of the two or three lightest quarks in the modern theory of strong interactions, quantum chromodynamics.

The implications of isotopic spin symmetry for reactions among strongly interacting particles can be worked out by the same familiar methods that were invented for deriving the implications of rotational invariance. In particular, for a two-body reaction  $A + B \rightarrow C + D$ , Eq. (3.3.33) requires that the  $S$ -matrix may be put in the form (suppressing all but isospin labels):

$$S_{t_{C3}t_{D3}, t_{A3}t_{B3}} = \sum_{T, t_3} C_{T_C T_D}(T t_3; t_{C3} t_{D3}) C_{T_A T_B}(T t_3; t_{A3} t_{B3}) S_T,$$

where  $C_{j_1 j_2}(j\sigma; \sigma_1 \sigma_2)$  is the usual Clebsch–Gordan coefficient [9] for forming a spin  $j$  with three-component  $\sigma$  from spins  $j_1$  and  $j_2$  with three-components  $\sigma_1$  and  $\sigma_2$ , respectively; and

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<sup>17</sup>Superscripts denote electric charges in units of the absolute value of the electronic charge. A ‘hyperon’ is any particle carrying non-zero strangeness and unit baryon number.

$S_T$  is a ‘reduced’  $S$ -matrix depending on  $T$  and on all the suppressed momentum and spin variables, but not on the isospin three-components  $t_{A3}, t_{B3}, t_{C3}, t_{D3}$ . Of course, this, like all of the consequences of isotopic spin invariance, is only approximate, because this symmetry is not respected by electromagnetic (and other) interactions, as shown for instance by the fact that different members of the same isospin multiplet like  $p$  and  $n$  have different electric charges and slightly different masses.

### Parity

To the extent that the symmetry under the transformation  $\mathbf{x} \rightarrow -\mathbf{x}$  is really valid, there must exist a unitary operator  $P$  under which both ‘in’ and ‘out’ states transform as a direct product of single-particle states:

$$P|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle_{\text{in/out}} = \eta_{n_1} \eta_{n_2} \cdots |\mathcal{P}p_1, \sigma_1, n_1; \mathcal{P}p_2, \sigma_2, n_2; \dots\rangle_{\text{in/out}}. \quad (3.3.41)$$

where  $\eta_n$  is the intrinsic parity of particles of species  $n$ , and  $\mathcal{P}$  reverses the space components of  $p^\mu$ . (This is for massive particles; the modification for massless particles is obvious.) The parity conservation condition for the  $S$ -matrix is then:

$$\begin{aligned} S_{p'_1 \sigma'_1 n'_1; p'_2 \sigma'_2 n'_2; \dots, p_1 \sigma_1 n_1; p_2 \sigma_2 n_2; \dots} \\ = \eta_{n'_1}^* \eta_{n'_2}^* \cdots \eta_{n_1} \eta_{n_2} \cdots S_{\mathcal{P}p'_1 \sigma'_1 n'_1; \mathcal{P}p'_2 \sigma'_2 n'_2; \dots, \mathcal{P}p_1 \sigma_1 n_1; \mathcal{P}p_2 \sigma_2 n_2; \dots} \end{aligned} \quad (3.3.42)$$

Just as for internal symmetries, an operator  $P$  satisfying Eq. (3.3.41) will actually exist if the operator  $P_0$  which is defined to act this way on free-particle states commutes with  $H_{\text{int}}$  as well as  $H_0$ .

The phases  $\eta_n$  may be inferred either from dynamical models or from experiment, but neither can provide a unique determination of the  $\eta$ s. This is because we are always free to redefine  $P$  by combining it with any conserved internal symmetry operator. For instance, if  $P$  is conserved, then so is

$$P' \equiv P \exp(i\alpha B + i\beta L + i\gamma Q),$$

where  $B$ ,  $L$ , and  $Q$  are respectively baryon number, lepton number, and electric charge, and  $\alpha$ ,  $\beta$ , and  $\gamma$  are arbitrary real phases; hence either  $P$  or  $P'$  could be called the parity operator. The neutron, proton, and electron have different combinations of values for  $B$ ,  $L$ , and  $Q$ , so by judicious choice of the phases  $\alpha$ ,  $\beta$ , and  $\gamma$  we can define the intrinsic parities of all three particles to be  $+1$ . However, once we have done this the intrinsic parities of other particles like the charged pion (which can be emitted in a transition  $n \rightarrow p + \pi^-$ ) are no longer arbitrary. Also, the intrinsic parity of any particle like the neutral pion  $\pi^0$  which carries no conserved quantum numbers is always meaningful.

The foregoing remarks help to clarify the question of whether intrinsic parities must always have the values  $\pm 1$ . It is easy to say that space inversion  $\mathcal{P}$  has the group multiplication law  $\mathcal{P}^2 = \mathbf{1}$ ; however, the parity operator that is conserved may not be this one, but rather may differ from it by a phase transformation of some sort. In any case, whether or not  $P^2 = \mathbf{1}$ , the operator  $P^2$  behaves just like an internal symmetry transformation:

$$P^2|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle_{\text{in/out}} = \eta_{n_1}^2 \eta_{n_2}^2 \cdots |p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle_{\text{in/out}}.$$

If this internal symmetry is part of a continuous symmetry group of phase transformations, such as the group of multiplication by the phases  $\exp(i\alpha B + i\beta L + i\gamma Q)$  with arbitrary values of  $\alpha$ ,  $\beta$ , and  $\gamma$ , then its inverse square root must also be a member of this group, say  $I_P$ , with  $I_P^2 P^2 = \mathbf{1}$  and  $[I_P, P] = 0$ . (For instance, if  $P^2 = \exp(i\alpha B + \dots)$ , then take  $I_P = \exp(-\frac{1}{2}i\alpha B + \dots)$ .) We can then define a new parity operator  $P' \equiv P I_P$  with  $P'^2 = \mathbf{1}$ . This is conserved to the same extent as  $P$ , so there is no reason why we should not call this *the* parity operator, in which case the intrinsic parities can only take the values  $\pm 1$ .

The only sort of theory in which it is not necessarily possible to define parity so that all intrinsic parities have the values  $\pm 1$  is one in which there is some discrete internal symmetry which is not a member of any continuous symmetry group of phase transformations [10]. For instance, it is a consequence of angular-momentum conservation that the total number  $F$  of all particles of half-integer spin can only change by even numbers, so the internal symmetry operator  $(-1)^F$  is conserved. All known particles of half-integer spin have odd values of the sum  $B + L$  of baryon number and lepton number, so as far as we know,  $(-1)^F = (-1)^{B+L}$ . If this is true, then  $(-1)^F$  is part of a continuous symmetry group, consisting of the operators  $\exp(i\alpha(B + L))$  with arbitrary real  $\alpha$ , and has an inverse square root  $\exp(-i\pi(B + L)/2)$ . In this case, if  $P^2 = (-1)^F$  then  $P$  can be redefined so that all intrinsic parities are  $\pm 1$ . However, if there were to be discovered a particle of half-integer spin and an even value of  $B + L$  (such as a so-called Majorana neutrino, with  $j = \frac{1}{2}$  and  $B + L = 0$ ), then it would be possible to have  $P^2 = (-1)^F$  without our being able to redefine the parity operator itself to have eigenvalues  $\pm 1$ . In this case, of course, we would have  $P^4 = \mathbf{1}$ , so all particles would have intrinsic parities either  $\pm 1$  or (like the Majorana neutrino)  $\pm i$ .

It follows from Eq. (3.3.42) that if the product of intrinsic parities in the final state is equal to the product of intrinsic parities in the initial state, or equal to minus this product, then the  $S$ -matrix must be respectively even or odd overall in the three-momenta. For instance, it was observed [11] in 1951 that a pion can be absorbed by a deuteron from the  $\ell = 0$  ground state of the  $\pi^- d$  atom, in the reaction  $\pi^- + d \rightarrow n + n$ . (As discussed in Section 3.7, the orbital angular-momentum quantum number  $\ell$  can be used in relativistic physics in the same way as in non-relativistic wave mechanics.) The initial state has total angular-momentum  $j = 1$  (the pion and deuteron having spins zero and one, respectively), so the final state must have orbital angular-momentum  $\ell = 1$  and total neutron spin  $s = 1$ . (The other possibilities  $\ell = 1, s = 0$ ;  $\ell = 0, s = 1$ ; and  $\ell = 2, s = 1$ , which are allowed by angular-momentum conservation, are forbidden by the requirement that the final state be antisymmetric in the two neutrons.) Because the final state has  $\ell = 1$ , the matrix element is odd under reversal of the direction of all three-momenta, so we can conclude that the intrinsic parities of the particles in this reaction must be related by:

$$\eta_d \eta_{\pi^-} = -\eta_n^2.$$

The deuteron is known to be a bound state of a proton and neutron with even orbital angular-momentum (chiefly  $\ell = 0$ ), and as we have seen we can take the neutron and proton to have the same intrinsic parity, so  $\eta_d = \eta_n^2$ , and we can conclude that  $\eta_{\pi^-} = -1$ ;

that is, the negative pion is a *pseudoscalar* particle. The  $\pi^+$  and  $\pi^0$  have also been found to have negative parity, as would be expected from the symmetry (isospin invariance) among these three particles.

The negative parity of the pion has some striking consequences. A spin zero particle that decays into three pions must have intrinsic parity  $\eta_\pi^3 = -1$ , because in the Lorentz frame in which the decaying particle is at rest, rotational invariance only allows the matrix element to depend on scalar products of the pion momenta with each other, all of which are even under reversal of all momenta. (The triple scalar product  $\mathbf{p}_1 \cdot (\mathbf{p}_2 \times \mathbf{p}_3)$  formed from the three pion momenta vanishes because  $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 = 0$ .) For the same reason, a spin zero particle that decays into two pions must have intrinsic parity  $\eta_\pi^2 = +1$ . In particular, among the strange particles discovered in the late 1940s there seemed to be two different particles of zero spin (inferred from the angular distribution of their decay products): one, the  $\tau$ , was identified by its decay into three pions, and hence was assigned a parity  $-1$ , while the other, the  $\theta$ , was identified by its decay into two pions and was assigned a parity  $+1$ . The trouble with all this was that as the  $\tau$  and  $\theta$  were studied in greater detail, they seemed increasingly to have identical masses and lifetimes. After many suggested solutions of this puzzle, Lee and Yang in 1956 finally cut the Gordian knot, and proposed that the  $\tau$  and  $\theta$  are the same particle, (now known as the  $K^\pm$ ) and that parity is simply not conserved in the weak interactions that lead to its decay [12].

As we shall see in detail in the next section of this chapter, the rate for a physical process  $\alpha \rightarrow \beta$  (with  $\alpha \neq \beta$ ) is proportional to  $|S_{\beta\alpha}|^2$ , with proportionality factors that are invariant under reversal of all three-momenta. As long as the states  $\alpha$  and  $\beta$  contain definite numbers of particles of each type, the phase factors in Eq. (3.3.42) have no effect on  $|S_{\beta\alpha}|^2$ , so Eq. (3.3.42) would imply that the rate for  $\alpha \rightarrow \beta$  is invariant under the reversal of direction of all three-momenta. As we have seen, this is a trivial consequence of rotational invariance for the decays of a  $K$  meson into two or three pions, but it is a non-trivial restriction on rates in more complicated processes. For example, following theoretical suggestions by Lee and Yang, Wu together with a group at the National Bureau of Standards measured the angular distribution of the electron in the final state of the beta decay  $\text{Co}^{60} \rightarrow \text{Ni}^{60} + e^- + \bar{\nu}$  with a polarized cobalt source [13]. (No attempt was made in this experiment to measure the momentum of the antineutrino or nickel nucleus.) The electrons were found to be preferentially emitted in a direction opposite to that of the spin of the decaying nucleus, which would, of course, be impossible if the decay rate were invariant under a reversal of all three-momenta. A similar result was found in the decay of a positive muon (polarized in its production in the process  $\pi^+ \rightarrow \mu^+ + \nu$ ) into a positron, neutrino, and antineutrino [14]. In this way, it became clear that parity is indeed not conserved in the weak interactions responsible for these decays. Nevertheless, for reasons discussed in Section 12.5, parity *is* conserved in the strong and electromagnetic interactions, and therefore continues to play an important part in theoretical physics.

## Time-Reversal

We saw in Section 2.6 that the time-reversal operator  $T$  acting on a one-particle state  $|p, \sigma, n\rangle$  gives a state  $|\mathcal{P}p, -\sigma, n\rangle$  with reversed spin and momentum, times a phase

$\zeta_n(-1)^{j-\sigma}$ . A multi-particle state transforms as usual as a direct product of one-particle states, except that since this is a time-reversal transformation, we expect ‘in’ and ‘out’ states to be interchanged:

$$\begin{aligned} T|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle_{\text{in/out}} \\ = \zeta_{n_1}(-1)^{j_1-\sigma_1} \zeta_{n_2}(-1)^{j_2-\sigma_2} \dots |\mathcal{P}p_1, -\sigma_1, n_1; \mathcal{P}p_2, -\sigma_2, n_2; \dots\rangle_{\text{out/in}}. \end{aligned} \quad (3.3.43)$$

(Again, this is for massive particles, with obvious modifications being required for massless particles.) It will be convenient to abbreviate this assumption as

$$T|\alpha\rangle_{\text{in/out}} = |\Theta\alpha\rangle_{\text{out/in}}, \quad (3.3.44)$$

where  $\Theta$  indicates a reversal of sign of three-momenta and spins as well as multiplication by the phase factors shown in Eq. (3.3.43). Because  $T$  is antiunitary, we have

$${}_{\text{out}}\langle\beta|\alpha\rangle_{\text{in}} = {}_{\text{out}}\langle\Theta\alpha|\Theta\beta\rangle_{\text{in}}, \quad (3.3.45)$$

so the time-reversal invariance condition for the  $S$ -matrix is

$$S_{\beta\alpha} = S_{\Theta\alpha, \Theta\beta} \quad (3.3.46)$$

or in more detail

$$\begin{aligned} S_{p'_1\sigma'_1n'_1; p'_2\sigma'_2n'_2; \dots, p_1\sigma_1n_1; p_2\sigma_2n_2; \dots} \\ = \zeta_{n'_1}(-1)^{j'_1-\sigma'_1} \zeta_{n'_2}(-1)^{j'_2-\sigma'_2} \dots \zeta_{n_1}^*(-1)^{j_1-\sigma_1} \zeta_{n_2}^*(-1)^{j_2-\sigma_2} \dots \\ \cdot S_{\mathcal{P}p_1, -\sigma_1, n_1; \mathcal{P}p_2, -\sigma_2, n_2; \dots, \mathcal{P}p'_1, -\sigma'_1, n'_1; \mathcal{P}p'_2, -\sigma'_2, n'_2; \dots}. \end{aligned} \quad (3.3.47)$$

Note that in addition to the reversal of momenta and spins, the role of initial and final states is interchanged, as would be expected for a symmetry involving the reversal of time.

The  $S$ -matrix will satisfy this transformation rule if the operator  $T_0$  that induces time-reversal transformations on free-particle states

$$T_0|\alpha\rangle_0 = |\Theta\alpha\rangle_0 \quad (3.3.48)$$

commutes not only with the free-particle Hamiltonian (which is automatic) but also with the interaction:

$$T_0^{-1}H_0T_0 = H_0, \quad (3.3.49)$$

$$T_0^{-1}H_{\text{int}}T_0 = H_{\text{int}}. \quad (3.3.50)$$

In this case we can take  $T = T_0$ , and use either (3.1.13) or (3.1.16) to show that time-reversal transformations do act as stated in Eq. (3.3.44). For instance, operating on the Lippmann–Schwinger equation (3.1.16) with  $T$  and using Eqs. (3.3.48)–(3.3.50), we have

$$T|\alpha\rangle_{\text{in/out}} = |\Theta\alpha\rangle_0 + \int d\gamma [E_\alpha - H_0 \mp i\epsilon]^{-1} H_{\text{int}} T|\alpha\rangle_{\text{in/out}}$$



with the sign of  $\pm i\epsilon$  reversed because  $T$  is antiunitary. This is just the Lippmann–Schwinger equation for  $|\Theta\alpha\rangle_{\text{out/in}}$ , thus justifying Eq. (3.3.44). Similarly, because  $T$  is antiunitary it changes the sign of the  $i$  in the exponent of  $\Omega(t)$ , so that

$$T\Omega(-\infty)|\alpha\rangle_0 = \Omega(\infty)|\Theta\alpha\rangle_0,$$

again leading to Eq. (3.3.44).

In contrast with the case of parity conservation, the time-reversal invariance condition (3.3.46) does *not* in general tell us that the rate for the process  $\alpha \rightarrow \beta$  is the same as for the process  $\Theta\alpha \rightarrow \Theta\beta$ . However, something like this is true in cases where the  $S$ -matrix takes the form

$$S_{\beta\alpha} = S_{\beta\alpha}^{(0)} + S_{\beta\alpha}^{(1)}, \quad (3.3.51)$$

where  $S^{(1)}$  is small, while  $S^{(0)}$  happens to have matrix element zero for some particular process of interest, though it generally has much larger matrix elements than  $S^{(1)}$ . (For instance, the process might be nuclear beta decay,  $N \rightarrow N' + e^- + \bar{\nu}$ , with  $S^{(0)}$  the  $S$ -matrix produced by the strong nuclear and electromagnetic interactions alone, and  $S^{(1)}$  the correction to the  $S$ -matrix produced by the weak interactions. Section 3.5 shows how the use of the ‘distorted-wave Born approximation’ leads to an  $S$ -matrix of the form (3.3.51) in cases of this type. In some cases  $S^{(0)}$  is simply the unit operator.) To first order in  $S^{(1)}$ , the unitarity condition for the  $S$ -operator reads

$$\mathbf{1} = S^\dagger S = S^{(0)\dagger} S^{(0)} + S^{(0)\dagger} S^{(1)} + S^{(1)\dagger} S^{(0)}.$$

Using the zeroth-order relation  $S^{(0)\dagger} S^{(0)} = \mathbf{1}$ , this gives a reality condition for  $S^{(1)}$ :

$$S^{(1)} = -S^{(0)} S^{(1)\dagger} S^{(0)}. \quad (3.3.52)$$

If  $S^{(1)}$  as well as  $S^{(0)}$  satisfies the time-reversal condition (3.3.46), then this can be put in the form

$$S_{\beta\alpha}^{(1)} = - \int d\gamma \int d\gamma' S_{\beta\gamma'}^{(0)} S_{\Theta\gamma', \Theta\gamma}^{(1)*} S_{\gamma\alpha}^{(0)}. \quad (3.3.53)$$

Since  $S^{(0)}$  is unitary, the rates for the processes  $\alpha \rightarrow \beta$  and  $\Theta\alpha \rightarrow \Theta\beta$  are thus the same if summed over sets  $\mathcal{J}$  and  $\mathcal{F}$  of final and initial states that are complete with respect to  $S^{(0)}$ . (By being ‘complete’ here is meant that if  $S_{\alpha'\alpha}^{(0)}$  is non-zero, and either  $\alpha$  or  $\alpha'$  are in  $\mathcal{J}$ , then both states are in  $\mathcal{J}$ ; and similarly for  $\mathcal{F}$ .) In the simplest case we have ‘complete’ sets  $\mathcal{J}$  and  $\mathcal{F}$  consisting of just one state each; that is, both the initial and the final states are eigenvectors of  $S^{(0)}$  with eigenvalues  $e^{2i\delta_\alpha}$  and  $e^{2i\delta_\beta}$ , respectively. (The  $\delta_\alpha$  and  $\delta_\beta$  are called ‘phase shifts’; they are real because  $S^{(0)}$  is unitary.) In this case, Eq. (3.3.53) becomes simply:

$$S_{\beta\alpha}^{(1)} = -e^{2i(\delta_\alpha + \delta_\beta)} S_{\Theta\beta, \Theta\alpha}^{(1)*}, \quad (3.3.54)$$

and it is clear that the absolute value of the  $S$ -matrix for the process  $\alpha \rightarrow \beta$  is the same as for the process  $\Theta\alpha \rightarrow \Theta\beta$ . This is the case for instance in nuclear beta decay (in the approximation in which we ignore the relatively weak Coulomb interaction between the electron and nucleus in the final state), because both the initial and final states are eigenstates of

the strong interaction  $S$ -matrix (with  $\delta_\alpha = \delta_\beta = 0$ ). Thus if time-reversal invariance is respected, the differential rate for a beta decay process should be unchanged if we reverse both the momenta and the spin  $z$ -components  $\sigma$  of all particles. This prediction was *not* contradicted in the 1956 experiments [13, 14] that discovered the non-conservation of parity; for instance, time-reversal invariance is consistent with the observation that electrons from the decay  $\text{Co}^{60} \rightarrow \text{Ni}^{60} + e^- + \bar{\nu}$  are preferentially emitted in a direction opposite to that of the  $\text{Co}^{60}$  spin. As described below, indirect evidence against time-reversal invariance did emerge in 1964, but it remains a useful approximate symmetry in weak as well as strong and electromagnetic interactions.

In some cases we can use a basis of states for which  $\Theta\alpha = \alpha$  and  $\Theta\beta = \beta$ , for which Eq. (3.3.54) reads

$$S_{\beta\alpha}^{(1)} = -e^{2i(\delta_\alpha + \delta_\beta)} S_{\beta\alpha}^{(1)*}, \quad (3.3.55)$$

which just says that  $iS_{\beta\alpha}^{(1)}$  has the phase  $\delta_\alpha + \delta_\beta \bmod \pi$ . This is known as *Watson's theorem* [15]. The phases in Eqs. (3.3.54) or (3.3.55) may be measured in processes where there is interference between different final states. For instance, in the decay of the spin 1/2 hyperon  $\Lambda$  into a nucleon and a pion, the final state can only have orbital angular-momentum  $\ell = 0$  or  $\ell = 1$ ; the angular distribution of the pion relative to the  $\Lambda$  spin involves the interference between these states, and hence according to Watson's theorem depends on the difference  $\delta_s - \delta_p$  of their phase shifts.

### $PT$

Although the 1957 experiments on parity violation did not rule out time-reversal invariance, they did show immediately that the product  $PT$  is not conserved. If conserved this operator would have to be antiunitary for the same reasons as for  $T$ , so in processes like nuclear beta decay its consequences would take the form of relations like Eq. (3.3.54):

$$S_{\beta\alpha}^{(1)} = -e^{i(\delta_\alpha + \delta_\beta)} S_{\mathcal{P}\Theta\beta, \mathcal{P}\Theta\alpha}^{(1)*},$$

where  $\mathcal{P}\Theta$  reverses the signs of all spin  $z$ -components but *not* any momenta. Neglecting the final-state Coulomb interaction, it would then follow that there could be no preference for the electron in the decay  $\text{Co}^{60} \rightarrow \text{Ni}^{60} + e^- + \bar{\nu}$  to be emitted in the same or the opposite direction to the  $\text{Co}^{60}$  spin, in contradiction with what was observed.

### $C$ , $CP$ , and $CPT$

As already mentioned, there is an internal symmetry transformation, known as charge-conjugation, which interchanges particles and antiparticles. Formally, this entails the existence of a unitary operator  $C$ , whose effect on multi-particle states is:

$$C|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \dots\rangle_{\text{in/out}} = \xi_{n_1} \xi_{n_2} \cdots |p_1, \sigma_1, n_1^c; p_2, \sigma_2, n_2^c; \dots\rangle_{\text{in/out}}, \quad (3.3.56)$$

where  $n^c$  is the antiparticle of particle type  $n$ , and  $\xi_n$  is yet another phase. If this is true for both 'in' and 'out' states then the  $S$ -matrix satisfies the invariance conditions

$$\begin{aligned} & S_{p'_1 \sigma'_1 n'_1; p'_2 \sigma'_2 n'_2; \dots, p_1 \sigma_1 n_1; p_2 \sigma_2 n_2; \dots} \\ &= \xi_{n'_1}^* \xi_{n'_2}^* \cdots \xi_{n_1} \xi_{n_2} \cdots S_{p'_1 \sigma'_1 n_1^c; p'_2 \sigma'_2 n_2^c; \dots, p_1 \sigma_1 n_1^c; p_2 \sigma_2 n_2^c; \dots}. \end{aligned} \quad (3.3.57)$$

As with other internal symmetries, the  $S$ -matrix will satisfy this condition if the operator  $C_0$  that is defined to act as stated in Eq. (3.3.56) on free-particle states commutes with the interaction  $H_{\text{int}}$  as well as  $H_0$ ; in this case, we take  $C = C_0$ .

The phases  $\xi_n$  are called charge-conjugation parities. Just as for the ordinary parities  $\eta_n$ , the  $\xi_n$  are in general not uniquely defined, because for any operator  $C$  that is defined to satisfy Eq. (3.3.56), we can find another such operator with different  $\xi_n$  by multiplying  $C$  with any internal symmetry phase transformation, such as  $\exp(i\alpha B + i\beta L + i\gamma Q)$ ; the only particles whose charge-conjugation parities are individually measurable are those completely neutral particles like the photon or the neutral pion that carry no conserved quantum numbers and are their own antiparticles. In reactions involving only completely neutral particles, Eq. (3.3.57) tells us that the product of the charge-conjugation parities in the initial and final states must be equal; for instance, as we shall see the photon is required by quantum electrodynamics to have charge-conjugation parity  $\eta_\gamma = -1$ , so the observation of the neutral pion decay  $\pi^0 \rightarrow 2\gamma$  requires that  $\eta_{\pi^0} = +1$ ; it then follows that the process  $\pi^0 \rightarrow 3\gamma$  should be forbidden, as is in fact known to be the case. For these two particles, the charge-conjugation parities are real, either  $+1$  or  $-1$ . Just as for ordinary parity, this will always be the case if all internal phase transformation symmetries are members of continuous groups of phase transformations, because then we can redefine  $C$  by multiplying by the inverse square root of the internal symmetry equal to  $C^2$ , with the result that the new  $C$  satisfies  $C^2 = \mathbf{1}$ .

For general reactions, Eq. (3.3.57) requires that the rate for a process equals the rate for the same process with particles replaced with their corresponding antiparticles. This was not directly contradicted by the 1957 experiments on parity non-conservation (it will be a long time before anyone is able to study the beta decay of anticobalt), but these experiments showed that  $C$  is not conserved in the *theory* of weak interactions as modified by Lee and Yang [12] to take account of parity non-conservation. (As we shall see below, the observed violation of  $TP$  conservation would imply a violation of  $C$  conservation in *any* field theory of weak interactions, not just in the particular theory considered by Lee and Yang.) It is understood today that  $C$  as well as  $P$  is not conserved in the weak interactions responsible for processes like beta decay and the decay of the pion and muon, though both  $C$  and  $P$  are conserved in the strong and electromagnetic interactions.

Although the early experiments on parity non-conservation indicated that neither  $C$  nor  $P$  are conserved in the weak interactions, they left open the possibility that their product  $CP$  is universally conserved. For some years it was expected (though not with complete confidence) that  $CP$  would be found to be generally conserved. This had particularly important consequences for the properties of the neutral  $K$  mesons. In 1954 Gell-Mann and Pais [16] had pointed out that because the  $K^0$  meson is not its own antiparticle (the  $K^0$  carries a non-zero value for the approximately conserved quantity known as strangeness) the particles with definite decay rates would be not  $K^0$  or  $\bar{K}^0$ , but the linear combinations  $K^0 \pm \bar{K}^0$ . This was originally explained in terms of  $C$  conservation, but with  $C$  not conserved in the weak interactions, the argument may be equally well be based on  $CP$  conservation.

If we arbitrarily define the phases in the  $CP$  operator and in the  $K^0$  and  $\bar{K}^0$  states so that

$$CP|K^0\rangle = |\bar{K}^0\rangle$$

and

$$CP|\bar{K}^0\rangle = |K^0\rangle$$

then we can define self-charge-conjugate one-particle states

$$|K_1^0\rangle \equiv \frac{1}{\sqrt{2}} [|K^0\rangle + |\bar{K}^0\rangle]$$

and

$$|K_2^0\rangle \equiv \frac{1}{\sqrt{2}} [|K^0\rangle - |\bar{K}^0\rangle],$$

which have  $CP$  eigenvalues  $+1$  and  $-1$ , respectively. The fastest available decay mode of these particles is into two-pion states, but  $CP$  conservation would allow this only<sup>18</sup> for the  $K_1$ , not the  $K_2$ . The  $K_2^0$  would thus be expected to decay only by slower modes, into three pions or into a pion, muon or electron, and neutrino. Nevertheless, it was found by Fitch and Cronin in 1964 that the long-lived neutral  $K$ -meson does have a small probability for decaying into two pions [17]. The conclusion was that  $CP$  is not exactly conserved in the weak interactions, although it seems more nearly conserved than  $C$  or  $P$  individually.

As we shall see in Chapter 5, there are good reasons to believe that although neither  $C$  nor  $CP$  is strictly conserved, the product  $CPT$  is exactly conserved in all interactions, at least in any quantum field theory. It is  $CPT$  that provides a precise correspondence between particles and antiparticles, and in particular it is the fact that  $CPT$  commutes with the Hamiltonian that tells us that stable particles and antiparticles have exactly the same mass. Because  $CPT$  is antiunitary, it relates the  $S$ -matrix for an arbitrary process to the  $S$ -matrix for the *inverse* process with all spin three-components reversed and particles replaced with antiparticles. However, in cases where the  $S$ -matrix can be divided into a weak term  $S^{(1)}$  that produces a given reaction and a strong term  $S^{(0)}$  that acts in the initial and final states, we can use the same arguments that were used above in studying the implications of  $T$  conservation to show that the rate for any process is equal to the rate for the same process with particles replaced with antiparticles and spin three-components reversed, provided that we sum over sets of initial and final states that are complete with respect to  $S^{(0)}$ . In particular, although the partial rates for decay of the particle into a pair of final states  $\beta_1, \beta_2$  with  $S_{\beta_1\beta_2}^{(0)} \neq 0$  may differ from the partial rates for the decay of the antiparticle into the corresponding final states  $CPT\beta_1$  and  $CPT\beta_2$ , we shall see in Section 3.5 that (without any approximations) the total decay rate of any particle is equal to that of its antiparticle.

We can now understand why the 1957 experiments on parity violation could be interpreted in the context of the existing theory of weak interactions as evidence that  $C$  conservation as well as  $P$  conservation are badly violated but  $CP$  is not. These theories

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<sup>18</sup>The neutral  $K$ -mesons have spin zero, so the two-pion final state has  $\ell = 0$ , and hence  $P = +1$ . Further,  $C = +1$  for two  $\pi^0$ s because the  $\pi^0$  has  $C = +1$ , and also for an  $\ell = 0$   $\pi^+\pi^-$  state because  $C$  interchanges the two pions.

were field theories, and therefore automatically conserved  $CPT$ . Since the experiments showed that  $PT$  conservation but not  $T$  conservation is badly violated in nuclear beta decay, *any* theory that was consistent with these experiments and in which  $CPT$  is conserved would have to also incorporate  $C$  but not  $CP$  non-conservation. Similarly, the observation in 1964 of small violations of  $CP$  conservation in the weak interactions together with the assumed invariance of all interactions under  $CPT$  allowed the immediate inference that the weak interactions also do not exactly conserve  $T$ . This has since been verified [18] by more detailed studies of the  $K^0$ - $\bar{K}^0$  system, but it has so far been impossible to find other direct evidence of the failure of invariance under time-reversal.

### 3.4 Rates and Cross-Sections

The  $S$ -matrix  $S_{\beta\alpha}$  is the probability amplitude for the transition  $\alpha \rightarrow \beta$ , but what does this have to do with the transition rates and cross-sections measured by experimentalists? In particular, (3.3.2) shows that  $S_{\beta\alpha}$  has a factor  $\delta^4(p_\beta - p_\alpha)$ , which ensures the conservation of the total energy and momentum, so what are we to make of the factor  $[\delta^4(p_\beta - p_\alpha)]^2$  in the transition probability  $|S_{\beta\alpha}|^2$ ? The proper way to approach these problems is by studying the way that experiments are actually done, using wave packets to represent particles localized far from each other before a collision, and then following the time-history of these superpositions of multi-particle states. In what follows we will instead give a quick and easy derivation of the main results, actually more a mnemonic than a derivation, with the excuse that (as far as I know) no interesting open problems in physics hinge on getting the fine points right regarding these matters.

We consider our whole system of physical particles to be enclosed in a large box with a macroscopic volume  $V$ . For instance, we can take this box as a cube, but with points on opposite sides identified, so that the single-valuedness of the spatial wave function requires the momenta to be quantized

$$\mathbf{p} = \frac{2\pi}{L}(n_1, n_2, n_3), \quad (3.4.1)$$

where the  $n_i$  are integers, and  $L^3 = V$ . Then all three-dimensional delta-functions become

$$\delta_V^3(\mathbf{p}' - \mathbf{p}) \equiv \frac{1}{(2\pi)^3} \int_V d^3x e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}} = \frac{V}{(2\pi)^3} \delta_{\mathbf{p}',\mathbf{p}}, \quad (3.4.2)$$

where  $\delta_{\mathbf{p}',\mathbf{p}}$  is an ordinary Kronecker delta symbol, equal to one if the subscripts are equal and zero otherwise. The normalization condition (3.1.2) thus implies that the states we have been using have scalar products in a box which are not just sums of products of Kronecker deltas, but also contain a factor  $[V/(2\pi)^3]^N$ , where  $N$  is the number of particles in the state. To calculate transition probabilities we should use states of unit norm, so let us introduce states normalized approximately for our box

$$|\alpha\rangle^{\text{Box}} \equiv [(2\pi)^3/V]^{N_\alpha/2} |\alpha\rangle, \quad (3.4.3)$$

with norm

$${}^{\text{Box}}\langle\beta|\alpha\rangle^{\text{Box}} = \delta_{\beta\alpha}, \quad (3.4.4)$$

where  $\delta_{\beta\alpha}$  is a product of Kronecker deltas, one for each three-momentum, spin, and species label, plus terms with particles permuted. Correspondingly, the  $S$ -matrix may be written

$$S_{\beta\alpha} = [V/(2\pi)^3]^{(N_\beta+N_\alpha)/2} S_{\beta\alpha}^{\text{Box}}, \quad (3.4.5)$$

where  $S_{\beta\alpha}^{\text{Box}}$  is calculated using the states (3.4.3).

Of course, if we just leave our particles in the box forever, then every possible transition will occur again and again. To calculate a meaningful transition probability we also have to put our system in a ‘time box’. We suppose that the interaction is turned on for only a time  $T$ . One immediate consequence is that the energy-conservation delta function is replaced with

$$\delta_T(E_\alpha - E_\beta) = \frac{1}{2\pi} \int_{-T/2}^{T/2} \exp(i(E_\alpha - E_\beta)t) dt. \quad (3.4.6)$$

The probability that a multi-particle system, which is in a state  $\alpha$  before the interaction is turned on, is found in a state  $\beta$  after the interaction is turned off, is

$$P(\alpha \rightarrow \beta) = |S_{\beta\alpha}^{\text{Box}}|^2 = [(2\pi)^3/V]^{N_\beta+N_\alpha} |S_{\beta\alpha}|^2. \quad (3.4.7)$$

This is the probability for a transition into one specific box state  $\beta$ . The number of one-particle box states in a momentum-space volume  $d^3p$  is  $Vd^3p/(2\pi)^3$ , because this is the number of triplets of integers  $n_1, n_2, n_3$  for which the momentum (3.4.1) lies in the momentum-space volume  $d^3p$  around  $\mathbf{p}$ . We shall define the final-state interval  $d\beta$  as a product of  $d^3p$  for each final particle, so the total number of states in this range is

$$d\mathcal{N}_\beta = [V/(2\pi)^3]^{N_\beta} d\beta. \quad (3.4.8)$$

Hence the total probability for the system to wind up in a range  $d\beta$  of final states is

$$dP(\alpha \rightarrow \beta) = P(\alpha \rightarrow \beta) d\mathcal{N}_\beta = [(2\pi)^3/V]^{N_\alpha} |S_{\beta\alpha}|^2 d\beta. \quad (3.4.9)$$

We will restrict our attention throughout this section to final states  $\beta$  that are not only different (however slightly) from the initial state  $\alpha$ , but that also satisfy the more stringent condition, that no subset of the particles in the state  $\beta$  (other than the whole state itself) has precisely the same four-momentum as some corresponding subset of the particles in the state  $\alpha$ . (In the language to be introduced in the next chapter, this means that we are considering only the connected part of the  $S$ -matrix.) For such states, we may define a delta function-free matrix element  $M_{\beta\alpha}$ :

$$S_{\beta\alpha} = -2\pi i \delta_V^3(\mathbf{p}_\beta - \mathbf{p}_\alpha) \delta_T(E_\beta - E_\alpha) M_{\beta\alpha}. \quad (3.4.10)$$

Our introduction of the box allows us to interpret the squares of delta functions in  $|S_{\beta\alpha}|^2$  for  $\beta \neq \alpha$  as

$$\begin{aligned} [\delta_V^3(\mathbf{p}_\beta - \mathbf{p}_\alpha)]^2 &= \delta_V^3(\mathbf{p}_\beta - \mathbf{p}_\alpha) \delta_V^3(0) = \delta_V^3(\mathbf{p}_\beta - \mathbf{p}_\alpha) V/(2\pi)^3, \\ [\delta_T(E_\beta - E_\alpha)]^2 &= \delta_T(E_\beta - E_\alpha) \delta_T(0) = \delta_T(E_\beta - E_\alpha) T/2\pi, \end{aligned}$$

so Eq. (3.4.9) gives a differential transition probability

$$dP(\alpha \rightarrow \beta) = (2\pi)^2 [(2\pi)^3/V]^{N_\alpha-1} (T/2\pi) |M_{\beta\alpha}|^2 \cdot \delta_V^3(\mathbf{p}_\beta - \mathbf{p}_\alpha) \delta_T(E_\beta - E_\alpha) d\beta.$$

If we let  $V$  and  $T$  be very large, the delta function product here may be interpreted as an ordinary four-dimensional delta function  $\delta^4(p_\beta - p_\alpha)$ . In this limit, the transition probability is simply proportional to the time  $T$  during which the interaction is acting, with a coefficient that may be interpreted as a differential transition *rate*:

$$\begin{aligned} d\Gamma(\alpha \rightarrow \beta) &\equiv dP(\alpha \rightarrow \beta)/T \\ &= (2\pi)^{3N_\alpha-2} V^{1-N_\alpha} |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta, \end{aligned} \quad (3.4.11)$$

where now

$$S_{\beta\alpha} = -2\pi i \delta^4(p_\beta - p_\alpha) M_{\beta\alpha}. \quad (3.4.12)$$

This is the master formula which is used to interpret calculations of  $S$ -matrix elements in terms of predictions for actual experiments. We will come back to the interpretation of the factor  $\delta^4(p_\alpha - p_\beta) d\beta$  later in this section.

There are two cases of special importance:

$N_\alpha = 1$  :

Here the volume  $V$  cancels in Eq. (3.4.11), which gives the transition rate for a single-particle state  $\alpha$  to decay into a general multi-particle state  $\beta$  as

$$d\Gamma(\alpha \rightarrow \beta) = 2\pi |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta. \quad (3.4.13)$$

Of course, this makes sense only if the time  $T$  during which the interaction acts is much less than the mean lifetime  $\tau_\alpha$  of the particle  $\alpha$ , so we cannot pass to the limit  $T \rightarrow \infty$  in  $\delta_T(E_\alpha - E_\beta)$ . There is an unremovable width  $\Delta E \simeq 1/T \gtrsim 1/\tau_\alpha$  in this delta function, so Eq. (3.4.13) is only useful if the total decay rate  $1/\tau_\alpha$  is much less than any of the characteristic energies of the process.

$N_\alpha = 2$  :

Here the rate (3.4.11) is proportional to  $1/V$ , or in other words, to the density of either particle at the position of the other one. Experimentalists generally report not the transition rate per density, but the *rate per flux*, also known as the *cross-section*. The flux of either particle at the position of the other particle is defined as the product of the density  $1/V$  and the relative velocity  $u_\alpha$ :

$$\Phi_\alpha = u_\alpha/V. \quad (3.4.14)$$

(A general definition of  $u_\alpha$  is given below; for the moment we will content ourselves with specifying that if either particle is at rest then  $u_\alpha$  is defined as the velocity of the other.) Thus the differential cross-section is

$$d\sigma(\alpha \rightarrow \beta) \equiv d\Gamma(\alpha \rightarrow \beta)/\Phi_\alpha = (2\pi)^4 u_\alpha^{-1} |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta. \quad (3.4.15)$$

Even though the cases  $N_\alpha = 1$  and  $N_\alpha = 2$  are the most important, transition rates for  $N_\alpha \geq 3$  are all measurable in principle, and some of them are very important in chemistry,



astrophysics, etc. (For instance, in one of the main reactions that release energy in the sun, two protons and an electron turn into a deuteron and a neutrino.) Section 3.6 presents an application of the master transition rate formula (3.4.11) for general numbers  $N_\alpha$  of initial particles.

We next take up the question of the Lorentz transformation properties of rates and cross-sections, which will help us to give a more general definition of the relative velocity  $u_\alpha$  in Eq. (3.4.15). The Lorentz transformation rule (3.3.1) for the  $S$ -matrix is complicated by the momentum-dependent matrices associated with each particle's spin. To avoid this complication, consider the absolute-value squared of (3.3.1) (after factoring out the Lorentz-invariant delta function in Eq. (3.4.12)), and sum over all spins. The unitarity of the matrices  $D^{(j)}(W)$  (or their analogs for zero mass) then shows that, apart from the energy factors in (3.3.1), the sum is Lorentz-invariant. That is, the quantity

$$\sum_{\text{spins}} |M_{\beta\alpha}|^2 \prod_{\beta} E \prod_{\alpha} E \equiv R_{\beta\alpha} \quad (3.4.16)$$

is a scalar function of the four-momenta of the particles in states  $\alpha$  and  $\beta$ . (By  $\prod_{\alpha} E$  and  $\prod_{\beta} E$  is meant the product of all the single-particle energies  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$  for the particles in the states  $\alpha$  and  $\beta$ .)

We can now write the spin-summed single-particle decay rate (3.4.13) as

$$\sum_{\text{spins}} d\Gamma(\alpha \rightarrow \beta) = 2\pi E_\alpha^{-1} R_{\beta\alpha} \delta^4(p_\beta - p_\alpha) d\beta / \prod_{\beta} E.$$

The factor  $d\beta / \prod_{\beta} E$  may be recognized as the product of the Lorentz-invariant momentum-space volume elements (2.5.15), so it is Lorentz-invariant. So also are  $R_{\beta\alpha}$  and  $\delta^4(p_\beta - p_\alpha)$ , leaving just the non-invariant factor  $1/E_\alpha$ , where  $E_\alpha$  is the energy of the single initial particle. Our conclusion then is that the decay rate has the same Lorentz transformation property as  $1/E_\alpha$ . This is, of course, just the usual special-relativistic time dilation—the faster the particle, the slower it decays.

Similarly, our result (3.4.15) for the spin-summed cross-section may be written as

$$\sum_{\text{spins}} d\sigma(\alpha \rightarrow \beta) = (2\pi)^4 u_\alpha^{-1} E_1^{-1} E_2^{-1} R_{\beta\alpha} \delta^4(p_\alpha - p_\beta) d\beta / \prod_{\beta} E,$$

where  $E_1$  and  $E_2$  are the energies of the two particles in the initial state  $\alpha$ . It is conventional to define the cross-section to be (when summed over spins) a Lorentz-invariant function of four-momenta. The factors  $R_{\beta\alpha}$ ,  $\delta^4(p_\beta - p_\alpha)$ , and  $d\beta / \prod_{\beta} E$  are already Lorentz-invariant, so this means that we must define the relative velocity  $u_\alpha$  in arbitrary inertial frames so that  $u_\alpha E_1 E_2$  is a scalar. We also mentioned earlier that in the Lorentz frame in which one particle (say, particle 1) is at rest,  $u_\alpha$  is the velocity of the other particle. This uniquely determines  $u_\alpha$  to have the value in general Lorentz frames

$$u_\alpha = \sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} / E_1 E_2, \quad (3.4.17)$$

where  $p_1, p_2$  and  $m_1, m_2$  are the four-momenta and mass of the two particles in the initial state  $\alpha$ .



As a bonus, we note that in the ‘center-of-mass’ frame, where the total three-momentum vanishes, we have

$$p_1 = (E_1, \mathbf{p}), \quad p_2 = (E_2, -\mathbf{p}),$$

and here Eq. (3.4.17) gives

$$u_\alpha = \frac{|\mathbf{p}|(E_1 + E_2)}{E_1 E_2} = \left| \frac{\mathbf{p}_1}{E_1} - \frac{\mathbf{p}_2}{E_2} \right| \quad (3.4.18)$$

as might have been expected for a relative velocity. However, in this frame  $u_\alpha$  is not really a physical velocity; in particular, Eq. (3.4.18) shows that for extremely relativistic particles, it can take values as large as 2.<sup>19</sup>

We now turn to the interpretation of the so-called phase-space factor  $\delta^4(p_\beta - p_\alpha)d\beta$ , which appears in the general formula (3.4.11) for transition rates, and also in Eqs. (3.4.13) and (3.4.15) for decay rates and cross-sections. We here specialize to the case of the ‘center-of-mass’ Lorentz frame, where the total three-momentum of the initial state vanishes

$$\mathbf{p}_\alpha = 0. \quad (3.4.19)$$

(For  $N_\alpha = 1$ , this is just the case of a particle decaying at rest.) If the final state consists of particles with momenta  $\mathbf{p}'_1, \mathbf{p}'_2, \dots$ , then

$$\delta^4(p_\beta - p_\alpha)d\beta = \delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 + \dots)\delta(E'_1 + E'_2 + \dots - E)d^3p'_1 d^3p'_2 \dots, \quad (3.4.20)$$

where  $E \equiv E_\alpha$  is the total energy of the initial state. Any one of the  $\mathbf{p}'_k$  integrals, say over  $\mathbf{p}'_1$ , can be done trivially by just dropping the momentum delta function

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta(E'_1 + E'_2 + \dots - E)d^3p'_2 \dots \quad (3.4.21)$$

with the understanding that wherever  $\mathbf{p}'_1$  appears (as in  $E'_1$ ) it must be replaced with

$$\mathbf{p}'_1 = -\mathbf{p}'_2 - \mathbf{p}'_3 - \dots. \quad (3.4.22)$$

We can similarly use the remaining delta function to eliminate any one of the remaining integrals.

In the simplest case, there are just two particles in the final state. Here (3.4.21) gives

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta(E'_1 + E'_2 - E)d^3p'_2.$$

In more detail, this is

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta\left(\sqrt{|\mathbf{p}'_1|^2 + m_1'^2} + \sqrt{|\mathbf{p}'_1|^2 + m_2'^2} - E\right)|\mathbf{p}'_1|^2 d|\mathbf{p}'_1| d\Omega, \quad (3.4.23)$$

where  $\mathbf{p}'_2 = -\mathbf{p}'_1$  and  $d\Omega \equiv \sin\theta d\theta d\phi$  is the solid-angle differential for  $\mathbf{p}'_1$ . This can be simplified by using the standard formula

$$\delta(f(x)) = \delta(x - x_0)/|f'(x_0)|,$$

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<sup>19</sup>Eq. (3.4.17) makes it obvious that  $E_1 E_2 u_\alpha$  is a scalar. Also, when particle 1 is at rest, we have  $\mathbf{p}_1 = 0$ ,  $E_1 = m_1$ , so  $p_1 \cdot p_2 = -m_1 E_2$ , and so Eq. (3.4.17) gives  $u_\alpha = \sqrt{E_2^2 - m_2^2}/E_2 = |\mathbf{p}_2|/E_2$ , which is just the velocity of particle 2.

where  $f(x)$  is an arbitrary real function with a single simple zero at  $x = x_0$ . In our case, the argument  $E'_1 + E'_2 - E$  of the delta function in Eq. (3.4.23) has a unique zero at  $|\mathbf{p}'_1| = k'$ , where

$$k' = \frac{\sqrt{(E^2 - m_1'^2 - m_2'^2)^2 - 4m_1'^2 m_2'^2}}{2E}, \quad (3.4.24)$$

$$E'_1 = \sqrt{k'^2 + m_1'^2} = \frac{E^2 - m_2'^2 + m_1'^2}{2E}, \quad (3.4.25)$$

$$E'_2 = \sqrt{k'^2 + m_2'^2} = \frac{E^2 - m_1'^2 + m_2'^2}{2E}, \quad (3.4.26)$$

with derivative

$$\left[ \frac{d}{d|\mathbf{p}'_1|} \left( \sqrt{|\mathbf{p}'_1|^2 + m_1'^2} + \sqrt{|\mathbf{p}'_1|^2 + m_2'^2} - E \right) \right]_{|\mathbf{p}'_1|=k'} = \frac{k'}{E'_1} + \frac{k'}{E'_2} = \frac{k'E}{E'_1 E'_2}. \quad (3.4.27)$$

We can thus drop the delta function and the differential  $d|\mathbf{p}'_1|$  in Eq. (3.4.23), by dividing by (3.4.27),

$$\delta^4(p_\beta - p_\alpha) d\beta \longrightarrow \frac{k'E'_1 E'_2}{E} d\Omega \quad (3.4.28)$$

with the understanding that  $k'$ ,  $E'_1$ , and  $E'_2$  are given everywhere by Eqs. (3.4.24)–(3.4.26). In particular, the differential rate (3.4.13) for decay of a one-particle state of zero momentum and energy  $E$  into two particles is

$$\frac{d\Gamma(\alpha \rightarrow \beta)}{d\Omega} = \frac{2\pi k'E'_1 E'_2}{E} |M_{\beta\alpha}|^2 \quad (3.4.29)$$

and the differential cross-section for the two-body scattering process  $12 \rightarrow 1'2'$  is given by Eq. (3.4.15) as

$$\frac{d\sigma(\alpha \rightarrow \beta)}{d\Omega} = \frac{(2\pi)^4 k'E'_1 E'_2}{Eu_\alpha} |M_{\beta\alpha}|^2 = \frac{(2\pi)^4 k'E'_1 E'_2 E_1 E_2}{E^2 k} |M_{\beta\alpha}|^2, \quad (3.4.30)$$

where  $k \equiv |\mathbf{p}_1| = |\mathbf{p}_2|$ .

The above case  $N_\beta = 2$  is particularly simple, but there is one nice result for  $N_\beta = 3$  that is also worth recording. For  $N_\beta = 3$ , Eq. (3.4.21) gives

$$\delta^4(p_\beta - p_\alpha) d\beta \longrightarrow d^3 p'_2 d^3 p'_3 \cdot \delta \left( \sqrt{|\mathbf{p}'_2 + \mathbf{p}'_3|^2 + m_1'^2} + \sqrt{|\mathbf{p}'_2|^2 + m_2'^2} + \sqrt{|\mathbf{p}'_3|^2 + m_3'^2} - E \right).$$

We write the momentum-space volume as

$$d^3 p'_2 d^3 p'_3 = |\mathbf{p}'_2|^2 d|\mathbf{p}'_2| |\mathbf{p}'_3|^2 d|\mathbf{p}'_3| d\Omega_3 d\phi_{23} d\cos\theta_{23},$$

where  $d\Omega_3$  is the differential element of solid angle for  $\mathbf{p}'_3$ , and  $\theta_{23}$  and  $\phi_{23}$  are the polar and azimuthal angles of  $\mathbf{p}'_2$  relative to the  $\mathbf{p}'_3$  direction. The orientation of the plane spanned by  $\mathbf{p}'_2$  and  $\mathbf{p}'_3$  is specified by  $\phi_{23}$  and the direction of  $\mathbf{p}'_3$ , with the remaining angle  $\theta_{23}$  fixed by the energy-conservation condition

$$\sqrt{|\mathbf{p}'_2|^2 + 2|\mathbf{p}'_2||\mathbf{p}'_3|\cos\theta_{23} + |\mathbf{p}'_3|^2 + m_1'^2} + \sqrt{|\mathbf{p}'_2|^2 + m_2'^2} + \sqrt{|\mathbf{p}'_3|^2 + m_3'^2} = E.$$

The derivative of the argument of the delta function with respect to  $\cos \theta_{23}$  is

$$\frac{\partial E'_1}{\partial \cos \theta_{23}} = \frac{|\mathbf{p}'_2||\mathbf{p}'_3|}{E'_1},$$

so we can do the integral over  $\cos \theta_{23}$  by just dropping the delta function and dividing by this derivative

$$\delta^4(p_\beta - p_\alpha) d\beta \longrightarrow |\mathbf{p}'_2| d|\mathbf{p}'_2| |\mathbf{p}'_3| d|\mathbf{p}'_3| E'_1 d\Omega_3 d\phi_{23}.$$

Replacing momenta with energies, this is finally

$$\delta^4(p_\beta - p_\alpha) d\beta \longrightarrow E'_1 E'_2 E'_3 dE'_2 dE'_3 d\Omega_3 d\phi_{23}. \quad (3.4.31)$$

But recall that the quantity (3.4.16), obtained by summing  $|M_{\beta\alpha}|^2$  over spins and multiplying with the product of energies, is a scalar function of four-momenta. If we approximate this scalar as a constant, then Eq. (3.4.31) tells us that for a fixed initial state, the distribution of events plotted in the  $E'_2, E'_3$  plane is uniform. Any departure from a uniform distribution of events in this plot thus provides a useful clue to the dynamics of the decay process, including possible centrifugal barriers or resonant intermediate states. This is known as a Dalitz plot [19], because of its use by Dalitz in 1953 to analyze the decay  $K^+ \rightarrow \pi^+ + \pi^+ + \pi^-$ .

### 3.5 Perturbation Theory

The technique that has historically been most useful in calculating the  $S$ -matrix is perturbation theory, an expansion in powers of the interaction term  $H_{\text{int}}$  in the Hamiltonian  $H = H_0 + H_{\text{int}}$ . Eqs. (3.2.7) and (3.1.18) give the  $S$ -matrix as

$$S_{\beta\alpha} = \delta(\beta - \alpha) - 2\pi i \delta(E_\beta - E_\alpha) T_{\beta\alpha}^+, \quad T_{\beta\alpha}^+ = \langle \beta | H_{\text{int}} | \alpha \rangle_{\text{in}},$$

where  $|\alpha\rangle_{\text{in}}$  satisfies the Lippmann–Schwinger equation (3.1.17):

$$|\alpha\rangle_{\text{in}} = |\alpha\rangle + \int d\gamma \frac{T_{\gamma\alpha}^+ |\gamma\rangle}{E_\alpha - E_\gamma + i\epsilon}.$$

Operating on this equation with  $H_{\text{int}}$  and taking the scalar product with  $|\beta\rangle$  yields an integral equation for  $T^+$

$$T_{\beta\alpha}^+ = (H_{\text{int}})_{\beta\alpha} + \int d\gamma \frac{(H_{\text{int}})_{\beta\gamma} T_{\gamma\alpha}^+}{E_\alpha - E_\gamma + i\epsilon}, \quad (3.5.1)$$

where

$$(H_{\text{int}})_{\beta\alpha} \equiv \langle \beta | H_{\text{int}} | \alpha \rangle. \quad (3.5.2)$$

The perturbation series for  $T_{\beta\alpha}^+$  is obtained by iteration from Eq. (3.5.1)

$$\begin{aligned} T_{\beta\alpha}^+ &= (H_{\text{int}})_{\beta\alpha} + \int d\gamma \frac{(H_{\text{int}})_{\beta\gamma} (H_{\text{int}})_{\gamma\alpha}}{E_\alpha - E_\gamma + i\epsilon} \\ &+ \int d\gamma d\gamma' \frac{(H_{\text{int}})_{\beta\gamma} (H_{\text{int}})_{\gamma\gamma'} (H_{\text{int}})_{\gamma'\alpha}}{(E_\alpha - E_\gamma + i\epsilon)(E_\alpha - E_{\gamma'} + i\epsilon)} + \dots \end{aligned} \quad (3.5.3)$$

The method of calculation based on Eq. (3.5.3), which dominated calculations of the  $S$ -matrix in the 1930s, is today known as *old-fashioned perturbation theory*. Its obvious drawback is that the energy denominators obscure the underlying Lorentz invariance of the  $S$ -matrix. It still has some uses, however, in clarifying the way that singularities of the  $S$ -matrix arise from various intermediate states. For the most part in this book, we will rely on a rewritten version of Eq. (3.5.3), known as *time-dependent perturbation theory*, which has the virtue of making Lorentz invariance much more transparent, while somewhat obscuring the contribution of individual intermediate states.

The easiest way to derive the time-ordered perturbation expansion is to use Eq. (3.2.5), which gives the  $S$ -operator as

$$S = U(\infty, -\infty),$$

where

$$U(\tau, \tau_0) \equiv \exp(iH_0\tau) \exp(-iH(\tau - \tau_0)) \exp(-iH_0\tau_0).$$

Differentiating this formula for  $U(\tau, \tau_0)$  with respect to  $\tau$  gives the differential equation

$$i \frac{d}{d\tau} U(\tau, \tau_0) = H_I(\tau) U(\tau, \tau_0), \quad (3.5.4)$$

where

$$H_I(t) \equiv \exp(iH_0t) H_{\text{int}} \exp(-iH_0t). \quad (3.5.5)$$

(Operators with this sort of time-dependence are said to be defined in the *interaction picture*, to distinguish their time-dependence from the time-dependence

$$O_H(t) = \exp(iHt) O_H \exp(-iHt)$$

required in the Heisenberg picture of quantum mechanics.) Eq. (3.5.4) as well as the initial condition  $U(\tau_0, \tau_0) = \mathbf{1}$  is obviously satisfied by the solution of the integral equation

$$U(\tau, \tau_0) = \mathbf{1} - i \int_{\tau_0}^{\tau} dt H_I(t) U(t, \tau_0). \quad (3.5.6)$$

By iteration of this integral equation, we obtain an expansion for  $U(\tau, \tau_0)$  in powers of  $H_{\text{int}}$

$$\begin{aligned} U(\tau, \tau_0) = & \mathbf{1} - i \int_{\tau_0}^{\tau} dt_1 H_I(t_1) + (-i)^2 \int_{\tau_0}^{\tau} dt_1 \int_{\tau_0}^{t_1} dt_2 H_I(t_1) H_I(t_2) \\ & + (-i)^3 \int_{\tau_0}^{\tau} dt_1 \int_{\tau_0}^{t_1} dt_2 \int_{\tau_0}^{t_2} dt_3 H_I(t_1) H_I(t_2) H_I(t_3) + \cdots \end{aligned} \quad (3.5.7)$$

Setting  $\tau = \infty$  and  $\tau_0 = -\infty$  then gives the perturbation expansion for the  $S$ -operator:

$$\begin{aligned} S = & \mathbf{1} - i \int_{-\infty}^{\infty} dt_1 H_I(t_1) + (-i)^2 \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 H_I(t_1) H_I(t_2) \\ & + (-i)^3 \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 \int_{-\infty}^{t_2} dt_3 H_I(t_1) H_I(t_2) H_I(t_3) + \cdots \end{aligned} \quad (3.5.8)$$

This can also be derived directly from the old-fashioned perturbation expansion (3.5.3), by using the Fourier representation of the energy factors in Eq. (3.5.3):

$$(E_\alpha - E_\gamma + i\epsilon)^{-1} = -i \int_0^\infty d\tau \exp(i(E_\alpha - E_\gamma)\tau) \quad (3.5.9)$$

with the understanding that such integrals are to be evaluated by inserting a convergence factor  $e^{-\epsilon\tau}$  in the integrand, with  $\epsilon \rightarrow 0+$ .

There is a way of rewriting Eq. (3.5.8) that proves very useful in carrying out manifestly Lorentz-invariant calculations. Define the *time-ordered product* of any time-dependent operators as the product with factors arranged so that the one with the latest time-argument is placed leftmost, the next-latest next to the leftmost, and so on. For instance,

$$T\{H_I(t)\} = H_I(t),$$

$$T\{H_I(t_1)H_I(t_2)\} = \theta(t_1 - t_2)H_I(t_1)H_I(t_2) + \theta(t_2 - t_1)H_I(t_2)H_I(t_1),$$

and so on, where  $\theta(\tau)$  is the step function, equal to +1 for  $\tau > 0$  and to zero for  $\tau < 0$ . The time-ordered product of  $n$   $H_I$ s is a sum over all  $n!$  permutations of the  $H_I$ s, each of which gives the same integral over all  $t_1 \cdots t_n$ , so Eq. (3.5.8) may be written

$$S = \mathbf{1} + \sum_{n=1}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 dt_2 \cdots dt_n T\{H_I(t_1) \cdots H_I(t_n)\}. \quad (3.5.10)$$

This is sometimes known as the Dyson series [20]. This series can be summed if the  $H_I(t)$  at different times all commute; the sum is then

$$S = \exp\left(-i \int_{-\infty}^{\infty} dt H_I(t)\right).$$

Of course, this is not usually the case; in general (3.5.10) does not even converge, and is at best an asymptotic expansion in whatever coupling constant factors appear in  $H_{\text{int}}$ . However Eq. (3.5.10) is sometimes written in the general case as

$$S = T \exp\left(-i \int_{-\infty}^{\infty} dt H_I(t)\right)$$

with  $T$  indicating here that the expression is to be evaluated by time-ordering each term in the series expansion for the exponential.

We can now readily find one large class of theories for which the  $S$ -matrix is manifestly Lorentz-invariant. Since the elements of the  $S$ -matrix are the matrix elements of the  $S$ -operator between free-particle states  $|\alpha\rangle$ ,  $|\beta\rangle$ , etc., what we want is that the  $S$ -operator should commute with the operator  $U_0(\Lambda, a)$  that produces Lorentz transformations on these free-particle states. Equivalently, the  $S$ -operator must commute with the generators of  $U_0(\Lambda, a)$ :  $H_0$ ,  $\mathbf{P}_0$ ,  $\mathbf{J}_0$ , and  $\mathbf{K}_0$ . To satisfy this requirement, let's try the hypothesis that  $H_I(t)$  is an integral over three-space

$$H_I(t) = \int d^3x \mathcal{H}_I(\mathbf{x}, t) \quad (3.5.11)$$

with  $\mathcal{H}_I(x)$  a scalar in the sense that

$$U_0(\Lambda, a)\mathcal{H}_I(x)U_0^{-1}(\Lambda, a) = \mathcal{H}_I(\Lambda x + a). \quad (3.5.12)$$

(By equating the coefficients of  $a^0$  for infinitesimal transformations it can be checked that  $\mathcal{H}_I(x)$  has a time-dependence consistent with Eq. (3.5.5).) Then  $S$  may be written as a sum of four-dimensional integrals

$$S = \mathbf{1} + \sum_{n=1}^{\infty} \frac{(-i)^n}{n!} \int d^4x_1 \cdots d^4x_n T\{\mathcal{H}_I(x_1) \cdots \mathcal{H}_I(x_n)\}. \quad (3.5.13)$$

Everything is now manifestly Lorentz invariant, except for the time-ordering of the operator product.

Now, the time-ordering of two spacetime points  $x_1, x_2$  is Lorentz-invariant unless  $x_1 - x_2$  is space-like, i.e., unless  $(x_1 - x_2)^2 > 0$ , so the time-ordering in Eq. (3.5.13) introduces no special Lorentz frame if (though not only if) the  $\mathcal{H}_I(x)$  all commute at space-like or light-like separations:

$$[\mathcal{H}_I(x), \mathcal{H}_I(x')] = 0 \quad \text{for} \quad (x - x')^2 \geq 0. \quad (3.5.14)$$

We can use the results of Section 3.3 to give a formal non-perturbative proof that an interaction (3.5.11) satisfying Eqs. (3.5.12) and (3.5.14) does lead to an  $S$ -matrix with the correct Lorentz transformation properties. For an infinitesimal boost, Eq. (3.5.12) gives

$$i[\mathbf{K}_0, \mathcal{H}_I(\mathbf{x}, t)] = t\nabla\mathcal{H}_I(\mathbf{x}, t) + \mathbf{x}\frac{\partial}{\partial t}\mathcal{H}_I(\mathbf{x}, t), \quad (3.5.15)$$

so integrating over  $\mathbf{x}$  and setting  $t = 0$ ,

$$[\mathbf{K}_0, H_{\text{int}}] = \left[ \mathbf{K}_0, \int d^3x \mathcal{H}_I(\mathbf{x}, 0) \right] = [H_0, \mathbf{W}], \quad (3.5.16)$$

where

$$\mathbf{W} \equiv \int d^3x \mathbf{x} \mathcal{H}_I(\mathbf{x}, 0). \quad (3.5.17)$$

If (as is usually the case) the matrix elements of  $\mathcal{H}_I(\mathbf{x}, 0)$  between eigenstates of  $H_0$  are smooth functions of the energy eigenvalues, then the same is true of  $H_{\text{int}}$ , as is necessary for the validity of scattering theory, and also true of  $\mathbf{W}$ , which is necessary in the proof of Lorentz invariance. The other condition for Lorentz invariance, the commutation relation (3.3.21), is also valid if and only if

$$0 = [\mathbf{W}, H_{\text{int}}] = \int d^3x \int d^3y \mathbf{x} [\mathcal{H}_I(\mathbf{x}, 0), \mathcal{H}_I(\mathbf{y}, 0)]. \quad (3.5.18)$$

This condition would follow from the ‘causality’ condition (3.5.14), but provides a somewhat less restrictive sufficient condition for Lorentz invariance of the  $S$ -matrix.

Theories of this class are not the only ones that are Lorentz invariant, but the most general Lorentz invariant theories are not very different. In particular, there is always a commutation condition something like (3.5.14) that needs to be satisfied. This condition

has no counterpart for non-relativistic systems, for which time-ordering is always Galilean-invariant. *It is this condition that makes the combination of Lorentz invariance and quantum mechanics so restrictive.* <sup>20</sup>

\* \* \*

The methods described so far in this section are useful when the interaction operator  $H_{\text{int}}$  is sufficiently small. There is also a modified version of this approximation, known as the *distorted-wave Born approximation*, that is useful when the interaction contains two terms

$$H_{\text{int}} = H_s + H_w \quad (3.5.19)$$

with  $H_w$  weak but  $H_s$  strong. We can define  $|\alpha; s\rangle_{\text{in/out}}$  as what the ‘in’ and ‘out’ states would be if  $H_s$  were the whole interaction

$$|\alpha; s\rangle_{\text{in/out}} = |\alpha\rangle + (E_\alpha - H_0 \pm i\epsilon)^{-1} H_s |\alpha; s\rangle_{\text{in/out}}. \quad (3.5.20)$$

We can then write (3.1.16) as

$$\begin{aligned} T_{\beta\alpha}^+ &= \langle\beta|H_{\text{int}}|\alpha\rangle_{\text{in}} \\ &= (\text{out}\langle\beta; s| - \text{out}\langle\beta; s|H_s(E_\beta - H_0 + i\epsilon)^{-1}) (H_s + H_w)|\alpha\rangle_{\text{in}} \\ &= \text{out}\langle\beta; s|H_w|\alpha\rangle_{\text{in}} \\ &\quad + \text{out}\langle\beta; s|[H_s - H_s(E_\beta - H_0 + i\epsilon)^{-1}(H_s + H_w)]|\alpha\rangle_{\text{in}}, \end{aligned}$$

and so

$$T_{\beta\alpha}^+ = \text{out}\langle\beta; s|H_w|\alpha\rangle_{\text{in}} + \text{out}\langle\beta; s|H_s|\alpha\rangle. \quad (3.5.21)$$

The second term on the right-hand side is just what  $T_{s,\beta\alpha}^+$  would be in the presence of strong interactions alone

$$T_{s,\beta\alpha}^+ = \langle\beta|H_s|\alpha; s\rangle_{\text{in}} = \text{out}\langle\beta; s|H_s|\alpha\rangle. \quad (3.5.22)$$

(For a proof of Eq. (3.5.22), just drop  $H_w$  everywhere in the derivation of Eq. (3.5.21).) Eq. (3.5.21) is most useful when this second term vanishes: that is, when the process  $\alpha \rightarrow \beta$  cannot be produced by the strong interactions alone. (For example, in nuclear beta decay we need a weak nuclear force to turn neutrons into protons, even though we cannot ignore the presence of the strong nuclear force acting in the initial and final nuclear states.) For such processes, the matrix element (3.5.22) vanishes, so Eq. (3.5.21) reads

$$T_{\beta\alpha}^+ = \text{out}\langle\beta; s|H_w|\alpha\rangle_{\text{in}}. \quad (3.5.23)$$

So far, this is all exact. However, this way of rewriting the  $T$ -matrix becomes worthwhile when  $H_w$  is so weak that we may neglect its effect on the state  $|\alpha\rangle_{\text{in}}$  in Eq. (3.5.23), and hence replace  $|\alpha\rangle_{\text{in}}$  with the state  $|\alpha; s\rangle_{\text{in}}$ , which takes account only of the strong interaction  $H_s$ . In this approximation, Eq. (3.5.23) becomes

$$T_{\beta\alpha}^+ \simeq \text{out}\langle\beta; s|H_w|\alpha; s\rangle_{\text{in}}. \quad (3.5.24)$$

---

<sup>20</sup>We write the condition on  $x$  and  $x'$  here as  $(x - x')^2 \geq 0$  instead of  $(x - x')^2 > 0$ , because as we shall see in Chapter 6, Lorentz invariance can be disturbed by troublesome singularities at  $x = x'$ .

This is valid to first order in  $H_w$ , but to all orders in  $H_s$ . This approximation is ubiquitous in physics; for instance, the  $S$ -matrix element for nuclear beta or gamma decay is calculated using Eq. (3.5.24) with  $H_s$  the strong nuclear interaction and  $H_w$  respectively either the weak nuclear interaction or the electromagnetic interaction, and with  $|\beta; s\rangle_{\text{out}}$  and  $|\alpha; s\rangle_{\text{in}}$  the final and initial nuclear states.

### 3.6 Implications of Unitarity

The unitarity of the  $S$ -matrix imposes an interesting and useful condition relating the amplitude  $M_{\alpha\alpha}$  for forward scattering in an arbitrary multi-particle state  $\alpha$  to the total rate for all reactions in that state. Recall that in the general case, where the state  $\beta$  may or may not be the same as the state  $\alpha$ , the  $S$ -matrix may be written as in (3.3.2):

$$S_{\beta\alpha} = \delta(\beta - \alpha) - 2\pi i \delta^4(p_\beta - p_\alpha) M_{\beta\alpha}.$$

The unitarity condition then gives

$$\begin{aligned} \delta(\gamma - \alpha) &= \int d\beta S_{\beta\gamma}^* S_{\beta\alpha} \\ &= \delta(\gamma - \alpha) - 2\pi i \delta^4(p_\gamma - p_\alpha) M_{\gamma\alpha} \\ &\quad + 2\pi i \delta^4(p_\gamma - p_\alpha) M_{\alpha\gamma}^* + 4\pi^2 \int d\beta \delta^4(p_\beta - p_\gamma) \delta^4(p_\beta - p_\alpha) M_{\beta\gamma}^* M_{\beta\alpha}. \end{aligned}$$

Cancelling the term  $\delta(\gamma - \alpha)$  and a factor  $2\pi \delta^4(p_\gamma - p_\alpha)$ , we find that for  $p_\gamma = p_\alpha$

$$0 = -i M_{\gamma\alpha} + i M_{\alpha\gamma}^* + 2\pi \int d\beta \delta^4(p_\beta - p_\alpha) M_{\beta\gamma}^* M_{\beta\alpha}. \quad (3.6.1)$$

This is most useful in the special case  $\alpha = \gamma$ , where it reads

$$\text{Im } M_{\alpha\alpha} = -\pi \int d\beta \delta^4(p_\beta - p_\alpha) |M_{\beta\alpha}|^2. \quad (3.6.2)$$

Using Eq. (3.4.11), this can be expressed as a formula for the total rate for all reactions produced by an initial state  $\alpha$  in a volume  $V$

$$\begin{aligned} \Gamma_\alpha &\equiv \int d\beta \frac{d\Gamma(\alpha \rightarrow \beta)}{d\beta} \\ &= (2\pi)^{3N_\alpha-2} V^{1-N_\alpha} \int d\beta \delta^4(p_\beta - p_\alpha) |M_{\beta\alpha}|^2 \\ &= -\frac{1}{\pi} (2\pi)^{3N_\alpha-2} V^{1-N_\alpha} \text{Im } M_{\alpha\alpha}. \end{aligned} \quad (3.6.3)$$

In particular, where  $\alpha$  is a two-particle state, this can be written

$$\text{Im } M_{\alpha\alpha} = -u_\alpha \sigma_\alpha / 16\pi^3, \quad (3.6.4)$$

where  $u_\alpha$  is the relative velocity (3.4.17) in state  $\alpha$ , and  $\sigma_\alpha$  is the *total* cross-section in this state, given by (3.4.15) as

$$\sigma_\alpha \equiv \int d\beta d\sigma(\alpha \rightarrow \beta) / d\beta = (2\pi)^4 u_\alpha^{-1} \int d\beta |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha). \quad (3.6.5)$$



This is usually expressed in a slightly different way, in terms of a *scattering amplitude*  $f(\alpha \rightarrow \beta)$ . Eq. (3.4.30) shows that the differential cross-section for *two-body* scattering in the center-of-mass frame is

$$\frac{d\sigma(\alpha \rightarrow \beta)}{d\Omega} = \frac{(2\pi)^4 k' E'_1 E'_2 E_1 E_2}{k E^2} |M_{\beta\alpha}|^2, \quad (3.6.6)$$

where  $k'$  and  $k$  are the magnitudes of the momenta in the initial and final states. We therefore define the scattering amplitude as<sup>21</sup>

$$f(\alpha \rightarrow \beta) \equiv -\frac{4\pi^2}{E} \sqrt{\frac{k' E'_1 E'_2 E_1 E_2}{k}} M_{\beta\alpha}, \quad (3.6.7)$$

so that the differential cross-section is simply

$$\frac{d\sigma(\alpha \rightarrow \beta)}{d\Omega} = |f(\alpha \rightarrow \beta)|^2. \quad (3.6.8)$$

In particular, for *elastic* two-body scattering, we have

$$f(\alpha \rightarrow \beta) \equiv -\frac{4\pi^2 E_1 E_2}{E} M_{\beta\alpha}. \quad (3.6.9)$$

Using (3.4.18) for the relative velocity  $u_\alpha$ , the unitarity prediction (3.6.3) now reads

$$\text{Im } f(\alpha \rightarrow \alpha) = \frac{k}{4\pi} \sigma_\alpha. \quad (3.6.10)$$

This form of the unitarity condition (3.6.3) is known as the *optical theorem* [22].

There is a pretty consequence of the optical theorem that tells much about the pattern of scattering at high energy. The scattering amplitude  $f$  may be expected to be a smooth function of angle, so there must be some solid angle  $\Delta\Omega$  within which  $|f|^2$  has nearly the same value (say, within a factor 2) as in the forward direction. The total cross-section is then bounded by

$$\sigma_\alpha \geq \int |f|^2 d\Omega \geq \frac{1}{2} |f(\alpha \rightarrow \alpha)|^2 \Delta\Omega \geq \frac{1}{2} |\text{Im } f(\alpha \rightarrow \alpha)|^2 \Delta\Omega.$$

Using Eq. (3.6.10) then yields an upper bound on  $\Delta\Omega$

$$\Delta\Omega \leq \frac{32\pi^2}{k^2 \sigma_\alpha}. \quad (3.6.11)$$

As we shall see in the next section, total cross-sections are usually expected to approach constants or grow slowly at high energy, so Eq. (3.6.11) shows that the solid angle around the forward direction within which the differential cross-section is roughly constant shrinks at least as fast as  $1/k^2$  for  $k \rightarrow \infty$ . This increasingly narrow peak in the forward direction at high energies is known as a *diffraction peak*.

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<sup>21</sup>The phase of  $f$  is conventional, and is motivated by the wave mechanical interpretation [21] of  $f$  as the coefficient of the outgoing wave in the solution of the time-independent Schrödinger equation. The normalization of  $f$  used here is slightly unconventional for inelastic scattering; usually  $f$  is defined so that a ratio of final and initial velocities appears in the formula for the differential cross-section.

Returning now to the general case of reactions involving arbitrary numbers of particles, we can use Eq. (3.6.2) together with CPT invariance to say something about the relations for total interaction rates of particles and antiparticles. Because CPT is antiunitary, its conservation does not in general imply any simple relation between a process  $\alpha \rightarrow \beta$  and the process with particles replaced with their antiparticles. Instead, it provides a relation between a process and the *inverse* process involving antiparticles: we can use the same arguments that allowed us to deduce (3.3.46) from time-reversal invariance to show that CPT invariance requires the  $S$ -matrix to satisfy the condition

$$S_{\beta,\alpha} = S_{\mathcal{CPT}\alpha,\mathcal{CPT}\beta}, \quad (3.6.12)$$

where  $\mathcal{CPT}$  indicates that we must reverse all spin  $z$ -components, change all particles into their corresponding antiparticles, and multiply the matrix element by various phase factors for the particles in the initial state and by their complex conjugates for the particles in the final state. Since CPT invariance also requires particles to have the same masses as their corresponding antiparticles, the same relation holds for the coefficient of  $\delta^4(p_\alpha - p_\beta)$  in  $S_{\beta\alpha}$ :

$$M_{\beta,\alpha} = M_{\mathcal{CPT}\alpha,\mathcal{CPT}\beta}. \quad (3.6.13)$$

In particular, when the initial and final states are the same the phase factors all cancel, and Eq. (3.6.13) says that

$$\begin{aligned} M_{p_1\sigma_1n_1; p_2\sigma_2n_2; \dots, p_1\sigma_1n_1; p_2\sigma_2n_2; \dots} \\ = M_{p_1, -\sigma_1, n_1^c; p_2, -\sigma_2, n_2^c; \dots, p_1, -\sigma_1, n_1^c; p_2, -\sigma_2, n_2^c; \dots}, \end{aligned} \quad (3.6.14)$$

where a superscript  $c$  on  $n$  indicates the antiparticle of  $n$ . The generalized optical theorem (3.6.2) then tells us that *the total reaction rate from an initial state consisting of some set of particles is the same as for an initial state consisting of the corresponding antiparticles with spins reversed*:

$$\Gamma_{p_1\sigma_1n_1; p_2\sigma_2n_2; \dots} = \Gamma_{p_1, -\sigma_1, n_1^c; p_2, -\sigma_2, n_2^c; \dots}. \quad (3.6.15)$$

In particular, applying this to one-particle states, we see that the decay rate of any particle equals the decay rate of the antiparticle with reversed spin. Rotational invariance does not allow particle decay rates to depend on the spin  $z$ -component of the decaying particle, so a special case of the general result (3.6.15) is that unstable particles and their corresponding antiparticles have precisely the same lifetimes.

\* \* \*

The same argument that led from the unitarity condition  $S^\dagger S = \mathbf{1}$  to our result (3.6.2) also allows us to use the other unitarity relation  $SS^\dagger = \mathbf{1}$  to derive the result

$$\text{Im } M_{\alpha\alpha} = -\pi \int d\beta \delta^4(p_\beta - p_\alpha) |M_{\alpha\beta}|^2. \quad (3.6.16)$$

Putting this together with Eq. (3.6.2) then yields the reciprocity relation

$$\int d\beta \delta^4(p_\beta - p_\alpha) |M_{\beta\alpha}|^2 = \int d\beta \delta^4(p_\beta - p_\alpha) |M_{\alpha\beta}|^2 \quad (3.6.17)$$

or in other words

$$\int d\beta \frac{d\Gamma(\alpha \rightarrow \beta)}{d\beta} = \int d\beta \frac{d\Gamma(\beta \rightarrow \alpha)}{d\alpha}. \quad (3.6.18)$$

This result can be used in deriving some of the most important results of kinetic theory [23]. If  $P_\alpha d\alpha$  is the probability of finding the system in a volume  $d\alpha$  of the space of multi-particle states  $|\alpha\rangle$ , then the rate of decrease in  $P_\alpha$  due to transitions to all other states is  $P_\alpha \int d\beta d\Gamma(\alpha \rightarrow \beta)/d\beta$ , while the rate of increase of  $P_\alpha$  due to transitions from all other states is  $\int d\beta P_\beta d\Gamma(\beta \rightarrow \alpha)/d\alpha$ ; the rate of change of  $P_\alpha$  is then

$$\frac{dP_\alpha}{dt} = \int d\beta P_\beta \frac{d\Gamma(\beta \rightarrow \alpha)}{d\alpha} - P_\alpha \int d\beta \frac{d\Gamma(\alpha \rightarrow \beta)}{d\beta}. \quad (3.6.19)$$

It follows immediately that  $\int P_\alpha d\alpha$  is time-independent. (Just interchange the labelling of the integration variables in the integral in the second term in Eq. (3.6.19).) On the other hand, the rate of change of the entropy  $-\int d\alpha P_\alpha \ln P_\alpha$  is

$$-\frac{d}{dt} \int d\alpha P_\alpha \ln P_\alpha = - \int d\alpha \int d\beta (\ln P_\alpha + 1) \left[ P_\beta \frac{d\Gamma(\beta \rightarrow \alpha)}{d\alpha} - P_\alpha \frac{d\Gamma(\alpha \rightarrow \beta)}{d\beta} \right].$$

Interchanging the labelling of the integration variables in the second term, this may be written

$$-\frac{d}{dt} \int d\alpha P_\alpha \ln P_\alpha = \int d\alpha \int d\beta P_\beta \ln \left( \frac{P_\beta}{P_\alpha} \right) \frac{d\Gamma(\beta \rightarrow \alpha)}{d\alpha}.$$

Now, for any positive probabilities  $P_\alpha$  and  $P_\beta$ , the function  $P_\beta \ln(P_\beta/P_\alpha)$  satisfies the inequality<sup>22</sup>

$$P_\beta \ln \left( \frac{P_\beta}{P_\alpha} \right) \geq P_\beta - P_\alpha.$$

The rate of change of the entropy is thus bounded by

$$-\frac{d}{dt} \int d\alpha P_\alpha \ln P_\alpha \geq \int d\alpha \int d\beta [P_\beta - P_\alpha] \frac{d\Gamma(\beta \rightarrow \alpha)}{d\alpha}$$

or interchanging variables of integration in the second term

$$-\frac{d}{dt} \int d\alpha P_\alpha \ln P_\alpha \geq \int d\alpha \int d\beta P_\beta \left[ \frac{d\Gamma(\beta \rightarrow \alpha)}{d\alpha} - \frac{d\Gamma(\alpha \rightarrow \beta)}{d\beta} \right].$$

But the unitarity relation (3.6.18) tells us that the integral over  $\alpha$  on the right-hand side vanishes, so we may conclude that the entropy always increases:

$$-\frac{d}{dt} \int d\alpha P_\alpha \ln P_\alpha \geq 0. \quad (3.6.20)$$

This is the ‘Boltzmann  $H$ -theorem.’ This theorem is often derived in statistical mechanics textbooks either by using the Born approximation, for which  $|M_{\beta\alpha}|^2$  is symmetric in  $\alpha$  and  $\beta$  so that  $d\Gamma(\beta \rightarrow \alpha)/d\alpha = d\Gamma(\alpha \rightarrow \beta)/d\beta$ , or by assuming time-reversal invariance,

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<sup>22</sup>The difference between the left- and right-hand sides approaches the positive quantity  $(P_\alpha - P_\beta)^2/2P_\beta$  for  $P_\alpha \rightarrow P_\beta$ , and has a derivative with respect to  $P_\alpha$  that is positive- or negative-definite for all  $P_\alpha > P_\beta$  or  $P_\alpha < P_\beta$ , respectively.

which would tell us that  $|M_{\beta\alpha}|^2$  is unchanged if we interchange  $\alpha$  and  $\beta$  and also reverse all momenta and spins. Of course, neither the Born approximation nor time-reversal invariance are exact, so it is a good thing that the unitarity result (3.6.18) is all we need in order to derive the  $H$ -theorem.

The increase of the entropy stops when the probability  $P_\alpha$  becomes a function only of conserved quantities such as the total energy and charge. In this case the conservation laws require  $d\Gamma(\beta \rightarrow \alpha)/d\alpha$  to vanish unless  $P_\alpha = P_\beta$ , so we can replace  $P_\beta$  with  $P_\alpha$  in the first term of Eq. (3.6.19). Using Eq. (3.6.18) again then shows that in this case,  $P_\alpha$  is time-independent. Here again, we need only the unitarity relation (3.6.18), not the Born approximation or time-reversal invariance.

### 3.7 Partial-Wave Expansions<sup>23</sup>

It is often convenient to work with the  $S$ -matrix in a basis of free-particle states in which all variables are discrete, except for the total momentum and energy. This is possible because the components of the momenta  $\mathbf{p}_1, \dots, \mathbf{p}_n$  in an  $n$ -particle state of definite total momentum  $\mathbf{p}$  and total energy  $E$  form a  $(3n - 4)$ -dimensional compact space; for instance, for  $n = 2$  particles in the center-of-mass frame with  $\mathbf{p} = 0$ , this space is a two-dimensional spherical surface. Any function on such a compact space may be expanded in a series of generalized ‘partial waves’, such as the spherical harmonics that are commonly used in representing functions on the two-sphere. We may thus define a basis for these  $n$ -particle states that apart from the continuous variables  $\mathbf{p}$  and  $E$  is discrete: we label the free-particle states in such a basis as  $|E, \mathbf{p}, N\rangle$ , with the index  $N$  incorporating all spin and species labels as well as whatever indices are used to label the generalized partial waves. These states may conveniently be chosen to be normalized so that their scalar products are:

$$\langle E', \mathbf{p}', N' | E, \mathbf{p}, N \rangle = \delta(E' - E) \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{N'N}. \quad (3.7.1)$$

The  $S$ -operator then has matrix elements in this basis of the form

$$\langle E', \mathbf{p}', N' | S | E, \mathbf{p}, N \rangle = \delta(E' - E) \delta^3(\mathbf{p}' - \mathbf{p}) S_{N'N}(E, \mathbf{p}), \quad (3.7.2)$$

where  $S_{N'N}$  is a unitary matrix. Similarly, the  $T$ -operator, whose free-particle matrix elements  $\langle \beta | T | \alpha \rangle$  are defined to be the quantities  $T_{\beta\alpha}^+$  defined by Eq. (3.1.18), may be expressed in our new basis (in accordance with Eq. (3.4.12)) as

$$\langle E', \mathbf{p}', N' | T | E, \mathbf{p}, N \rangle = \delta^3(\mathbf{p}' - \mathbf{p}) M_{N'N}(E, \mathbf{p}) \quad (3.7.3)$$

and the relation (3.2.7) is now an ordinary matrix equation:

$$S_{N'N}(E, \mathbf{p}) = \delta_{N'N} - 2\pi i M_{N'N}(E, \mathbf{p}). \quad (3.7.4)$$

We shall use this general formalism in the following section; for the present we will concentrate on reactions in which the initial state involves just two particles.

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<sup>23</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

For example, consider a state consisting of two non-identical particles of species  $n_1, n_2$  with non-zero masses  $M_1, M_2$  and arbitrary spins  $s_1, s_2$ . In this case, the states may be labelled by their total momentum  $\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2$ , the energy  $E$ , the species labels  $n_1, n_2$ , the spin  $z$ -components  $\sigma_1, \sigma_2$ , and a pair of integers  $\ell, m$  (with  $|m| \leq \ell$ ) that specify the dependence of the state on the directions of, say,  $\mathbf{p}_1$ . Alternatively, we can form a convenient discrete basis by using Clebsch–Gordan coefficients [9] to combine the two spins to give a total spin  $s$  with  $z$ -component  $\mu$ , and then using Clebsch–Gordan coefficients again to combine this with the orbital angular momentum  $\ell$  with three-component  $m$  to form a total angular momentum  $j$  with three-component  $\sigma$ . This gives a basis of states  $|E, \mathbf{p}, j, \sigma, \ell, s, n\rangle$  (with  $n$  a ‘channel index’ labelling the two particle species  $n_1, n_2$ ), defined by their scalar products with the states of definite individual momenta and spin three-components:

$$\begin{aligned} \langle \mathbf{p}_1, \sigma_1; \mathbf{p}_2, \sigma_2; n' | E, \mathbf{p}, j, \sigma, \ell, s, n \rangle &= \left( \frac{|\mathbf{p}_1| E_1 E_2}{E} \right)^{-1/2} \delta^3(\mathbf{p} - \mathbf{p}_1 - \mathbf{p}_2) \\ &\cdot \delta \left( E - \sqrt{\mathbf{p}_1^2 + M_1^2} - \sqrt{\mathbf{p}_2^2 + M_2^2} \right) \delta_{n'n} \sum_{m, \mu} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{\ell s}(j, \sigma; m, \mu) Y_\ell^m(\hat{\mathbf{p}}_1). \end{aligned} \quad (3.7.5)$$

where  $Y_\ell^m$  are the usual spherical harmonics [24]. The factor  $(|\mathbf{p}_1| E_1 E_2 / E)^{-1/2}$  is inserted so that in the center-of-mass frame these states will be properly normalized:

$$\begin{aligned} \langle E', \mathbf{p}', j', \sigma', \ell', s', n' | E, \mathbf{0}, j, \sigma, \ell, s, n \rangle \\ = \delta^3(\mathbf{p}') \delta(E' - E) \delta_{j'j} \delta_{\sigma'\sigma} \delta_{\ell'\ell} \delta_{s's} \delta_{n'n}. \end{aligned} \quad (3.7.6)$$

For identical particles to avoid double counting we must integrate only over half of the two-particle momentum space, so an extra factor  $\sqrt{2}$  should appear in the scalar product (3.7.6).

In the center-of-mass frame the matrix elements of any momentum-conserving and rotationally-invariant operator  $O$  must take the form:

$$\langle E, \mathbf{p}', j', \sigma', \ell', s', n' | O | E, \mathbf{0}, j, \sigma, \ell, s, n \rangle = \delta^3(\mathbf{p}') O_{\ell' s' n', \ell s n}^j(E) \delta_{j'j} \delta_{\sigma'\sigma}. \quad (3.7.7)$$

(The fact that this is diagonal in  $j$  and  $\sigma$  follows from the commutation of  $O$  with  $\mathbf{J}^2$  and  $J_3$ , and the further fact that the coefficient of  $\delta_{\sigma\sigma'}$  is independent of  $\sigma$  follows from the commutation of  $O$  with  $J_1 \pm iJ_2$ . This is a special case of a general result known as the Wigner–Eckart theorem [25].) Applying this to the operator  $M$  whose matrix elements are the quantities  $M_{\beta\alpha}$ , it follows that the scattering amplitude (3.6.7) in the center-of-mass system takes the form

$$\begin{aligned} f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \rightarrow \mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n') \\ \equiv -4\pi^2 \sqrt{\frac{k' E'_1 E'_2 E_1 E_2}{E^2 k}} M_{\mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n'; \mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n} \\ = -\frac{4\pi^2}{k} \sum_{j\sigma\ell m\ell' m' s\mu s' \mu'} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{\ell s}(j, \sigma; m, \mu) \\ \cdot C_{s'_1 s'_2}(s', \mu'; \sigma'_1, \sigma'_2) C_{\ell' s'}(j, \sigma; m', \mu') Y_{\ell'}^{m'*}(\hat{\mathbf{k}}') Y_\ell^m(\hat{\mathbf{k}}) M_{\ell' s' n', \ell s n}^j(E). \end{aligned} \quad (3.7.8)$$

The differential scattering cross-section is  $|f|^2$ . We will take the direction of the initial momentum  $\mathbf{k}$  to be along the three-direction, in which case

$$Y_\ell^m(\hat{\mathbf{k}}) = \delta_{m0} \sqrt{\frac{2\ell+1}{4\pi}}. \quad (3.7.9)$$

Integrating  $|f|^2$  over the direction of the final momentum  $\mathbf{k}'$  and respectively summing and averaging over final and initial spin three-components gives the total cross-section<sup>24</sup> for transitions from channel  $n$  to  $n'$ :

$$\begin{aligned} \sigma(n \rightarrow n'; E) &= \frac{\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s\ell' s'} (2j+1) \\ &\cdot \left| \delta_{\ell'\ell} \delta_{s's} \delta_{n'n} - S_{\ell' s' n', \ell s n}^j(E) \right|^2. \end{aligned} \quad (3.7.10)$$

Summing (3.7.10) over all two-body channels gives the total cross-section for all elastic or inelastic two-body reactions:

$$\begin{aligned} \sum_{n'} \sigma(n \rightarrow n'; E) &= \frac{\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s} (2j+1) \\ &\cdot [(\mathbf{1} - S^j(E))^\dagger (\mathbf{1} - S^j(E))]_{\ell s n, \ell s n}. \end{aligned} \quad (3.7.11)$$

For comparison, Eqs. (3.7.8), (3.7.9), (3.7.4), and the Clebsch–Gordan sum rules give the spin-averaged forward scattering amplitude as

$$f(n; E) = \frac{i}{2k(2s_1+1)(2s_2+1)} \sum_{j\ell s} (2j+1) [\mathbf{1} - S^j(E)]_{\ell s n, \ell s n}.$$

The optical theorem (3.6.10) then gives the total cross-section as

$$\sigma_{\text{total}}(n; E) = \frac{2\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s} (2j+1) \text{Re}[\mathbf{1} - S^j(E)]_{\ell s n, \ell s n}. \quad (3.7.12)$$

If only two-body channels can be reached from the channel  $n$  at energy  $E$ , then the matrix  $S^j(E)$  (or at least some submatrix that includes channel  $n$ ) is unitary, and thus

$$[(\mathbf{1} - S^j(E))^\dagger (\mathbf{1} - S^j(E))]_{\ell s n, \ell s n} = 2 \text{Re}[\mathbf{1} - S^j(E)]_{\ell s n, \ell s n}, \quad (3.7.13)$$

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<sup>24</sup>In deriving this result, we use standard sum rules [9] for the Clebsch–Gordan coefficients: first

$$\sum_{\sigma_1 \sigma_2} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{s_1 s_2}(\bar{s}, \bar{\mu}; \sigma_1, \sigma_2) = \delta_{s\bar{s}} \delta_{\mu\bar{\mu}},$$

and the same with primes; then

$$\sum_{m\sigma} C_{\ell s}(j, \sigma; m, \bar{\sigma}) C_{\ell' s}(\bar{j}, \bar{\sigma}; m, \bar{\sigma}) = \delta_{j\bar{j}} \delta_{\sigma\bar{\sigma}},$$

and finally

$$\sum_{\sigma\mu} C_{\ell s}(j, \sigma; 0, \mu) C_{\ell' s}(j, \sigma; 0, \mu) = \frac{2j+1}{2\ell+1} \delta_{\ell\ell'}.$$

so (3.7.12) and (3.7.11) are equal. On the other hand, if channels involving three or more particles are open, then the difference of (3.7.12) and (3.7.11) gives the total cross-section for producing extra particles:

$$\sigma_{\text{production}}(n; E) = \frac{\pi}{k^2(2s_1 + 1)(2s_2 + 1)} \sum_{j\ell s} (2j + 1) [\mathbf{1} - S^j(E)^\dagger S^j(E)]_{\ell sn, \ell sn}, \quad (3.7.14)$$

and this must be positive.

The partial wave expansion is particularly useful when applied to processes where the relevant part of the  $S$ -matrix is diagonal. This is the case, for instance, if the initial channel  $n$  contains just two spinless particles, and no other channels are open at this energy, as in  $\pi^+ - \pi^+$  or  $\pi^+ - \pi^0$  scattering at energies below the threshold for producing extra pions (provided one ignores weak and electromagnetic interactions). For a pair of spinless particles we have  $j = \ell$ , and angular momentum conservation keeps the  $S$ -matrix diagonal. It is also possible for the  $S$ -matrix to be diagonal in certain processes involving particles with spin; for instance in pion–nucleon scattering we can have  $j = \ell + \frac{1}{2}$  or  $j = \ell - \frac{1}{2}$ , but for a given  $j$  these two states have opposite parity, so they cannot be connected by non-zero  $S$ -matrix elements. In any case, if for some  $n$  and  $E$  the  $S$ -matrix elements  $S_{N', \ell sn}^j(E, \mathbf{0})$  all vanish unless  $N'$  is the two-body state  $j, \ell, s, n$ , then unitarity requires that

$$S_{\ell' s' n', \ell sn}^j(E) = \exp[2i\delta_{j\ell sn}(E)] \delta_{\ell' \ell} \delta_{s' s} \delta_{n' n}, \quad (3.7.15)$$

where  $\delta_{j\ell sn}(E)$  is a real phase, commonly known as the *phase shift*. This formula is also often used where the two-body part of the  $S$ -matrix is diagonal but channels containing three or more particles are also open; in such cases the phase shift must have a positive imaginary part, to keep (3.7.14) positive. For real phase shifts, the elastic and total cross-sections are then given by Eq. (3.7.10) or Eq. (3.7.12) as:

$$\sigma(n \rightarrow n; E) = \sigma_{\text{total}}(n; E) = \frac{4\pi}{k^2(2s_1 + 1)(2s_2 + 1)} \sum_{j\ell s} (2j + 1) \sin^2 \delta_{j\ell sn}(E). \quad (3.7.16)$$

This familiar result is usually derived in non-relativistic quantum mechanics by studying the coordinate space wave function for a particle in a potential. The derivation given here is offered both to show that the partial wave expansion applies for elastic scattering even at relativistic velocities, and also to emphasize that it depends on no particular dynamical assumptions, only on unitarity and invariance principles.

It is also often useful to introduce phase shifts in dealing with problems where several channels are open, forming a few irreducible representations of some internal symmetry group. The classic example of such an internal symmetry is isotopic spin symmetry, for which the channel index  $n$  includes a specification of the isospins  $T_1, T_2$  of the two particles together with their three-components  $t_1, t_2$ ; the states in channel  $n$  may be expressed as linear combinations of the  $t$ th components of irreducible representations  $T$ , with coefficients given by the familiar Clebsch–Gordan coefficients  $C_{T_1 T_2}(T, t; t_1, t_2)$ . Suppose that for the channels and energy of interest the  $S$ -matrix is diagonal in  $\ell$  and  $s$  as well as  $j, T$ , and  $t$ . Unitarity and isospin symmetry then allow us to write the  $S$ -matrix as

$$S_{\ell' s' T' t', \ell s T t}^j = \exp[2i\delta_{j\ell s T}(E)] \delta_{\ell' \ell} \delta_{s' s} \delta_{T' T} \delta_{t' t}, \quad (3.7.17)$$

with  $\delta_{j\ell sT}(E)$  a real phase shift,  $t$ -independent according to the Wigner–Eckart theorem. The partial cross-sections can again be calculated from Eq. (3.7.10), and the total cross-section is given by Eq. (3.7.12) as

$$\sigma_{\text{total}}(t_1, t_2; E) = \frac{4\pi}{k^2(2s_1 + 1)(2s_2 + 1)} \cdot \sum_{j\ell sTt} (2j + 1) C_{T_1 T_2}(T, t; t_1, t_2)^2 \sin^2 \delta_{j\ell sT}(E). \quad (3.7.18)$$

For instance, in pion–pion scattering we have phase shifts  $\delta_{\ell\ell 0T}(E)$  with  $T = 0$  or  $T = 2$  for each even  $\ell$  and  $T = 1$  for each odd  $\ell$ , while for pion–nucleon scattering we have phase shifts  $\delta_{j,j\pm\frac{1}{2},\frac{1}{2},T}$  with  $T = \frac{1}{2}$  or  $T = \frac{3}{2}$ .

We can gain some useful insight about the threshold behavior of the scattering amplitudes and phase shifts from considerations of analyticity that are nearly independent of any dynamical assumptions. Unless there are special circumstances that would produce singularities in momentum space, we would expect the matrix element  $M_{\mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n'; \mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n}$  to be an analytic function<sup>25</sup> of the three-momenta  $\mathbf{k}$  and  $\mathbf{k}'$  near  $k = 0$  or  $k' = 0$  or (for elastic scattering)  $k = k' = 0$ . Turning to the partial wave expansion (3.7.8) for  $M$ , we note that  $k^\ell Y_\ell^m(\hat{\mathbf{k}})$  is a simple polynomial function of the three-vector  $\mathbf{k}$ , so in order for the matrix element to be an analytic function of the three-momenta  $\mathbf{k}$  and  $\mathbf{k}'$  near  $k = 0$  or  $k' = 0$ , the coefficients  $M_{\ell's'n', \ell sn}^j$  or equivalently  $\delta_{\ell'\ell} \delta_{s's} \delta_{n'n} - S_{\ell's'n', \ell sn}^j$  must go as  $k^{\ell+1/2} k'^{\ell'+1/2}$  when  $k$  and/or  $k'$  go to zero. Hence for small  $k$  and/or  $k'$ , it is only the lowest partial waves in the initial and/or final state that contribute appreciably to the scattering amplitude. We have three possible cases:

#### *Exothermic reactions*

Here  $k'$  approaches a finite value as  $k \rightarrow 0$ , and in this limit  $\delta_{\ell'\ell} \delta_{s's} \delta_{n'n} - S_{\ell's'n', \ell sn}^j$  goes as  $k^{\ell+1/2}$ . The cross-section (3.7.11) in this case goes as  $k^{2\ell-1}$ , where  $\ell$  is here the *lowest* orbital angular momentum that can lead to the reaction. In the most usual case  $\ell = 0$ , so the reaction cross-section goes as  $1/k$ . (This is the case, for instance, in the absorption of slow neutrons by complex nuclei, or for the annihilation of electron–positron pairs into photons at low energy, aside from the higher-order effects of Coulomb forces.) The reaction rate is the cross-section times the flux, which goes like  $k$ , so the rate for an exothermic reaction behaves like a constant for  $k \rightarrow 0$ . However, it is the cross-section rather than the reaction rate that determines the probability of absorption when a beam crosses a given thickness of target material, and the factor  $1/k$  makes this probability very high for slow neutrons in an absorbing material like boron.

#### *Endothermic Reactions*

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<sup>25</sup>For instance, in the Born approximation (3.2.8),  $M$  is proportional to the Fourier transform of the coordinate space matrix elements of the interaction, and hence is analytic at zero momenta as long as these matrix elements fall off sufficiently rapidly at large separations. The chief exception is for scattering involving long-distance forces, such as the Coulomb force.



Here the reaction is forbidden until  $k$  reaches a finite threshold value, where  $k' = 0$ . Just above this threshold  $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S_{\ell's'n',\ell sn}^j$  goes as  $(k')^{\ell'+1/2}$ . The cross-section (3.7.11) in this case goes as  $(k')^{2\ell'+1}$ , where  $\ell'$  is here the *lowest* orbital angular momentum that can be produced at threshold. In the most usual case  $\ell' = 0$ , so the reaction cross-section rises above threshold like  $k'$ , and hence like  $\sqrt{E - E_{\text{threshold}}}$ . (This is the case, for instance, in the associated production of strange particles, or the production of electron-positron pairs in the scattering of photons.)

### Elastic Reactions

Here  $k = k'$ , so  $k$  and  $k'$  go to zero together. (This is the case where  $n' = n$ , or where  $n'$  consists of particles in the same isotopic spin multiplets as are those in  $n$ .) In elastic scattering the partial waves with  $\ell = \ell' = 0$  are always present, so in the limit  $k \rightarrow 0$  the scattering amplitude (3.7.8) becomes a constant:

$$\begin{aligned} f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \rightarrow \mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n') \\ \rightarrow \sum_{s\sigma} C_{s_1 s_2}(s, \sigma; \sigma_1, \sigma_2) C_{s'_1 s'_2}(s, \sigma; \sigma'_1, \sigma'_2) a_s(n \rightarrow n'), \end{aligned} \quad (3.7.19)$$

where  $a$  is a constant, known as the *scattering length*, defined by the limit

$$S_{0sn',0sn}^s \rightarrow \delta_{n'n} + 2ika_s(n \rightarrow n') \quad (k = k' \rightarrow 0). \quad (3.7.20)$$

Summing  $4\pi|f|^2$  over final spins and averaging over initial spins gives the total cross-section for the transition  $n \rightarrow n'$  at  $k = k' = 0$ :

$$\sigma(n \rightarrow n'; k = 0) = \frac{4\pi}{(2s_1 + 1)(2s_2 + 1)} \sum_s (2s + 1) |a_s(n \rightarrow n')|^2. \quad (3.7.21)$$

The classic instance of the use of this formula is in neutron-proton scattering, where there are two scattering lengths, with the spin singlet length  $a_0$  considerably larger than the spin triplet length  $a_1$ .

The partial wave expansion can also be used to make a crude guess about the behavior of cross-sections at high energy. With decreasing wavelengths, we may expect scattering to be described more or less classically: a particle of momentum  $k$  and orbital angular momentum  $\ell$  would have an impact parameter  $\ell/k$ , and will therefore strike a disk of radius  $R$  if  $\ell \leq kR$ . This can be interpreted as a statement about  $S$ -matrix elements:

$$S_{\ell's'n',\ell sn}^j \rightarrow \begin{cases} 0, & \ell \ll kR_n, \\ 1, & \ell \gg kR_n, \end{cases} \quad (3.7.22)$$

where  $R_n$  is some sort of interaction radius for channel  $n$ . For a given  $\ell \gg s$ , there are  $2s + 1$  values of  $j$ , all close enough to  $\ell$  to approximate  $2j + 1 \simeq 2\ell + 1$ , so the sum over  $j$  and  $s$  in Eq. (3.7.12) merely gives a factor of order

$$\sum_{js} (2j + 1) = (2\ell + 1) \sum_s (2s + 1) = (2\ell + 1)(2s_1 + 1)(2s_2 + 1).$$

The total cross-section is then given for  $k \gg 1/R_n$  by Eq. (3.7.12) as

$$\sigma_{\text{total}}(n; E) \longrightarrow \frac{2\pi}{k^2} \sum_{\ell \lesssim kR_n} (2\ell + 1) \longrightarrow 2\pi R_n^2. \quad (3.7.23)$$

In exactly the same way, Eq. (3.7.10) gives the elastic scattering cross-section

$$\sigma(n \rightarrow n; E) \longrightarrow \pi R_n^2. \quad (3.7.24)$$

The difference between Eqs. (3.7.23) and (3.7.24) gives an inelastic cross-section  $\pi R_n^2$ , which is what we would expect for collisions with an opaque disk of radius  $R_n$ . (The somewhat surprising elastic scattering cross-section  $\pi R_n^2$  may be attributed to diffraction by the disk.) On the other hand, if we assume along with Eq. (3.7.22) that  $S_{\ell's'n', \ell sn}^j$  is complex only for impact parameters  $\ell/k$  within a small range of width  $\Delta_n \ll R_n$  around  $\ell/k = R_n$ , then using the inequality  $|\text{Im}(1 - S_{\ell's'n', \ell sn}^j)| \leq 2$ , the same analysis gives a bound on the real part of the forward scattering amplitude

$$|\text{Re } f(n; E)| \leq 2kR_n\Delta_n \ll |\text{Im } f(n; E)|. \quad (3.7.25)$$

The smallness of the real part of the forward scattering amplitude at high energy is confirmed by experiment.

So far we have not said anything about whether the interaction radius  $R_n$  itself may depend on energy. As a very crude guess, we may take  $R_n$  as the distance at which the factor  $\exp(-\mu r)$  in the Yukawa potential (1.2.74) takes a value proportional to some unknown power of  $E$ , in which case  $R_n$  goes as  $\log E$  for  $E \rightarrow \infty$ , and the cross-sections go as  $(\log E)^2$ . As it happens, it has been rigorously shown [26] on the basis of very general assumptions that the total cross-section can grow no faster than  $(\log E)^2$  for  $E \rightarrow \infty$ , and in fact the observed proton-proton total cross-section rises something like  $(\log E)^2$  at high energies, so this rough picture of high-energy scattering does seem to have some correspondence with reality.

### 3.8 Resonances<sup>26</sup>

It often happens that the particles participating in a multi-particle collision can form an intermediate state consisting of a *single* unstable particle  $R$ , that eventually decays into the particles observed as the final state. If the total decay rate of  $R$  is small the cross-section exhibits a rapid variation (usually a peak), known as a *resonance*, at the energy of the intermediate state  $R$ .

We shall see that the behavior of the cross-sections near a resonance is pretty much prescribed by the unitarity condition alone, which is a good thing since there are a number of very different mechanisms that can produce a nearly stable state:

- (a) The simplest possibility is that the Hamiltonian can be decomposed into two terms, a ‘strong’ Hamiltonian  $H_0 + H_s$ , which has the particle  $R$  as an eigenstate, plus a

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<sup>26</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

weak perturbation  $H_w$ , which allows  $R$  to decay into various states, including the initial and final states  $\alpha, \beta$  of our collision process. For instance there is a neutral particle, the  $Z^0$ , with  $j = 1$  and mass 91 GeV, that would be stable in the absence of the electroweak interactions. These interactions allow the  $Z^0$  to decay into electron–positron pairs, muon–antimuon pairs, etc., but with a total decay rate that is much less than the  $Z^0$  mass. In 1989 the  $Z^0$  particle was seen as a resonance<sup>27</sup> in electron–positron collisions at CERN and Stanford, in the reactions  $e^+ + e^- \rightarrow Z^0 \rightarrow e^+ + e^-$ ,  $e^+ + e^- \rightarrow Z^0 \rightarrow \mu^+ + \mu^-$ , etc.

- (b) In some cases a particle is long-lived because there is a potential barrier that nearly prevents its constituents from escaping. The classic example is nuclear alpha decay: it may be energetically possible for a nucleus to emit an alpha particle (a  $\text{He}^4$  nucleus) but the strong electrostatic repulsion between the alpha particle and the nucleus creates a barrier region around the daughter nucleus, which the alpha particle is classically forbidden to enter. The decay then can proceed only by quantum mechanical barrier penetration, and is exponentially slow. Such an unstable state shows up as a resonance in the scattering of the alpha particle on the daughter nucleus. For instance, the lowest-energy state of the  $\text{Be}^8$  nucleus is unstable against decay into two alpha particles, and is seen as a resonance in  $\text{He}^4$ – $\text{He}^4$  scattering. (In addition to Coulomb barriers, there are also centrifugal barriers that help lengthen the life of alpha-, beta-, and gamma-unstable nuclei of high spin.)
- (c) It is possible for complicated systems to be nearly unstable for statistical reasons, without the presence of any potential barriers or weak interactions. For instance, an excited state of a heavy nucleus may be able to decay only if, through a statistical fluctuation, a large part of its energy is concentrated on a single neutron. This state will then show up as a resonance in the scattering of a neutron on the daughter nucleus.

These mechanisms for producing long-lived states are so different that it is truly fortunate that most of the properties of resonances follow from unitarity alone, without regard to the dynamical mechanism that produces the resonance.

First, let's consider the energy dependence of the matrix element for a reaction near a resonance. A wave packet  $\int d\alpha g(\alpha)|\alpha\rangle_{\text{in}} \exp(-iE_\alpha t)$  of 'in' states has a time-dependence given by (3.1.19)

$$\int d\alpha g(\alpha)|\alpha\rangle_{\text{in}} e^{-iE_\alpha t} = \int d\alpha g(\alpha)|\alpha\rangle_0 e^{-iE_\alpha t} + \int d\beta |\beta\rangle_0 \int d\alpha \frac{e^{-iE_\alpha t} g(\alpha) T_{\beta\alpha}^+}{E_\alpha - E_\beta + i\epsilon}.$$

As mentioned in Section 3.1, a pole in the function  $T_{\beta\alpha}^+$  in the lower-half complex  $E_\alpha$  plane would make a contribution to the second term that decays exponentially as  $t \rightarrow \infty$ . Specifically, a pole at  $E_\alpha = E_R - i\Gamma/2$  yields a term in the *amplitude* that behaves like

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<sup>27</sup>Incidentally, this example shows that a resonant state only needs to decay *relatively* slowly; the  $Z^0$  lifetime is  $2.6 \cdot 10^{-25}$  seconds, which is not long enough for a  $Z^0$  travelling near the speed of light to cross an atomic nucleus. What is important is that the decay rate is 36 times smaller than the rate  $M_Z/\hbar$  of oscillation of the  $Z^0$  wave function in its rest frame.

$\exp(-iE_R t - \Gamma t/2)$ , so it corresponds to a state whose *probability* decays like  $\exp(-\Gamma t)$ . We conclude then that a long-lived state of energy  $E_R$  with a slow decay rate  $\Gamma$  produces a term in the scattering amplitude that varies as

$$T_{\beta\alpha}^+ \sim (E_\alpha - E_R + i\Gamma/2)^{-1} + \text{constant}. \quad (3.8.1)$$

To go further, it will be convenient to adopt as a basis the orthonormal discrete multi-particle states  $|\mathbf{p}, E, N\rangle$  discussed in the previous section;  $\mathbf{p}$  and  $E$  are the total momentum and energy, and  $N$  is an index that takes only discrete (though infinitely many) values. In this basis, the  $S$ -matrix may be written

$$S_{\mathbf{p}'E'N', \mathbf{p}EN} = \delta^3(\mathbf{p}' - \mathbf{p})\delta(E' - E)S_{N'N}(\mathbf{p}, E). \quad (3.8.2)$$

Near a resonance, we expect the center-of-mass frame amplitude  $S(\mathbf{0}, E) \equiv \mathcal{S}(E)$  to have the form

$$\mathcal{S}_{N'N}(E) \equiv S_{N'N}(\mathbf{0}, E) = \mathcal{S}_{0N'N} + \frac{\mathcal{R}_{N'N}}{E - E_R + i\Gamma/2}, \quad (3.8.3)$$

where  $\mathcal{S}_0$  and  $\mathcal{R}$  are approximately constant at least over the relatively small range of energies  $|E - E_R| \lesssim \Gamma$ .

In this basis, the unitarity of the  $S$ -matrix is an ordinary matrix equation

$$\mathcal{S}(E)^\dagger \mathcal{S}(E) = \mathbf{1}. \quad (3.8.4)$$

Applied to Eq. (3.8.3), this tells us that the non-resonant background  $S$ -matrix is unitary

$$\mathcal{S}_0^\dagger \mathcal{S}_0 = \mathbf{1}, \quad (3.8.5)$$

and also that the residue matrix  $\mathcal{R}$  satisfies the two conditions

$$\mathcal{S}_0^\dagger \mathcal{R} - \mathcal{R}^\dagger \mathcal{S}_0 = 0, \quad (3.8.6)$$

$$-\frac{i}{2}\Gamma \mathcal{S}_0^\dagger \mathcal{R} + \frac{i}{2}\Gamma \mathcal{R}^\dagger \mathcal{S}_0 + \mathcal{R}^\dagger \mathcal{R} = 0. \quad (3.8.7)$$

These conditions can be put in a more transparent form by setting

$$\mathcal{R} \equiv -i\Gamma \mathcal{A} \mathcal{S}_0. \quad (3.8.8)$$

The unitarity conditions on the matrix  $\mathcal{A}$  are then simply

$$\mathcal{A}^\dagger = \mathcal{A}, \quad \mathcal{A}^2 = \mathcal{A}. \quad (3.8.9)$$

Any such Hermitian idempotent matrix is called a *projection matrix*. Such matrices can always be expressed as a sum of dyads of orthonormal vectors  $u^{(r)}$ :

$$\mathcal{A}_{N'N} = \sum_r u_{N'}^{(r)} u_N^{(r)*}, \quad \sum_N u_N^{(r)*} u_N^{(s)} = \delta_{rs}. \quad (3.8.10)$$

The discrete part of the  $S$ -matrix is then

$$\mathcal{S}_{N'N}(E) = \sum_{N''} \left[ \delta_{N'N''} - i \frac{\Gamma}{E - E_R + i\Gamma/2} \sum_r u_{N'}^{(r)} u_{N''}^{(r)*} \right] \mathcal{S}_{0N''N}. \quad (3.8.11)$$

Each term in the sum over  $r$  can be thought of as arising from a different resonant state, all these states having the same values for  $E_R$  and  $\Gamma$ .

What has this to do with rates and cross-sections? For simplicity let's now ignore the non-resonant background scattering, setting  $\mathcal{S}_{0N'N}$  equal to  $\delta_{N'N}$ ; we will come back to the more general case a little later. Then for the two-body discrete center-of-mass states described in the previous section, Eq. (3.8.11) reads:

$$\begin{aligned} \mathcal{S}_{j'\sigma'\ell's'n',j\sigma\ell sn}(E) &= \delta_{j'j}\delta_{\sigma'\sigma}\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} \\ &\quad - i \frac{\Gamma}{E - E_R + i\Gamma/2} \sum_r u_{j'\sigma'\ell's'n'}^{(r)} u_{j\sigma\ell sn}^{(r)*}. \end{aligned} \quad (3.8.12)$$

In all cases the label  $r$  will include an index  $\sigma_R$  giving the  $z$ -component of the total angular momentum of the resonant state; for a resonant state of total angular momentum  $j_R$ ,  $\sigma_R$  takes  $2j_R + 1$  values. If there is no other degeneracy, then  $r$  just labels the value of  $\sigma_R$ , and

$$u_{j\sigma\ell sn}^{(\sigma_R)} = \delta_{jj_R}\delta_{\sigma\sigma_R} u_{\ell sn}, \quad (3.8.13)$$

where  $u_{\ell sn}$  are a set of complex amplitudes that (because of the Wigner–Eckart theorem) are independent of  $\sigma$ . Now Eq. (3.8.12) gives the amplitude  $S^j$  defined by Eq. (3.7.7) as

$$S_{\ell's'n',\ell sn}^j(E) = \delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - i\delta_{jj_R} \frac{\Gamma}{E - E_R + i\Gamma/2} u_{\ell's'n'} u_{\ell sn}^*. \quad (3.8.14)$$

Also, Eq. (3.8.10) now reads

$$\sum_{\ell sn} |u_{\ell sn}|^2 + \dots = 1 \quad (3.8.15)$$

with the dots representing the positive contribution of any states containing three or more particles. As we shall see, the quantities  $|u_{\ell sn}|^2$  have the interpretation of branching ratios for the decay of the resonant state into the various accessible two-body states.

Eq. (3.7.12) now gives the total cross-section for all reactions in channel  $n$ :

$$\sigma_{\text{total}}(n; E) = \frac{\pi(2j_R + 1)}{k^2(2s_1 + 1)(2s_2 + 1)} \frac{\Gamma\Gamma_n}{(E - E_R)^2 + \Gamma^2/4}, \quad (3.8.16)$$

where

$$\Gamma_n \equiv \Gamma \sum_{\ell s} |u_{\ell sn}|^2. \quad (3.8.17)$$

This is a version of the celebrated Breit–Wigner single-level formula [27]. We can also use these results to calculate the cross-section for resonant scattering from an initial two-body channel  $n$  to a final two-body channel  $n'$ . Using Eq. (3.8.14) in Eq. (3.7.10) gives

$$\sigma(n \rightarrow n'; E) = \frac{\pi(2j_R + 1)}{k^2(2s_1 + 1)(2s_2 + 1)} \frac{\Gamma_n \Gamma_{n'}}{(E - E_R)^2 + \Gamma^2/4}. \quad (3.8.18)$$

This shows that the probabilities that the resonant state will decay into any one of the final two-body channels  $n'$  are proportional to the  $\Gamma_{n'}$ . According to Eq. (3.8.15), the sum of the  $\Gamma_n$  (including contributions from final states containing three or more particles) is just

equal to the total decay rate  $\Gamma$ , so we can conclude that  $\Gamma_n$  is just the rate for the decay of the resonant state into channel  $n$ .

We see in Eqs. (3.8.16) and (3.8.18) the characteristic resonant peak at energy  $E_R$ , with a width (the full width at half maximum) equal to the decay rate  $\Gamma$ . (The individual  $\Gamma_n$  are often called partial widths.) Since  $\Gamma_n \leq \Gamma$ , the total cross-section at the peak of the resonance is roughly bounded by one square wavelength,  $(2\pi/k)^2$ . This rule, that cross-sections at a single resonance are roughly bounded by a square wavelength, is universally applicable even in classical physics (where energy conservation plays the role played here by unitarity), as for instance in the resonant interaction of sound waves with bubbles in the sea, or gravitational waves with gravitational wave antennae. (In the latter case, the branching ratio for oscillations in any laboratory mass to lose their energy through gravitational radiation is tiny, so the cross-section even at a resonance peak is vastly less than a square wavelength [28].)

Incidentally, it often happens that a resonance is detected, but energy measurements are insufficiently precise to resolve its width. In this case, what is measured experimentally is the integral of the cross-section over the resonant peak. For the total cross-section (3.8.14), this is

$$\int \sigma_{\text{total}}(n; E) dE = \frac{2\pi^2(2j_R + 1)\Gamma_n}{k_R^2(2s_1 + 1)(2s_2 + 1)}. \quad (3.8.19)$$

Such experiments can reveal only the partial width for decay of the resonant state into the initial particles, not the total width or branching ratios.

This formalism can also be applied when the resonant states with a given spin  $z$ -component form a multiplet related by some symmetry group. For instance, to the extent that isospin symmetry is respected, for a resonance of total isospin  $T_R$  the index  $r$  labelling resonant states includes a specification not only of the angular-momentum  $z$ -component  $\sigma_R$ , but also of an isospin three-component  $t_R$ , taking values  $-T_R, -T_R + 1, \dots, T_R$ . In this case there is no change in the above results for the total and partial cross-sections, because each two-body channel  $n$  has definite values  $t_1, t_2$  of the isospin  $z$ -components of the two particles, and hence can only couple to the resonant state with the single value  $t_1 + t_2$  for  $t_R$ . The partial widths  $\Gamma_n$  here depend on  $t_1$  and  $t_2$  only through factors  $C_{T_1 T_2}(T_R, t_R; t_1, t_2)^2$ .

The presence of a resonance shows itself in a characteristic behavior of phase shifts near the resonance. Returning to the general formula (3.8.11) (but still taking  $\mathcal{S}_0 = \mathbf{1}$ ), we see from Eq. (3.8.10) that for each individual resonant state  $r$ , there is an eigenvector  $u_N^{(r)}$  of  $\mathcal{S}_{N'N}(E)$  with eigenvalue

$$\exp[2i\delta^{(r)}(E)] = 1 - i \frac{\Gamma}{E - E_R + i\Gamma/2}$$

or, in other words,

$$\tan \delta^{(r)}(E) = - \frac{\Gamma/2}{E - E_R}. \quad (3.8.20)$$

We see that over an energy range of order  $\Gamma$  centered around the resonant energy, the ‘eigenphase’  $\delta^{(r)}(E)$  jumps from a value  $\nu\pi$  (with  $\nu$  a positive or negative integer) below the resonance to  $(\nu + 1)\pi$  above it. However, in order to use this result to say something about

reaction rates, we need to know the eigenvectors  $u_N^{(r)}$ , which, in general, have components with arbitrary numbers of particles with various momenta, spin, and species.

These results are much more useful in those special cases when the particles in a particular channel  $N$  are forbidden (usually by conservation laws) from making a transition to any other channel. With this assumption, it is not difficult to include the effects of a non-resonant background scattering matrix  $\mathcal{S}_0$  in the general result (3.8.11). In order for  $\mathcal{S}_{N'N}$  to vanish for some particular  $N$  and all  $N' \neq N$ , it is necessary that the same is true of  $\mathcal{S}_{0N'N}$ , and also true of  $u_{N'}^{(r)}$  for any  $r$  for which  $u_N^{(r)} \neq 0$ . The unitarity requirement (3.8.5) then requires that for this  $N$

$$\mathcal{S}_{0N'N} = \exp(2i\delta_{0N})\delta_{N'N}$$

and Eq. (3.8.10) requires that

$$u_N^{(r)*} u_N^{(s)} = \delta_{rs},$$

so that there can be only one term  $r$  in Eq. (3.8.11) for which  $u_N^{(r)} \neq 0$ . In this case, Eq. (3.8.11) gives

$$\mathcal{S}_{N'N}(E) = \delta_{N'N} \left[ 1 - \frac{i\Gamma}{E - E_R + i\Gamma/2} \right] \exp(2i\delta_{0N}) \equiv \delta_{N'N} \exp[2i\delta_N(E)]$$

with total phase shift

$$\delta_N(E) = \delta_{0N} - \arctan\left(\frac{\Gamma/2}{E - E_R}\right). \quad (3.8.21)$$

We see that over an energy range of order  $\Gamma$  centered around the resonant energy  $E_R$ , the phase shift  $\delta_N(E)$  jumps from a value  $\delta_{0N}$  below the

resonance to  $\delta_{0N} + \pi$  above it. For instance, as we saw in the previous section, these assumptions are satisfied in various two-body reactions such as pion–pion and pion–nucleon scattering at energies below the threshold for producing extra pions, with  $N$  incorporating the total and orbital angular-momenta  $j, \ell$  (with  $j = \ell$  for pion–pion scattering) and the total angular-momentum  $z$ -component  $\sigma$ , as well as the total isospin  $T$  and its three-component  $t$ . The Wigner–Eckart theorem allows the phase shifts to depend only on  $j, \ell$ , and  $T$ , not on  $t$  or  $\sigma$ . There are famous resonances in these channels: in pion–pion scattering there is a resonance at 770 MeV called  $\rho$  with  $j = \ell = 1$ ,  $T = 1$ , and  $\Gamma = 150$  MeV; in pion–nucleon scattering there is a resonance at 1232 MeV called  $\Delta$  with  $j = \frac{3}{2}$ ,  $\ell = 1$ ,  $T = \frac{3}{2}$  and  $\Gamma = 110$  to 120 MeV.

Inspection of Eq. (3.7.12) or Eq. (3.7.18) shows that the total cross-section reaches a peak when the resonant phase shift passes through  $\pi/2$  (or odd-integer multiples of  $\pi/2$ .) The non-resonant phase shifts are typically rather small, so as we saw earlier,  $\sigma_{\text{total}}$  will exhibit a sharp peak when the phase shift  $\delta_\ell$  goes through  $\pi/2$ , at an energy close to  $E_R$ . However, it sometimes happens that the non-resonant background phase shift  $\delta_{0N}$  is near  $\pi/2$ , in which case the cross-section will exhibit a sharp *dip* as the phase shift rises through  $\pi$  near  $E_R$ , due to destructive interference between the resonance and the non-resonant background amplitude. Such dips were first observed by Ramsauer and Townsend [29] in 1922, in the scattering of electrons by noble gas atoms.

## Weinberg Exercises

**W.3.1.** Consider a theory with a separable interaction; that is,

$$\langle \beta | H_{\text{int}} | \alpha \rangle = g u_\beta u_\alpha^*,$$

where  $g$  is a real coupling constant, and  $u_\alpha$  is a set of complex quantities with

$$\sum_\alpha |u_\alpha|^2 = 1.$$

Use the Lippmann–Schwinger equation (3.1.16) to find explicit solutions for the ‘in’ and ‘out’ states and the  $S$ -matrix.

**W.3.2.** Suppose that a resonance of spin one is discovered in  $e^+e^-$  scattering at a total energy of 150 GeV and with a cross-section (in the center-of-mass frame, averaged over initial spins, and summed over final spins) for elastic  $e^+e^-$  scattering at the peak of the resonance equal to  $10^{-34} \text{ cm}^2$ . What is the branching ratio for the decay of the resonant state  $R$  by the mode  $R \rightarrow e^- + e^+$ ? What is the total cross-section for  $e^+e^-$  scattering at the peak of the resonance? (In answering both questions, ignore the non-resonant background scattering.)

**W.3.3.** Express the differential cross-section for two-body scattering in the *laboratory* frame, in which one of the two particles is initially at rest, in terms of kinematic variables and the matrix element  $M_{\beta\alpha}$ . (Derive the result directly, without using the results derived in this chapter for the differential cross-section in the center-of-mass frame.)

**W.3.4.** Derive the perturbation expansion (3.5.8) directly from the expansion (3.5.3) of old-fashioned perturbation theory.

**W.3.5.** We can define ‘standing wave’ states  $|\alpha\rangle^0$  by a modified version of the Lippmann–Schwinger equation

$$|\alpha\rangle^0 = |\alpha\rangle_0 + \text{P. V.} \frac{1}{E_\alpha - H_0} H_{\text{int}} |\alpha\rangle^0.$$

Show that the matrix  $K_{\beta\alpha} = \pi \delta(E_\beta - E_\alpha) \langle \beta | H_{\text{int}} | \alpha \rangle^0$  is Hermitian. Show how to express the  $S$ -matrix in terms of the  $K$ -matrix.

**W.3.6.** Express the differential cross-section for elastic  $\pi^+$ -proton and  $\pi^-$ -proton scattering in terms of the phase shifts for states of definite total angular momentum, parity, and isospin.

**W.3.7.** Show that the states  $|E, \mathbf{p}, j, \sigma, \ell, s, n\rangle$  defined by Eq. (3.7.5) are correctly normalized to have the scalar products (3.7.6).



### Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

**W.3.1.** Introduce the boundary-value functions

$$F^\pm(E) = \sum_\gamma \frac{|u_\gamma|^2}{E - E_\gamma \pm i0}.$$

The Lippmann–Schwinger equation and the separable form of the interaction give

$$\langle \beta | \alpha^\pm \rangle = \delta_{\beta\alpha} \frac{g u_\beta}{E_\alpha - E_\beta \pm i0} \sum_\gamma u_\gamma^* \langle \gamma | \alpha^\pm \rangle.$$

Multiplying by  $u_\beta^*$ , summing over  $\beta$ , and solving the resulting scalar equation yields

$$\sum_\gamma u_\gamma^* \langle \gamma | \alpha^\pm \rangle = \frac{u_\alpha^*}{1 - g F^\pm(E_\alpha)}.$$

Thus the explicit scattering states are

$$\begin{aligned} |\alpha\rangle_{\text{in}} &= |\alpha\rangle_0 + \frac{g u_\alpha^*}{1 - g F^+(E_\alpha)} \sum_\beta \frac{u_\beta |\beta\rangle_0}{E_\alpha - E_\beta + i0}, \\ |\alpha\rangle_{\text{out}} &= |\alpha\rangle_0 + \frac{g u_\alpha^*}{1 - g F^-(E_\alpha)} \sum_\beta \frac{u_\beta |\beta\rangle_0}{E_\alpha - E_\beta - i0}. \end{aligned}$$

In particular,

$$T_{\beta\alpha}^+ = \langle \beta | H_{\text{int}} | \alpha \rangle_{\text{in}} = \frac{g u_\beta u_\alpha^*}{1 - g F^+(E_\alpha)}.$$

With the convention of Eq. (3.1.14), the answer for the  $S$ -matrix is therefore

$${}_{\text{out}}\langle \beta | | \alpha \rangle_{\text{in}} = \delta(\beta - \alpha) - 2\pi i \delta(E_\beta - E_\alpha) \frac{g u_\beta u_\alpha^*}{1 - g F^+(E_\alpha)}.$$

Since  $F^-(E) = F^+(E)^*$  for real  $E$  and real  $g$ , this expression also displays the relation between the incoming and outgoing boundary values required by unitarity.

**W.3.2.** At the peak of an isolated resonance, Eq. (3.8.14) gives, for unpolarized  $e^+e^-$  beams and  $j_R = 1$ ,

$$\sigma_{\text{el}}^{\text{peak}} = \frac{4\pi}{k^2} \frac{2j_R + 1}{(2s_{e^-} + 1)(2s_{e^+} + 1)} B_e^2 = \frac{3\pi}{k^2} B_e^2,$$

where  $B_e = \Gamma(R \rightarrow e^+e^-)/\Gamma$  and  $k = E_{\text{cm}}/2 = 75 \text{ GeV}$ . Using

$$1 \text{ GeV}^{-2} = 0.389379 \times 10^{-27} \text{ cm}^2$$

gives

$$\frac{3\pi}{k^2} = 1.6755 \times 10^{-3} \text{ GeV}^{-2} = 6.523 \times 10^{-31} \text{ cm}^2.$$

Consequently

$$B_e = \left( \frac{10^{-34}}{6.523 \times 10^{-31}} \right)^{1/2} = 1.24 \times 10^{-2}.$$

The inclusive peak cross section contains only one power of the entrance branching fraction:

$$\sigma_{\text{tot}}^{\text{peak}} = \frac{3\pi}{k^2} B_e = 8.08 \times 10^{-33} \text{ cm}^2.$$

Thus the electronic branching ratio is about 1.24%, and the total peak cross section is about  $8.1 \times 10^{-33} \text{ cm}^2$ .

**W.3.3.** Let particle 2 be initially at rest, and choose the incident momentum along  $z$ :

$$p_1 = (E_1, 0, 0, p), \quad p_2 = (m_2, \mathbf{0}), \quad p_3 = (E_3, q\hat{\mathbf{n}}), \quad \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} = \cos \theta.$$

Momentum conservation fixes  $\mathbf{p}_4 = \mathbf{p} - \mathbf{q}$ , so

$$E_3 = \sqrt{q^2 + m_3^2}, \quad E_4 = \sqrt{p^2 + q^2 - 2pq \cos \theta + m_4^2}.$$

The positive physical root  $q = q(\theta)$  is determined by

$$E_1 + m_2 = E_3 + E_4.$$

In the laboratory frame the relative speed is  $u_\alpha = p/E_1$ . After using the momentum delta function in Eq. (3.4.15), the remaining energy delta function has the Jacobian

$$\left| \frac{d(E_3 + E_4)}{dq} \right| = \left| \frac{q}{E_3} + \frac{q - p \cos \theta}{E_4} \right|.$$

It follows directly that

$$\boxed{\frac{d\sigma}{d\Omega_{\text{lab}}} = \frac{(2\pi)^4 E_1}{p} |M_{\beta\alpha}|^2 \frac{q^2}{|q/E_3 + (q - p \cos \theta)/E_4|}}$$

or, equivalently,

$$\frac{d\sigma}{d\Omega_{\text{lab}}} = \frac{(2\pi)^4 E_1}{p} |M_{\beta\alpha}|^2 \frac{q^2 E_3 E_4}{|q(E_3 + E_4) - p E_3 \cos \theta|}.$$

The usual average over unresolved initial spins and sum over final spins is understood. If the final particles are identical, the result is integrated over only one copy of phase space (or multiplied by  $1/2!$  when the full solid angle is used).

**W.3.4.** For every denominator in Eq. (3.5.3), use

$$\frac{1}{E_\alpha - E_\gamma + i0} = -i \int_0^\infty d\tau e^{i(E_\alpha - E_\gamma)\tau},$$

and represent the energy-conserving factor in the  $S$ -matrix by

$$2\pi\delta(E_\beta - E_\alpha) = \int_{-\infty}^\infty dt e^{i(E_\beta - E_\alpha)t}.$$

At second order, setting  $t_1 = t$  and  $t_2 = t - \tau$  gives

$$\begin{aligned} & -2\pi i \delta(E_\beta - E_\alpha) \sum_\gamma \frac{(H_{\text{int}})_{\beta\gamma} (H_{\text{int}})_{\gamma\alpha}}{E_\alpha - E_\gamma + i0} \\ & = (-i)^2 \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 \langle \beta | H_I(t_1) H_I(t_2) | \alpha \rangle. \end{aligned}$$

For the  $n$ th term introduce

$$t_1 = t, \quad t_{r+1} = t_r - \tau_r, \quad \tau_r \geq 0.$$

The domain is exactly  $t_1 \geq t_2 \geq \dots \geq t_n$ . Inserting a complete set of  $H_0$ -eigenstates between consecutive factors of  $H_I$  reproduces all the exponentials from the energy denominators. Hence the  $n$ th old-fashioned term becomes

$$(-i)^n \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 \dots \int_{-\infty}^{t_{n-1}} dt_n \langle \beta | H_I(t_1) \dots H_I(t_n) | \alpha \rangle.$$

Adding the identity term and suppressing the external matrix elements gives precisely Eq. (3.5.8).

**W.3.5.** Write

$$G_P(E) = \text{P.V.} \frac{1}{E - H_0}.$$

Both  $H_{\text{int}}$  and  $G_P(E)$  are Hermitian. The standing-wave transition operator satisfies

$$T^0 = H_{\text{int}} + H_{\text{int}} G_P T^0 = (1 - H_{\text{int}} G_P)^{-1} H_{\text{int}}.$$

Using  $(1 - AB)^{-1}A = A(1 - BA)^{-1}$ , one also has

$$T^0 = H_{\text{int}}(1 - G_P H_{\text{int}})^{-1} = (T^0)^\dagger.$$

Restriction to equal-energy states and multiplication by the real kernel  $\pi\delta(E_\beta - E_\alpha)$  therefore makes

$$K_{\beta\alpha} = \pi\delta(E_\beta - E_\alpha) T_{\beta\alpha}^0$$

Hermitian.

The identity

$$\frac{1}{x + i0} = \text{P.V.} \frac{1}{x} - i\pi\delta(x)$$

relates the outgoing resolvent to the principal-value resolvent. If  $X = \pi\delta(E - H_0)T^+$  on the energy shell, its integral equation is

$$X = K - iKX, \quad X = (1 + iK)^{-1}K.$$

Since  $X$  is a function of  $K$ , the order is immaterial, and

$$S = 1 - 2iX = (1 - iK)(1 + iK)^{-1}.$$

Hermiticity of  $K$  makes the Cayley transform manifestly unitary.

**W.3.6.** For pion–nucleon scattering, couple the pion orbital angular momentum  $\ell$  to the nucleon spin  $1/2$ . Denote

$$S_{\ell\pm}^I = e^{2i\delta_{\ell\pm}^I}, \quad j = \ell \pm \frac{1}{2}, \quad P = (-1)^{\ell+1},$$

where  $I = 1/2, 3/2$ . The spin-nonflip and spin-flip amplitudes in a definite isospin channel may be written

$$f_I(\theta) = \frac{1}{2ik} \sum_{\ell=0}^{\infty} [(\ell+1)(S_{\ell+}^I - 1) + \ell(S_{\ell-}^I - 1)] P_{\ell}(\cos\theta),$$

$$g_I(\theta) = \frac{1}{2ik} \sum_{\ell=1}^{\infty} (S_{\ell+}^I - S_{\ell-}^I) P_{\ell}^1(\cos\theta).$$

(Changing the convention for  $P_{\ell}^1$  reverses the irrelevant overall sign of  $g_I$ .) Averaging over the initial proton spin and summing over the final spin gives

$$\frac{d\sigma_I}{d\Omega} = |f_I|^2 + |g_I|^2.$$

The state  $\pi^+p$  has  $I_3 = 3/2$ , and hence is pure  $I = 3/2$ :

$$\frac{d\sigma(\pi^+p)}{d\Omega} = |f_{3/2}|^2 + |g_{3/2}|^2.$$

Clebsch–Gordan decomposition of the elastic  $\pi^-p$  channel gives

$$f_{\pi^-p} = \frac{2f_{1/2} + f_{3/2}}{3}, \quad g_{\pi^-p} = \frac{2g_{1/2} + g_{3/2}}{3},$$

and therefore

$$\frac{d\sigma(\pi^-p)}{d\Omega} = \left| \frac{2f_{1/2} + f_{3/2}}{3} \right|^2 + \left| \frac{2g_{1/2} + g_{3/2}}{3} \right|^2.$$

These formulas express the answer entirely through phase shifts of definite  $j$ , parity, and isospin.

**W.3.7.** Insert a complete set of two-particle momentum–spin states between the bra and ket and use Eq. (3.7.5) twice. The two total-momentum delta functions give  $\delta^3(\mathbf{p}' - \mathbf{p})$ . In the center-of-mass frame write  $\mathbf{p}_2 = -\mathbf{p}_1$ ,  $k = |\mathbf{p}_1|$ , and  $E = E_1(k) + E_2(k)$ . The radial energy delta function has derivative

$$\frac{d(E_1 + E_2)}{dk} = k \left( \frac{1}{E_1} + \frac{1}{E_2} \right) = \frac{kE}{E_1 E_2}.$$

Thus its Jacobian converts the radial measure as

$$k^2 dk \delta(E - E_1 - E_2) \delta(E' - E_1 - E_2) = \frac{kE_1 E_2}{E} \delta(E' - E).$$

This factor is cancelled exactly by the product of the two normalization factors  $(kE_1 E_2 / E)^{-1/2}$  in Eq. (3.7.5).

The angular integral gives  $\int d\Omega Y_{\ell'}^{m'*} Y_{\ell}^m = \delta_{\ell'\ell} \delta_{m'm}$ . Successive use of the two Clebsch–Gordan orthogonality relations then produces  $\delta_{s's} \delta_{j'j} \delta_{\sigma'\sigma}$ , while the species labels give  $\delta_{n'n}$ . Altogether,

$$\langle E', \mathbf{p}', j', \sigma', \ell', s', n' | E, \mathbf{0}, j, \sigma, \ell, s, n \rangle = \delta^3(\mathbf{p}') \delta(E' - E) \delta_{j'j} \delta_{\sigma'\sigma} \delta_{\ell'\ell} \delta_{s's} \delta_{n'n},$$

which is Eq. (3.7.6). For identical particles the independent integration region is only half of momentum space; the additional  $\sqrt{2}$  in each state restores the same normalization.

## Supplementary Exercises

### S.3.1. Møller Operators and Scattering States and $S$ -Matrix as an In–Out Overlap

(Inspired by D. Harlow, Quantum Field Theory I, 2024, Section 8.3 / D. Harlow, Quantum Field Theory I, 2024, Sections 8.3–8.4.)

- Define  $\Omega_{\pm} = \lim_{t \rightarrow \mp\infty} e^{iHt} e^{-iH_0 t}$ , whenever the strong limits exist. Show that  $H\Omega_{\pm} = \Omega_{\pm}H_0$ , identify  $|\alpha\rangle_{\text{in}}$  and  $|\alpha\rangle_{\text{out}}$ , and write the scattering operator in terms of  $\Omega_{\pm}$ .
- Starting from translation covariance, prove that  ${}_{\text{out}}\langle\beta||\alpha\rangle_{\text{in}}$  can be nonzero only when the total four-momenta agree. Separate the identity contribution and explain why the remaining connected matrix element contains a four-momentum delta function.

### S.3.2. Completeness on the Scattering Subspace

(Adapted from D. Harlow, Quantum Field Theory I, 2024, Section 8.6, Problem 3.)

For a free identical scalar, let  $|\mathbf{p}, \mathbf{q}\rangle = a^{\dagger}(\mathbf{p})a^{\dagger}(\mathbf{q})|\Omega\rangle$  and use  $[a(\mathbf{p}), a^{\dagger}(\mathbf{q})] = \delta^3(\mathbf{p} - \mathbf{q})$ . Verify explicitly on an arbitrary two-particle state that the two-particle part of the source resolution of the identity,

$$I_2 = \frac{1}{2!} \int d^3\mathbf{p} d^3\mathbf{q} |\mathbf{p}, \mathbf{q}\rangle \langle \mathbf{p}, \mathbf{q}|,$$

acts as the identity. Explain the origin of the  $1/2!$ , generalize the symmetry factor to repeated species in a multiparticle basis, and state how asymptotic completeness transports the resolution to the in and out scattering bases.

### S.3.3. Invariant Flux, Two-Body Phase Space, and Decay Widths

(Inspired by D. Harlow, Quantum Field Theory I, 2024, Sections 8.4 and 8.6, Problem 6 / D. Harlow, Quantum Field Theory I, 2024, Section 8.4.)

- For incident four-momenta  $p_1, p_2$ , show that

$$u = \frac{\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}}{E_1 E_2}$$

in the mostly-plus metric. Verify that it becomes the ordinary speed of particle 1 when particle 2 is at rest and becomes  $|\mathbf{v}_1 - \mathbf{v}_2|$  for collinear velocities.

- (b) Starting with the four-dimensional delta function in Eq. (3.4.15), integrate the two-body final-state phase space in the center-of-mass frame and rederive Eq. (3.4.28), including the unequal-mass expression for  $k'$ .
- (c) A spinless particle of mass  $M$  at rest decays to two nonidentical spinless particles of masses  $m_1, m_2$ , with angle-independent Weinberg-normalized matrix element  $M_{fi}$ . Derive the total width and state the change for two identical daughters.

#### S.3.4. The Optical Theorem and Forward Scattering

(Inspired by D. Harlow, Quantum Field Theory I, 2024, Section 8.5.)

- (a) Write  $S = 1 - i\mathcal{T}$ . Derive the off-diagonal unitarity relation for  $\mathcal{T}$ , then set the external states equal and obtain the optical theorem. Explain why its right-hand side is nonnegative.
- (b) Combine the optical theorem with Eq. (3.4.15) to express the total cross section for a two-particle in-state in terms of the imaginary part of its forward invariant amplitude. Keep the book's state normalization explicit.

#### S.3.5. Exact Scattering from a Delta Potential

(Adapted from D. Harlow, Quantum Field Theory I, 2024, Section 8.6, Problem 4.)

For  $H = p^2/(2m) + V_0\delta(x)$ ,  $V_0 > 0$ , construct explicit stationary left- and right-incident in-states and the corresponding out basis. Determine the reflection and transmission amplitudes and the  $2 \times 2$   $S$ -matrix, and verify its unitarity. As emphasized in the source hint, both incidence directions are needed for completeness, and every state used must be an eigenstate of the full Hamiltonian rather than of the free Hamiltonian alone.

#### S.3.6. Disconnected Spectators

(Inspired by D. Harlow, Quantum Field Theory I, 2024, Section 8.4.)

Suppose one outgoing particle has exactly the same species and momentum as one incoming particle and does not interact. Show how a spectator delta function factors from the corresponding  $S$ -matrix element. Explain why connected amplitudes remove these extra singular distributions.

#### S.3.7. Mandelstam Variables in Massless Kinematics

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 5, 2022, Problem 1.)

- (a) In the mostly-plus metric define

$$s = -(p_1 + p_2)^2, \quad t = -(p_1 - p_3)^2, \quad u = -(p_1 - p_4)^2$$

for  $1 + 2 \rightarrow 3 + 4$ . Prove  $s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2$ .

- (b) For massless  $2 \rightarrow 2$  scattering in the center-of-mass frame, express  $s, t, u$  in terms of  $E_{\text{cm}}$  and the scattering angle. Determine the physical ranges of  $t$  and  $u$ .

### S.3.8. Yukawa Decay, Exchange, and Heavy-Field Matching

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 5, 2022, Problem 2.)

- (a) Let a real scalar  $\phi$  of mass  $M$  and a complex scalar  $\psi$  of mass  $m$  have

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}M^2\phi^2 - \partial_\mu\psi^*\partial^\mu\psi - m^2\psi^*\psi - g\psi^*\psi\phi.$$

The source supplies the usual scalar rules; in this volume's mostly-plus convention they are

$$D_\phi(p) = \frac{-i}{p^2 + M^2 - i0}, \quad D_\psi(p) = \frac{-i}{p^2 + m^2 - i0}, \quad V_{\phi\psi\psi^*} = -ig.$$

For  $M > 2m$ , calculate  $\Gamma(\phi \rightarrow \psi\psi^*)$  using covariantly normalized states.

- (b) Find the leading invariant amplitude for  $\psi(p_1)\psi^*(p_2) \rightarrow \psi(p_3)\psi^*(p_4)$ . Identify the two channels, define the Mandelstam variables used, and give the low-energy limit when every momentum component is small compared with  $M$ .
- (c) Use the classical equation of motion for the heavy field in the preceding part to derive the leading local four- $\psi$  interaction. Check that its tree amplitude agrees with the low-energy expansion of heavy exchange, including the combinatorial factor, and determine the  $\lambda$  for which  $\mathcal{L}_{\text{eff,int}} = -\lambda(\psi^*\psi)^2$ .

### S.3.9. The Linear $O(N)$ Sigma Model

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 5, 2022, Problem 4.)

- (a) Start from

$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\Phi_i\partial^\mu\Phi_i - \frac{1}{2}m^2\Phi_i\Phi_i, \quad [\Phi_i(\mathbf{x}, 0), \Pi_j(\mathbf{y}, 0)] = i\delta_{ij}\delta^3(\mathbf{x} - \mathbf{y}).$$

Quantize the  $N$  equal-mass real fields and show that  $\langle 0|T\{\Phi_i(x)\Phi_j(y)\}|0\rangle = \delta_{ij}D_F(x - y)$ , including its momentum-space rule.

- (b) Add  $\mathcal{L}_{\text{int}} = -(\lambda/4)(\Phi_i\Phi_i)^2$ , and obtain the four-scalar rule by differentiating with respect to labeled fields. As in the source hint, check separately the all-equal-flavor case and the case of two equal-flavor pairs.
- (c) Calculate the center-of-mass total cross sections for  $\Phi_1\Phi_2 \rightarrow \Phi_1\Phi_2$ ,  $\Phi_1\Phi_1 \rightarrow \Phi_2\Phi_2$ , and  $\Phi_1\Phi_1 \rightarrow \Phi_1\Phi_1$ , in terms of  $E_{\text{cm}}$ . Start from the invariant two-body cross-section formula and include the final-state symmetry factors highlighted in the source hint.

(d) The uniform-field potential is

$$V(\Phi) = \frac{1}{2}m^2\Phi_i\Phi_i + \frac{\lambda}{4}(\Phi_i\Phi_i)^2.$$

Take  $m^2 = -\mu^2 < 0$ . Find the vacuum manifold, choose a vacuum in the  $N$ -direction, and define

$$\sigma = \Phi_N - v, \quad \pi_a = \Phi_a, \quad a = 1, \dots, N-1.$$

(e) Expand about that vacuum and put the answer in the supplied form

$$\begin{aligned} \mathcal{L} = & -\frac{1}{2}(\partial\pi)^2 - \frac{1}{2}(\partial\sigma)^2 - \frac{1}{2}m_\sigma^2\sigma^2 - \frac{\lambda_{4\pi}}{4}(\pi^2)^2 - \lambda_{3\sigma}\sigma^3 - \frac{\lambda_{4\sigma}}{4}\sigma^4 \\ & - \lambda_{\pi\sigma 1}\pi^2\sigma - \lambda_{\pi\sigma 2}\pi^2\sigma^2 + C. \end{aligned}$$

Determine  $m_\sigma$ , the pion mass,  $C$ , and every displayed cubic and quartic coupling in terms of  $\mu, \lambda$ .

- (f) List all propagators and vertices in the broken parameterization, with their flavor tensors and numerical factors. No rule corresponds to the constant  $C$ ; use the source's pattern-recognition hint rather than expanding every flavor assignment separately.
- (g) Find the total width for  $\sigma$  to decay into every allowed two-pion state. Evaluate the lifetime for  $N = 3$ ,  $\lambda = 0.1$ , and  $\mu = 10 \text{ GeV}$ .
- (h) Sum the contact and three  $\sigma$ -exchange graphs for  $\pi_i\pi_j \rightarrow \pi_k\pi_l$ , writing the result in terms of  $s, t, u$ .
- (i) First show, as asked in the source, that the pion amplitude vanishes when all external spatial momenta tend to zero. Then prove the stronger single-soft on-shell limit. In the preceding part there should be four diagrams: one contact graph and three exchange graphs.
- (j) Add  $+a\Phi_N$  to the Lagrangian. Determine the shifted-vacuum condition and the pion mass through first order in  $a$ , keeping only the lowest useful order in  $a$  as suggested by the source.

### S.3.10. A Connected $3 \rightarrow 3$ Tree Amplitude

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 4, 2007, Problem 1.)

For the real scalar theory

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4,$$

use the Dyson series and Wick's theorem to identify the connected  $3 \rightarrow 3$  contribution at order  $\lambda^2$ . For a chosen three-leg partition  $A$  of six all-incoming external momenta, with  $q_A = \sum_{a \in A} k_a$ , show that the graph contains

$$(-i\lambda)^2 \frac{-i}{q_A^2 + m^2 - i0}.$$



Check this against the Feynman rules and enumerate all other distinct labeled tree graphs contributing to the process.

### S.3.11. Exponentiation of Vacuum Bubbles

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 4, 2007, Problem 2.)

For

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4,$$

examine  $\langle 0|S|0\rangle$  through order  $\lambda^2$ . Use Wick's theorem to identify the one-vertex double-tadpole, the connected two-vertex four-line “basketball,” the connected two-vertex graph with one self-contraction at each vertex and two lines joining the vertices, and the disconnected pair of one-vertex bubbles. Show from their combinatorial factors that the result is the beginning of the exponential of the sum of connected vacuum-bubble types. Explain why normalized scattering amplitudes contain no vacuum-bubble factor.

### S.3.12. An $O(3)$ -Symmetric Scattering Theory

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 4, 2007, Problem 3.)

- (a) Start from the complete  $O(3)$ -invariant Lagrangian

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi_i\partial^\mu\phi_i - \frac{1}{2}m^2\phi_i\phi_i - \frac{\lambda}{8}(\phi_i\phi_i)^2, \quad i = 1, 2, 3.$$

In the free theory derive  $\langle 0|T\{\phi_i(x)\phi_j(y)\}|0\rangle = \delta_{ij}D_F(x-y)$ .

- (b) Give all free and interacting Feynman rules, including the invariant rank-four flavor tensor and every numerical factor.
- (c) Evaluate  $\phi_i\phi_j \rightarrow \phi_k\phi_l$  at tree level.

### S.3.13. Fermion and Antifermion Yukawa Scattering

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 4, 2007, Problem 4.)

For

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}\mu^2\phi^2 - \bar{\psi}(\gamma^\mu\partial_\mu + m)\psi - \lambda\phi\bar{\psi}\psi,$$

use covariantly normalized external states. For example,

$$|i_{\psi\psi}\rangle = \sqrt{4E_{\mathbf{p}}E_{\mathbf{q}}}b_{\mathbf{p}s}^\dagger b_{\mathbf{q}r}^\dagger|0\rangle, \quad |f_{\psi\psi}\rangle = \sqrt{4E_{\mathbf{p}'}E_{\mathbf{q}'}}b_{\mathbf{p}'s'}^\dagger b_{\mathbf{q}'r'}^\dagger|0\rangle,$$

and replace the second  $b^\dagger$  by  $c^\dagger$  for a fermion–antifermion state.

- (a) Starting from the Dyson formula and Wick's theorem, compute  $\psi\psi \rightarrow \psi\psi$ , retaining both scalar-exchange graphs and deriving their relative sign from the external Fock states. In local Mandelstam conventions, show that the answer has the form

$$\mathcal{M}_{\psi\psi} = -\lambda^2 \left[ \frac{(\bar{u}_3 u_1)(\bar{u}_4 u_2)}{t - \mu^2 + i0} - \frac{(\bar{u}_4 u_1)(\bar{u}_3 u_2)}{u - \mu^2 + i0} \right].$$

- (b) Compute  $\psi\bar{\psi} \rightarrow \psi\bar{\psi}$ , including elastic exchange and annihilation, and show

$$\mathcal{M}_{\psi\bar{\psi}} = -\lambda^2 \left[ \frac{(\bar{u}_3 u_1)(\bar{v}_2 v_4)}{t - \mu^2 + i0} - \frac{(\bar{v}_2 u_1)(\bar{u}_3 v_4)}{s - \mu^2 + i0} \right]$$

up to the common overall  $S$ -matrix convention. Display the spinor order and Mandelstam channel of each term and account for every minus sign.

### S.3.14. Three Routes to a Half-Line Fourier Transform

(Adapted from A. Guth, MIT 8.323, Problem Set 4, 2008, Problem 2.)

- (a) Regard  $g(t) = \theta(t)e^{-i\omega_0 t}$  as a tempered distribution. Pair its Fourier transform with

$$\varphi_\sigma(\omega) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(\omega - \omega_1)^2 / (2\sigma^2)}$$

by transforming the test function first, and evaluate the half-line integral. As in the source, the answer may be left as a definite integral or expressed with the error function.

- (b) Replace  $g(t)$  by  $g_\epsilon(t) = g(t)e^{-\epsilon t}$ . Perform the Gaussian pairing before taking  $\epsilon \rightarrow 0^+$ . Use the source hint that absolute convergence permits the order of integration to be exchanged.
- (c) Instead truncate the time integral at  $T$ . Pair its ordinary Fourier transform with the Gaussian and take  $T \rightarrow \infty$ . Prove that all three routes give the same principal-value-plus-delta distribution. Relate this to the source conditions  $|g_\epsilon(t)| \leq |g(t)|$  and  $g_\epsilon(t) \rightarrow g(t)$  pointwise.
- (d) The preceding regulators are pointwise convergent and uniformly dominated. Construct a family  $g_\epsilon(t)$  that still converges to  $g(t)$  at every fixed  $t$  but violates domination so strongly that its pairing with the same Gaussian has a different limit. It is enough to carry the construction far enough to exhibit the nonzero extra term.

### S.3.15. Partial-Wave Unitarity and Cross-Section Bounds

(Inspired by D. Harlow, Quantum Field Theory I, 2024, Section 8.5 / D. Harlow, Quantum Field Theory I, 2024, Sections 8.4–8.5.)

- (a) For spinless elastic scattering with

$$f(\theta) = \sum_{\ell=0}^{\infty} (2\ell + 1) f_\ell P_\ell(\cos \theta),$$

use  $S^\dagger S = 1$  to prove  $\text{Im } f_\ell = k|f_\ell|^2$ . Parameterize the general elastic solution by a real phase shift.

(b) Using the phase-shift form of the preceding part, derive

$$\sigma_{\text{tot}} = \frac{4\pi}{k^2} \sum_{\ell=0}^{\infty} (2\ell + 1) \sin^2 \delta_{\ell}$$

and the maximum elastic cross section carried by one partial wave.

### S.3.16. The $K$ -Matrix and Inelastic Partial Waves

(Inspired by D. Harlow, Quantum Field Theory I, 2024, Section 8.5.)

- (a) For a single elastic partial wave, combine the Cayley transform  $S = (1 - iK)(1 + iK)^{-1}$  with  $S = e^{2i\delta}$ . Find  $K$  in terms of  $\delta$ , locate the phase-shift resonance, and explain why a real  $K$  preserves unitarity.
- (b) When other channels are open, write an elastic diagonal element as  $S_{\ell} = \eta_{\ell} e^{2i\delta_{\ell}}$ . Prove  $0 \leq \eta_{\ell} \leq 1$  and derive the elastic, reaction, and total partial-wave cross sections.

### S.3.17. Breit–Wigner Phase Motion

(Inspired by B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 5, 2022, Problems 1–2.)

Take

$$S_{\ell}(E) = \frac{E - E_R - i\Gamma/2}{E - E_R + i\Gamma/2}.$$

Show that it is unitary on the real axis, determine a continuous phase shift, and derive the Lorentzian form and narrow-width limit of  $\sin^2 \delta_{\ell}$ .

## Solutions to Supplementary Exercises

### S.3.1. Møller Operators and Scattering States and $S$ -Matrix as an In–Out Overlap

- (a) For fixed  $s$ , shifting the limiting variable gives

$$e^{iHs}\Omega_{\pm}e^{-iH_0s} = \lim_{t \rightarrow \mp\infty} e^{iH(t+s)}e^{-iH_0(t+s)} = \Omega_{\pm}.$$

Differentiation at  $s = 0$  yields the intertwining relation

$$H\Omega_{\pm} = \Omega_{\pm}H_0.$$

Thus an  $H_0$ -eigenstate of energy  $E_{\alpha}$  is mapped to a generalized  $H$ -eigenstate of the same energy. With the signs in the question,

$$|\alpha\rangle_{\text{in}} = \Omega_{+}|\alpha\rangle_0, \quad |\alpha\rangle_{\text{out}} = \Omega_{-}|\alpha\rangle_0.$$

On the scattering subspace the wave operators are isometries. The operator acting on free asymptotic labels is

$$S = \Omega_-^\dagger \Omega_+, \quad \langle \beta | S \alpha \rangle_0 = {}_{\text{out}} \langle \beta | | \alpha \rangle_{\text{in}}.$$

If their ranges exhaust the continuous spectral subspace,  $S$  is unitary there; bound states lie outside this scattering subspace.

- (b) The total momentum acts on the asymptotic states with eigenvalues  $p_\alpha$  and  $p_\beta$ . Translation by  $a$  therefore gives

$${}_{\text{out}} \langle \beta | | \alpha \rangle_{\text{in}} = e^{i(p_\beta - p_\alpha) \cdot a} {}_{\text{out}} \langle \beta | | \alpha \rangle_{\text{in}}$$

for every  $a$ . The overlap consequently has support only at  $p_\beta = p_\alpha$ . Isolating free propagation in the book's normalization gives

$${}_{\text{out}} \langle \beta | | \alpha \rangle_{\text{in}} = \delta(\beta - \alpha) - 2\pi i \delta^4(p_\beta - p_\alpha) M_{\beta\alpha}.$$

The identity term contains delta functions matching every outgoing particle with an incoming particle. A connected matrix element retains only the single delta function required by conservation of the total four-momentum. Extra delta functions matching proper subsets signal disconnected spectator pieces.

### S.3.2. Completeness on the Scattering Subspace

Symmetrized two-boson states obey

$$\langle \mathbf{k}, \mathbf{l} | \mathbf{p}, \mathbf{q} \rangle = \delta^3(\mathbf{k} - \mathbf{p}) \delta^3(\mathbf{l} - \mathbf{q}) + \delta^3(\mathbf{k} - \mathbf{q}) \delta^3(\mathbf{l} - \mathbf{p}).$$

Hence

$$I_2 = \frac{1}{2} \int d^3p d^3q |\mathbf{p}, \mathbf{q}\rangle \langle \mathbf{p}, \mathbf{q}|$$

acts on  $|\mathbf{k}, \mathbf{l}\rangle$  as one half times two equal terms, and is the identity. Equivalently one may integrate over only one ordering of the two momenta and omit  $1/2!$ .

Applying the wave operators gives

$$I_{2,\text{in}} = \frac{1}{2} \int d^3p d^3q |\mathbf{p}, \mathbf{q}\rangle_{\text{in}} {}_{\text{in}} \langle \mathbf{p}, \mathbf{q}|, \quad I_{2,\text{out}} = \frac{1}{2} \int d^3p d^3q |\mathbf{p}, \mathbf{q}\rangle_{\text{out}} {}_{\text{out}} \langle \mathbf{p}, \mathbf{q}|.$$

These are identities on the corresponding two-particle asymptotic subspaces. Asymptotic completeness asserts that the sum over every open particle-number channel supplies the full scattering projector. For a general multiparticle state the divisor is

$$S(n) = \prod_a N_a!,$$

where  $N_a$  is the number of identical particles of species and polarization type  $a$ . It removes the permutations already generated by the creation-operator algebra.

### S.3.3. Invariant Flux, Two-Body Phase Space, and Decay Widths

- (a) With  $p_i^2 = -m_i^2$ , the numerator

$$F = \sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}$$

is invariant. The ratio  $F/(E_1 E_2)$  is the relative velocity that combines with the density factors implicit in the book's  $\delta^3$ -normalized states. If  $p_2 = (m_2, \mathbf{0})$ , then

$$F = m_2 |\mathbf{p}_1|, \quad u = \frac{|\mathbf{p}_1|}{E_1} = |\mathbf{v}_1|.$$

For collinear velocities,

$$\frac{F^2}{E_1^2 E_2^2} = (1 - v_1 v_2)^2 - (1 - v_1^2)(1 - v_2^2) = (v_1 - v_2)^2.$$

Thus  $u = |v_1 - v_2|$ , including the case of oppositely directed particles through the signs of  $v_1, v_2$ .

- (b) Momentum conservation sets  $\mathbf{p}'_2 = -\mathbf{p}'_1$ . Write  $q = |\mathbf{p}'_1|$ . The remaining energy delta function is

$$\delta(E - E'_1(q) - E'_2(q)), \quad \frac{d(E'_1 + E'_2)}{dq} = \frac{qE}{E'_1 E'_2}.$$

Consequently

$$\delta^4(p_\beta - p_\alpha) d\beta \longrightarrow \frac{k' E'_1 E'_2}{E} d\Omega,$$

which is Eq. (3.4.28). Squaring  $E = E'_1 + E'_2$  and solving for the positive momentum gives

$$k' = \frac{\sqrt{[E^2 - (m'_1 + m'_2)^2][E^2 - (m'_1 - m'_2)^2]}}{2E}.$$

The expression vanishes at threshold and is real precisely when the final channel is open.

- (c) Equation (3.4.29) gives

$$\frac{d\Gamma}{d\Omega} = \frac{2\pi k' E'_1 E'_2}{M} |M_{fi}|^2.$$

For an angle-independent matrix element,

$$\Gamma = \frac{8\pi^2 k' E'_1 E'_2}{M} |M_{fi}|^2,$$

where

$$k' = \frac{\sqrt{[M^2 - (m_1 + m_2)^2][M^2 - (m_1 - m_2)^2]}}{2M}, \quad E'_{1,2} = \frac{M^2 + m_{1,2}^2 - m_{2,1}^2}{2M}.$$

These factors are specific to the book's  $\delta^3$  state normalization. For two identical daughters, integrating over the full labeled phase space double-counts every physical state, so the result is divided by  $2!$ .

### S.3.4. The Optical Theorem and Forward Scattering

(a) Substitution of  $S = 1 - i\mathcal{T}$  into unitarity yields

$$i(\mathcal{T} - \mathcal{T}^\dagger) = \mathcal{T}^\dagger \mathcal{T}.$$

Between arbitrary states,

$$i(\mathcal{T}_{fi} - \mathcal{T}_{if}^*) = \sum_n \mathcal{T}_{nf}^* \mathcal{T}_{ni},$$

where the sum includes all discrete labels and phase-space integrals. For  $f = i$ ,

$$2 \operatorname{Im} \mathcal{T}_{ii} = \sum_n |\mathcal{T}_{ni}|^2.$$

Every term on the right is a squared modulus, so the absorptive part of the forward matrix element is nonnegative. This is the optical theorem before the common energy-momentum delta functions are removed.

(b) In the book's convention,

$$\mathcal{T}_{\beta\alpha} = 2\pi\delta^4(p_\beta - p_\alpha)M_{\beta\alpha}.$$

Removing the common factor  $\delta^4(0)$  from the forward optical theorem gives

$$\operatorname{Im} M_{\alpha\alpha} = \pi \int d\beta \delta^4(p_\beta - p_\alpha) |M_{\beta\alpha}|^2.$$

On the other hand, integration of Eq. (3.4.15) gives

$$\sigma_{\text{tot}}(\alpha) = \frac{(2\pi)^4}{u_\alpha} \int d\beta \delta^4(p_\beta - p_\alpha) |M_{\beta\alpha}|^2.$$

Therefore

$$\sigma_{\text{tot}}(\alpha) = \frac{16\pi^3}{u_\alpha} \operatorname{Im} M_{\alpha\alpha}.$$

The numerical factor changes when covariantly normalized one-particle states and the more common invariant amplitude are used; the content of the theorem does not.

### S.3.5. Exact Scattering from a Delta Potential

For incidence from the left, take

$$\psi_L(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0, \\ te^{ikx}, & x > 0. \end{cases}$$

Continuity gives  $t = 1 + r$ , while  $\psi'(0^+) - \psi'(0^-) = 2mV_0\psi(0)$  gives, with  $\lambda = mV_0/k$ ,

$$t = \frac{1}{1 + i\lambda}, \quad r = \frac{-i\lambda}{1 + i\lambda}.$$

Reflection symmetry gives the right-incident solution by  $x \mapsto -x$ . For this real, time-reversal-invariant potential, complex conjugating the two incoming eigenfunctions gives the corresponding outgoing basis. Thus all four boundary-condition states solve the full stationary Schrödinger equation; neither a lone left-incident state nor free plane waves form a complete scattering basis. In the basis of incoming waves from the left and right,

$$S = \begin{pmatrix} t & r \\ r & t \end{pmatrix}.$$

Direct calculation gives

$$|t|^2 + |r|^2 = 1, \quad tr^* + rt^* = 0,$$

so  $S^\dagger S = 1$ . As  $V_0 \rightarrow 0$ ,  $t \rightarrow 1$ ; as  $V_0 \rightarrow \infty$ ,  $r \rightarrow -1$ , as expected.

### S.3.6. Disconnected Spectators

Let  $s$  denote the spectator and  $\hat{\alpha}, \hat{\beta}$  the interacting parts. With the book's normalization, a disconnected term has the form

$$\text{out}\langle \mathbf{p}'_s, s; \hat{\beta} | \mathbf{p}_s, s; \hat{\alpha} \rangle_{\text{in}}|_{\text{disc}} = \delta^3(\mathbf{p}'_s - \mathbf{p}_s) \text{out}\langle \hat{\beta} | \hat{\alpha} \rangle_{\text{in}},$$

with an additional Kronecker delta for the spectator's spin and species. The reduced overlap contains its own conservation delta function. Thus the full term is more singular than a connected amplitude, because a proper subset of momenta is separately conserved. Defining connected matrix elements by cumulant subtraction removes every such factorization, leaving only the one delta function for total four-momentum conservation.

### S.3.7. Mandelstam Variables in Massless Kinematics

(a) Momentum conservation gives  $p_2 = p_3 + p_4 - p_1$ . Expanding the definitions,

$$\begin{aligned} s + t + u &= -3p_1^2 - 2p_1 \cdot (p_2 - p_3 - p_4) - p_2^2 \\ &= -p_1^2 - p_2^2 - p_3^2 - p_4^2, \end{aligned}$$

where the second line follows after using conservation once more. Because the mostly-plus mass shells obey  $p_i^2 = -m_i^2$ ,

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2.$$

This proof does not require any particular Lorentz frame.

(b) Choose

$$p_1 = \frac{E}{2}(1, 0, 0, 1), \quad p_2 = \frac{E}{2}(1, 0, 0, -1), \quad p_3 = \frac{E}{2}(1, \sin \theta, 0, \cos \theta),$$

with  $p_4$  opposite to  $p_3$  and  $E = E_{\text{cm}}$ . In the definitions of S.3.7,

$$s = E^2, \quad t = -\frac{E^2}{2}(1 - \cos \theta), \quad u = -\frac{E^2}{2}(1 + \cos \theta).$$

They sum to zero. Since  $-1 \leq \cos \theta \leq 1$ ,

$$-s \leq t \leq 0, \quad -s \leq u \leq 0,$$

and  $t + u = -s$ .

### S.3.8. Yukawa Decay, Exchange, and Heavy-Field Matching

(a) The supplied mostly-plus rules are

$$D_\phi(p) = \frac{-i}{p^2 + M^2 - i0}, \quad D_\psi(p) = \frac{-i}{p^2 + m^2 - i0}, \quad V_{\phi\psi\psi^*} = -ig.$$

The vertex gives the covariant invariant amplitude  $\mathcal{M} = -g$ , so  $|\mathcal{M}|^2 = g^2$ . In the rest frame of  $\phi$ ,

$$|\mathbf{p}| = \frac{M}{2} \sqrt{1 - \frac{4m^2}{M^2}}.$$

The standard two-body phase-space integral gives

$$\Gamma(\phi \rightarrow \psi\psi^*) = \frac{|\mathbf{p}|}{8\pi M^2} |\mathcal{M}|^2 = \frac{g^2}{16\pi M} \sqrt{1 - \frac{4m^2}{M^2}}.$$

The daughters have opposite charge and are distinguishable, so there is no  $1/2!$  factor.

(b) Charge flow permits annihilation through the  $s$ -channel and exchange through the  $t$ -channel. In the usual covariant amplitude convention,

$$\mathcal{M} = -g^2 \left( \frac{1}{s - M^2 + i0} + \frac{1}{t - M^2 + i0} \right).$$

There is no  $u$ -channel graph for these assigned charges. When  $|s|, |t| \ll M^2$ ,

$$\mathcal{M} = \frac{2g^2}{M^2} + O\left(\frac{s+t}{M^4}\right).$$

The leading term is momentum independent and hence can be reproduced by a local four-field interaction.

(c) Write  $X = \psi^*\psi$ . At momenta small compared with  $M$ , the heavy field equation is

$$(\square - M^2)\phi = gX, \quad \phi = -\frac{g}{M^2}X + O(M^{-4}\square X).$$

Substitution into the mass and interaction terms gives

$$\mathcal{L}_{\text{eff}} = \frac{g^2}{2M^2} X^2 + O(M^{-4}) = -\lambda_{\text{eff}}(\psi^*\psi)^2 + \cdots, \quad \lambda_{\text{eff}} = -\frac{g^2}{2M^2}.$$

Differentiating  $X^2$  with respect to two  $\psi$ 's and two  $\psi^*$ 's supplies a factor  $2!2! = 4$ . The contact vertex is  $-4i\lambda_{\text{eff}} = 2ig^2/M^2$ , exactly  $i\mathcal{M}$  from the leading heavy-exchange result.



### S.3.9. The Linear $O(N)$ Sigma Model

(a) Canonical quantization gives

$$[a_i(\mathbf{p}), a_j^\dagger(\mathbf{q})] = (2\pi)^3 2E_{\mathbf{p}} \delta_{ij} \delta^3(\mathbf{p} - \mathbf{q}), \quad \langle 0 | T \Phi_i(x) \Phi_j(y) | 0 \rangle = \delta_{ij} D_F(x - y).$$

(b) Four differentiations of the interaction give

$$-2i\lambda(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

(c) The three invariant amplitudes are  $-2\lambda, -2\lambda, -6\lambda$ . Since  $d\sigma/d\Omega = |\mathcal{M}|^2/(64\pi^2 s)$ , and identical final particles contribute  $1/2!$ ,

$$\sigma_{12 \rightarrow 12} = \frac{\lambda^2}{4\pi s}, \quad \sigma_{11 \rightarrow 22} = \frac{\lambda^2}{8\pi s}, \quad \sigma_{11 \rightarrow 11} = \frac{9\lambda^2}{8\pi s}.$$

(d) The minima obey  $\Phi_i \Phi_i = v^2$ ,  $v = \mu/\sqrt{\lambda}$ . Choose  $\Phi_N = v + \sigma$ ,  $\Phi_a = \pi_a$ ,  $a = 1, \dots, N-1$ .

(e) With  $\pi^2 = \pi_a \pi_a$ , the expanded theory is

$$\mathcal{L} = -\frac{1}{2}(\partial\sigma)^2 - \frac{1}{2}(\partial\pi_a)^2 - \mu^2\sigma^2 - \lambda v\sigma^3 - \frac{\lambda}{4}\sigma^4 - \lambda v\sigma\pi^2 - \frac{\lambda}{2}\sigma^2\pi^2 - \frac{\lambda}{4}(\pi^2)^2 + \frac{\mu^4}{4\lambda}.$$

Thus  $m_\sigma^2 = 2\mu^2$  and  $m_\pi^2 = 0$ , and the displayed coefficients exhaust the cubic and quartic interactions.

(f) The  $\sigma$  and pion propagators are  $-i/(p^2 + m_\sigma^2 - i0)$  and  $-i\delta_{ab}/(p^2 - i0)$ . The  $\sigma^3, \sigma^4, \pi_a \pi_b \sigma, \pi_a \pi_b \sigma \sigma$  rules are

$$-6i\lambda v, \quad -6i\lambda, \quad -2i\lambda v \delta_{ab}, \quad -2i\lambda \delta_{ab},$$

and the four-pion rule is  $-2i\lambda(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})$ .

(g) For each flavor the decay rule is  $-2i\lambda v$ ; after the identical-particle factor and flavor sum,

$$\Gamma_\sigma = (N-1) \frac{(2\lambda v)^2}{32\pi m_\sigma} = \frac{(N-1)\lambda\mu}{8\pi\sqrt{2}}.$$

At the stated values this is  $5.63 \times 10^{-2} \text{ GeV}$ , so  $\tau = \hbar/\Gamma_\sigma = 1.17 \times 10^{-23} \text{ s}$ .

(h) Using  $2\lambda v^2 = m_\sigma^2$ , contact and pole terms combine into

$$\mathcal{M}_{ij \rightarrow kl} = -2\lambda \left[ \delta_{ij}\delta_{kl} \frac{s}{s - m_\sigma^2} + \delta_{ik}\delta_{jl} \frac{t}{t - m_\sigma^2} + \delta_{il}\delta_{jk} \frac{u}{u - m_\sigma^2} \right].$$

(i) At vanishing external spatial momenta the pions are massless, so every external four-momentum tends to zero and hence  $s, t, u \rightarrow 0$ ; the amplitude vanishes. More strongly, taking only one external pion soft on shell also sends the three channel invariants multiplying its possible flavor tensors to zero. Every term in the last expression therefore vanishes, which is the Adler zero.

(j) The shifted vacuum  $v_a$  obeys

$$-\mu^2 v_a + \lambda v_a^3 - a = 0.$$

Its transverse curvature is

$$m_\pi^2 = -\mu^2 + \lambda v_a^2 = \frac{a}{v_a} = \frac{a}{v} + O(a^2).$$

### S.3.10. A Connected $3 \rightarrow 3$ Tree Amplitude

The connected contribution first occurs in the second Dyson term:

$$\frac{(-i)^2}{2!} \left( \frac{\lambda}{4!} \right)^2 \int d^4x d^4y T\{\phi^4(x)\phi^4(y)\}.$$

Three external legs contract at each vertex and the remaining field at each vertex forms the internal line. Taking all six external momenta  $k_a$  incoming, a chosen split  $A \cup A^c$ ,  $|A| = 3$ , contributes

$$i\mathcal{M}_A = (-i\lambda)^2 \frac{-i}{q_A^2 + m^2 - i0}, \quad q_A = \sum_{a \in A} k_a$$

in the volume's mostly-plus convention. The replacement  $A \leftrightarrow A^c$  describes the same graph, so the number of distinct labeled trees is

$$\frac{1}{2} \binom{6}{3} = 10.$$

The Dyson  $1/2!$  is cancelled by interchanging the two vertices, and the  $4!$  factors are cancelled by their Wick contractions.

### S.3.11. Exponentiation of Vacuum Bubbles

At one vertex the three pairings of four fields give the double-tadpole with symmetry factor 8. At two vertices,  $4! = 24$  contractions join all four legs between the vertices, giving the four-line basketball with symmetry factor 48. Choosing two legs at each vertex and pairing them across gives  $\binom{4}{2}^2 2! = 72$  contractions; the remaining two legs at each vertex self-contract, so this connected topology has symmetry factor 16. Finally, the  $3^2 = 9$  contractions that self-pair both vertices form a disconnected pair of one-vertex bubbles.

Let  $C_1$  denote the connected one-vertex contribution and  $C_2$  the sum of the two connected two-vertex topologies, including their integrals and the symmetry factors just found. The Dyson  $1/2!$  makes the disconnected 9-contraction term exactly  $\frac{1}{2}C_1^2$ . Wick's theorem therefore organizes the vacuum amplitude as

$$\langle 0|S|0\rangle = 1 + C_1 + \left( C_2 + \frac{1}{2}C_1^2 \right) + O(\lambda^3).$$

The  $1/2$  divides permutations of the two identical disconnected components. This is precisely

$$\exp(C_1 + C_2 + \dots) = 1 + C_1 + C_2 + \frac{1}{2}C_1^2 + \dots.$$

The same component-counting argument works at every order: a collection of  $n_r$  connected bubbles of type  $r$  carries  $1/n_r!$ . Normalized matrix elements divide by  $\langle 0|S|0\rangle$ , so the common vacuum exponential cancels and no vacuum bubbles remain in scattering amplitudes.

### S.3.12. An $O(3)$ -Symmetric Scattering Theory

- (a) The canonical commutator is diagonal in flavor, so

$$\langle 0|T\phi_i(x)\phi_j(y)|0\rangle = \delta_{ij}D_F(x-y).$$

- (b) An internal scalar line therefore carries  $-i\delta_{ij}/(p^2 + m^2 - i0)$ . An invariant rank-four tensor with the required Bose symmetries is a multiple of

$$\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}.$$

Four differentiation of  $-(\lambda/8)(\phi_a\phi_a)^2$  fixes the vertex to

$$-i\lambda(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

- (c) Thus

$$\mathcal{M}_{ij\rightarrow kl} = -\lambda(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

The tensor is unchanged by any  $O(3)$  rotation and is symmetric under interchange within either the incoming or outgoing pair.

### S.3.13. Fermion and Antifermion Yukawa Scattering

- (a) An ordered two-fermion out-state is created by  $b_3^\dagger b_4^\dagger|0\rangle$ . Interchanging its labels gives

$$b_4^\dagger b_3^\dagger|0\rangle = -b_3^\dagger b_4^\dagger|0\rangle.$$

Accordingly the two contractions that attach the outgoing fermions to the incoming lines enter with opposite signs. For scalar Yukawa exchange one may write

$$\mathcal{M}_{\psi\psi} = -\lambda^2 \left[ \frac{(\bar{u}_3 u_1)(\bar{u}_4 u_2)}{t - \mu^2 + i0} - \frac{(\bar{u}_4 u_1)(\bar{u}_3 u_2)}{u - \mu^2 + i0} \right].$$

Under  $3 \leftrightarrow 4$ , the two terms exchange and the amplitude changes sign. This antisymmetry, rather than a manually appended rule, is the direct consequence of the external fermionic Fock states.

(b) For  $\psi(p_1)\bar{\psi}(p_2) \rightarrow \psi(p_3)\bar{\psi}(p_4)$ ,

$$\mathcal{M}_{\psi\bar{\psi}} = -\lambda^2 \left[ \frac{(\bar{u}_3 u_1)(\bar{v}_2 v_4)}{t - \mu^2 + i0} - \frac{(\bar{v}_2 u_1)(\bar{u}_3 v_4)}{s - \mu^2 + i0} \right],$$

up to the common overall sign of the chosen  $S$ -matrix convention. The first term is  $t$ -channel elastic exchange and the second is  $s$ -channel annihilation. Reordering the external fermion operators into the fixed canonical order supplies their relative sign.

### S.3.14. Three Routes to a Half-Line Fourier Transform

(a) Set  $\Delta = \omega_1 - \omega_0$ . Transforming the Gaussian first gives the well-defined pairing

$$\langle \hat{g}, \varphi_\sigma \rangle = \int_0^\infty dt e^{i\Delta t - \sigma^2 t^2/2} = \frac{\sqrt{\pi}}{\sqrt{2}\sigma} e^{-\Delta^2/(2\sigma^2)} \left[ 1 + \operatorname{erf} \left( \frac{i\Delta}{\sqrt{2}\sigma} \right) \right].$$

(b) For  $\epsilon > 0$  the ordinary transform is

$$\hat{g}_\epsilon(\omega) = \frac{i}{\omega - \omega_0 + i\epsilon}.$$

Its Gaussian pairing is equivalently  $\int_0^\infty dt e^{i\Delta t - \sigma^2 t^2/2 - \epsilon t}$ . Dominated convergence produces exactly the result of part 1.

(c) With a sharp endpoint,

$$\hat{g}_T(\omega) = \frac{e^{i(\omega - \omega_0)T} - 1}{i(\omega - \omega_0)}.$$

After pairing this becomes  $\int_0^T dt e^{i\Delta t - \sigma^2 t^2/2}$ , which tends to the same answer as  $T \rightarrow \infty$ . Sokhotski–Plemelj therefore identifies the common distribution as

$$\hat{g}(\omega) = \pi\delta(\omega - \omega_0) + i \text{ P.V. } \frac{1}{\omega - \omega_0}.$$

(d) Fix the Gaussian in the question and put  $\Delta = \omega_1 - \omega_0$ . Let

$$T_\epsilon = \epsilon^{-1}, \quad w_\epsilon = \epsilon^3, \quad A_\epsilon = w_\epsilon^{-1} \exp\left(\frac{\sigma^2 T_\epsilon^2}{2} - i\Delta T_\epsilon\right),$$

and define

$$g_\epsilon(t) = g(t) [1 + A_\epsilon \mathbf{1}_{[T_\epsilon, T_\epsilon + w_\epsilon]}(t)].$$

For every fixed  $t$ , the moving interval eventually lies to its right, so  $g_\epsilon(t) \rightarrow g(t)$ . The extra pulse is not uniformly dominated. Its Gaussian pairing is

$$A_\epsilon \int_{T_\epsilon}^{T_\epsilon + w_\epsilon} dt e^{i\Delta t - \sigma^2 t^2/2} = 1 + o(1).$$

Thus this regulator adds a nonzero constant to the limiting functional, even though it converges pointwise. This exhibits why a domination hypothesis, not pointwise convergence alone, is needed.

### S.3.15. Partial-Wave Unitarity and Cross-Section Bounds

- (a) Orthogonality of angular momentum channels makes unitarity diagonal:  $|S_\ell| = 1$ . With

$$S_\ell = 1 + 2ikf_\ell,$$

the identity  $S_\ell^* S_\ell = 1$  gives

$$\text{Im } f_\ell = k|f_\ell|^2.$$

Every unit-modulus complex number can be written  $S_\ell = e^{2i\delta_\ell}$  with real  $\delta_\ell$ . Hence

$$f_\ell = \frac{e^{2i\delta_\ell} - 1}{2ik} = \frac{e^{i\delta_\ell} \sin \delta_\ell}{k}.$$

This form satisfies the unitarity relation identically and makes the phase-shift ambiguity  $\delta_\ell \sim \delta_\ell + \pi$  explicit.

- (b) Using  $\int d\Omega P_\ell(\cos \theta) P_{\ell'}(\cos \theta) = 4\pi \delta_{\ell\ell'}/(2\ell + 1)$ , one finds

$$\sigma_{\text{el}} = 4\pi \sum_{\ell} (2\ell + 1) |f_\ell|^2 = \frac{4\pi}{k^2} \sum_{\ell} (2\ell + 1) \sin^2 \delta_\ell.$$

For purely elastic scattering this also equals the total cross section, as follows independently from the forward optical theorem. Since  $\sin^2 \delta_\ell \leq 1$ , one partial wave obeys

$$\sigma_\ell \leq \frac{4\pi}{k^2} (2\ell + 1).$$

The bound is saturated at  $\delta_\ell = \pi/2 \pmod{\pi}$ .

### S.3.16. The $K$ -Matrix and Inelastic Partial Waves

- (a) Equating the two representations,

$$e^{2i\delta} = \frac{1 - iK}{1 + iK},$$

and solving gives

$$K = -\tan \delta.$$

The minus sign follows from the chapter's convention  $S = 1 - 2iK + \dots$ ; the opposite convention for the transition operator would give  $K = \tan \delta$ . At a resonance the phase passes through  $\delta = \pi/2$ , so  $K$  has a real-axis pole while  $S = -1$  remains finite. If  $K = K^*$ , then

$$S^* = \frac{1 + iK}{1 - iK} = S^{-1},$$

which proves one-channel unitarity without further constraints.

- (b) Unitarity of the full multichannel matrix implies that the squared norm of any one diagonal entry cannot exceed one. Thus

$$S_\ell = \eta_\ell e^{2i\delta_\ell}, \quad 0 \leq \eta_\ell \leq 1.$$

With  $f_\ell = (S_\ell - 1)/(2ik)$ , the three standard partial-wave cross sections are

$$\begin{aligned} \sigma_{\text{el},\ell} &= \frac{\pi}{k^2} (2\ell + 1) |1 - \eta_\ell e^{2i\delta_\ell}|^2, \\ \sigma_{\text{reac},\ell} &= \frac{\pi}{k^2} (2\ell + 1) (1 - \eta_\ell^2), \\ \sigma_{\text{tot},\ell} &= \frac{2\pi}{k^2} (2\ell + 1) (1 - \eta_\ell \cos 2\delta_\ell). \end{aligned}$$

Direct expansion verifies  $\sigma_{\text{tot}} = \sigma_{\text{el}} + \sigma_{\text{reac}}$ .

### S.3.17. Breit–Wigner Phase Motion

For real  $E$ , numerator and denominator are complex conjugates, so  $|S_\ell| = 1$ . Put  $x = E - E_R$  and  $\gamma = \Gamma/2$ . A continuous choice increasing from 0 to  $\pi$  obeys

$$\tan \delta_\ell = -\frac{\gamma}{x}, \quad \delta_\ell(E_R) = \frac{\pi}{2}.$$

Therefore

$$\sin^2 \delta_\ell = \frac{\gamma^2}{x^2 + \gamma^2} = \frac{\Gamma^2/4}{(E - E_R)^2 + \Gamma^2/4}.$$

Using  $\lim_{\gamma \rightarrow 0+} \gamma/(x^2 + \gamma^2) = \pi \delta(x)$ ,

$$\sin^2 \delta_\ell \longrightarrow \pi \gamma \delta(E - E_R) = \frac{\pi \Gamma}{2} \delta(E - E_R).$$

The peak reaches the elastic unitarity limit independently of its width, while its integrated area is proportional to  $\Gamma$ .

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## 4 The Cluster Decomposition Principle

Up to this point we have not had much to say about the detailed structure of the Hamiltonian operator  $H$ . This operator can be defined by giving all its matrix elements between states with arbitrary numbers of particles. Equivalently, as we shall show here, any such operator may be expressed as a function of certain operators that create and destroy single particles. We saw in Chapter 1 that such creation and annihilation operators were first encountered in the canonical quantization of the electromagnetic field and other fields in the early days of quantum mechanics. They provided a natural formalism for theories in which massive particles as well as photons can be produced and destroyed, beginning in the early 1930s with Fermi's theory of beta decay.

However, there is a deeper reason for constructing the Hamiltonian out of creation and annihilation operators, which goes beyond the need to quantize any pre-existing field theory like electrodynamics, and has nothing to do with whether particles can actually be produced or destroyed. The great advantage of this formalism is that if we express the Hamiltonian as a sum of products of creation and annihilation operators, with suitable non-singular coefficients, then the  $S$ -matrix will automatically satisfy a crucial physical requirement, the cluster decomposition principle [1], which says in effect that distant experiments yield uncorrelated results. Indeed, it is for this reason that the formalism of creation and annihilation operators is widely used in non-relativistic quantum statistical mechanics, where the number of particles is typically fixed. In relativistic quantum theories, the cluster decomposition principle plays a crucial part in making field theory inevitable. There have been many attempts to formulate a relativistically invariant theory that would not be a local field theory, and it is indeed possible to construct theories that are not field theories and yet yield a Lorentz-invariant  $S$ -matrix for two-particle scattering [2], but such efforts have always run into trouble in sectors with more than two particles: either the three-particle  $S$ -matrix is not Lorentz-invariant, or else it violates the cluster decomposition principle.

In this chapter we will first discuss the basis of states containing arbitrary numbers of bosons and fermions, then define the creation and annihilation operators, and finally show how their use facilitates the construction of Hamiltonians that yield  $S$ -matrices satisfying the cluster decomposition condition.

### 4.1 Bosons and Fermions

The Hilbert space of physical states is spanned by states containing 0, 1, 2,  $\dots$  free particles. These can be free-particle states, or 'in' states, or 'out' states; for definiteness we shall deal here with the free-particle states  $|\mathbf{p}_1\sigma_1n_1; \mathbf{p}_2\sigma_2n_2; \dots\rangle$ , but all our results will apply equally to 'in' or 'out' states. As usual,  $\sigma$  labels spin  $z$ -components (or helicities, for massless particles) and  $n$  labels particle species.

We must now go into a matter that has been passed over in Chapter 3; the symmetry properties of these states. As far as we know, all particles are either *bosons* or *fermions*, the difference being that a state is unchanged by the interchange of two identical bosons,

and changes sign under the interchange of two identical fermions. That is

$$|\dots \mathbf{p}\sigma n \dots \mathbf{p}'\sigma' n \dots\rangle = \pm |\dots \mathbf{p}'\sigma' n \dots \mathbf{p}\sigma n \dots\rangle \quad (4.1.1)$$

with an upper or lower sign if  $n$  is a boson or a fermion, respectively, and dots representing other particles that may be present in the state. (Equivalently, this could be stated as a condition on the ‘wave functions,’ the coefficients of these multi-particle basis vectors in physically allowable state-vectors.) These two cases are often referred to as Bose or Fermi ‘statistics.’ We will see in the next chapter that Bose and Fermi statistics are only possible for particles that have integer or half-integer spins, respectively, but we shall not need this information in the present chapter. In this section we shall offer a non-rigorous argument that all particles must be either bosons or fermions, and then set up normalization conditions for multi-boson or multi-fermion states.

First note that if two particles with spins and momenta  $\mathbf{p}, \sigma$  and  $\mathbf{p}', \sigma'$  belong to identical species  $n$ , then the state-vectors  $|\dots \mathbf{p}\sigma n \dots \mathbf{p}'\sigma' n \dots\rangle$  and  $|\dots \mathbf{p}'\sigma' n \dots \mathbf{p}\sigma n \dots\rangle$  represent the same physical state; if this were not the case then the particles would be distinguished by their order in the labelling of the state-vector, and the first listed would not be identical with the second. Since the two state-vectors are physically indistinguishable, they must belong to the same ray, and so

$$|\dots \mathbf{p}\sigma n \dots \mathbf{p}'\sigma' n \dots\rangle = \alpha_n |\dots \mathbf{p}'\sigma' n \dots \mathbf{p}\sigma n \dots\rangle, \quad (4.1.2)$$

where  $\alpha_n$  is a complex number of unit absolute value. We may regard this as part of the definition of what we mean by identical particles.

The crux of the matter is to decide on what the phase factor  $\alpha_n$  may depend. *If* it depends only on the species index  $n$ , then we are nearly done. Interchanging the two particles in Eq. (4.1.2) again, we find

$$|\dots \mathbf{p}\sigma n \dots \mathbf{p}'\sigma' n \dots\rangle = \alpha_n^2 |\dots \mathbf{p}\sigma n \dots \mathbf{p}'\sigma' n \dots\rangle$$

so that  $\alpha_n^2 = 1$ , yielding Eq. (4.1.1) as the only two possibilities.

On what else could  $\alpha_n$  depend? It might depend on the numbers and species of the other particles in the state (indicated by dots in Eqs. (4.1.1) and (4.1.2)), but this would lead to the uncomfortable result that the symmetry of state-vectors under interchange of particles here on earth may depend on the presence of particles elsewhere in the universe. This is the sort of thing that is ruled out by the cluster decomposition principle, to be discussed later in this chapter. The phase  $\alpha_n$  cannot have any non-trivial dependence on the spins of the two particles that are interchanged, because then these spin-dependent phase factors would have to furnish a representation of the rotation group, and there are no non-trivial representations of the three-dimensional rotation group that are one-dimensional—that is, by phase factors. The phase  $\alpha_n$  might conceivably depend on the momenta of the two particles that are interchanged, but Lorentz invariance would require  $\alpha_n$  to depend only on the scalar  $p_1^\mu p_{2\mu}$ ; this is symmetric under interchange of particles 1 and 2, and therefore such dependence would not change the argument leading to the conclusion that  $\alpha_n^2 = 1$ .

The logical gap in the above argument is that (although our notation hides the fact) the states  $|\mathbf{p}_1\sigma_1n; \mathbf{p}_2\sigma_2n; \dots\rangle$  may carry a phase factor that depends on the path through momentum space by which the momenta of the particles are brought to the values  $p_1$ ,  $p_2$ , etc. In this case the interchange of two particles twice might change the state by a phase factor, so that  $\alpha_n^2 \neq 1$ . We will see in Section 9.7 that this is a real possibility in two-dimensional space, but not for three or more spatial dimensions.

What about interchanges of particles belonging to different species? If we like, we can avoid this question by simply agreeing from the beginning to label the state-vector by listing all photon momenta and helicities first, then all electron momenta and spin  $z$ -components, and so on through the table of elementary particle types. Alternatively, we can allow the particle labels to appear in any order, and *define* the state-vectors with particle labels in an arbitrary order as equal to the state-vector with particle labels in some standard order times phase factors, whose dependence on the interchange of particles of different species can be anything we like. In order to deal with symmetries like isospin invariance that relate particles of different species, it is convenient to adopt a convention that generalizes Eq. (4.1.1): the state-vector will be taken to be symmetric under interchange of any bosons with each other, or any bosons with any fermions, and antisymmetric with respect to interchange of any two fermions with each other, in all cases, whether the particles are of the same species or not.<sup>28</sup>

The normalization of these states must be defined in consistency with these symmetry conditions. To save writing, we will use a label  $q$  to denote all the quantum numbers of a single particle: its momentum,  $\mathbf{p}$ , spin  $z$ -component (or, for massless particles, helicity)  $\sigma$ , and species  $n$ . The  $N$ -particle states are thus labelled  $|q_1 \cdots q_N\rangle$  (with  $N = 0$  for the vacuum state  $|\Omega\rangle$ .) For  $N = 0$  and  $N = 1$  the question of symmetry does not arise: here we have

$$\langle\Omega|\Omega\rangle = 1 \quad (4.1.3)$$

and

$$\langle q'|q\rangle = \delta(q' - q), \quad (4.1.4)$$

where  $\delta(q' - q)$  is a product of all the delta functions and Kronecker deltas for the particle's quantum numbers,

$$\delta(q' - q) = \delta^3(\mathbf{p}' - \mathbf{p})\delta_{\sigma'\sigma}\delta_{n'n}. \quad (4.1.5)$$

On the other hand, for  $N = 2$  the states  $|q'_1q'_2\rangle$  and  $|q'_2q'_1\rangle$  are physically the same, so here we must take

$$\langle q'_1q'_2|q_1q_2\rangle = \delta(q'_1 - q_1)\delta(q'_2 - q_2) \pm \delta(q'_2 - q_1)\delta(q'_1 - q_2) \quad (4.1.6)$$

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<sup>28</sup>In fact, by the same reasoning, the symmetry or antisymmetry of the state-vector under interchange of particles of the same species but different helicities or spin  $z$ -components is purely conventional, because we could have agreed from the beginning to list first the momenta of photons of helicity  $+1$ , then the momenta of all photons of helicity  $-1$ , then the momenta of all electrons of spin  $z$ -component  $+\frac{1}{2}$ , and so on. We adopt the *convention* that the state-vector is symmetric or antisymmetric under interchange of identical bosons or fermions of different helicities or spin  $z$ -components in order to facilitate the use of rotational invariance.

the sign  $\pm$  being  $-$  if both particles are fermions and  $+$  otherwise. This obviously is consistent with the above stated symmetry properties of the states. More generally,

$$\langle q'_1 q'_2 \cdots q'_M | q_1 q_2 \cdots q_N \rangle = \delta_{NM} \sum_{\mathcal{P}} \delta_{\mathcal{P}} \prod_i \delta(q_i - q'_{\mathcal{P}i}). \quad (4.1.7)$$

The sum here is over all permutations  $\mathcal{P}$  of the integers  $1, 2, \dots, N$ . (For instance, in the first term in Eq. (4.1.6),  $\mathcal{P}$  is the identity,  $\mathcal{P}1 = 1$ ,  $\mathcal{P}2 = 2$ , while in the second term  $\mathcal{P}1 = 2$ ,  $\mathcal{P}2 = 1$ .) Also,  $\delta_{\mathcal{P}}$  is a sign factor equal to  $-1$  if  $\mathcal{P}$  involves an odd permutation of fermions (an odd number of fermion interchanges) and  $+1$  otherwise. It is easy to see that Eq. (4.1.7) has the desired symmetry or antisymmetry properties under interchange of the  $q_i$ , and also under interchange of the  $q'_j$ .

## 4.2 Creation and Annihilation Operators

Creation and annihilation operators may be defined in terms of their effect on the normalized multi-particle states discussed in the previous section. The *creation operator*  $a_q^\dagger$  (or in more detail,  $a_{\mathbf{p}, \sigma, n}^\dagger$ ) is defined as the operator that simply adds a particle with quantum numbers  $q$  at the front of the list of particles in the state

$$a_q^\dagger |q_1 q_2 \cdots q_N\rangle = |q q_1 q_2 \cdots q_N\rangle. \quad (4.2.1)$$

In particular, the  $N$ -particle state can be obtained by acting on the vacuum with  $N$  creation operators

$$a_{q_1}^\dagger a_{q_2}^\dagger \cdots a_{q_N}^\dagger |\Omega\rangle = |q_1 \cdots q_N\rangle. \quad (4.2.2)$$

It is conventional for this operator to be called  $a_q^\dagger$ ; its adjoint, which is then called  $a_q$ , may be calculated from Eq. (4.1.7). As we shall now show,  $a_q$  removes a particle from any state on which it acts, and is therefore known as an *annihilation operator*. In particular, when the particles  $q, q_1, \dots, q_N$  are either all bosons or all fermions, we have

$$a_q |q_1 q_2 \cdots q_N\rangle = \sum_{r=1}^N (\pm)^{r+1} \delta(q - q_r) |q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle, \quad (4.2.3)$$

with a  $+1$  or  $-1$  sign for bosons or fermions, respectively. (Here is the proof. We want to calculate the scalar product of  $a_q |q_1 q_2 \cdots q_N\rangle$  with an arbitrary state  $|q'_1 \cdots q'_M\rangle$ . Using Eq. (4.2.1), this is

$$\langle q'_1 \cdots q'_M | a_q |q_1 \cdots q_N\rangle = (a_q^\dagger |q'_1 \cdots q'_M\rangle)^\dagger |q_1 \cdots q_N\rangle = \langle q q'_1 \cdots q'_M | q_1 \cdots q_N\rangle.$$

We now use Eq. (4.1.7). The sum over permutations  $\mathcal{P}$  of  $1, 2, \dots, N$  can be written as a sum over the integer  $r$  that is permuted into the first place, i.e.  $\mathcal{P}r = 1$ , and over mappings  $\bar{\mathcal{P}}$  of the remaining integers  $1, \dots, r-1, r+1, \dots, N$  into  $1, \dots, N-1$ . Furthermore, the sign factor is

$$\delta_{\mathcal{P}} = (\pm)^{r-1} \delta_{\bar{\mathcal{P}}}$$

with upper and lower signs for bosons and fermions, respectively. Hence, using Eq. (4.1.7) twice,

$$\begin{aligned}\langle q'_1 \cdots q'_M | a_q | q_1 \cdots q_N \rangle &= \delta_{N,M+1} \sum_{r=1}^N \sum_{\overline{p}} (\pm)^{r-1} \delta_{\overline{p}} \delta(q - q_r) \prod_{i=1}^M \delta(q'_i - q_{\overline{p}_i}) \\ &= \delta_{N,M+1} \sum_{r=1}^N (\pm)^{r-1} \langle q'_1 \cdots q'_M | q_1 \cdots q_{r-1} q_{r+1} \cdots q_N \rangle.\end{aligned}$$

Both sides of Eq. (4.2.3) thus have the same matrix element with any state  $|q'_1 \cdots q'_M\rangle$ , and are therefore equal, as was to be shown.) As a special case of Eq. (4.2.3), we note that for both bosons and fermions,  $a_q$  annihilates the vacuum

$$a_q |\Omega\rangle = 0. \quad (4.2.4)$$

As defined here, the creation and annihilation operators satisfy an important commutation or anticommutation relation. Applying the operator  $a_{q'}$  to Eq. (4.2.1) and using Eq. (4.2.3) gives

$$\begin{aligned}a_{q'} a_q^\dagger |q_1 \cdots q_N\rangle &= \delta(q' - q) |q_1 \cdots q_N\rangle \\ &\quad + \sum_{r=1}^N (\pm)^{r+2} \delta(q' - q_r) |qq_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle.\end{aligned}$$

(The sign in the second term is  $(\pm)^{r+2}$  because  $q_r$  is in the  $(r+1)$ -th place in  $|qq_1 \cdots q_N\rangle$ .) On the other hand, applying the operator  $a_q^\dagger$  to Eq. (4.2.3) gives

$$a_q^\dagger a_{q'} |q_1 \cdots q_N\rangle = \sum_{r=1}^N (\pm)^{r+1} \delta(q' - q_r) |qq_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle.$$

Subtracting or adding, we have then

$$[a_{q'} a_q^\dagger \mp a_q^\dagger a_{q'}] |q_1 \cdots q_N\rangle = \delta(q' - q) |q_1 \cdots q_N\rangle.$$

But this holds for all states  $|q_1 \cdots q_N\rangle$  (and may easily be seen to hold also for states containing both bosons and fermions) and therefore implies the operator relation

$$a_{q'} a_q^\dagger \mp a_q^\dagger a_{q'} = \delta(q' - q). \quad (4.2.5)$$

In addition, Eq. (4.2.2) gives immediately

$$a_{q'}^\dagger a_q^\dagger \mp a_q^\dagger a_{q'}^\dagger = 0 \quad (4.2.6)$$

and so also

$$a_{q'} a_q \mp a_q a_{q'} = 0. \quad (4.2.7)$$

As always, the top and bottom signs apply for bosons and fermions, respectively. According to the conventions discussed in the previous section, the creation and/or annihilation

operators for particles of two different species commute if either particle is a boson, and anticommute if both are fermions.

The above discussion could have been presented in reverse order (and in most textbooks usually is). That is, we could have started with the commutation or anticommutation relations Eqs. (4.2.5)–(4.2.7), derived from the canonical quantization of some given field theory. Multi-particle states would have then been defined by Eq. (4.2.2), and their scalar products Eq. (4.1.7) derived from the commutation or anticommutation relations. In fact, as discussed in Chapter 1, such a treatment would be much closer to the way that this formalism developed historically. We have followed an unhistorical approach here because we want to free ourselves from any dependence on pre-existing field theories, and rather wish to understand why field theories are the way they are.

We will now prove the fundamental theorem quoted at the beginning of this chapter: *any operator  $\mathcal{O}$  may be expressed as a sum of products of creation and annihilation operators*

$$\begin{aligned} \mathcal{O} = \sum_{N=0}^{\infty} \sum_{M=0}^{\infty} \int dq'_1 \cdots dq'_N dq_1 \cdots dq_M \\ \cdot a_{q'_1}^\dagger \cdots a_{q'_N}^\dagger a_{q_M} \cdots a_{q_1} \\ \cdot C_{NM}(q'_1 \cdots q'_N q_1 \cdots q_M). \end{aligned} \quad (4.2.8)$$

That is, we want to show that the  $C_{NM}$  coefficients can be chosen to give the matrix elements of this expression any desired values. We do this by mathematical induction. First, it is trivial that by choosing  $C_{00}$  properly, we can give  $\langle \Omega | \mathcal{O} | \Omega \rangle$  any desired value, irrespective of the values of  $C_{NM}$  with  $N > 0$  and/or  $M > 0$ . We need only use Eq. (4.2.4) to see that Eq. (4.2.8) has the vacuum expectation value

$$\langle \Omega | \mathcal{O} | \Omega \rangle = C_{00}.$$

Now suppose that the same is true for all matrix elements of  $\mathcal{O}$  between  $N$ - and  $M$ -particle states, with  $N < L, M \leq K$  or  $N \leq L, M < K$ ; that is, that these matrix elements have been given some desired values by an appropriate choice of the corresponding coefficients  $C_{NM}$ . To see that the same is then also true of matrix elements of  $\mathcal{O}$  between any  $L$ - and  $K$ -particle states, use Eq. (4.2.8) to evaluate

$$\begin{aligned} \langle q'_1 \cdots q'_L | \mathcal{O} | q_1 \cdots q_K \rangle = L!K!C_{LK}(q'_1 \cdots q'_L q_1 \cdots q_K) \\ + \text{terms involving } C_{NM} \text{ with } N < L, M \leq K \text{ or } N \leq L, M < K. \end{aligned}$$

Whatever values have already been given to  $C_{NM}$  with  $N < L, M \leq K$  or  $N \leq L, M < K$ , there is clearly some choice of  $C_{LK}$  which gives this matrix element any desired value.

Of course, an operator need not be expressed in the form (4.2.8), with all creation operators to the left of all annihilation operators. (This is often called the ‘normal’ order of the operators.) However, if the formula for some operator has the creation and annihilation operators in some other order, we can always bring the creation operators to the left of the annihilation operators by repeated use of the commutation or anticommutation relations, picking up new terms from the delta function in Eq. (4.2.5).

For instance, consider any sort of additive operator  $F$  (like momentum, charge, etc.) for which

$$F|q_1 \cdots q_N\rangle = (f(q_1) + \cdots + f(q_N))|q_1 \cdots q_N\rangle. \quad (4.2.9)$$

Such an operator can be written as in Eq. (4.2.8), but using only the term with  $N = M = 1$ :

$$F = \int dq a_q^\dagger a_q f(q). \quad (4.2.10)$$

In particular, the free-particle Hamiltonian is always

$$H_0 = \int dq a_q^\dagger a_q E(q) \quad (4.2.11)$$

where  $E(q)$  is the single-particle energy

$$E(\mathbf{p}, \sigma, n) = \sqrt{\mathbf{p}^2 + m_n^2}.$$

We will need the transformation properties of the creation and annihilation operators for various symmetries. First, let's consider inhomogeneous proper orthochronous Lorentz transformations. Recall that the  $N$ -particle states have the Lorentz transformation property

$$\begin{aligned} & U_0(\Lambda, a)|\mathbf{p}_1\sigma_1n_1; \mathbf{p}_2\sigma_2n_2; \dots\rangle \\ &= e^{-i(\Lambda p_1)\cdot a} e^{-i(\Lambda p_2)\cdot a} \dots \sqrt{\frac{(\Lambda p_1)^0(\Lambda p_2)^0 \cdots}{p_1^0 p_2^0 \cdots}} \\ &\quad \cdot \sum_{\bar{\sigma}_1 \bar{\sigma}_2 \cdots} D_{\bar{\sigma}_1 \sigma_1}^{(j_1)}(W(\Lambda, p_1)) D_{\bar{\sigma}_2 \sigma_2}^{(j_2)}(W(\Lambda, p_2)) \cdots \\ &\quad \cdot |\mathbf{p}_{1\Lambda} \bar{\sigma}_1 n_1; \mathbf{p}_{2\Lambda} \bar{\sigma}_2 n_2; \dots\rangle. \end{aligned}$$

Here  $\mathbf{p}_\Lambda$  is the three-vector part of  $\Lambda p$ ,  $D_{\bar{\sigma}\sigma}^{(j)}(R)$  is the same unitary spin- $j$  representation of the three-dimensional rotation group as used in Section 2.5, and  $W(\Lambda, p)$  is the particular rotation

$$W(\Lambda, p) = L^{-1}(\Lambda p) \Lambda L(p),$$

where  $L(p)$  is the standard ‘boost’ that takes a particle of mass  $m$  from rest to four-momentum  $p^\mu$ . (Of course,  $m$  and  $j$  depend on the species label  $n$ . This is all for  $m \neq 0$ ; we will return to the massless particle case in the following chapter.) Now, these states can be expressed as in Eq. (4.2.2)

$$|\mathbf{p}_1\sigma_1n_1; \mathbf{p}_2\sigma_2n_2; \dots\rangle = a_{\mathbf{p}_1, \sigma_1, n_1}^\dagger a_{\mathbf{p}_2, \sigma_2, n_2}^\dagger \cdots |\Omega\rangle,$$

where  $|\Omega\rangle$  is the Lorentz-invariant vacuum state

$$U_0(\Lambda, a)|\Omega\rangle = |\Omega\rangle.$$

In order that the state (4.2.2) should transform properly, it is necessary and sufficient that the creation operator have the transformation rule

$$\begin{aligned} & U_0(\Lambda, a) a_{\mathbf{p}, \sigma, n}^\dagger U_0^{-1}(\Lambda, a) \\ &= e^{-i(\Lambda p)\cdot a} \sqrt{\frac{(\Lambda p)^0}{p^0}} \sum_{\bar{\sigma}} D_{\bar{\sigma}\sigma}^{(j)}(W(\Lambda, p)) a_{\mathbf{p}_\Lambda, \bar{\sigma}, n}^\dagger. \end{aligned} \quad (4.2.12)$$

In the same way, the operators  $C$ ,  $P$ , and  $T$ , that induce charge-conjugation, space inversion, and time-reversal transformations on free particle states<sup>29</sup> transform the creation operators as:

$$Ca_{\mathbf{p},\sigma,n}^\dagger C^{-1} = \xi_n a_{\mathbf{p},\sigma,n^c}^\dagger, \quad (4.2.13)$$

$$Pa_{\mathbf{p},\sigma,n}^\dagger P^{-1} = \eta_n a_{-\mathbf{p},\sigma,n}^\dagger, \quad (4.2.14)$$

$$Ta_{\mathbf{p},\sigma,n}^\dagger T^{-1} = \zeta_n (-1)^{j-\sigma} a_{-\mathbf{p},-\sigma,n}^\dagger. \quad (4.2.15)$$

As mentioned in the previous section, although we have been dealing with operators that create and annihilate particles in free-particle states, the whole formalism can be applied to ‘in’ and ‘out’ states, in which case we would introduce operators  $a_{\text{in}}$  and  $a_{\text{out}}$  defined in the same way by their action on these states. These operators satisfy a Lorentz transformation rule just like Eq. (4.2.12), but with the true Lorentz transformation operator  $U(\Lambda, a)$  instead of the free-particle operator  $U_0(\Lambda, a)$ .

### 4.3 Cluster Decomposition and Connected Amplitudes

It is one of the fundamental principles of physics (indeed, of all science) that experiments that are sufficiently separated in space have unrelated results. The probabilities for various collisions measured at Fermilab should not depend on what sort of experiments are being done at CERN at the same time. If this principle were not valid, then we could never make any predictions about any experiment without knowing everything about the universe.

In  $S$ -matrix theory, the cluster decomposition principle states that if multi-particle processes  $\alpha_1 \rightarrow \beta_1$ ,  $\alpha_2 \rightarrow \beta_2$ , ...,  $\alpha_N \rightarrow \beta_N$  are studied in  $N$  very distant laboratories, then the  $S$ -matrix element for the overall process factorizes. That is,<sup>30</sup>

$$S_{\beta_1+\beta_2+\dots+\beta_N, \alpha_1+\alpha_2+\dots+\alpha_N} \longrightarrow S_{\beta_1\alpha_1} S_{\beta_2\alpha_2} \cdots S_{\beta_N\alpha_N} \quad (4.3.1)$$

if for all  $i \neq j$ , *all* of the particles in states  $\alpha_i$  and  $\beta_i$  are at a great spatial distance from *all* of the particles in states  $\alpha_j$  and  $\beta_j$ . This factorization of  $S$ -matrix elements will ensure a factorization of the corresponding transition probabilities, corresponding to uncorrelated experimental results.

There is a combinatoric trick that allows us to rewrite Eq. (4.3.1) in a more transparent way. Suppose we define the *connected* part of the  $S$ -matrix,  $S_{\beta\alpha}^C$ , by the formula<sup>31</sup>

$$S_{\beta\alpha} = \sum_{\text{PART}} (\pm) S_{\beta_1\alpha_1}^C S_{\beta_2\alpha_2}^C \cdots \quad (4.3.2)$$

<sup>29</sup>We omit the subscript ‘0’ on these operators, because in virtually all cases where  $C$ ,  $P$ , and/or  $T$  are conserved, the operators that induce these transformations on ‘in’ and ‘out’ states are the same as those defined by their action on free-particle states. This is not the case for continuous Lorentz transformations, for which it is necessary to distinguish between the operators  $U(\Lambda, a)$  and  $U_0(\Lambda, a)$ .

<sup>30</sup>We are here returning to the notation used in Chapter 3; Greek letters  $\alpha$  or  $\beta$  stand for a collection of particles, including for each particle a specification of its momentum, spin, and species. Also,  $\alpha_1 + \alpha_2 + \dots + \alpha_N$  is the state formed by combining all the particles in the states  $\alpha_1, \alpha_2, \dots$ , and  $\alpha_N$ , and likewise for  $\beta_1 + \beta_2 + \dots + \beta_N$ .

<sup>31</sup>This decomposition has been used in classical statistical mechanics by Ursell, Mayer, and others, and in quantum statistical mechanics by Lee and Yang and others [3]. It has also been used to calculate many-body ground state energies by Goldstone [4] and Hugenholtz [5]. In all of these applications the purpose of isolating the connected parts of Green’s functions, partition functions, resolvents, etc., is to deal with



Here the sum is over all different ways of partitioning the particles in the state  $\alpha$  into clusters  $\alpha_1, \alpha_2, \dots$ , and likewise a sum over all ways of partitioning the particles in the state  $\beta$  into clusters  $\beta_1, \beta_2, \dots$ ; not counting as different those that merely arrange particles within a given cluster or permute whole clusters. The sign is  $+$  or  $-$  according to whether the rearrangements  $\alpha \rightarrow \alpha_1 \alpha_2 \dots$  and  $\beta \rightarrow \beta_1 \beta_2 \dots$  involve altogether an even or an odd number of fermion interchanges, respectively. The term ‘connected’ is used because of the interpretation of  $S_{\beta\alpha}^C$  in terms of diagrams representing different contributions in perturbation theory, to be discussed in the next section.

This is a recursive definition. For each  $\alpha$  and  $\beta$ , the sum on the right-hand side of Eq. (4.3.2) consists of a term  $S_{\beta\alpha}^C$ , plus a sum  $\sum'$  over products of two or more  $S^C$ -matrix elements, with a total number of particles in each of the states  $\alpha_j$  and  $\beta_j$  that is *less* than the number of particles in the states  $\alpha$  and  $\beta$

$$S_{\beta\alpha} = S_{\beta\alpha}^C + \sum_{\text{PART}}' (\pm) S_{\beta_1\alpha_1}^C S_{\beta_2\alpha_2}^C \dots$$

Suppose that the  $S^C$ -matrix elements in this sum have already been chosen in such a way that Eq. (4.3.2) is satisfied for states  $\beta, \alpha$  containing together fewer than, say,  $N$  particles. Then no matter what values are found in this way for the  $S^C$ -matrix elements appearing in the sum  $\sum'$ , we can always choose the remaining term  $S_{\beta\alpha}^C$  so that Eq. (4.3.2) is also satisfied for states  $\alpha, \beta$  containing a total of  $N$  particles.<sup>32</sup> Thus Eq. (4.3.2) contains no information in itself; it is merely a definition of  $S^C$ .

If the states  $\alpha$  and  $\beta$  each consist of just a single particle, say with quantum numbers  $q$  and  $q'$  respectively, then the only term on the right-hand-side of Eq. (4.3.2) is just  $S_{\beta\alpha}^C$  itself, so for one-particle states

$$S_{q'q}^C = S_{q'q} = \delta(q' - q). \quad (4.3.3)$$

(Apart from possible degeneracies, the fact that  $S_{q'q}$  is proportional to  $\delta(q' - q)$  follows from conservation laws. The absence of any proportionality factor in Eq. (4.3.3) is based on a suitable choice of the relative phase of ‘in’ and ‘out’ states.) We are here assuming that single-particle states are stable, so that there are no transitions between single-particle states and any others, such as the vacuum.

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objects with a simple volume dependence. This is essentially our purpose too, because as we shall see, the crucial property of the connected parts of the  $S$ -matrix is that they are proportional to a single momentum-conservation delta function, and in a box the delta function becomes a Kronecker delta times the volume. The cluster decomposition is also the same formal device as that used in the theory of noise [6] to decompose the correlation function of several random variables into its ‘cumulants’; if the random variable receives contributions from a large number  $N$  of independent fluctuations, then each cumulant is proportional to  $N$ .

<sup>32</sup>A technicality should be mentioned here. This argument works only if we neglect the possibility that for one or more of the connected  $S$ -matrix elements in Eq. (4.3.2), the states  $\alpha_j$  and  $\beta_j$  both contain no particles at all. We must therefore define the connected vacuum-vacuum element  $S_{0,0}^C$  to be zero. We do not use Eq. (4.3.2) for the vacuum-vacuum  $S$ -matrix  $S_{0,0}$ , which in the absence of time-varying external fields is simply defined to be unity,  $S_{0,0} = 1$ . We will have more to say about the vacuum-vacuum amplitude in the presence of external fields in Volume II.

For transitions between two-particle states, Eq. (4.3.2) reads

$$S_{q'_1 q'_2, q_1 q_2} = S_{q'_1 q'_2, q_1 q_2}^C + \delta(q'_1 - q_1) \delta(q'_2 - q_2) \pm \delta(q'_1 - q_2) \delta(q'_2 - q_1). \quad (4.3.4)$$

(We are here using Eq. (4.3.3).) The sign  $\pm$  is  $-$  if both particles are fermions, and otherwise  $+$ . We recognize that the two delta function terms just add up to the norm (4.1.6), so here  $S_{\beta\alpha}^C$  is just  $(S - \mathbf{1})_{\beta\alpha}$ . But the general case is more complicated.

For transitions between three-particle or four-particle states, Eq. (4.3.2) reads

$$\begin{aligned} S_{q'_1 q'_2 q'_3, q_1 q_2 q_3} &= S_{q'_1 q'_2 q'_3, q_1 q_2 q_3}^C \\ &+ \delta(q'_1 - q_1) S_{q'_2 q'_3, q_2 q_3}^C \pm \text{permutations} \\ &+ \delta(q'_1 - q_1) \delta(q'_2 - q_2) \delta(q'_3 - q_3) \pm \text{permutations} \end{aligned} \quad (4.3.5)$$

and

$$\begin{aligned} S_{q'_1 q'_2 q'_3 q'_4, q_1 q_2 q_3 q_4} &= S_{q'_1 q'_2 q'_3 q'_4, q_1 q_2 q_3 q_4}^C \\ &+ S_{q'_1 q'_2, q_1 q_2}^C S_{q'_3 q'_4, q_3 q_4}^C \pm \text{permutations} \\ &+ \delta(q'_1 - q_1) S_{q'_2 q'_3 q'_4, q_2 q_3 q_4}^C \pm \text{permutations} \\ &+ \delta(q'_1 - q_1) \delta(q'_2 - q_2) S_{q'_3 q'_4, q_3 q_4}^C \pm \text{permutations} \\ &+ \delta(q'_1 - q_1) \delta(q'_2 - q_2) \delta(q'_3 - q_3) \delta(q'_4 - q_4) \pm \text{permutations}. \end{aligned} \quad (4.3.6)$$

(Taking account of all permutations, there are a total of  $1 + 9 + 6 = 16$  terms in Eq. (4.3.5) and  $1 + 18 + 16 + 72 + 24 = 131$  terms in Eq. (4.3.6). If we had not assumed that one-particle states are stable, there would be even more terms.) As explained previously, the definition of  $S_{\beta\alpha}^C$  is recursive: we use Eq. (4.3.4) to define  $S_{\beta\alpha}^C$  for two-particle states, then use this definition in Eq. (4.3.5) when we define  $S_{\beta\alpha}^C$  for three-particle states, then use both of these definitions in Eq. (4.3.6) to obtain the definition of  $S_{\beta\alpha}^C$  for four-particle states, and so on.

The point of this definition of the connected part of the  $S$ -matrix is that the cluster decomposition principle is equivalent to the requirement that  $S_{\beta\alpha}^C$  must vanish when any one or more of the particles in the states  $\beta$  and/or  $\alpha$  are far away in space from the others.<sup>33</sup> To see this, suppose that the particles in the states  $\beta$  and  $\alpha$  are grouped into clusters  $\beta_1, \beta_2, \dots$  and  $\alpha_1, \alpha_2, \dots$ , and that all particles in the set  $\alpha_i + \beta_i$  are far from all particles in the set  $\alpha_j + \beta_j$  for any  $j \neq i$ . Then if  $S_{\beta'\alpha'}^C$  vanishes if any particles in  $\beta'$  or  $\alpha'$  are far from the others, it vanishes if any particles in these states are in different clusters, so the definition Eq. (4.3.2) yields

$$\begin{aligned} S_{\beta\alpha} &\longrightarrow \sum_{\text{PART}}^{(1)} (\pm) S_{\beta_{11}\alpha_{11}}^C S_{\beta_{12}\alpha_{12}}^C \cdots \\ &\quad \cdot \sum_{\text{PART}}^{(2)} (\pm) S_{\beta_{21}\alpha_{21}}^C S_{\beta_{22}\alpha_{22}}^C \cdots \cdots \cdots, \end{aligned} \quad (4.3.7)$$

<sup>33</sup>In order to give a meaning to ‘far,’ we will have to Fourier transform  $S^C$ , so that each three-momentum label  $\mathbf{p}$  is replaced with a spatial coordinate three-vector  $\mathbf{x}$ .

where  $\sum^{(j)}$  is a sum over all different ways of partitioning the clusters  $\beta_j$  and  $\alpha_j$  into subclusters  $\beta_{j1}, \beta_{j2}, \dots$  and  $\alpha_{j1}, \alpha_{j2}, \dots$ . But referring back to Eq. (4.3.2), this is just the desired factorization property (4.3.1).

For instance, suppose that in the four-particle reaction  $1234 \rightarrow 1'2'3'4'$ , we let particles 1, 2, 1', and 2' be very far from 3, 4, 3', and 4'. Then if  $S_{\beta\alpha}^C$  vanishes when any particles in  $\beta$  and/or  $\alpha$  are far from the others, the only terms in Eq. (4.3.6) that survive (in an even more abbreviated notation) are

$$\begin{aligned} S_{1'2'3'4', 1234} &\longrightarrow S_{1'2', 12}^C S_{3'4', 34}^C \\ &\quad + (\delta_{1'1} \delta_{2'2} \pm \delta_{1'2} \delta_{2'1}) S_{3'4', 34}^C \\ &\quad + (\delta_{3'3} \delta_{4'4} \pm \delta_{3'4} \delta_{4'3}) S_{1'2', 12}^C \\ &\quad + (\delta_{1'1} \delta_{2'2} \pm \delta_{1'2} \delta_{2'1}) (\delta_{3'3} \delta_{4'4} \pm \delta_{3'4} \delta_{4'3}). \end{aligned}$$

Comparison with Eq. (4.3.4) shows that this is just the required factorization condition (4.3.1)

$$S_{1'2'3'4', 1234} \longrightarrow S_{1'2', 12} S_{3'4', 34}.$$

We have formulated the cluster decomposition principle in coordinate space, as the condition that  $S_{\beta\alpha}^C$  vanishes if any particles in the states  $\beta$  or  $\alpha$  are far from any others. It is convenient for us to reexpress this in momentum space. The coordinate space matrix elements are defined as a Fourier transform

$$\begin{aligned} S_{\mathbf{x}'_1 \mathbf{x}'_2 \dots, \mathbf{x}_1 \mathbf{x}_2 \dots}^C &\equiv \int d^3 p'_1 d^3 p'_2 \dots d^3 p_1 d^3 p_2 \dots S_{\mathbf{p}'_1 \mathbf{p}'_2 \dots, \mathbf{p}_1 \mathbf{p}_2 \dots}^C \\ &\quad \cdot e^{i\mathbf{p}'_1 \cdot \mathbf{x}'_1} e^{i\mathbf{p}'_2 \cdot \mathbf{x}'_2} \dots e^{-i\mathbf{p}_1 \cdot \mathbf{x}_1} e^{-i\mathbf{p}_2 \cdot \mathbf{x}_2} \dots \end{aligned} \quad (4.3.8)$$

(We are here temporarily dropping spin and species labels, which just go along with the momentum or coordinate labels.) If  $|S_{\mathbf{p}'_1 \mathbf{p}'_2 \dots, \mathbf{p}_1 \mathbf{p}_2 \dots}^C|$  were sufficiently well behaved (to be specific, if it were Lebesgue integrable) then according to the Riemann–Lebesgue theorem [7] the integral (4.3.8) would vanish when any combination of spatial coordinates goes to infinity. Now, this is certainly too strong a requirement. Translational invariance tells us that the connected part of the  $S$ -matrix, like the  $S$ -matrix itself, can only depend on differences of coordinate vectors, and therefore does not change at all if all of the  $x_i$  and  $x'_j$  vary together, with their differences held constant. This requires that the elements of  $S^C$  in a momentum basis must, like those of  $S$ , be proportional to a three-dimensional delta function that ensures momentum conservation (and makes  $|S_{\mathbf{p}'_1 \mathbf{p}'_2 \dots, \mathbf{p}_1 \mathbf{p}_2 \dots}^C|$  *not* Lebesgue integrable), as well as the energy-conservation delta function required by scattering theory. That is, we can write

$$\begin{aligned} S_{\mathbf{p}'_1 \mathbf{p}'_2 \dots, \mathbf{p}_1 \mathbf{p}_2 \dots}^C &= \delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 + \dots - \mathbf{p}_1 - \mathbf{p}_2 - \dots) \\ &\quad \cdot \delta(E'_1 + E'_2 + \dots - E_1 - E_2 - \dots) \\ &\quad \cdot C_{\mathbf{p}'_1 \mathbf{p}'_2 \dots, \mathbf{p}_1 \mathbf{p}_2 \dots} \end{aligned} \quad (4.3.9)$$

This is no problem: the cluster decomposition principle only requires that Eq. (4.3.8) vanish when the *differences* among some of the  $x_i$  and/or  $x'_i$  become large. However, if

$C$  itself in Eq. (4.3.9) contained additional delta functions of linear combinations of the three-momenta, then this principle would not be satisfied. For instance, suppose that there were a delta function in  $C$  that required that the sum of the  $\mathbf{p}'_i$  and  $-\mathbf{p}_i$  for some subset of the particles vanished. Then Eq. (4.3.8) would not vary if all of the  $\mathbf{x}'_i$  and  $\mathbf{x}_i$  for the particles in that subset moved together (with constant differences) away from all the other  $\mathbf{x}'_k$  and  $\mathbf{x}_k$ , in contradiction to the cluster decomposition principle. Loosely speaking then, the cluster decomposition principle simply says that *the connected part of the  $S$ -matrix, unlike the  $S$ -matrix itself, contains just a single momentum-conservation delta function.*

In order to put this a bit more precisely, we can say that the coefficient function  $C_{\mathbf{p}'_1 \mathbf{p}'_2 \dots, \mathbf{p}_1 \mathbf{p}_2 \dots}$  in Eq. (4.3.9) is a smooth function of its momentum labels. But how smooth? It would be most straightforward if we could simply require that  $C_{\mathbf{p}'_1 \mathbf{p}'_2 \dots, \mathbf{p}_1 \mathbf{p}_2 \dots}$  be analytic in all of the momenta at  $\mathbf{p}'_1 = \mathbf{p}'_2 = \dots = \mathbf{p}_1 = \mathbf{p}_2 = \dots = 0$ . This requirement would indeed guarantee that  $S^C_{\mathbf{x}'_1 \mathbf{x}'_2 \dots, \mathbf{x}_1 \mathbf{x}_2 \dots}$  vanishes exponentially fast when any of the  $\mathbf{x}$  and  $\mathbf{x}'$  is very distant from any of the other  $\mathbf{x}$  and  $\mathbf{x}'$ . However, an exponential fall-off of  $S^C$  is not an essential part of the cluster decomposition principle, and, in fact, the requirement of analyticity is not met in all theories. Most notably, in theories with massless particles,  $S^C$  can have poles at certain values of the  $\mathbf{p}$  and  $\mathbf{p}'$ . For instance, as we will see in Chapter 10, if a massless particle can be emitted in the transition  $1 \rightarrow 3$  and absorbed in the transition  $2 \rightarrow 4$ , then  $S^C_{34,12}$  will have a term proportional to  $1/(\mathbf{p}_1 - \mathbf{p}_3)^2$ . After Fourier transforming, such poles yield terms in  $S^C_{\mathbf{x}'_1 \mathbf{x}'_2 \dots, \mathbf{x}_1 \mathbf{x}_2 \dots}$  that fall off only as negative powers of coordinate differences [1]. There is no need to formulate the cluster decomposition principle so stringently that such behavior is ruled out. Thus the ‘smoothness’ condition on  $S^C$  should be understood to allow various poles and branch-cuts at certain values of the  $\mathbf{p}$  and  $\mathbf{p}'$ , but not singularities as severe as delta functions.

#### 4.4 Structure of the Interaction

We now ask, what sort of Hamiltonian will yield an  $S$ -matrix that satisfies the cluster decomposition principle? It is here that the formalism of creation and annihilation operators comes into its own. The answer is contained in the theorem that the  $S$ -matrix satisfies the cluster decomposition principle if (and as far as I know, only if) the Hamiltonian can be expressed as in Eq. (4.2.8):

$$\begin{aligned}
H = \sum_{N=0}^{\infty} \sum_{M=0}^{\infty} \int & dq'_1 \cdots dq'_N dq_1 \cdots dq_M \\
& \cdot a_{q'_1}^\dagger \cdots a_{q'_N}^\dagger a_{q_M} \cdots a_{q_1} \\
& \cdot h_{NM}(q'_1 \cdots q'_N, q_1 \cdots q_M).
\end{aligned} \tag{4.4.1}$$

with coefficient functions  $h_{NM}$  that contain just a *single* three-dimensional momentum-conservation delta function (returning here briefly to a more explicit notation)

$$\begin{aligned}
& h_{NM}(\mathbf{p}'_1 \sigma'_1 n'_1 \cdots \mathbf{p}'_N \sigma'_N n'_N, \mathbf{p}_1 \sigma_1 n_1 \cdots \mathbf{p}_M \sigma_M n_M) \\
& = \delta^3(\mathbf{p}'_1 + \cdots + \mathbf{p}'_N - \mathbf{p}_1 - \cdots - \mathbf{p}_M) \\
& \cdot \tilde{h}_{NM}(\mathbf{p}'_1 \sigma'_1 n'_1 \cdots \mathbf{p}'_N \sigma'_N n'_N, \mathbf{p}_1 \sigma_1 n_1 \cdots \mathbf{p}_M \sigma_M n_M),
\end{aligned} \tag{4.4.2}$$

where  $\tilde{h}_{NM}$  contains no delta function factors. Note that Eq. (4.4.1) by itself has no content—we saw in Section 4.2 that *any* operator can be put in this form. It is only Eq. (4.4.1) combined with the requirement that  $h_{NM}$  has only the single delta function shown in Eq. (4.4.2) that guarantees that the  $S$ -matrix satisfies the cluster decomposition principle.

The validity of this theorem in perturbation theory will become obvious when we develop the Feynman diagram formalism in Chapter 6. The trusting reader may prefer to skip the rest of the present chapter, and move on to consider the implications of this theorem in Chapter 5. However, the proof has some instructive features, and will help to clarify in what sense the field theory of the next chapter is inevitable.

To prove this theorem, we make use of perturbation theory in its time-dependent form. (One of the advantages of time-dependent perturbation theory is that it makes the combinatorics underlying the cluster decomposition principle much more transparent; if  $E$  is a sum of one-particle energies then  $e^{-iEt}$  is a product of functions of the individual energies, while  $[E - E_\alpha + i\epsilon]^{-1}$  is not.) The  $S$ -matrix is given by Eq. (3.5.10) as<sup>34</sup>

$$S_{\beta\alpha} = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 \cdots dt_n \langle \beta | T \{ H_I(t_1) \cdots H_I(t_n) \} | \alpha \rangle, \quad (4.4.3)$$

where the Hamiltonian is split into a free-particle part  $H_0$  and an interaction  $H_{\text{int}}$ , and

$$H_I(t) \equiv \exp(iH_0 t) H_{\text{int}} \exp(-iH_0 t). \quad (4.4.4)$$

Now, the states  $|\alpha\rangle$  and  $|\beta\rangle$  may be expressed as in Eq. (4.2.2) as products of creation operators acting on the vacuum  $|\Omega\rangle$ , and  $H_I(t)$  is itself a sum of products of creation and annihilation operators, so each term in the sum (4.4.3) may be written as a sum of vacuum expectation values of products of creation and annihilation operators. By using the commutation or anticommutation relations (4.2.5) we may move each annihilation operator in turn to the right past all the creation operators. For each annihilation operator moved to the right past a creation operator we have two terms, as shown by writing Eq. (4.2.5) in the form

$$a_{q'} a_q^\dagger = \pm a_q^\dagger a_{q'} + \delta(q' - q).$$

Moving other creation operators past the annihilation operator in the first term generates yet more terms. But Eq. (4.2.4) shows that any annihilation operator that moves all the way to the right and acts on  $|\Omega\rangle$  gives zero, so in the end all we have left is the delta functions. In this way, the vacuum expectation value of a product of creation and annihilation operators is given by a sum of different terms, each term equal to a product of delta functions and  $\pm$  signs from the commutators or anticommutators. It follows that each term in Eq. (4.4.3) may be expressed as a sum of terms, each term equal to a product of delta functions and  $\pm$  signs from the commutators or anticommutators and whatever factors are contributed by  $H_I(t)$ , integrated over all the times and integrated and summed over the momenta, spins, and species in the arguments of the delta functions.

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<sup>34</sup>We are now adopting the convention that for  $n = 0$ , the time-ordered product in Eq. (4.4.3) is taken as the unit operator, so the  $n = 0$  term in the sum just yields the term  $\delta(\beta - \alpha)$  in  $S_{\beta\alpha}$ .

Each of the terms generated in this way may be symbolized by a diagram. (This is not yet the full Feynman diagram formalism, because we are not yet going to associate numerical quantities with the ingredients in the diagrams; we are using the diagrams here only as a way of keeping track of three-momentum delta functions.) Draw  $n$  points, called *vertices*, one for each  $H_I(t)$  operator. For each delta function produced when an annihilation operator in one of these  $H_I(t)$  operators moves past a creation operator in the initial state  $|\alpha\rangle$ , draw a line coming into the diagram from below that ends at the corresponding vertex. For each delta function produced when an annihilation operator in the adjoint of the final state  $\langle\beta|$  moves past a creation operator in one of the  $H_I(t)$ , draw a line from the corresponding vertex upwards out of the diagram. For each delta function produced when an annihilation operator in one  $H_I(t)$  moves past a creation operator in another  $H_I(t)$  draw a line between the two corresponding vertices. Finally, for each delta function produced when an annihilation operator in the adjoint of the final state moves past a creation operator in the initial state, draw a line from bottom to top, right through the diagrams. Each of the delta functions associated with one of these lines enforces the equality of the momentum arguments of the pair of creation and annihilation operators represented by the line. There is also at least one delta function contributed by each of the vertices, which enforces the conservation of the total three-momentum at the vertex.

Such a diagram may be connected (every point connected to every other by a set of lines) and if not connected, it breaks up into a number of connected pieces. The  $H_I(t)$  operator associated with a vertex in one connected component effectively commutes with the  $H_I(t)$  associated with any vertex in any other connected component, because for this diagram, we are not including any terms in which an annihilation operator in one vertex destroys a particle that is produced by a creation operator in the other vertex—if we did, the two vertices would be in the same connected component. Thus the matrix element in Eq. (4.4.3) can be expressed as a sum over *products* of contributions, one from each connected component:

$$\begin{aligned} & \langle\beta|T\{H_I(t_1)\cdots H_I(t_n)\}|\alpha\rangle \\ &= \sum_{\text{clustering}} (\pm) \prod_{j=1}^{\nu} \left( \langle\beta_j|T\{H_I(t_{j1})\cdots H_I(t_{jn_j})\}|\alpha_j\rangle \right)_C. \end{aligned} \quad (4.4.5)$$

Here the sum is over all ways of splitting up the incoming and outgoing particles and  $H_I(t)$  operators into  $\nu$  clusters (including a sum over  $\nu$  from 1 to  $n$ ) with the  $n_j$  operators  $H_I(t_{j1})\cdots H_I(t_{jn_j})$  and the subsets of initial particles  $\alpha_j$  and final particles  $\beta_j$  all in the  $j$ th cluster. Of course, this means that

$$n = n_1 + \cdots + n_\nu$$

and also the set  $\alpha$  is the union of all the particles in the subsets  $\alpha_1, \alpha_2, \dots, \alpha_\nu$ , and likewise for the final state. Some of the clusters in Eq. (4.4.5) may contain no vertices at all, i.e.,  $n_j = 0$ ; for these factors, we must take the matrix element factor in Eq. (4.4.5) to vanish unless  $\beta_j$  and  $\alpha_j$  are both one-particle states (in which case it is just a delta function  $\delta(\alpha_j - \beta_j)$ ), because the only connected diagrams without vertices consist of a single line running

through the diagram from bottom to top. Most important, the subscript  $C$  in Eq. (4.4.5) means that we exclude any contributions corresponding to disconnected diagrams, that is, any contributions in which any  $H_I(t)$  operator or any initial or final particle is not connected to every other by a sequence of particle creations and annihilations.

Now let us use Eq. (4.4.5) in the sum (4.4.3). Every time variable is integrated from  $-\infty$  to  $+\infty$ , so it makes no difference which of the  $t_1, \dots, t_n$  are sorted out into each cluster. The sum over clusterings therefore yields a factor  $n!/n_1!n_2! \cdots n_\nu!$ , equal to the number of ways of sorting out  $n$  vertices into  $\nu$  clusters, each containing  $n_1, n_2, \dots$  vertices:

$$\begin{aligned} & \int_{-\infty}^{\infty} dt_1 \cdots dt_n \langle \beta | T \{ H_I(t_1) \cdots H_I(t_n) \} | \alpha \rangle \\ &= \frac{n!}{n_1!n_2! \cdots n_\nu!} \sum_{\text{PART}} (\pm) \sum_{n_1 + \cdots + n_\nu = n} \prod_{j=1}^{\nu} \int_{-\infty}^{\infty} dt_{j1} \cdots dt_{jn_j} \\ & \quad \cdot \left( \langle \beta_j | T \{ H_I(t_{j1}) \cdots H_I(t_{jn_j}) \} | \alpha_j \rangle \right)_C. \end{aligned}$$

The first sum here is over all ways of partitioning the particles in the initial and final states into clusters  $\alpha_1 \cdots \alpha_\nu$  and  $\beta_1 \cdots \beta_\nu$  (including a sum over the number  $\nu$  of clusters). The factor  $n!$  here cancels the  $1/n!$  in Eq. (4.4.3), and the factor  $(-i)^n$  in the perturbation series for (4.4.5) can be written as a product  $(-i)^{n_1} \cdots (-i)^{n_\nu}$ , so instead of summing over  $n$  and then summing separately over  $n_1, \dots, n_\nu$ , constrained by  $n_1 + \cdots + n_\nu = n$ , we can simply sum independently over each  $n_1, \dots, n_\nu$ . This gives finally

$$\begin{aligned} S_{\beta\alpha} &= \sum_{\text{PART}} (\pm) \prod_{j=1}^{\nu} \sum_{n_j=0}^{\infty} \frac{(-i)^{n_j}}{n_j!} \int_{-\infty}^{\infty} dt_{j1} \cdots dt_{jn_j} \\ & \quad \cdot \left( \langle \beta_j | T \{ H_I(t_{j1}) \cdots H_I(t_{jn_j}) \} | \alpha_j \rangle \right)_C. \end{aligned}$$

Comparing this with the definition (4.2.2) of the connected matrix elements  $S_{\beta\alpha}^C$ , we see that these matrix elements are just given by the factors in the product here

$$S_{\beta\alpha}^C = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 \cdots dt_n \left( \langle \beta | T \{ H_I(t_1) \cdots H_I(t_n) \} | \alpha \rangle \right)_C. \quad (4.4.6)$$

(The subscript  $j$  is dropped on all the  $ts$  and  $ns$ , as these are now mere integration and summation variables.) We see that  $S_{\beta\alpha}^C$  is calculated by a very simple prescription:  *$S_{\beta\alpha}^C$  is the sum of all contributions to the  $S$ -matrix that are connected, in the sense that we drop all terms in which any initial or final particle or any operator  $H_I(t)$  is not connected to all the others by a sequence of particle creations and annihilations.* This justifies the adjective ‘connected’ for  $S^C$ .

As we have seen, momentum is conserved at each vertex and along every line, so the connected parts of the  $S$ -matrix individually conserve momentum:  $S_{\beta\alpha}^C$  contains a factor  $\delta^3(\mathbf{p}_\beta - \mathbf{p}_\alpha)$ . What we want to prove is that  $S_{\beta\alpha}^C$  contains no other delta functions.

We now make the assumption that the coefficient fractions  $h_{NM}$  in the expansion (4.4.1) of the Hamiltonian in terms of creation and annihilation operators are proportional to



a *single* three-dimensional delta function, that ensures momenta conservation. This is automatically true for the free-particle Hamiltonian  $H_0$ , so it is also then true separately for the interaction  $H_{\text{int}}$ . Returning to the graphical interpretation of the matrix elements that we have been using, this means that each vertex contributes one three-dimensional delta function. (The other delta functions in matrix elements  $(H_{\text{int}})_{\gamma\delta}$  simply keep the momentum of any particle that is not created or annihilated at the corresponding vertex unchanged.) Now, most of these delta functions simply go to fix the momentum of intermediate particles. The only momenta that are left unfixed by such delta functions are those that circulate in loops of internal lines. (Any line which if cut leaves the diagram disconnected carries a momentum that is fixed by momentum conservation as some linear combination of the momenta of the lines coming into or going out of the diagram. If the diagram has  $L$  lines that can all be cut at the same time without the diagram becoming disconnected, then we say it has  $L$  independent loops, and there are  $L$  momenta that are not fixed by momentum conservation.) With  $V$  vertices,  $I$  internal lines, and  $L$  loops, there are  $V$  delta functions, of which  $I - L$  go to fix internal momenta, leaving  $V - I + L$  delta functions relating the momenta of incoming or outgoing particles. But a well-known topological identity<sup>35</sup> tells that for any graph consisting of  $C$  connected pieces, the numbers of vertices, internal lines, and loops are related by

$$V - I + L = C. \quad (4.4.7)$$

Hence for a connected matrix element like  $S_{\beta\alpha}^C$ , which arises from graphs with  $C = 1$ , we find just a single three-dimensional delta function  $\delta^3(\mathbf{p}_\beta - \mathbf{p}_\alpha)$ , as was to be proved.

It was not important in the above argument that the time variables were integrated from  $-\infty$  to  $+\infty$ . Thus exactly the same arguments can be used to show that if the coefficients  $h_{N,M}$  in the Hamiltonian contain just single delta functions, then  $U(t, t_0)$  can also be decomposed into connected parts, each containing a single momentum-conservation delta function factor. On the other hand, the connected part of the  $S$ -matrix also contains an energy-conservation delta function, and when we come to Feynman diagrams in Chapter 6 we shall see that  $S_{\beta\alpha}^C$  contains only a single energy-conservation delta function,  $\delta(E_\beta - E_\alpha)$ , while  $U(t, t_0)$  contains no energy-conservation delta functions at all.

It should be emphasized that the requirement that  $h_{NM}$  in Eq. (4.4.1) should have only a single three-dimensional momentum conservation delta function factor is very far from trivial, and has far-reaching implications. For instance, assume that  $H_{\text{int}}$  has non-vanishing matrix elements between two-particle states. Then Eq. (4.4.1) must contain a term with  $N = M = 2$ , and coefficient

$$v_{2,2}(\mathbf{p}'_1 \mathbf{p}'_2, \mathbf{p}_1 \mathbf{p}_2) = (H_{\text{int}})_{\mathbf{p}'_1 \mathbf{p}'_2; \mathbf{p}_1 \mathbf{p}_2}. \quad (4.4.8)$$

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<sup>35</sup>A graph consisting of a single vertex has  $V = 1$ ,  $L = 0$ , and  $C = 1$ . If we add  $V - 1$  vertices with just enough internal lines to keep the graph connected, we have  $I = V - 1$ ,  $L = 0$ , and  $C = 1$ . Any additional internal lines attached (without new vertices) to the same connected graph produce an equal number of loops, so  $I = V + L - 1$  and  $C = 1$ . If a disconnected graph consists of  $C$  such connected parts, the sums of  $I$ ,  $V$ , and  $L$  in each connected part will then satisfy  $\sum I = \sum V + \sum L - C$ .



(We are here temporarily dropping spin and species labels.) But then the matrix element of the interaction between three-particle states is

$$(H_{\text{int}})_{\mathbf{p}'_1 \mathbf{p}'_2 \mathbf{p}'_3; \mathbf{p}_1 \mathbf{p}_2 \mathbf{p}_3} = v_{3,3}(\mathbf{p}'_1 \mathbf{p}'_2 \mathbf{p}'_3, \mathbf{p}_1 \mathbf{p}_2 \mathbf{p}_3) + v_{2,2}(\mathbf{p}'_1 \mathbf{p}'_2, \mathbf{p}_1 \mathbf{p}_2) \delta^3(\mathbf{p}'_3 - \mathbf{p}_3) \pm \text{permutations.} \quad (4.4.9)$$

As mentioned at the beginning of this chapter, we might try to make a relativistic quantum theory that is not a field theory by choosing  $v_{2,2}$  so that the two-body  $S$ -matrix is Lorentz-invariant, and adjusting the rest of the Hamiltonian so that there is no scattering in states containing three or more particles. We would then have to take  $v_{3,3}$  to cancel the other terms in Eq. (4.4.9)

$$v_{3,3}(\mathbf{p}'_1 \mathbf{p}'_2 \mathbf{p}'_3, \mathbf{p}_1 \mathbf{p}_2 \mathbf{p}_3) = -v_{2,2}(\mathbf{p}'_1 \mathbf{p}'_2, \mathbf{p}_1 \mathbf{p}_2) \delta^3(\mathbf{p}'_3 - \mathbf{p}_3) \mp \text{permutations.} \quad (4.4.10)$$

However, this would mean that each term in  $v_{3,3}$  contains *two* delta function factors (recall that  $v_{2,2}(\mathbf{p}'_1 \mathbf{p}'_2, \mathbf{p}_1 \mathbf{p}_2)$  has a factor  $\delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 - \mathbf{p}_1 - \mathbf{p}_2)$ ) and this would violate the cluster decomposition principle. Thus in a theory satisfying the cluster decomposition principle, the existence of scattering processes involving two particles makes processes involving three or more particles inevitable.

\* \* \*

When we set out to solve three-body problems in quantum theories that satisfy the cluster decomposition principle, the term  $v_{3,3}$  in Eq. (4.4.9) gives no particular trouble, but the extra delta function in the other terms makes the Lippmann–Schwinger equation difficult to solve directly. The problem is that these delta functions make the kernel  $[E_\alpha - E_\beta + i\epsilon]^{-1} (H_{\text{int}})_{\beta\alpha}$  of this equation not square-integrable, even after we factor out an overall momentum conservation delta function. In consequence, it cannot be approximated by a finite matrix, even one of very large rank. To solve problems involving three or more particles, it is necessary to replace the Lippmann–Schwinger equation with one that has a connected right-hand side. Such equations have been developed for the scattering of three or more particles [8, 9], and in non-relativistic scattering problems they can be solved recursively, but they have not turned out to be useful in relativistic theories and so will not be described in detail here.

However, recasting the Lippmann–Schwinger equation in this manner is useful in another way. Our arguments in this section have so far relied on perturbation theory. I do not know of any non-perturbative proof of the main theorem of this section, but it has been shown [9] that these reformulated non-perturbative dynamical equations are *consistent* with the requirement that  $U^C(t, t_0)$  (and hence  $S^C$ ) should also contain only a single momentum-conservation delta function, as required by the cluster decomposition principle, provided that the Hamiltonian satisfies our condition that the coefficient functions  $h_{N,M}$  each contain only a single momentum-conservation delta function.

## Weinberg Exercises

**W.4.1.** Define generating functionals for the  $S$ -matrix and its connected part:

$$F[v] \equiv 1 + \sum_{N=1}^{\infty} \sum_{M=1}^{\infty} \frac{1}{N!M!} \int v^*(q'_1) \cdots v^*(q'_N) v(q_1) \cdots v(q_M) \\ \cdot S_{q'_1 \cdots q'_N, q_1 \cdots q_M} dq'_1 \cdots dq'_N dq_1 \cdots dq_M, \\ F^C[v] \equiv \sum_{N=1}^{\infty} \sum_{M=1}^{\infty} \frac{1}{N!M!} \int v^*(q'_1) \cdots v^*(q'_N) v(q_1) \cdots v(q_M) \\ \cdot S_{q'_1 \cdots q'_N, q_1 \cdots q_M}^C dq'_1 \cdots dq'_N dq_1 \cdots dq_M.$$

Derive a formula relating  $F[v]$  and  $F^C[v]$ . (You may consider the purely bosonic case.)

**W.4.2.** Consider an interaction

$$H_{\text{int}} = g \int d^3\mathbf{p}_1 d^3\mathbf{p}_2 d^3\mathbf{p}_3 d^3\mathbf{p}_4 \delta^3(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_3 - \mathbf{p}_4) \\ \cdot a_{\mathbf{p}_1}^\dagger a_{\mathbf{p}_2}^\dagger a_{\mathbf{p}_3} a_{\mathbf{p}_4},$$

where  $g$  is a real constant and  $a_{\mathbf{p}}$  is the annihilation operator of a spinless boson of mass  $M > 0$ . Use perturbation theory to calculate the  $S$ -matrix element for scattering of these particles in the center-of-mass frame to order  $g^2$ . What is the corresponding differential cross-section?

**W.4.3.** A coherent state  $|\lambda\rangle$  is defined to be an eigenstate of the annihilation operators  $a_q$  with eigenvalues  $\lambda(q)$ . Construct such a state as a superposition of the multi-particle states  $|q_1, q_2, \dots, q_N\rangle$ .

## Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

**W.4.1.** For bosons, substitute the partition expansion Eq. (4.3.2) into  $F[v]$ . A term containing  $r_k$  connected components of type  $k$  has the product of their connected matrix elements and the combinatorial factor  $1/r_k!$  for each set of identical components. The factors  $N!M!$  in the definition of  $F$  cancel the ways of distributing the outgoing and incoming labels among the components. Thus

$$F[v] = \prod_k \sum_{r_k=0}^{\infty} \frac{(F_k^C[v])^{r_k}}{r_k!},$$

where  $F_k^C$  denotes one possible connected contribution, including all its label integrals. Multiplying the exponentials and summing over all component types gives

$$F[v] = \exp(F^C[v]), \quad F^C[v] = \log F[v].$$

The constant term is correct because  $F[0] = 1$  and  $F^C[0] = 0$ .

One can also verify the result by functional differentiation. Every derivative of  $\exp(F^C)$  is a sum over set partitions of the derivatives of  $F^C$ ; after setting  $v = 0$ , these derivatives are the connected matrix elements, and the sum over partitions is exactly Eq. (4.3.2).

**W.4.2.** Let

$$|i\rangle = |\mathbf{k}, -\mathbf{k}\rangle, \quad |f\rangle = |\mathbf{k}', -\mathbf{k}'\rangle, \quad \varepsilon_k = \sqrt{k^2 + M^2}.$$

Commuting the two annihilation and two creation operators through the external states gives

$$\langle f | H_{\text{int}} | i \rangle = 4g \delta^3(\mathbf{p}_f - \mathbf{p}_i).$$

At second order, the two-particle completeness relation contributes  $1/2!$ . The reduced transition matrix element is therefore

$$\mathcal{M}(k) = 4g + 8g^2 I(k) + O(g^3), \quad I(k) = \int d^3\mathbf{q} \frac{1}{2\varepsilon_k - 2\varepsilon_q + i0}.$$

The integral is ultraviolet divergent, as expected for an unregulated three-dimensional contact interaction. With a momentum cutoff  $\Lambda$ ,

$$I_\Lambda(k) = 4\pi \int_0^\Lambda dq \frac{q^2}{2\varepsilon_k - 2\varepsilon_q + i0}, \quad \text{Im } I_\Lambda(k) = -2\pi^2 k \varepsilon_k$$

for  $\Lambda > k$ . The divergent real part must be absorbed into a definition of the physical coupling.

Including the identity term, the center-of-mass  $S$ -matrix through order  $g^2$  is

$$\begin{aligned} {}_{\text{out}}\langle \mathbf{k}', -\mathbf{k}' | | \mathbf{k}, -\mathbf{k} \rangle_{\text{in}} &= \langle \mathbf{k}', -\mathbf{k}' | \mathbf{k}, -\mathbf{k} \rangle \\ &\quad - 2\pi i \delta(2\varepsilon_{k'} - 2\varepsilon_k) \delta^3(\mathbf{p}_f - \mathbf{p}_i) [4g + 8g^2 I_\Lambda(k)] + O(g^3). \end{aligned}$$

Equation (3.4.30), with the usual  $1/2!$  for identical final particles when the full labeled solid angle is used, gives

$$\frac{d\sigma}{d\Omega} = \frac{(2\pi)^4 \varepsilon_k^2}{8} |4g + 8g^2 I_\Lambda(k)|^2.$$

In particular, the leading cross section is

$$\frac{d\sigma}{d\Omega} = 2(2\pi)^4 g^2 \varepsilon_k^2 + O(g^3)$$

over the full sphere. Equivalently, one may omit the  $1/2!$  and integrate over only one hemisphere.

**W.4.3.** Put

$$\|\lambda\|^2 = \int dq |\lambda(q)|^2, \quad A^\dagger(\lambda) = \int dq \lambda(q) a_q^\dagger.$$

The normalized coherent state is

$$|\lambda\rangle = e^{-\|\lambda\|^2/2} e^{A^\dagger(\lambda)} |\Omega\rangle = e^{-\|\lambda\|^2/2} \sum_{N=0}^{\infty} \frac{1}{N!} \int dq_1 \cdots dq_N \prod_{r=1}^N \lambda(q_r) |q_1 \cdots q_N\rangle.$$

Indeed,

$$[a_q, A^\dagger(\lambda)] = \lambda(q)$$

is a c-number, so

$$a_q e^{A^\dagger(\lambda)} = e^{A^\dagger(\lambda)} (a_q + \lambda(q)).$$

Since  $a_q |\Omega\rangle = 0$ , this proves  $a_q |\lambda\rangle = \lambda(q) |\lambda\rangle$ .

Finally,

$$\langle \Omega | e^{A(\lambda)} e^{A^\dagger(\lambda)} | \Omega \rangle = e^{\|\lambda\|^2},$$

so the prefactor makes the norm unity. The probability of finding  $N$  particles is

$$P_N = e^{-\|\lambda\|^2} \frac{\|\lambda\|^{2N}}{N!},$$

which also checks the normalization by summing the Poisson series.

## Supplementary Exercises

### S.4.1. A Complex Gaussian and Its Pairings

(Adapted from D. Harlow, Quantum Field Theory I, 2024, Section 7.8, Homework Problem 5.)

Let  $x$  be an  $n$ -component complex vector, let  $A$  be a positive real symmetric matrix, and let  $B$  and  $\tilde{B}$  be independent complex vectors. With the componentwise convention

$$dx \, dx^* \equiv \prod_{a=1}^n dx_a \, dx_a^*, \quad dx_a \, dx_a^* = -2i \, d(\operatorname{Re} x_a) \, d(\operatorname{Im} x_a),$$

show that

$$\int dx \, dx^* e^{-x^\dagger A x + B^T x + \tilde{B}^T x^*} = \frac{1}{\det\left(\frac{iA}{2\pi}\right)} e^{\tilde{B}^T A^{-1} B}.$$

Use this identity to evaluate the normalized Gaussian correlators

$$\langle x_{i_1} \cdots x_{i_M} x_{j_1}^* \cdots x_{j_N}^* \rangle_A \equiv \frac{\int dx \, dx^* e^{-x^\dagger A x} x_{i_1} \cdots x_{i_M} x_{j_1}^* \cdots x_{j_N}^*}{\int dx \, dx^* e^{-x^\dagger A x}}.$$

Prove that the answer vanishes unless  $M = N$ , and that when  $M = N$  it is the sum over every pairing in which each unstarred variable is joined to one starred variable. State the factor contributed by each joined pair.

The source prints  $j_M$ , rather than  $j_N$ , as the final starred index on the right-hand side of its correlator formula even though its left-hand side contains  $N$  starred variables. Diagnose this mismatch before giving the corrected general formula. For  $M = N = 2$ , the two allowed arrow patterns are

$$\begin{array}{ccc} x_{i_1} & \longrightarrow & x_{j_1}^* \\ x_{i_2} & \longrightarrow & x_{j_2}^* \end{array} + \begin{array}{ccc} x_{i_1} & \searrow & x_{j_1}^* \\ & \nearrow & \\ x_{i_2} & \nearrow & x_{j_2}^* \end{array}$$

#### S.4.2. Generating Connected Contributions and Their Volume Scaling

(Inspired by D. Harlow, Quantum Field Theory I, 2024, Section 7.4 / D. Harlow, Quantum Field Theory I, 2024, Section 7.5 / D. Harlow, Quantum Field Theory I, 2024, Sections 7.4–7.5.)

- Suppose each connected diagram type  $C$  has value  $V_C$ , and a disconnected diagram contains  $m_C$  identical copies of each type. Show that its combinatorial weight is  $\prod_C V_C^{m_C}/m_C!$ , and prove that the sum over all diagrams is  $\exp(\sum_C V_C)$ .
- Let  $Z[J]$  be normalized by  $Z[0] = 1$ , and define  $W[J] = \log Z[J]$ . Show through fourth order in  $J$  that functional derivatives of  $W$  at  $J = 0$  give connected correlation functions, while derivatives of  $Z$  give the full functions.
- In a translationally invariant theory in a periodic box of volume  $V$ , show that a connected momentum-space amplitude carries one momentum Kronecker delta and scales as  $V$ , whereas a contribution with  $C$  disconnected components scales as  $V^C$ . Explain why the logarithm is extensive.

#### S.4.3. Vacuum Cancellation, Connectedness, and Momentum Delta Functions

(Inspired by D. Tong, Lectures on Quantum Field Theory, 2006, Section 3.7.1 / D. Tong, Lectures on Quantum Field Theory, 2006, Sections 3.5.4 and 3.7.1 / D. Tong, Lectures on Quantum Field Theory, 2006, Sections 3.4.1 and 3.5.4.)

- Use the Gell-Mann–Low ratio for an  $n$ -point function to prove that all vacuum bubbles cancel between numerator and denominator. Distinguish a vacuum bubble from a diagram disconnected only with respect to external points.
- Give graph-theoretic definitions of a connected graph and a one-particle-irreducible graph. Construct a connected two-point graph that becomes disconnected when one internal line is cut, and explain why it contributes to a connected correlator.
- For a diagram with  $V$  vertices,  $I$  internal lines,  $L$  loop momenta, and  $C$  connected components, prove  $V - I + L = C$ . Use momentum conservation at every vertex to show that exactly  $C$  independent momentum delta functions remain.

#### S.4.4. Microcausality, Statistics, and Cluster Decomposition

(Adapted from D. Harlow, Quantum Field Theory I, 2024, Section 6.6, Homework Problem 5.)

Suppose that at spacelike separation a nontrivial local field obeys

$$\Phi(x)\Phi^\dagger(y) \pm \Phi^\dagger(y)\Phi(x) = 0.$$

Show that the same choice of sign must occur in both of the relations

$$\Phi(x)\Phi(y) \pm \Phi(y)\Phi(x) = 0, \quad \Phi^\dagger(x)\Phi^\dagger(y) \pm \Phi^\dagger(y)\Phi^\dagger(x) = 0$$

at spacelike separation.

*Hint.* Write the first same-field relation provisionally as

$$\Phi(x)\Phi(y) + s\Phi(y)\Phi(x) = 0, \quad s = \pm 1,$$

and determine  $s$  by considering the norm of  $\Phi(x)\Phi(y)|\Omega\rangle$ . You will need the Reeh–Schlieder property and the large-separation limit

$$\langle \Phi^\dagger(x)\Phi(x)\Phi^\dagger(y)\Phi(y) \rangle \longrightarrow \langle \Phi^\dagger(x)\Phi(x) \rangle \langle \Phi^\dagger(y)\Phi(y) \rangle \quad \text{as } (x-y)^2 \rightarrow +\infty.$$

This is cluster decomposition: connected correlators of local operators decay at large separation. Here it is being applied to the connected two-point function of the composite local operator  $\Phi^\dagger\Phi$ .

#### S.4.5. Wick’s Theorem for Three Fields

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 4, 2022, Problem 1.)

You may use Wick’s theorem for a product of two free real scalar fields.

- (a) For a free real scalar, show separately that  $T\{\phi(x_1)\phi(x_2)\}$  and  $:\phi(x_1)\phi(x_2):$  are unchanged under  $x_1 \leftrightarrow x_2$ . Use

$$T\{\phi(x_1)\phi(x_2)\} = :\phi(x_1)\phi(x_2): + D_F(x_1 - x_2)$$

to prove  $D_F(x_1 - x_2) = D_F(x_2 - x_1)$ .

- (b) By separating each field into creation and annihilation parts, prove

$$\begin{aligned} T\{\phi(x_1)\phi(x_2)\phi(x_3)\} = & :\phi(x_1)\phi(x_2)\phi(x_3): + \phi(x_1)D_F(x_2 - x_3) \\ & + \phi(x_2)D_F(x_3 - x_1) + \phi(x_3)D_F(x_1 - x_2). \end{aligned}$$

#### S.4.6. Cubic-Theory Vacuum Graphs and Normalized Correlators

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 4, 2022, Problem 2.)

Consider a real scalar field of mass  $m$  with

$$\mathcal{L}_0 = -\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2, \quad \mathcal{L}_{\text{int}} = \frac{\lambda}{3!}\phi^3.$$

Let  $|0\rangle$  be the free vacuum,  $\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}}$ , and  $H_I(t)$  the interaction-picture Hamiltonian.

- (a) Use Wick's theorem to evaluate

$$\langle 0|T \exp\left[-i \int dt H_I(t)\right] |0\rangle$$

through order  $\lambda^2$ , as an explicit sum of position-space propagator integrals. Terms containing  $D_F(0)$  should be retained.

- (b) Draw every distinct vacuum diagram at orders  $\lambda^3$  and  $\lambda^4$ . No evaluation or symmetry-factor calculation is required in this part.
- (c) For the interacting-vacuum two-point function

$$\langle \Omega|T\{\phi_H(x)\phi_H(y)\}|\Omega\rangle = \frac{\langle 0|T\{\phi_I(x)\phi_I(y)\exp[-i \int dt H_I(t)]\}|0\rangle}{\langle 0|T \exp[-i \int dt H_I(t)]|0\rangle},$$

draw every contributing diagram through order  $\lambda^4$  and give its symmetry factor. Include diagrams with more than one component when every component touches  $x$  or  $y$ , but omit vacuum components. As a check, thirteen diagrams in the complete list are fully connected, meaning that every field is connected to both external points.

#### S.4.7. A Classical Source and Coherent In–Out States

(Adapted from A. Guth, MIT 8.323, Problem Set 5, 2008, Problem 3.)

For a quantized scalar field driven by a fixed classical source  $j(x)$ , suppose the in- and out-fields obey

$$\phi_{\text{out}}(x) = S^{-1}\phi_{\text{in}}(x)S, \quad a_{\text{out}}(\mathbf{p}) = S^{-1}a_{\text{in}}(\mathbf{p})S.$$

Write

$$\mu = \int \frac{d^3p}{(2\pi)^3} \frac{|\tilde{j}(p)|^2}{2E_{\mathbf{p}}}, \quad \tilde{j}(p) = \int d^4y e^{ip\cdot y} j(y),$$

and take

$$S = e^{-\mu/2} e^F e^G,$$

$$F = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \tilde{j}(p) a_{\text{in}}^\dagger(\mathbf{p}), \quad G = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \tilde{j}(-p) a_{\text{in}}(\mathbf{p}).$$

- (a) Define  $F'$  and  $G'$  by replacing the in-operators in  $F, G$  with out-operators. Prove

$$S = e^{-\mu/2} e^{F'} e^{G'}$$

and hence

$$|0\rangle_{\text{in}} = e^{-\mu/2} e^{F'} |0\rangle_{\text{out}}.$$

(b) For one harmonic oscillator,

$$H = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2},$$

make the canonical rescaling  $p = \sqrt{m\omega} \bar{p}$ ,  $q = \bar{q}/\sqrt{m\omega}$ , drop the bars, and use

$$a^\dagger = \frac{q - ip}{\sqrt{2}}, \quad a = \frac{q + ip}{\sqrt{2}}.$$

For the unnormalized coherent state  $|z\rangle = e^{za^\dagger}|0\rangle$ , show that it is an eigenstate of  $a$  and determine the eigenvalue.

(c) Compute  $\langle z_2 | z_1 \rangle$ .

(d) Using expectation values normalized by  $\langle z | z \rangle$ , find  $\langle q \rangle_z$  and  $\langle p \rangle_z$ .

(e) Compute  $(\Delta q)^2$  and  $(\Delta p)^2$ , and show  $\Delta q \Delta p = 1/2$ .

#### S.4.8. Recovering a Classical Field from an Impulse

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 3, 2022, Problem 2.)

Let a free real scalar field couple to a classical source through

$$H(t) = H_0 - \int d^3x \phi(\mathbf{x}, t) J(\mathbf{x}, t).$$

For an operator with no explicit Schrödinger-picture time dependence, use the Heisenberg equation  $\dot{O} = i[H, O]$ .

(a) Derive the sourced Klein–Gordon equation in the local mostly-plus convention,

$$(\square - m^2)\phi(x) = -J(x).$$

(b) Starting in the vacuum, apply

$$J(\mathbf{x}, t) = \delta(t)j(\mathbf{x}).$$

With  $0^\pm$  denoting the limits just after and before the impulse, show

$$\phi(\mathbf{x}, 0^+) = \phi(\mathbf{x}, 0^-), \quad \pi(\mathbf{x}, 0^+) = \pi(\mathbf{x}, 0^-) + j(\mathbf{x}).$$

(c) For

$$\phi(\mathbf{x}, 0) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left[ a(\mathbf{p}, 0) e^{i\mathbf{p}\cdot\mathbf{x}} + a^\dagger(\mathbf{p}, 0) e^{-i\mathbf{p}\cdot\mathbf{x}} \right],$$

show

$$a(\mathbf{p}, 0^+) = a(\mathbf{p}, 0^-) + \frac{i\tilde{j}(\mathbf{p})}{\sqrt{2E_{\mathbf{p}}}}, \quad \tilde{j}(\mathbf{p}) = \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} j(\mathbf{x}).$$

(d) Show that the state is a multimode coherent state with respect to  $a(\mathbf{p}, 0^+)$ , and evaluate its produced-particle momentum density  $\langle a^\dagger(\mathbf{p}, 0^+) a(\mathbf{q}, 0^+) \rangle$ .



### S.4.9. Relativistic Propagators and Causal Support

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 3, 2022, Problem 1.)

For a free real scalar, define

$$D(z) = \langle 0 | \phi(x) \phi(y) | 0 \rangle, \quad z = x - y.$$

- (a) Starting from the free-field expansion, show

$$D(z) = \int \frac{d^4 p}{(2\pi)^3} \delta(p^2 + m^2) \theta(p^0) e^{ip \cdot z}.$$

- (b) For a proper orthochronous Lorentz transformation, prove  $D(\Lambda z) = D(z)$ . In particular, verify the invariance of  $d^4 p$  and  $\delta(p^2 + m^2) \theta(p^0)$ .  
(c) At spacelike separation  $z^2 > 0$ , derive

$$D(z) = \frac{m}{4\pi^2 \sqrt{z^2}} K_1(m\sqrt{z^2})$$

or verify it by numerical integration, and extract the large-distance exponential falloff. Standard tables or computer algebra may be used.

Although  $D(z)$  is nonzero at spacelike separation, relativistic locality concerns the propagation of a change in the state, not the existence of vacuum correlations.

- (d) Show

$$\langle 0 | [\phi(x), \phi(y)] | 0 \rangle = 0$$

when  $x - y$  is spacelike.

- (e) Now take

$$\mathcal{L} = -\frac{1}{2} (\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2 + \phi(x) J(x).$$

Derive

$$(\square - m^2) \phi(x) = -J(x).$$

- (f) Show that a particular solution is

$$\phi(x) = i \int d^4 y G(x - y) J(y),$$

provided

$$(\square - m^2) G(z) = i \delta^4(z),$$

and state the freedom to add a homogeneous solution.

- (g) Using the residue theorem for the  $p^0$  integral, evaluate

$$\begin{aligned} G_R(z) &= \lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4} \frac{-i e^{ip \cdot z}}{(p^0 + i\epsilon)^2 + \mathbf{p}^2 + m^2}, \\ G_A(z) &= \lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4} \frac{-i e^{ip \cdot z}}{(p^0 - i\epsilon)^2 + \mathbf{p}^2 + m^2}, \\ G_F(z) &= \lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4} \frac{-i e^{ip \cdot z}}{p^2 + m^2 - i\epsilon}, \end{aligned}$$

and prove

$$\begin{aligned} G_R(z) &= \theta(z^0) [D(z) - D(-z)], \\ G_A(z) &= \theta(-z^0) [D(-z) - D(z)], \\ G_F(z) &= \theta(z^0) D(z) + \theta(-z^0) D(-z). \end{aligned}$$

- (h) For a field that vanishes before the source acts, use  $G_R$  to show that the response vanishes at spacelike separation from the source.
- (i) Does  $G_F$  vanish at spacelike separation? Explain why its answer is consistent with causal signaling.

#### S.4.10. The Force Between Static Sources

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 4, 2022, Problem 4.)

Consider a free real scalar field coupled to a source,

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2 + \phi(x)J(x).$$

- (a) Using Wick's theorem, prove

$$\langle 0|T \exp\left(i \int d^4x \phi(x)J(x)\right)|0\rangle = \exp\left[-\frac{1}{2} \int d^4x d^4y J(x)D_F(x-y)J(y)\right].$$

- (b) Take two point sources, separated by  $\mathbf{R}$ , that remain on for a time  $T$ :

$$J(x) = gf(t) [\delta^3(\mathbf{x}) + \delta^3(\mathbf{x} - \mathbf{R})], \quad f(t) = \begin{cases} 1, & -T/2 < t < T/2, \\ 0, & \text{otherwise.} \end{cases}$$

Discard the  $D_F(0)$  self-energy terms and evaluate the remaining vacuum amplitude when  $T \gg R \gg m^{-1}$ .

- (c) Match the result to  $\exp[-iV(R)T]$ , determine  $V(R)$ , and decide whether the force is attractive or repulsive.

### Solutions to Supplementary Exercises

#### S.4.1. A Complex Gaussian and Its Pairings

Because  $A$  is positive and symmetric, an orthogonal matrix diagonalizes it:  $A = O^T D O$ , with  $D = \text{diag}(a_1, \dots, a_n)$  and  $a_r > 0$ . The simultaneous change of variables  $x \mapsto O x$ ,  $x^* \mapsto O x^*$  has unit Jacobian. For one complex component,

$$\int dx dx^* e^{-ax^*x} = -2i \int_{\mathbb{R}^2} du dv e^{-a(u^2+v^2)} = -\frac{2\pi i}{a} = \frac{2\pi}{ia}.$$

The product of these one-component integrals is therefore

$$Z_A[0,0] = \prod_{r=1}^n \frac{2\pi}{ia_r} = \frac{1}{\det(iA/2\pi)}.$$

Since  $A = A^T$ , completing the square gives

$$-x^{*T}Ax + B^T x + \tilde{B}^T x^* = -\left(x^* - A^{-1}B\right)^T A \left(x - A^{-1}\tilde{B}\right) + \tilde{B}^T A^{-1}B.$$

The contours may be translated because the Gaussian is entire and decays in the required directions. Thus

$$Z_A[B, \tilde{B}] = \frac{1}{\det(iA/2\pi)} e^{\tilde{B}^T A^{-1}B},$$

as claimed.

After division by  $Z_A[0, 0]$ , insertions are generated by

$$\left. \frac{\partial}{\partial B_{i_1}} \cdots \frac{\partial}{\partial B_{i_M}} \frac{\partial}{\partial \tilde{B}_{j_1}} \cdots \frac{\partial}{\partial \tilde{B}_{j_N}} e^{\tilde{B}^T A^{-1}B} \right|_{B=\tilde{B}=0}.$$

Every term in the exponential contains one  $B$  and one  $\tilde{B}$ . Consequently the derivative vanishes unless  $M = N$ . For  $M = N = r$ , it is

$$\boxed{\langle x_{i_1} \cdots x_{i_r} x_{j_1}^* \cdots x_{j_r}^* \rangle_A = \sum_{\pi \in S_r} \prod_{a=1}^r (A^{-1})_{i_a j_{\pi(a)}}.}$$

Thus an arrow joining  $x_i$  to  $x_j^*$  contributes  $(A^{-1})_{ij}$ , while  $xx$  and  $x^*x^*$  contractions vanish. For two pairs the formula gives

$$(A^{-1})_{i_1 j_1} (A^{-1})_{i_2 j_2} + (A^{-1})_{i_1 j_2} (A^{-1})_{i_2 j_1},$$

corresponding to the direct and crossed arrow patterns in the question.

The source's final  $j_M$  is harmless only after it has established  $M = N$ ; as a purported formula before that restriction, it is an index mismatch. The corrected general left-hand side ends at  $j_N$ , and the nonzero formula above then sets  $M = N = r$ .

#### S.4.2. Generating Connected Contributions and Their Volume Scaling

- (a) If a disconnected graph contains  $m_C$  indistinguishable copies of component  $C$ , permutations of these copies do not produce a new graph. Its value is therefore

$$V_D = \prod_C \frac{V_C^{m_C}}{m_C!}.$$

Summing independently over every nonnegative  $m_C$ ,

$$\sum_D V_D = \prod_C \sum_{m_C=0}^{\infty} \frac{V_C^{m_C}}{m_C!} = \prod_C e^{V_C} = \exp\left(\sum_C V_C\right).$$

Taking the logarithm removes all products and leaves exactly one copy of each connected component value.

(b) Use condensed indices and write

$$Z[J] = 1 + Z_1 + \frac{1}{2!}Z_2 + \frac{1}{3!}Z_3 + \frac{1}{4!}Z_4 + \cdots,$$

where  $Z_n$  is homogeneous of degree  $n$  in  $J$ . Expanding  $\log Z$  gives

$$\begin{aligned} W_1 &= Z_1, & W_2 &= Z_2 - Z_1^2, \\ W_3 &= Z_3 - 3Z_2Z_1 + 2Z_1^3, \end{aligned}$$

with the products understood as sums over assignments of labeled source indices. At fourth order,

$$W_4 = Z_4 - 4Z_3Z_1 - 3Z_2^2 + 12Z_2Z_1^2 - 6Z_1^4.$$

Functional differentiation assigns the external labels to every displayed factor and reproduces the partition formulas of S.4.1. Consequently

$$\left. \frac{\delta^n W}{\delta J_1 \cdots \delta J_n} \right|_{J=0}$$

is the connected  $n$ -point function, while the same derivative of  $Z$  is the full one.

(c) In a periodic box,

$$\int_V d^3x e^{i(\mathbf{P}_f - \mathbf{P}_i) \cdot \mathbf{x}} = V \delta_{\mathbf{P}_f, \mathbf{P}_i}.$$

A connected component has one freely translatable center and therefore one such factor. Its remaining integrals involve relative coordinates and do not acquire another overall power of  $V$ . A graph with  $C$  disconnected components has  $C$  independently translatable centers, so its leading volume dependence is  $V^C$ .

Since the full generating object is the exponential of its connected pieces, its logarithm contains only one connected component at a time and is proportional to  $V$ . This is the linked-cluster origin of extensivity.

#### S.4.3. Vacuum Cancellation, Connectedness, and Momentum Delta Functions

(a) Let  $B$  be the sum of connected vacuum bubbles. Diagram exponentiation gives

$$\langle 0|S|0\rangle = e^B.$$

In the numerator with external fields, every diagram uniquely splits into vacuum components and a remainder in which each component touches at least one external point. Hence

$$\langle 0|T\{\phi_1 \cdots \phi_n S\}|0\rangle = e^B \mathcal{N}_n.$$

The Gell-Mann–Low ratio is therefore  $\mathcal{N}_n$ : the factor  $e^B$  cancels exactly.

A vacuum bubble touches no external point. By contrast, a diagram may have two separate components, each attached to some external points. Such a diagram survives vacuum normalization but is removed when one passes from the full correlator to its connected part.

- (b) A graph is connected if every pair of vertices is joined by some path. It is one-particle irreducible if it remains connected after any one internal line is cut. For example, take two interaction subgraphs joined by a single internal propagator, with one external leg attached at each end. The complete graph is connected, but cutting the joining propagator separates it into two pieces.

Connected correlation functions sum all connected graphs, including this one-particle-reducible example. One-particle-irreducible graphs form a smaller set used to build effective vertices; chains of them reconstruct general connected graphs.

- (c) Choose a spanning forest, one spanning tree in each connected component. It contains  $V - C$  edges. Each of the remaining

$$L = I - (V - C)$$

internal edges creates one independent cycle. Rearrangement gives

$$V - I + L = C.$$

Every vertex initially supplies one momentum delta function. Of the  $I$  internal momenta,  $I - L$  are fixed by these constraints; the  $L$  loop momenta remain to be integrated. The number of unused delta functions is therefore

$$V - (I - L) = V - I + L = C.$$

They impose conservation of the total momentum entering each connected component. A connected graph has  $C = 1$  and hence only the overall conservation delta function.

#### S.4.4. Microcausality, Statistics, and Cluster Decomposition

Let  $\sigma = \pm 1$  denote the sign in the mixed relation, so that

$$\Phi(x)\Phi^\dagger(y) + \sigma\Phi^\dagger(y)\Phi(x) = 0$$

at spacelike separation. Suppose provisionally that

$$\Phi(x)\Phi(y) + s\Phi(y)\Phi(x) = 0, \quad s = \pm 1.$$

Taking the adjoint shows that the same provisional sign  $s$  occurs in the  $\Phi^\dagger\Phi^\dagger$  relation.

For rigor one smears the fields in two spacelike-separated regions; we use point notation to keep the algebra readable. Positivity gives

$$\begin{aligned} 0 \leq N(x, y) &\equiv \|\Phi(x)\Phi(y)|\Omega\rangle\|^2 \\ &= \langle \Phi^\dagger(y)\Phi^\dagger(x)\Phi(x)\Phi(y) \rangle. \end{aligned}$$

Move  $\Phi^\dagger(y)$  first past  $\Phi^\dagger(x)$ , using the adjoint same-field relation, and then past  $\Phi(x)$ , using the mixed relation. The two exchanges give

$$N(x, y) = s\sigma \left\langle \Phi^\dagger(x)\Phi(x)\Phi^\dagger(y)\Phi(y) \right\rangle.$$

At large spacelike separation, cluster decomposition makes the last expectation value tend to

$$\left\langle \Phi^\dagger(x)\Phi(x) \right\rangle \left\langle \Phi^\dagger(y)\Phi(y) \right\rangle.$$

Each factor is a nonnegative norm. It is strictly positive for a suitable smearing of any nontrivial local field: if every such  $\Phi(f)|\Omega\rangle$  vanished, the separating property supplied by Reeh–Schlieder would make  $\Phi$  the zero operator. Hence the large-separation limit is positive. Since  $N(x, y) \geq 0$ , it follows that

$$s\sigma = +1, \quad \text{and therefore} \quad s = \sigma.$$

The same conclusion for the adjoint relation follows by Hermitian conjugation. Thus the mixed, unstarred, and starred spacelike relations all necessarily carry the same statistics sign.

#### S.4.5. Wick’s Theorem for Three Fields

- (a) For bosonic fields, interchanging  $x_1$  and  $x_2$  merely interchanges the two terms

$$\theta(x_1^0 - x_2^0)\phi(x_1)\phi(x_2) \quad \text{and} \quad \theta(x_2^0 - x_1^0)\phi(x_2)\phi(x_1)$$

in the time-ordered product. The equal-time ambiguity is harmless because scalar fields commute at equal time. In the normal-ordered product, creation parts commute with creation parts and annihilation parts commute with annihilation parts; the two mixed terms exchange places. Thus both products are symmetric. Subtracting them in

$$D_F(x_1 - x_2) = T\{\phi(x_1)\phi(x_2)\} - :\phi(x_1)\phi(x_2):$$

proves  $D_F(x_1 - x_2) = D_F(x_2 - x_1)$ .

- (b) Write  $\phi_i = \phi_i^{(+)} + \phi_i^{(-)}$ , with the positive-frequency part annihilating the vacuum and the negative-frequency part creating quanta. Moving every annihilation part to the right produces one commutator for each interchange with a creation part. Since three fields cannot be completely paired, the possibilities are no contraction or one contraction:

$$\begin{aligned} T\{\phi_1\phi_2\phi_3\} &= :\phi_1\phi_2\phi_3: + D_F(x_1 - x_2)\phi_3 \\ &\quad + D_F(x_1 - x_3)\phi_2 + D_F(x_2 - x_3)\phi_1. \end{aligned}$$

The propagators are c-numbers, so their position relative to the uncontracted field is immaterial. The result is symmetric in all three points, as required by time ordering for a scalar field.

#### S.4.6. Cubic-Theory Vacuum Graphs and Normalized Correlators

- (a) The zeroth-order term is one, and the first-order term vanishes because it contains three free fields. With the vertex factor denoted  $i\lambda$ , the second-order term is

$$\frac{(i\lambda)^2}{2(3!)^2} \int d^4x d^4y \langle 0|T\{\phi(x)^3\phi(y)^3\}|0\rangle.$$

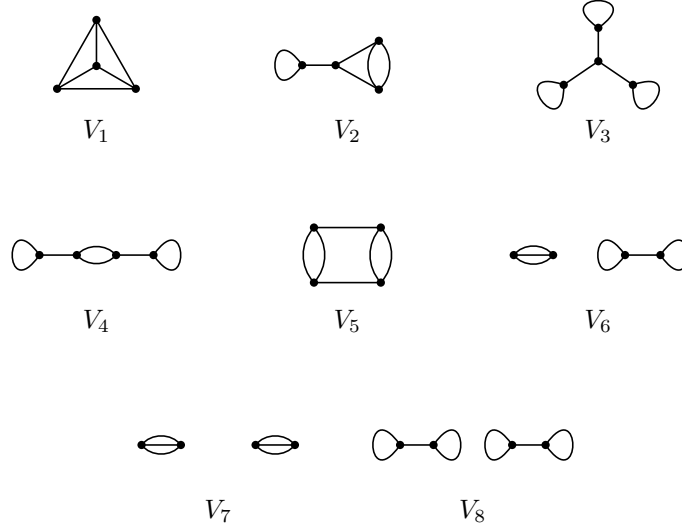
There are  $3! = 6$  contractions joining every field at  $x$  to one at  $y$ , and  $3 \cdot 3 = 9$  contractions with one joining line and one tadpole at each vertex. Thus

$$\langle 0|S|0\rangle = 1 + (i\lambda)^2 \int d^4x d^4y \left[ \frac{D_F(x-y)^3}{12} + \frac{D_F(x-y)D_F(0)^2}{8} \right] + O(\lambda^4).$$

The two connected vacuum graphs have symmetry factors 12 and 8.

- (b) At order  $n$ , cubic theory contains  $3n$  free fields. A vacuum expectation value is a sum over complete pairings and hence vanishes when  $3n$  is odd. Since  $3n$  has the parity of  $n$ , every odd perturbative order vanishes.

At fourth order there are eight distinct vacuum topologies. Label the four cubic vertices  $1, \dots, 4$ ; write  $ij^r$  for  $r$  parallel lines between distinct vertices and  $ii$  for a tadpole at  $i$ . Relabelings of  $1, \dots, 4$  do not make a new diagram. The first five are connected and the last three are not:



In edge notation, with a semicolon separating components, the connected topologies are

|       | edge multiset                              | $S_\Gamma$ |
|-------|--------------------------------------------|------------|
| $V_1$ | 12, 13, 14, 23, 24, 34                     | 24         |
| $V_2$ | 12, 13, 14, 23 <sup>2</sup> , 44           | 8          |
| $V_3$ | 12, 13, 14, 22, 33, 44                     | 48         |
| $V_4$ | 12 <sup>2</sup> , 14, 23, 33, 44           | 16         |
| $V_5$ | 13, 14 <sup>2</sup> , 23 <sup>2</sup> , 24 | 16         |

and the disconnected ones are

|       |                          |      |
|-------|--------------------------|------|
| $V_6$ | $12^3; 33, 34, 44$       | 96   |
| $V_7$ | $14^3; 23^3$             | 288  |
| $V_8$ | $11, 14, 44; 22, 23, 33$ | 128. |

The last column, though not required in this part, supplies a check and will be useful below. Each factor is read off the drawing:  $V_1$  is  $K_4$  with  $|\text{Aut}| = 24$ ;  $V_8$  is a pair of dumbbells, giving  $8 \cdot 8 \cdot 2! = 128$ .

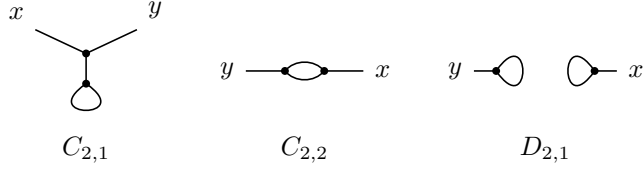
- (c) Extend the notation by writing  $xi$  and  $yi$  for the two labeled external attachments. Each row below denotes the corresponding position-space propagator product integrated over all numbered vertices; its coefficient is  $(i\lambda)^n/S_\Gamma$ , where  $n$  is the number of cubic vertices. The free diagram is

$$x \text{ ————— } y$$

$$C_0$$

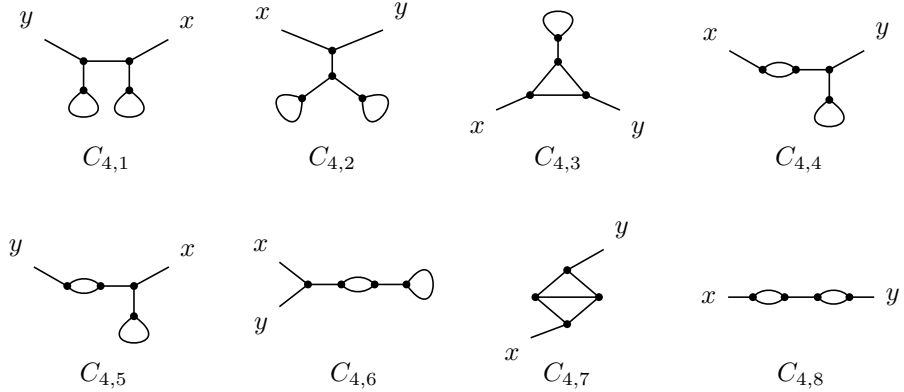
$$C_0|xy|1.$$

At order  $\lambda^2$ , the two connected diagrams and the one two-component diagram are

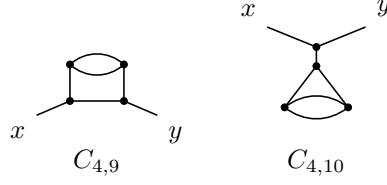


|           |                  |    |
|-----------|------------------|----|
| $C_{2,1}$ | $x1, y1, 12, 22$ | 2  |
| $C_{2,2}$ | $x2, y1, 12^2$   | 2  |
| $D_{2,1}$ | $x2, y1, 11, 22$ | 4. |

At order  $\lambda^4$ , the ten connected diagrams are

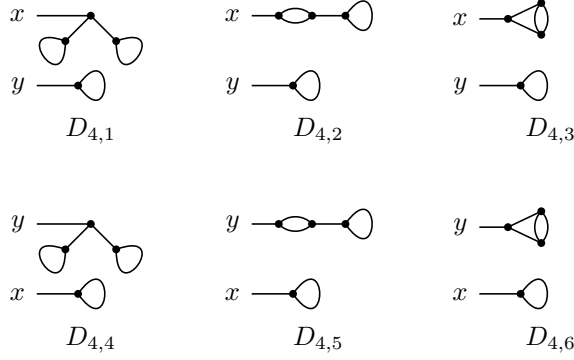






|            |                              |    |
|------------|------------------------------|----|
| $C_{4,1}$  | $x2, y1, 12, 14, 23, 33, 44$ | 4  |
| $C_{4,2}$  | $x2, y2, 12, 13, 14, 33, 44$ | 8  |
| $C_{4,3}$  | $x3, y2, 12, 13, 14, 23, 44$ | 2  |
| $C_{4,4}$  | $x3, y2, 12, 13^2, 24, 44$   | 4  |
| $C_{4,5}$  | $x3, y2, 12^2, 13, 34, 44$   | 4  |
| $C_{4,6}$  | $x3, y3, 12^2, 14, 23, 44$   | 4  |
| $C_{4,7}$  | $x4, y3, 12, 13, 14, 23, 24$ | 2  |
| $C_{4,8}$  | $x4, y3, 12, 14^2, 23^2$     | 4  |
| $C_{4,9}$  | $x4, y3, 12^2, 14, 23, 34$   | 2  |
| $C_{4,10}$ | $x4, y4, 12, 13, 14, 23^2$   | 4, |

and the six diagrams with two components are



|           |                              |    |
|-----------|------------------------------|----|
| $D_{4,1}$ | $x1, y4, 12, 13, 22, 33, 44$ | 16 |
| $D_{4,2}$ | $x2, y4, 12^2, 13, 33, 44$   | 8  |
| $D_{4,3}$ | $x3, y4, 12^2, 13, 23, 44$   | 8  |
| $D_{4,4}$ | $x4, y1, 12, 13, 22, 33, 44$ | 16 |
| $D_{4,5}$ | $x4, y2, 12^2, 13, 33, 44$   | 8  |
| $D_{4,6}$ | $x4, y3, 12^2, 13, 23, 44$   | 8. |

The free row, the two  $C_2$  rows, and the ten  $C_4$  rows give the thirteen fully connected diagrams advertised in the question. The seven  $D$  rows are not vacuum bubbles: each of their two components touches one external point and therefore survives the vacuum normalization.

For completeness, the tabulated factors follow from

$$S_\Gamma = |\text{Aut}_{x,y} \Gamma| \prod_i 2^{\ell_i} \ell_i! \prod_{i < j} m_{ij}!,$$

where  $\ell_i$  is the number of tadpoles at vertex  $i$ ,  $m_{ij}$  is the number of parallel  $ij$ -lines, and automorphisms fix  $x, y$ . This is equivalent to dividing the number

of labeled Wick pairings by the Dyson factor  $n!(3!)^n$ , so the list contains both every requested topology and its contraction weight.

#### S.4.7. A Classical Source and Coherent In–Out States

(a) The in–out relation gives

$$F' = S^{-1}FS, \quad G' = S^{-1}GS,$$

and therefore

$$e^{F'} = S^{-1}e^F S, \quad e^{G'} = S^{-1}e^G S.$$

Because  $\mu$  is a c-number,

$$e^{-\mu/2}e^{F'}e^{G'} = S^{-1}\left(e^{-\mu/2}e^F e^G\right)S = S^{-1}SS = S.$$

With the convention  $|0\rangle_{\text{in}} = S|0\rangle_{\text{out}}$ , and since  $G'|0\rangle_{\text{out}} = 0$ ,

$$|0\rangle_{\text{in}} = S|0\rangle_{\text{out}} = e^{-\mu/2}e^{F'}|0\rangle_{\text{out}}.$$

Thus the in-vacuum is a normalized multimode coherent state of out-particles.

(b) Since  $[a, za^\dagger] = z$ ,  $ae^{za^\dagger} = e^{za^\dagger}(a+z)$ , so  $a|z\rangle = z|z\rangle$ . Equivalently,

$$|z\rangle = \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n!}}|n\rangle,$$

so the eigenvalue is  $z$ .

(c) Using the number-state expansion,

$$\langle z_2|z_1\rangle = \sum_{n=0}^{\infty} \frac{(z_2^* z_1)^n}{n!} = e^{z_2^* z_1}.$$

In particular,  $\langle z|z\rangle = e^{|z|^2}$ ; the state in this problem is deliberately unnormalized.

(d) Since  $a|z\rangle = z|z\rangle$  and  $\langle z|a^\dagger = z^*\langle z|$ ,

$$\langle q\rangle = \frac{z+z^*}{\sqrt{2}} = \sqrt{2} \operatorname{Re} z, \quad \langle p\rangle = \frac{z-z^*}{i\sqrt{2}} = \sqrt{2} \operatorname{Im} z.$$

(e) Normal ordering  $q^2$  and  $p^2$  and using  $[a, a^\dagger] = 1$  gives

$$\langle q^2\rangle = \langle q\rangle^2 + \frac{1}{2}, \quad \langle p^2\rangle = \langle p\rangle^2 + \frac{1}{2}.$$

Hence

$$\Delta q = \Delta p = \frac{1}{\sqrt{2}}, \quad \Delta q \Delta p = \frac{1}{2}.$$

The displacement changes the means but leaves the vacuum fluctuations unchanged.

#### S.4.8. Recovering a Classical Field from an Impulse

- (a) The equal-time algebra and the free Hamiltonian give

$$\dot{\phi} = i[H, \phi] = \pi, \quad \dot{\pi} = i[H, \pi] = (\nabla^2 - m^2)\phi + J.$$

Consequently

$$\ddot{\phi} = (\nabla^2 - m^2)\phi + J,$$

which is precisely  $(\square - m^2)\phi = -J$  for  $\square = \nabla^2 - \partial_t^2$ .

- (b) Integrating the equation across an infinitesimal interval containing  $t = 0$ , the finite spatial and mass terms vanish in the limit, while  $\int dt J = j$ . Hence

$$\dot{\phi}(0^+) - \dot{\phi}(0^-) = j.$$

The field itself cannot jump without producing a derivative of a delta function, so

$$\phi(0^+) = \phi(0^-), \quad \pi(0^+) = \pi(0^-) + j.$$

- (c) The inverse of the displayed mode expansion is

$$a(\mathbf{p}, 0) = \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} \left[ \sqrt{\frac{E_{\mathbf{p}}}{2}} \phi(\mathbf{x}, 0) + \frac{i}{\sqrt{2E_{\mathbf{p}}}} \pi(\mathbf{x}, 0) \right].$$

Only the momentum jump contributes, giving

$$a(\mathbf{p}, 0^+) - a(\mathbf{p}, 0^-) = \frac{i \tilde{j}(\mathbf{p})}{\sqrt{2E_{\mathbf{p}}}}.$$

- (d) Let  $\alpha(\mathbf{p}) = \tilde{j}(\mathbf{p})/\sqrt{2E_{\mathbf{p}}}$ . Since the pre-impulse vacuum obeys  $a(\mathbf{p}, 0^-)|0\rangle = 0$ ,

$$a(\mathbf{p}, 0^+)|0\rangle = \alpha(\mathbf{p})|0\rangle.$$

Thus the same state is a multimode coherent state for the post-impulse annihilation operators. Its one-particle density matrix is

$$\langle 0|a^\dagger(\mathbf{p}, 0^+)a(\mathbf{q}, 0^+)|0\rangle = \alpha(\mathbf{p})^* \alpha(\mathbf{q}),$$

so the diagonal momentum density is  $|\tilde{j}(\mathbf{p})|^2/(2E_{\mathbf{p}})$ , with the  $(2\pi)^3$  measure fixed by the mode convention in the question.

#### S.4.9. Relativistic Propagators and Causal Support

- (a) The positive-frequency contraction is

$$D(z) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2E_{\mathbf{p}}} e^{ip\cdot z} = \int \frac{d^4p}{(2\pi)^3} \delta(p^2 + m^2) \theta(p^0) e^{ip\cdot z}.$$

- (b) The measure, mass-shell delta, positive-energy sheet, and scalar product are invariant under the proper orthochronous Lorentz group; hence  $D(\Lambda z) = D(z)$ .
- (c) For spacelike  $z$ , choose  $z^0 = 0$  and set  $r = \sqrt{z^2}$ . The radial integral gives

$$D(z) = \frac{m}{4\pi^2 r} K_1(mr) \sim \frac{\sqrt{m}}{2(2\pi)^{3/2} r^{3/2}} e^{-mr}.$$

- (d)  $[\phi(x), \phi(y)] = D(x - y) - D(y - x)$ . Spacelike-separated points have a frame in which they are simultaneous, so the two terms coincide and the vacuum expectation value vanishes. For a free field the commutator is a c-number, so the operator commutator vanishes as well.
- (e) For  $\mathcal{L} = \mathcal{L}_0 + J\phi$ , variation gives  $(\square - m^2)\phi = -J$ .
- (f) Applying the Klein–Gordon operator gives

$$(\square - m^2) \left[ i \int d^4 y G(x - y) J(y) \right] = i^2 J(x) = -J(x).$$

Thus

$$\phi(x) = \phi_{\text{hom}}(x) + i \int d^4 y G(x - y) J(y), \quad (\square - m^2)\phi_{\text{hom}} = 0.$$

- (g) For  $z^0 > 0$ , close the  $p^0$  contour in the lower half-plane; for  $z^0 < 0$ , close it in the upper half-plane. The retarded prescription places both poles below the real axis, the advanced prescription places both above it, and the Feynman prescription places the positive-energy pole below and the negative-energy pole above. Summing residues gives

$$\begin{aligned} G_R(z) &= \theta(z^0)[D(z) - D(-z)], \\ G_A(z) &= \theta(-z^0)[D(-z) - D(z)], \\ G_F(z) &= \theta(z^0)D(z) + \theta(-z^0)D(-z). \end{aligned}$$

- (h) Since  $G_R(z) = \theta(z^0)[D(z) - D(-z)]$ , part 4 makes it vanish outside the future light cone. The retarded source response is causal.
- (i) At spacelike separation  $G_F = D$ , which is nonzero but exponentially small. It encodes vacuum correlation; signaling is governed by the retarded commutator and therefore remains causal.

#### S.4.10. The Force Between Static Sources

- (a) Expand the exponential and use Wick’s theorem. Odd terms vanish. At order  $2n$ , complete pairings contribute  $(2n)!/(2^n n!)$  copies of the same product, so

$$\begin{aligned} Z[J] &= \sum_{n=0}^{\infty} \frac{i^{2n}}{(2n)!} \frac{(2n)!}{2^n n!} \left[ \int d^4 x d^4 y J(x) D_F(x - y) J(y) \right]^n \\ &= \exp \left[ -\frac{1}{2} \int d^4 x d^4 y J(x) D_F(x - y) J(y) \right]. \end{aligned}$$

This is the exact Gaussian result.

- (b) Write  $J = J_1 + J_2$ . The terms  $J_1 D_F J_1$  and  $J_2 D_F J_2$  are the discarded point-source self-energies. The factor  $1/2$  cancels the two equal cross terms, leaving

$$\log Z_{\text{int}} = -g^2 \int_{-T/2}^{T/2} dt dt' D_F(t - t', \mathbf{R}).$$

For  $T$  much longer than every other scale, set  $\tau = t - t'$  and replace the overlap length by  $T$ :

$$\log Z_{\text{int}} = -g^2 T \int_{-\infty}^{\infty} d\tau D_F(\tau, \mathbf{R}) + O(T^0).$$

With the mostly-plus propagator convention,

$$D_F(x) = \int \frac{d^4 p}{(2\pi)^4} \frac{-i e^{ip \cdot x}}{p^2 + m^2 - i\epsilon},$$

the time integral sets  $p^0 = 0$ , and therefore

$$\int d\tau D_F(\tau, \mathbf{R}) = -i \int \frac{d^3 p}{(2\pi)^3} \frac{e^{i\mathbf{p} \cdot \mathbf{R}}}{\mathbf{p}^2 + m^2} = -\frac{i}{4\pi R} e^{-mR}.$$

Consequently

$$Z_{\text{int}} = \exp \left[ \frac{ig^2 T}{4\pi R} e^{-mR} \right]$$

up to the omitted self-energy phase and endpoint corrections.

- (c) Comparison with  $Z_{\text{int}} = e^{-iV(R)T}$  gives the Yukawa potential

$$V(R) = -\frac{g^2}{4\pi R} e^{-mR}.$$

Since

$$-\frac{dV}{dR} = -\frac{g^2 e^{-mR}}{4\pi} \left( \frac{m}{R} + \frac{1}{R^2} \right) < 0$$

when the positive radial direction points from one source away from the other, the force is attractive.

## References

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## 5 Quantum Fields and Antiparticles

We now have all the pieces needed to motivate the introduction of quantum fields [1]. In the course of this construction, we shall encounter some of the most remarkable and universal consequences of the union of relativity with quantum mechanics: the connection between spin and statistics, the existence of antiparticles, and various relationships between particles and antiparticles, including the celebrated CPT theorem.

### 5.1 Free Fields

We have seen in Chapter 3 that the  $S$ -matrix will be Lorentz-invariant if the interaction can be written as

$$H_I(t) = \int d^3x \mathcal{H}_I(\mathbf{x}, t), \quad (5.1.1)$$

where  $\mathcal{H}_I$  is a scalar, in the sense that

$$U_0(\Lambda, a) \mathcal{H}_I(x) U_0^{-1}(\Lambda, a) = \mathcal{H}_I(\Lambda x + a), \quad (5.1.2)$$

and satisfies the additional condition:

$$[\mathcal{H}_I(x), \mathcal{H}_I(x')] = 0 \quad \text{for} \quad (x - x')^2 \geq 0. \quad (5.1.3)$$

As we shall see, there are more general possibilities, but none of them are very different from this. (For the present we are leaving it as an open question whether  $\Lambda$  here is restricted to a proper orthochronous Lorentz transformation, or can also include space inversions.) In order to facilitate also satisfying the cluster decomposition principle we are going to construct  $\mathcal{H}_I(x)$  out of creation and annihilation operators, but here we face a problem: as shown by Eq. (4.2.12), under Lorentz transformations each such operator is multiplied by a matrix that depends on the momentum carried by that operator. How can we couple such operators together to make a scalar? The solution is to build  $\mathcal{H}_I(x)$  out of fields—both annihilation fields  $\psi_\ell^{(+)}(x)$  and creation fields  $\psi_\ell^{(-)}(x)$ :

$$\psi_\ell^{(+)}(x) = \sum_{\sigma n} \int d^3p u_\ell(x; \mathbf{p}, \sigma, n) a_{\mathbf{p}, \sigma, n}, \quad (5.1.4)$$

$$\psi_\ell^{(-)}(x) = \sum_{\sigma n} \int d^3p v_\ell(x; \mathbf{p}, \sigma, n) a_{\mathbf{p}, \sigma, n}^\dagger, \quad (5.1.5)$$

with coefficients<sup>36</sup>  $u_\ell(x; \mathbf{p}, \sigma, n)$  and  $v_\ell(x; \mathbf{p}, \sigma, n)$  chosen so that under Lorentz transformations each field is multiplied with a position-independent matrix:

$$U_0(\Lambda, a) \psi_\ell^{(+)}(x) U_0^{-1}(\Lambda, a) = \sum_{\bar{\ell}} D_{\ell \bar{\ell}}(\Lambda^{-1}) \psi_{\bar{\ell}}^{(+)}(\Lambda x + a), \quad (5.1.6)$$

$$U_0(\Lambda, a) \psi_\ell^{(-)}(x) U_0^{-1}(\Lambda, a) = \sum_{\bar{\ell}} D_{\ell \bar{\ell}}(\Lambda^{-1}) \psi_{\bar{\ell}}^{(-)}(\Lambda x + a). \quad (5.1.7)$$

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<sup>36</sup>A reminder: the labels  $n$  and  $\sigma$  run over all different particle species and spin  $z$ -components, respectively.

(We might, in principle, have different transformation matrices  $D^\pm$  for the annihilation and creation fields, but as we shall see, it is always possible to choose the fields so that these matrices are the same.) By applying a second Lorentz transformation  $\bar{\Lambda}$ , we find that

$$D(\Lambda^{-1})D(\bar{\Lambda}^{-1}) = D((\bar{\Lambda}\Lambda)^{-1}),$$

so taking  $\Lambda_1 = (\Lambda)^{-1}$  and  $\Lambda_2 = (\bar{\Lambda})^{-1}$ , we see that the  $D$ -matrices furnish a *representation* of the homogeneous Lorentz group:

$$D(\Lambda_1)D(\Lambda_2) = D(\Lambda_1\Lambda_2). \quad (5.1.8)$$

There are many such representations, including the scalar  $D(\Lambda) = \mathbf{1}$ , the vector  $D(\Lambda)^\mu{}_\nu = \Lambda^\mu{}_\nu$ , and a host of tensor and spinor representations. These particular representations are irreducible, in the sense that it is not possible by a choice of basis to reduce all  $D(\Lambda)$  to the same block-diagonal form, with two or more blocks. However, we do not require at this point that  $D(\Lambda)$  be irreducible; in general it is a block-diagonal matrix with an arbitrary array of irreducible representations in the blocks. That is, the index  $\ell$  here includes a label that runs over the types of particle described and the irreducible representations in the different blocks, as well as another that runs over the components of the individual irreducible representations. Later we will separate these fields into irreducible fields that each describe only a single particle species (and its antiparticle) and transform irreducibly under the Lorentz group.

Once we have learned how to construct fields satisfying the Lorentz transformation rules (5.1.6) and (5.1.7), we will be able to construct the interaction density as

$$\begin{aligned} \mathcal{H}_I(x) = & \sum_{NM} \sum_{\ell'_1 \dots \ell'_N} \sum_{\ell_1 \dots \ell_M} g_{\ell'_1 \dots \ell'_N, \ell_1 \dots \ell_M} \\ & \cdot \psi_{\ell'_1}^{(-)}(x) \dots \psi_{\ell'_N}^{(-)}(x) \psi_{\ell_1}^{(+)}(x) \dots \psi_{\ell_M}^{(+)}(x), \end{aligned} \quad (5.1.9)$$

and this will be a scalar in the sense of Eq. (5.1.2) if the constant coefficients  $g_{\ell'_1 \dots \ell'_N, \ell_1 \dots \ell_M}$  are chosen to be Lorentz covariant, in the sense that for all  $\Lambda$ :

$$\begin{aligned} & \sum_{\ell'_1 \dots \ell'_N} \sum_{\ell_1 \dots \ell_M} D_{\ell'_1 \bar{\ell}'_1}(\Lambda^{-1}) \dots D_{\ell'_N \bar{\ell}'_N}(\Lambda^{-1}) D_{\ell_1 \bar{\ell}_1}(\Lambda^{-1}) \dots D_{\ell_M \bar{\ell}_M}(\Lambda^{-1}) \\ & \cdot g_{\ell'_1 \dots \ell'_N, \ell_1 \dots \ell_M} = g_{\bar{\ell}'_1 \dots \bar{\ell}'_N, \bar{\ell}_1 \dots \bar{\ell}_M}. \end{aligned} \quad (5.1.10)$$

(Note that we do not include derivatives here, because we regard the derivatives of components of these fields as just additional sorts of field components.) The task of finding coefficients  $g_{\ell'_1 \dots \ell'_N, \ell_1 \dots \ell_M}$  that satisfy Eq. (5.1.10) is no different in principle (and not much more difficult in practice) than that of using Clebsch–Gordan coefficients to couple together various representations of the three-dimensional rotation group to form rotational scalars. Later we will be able to combine creation and annihilation fields so that this density also commutes with itself at space-like separations.

Now, what shall we take as the coefficient functions  $u_\ell(x; \mathbf{p}, \sigma, n)$  and  $v_\ell(x; \mathbf{p}, \sigma, n)$ ? Eq. (4.2.12) and its adjoint give the transformation rules<sup>37</sup> for the annihilation and creation

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<sup>37</sup>This is for massive particles. The case of zero mass will be taken up in Section 5.9.



operators

$$\begin{aligned}
U_0(\Lambda, b) a_{\mathbf{p}, \sigma, n} U_0^{-1}(\Lambda, b) &= \exp(i(\Lambda p) \cdot b) \sqrt{\frac{(\Lambda p)^0}{p^0}} \\
&\cdot \sum_{\bar{\sigma}} D_{\sigma \bar{\sigma}}^{(j_n)}(W^{-1}(\Lambda, p)) a_{\mathbf{p}_\Lambda, \bar{\sigma}, n},
\end{aligned} \tag{5.1.11}$$

$$\begin{aligned}
U_0(\Lambda, b) a_{\mathbf{p}, \sigma, n}^\dagger U_0^{-1}(\Lambda, b) &= \exp(-i(\Lambda p) \cdot b) \sqrt{\frac{(\Lambda p)^0}{p^0}} \\
&\cdot \sum_{\bar{\sigma}} D_{\sigma \bar{\sigma}}^{(j_n)*}(W^{-1}(\Lambda, p)) a_{\mathbf{p}_\Lambda, \bar{\sigma}, n}^\dagger,
\end{aligned} \tag{5.1.12}$$

where  $j_n$  is the spin of particles of species  $n$ , and  $\mathbf{p}_\Lambda$  is the three-vector part of  $\Lambda p$ . (We have used the unitarity of the rotation matrices  $D^{(j_n)}$  to put both Eqs. (5.1.11) and (5.1.12) in the form shown here.) Also, as we saw in Section 2.5 the volume element  $d^3 p/p^0$  is Lorentz-invariant, so we can replace  $d^3 p$  in Eqs. (5.1.4) and (5.1.5) with  $d^3(\Lambda p)p^0/(\Lambda p)^0$ . Putting this all together, we find

$$\begin{aligned}
U_0(\Lambda, b) \psi_\ell^{(+)}(x) U_0^{-1}(\Lambda, b) &= \sum_{\sigma \bar{\sigma} n} \int d^3(\Lambda p) u_\ell(x; \mathbf{p}, \sigma, n) \\
&\cdot \exp(i(\Lambda p) \cdot b) D_{\sigma \bar{\sigma}}^{(j_n)}(W^{-1}(\Lambda, p)) \sqrt{\frac{p^0}{(\Lambda p)^0}} a_{\mathbf{p}_\Lambda, \bar{\sigma}, n}
\end{aligned}$$

and

$$\begin{aligned}
U_0(\Lambda, b) \psi_\ell^{(-)}(x) U_0^{-1}(\Lambda, b) &= \sum_{\sigma \bar{\sigma} n} \int d^3(\Lambda p) v_\ell(x; \mathbf{p}, \sigma, n) \\
&\cdot \exp(-i(\Lambda p) \cdot b) D_{\sigma \bar{\sigma}}^{(j_n)*}(W^{-1}(\Lambda, p)) \sqrt{\frac{p^0}{(\Lambda p)^0}} a_{\mathbf{p}_\Lambda, \bar{\sigma}, n}^\dagger.
\end{aligned}$$

We see that in order for the fields to satisfy the Lorentz transformation rules (5.1.6) and (5.1.7), it is necessary and sufficient that

$$\begin{aligned}
\sum_{\bar{\ell}} D_{\ell \bar{\ell}}(\Lambda^{-1}) u_{\bar{\ell}}(\Lambda x + b; \mathbf{p}_\Lambda, \sigma, n) &= \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\bar{\sigma}} D_{\sigma \bar{\sigma}}^{(j_n)}(W^{-1}(\Lambda, p)) \\
&\cdot \exp(+i(\Lambda p) \cdot b) u_\ell(x; \mathbf{p}, \bar{\sigma}, n)
\end{aligned}$$

and

$$\begin{aligned}
\sum_{\bar{\ell}} D_{\ell \bar{\ell}}(\Lambda^{-1}) v_{\bar{\ell}}(\Lambda x + b; \mathbf{p}_\Lambda, \sigma, n) &= \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\bar{\sigma}} D_{\sigma \bar{\sigma}}^{(j_n)*}(W^{-1}(\Lambda, p)) \\
&\cdot \exp(-i(\Lambda p) \cdot b) v_\ell(x; \mathbf{p}, \bar{\sigma}, n),
\end{aligned}$$

or somewhat more conveniently

$$\sum_{\bar{\sigma}} u_{\ell}(\Lambda x + b; \mathbf{p}_{\Lambda}, \bar{\sigma}, n) D_{\bar{\sigma}\sigma}^{(j_n)}(W(\Lambda, p)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \cdot \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(\Lambda) \exp(i(\Lambda p) \cdot b) u_{\bar{\ell}}(x; \mathbf{p}, \sigma, n), \quad (5.1.13)$$

and

$$\sum_{\bar{\sigma}} v_{\ell}(\Lambda x + b; \mathbf{p}_{\Lambda}, \bar{\sigma}, n) D_{\bar{\sigma}\sigma}^{(j_n)*}(W(\Lambda, p)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \cdot \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(\Lambda) \exp(-i(\Lambda p) \cdot b) v_{\bar{\ell}}(x; \mathbf{p}, \sigma, n). \quad (5.1.14)$$

These are the fundamental requirements that will allow us to calculate the  $u_{\ell}$  and  $v_{\ell}$  coefficient functions in terms of a finite number of free parameters.

### Translations

We will use Eqs. (5.1.13) and (5.1.14) in three steps, considering in turn three different types of proper orthochronous Lorentz transformation. First we consider Eqs. (5.1.13) and (5.1.14) with  $\Lambda = \mathbf{1}$  and  $b$  arbitrary. We see immediately that  $u_{\ell}(x; \mathbf{p}, \sigma, n)$  and  $v_{\ell}(x; \mathbf{p}, \sigma, n)$  must take the form

$$u_{\ell}(x; \mathbf{p}, \sigma, n) = (2\pi)^{-3/2} e^{ip \cdot x} u_{\ell}(\mathbf{p}, \sigma, n), \quad (5.1.15)$$

$$v_{\ell}(x; \mathbf{p}, \sigma, n) = (2\pi)^{-3/2} e^{-ip \cdot x} v_{\ell}(\mathbf{p}, \sigma, n), \quad (5.1.16)$$

so the fields are Fourier transforms:

$$\psi_{\ell}^{(+)}(x) = \sum_{\sigma, n} (2\pi)^{-3/2} \int d^3 p u_{\ell}(\mathbf{p}, \sigma, n) e^{ip \cdot x} a_{\mathbf{p}, \sigma, n}, \quad (5.1.17)$$

and

$$\psi_{\ell}^{(-)}(x) = \sum_{\sigma, n} (2\pi)^{-3/2} \int d^3 p v_{\ell}(\mathbf{p}, \sigma, n) e^{-ip \cdot x} a_{\mathbf{p}, \sigma, n}^{\dagger}. \quad (5.1.18)$$

(The factors  $(2\pi)^{-3/2}$  could be absorbed into the definition of  $u_{\ell}$  and  $v_{\ell}$ , but it is conventional to show them explicitly in these Fourier integrals.) Using Eqs. (5.1.15) and (5.1.16), we see that Eqs. (5.1.13) and (5.1.14) are satisfied if and only if

$$\sum_{\bar{\sigma}} u_{\ell}(\mathbf{p}_{\Lambda}, \bar{\sigma}, n) D_{\bar{\sigma}\sigma}^{(j_n)}(W(\Lambda, p)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(\Lambda) u_{\bar{\ell}}(\mathbf{p}, \sigma, n) \quad (5.1.19)$$

and

$$\sum_{\bar{\sigma}} v_{\ell}(\mathbf{p}_{\Lambda}, \bar{\sigma}, n) D_{\bar{\sigma}\sigma}^{(j_n)*}(W(\Lambda, p)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(\Lambda) v_{\bar{\ell}}(\mathbf{p}, \sigma, n) \quad (5.1.20)$$

for arbitrary homogeneous Lorentz transformations  $\Lambda$ .

## Boosts

Next take  $\mathbf{p} = 0$  in Eqs. (5.1.19) and (5.1.20), and let  $\Lambda$  be the standard boost  $L(q)$  that takes a particle of mass  $m$  from rest to some four-momentum  $q^\mu$ . Then  $L(p) = \mathbf{1}$ , and

$$W(\Lambda, p) \equiv L^{-1}(\Lambda p) \Lambda L(p) = L^{-1}(q) L(q) = \mathbf{1}.$$

Hence in this special case, Eqs. (5.1.19) and (5.1.20) give

$$u_\ell(\mathbf{q}, \sigma, n) = (m/q^0)^{1/2} \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(L(q)) u_{\bar{\ell}}(0, \sigma, n) \quad (5.1.21)$$

and

$$v_\ell(\mathbf{q}, \sigma, n) = (m/q^0)^{1/2} \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(L(q)) v_{\bar{\ell}}(0, \sigma, n). \quad (5.1.22)$$

In other words, if we know the quantities  $u_\ell(0, \sigma, n)$  and  $v_\ell(0, \sigma, n)$  for zero momentum, then for a given representation  $D(\Lambda)$  of the homogeneous Lorentz group, we know the functions  $u_\ell(\mathbf{p}, \sigma, n)$  and  $v_\ell(\mathbf{p}, \sigma, n)$  for all  $\mathbf{p}$ . (Explicit formulas for the matrices  $D_{\ell\bar{\ell}}(L(q))$  will be given for arbitrary representations of the homogeneous Lorentz group in Section 5.7.)

## Rotations

Next, take  $\mathbf{p} = 0$ , but this time let  $\Lambda$  be a Lorentz transformation with  $\mathbf{p}_\Lambda = 0$ ; that is, take  $\Lambda$  as a rotation  $R$ . Here obviously  $W(\Lambda, p) = R$ , and so Eqs. (5.1.19) and (5.1.20) read

$$\sum_{\bar{\sigma}} u_\ell(0, \bar{\sigma}, n) D_{\bar{\sigma}\sigma}^{(j_n)}(R) = \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(R) u_{\bar{\ell}}(0, \sigma, n) \quad (5.1.23)$$

and

$$\sum_{\bar{\sigma}} v_\ell(0, \bar{\sigma}, n) D_{\bar{\sigma}\sigma}^{(j_n)*}(R) = \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(R) v_{\bar{\ell}}(0, \sigma, n), \quad (5.1.24)$$

or equivalently

$$\sum_{\bar{\sigma}} u_\ell(0, \bar{\sigma}, n) J_{\bar{\sigma}\sigma}^{(j_n)} = \sum_{\bar{\ell}} \mathcal{J}_{\ell\bar{\ell}} u_{\bar{\ell}}(0, \sigma, n) \quad (5.1.25)$$

and

$$\sum_{\bar{\sigma}} v_\ell(0, \bar{\sigma}, n) J_{\bar{\sigma}\sigma}^{(j_n)*} = - \sum_{\bar{\ell}} \mathcal{J}_{\ell\bar{\ell}} v_{\bar{\ell}}(0, \sigma, n), \quad (5.1.26)$$

where  $\mathbf{J}^{(j)}$  and  $\mathcal{J}$  are the angular-momentum matrices in the representations  $D^{(j)}(R)$  and  $D(R)$ , respectively. Any representation  $D(\Lambda)$  of the homogeneous Lorentz group obviously yields a representation of the rotation group when  $\Lambda$  is restricted to rotations  $R$ ; Eqs. (5.1.25) and (5.1.26) tell us that if the field  $\psi_\ell^{(\pm)}(x)$  is to describe particles of some particular spin  $j$ , then this representation  $D(R)$  must contain among its irreducible components the spin- $j$  representation  $D^{(j)}(R)$ , with the coefficients  $u_\ell(0, \sigma, n)$  and  $v_\ell(0, \sigma, n)$  simply describing how the spin- $j$  representation of the rotation group is embedded in  $D(R)$ . We shall see in Section 5.5 that each irreducible representation of the proper orthochronous Lorentz group contains any given irreducible representation of the rotation group at most once, so that if the fields  $\psi_\ell^{(+)}(x)$  and  $\psi_\ell^{(-)}(x)$  transform irreducibly, then they are unique

up to overall scale. More generally, the number of free parameters in the annihilation or creation fields (including their overall scales) is equal to the number of irreducible representations in the field.

It is straightforward to show that coefficient functions  $u_\ell(\mathbf{p}, \sigma, n)$  and  $v_\ell(\mathbf{p}, \sigma, n)$  given by Eqs. (5.1.21) and (5.1.22), with  $u_\ell(0, \sigma, n)$  and  $v_\ell(0, \sigma, n)$  satisfying Eqs. (5.1.23) and (5.1.24), will automatically satisfy the more general requirements (5.1.19) and (5.1.20). This is left as an exercise for the reader.

Let us now return to the cluster decomposition principle. Inserting Eqs. (5.1.17) and (5.1.18) in Eq. (5.1.9) and integrating over  $\mathbf{x}$ , the interaction Hamiltonian is

$$\begin{aligned} H_{\text{int}} = & \sum_{NM} \int d^3 p'_1 \cdots d^3 p'_N d^3 p_1 \cdots d^3 p_M \sum_{\sigma'_1 \cdots \sigma'_N} \sum_{\sigma_1 \cdots \sigma_M} \sum_{n'_1 \cdots n'_N} \sum_{n_1 \cdots n_M} \\ & \cdot a_{\mathbf{p}'_1, \sigma'_1, n'_1}^\dagger \cdots a_{\mathbf{p}'_N, \sigma'_N, n'_N}^\dagger a_{\mathbf{p}_M, \sigma_M, n_M} \cdots a_{\mathbf{p}_1, \sigma_1, n_1} \\ & \cdot \mathcal{V}_{NM}(\mathbf{p}'_1 \sigma'_1 n'_1 \cdots \mathbf{p}'_N \sigma'_N n'_N, \mathbf{p}_1 \sigma_1 n_1 \cdots \mathbf{p}_M \sigma_M n_M) \end{aligned} \quad (5.1.27)$$

with coefficient functions given by

$$\begin{aligned} \mathcal{V}_{NM}(\mathbf{p}'_1 \sigma'_1 n'_1 \cdots \mathbf{p}_1 \sigma_1 n_1 \cdots) &= \delta^3(\mathbf{p}'_1 + \cdots - \mathbf{p}_1 - \cdots) \\ &\cdot \tilde{\mathcal{V}}_{NM}(\mathbf{p}'_1 \sigma'_1 n'_1 \cdots \mathbf{p}_1 \sigma_1 n_1 \cdots), \end{aligned} \quad (5.1.28)$$

where

$$\begin{aligned} \tilde{\mathcal{V}}_{NM}(\mathbf{p}'_1 \sigma'_1 n'_1 \cdots \mathbf{p}'_N \sigma'_N n'_N, \mathbf{p}_1 \sigma_1 n_1 \cdots \mathbf{p}_M \sigma_M n_M) &= (2\pi)^{3-3N/2-3M/2} \\ &\cdot \sum_{\ell'_1 \cdots \ell'_N} \sum_{\ell_1 \cdots \ell_M} g_{\ell'_1 \cdots \ell'_N, \ell_1 \cdots \ell_M} v_{\ell'_1}(\mathbf{p}'_1, \sigma'_1, n'_1) \cdots v_{\ell'_N}(\mathbf{p}'_N, \sigma'_N, n'_N) \\ &\cdot u_{\ell_1}(\mathbf{p}_1, \sigma_1, n_1) \cdots u_{\ell_M}(\mathbf{p}_M, \sigma_M, n_M). \end{aligned} \quad (5.1.29)$$

This interaction is manifestly of the form that will guarantee that the  $S$ -matrix satisfies the cluster decomposition principle:  $\mathcal{V}_{NM}$  has a single delta function factor, with a coefficient  $\tilde{\mathcal{V}}_{NM}$  that (at least for a finite number of field types) has at most branch point singularities at zero particle momenta. In fact, we could turn this argument around; any operator can be written as in Eq. (5.1.27), and the cluster decomposition principle requires that the coefficient  $\mathcal{V}_{NM}$  may be written as in Eq. (5.1.28) as the product of a single momentum-conservation delta function times a smooth coefficient function. Any sufficiently smooth function (but not one containing additional delta functions) can be expressed as in Eq. (5.1.29).<sup>38</sup> *The cluster decomposition principle together with Lorentz invariance thus makes it natural that the interaction density should be constructed out of the annihilation and creation fields.*

If all we needed were to construct a scalar interaction density that satisfied the cluster decomposition principle, then we could combine annihilation and creation operators in

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<sup>38</sup>For general functions the indices  $\ell$  and  $\ell'$  may have to run over an infinite range. The reasons for restricting  $\ell$  and  $\ell'$  to a finite range have to do with the principle of renormalizability, discussed in Chapter 12.

arbitrary polynomials (5.1.9), with coupling coefficients  $g_{\ell'_1 \dots \ell'_N, \ell_1 \dots \ell_M}$  subject only to the invariance condition (5.1.10) (and a suitable reality condition). However, for the Lorentz invariance of the  $S$ -matrix it is necessary also that the interaction density satisfy the commutation condition (5.1.3). This condition is not satisfied for arbitrary functions of the creation and annihilation fields because

$$[\psi_\ell^{(+)}(x), \psi_{\ell'}^{(-)}(y)]_{\mp} = (2\pi)^{-3} \sum_{\sigma n} \int d^3p u_\ell(\mathbf{p}, \sigma, n) v_{\ell'}(\mathbf{p}, \sigma, n) e^{ip \cdot (x-y)} \quad (5.1.30)$$

(with the sign  $\mp$  indicating a commutator or anticommutator if the particles destroyed and created by the components  $\psi_\ell^{(+)}$  and  $\psi_{\ell'}^{(-)}$  are bosons or fermions, respectively,) and in general this does not vanish even for  $x - y$  space-like. It is obviously not possible to avoid this problem by making the interaction density out of creation or annihilation fields alone, for then the interaction could not be Hermitian. The only way out of this difficulty is to combine annihilation and creation fields in linear combinations:

$$\psi_\ell(x) \equiv \kappa \psi_\ell^{(+)}(x) + \lambda \psi_\ell^{(-)}(x), \quad (5.1.31)$$

with the constants  $\kappa$  and  $\lambda$  and any other arbitrary constants in the fields adjusted so that for  $x - y$  space-like

$$[\psi_\ell(x), \psi_{\ell'}(y)]_{\mp} = [\psi_\ell(x), \psi_{\ell'}^\dagger(y)]_{\mp} = 0. \quad (5.1.32)$$

We will see in subsequent sections of this chapter how to do this for various irreducibly transforming fields. (By including explicit constants  $\kappa$  and  $\lambda$  in Eq. (5.1.31) we are leaving ourselves free to choose the overall scale of the annihilation and creation fields in any way that seems convenient.) The Hamiltonian density  $\mathcal{H}_I(x)$  will satisfy the commutation condition (5.1.3) if it is constructed out of such fields and their adjoints, with an even number of any field components that destroy and create fermions.

The condition (5.1.32) is often described as a *causality* condition, because if  $x - y$  is space-like then no signal can reach  $y$  from  $x$ , so that a measurement of  $\psi_\ell$  at point  $x$  should not be able to interfere with a measurement of  $\psi_{\ell'}$  or  $\psi_{\ell'}^\dagger$  at point  $y$ . Such considerations of causality are plausible for the electromagnetic field, any one of whose components may be measured at a given spacetime point, as shown in a classic paper of Bohr and Rosenfeld [2]. However, we will be dealing here with fields like the Dirac field of the electron that do not seem in any sense measurable. The point of view taken here is that Eq. (5.1.32) is needed for the Lorentz invariance of the  $S$ -matrix, without any ancillary assumptions about measurability or causality.

There is an obstacle to the construction of fields (5.1.31) satisfying (5.1.32). It may be that the particles that are destroyed and created by these fields carry non-zero values of one or more conserved quantum numbers like the electric charge. For instance, if particles of species  $n$  carry a value  $q(n)$  for the electric charge  $Q$ , then

$$[Q, a_{\mathbf{p}, \sigma, n}] = -q(n) a_{\mathbf{p}, \sigma, n},$$

$$[Q, a_{\mathbf{p}, \sigma, n}^\dagger] = +q(n) a_{\mathbf{p}, \sigma, n}^\dagger.$$

In order that  $\mathcal{H}_I(x)$  should commute with the charge operator  $Q$  (or some other symmetry operator) it is necessary that it be formed out of fields that have simple commutation relations with  $Q$ :

$$[Q, \psi_\ell(x)] = -q_\ell \psi_\ell(x) \quad (5.1.33)$$

for then we can make  $\mathcal{H}_I(x)$  commute with  $Q$  by constructing it as a sum of products of fields  $\psi_{\ell_1}, \psi_{\ell_2}, \dots$  and adjoints  $\psi_{m_1}^\dagger, \psi_{m_2}^\dagger, \dots$  such that

$$q_{\ell_1} + q_{\ell_2} + \dots - q_{m_1} - q_{m_2} - \dots = 0.$$

Now, Eq. (5.1.33) is satisfied for one particular component  $\psi_\ell^{(+)}(x)$  of the annihilation field if and only if all particle species  $n$  that are destroyed by the field carry the same charge  $q(n) = q_\ell$ , and it is satisfied for one particular component  $\psi_\ell^{(-)}(x)$  of the creation field if and only if all particle species  $\bar{n}$  that are created by the field carry the charge  $q(\bar{n}) = -q_\ell$ . We see that in order for such a theory to conserve quantum numbers like electric charge, there must be a doubling of particle species carrying non-zero values of such quantum numbers: if a particular component of the annihilation field destroys a particle of species  $n$ , then the same component of the creation field must create particles of a species  $\bar{n}$ , known as the *antiparticles* of the particles of species  $n$ , which have opposite values of all conserved quantum numbers. *This is the reason for antiparticles.*

If the representation  $D(\Lambda)$  is not irreducible, then we can adopt a basis for the fields in which  $D(\Lambda)$  breaks up into blocks along the main diagonal, such that fields that belong to different blocks do not transform into each other under Lorentz transformations. Also, Lorentz transformations have no effect on the particle species. Therefore, instead of considering one big field, including many irreducible components and many particle species, we shall from now on restrict our attention to fields that destroy only a single type of particle (dropping the label  $n$ ) and create only the corresponding antiparticle, and that transform irreducibly under the Lorentz group (which as mentioned above may or may not be supposed to include space inversion), with the understanding that, in general, we shall have to consider many different such fields, some perhaps formed as the derivatives of other fields. In the following sections we are going to finish the determination of the coefficient functions  $u_\ell(\mathbf{p}, \sigma)$  and  $v_\ell(\mathbf{p}, \sigma)$ , fix the relative values of the constants  $\kappa$  and  $\lambda$ , and deduce the relations between the properties of particles and antiparticles, first for fields that belong to the simplest irreducible representations of the Lorentz group, the scalar, vector, and Dirac spinor representations. After that, we will repeat the analysis for a completely general irreducible representation.

A word about field equations. Inspection of Eqs. (5.1.31), (5.1.17), and (5.1.18) shows that all the components of a field of definite mass  $m$  satisfy the Klein–Gordon equation:

$$(\square - m^2)\psi_\ell(x) = 0. \quad (5.1.34)$$

Some fields satisfy other field equations as well, depending on whether or not there are more field components than independent particle states. Traditionally in quantum field theory one begins with such field equations, or with the Lagrangian from which they are derived, and then uses them to derive the expansion of the fields in terms of one-particle

annihilation and creation operators. In the approach followed here, we start with the particles, and derive the fields according to the dictates of Lorentz invariance, with the field equations arising almost incidentally as a byproduct of this construction.

\* \* \*

A technicality must be mentioned here. According to the theorem proved in Section 4.4, the condition that guarantees that a theory will satisfy the cluster decomposition principle is that the interaction can be expressed as a sum of products of creation and annihilation operators, with all creation operators to the left of all annihilation operators, and with coefficients that contain only a single momentum-conservation delta function. For this reason, we should write the interaction in the ‘normal ordered’ form

$$H_{\text{int}} = \int d^3x : \mathcal{H}_I(\psi(x), \psi^\dagger(x)) : \quad (5.1.35)$$

the colons indicating that the enclosed expression is to be rewritten (ignoring non-vanishing commutators or anticommutators, but including minus signs for permutations of fermionic operators) so that all creation operators stand to the left of all annihilation operators. By using the commutation or anticommutation relations of the fields, any such normal-ordered function of the fields can just as well be written as a sum of ordinary products of the fields with c-number coefficients. Rewriting  $:\mathcal{H}_I:$  in this way makes it obvious that despite the normal ordering,  $:\mathcal{H}_I(\psi(x), \psi^\dagger(x)):$  will commute with  $:\mathcal{H}_I(\psi(y), \psi^\dagger(y)):$  when  $x - y$  is space-like, if it is constructed out of fields that satisfy Eq. (5.1.32), with even numbers of any fermionic field components.

## 5.2 Causal Scalar Fields

We first consider one-component annihilation and creation fields  $\phi^{(+)}(x)$  and  $\phi^{(-)}(x)$  that transform as the simplest of all representations of the Lorentz group, the scalar, with  $D(\Lambda) = \mathbf{1}$ . Restricted to rotations, this is just the scalar representation of the rotation group, for which  $\mathcal{J} = 0$ , so Eqs. (5.1.25) and (5.1.26) have no solutions except for  $j = 0$ , in which case  $\sigma, \bar{\sigma}$  take only the value zero. Thus a scalar field can only describe particles of zero spin. Assuming also for the moment that the field describes only a single species of particle, with no distinct antiparticle (and dropping the species label  $n$  as well as the spin label  $\sigma$  and the field label  $\ell$ ), the quantities  $u_\ell(0\sigma n)$  and  $v_\ell(0\sigma n)$  are here just the numbers  $u(0)$  and  $v(0)$ . It is conventional to adjust the overall scales of the annihilation and creation fields so that these constants both have the values  $(2m)^{-1/2}$ . Eqs. (5.1.21) and (5.1.22) then give simply

$$u(\mathbf{p}) = (2p^0)^{-1/2} \quad (5.2.1)$$

and

$$v(\mathbf{p}) = (2p^0)^{-1/2}. \quad (5.2.2)$$

The fields (5.1.17) and (5.1.18) are then, in the scalar case,

$$\phi^{(+)}(x) = \int d^3p (2\pi)^{-3/2} (2p^0)^{-1/2} a_{\mathbf{p}} e^{ip \cdot x} \quad (5.2.3)$$

and

$$\phi^{(-)}(x) = \int d^3p (2\pi)^{-3/2} (2p^0)^{-1/2} a_{\mathbf{p}}^\dagger e^{-ip \cdot x} = \phi^{(+)\dagger}(x). \quad (5.2.4)$$

A Hamiltonian density  $\mathcal{H}_I(x)$  that is formed as a polynomial in  $\phi^{(+)}(x)$  and  $\phi^{(-)}(x)$  will automatically satisfy the requirement (5.1.9), that it transform as a scalar. It remains to satisfy the other condition for the Lorentz invariance of the  $S$ -matrix, that  $\mathcal{H}_I(x)$  commute with  $\mathcal{H}_I(y)$  at space-like separations  $x - y$ . If  $\mathcal{H}_I(x)$  were a polynomial in  $\phi^{(+)}(x)$  alone, there would be no problem. All annihilation operators commute or anticommute, so  $\phi^{(+)}(x)$  either commutes or anticommutes with  $\phi^{(+)}(y)$  for all  $x$  and  $y$ , according to whether the particle is a boson or fermion, respectively:

$$[\phi^{(+)}(x), \phi^{(+)}(y)]_{\mp} = 0. \quad (5.2.5)$$

Hence any  $\mathcal{H}_I(x)$  formed as a polynomial in  $\phi^{(+)}(x)$  (or, for fermions, any such even polynomial) will commute with  $\mathcal{H}_I(y)$  for all  $x$  and  $y$ . The problem, of course, is that, in order to be Hermitian,  $\mathcal{H}_I(x)$  must involve  $\phi^{(+)\dagger}(x) = \phi^{(-)}(x)$  as well as  $\phi^{(+)}(x)$ , and  $\phi^{(+)}(x)$  does not commute or anticommute with  $\phi^{(-)}(y)$  for general space-like separations. Using the commutation (for bosons) or anticommutation (for fermions) relations (4.2.5), we have

$$[\phi^{(+)}(x), \phi^{(-)}(y)]_{\mp} = \int \frac{d^3p d^3p'}{(2\pi)^3 (2p^0 \cdot 2p'^0)^{1/2}} e^{ip \cdot x} e^{-ip' \cdot y} \delta^3(\mathbf{p} - \mathbf{p}'),$$

which collapses to the single integral

$$[\phi^{(+)}(x), \phi^{(-)}(y)]_{\mp} = \Delta_+(x - y), \quad (5.2.6)$$

where  $\Delta_+$  is a standard function:

$$\Delta_+(x) \equiv \frac{1}{(2\pi)^3} \int \frac{d^3p}{2p^0} e^{ip \cdot x}. \quad (5.2.7)$$

This is manifestly Lorentz-invariant, and therefore for space-like  $x$  it can depend only on the invariant square  $x^2 > 0$ . We can thus evaluate  $\Delta_+(x)$  for space-like  $x$  by choosing the coordinate system so that

$$x^0 = 0, \quad |\mathbf{x}| = \sqrt{x^2}.$$

Eq. (5.2.7) then gives

$$\begin{aligned} \Delta_+(x) &= \frac{1}{(2\pi)^3} \int \frac{d^3p}{2\sqrt{\mathbf{p}^2 + m^2}} e^{i\mathbf{p} \cdot \mathbf{x}} \\ &= \frac{4\pi}{(2\pi)^3} \int_0^\infty \frac{p^2 dp}{2\sqrt{p^2 + m^2}} \frac{\sin(p\sqrt{x^2})}{p\sqrt{x^2}}. \end{aligned}$$

Changing the variable of integration to  $u = p/m$ , this is

$$\Delta_+(x) = \frac{m}{4\pi^2 \sqrt{x^2}} \int_0^\infty \frac{u du}{\sqrt{u^2 + 1}} \sin(m\sqrt{x^2} u) \quad (5.2.8)$$



or, in terms of a standard Hankel function,

$$\Delta_+(x) = \frac{m}{4\pi^2\sqrt{x^2}} K_1(m\sqrt{x^2}). \quad (5.2.9)$$

This isn't zero, so what are we to do with it? Note that even though  $\Delta_+(x)$  is not zero, for  $x^2 > 0$  it is even in  $x^\mu$ . Instead of using only  $\phi^{(+)}(x)$ , suppose we try to construct  $\mathcal{H}_I(x)$  out of a linear combination

$$\phi(x) = \kappa\phi^{(+)}(x) + \lambda\phi^{(-)}(x).$$

Using Eq. (5.2.6), we have then for  $x - y$  space-like

$$\begin{aligned} [\phi(x), \phi^\dagger(y)]_\mp &= |\kappa|^2 [\phi^{(+)}(x), \phi^{(-)}(y)]_\mp + |\lambda|^2 [\phi^{(-)}(x), \phi^{(+)}(y)]_\mp \\ &= (|\kappa|^2 \mp |\lambda|^2) \Delta_+(x - y), \end{aligned}$$

$$\begin{aligned} [\phi(x), \phi(y)]_\mp &= \kappa\lambda ([\phi^{(+)}(x), \phi^{(-)}(y)]_\mp + [\phi^{(-)}(x), \phi^{(+)}(y)]_\mp) \\ &= \kappa\lambda(1 \mp 1) \Delta_+(x - y). \end{aligned}$$

Both of these will vanish if and only if the particle is a *boson* (i.e., it is the top sign that applies) and  $\kappa$  and  $\lambda$  are equal in magnitude

$$|\kappa| = |\lambda|.$$

We can change the relative phase of  $\kappa$  and  $\lambda$  by redefining the phases of the states so that  $a_{\mathbf{p}} \rightarrow e^{i\alpha} a_{\mathbf{p}}$ ,  $a_{\mathbf{p}}^\dagger \rightarrow e^{-i\alpha} a_{\mathbf{p}}^\dagger$ , and hence  $\kappa \rightarrow \kappa e^{i\alpha}$ ,  $\lambda \rightarrow \lambda e^{-i\alpha}$ . Taking  $\alpha = \frac{1}{2} \text{Arg}(\lambda/\kappa)$ , we can in this way make  $\kappa$  and  $\lambda$  equal in phase, and hence equal.

Redefining  $\phi(x)$  to absorb the overall factor  $\kappa = \lambda$ , we have then

$$\phi(x) = \phi^{(+)}(x) + \phi^{(+)\dagger}(x) = \phi^\dagger(x). \quad (5.2.10)$$

The interaction density  $\mathcal{H}_I(x)$  will commute with  $\mathcal{H}_I(y)$  at space-like separations  $x - y$  if formed as a normal-ordered polynomial in the self-adjoint scalar field  $\phi(x)$ .

Even though the choice of the relative phase of the two terms in Eq. (5.2.10) is a matter of convention, it is a convention that once adopted must be used wherever a scalar field for this particle appears in the interaction Hamiltonian density. For instance, suppose that the interaction density involved not only the field (5.2.10), but also another scalar field for the same particle

$$\tilde{\phi}(x) = e^{i\alpha} \phi^{(+)}(x) + e^{-i\alpha} \phi^{(+)\dagger}(x)$$

with  $\alpha$  an arbitrary phase. This  $\tilde{\phi}$ , like  $\phi$ , would be causal in the sense that  $\tilde{\phi}(x)$  commutes with  $\tilde{\phi}(y)$  when  $x - y$  is space-like, but  $\tilde{\phi}(x)$  would not commute with  $\phi(y)$  at space-like separations, and therefore we cannot have both of these fields appearing in the same theory.

If the particles that are destroyed and created by  $\phi(x)$  carry some conserved quantum number like electric charge, then  $\mathcal{H}_I(x)$  will conserve the quantum number if and only if each term in  $\mathcal{H}_I(x)$  contains equal numbers of operators  $a_{\mathbf{p}}$  and  $a_{\mathbf{p}}^\dagger$ . But this is impossible

if  $\mathcal{H}_I(x)$  is formed as a polynomial in  $\phi(x) = \phi^{(+)}(x) + \phi^{(+)\dagger}(x)$ . To put this another way, in order that  $\mathcal{H}_I(x)$  should commute with the charge operator  $Q$  (or some other symmetry generator) it is necessary that it be formed out of fields that have simple commutation relations with  $Q$ . This is true for  $\phi^{(+)}(x)$  and its adjoint, for which

$$[Q, \phi^{(+)}(x)]_- = -q\phi^{(+)}(x),$$

$$[Q, \phi^{(+)\dagger}(x)]_- = +q\phi^{(+)\dagger}(x),$$

but not for the self-adjoint field (5.2.10).

In order to deal with this problem, we must suppose that there are *two* spinless bosons, with the same mass  $m$ , but charges  $+q$  and  $-q$ , respectively. Let  $\phi^{(+)}(x)$  and  $\phi^{(+c)}(x)$  denote the annihilation fields for these two particles, so that<sup>39</sup>

$$[Q, \phi^{(+)}(x)]_- = -q\phi^{(+)}(x),$$

$$[Q, \phi^{(+c)}(x)]_- = +q\phi^{(+c)}(x).$$

Define  $\phi(x)$  as the linear combination

$$\phi(x) = \kappa\phi^{(+)}(x) + \lambda\phi^{(+c)\dagger}(x),$$

which manifestly has the same commutator with  $Q$  as  $\phi^{(+)}(x)$  alone

$$[Q, \phi(x)]_- = -q\phi(x).$$

The commutator or anticommutator of  $\phi(x)$  with its adjoint is then, at space-like separation

$$\begin{aligned} [\phi(x), \phi^\dagger(y)]_\mp &= |\kappa|^2 [\phi^{(+)}(x), \phi^{(+)\dagger}(y)]_\mp + |\lambda|^2 [\phi^{(+c)\dagger}(x), \phi^{(+c)}(y)]_\mp \\ &= (|\kappa|^2 \mp |\lambda|^2) \Delta_+(x - y), \end{aligned}$$

while  $\phi(x)$  and  $\phi(y)$  automatically commute or anticommute with each other for all  $x$  and  $y$  because  $\phi^{(+)}$  and  $\phi^{(+c)\dagger}$  destroy and create different particles. In deriving this result, we have tacitly assumed that the particle and antiparticle have the same mass, so that the commutators or anticommutators involve the same function  $\Delta_+(x - y)$ . Fermi statistics is again ruled out here, because it is not possible that  $\phi(x)$  should anticommute with  $\phi^\dagger(y)$  at space-like separations unless  $\kappa = \lambda = 0$ , in which case the fields simply vanish. So a spinless particle must be a boson.

For Bose statistics, in order that a complex  $\phi(x)$  should commute with  $\phi^\dagger(y)$  at space-like separations, it is necessary and sufficient that  $|\kappa|^2 = |\lambda|^2$ , as well as for the particle and antiparticle to have the same mass. By redefining the relative phase of states of these two particles, we can again give  $\kappa$  and  $\lambda$  the same phase, in which case  $\kappa = \lambda$ . This common factor can again be eliminated by a redefinition of the field  $\phi$ , so that

$$\phi(x) = \phi^{(+)}(x) + \phi^{(+c)\dagger}(x)$$

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<sup>39</sup>The label ‘ $c$ ’ denotes ‘charge conjugate.’ It should be kept in mind that a particle that carries no conserved quantum numbers may or may not be its own antiparticle, with  $a_{\mathbf{p}}^c = a_{\mathbf{p}}$ .

or in more detail

$$\phi(x) = \int \frac{d^3p}{(2\pi)^{3/2}(2p^0)^{1/2}} \left[ a_{\mathbf{p}} e^{ip \cdot x} + a_{\mathbf{p}}^{\dagger} e^{-ip \cdot x} \right]. \quad (5.2.11)$$

This is the essentially unique causal scalar field. This formula can be used both for purely neutral spinless particles that are their own antiparticles (in which case we take  $a_{\mathbf{p}}^c = a_{\mathbf{p}}$ ), and for particles with distinct antiparticles (for which  $a_{\mathbf{p}}^c \neq a_{\mathbf{p}}$ ).

For future use, we note here that the commutator of the complex scalar field with its adjoint is

$$[\phi(x), \phi^{\dagger}(y)] = \Delta(x - y), \quad (5.2.12)$$

where

$$\Delta(x - y) \equiv \Delta_+(x - y) - \Delta_+(y - x) \equiv \int \frac{d^3p}{2p^0(2\pi)^3} \left[ e^{ip \cdot (x-y)} - e^{-ip \cdot (x-y)} \right]. \quad (5.2.13)$$

Let's now consider the effect of the various inversion symmetries on this field. First, from the results of Section 4.2, we can readily see that the effect of the space-inversion operator on the annihilation and creation operators is:<sup>40</sup>

$$Pa_{\mathbf{p}}P^{-1} = \eta^* a_{-\mathbf{p}}, \quad (5.2.14)$$

$$Pa_{\mathbf{p}}^{\dagger}P^{-1} = \eta^c a_{-\mathbf{p}}^{\dagger}, \quad (5.2.15)$$

where  $\eta$  and  $\eta^c$  are the intrinsic parities of the particle and antiparticle, respectively. Applying these results to the annihilation field (5.2.3) and the charge-conjugate of the creation field (5.2.4), and changing the variable of integration from  $\mathbf{p}$  to  $-\mathbf{p}$ , we see that

$$P\phi^{(+)}(x)P^{-1} = \eta^* \phi^{(+)}(\mathcal{P}x) \quad (5.2.16)$$

$$P\phi^{(+)\dagger}(x)P^{-1} = \eta^c \phi^{(+)\dagger}(\mathcal{P}x), \quad (5.2.17)$$

where as before  $\mathcal{P}x = (-\mathbf{x}, x^0)$ . We see that in general applying the space inversion to the scalar field  $\phi(x) = \phi^{(+)}(x) + \phi^{(+)\dagger}(x)$  would give a different field  $\phi_P = \eta^* \phi^{(+)} + \eta^c \phi^{(+)\dagger}$ . Both fields are separately causal, but if  $\phi$  and  $\phi_P$  appear in the same interaction then we are in trouble, because in general they do not commute at space-like separations. The only way to preserve Lorentz invariance as well as parity conservation and the hermiticity of the interaction is to require that  $\phi_P$  be proportional to  $\phi$ , and hence that

$$\eta^c = \eta^*. \quad (5.2.18)$$

That is, *the intrinsic parity  $\eta\eta^c$  of a state containing a spinless particle and its antiparticle is even*. We have now simply

$$P\phi(x)P^{-1} = \eta^* \phi(\mathcal{P}x). \quad (5.2.19)$$

These results also apply when the spinless particle is its own antiparticle, for which  $\eta^c = \eta$ , and imply that the intrinsic parity of such a particle is real:  $\eta = \pm 1$ .

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<sup>40</sup>We are omitting the subscript 0 on inversion operators  $P$ ,  $C$ , and  $T$ , because in virtually all cases where these inversions are good symmetries, the same operators induce inversion transformations on ‘in’ and ‘out’ states and on free-particle states.

Charge-conjugation can be handled in much the same way. From the results of Section 4.2, we have

$$Ca_{\mathbf{p}}C^{-1} = \xi^* a_{\mathbf{p}}^c, \quad (5.2.20)$$

$$Ca_{\mathbf{p}}^{c\dagger}C^{-1} = \xi^c a_{\mathbf{p}}^\dagger, \quad (5.2.21)$$

where  $\xi$  and  $\xi^c$  are the phases associated with the operation of charge-conjugation on one-particle states. It follows then that

$$C\phi^{(+)}(x)C^{-1} = \xi^* \phi^{(+c)}(x), \quad (5.2.22)$$

$$C\phi^{(+c)\dagger}(x)C^{-1} = \xi^c \phi^{(+)\dagger}(x). \quad (5.2.23)$$

In order that  $C\phi(x)C^{-1}$  should be proportional to the field  $\phi^\dagger(x)$  with which it commutes at space-like separations, it is evidently necessary that

$$\xi^c = \xi^*. \quad (5.2.24)$$

Just as for ordinary parity, the intrinsic charge-conjugation parity  $\xi\xi^c$  of a state consisting of a spinless particle and its antiparticle is even. We now have simply

$$C\phi(x)C^{-1} = \xi^* \phi^\dagger(x). \quad (5.2.25)$$

Again, these results apply also in the case where the particle is its own antiparticle, where  $\xi^c = \xi$ . In this case the charge-conjugation parity like the ordinary parity must be real,  $\xi = \pm 1$ .

Finally we come to time-reversal. From Section 4.2 we have

$$Ta_{\mathbf{p}}T^{-1} = \zeta^* a_{-\mathbf{p}}, \quad (5.2.26)$$

$$Ta_{\mathbf{p}}^{c\dagger}T^{-1} = \zeta^c a_{-\mathbf{p}}^{c\dagger}. \quad (5.2.27)$$

Recalling that  $T$  is antiunitary, and again changing the variable of integration from  $\mathbf{p}$  to  $-\mathbf{p}$ , we find that

$$T\phi^{(+)}(x)T^{-1} = \zeta^* \phi^{(+)}(-\mathcal{P}x) \quad (5.2.28)$$

$$T\phi^{(+c)\dagger}(x)T^{-1} = \zeta^c \phi^{(+c)\dagger}(-\mathcal{P}x). \quad (5.2.29)$$

In order for  $T\phi(x)T^{-1}$  to be simply related to the field  $\phi$  at the time-reversed point  $-\mathcal{P}x$ , we must have

$$\zeta^c = \zeta^* \quad (5.2.30)$$

and then

$$T\phi(x)T^{-1} = \zeta^* \phi(-\mathcal{P}x). \quad (5.2.31)$$

### 5.3 Causal Vector Fields

We now take up the next simplest kind of field, which transforms as a four-vector, the simplest non-trivial representation of the homogeneous Lorentz group. There are massive particles, the  $W^\pm$  and  $Z^0$ , that at low energies are described by such fields and that play an increasing role in modern elementary particle physics, so this example is not merely of pedagogical interest. (Also, although we are here considering only massive particles, one approach to quantum electrodynamics is to describe the photon in terms of a massive vector field in the limit of very small mass). For the moment we will suppose that only one species of particle is described by this field (dropping the species label  $n$ ); then we shall consider the possibility that the field describes both a particle and a distinct antiparticle.

In the four-vector representation of the Lorentz group, the rows and columns of the representation matrices  $D(\Lambda)$  are labelled with four-component indices  $\mu, \nu$ , etc., with

$$D(\Lambda)^\mu{}_\nu = \Lambda^\mu{}_\nu. \quad (5.3.1)$$

The annihilation and creation parts of the vector field are written:

$$\phi^{(+)\mu}(x) = \sum_{\sigma} (2\pi)^{-3/2} \int d^3p u^\mu(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma} e^{ip \cdot x}, \quad (5.3.2)$$

$$\phi^{(-)\mu}(x) = \sum_{\sigma} (2\pi)^{-3/2} \int d^3p v^\mu(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma}^\dagger e^{-ip \cdot x}. \quad (5.3.3)$$

The coefficient functions  $u^\mu(\mathbf{p}, \sigma)$  and  $v^\mu(\mathbf{p}, \sigma)$  for arbitrary momentum are given in terms of those for zero momentum by Eqs. (5.1.21) and (5.1.22), which here read:

$$u^\mu(\mathbf{p}, \sigma) = (m/p^0)^{1/2} L(\mathbf{p})^\mu{}_\nu u^\nu(0, \sigma), \quad (5.3.4)$$

$$v^\mu(\mathbf{p}, \sigma) = (m/p^0)^{1/2} L(\mathbf{p})^\mu{}_\nu v^\nu(0, \sigma). \quad (5.3.5)$$

(We are using the usual summation convention for spacetime indices  $\mu, \nu$ , etc.) Also, the coefficient functions at zero momentum are subject to the conditions (5.1.25) and (5.1.26):

$$\sum_{\bar{\sigma}} u^\mu(0, \bar{\sigma}) J_{\bar{\sigma}\sigma}^{(j)} = \mathcal{J}^\mu{}_\nu u^\nu(0, \sigma) \quad (5.3.6)$$

and

$$-\sum_{\bar{\sigma}} v^\mu(0, \bar{\sigma}) J_{\bar{\sigma}\sigma}^{(j)*} = \mathcal{J}^\mu{}_\nu v^\nu(0, \sigma). \quad (5.3.7)$$

The rotation generators  $\mathcal{J}^\mu{}_\nu$  in the four-vector representation are given by Eq. (5.3.1) as

$$(\mathcal{J}_k)^0{}_0 = (\mathcal{J}_k)^0{}_i = (\mathcal{J}_k)^i{}_0 = 0, \quad (5.3.8)$$

$$(\mathcal{J}_k)^i{}_j = -i\epsilon_{ijk}, \quad (5.3.9)$$

with  $i, j, k$  here running over the values 1, 2, and 3. We note in particular that  $\mathcal{J}^2$  takes the form

$$(\mathcal{J}^2)^0{}_0 = (\mathcal{J}^2)^0{}_i = (\mathcal{J}^2)^i{}_0 = 0, \quad (5.3.10)$$

$$(\mathcal{J}^2)^i_j = 2\delta^i_j. \quad (5.3.11)$$

From Eqs. (5.3.6) and (5.3.7) it follows then that

$$\sum_{\bar{\sigma}} u^0(0, \bar{\sigma}) (\mathbf{J}^{(j)})_{\bar{\sigma}\sigma}^2 = 0, \quad (5.3.12)$$

$$\sum_{\bar{\sigma}} u^i(0, \bar{\sigma}) (\mathbf{J}^{(j)})_{\bar{\sigma}\sigma}^2 = 2u^i(0, \sigma) \quad (5.3.13)$$

and

$$\sum_{\bar{\sigma}} v^0(0, \bar{\sigma}) (\mathbf{J}^{(j)*})_{\bar{\sigma}\sigma}^2 = 0, \quad (5.3.14)$$

$$\sum_{\bar{\sigma}} v^i(0, \bar{\sigma}) (\mathbf{J}^{(j)*})_{\bar{\sigma}\sigma}^2 = 2v^i(0, \sigma). \quad (5.3.15)$$

Also, we recall the familiar result that  $(\mathbf{J}^{(j)})_{\bar{\sigma}\sigma}^2 = j(j+1)\delta_{\bar{\sigma}\sigma}$ . From Eqs. (5.3.12)–(5.3.15) we see that there are just two possibilities for the spin of the particle described by the vector field: either  $j = 0$ , for which at  $\mathbf{p} = 0$  only  $u^0$  and  $v^0$  are non-zero, or else  $j = 1$  (so that  $j(j+1) = 2$ ), for which at  $\mathbf{p} = 0$  only the space-components  $u^i$  and  $v^i$  are non-zero. Let us look in a little more detail at each of these two possibilities.

### Spin Zero

By an appropriate choice of normalization of the fields, we can take the only non-vanishing component of  $u^\mu(0)$  and  $v^\mu(0)$  to have the conventional values:

$$u^0(0) = i(m/2)^{1/2}, \quad v^0(0) = -i(m/2)^{1/2}.$$

(The label  $\sigma$  here takes only the single value zero, and is therefore dropped.) Then Eqs. (5.3.4) and (5.3.5) yield for general momenta

$$u^\mu(\mathbf{p}) = ip^\mu(2p^0)^{-1/2} \quad (5.3.16)$$

and

$$v^\mu(\mathbf{p}) = -ip^\mu(2p^0)^{-1/2}. \quad (5.3.17)$$

The vector annihilation and creation fields here are nothing but the derivatives of the scalar annihilation and creation fields  $\phi^{(\pm)}$  for a spinless particle that were defined in the previous section:

$$\phi^{(+)\mu}(x) = \partial^\mu \phi^{(+)}(x), \quad \phi^{(-)\mu}(x) = \partial^\mu \phi^{(-)}(x). \quad (5.3.18)$$

It is obvious that the causal vector field for a spinless particle is also just the derivative of the causal scalar field:

$$\phi^\mu(x) = \phi^{(+)\mu}(x) + \phi^{(-)\mu}(x) = \partial^\mu \phi(x). \quad (5.3.19)$$

Hence we need not explore this case any further here.

### Spin One

From Eqs. (5.3.6) and (5.3.7) we see immediately that the vectors  $u^i(0,0)$  and  $v^i(0,0)$  for  $\sigma = 0$  are in the 3-direction. By a suitable normalization of the fields, we can take these vectors to have the values

$$u^\mu(0,0) = v^\mu(0,0) = (2m)^{-1/2} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (5.3.20)$$

with four-vector components listed in the modern order 0, 1, 2, 3. To find the other components, we use Eqs. (5.3.6), (5.3.7), and (5.3.9) to calculate the effect of the raising and lowering operators  $J_1^{(1)} \pm iJ_2^{(1)}$  on  $u$  and  $v$ . This gives:

$$u^\mu(0,+1) = -v^\mu(0,-1) = -\frac{1}{\sqrt{2}}(2m)^{-1/2} \begin{bmatrix} 0 \\ 1 \\ +i \\ 0 \end{bmatrix}, \quad (5.3.21)$$

$$u^\mu(0,-1) = -v^\mu(0,+1) = \frac{1}{\sqrt{2}}(2m)^{-1/2} \begin{bmatrix} 0 \\ 1 \\ -i \\ 0 \end{bmatrix}. \quad (5.3.22)$$

Applying Eqs. (5.3.4) and (5.3.5) now yields

$$u^\mu(\mathbf{p},\sigma) = v^{\mu*}(\mathbf{p},\sigma) = (2p^0)^{-1/2} e^\mu(\mathbf{p},\sigma) \quad (5.3.23)$$

where

$$e^\mu(\mathbf{p},\sigma) \equiv L^\mu{}_\nu(\mathbf{p}) e^\nu(0,\sigma) \quad (5.3.24)$$

with

$$e^\mu(0,0) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad e^\mu(0,+1) = -\frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ +i \\ 0 \end{bmatrix}, \quad e^\mu(0,-1) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -i \\ 0 \end{bmatrix}. \quad (5.3.25)$$

The annihilation and creation fields (5.3.2) and (5.3.3) here are

$$\phi^{(+)\mu}(x) = \phi^{(-)\mu\dagger}(x) = (2\pi)^{-3/2} \sum_\sigma \int \frac{d^3p}{\sqrt{2p^0}} e^\mu(\mathbf{p},\sigma) a_{\mathbf{p},\sigma} e^{ip \cdot x}. \quad (5.3.26)$$

The fields  $\phi^{(+)\mu}(x)$  and  $\phi^{(+)\nu}(y)$  of course commute (or anticommute) for all  $x$  and  $y$ , but  $\phi^{(+)\mu}(x)$  and  $\phi^{(-)\nu}(y)$  do not. Their commutator (for bosons) or anticommutator (for fermions) is

$$[\phi^{(+)\mu}(x), \phi^{(-)\nu}(y)]_\mp = \int \frac{d^3p}{(2\pi)^3 2p^0} e^{ip \cdot (x-y)} \Pi^{\mu\nu}(\mathbf{p}) \quad (5.3.27)$$

where

$$\Pi^{\mu\nu}(\mathbf{p}) \equiv \sum_{\sigma} e^{\mu}(\mathbf{p}, \sigma) e^{\nu*}(\mathbf{p}, \sigma). \quad (5.3.28)$$

A straightforward calculation using Eq. (5.3.25) shows that  $\Pi^{\mu\nu}(0)$  is the projection matrix on the space orthogonal to the time-direction, and Eq. (5.3.24) then shows that  $\Pi^{\mu\nu}(\mathbf{p})$  is the projection matrix on the space orthogonal to the four-vector  $p^{\mu}$ :

$$\Pi^{\mu\nu}(\mathbf{p}) = \eta^{\mu\nu} + p^{\mu} p^{\nu} / m^2. \quad (5.3.29)$$

The commutator (or anticommutator) (5.3.27) may then be written in terms of the  $\Delta_+$  function defined in the previous section, as

$$[\phi^{(+)\mu}(x), \phi^{(-)\nu}(y)]_{\mp} = \left[ \eta^{\mu\nu} - \frac{\partial^{\mu} \partial^{\nu}}{m^2} \right] \Delta_+(x - y). \quad (5.3.30)$$

For our present purposes, the important thing about this expression is that for  $x - y$  space-like it does not vanish and is *even* in  $x - y$ . We can therefore repeat the reasoning of the previous section in seeking to construct a causal field; we form a linear combination of annihilation and creation fields

$$v^{\mu}(x) = \kappa \phi^{(+)\mu}(x) + \lambda \phi^{(-)\mu}(x)$$

for which, for  $x - y$  spacelike,

$$[v^{\mu}(x), v^{\nu}(y)]_{\mp} = \kappa \lambda [1 \mp 1] \left[ \eta^{\mu\nu} - \frac{\partial^{\mu} \partial^{\nu}}{m^2} \right] \Delta_+(x - y)$$

and

$$[v^{\mu}(x), v^{\nu\dagger}(y)]_{\mp} = (|\kappa|^2 \mp |\lambda|^2) \left[ \eta^{\mu\nu} - \frac{\partial^{\mu} \partial^{\nu}}{m^2} \right] \Delta_+(x - y).$$

In order for both to vanish for space-like  $x - y$ , it is necessary and sufficient that the spin one particles be bosons and that  $|\kappa| = |\lambda|$ . By a suitable choice of phase of the one-particle states we can give  $\kappa$  and  $\lambda$  the same phase, so that  $\kappa = \lambda$ , and then drop the common factor  $\kappa$  by redefining the overall normalization of the field. After all this, we find that the causal vector field for a massive particle of spin one is

$$v^{\mu}(x) = \phi^{(+)\mu}(x) + \phi^{(+)\mu\dagger}(x). \quad (5.3.31)$$

We note that this is real:

$$v^{\mu}(x) = v^{\mu\dagger}(x). \quad (5.3.32)$$

However, if the particles it describes carry a non-zero value of some conserved quantum number  $Q$ , then we cannot construct an interaction that conserves  $Q$  out of such a field. Instead, we must suppose that there is another boson of the same mass and spin which carries an opposite value of  $Q$ , and construct the causal field as

$$v^{\mu}(x) = \phi^{(+)\mu}(x) + \phi^{(+)\mu\dagger}(x) \quad (5.3.33)$$



or in more detail

$$v^\mu(x) = (2\pi)^{-3/2} \sum_\sigma \int \frac{d^3p}{\sqrt{2p^0}} \left[ e^\mu(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma} e^{ip \cdot x} + e^{\mu*}(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma}^{\dagger c} e^{-ip \cdot x} \right], \quad (5.3.34)$$

where the superscript  $c$  indicates operators that create the antiparticle that is charge-conjugate to the particle annihilated by  $\phi^{(+)\mu}(x)$ . This again is a causal field, but no longer real. We can also use this formula for the case of a purely neutral, spin one particle that is its own antiparticle, by simply setting  $a_{\mathbf{p}}^c = a_{\mathbf{p}}$ . In either case, the commutator of a vector field with its adjoint is

$$[v^\mu(x), v^{\nu\dagger}(y)] = \left[ \eta^{\mu\nu} - \frac{\partial^\mu \partial^\nu}{m^2} \right] \Delta(x - y), \quad (5.3.35)$$

where  $\Delta(x - y)$  is the function (5.2.13).

The real and complex fields we have constructed for a massive, spin one particle satisfy interesting field equations. First, since  $p^\mu$  in the exponential in Eq. (5.3.26) satisfies  $p^2 = -m^2$ , the field satisfies the Klein–Gordon equation:

$$(\square - m^2)v^\mu(x) = 0, \quad (5.3.36)$$

just as for the scalar field. In addition, since Eq. (5.3.24) shows that

$$e^\mu(\mathbf{p}, \sigma) p_\mu = 0, \quad (5.3.37)$$

we now have another equation

$$\partial_\mu v^\mu(x) = 0. \quad (5.3.38)$$

In the limit of small mass, Eqs. (5.3.36) and (5.3.38) are just the equations for the potential four-vector of electrodynamics in what is called Lorentz gauge.

However, we cannot obtain electrodynamics from just any theory of massive spin one particles by letting the mass go to zero. The trouble can be seen by considering the rate of production of a spin one particle by an interaction density  $\mathcal{H}_I = J_\mu v^{\mu\dagger}$ , where  $J_\mu$  is an arbitrary four-vector current. Squaring the matrix element and summing over the spin  $z$ -components of the spin one particle gives a rate proportional to

$$\sum_\sigma |\langle J_\mu \rangle e^\mu(\mathbf{p}, \sigma)^*|^2 = \langle J_\mu \rangle \langle J_\nu \rangle^* \Pi^{\mu\nu}(\mathbf{p}),$$

where  $\mathbf{p}$  is the momentum of the emitted spin one particle, and  $\langle J_\mu \rangle$  is the matrix element of the current (say, at  $x = 0$ ) between the initial and final states of all other particles. The term  $p^\mu p^\nu / m^2$  in  $\Pi^{\mu\nu}(\mathbf{p})$  will, in general, cause the emission rate to blow up when  $m \rightarrow 0$ . The only way to avert this catastrophe is to suppose that  $\langle J_\mu \rangle p^\mu$  vanishes, which in coordinate space is just the statement that the current  $J^\mu$  must be *conserved*, in the sense that  $\partial_\mu J^\mu = 0$ . Indeed, the need for conservation of the current can be seen by simply counting states. A massive spin one particle has three spin states, which can be

taken as the states with helicity  $+1$ ,  $0$ , and  $-1$ , while any massless, spin one particle like the photon can only have helicities  $+1$  and  $-1$ : the current conservation condition just ensures that the helicity zero states of the spin one particle are not emitted in the limit of zero mass.

The inversions can be dealt with in much the same way as for the scalar field discussed in the previous section. To evaluate the effect of space inversion, we need a formula for  $e^\mu(-\mathbf{p}, \sigma)$ . Using  $L^\mu_\nu(-\mathbf{p}) = \mathcal{P}^\mu_\rho L^\rho_\tau(\mathbf{p}) \mathcal{P}^\tau_\nu$  and Eq. (5.3.24), we have

$$e^\mu(-\mathbf{p}, \sigma) = -\mathcal{P}^\mu_\nu e^\nu(\mathbf{p}, \sigma). \quad (5.3.39)$$

Also, to evaluate the effect of time-reversal we need a formula for  $(-1)^{1+\sigma} e^{\mu*}(-\mathbf{p}, -\sigma)$ . Using  $e^{\mu*}(-\sigma) = -e^\mu(\sigma)$  and the above formula for  $L^\mu_\nu(-\mathbf{p})$ , we find

$$(-1)^{1+\sigma} e^{\mu*}(-\mathbf{p}, -\sigma) = \mathcal{P}^\mu_\nu e^\nu(\mathbf{p}, \sigma). \quad (5.3.40)$$

Using these results and the transformation properties of the annihilation and creation operators given in Section 4.2, it is straightforward to work out the inversion transformation properties of the annihilation and creation fields. Once again we find that, in order for causal fields to be transformed into other fields with which they commute at space-like separations, it is necessary that the intrinsic space inversion, charge-conjugation, and time-reversal phases for spin one particles and their antiparticles be related by

$$\eta^c = \eta^*, \quad (5.3.41)$$

$$\xi^c = \xi^*, \quad (5.3.42)$$

$$\zeta^c = \zeta^*. \quad (5.3.43)$$

(In particular all phases must be real if the spin one particle is its own antiparticle.) With these phase conditions satisfied, our causal vector field (5.3.34) has the inversion transformation properties

$$P v^\mu(x) P^{-1} = -\eta^* \mathcal{P}^\mu_\nu v^\nu(\mathcal{P}x), \quad (5.3.44)$$

$$C v^\mu(x) C^{-1} = \xi^* v^{\mu\dagger}(x), \quad (5.3.45)$$

$$T v^\mu(x) T^{-1} = \zeta^* \mathcal{P}^\mu_\nu v^\nu(-\mathcal{P}x). \quad (5.3.46)$$

In particular, the minus sign in Eq. (5.3.44) means that a vector field that transforms as a polar vector, with no extra phases or signs accompanying the matrix  $\mathcal{P}^\mu_\nu$ , describes a spin one particle with intrinsic parity  $\eta = -1$ .

## 5.4 The Dirac Formalism

Among all the representations of the homogeneous Lorentz group, there is one that plays a special role in physics. As we saw in Section 1.1, this representation was introduced into the theory of the electron by Dirac [3], but as so often happens it was already known to mathematicians [4], because it provides the basis of one of the two broad classes of representations of the rotation or Lorentz groups (actually, of their covering groups—see

Section 2.7) in any number of dimensions. From the point of view we are following here, the structure and properties of any quantum field are dictated by the representation of the homogeneous Lorentz group under which it transforms, so it will be natural for us to describe the Dirac formalism as it first appeared in mathematics, rather than as it was introduced by Dirac.

By a representation of the homogeneous Lorentz group, we mean a set of matrices  $D(\Lambda)$  satisfying the group multiplication law

$$D(\bar{\Lambda})D(\Lambda) = D(\bar{\Lambda}\Lambda).$$

Just as for the unitary operators  $U(\Lambda)$ , we can study the properties of these matrices by considering the infinitesimal case,

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad (5.4.1)$$

$$\omega_{\mu\nu} = -\omega_{\nu\mu}, \quad (5.4.2)$$

for which

$$D(\Lambda) = \mathbf{1} + \frac{i}{2}\omega_{\mu\nu}\mathcal{J}^{\mu\nu}, \quad (5.4.3)$$

with  $\mathcal{J}^{\mu\nu} = -\mathcal{J}^{\nu\mu}$  a set of matrices satisfying the commutation relations (2.4.12):

$$i[\mathcal{J}^{\mu\nu}, \mathcal{J}^{\rho\sigma}] = \eta^{\nu\rho}\mathcal{J}^{\mu\sigma} - \eta^{\mu\rho}\mathcal{J}^{\nu\sigma} - \eta^{\nu\sigma}\mathcal{J}^{\mu\rho} + \eta^{\mu\sigma}\mathcal{J}^{\nu\rho}. \quad (5.4.4)$$

To find such a set of matrices, suppose we first construct matrices  $\gamma^\mu$  that satisfy the anticommutation relations

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \quad (5.4.5)$$

and tentatively define

$$\mathcal{J}^{\mu\nu} = -\frac{i}{4}[\gamma^\mu, \gamma^\nu]. \quad (5.4.6)$$

It is elementary, using Eq. (5.4.5), to show that

$$[\mathcal{J}^{\mu\nu}, \gamma^\rho] = -i\gamma^\mu\eta^{\nu\rho} + i\gamma^\nu\eta^{\mu\rho} \quad (5.4.7)$$

and from this we easily see that Eq. (5.4.6) does indeed satisfy the desired commutation relation Eq. (5.4.4). We shall further assume that the matrices  $\gamma_\mu$  are *irreducible*; that is, that there is no proper subspace that is left invariant by all these matrices. Otherwise we could choose some smaller set of field components, which would transform as in Eqs. (5.4.3) and (5.4.6), with an irreducible set of  $\gamma_\mu$ s.

Any set of matrices satisfying a relation like Eq. (5.4.5) (or its Euclidean analog, with  $\eta_{\mu\nu}$  replaced with a Kronecker delta) is called a *Clifford algebra*. The importance in mathematics of this particular representation of the homogeneous Lorentz group (or, more accurately, its covering group) arises from the fact (shown in Section 5.6) that the most general irreducible representation of the Lorentz group is either a tensor, or a spinor transforming as in Eqs. (5.4.3) and (5.4.6), or a direct product of a spinor and a tensor.

The commutation relation (5.4.7) can be summarized by saying that  $\gamma^\rho$  is a *vector*, in the sense that Eq. (5.4.3) satisfies

$$D(\Lambda)\gamma^\rho D^{-1}(\Lambda) = \Lambda_\sigma{}^\rho \gamma^\sigma. \quad (5.4.8)$$

In the same sense, the unit matrix is trivially a *scalar*

$$D(\Lambda)\mathbf{1}D^{-1}(\Lambda) = \mathbf{1} \quad (5.4.9)$$

and Eq. (5.4.4) shows that  $\mathcal{J}^{\rho\sigma}$  is an antisymmetric *tensor*

$$D(\Lambda)\mathcal{J}^{\rho\sigma}D^{-1}(\Lambda) = \Lambda_\mu{}^\rho \Lambda_\nu{}^\sigma \mathcal{J}^{\mu\nu}. \quad (5.4.10)$$

The matrices  $\gamma^\mu$  can be used to construct other totally antisymmetric tensors

$$\mathcal{A}^{\rho\sigma\tau} \equiv \gamma^{[\rho}\gamma^\sigma\gamma^{\tau]}, \quad (5.4.11)$$

$$\mathcal{P}^{\mu\rho\sigma\tau} \equiv \gamma^{[\mu}\gamma^\rho\gamma^\sigma\gamma^{\tau]}. \quad (5.4.12)$$

The brackets here are a standard notation, indicating that we are to sum over all permutations of the indices within the brackets, with a plus or minus sign for even or odd permutations, respectively. For instance, Eq. (5.4.11) is shorthand for

$$\begin{aligned} \mathcal{A}^{\rho\sigma\tau} \equiv & \gamma^\rho\gamma^\sigma\gamma^\tau - \gamma^\rho\gamma^\tau\gamma^\sigma - \gamma^\sigma\gamma^\rho\gamma^\tau \\ & + \gamma^\sigma\gamma^\tau\gamma^\rho + \gamma^\tau\gamma^\rho\gamma^\sigma - \gamma^\tau\gamma^\sigma\gamma^\rho. \end{aligned}$$

By repeated use of Eq. (5.4.5) we can write any product of  $\gamma$ s as a sum of antisymmetrized products of  $\gamma$ s times a product of metric tensors, so the totally antisymmetric tensors form a complete basis for the set of all matrices that can be constructed from the Dirac matrices.

This formalism automatically contains a parity transformation, conventionally taken as

$$\beta \equiv i\gamma^0. \quad (5.4.13)$$

Applied to the Dirac matrices, this gives

$$\beta\gamma^i\beta^{-1} = -\gamma^i, \quad \beta\gamma^0\beta^{-1} = +\gamma^0. \quad (5.4.14)$$

(We here label indices so that  $\mu$  runs over values  $0, 1, 2, \dots$ .) The same similarity transformation, applied to any product of  $\gamma$ -matrices, then yields just a plus or minus sign, according to whether the product contains an even or an odd number of  $\gamma$ s with space-like indices, respectively. In particular,

$$\beta\mathcal{J}^{ij}\beta^{-1} = \mathcal{J}^{ij}, \quad (5.4.15)$$

$$\beta\mathcal{J}^{i0}\beta^{-1} = -\mathcal{J}^{i0}. \quad (5.4.16)$$

Everything so far in this section applies in any number of spacetime dimensions and for any ‘metric’  $\eta_{\mu\nu}$ . In four spacetime dimensions, however, there is a special feature, that no totally antisymmetric tensor can have more than four indices, so the sequence of tensors  $\mathbf{1}$ ,

$\gamma^\rho$ ,  $\mathcal{J}^{\rho\sigma}$ ,  $\mathcal{A}^{\rho\sigma\tau}$ , ... terminates with the tensor (5.4.12). Furthermore, each of these tensors transforms differently under Lorentz and/or parity transformations so they are all linearly independent.<sup>41</sup> The number of linearly independent components of these tensors is one for  $\mathbf{1}$ , four for  $\gamma^\rho$ , six for  $\mathcal{J}^{\rho\sigma}$ , four for  $\mathcal{A}^{\rho\sigma\tau}$ , and one for  $\mathcal{P}^{\mu\nu\rho\sigma}$ , or 16 independent components in all. (The general rule is that a totally antisymmetric tensor with  $n$  indices in  $d$  dimensions has a number of independent components equal to the binomial coefficient  $d!/n!(d-n)!$ .) There are at most  $v^2$  independent  $v \cdot v$  matrices, so they must have at least  $\sqrt{16} = 4$  rows and columns. Dirac matrices of the minimum dimensionality are necessarily irreducible; if reducible, the subspace left invariant by these matrices would furnish a representation of lower dimensionality. We shall therefore take the  $\gamma^\mu$  to be  $4 \times 4$  matrices.

(More generally, in any even number  $d$  of spacetime dimensions, one can form antisymmetric tensors with  $0, 1, \dots, d$  indices, which contain altogether a number of independent components equal to

$$\sum_{n=0}^d \frac{d!}{n!(d-n)!} = 2^d,$$

so the  $\gamma$ -matrices must have at least  $2^{d/2}$  rows and columns. In spaces or spacetimes with odd dimensionality, the totally antisymmetric tensors of rank  $n$  and  $d-n$  can be linearly related by the conditions<sup>42</sup>

$$\gamma^{[\mu_1 \gamma^{\mu_2} \dots \gamma^{\mu_r}] \propto \epsilon^{\mu_1 \mu_2 \dots \mu_d} \gamma_{[\mu_{r+1} \gamma_{\mu_{r+2}} \dots \gamma_{\mu_d}],}$$

for  $r = 0, 1, 2, \dots, d-1$ , with  $\epsilon^{\mu_1 \mu_2 \dots \mu_d}$  totally antisymmetric, and the left-hand side taken as the unit matrix for  $r = 0$ . Under these conditions there are only  $2^{d-1}$  independent tensors, requiring  $\gamma$ -matrices of dimensionality at least  $2^{(d-1)/2}$ .)

Returning now to four spacetime dimensions, we shall choose an explicit set of  $4 \times 4$   $\gamma$ -matrices. One very convenient choice is

$$\gamma^0 = -i \begin{bmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{bmatrix}, \quad \gamma = -i \begin{bmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{bmatrix}, \quad (5.4.17)$$

where  $\mathbf{1}$  is the unit  $2 \times 2$  matrix, and the components of  $\boldsymbol{\sigma}$  are the usual Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (5.4.18)$$

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<sup>41</sup>Alternatively, these matrices can be shown to be linearly independent by noting that they form an orthogonal set, with the scalar product of two matrices defined by the trace of their product. Note that none of these matrices can vanish, because each component of each of these tensors is proportional to a product of different  $\gamma$ -matrices, and such a product has a square equal to plus or minus the product of the corresponding squares, and hence equal to  $\pm 1$ .

<sup>42</sup>This constraint does not interfere with the inclusion of space inversion in the Dirac representation of the Lorentz group in odd-dimensional spacetime, because here the tensor  $\epsilon^{\mu_1 \mu_2 \dots \mu_d}$  is even under inversion of space coordinates. If we don't care about space inversion, we can also construct  $2^{(d-2)/2}$ -dimensional irreducible representations of the proper orthochronous Lorentz group in even spacetime dimensions by imposing the above condition relating antisymmetrized products of  $r$  and  $d-r$  Dirac matrices. An example is provided by the submatrices in Eqs. (5.4.19) and (5.4.20) below.

(The  $\sigma_i$  are just the  $2 \times 2$   $\gamma$ -matrices in three dimensions.) It can be shown [5] that any other irreducible set of  $\gamma$ -matrices are related to these by a similarity transformation. From Eq. (5.4.17), we can easily calculate the Lorentz group generators (5.4.6):

$$\mathcal{J}^{ij} = \frac{1}{2} \epsilon_{ijk} \begin{bmatrix} \sigma_k & 0 \\ 0 & \sigma_k \end{bmatrix}, \quad (5.4.19)$$

$$\mathcal{J}^{i0} = +\frac{i}{2} \begin{bmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{bmatrix}. \quad (5.4.20)$$

(Here  $\epsilon_{ijk}$  is the totally antisymmetric tensor in three dimensions, with  $\epsilon_{123} = +1$ .) We note that these are block-diagonal, so the Dirac matrices provide a *reducible* representation of the proper orthochronous Lorentz group, the direct sum of two irreducible representation with  $\mathcal{J}^{ij} = \pm i \epsilon_{ijk} \mathcal{J}^{k0}$ .

It is convenient to write the totally antisymmetric tensors (5.4.11) and (5.4.12) in a somewhat simpler way. The matrix (5.4.12) is totally antisymmetric, and therefore proportional to the pseudotensor  $\epsilon^{\rho\sigma\tau\eta}$ , defined as a totally antisymmetric quantity with  $\epsilon^{0123} = +1$ . Setting  $\rho, \sigma, \tau, \eta$  equal to 0, 1, 2, 3, respectively, we see that

$$\mathcal{P}^{\rho\sigma\tau\eta} = 4! i \epsilon^{\rho\sigma\tau\eta} \gamma_5, \quad (5.4.21)$$

where

$$\gamma_5 \equiv -i \gamma^0 \gamma^1 \gamma^2 \gamma^3. \quad (5.4.22)$$

The matrix  $\gamma_5$  is a pseudoscalar in the sense that

$$[\mathcal{J}^{\rho\sigma}, \gamma_5] = 0, \quad (5.4.23)$$

$$\beta \gamma_5 \beta^{-1} = -\gamma_5. \quad (5.4.24)$$

Similarly,  $\mathcal{A}^{\rho\sigma\tau}$  must be proportional to  $\epsilon^{\rho\sigma\tau\eta}$  contracted with some matrix  $\mathcal{A}_\eta$ , and by setting  $\rho, \sigma, \tau$  equal in turn to 0, 1, 2 or 0, 1, 3 or 0, 2, 3 or 1, 2, 3, we find

$$\mathcal{A}^{\rho\sigma\tau} = 3! i \epsilon^{\rho\sigma\tau\eta} \gamma_5 \gamma_\eta. \quad (5.4.25)$$

The 16 independent  $4 \times 4$  matrices can therefore be taken as the components of the scalar  $\mathbf{1}$ , the vector  $\gamma^\rho$ , the antisymmetric tensor  $\mathcal{J}^{\rho\sigma}$ , the ‘axial’ vector  $\gamma_5 \gamma_\eta$ , and the pseudoscalar  $\gamma_5$ . It is easy to see that the matrix  $\gamma_5$  has unit square

$$(\gamma_5)^2 = \mathbf{1} \quad (5.4.26)$$

and anticommutes with all  $\gamma^\mu$

$$\{\gamma_5, \gamma^\mu\} = 0. \quad (5.4.27)$$

The notation  $\gamma_5$  is particularly appropriate, because the anticommutation relations (5.4.26) and (5.4.27), together with Eq. (5.4.5) show that  $\gamma^0, \gamma^1, \gamma^2, \gamma^3, \gamma_5$  provide a Clifford algebra

in five spacetime dimensions. For the particular  $4 \times 4$  representation (5.4.17) of the  $\gamma$ -matrices, the matrix  $\gamma_5$  is

$$\gamma_5 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}. \quad (5.4.28)$$

This representation is convenient because it reduces  $\mathcal{J}^{\rho\sigma}$  and  $\gamma_5$  to block-diagonal form. As we shall see, this makes it particularly useful for dealing with particles in the ultra-relativistic limit,  $v \rightarrow c$ . (It is not, however, the representation described in Section 1.1 that was originally introduced by Dirac, because Dirac was mostly interested in electrons in atoms where  $v \ll c$ , and in this case it is more convenient to adopt a representation for which  $\gamma^0$  rather than  $\gamma_5$  is diagonal.)

The representation of the homogeneous Lorentz group we have constructed here is not unitary, because the generators  $\mathcal{J}^{\rho\sigma}$  are not all represented by Hermitian matrices. In particular, in the representation (5.4.17) we have  $\mathcal{J}^{ij}$  Hermitian, but  $\mathcal{J}^{i0}$  is anti-Hermitian. Such reality conditions can conveniently be written in a manifestly Lorentz-invariant fashion by introducing the matrix  $\beta = i\gamma^0$  of Eq. (5.4.13), which in the representation (5.4.17) takes the form

$$\beta = \begin{bmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{bmatrix}. \quad (5.4.29)$$

Inspection of Eq. (5.4.17) shows that

$$\beta \gamma^{\mu\dagger} \beta = -\gamma^\mu \quad (5.4.30)$$

and it follows then that

$$\beta \mathcal{J}^{\rho\sigma\dagger} \beta = \mathcal{J}^{\rho\sigma}. \quad (5.4.31)$$

Hence, though not unitary, the matrices  $D(\Lambda)$  satisfy the pseudounitariness relation

$$\beta D(\Lambda)^\dagger \beta = D(\Lambda)^{-1}. \quad (5.4.32)$$

Also,  $\gamma_5$  is Hermitian and anticommutes with  $\beta$  so

$$\beta (\gamma_5)^\dagger \beta = -\gamma_5 \quad (5.4.33)$$

and it follows that

$$\beta (\gamma_5 \gamma_\mu)^\dagger \beta = -\gamma_5 \gamma_\mu. \quad (5.4.34)$$

The Dirac and related matrices also have important symmetry properties. Inspection of Eqs. (5.4.17) and (5.4.18) shows that  $\gamma_\mu$  is symmetric for  $\mu = 0, 2$  and antisymmetric for  $\mu = 1, 3$ , so

$$\gamma_\mu^T = -\mathcal{C} \gamma_\mu \mathcal{C}^{-1}, \quad (5.4.35)$$

where T denotes a transpose, and

$$\mathcal{C} \equiv \gamma_2 \beta = -i \begin{bmatrix} \sigma_2 & 0 \\ 0 & -\sigma_2 \end{bmatrix}. \quad (5.4.36)$$

It follows immediately that

$$\mathcal{J}_{\mu\nu}^\top = -\mathcal{C}\mathcal{J}_{\mu\nu}\mathcal{C}^{-1}, \quad (5.4.37)$$

$$(\gamma_5)^\top = +\mathcal{C}\gamma_5\mathcal{C}^{-1}, \quad (5.4.38)$$

$$(\gamma_5\gamma_\mu)^\top = +\mathcal{C}\gamma_5\gamma_\mu\mathcal{C}^{-1}. \quad (5.4.39)$$

These signs will prove significant when we consider the charge-conjugation properties of various currents in the next section. Of course, we can combine our results for adjoints and transposes to obtain the complex conjugates of the Dirac and allied matrices:

$$\gamma_\mu^* = \beta\mathcal{C}\gamma_\mu\mathcal{C}^{-1}\beta, \quad (5.4.40)$$

$$\mathcal{J}_{\mu\nu}^* = -\beta\mathcal{C}\mathcal{J}_{\mu\nu}\mathcal{C}^{-1}\beta, \quad (5.4.41)$$

$$(\gamma_5)^* = -\beta\mathcal{C}\gamma_5\mathcal{C}^{-1}\beta, \quad (5.4.42)$$

$$(\gamma_5\gamma_\mu)^* = -\beta\mathcal{C}\gamma_5\gamma_\mu\mathcal{C}^{-1}\beta. \quad (5.4.43)$$

## 5.5 Causal Dirac Fields

We now want to construct particle annihilation and antiparticle creation fields that transform under the Lorentz group according to the Dirac representation of this group, discussed in the previous section. In general these take the form given in Eqs. (5.1.17) and (5.1.18):

$$\psi_\ell^{(+)}(x) = (2\pi)^{-3/2} \sum_\sigma \int d^3p u_\ell(\mathbf{p}, \sigma) e^{ip \cdot x} a_{\mathbf{p}, \sigma} \quad (5.5.1)$$

and

$$\psi_\ell^{(-c)}(x) = (2\pi)^{-3/2} \sum_\sigma \int d^3p v_\ell(\mathbf{p}, \sigma) e^{-ip \cdot x} a_{\mathbf{p}, \sigma}^{c\dagger}, \quad (5.5.2)$$

with the particle species label omitted here. In order to calculate the coefficient functions  $u_\ell(\mathbf{p}, \sigma)$  and  $v_\ell(\mathbf{p}, \sigma)$  appearing in these formulas, we must first use Eqs. (5.1.25) and (5.1.26) to find  $u_\ell$  and  $v_\ell$  for zero momentum, and then apply Eqs. (5.1.21) and (5.1.22) to calculate them for arbitrary momenta, with  $D_{\ell\bar{\ell}}(\Lambda)$  in both cases taken as the  $4 \times 4$  Dirac representation of the homogeneous Lorentz group discussed in the previous section.

Using Eq. (5.4.19), the zero-momentum conditions (5.1.25) and (5.1.26) read<sup>43</sup>

$$\sum_{\bar{\sigma}} u_{m,\pm}(\mathbf{0}, \bar{\sigma}) J_{\bar{\sigma}\sigma}^{(j)} = \sum_{\bar{m}} \frac{1}{2} \sigma_{m\bar{m}} u_{\bar{m},\pm}(\mathbf{0}, \sigma)$$

and

$$-\sum_{\bar{\sigma}} v_{m,\pm}(\mathbf{0}, \bar{\sigma}) J_{\bar{\sigma}\sigma}^{(j)*} = \sum_{\bar{m}} \frac{1}{2} \sigma_{m\bar{m}} v_{\bar{m},\pm}(\mathbf{0}, \sigma).$$

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<sup>43</sup>We are here dropping the species label  $n$ , and replacing the four-component index  $\ell$  with a pair of indices, one 2-valued index  $m$  labelling the rows and columns of the submatrices in Eqs. (5.4.19) and (5.4.20), and a second index taking values  $\pm$ , labelling the rows and columns of the supermatrix in Eqs. (5.4.19) and (5.4.20).



In other words, if we regard  $u_{m,\pm}(\mathbf{0}, \sigma)$  and  $v_{m,\pm}(\mathbf{0}, \sigma)$  as the  $m, \sigma$  elements of matrices  $U_{\pm}$  and  $V_{\pm}$ , we have in matrix notation

$$U_{\pm} \mathbf{J}^{(j)} = \frac{1}{2} \boldsymbol{\sigma} U_{\pm} \quad (5.5.3)$$

and

$$-V_{\pm} \mathbf{J}^{(j)*} = \frac{1}{2} \boldsymbol{\sigma} V_{\pm}. \quad (5.5.4)$$

Now, the  $(2j + 1)$ -dimensional matrices  $\mathbf{J}^{(j)}$  and  $-\mathbf{J}^{(j)*}$  and the  $2 \times 2$  matrices  $\frac{1}{2} \boldsymbol{\sigma}$  all provide irreducible representations of the Lie algebra of the rotation group. A general theorem of group theory known as Schur's lemma [6] tells us that when a matrix like  $U_{\pm}$  or  $V_{\pm}$  connects two such representations as in Eqs. (5.5.3) and (5.5.4), the matrix must either vanish (a possibility of no interest here) or else be square and non-singular. Hence the Dirac field can only describe particles of spin  $j = \frac{1}{2}$  (so that  $2j + 1 = 2$ ) and the matrices  $\mathbf{J}^{(1/2)}$  and  $-\mathbf{J}^{(1/2)*}$  must be the same as  $\frac{1}{2} \boldsymbol{\sigma}$  up to a similarity transformation. In fact, in the standard representation (2.5.21), (2.5.22) of the rotation generators, we have  $\mathbf{J}^{(1/2)} = \frac{1}{2} \boldsymbol{\sigma}$  and  $-\mathbf{J}^{(1/2)*} = \frac{1}{2} \sigma_2 \boldsymbol{\sigma} \sigma_2$ . It follows then that  $U_{\pm}$  and  $V_{\pm} \sigma_2$  must commute with  $\boldsymbol{\sigma}$ , and hence must be proportional to the unit matrix:

$$u_{m,\pm}(\mathbf{0}, \sigma) = c_{\pm} \delta_{m\sigma}, \quad v_{m,\pm}(\mathbf{0}, \sigma) = id_{\pm} (\sigma_2)_{m\sigma}. \quad (5.5.5)$$

In other words

$$u(\mathbf{0}, \tfrac{1}{2}) = \begin{bmatrix} c_+ \\ 0 \\ c_- \\ 0 \end{bmatrix}, \quad u(\mathbf{0}, -\tfrac{1}{2}) = \begin{bmatrix} 0 \\ c_+ \\ 0 \\ c_- \end{bmatrix},$$

$$v(\mathbf{0}, \tfrac{1}{2}) = \begin{bmatrix} 0 \\ d_+ \\ 0 \\ d_- \end{bmatrix}, \quad v(\mathbf{0}, -\tfrac{1}{2}) = - \begin{bmatrix} d_+ \\ 0 \\ d_- \\ 0 \end{bmatrix},$$

and the spinors at finite momentum are

$$u(\mathbf{p}, \sigma) = \sqrt{m/p^0} D(L(\mathbf{p})) u(\mathbf{0}, \sigma), \quad (5.5.6)$$

$$v(\mathbf{p}, \sigma) = \sqrt{m/p^0} D(L(\mathbf{p})) v(\mathbf{0}, \sigma). \quad (5.5.7)$$

It now only remains to say something about the constants  $c_{\pm}$  and  $d_{\pm}$ . In general, these are quite arbitrary—we could even choose  $c_-$  and  $d_-$  or  $c_+$  and  $d_+$  to be zero if we liked, so that the Dirac field would have only two non-vanishing components. The only physical principle that could tell us anything about the relative values of the  $c_{\pm}$  or the  $d_{\pm}$  is the conservation of parity. We recall that under a space inversion, the particle annihilation and antiparticle creation operators undergo the transformations:

$$P a_{\mathbf{p},\sigma} P^{-1} = \eta^* a_{-\mathbf{p},\sigma} \quad (5.5.8)$$

$$P a_{\mathbf{p},\sigma}^{\dagger} P^{-1} = \eta^c a_{-\mathbf{p},\sigma}^{\dagger} \quad (5.5.9)$$

and so

$$P\psi_\ell^{(+)}(x)P^{-1} = \eta^*(2\pi)^{-3/2} \sum_\sigma \int d^3p u_\ell(-\mathbf{p}, \sigma) e^{ip \cdot \mathcal{P}x} a_{\mathbf{p}, \sigma}, \quad (5.5.10)$$

$$P\psi_\ell^{(-c)}(x)P^{-1} = \eta^c(2\pi)^{-3/2} \sum_\sigma \int d^3p v_\ell(-\mathbf{p}, \sigma) e^{-ip \cdot \mathcal{P}x} a_{\mathbf{p}, \sigma}^\dagger. \quad (5.5.11)$$

Also, Eqs. (5.4.16), (5.1.21), and (5.1.22) give

$$u(-\mathbf{p}, \sigma) = \sqrt{m/p^0} \beta D(L(\mathbf{p})) \beta u(\mathbf{0}, \sigma) \quad (5.5.12)$$

$$v(-\mathbf{p}, \sigma) = \sqrt{m/p^0} \beta D(L(\mathbf{p})) \beta v(\mathbf{0}, \sigma). \quad (5.5.13)$$

(Since  $\beta^2 = \mathbf{1}$ , we are no longer making a distinction between  $\beta$  and  $\beta^{-1}$ .) In order that the parity operator should transform the annihilation and creation fields at the point  $x$  into something proportional to these fields at  $\mathcal{P}x$ , it is necessary that  $\beta u(\mathbf{0}, \sigma)$  and  $\beta v(\mathbf{0}, \sigma)$  be proportional to  $u(\mathbf{0}, \sigma)$  and  $v(\mathbf{0}, \sigma)$ , respectively:

$$\beta u(\mathbf{0}, \sigma) = b_u u(\mathbf{0}, \sigma), \quad \beta v(\mathbf{0}, \sigma) = b_v v(\mathbf{0}, \sigma), \quad (5.5.14)$$

where  $b_u$  and  $b_v$  are sign factors,  $b_u^2 = b_v^2 = 1$ . In this case, the fields have the simple space-inversion properties:

$$P\psi^{(+)}(x)P^{-1} = \eta^* b_u \beta \psi^{(+)}(\mathcal{P}x), \quad (5.5.15)$$

$$P\psi^{(-c)}(x)P^{-1} = \eta^c b_v \beta \psi^{(-c)}(\mathcal{P}x). \quad (5.5.16)$$

By adjusting the overall scales of the fields, we can choose the coefficient functions at zero momentum to have the form:

$$u(\mathbf{0}, \tfrac{1}{2}) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ b_u \\ 0 \end{bmatrix}, \quad u(\mathbf{0}, -\tfrac{1}{2}) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ b_u \end{bmatrix}, \quad (5.5.17)$$

$$v(\mathbf{0}, \tfrac{1}{2}) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ b_v \end{bmatrix}, \quad v(\mathbf{0}, -\tfrac{1}{2}) = -\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ b_v \\ 0 \end{bmatrix}. \quad (5.5.18)$$

Now let's try to put together the annihilation and creation fields in a linear combination

$$\psi(x) = \kappa \psi^{(+)}(x) + \lambda \psi^{(-c)}(x) \quad (5.5.19)$$

that commutes or anticommutes with itself and its adjoint at space-like separations. A straightforward calculation gives

$$[\psi_\ell(x), \psi_\ell^\dagger(y)]_\mp = \int d^3p [|\kappa|^2 N_{\ell\bar{\ell}}(\mathbf{p}) e^{ip \cdot (x-y)} \mp |\lambda|^2 M_{\ell\bar{\ell}}(\mathbf{p}) e^{-ip \cdot (x-y)}], \quad (5.5.20)$$

where

$$N_{\ell\bar{\ell}}(\mathbf{p}) \equiv \sum_{\sigma} u_{\ell}(\mathbf{p}, \sigma) u_{\bar{\ell}}^*(\mathbf{p}, \sigma), \quad (5.5.21)$$

$$M_{\ell\bar{\ell}}(\mathbf{p}) \equiv \sum_{\sigma} v_{\ell}(\mathbf{p}, \sigma) v_{\bar{\ell}}^*(\mathbf{p}, \sigma). \quad (5.5.22)$$

By using either the eigenvalue conditions (5.5.14) or the explicit formulas (5.5.17) and (5.5.18), we find at zero momentum:

$$N(\mathbf{0}) = \frac{1 + b_u \beta}{2(2\pi)^3}, \quad M(\mathbf{0}) = \frac{1 + b_v \beta}{2(2\pi)^3}. \quad (5.5.23)$$

From Eqs. (5.5.6) and (5.5.7) we have then

$$N(\mathbf{p}) = \frac{m}{2p^0} D(L(\mathbf{p})) [1 + b_u \beta] D^{\dagger}(L(\mathbf{p})), \quad (5.5.24)$$

$$M(\mathbf{p}) = \frac{m}{2p^0} D(L(\mathbf{p})) [1 + b_v \beta] D^{\dagger}(L(\mathbf{p})). \quad (5.5.25)$$

The pseudounitariness condition (5.4.32) yields

$$D(L(\mathbf{p})) \beta D^{\dagger}(L(\mathbf{p})) = \beta$$

and

$$D(L(\mathbf{p})) D^{\dagger}(L(\mathbf{p})) = D(L(\mathbf{p})) \beta D^{-1}(L(\mathbf{p})) \beta.$$

We also recall that  $\beta = -i\gamma^0$ , so by using the Lorentz transformation rule (5.4.8) we have

$$D(L(\mathbf{p})) \beta D^{-1}(L(\mathbf{p})) = -iL_{\mu}^0(\mathbf{p}) \gamma^{\mu} = -ip^{\mu} \gamma_{\mu} / m. \quad (5.5.26)$$

Putting this together, we find<sup>44</sup>

$$N(\mathbf{p}) = \frac{1}{2p^0} [-ip^{\mu} \gamma_{\mu} + b_u m] \beta, \quad (5.5.27)$$

$$M(\mathbf{p}) = \frac{1}{2p^0} [-ip^{\mu} \gamma_{\mu} + b_v m] \beta. \quad (5.5.28)$$

Using this in Eq. (5.5.20) yields finally

$$\begin{aligned} [\psi_{\ell}(x), \psi_{\bar{\ell}}^{\dagger}(y)]_{\mp} &= (|\kappa|^2 [-\gamma^{\mu} \partial_{\mu} + b_u m] \beta \Delta_{+}(x - y) \\ &\quad \mp |\lambda|^2 [-\gamma^{\mu} \partial_{\mu} + b_v m] \beta \Delta_{+}(y - x))_{\ell\bar{\ell}}, \end{aligned} \quad (5.5.29)$$

where  $\Delta_{+}$  is the function introduced in Section 5.2:

$$\Delta_{+}(x) \equiv \int \frac{d^3 p}{2p^0 (2\pi)^3} e^{ip \cdot x}.$$

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<sup>44</sup>Sometimes an extra factor  $\sqrt{p^0/m}$  is included in the Dirac spinors, so that  $m$  appears in place of  $p^0$  in the denominators of the spin sums (5.5.27) and (5.5.28). The normalization convention used here has the advantage that it goes smoothly over to the case  $m = 0$ .

We saw in Section 5.2 that for  $x - y$  space-like  $\Delta_+(x - y)$  is an even function of  $x - y$  and, of course, this implies that its first derivatives are odd functions of  $x - y$ . Hence, in order that both the derivative and the non-derivative terms in the commutator or anticommutator should vanish at space-like separations, it is necessary and sufficient that

$$|\kappa|^2 = \mp |\lambda|^2 \quad (5.5.30)$$

and

$$|\kappa|^2 b_u = \pm |\lambda|^2 b_v. \quad (5.5.31)$$

Clearly Eq. (5.5.30) is only possible if we choose the bottom sign,  $\mp = +$ ; that is, *the particles described by a Dirac field must be fermions*. It is also then necessary that  $|\kappa|^2 = |\lambda|^2$  and  $b_u = -b_v$ . Just as for scalars we have the freedom to redefine the relative phase of the creation and annihilation operators to make the ratio  $\kappa/\lambda$  real, in which case  $\kappa = \lambda$ , and by adjusting the overall scale and phase of the field  $\psi$  we may then take

$$\kappa = \lambda = 1. \quad (5.5.32)$$

Finally if we like we can replace  $\psi$  with  $\gamma_5 \psi$ , which changes the sign of both  $b_u$  and  $b_v$ , so we can always take

$$b_u = -b_v = +1. \quad (5.5.33)$$

For future use, we record here that the Dirac field is now

$$\psi_\ell(x) = (2\pi)^{-3/2} \sum_{\sigma} \int d^3p \left[ u_\ell(\mathbf{p}, \sigma) e^{ip \cdot x} a_{\mathbf{p}, \sigma} + v_\ell(\mathbf{p}, \sigma) e^{-ip \cdot x} a_{\mathbf{p}, \sigma}^\dagger \right] \quad (5.5.34)$$

while the coefficient functions at zero momentum are

$$u(\mathbf{0}, \tfrac{1}{2}) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad u(\mathbf{0}, -\tfrac{1}{2}) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad (5.5.35)$$

$$v(\mathbf{0}, \tfrac{1}{2}) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \quad v(\mathbf{0}, -\tfrac{1}{2}) = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}. \quad (5.5.36)$$

The spin sums are

$$N(\mathbf{p}) = \frac{1}{2p^0} [-ip^\mu \gamma_\mu + m] \beta, \quad (5.5.37)$$

$$M(\mathbf{p}) = \frac{1}{2p^0} [-ip^\mu \gamma_\mu - m] \beta, \quad (5.5.38)$$

so the anticommutator is given by Eq. (5.5.20) as

$$[\psi_\ell(x), \psi_\ell^\dagger(y)]_+ = \{[-\gamma^\mu \partial_\mu + m] \beta\}_{\ell\bar{\ell}} \Delta(x - y). \quad (5.5.39)$$

Now let's return to the requirement that under a space inversion the field  $\psi(x)$  must transform into something proportional to  $\psi(\mathcal{P}x)$ . For this to be possible the phases in Eqs. (5.5.15) and (5.5.16) must be equal, and so the intrinsic parities of particles and their antiparticles must be related by

$$\eta^c = -\eta^*. \quad (5.5.40)$$

That is, *the intrinsic parity  $\eta\eta^c$  of a state consisting of a spin  $\frac{1}{2}$  particle and its antiparticle is odd*. It is for this reason that negative parity mesons like the  $\rho^0$  and  $J/\psi$  can be interpreted as s-wave bound states of quark–antiquark pairs. Eqs. (5.5.15) and (5.5.16) now give the transformation of the causal Dirac field under space inversion as

$$P\psi(x)P^{-1} = \eta^*\beta\psi(\mathcal{P}x). \quad (5.5.41)$$

Before going on to the other inversions, this is a good place to mention that Eqs. (5.5.14), (5.5.33), and (5.5.26) show that  $u(\mathbf{p}, \sigma)$  and  $v(\mathbf{p}, \sigma)$  are eigenvectors of  $-ip^\mu\gamma_\mu/m$  with eigenvalues  $+1$  and  $-1$ , respectively:

$$(ip^\mu\gamma_\mu + m)u(\mathbf{p}, \sigma) = 0, \quad (-ip^\mu\gamma_\mu + m)v(\mathbf{p}, \sigma) = 0. \quad (5.5.42)$$

It follows then that the field (5.5.33) satisfies the differential equation

$$(\gamma^\mu\partial_\mu + m)\psi(x) = 0. \quad (5.5.43)$$

This is the celebrated Dirac equation for a free particle of spin  $\frac{1}{2}$ . From the point of view adopted here, the free-particle Dirac equation is nothing but a Lorentz-invariant record of the convention that we have used in putting together the two irreducible representations of the proper orthochronous Lorentz group to form a field that transforms simply also under space inversion.

In order to work out the charge-conjugation and time-reversal properties of the Dirac field, we will need expressions for the complex-conjugates of the  $u$  and  $v$  coefficient functions. These functions are real for zero momentum, but to obtain the coefficient functions at finite momentum we have to multiply with the complex matrix  $D(L(\mathbf{p}))$ . From Eq. (5.4.41) we see that for general real  $\omega_{\mu\nu}$ :

$$[\exp(\frac{1}{2}i\mathcal{J}^{\mu\nu}\omega_{\mu\nu})]^* = \beta\mathcal{C}\exp(\frac{1}{2}i\mathcal{J}^{\mu\nu}\omega_{\mu\nu})\mathcal{C}^{-1}\beta$$

and so in particular

$$D(L(\mathbf{p}))^* = \beta\mathcal{C}D(L(\mathbf{p}))\mathcal{C}^{-1}\beta.$$

We also note that  $\mathcal{C}^{-1}\beta u(\mathbf{0}, \sigma) = -v(\mathbf{0}, \sigma)$  and  $\mathcal{C}^{-1}\beta v(\mathbf{0}, \sigma) = -u(\mathbf{0}, \sigma)$ , so

$$u_\ell^*(\mathbf{p}, \sigma) = -(\beta\mathcal{C}v(\mathbf{p}, \sigma))_\ell, \quad (5.5.44)$$

$$v_\ell^*(\mathbf{p}, \sigma) = -(\beta\mathcal{C}u(\mathbf{p}, \sigma))_\ell. \quad (5.5.45)$$

In order for the field to transform under charge-conjugation into another field with which it commutes at space-like separations, it is necessary again that the charge conjugation parities of the particle and antiparticle be related by

$$\xi^c = \xi^*. \quad (5.5.46)$$

In this case, the field transforms as

$$C\psi(x)C^{-1} = -\xi^*\beta\mathcal{C}\psi^*(x). \quad (5.5.47)$$

(We are calling the Hermitian adjoint of the field on the right-hand side  $\psi^*$  instead of  $\psi^\dagger$  to emphasize that this is still a column vector, not a row.)

Although we have been distinguishing particles from their antiparticles, we have not ruled out the possibility that the two are actually identical. Such spin  $\frac{1}{2}$  particles are called *Majorana fermions*. Following the same reasoning that led to Eq. (5.5.47), the Dirac field of such a particle must satisfy the reality condition

$$\psi(x) = -\beta\mathcal{C}\psi^*(x). \quad (5.5.48)$$

For Majorana fermions the intrinsic space-inversion parity must be imaginary,  $\eta = \pm i$ , while the charge-conjugation parity must be real,  $\xi = \pm 1$ .

There is an important difference between fermions and bosons in the intrinsic charge-conjugation phase of states consisting of a particle and its antiparticle. Such a state may be written

$$|\Phi\rangle = \sum_{\sigma,\sigma'} \int d^3p \int d^3p' \chi(\mathbf{p}, \sigma; \mathbf{p}', \sigma') a_{\mathbf{p},\sigma}^\dagger a_{\mathbf{p}',\sigma'}^{c\dagger} |\Omega\rangle,$$

where  $|\Omega\rangle$  is the vacuum state. Under charge-conjugation, this state is transformed into

$$C|\Phi\rangle = \xi^*\xi^{c*} \sum_{\sigma,\sigma'} \int d^3p \int d^3p' \chi(\mathbf{p}, \sigma; \mathbf{p}', \sigma') a_{\mathbf{p},\sigma}^{c\dagger} a_{\mathbf{p}',\sigma'}^\dagger |\Omega\rangle.$$

Interchanging the variables of integration and summation and using the anticommutation of the creation operators and Eq. (5.5.46), we can rewrite this as

$$C|\Phi\rangle = - \sum_{\sigma,\sigma'} \int d^3p \int d^3p' \chi(\mathbf{p}', \sigma'; \mathbf{p}, \sigma) a_{\mathbf{p},\sigma}^\dagger a_{\mathbf{p}',\sigma'}^{c\dagger} |\Omega\rangle.$$

That is, *the intrinsic charge-conjugation parity of a state consisting of a particle described by a Dirac field and its antiparticle is odd*, in the sense that if the wave function  $\chi$  of the state is even or odd under interchange of the momenta and spins of the particle and antiparticle, then the charge-conjugation operator applied to such a state gives a sign  $-1$  or  $+1$ , respectively. The classic example here is positronium, the bound state of an electron and a positron. The two lowest states are a pair of nearly degenerate s-wave states with total spin  $s = 0$  and  $s = 1$ , known respectively as para- and ortho-positronium. The wave function for these two states is even under interchange of momenta and odd or even respectively under interchange of spin  $z$ -components, so para- and ortho-positronium have  $C = +1$  and  $C = -1$ , respectively. These values are dramatically confirmed in the decay modes of positronium: para-positronium decays rapidly into a pair of photons (each of which has  $C = -1$ ), while ortho-positronium can only decay much more slowly into three or more photons. In the same way, single  $\rho^0$  and  $\omega^0$  mesons are produced as resonances in high-energy electron-positron annihilation through a one-photon intermediate state, so

they must have  $C = -1$ , which is consistent with their interpretation as quark–antiquark bound states with orbital angular momentum zero and total quark spin one.

Now we come to time-reversal. Recall the transformation properties of the particle annihilation and antiparticle creation operators given by Eq. (4.2.15):

$$T a_{\mathbf{p},\sigma} T^{-1} = \zeta^* (-1)^{1/2-\sigma} a_{-\mathbf{p},-\sigma}, \quad (5.5.49)$$

$$T a_{\mathbf{p},\sigma}^{c\dagger} T^{-1} = \zeta^c (-1)^{1/2-\sigma} a_{-\mathbf{p},-\sigma}^{c\dagger}. \quad (5.5.50)$$

Time-reversal of the field (5.5.34) thus gives

$$\begin{aligned} T \psi_\ell(x) T^{-1} &= (2\pi)^{-3/2} \sum_\sigma \int d^3p (-1)^{1/2-\sigma} \\ &\cdot [\zeta^* u_\ell^*(\mathbf{p}, \sigma) e^{-ip \cdot x} a_{-\mathbf{p},-\sigma} + \zeta^c v_\ell^*(\mathbf{p}, \sigma) e^{ip \cdot x} a_{-\mathbf{p},-\sigma}^{c\dagger}]. \end{aligned}$$

In order to put this back in the form given for  $\psi$ , we shall redefine the variables of integration and summation as  $-\mathbf{p}$  and  $-\sigma$ , so we need formulas for  $u^*(-\mathbf{p}, -\sigma)$  and  $v^*(-\mathbf{p}, -\sigma)$  in terms of  $u(\mathbf{p}, \sigma)$  and  $v(\mathbf{p}, \sigma)$ , respectively. For this purpose, we can use the fact that  $\mathcal{J}^{i0}$  anticommutes with  $\beta$  and commutes with  $\gamma_5$  together with our former result for  $D(L(\mathbf{p}))^*$  to write

$$D^*(L(-\mathbf{p})) = \gamma_5 \beta D^*(L(\mathbf{p})) \beta \gamma_5 = \gamma_5 \mathcal{C} D(L(\mathbf{p})) \mathcal{C}^{-1} \gamma_5.$$

Also, Eqs. (5.4.36) and (5.5.35)–(5.5.36) give

$$\gamma_5 \mathcal{C}^{-1} u(\mathbf{0}, -\sigma) = (-1)^{1/2-\sigma} u(\mathbf{0}, \sigma),$$

$$\gamma_5 \mathcal{C}^{-1} v(\mathbf{0}, -\sigma) = (-1)^{1/2-\sigma} v(\mathbf{0}, \sigma),$$

so

$$(-1)^{1/2+\sigma} u^*(-\mathbf{p}, -\sigma) = -\gamma_5 \mathcal{C} u(\mathbf{p}, \sigma), \quad (5.5.51)$$

$$(-1)^{1/2+\sigma} v^*(-\mathbf{p}, -\sigma) = -\gamma_5 \mathcal{C} v(\mathbf{p}, \sigma). \quad (5.5.52)$$

We see then that in order for time-reversal to take the Dirac field into something proportional to itself at the time-reversed point (with which it would anticommute at space-like separations) it is necessary that the intrinsic time-reversal phases be related by

$$\zeta^c = \zeta^* \quad (5.5.53)$$

and in this case

$$T \psi(x) T^{-1} = -\zeta^* \gamma_5 \mathcal{C} \psi(-\mathcal{T}x). \quad (5.5.54)$$

Now let us consider how to construct scalar interaction densities out of the Dirac fields and their adjoints. As already mentioned the Dirac representation is not unitary, so  $\psi^\dagger \psi$  is not a scalar. To deal with this complication it is convenient to define a new sort of adjoint:

$$\bar{\psi} \equiv \psi^\dagger \beta. \quad (5.5.55)$$

Using the pseudounitariness condition (5.4.32), we see that the fermion bilinears constructed with  $\bar{\psi}$  have the Lorentz transformation property

$$U_0(\Lambda)[\bar{\psi}(x)M\psi(x)]U_0^{-1}(\Lambda) = \bar{\psi}(\Lambda x)D^{-1}(\Lambda)MD(\Lambda)\psi(\Lambda x). \quad (5.5.56)$$

Also, under a space inversion

$$P[\bar{\psi}(x)M\psi(x)]P^{-1} = \bar{\psi}(\mathcal{P}x)\beta M\beta\psi(\mathcal{P}x). \quad (5.5.57)$$

Taking the matrix  $M$  as  $\mathbf{1}$ ,  $\gamma^\mu$ ,  $\mathcal{J}^{\mu\nu}$ ,  $\gamma_5\gamma^\mu$ , or  $\gamma_5$  yields a bilinear  $\bar{\psi}M\psi$  that transforms as a scalar, vector, tensor, axial vector, and pseudoscalar, respectively. (The terms ‘axial’ and ‘pseudo’ indicate that these have space-inversion properties opposite to those of ordinary vectors and scalars: a pseudoscalar has negative parity, while the space and time components of an axial vector have positive and negative parity, respectively.) These results apply also when the two fermion fields in the bilinear refer to different particle species, except that in this case a space inversion also yields a ratio of the intrinsic parities.

For instance, the original Fermi theory of beta decay involved an interaction density proportional to  $\bar{\psi}_p\gamma^\mu\psi_n\bar{\psi}_e\gamma_\mu\psi_\nu$ . Later it was realized that the most general Lorentz-invariant and parity-conserving non-derivative beta decay interaction takes the form of a linear combination of products like this, with  $\gamma_\mu$  replaced with any one of the five covariant types of  $4 \times 4$  matrices  $\mathbf{1}$ ,  $\gamma^\mu$ ,  $\mathcal{J}^{\mu\nu}$ ,  $\gamma_5\gamma^\mu$ , or  $\gamma_5$ . (As discussed in Chapter 2, we are defining the space-inversion operator so that the proton, neutron, and electron all have parity  $+1$ . If the neutrino is massless then its parity may also be defined as  $+1$ , if necessary by replacing the neutrino field with  $\gamma_5\psi_\nu$ .) When Lee and Yang [7] called parity conservation into question in 1956, they expanded the list of possible non-derivative interactions to include ten terms proportional to  $\bar{\psi}_pM\psi_n\bar{\psi}_eM\psi_\nu$  and also  $\bar{\psi}_pM\psi_n\bar{\psi}_eM\gamma_5\psi_\nu$ , with  $M$  running over the matrices  $\mathbf{1}$ ,  $\gamma^\mu$ ,  $\mathcal{J}^{\mu\nu}$ ,  $\gamma_5\gamma^\mu$ , or  $\gamma_5$ .

It is also of some interest to study the charge-conjugation properties of these bilinears. Using Eqs. (5.5.47) and (5.4.35)–(5.4.39), we have

$$\begin{aligned} C(\bar{\psi}M\psi)C^{-1} &= (\beta\mathcal{C}\psi)^\top\beta M(\beta\mathcal{C}\psi^*) = -(\beta\mathcal{C}\psi^*)^\top M^\top\mathcal{C}\psi \\ &= \bar{\psi}\mathcal{C}^{-1}M^\top\mathcal{C}\psi = \pm\bar{\psi}M\psi, \end{aligned} \quad (5.5.58)$$

the sign in the last expression being  $+$  for the matrices  $\mathbf{1}$ ,  $\gamma_5\gamma_\mu$ , and  $\gamma_5$ , and  $-$  for  $\gamma_\mu$  and  $\mathcal{J}_{\mu\nu}$ . (The minus sign in the first line arises from Fermi statistics. We ignore a c-number anticommutator.) A boson field that interacts with the current  $\bar{\psi}M\psi$  must therefore have  $C = +1$  for scalars, pseudoscalars, or axial vectors, and  $C = -1$  for vectors or antisymmetric tensors. This is one way of seeing that the  $\pi^0$  (which couples to pseudoscalar or axial-vector nucleon currents) has  $C = +1$ , while the photon has  $C = -1$ .

## 5.6 General Irreducible Representations of the Homogeneous Lorentz Group<sup>45</sup>

<sup>45</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.



We shall now generalize from the special cases of vector and Dirac fields to the case of a field that transforms according to a general irreducible representation of the homogeneous Lorentz group. All fields may be constructed as direct sums of these irreducible fields.

A general representation of the proper orthochronous homogeneous Lorentz group (or, more properly, its infinitesimal part) is provided by a set of matrices  $\mathcal{J}_{\mu\nu}$  satisfying the same commutation relations (5.4.4) as the generators of the group

$$[\mathcal{J}_{\mu\nu}, \mathcal{J}_{\rho\sigma}] = i(\mathcal{J}_{\rho\nu}\eta_{\sigma\mu} + \mathcal{J}_{\mu\sigma}\eta_{\nu\rho} - \mathcal{J}_{\sigma\nu}\eta_{\rho\mu} - \mathcal{J}_{\mu\rho}\eta_{\nu\sigma}). \quad (5.6.1)$$

(Of course,  $\mathcal{J}_{\mu\nu} = -\mathcal{J}_{\nu\mu}$ , and indices on  $\mathcal{J}_{\mu\nu}$  are as usual raised or lowered by contraction with  $\eta^{\mu\nu}$  or  $\eta_{\mu\nu}$ .) To see how to construct such matrices, first divide the six independent components of  $\mathcal{J}_{\mu\nu}$  into two three-vectors: an angular momentum matrix

$$\mathcal{J}_1 = \mathcal{J}_{23}, \quad \mathcal{J}_2 = \mathcal{J}_{31}, \quad \mathcal{J}_3 = \mathcal{J}_{12} \quad (5.6.2)$$

and a boost

$$\mathcal{K}_1 = \mathcal{J}_{10}, \quad \mathcal{K}_2 = \mathcal{J}_{20}, \quad \mathcal{K}_3 = \mathcal{J}_{30}. \quad (5.6.3)$$

Eq. (5.6.1) then reads

$$[\mathcal{J}_i, \mathcal{J}_j] = i\epsilon_{ijk}\mathcal{J}_k, \quad (5.6.4)$$

$$[\mathcal{J}_i, \mathcal{K}_j] = i\epsilon_{ijk}\mathcal{K}_k, \quad (5.6.5)$$

$$[\mathcal{K}_i, \mathcal{K}_j] = -i\epsilon_{ijk}\mathcal{J}_k, \quad (5.6.6)$$

where  $i, j, k$  run over the values 1, 2, 3, and  $\epsilon_{ijk}$  is the totally antisymmetric quantity with  $\epsilon_{123} = +1$ . Eq. (5.6.4) just says that the matrices  $\mathcal{J}$  generate a representation of the rotation subgroup of the Lorentz group, and Eq. (5.6.5) just represents the fact that  $\mathcal{K}$  is a three-vector. The minus sign in the right-hand side of Eq. (5.6.6) arises from the fact that  $\eta_{00} = -1$ , and plays a crucial role in what follows.

It is very convenient to replace the matrices  $\mathcal{J}$  and  $\mathcal{K}$  with two decoupled spin three-vectors, writing

$$\mathcal{A} \equiv \frac{1}{2}(\mathcal{J} + i\mathcal{K}), \quad (5.6.7)$$

$$\mathcal{B} \equiv \frac{1}{2}(\mathcal{J} - i\mathcal{K}). \quad (5.6.8)$$

It is easy to see that the commutation relations (5.6.4)–(5.6.6) are equivalent to

$$[\mathcal{A}_i, \mathcal{A}_j] = i\epsilon_{ijk}\mathcal{A}_k, \quad (5.6.9)$$

$$[\mathcal{B}_i, \mathcal{B}_j] = i\epsilon_{ijk}\mathcal{B}_k, \quad (5.6.10)$$

$$[\mathcal{A}_i, \mathcal{B}_j] = 0. \quad (5.6.11)$$

We find matrices satisfying Eqs. (5.6.9)–(5.6.11) in the same way that we find matrices representing the spins of a pair of uncoupled particles—as a direct sum. That is, we label the rows and columns of these matrices with a pair of integers and/or half-integers  $a, b$ , running over the values

$$a = -A, -A + 1, \dots, +A, \quad (5.6.12)$$

$$b = -B, -B + 1, \dots, +B \quad (5.6.13)$$

and take<sup>46</sup>

$$(\mathcal{A})_{a'b',ab} = \delta_{b'b} \mathbf{J}_{a'a}^{(A)}, \quad (5.6.14)$$

$$(\mathcal{B})_{a'b',ab} = \delta_{a'a} \mathbf{J}_{b'b}^{(B)}, \quad (5.6.15)$$

where  $\mathbf{J}^{(A)}$  and  $\mathbf{J}^{(B)}$  are the standard spin matrices for spins  $A$  or  $B$ :

$$(\mathbf{J}_3^{(A)})_{a'a} = a \delta_{a'a}, \quad (5.6.16)$$

$$(\mathbf{J}_1^{(A)} \pm i \mathbf{J}_2^{(A)})_{a'a} = \delta_{a',a \pm 1} \sqrt{(A \mp a)(A \pm a + 1)}, \quad (5.6.17)$$

and likewise for  $\mathbf{J}^{(B)}$ . The representation is labelled by the values of the positive integers and/or half-integers  $A$  and  $B$ . We see that the  $(A, B)$  representation has dimensionality  $(2A + 1)(2B + 1)$ .

The finite-dimensional representations of the homogeneous Lorentz group are not unitary, because  $\mathcal{A}$  and  $\mathcal{B}$  are Hermitian, and therefore  $\mathcal{J}$  is Hermitian but  $\mathcal{K}$  is *anti-Hermitian*. This is because of the  $i$  in Eqs. (5.6.7) and (5.6.8), which is required by the minus sign in (5.6.6), and hence stems from the fact that the homogeneous Lorentz group is not the same as the four-dimensional rotation group  $\text{SO}(4)$ , a compact group, but instead is the non-compact group known as  $\text{SO}(3, 1)$ . It is only compact groups that can have finite-dimensional unitary representations (aside from representations in which the non-compact part is represented trivially, by the identity). There is no problem in working with non-unitary representations, because the objects we are now concerned with are fields, not wave functions, and do not need to have a Lorentz-invariant positive norm.

In contrast, the rotation group is represented unitarily, with its generators represented by the Hermitian matrices

$$\mathcal{J} = \mathcal{A} + \mathcal{B}. \quad (5.6.18)$$

By the usual rules of vector addition, we can see that a field that transforms according to the  $(A, B)$  representation of the homogeneous Lorentz group has components that rotate like objects of spin  $j$ , with

$$j = A + B, A + B - 1, \dots, |A - B|.$$

This is enough to identify the  $(A, B)$  representations with the perhaps more familiar tensors and spinors. For instance, a  $(0, 0)$  field is obviously scalar, with only a single  $j = 0$  component. A  $(1/2, 0)$  or  $(0, 1/2)$  field can only have  $j = 1/2$ ; these are the top (i.e.,  $\gamma_5 = +1$ ) and bottom ( $\gamma_5 = -1$ ) two components of the Dirac spinor. A  $(1/2, 1/2)$  field has components with  $j = 1$  and  $j = 0$ , corresponding to the spatial part  $\mathbf{v}$  and time-component  $v^0$  of a four-vector  $v^\mu$ . More generally, an  $(A, A)$  field contains terms with only integer spins

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<sup>46</sup>There is an alternative formalism [8], based on the fact that the spin  $j$  representation of the rotation group can be written as the symmetrized direct product of  $2j$  spin  $1/2$  representations—i.e., as a symmetric  $\text{SU}(2)$  tensor with  $2j$  two-valued indices. We can therefore write fields belonging to the  $(A, B)$  representation with  $2A$  two-valued  $(1/2, 0)$  indices and  $2B$  two-valued  $(0, 1/2)$  indices, the latter written with dots to distinguish them from the former.

$2A, 2A - 1, \dots, 0$ , and corresponds to a traceless symmetric tensor of rank  $2A$ . (Note that the number of independent components of a symmetric tensor of rank  $2A$  in four dimensions is

$$\frac{4 \cdot 5 \cdots (4 + 2A - 1)}{(2A)!} = \frac{(3 + 2A)!}{6(2A)!}$$

and the tracelessness condition reduces this to

$$\frac{(3 + 2A)!}{6(2A)!} - \frac{(1 + 2A)!}{6(2A - 2)!} = (2A + 1)^2$$

as expected for an  $(A, A)$  field.) One more example: a  $(1, 0)$  or  $(0, 1)$  field can only have  $j = 1$ , and corresponds to an antisymmetric tensor  $F^{\mu\nu}$  that satisfies the further irreducibility ‘duality’ conditions

$$F^{\mu\nu} = \pm \frac{i}{2} \epsilon^{\mu\nu\lambda\rho} F_{\lambda\rho}$$

for  $(1, 0)$  and  $(0, 1)$  fields, respectively. Of course, it is only in four dimensions that an antisymmetric two-index tensor  $F^{\mu\nu}$  can be divided into such ‘self-dual’ and ‘anti-self-dual’ parts.

A general tensor of rank  $N$  transforms as the direct product of  $N$   $(1/2, 1/2)$  four-vector representations. It can therefore be decomposed (by suitable symmetrizations and antisymmetrizations and extracting traces) into irreducible terms  $(A, B)$  with  $A = N/2, N/2 - 1, \dots$  and  $B = N/2, N/2 - 1, \dots$ . In this way, we can construct any irreducible representation  $(A, B)$  for which  $A+B$  is an integer. The spin representations, for which  $A+B$  is half an odd integer, can similarly be constructed from the direct product of these tensor representations and the Dirac representation  $(1/2, 0) \oplus (0, 1/2)$ .

For instance, taking the direct product of the vector  $(1/2, 1/2)$  representation and the Dirac  $(1/2, 0) \oplus (0, 1/2)$  representation gives a spinor-vector  $\psi^\mu$ , that transforms according to the reducible representation

$$(\tfrac{1}{2}, \tfrac{1}{2}) \otimes [(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2})] = (\tfrac{1}{2}, 1) \oplus (\tfrac{1}{2}, 0) \oplus (1, \tfrac{1}{2}) \oplus (0, \tfrac{1}{2}).$$

The quantity  $\gamma_\mu \psi^\mu$  would transform as an ordinary  $(1/2, 0) \oplus (0, 1/2)$  Dirac field, so we can isolate the  $(1/2, 1) \oplus (1, 1/2)$  representation<sup>47</sup> by requiring that  $\gamma_\mu \psi^\mu = 0$ . This is the *Rarita–Schwinger field* [9].

So far in this section we have only considered the representations of the proper orthochronous Lorentz group. In any representation of the Lorentz group including space inversion, there must be a matrix  $\beta$  which reverses the signs of tensors with odd numbers of space indices, and in particular

$$\beta \mathcal{J} \beta^{-1} = +\mathcal{J}, \quad \beta \mathcal{K} \beta^{-1} = -\mathcal{K}. \quad (5.6.19)$$

In terms of the matrices (5.6.7) and (5.6.8), this is

$$\beta \mathcal{A} \beta^{-1} = \mathcal{B}, \quad \beta \mathcal{B} \beta^{-1} = \mathcal{A}. \quad (5.6.20)$$

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<sup>47</sup> According to Eq. (5.6.18), such a field transforms under ordinary rotations as a direct sum of two  $j = 3/2$  and two  $j = 1/2$  components. The doubling is eliminated by imposing the Dirac equation  $[\gamma^\nu \partial_\nu + m]\psi^\mu = 0$ , and the remaining  $j = 1/2$  component is eliminated by requiring that  $\partial_\mu \psi^\mu = 0$ . With these conditions, the field describes a single particle of spin  $j = 3/2$ .

Thus an irreducible  $(A, B)$  representation of the proper orthochronous homogeneous Lorentz group does not provide a representation of the Lorentz group including space inversion unless  $A = B$ . As we have seen, these  $(A, A)$  representations are the scalar, the vector, and the symmetric traceless tensors. For  $A \neq B$ , the irreducible representations of the Lorentz group including space inversion are the direct sums  $(A, B) \oplus (B, A)$ , of dimensionality  $2(2A + 1)(2B + 1)$ . One of these is the  $(1/2, 0) \oplus (0, 1/2)$  Dirac representation discussed in Section 5.4. The  $4 \times 4$  matrix (5.4.29) provides the  $\beta$ -matrix for this representation. Another familiar example is the  $(1, 0) \oplus (0, 1)$  representation, which as we have seen is just the antisymmetric tensor of second rank, including both self-dual and anti-self-dual parts.

## 5.7 General Causal Fields<sup>48</sup>

We now proceed to construct causal fields that transform according to the general irreducible  $(A, B)$  representations described in the previous section. The index  $\ell$  is replaced here with a pair of indices  $a, b$ , running over the ranges (5.6.12), (5.6.13), so the fields are now written as

$$\psi_{ab}(x) = (2\pi)^{-3/2} \sum_{\sigma} \int d^3p \left[ \kappa a_{\mathbf{p},\sigma} e^{ip \cdot x} u_{ab}(\mathbf{p}, \sigma) + \lambda a_{\mathbf{p},\sigma}^{\dagger} e^{-ip \cdot x} v_{ab}(\mathbf{p}, \sigma) \right]. \quad (5.7.1)$$

with  $\kappa$  and  $\lambda$  arbitrary constants. We are here leaving open the possibility that this particle is its own antiparticle, in which case  $a_{\mathbf{p},\sigma}^{\dagger} = a_{\mathbf{p},\sigma}$ .

Our first task is to find the zero-momentum coefficient functions  $u_{ab}(0, \sigma)$  and  $v_{ab}(0, \sigma)$ . The fundamental conditions (5.1.25)–(5.1.26) on  $u(0, \sigma)$  and  $v(0, \sigma)$  read here

$$\begin{aligned} \sum_{\bar{\sigma}} u_{\bar{a}\bar{b}}(0, \bar{\sigma}) J_{\bar{\sigma}\sigma}^{(j)} &= \sum_{a,b} \mathcal{J}_{\bar{a}\bar{b},ab} u_{ab}(0, \sigma), \\ - \sum_{\bar{\sigma}} v_{\bar{a}\bar{b}}(0, \bar{\sigma}) J_{\bar{\sigma}\sigma}^{(j)*} &= \sum_{a,b} \mathcal{J}_{\bar{a}\bar{b},ab} v_{ab}(0, \sigma), \end{aligned}$$

or using Eqs. (5.6.14)–(5.6.15)

$$\sum_{\bar{\sigma}} u_{\bar{a}\bar{b}}(0, \bar{\sigma}) J_{\bar{\sigma}\sigma}^{(j)} = \sum_a \mathbf{J}_{\bar{a}\bar{a}}^{(A)} u_{\bar{a}\bar{b}}(0, \sigma) + \sum_b \mathbf{J}_{\bar{b}\bar{b}}^{(B)} u_{\bar{a}\bar{b}}(0, \sigma), \quad (5.7.2)$$

$$- \sum_{\bar{\sigma}} v_{\bar{a}\bar{b}}(0, \bar{\sigma}) J_{\bar{\sigma}\sigma}^{(j)*} = \sum_a \mathbf{J}_{\bar{a}\bar{a}}^{(A)} v_{\bar{a}\bar{b}}(0, \sigma) + \sum_b \mathbf{J}_{\bar{b}\bar{b}}^{(B)} v_{\bar{a}\bar{b}}(0, \sigma). \quad (5.7.3)$$

But Eq. (5.7.2) is the defining condition for the Clebsch–Gordan coefficients  $C_{AB}(j\sigma; ab)!$  These coefficients are defined by the requirement that if  $\Psi_{ab}$  are states that under an infinitesimal rotation transform as

$$\delta \Psi_{\bar{a}\bar{b}} = i \sum_a \boldsymbol{\theta} \cdot \mathbf{J}_{\bar{a}\bar{a}}^{(A)} \Psi_{\bar{a}\bar{b}} + i \sum_b \boldsymbol{\theta} \cdot \mathbf{J}_{\bar{b}\bar{b}}^{(B)} \Psi_{\bar{a}\bar{b}},$$

<sup>48</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

then, under the same rotation, the state

$$\Psi_\sigma^j \equiv \sum_{ab} C_{AB}(j\sigma; ab) \Psi_{ab}$$

transforms as

$$\delta \Psi_\sigma^j = i \sum_{\bar{\sigma}} \boldsymbol{\theta} \cdot \mathbf{J}_{\bar{\sigma}\sigma}^{(j)} \Psi_{\bar{\sigma}}^j.$$

Inspection of Eq. (5.7.2) shows that this requirement is satisfied by the coefficients  $u_{ab}(0, \sigma)$ , and therefore, up to a possible proportionality factor,  $u_{ab}(0, \sigma)$  is just  $C_{AB}(j\sigma; ab)$ . This constant is conventionally chosen so that

$$u_{ab}(0, \sigma) = (2m)^{-1/2} C_{AB}(j\sigma; ab). \quad (5.7.4)$$

This result is unique because each irreducible  $(A, B)$  representation of the homogeneous Lorentz group contains a given spin  $j$  representation of the rotation group at most *once*. Similarly, inspection of Eqs. (5.6.16)–(5.6.17) shows that the complex conjugates of the angular momentum matrices are

$$-J_{\sigma\sigma'}^{(j)*} = (-1)^{\sigma-\sigma'} J_{-\sigma, -\sigma'}^{(j)}. \quad (5.7.5)$$

Therefore if we write Eq. (5.7.3) in terms of  $(-1)^{j-\sigma} v_{ab}(\mathbf{p}, -\sigma)$ , it takes the same form as Eq. (5.7.2). With a suitable adjustment of a constant factor, the unique solution for  $v(0, \sigma)$  is

$$v_{ab}(0, \sigma) = (-1)^{j+\sigma} u_{ab}(0, -\sigma). \quad (5.7.6)$$

We must now perform a boost to calculate the coefficient functions for finite momentum. For a fixed direction  $\hat{\mathbf{p}} \equiv \mathbf{p}/|\mathbf{p}|$ , we can write the boost (2.5.24) as a function of a parameter  $\theta$  defined by

$$\cosh \theta = \sqrt{\mathbf{p}^2 + m^2}/m, \quad \sinh \theta = |\mathbf{p}|/m \quad (5.7.7)$$

and write  $L^\mu{}_\nu(\theta)$  in place of  $L^\mu{}_\nu(p)$ , where

$$\begin{aligned} L^i{}_k(\theta) &= \delta_{ik} + (\cosh \theta - 1) \hat{p}_i \hat{p}_k, \\ L^i{}_0(\theta) &= L^0{}_i(\theta) = \hat{p}_i \sinh \theta, \\ L^0{}_0(\theta) &= \cosh \theta. \end{aligned} \quad (5.7.8)$$

The advantage of this parameterization is that

$$L(\bar{\theta})L(\theta) = L(\bar{\theta} + \theta). \quad (5.7.9)$$

For infinitesimal  $\theta$ , we have  $[L(\theta)]^\mu{}_\nu \rightarrow \delta^\mu{}_\nu + \omega^\mu{}_\nu$ , where  $\omega^i{}_0 = \omega^0{}_i = \hat{p}_i \theta$  and  $\omega^i{}_j = \omega^j{}_i = 0$ . Following the same reasoning that led from Eq. (2.2.24) to Eq. (2.2.26), it follows then that

$$D(L(p)) = \exp(-i\hat{\mathbf{p}} \cdot \boldsymbol{\mathcal{K}}\theta). \quad (5.7.10)$$

This is for any representation of the homogeneous Lorentz group; for the irreducible  $(A, B)$  representations, Eqs. (5.6.7) and (5.6.8) give

$$i\boldsymbol{\mathcal{K}} = \boldsymbol{\mathcal{A}} - \boldsymbol{\mathcal{B}} \quad (5.7.11)$$

and since  $\mathcal{A}$  and  $\mathcal{B}$  are commuting matrices

$$D(L(p)) = \exp(-\hat{\mathbf{p}} \cdot \mathcal{A}\theta) \exp(+\hat{\mathbf{p}} \cdot \mathcal{B}\theta). \quad (5.7.12)$$

In more detail, using Eqs. (5.6.14) and (5.6.15)

$$D(L(p))_{a'b',ab} = (\exp(-\hat{\mathbf{p}} \cdot \mathbf{J}^{(A)}\theta))_{a'a} (\exp(+\hat{\mathbf{p}} \cdot \mathbf{J}^{(B)}\theta))_{b'b}. \quad (5.7.13)$$

Eqs. (5.7.4) and (5.7.6) then give the coefficient functions at finite momentum as

$$u_{ab}(\mathbf{p}, \sigma) = \frac{1}{\sqrt{2p^0}} \sum_{a'b'} (\exp(-\hat{\mathbf{p}} \cdot \mathbf{J}^{(A)}\theta))_{aa'} (\exp(+\hat{\mathbf{p}} \cdot \mathbf{J}^{(B)}\theta))_{bb'} \cdot C_{AB}(j\sigma; a'b') \quad (5.7.14)$$

and

$$v_{ab}(\mathbf{p}, \sigma) = (-1)^{j+\sigma} u_{ab}(\mathbf{p}, -\sigma). \quad (5.7.15)$$

These results give the field explicitly for a given transformation type  $(A, B)$ , so the field (5.1.31) of this type is unique up to the choice of the constant factors  $\kappa$  and  $\lambda$ .

It is very easy in this formalism to construct Lorentz scalar interaction densities. The  $(A, B)$  representation of the homogeneous Lorentz group is just the direct product of the  $(A, 0)$  and  $(0, B)$  representations, so the general Lorentz transformation rules (5.1.6), (5.1.7) read here

$$U_0(\Lambda)\psi_{ab}(x)U_0^{-1}(\Lambda) = \sum_{a'b'} D_{aa'}^{A0}(\Lambda^{-1}) D_{bb'}^{0B}(\Lambda^{-1}) \psi_{a'b'}(\Lambda x). \quad (5.7.16)$$

Furthermore, Eqs. (5.6.14) and (5.6.15) show that the matrix generators of the  $(A, 0)$  and  $(0, B)$  representations are just the spin matrices for spin  $A$  and  $B$ , respectively. Thus we can construct scalars of the form

$$\sum_{a_1 a_2 \dots a_n} \sum_{b_1 b_2 \dots b_n} g_{a_1 a_2 \dots a_n; b_1 b_2 \dots b_n} \psi_{a_1 b_1}^{(1)}(x) \psi_{a_2 b_2}^{(2)}(x) \dots \psi_{a_n b_n}^{(n)}(x) \quad (5.7.17)$$

by simply taking  $g_{a_1 a_2 \dots a_n; b_1 b_2 \dots b_n}$  as the product of a coefficient for coupling spins  $A_1, A_2, \dots, A_n$  to make a scalar and a coefficient for coupling spins  $B_1, B_2, \dots, B_n$  to make a scalar. (Even though we do not explicitly consider interactions involving derivatives, we will in this way obtain the most general interaction involving  $n$  fields, because the derivative of a field of type  $(A, B)$  can always be decomposed into fields of other types without derivatives.) For instance, the most general Lorentz scalar formed from a product of three fields of transformation types  $(A_1, B_1)$ ,  $(A_2, B_2)$ , and  $(A_3, B_3)$  is

$$g \sum_{a_1 a_2 a_3} \sum_{b_1 b_2 b_3} \begin{pmatrix} A_1 & A_2 & A_3 \\ a_1 & a_2 & a_3 \end{pmatrix} \begin{pmatrix} B_1 & B_2 & B_3 \\ b_1 & b_2 & b_3 \end{pmatrix} \psi_{a_1 b_1}^{(1)} \psi_{a_2 b_2}^{(2)} \psi_{a_3 b_3}^{(3)}. \quad (5.7.18)$$

with a single free parameter  $g$ . This is the most general three-field interaction. (The brackets in (5.7.17) denote the Wigner “three- $j$ ” symbols [10]

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \equiv \sum_{m'_3} C_{j_1 j_2}(j_3 m'_3; m_1 m_2) C_{j_3 j_3}(00; m'_3 m_3)$$

which describe the coupling of three spins to make a rotational scalar.)

For the  $S$ -matrix to be Lorentz-invariant it is not enough that the interaction density  $\mathcal{H}_I(x)$  be a scalar like (5.7.18); it is also necessary that  $\mathcal{H}_I(x)$  should commute with  $\mathcal{H}_I(y)$  at space-like separations  $x - y$ . To see how to satisfy this condition, consider the commutator or anticommutator of two fields for the same particle species, a field  $\psi$  of type  $(A, B)$ , and the adjoint  $\tilde{\psi}^\dagger$  of a field  $\tilde{\psi}$  of type  $(\tilde{A}, \tilde{B})$ . We find

$$[\psi_{ab}(x), \tilde{\psi}_{\tilde{a}\tilde{b}}^\dagger(y)]_{\mp} = (2\pi)^{-3/2} \int d^3p (2p^0)^{-1} \pi_{ab, \tilde{a}\tilde{b}}(\mathbf{p}) \cdot [\kappa \tilde{\kappa}^* e^{ip \cdot (x-y)} \mp \lambda \tilde{\lambda}^* e^{-ip \cdot (x-y)}], \quad (5.7.19)$$

where  $\pi(\mathbf{p})$  is the spin sum

$$(2p^0)^{-1} \pi_{ab, \tilde{a}\tilde{b}}(\mathbf{p}) = \sum_{\sigma} u_{ab}(\mathbf{p}, \sigma) \tilde{u}_{\tilde{a}\tilde{b}}^*(\mathbf{p}, \sigma) = \sum_{\sigma} v_{ab}(\mathbf{p}, \sigma) \tilde{v}_{\tilde{a}\tilde{b}}^*(\mathbf{p}, \sigma) \quad (5.7.20)$$

and as usual, the top and bottom signs are for bosons and fermions, respectively. (We allow here for different coefficients  $\tilde{\kappa}$  and  $\tilde{\lambda}$  in the  $\tilde{\psi}$  field.) In more detail

$$\begin{aligned} \pi_{ab, \tilde{a}\tilde{b}}(\mathbf{p}) &= \sum_{a'b'} \sum_{\tilde{a}'\tilde{b}'} \sum_{\sigma} C_{AB}(j\sigma; a'b') C_{\tilde{A}\tilde{B}}(j\sigma; \tilde{a}'\tilde{b}') \\ &\quad \cdot (\exp(-\hat{\mathbf{p}} \cdot \mathbf{J}^{(A)}\theta))_{aa'} (\exp(+\hat{\mathbf{p}} \cdot \mathbf{J}^{(B)}\theta))_{bb'} \\ &\quad \cdot (\exp(-\hat{\mathbf{p}} \cdot \mathbf{J}^{(\tilde{A})}\theta))_{\tilde{a}\tilde{a}'}^* (\exp(+\hat{\mathbf{p}} \cdot \mathbf{J}^{(\tilde{B})}\theta))_{\tilde{b}\tilde{b}'}^*. \end{aligned} \quad (5.7.21)$$

The function  $\pi(\mathbf{p})$  has been calculated explicitly [11]. What concerns us here is the fact that it turns out to be the mass-shell value of a *polynomial* function  $P$  of  $\mathbf{p}$  and  $p^0$ :

$$\pi_{ab, \tilde{a}\tilde{b}}(\mathbf{p}) = P_{ab, \tilde{a}\tilde{b}}(\mathbf{p}, \sqrt{\mathbf{p}^2 + m^2}) \quad (5.7.22)$$

and that  $P$  is even or odd according to whether  $2A + 2\tilde{B}$  is an even or odd integer

$$P(-\mathbf{p}, -p^0) = (-1)^{2A+2\tilde{B}} P(\mathbf{p}, p^0). \quad (5.7.23)$$

We shall check this here for just one particular direction of  $\mathbf{p}$ . Taking  $\mathbf{p}$  in the three-direction, (5.7.21) gives

$$\pi_{ab, \tilde{a}\tilde{b}}(\mathbf{p}) = \sum_{\sigma} C_{AB}(j\sigma; ab) C_{\tilde{A}\tilde{B}}(j\sigma; \tilde{a}\tilde{b}) \exp([-a + b - \tilde{a} + \tilde{b}]\theta).$$

The Clebsch–Gordan coefficients vanish unless  $\sigma = a + b$  and  $\sigma = \tilde{a} + \tilde{b}$ , so we can replace

$$-a + b - \tilde{a} + \tilde{b} = -2a + \sigma + 2\tilde{b} - \sigma = 2\tilde{b} - 2a.$$

We can write  $\exp(\pm\theta)$  as  $(p^0 \pm p^3)/m$ , so here

$$\pi_{ab, \tilde{a}\tilde{b}}(\mathbf{p}) = \sum_{\sigma} C_{AB}(j\sigma; ab) C_{\tilde{A}\tilde{B}}(j\sigma; \tilde{a}\tilde{b}) \begin{cases} [(p^0 + p^3)/m]^{2\tilde{b}-2a}, & \tilde{b} \geq a, \\ [(p^0 - p^3)/m]^{2a-2\tilde{b}}, & a \geq \tilde{b}. \end{cases}$$

where  $p^0 \equiv \sqrt{\mathbf{p}^2 + m^2}$ . We see that  $\pi(\mathbf{p})$  can indeed be written as the mass-shell value of a polynomial  $P(\mathbf{p}, p^0)$ . Also,  $2\tilde{b} - 2a$  equals  $2\tilde{B} + 2A$  minus an even integer, so the polynomial satisfies the reflection condition (5.7.23).

Any polynomial in  $\mathbf{p}$  and  $\sqrt{\mathbf{p}^2 + m^2}$  can be written in a form *linear* in  $\sqrt{\mathbf{p}^2 + m^2}$  (by expressing even powers of  $\sqrt{\mathbf{p}^2 + m^2}$  in terms of  $\mathbf{p}$ ) so  $\pi(\mathbf{p})$  can be written

$$\pi_{ab,\tilde{a}\tilde{b}}(\mathbf{p}) = P_{ab,\tilde{a}\tilde{b}}(\mathbf{p}) + 2\sqrt{\mathbf{p}^2 + m^2} Q_{ab,\tilde{a}\tilde{b}}(\mathbf{p}), \quad (5.7.24)$$

where  $P$  and  $Q$  are now polynomials in  $\mathbf{p}$  alone, with

$$P(-\mathbf{p}) = (-1)^{2A+2\tilde{B}} P(\mathbf{p}), \quad (5.7.25)$$

$$Q(-\mathbf{p}) = -(-1)^{2A+2\tilde{B}} Q(\mathbf{p}). \quad (5.7.26)$$

For  $x - y$  space-like, we can adopt a Lorentz frame in which  $x^0 = y^0$ , and write Eq. (5.7.19) as

$$\begin{aligned} [\psi_{ab}(x), \tilde{\psi}_{\tilde{a}\tilde{b}}^\dagger(y)]_\mp &= [\kappa\tilde{\kappa}^* \mp (-1)^{2A+2\tilde{B}} \lambda\tilde{\lambda}^*] P_{ab,\tilde{a}\tilde{b}}(-i\nabla) \Delta_+(\mathbf{x} - \mathbf{y}, 0) \\ &\quad + [\kappa\tilde{\kappa}^* \pm (-1)^{2A+2\tilde{B}} \lambda\tilde{\lambda}^*] Q_{ab,\tilde{a}\tilde{b}}(-i\nabla) \delta^3(\mathbf{x} - \mathbf{y}). \end{aligned}$$

In order that this should vanish when  $\mathbf{x} \neq \mathbf{y}$ , we must have

$$\kappa\tilde{\kappa}^* = \pm (-1)^{2A+2\tilde{B}} \lambda\tilde{\lambda}^*. \quad (5.7.27)$$

Now let us consider the special case where  $\psi$  and  $\tilde{\psi}$  are the same, so in particular  $A = \tilde{A}$  and  $B = \tilde{B}$ . (It is unavoidable that such commutators or anticommutators will appear in  $[\mathcal{H}_I(x), \mathcal{H}_I(y)]$ , because the hermiticity of the Hamiltonian requires that if  $\mathcal{H}_I(x)$  involves  $\psi$ , it also involves  $\psi^\dagger$ .) In this case, Eqs. (5.7.27) gives

$$|\kappa|^2 = \pm (-1)^{2A+2B} |\lambda|^2.$$

This is possible if and only if

$$\pm (-1)^{2A+2B} = +1 \quad (5.7.28)$$

and

$$|\kappa|^2 = |\lambda|^2. \quad (5.7.29)$$

Of course,  $2A + 2B$  differs from  $2j$  by an *even* integer, so Eq. (5.7.28) says that *our particle is a boson or fermion according to whether  $2j$  is even or odd*. This is the general relation between spin and statistics [12], of which we have already seen special examples for particles described by scalar, vector, or Dirac fields.

Now let's return to the general case, where the fields  $\psi$  and  $\tilde{\psi}$  may be different. Using Eq. (5.7.27), and dividing both sides by  $|\tilde{\kappa}|^2 = |\tilde{\lambda}|^2$ , we have

$$\frac{\kappa}{\tilde{\kappa}} = (-1)^{2B+2\tilde{B}} \frac{\lambda}{\tilde{\lambda}}.$$

It follows that, for any field,

$$\lambda = (-1)^{2B} c \kappa, \quad (5.7.30)$$



where  $c$  is the same factor for all fields of a given particle. Furthermore, Eq. (5.7.29) shows that  $c$  is just a phase,  $|c| = 1$ . We can therefore eliminate  $c$  for all fields by a redefinition of the relative phase of the operators  $a_{\mathbf{p},\sigma}$  and  $a_{\mathbf{p},\sigma}^\dagger$ , so that  $c = 1$ , and hence  $\lambda = (-1)^{2B}\kappa$ . Also, the factor  $\kappa$  for each field type may be eliminated by a redefinition of the over-all scale of the field. We emerge from all this with a formula for the  $(A, B)$  field of a given particle, that is unique up to overall scale

$$\begin{aligned} \psi_{ab}(x) = (2\pi)^{-3/2} \sum_{\sigma} \int d^3p \left[ u_{ab}(\mathbf{p}, \sigma) a_{\mathbf{p},\sigma} e^{ip \cdot x} \right. \\ \left. + (-1)^{2B} u_{ab}(\mathbf{p}, \sigma) a_{\mathbf{p},-\sigma}^\dagger e^{-ip \cdot x} \right]. \end{aligned} \quad (5.7.31)$$

The different fields for a given particle do not really represent possibilities that are physically distinct. For instance, the possible fields for  $j = 0$  are those of type  $(A, A)$  (because the triangle inequality  $|A - B| \leq j \leq A + B$  here requires  $A = B$ ). Starting with a  $(0, 0)$  scalar field  $\phi$ , we can easily construct such  $(A, A)$  fields from the  $2A$ th derivative

$$\{\partial_{\mu_1} \cdots \partial_{\mu_{2A}}\} \phi, \quad (5.7.32)$$

where  $\{\}$  here denotes the traceless part; for instance

$$\{\partial_{\mu} \partial_{\nu}\} \equiv \frac{\partial^2}{\partial x^{\mu} \partial x^{\nu}} - \frac{1}{4} \eta_{\mu\nu} \square.$$

(Recall that a traceless symmetric tensor of rank  $N$  transforms according to the  $(N/2, N/2)$  representation.) But Eq. (5.7.31) represents the *unique* causal  $(A, B)$  field for a given particle of spin  $j$ , so the  $(A, A)$  fields (5.7.31) for  $j = 0$  can be nothing but linear combinations of the  $2A$ th derivatives (5.7.32) of a scalar field.

More generally, *any* field  $(A, B)$  for a given particle of spin  $j$  can be expressed as a differential operator of rank  $2B$  acting on the field<sup>49</sup>  $\phi_{\sigma}(x)$  of type  $(j, 0)$  (or a differential operator of rank  $2A$  acting on the field of type  $(0, j)$ ). To see this, consider the field

$$\{\partial_{\mu_1} \cdots \partial_{\mu_{2B}}\} \phi_{\sigma}. \quad (5.7.33)$$

This transforms as the direct product of the representations  $(B, B)$  and  $(j, 0)$ , and hence by the usual rules of vector addition, it can be decomposed into fields transforming according to all the irreducible representations  $(A, B)$  with  $|j - B| \leq A \leq j + B$ , or equivalently  $|A - B| \leq j \leq A + B$ . Since Eq. (5.7.31) represents the unique field of type  $(A, B)$  for a given particle of spin  $j$ , it can be nothing but the  $(A, B)$  field obtained from the derivatives (5.7.33).

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<sup>49</sup>The only possible flaw in this argument would be if some of the  $(A, B)$  fields obtained in this way actually vanished. But in this case, the  $(j, 0)$  field  $\phi_{\sigma}$  would satisfy a field equation  $\sum_{\bar{\sigma}} M_{\bar{\sigma}}(i\partial/\partial x) \phi_{\bar{\sigma}}(x) = 0$  and hence, for each  $\sigma$ ,  $\sum_{\bar{\sigma}} M_{\bar{\sigma}}(ip) u_{\bar{\sigma}}(\mathbf{p}, \sigma) = 0$ . For the  $(j, 0)$  representation the Clebsch–Gordan coefficient  $C_{j0}(j\sigma; \bar{\sigma}0)$  is just the Kronecker symbol  $\delta_{\bar{\sigma}\sigma}$ , so this would require  $\sum_{\bar{\sigma}} M_{\bar{\sigma}}(ip) D_{\bar{\sigma}\sigma}(L(p)) = 0$ , which is impossible unless all the  $M_{\sigma}(ip)$  vanish, since  $D(\Lambda)$  has an inverse  $D(\Lambda^{-1})$ . The  $(j, 0)$  fields  $\phi_{\sigma}(x)$  thus satisfy no field equation other than the Klein–Gordon equation  $(\square - m^2)\phi_{\sigma}(x) = 0$ , and therefore none of the  $(A, B)$  fields obtained from (5.7.33) can vanish.

Now let us consider the behavior of these fields under inversions, beginning with space inversion. Using the results of Section 4.2, the space-inversion properties of the particle annihilation and antiparticle creation operators are

$$Pa_{\mathbf{p},\sigma}P^{-1} = \eta^* a_{-\mathbf{p},\sigma}, \quad (5.7.34)$$

$$Pa_{\mathbf{p},\sigma}^{c\dagger}P^{-1} = \eta^c a_{-\mathbf{p},\sigma}^{c\dagger}, \quad (5.7.35)$$

where  $\eta$  and  $\eta^c$  are the intrinsic parities for the particle and antiparticle, respectively. The general causal  $(A, B)$  field (5.7.31) thus transforms under the parity operator  $P$  into

$$\begin{aligned} P\psi_{ab}^{AB}(x)P^{-1} &= (2\pi)^{-3/2} \sum_{\sigma} \int d^3p \left[ \eta^* a_{-\mathbf{p},\sigma} e^{ip \cdot x} u_{ab}^{AB}(\mathbf{p}, \sigma) \right. \\ &\quad \left. + \eta^c (-1)^{2B} a_{-\mathbf{p},\sigma}^{c\dagger} e^{-ip \cdot x} v_{ab}^{AB}(\mathbf{p}, \sigma) \right]. \end{aligned} \quad (5.7.36)$$

We want to change the integration variable from  $\mathbf{p}$  to  $-\mathbf{p}$ , and for this purpose we will need to evaluate  $u_{ab}(-\mathbf{p}, \sigma)$  and  $v_{ab}(-\mathbf{p}, \sigma)$ . To do this, we need only glance back at (5.7.14) and (5.7.15), and use the symmetry property of the Clebsch–Gordan coefficient [14]

$$C_{AB}(j\sigma; ab) = (-1)^{A+B-j} C_{BA}(j\sigma; ba). \quad (5.7.37)$$

This gives

$$u_{ab}^{AB}(-\mathbf{p}, \sigma) = (-1)^{A+B-j} u_{ba}^{BA}(\mathbf{p}, \sigma), \quad (5.7.38)$$

$$v_{ab}^{AB}(-\mathbf{p}, \sigma) = (-1)^{A+B-j} v_{ba}^{BA}(\mathbf{p}, \sigma), \quad (5.7.39)$$

so

$$\begin{aligned} P\psi_{ab}^{AB}(x)P^{-1} &= (2\pi)^{-3/2} \sum_{\sigma} \int d^3p (-1)^{A+B-j} \\ &\quad \cdot \left[ \eta^* a_{\mathbf{p},\sigma} e^{ip \cdot \mathcal{P}x} u_{ba}^{BA}(\mathbf{p}, \sigma) \right. \\ &\quad \left. + \eta^c (-1)^{2B} a_{\mathbf{p},\sigma}^{c\dagger} e^{-ip \cdot \mathcal{P}x} v_{ba}^{BA}(\mathbf{p}, \sigma) \right], \end{aligned} \quad (5.7.40)$$

where, as before,  $\mathcal{P}x \equiv (-\mathbf{x}, x^0)$ . This is the causal field  $\psi_{ba}^{BA}$  evaluated at  $\mathcal{P}x$ , except that the coefficients of the annihilation and creation terms may not be the same as called for in Eq. (5.7.31). But these coefficients *must* be the same up to an overall constant factor as in Eq. (5.7.31) because, aside from scale, Eq. (5.7.31) is the unique causal field of any type. Hence the ratio of the coefficients of the two terms in Eq. (5.7.40) must be the same as in Eq. (5.7.31) (but with  $B$  replaced with  $A$  because this is supposed to be a  $(B, A)$  field):

$$\eta^c (-1)^{2B} / \eta^* = (-1)^{2A}. \quad (5.7.41)$$

But  $A - B$  differs from the spin  $j$  by only an integer, so this gives

$$\eta^c = \eta^* (-1)^{2j}. \quad (5.7.42)$$

We saw special cases of this result in Sections 5.2, 5.3, and 5.5, where  $j = 0$ ,  $j = 1$ , and  $j = \frac{1}{2}$ , respectively. We now see that the result is general; *the intrinsic parity  $\eta^c \eta$  of a*

particle–antiparticle pair is +1 for bosons, and −1 for fermions. Using Eq. (5.7.42) in Eq. (5.7.40), our final result for space inversion is

$$P\psi_{ab}^{AB}(x)P^{-1} = \eta^*(-1)^{A+B-j}\psi_{ba}^{BA}(-\mathbf{x}, x^0). \quad (5.7.43)$$

Let's see how this applies to the Dirac field. For the top  $(\frac{1}{2}, 0)$  and bottom  $(0, \frac{1}{2})$  components of the Dirac field, the sign  $(-1)^{A+B-j}$  is just +1, so the parity operator simply takes  $\mathbf{x}$  into  $-\mathbf{x}$ ; reverses the top and bottom components; and multiplies the field with  $\eta^*$ . The reversal of the top and bottom components of the Dirac field is accomplished by the matrix  $\beta$  in (5.5.41).

Now let us consider charge-conjugation. Its effect on the particle annihilation and antiparticle creation operators is

$$Ca_{\mathbf{p},\sigma}C^{-1} = \xi^*a_{\mathbf{p},\sigma}^c, \quad (5.7.44)$$

$$Ca_{\mathbf{p},\sigma}^{c\dagger}C^{-1} = \xi^c a_{\mathbf{p},\sigma}^{\dagger}, \quad (5.7.45)$$

where  $\xi$  and  $\xi^c$  are the charge-conjugation parities of the particle and antiparticle, respectively. Applying this transformation to the field (5.7.31), we find

$$\begin{aligned} C\psi_{ab}^{AB}(x)C^{-1} &= (2\pi)^{-3/2} \sum_{\sigma} \int d^3p u_{ab}^{AB}(\mathbf{p}, \sigma) \\ &\cdot [\xi^* a_{\mathbf{p},\sigma}^c e^{ip \cdot x} + \xi^c (-1)^{2B} a_{\mathbf{p},-\sigma}^{\dagger} (-1)^{j-\sigma} e^{-ip \cdot x}]. \end{aligned} \quad (5.7.46)$$

It is useful to compare this formula for the charge-conjugate of an  $(A, B)$  field with the adjoint of the  $(B, A)$  field for the same particle:

$$\begin{aligned} \psi_{ba}^{BA\dagger}(x) &= (2\pi)^{-3/2} \sum_{\sigma} \int d^3p u_{ba}^{BA*}(\mathbf{p}, \sigma) \\ &\cdot [(-1)^{2A} (-1)^{j-\sigma} a_{\mathbf{p},-\sigma}^c e^{ip \cdot x} + a_{\mathbf{p},\sigma}^{\dagger} e^{-ip \cdot x}]. \end{aligned} \quad (5.7.47)$$

To calculate the  $u^*$ , we use our previous result

$$\mathbf{J}^{(j)*} = -\mathcal{C}_j \mathbf{J}^{(j)} \mathcal{C}_j^{-1}, \quad (\mathcal{C}_j)_{\bar{\sigma}\sigma} \propto (-1)^{j-\sigma} \delta_{\bar{\sigma}, -\sigma}.$$

The Clebsch–Gordan coefficient in Eq. (5.7.14) is real, so

$$\begin{aligned} u_{ba}^{BA}(\mathbf{p}, \sigma)^* &= \frac{1}{\sqrt{2p^0}} \sum_{a'b'} (\exp(-\hat{\mathbf{p}} \cdot \mathbf{J}^{(A)} \theta))_{-a, -a'} (\exp(+\hat{\mathbf{p}} \cdot \mathbf{J}^{(B)} \theta))_{-b, -b'} \\ &\cdot (-1)^{a'-a} (-1)^{b'-b} C_{BA}(j\sigma; b'a'). \end{aligned}$$

We use the reflection property of the Clebsch–Gordan coefficients [14]

$$C_{BA}(j, -\sigma; -b', -a') = C_{AB}(j\sigma; a'b') \quad (5.7.48)$$

and the fact that those coefficients vanish unless  $a' + b' = \sigma$ , to write

$$u_{-b, -a}^{BA}(\mathbf{p}, -\sigma)^* = (-1)^{a+b-\sigma} u_{ab}^{AB}(\mathbf{p}, \sigma). \quad (5.7.49)$$

The field adjoint (5.7.47) is then (replacing  $a \rightarrow -a$ ,  $b \rightarrow -b$ ,  $\sigma \rightarrow -\sigma$ )

$$\begin{aligned} \psi_{-b,-a}^{BA\dagger}(x) &= (2\pi)^{-3/2} \sum_{\sigma} \int d^3p (-1)^{a+b-\sigma} u_{ab}^{AB}(\mathbf{p}, \sigma) \\ &\quad \cdot [(-1)^{2A}(-1)^{j+\sigma} a_{\mathbf{p},\sigma}^c e^{ip \cdot x} + a_{\mathbf{p},-\sigma}^{\dagger} e^{-ip \cdot x}]. \end{aligned}$$

Using the sign relation  $(-1)^{-2A-j} = (-1)^{2B+j}$ , this is

$$\begin{aligned} (-1)^{-2A-a-b-j} \psi_{-b,-a}^{BA\dagger}(x) &= (2\pi)^{-3/2} \sum_{\sigma} \int d^3p u_{ab}^{AB}(\mathbf{p}, \sigma) \\ &\quad \cdot [a_{\mathbf{p},\sigma}^c e^{ip \cdot x} + (-1)^{j-\sigma+2B} a_{\mathbf{p},-\sigma}^{\dagger} e^{-ip \cdot x}]. \end{aligned} \quad (5.7.50)$$

In order that  $C\psi_{ab}^{AB}(x)C^{-1}$  should commute or anticommute with all ordinary fields at space-like separations, it is necessary that it be proportional to  $\psi_{-b,-a}^{BA\dagger}(x)$ , because this is the adjoint of the unique causal field of transformation type  $(B, A)$ . Comparing Eq. (5.7.50) with Eq. (5.7.46), we see that this is only possible if the charge-conjugation parities are related by

$$\xi^* = \xi^c, \quad (5.7.51)$$

in which case

$$C\psi_{ab}^{AB}(x)C^{-1} = \xi^*(-1)^{-2A-a-b-j} \psi_{-b,-a}^{BA\dagger}(x). \quad (5.7.52)$$

We have already encountered the relation (5.7.51) for spins 0, 1, and  $\frac{1}{2}$  in Sections 5.2, 5.3, and 5.5, and noted some of its implications for electron-positron and quark-antiquark states in Section 5.5.

In particular, for a particle that is its own antiparticle, Eq. (5.7.52) is satisfied without any charge-conjugation operator on the left-hand side or phase  $\xi^*$  on the right:

$$\psi_{ab}^{AB}(x) = (-1)^{-2A-a-b-j} \psi_{-b,-a}^{BA\dagger}(x). \quad (5.7.53)$$

We have already seen an example of this sort of reality condition for Majorana spin  $\frac{1}{2}$  particles in Section 5.5.

Finally we come to time-reversal. Applied to particle annihilation and antiparticle creation operators, this gives

$$Ta_{\mathbf{p},\sigma}T^{-1} = \zeta^*(-1)^{j-\sigma} a_{-\mathbf{p},-\sigma}, \quad (5.7.54)$$

$$Ta_{\mathbf{p},\sigma}^{\dagger}T^{-1} = \zeta^c(-1)^{j-\sigma} a_{-\mathbf{p},-\sigma}^{\dagger}. \quad (5.7.55)$$

The irreducible field (5.7.31) thus has the transformation property

$$\begin{aligned} T\psi_{ab}^{AB}(x)T^{-1} &= (2\pi)^{-3/2} \sum_{\sigma} \int d^3p u_{ab}^{AB*}(\mathbf{p}, \sigma) (-1)^{j-\sigma} \\ &\quad \cdot [\zeta^* a_{-\mathbf{p},-\sigma} e^{-ip \cdot x} + \zeta^c (-1)^{2B} a_{-\mathbf{p},-\sigma}^{\dagger} e^{ip \cdot x}]. \end{aligned} \quad (5.7.56)$$

To calculate the complex conjugate of the coefficient function, we use Eq. (5.7.14) and the standard formula [14]

$$C_{AB}(j\sigma; a, b) = (-1)^{A+B-j} C_{AB}(j, -\sigma; -a, -b) \quad (5.7.57)$$

and find:

$$u_{ab}^{AB*}(-\mathbf{p}, -\sigma) = (-1)^{a+b+\sigma+A+B-j} u_{-a,-b}^{AB}(\mathbf{p}, \sigma). \quad (5.7.58)$$

Changing the variables of integration and summation in Eq. (5.7.56) to  $-\mathbf{p}$  and  $-\sigma$ , we find that in order for an  $(A, B)$  field to be transformed by time-reversal into something proportional to another  $(A, B)$  field, it is necessary that

$$\zeta^c = \zeta^*, \quad (5.7.59)$$

in which case

$$T\psi_{ab}^{AB}(x)T^{-1} = (-1)^{a+b+A+B-2j}\zeta^*\psi_{-a,-b}^{AB}(\mathbf{x}, -x^0). \quad (5.7.60)$$

\* \* \*

It should be mentioned that from time to time various difficulties have been reported [15] in the field theory of particles with spin  $j \geq 3/2$ . Generally, these are encountered in the study of the propagation of a higher spin field in the presence of a c-number external field. Depending on the details of the theory, the difficulties encountered include non-causality, inconsistency, unphysical mass states, and violation of unitarity. I will not go into details about these problems here, because it seems to me that they are not relevant to the calculational scheme described in this chapter, for the following reasons:

(1) The fields  $\psi_{ab}(x)$  have been constructed here directly from the creation and annihilation operators for physical particles, so no question of inconsistency or unphysical mass states can arise. These are free fields, but by incorporating them into an interaction Hamiltonian density in the interaction picture, we can use perturbation theory to calculate  $S$ -matrix elements that automatically satisfy the cluster decomposition principle. As long as the interaction Hamiltonian is Hermitian, there can be no difficulty with unitarity. Lorentz invariance is guaranteed in perturbation theory as long as we add appropriate local but non-covariant terms in the Hamiltonian density; though a rigorous proof is lacking, there is no reason to doubt that this is always possible. Thus any difficulties with higher spin can only arise when we try to go beyond perturbation theory.

(2) As discussed in Section 13.6, the solution of field equations in the presence of a c-number background field (the context where all the problems with higher spin have been found) does go beyond perturbation theory, in that the results correspond to summing an infinite subset of terms in the perturbation series. This partial summation is justified, even for weak external fields, if the fields are sufficiently slowly varying, the smallness of energy denominators making up for the weakness of the fields. But the results obtained in this way depend on all the details of the interaction of the high-spin particle with the external fields: not only the multipole moments of the particle but also possible terms in the interaction that are non-linear in the external fields. The problems reported [15] with higher spin have been encountered only for higher-spin particles that have been arbitrarily assumed to have only very simple interactions with external fields. No one has shown that the problems persist for arbitrary interactions, and as we shall see in Chapter 12, particles of higher spin are expected to have interactions of all possible types allowed by symmetry principles.

(3) In fact, there are good reasons to believe that the problems with higher spin disappear if the interaction with external fields is sufficiently complicated. For one thing, there is no doubt about the existence of higher-spin *particles*, including various stable nuclei and hadronic resonances. If there is any problem with higher spin, it can only be for ‘point’ particles, that is, those whose interactions with external fields are particularly simple. It should be kept in mind that the requirement of simplicity depends on the choice of which field we choose to represent the higher-spin particle. Remember that any free field types for a given particle can be expressed as a derivative operator acting on any other field type, so in the interaction picture any interaction with external fields may be written in terms of any field types we like, but interactions that are simple when expressed in terms of a field of one type may look complicated when expressed in terms of a field of another type. So the requirement of simplicity does not seem to have any objective content.

(4) Also, both higher-dimensional ‘Kaluza–Klein’ theories and string theories provide examples of consistent theories of charged massive particles of spin two interacting with an electromagnetic background field [16]. (It was found that the consistency of the theory depends on the assumption of realistic external fields that satisfy the field equations, a point generally neglected in earlier work.) Reformulating this work in the interaction picture, the spin two particle is represented by a  $(1, 1)$  free field, but as mentioned above, the interactions may be reexpressed in the interaction picture in terms of any field type  $(A, B)$  that contains the  $j = 2$  representation of the rotation group.

## 5.8 The CPT Theorem

We have seen that the demands of relativity combined with quantum mechanics require the existence of antiparticles. Not only is it necessary that every particle have an antiparticle (which may for a purely neutral particle be itself); there is a precise relation between the properties of particles and antiparticles, that can be summarized in the statement that *for an appropriate choice of inversion phases, the product CPT of all the inversions is conserved*. This is the celebrated CPT theorem.<sup>50</sup>

As a first step in the proof, let us work out the effect of the product CPT on free fields of various types. For a scalar, vector, or Dirac field the results of Sections 5.2, 5.3, and 5.5 give

$$CPT \phi(x)[CPT]^{-1} = \zeta^* \xi^* \eta^* \phi^\dagger(-x), \quad (5.8.1)$$

$$CPT \phi_\mu(x)[CPT]^{-1} = -\zeta^* \xi^* \eta^* \phi_\mu^\dagger(-x), \quad (5.8.2)$$

$$CPT \psi(x)[CPT]^{-1} = -\zeta^* \xi^* \eta^* \gamma_5 \psi^*(-x). \quad (5.8.3)$$

(Of course, the phases  $\zeta$ ,  $\xi$ , and  $\eta$  depend on the species of particle described by each field.) We are going to choose the phases so that for all particles

$$\zeta \xi \eta = 1. \quad (5.8.4)$$

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<sup>50</sup>The original proofs of this theorem were by Lüders and Pauli [17]. It has been proved rigorously in axiomatic field theory [18], by using commutativity assumptions to extend the Lorentz invariance of the theory to the complex Lorentz group, then using complex Lorentz transformations to prove a reflection property of vacuum expectation values of products of fields, and then using this reflection property to infer the existence of an antiunitary operator that induces CPT transformations on the fields.

Then any tensor  $\phi_{\mu_1 \dots \mu_n}$  formed from any set of scalar and vector fields and their derivatives transforms into

$$CPT \phi_{\mu_1 \dots \mu_n}(x) [CPT]^{-1} = (-1)^n \phi_{\mu_1 \dots \mu_n}^\dagger(-x). \quad (5.8.5)$$

(Any complex numerical coefficient appearing in these tensors is transformed into its complex conjugate because CPT is antiunitary.) We can easily see that the same transformation rule applies to tensors formed from bilinear combinations of Dirac fields. Applying Eq. (5.8.3) to such a bilinear gives

$$\begin{aligned} CPT[\bar{\psi}_1(x) M \psi_2(x)] [CPT]^{-1} &= \psi_1^\dagger(-x) \gamma_5 \beta M^* \gamma_5 \psi_2^*(-x) \\ &= [\bar{\psi}_1(-x) \gamma_5 M \gamma_5 \psi_2(-x)]^\dagger. \end{aligned} \quad (5.8.6)$$

(A minus sign from the anticommutation of  $\beta$  and  $\gamma_5$  is cancelled by the minus sign from the anticommutation of fermionic operators.) If the bilinear is a tensor of rank  $n$ , then  $M$  is a product of  $n$  modulo 2 Dirac matrices, so  $\gamma_5 M \gamma_5 = (-1)^n M$ , and the bilinear therefore satisfies Eq. (5.8.5).

A Hermitian scalar interaction density  $\mathcal{H}_I(x)$  must be formed from tensors with an *even* total number of spacetime indices, and therefore

$$CPT \mathcal{H}_I(x) [CPT]^{-1} = \mathcal{H}_I(-x). \quad (5.8.7)$$

More generally (and somewhat more easily) we can see that the same is true for Hermitian scalars formed from the fields  $\psi_{ab}^{AB}(x)$  belonging to one or more of the general irreducible representations of the homogeneous Lorentz group. Putting together our results in the previous section for the effects of inversions on such fields, we find

$$CPT \psi_{ab}^{AB}(x) [CPT]^{-1} = (-1)^{2B} \psi_{ba}^{AB\dagger}(-x). \quad (5.8.8)$$

(For the Dirac field the factor  $(-1)^{2B}$  is supplied by the matrix  $\gamma_5$  in Eq. (5.8.3).) In order to couple together a product  $\psi_{a_1 b_1}^{A_1 B_1}(x) \psi_{a_2 b_2}^{A_2 B_2}(x) \dots$  to form a scalar  $\mathcal{H}_I(x)$ , it is necessary that both  $A_1 + A_2 + \dots$  and  $B_1 + B_2 + \dots$  be integers, so  $(-1)^{2B_1 + 2B_2 + \dots} = 1$ , and so a Hermitian scalar  $\mathcal{H}_I(x)$  will automatically satisfy Eq. (5.8.7).

From Eq. (5.8.7) it follows immediately that CPT commutes with the interaction  $H_{\text{int}} \equiv \int d^3x \mathcal{H}_I(\mathbf{x}, 0)$ :

$$CPT H_{\text{int}} [CPT]^{-1} = H_{\text{int}}. \quad (5.8.9)$$

Also, in any theory CPT commutes with the free-particle Hamiltonian  $H_0$ . Thus the operator CPT, which has been defined here by its operation on free-particle operators, acts on ‘in’ and ‘out’ states in the way described in Section 3.3. The physical consequences of this symmetry principle have already been discussed in Sections 3.3 and 3.6.

## 5.9 Massless Particle Fields

Up to this point we have dealt only with the fields of massive particles. For some of these fields, such as the scalar and Dirac fields discussed in Sections 5.2 and 5.5, there is no special problem in passing to the limit of zero mass. On the other hand, we saw in Section

5.3 that there is a difficulty in taking the zero-mass limit of the vector field for a particle of spin one: at least one of the polarization vectors blows up in this limit. In fact, we shall see in this section that the creation and annihilation operators for physical massless particles of spin  $j \geq 1$  cannot be used to construct all of the irreducible  $(A, B)$  fields that can be constructed for finite mass. This peculiar limitation on field types will lead us naturally to the introduction of gauge invariance.

Just as we did for massive particles, let us attempt to construct a general free field for a massless particle as a linear combination of the annihilation operators  $a_{\mathbf{p},\sigma}$  for particles of momentum  $\mathbf{p}$  and helicity  $\sigma$ , and the corresponding creation operators  $a_{\mathbf{p},\sigma}^{c\dagger}$  for the antiparticles:<sup>51</sup>

$$\psi_\ell(x) = (2\pi)^{-3/2} \int d^3p \sum_\sigma [\kappa a_{\mathbf{p},\sigma} u_\ell(\mathbf{p}, \sigma) e^{ip \cdot x} + \lambda a_{\mathbf{p},\sigma}^{c\dagger} v_\ell(\mathbf{p}, \sigma) e^{-ip \cdot x}] \quad (5.9.1)$$

where now  $p^0 \equiv |\mathbf{p}|$ . The creation operators transform just like the one-particle states in Eq. (2.5.42)

$$U(\Lambda) a_{\mathbf{p},\sigma}^\dagger U^{-1}(\Lambda) = \sqrt{\frac{(\Lambda p)^0}{p^0}} \exp(i\sigma\theta(p, \Lambda)) a_{\mathbf{p}_\Lambda, \sigma}^\dagger, \quad (5.9.2)$$

$$U(\Lambda) a_{\mathbf{p},\sigma}^{c\dagger} U^{-1}(\Lambda) = \sqrt{\frac{(\Lambda p)^0}{p^0}} \exp(i\sigma\theta(p, \Lambda)) a_{\mathbf{p}_\Lambda, \sigma}^{c\dagger}, \quad (5.9.3)$$

and hence also

$$U(\Lambda) a_{\mathbf{p},\sigma} U^{-1}(\Lambda) = \sqrt{\frac{(\Lambda p)^0}{p^0}} \exp(-i\sigma\theta(p, \Lambda)) a_{\mathbf{p}_\Lambda, \sigma}, \quad (5.9.4)$$

where  $p_\Lambda \equiv \Lambda p$ , and  $\theta$  is the angle defined by Eqs. (2.5.43). Hence if we want the field to transform according to some representation  $D(\Lambda)$  of the homogeneous Lorentz group

$$U(\Lambda) \psi_\ell(x) U^{-1}(\Lambda) = \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(\Lambda^{-1}) \psi_{\bar{\ell}}(\Lambda x), \quad (5.9.5)$$

then we must take the coefficient functions  $u$  and  $v$  to satisfy the relations

$$u_{\bar{\ell}}(\mathbf{p}_\Lambda, \sigma) \exp(i\sigma\theta(p, \Lambda)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\ell} D_{\bar{\ell}\ell}(\Lambda) u_\ell(\mathbf{p}, \sigma), \quad (5.9.6)$$

$$v_{\bar{\ell}}(\mathbf{p}_\Lambda, \sigma) \exp(-i\sigma\theta(p, \Lambda)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\ell} D_{\bar{\ell}\ell}(\Lambda) v_\ell(\mathbf{p}, \sigma) \quad (5.9.7)$$

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<sup>51</sup>We deal here with only a single species of particle, and drop the species label  $n$ . Also,  $\kappa$  and  $\lambda$  are constant coefficients to be determined by the requirement of causality with some convenient choice of normalization of the coefficient functions  $u_\ell$  and  $v_\ell$ .



in place of Eqs. (5.1.19) and (5.1.20). (Again,  $p_\Lambda \equiv \Lambda p$ .) As in the massive particle case, we can satisfy these requirements by setting (in place of Eqs. (5.1.21) and (5.1.22))

$$u_{\bar{\ell}}(\mathbf{p}, \sigma) = \sqrt{\frac{|\mathbf{k}|}{p^0}} \sum_{\ell} D_{\bar{\ell}\ell}(\mathcal{L}(p)) u_{\ell}(\mathbf{k}, \sigma), \quad (5.9.8)$$

$$v_{\bar{\ell}}(\mathbf{p}, \sigma) = \sqrt{\frac{|\mathbf{k}|}{p^0}} \sum_{\ell} D_{\bar{\ell}\ell}(\mathcal{L}(p)) v_{\ell}(\mathbf{k}, \sigma), \quad (5.9.9)$$

where  $\mathbf{k}$  is a standard momentum, say  $(0, 0, k)$ , and  $\mathcal{L}(p)$  is a standard Lorentz transformation that takes a massless particle from momentum  $\mathbf{k}$  to momentum  $\mathbf{p}$ . Also, in place of Eqs. (5.1.23) and (5.1.24), the coefficient functions at the standard momentum must satisfy

$$u_{\bar{\ell}}(\mathbf{k}, \sigma) \exp(i\sigma\theta(k, W)) = \sum_{\ell} D_{\bar{\ell}\ell}(W) u_{\ell}(\mathbf{k}, \sigma), \quad (5.9.10)$$

$$v_{\bar{\ell}}(\mathbf{k}, \sigma) \exp(-i\sigma\theta(k, W)) = \sum_{\ell} D_{\bar{\ell}\ell}(W) v_{\ell}(\mathbf{k}, \sigma) \quad (5.9.11)$$

where  $W^\mu_\nu$  is an arbitrary element of the ‘little group’ for four-momentum  $k = (\mathbf{k}, |\mathbf{k}|)$ , i.e., an arbitrary Lorentz transformation that leaves this four-momentum invariant.

We can extract the content of Eqs. (5.9.10) and (5.9.11) by considering separately the two kinds of little-group elements in Eq. (2.5.28). For a rotation  $R(\theta)$  by an angle  $\theta$  around the  $z$ -axis, given by Eq. (2.5.27),

$$R^\mu_\nu(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

we find from Eqs. (5.9.10) and (5.9.11)

$$u_{\bar{\ell}}(\mathbf{k}, \sigma) e^{i\sigma\theta} = \sum_{\ell} D_{\bar{\ell}\ell}(R(\theta)) u_{\ell}(\mathbf{k}, \sigma), \quad (5.9.12)$$

$$v_{\bar{\ell}}(\mathbf{k}, \sigma) e^{-i\sigma\theta} = \sum_{\ell} D_{\bar{\ell}\ell}(R(\theta)) v_{\ell}(\mathbf{k}, \sigma). \quad (5.9.13)$$

For combined rotations and boosts  $S(\alpha, \beta)$  in the  $x - y$  plane, given by (2.5.26),

$$S^\mu_\nu(\alpha, \beta) = \begin{bmatrix} 1 & 0 & -\alpha & \alpha \\ 0 & 1 & -\beta & \beta \\ \alpha & \beta & 1 - \gamma & \gamma \\ \alpha & \beta & -\gamma & 1 + \gamma \end{bmatrix}, \quad \gamma \equiv (\alpha^2 + \beta^2)/2,$$

Eqs. (5.9.10) and (5.9.11) give

$$u_{\bar{\ell}}(\mathbf{k}, \sigma) = \sum_{\ell} D_{\bar{\ell}\ell}(S(\alpha, \beta)) u_{\ell}(\mathbf{k}, \sigma), \quad (5.9.14)$$

$$v_{\bar{\ell}}(\mathbf{k}, \sigma) = \sum_{\ell} D_{\bar{\ell}\ell}(S(\alpha, \beta)) v_{\ell}(\mathbf{k}, \sigma). \quad (5.9.15)$$

Eqs. (5.9.12)–(5.9.15) are the conditions that determine the coefficient functions  $u$  and  $v$  at the standard momentum  $\mathbf{k}$ ; Eqs. (5.9.8) and (5.9.9) then give them at arbitrary momenta. The equations for  $v$  are just the complex conjugates of the equations for  $u$ , so with a suitable adjustment of the constants  $\kappa$  and  $\lambda$  we may normalize the coefficient functions so that

$$v_\ell(\mathbf{p}, \sigma) = u_\ell(\mathbf{p}, \sigma)^*. \quad (5.9.16)$$

The problem is that we cannot find a  $u_\ell$  that satisfies Eq. (5.9.14) for general representations of the homogeneous Lorentz group, even for those representations for which it is possible to construct fields for particles of a given helicity in the case  $m \neq 0$ .

To see what goes wrong here, let's try to construct the four-vector  $[(\frac{1}{2}, \frac{1}{2})]$  field for a massless particle of helicity  $\pm 1$ . In the four-vector representation, we have simply

$$D^\mu{}_\nu(\Lambda) = \Lambda^\mu{}_\nu.$$

It is conventional to write the coefficient function  $u_\mu$  here in terms of a ‘polarization vector’  $e_\mu$ :

$$u_\mu(\mathbf{p}, \sigma) \equiv (2p^0)^{-1/2} e_\mu(\mathbf{p}, \sigma), \quad (5.9.17)$$

so that Eq. (5.9.8) gives

$$e^\mu(\mathbf{p}, \sigma) = \mathcal{L}(p)^\mu{}_\nu e^\nu(\mathbf{k}, \sigma). \quad (5.9.18)$$

Also, Eqs. (5.9.12) and (5.9.14) read here

$$e^\mu(\mathbf{k}, \sigma) e^{i\sigma\theta} = R(\theta)^\mu{}_\nu e^\nu(\mathbf{k}, \sigma), \quad (5.9.19)$$

$$e^\mu(\mathbf{k}, \sigma) = S(\alpha, \beta)^\mu{}_\nu e^\nu(\mathbf{k}, \sigma). \quad (5.9.20)$$

Eq. (5.9.19) requires that (up to a constant which can be absorbed into the coefficients  $\kappa$  and  $\lambda$ ),

$$e^\mu(\mathbf{k}, \pm 1) = (1, \pm i, 0, 0)/\sqrt{2}. \quad (5.9.21)$$

But then Eq. (5.9.20) would require also that  $\alpha + i\beta = 0$ , which is impossible for general real  $\alpha, \beta$ . We therefore cannot satisfy the fundamental requirement (5.9.14) or (5.9.10); instead, we have here

$$\begin{aligned} D^\mu{}_\nu(W(\theta, \alpha, \beta)) e^\nu(\mathbf{k}, \pm 1) &= S^\mu{}_\lambda(\alpha, \beta) R^\lambda{}_\nu(\theta) e^\nu(\mathbf{k}, \pm 1) \\ &= \exp(\pm i\theta) \left\{ e^\mu(\mathbf{k}, \pm 1) + \frac{\alpha \pm i\beta}{\sqrt{2}|\mathbf{k}|} k^\mu \right\}. \end{aligned} \quad (5.9.22)$$

We have thus come to the conclusion that no four-vector field can be constructed from the annihilation and creation operators for a particle of mass zero and helicity  $\pm 1$ .

Let's temporarily close our eyes to this difficulty, and go ahead anyway, using Eqs. (5.9.18) and (5.9.21) to define a polarization vector for arbitrary momentum, and take the field as

$$a_\mu(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2p^0}} \cdot \sum_{\sigma=\pm 1} [e_\mu(\mathbf{p}, \sigma) e^{ip \cdot x} a_{\mathbf{p}, \sigma} + e_\mu(\mathbf{p}, \sigma)^* e^{-ip \cdot x} a_{\mathbf{p}, -\sigma}^\dagger]. \quad (5.9.23)$$

We will come back later to consider how such a field can be used as an ingredient in a physical theory.

The field (5.9.23) of course satisfies

$$\square a^\mu(x) = 0. \quad (5.9.24)$$

Other properties of the field follow from those of the polarization vector. (We shall need these properties of the polarization vector later when we come to quantum electrodynamics.) Note that the Lorentz transformation  $\mathcal{L}(p)$  that takes a massless particle momentum from  $\mathbf{k}$  to  $\mathbf{p}$  may be written as a ‘boost’  $\mathcal{B}(|\mathbf{p}|)$  along the  $z$ -axis which takes the particle from energy  $|\mathbf{k}|$  to energy  $|\mathbf{p}|$ , followed by a standardized rotation  $R(\hat{\mathbf{p}})$  that takes the  $z$ -axis into the direction of  $\mathbf{p}$ . Since  $e^\nu(\mathbf{k}, \pm 1)$  is a purely spatial vector with only  $x$  and  $y$  components, it is unaffected by the boost along the  $z$ -axis, and so

$$e^\mu(\mathbf{p}, \pm 1) = R(\hat{\mathbf{p}})^\mu{}_\nu e^\nu(\mathbf{k}, \pm 1). \quad (5.9.25)$$

In particular,  $e^0(\mathbf{k}, \pm 1) = 0$  and  $\mathbf{k} \cdot \mathbf{e}(\mathbf{k}, \pm 1) = 0$  so

$$e^0(\mathbf{p}, \pm 1) = 0 \quad (5.9.26)$$

and

$$\mathbf{p} \cdot \mathbf{e}(\mathbf{p}, \pm 1) = 0. \quad (5.9.27)$$

It follows that

$$a^0(x) = 0 \quad (5.9.28)$$

and

$$\nabla \cdot \mathbf{a}(x) = 0. \quad (5.9.29)$$

As we shall see in Chapter 9, these are the conditions satisfied by the vacuum vector potential of electrodynamics in what is called Coulomb or radiation gauge.

The fact that  $a^0$  vanishes in all Lorentz frames shows vividly that  $a^\mu$  cannot be a four-vector. Instead, Eq. (5.9.22) shows that for a general momentum  $\mathbf{p}$  and a general Lorentz transformation  $\Lambda$ , in place of Eq. (5.9.6) we have

$$e^\mu(\mathbf{p}_\Lambda, \pm 1) \exp(\pm i\theta(p, \Lambda)) = D^\mu{}_\nu(\Lambda) e^\nu(\mathbf{p}, \pm 1) + p^\mu \Omega_\pm(p, \Lambda), \quad (5.9.30)$$

so that under a general Lorentz transformation

$$U(\Lambda) a_\mu(x) U^{-1}(\Lambda) = \Lambda^\nu{}_\mu a_\nu(\Lambda x) + \partial_\mu \Omega(x, \Lambda), \quad (5.9.31)$$

where  $\Omega(x, \Lambda)$  is a linear combination of annihilation and creation operators, whose precise form will not concern us here. As we will see in more detail in Chapter 8, we will be able to use a field like  $a^\mu(x)$  as an ingredient in Lorentz-invariant physical theories if the couplings of  $a^\mu(x)$  are not only formally Lorentz-invariant (that is, invariant under formal

Lorentz transformations under which  $a^\mu \rightarrow \Lambda^\mu_\nu a^\nu$ ), but are also invariant under the ‘gauge’ transformations  $a_\mu \rightarrow a_\mu + \partial_\mu \Omega$ . This is accomplished by taking the couplings of  $a_\mu$  to be of the form  $a_\mu j^\mu$ , where  $j^\mu$  is a four-vector current with  $\partial_\mu j^\mu = 0$ .

Although there is no ordinary four-vector field for massless particles of helicity  $\pm 1$ , there is no problem in constructing an antisymmetric tensor field for such particles. From Eq. (5.9.22) and the invariance of  $k^\mu$  under the little group we see immediately that

$$\begin{aligned} D^\mu{}_\rho(W(\theta, \alpha, \beta)) D^\nu{}_\sigma(W(\theta, \alpha, \beta)) (k^\rho e^\sigma(\mathbf{k}, \pm 1) - k^\sigma e^\rho(\mathbf{k}, \pm 1)) \\ = e^{\pm i\theta} (k^\mu e^\nu(\mathbf{k}, \pm 1) - k^\nu e^\mu(\mathbf{k}, \pm 1)). \end{aligned} \quad (5.9.32)$$

This shows that the coefficient function that satisfies Eq. (5.9.6) for the antisymmetric tensor representation of the homogeneous Lorentz group is (with an appropriate choice of normalization)

$$u_{\mu\nu}(\mathbf{p}, \pm 1) = \frac{i}{\sqrt{(2\pi)^3 2p^0}} [p^\mu e^\nu(\mathbf{p}, \pm 1) - p^\nu e^\mu(\mathbf{p}, \pm 1)], \quad (5.9.33)$$

where  $e^\mu(\mathbf{p}, \pm 1)$  is given by Eq. (5.9.25). Using this together with Eq. (5.9.23) gives the general antisymmetric tensor field for massless particles of helicity  $\pm 1$  in the form

$$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu. \quad (5.9.34)$$

Note that this is a tensor even though  $a^\mu$  is not a four-vector, because the extra term in Eq. (5.9.31) drops out in Eq. (5.9.34). Note also that Eqs. (5.9.34), (5.9.24), (5.9.28), and (5.9.29) show that  $f^{\mu\nu}$  satisfies the vacuum Maxwell equations:

$$\partial_\mu f^{\mu\nu} = 0, \quad (5.9.35)$$

$$\epsilon^{\rho\sigma\mu\nu} \partial_\sigma f_{\mu\nu} = 0. \quad (5.9.36)$$

To calculate the commutation relations for the tensor fields we need sums over helicities of the bilinears  $e^\mu e^{\nu*}$ . The explicit formula (5.9.21) gives

$$\sum_{\sigma=\pm 1} e^i(\mathbf{k}, \sigma) e^j(\mathbf{k}, \sigma)^* = \delta_{ij} - \frac{k^i k^j}{|\mathbf{k}|^2}$$

and so, using Eq. (5.9.25),

$$\sum_{\sigma=\pm 1} e^i(\mathbf{p}, \sigma) e^j(\mathbf{p}, \sigma)^* = \delta_{ij} - \frac{p^i p^j}{|\mathbf{p}|^2}. \quad (5.9.37)$$

A straightforward calculation gives then

$$\begin{aligned} [f_{\mu\nu}(x), f_{\rho\sigma}(y)^\dagger] &= (2\pi)^{-3} [-\eta_{\mu\rho} \partial_\nu \partial_\sigma + \eta_{\nu\rho} \partial_\mu \partial_\sigma + \eta_{\mu\sigma} \partial_\nu \partial_\rho - \eta_{\nu\sigma} \partial_\mu \partial_\rho] \\ &\quad \cdot \int d^3p (2p^0)^{-1} [|\kappa|^2 e^{ip \cdot (x-y)} - |\lambda|^2 e^{-ip \cdot (x-y)}]. \end{aligned} \quad (5.9.38)$$

This clearly vanishes for  $x^0 = y^0$  if and only if

$$|\kappa|^2 = |\lambda|^2 \quad (5.9.39)$$

in which case since  $f_{\mu\nu}$  is a tensor the commutator also vanishes for all space-like separations. Eq. (5.9.39) also implies that the commutator of the  $a^\mu$  vanishes at equal times, and as we shall see in Chapter 8 this is enough to yield a Lorentz-invariant  $S$ -matrix. The relative phase of the creation and annihilation operators can be adjusted so that  $\kappa = \lambda$ ; the fields are then Hermitian if the particles are their own charge-conjugates, as is the case for the photon.

Why should we want to use fields like  $a^\mu(x)$  in constructing theories of massless particles of spin one, rather than being content with fields like  $f^{\mu\nu}(x)$  with simple Lorentz transformation properties? The presence of the derivatives in Eq. (5.9.34) means that an interaction density constructed solely from  $f_{\mu\nu}$  and its derivatives will have matrix elements that vanish more rapidly for small massless particle energy and momentum than one that uses the vector field  $a_\mu$ . Interactions in such a theory will have a correspondingly rapid fall-off at large distances, faster than the usual inverse-square law. This is perfectly possible, but gauge-invariant theories that use vector fields for massless spin one particles represent a more general class of theories, including those that are actually realized in nature.

Parallel remarks apply to gravitons, massless particles of helicity  $\pm 2$ . From the annihilation and creation operators for such particles we can construct a tensor  $R_{\mu\nu\rho\sigma}$  with the algebraic properties of the Riemann–Christoffel curvature tensor: antisymmetric within the pairs  $\mu, \nu$  and  $\rho, \sigma$ , and symmetric between the pairs. However, in order to incorporate the usual inverse-square gravitational interactions we need to introduce a field  $h_{\mu\nu}$  that transforms as a symmetric tensor, up to gauge transformations of the sort associated in general relativity with general coordinate transformations. Thus in order to construct a theory of massless particles of helicity  $\pm 2$  that incorporates long-range interactions, it is necessary for it to have a symmetry something like general covariance. As in the case of electromagnetic gauge invariance, this is achieved by coupling the field to a conserved ‘current’  $\theta^{\mu\nu}$ , now with two spacetime indices, satisfying  $\partial_\mu \theta^{\mu\nu} = 0$ . The only such conserved tensor is the energy-momentum tensor, aside from possible total derivative terms that do not affect the long-range behavior of the force produced.<sup>52</sup> The fields of massless particles of spin  $j \geq 3$  would have to couple to conserved tensors with three or more spacetime indices, but aside from total derivatives there are none, so *high-spin massless particles cannot produce long-range forces*.

\* \* \*

The problems we have encountered in constructing four-vector fields for helicities  $\pm 1$  or symmetric tensor fields for helicity  $\pm 2$  are just special cases of a more general limitation. To see this, let’s consider how to construct fields for massless particles belonging to arbitrary representations of the homogeneous Lorentz group. As we saw in Section 5.6, any representation  $D(\Lambda)$  of the homogeneous Lorentz group can be decomposed into  $(2A + 1)(2B + 1)$ -dimensional representations  $(A, B)$ , for which the generators of the homogeneous Lorentz

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<sup>52</sup>If  $\theta^{\mu_1 \cdots \mu_N}$  is a tensor current satisfying  $\partial_{\mu_1} \theta^{\mu_1 \cdots \mu_N} = 0$ , then  $\int d^3x \theta^{0\mu_2 \cdots \mu_N}$  is a conserved quantity that transforms like a tensor of rank  $N - 1$ . The only such conserved tensors are the scalar ‘charges’ associated with various continuous symmetries, and the energy-momentum four-vector. The conservation of any other four-vector, or any tensor of higher rank, would forbid all but forward collisions.

group are represented by

$$(\mathcal{J}_{ij})_{a'b',ab} = \epsilon_{ijk}[(J_k^{(A)})_{a'a}\delta_{b'b} + (J_k^{(B)})_{b'b}\delta_{a'a}],$$

$$(\mathcal{J}_{i0})_{a'b',ab} = -i[(J_i^{(A)})_{a'a}\delta_{b'b} - (J_i^{(B)})_{b'b}\delta_{a'a}],$$

where  $\mathbf{J}^{(j)}$  are the angular-momentum matrices for spin  $j$ . For  $\theta$  infinitesimal,  $D(R(\theta)) = 1 + i\mathcal{J}_{23}\theta$ , so Eqs. (5.9.12) and (5.9.13) give

$$\sigma u_{ab}(\mathbf{k}, \sigma) = (a + b)u_{ab}(\mathbf{k}, \sigma),$$

$$-\sigma v_{ab}(\mathbf{k}, \sigma) = (a + b)v_{ab}(\mathbf{k}, \sigma),$$

and so  $u_{ab}(\mathbf{k}, \sigma)$  and  $v_{ab}(\mathbf{k}, \sigma)$  must vanish unless  $\sigma = a + b$  and  $\sigma = -a - b$ , respectively. Also, letting  $\alpha$  and  $\beta$  become infinitesimal in Eq. (5.9.14) gives

$$\begin{aligned} 0 &= (\mathcal{J}_{31} + \mathcal{J}_{01})_{ab,a'b'}u_{a'b'}(\mathbf{k}, \sigma) \\ &= (J_2^{(A)} + iJ_1^{(A)})_{aa'}u_{a'b}(\mathbf{k}, \sigma) + (J_2^{(B)} - iJ_1^{(B)})_{bb'}u_{ab'}(\mathbf{k}, \sigma), \end{aligned}$$

$$\begin{aligned} 0 &= (\mathcal{J}_{32} + \mathcal{J}_{02})_{ab,a'b'}u_{a'b'}(\mathbf{k}, \sigma) \\ &= (-J_1^{(A)} + iJ_2^{(A)})_{aa'}u_{a'b}(\mathbf{k}, \sigma) + (-J_1^{(B)} - iJ_2^{(B)})_{bb'}u_{ab'}(\mathbf{k}, \sigma), \end{aligned}$$

or more simply

$$\begin{aligned} (J_1^{(A)} - iJ_2^{(A)})_{aa'}u_{a'b}(\mathbf{k}, \sigma) &= 0, \\ (J_1^{(B)} + iJ_2^{(B)})_{bb'}u_{ab'}(\mathbf{k}, \sigma) &= 0. \end{aligned}$$

These require that  $u_{ab}(\mathbf{k}, \sigma)$  vanishes unless

$$a = -A, \quad b = +B \tag{5.9.40}$$

and the same is obviously also true of  $v_{ab}(\mathbf{k}, \sigma)$ . Putting this together, we see that a field of type  $(A, B)$  can be formed only from the annihilation operators for a massless particle of helicity  $\sigma$  and the creation operators for the antiparticle of helicity  $-\sigma$ , where

$$\sigma = B - A. \tag{5.9.41}$$

For instance, the  $(\frac{1}{2}, 0)$  and  $(0, \frac{1}{2})$  parts of the Dirac field for a massless particle can only destroy particles of helicity  $-\frac{1}{2}$  and  $+\frac{1}{2}$  respectively, and create antiparticles of helicity  $+\frac{1}{2}$  and  $-\frac{1}{2}$ , respectively. In the ‘two-component’ theory of the neutrino, there is only a  $(\frac{1}{2}, 0)$  field and its adjoint, so neutrinos have helicity  $-\frac{1}{2}$  and antineutrinos helicity  $+\frac{1}{2}$  in this theory.

By the same methods as in Section 5.7, it can be shown that the  $(j, 0)$  and  $(0, j)$  fields for massless particles of spin  $j$  (i.e., helicity  $\mp j$ ) commute with each other and their adjoints at space-like separations if the coefficients of the annihilation and creation terms in Eq. (5.9.1) satisfy Eq. (5.9.39). The relative phase of the annihilation and creation operators may then be adjusted so that these coefficients are equal. It is easy to see that

the fields for a massless particle of spin  $j$  of type  $(A, A + j)$  or  $(B + j, B)$  are just the  $2A$ th or  $2B$ th derivatives of fields of type  $(0, j)$  or  $(j, 0)$ , respectively, so these more general fields do not need to be considered separately here.

We can now see why it was impossible to construct a vector field for massless particles of helicity  $\pm 1$ . A vector field transforms according to the  $(\frac{1}{2}, \frac{1}{2})$  representation, and hence according to Eq. (5.9.19) can only describe helicity zero. (It is, of course, possible to construct a vector field for helicity zero—just take the derivative  $\partial_\mu \phi$  of a massless scalar field  $\phi$ .) The simplest covariant massless field for helicity  $\pm 1$  has the Lorentz transformation type  $(1, 0) \oplus (0, 1)$ ; that is, it is an antisymmetric tensor  $f_{\mu\nu}$ . Similarly, the simplest covariant massless field for helicity  $\pm 2$  has the Lorentz transformation type  $(2, 0) \oplus (0, 2)$ : a fourth rank tensor which like the Riemann–Christoffel curvature tensor is antisymmetric within each pair of indices and symmetric between the two pairs.

The discussion of the inversions  $P, C, T$  given in the previous section can be carried over to the case of zero mass with only obvious modifications.

### Weinberg Exercises

**W.5.1.** Show that if the zero-momentum coefficient functions satisfy the conditions (5.1.23) and (5.1.24), then the coefficient functions (5.1.21) and (5.1.22) for arbitrary momentum satisfy the defining conditions Eqs. (5.1.19) and (5.1.20).

**W.5.2.** Consider a free field  $\psi_\ell^\mu(x)$  which annihilates and creates a self-charge-conjugate particle of spin  $\frac{3}{2}$  and mass  $m \neq 0$ . Show how to calculate the coefficient functions  $u_\ell^\mu(\mathbf{p}, \sigma)$ , which multiply the annihilation operators  $a_{\mathbf{p}, \sigma}$  in this field, in such a way that the field transforms under Lorentz transformations like a Dirac field  $\psi_\ell$  with an extra four-vector index  $\mu$ . What field equations and algebraic and reality conditions does this field satisfy? Evaluate the matrix  $P_{\ell m}^{\mu\nu}(p)$ , defined (for  $p^2 = -m^2$ ) by

$$\sum_{\sigma} u_\ell^\mu(\mathbf{p}, \sigma) u_m^{\nu*}(\mathbf{p}, \sigma) \equiv (2p^0)^{-1} P_{\ell m}^{\mu\nu}(p).$$

What are the commutation relations of this field? How does the field transform under the inversions  $P, C, T$ ?

**W.5.3.** Consider a free field  $h^{\mu\nu}(x)$  satisfying  $h^{\mu\nu}(x) = h^{\nu\mu}(x)$  and  $h^\mu{}_\mu(x) = 0$ , which annihilates and creates a particle of spin two and mass  $m \neq 0$ . Show how to calculate the coefficient functions  $u^{\mu\nu}(\mathbf{p}, \sigma)$ , which multiply the annihilation operators  $a_{\mathbf{p}, \sigma}$  in this field, in such a way that the field transforms under Lorentz transformations like a tensor. What field equations does this field satisfy? Evaluate the function  $P^{\mu\nu, \kappa\lambda}(p)$ , defined by

$$\sum_{\sigma} u^{\mu\nu}(\mathbf{p}, \sigma) u^{\kappa\lambda*}(\mathbf{p}, \sigma) \equiv (2p^0)^{-1} P^{\mu\nu, \kappa\lambda}(p).$$

What are the commutation relations of this field? How does the field transform under the inversions  $P, C, T$ ?

**W.5.4.** Show that the fields for a massless particle of spin  $j$  of type  $(A, A + j)$  or  $(B + j, B)$  are the  $2A$ th or  $2B$ th derivatives of fields of type  $(0, j)$  or  $(j, 0)$ , respectively.

- W.5.5.** Work out the transformation properties of fields of transformation type  $(j, 0) + (0, j)$  for massless particles of helicity  $\pm j$  under the inversions  $P, C, T$ .
- W.5.6.** Consider a generalized Dirac field  $\psi$  that transforms according to the  $(j, 0) + (0, j)$  representation of the homogeneous Lorentz group. List the tensors that can be formed from products of the components of  $\psi$  and  $\psi^\dagger$ . Check your result against what we found for  $j = \frac{1}{2}$ .
- W.5.7.** Consider a general field  $\psi_{ab}$  describing particles of spin  $j$  and mass  $m \neq 0$ , that transforms according to the  $(A, B)$  representation of the homogeneous Lorentz group. Suppose it has an interaction Hamiltonian of the form

$$H_{\text{int}} = \int d^3x [\psi_{ab}(x) J^{ab}(x) + J^{ab\dagger}(x) \psi_{ab}^\dagger(x)],$$

where  $J^{ab}$  is an external c-number current. What is the asymptotic behavior of the matrix element for emitting these particles for energy  $E \gg m$  and definite helicity? (Assume that the Fourier transform of the current has values for different  $a, b$  that are of the same order of magnitude, and that do not depend strongly on  $E$ .)

### Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

- W.5.1.** Write  $k = (m, \mathbf{0})$ ,  $q = \Lambda p$ , and

$$W(\Lambda, p) = L^{-1}(q) \Lambda L(p).$$

The defining identity is therefore

$$L(q) W(\Lambda, p) = \Lambda L(p).$$

Starting from Eq. (5.1.21) and using the zero-momentum intertwining relation (5.1.23), we obtain

$$\begin{aligned} \sum_{\bar{\sigma}} u_\ell(\mathbf{q}, \bar{\sigma}) D_{\bar{\sigma}\sigma}^{(j)}(W) &= \sqrt{\frac{m}{q^0}} \left[ D(L(q)) u(\mathbf{0}) D^{(j)}(W) \right]_{\ell\sigma} \\ &= \sqrt{\frac{m}{q^0}} [D(L(q)) D(W) u(\mathbf{0})]_{\ell\sigma} \\ &= \sqrt{\frac{m}{q^0}} [D(\Lambda) D(L(p)) u(\mathbf{0})]_{\ell\sigma} \\ &= \sqrt{\frac{p^0}{q^0}} \sum_{\bar{\ell}} D_{\ell\bar{\ell}}(\Lambda) u_{\bar{\ell}}(\mathbf{p}, \sigma). \end{aligned}$$

This is precisely Eq. (5.1.19). The same calculation, with  $v(\mathbf{0}) D^{(j)*}(W) = D(W) v(\mathbf{0})$ , proves Eq. (5.1.20). Thus the rest-frame rotation intertwiners and one standard boost determine coefficient functions satisfying the full Lorentz covariance conditions.



**W.5.2.** Let  $u_\ell(\mathbf{p}, s)$  be the spin- $\frac{1}{2}$  Dirac coefficient functions of Section 5.5, and let  $e^\mu(\mathbf{p}, r)$  be the massive spin-one polarization vectors of Section 5.3. Couple the two rotation representations at rest and then boost:

$$u_\ell^\mu(\mathbf{p}, \sigma) = \sum_{r=-1}^1 \sum_{s=\pm 1/2} \langle 1r, \frac{1}{2}s | \frac{3}{2}\sigma \rangle e^\mu(\mathbf{p}, r) u_\ell(\mathbf{p}, s).$$

The Clebsch–Gordan coefficient projects  $1 \otimes \frac{1}{2}$  onto spin  $\frac{3}{2}$ ; covariance of the two factors then gives the vector-spinor transformation law. The creation coefficients are fixed by the Majorana relation, so the causal field may be written

$$\psi_\ell^\mu(x) = \frac{1}{(2\pi)^{3/2}} \sum_\sigma \int d^3p \left[ u_\ell^\mu(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma} e^{ip \cdot x} + v_\ell^\mu(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma}^\dagger e^{-ip \cdot x} \right],$$

with  $v$  the charge conjugate of  $u$ . It obeys

$$(\gamma^\rho \partial_\rho + m)\psi^\mu = 0, \quad \gamma_\mu \psi^\mu = 0, \quad \partial_\mu \psi^\mu = 0,$$

where the last equation follows from the first two. Self-charge-conjugacy gives the reality condition

$$\psi^\mu = -\beta \mathcal{C} \psi^{\mu*}.$$

For a compact form of the spin sum define, on  $p^2 = -m^2$ ,

$$\Theta^{\mu\nu}(p) = \eta^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}, \quad \Gamma^\mu(p) = \Theta^\mu{}_\rho(p) \gamma^\rho,$$

$$Q_{3/2}^{\mu\nu}(p) = \Theta^{\mu\nu}(p) - \frac{1}{3} \Gamma^\mu(p) \Gamma^\nu(p).$$

This is transverse, gamma-traceless on the physical Dirac subspace, and at rest is the familiar projector  $\delta^{ij} - \sigma^i \sigma^j / 3$ . The requested answer is

$$P_{\ell m}^{\mu\nu}(p) = \left\{ Q_{3/2}^{\mu\nu}(p) [-ip^\rho \gamma_\rho + m] \beta \right\}_{\ell m}.$$

Equivalently, suppressing spinor indices, ordinary matrix multiplication is understood:

$$P^{\mu\nu} = Q_{3/2}^{\mu\nu} [-i\not{p} + m] \beta.$$

It immediately satisfies  $p_\mu P^{\mu\nu} = \gamma_\mu P^{\mu\nu} = 0$ .

The field is fermionic. If

$$\hat{\Theta}^{\mu\nu} = \eta^{\mu\nu} - \frac{\partial^\mu \partial^\nu}{m^2}, \quad \hat{\Gamma}^\mu = \gamma^\mu - \frac{\partial^\mu \gamma^\rho \partial_\rho}{m^2}, \quad \hat{Q}_{3/2}^{\mu\nu} = \hat{\Theta}^{\mu\nu} - \frac{1}{3} \hat{\Gamma}^\mu \hat{\Gamma}^\nu,$$

then its causal anticommutator is

$$\{\psi_\ell^\mu(x), \psi_m^{\nu\dagger}(y)\} = \left\{ \hat{Q}_{3/2}^{\mu\nu} [-\gamma^\rho \partial_\rho + m] \beta \right\}_{\ell m} \Delta(x - y),$$

and anticommutators without an adjoint are fixed by the Majorana condition. They vanish at spacelike separation.

Up to the conventional intrinsic phases, the inversions are the tensor product of the vector and Dirac rules:

$$\begin{aligned} P\psi^\mu(x)P^{-1} &= -\eta^* \mathcal{P}^\mu{}_\nu \beta \psi^\nu(\mathcal{P}x), \\ C\psi^\mu(x)C^{-1} &= -\xi^* \beta \mathcal{C} \psi^{\mu*}(x), \\ T\psi^\mu(x)T^{-1} &= \zeta^* \mathcal{P}^\mu{}_\nu \gamma_5 \mathcal{C} \psi^\nu(-\mathcal{T}x). \end{aligned}$$

A rephasing can move the overall sign in the last line into  $\zeta$ . For a self-conjugate field,  $\xi = \pm 1$  and  $\eta = \pm i$ ; the usual relations  $\xi^c = \xi^*$ ,  $\zeta^c = \zeta^*$ , and  $\eta^c = -\eta^*$  hold.

**W.5.3.** Use the spin-one polarization vectors and project the product  $1 \otimes 1$  onto its spin-two part:

$$e^{\mu\nu}(\mathbf{p}, \sigma) = \sum_{r,s=-1}^1 \langle 1r, 1s | 2\sigma \rangle e^\mu(\mathbf{p}, r) e^\nu(\mathbf{p}, s), \quad u^{\mu\nu}(\mathbf{p}, \sigma) = \frac{e^{\mu\nu}(\mathbf{p}, \sigma)}{\sqrt{2p^0}}.$$

The Clebsch–Gordan coefficient makes this tensor symmetric and traceless. Since every vector polarization is transverse, the resulting free field satisfies

$$h^{\mu\nu} = h^{\nu\mu}, \quad h^\mu{}_\mu = 0, \quad \partial_\mu h^{\mu\nu} = 0, \quad (\square - m^2)h^{\mu\nu} = 0.$$

These conditions leave  $10 - 1 - 4 = 5$  independent components, the required  $2j + 1$  polarizations.

With  $\Theta^{\mu\nu} = \eta^{\mu\nu} + p^\mu p^\nu / m^2$ , completeness of the spin-two polarization tensors gives

$$P^{\mu\nu, \kappa\lambda}(p) = \frac{1}{2} \left( \Theta^{\mu\kappa} \Theta^{\nu\lambda} + \Theta^{\mu\lambda} \Theta^{\nu\kappa} \right) - \frac{1}{3} \Theta^{\mu\nu} \Theta^{\kappa\lambda}.$$

It is symmetric within and between the index pairs, transverse in every index, traceless, and has trace five. The field is bosonic, and

$$[h^{\mu\nu}(x), h^{\kappa\lambda\dagger}(y)] = \hat{P}^{\mu\nu, \kappa\lambda}(\partial) \Delta(x - y),$$

where

$$\hat{\Theta}^{\mu\nu} = \eta^{\mu\nu} - \frac{\partial^\mu \partial^\nu}{m^2}$$

is inserted in the preceding expression for  $P$ . This commutator vanishes at spacelike separation. For a self-charge-conjugate particle the field can be chosen Hermitian.

The inversion laws, with intrinsic phases left explicit, are

$$\begin{aligned} Ph^{\mu\nu}(x)P^{-1} &= \eta^* \mathcal{P}^\mu{}_\rho \mathcal{P}^\nu{}_\sigma h^{\rho\sigma}(\mathcal{P}x), \\ Ch^{\mu\nu}(x)C^{-1} &= \xi^* h^{\mu\nu\dagger}(x), \\ Th^{\mu\nu}(x)T^{-1} &= \zeta^* \mathcal{P}^\mu{}_\rho \mathcal{P}^\nu{}_\sigma h^{\rho\sigma}(-\mathcal{P}x). \end{aligned}$$

Here  $\eta^c = \eta^*$ ,  $\xi^c = \xi^*$ , and  $\zeta^c = \zeta^*$ ; all three phases may be chosen real for a neutral self-conjugate spin-two particle.

**W.5.4.** It is clearest to use two-component spinor indices. A derivative  $\partial_{\alpha\dot{\alpha}}$  transforms as  $(\frac{1}{2}, \frac{1}{2})$ . Let  $\phi_{\dot{\beta}_1 \dots \dot{\beta}_{2j}}$  be a totally symmetric  $(0, j)$  field. The completely irreducible part of

$$\partial_{\alpha_1 \dot{\alpha}_1} \cdots \partial_{\alpha_{2A} \dot{\alpha}_{2A}} \phi_{\dot{\beta}_1 \dots \dot{\beta}_{2j}}$$

that is symmetric separately in all undotted and all dotted indices has  $2A$  undotted indices and  $2A + 2j$  dotted indices. It therefore transforms as  $(A, A + j)$ . On each plane wave the derivatives merely multiply the coefficient function by  $(ip_{\alpha\dot{\alpha}})^{2A}$ , so they neither introduce a new particle species nor change the helicity. Uniqueness of the covariant field of a fixed Lorentz type then shows that every  $(A, A + j)$  field is proportional to this derivative field.

Starting instead with a symmetric  $(j, 0)$  field and taking the separately symmetric irreducible part of  $2B$  derivatives gives  $2B + 2j$  undotted and  $2B$  dotted indices, hence type  $(B + j, B)$ . This proves both statements.

**W.5.5.** Write the two chiral fields as

$$\Psi(x) = \begin{pmatrix} \phi_L(x) \\ \phi_R(x) \end{pmatrix}, \quad \phi_L \sim (j, 0), \quad \phi_R \sim (0, j).$$

Equation (5.9.41) says that  $\phi_L$  annihilates helicity  $-j$  and creates the opposite-helicity antiparticle, whereas  $\phi_R$  annihilates helicity  $+j$  and creates its opposite-helicity antiparticle. Let

$$(\mathbb{C}_j)_{mn} = (-1)^{j-m} \delta_{m, -n}$$

be the spin- $j$  reversal matrix. With harmless convention-dependent signs absorbed into the intrinsic phases, the inversion laws can be written

$$\begin{aligned} P\Psi(x)P^{-1} &= \eta^* \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix} \Psi(\mathcal{P}x), \\ C\Psi(x)C^{-1} &= \xi^* \begin{pmatrix} 0 & \mathbb{C}_j \\ (-1)^{2j} \mathbb{C}_j & 0 \end{pmatrix} \Psi^*(x), \\ T\Psi(x)T^{-1} &= \zeta^* \begin{pmatrix} \mathbb{C}_j & 0 \\ 0 & \mathbb{C}_j \end{pmatrix} \Psi(\mathbf{x}, -x^0). \end{aligned}$$

The operator  $T$  is antiunitary, so the numerical coefficients in the last line are complex conjugated when the rule is used.

These formulas display the physics independently of phase convention:  $P$  reverses momentum but not spin and therefore reverses helicity, so it interchanges  $(j, 0)$  and  $(0, j)$ ;  $C$  exchanges particles and antiparticles; and  $T$  reverses both momentum and spin, preserving helicity and each chiral transformation type. The particle and antiparticle phases obey

$$\eta^c = (-1)^{2j} \eta^*, \quad \xi^c = \xi^*, \quad \zeta^c = \zeta^*.$$

**W.5.6.** Decompose the generalized Dirac field as  $\psi = (\psi_L, \psi_R)$ , with  $\psi_L \sim (j, 0)$  and  $\psi_R \sim (0, j)$ . The four kinds of bilinear products transform as

| bilinear                | Lorentz content                                       |
|-------------------------|-------------------------------------------------------|
| $\psi_L^\dagger \psi_L$ | $(j, j)$                                              |
| $\psi_R^\dagger \psi_R$ | $(j, j)$                                              |
| $\psi_R^\dagger \psi_L$ | $(j, 0) \otimes (j, 0) = \bigoplus_{k=0}^{2j} (k, 0)$ |
| $\psi_L^\dagger \psi_R$ | $(0, j) \otimes (0, j) = \bigoplus_{k=0}^{2j} (0, k)$ |

Thus the complete list is

$$2(j, j) \oplus \bigoplus_{k=0}^{2j} [(k, 0) \oplus (0, k)].$$

For each pair related by parity one may take a sum and a difference to form ordinary and pseudo tensors. Dimension counting checks completeness:

$$2(2j+1)^2 + 2 \sum_{k=0}^{2j} (2k+1) = 4(2j+1)^2,$$

which is the dimension of the full matrix algebra acting on the  $2(2j+1)$ -component field.

For  $j = \frac{1}{2}$ , the two  $(0, 0)$ 's are the scalar and pseudoscalar, the two  $(\frac{1}{2}, \frac{1}{2})$ 's are the vector and axial vector, and  $(1, 0) \oplus (0, 1)$  is the antisymmetric tensor. These are exactly the sixteen standard Dirac bilinears  $\bar{\psi}\psi$ ,  $\bar{\psi}\gamma_5\psi$ ,  $\bar{\psi}\gamma^\mu\psi$ ,  $\bar{\psi}\gamma^\mu\gamma_5\psi$ , and  $\bar{\psi}[\gamma^\mu, \gamma^\nu]\psi$ .

**W.5.7.** Choose the emitted momentum along the 3-axis. Equation (5.7.14) then reduces to

$$u_{ab}(\mathbf{p}, \sigma) = \frac{C_{AB}(j\sigma; ab)}{\sqrt{2E}} e^{(-a+b)\theta}, \quad a+b=\sigma,$$

where

$$e^\theta = \frac{E + |\mathbf{p}|}{m} = \frac{2E}{m} \left[ 1 + O\left(\frac{m^2}{E^2}\right) \right].$$

Because the Fourier components  $J^{ab}(p)$  are assumed to be comparable, the largest exponent  $-a+b$  among nonzero Clebsch–Gordan coefficients sets the generic high-energy behavior. Subject to  $-A \leq a \leq A$ ,  $-B \leq b \leq B$ , and  $a+b=\sigma$ , it is

$$r_\sigma = \max(-a+b) = \min(2A+\sigma, 2B-\sigma).$$

Consequently a definite-helicity emission matrix element behaves as

$$\mathcal{A}_\sigma(E) \sim E^{-1/2} \left( \frac{E}{m} \right)^{r_\sigma}$$

up to an  $E$ -independent current component, a numerical Clebsch–Gordan coefficient, and the overall spacetime momentum-conservation factor. Accidental cancellations or current conservation can soften this leading power.

As checks, a scalar has  $r_0 = 0$ ; for a four-vector  $(A, B) = (\frac{1}{2}, \frac{1}{2})$ , the transverse helicities have  $r_{\pm 1} = 0$  while the longitudinal helicity has  $r_0 = 1$ , producing the familiar  $E/m$  enhancement; and for a chiral  $(j, 0)$  field the helicity  $-j$  is the unsuppressed massless component while the opposite helicity is power suppressed.

## Supplementary Exercises

### S.5.1. The Klein–Gordon Current Is Not a Probability

(Adapted from H. Liu, MIT 8.323, Assignment 2, 2023, Problem 1.)

- Derive the probability continuity equation and current from the nonrelativistic Schrödinger equation.
- Find both energy branches of the relativistic scalar plane-wave dispersion relation.
- For a complex Klein–Gordon solution, derive its conserved current and evaluate the density on each energy branch.
- Explain precisely why this density cannot be interpreted as a one-particle probability, and why it is instead a charge density.

### S.5.2. Scalar Mode Recovery, Four-Momentum, and One-Particle Matrix Elements

(Inspired by H. Liu, MIT 8.323, Assignment 2, 2023, Problem 2(b)–(c), pp. 3–4 / H. Liu, MIT 8.323, Assignment 2, 2023, Problem 3(a)–(b), p. 4 / H. Liu, MIT 8.323, Assignment 2, 2023, Problem 3(c), p. 4 / H. Liu, MIT 8.323, Assignment 2, 2023, Problems 2–3, pp. 3–4.)

- At  $t = 0$ , invert the real scalar mode expansion to express  $a(\mathbf{p})$  and  $a^\dagger(\mathbf{p})$  in terms of the spatial Fourier transforms of  $\phi$  and its conjugate momentum. Derive the oscillator commutator from the canonical equal-time commutators.
- Starting from the symmetric stress tensor of a real scalar field, show after normal ordering that

$$P^\mu = \int \frac{d^3p}{(2\pi)^3} p^\mu a^\dagger(\mathbf{p}) a(\mathbf{p}).$$

Verify its action on a one-particle state.

- Use the mode expression for  $P^\mu$  to prove  $[P^\mu, \phi(x)] = -i\partial^\mu \phi(x)$ , and exponentiate the commutator to obtain the finite translation law.
- With relativistically normalized states  $|p\rangle = \sqrt{2p^0} a^\dagger(\mathbf{p})|\Omega\rangle$ , compute  $\langle\Omega|\phi(x)|p\rangle$ . Check its Klein–Gordon equation and Lorentz transformation law.

### S.5.3. Orbital Lorentz Charge and Spin Zero

(Adapted from H. Liu, MIT 8.323, Assignment 2, 2023, Problem 4.)

- (a) For an infinitesimal transformation  $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$ , with  $\omega_{\mu\nu} = -\omega_{\nu\mu}$ , verify directly to first order that

$$\Lambda^\mu{}_\rho \Lambda^\nu{}_\lambda \eta^{\rho\lambda} = \eta^{\mu\nu}.$$

- (b) Use  $\phi'(x') = \phi(x)$  for a real scalar to derive the Lorentz Noether current

$$M^{\rho\mu\nu} = x^\mu T^{\rho\nu} - x^\nu T^{\rho\mu}.$$

This part should follow from the transformation law and Noether's theorem without a long component calculation.

- (c) Using symmetry and conservation of  $T^{\mu\nu}$ , verify  $\partial_\rho M^{\rho\mu\nu} = 0$ .  
(d) For the free scalar theory

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2,$$

write all six charges

$$M^{\mu\nu} = \int d^3x M^{0\mu\nu}$$

in terms of  $a(\mathbf{p})$  and  $a^\dagger(\mathbf{p})$ . This final, calculation-heavy part is optional in the source assignment.

### S.5.4. The Lorentz Algebra in the Vector Representation

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 2, 2022, Problem 1.)

- (a) Starting from preservation of inner products under  $x'^\mu = \Lambda^\mu{}_\nu x^\nu$ , prove

$$\eta_{\mu\nu} = \eta_{\rho\sigma} \Lambda^\rho{}_\mu \Lambda^\sigma{}_\nu.$$

- (b) For  $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \varepsilon\omega^\mu{}_\nu$ , prove  $\omega_{\mu\nu} = -\omega_{\nu\mu}$ . Explain why proper, orthochronous Lorentz transformations can be built from such infinitesimal transformations.  
(c) Write the  $4 \times 4$  matrix  $iJ_3$  for an infinitesimal rotation  $\varepsilon = d\theta$  about the 3-axis, and identify the exponential that gives a finite rotation through  $\theta$ .  
(d) Write the matrix  $iK_1$  for an infinitesimal boost  $\varepsilon = dv$  along the 1-axis. Express the finite boost of velocity  $v$  as an exponential of  $K_1$ .  
(e) Define the remaining generators analogously. The Lorentz algebra is

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \quad [J_i, K_j] = i\epsilon_{ijk}K_k, \quad [K_i, K_j] = -i\epsilon_{ijk}J_k.$$

Verify these relations explicitly for  $(i, j) = (1, 1)$  and  $(i, j) = (1, 2)$ ; the remaining cases follow in the same way. Interpret the first two commutators as the vector transformation laws for angular velocity and velocity, and the third as the possibility of obtaining a rotation by composing non-collinear boosts.

### S.5.5. Quantization and Noether Charges of a Complex Scalar

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 2, 2022, Problem 2.)

Consider

$$\mathcal{L} = -\partial_\mu \Phi^\dagger \partial^\mu \Phi - m^2 \Phi^\dagger \Phi, \quad E_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

At  $t = 0$ , use

$$\begin{aligned} \Phi(\mathbf{x}) &= \int \frac{d^3 k}{(2\pi)^3 \sqrt{2E_{\mathbf{k}}}} \left[ a(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{x}} + b^\dagger(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{x}} \right], \\ \Phi^\dagger(\mathbf{x}) &= \int \frac{d^3 k}{(2\pi)^3 \sqrt{2E_{\mathbf{k}}}} \left[ b(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{x}} + a^\dagger(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{x}} \right], \\ \Pi_\Phi(\mathbf{x}) &= -i \int \frac{d^3 k}{(2\pi)^3} \sqrt{\frac{E_{\mathbf{k}}}{2}} \left[ b(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{x}} - a^\dagger(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{x}} \right], \\ \Pi_{\Phi^\dagger}(\mathbf{x}) &= -i \int \frac{d^3 k}{(2\pi)^3} \sqrt{\frac{E_{\mathbf{k}}}{2}} \left[ a(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{x}} - b^\dagger(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{x}} \right]. \end{aligned}$$

Quantize with

$$[\Phi(\mathbf{x}), \Pi_\Phi(\mathbf{y})] = [\Phi^\dagger(\mathbf{x}), \Pi_{\Phi^\dagger}(\mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}),$$

all other canonical commutators being zero. The vacuum obeys  $a(\mathbf{k})|0\rangle = b(\mathbf{k})|0\rangle = 0$ . As in the source assignment, parallel calculations need not be repeated once the common argument has been made explicit.

- Derive the commutators of  $a, a^\dagger, b, b^\dagger$ , including the vanishing mixed commutators.
- For  $\Phi \rightarrow e^{i\alpha}\Phi$ ,  $\Phi^\dagger \rightarrow e^{-i\alpha}\Phi^\dagger$ , derive the Noether current and conserved charge  $Q$ .
- Normal order  $Q$ , impose  $Q|0\rangle = 0$ , and find the charge of  $a^\dagger(\mathbf{k})|0\rangle$ ,  $b^\dagger(\mathbf{k})|0\rangle$ , and  $a^\dagger(\mathbf{k}_1)a^\dagger(\mathbf{k}_2)|0\rangle$ .
- Prove

$$e^{i\alpha Q}\Phi(x)e^{-i\alpha Q} = e^{i\alpha}\Phi(x), \quad e^{i\alpha Q}\Phi^\dagger(x)e^{-i\alpha Q} = e^{-i\alpha}\Phi^\dagger(x),$$

first by deriving the infinitesimal commutators with  $Q$ .

- Use Noether's theorem to derive  $T^{\mu\nu}$  and  $P^\mu = \int d^3x T^{0\mu}$ .
- Normal order  $H = P^0$  and  $\mathbf{P}$  in oscillator variables, set  $P^\mu|0\rangle = 0$ , and find the four-momentum of the three states in part (c).
- Show

$$e^{ia \cdot P}\Phi(x)e^{-ia \cdot P} = \Phi(x - a), \quad [P^i, \Phi(x)] = i\partial^i\Phi(x),$$

and verify the commutator from the oscillator expression for  $\mathbf{P}$ .

### S.5.6. Boost Charges on Scalar Creation Operators

(Adapted from B. Mistlberger and K. Zhou, Stanford PHYS-330, Problem Set 2, 2022, Problem 3.)

Under a Lorentz transformation, a scalar profile obeys

$$\phi'(x') = \phi(x), \quad \phi'(x) = \phi(\Lambda^{-1}x).$$

For  $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \varepsilon\omega^\mu{}_\nu$ , this gives, through first order,

$$\phi'(x) = \phi(x) - \varepsilon\omega^\mu{}_\nu x^\nu \partial_\mu \phi(x).$$

Work with a real scalar field. Equivalently, one may obtain the same real field by identifying the two oscillator families in a complex-scalar mode expansion,  $a(\mathbf{k}) = b(\mathbf{k})$ .

- (a) For a real scalar, derive

$$M^{\mu\nu\rho} = x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu}$$

and the associated charges  $M^{\nu\rho}$ .

- (b) The six antisymmetric charges split into  $M^{ij}$  and  $M^{0i}$ . Explain the physical meaning of the three  $M^{0i}$ .
- (c) After normal ordering, prove

$$M^{ij} = i \int \frac{d^3k}{(2\pi)^3} a^\dagger(\mathbf{k}) \left( k^j \frac{\partial}{\partial k_i} - k^i \frac{\partial}{\partial k_j} \right) a(\mathbf{k}),$$

and explain why this shows that scalar quanta have no intrinsic angular momentum.

### S.5.7. Representation-Independent Clifford Identities

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 3, 2006, Problem 3.)

Work in four dimensions with

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \quad \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3, \quad \not{p} = p_\mu\gamma^\mu, \quad S^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu].$$

Useful methods include cyclicity of the trace and inserting  $\gamma_5^2 = \mathbf{1}$  into a trace. Without choosing a representation, prove:

- (a)  $\text{tr } \gamma^\mu = 0$ ;  
 (b)  $\text{tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}$ ;  
 (c)  $\text{tr}(\gamma^\mu\gamma^\nu\gamma^\rho) = 0$ ;  
 (d)  $\gamma_5^2 = \mathbf{1}$ ;  
 (e)  $\text{tr } \gamma_5 = 0$ ;



(f)  $\not{p}\not{q} = 2p \cdot q - \not{q}\not{p} = p \cdot q + 2S^{\mu\nu}p_\mu q_\nu$ ;

(g)  $\text{tr}(\not{p}\not{q}) = 4p \cdot q$ ;

(h)  $\text{tr}(\not{p}_1 \cdots \not{p}_n) = 0$  for odd  $n$ ;

(i)

$$\text{tr}(\not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = 4[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_3)(p_2 \cdot p_4)];$$

(j)  $\text{tr}(\gamma_5 \not{p}\not{q}) = 0$ ;

(k)  $\gamma_\mu \not{p} \gamma^\mu = -2\not{p}$ ;

(l)  $\gamma_\mu \not{p}_1 \not{p}_2 \gamma^\mu = 4p_1 \cdot p_2$ ;

(m)  $\gamma_\mu \not{p}_1 \not{p}_2 \not{p}_3 \gamma^\mu = -2\not{p}_3 \not{p}_2 \not{p}_1$ ;

(n)

$$\text{tr}(\gamma_5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = -4i\epsilon^{\mu\nu\rho\sigma} p_{1\mu} p_{2\nu} p_{3\rho} p_{4\sigma}.$$

### S.5.8. Finite Rotations and Boosts of Dirac Spinors

(Adapted from *University of Cambridge, Part III Quantum Field Theory, Paper 301, 2023, Question 3.*)

Define the Dirac-spinor representation of the proper orthochronous Lorentz group. In particular, give the spinor representative  $S[\Lambda]$  when

$$\Lambda = \exp\left(\frac{1}{2}\Omega_{\rho\sigma}M^{\rho\sigma}\right), \quad \Omega_{\rho\sigma} = -\Omega_{\sigma\rho},$$

and the vector-representation generators are

$$(M^{\rho\sigma})^{\mu\nu} = \eta^{\rho\mu}\eta^{\sigma\nu} - \eta^{\sigma\mu}\eta^{\rho\nu}.$$

Using a suitable representation of the Dirac matrices, calculate  $S[\Lambda]$  explicitly for

$$R(\varphi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \varphi & \sin \varphi & 0 \\ 0 & -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad B(\beta) = \begin{pmatrix} \cosh \beta & \sinh \beta & 0 & 0 \\ \sinh \beta & \cosh \beta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Find  $S[R(2\pi)]$  and explain the significance of the result.

For the final part, use the local mostly-plus Weyl basis

$$\gamma^0 = -i \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \quad \gamma^i = -i \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}.$$

A positive-frequency spinor may then be written as

$$u(p) = \begin{pmatrix} \sqrt{E\mathbf{1} - \mathbf{p} \cdot \boldsymbol{\sigma}} \xi \\ \sqrt{E\mathbf{1} + \mathbf{p} \cdot \boldsymbol{\sigma}} \xi \end{pmatrix}, \quad \xi \in \mathbb{C}^2.$$

Compare this spinor in the rest frame  $p_1 = (m, 0, 0, 0)$  and in the boosted frame  $p_2 = (m \cosh \beta, m \sinh \beta, 0, 0)$ . Show directly that

$$u(p_2) = S[B(\beta)]u(p_1).$$

The source uses the mostly-minus metric; all metric, gamma-matrix, and factor-of- $i$  conventions above have been translated to those of this volume, while its two Lorentz matrices and all requested calculations are unchanged.

### S.5.9. Dirac Spinor Norms and Orthogonality

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 3, 2006, Problem 4.)

Use the local Weyl basis

$$\gamma^0 = -i \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \quad \gamma^i = -i \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \beta = i\gamma^0,$$

and let  $\xi^s$ ,  $s = 1, 2$ , be orthonormal two-component spinors. For  $p^0 = E_{\mathbf{p}} > 0$ , define

$$p \cdot \sigma = E_{\mathbf{p}} \mathbf{1} - \mathbf{p} \cdot \boldsymbol{\sigma}, \quad p \cdot \bar{\sigma} = E_{\mathbf{p}} \mathbf{1} + \mathbf{p} \cdot \boldsymbol{\sigma},$$

$$u^s(\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix}, \quad v^s(\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ -\sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix}.$$

Prove

$$\begin{aligned} u^r(\mathbf{p})^\dagger u^s(\mathbf{p}) &= 2E_{\mathbf{p}} \delta^{rs}, & \bar{u}^r(\mathbf{p}) u^s(\mathbf{p}) &= 2m \delta^{rs}, \\ v^r(\mathbf{p})^\dagger v^s(\mathbf{p}) &= 2E_{\mathbf{p}} \delta^{rs}, & \bar{v}^r(\mathbf{p}) v^s(\mathbf{p}) &= -2m \delta^{rs}. \end{aligned}$$

as well as

$$\bar{u}^s(\mathbf{p}) v^r(\mathbf{p}) = 0, \quad u^r(\mathbf{p})^\dagger v^s(-\mathbf{p}) = 0.$$

### S.5.10. Completeness of Free Dirac Spinors

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 3, 2006, Problem 5.)

Using exactly the notation and normalization of S.5.9, prove the matrix identities

$$\sum_{s=1}^2 u^s(\mathbf{p}) \bar{u}^s(\mathbf{p}) = -i\not{p} + m, \quad \sum_{s=1}^2 v^s(\mathbf{p}) \bar{v}^s(\mathbf{p}) = -i\not{p} - m.$$

The spinors on each left-hand side are outer products, not contracted spinor indices.

### S.5.11. Equal-Time Dirac Anticommutators

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 3, 2006, Problem 6.)

At fixed time, take

$$\psi(\mathbf{x}) = \sum_{s=1}^2 \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left[ b_{\mathbf{p}}^s u^s(\mathbf{p}) e^{i\mathbf{p} \cdot \mathbf{x}} + c_{\mathbf{p}}^{s\dagger} v^s(\mathbf{p}) e^{-i\mathbf{p} \cdot \mathbf{x}} \right],$$

$$\psi^\dagger(\mathbf{x}) = \sum_{s=1}^2 \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left[ b_{\mathbf{p}}^{s\dagger} u^s(\mathbf{p})^\dagger e^{-i\mathbf{p} \cdot \mathbf{x}} + c_{\mathbf{p}}^s v^s(\mathbf{p})^\dagger e^{i\mathbf{p} \cdot \mathbf{x}} \right].$$

Assume

$$\{b_{\mathbf{p}}^r, b_{\mathbf{q}}^{s\dagger}\} = \{c_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}\} = (2\pi)^3 \delta^{rs} \delta^3(\mathbf{p} - \mathbf{q}),$$

with every other oscillator anticommutator zero. Prove

$$\{\psi_\alpha(\mathbf{x}), \psi_\beta(\mathbf{y})\} = \{\psi_\alpha^\dagger(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} = 0,$$

$$\{\psi_\ell(\mathbf{x}), \psi_m^\dagger(\mathbf{y})\} = \delta_{\ell m} \delta^3(\mathbf{x} - \mathbf{y}),$$

using the two completeness identities in S.5.10.

### S.5.12. What Fails with Bosonic Spinors

(Adapted from D. Tong, Quantum Field Theory: Example Sheet 3, 2006, Problem 8.)

Use the mode expansion of S.5.11, but impose bosonic commutators

$$[b_{\mathbf{p}}^r, b_{\mathbf{q}}^{s\dagger}] = (2\pi)^3 \delta^{rs} \delta^3(\mathbf{p} - \mathbf{q}), \quad [c_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}] = -(2\pi)^3 \delta^{rs} \delta^3(\mathbf{p} - \mathbf{q}),$$

with all others zero.

(a) Repeat S.5.11 and show

$$[\psi_\alpha(\mathbf{x}), \psi_\beta(\mathbf{y})] = [\psi_\alpha^\dagger(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})] = 0, \quad [\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})] = \delta_{\alpha\beta} \delta^3(\mathbf{x} - \mathbf{y}).$$

(b) Starting from

$$H = \int d^3x \bar{\psi}(\gamma^i \partial_i + m)\psi,$$

normal order the Hamiltonian and prove

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 \left( b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s \right).$$

Explain why this bosonic spin- $\frac{1}{2}$  theory is inconsistent.

### S.5.13. Massive-Vector Constraints and Polarizations

(Inspired by D. Tong, Lectures on Quantum Field Theory, 2006, Section 6.1 / D. Tong, Lectures on Quantum Field Theory, 2006, Sections 6.1–6.2.)

- (a) Starting from the Proca equation  $\partial_\mu F^{\mu\nu} - m^2 A^\nu = 0$  in mostly-plus convention, derive the transversality condition and the Klein–Gordon equation. Count the propagating polarizations.
- (b) Construct three polarization vectors for a massive vector with momentum along the 3-axis and prove

$$\sum_{\sigma=-1}^1 e^\mu(\mathbf{p}, \sigma) e^{\nu*}(\mathbf{p}, \sigma) = \eta^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}.$$

- (c) Find the high-energy behavior of the longitudinal massive-vector polarization. Show that its coupling to  $J_\mu$  has a smooth massless limit when  $p_\mu J^\mu = 0$ , and determine the leading residual behavior.

#### S.5.14. Helicity, Gauge Shifts, and the Covariant Field Strength

(Inspired by D. Tong, Lectures on Quantum Field Theory, 2006, Section 6.2 / D. Tong, Lectures on Quantum Field Theory, 2006, Sections 6.1–6.2.)

- (a) For a massless spin-one polarization, show that the translation part of the little group changes  $e^\mu$  by a multiple of  $p^\mu$ . Prove that a Lorentz-covariant emission amplitude is insensitive to this change exactly when the current is conserved.
- (b) Define  $f^{\mu\nu}(p, \sigma) = i(p^\mu e^\nu - p^\nu e^\mu)$  for helicity  $\sigma = \pm 1$ . Show that it is invariant under  $e^\mu \rightarrow e^\mu + \alpha p^\mu$ , obeys the source-free Maxwell identities, and carries the two helicities in the  $(1, 0) \oplus (0, 1)$  representation.
- (c) In Coulomb gauge, derive

$$[A_i(\mathbf{x}, t), E_j(\mathbf{y}, t)] = i \left( \delta_{ij} - \frac{\partial_i \partial_j}{\nabla^2} \right) \delta^3(\mathbf{x} - \mathbf{y}).$$

Explain how this projector implements the two physical photon degrees of freedom.

#### S.5.15. Antiparticles and the Locality of Charged Fields

(Inspired by D. Harlow, Lecture Notes for Physics 8.323, 2024, Section 3.3, pp. 35–36 / D. Harlow, Lecture Notes for Physics 8.323, 2024, Section 3.5, pp. 37–38 / D. Harlow, Lecture Notes for Physics 8.323, 2024, Section 3.4, pp. 35–37.)

- (a) At fixed time, solve for the positive-frequency part  $\phi^{(+)}$  of a real scalar field in terms of  $\phi$  and  $\pi$ . Show that the operator  $\sqrt{m^2 - \nabla^2}$  makes this separation nonlocal in space even though the full field is local.
- (b) Try to build a charged scalar field from particle annihilation operators and their adjoints alone. Show why a local field of definite charge instead requires an independent antiparticle oscillator with the opposite charge.
- (c) Write the free scalar commutator as the Pauli–Jordan integral. Prove its vanishing at spacelike separation by evaluating it at equal time in a suitable Lorentz frame, and explain why the separate positive-frequency function does not vanish.

#### S.5.16. Wigner’s Characterization of Antiunitary Operators

(Adapted from D. Harlow, Quantum Field Theory I, 2024, Section 4.7, Homework Problem 7.)

Let  $\Theta$  be an antiunitary operator. First show that  $\Theta^2$  is unitary. There is consequently a basis  $\{|i\rangle\}$  in which

$$\Theta^2|i\rangle = e^{-2i\theta_i}|i\rangle, \quad (\Theta^\dagger)^2|i\rangle = (\Theta^2)^\dagger|i\rangle = e^{2i\theta_i}|i\rangle, \quad \theta_i \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

Determine how  $\Theta$  and  $\Theta^\dagger$  act in such a basis, and then use antilinearity to determine their action on arbitrary superpositions.

*Hint.* Up to rephasing, the basis can be chosen to consist of states that are invariant under  $\Theta$  and pairs of states that  $\Theta$  exchanges up to a phase. Begin by proving that  $\Theta|i\rangle$  is also an eigenstate of  $\Theta^2$ .

## Solutions to Supplementary Exercises

### S.5.1. The Klein–Gordon Current Is Not a Probability

- (a) The Schrödinger equation and its conjugate imply

$$\partial_t |\psi|^2 + \nabla \cdot \frac{1}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) = 0.$$

Here  $|\psi|^2 \geq 0$ , so the conserved integral has a probability interpretation.

- (b) Substitution of  $e^{-iEt+i\mathbf{p}\cdot\mathbf{x}}$  in the Klein–Gordon equation gives

$$E = \pm \sqrt{\mathbf{p}^2 + m^2}.$$

- (c) Multiply the field equation by  $\phi^*$ , subtract its complex conjugate multiplied by  $\phi$ , and use the product rule:

$$0 = \phi^* \square \phi - \phi \square \phi^* = \partial_\mu (\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*).$$

Thus, with a sign chosen so positive-frequency modes have positive charge,

$$j^\mu = -i (\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*), \quad \partial_\mu j^\mu = 0.$$

For  $\phi = Ae^{ip\cdot x}$ ,  $p^0 = E > 0$ , this gives  $j^0 = 2E|A|^2$ . For the negative-frequency solution  $Ae^{-ip\cdot x}$ , it gives  $j^0 = -2E|A|^2$ . A probability density cannot change sign between allowed solutions.

- (d) In field theory the two signs instead label opposite conserved charges, so  $j^\mu$  is a charge current.

### S.5.2. Scalar Mode Recovery, Four-Momentum, and One-Particle Matrix Elements

- (a) Use

$$\begin{aligned} \phi(\mathbf{x}, 0) &= \int \frac{d^3 p}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left[ a(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}} + a^\dagger(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x}} \right], \\ \pi(\mathbf{x}, 0) &= -i \int \frac{d^3 p}{(2\pi)^3} \sqrt{\frac{E_{\mathbf{p}}}{2}} \left[ a(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}} - a^\dagger(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x}} \right]. \end{aligned}$$

For the Fourier convention  $\tilde{\phi}(\mathbf{p}) = \int d^3 x e^{-i\mathbf{p}\cdot\mathbf{x}} \phi(\mathbf{x}, 0)$ , one finds

$$a(\mathbf{p}) = \sqrt{\frac{E_{\mathbf{p}}}{2}} \tilde{\phi}(\mathbf{p}) + \frac{i}{\sqrt{2E_{\mathbf{p}}}} \tilde{\pi}(\mathbf{p}),$$

$$a^\dagger(\mathbf{p}) = \sqrt{\frac{E_{\mathbf{p}}}{2}} \tilde{\phi}(-\mathbf{p}) - \frac{i}{\sqrt{2E_{\mathbf{p}}}} \tilde{\pi}(-\mathbf{p}).$$

Fourier transformation of  $[\phi(\mathbf{x}), \pi(\mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y})$  gives

$$[\tilde{\phi}(\mathbf{p}), \tilde{\pi}(\mathbf{q})] = i(2\pi)^3 \delta^3(\mathbf{p} + \mathbf{q}).$$

Substitution in the inverse formulas yields

$$[a(\mathbf{p}), a^\dagger(\mathbf{q})] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}),$$

and the  $aa$  and  $a^\dagger a^\dagger$  commutators cancel.

(b) For

$$\mathcal{L} = -\frac{1}{2} \partial_\rho \phi \partial^\rho \phi - \frac{1}{2} m^2 \phi^2,$$

the symmetric stress tensor is

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi + \eta^{\mu\nu} \mathcal{L}.$$

Insert the mode expansion into  $P^\mu = \int d^3x T^{0\mu}$ . The spatial integral sets the two momenta equal. Terms  $aa$  and  $a^\dagger a^\dagger$  cancel between the kinetic, gradient, and mass pieces by  $E_{\mathbf{p}}^2 = \mathbf{p}^2 + m^2$ . After normal ordering,

$$P^\mu = \int \frac{d^3p}{(2\pi)^3} p^\mu a^\dagger(\mathbf{p}) a(\mathbf{p}).$$

Hence

$$[P^\mu, a^\dagger(\mathbf{q})] = q^\mu a^\dagger(\mathbf{q}), \quad P^\mu a^\dagger(\mathbf{q}) |\Omega\rangle = q^\mu a^\dagger(\mathbf{q}) |\Omega\rangle,$$

after setting the vacuum four-momentum to zero.

(c) The oscillator expression gives

$$[P^\mu, a(\mathbf{p})] = -p^\mu a(\mathbf{p}), \quad [P^\mu, a^\dagger(\mathbf{p})] = p^\mu a^\dagger(\mathbf{p}).$$

Applying these relations term by term to the field expansion,

$$[P^\mu, \phi(x)] = -i\partial^\mu \phi(x).$$

Let  $F(s) = e^{isa \cdot P} \phi(x) e^{-isa \cdot P}$ . Then

$$\frac{dF}{ds} = ie^{isa \cdot P} [a \cdot P, \phi(x)] e^{-isa \cdot P} = a^\mu \partial_\mu F.$$

With  $F(0) = \phi(x)$ , the solution is

$$e^{ia \cdot P} \phi(x) e^{-ia \cdot P} = \phi(x + a).$$

Changing the sign in the definition of the translation operator changes both displayed signs but not the physical statement.

- (d) Only the annihilation part of  $\phi(x)$  contributes between  $\langle\Omega|$  and the one-particle ket. The oscillator commutator cancels the normalization factors, giving

$$\langle\Omega|\phi(x)|p\rangle = e^{ip\cdot x}$$

in the normalization stated in the problem. Since  $p^2 = -m^2$ ,

$$(\square - m^2)e^{ip\cdot x} = (-p^2 - m^2)e^{ip\cdot x} = 0.$$

For a scalar field,  $U(\Lambda)\phi(x)U(\Lambda)^{-1} = \phi(\Lambda x)$ , while the relativistically normalized state transforms as  $U(\Lambda)|p\rangle = |\Lambda p\rangle$ . Vacuum invariance then gives

$$\langle\Omega|\phi(x)|\Lambda p\rangle = \langle\Omega|\phi(\Lambda^{-1}x)|p\rangle = e^{ip\cdot\Lambda^{-1}x} = e^{i(\Lambda p)\cdot x},$$

so the result is covariant.

### S.5.3. Orbital Lorentz Charge and Spin Zero

- (a) Expanding the Lorentz condition to first order gives

$$\begin{aligned}\Lambda^\mu{}_\rho \Lambda^\nu{}_\lambda \eta^{\rho\lambda} &= (\delta^\mu{}_\rho + \omega^\mu{}_\rho)(\delta^\nu{}_\lambda + \omega^\nu{}_\lambda)\eta^{\rho\lambda} + O(\omega^2) \\ &= \eta^{\mu\nu} + \omega^{\mu\nu} + \omega^{\nu\mu} + O(\omega^2) = \eta^{\mu\nu} + O(\omega^2),\end{aligned}$$

where the final equality uses  $\omega^{\mu\nu} = -\omega^{\nu\mu}$ .

- (b) At the same spacetime point, the scalar transformation law reads

$$\delta\phi(x) = -\frac{1}{2}\omega_{\mu\nu}(x^\mu\partial^\nu - x^\nu\partial^\mu)\phi(x).$$

Applying Noether's theorem to each independent antisymmetric parameter  $\omega_{\mu\nu}$  gives

$$M^{\rho\mu\nu} = x^\mu T^{\rho\nu} - x^\nu T^{\rho\mu}.$$

- (c) Its divergence is

$$\partial_\rho M^{\rho\mu\nu} = T^{\mu\nu} - T^{\nu\mu} + x^\mu\partial_\rho T^{\rho\nu} - x^\nu\partial_\rho T^{\rho\mu} = 0.$$

The first two terms cancel because the scalar stress tensor is symmetric, and the remaining terms vanish by stress-tensor conservation.

- (d) After inserting the free-field expansion and normal ordering, the three rotation charges are

$$M^{ij} = i \int \frac{d^3p}{(2\pi)^3} a^\dagger(\mathbf{p}) \left( p^j \frac{\partial}{\partial p_i} - p^i \frac{\partial}{\partial p_j} \right) a(\mathbf{p}), \quad 1 \leq i < j \leq 3,$$

and the three boost charges are

$$M^{0i} = i \int \frac{d^3p}{(2\pi)^3} a^\dagger(\mathbf{p}) \left( E_{\mathbf{p}} \frac{\partial}{\partial p_i} + \frac{p_i}{2E_{\mathbf{p}}} \right) a(\mathbf{p}), \quad i = 1, 2, 3.$$

The  $p_i/(2E_{\mathbf{p}})$  term makes the boost operator Hermitian for the  $d^3p$  oscillator normalization used here. These formulas list all six independent charges. In particular,  $M^{ij}$  contains only an orbital differential operator and no momentum-independent matrix acting on an internal index. Thus the one-particle quanta have spin zero.

#### S.5.4. The Lorentz Algebra in the Vector Representation

- (a) In matrix notation, preservation of every inner product is

$$x^\top \Lambda^\top \eta \Lambda y = x^\top \eta y \quad \text{for all } x, y.$$

Hence  $\Lambda^\top \eta \Lambda = \eta$ , or in components,

$$\eta_{\mu\nu} = \eta_{\rho\sigma} \Lambda^\rho{}_\mu \Lambda^\sigma{}_\nu.$$

- (b) Writing  $\Lambda = \mathbf{1} + \varepsilon \omega$  and retaining first order in  $\varepsilon$  gives

$$\omega^\top \eta + \eta \omega = 0.$$

After lowering the first index this is  $\omega_{\mu\nu} + \omega_{\nu\mu} = 0$ . The proper, orthochronous component contains the identity and is connected, so a continuous path from the identity can be partitioned into infinitesimal transformations; their ordered product reconstructs the finite transformation.

- (c) Choose the orientation in which

$$iJ_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Then

$$R_3(\theta) = e^{i\theta J_3} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & \sin \theta & 0 \\ 0 & -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Changing the orientation of positive rotations changes both signs in the off-diagonal entries and amounts to  $J_3 \mapsto -J_3$ .

- (d) For a boost along the 1-axis, take

$$iK_1 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$



With rapidity  $\chi = \text{arctanh } v$ ,

$$B_1(v) = e^{i\chi K_1} = \begin{pmatrix} \cosh \chi & -\sinh \chi & 0 & 0 \\ -\sinh \chi & \cosh \chi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

(e) The convention above is generated by

$$(J_i)^0{}_\nu = (J_i)^\mu{}_0 = 0, \quad (J_i)^j{}_k = -i\epsilon_{ijk},$$

$$(K_i)^0{}_j = (K_i)^j{}_0 = i\delta_{ij},$$

with all other entries of  $K_i$  zero; reversing all  $K_i$  signs is the opposite boost convention. Direct matrix multiplication, together with  $\epsilon_{iab}\epsilon_{jbc} = \delta_{ic}\delta_{aj} - \delta_{ij}\delta_{ac}$ , gives

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \quad [J_i, K_j] = i\epsilon_{ijk}K_k,$$

$$[K_i, K_j] = -i\epsilon_{ijk}J_k.$$

For  $(i, j) = (1, 1)$ , all three equations reduce to a vanishing commutator. For  $(i, j) = (1, 2)$ , direct multiplication of the displayed matrices and their cyclic analogues gives

$$[J_1, J_2] = iJ_3,$$

$$[J_1, K_2] = iK_3,$$

$$[K_1, K_2] = -iJ_3,$$

which are precisely the three nontrivial checks requested. The first two lines say that rotations and boosts each transform as spatial vectors. The minus sign in the last line is the algebraic origin of the rotation produced by composing non-collinear boosts.

### S.5.5. Quantization and Noether Charges of a Complex Scalar

(a) Insert the four supplied mode expansions in the two nonzero canonical commutators. Fourier orthogonality gives

$$[a(\mathbf{p}), a^\dagger(\mathbf{q})] = [b(\mathbf{p}), b^\dagger(\mathbf{q})] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}),$$

with every  $aa$ ,  $bb$ ,  $ab$ ,  $ab^\dagger$ , and conjugate commutator equal to zero. Conversely, these oscillator relations reproduce all the stated canonical commutators.

(b) For  $\delta\Phi = i\alpha\Phi$  and  $\delta\Phi^\dagger = -i\alpha\Phi^\dagger$ , Noether's theorem gives

$$j^\mu = i \left( \Phi^\dagger \partial^\mu \Phi - (\partial^\mu \Phi^\dagger) \Phi \right).$$

Thus

$$Q = \int d^3x j^0 = i \int d^3x \left( \dot{\Phi}^\dagger \Phi - \Phi^\dagger \dot{\Phi} \right), \quad \dot{Q} = 0.$$

- (c) Substitution of the modes gives an ordering-dependent vacuum constant. Normal ordering and  $Q|0\rangle = 0$  fix it:

$$Q = \int \frac{d^3p}{(2\pi)^3} \left[ b^\dagger(\mathbf{p})b(\mathbf{p}) - a^\dagger(\mathbf{p})a(\mathbf{p}) \right].$$

Consequently the charges of

$$a^\dagger(\mathbf{k})|0\rangle, \quad b^\dagger(\mathbf{k})|0\rangle, \quad a^\dagger(\mathbf{k}_1)a^\dagger(\mathbf{k}_2)|0\rangle$$

are respectively  $-1, +1, -2$ .

- (d) The oscillator expression immediately yields

$$[Q, \Phi] = \Phi, \quad [Q, \Phi^\dagger] = -\Phi^\dagger.$$

If  $F(\alpha) = e^{i\alpha Q}\Phi e^{-i\alpha Q}$ , then  $F'(\alpha) = iF(\alpha)$ , and hence

$$e^{i\alpha Q}\Phi e^{-i\alpha Q} = e^{i\alpha}\Phi.$$

The adjoint calculation similarly gives the phase  $e^{-i\alpha}$ .

- (e) The symmetric stress tensor and conserved four-momentum are

$$\begin{aligned} T^{\mu\nu} &= (\partial^\mu \Phi^\dagger)(\partial^\nu \Phi) + (\partial^\nu \Phi^\dagger)(\partial^\mu \Phi) + \eta^{\mu\nu} \mathcal{L}, \\ P^\mu &= \int d^3x T^{0\mu}. \end{aligned}$$

- (f) After normal ordering and setting the vacuum four-momentum to zero,

$$P^\mu = \int \frac{d^3p}{(2\pi)^3} p^\mu \left[ a^\dagger(\mathbf{p})a(\mathbf{p}) + b^\dagger(\mathbf{p})b(\mathbf{p}) \right].$$

In particular,  $H = P^0$  is a sum of positive number operators. Both one-particle states have four-momentum  $k^\mu$ , while the two- $a$  state has four-momentum  $k_1^\mu + k_2^\mu$ .

- (g) From the number-operator formula,

$$[P^i, a(\mathbf{k})] = -k^i a(\mathbf{k}), \quad [P^i, b^\dagger(\mathbf{k})] = k^i b^\dagger(\mathbf{k}).$$

Applying these relations to the field expansion gives

$$[P^i, \Phi(x)] = i\partial^i \Phi(x).$$

For  $F(s) = e^{i\mathbf{s}\cdot\mathbf{P}}\Phi(x)e^{-i\mathbf{s}\cdot\mathbf{P}}$ , this commutator implies  $F'(s) = -\mathbf{a}\cdot\nabla F(s)$ , and therefore

$$e^{i\mathbf{a}\cdot\mathbf{P}}\Phi(x)e^{-i\mathbf{a}\cdot\mathbf{P}} = \Phi(x - \mathbf{a}).$$

The sign differs from the source's upper-index formula only because its mostly-minus metric has been translated to the present mostly-plus convention.

### S.5.6. Boost Charges on Scalar Creation Operators

- (a) The scalar variation  $\delta\phi = -\frac{1}{2}\omega_{\nu\rho}(x^\nu\partial^\rho - x^\rho\partial^\nu)\phi$  and Noether's theorem give

$$M^{\rho\mu\nu} = x^\mu T^{\rho\nu} - x^\nu T^{\rho\mu}, \quad \partial_\rho M^{\rho\mu\nu} = 0,$$

and the charges are

$$M^{\mu\nu} = \int d^3x M^{0\mu\nu}.$$

- (b) The three  $M^{ij}$  generate rotations. The remaining  $M^{0i}$  generate Lorentz boosts, mixing energy with the  $i$ -th momentum component. For example, with the oscillator normalization in the problem, their normal-ordered form is

$$M^{0i} = i \int \frac{d^3p}{(2\pi)^3} a^\dagger(\mathbf{p}) \left( E_{\mathbf{p}} \frac{\partial}{\partial p_i} + \frac{p_i}{2E_{\mathbf{p}}} \right) a(\mathbf{p}).$$

The  $p_i/(2E_{\mathbf{p}})$  term is the measure term required for Hermiticity with respect to  $d^3p$ .

- (c) Inserting the free scalar modes into  $M^{ij} = \int d^3x (x^i T^{0j} - x^j T^{0i})$ , using  $x^i e^{i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}} = -i\partial_{p_i} e^{i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}}$ , and integrating by parts gives

$$M^{ij} = i \int \frac{d^3p}{(2\pi)^3} a^\dagger (p^j \partial_{p_i} - p^i \partial_{p_j}) a.$$

It contains no momentum-independent matrix, so it is purely orbital and the scalar one-particle representation has spin zero.

### S.5.7. Representation-Independent Clifford Identities

First note that  $\gamma_5^2 = 1$  and  $\{\gamma_5, \gamma^\mu\} = 0$ : moving the four gamma matrices in the second  $\gamma_5$  through the first four gives six interchanges, while  $(\gamma^0)^2 = -1$  and  $(\gamma^i)^2 = 1$ . These facts, the Clifford algebra, and cyclicity suffice for the entire list.

- (a)  $\text{tr } \gamma^\mu = \text{tr}(\gamma_5 \gamma^\mu \gamma_5) = -\text{tr } \gamma^\mu$ , so the trace is zero.  
(b) Taking the trace of the anticommutator gives

$$2 \text{tr}(\gamma^\mu \gamma^\nu) = 2\eta^{\mu\nu} \text{tr } \mathbf{1} = 8\eta^{\mu\nu}.$$

- (c) Conjugating three gamma matrices by  $\gamma_5$  changes their sign, so their trace vanishes.  
(d) The preliminary calculation gives  $\gamma_5^2 = 1$ .  
(e) Choose any invertible  $\gamma^\mu$ . Conjugating  $\gamma_5$  by it changes its sign, so trace invariance gives  $\text{tr } \gamma_5 = 0$ .  
(f) The Clifford algebra gives

$$\not{p}\not{q} + \not{q}\not{p} = 2p \cdot q.$$

Separating the symmetric and antisymmetric gamma products yields

$$\not{p}\not{q} = p \cdot q + \frac{1}{2} p_\mu q_\nu [\gamma^\mu, \gamma^\nu] = p \cdot q + 2S^{\mu\nu} p_\mu q_\nu.$$

- (g) Taking the trace of part 6 and using  $\text{tr } S^{\mu\nu} = 0$  gives  $\text{tr}(\not{p}\not{q}) = 4p \cdot q$ .
- (h) For odd  $n$ , insert  $\gamma_5^2$ , move one  $\gamma_5$  through all  $n$  slashed vectors, and use cyclicity. The trace equals its negative.
- (i) Successive anticommutation and cyclicity give

$$\text{tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(\eta^{\mu\nu} \eta^{\rho\sigma} - \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}).$$

Contracting with the four momenta gives the displayed result.

- (j)  $\text{tr}(\gamma_5 \gamma^\mu \gamma^\nu)$  is antisymmetric in  $\mu, \nu$ , while inserting and contracting a further Clifford pair shows it equals its negative; hence it vanishes. Contracting with  $p_\mu q_\nu$  proves the claim.
- (k) Move the left  $\gamma_\mu$  through  $\not{p}$ :

$$\gamma_\mu \not{p} \gamma^\mu = 2p_\mu \gamma^\mu - \not{p} \gamma_\mu \gamma^\mu = 2\not{p} - 4\not{p} = -2\not{p}.$$

- (l) Applying part 11 twice,

$$\gamma_\mu \not{p}_1 \not{p}_2 \gamma^\mu = 2\not{p}_2 \not{p}_1 + 2\not{p}_1 \not{p}_2 = 4p_1 \cdot p_2.$$

- (m) The same anticommutation step, followed by part 12, gives

$$\gamma_\mu \not{p}_1 \not{p}_2 \not{p}_3 \gamma^\mu = -2\not{p}_3 \not{p}_2 \not{p}_1.$$

- (n) The trace is totally antisymmetric in the four indices, so it is a constant times  $\epsilon^{\mu\nu\rho\sigma}$ . Evaluating 0123,

$$\text{tr}(\gamma_5 \gamma^0 \gamma^1 \gamma^2 \gamma^3) = i \text{tr}[(\gamma^0 \gamma^1 \gamma^2 \gamma^3)^2] = -4i.$$

Thus

$$\text{tr}(\gamma_5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = -4i \epsilon^{\mu\nu\rho\sigma},$$

and contraction gives the requested slash identity.

### S.5.8. Finite Rotations and Boosts of Dirac Spinors

For the conventions of this volume, introduce

$$\mathcal{J}^{\mu\nu} = -\frac{i}{4}[\gamma^\mu, \gamma^\nu].$$

These matrices obey the Lorentz algebra and satisfy  $[\mathcal{J}^{\mu\nu}, \gamma^\rho] = -i\gamma^\mu \eta^{\nu\rho} + i\gamma^\nu \eta^{\mu\rho}$ . It follows that

$$S[\Lambda] = \exp\left(\frac{i}{2}\Omega_{\mu\nu}\mathcal{J}^{\mu\nu}\right) = \exp\left(\frac{1}{8}\Omega_{\mu\nu}[\gamma^\mu, \gamma^\nu]\right)$$

implements  $S[\Lambda]\gamma^\rho S[\Lambda]^{-1} = \Lambda_\sigma{}^\rho \gamma^\sigma$ . The factor  $1/2$  compensates for summing both orders of the antisymmetric pair.

In the Weyl basis displayed in the question, direct multiplication gives

$$\mathcal{J}^{12} = \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix}, \quad \mathcal{J}^{10} = \frac{i}{2} \begin{pmatrix} \sigma^1 & 0 \\ 0 & -\sigma^1 \end{pmatrix}.$$

For the rotation matrix in the question,  $\Omega_{12} = \varphi$ , and hence

$$S[R(\varphi)] = \begin{pmatrix} e^{i\varphi\sigma^3/2} & 0 \\ 0 & e^{i\varphi\sigma^3/2} \end{pmatrix}.$$

For the displayed boost,  $\Omega_{10} = \beta$ , and

$$S[B(\beta)] = \begin{pmatrix} e^{-\beta\sigma^1/2} & 0 \\ 0 & e^{\beta\sigma^1/2} \end{pmatrix}.$$

These expressions may also be checked directly in the gamma-matrix transformation law, so their signs are tied to the two explicit Lorentz matrices rather than to a convention for naming the parameters.

Since  $e^{i\pi\sigma^3} = -\mathbf{1}_2$ ,

$$S[R(2\pi)] = -\mathbf{1}_4.$$

Thus the Dirac representation is a representation of the double cover  $\text{Spin}^+(1,3)$ , not a single-valued representation of  $SO^+(1,3)$ : a  $2\pi$  rotation changes a spinor's sign, while a  $4\pi$  rotation acts trivially.

Finally,

$$u(p_1) = \sqrt{m} \begin{pmatrix} \xi \\ \xi \end{pmatrix}.$$

For  $p_2 = (m \cosh \beta, m \sinh \beta, 0, 0)$ , use

$$\cosh \beta \mathbf{1} \mp \sinh \beta \sigma^1 = e^{\mp \beta \sigma^1}$$

and take the positive Hermitian square roots. This gives

$$u(p_2) = \sqrt{m} \begin{pmatrix} e^{-\beta\sigma^1/2} \xi \\ e^{\beta\sigma^1/2} \xi \end{pmatrix} = S[B(\beta)]u(p_1),$$

which is the required finite-spinor transformation.

### S.5.9. Dirac Spinor Norms and Orthogonality

Set

$$A = \sqrt{p \cdot \sigma}, \quad B = \sqrt{p \cdot \bar{\sigma}}.$$

The two Hermitian matrices are functions of  $\mathbf{p} \cdot \boldsymbol{\sigma}$ , so they commute. Moreover,

$$A^2 + B^2 = 2E_{\mathbf{p}} \mathbf{1}, \quad A^2 B^2 = (E_{\mathbf{p}}^2 - \mathbf{p}^2) \mathbf{1} = m^2 \mathbf{1},$$

and hence  $AB = BA = m\mathbf{1}$ . It follows immediately that

$$\begin{aligned} u^{r\dagger} u^s &= \xi^{r\dagger} (A^2 + B^2) \xi^s = 2E_{\mathbf{p}} \delta^{rs}, \\ v^{r\dagger} v^s &= \xi^{r\dagger} (A^2 + B^2) \xi^s = 2E_{\mathbf{p}} \delta^{rs}. \end{aligned}$$

In the displayed Weyl basis  $\beta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}$ . Therefore

$$\bar{u}^r = (\xi^{r\dagger} B, \xi^{r\dagger} A), \quad \bar{v}^r = (-\xi^{r\dagger} B, \xi^{r\dagger} A),$$

which gives

$$\bar{u}^r u^s = 2m \delta^{rs}, \quad \bar{v}^r v^s = -2m \delta^{rs}, \quad \bar{u}^r v^s = 0.$$

Finally  $A(-\mathbf{p}) = B(\mathbf{p})$  and  $B(-\mathbf{p}) = A(\mathbf{p})$ , so

$$u^{r\dagger}(\mathbf{p}) v^s(-\mathbf{p}) = \xi^{r\dagger} (AB - BA) \xi^s = 0.$$

### S.5.10. Completeness of Free Dirac Spinors

Completeness of the two-component basis gives  $\sum_s \xi^s \xi^{s\dagger} = \mathbf{1}$ . With the  $A, B$  notation of S.5.9,

$$\sum_s u^s \bar{u}^s = \begin{pmatrix} AB & A^2 \\ B^2 & BA \end{pmatrix} = \begin{pmatrix} m\mathbf{1} & p \cdot \sigma \\ p \cdot \bar{\sigma} & m\mathbf{1} \end{pmatrix}.$$

For the local Weyl gamma matrices, the off-diagonal matrix is  $-i\not{p}$ , so this is  $-i\not{p} + m$ . Similarly,

$$\sum_s v^s \bar{v}^s = \begin{pmatrix} -AB & A^2 \\ B^2 & -BA \end{pmatrix} = -i\not{p} - m.$$

These are outer-product identities in the  $4 \times 4$  spinor space. They also agree with the on-shell equations  $(i\not{p} + m)u = 0$  and  $(-i\not{p} + m)v = 0$ .

### S.5.11. Equal-Time Dirac Anticommutators

Insert the field expansion and use the oscillator anticommutators. Mixed particle–antiparticle terms vanish. The remaining terms are

$$\begin{aligned} \{\psi(\mathbf{x}, t), \psi^\dagger(\mathbf{y}, t)\} &= \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} ([-i\not{p} + m] \beta e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} \\ &\quad + [-i\not{p} - m] \beta e^{-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})}). \end{aligned}$$

In the second term change  $\mathbf{p} \rightarrow -\mathbf{p}$ . The spatial-gamma and mass pieces cancel; the time-gamma piece equals  $2E_{\mathbf{p}}\mathbf{1}$ . Therefore

$$\{\psi_\ell(\mathbf{x}, t), \psi_m^\dagger(\mathbf{y}, t)\} = \delta_{\ell m} \int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} = \delta_{\ell m} \delta^3(\mathbf{x} - \mathbf{y}).$$

The  $bb$ ,  $cc$ , and mixed anticommutators vanish, proving the other equal-time relations.

### S.5.12. What Fails with Bosonic Spinors

- (a) In the particle term, the commutator calculation is identical to the anticommutator calculation in S.5.11. Interchanging the two antiparticle operators would normally introduce the opposite sign; the explicit minus sign in  $[c, c^\dagger]$  cancels it. Thus the same spin-sum combination appears and gives

$$[\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})] = \delta_{\alpha\beta} \delta^3(\mathbf{x} - \mathbf{y}).$$

The  $bb$ ,  $cc$ , and mixed commutators vanish, proving the other two relations.

- (b) Substitution of the modes into the Hamiltonian uses  $(i\not{p}+m)u=0$ ,  $(-i\not{p}+m)v=0$ , and the norms in S.5.9. Before normal ordering, the particle term is  $E_{\mathbf{p}} b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s$ , while the antiparticle term is  $-E_{\mathbf{p}} c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger}$ . Moving  $c^\dagger$  to the left with the negative commutator and dropping the c-number gives

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 \left( b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s \right).$$

Adding  $c$ -quanta lowers the energy without bound. Independently,

$$\|c_{\mathbf{p}}^{s\dagger}|0\rangle\|^2 = \langle 0|c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger}|0\rangle < 0$$

because of the same minus-sign commutator. The construction therefore has both an indefinite norm and an energy spectrum unbounded below: bosonic statistics is incompatible with a healthy spin- $\frac{1}{2}$  field.

### S.5.13. Massive-Vector Constraints and Polarizations

- (a) Take the divergence of

$$\partial_\mu F^{\mu\nu} - m^2 A^\nu = 0.$$

The first term vanishes because  $F^{\mu\nu}$  is antisymmetric while  $\partial_\mu \partial_\nu$  is symmetric. For  $m \neq 0$ ,

$$\partial_\nu A^\nu = 0.$$

Expanding  $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$  in the equation of motion then gives

$$(\square - m^2)A^\nu = 0.$$

A four-vector begins with four components. Transversality removes one on-shell component, leaving three independent polarizations. In the rest frame they are simply the three spatial unit vectors, the spin-one representation of rotations.

- (b) For  $p^\mu = (E, 0, 0, k)$ , choose

$$e^\mu(\pm 1) = \frac{1}{\sqrt{2}}(0, 1, \pm i, 0), \quad e^\mu(0) = \frac{1}{m}(k, 0, 0, E),$$

up to helicity phase conventions. They satisfy

$$p_\mu e^\mu(\sigma) = 0, \quad e_\mu(\sigma) e^{\mu*}(\sigma') = \delta_{\sigma\sigma'}.$$

The sum is a Lorentz tensor, is transverse in either index, and acts as the identity on the three-dimensional subspace orthogonal to  $p^\mu$ . The unique tensor with these properties is

$$\Pi^{\mu\nu}(p) = \eta^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}.$$

Direct substitution of the three displayed vectors verifies every component and fixes the coefficient of  $p^\mu p^\nu$ .

- (c) For  $p^\mu = (E, 0, 0, k)$ ,

$$e_L^\mu = \frac{1}{m}(k, 0, 0, E) = \frac{p^\mu}{m} + \frac{1}{m}(k - E, 0, 0, E - k).$$

Since  $E - k = m^2/(E + k)$ ,

$$e_L^\mu = \frac{p^\mu}{m} + O\left(\frac{m}{E}\right).$$

The emission factor is

$$J_\mu e_L^\mu = \frac{J_\mu p^\mu}{m} + O\left(\frac{m}{E} J\right).$$

For a generic current the first term diverges as  $m \rightarrow 0$ . If  $p_\mu J^\mu = 0$ , it vanishes exactly and the remaining longitudinal coupling is  $O(m/E)$ . Current conservation therefore decouples the helicity-zero state and leaves the two transverse massless helicities.

#### S.5.14. Helicity, Gauge Shifts, and the Covariant Field Strength

- (a) At a standard null momentum  $k$ , the rotation part of the little group multiplies a helicity polarization by  $e^{i\sigma\theta}$ . Its two translation generators instead give

$$e^\mu(k, \sigma) \mapsto e^{i\sigma\theta} [e^\mu(k, \sigma) + \alpha_\sigma k^\mu].$$

Conjugating by the standard Lorentz transformation that takes  $k$  to  $p$  yields

$$e^\mu(p, \sigma) \mapsto e^{i\sigma\theta} [e^\mu(p, \sigma) + \alpha_\sigma p^\mu].$$

An emission amplitude  $\mathcal{M} = J_\mu e^\mu$  changes by  $\delta\mathcal{M} = \alpha_\sigma J_\mu p^\mu$ . It is therefore independent of the arbitrary little-group translation exactly when

$$p_\mu J^\mu = 0,$$

the momentum-space conservation law. In position space the shift is  $A^\mu \rightarrow A^\mu + \partial^\mu \alpha$ .

- (b) Under  $e^\mu \rightarrow e^\mu + \alpha p^\mu$ ,

$$\delta f^{\mu\nu} = i\alpha(p^\mu p^\nu - p^\nu p^\mu) = 0.$$

For  $p^2 = 0$  and  $p \cdot e = 0$ ,

$$p_\mu f^{\mu\nu} = i(p^2 e^\nu - p^\nu p \cdot e) = 0.$$



The cyclic identity also holds:

$$p_{[\rho} f_{\mu\nu]} = 0,$$

because every term contains a symmetric product of two momenta contracted with an antisymmetrization. These are the source-free Maxwell equations in momentum space.

The self-dual and anti-self-dual combinations

$$f_{\mu\nu}^{\pm} = \frac{1}{2}(f_{\mu\nu} \mp i\tilde{f}_{\mu\nu})$$

transform as  $(1, 0)$  and  $(0, 1)$ , respectively. On a plane wave they select helicity  $-1$  and  $+1$  (the assignment reverses if the duality sign convention is reversed). Together they contain precisely the two photon helicities.

- (c) In Coulomb gauge the mode expansion contains only polarization vectors with  $\mathbf{p} \cdot \mathbf{e}_r = 0$ . Their spatial completeness relation is

$$\sum_{r=1}^2 e_{ri}(\mathbf{p}) e_{rj}^*(\mathbf{p}) = \delta_{ij} - \frac{p_i p_j}{\mathbf{p}^2}.$$

Using the oscillator commutator in the equal-time  $A_i, E_j$  bracket gives

$$\begin{aligned} [A_i(\mathbf{x}, t), E_j(\mathbf{y}, t)] &= i \int \frac{d^3 p}{(2\pi)^3} \left( \delta_{ij} - \frac{p_i p_j}{\mathbf{p}^2} \right) e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} \\ &= i \left( \delta_{ij} - \frac{\partial_i \partial_j}{\nabla^2} \right) \delta^3(\mathbf{x} - \mathbf{y}). \end{aligned}$$

Taking a divergence in either index gives zero. Thus longitudinal canonical data have zero bracket with everything and are removed; the projector has rank two for every nonzero momentum.

### S.5.15. Antiparticles and the Locality of Charged Fields

- (a) Let

$$\Omega = \sqrt{m^2 - \nabla^2}.$$

At a fixed time, comparison with the oscillator expansion gives

$$\phi^{(+)} = \frac{1}{2} (\phi + i\Omega^{-1}\pi), \quad \phi^{(-)} = \frac{1}{2} (\phi - i\Omega^{-1}\pi).$$

In momentum space  $\Omega^{-1}$  multiplies by  $(\mathbf{p}^2 + m^2)^{-1/2}$ . Its position-space kernel is not supported at a point:

$$(\Omega^{-1}\pi)(\mathbf{x}) = \int d^3 y K_m(\mathbf{x} - \mathbf{y}) \pi(\mathbf{y}),$$

with  $K_m$  decaying on the scale  $m^{-1}$  but nonzero away from the origin. Hence separating annihilation from creation frequency is spatially nonlocal. Their sum  $\phi = \phi^{(+)} + \phi^{(-)}$  is nevertheless a local field whose commutator vanishes at spacelike separation.

- (b) An annihilation field made only from  $a(\mathbf{p})e^{ip \cdot x}$  has a nonvanishing positive-frequency commutator at spacelike separation. Adding its Hermitian adjoint cancels that commutator, but the result is Hermitian and therefore neutral: its annihilation and creation pieces carry opposite operator charges but refer to the same species.

For a field of definite charge  $-q$ , one needs

$$\Phi(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left[ a(\mathbf{p})e^{ip \cdot x} + b^\dagger(\mathbf{p})e^{-ip \cdot x} \right],$$

where

$$[Q, a] = -q a, \quad [Q, b^\dagger] = -q b^\dagger.$$

Thus  $a^\dagger$  creates charge  $+q$  and  $b^\dagger$  creates charge  $-q$ . The two frequency pieces now have the same field charge, while their commutators combine into the causal Pauli–Jordan function. The independent  $b$  oscillator is the antiparticle.

- (c) For a real scalar,

$$[\phi(x), \phi(y)] = \Delta(x - y) = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} \left[ e^{ip \cdot (x-y)} - e^{-ip \cdot (x-y)} \right].$$

The mass-shell measure is invariant, so  $\Delta$  is a Lorentz scalar. If  $x - y$  is spacelike, a proper Lorentz transformation brings the two points to equal time. There the two terms cancel after  $\mathbf{p} \rightarrow -\mathbf{p}$ , proving  $\Delta(x - y) = 0$ .

By contrast,

$$\Delta_+(x - y) = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} e^{ip \cdot (x-y)}$$

contains only one frequency sign. There is no second term to cancel it, and it is generally nonzero at spacelike separation. Locality belongs to the full field, not to either frequency part separately.

### S.5.16. Wigner’s Characterization of Antiunitary Operators

The product of two antilinear maps is linear. Moreover, antiunitarity gives

$$\langle \Theta\psi \mid \Theta\phi \rangle = \langle \phi \mid \psi \rangle,$$

and applying this twice yields

$$\langle \Theta^2\psi \mid \Theta^2\phi \rangle = \langle \psi \mid \phi \rangle.$$

Thus  $U \equiv \Theta^2$  is unitary.

As maps,  $U\Theta = \Theta U$ . If  $U|i\rangle = \lambda_i|i\rangle$ , however, antilinearity complex-conjugates the eigenvalue:

$$U\Theta|i\rangle = \Theta U|i\rangle = \lambda_i^* \Theta|i\rangle.$$

Therefore  $\Theta$  pairs the  $e^{-2i\theta}$  eigenspace of  $U$  with its  $e^{2i\theta}$  eigenspace. For each such pair choose orthonormal vectors  $|i\rangle, |\bar{i}\rangle$  and phases so that

$$\boxed{\Theta|i\rangle = e^{i\theta_i}|\bar{i}\rangle, \quad \Theta|\bar{i}\rangle = e^{-i\theta_i}|i\rangle.}$$

Because the first coefficient is conjugated on the second application, these equations give

$$\Theta^2|i\rangle = e^{-2i\theta_i}|i\rangle, \quad \Theta^2|\bar{i}\rangle = e^{2i\theta_i}|\bar{i}\rangle.$$

Since  $\Theta^\dagger = \Theta^{-1}$ , inversion gives

$$\boxed{\Theta^\dagger|i\rangle = e^{-i\theta_i}|\bar{i}\rangle, \quad \Theta^\dagger|\bar{i}\rangle = e^{i\theta_i}|i\rangle.}$$

There are two self-conjugate eigenvalues requiring separate mention. If  $\Theta^2 = +1$ , a real orthonormal basis can be chosen with  $\Theta|a\rangle = |a\rangle$ . If  $\Theta^2 = -1$ , no nonzero state can be fixed by  $\Theta$ ; the states occur in orthogonal Kramers pairs, which may be phased so that

$$\Theta|a\rangle = |b\rangle, \quad \Theta|b\rangle = -|a\rangle.$$

This is the  $\theta = \pi/2$  case of the paired construction after a rephasing.

Finally, for arbitrary coefficients, antilinearity supplies the complete action:

$$\begin{aligned} \Theta \sum_i (c_i|i\rangle + d_i|\bar{i}\rangle) &= \sum_i \left( c_i^* e^{i\theta_i}|\bar{i}\rangle + d_i^* e^{-i\theta_i}|i\rangle \right), \\ \Theta^\dagger \sum_i (c_i|i\rangle + d_i|\bar{i}\rangle) &= \sum_i \left( c_i^* e^{-i\theta_i}|\bar{i}\rangle + d_i^* e^{i\theta_i}|i\rangle \right). \end{aligned}$$

Together with invariant states in the  $+1$  sector and Kramers pairs in the  $-1$  sector, these blocks characterize a general antiunitary operator up to unitary changes of basis within degenerate eigenspaces.

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## 6 The Feynman Rules

In previous chapters the use of covariant free fields in the construction of the Hamiltonian density has been motivated by the requirement that the  $S$ -matrix satisfy Lorentz invariance and cluster decomposition conditions. With the Hamiltonian density constructed in this way, it makes no difference which form of perturbation theory we use to calculate the  $S$ -matrix; the results will automatically satisfy these invariance and clustering conditions in each order in the interaction density. Nevertheless, there are obvious practical advantages in using a version of perturbation theory in which the Lorentz invariance and cluster decomposition properties of the  $S$ -matrix are kept manifest at every stage in the calculation. This was not true for the perturbation theory used in the 1930s, now known as ‘old-fashioned perturbation theory,’ described at the beginning of Section 3.5. The great achievement of Feynman, Schwinger, and Tomonaga in the late 1940s was to develop perturbative techniques for calculating the  $S$ -matrix, in which Lorentz invariance and cluster decomposition properties are transparent throughout. This chapter will outline the diagrammatic calculational technique first described by Feynman at the Poconos Conference in 1948. Feynman was led to these diagrammatic rules in part through his development of a path-integral approach, which will be the subject of Chapter 9. In this chapter, we shall use the approach described by Dyson [1] in 1949, which until the 1970s was the basis of almost all analyses of perturbation theory in quantum field theory, and still provides a particularly transparent introduction to the Feynman rules.

### 6.1 Derivation of the Rules

Our starting point is a formula for the  $S$ -matrix, obtained by putting together the Dyson series (3.5.10) with expression (4.2.2) for the free-particle states:

$$\begin{aligned}
 S_{\mathbf{p}'_1 \sigma'_1 n'_1; \mathbf{p}'_2 \sigma'_2 n'_2; \dots, \mathbf{p}_1 \sigma_1 n_1; \mathbf{p}_2 \sigma_2 n_2; \dots} \\
 = \sum_{N=0}^{\infty} \frac{(-i)^N}{N!} \int d^4 x_1 \cdots d^4 x_N \langle \Omega | \cdots a_{\mathbf{p}'_2 \sigma'_2 n'_2} a_{\mathbf{p}'_1 \sigma'_1 n'_1} \\
 \cdot T \{ \mathcal{H}_I(x_1) \cdots \mathcal{H}_I(x_N) \} a_{\mathbf{p}_1 \sigma_1 n_1}^\dagger a_{\mathbf{p}_2 \sigma_2 n_2}^\dagger \cdots | \Omega \rangle.
 \end{aligned} \tag{6.1.1}$$

As a reminder:  $\mathbf{p}$ ,  $\sigma$ , and  $n$  label particle momenta, spin, and species; primes denote labels for particles in the final state;  $|\Omega\rangle$  is the free-particle vacuum state;  $a$  and  $a^\dagger$  are annihilation and creation operators;  $T$  indicates a time-ordering, which puts the  $\mathcal{H}_I(x)$  in an order in which the arguments  $x^0$  decrease from left to right; and  $\mathcal{H}_I(x)$  is the interaction Hamiltonian density, taken as a polynomial in the fields and their adjoints

$$\mathcal{H}_I(x) = \sum_i g_i \mathcal{H}_{I,i}(x), \tag{6.1.2}$$

each term  $\mathcal{H}_{I,i}$  being a product of definite numbers of fields and field adjoints of each type. The field of a particle of species  $n$  that transforms under a particular representation of the

homogeneous Lorentz group (with or without space inversions) is given by

$$\psi_\ell(x) = \sum_\sigma (2\pi)^{-3/2} \int d^3p \left[ u_\ell(\mathbf{p}, \sigma, n) a_{\mathbf{p}, \sigma, n} e^{ip \cdot x} + v_\ell(\mathbf{p}, \sigma, n) a_{\mathbf{p}, \sigma, n^c}^\dagger e^{-ip \cdot x} \right]. \quad (6.1.3)$$

Here  $n^c$  denotes the antiparticle of the species  $n$ , and  $\exp(\pm ip \cdot x)$  is calculated with  $p^0$  set equal to  $\sqrt{\mathbf{p}^2 + M_n^2}$ . The coefficient functions  $u_\ell$  and  $v_\ell$  depend on the Lorentz transformation properties of the field and the spin of the particle it describes; they were calculated in Chapter 5. (For instance, in the scalar field the  $u_\ell$  for a particle of energy  $E$  is simply  $(2E)^{-1/2}$ , while in a Dirac field  $u_\ell$  and  $v_\ell$  are the normalized Dirac spinors introduced in Section 5.5.) The index  $\ell$  on the field should here be understood to indicate the particle type and the representation of the Lorentz group by which the field transforms, as well as including a running index labelling the components in this representation. There is no need to deal separately with interactions that involve derivatives of fields; from our point of view, the derivative of a field (6.1.3) is just another field described by (6.1.3), with different  $u_\ell$  and  $v_\ell$ . We will here make a distinction between some particle species that we arbitrarily call ‘particles,’ for instance electrons, protons, etc., and those we call ‘antiparticles,’ such as positrons and antiprotons. The field operators that destroy particles and create antiparticles are called simply ‘fields’; their adjoints, which destroy antiparticles and create particles, are called ‘field adjoints.’ Of course, some particle species like the photon and  $\pi^0$  are their own antiparticles; for these the field adjoints are proportional to the fields.

We now proceed to move all annihilation operators to the right in Eq. (6.1.1), repeatedly using for this purpose the commutation or anticommutation relations:

$$a_{\mathbf{p}\sigma n} a_{\mathbf{p}'\sigma' n'}^\dagger = \pm a_{\mathbf{p}'\sigma' n'}^\dagger a_{\mathbf{p}\sigma n} + \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma'\sigma} \delta_{n'n}, \quad (6.1.4)$$

$$a_{\mathbf{p}\sigma n} a_{\mathbf{p}'\sigma' n'} = \pm a_{\mathbf{p}'\sigma' n'} a_{\mathbf{p}\sigma n}, \quad (6.1.5)$$

$$a_{\mathbf{p}\sigma n}^\dagger a_{\mathbf{p}'\sigma' n'}^\dagger = \pm a_{\mathbf{p}'\sigma' n'}^\dagger a_{\mathbf{p}\sigma n}^\dagger \quad (6.1.6)$$

(and likewise for antiparticles), the  $\pm$  sign on the right being  $-$  if both particles  $n, n'$  are fermions, and  $+$  if either or both are bosons. Whenever an annihilation operator appears on the extreme right (or a creation operator on the extreme left), the corresponding contribution to Eq. (6.1.1) vanishes, because these operators annihilate the vacuum state:

$$a_{\mathbf{p}\sigma n} |\Omega\rangle = 0, \quad (6.1.7)$$

$$\langle \Omega | a_{\mathbf{p}\sigma n}^\dagger = 0. \quad (6.1.8)$$

The remaining contributions to Eq. (6.1.1) are those arising from the delta function terms on the right-hand side of Eq. (6.1.4), with every creation and annihilation operator in the initial or final states or in the interaction Hamiltonian density paired in this way with some other annihilation or creation operator.

In this way, the contribution to Eq. (6.1.1) of a given order in each of the terms  $\mathcal{H}_{I,i}$  in the polynomial  $\mathcal{H}_I(\psi(x), \psi^\dagger(x))$  is given by a sum, over all ways of pairing creation and annihilation operators [2], of the integrals of products of factors, as follows:

(a) Pairing of a final particle having quantum numbers  $\mathbf{p}', \sigma', n'$  with a field adjoint  $\psi_\ell^\dagger(x)$  in  $\mathcal{H}_{I,i}(x)$  yields a factor

$$\left[ a_{\mathbf{p}'\sigma'n'}, \psi_\ell^\dagger(x) \right]_{\mp} = (2\pi)^{-3/2} e^{-ip' \cdot x} u_\ell^*(\mathbf{p}', \sigma', n'). \quad (6.1.9)$$

(b) Pairing of a final antiparticle having quantum numbers  $\mathbf{p}', \sigma', n'^c$  with a field  $\psi_\ell(x)$  in  $\mathcal{H}_{I,i}(x)$  yields a factor

$$\left[ a_{\mathbf{p}'\sigma'n'^c}, \psi_\ell(x) \right]_{\mp} = (2\pi)^{-3/2} e^{-ip' \cdot x} v_\ell(\mathbf{p}', \sigma', n'). \quad (6.1.10)$$

(c) Pairing of an initial particle having quantum numbers  $\mathbf{p}, \sigma, n$  with a field  $\psi_\ell(x)$  in  $\mathcal{H}_{I,i}(x)$  yields a factor

$$\left[ \psi_\ell(x), a_{\mathbf{p}\sigma n}^\dagger \right]_{\mp} = (2\pi)^{-3/2} e^{ip \cdot x} u_\ell(\mathbf{p}, \sigma, n). \quad (6.1.11)$$

(d) Pairing of an initial antiparticle having quantum numbers  $\mathbf{p}, \sigma, n^c$  with a field adjoint  $\psi_\ell^\dagger(x)$  in  $\mathcal{H}_{I,i}(x)$  yields a factor

$$\left[ \psi_\ell^\dagger(x), a_{\mathbf{p}\sigma n^c}^\dagger \right]_{\mp} = (2\pi)^{-3/2} e^{ip \cdot x} v_\ell^*(\mathbf{p}, \sigma, n). \quad (6.1.12)$$

(e) Pairing of a final particle (or antiparticle) having numbers  $\mathbf{p}', \sigma', n'$  with an initial particle (or antiparticle) having quantum numbers  $\mathbf{p}, \sigma, n$  yields a factor

$$\left[ a_{\mathbf{p}'\sigma'n'}, a_{\mathbf{p}\sigma n}^\dagger \right]_{\mp} = \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma'\sigma} \delta_{n'n}. \quad (6.1.13)$$

(f) Pairing of a field  $\psi_\ell(x)$  in  $\mathcal{H}_{I,i}(x)$  with a field adjoint  $\psi_m^\dagger(y)$  in  $\mathcal{H}_{I,j}(y)$  yields a factor<sup>53</sup>

$$\begin{aligned} \theta(x - y) \left[ \psi_\ell^{(+)}(x), \psi_m^{(+)\dagger}(y) \right]_{\mp} \pm \theta(y - x) \left[ \psi_m^{(-)\dagger}(y), \psi_\ell^{(-)}(x) \right]_{\mp} \\ \equiv -i\Delta_{\ell m}(x, y), \end{aligned} \quad (6.1.14)$$

where  $\psi^{(+)}$  and  $\psi^{(-)}$  are the terms in  $\psi$  that destroy particles and create antiparticles, respectively:

$$\psi_\ell^{(+)}(x) = (2\pi)^{-3/2} \int d^3p \sum_{\sigma} u_\ell(\mathbf{p}\sigma n) e^{ip \cdot x} a_{\mathbf{p}\sigma n}, \quad (6.1.15)$$

$$\psi_\ell^{(-)}(x) = (2\pi)^{-3/2} \int d^3p \sum_{\sigma} v_\ell(\mathbf{p}\sigma n) e^{-ip \cdot x} a_{\mathbf{p}\sigma n^c}^\dagger. \quad (6.1.16)$$

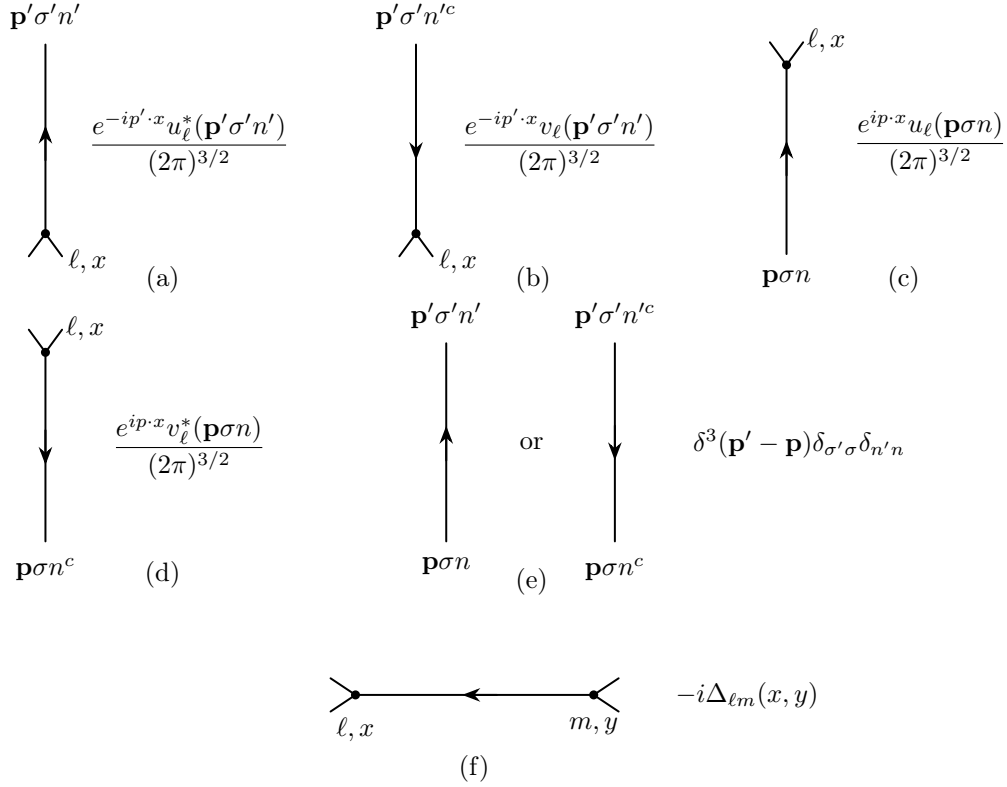
Recall that  $\theta(x - y)$  is a step function, equal to +1 for  $x^0 > y^0$  and zero for  $x^0 < y^0$ . These step functions appear in Eq. (6.1.14) because of the time-ordering in Eq. (6.1.1); we can encounter a pairing of an annihilation field  $\psi^{(+)}(x)$  in  $\mathcal{H}_I(x)$  with a creation field  $\psi^{(+)\dagger}(y)$  in  $\mathcal{H}_I(y)$  only if  $\mathcal{H}_I(x)$  was initially to the left of  $\mathcal{H}_I(y)$  in Eq. (6.1.1), i.e., if  $x^0 > y^0$ ;

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<sup>53</sup>If the interaction  $\mathcal{H}_I(x)$  is written in the normal-ordered form, as in Eq. (5.1.33), then there is no pairing of fields and field adjoints in the *same* interaction. Otherwise some sort of regularization is needed to give meaning to  $\Delta_{\ell m}(0)$ .

similarly, we encounter a pairing of an annihilation field  $\psi^{(-)\dagger}(y)$  in  $\mathcal{H}_I(y)$  with a creation field  $\psi^{(-)}(x)$  in  $\mathcal{H}_I(x)$  only if  $\mathcal{H}_I(y)$  was initially to the left of  $\mathcal{H}_I(x)$  in Eq. (6.1.1), i.e., if  $y^0 > x^0$ . (The  $\pm$  sign in the second term in (6.1.14) will be explained a little later.) The quantity (6.1.14) is known as a *propagator*; it is calculated in the following section.

The  $S$ -matrix is obtained by multiplying these factors together, along with additional numerical factors to be discussed below, then integrating over  $x_1 \cdots x_N$ , then summing over all pairings, and then over the numbers of interaction of each type. Before filling in all the details, it will be convenient first to describe a diagrammatic formalism for keeping track of all these pairings.



**Figure 6.1.** Graphical representation of pairings of operators arising in the coordinate-space evaluation of the  $S$ -matrix. The expressions on the right are the factors that must be included in the coordinate-space integrand of the  $S$ -matrix for each line of the Feynman diagram.

The rules for calculating the  $S$ -matrix are conveniently summarized in terms of *Feynman diagrams*. (See Figure 6.1.) The diagrams consist of points called *vertices*, each representing one of the  $\mathcal{H}_{I,i}(x)$ , and *lines*, each representing the pairing of a creation with an annihilation operator. More specifically:

(a) The pairing of a final particle with a field adjoint in one of the  $\mathcal{H}_{I,i}(x)$  is represented by a line running from the vertex representing that  $\mathcal{H}_{I,i}(x)$  upwards out of the diagram, carrying an arrow pointed upwards.

(b) The pairing of a final antiparticle with a field in one of the  $\mathcal{H}_{I,i}(x)$  is also represented



by a line running from the vertex representing that  $\mathcal{H}_{I,i}(x)$  upwards out of the diagram, but carrying an arrow pointed downwards. (Arrows are omitted throughout for particles like  $\gamma$ ,  $\pi^0$ , etc. that are their own antiparticles.)

(c) The pairing of an initial particle with a field in one of the  $\mathcal{H}_{I,i}(x)$  is represented by a line running into the diagram from below, ending in the vertex representing that  $\mathcal{H}_{I,i}(x)$ , carrying an arrow pointed upwards.

(d) The pairing of an initial antiparticle with a field in one of the  $\mathcal{H}_{I,i}(x)$  is also represented by a line running into the diagram from below, ending in the vertex representing that  $\mathcal{H}_{I,i}(x)$ , but carrying an arrow pointed downwards.

(e) The pairing of a final particle or antiparticle with an initial particle or antiparticle is represented by a line running clear through the diagram from bottom to top, not touching any vertex, with an arrow pointed upwards or downwards for particles or antiparticles, respectively.

(f) The pairing of a field in  $\mathcal{H}_{I,i}(x)$  with a field adjoint in  $\mathcal{H}_{I,j}(y)$  is represented by a line joining the vertices representing  $\mathcal{H}_{I,i}(x)$  and  $\mathcal{H}_{I,j}(y)$ , carrying an arrow pointing from  $y$  to  $x$ .

Note that arrows always point in the direction a particle is moving, and opposite to the direction an antiparticle is moving. (As mentioned above, arrows should be omitted for particles like photons that are their own antiparticles.) The arrow direction indicated in rule (f) is consistent with this convention because a field adjoint in  $\mathcal{H}_{I,j}(y)$  can either create a particle destroyed by a field in  $\mathcal{H}_{I,i}(x)$ , or destroy an antiparticle created by a field in  $\mathcal{H}_{I,i}(x)$ . Note also that since every field or field adjoint in  $\mathcal{H}_{I,i}(x)$  must be paired with something, the total number of lines at a vertex of type  $i$ , corresponding to a term  $\mathcal{H}_{I,i}(x)$  in Eq. (6.1.2), is just equal to the total number of field or field adjoint factors in  $\mathcal{H}_{I,i}(x)$ . Of these lines, the number with arrows pointed into the vertex or out of it equals the number of fields or field adjoints respectively in the corresponding interaction term.

To calculate the contribution to the  $S$ -matrix for a given process, of a given order  $N_i$  in each of the interaction terms  $\mathcal{H}_{I,i}(x)$  in Eq. (6.1.2), we must carry out the following steps:

(i) Draw all Feynman diagrams containing  $N_i$  vertices of each type  $i$ , and containing a line coming into the diagrams from below for each particle or antiparticle in the initial state, and a line going upwards out of the diagram for every particle or antiparticle in the final state, together with any number of internal lines running from one vertex to another, as required to give each vertex the proper number of attached lines. The lines carry arrows as described above, each of which may point upwards or downwards. Each vertex is labelled with an interaction type  $i$  and spacetime coordinate  $x^\mu$ . Each internal or external line is labelled at the end where it runs into a vertex with a field type  $\ell$  (corresponding to the field  $\psi_\ell(x)$  or  $\psi_\ell^\dagger(x)$  that creates or destroys the particle or antiparticle at that vertex), and each external line where it enters or leaves the diagram is labelled with the quantum numbers  $\mathbf{p}, \sigma, n$  or  $\mathbf{p}', \sigma', n'$  of the initial or final particle (or antiparticle).

(ii) For each vertex of type  $i$ , include a factor  $-i$  (from the  $(-i)^N$  in Eq. (6.1.1)) and a factor  $g_i$  (the coupling constant multiplying the product of fields in  $\mathcal{H}_{I,i}(x)$ ). For each line running upwards out of the diagram, include a factor (6.1.9) or (6.1.10), depending on whether the arrow is pointing up or down. For each line running from below into the

diagram, include a factor (6.1.11) or (6.1.12), again depending on the arrow direction. For each line running straight through the diagram include a factor (6.1.13). For each internal line connecting two vertices include a factor (6.1.14).

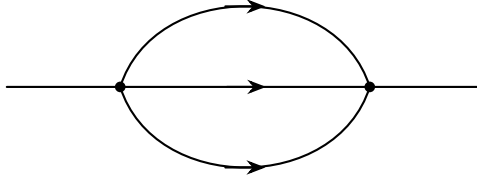
(iii) Integrate the product of all these factors over the coordinates  $x_1, x_2, \dots$  of each vertex.

(iv) Add up the results obtained in this way from each Feynman diagram. The complete perturbation series for the  $S$ -matrix is obtained by adding up the contributions of each order in each interaction type, up to whatever order our strength permits.

Note that we have not included the factor  $1/N!$  from Eq. (6.1.1) in these rules, because the time-ordered product in Eq. (6.1.1) is a sum over the  $N!$  permutations of  $x_1 x_2 \cdots x_N$ , each permutation giving the same contribution to the final result. To put this another way, a Feynman diagram with  $N$  vertices is one of  $N!$  identical diagrams, which differ only in permutations of the labels on the vertices, and this yields a factor of  $N!$  which cancels the  $1/N!$  in Eq. (6.1.1). (There are exceptions to this rule, discussed below.) For this reason, henceforth we do *not* include more than one of a set of Feynman diagrams that differ only by relabelling the vertices.

In some cases, there are additional combinatoric factors or signs that must be included in the contribution of individual Feynman diagrams:

(v) Suppose that an interaction  $\mathcal{H}_{I,i}(x)$  contains (among other fields and field adjoints)  $M$  factors of the same field. Suppose that each of these fields is paired with a field adjoint in a different interaction (different for each one), or in the initial or final state. The first of these field adjoints



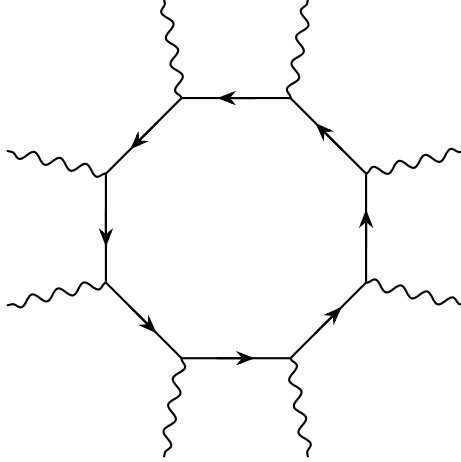
**Figure 6.2.** Example of a graph requiring extra combinatoric factors in the  $S$ -matrix. For an interaction involving, say, three factors of some field (as well as other fields) we usually include a factor  $1/3!$  in the interaction Hamiltonian, to cancel factors arising from sums over ways of pairing these fields with their adjoints in other interactions. But in this diagram there are two such factors of  $1/3!$  and only  $3!$  different pairings, so we are left with an extra factor of  $1/3!$ .

can be paired with any one of the  $M$  identical fields in  $\mathcal{H}_{I,i}(x)$ ; the second with any one of the remaining  $M - 1$  identical fields; and so on, yielding an extra factor of  $M!$ . To compensate for this, it is conventional to define the coupling constants  $g_i$  so that an explicit factor  $1/M!$  appears in any  $\mathcal{H}_{I,i}(x)$  containing  $M$  identical fields (or field adjoints.) For instance, the interaction of  $M$ th order in a scalar field  $\phi(x)$  would be written  $g\phi^M/M!$ . (More generally, one often also displays an explicit factor of  $1/M!$  when the interaction involves a sum of  $M$  factors of fields from the same symmetry multiplet, or when for this or any other reason the coupling coefficient is totally symmetric or antisymmetric under permutations of  $M$  boson or fermion fields.)

However, this cancellation of  $M!$  factors is not always complete. For instance, consider a Feynman diagram in which the  $M$  identical fields in one interaction  $\mathcal{H}_{I,i}(x)$  are paired with  $M$  corresponding field adjoints in a *single* other interaction  $\mathcal{H}_{I,j}(y)$ . (See Figure 6.2.) Then by following the above analysis, we find only  $M!$  different pairings (since it makes no difference which of the field adjoints we call the first, second,  $\dots$ ), cancelling only one of the two factors of  $1/M!$  in the two different interactions. In this case, we would have to insert an extra factor of  $1/M!$  ‘by hand’ into the contribution of such a Feynman diagram.

Other combinatoric factors arise when some of the permutations of vertices have no effect on the Feynman diagram. We noted earlier that the factor  $1/N!$  in the series (6.1.1) is usually cancelled by the sum over the  $N!$  diagrams that differ only in the labelling of the  $N$  vertices. However, this cancellation is incomplete when relabelling the vertices does not yield a new diagram. This happens most commonly in the calculation of vacuum-to-vacuum  $S$ -matrix elements in a theory with a quadratic interaction  $\mathcal{H}_I = \psi_\ell^\dagger M_{\ell\ell'} \psi_{\ell'}$ , where  $M$  may depend on external fields. (The physical significance of such vacuum fluctuation diagrams is discussed in detail in Volume II.) The Feynman diagram of  $N$ th order in  $\mathcal{H}_I$  is a ring with  $N$  corners. (See Figure 6.3.) There are only  $(N-1)!$  different diagrams here because a permutation of labels that moves each label to the next vertex around the ring yields the same diagram. Hence such a graph is accompanied with a factor

$$\frac{(N-1)!}{N!} = \frac{1}{N}. \quad (6.1.17)$$



**Figure 6.3.** An eighth-order graph for the vacuum-to-vacuum amplitude with particles interacting only with an external field. In this diagram the external field is represented by wiggly lines. There are  $7!$  such diagrams, differing only by relabelling the vertices, and not counting as different those labellings that simply rotate the ring. The factor  $1/8!$  from the Dyson formula (6.1.1) is therefore not entirely cancelled here, leaving us with an extra factor  $1/8$ .

(vi) In theories involving fermion fields, the use of Eqs. (6.1.4)–(6.1.6) to move annihilation and creation operators to the right and left introduces minus signs into the contribution of various pairings. To be specific, we get a minus sign wherever the permutation of the

operators in Eq. (6.1.1) that is required to put all paired operators adjacent to one another (with annihilation operators just to the left of the paired creation operators) involves an odd number of interchanges of fermion operators. (This is because to compute the contribution of a certain pairing, we can first permute all operators in Eq. (6.1.1) so that each annihilation operator is just to the left of the creation operator with which it is paired, ignoring all commutators and anticommutators of unpaired operators, and then replace each product of paired operators with their commutators or anticommutators.) One immediate consequence is to produce the minus sign in the relative sign of the two terms in Eq. (6.1.14) for the fermion propagator. Whatever permutation puts the annihilation part  $\psi^{(+)}(x)$  of a field in  $\mathcal{H}_I(x)$  just to the left of the creation part  $\psi^{(+)\dagger}(y)$  of a field adjoint in  $\mathcal{H}_I(y)$ , the permutation that puts the annihilation part  $\psi^{(-)\dagger}(y)$  of the field adjoint just to the left of the creation part  $\psi^{(-)}(x)$  of the field involves one extra interchange of fermion operators, yielding the minus sign in the second term of Eq. (6.1.14) for fermions.

In addition, minus signs can arise in the contribution of whole Feynman diagrams. As an example, let us take up a theory in which the sole interaction of fermions takes the form

$$\mathcal{H}_I(x) = \sum_{\ell mk} g_{\ell mk} \psi_{\ell}^{\dagger}(x) \psi_m(x) \phi_k(x), \quad (6.1.18)$$

where  $g_{\ell mk}$  are general constants,  $\psi_{\ell}(x)$  are a set of complex fermion fields, and  $\phi_m(x)$  are a set of real bosonic (but not necessarily scalar) fields. (Not only quantum electrodynamics, but the whole ‘standard model’ of weak, electromagnetic, and strong interactions, has fermionic interactions that can all be put in this form.) Let us first take up the process of fermion–fermion scattering,  $12 \rightarrow 1'2'$ , to second order in  $\mathcal{H}_I$ . The fermion operators in the second-order term in Eq. (6.1.1) appear in the order (with obvious abbreviations)

$$a_{2'} a_{1'} \psi^{\dagger}(x) \psi(x) \psi^{\dagger}(y) \psi(y) a_1^{\dagger} a_2^{\dagger}. \quad (6.1.19)$$

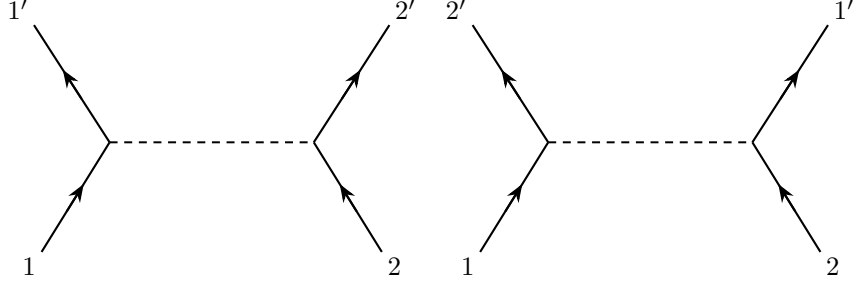
There are two connected diagrams to this order, corresponding to the pairings

$$[a_{2'} \psi^{\dagger}(x)] [a_{1'} \psi^{\dagger}(y)] [\psi(y) a_1^{\dagger}] [\psi(x) a_2^{\dagger}] \quad (6.1.20)$$

and

$$[a_{1'} \psi^{\dagger}(x)] [a_{2'} \psi^{\dagger}(y)] [\psi(y) a_1^{\dagger}] [\psi(x) a_2^{\dagger}]. \quad (6.1.21)$$

(See Figure 6.4.) To go from (6.1.19) to (6.1.20) requires an even permutation of fermionic operators. (For instance, move  $\psi(x)$  past three operators to the right, and then move  $a_{1'}$  past one operator to the right.) Thus there is no extra minus sign in the contribution of the pairing (6.1.20). This in itself is not so important; the *overall* sign of the  $S$ -matrix does not matter in transition rates, and in any case depends on sign conventions for the initial and final states. What is important is that the contributions of pairings (6.1.20) and (6.1.21) have opposite sign, as can be seen most easily by noting that the only difference between these two pairings is the interchange of two fermionic operators,  $a_{1'}$  and  $a_{2'}$ . In fact, this relative minus sign is just what is required by Fermi statistics: it makes the scattering amplitude antisymmetric under the interchange of particles  $1'$  and  $2'$  (or 1 and 2).



**Figure 6.4.** The connected second-order diagrams for fermion–fermion scattering in a theory with interaction (6.1.18). Here straight lines represent fermions; dotted lines are neutral bosons. There is a minus sign difference in the contributions of these two diagrams, arising from an extra interchange of fermion operators in the pairings represented by the second diagram.

However, it must not be thought that all sign factors can be related in such a simple way to the antisymmetry of the final or initial states, even in the lowest order of perturbation theory. To illustrate this point, let’s now consider fermion–antifermion scattering,  $1, 2^c \rightarrow 1' 2'^c$ , to second order in the same interaction (6.1.18). The fermionic operators in the second-order term in Eq. (6.1.1) appear in the order:

$$a_{2'^c} a_{1'} \psi^\dagger(x) \psi(x) \psi^\dagger(y) \psi(y) a_1^\dagger a_{2^c}^\dagger. \quad (6.1.22)$$

Here again there are two Feynman diagrams to this order, corresponding to the pairings

$$[a_{2'^c} \psi(x)] [a_{1'} \psi^\dagger(x)] [\psi(y) a_1^\dagger] [\psi^\dagger(y) a_{2^c}^\dagger] \quad (6.1.23)$$

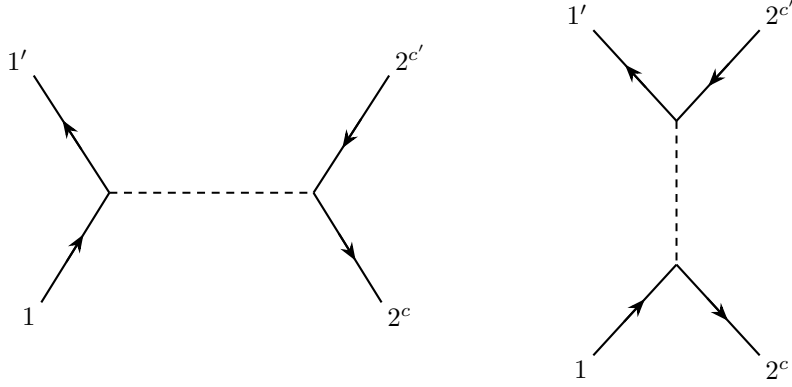
and

$$[a_{2'^c} \psi(x)] [a_{1'} \psi^\dagger(y)] [\psi(y) a_1^\dagger] [\psi^\dagger(x) a_{2^c}^\dagger]. \quad (6.1.24)$$

(See Figure 6.5.) To go from (6.1.22) to (6.1.23) requires an even permutation of fermionic operators (for instance, move  $\psi(x)$  past two operators to the left and move  $\psi^\dagger(y)$  past two operators to the right) so there is no extra minus sign in the contribution of the pairing (6.1.23). On the other hand, to go from (6.1.22) to (6.1.24) requires an odd permutation of fermionic operators (the same as for (6.1.23), *plus* the interchange of  $\psi^\dagger(x)$  and  $\psi^\dagger(y)$ ) so the contribution of this pairing does come with an extra minus sign.<sup>54</sup>

Additional signs are encountered when we consider contributions of higher order. In theories of the type considered here, in which the interactions of fermions all take the form

<sup>54</sup>Actually, this sign is not wholly unrelated to the requirements of Fermi statistics. The same field can destroy a particle and create an antiparticle, so there is a relation, known as ‘crossing symmetry’, between processes in which initial particles or antiparticles are exchanged with final antiparticles or particles. In particular the amplitudes for the process  $1, 2^c \rightarrow 1' 2'^c$  are related to those for the ‘crossed’ process  $1, 2' \rightarrow 1' 2$ ; the two pairings (6.1.23) and (6.1.24) just correspond to the two diagrams for this process, which differ by an interchange of 1 and 2' (or 1' and 2), so the antisymmetry of the scattering amplitude under interchange of initial (or final) particles naturally requires a minus sign in the relative contribution of these two pairings. However, crossing symmetry is not an ordinary symmetry (it involves an analytic continuation in kinematic variables) and it is difficult to use it with any precision for general processes.



**Figure 6.5.** The connected second-order diagrams for fermion–antifermion scattering in a theory with interaction (6.1.18). Here straight lines represent fermions or antifermions, depending on the arrow direction; dotted lines are neutral bosons. There is again a minus sign difference in the contributions of these two diagrams, arising from an extra interchange of fermion operators in the pairings represented by the second diagram.

(6.1.18), the fermion lines in general Feynman diagrams form either chains of lines that pass through the diagram with arbitrary numbers of interactions with the boson fields, as in Figure 6.6, or else fermionic *loops*, like that shown in Figure 6.7. Consider the effect of adding a fermionic loop with  $M$  corners to the Feynman diagram for any process. This corresponds to the pairing of fermionic operators

$$[\psi(x_1)\psi^\dagger(x_2)][\psi(x_2)\psi^\dagger(x_3)] \cdots [\psi(x_M)\psi^\dagger(x_1)]. \quad (6.1.25)$$

On the other hand, these operators appear in Eq. (6.1.1) in the order

$$\psi^\dagger(x_1)\psi(x_1)\psi^\dagger(x_2)\psi(x_2) \cdots \psi^\dagger(x_M)\psi(x_M). \quad (6.1.26)$$

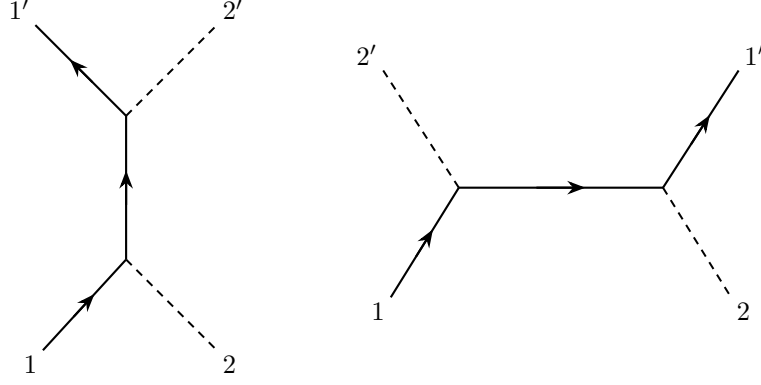
To go from (6.1.26) to (6.1.25) requires an odd permutation of fermionic operators (move  $\psi^\dagger(x_1)$  to the right past  $2M - 1$  operators) so the contribution of each such fermionic loop is accompanied with a minus sign.

These rules yield the full  $S$ -matrix, including contributions from processes in which various clusters of particles interact in widely separated regions of spacetime. As discussed in Chapter 4, to calculate the part of the  $S$ -matrix that excludes such contributions, we should include only *connected* Feynman diagrams. In particular, this excludes lines passing clean through the diagram without interacting, which would yield the factors (6.1.13).

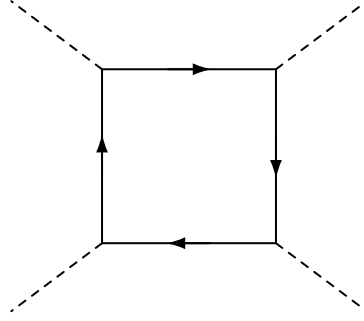
To make the Feynman rules perfectly clear, we will calculate the low-order contributions to the  $S$ -matrix for particle scattering in two different theories.

### Theory I

Consider the theory of fermions and self-charge-conjugate bosons with interaction (6.1.18). The lowest-order connected diagrams for fermion–boson scattering are shown in



**Figure 6.6.** The connected second-order diagrams for boson–fermion scattering in a theory with interaction (6.1.18). Straight lines are fermions; dashed lines are neutral bosons.



**Figure 6.7.** The lowest-order connected diagram for boson–boson scattering in a theory with interaction (6.1.18). Such fermion loop graphs yield an extra minus sign, arising from permutations of the paired fermion fields.

Figure 6.6. Following the rules outlined in Figure 6.1, the corresponding  $S$ -matrix element is

$$\begin{aligned}
 & S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\
 &= (2\pi)^{-6} \sum_{\ell' m' k' \ell m k} (-i)^2 g_{\ell' m' k'} g_{\ell m k} u_{\ell'}^*(\mathbf{p}'_1 \sigma'_1 n'_1) u_{\ell}(\mathbf{p}_1 \sigma_1 n_1) \\
 &\quad \cdot \int d^4 x \int d^4 y (-i \Delta_{m' m}(y - x)) e^{-ip'_1 \cdot y} e^{ip_1 \cdot x} \\
 &\quad \cdot \left[ e^{-ip'_2 \cdot y} u_{k'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) e^{ip_2 \cdot x} u_k(\mathbf{p}_2 \sigma_2 n_2) \right. \\
 &\quad \quad \left. + e^{-ip'_2 \cdot x} u_{k'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) e^{ip_2 \cdot y} u_k(\mathbf{p}_2 \sigma_2 n_2) \right].
 \end{aligned} \tag{6.1.27}$$

(The labels 1 and 2 are used here for fermions and bosons, respectively.) For fermion–fermion scattering there are also two second-order diagrams, shown in Figure 6.4. They

yield the  $S$ -matrix element

$$\begin{aligned}
& S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\
&= (2\pi)^{-6} \sum_{k' \ell' m' k \ell m} (-i)^2 g_{m' \ell k'} g_{\ell' m k} u_{m'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_{\ell'}^*(\mathbf{p}'_1 \sigma'_1 n'_1) u_m(\mathbf{p}_2 \sigma_2 n_2) u_\ell(\mathbf{p}_1 \sigma_1 n_1) \\
&\quad \cdot \int d^4 x \int d^4 y e^{-ip'_2 \cdot x} e^{-ip'_1 \cdot y} e^{ip_2 \cdot x} e^{ip_1 \cdot y} (-i \Delta_{k'k}(x-y)) - [1' \rightleftharpoons 2']. \tag{6.1.28}
\end{aligned}$$

with the last term indicating subtraction of the preceding term with interchange of particles  $1'$  and  $2'$  (or equivalently  $1$  and  $2$ ). There are no second-order graphs for boson-boson scattering in this theory; the lowest-order graphs are of fourth order, such as that shown in Figure 6.7. More specific examples of formulas like Eqs. (6.1.27) and (6.1.28) will be given in Section 6.3, after we have had a chance to evaluate the propagators and go over to momentum space.

In the interaction (6.1.18) of the preceding example, the three fields are all different. It is instructive also to look at an example with a trilinear interaction in which all three fields are the same, or at least enter into the interaction in a symmetric way.

### Theory II

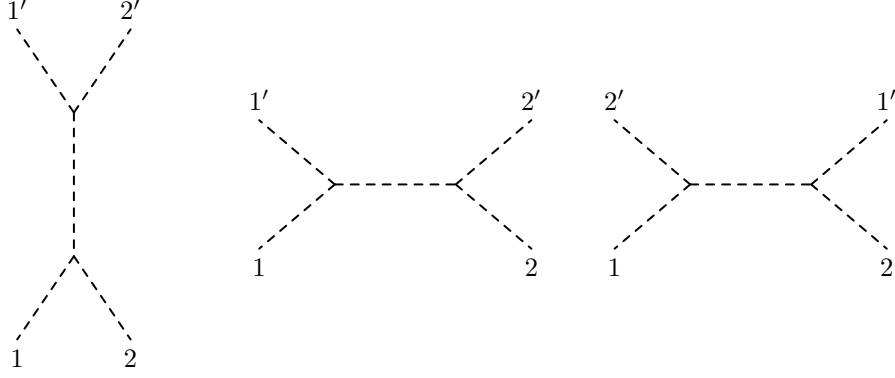
Now take the interaction density to be a sum of terms that are trilinear in a set of *real* bosonic fields  $\phi_\ell(x)$ :

$$\mathcal{H}_I(x) = \frac{1}{3!} \sum_{\ell mn} g_{\ell mn} \phi_\ell(x) \phi_m(x) \phi_n(x) \tag{6.1.29}$$

with  $g_{\ell mn}$  a real totally symmetric coupling coefficient. Suppose we want to consider a scattering process  $1, 2 \rightarrow 1'2'$  to second order in this interaction. Each of the two vertices must have two of the four external lines attached to it. (The only other possibility is that one of the external lines is attached to one vertex and three to the other vertex, but the vertex with three external lines attached to it would have no remaining lines to connect it to the other vertex so this would be a disconnected contribution.) The additional line required at each vertex must then just serve to connect the two vertices to each other. There are three graphs of this type, differing in whether the other external line that is attached to the same vertex as line 1 is line 2 or  $1'$  or  $2'$ . (See Figure 6.8.) Following the rules given above, the contribution to the  $S$ -matrix from those three diagrams is

$$\begin{aligned}
& S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\
&= (-i)^2 (2\pi)^{-6} \sum_{\ell \ell' \ell'' m m' m''} g_{\ell \ell' \ell''} g_{m m' m''} \int d^4 x \int d^4 y (-i \Delta_{\ell'' m''}(x, y)) \\
&\quad \cdot \left[ u_\ell^*(\mathbf{p}'_1 \sigma'_1 n'_1) e^{-ip'_1 \cdot x} u_{\ell'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) e^{-ip'_2 \cdot x} u_m(\mathbf{p}_1 \sigma_1 n_1) e^{ip_1 \cdot y} u_{m'}(\mathbf{p}_2 \sigma_2 n_2) e^{ip_2 \cdot y} \right. \\
&\quad + u_\ell^*(\mathbf{p}'_1 \sigma'_1 n'_1) e^{-ip'_1 \cdot x} u_{\ell'}(\mathbf{p}_1 \sigma_1 n_1) e^{ip_1 \cdot x} u_{m'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) e^{-ip'_2 \cdot y} u_m(\mathbf{p}_2 \sigma_2 n_2) e^{ip_2 \cdot y} \\
&\quad \left. + u_\ell^*(\mathbf{p}'_2 \sigma'_2 n'_2) e^{-ip'_2 \cdot x} u_{\ell'}(\mathbf{p}_1 \sigma_1 n_1) e^{ip_1 \cdot x} u_{m'}^*(\mathbf{p}'_1 \sigma'_1 n'_1) e^{-ip'_1 \cdot y} u_m(\mathbf{p}_2 \sigma_2 n_2) e^{ip_2 \cdot y} \right]. \tag{6.1.30}
\end{aligned}$$





**Figure 6.8.** The connected-second order diagrams for boson–boson scattering in a theory with interaction (6.1.29).

To be even more specific, if the bosons in this theory are spinless particles of a single species, then we write the interaction (6.1.29) in the form

$$\mathcal{H}_I = g\phi^3/3! \quad (6.1.31)$$

and the  $S$ -matrix element (6.1.30) for scalar–scalar scattering is

$$\begin{aligned} S_{\mathbf{p}'_1 \mathbf{p}'_2, \mathbf{p}_1 \mathbf{p}_2} = & \frac{ig^2}{(2\pi)^6 \sqrt{16E'_1 E'_2 E_1 E_2}} \int d^4x \int d^4y \Delta_F(x-y) \\ & \cdot \left[ \exp(-i(p'_1 + p'_2) \cdot x) \exp(i(p_1 + p_2) \cdot y) \right. \\ & + \exp(i(p_1 - p'_1) \cdot x) \exp(i(p_2 - p'_2) \cdot y) \\ & \left. + \exp(i(p_1 - p'_2) \cdot x) \exp(i(p_2 - p'_1) \cdot y) \right], \end{aligned}$$

where  $\Delta_F(x-y)$  is the scalar field propagator, calculated in the next section. There are no terms of third order in  $\mathcal{H}_I(x)$ , or of any odd order in  $\mathcal{H}_I(x)$ .

## 6.2 Calculation of the Propagator

We now turn to a calculation of the propagator (6.1.14), an essential ingredient in the Feynman rules that arises in the pairing of a field  $\psi_\ell(x)$  with a field adjoint  $\psi_m^\dagger(y)$ . Inserting Eqs. (6.1.15) and (6.1.16) in Eq. (6.1.14), and using the commutation or anticommutation relations for annihilation and creation operators, we have immediately

$$\begin{aligned} -i\Delta_{\ell m}(x, y) = & \theta(x-y)(2\pi)^{-3/2} \int d^3p \sum_{\sigma} u_{\ell}(\mathbf{p}\sigma n) u_m^*(\mathbf{p}\sigma n) e^{ip \cdot (x-y)} \\ & \pm \theta(y-x)(2\pi)^{-3/2} \int d^3p \sum_{\sigma} v_m^*(\mathbf{p}\sigma n) v_{\ell}(\mathbf{p}\sigma n) e^{ip \cdot (y-x)}. \end{aligned} \quad (6.2.1)$$

In the course of calculating commutators and anticommutators in Chapter 5, we showed that

$$\sum_{\sigma} u_{\ell}(\mathbf{p}\sigma n) u_m^*(\mathbf{p}\sigma n) = \left(2\sqrt{\mathbf{p}^2 + m_n^2}\right)^{-1} P_{\ell m}(\mathbf{p}, \sqrt{\mathbf{p}^2 + m_n^2}), \quad (6.2.2)$$

$$\sum_{\sigma} v_{\ell}(\mathbf{p}\sigma n) v_m^*(\mathbf{p}\sigma n) = \pm \left(2\sqrt{\mathbf{p}^2 + m_n^2}\right)^{-1} P_{\ell m}(-\mathbf{p}, -\sqrt{\mathbf{p}^2 + m_n^2}), \quad (6.2.3)$$

where  $P_{\ell m}(\mathbf{p}, \omega)$  is a polynomial in  $\mathbf{p}$  and  $\omega$ . (Here as in Eq. (6.2.1), the top and bottom signs refer to bosonic and fermionic fields, respectively.) For instance, if  $\psi_{\ell}(x)$  and  $\psi_m(y)$  are scalar fields  $\phi(x)$  and  $\phi(y)$  for a particle of spin zero, then we have simply

$$P(p) = 1. \quad (6.2.4)$$

If  $\psi_{\ell}(x)$  and  $\psi_m(y)$  are Dirac fields for a particle of spin  $\frac{1}{2}$ , then

$$P_{\ell m}(p) = [(-i\gamma_{\mu} p^{\mu} + m)\beta]_{\ell m}, \quad (6.2.5)$$

where  $\ell$  and  $m$  are here four-valued Dirac indices. (The matrix  $\beta$  appears here because we are considering the pairing of  $\psi_{\ell}(x)$  with  $\psi_m^{\dagger}(y)$ . It is absent in the pairing of  $\psi_{\ell}(x)$  with  $\bar{\psi}_m(y) = \psi_m^{\dagger}(y)\beta$ .) If  $\psi_{\ell}(x)$  and  $\psi_m(y)$  are vector fields  $V_{\mu}(x)$  and  $V_{\nu}(y)$  for a particle of spin one, then

$$P_{\mu\nu}(p) = \eta_{\mu\nu} + m^{-2} p_{\mu} p_{\nu}. \quad (6.2.6)$$

More generally, if  $\psi_{\ell}(x)$  and  $\psi_m(y)$  are components of fields  $\psi_{ab}(x)$  and  $\psi_{\bar{a}\bar{b}}(y)$  for a particle of spin  $j$ , in the irreducible  $(A, B)$  and  $(\bar{A}, \bar{B})$  representations of the homogeneous Lorentz group, then

$$\begin{aligned} P_{ab, \bar{a}\bar{b}}(p) = & \sum_{a'b'} \sum_{\bar{a}'\bar{b}'} \sum_{\sigma} C_{AB}(j\sigma, a'b') C_{\bar{A}\bar{B}}(j\sigma, \bar{a}'\bar{b}') \\ & \cdot \left[ \exp(-\theta \hat{\mathbf{p}} \cdot \mathbf{J}^{(A)}) \right]_{aa'} \left[ \exp(+\theta \hat{\mathbf{p}} \cdot \mathbf{J}^{(B)}) \right]_{bb'} \\ & \cdot \left[ \exp(-\theta \hat{\mathbf{p}} \cdot \mathbf{J}^{(\bar{A})}) \right]_{\bar{a}\bar{a}'} \left[ \exp(+\theta \hat{\mathbf{p}} \cdot \mathbf{J}^{(\bar{B})}) \right]_{\bar{b}\bar{b}'} . \end{aligned} \quad (6.2.7)$$

where  $\sinh \theta = |\mathbf{p}|/m$ , while  $a, b, \bar{a}, \bar{b}$  run by unit steps from  $-A$  to  $+A$ ,  $-B$  to  $+B$ ,  $-\bar{A}$  to  $+\bar{A}$ , and  $-\bar{B}$  to  $+\bar{B}$ , respectively, and likewise for the running indices  $a', b', \bar{a}'$ , and  $\bar{b}'$ .

Inserting Eqs. (6.2.2) and (6.2.3) in Eq. (6.2.1) yields

$$\begin{aligned} -i\Delta_{\ell m}(x, y) = & \theta(x - y) P_{\ell m} \left( -i \frac{\partial}{\partial x} \right) \Delta_+(x - y) \\ & + \theta(y - x) P_{\ell m} \left( -i \frac{\partial}{\partial x} \right) \Delta_+(y - x), \end{aligned} \quad (6.2.8)$$

where  $\Delta_+(x)$  is the function introduced in Chapter 5

$$\Delta_+(x) \equiv (2\pi)^{-3} \int d^3 p (2p^0)^{-1} e^{ip \cdot x} \quad (6.2.9)$$

in which  $p^0$  is taken as  $+\sqrt{\mathbf{p}^2 + m^2}$ .

To go further, we must say a bit about how to extend the definition of the polynomial  $P(p)$ . Eqs. (6.2.2) and (6.2.3) only define  $P(p)$  for four-momenta ‘on the mass shell’, i.e., with  $p^0 = \pm\sqrt{\mathbf{p}^2 + m^2}$ . Any polynomial function of such four-momentum can always be taken as linear in  $p^0$ , because any power  $(p^0)^{2\nu}$  or  $(p^0)^{2\nu+1}$  can be written as  $(\mathbf{p}^2 + m^2)^{\nu}$

or  $p^0(\mathbf{p}^2 + m^2)^\nu$ , respectively. Thus we can define a polynomial  $P^{(L)}(q)$  by the conditions that

$$\begin{aligned} P^{(L)}(p) &= P(p) \quad \left( \text{for } p^0 = \pm\sqrt{\mathbf{p}^2 + m^2} \right), \\ P^{(L)}(q) &= P^{(0)}(\mathbf{q}) + q^0 P^{(1)}(\mathbf{q}) \quad (\text{for general } q^\mu), \end{aligned} \quad (6.2.10)$$

where  $P^{(0,1)}$  are polynomials depending only on  $\mathbf{q}$ . We can now use the relations

$$\frac{\partial}{\partial x^0} \theta(x^0 - y^0) = -\frac{\partial}{\partial x^0} \theta(y^0 - x^0) = \delta(x^0 - y^0) \quad (6.2.11)$$

(recall that  $\theta(x)$  has a unit step at  $x^0$ , and is otherwise constant) to move the derivative operators to the left of the  $\theta$  functions in Eq. (6.2.8)

$$\begin{aligned} \Delta_{\ell m}(x, y) &= P_{\ell m}^{(L)} \left( -i \frac{\partial}{\partial x} \right) \Delta_F(x - y) \\ &\quad + \delta(x^0 - y^0) P_{\ell m}^{(1)}(-i\nabla) [\Delta_+(x - y) - \Delta_+(y - x)], \end{aligned} \quad (6.2.12)$$

where  $\Delta_F$  is the ‘Feynman propagator’

$$-i\Delta_F(x) \equiv \theta(x)\Delta_+(x) + \theta(-x)\Delta_+(-x). \quad (6.2.13)$$

However, for  $x^0 = 0$  the function  $\Delta_+(x)$  is even in  $\mathbf{x}$ , since a change  $\mathbf{x} \rightarrow -\mathbf{x}$  in Eq. (6.2.9) can be compensated by a change  $\mathbf{p} \rightarrow -\mathbf{p}$  in the integration variable. We can therefore drop the second term in Eq. (6.2.12), and write simply

$$\Delta_{\ell m}(x, y) = P_{\ell m}^{(L)} \left( -i \frac{\partial}{\partial x} \right) \Delta_F(x - y). \quad (6.2.14)$$

It will be most useful to use the expression of the Feynman propagator as a Fourier integral. The step functions in Eq. (6.2.13) have the Fourier representation<sup>55</sup>

$$\theta(t) = \frac{-1}{2\pi i} \int_{-\infty}^{\infty} \frac{\exp(-ist)}{s + i\epsilon} ds. \quad (6.2.15)$$

This can be combined with the Fourier integral (6.2.9) for  $\Delta_+(x)$ . We introduce new integration variables,  $\mathbf{q} = \mathbf{p}$ ,  $q^0 = p^0 + s$  in the first term of Eq. (6.2.13), yielding

$$\begin{aligned} -i\Delta_F(x) &= \frac{-1}{2\pi i} \int d^3 q \int_{-\infty}^{\infty} dq^0 \frac{\exp(i\mathbf{q} \cdot \mathbf{x} - iq^0 x^0)}{(2\pi)^3 2\sqrt{\mathbf{q}^2 + m^2}} \\ &\quad \cdot \left[ \left( q^0 - \sqrt{\mathbf{q}^2 + m^2} - i\epsilon \right)^{-1} + \left( -q^0 - \sqrt{\mathbf{q}^2 + m^2} - i\epsilon \right)^{-1} \right]. \end{aligned}$$

Combining denominators and adopting a four-dimensional notation, we have simply

$$\Delta_F(x) = (2\pi)^{-4} \int d^4 q \frac{\exp(iq \cdot x)}{q^2 + m^2 - i\epsilon}, \quad (6.2.16)$$

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<sup>55</sup>To prove this, note that if  $t > 0$  then the contour of integration can be closed with a large clockwise semi-circle in the lower half-plane, so the integral picks up a contribution of  $-2\pi i$  from the pole at  $s = -i\epsilon$ . If  $t < 0$  then the contour can be closed with a large counter-clockwise semi-circle in the upper half-plane, where the integrand is analytic, giving an integral equal to zero.

where  $q^2 \equiv \mathbf{q}^2 - (q^0)^2$ . (In the denominator we have replaced  $2\epsilon\sqrt{\mathbf{q}^2 + m^2}$  with  $\epsilon$ , because the only important thing about this quantity is that it is a positive infinitesimal.) This shows incidentally that  $\Delta_F$  is a Green's function for the Klein–Gordon differential operator, in the sense that

$$(\square - m^2)\Delta_F(x) = -\delta^4(x) \quad (6.2.17)$$

with boundary conditions specified by the  $-i\epsilon$  in the denominator: as shown by Eq. (6.2.13),  $\Delta_F(x)$  for  $x^0 \rightarrow +\infty$  or  $x^0 \rightarrow -\infty$  involves only positive or negative frequency terms,  $\exp(-ix^0\sqrt{\mathbf{p}^2 + m^2})$  or  $\exp(+ix^0\sqrt{\mathbf{p}^2 + m^2})$ , respectively.

Inserting Eq. (6.2.16) in Eq. (6.2.14) now gives the propagator as

$$\Delta_{\ell m}(x, y) = (2\pi)^{-4} \int d^4q \frac{P_{\ell m}^{(L)}(q) e^{iq \cdot (x-y)}}{q^2 + m^2 - i\epsilon}. \quad (6.2.18)$$

There is one obvious problem with this expression. The polynomial  $P(p)$  is Lorentz-covariant when  $p$  is on the mass shell,  $p^2 = -m^2$ , but in Eq. (6.2.18) we integrate over all  $q^\mu$ , not restricted to the mass shell. The polynomial  $P^{(L)}(q)$  is defined for general  $q^\mu$  to be linear in  $q^0$ , a condition that clearly does not respect Lorentz covariance unless the polynomial is also linear in each spatial component  $q^i$  as well. We can instead always define our extension of the polynomial  $P(p)$  to general four-momenta  $q^\mu$ , which we shall call simply  $P(q)$ , in such a way that  $P(q)$  is Lorentz-covariant for general  $q^\mu$ , in the sense that

$$P_{\ell m}(\Lambda q) = D_{\ell \ell'}(\Lambda) D_{mm'}^*(\Lambda) P_{\ell' m'}(q),$$

where  $\Lambda^\mu{}_\nu$  is a general Lorentz transformation, and  $D(\Lambda)$  is the appropriate representation of the Lorentz group. For instance, for scalar, Dirac, and four-vector fields, these covariant extensions are obviously provided by just replacing  $p^\mu$  with a general four-momentum  $q^\mu$  in Eqs. (6.2.4), (6.2.5), and (6.2.6). For the scalar and Dirac fields, these are already linear in  $q^0$ , so here there is no difference between  $P^{(L)}(q)$  and  $P(q)$ :

$$P_{\ell m}^{(L)}(q) = P_{\ell m}(q) \quad (\text{scalar, Dirac fields}). \quad (6.2.19)$$

On the other hand, for the vector field of a spin one particle, the 00 components of the covariant polynomial  $P_{\mu\nu}(q) = \eta_{\mu\nu} + m^{-2}q_\mu q_\nu$  are quadratic in  $q^0$ , so here there is a difference:

$$\begin{aligned} P_{\mu\nu}^{(L)}(q) &= \eta_{\mu\nu} + m^{-2} [q_\mu q_\nu - \delta_\mu^0 \delta_\nu^0 (q_0^2 - \mathbf{q}^2 - m^2)] \\ &= P_{\mu\nu}(q) + m^{-2} (q^2 + m^2) \delta_\mu^0 \delta_\nu^0. \end{aligned} \quad (6.2.20)$$

(The extra term here is fixed by the two conditions that it must cancel the  $(q_0)^2$  term in  $P_{00}(q)$ , and must vanish when  $q^\mu$  is on the mass shell.) Inserting this in Eq. (6.2.18) gives the propagator of a vector field as

$$\Delta_{\mu\nu}(x, y) = (2\pi)^{-4} \int d^4q \frac{P_{\mu\nu}(q) e^{iq \cdot (x-y)}}{q^2 + m^2 - i\epsilon} + m^{-2} \delta^4(x - y) \delta_\mu^0 \delta_\nu^0. \quad (6.2.21)$$

The first term is manifestly covariant, and the second term, though not covariant, is local, so it can be cancelled by adding a local non-covariant term to the Hamiltonian density.

Specifically, if  $V_\mu(x)$  interacts with other fields through a term  $V_\mu(x)J^\mu(x)$  in  $\mathcal{H}_I(x)$ , then the effect of the second term in Eq. (6.2.21) is to produce an effective interaction

$$-i\mathcal{H}_{\text{eff}}(x) = \frac{1}{2}[-iJ^\mu(x)][-iJ^\nu(x)][-im^{-2}\delta_\mu^0\delta_\nu^0].$$

(The factors  $-i$  are the usual ones which always accompany vertices and propagators. The factor  $\frac{1}{2}$  is needed because there are two ways to pair other fields with  $\mathcal{H}_{\text{eff}}(x)$ , differing in the interchange of  $J^\mu$  and  $J^\nu$ .) Thus the effect of the non-covariant second term in Eq. (6.2.21) can be cancelled by adding to  $\mathcal{H}_I(x)$  the non-covariant term

$$\mathcal{H}_{\text{NC}}(x) = -\mathcal{H}_{\text{eff}}(x) = \frac{1}{2m^2}[J^0(x)]^2. \quad (6.2.22)$$

It is the singularity of the equal-time commutators of vector fields at zero separation that requires us to employ a wider class of interactions than those with a scalar density. A detailed non-perturbative proof of the Lorentz invariance of the  $S$ -matrix in this theory will be given in the next chapter.

It should not be thought that this is solely a phenomenon associated with spins  $j \geq 1$ . For instance, consider the vector field associated with a particle of spin  $j = 0$ , equal (as discussed in Chapter 5) to the derivative  $\partial_\lambda\phi(x)$  of a scalar field. For the pairing of this field with a scalar  $\phi^\dagger(y)$ , the polynomial  $P(p)$  on the mass shell is

$$P_\lambda(p) = ip_\lambda, \quad (6.2.23)$$

while the pairing of  $\partial_\lambda\phi(x)$  with  $\partial_\eta\phi^\dagger(y)$  yields a polynomial

$$P_{\lambda\eta}(p) = p_\lambda p_\eta. \quad (6.2.24)$$

The covariant polynomials for general off-shell four-momenta  $q^\mu$  are again obtained by just substituting  $q^\mu$  for  $p^\mu$  in Eqs. (6.2.23) and (6.2.24). Eq. (6.2.23) shows that  $P_\lambda(q)$  is already linear in  $q_0$ , so here there is no difference between  $P_\lambda(q)$  and  $P_\lambda^{(L)}(q)$ . However, for Eq. (6.2.24) there is a difference:

$$\begin{aligned} P_{\lambda\eta}^{(L)}(q) &= q_\lambda q_\eta - (q_0^2 - \mathbf{q}^2 - m^2)\delta_\lambda^0\delta_\eta^0 \\ &= P_{\lambda\eta}(q) + (q^2 + m^2)\delta_\lambda^0\delta_\eta^0, \end{aligned} \quad (6.2.25)$$

so here the propagator is

$$\Delta_{\lambda\eta}(x, y) = (2\pi)^{-4} \int d^4q \frac{q_\lambda q_\eta e^{iq \cdot x}}{q^2 + m^2 - i\epsilon} + \delta_\lambda^0\delta_\eta^0\delta^4(x - y). \quad (6.2.26)$$

Just as before, the non-covariant effects of the second term may be removed by adding to the interaction a non-covariant term

$$\mathcal{H}_{\text{NC}}(x) = \frac{1}{2}[J^0(x)]^2, \quad (6.2.27)$$

where  $J^\mu(x)$  is here the current which multiplies  $\partial_\mu\phi(x)$  in the covariant part of  $\mathcal{H}_I(x)$ .

It should be clear that (at least for massive particles) the effects of non-covariant parts of the propagator can always be cancelled in this way by adding non-covariant local terms to the Hamiltonian density. This is because the numerator  $P_{\ell m}^{(L)}(q)$  in the propagator must equal the covariant polynomial  $P_{\ell m}(q)$  when  $q^\mu$  is on the mass shell, so the difference between  $P_{\ell m}^{(L)}(q)$  and  $P_{\ell m}(q)$  must contain a factor  $q^2 + m^2$ . This factor cancels the denominator  $(q^2 + m^2 - i\epsilon)$  in the contribution of this difference to Eq. (6.2.18), so Eq. (6.2.18) always equals a covariant term plus a term proportional to the delta function  $\delta^4(x - y)$  or its derivatives. The effect of the latter term may be cancelled by adding to the interaction a term quadratic in the currents to which the paired fields couple, or in their derivatives. In what follows, it will be assumed tacitly that such a term has been included in the interaction, and in consequence we shall use the *covariant* polynomial  $P_{\ell m}(q)$  in the propagator (6.2.18), and will thus henceforth drop the label ‘ $L$ ’.

It may seem that this is a rather *ad hoc* procedure. Fortunately, in the canonical formalism discussed in the following chapter, the non-covariant term in the Hamiltonian density needed to cancel non-covariant terms in the propagator arises automatically. This, in fact, forms part of the motivation for introducing the canonical formalism.

\* \* \*

Before closing this section, it may be useful to mention some other definitions of the propagator, equivalent to Eq. (6.2.1), that appear commonly in the literature. First, taking the vacuum expectation value of Eq. (6.1.14) gives

$$\begin{aligned} -i\Delta_{\ell m}(x, y) &= \theta(x - y) \left\langle [\psi_\ell^{(+)}(x), \psi_m^{(+)\dagger}(y)]_{\mp} \right\rangle_0 \\ &\quad \pm \theta(y - x) \left\langle [\psi_m^{(-)\dagger}(y), \psi_\ell^{(-)}(x)]_{\mp} \right\rangle_0. \end{aligned} \quad (6.2.28)$$

(Here  $\langle AB \dots \rangle_0$  denotes the vacuum expectation value  $\langle \Omega | AB \dots | \Omega \rangle$ .) Both  $\psi_\ell^{(+)}(x)$  and  $\psi_m^{(-)\dagger}(y)$  annihilate the vacuum, so only one term in each commutator or anticommutator in Eq. (6.2.28) actually contributes to the propagator:

$$\begin{aligned} -i\Delta_{\ell m}(x, y) &= \theta(x - y) \left\langle \psi_\ell^{(+)}(x) \psi_m^{(+)\dagger}(y) \right\rangle_0 \\ &\quad \pm \theta(y - x) \left\langle \psi_m^{(-)\dagger}(y) \psi_\ell^{(-)}(x) \right\rangle_0. \end{aligned} \quad (6.2.29)$$

Also,  $\psi^{(-)\dagger}$  and  $\psi^{(+)}$  would annihilate a vacuum state on the right, and  $\psi^{(-)}$  and  $\psi^{(+)\dagger}$  would annihilate the vacuum state on the left, so  $\psi^{(+)}$  and  $\psi^{(-)}$  may be replaced everywhere in Eq. (6.2.29) with the complete field  $\psi = \psi^{(+)} + \psi^{(-)}$ :

$$-i\Delta_{\ell m}(x, y) = \theta(x - y) (\psi_\ell(x) \psi_m^\dagger(y))_0 \pm \theta(y - x) (\psi_m^\dagger(y) \psi_\ell(x))_0. \quad (6.2.30)$$

This is often written

$$-i\Delta_{\ell m}(x, y) = (T\{\psi_\ell(x) \psi_m^\dagger(y)\})_0, \quad (6.2.31)$$

where  $T$  is a time-ordered product, whose definition is now extended to all fields, with a minus sign for any odd permutation of fermionic operators.<sup>56</sup>

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<sup>56</sup>This is not inconsistent with our previous definition of the time-ordered product of Hamiltonian densities in Chapter 3, because the Hamiltonian density can only contain even numbers of fermionic field factors.

### 6.3 Momentum Space Rules

The Feynman rules outlined in Section 6.1 specify how to calculate the contribution to the  $S$ -matrix of a given  $N$ th order diagram, as the integral over  $N$  spacetime coordinates of a product of spacetime-dependent factors. For a final particle (or antiparticle) line with momentum  $p'^\mu$  leaving a vertex with spacetime coordinate  $x^\mu$ , we get a factor proportional to  $\exp(-ip' \cdot x)$ , and for an initial particle line with momentum  $p^\mu$  entering a vertex with spacetime coordinate  $x^\mu$ , we get a factor proportional to  $\exp(+ip \cdot x)$ . In Section 6.2 we saw that the factor associated with an internal line running from  $y$  to  $x$  can be expressed as a Fourier integral, over off-shell four-momenta  $q^\mu$ , of an integrand proportional to  $\exp(iq \cdot (x - y))$ . We can think of  $q^\mu$  as the four-momentum flowing along the internal line in the direction of the arrow from  $y$  to  $x$ . Hence the integral over each vertex's spacetime position merely yields a factor

$$(2\pi)^4 \delta^4 \left( \sum p + \sum q - \sum p' - \sum q' \right), \quad (6.3.1)$$

where  $\sum p'$  and  $\sum p$  denote the total four-momentum of all the final or initial particles leaving or entering the vertex, and  $\sum q'$  and  $\sum q$  denote the total four-momentum of all the internal lines with arrows leaving or entering the vertex, respectively. Of course, in place of these integrals over  $x^\mu$ s, we now have to do integrals over the Fourier variables  $q^\mu$ , one for each internal line.

These considerations can be encapsulated in a new set of Feynman rules

(see Figure 6.9) for calculating contribution to the  $S$ -matrix as integrals over momentum variables:

- (i) Draw all Feynman diagrams of the desired order, just as described in Section 6.1. However, instead of labelling each vertex with a spacetime coordinate, each internal line is now labelled with an off-mass-shell four-momentum, considered conventionally to flow in the direction of the arrow (or in either direction for neutral particle lines without arrows.)
- (ii) For each vertex of type  $j$ , include a factor

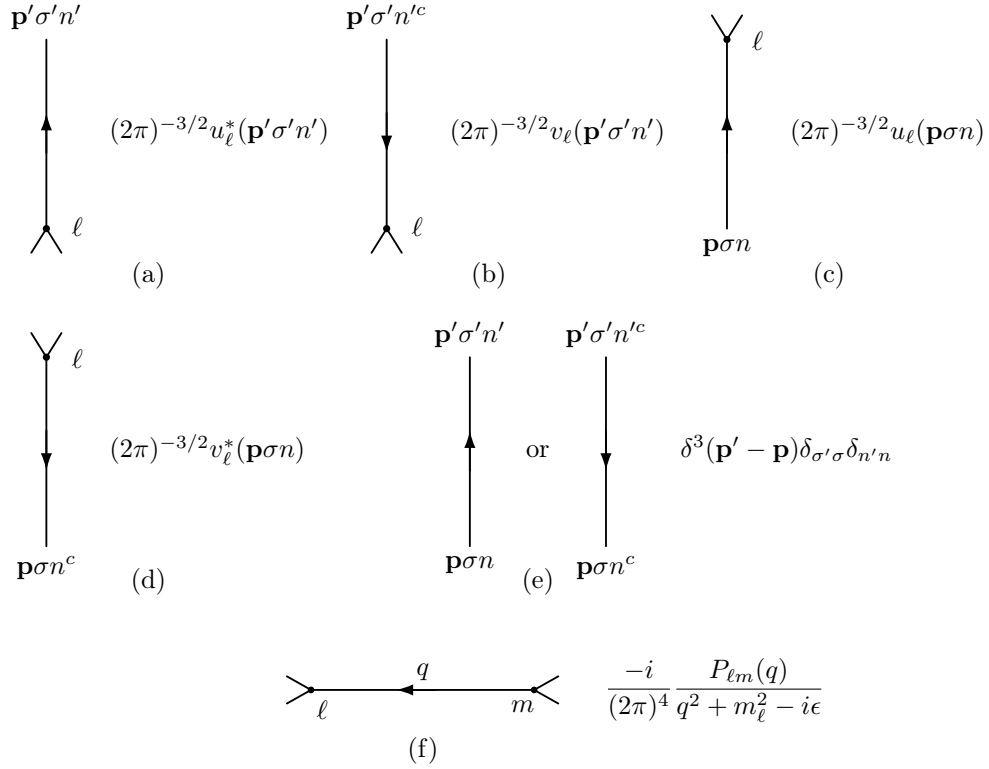
$$-i(2\pi)^4 g_j \delta^4 \left( \sum p + \sum q - \sum p' - \sum q' \right) \quad (6.3.2)$$

with the momentum sums having the same meaning as in (6.3.1). This delta function ensures that the four-momentum is conserved at every point in the diagram. For each external line running upwards out of the diagram, include a factor

$$(2\pi)^{-3/2} u_\ell^*(\mathbf{p}' \sigma' n') \quad \text{or} \quad (2\pi)^{-3/2} v_\ell(\mathbf{p}' \sigma' n'),$$

for arrows pointing upwards or downwards, respectively. For each external line running from below into the diagram, include a factor

$$(2\pi)^{-3/2} u_\ell(\mathbf{p} \sigma n) \quad \text{or} \quad (2\pi)^{-3/2} v_\ell^*(\mathbf{p} \sigma n),$$



**Figure 6.9.** Graphical representation of pairings of operators arising in the momentum-space evaluation of the  $S$ -matrix. The expressions on the right are the factors that must be included in the momentum space integrand of the  $S$ -matrix for each line of the Feynman diagram.

for arrows pointing upwards or downwards, respectively. For each internal line with ends labelled  $\ell$  and  $m$ , the arrow pointing from  $m$  to  $\ell$ , and carrying a momentum label  $q^\mu$ , include as a factor the integrand of the integral for  $-i\Delta_{\ell m}(q)$ :

$$-i(2\pi)^{-4} \frac{P_{\ell m}(q)}{q^2 + m_\ell^2 - i\epsilon}. \quad (6.3.3)$$

A reminder: for scalars or antiscalars of four-momentum  $q$ , the  $us$  and  $vs$  are simply  $(2q^0)^{-1/2}$ , while the polynomial  $P(q)$  is unity. For Dirac spinors of four-momentum  $p$  and mass  $M$ , the  $us$  and  $vs$  are the normalized Dirac spinors described in Section 5.5, and the polynomial  $P(p)$  is the matrix  $(-i\gamma_\mu p^\mu + M)\beta$ .

- (iii) Integrate the product of all these factors over the four-momenta carried by internal lines, and sum over all field indices  $\ell, m$ , etc.
- (iv) Add up the results obtained in this way from each Feynman diagram.

Additional combinatoric factors and fermionic signs may need to be included, as described in parts (v) and (vi) of Section 6.1. Examples will be given at the end of this section.



We have a four-momentum integration variable for every internal line, but many of these are eliminated by the delta functions associated with vertices. Since energy and momentum are separately conserved for each connected part of a Feynman diagram, there will be  $C$  delta functions left over in a graph with  $C$  connected parts. Hence in a diagram with  $I$  internal lines and  $V$  vertices, the number of independent four-momenta that are not fixed by the delta functions is  $I - [V - C]$ . This is clearly also the number  $L$  of independent loops:

$$L = I - V + C, \quad (6.3.4)$$

which is defined as the maximum number of internal lines that can be cut without disconnecting the diagram, because any such and only such internal lines can be assigned an independent four-momentum. We can think of the independent momentum variables as characterizing the momenta that circulate in each loop. In particular a *tree* graph is one without loops; after taking the delta functions into account there are no momentum-space integrals left for such graphs.

For instance, in a theory with interaction (6.1.18), the  $S$ -matrix (6.1.27) for fermion–boson scattering is given by the momentum-space Feynman rules as

$$\begin{aligned} & S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\ &= \sum_{k' \ell' m' k \ell m} (-i)^2 (2\pi)^8 g_{\ell' m' k'} g_{m \ell k} u_{\ell'}^*(\mathbf{p}'_1 \sigma'_1 n'_1) u_{\ell}(\mathbf{p}_1 \sigma_1 n_1) \\ &\quad \cdot \int d^4 q \left( -i (2\pi)^{-4} \frac{P_{m'm}(q)}{q^2 + m_m^2 - i\epsilon} \right) (2\pi)^{-6} \\ &\quad \cdot \left[ u_{k'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_k(\mathbf{p}_2 \sigma_2 n_2) \delta^4(p_2 - p'_2 - q) \delta^4(p_1 + q - p'_1) \right. \\ &\quad \left. + u_k^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_{k'}(\mathbf{p}_2 \sigma_2 n_2) \delta^4(p_2 - p'_1 - q) \delta^4(p_1 - p'_2 + q) \right], \end{aligned}$$

with labels 1 and 2 here denoting fermions and bosons, respectively. The momentum-space integral here is trivial, and gives

$$\begin{aligned} & S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\ &= i (2\pi)^{-2} \delta^4(p_1 + p_2 - p'_1 - p'_2) \sum_{k' \ell' m' k \ell m} g_{\ell' m' k'} g_{m \ell k} u_{\ell'}^*(\mathbf{p}'_1 \sigma'_1 n'_1) u_{\ell}(\mathbf{p}_1 \sigma_1 n_1) \\ &\quad \cdot \left[ \frac{P_{m'm}(p'_1 - p_1)}{(p'_1 - p_1)^2 + m_m^2 - i\epsilon} u_{k'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_k(\mathbf{p}_2 \sigma_2 n_2) \right. \\ &\quad \left. + \frac{P_{m'm}(p'_2 - p_1)}{(p'_2 - p_1)^2 + m_m^2 - i\epsilon} u_k^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_{k'}(\mathbf{p}_2 \sigma_2 n_2) \right]. \quad (6.3.5) \end{aligned}$$

In the same way, the  $S$ -matrix element (6.1.28) for fermion–fermion scattering in the same theory is

$$\begin{aligned} & S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\ &= i (2\pi)^{-2} \delta^4(p_1 + p_2 - p'_1 - p'_2) \sum_{k' \ell' m' k \ell m} g_{\ell m' k'} g_{\ell' m k} \frac{P_{k'k}(p'_1 - p_1)}{(p'_1 - p_1)^2 + m_k^2 - i\epsilon} \\ &\quad \cdot u_{\ell'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_{\ell'}^*(\mathbf{p}'_1 \sigma'_1 n'_1) u_{m'}(\mathbf{p}_2 \sigma_2 n_2) u_m(\mathbf{p}_1 \sigma_1 n_1) - [1' \rightleftharpoons 2']. \quad (6.3.6) \end{aligned}$$

These results illustrate the need for a more compact notation. We may define a fermion–boson coupling matrix

$$[\Gamma_k]_{\ell m} \equiv g_{\ell m k}. \quad (6.3.7)$$

The matrix elements (6.3.5) and (6.3.6) for fermion–boson and fermion–fermion scattering can then be rewritten in matrix notation as

$$\begin{aligned} & S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\ &= i(2\pi)^{-2} \delta^4(p_1 + p_2 - p'_1 - p'_2) \sum_{k'k} \left[ \left( u^\dagger(\mathbf{p}'_1 \sigma'_1 n'_1) \Gamma_{k'} \frac{P(p'_1 - p_1)}{(p'_1 - p_1)^2 + M^2 - i\epsilon} \Gamma_k u(\mathbf{p}_1 \sigma_1 n_1) \right) \right. \\ & \quad \cdot u_{k'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_k(\mathbf{p}_2 \sigma_2 n_2) \\ & \quad + \left( u^\dagger(\mathbf{p}'_1 \sigma'_1 n'_1) \Gamma_{k'} \frac{P(p'_2 - p_1)}{(p'_2 - p_1)^2 + M^2 - i\epsilon} \Gamma_k u(\mathbf{p}_1 \sigma_1 n_1) \right) \\ & \quad \cdot u_k^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_{k'}(\mathbf{p}_2 \sigma_2 n_2) \left. \right]. \end{aligned} \quad (6.3.8)$$

and

$$\begin{aligned} & S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\ &= i(2\pi)^{-2} \delta^4(p_1 + p_2 - p'_1 - p'_2) \sum_{k'k} \frac{P_{k'k}(p'_1 - p_1)}{(p'_1 - p_1)^2 + m_k^2 - i\epsilon} \\ & \quad \cdot \left( u^\dagger(\mathbf{p}'_2 \sigma'_2 n'_2) \Gamma_{k'} u(\mathbf{p}'_1 \sigma'_1 n'_1) \right) \left( u^\dagger(\mathbf{p}_2 \sigma_2 n_2) \Gamma_k u(\mathbf{p}_1 \sigma_1 n_1) \right) - [1' \rightleftharpoons 2']. \end{aligned} \quad (6.3.9)$$

where  $M^2$  and  $m^2$  are the diagonal mass matrices of the fermions and bosons in Eqs. (6.3.8) and (6.3.9), respectively. The general rule is that in using matrix notation, one writes coefficient functions, coupling matrices, and propagators in an order dictated by following lines *backwards* from the order indicated by the arrows. In the same notation, the  $S$ -matrix for boson–boson scattering in the same theory would be given by a sum of one-loop diagrams, shown in Figure 6.7:

$$\begin{aligned} & S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\ &= -\delta^4(p_1 + p_2 - p'_1 - p'_2) \sum_{k_1 k_2 k'_1 k'_2} u_{k'_1}^*(\mathbf{p}'_1 \sigma'_1 n'_1) u_{k'_2}^*(\mathbf{p}'_2 \sigma'_2 n'_2) u_{k_1}(\mathbf{p}_1 \sigma_1 n_1) u_{k_2}(\mathbf{p}_2 \sigma_2 n_2) \\ & \quad \cdot \int d^4 q \operatorname{Tr} \left\{ \Gamma_{k'_2} \frac{P(q)}{q^2 + M^2 - i\epsilon} \Gamma_{k'_1} \frac{P(q + p'_1)}{(q + p'_1)^2 + M^2 - i\epsilon} \right. \\ & \quad \cdot \Gamma_{k_1} \frac{P(q + p'_1 - p_1)}{(q + p'_1 - p_1)^2 + M^2 - i\epsilon} \Gamma_{k_2} \frac{P(q - p'_2)}{(q - p'_2)^2 + M^2 - i\epsilon} \left. \right\} + \cdots. \end{aligned} \quad (6.3.10)$$

where the ellipsis in the last line indicates terms obtained by permuting bosons 1', 2', and 2. The minus sign at the beginning of the right-hand side is the extra minus sign associated with fermionic loops. Note that after elimination of delta functions there is just one momentum-space integral here, as appropriate for a diagram with one loop. We shall see how to do this sort of momentum-space integral in Chapter 11.

To make this more specific, consider a theory with a Dirac spinor field  $\psi(x)$  of mass  $M$  and a pseudoscalar field  $\phi(x)$  of mass  $m$ , interacting through the interaction  $-ig\bar{\psi}\gamma_5\psi\phi$ . (The factor  $-i$  is inserted to make this interaction Hermitian for real coupling constants  $g$ .) Recall that the polynomial  $P(q)$  for the scalar is just unity, while for the spinor it is  $[-i\gamma_\mu q^\mu + M]\beta$ . Also, the  $u$  for a scalar of energy  $E$  is  $(2E)^{-1/2}$ , while for the spinor  $u$  is the conventionally normalized Dirac spinor discussed in Section 5.5. Eqs. (6.3.8), (6.3.9), and (6.3.10) give the lowest-order connected  $S$ -matrix elements for fermion–boson scattering, fermion–fermion scattering, and boson–boson scattering:

$$\begin{aligned}
S_{\mathbf{p}'_1\sigma'_1\mathbf{p}'_2,\mathbf{p}_1\sigma_1\mathbf{p}_2} &= -i(2\pi)^{-2}g^2(4E'_2E_2)^{-1/2}\delta^4(p_1+p_2-p'_1-p'_2) \\
&\cdot \left[ \bar{u}(\mathbf{p}'_1\sigma'_1)\gamma_5 \frac{-i\gamma_\mu(p'_1-p_1)^\mu + M}{(p'_1-p_1)^2 + m^2 - i\epsilon} \gamma_5 u(\mathbf{p}_1\sigma_1) \right. \\
&\quad \left. + \bar{u}(\mathbf{p}'_1\sigma'_1)\gamma_5 \frac{-i\gamma_\mu(p'_2-p_1)^\mu + M}{(p'_2-p_1)^2 + m^2 - i\epsilon} \gamma_5 u(\mathbf{p}_1\sigma_1) \right], \\
S_{\mathbf{p}'_1\sigma'_1\mathbf{p}'_2\sigma'_2,\mathbf{p}_1\sigma_1\mathbf{p}_2\sigma_2} &= i(2\pi)^{-2}g^2\delta^4(p_1+p_2-p'_1-p'_2) \\
&\cdot (\bar{u}(\mathbf{p}'_2\sigma'_2)\gamma_5 u(\mathbf{p}'_1\sigma'_1)) (\bar{u}(\mathbf{p}_2\sigma_2)\gamma_5 u(\mathbf{p}_1\sigma_1)) \\
&\cdot \frac{1}{(p'_1-p_1)^2 + m^2 - i\epsilon} - [1' \leftrightarrow 2'], \\
S_{\mathbf{p}'_1\mathbf{p}'_2,\mathbf{p}_1\mathbf{p}_2} &= -(2\pi)^{-6}g^2(16E_1E_2E'_1E'_2)^{-1/2}\delta^4(p_1+p_2-p'_1-p'_2) \\
&\cdot \int d^4q \operatorname{Tr} \left\{ \gamma_5 \frac{-i\gamma_\mu q^\mu + M}{q^2 + M^2 - i\epsilon} \gamma_5 \frac{-i\gamma_\mu(q+p'_1)^\mu + M}{(q+p'_1)^2 + M^2 - i\epsilon} \right. \\
&\quad \left. \cdot \gamma_5 \frac{-i\gamma_\mu(q+p'_1-p_1)^\mu + M}{(q+p'_1-p_1)^2 + M^2 - i\epsilon} \gamma_5 \frac{-i\gamma_\mu(q-p'_2)^\mu + M}{(q-p'_2)^2 + M^2 - i\epsilon} \right\} + \dots,
\end{aligned}$$

where in the last formula the ellipsis indicates a sum over permutations of particles 2, 1', 2'. The factors  $\beta$  in the fermion propagator numerators have been used to replace  $u^\dagger$  with  $\bar{u}$ .

\* \* \*

Another useful topological result expresses a sort of conservation law of lines. For the moment we can think of all internal and external lines as being created at vertices and destroyed in pairs at the centers of internal lines or when external lines leave the diagram. (This has nothing to do with the directions of the arrows carried by these lines.) Equating the numbers of lines that are created and destroyed then gives

$$2I + E = \sum_i n_i V_i, \quad (6.3.11)$$

where  $I$  and  $E$  are the numbers of internal and external lines,  $V_i$  are the numbers of vertices of various types labelled  $i$ , and  $n_i$  are the number of lines attached to each vertex. (This

also holds separately for fields of each type.) In particular, if all interactions involve the same number  $n_i = n$  of fields, then this reads

$$2I + E = nV, \quad (6.3.12)$$

where  $V$  is the total number of all vertices. In this case, we can eliminate  $I$  from Eqs. (6.3.4) and (6.3.11), and find that for a connected (i.e.,  $C = 1$ ) graph the number of vertices is given by

$$V = \frac{2L + E - 2}{n - 2}. \quad (6.3.13)$$

For instance for a trilinear interaction the diagrams for a scattering process ( $E = 4$ ) with  $L = 0, 1, 2, \dots$  has  $V = 2, 4, 6, \dots$  vertices. In general, the expansion in powers of the coupling constants is an expansion in increasing numbers of loops.

#### 6.4 Off the Mass Shell

In the Feynman diagrams for any  $S$ -matrix element all external lines are “on the mass shell”; that is, the four-momentum associated with an external line for a particle of mass  $m$  is constrained to satisfy  $p_\mu p^\mu = -m^2$ . It is often important also to consider Feynman diagrams “off the mass shell”, for which the external line energies like the energies associated with internal lines are free variables, unrelated to any three-momenta. For one thing, these arise as parts of larger Feynman diagrams; for instance, a loop appearing as an insertion in some internal line of a diagram could be regarded as a Feynman diagram with two external lines, both off the mass shell.

Of course, once we calculate the contribution of a given Feynman diagram off the mass shell, it is easy to calculate the associated  $S$ -matrix elements by going to the mass shell, taking the four-momentum  $p^\mu$  flowing along the line *into* the diagram to have  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$  for particles in the initial state and  $p^0 = -\sqrt{\mathbf{p}^2 + m^2}$  for particles in the final state, and including the appropriate external line factors  $(2\pi)^{-3/2}u_\ell$  or  $(2\pi)^{-3/2}v_\ell^*$  for initial particles or antiparticles and  $(2\pi)^{-3/2}u_\ell^*$  or  $(2\pi)^{-3/2}v_\ell$  for final particles or antiparticles. Indeed, when we come to the path integral approach in Chapter 8 we shall find it easiest first to derive the Feynman rules for diagrams with *all* external lines off the mass shell, and then obtain  $S$ -matrix elements by letting the momenta associated with external lines approach their appropriate mass shells.

Feynman graphs with lines off the mass shell are just a special case of a wider generalization of the Feynman rules that takes into account the effects of various possible external fields. Suppose we add a sum of terms involving external fields  $\epsilon_a(x)$  to the Hamiltonian, so that the interaction  $H_I(t)$  that is used in the Dyson series (3.5.10) for the  $S$ -matrix is replaced with

$$H_{I,\epsilon}(t) = H_I(t) + \sum_a \int d^3x \epsilon_a(\mathbf{x}, t) o_a(\mathbf{x}, t). \quad (6.4.1)$$

The “currents”  $o_a(t)$  have the usual time-dependence of the interaction picture

$$o_a(t) = \exp(iH_0 t) o_a(0) \exp(-iH_0 t), \quad (6.4.2)$$

but are otherwise quite arbitrary operators. The  $S$ -matrix for any given transition  $\alpha \rightarrow \beta$  then becomes a functional  $S_{\beta\alpha}[\epsilon]$  of the c-number functions  $\epsilon_a(t)$ . The Feynman rules for computing this functional are given by an obvious extension of the usual Feynman rules. In addition to the usual vertices obtained from  $H_I(t)$ , we must include additional vertices: if  $o_a(x)$  is a product of  $n_a$  field factors, then any  $o_a$  vertex with position label  $x$  must have  $n_a$  lines of corresponding types attached, and makes a contribution to the position-space Feynman rules equal to  $-i\epsilon_a(x)$  times whatever numerical factors appear in  $o_a(x)$ . It follows then that the  $r$ th variational derivative of  $S_{\beta\alpha}[\epsilon]$  with respect to  $\epsilon_a(x), \epsilon_b(y) \dots$  at  $\epsilon = 0$  is given by position space diagrams with  $r$  additional vertices, to which are attached respectively  $n_a, n_b, \dots$  internal lines, and no external lines. These vertices carry position labels  $x, y \dots$  over which we do *not* integrate; each such vertex makes a contribution equal to  $-i$  times whatever numerical factors appear in the associated current  $o_a$ .

In particular, in the case where these currents are all single field factors, i.e.,

$$H_{I,\epsilon}(t) = H_I(t) + \sum_{\ell} \int d^3x \epsilon_{\ell}(\mathbf{x}, t) \psi_{\ell}(\mathbf{x}, t),$$

the  $r$ th variational derivative of  $S_{\beta\alpha}[\epsilon]$  with respect to  $\epsilon_{\ell}(x), \epsilon_m(y), \dots$  at  $\epsilon = 0$  is given by position space diagrams with  $r$  additional vertices carrying spacetime labels  $x, y \dots$ , to each of which is attached a single internal particle line of type  $\ell, m \dots$ . These can be thought of as off-shell external lines, with the difference that their contribution to the matrix element is not a coefficient function like  $(2\pi)^{-3/2} u_{\ell}(\mathbf{p}, \sigma) e^{ip \cdot x}$  or  $(2\pi)^{-3/2} u_{\ell}^*(\mathbf{p}, \sigma) e^{-ip \cdot x}$  but a *propagator*, as well as a factor  $-i$  from the vertex at the end of the line. We obtain a momentum space Feynman diagram with particles in states  $\alpha$  and  $\beta$  on the mass shell plus  $r$  external lines of type  $\ell, m \dots$  carrying momenta  $p, p' \dots$  from the variational derivative

$$\left[ \frac{\delta^r S_{\beta\alpha}[\epsilon]}{\delta \epsilon_{\ell}(x) \delta \epsilon_m(y) \dots} \right]_{\epsilon=0}$$

by stripping away the propagators on each of the off-shell lines and then taking the appropriate Fourier transforms and multiplying with appropriate coefficient functions  $u_{\ell}, u_{\ell}^*$ , etc. and a factor  $(-i)^r$ .

It is very useful for a number of purposes to recognize that there is a simple relation between the sum of contributions from all perturbation theory diagrams for any off-shell amplitude and a matrix element, between eigenstates of the full Hamiltonian, of a time-ordered product of corresponding operators in the Heisenberg picture. This relation is provided by a theorem [3], which states that to all orders of perturbation theory<sup>57</sup>

$$\left[ \frac{\delta^r S_{\beta\alpha}[\epsilon]}{\delta \epsilon_a(x) \delta \epsilon_b(y) \dots} \right]_{\epsilon=0} = {}_{\text{out}} \langle \beta | T \{ -i O_a(x), -i O_b(y), \dots \} | \alpha \rangle_{\text{in}}, \quad (6.4.3)$$

where  $O_a(x)$ , etc. are the counterparts of  $o_a(x)$  in the Heisenberg picture

$$O_a(\mathbf{x}, t) = \exp(iHt) o_a(\mathbf{x}, 0) \exp(-iHt) = \Omega(t) o_a(\mathbf{x}, t) \Omega^{-1}(t), \quad (6.4.4)$$

---

<sup>57</sup>For a single  $O$  operator, this is a version of the Schwinger action principle [4].

$$\Omega(t) \equiv e^{iHt} e^{-iH_0 t}, \quad (6.4.5)$$

and  $|\beta\rangle_{\text{in}}$  and  $|\beta\rangle_{\text{out}}$  are “in” and “out” eigenstates of the full Hamiltonian  $H$ , respectively.

Here is the proof. From Eq. (3.5.10), we see immediately that the left-hand side of Eq. (6.4.3) is

$$\begin{aligned} & \left[ \frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1) \cdots \delta\epsilon_{a_r}(x_r)} \right]_{\epsilon=0} \\ &= \sum_{N=0}^{\infty} \frac{(-i)^{N+r}}{N!} \int_{-\infty}^{\infty} d\tau_1 \cdots d\tau_N \langle \beta | T \{ H_I(\tau_1) \cdots H_I(\tau_N) o_{a_1}(x_1) \cdots o_{a_r}(x_r) \} | \alpha \rangle. \end{aligned} \quad (6.4.6)$$

For definiteness, suppose that  $x_1^0 \geq x_2^0 \geq \cdots \geq x_r^0$ . Then we can denote by  $\tau_{01} \cdots \tau_{0N_0}$  all  $\tau$ s greater than  $x_1^0$ ; by  $\tau_{11} \cdots \tau_{1N_1}$  all  $\tau$ s between  $x_1^0$  and  $x_2^0$ , and so on; finally denoting by  $\tau_{r1} \cdots \tau_{rN_r}$  all  $\tau$ s that are less than  $x_r^0$ . Eq. (6.4.6) then becomes:

$$\begin{aligned} & \left[ \frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1) \cdots \delta\epsilon_{a_r}(x_r)} \right]_{\epsilon=0} \\ &= \sum_{N=0}^{\infty} \frac{(-i)^{N+r}}{N!} \sum_{N_0, N_1, \dots, N_r} \frac{N! \delta_{N, N_0 + N_1 + \dots + N_r}}{N_0! N_1! \cdots N_r!} \\ & \quad \cdot \int_{x_1^0}^{\infty} d\tau_{01} \cdots d\tau_{0N_0} \int_{x_2^0}^{x_1^0} d\tau_{11} \cdots d\tau_{1N_1} \cdots \int_{-\infty}^{x_r^0} d\tau_{r1} \cdots d\tau_{rN_r} \\ & \quad \cdot \langle \beta | T \{ H_I(\tau_{01}) \cdots H_I(\tau_{0N_0}) \} o_{a_1}(x_1) T \{ H_I(\tau_{11}) \cdots H_I(\tau_{1N_1}) \} o_{a_2}(x_2) \cdots \\ & \quad \cdots o_{a_r}(x_r) T \{ H_I(\tau_{r1}) \cdots H_I(\tau_{rN_r}) \} | \alpha \rangle. \end{aligned}$$

The factor  $N!/N_0!N_1!\cdots N_r!$  is the number of ways of sorting  $N$   $\tau$ s into  $r+1$  subsets, each containing  $N_0, N_1, \dots, N_r$  of these  $\tau$ s. Instead of summing over  $N_0, N_1, \dots, N_r$ , subject to the condition  $N_0 + N_1 + \cdots + N_r = N$ , and then summing over  $N$ , we can just sum independently over  $N_0, N_1, \dots, N_r$ , setting  $N$  where it appears in  $(-i)^N$  equal to  $N_0 + N_1 + \cdots + N_r$ . This gives

$$\begin{aligned} & \left[ \frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1^0) \cdots \delta\epsilon_{a_r}(x_r^0)} \right]_{\epsilon=0} \\ &= (-i)^r \langle \beta | U(\infty, x_1^0) o_{a_1}(x_1) U(x_1^0, x_2^0) o_{a_2}(x_2) U(x_2^0, x_3^0) \cdots o_{a_r}(x_r) U(x_r^0, -\infty) | \alpha \rangle. \end{aligned} \quad (6.4.7)$$

where

$$U(t', t) = \sum_{N=0}^{\infty} \frac{(-i)^N}{N!} \int_t^{t'} d\tau_1 \cdots d\tau_N T \{ H_I(\tau_1) \cdots H_I(\tau_N) \}. \quad (6.4.8)$$

The operator  $U(t', t)$  satisfies the differential equation

$$\frac{d}{dt'} U(t', t) = -i H_I(t') U(t', t) \quad (6.4.9)$$

with the obvious initial condition

$$U(t, t) = \mathbf{1}. \quad (6.4.10)$$

This has the solution

$$U(t', t) = \exp(iH_0 t') \exp\{-iH(t' - t)\} \exp(-iH_0 t) = \Omega^{-1}(t') \Omega(t) \quad (6.4.11)$$

with  $\Omega$  given by Eq. (6.4.5). Inserting Eq. (6.4.11) in Eq. (6.4.7) and using Eq. (6.4.4), we have

$$\left[ \frac{\delta S[\epsilon]}{\delta \epsilon_{a_1}(x_1) \cdots \delta \epsilon_{a_r}(x_r)} \right]_{\epsilon=0} = (-i)^r \langle \beta | \Omega_{\infty}^{\dagger} O_{a_1}(x_1) \cdots O_{a_r}(x_r) \Omega(-\infty) | \alpha \rangle. \quad (6.4.12)$$

In deriving this result we supposed that  $x_1^0 \geq x_2^0 \geq \cdots \geq x_r^0$ , so we could just as well replace the product of operators on the right-hand-side with the time-ordered product:

$$\left[ \frac{\delta S[\epsilon]}{\delta \epsilon_{a_1}(x_1) \cdots \delta \epsilon_{a_r}(x_r)} \right]_{\epsilon=0} = (-i)^r \langle \beta | \Omega_{\infty}^{\dagger} T\{O_{a_1}(x_1) \cdots O_{a_r}(x_r)\} \Omega(-\infty) | \alpha \rangle. \quad (6.4.13)$$

But now both sides are entirely symmetric (or for fermions antisymmetric) in the  $a$ s and  $x$ s, so this relation holds whatever the order of the times  $x_1^0 \dots x_r^0$ . Also, we saw in Section 3.1 that (in the sense of Eq. (3.1.12))

$$|\beta\rangle_{\text{in/out}} = \Omega(\mp\infty)|\beta\rangle. \quad (6.4.14)$$

Hence Eq. (6.4.13) is the desired result (6.4.3).

### Weinberg Exercises

- W.6.1.** Consider the theory of a real scalar field  $\phi$ , with interaction (in the interaction picture)  $H_I = g \int d^3x \phi(x)^3/3!$ . Calculate the connected  $S$ -matrix element for scalar-scalar scattering to second order in  $g$ , doing all integrals. Use the results to calculate the differential cross-section for scalar-scalar scattering in the center-of-mass system.
- W.6.2.** Consider a theory involving a neutral scalar field  $\phi(x)$  for a boson  $B$  and a complex Dirac field  $\psi(x)$  for a fermion  $F$ , with interaction (in the interaction picture)  $H_I = ig \int d^3x \bar{\psi}(x) \gamma_5 \psi(x) \phi(x)$ . Draw all the connected order- $g^2$  Feynman diagrams and calculate the corresponding  $S$ -matrix elements for the processes  $F^c + B \rightarrow F^c + B$ ,  $F + F^c \rightarrow F + F^c$ , and  $F^c + F \rightarrow B + B$  (where  $F^c$  is the antiparticle of  $F$ ). Do all integrals.
- W.6.3.** Consider the theory of a real scalar field  $\phi(x)$ , with interaction  $H_I = g \int d^3x \phi(x)^4/4!$ . Calculate the  $S$ -matrix for scalar-scalar scattering to order  $g$ , and use the result to calculate the differential scattering cross-section. Calculate the correction terms in the  $S$ -matrix for scalar-scalar scattering to order  $g^2$ , expressing the result as an integral over a single four-momentum, but do all  $x$ -integrals.
- W.6.4.** What is the contribution in Feynman diagrams from the contraction of the derivative  $\partial_{\mu} \psi_{\ell}(x)$  of a Dirac field with the adjoint  $\bar{\psi}_m(y)$  of the field?
- W.6.5.** Use the theorem of Section 6.4 to give expressions for the vacuum expectation values of Heisenberg picture operators  $\langle \Omega | \Phi(x) | \Omega \rangle$  and  $\langle \Omega | T\{\Phi(x), \Phi(y)\} | \Omega \rangle$  in the theory of Problem 1, to orders  $g$  and  $g^2$ , respectively.

## Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

### W.6.1. Put

$$s = -(p_1 + p_2)^2, \quad t = -(p_1 - p'_1)^2, \quad u = -(p_1 - p'_2)^2, \quad s + t + u = 4m^2.$$

There are three connected second-order contractions, according as the internal scalar carries  $p_1 + p_2$ ,  $p_1 - p'_1$ , or  $p_1 - p'_2$ . Using Eq. (6.2.16) in the position-space result following Eq. (6.1.31), both vertex integrals can be done:

$$S_{fi}^c = \frac{i(2\pi)^{-2}g^2\delta^4(p_1 + p_2 - p'_1 - p'_2)}{\sqrt{16E_1E_2E'_1E'_2}} \cdot \left[ \frac{1}{m^2 - s - i0} + \frac{1}{m^2 - t - i0} + \frac{1}{m^2 - u - i0} \right].$$

Thus, up to the immaterial overall sign convention for  $\mathcal{M}$ ,

$$|\mathcal{M}|^2 = g^4 \left| \frac{1}{m^2 - s - i0} + \frac{1}{m^2 - t - i0} + \frac{1}{m^2 - u - i0} \right|^2.$$

For elastic scattering in the center-of-mass frame,

$$t = -2k^2(1 - \cos \theta), \quad u = -2k^2(1 + \cos \theta), \quad k^2 = \frac{s}{4} - m^2.$$

The two final particles are identical. If the full solid angle is used, the differential cross-section therefore includes  $1/2!$ :

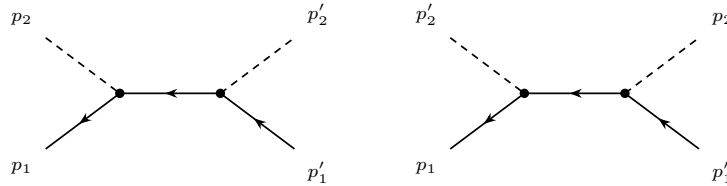
$$\boxed{\frac{d\sigma}{d\Omega} = \frac{g^4}{128\pi^2s} \left| \frac{1}{m^2 - s - i0} + \frac{1}{m^2 + 2k^2(1 - \cos \theta)} + \frac{1}{m^2 + 2k^2(1 + \cos \theta)} \right|^2}.$$

Equivalently one may omit  $1/2!$  and integrate over only one copy of the identical-particle phase space.

### W.6.2. It is useful to display the answer with the common fermion numerator

$$N(q) = -i\gamma_\mu q^\mu + M, \quad D_F(q) = \frac{N(q)}{q^2 + M^2 - i0}.$$

The factors  $\beta = i\gamma^0$  in the propagator convert the external  $u^\dagger, v^\dagger$  into barred spinors. Apart from the common momentum delta function and the scalar external-line factors, the two connected graphs for antifermion–boson scattering are

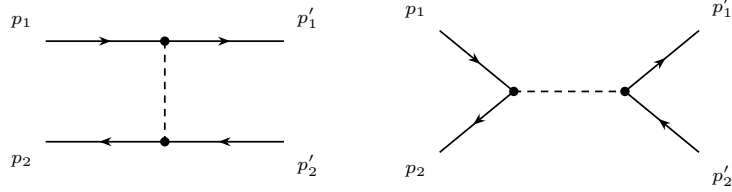




They give

$$\mathcal{A}_{\bar{F}B} = g^2 \bar{v}(p_1) \gamma_5 [D_F(-p_1 - p_2) + D_F(p'_2 - p_1)] \gamma_5 v(p'_1).$$

The first term is the annihilation-ordering (the crossed version of the fermion  $s$ -channel graph), and the second is the exchange ordering. For fermion–antifermion elastic scattering the connected scalar exchange can be either annihilating or nonannihilating:

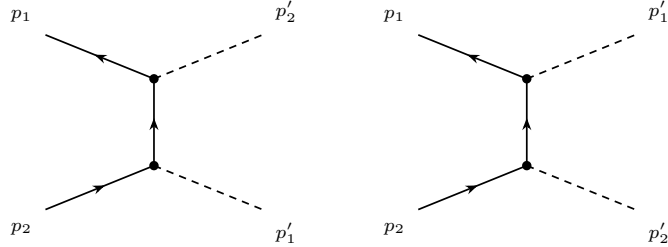


These give

$$\begin{aligned} \mathcal{A}(F(p_1) + F^c(p_2) \rightarrow F(p'_1) + F^c(p'_2)) &= g^2 \frac{\bar{u}(p'_1) \gamma_5 u(p_1) \bar{v}(p_2) \gamma_5 v(p'_2)}{(p_1 - p'_1)^2 + m^2 - i0} \\ &\quad - g^2 \frac{\bar{v}(p_2) \gamma_5 u(p_1) \bar{u}(p'_1) \gamma_5 v(p'_2)}{(p_1 + p_2)^2 + m^2 - i0}. \end{aligned}$$

The relative minus sign follows by putting the external fermionic operators into the same reference order.

Finally,  $F^c + F \rightarrow B + B$  has two fermion-exchange graphs:



They give

$$\mathcal{A}_{\bar{F}F \rightarrow BB} = g^2 \bar{v}(p_1) \gamma_5 [D_F(p_2 - p'_1) + D_F(p_2 - p'_2)] \gamma_5 u(p_2).$$

With the chapter's delta-normalized states, the connected matrix element for any of these processes is

$$S_{fi}^c = -i(2\pi)^{-2} \delta^4(P_f - P_i) \prod_{\text{external } B} (2E_B)^{-1/2} \mathcal{A},$$

where the displayed spinors already contain the fermionic external-line normalizations. These formulas contain no remaining position or internal momentum integrals.

**W.6.3.** At first order, all  $4!$  attachments of the four external scalars cancel the  $1/4!$  in  $H_I$ . Hence

$$S_{fi}^{(1)c} = \frac{-i(2\pi)^{-2} g \delta^4(p_1 + p_2 - p'_1 - p'_2)}{\sqrt{16E_1 E_2 E'_1 E'_2}}, \quad \mathcal{M}^{(1)} = -g.$$

The center-of-mass differential cross-section is consequently

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{g^2}{128\pi^2 s}}$$

when the full solid angle is retained for the identical final particles.

At order  $g^2$ , vacuum bubbles and disconnected spectator pieces do not contribute to the connected  $2 \rightarrow 2$  matrix element. The three remaining bubble diagrams have identical symmetry factor  $1/2$ . Defining

$$I(P) = \int \frac{d^4 q}{(2\pi)^4} \frac{1}{(q^2 + m^2 - i0)((q + P)^2 + m^2 - i0)},$$

the correction can be written, with all  $x$ -integrals done, as

$$\mathcal{M}^{(2)} = \frac{ig^2}{2} [I(p_1 + p_2) + I(p_1 - p'_1) + I(p_1 - p'_2)],$$

up to the same overall  $i\mathcal{M}$  convention used at first order. Equivalently,

$$S_{fi}^{(2)c} = \frac{-i(2\pi)^{-2} \delta^4(P_f - P_i)}{\sqrt{16E_1 E_2 E'_1 E'_2}} \mathcal{M}^{(2)}.$$

Each term contains precisely one independent four-momentum, in agreement with  $L = I - V + 1 = 2 - 2 + 1 = 1$ . The integrals are ultraviolet divergent; the problem asks only for their unrenormalized one-momentum representation.

**W.6.4.** Differentiation commutes with forming the contraction, provided the derivative acts on the  $x$  argument:

$$(\partial_\mu \psi_\ell(x) \bar{\psi}_m(y))_0 = \partial_{x^\mu} (\psi_\ell(x) \bar{\psi}_m(y))_0 = -i \partial_{x^\mu} \Delta_{\ell m}(x - y).$$

For a Dirac field,

$$\Delta_{\ell m}(x - y) = \int \frac{d^4 q}{(2\pi)^4} \frac{[(-i\gamma_\nu q^\nu + M)\beta]_{\ell m} e^{iq \cdot (x-y)}}{q^2 + M^2 - i0}.$$

Therefore the line with the differentiated end contributes

$$-i \int \frac{d^4 q}{(2\pi)^4} \frac{iq_\mu [(-i\gamma_\nu q^\nu + M)\beta]_{\ell m}}{q^2 + M^2 - i0} e^{iq \cdot (x-y)}.$$

Thus the ordinary Dirac propagator factor is simply multiplied by  $iq_\mu$  when the momentum  $q$  flows from  $y$  to  $x$ . No extra contact term occurs for one derivative; contact terms first arise when enough time derivatives act on the step functions.

**W.6.5.** Let  $\phi$  denote the free interaction-picture field and write

$$D(x-y) \equiv \langle \Omega | T\{\phi(x)\phi(y)\} | \Omega \rangle = -i\Delta_F(x-y).$$

The theorem of Section 6.4, with vacuum-bubble normalization understood, gives the Gell-Mann–Low ratio

$$\langle \Omega | T\{\Phi(x_1) \cdots \Phi(x_n)\} | \Omega \rangle = \frac{\langle \Omega | T\{\phi(x_1) \cdots \phi(x_n) e^{-i(g/3!) \int d^4z \phi(z)^3} \} | \Omega \rangle}{\langle \Omega | T\{e^{-i(g/3!) \int d^4z \phi(z)^3} \} | \Omega \rangle}.$$

At order  $g$ , three equivalent contractions attach  $x$  to the vertex and contract the remaining two fields together. Hence

$$\langle \Omega | \Phi(x) | \Omega \rangle = -\frac{ig}{2} \int d^4z D(x-z)D(0) + O(g^3).$$

The absence of an order- $g^2$  term follows from the odd total number of free fields.

For the time-ordered two-point function, the denominator cancels every vacuum bubble. Through order  $g^2$  the remaining Wick contractions are

$$\begin{aligned} \langle \Omega | T\{\Phi(x)\Phi(y)\} | \Omega \rangle &= D(x-y) \\ &+ \frac{(-ig)^2}{2} \int d^4z d^4w D(x-z)D(z-w)^2D(w-y) \\ &+ \frac{(-ig)^2}{2} \int d^4z d^4w D(x-z)D(y-z)D(z-w)D(0) \\ &+ \frac{(-ig)^2}{4} \int d^4z d^4w D(x-z)D(0)D(y-w)D(0) + O(g^4). \end{aligned}$$

The second line is the one-loop self-energy graph, the third is the connected tadpole insertion, and the fourth is the disconnected product  $\langle \Omega | \Phi(x) | \Omega \rangle \langle \Omega | \Phi(y) | \Omega \rangle$  to this order. The coefficients  $1/2, 1/2, 1/4$  are the corresponding Wick-contraction symmetry factors.

## Supplementary Exercises

### S.6.1. Properties of Wightman and Feynman Functions

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Relativistic Quantum Field Theory I, Spring 2023, Assignment 4, Notes 1–3 and Problem 1, pp. 1–3.)

For this problem use the following definitions and transformation laws. For a complex scalar field  $\phi$ , define

$$\begin{aligned} G_+(x, x') &= \langle \Omega | \phi(x)\phi^\dagger(x') | \Omega \rangle, \\ G_R(x, x') &= \theta(x^0 - x'^0)\Delta(x, x'), & \Delta(x, x') &= \langle \Omega | [\phi(x), \phi^\dagger(x')] | \Omega \rangle, \\ G_A(x, x') &= -\theta(x'^0 - x^0)\Delta(x, x'), \end{aligned}$$

and

$$\begin{aligned} G_F(x, x') &= \langle \Omega | T \{ \phi(x) \phi^\dagger(x') \} | \Omega \rangle \\ &= \theta(x^0 - x'^0) \langle \Omega | \phi(x) \phi^\dagger(x') | \Omega \rangle + \theta(x'^0 - x^0) \langle \Omega | \phi^\dagger(x') \phi(x) | \Omega \rangle. \end{aligned}$$

These definitions can differ from those in other texts by an overall factor of  $i$  or by a minus sign.

The unitary operator generating a spacetime translation  $y^\mu$  is

$$U_y = e^{-iP^\mu y_\mu} = e^{iHy^0 - iP^i y_i}, \quad P^\mu = (H, \mathbf{P}),$$

where the vacuum energy is understood to have been subtracted from  $H$ . For an on-shell momentum  $k^\mu = (\omega_{\mathbf{k}}, \mathbf{k})$ , the free complex field and its adjoint have the mode expansions

$$\begin{aligned} \phi(x) &= \int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} \left[ a(\mathbf{k}) e^{ik \cdot x} + b^\dagger(\mathbf{k}) e^{-ik \cdot x} \right], \\ \phi^\dagger(x) &= \int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} \left[ b(\mathbf{k}) e^{ik \cdot x} + a^\dagger(\mathbf{k}) e^{-ik \cdot x} \right], \quad \omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}. \end{aligned}$$

The antiparticle oscillators transform in the same way as the particle oscillators. Explicitly,

$$\begin{aligned} U_y a(\mathbf{k}) U_y^\dagger &= e^{ik \cdot y} a(\mathbf{k}), & U_y a^\dagger(\mathbf{k}) U_y^\dagger &= e^{-ik \cdot y} a^\dagger(\mathbf{k}), \\ U_y b(\mathbf{k}) U_y^\dagger &= e^{ik \cdot y} b(\mathbf{k}), & U_y b^\dagger(\mathbf{k}) U_y^\dagger &= e^{-ik \cdot y} b^\dagger(\mathbf{k}). \end{aligned}$$

The operator generating a Lorentz transformation is

$$U_\Lambda = e^{\frac{i}{2} \omega_{\mu\nu} M^{\mu\nu}}, \quad \omega_{\mu\nu} = -\omega_{\nu\mu},$$

where the  $\omega_{\mu\nu}$  are finite constants. It acts as

$$\begin{aligned} U_\Lambda a(\mathbf{k}) U_\Lambda^\dagger &= \sqrt{\frac{\omega_{\Lambda\mathbf{k}}}{\omega_{\mathbf{k}}}} a(\Lambda\mathbf{k}), \\ U_\Lambda a^\dagger(\mathbf{k}) U_\Lambda^\dagger &= \sqrt{\frac{\omega_{\Lambda\mathbf{k}}}{\omega_{\mathbf{k}}}} a^\dagger(\Lambda\mathbf{k}), \\ U_\Lambda b(\mathbf{k}) U_\Lambda^\dagger &= \sqrt{\frac{\omega_{\Lambda\mathbf{k}}}{\omega_{\mathbf{k}}}} b(\Lambda\mathbf{k}), \\ U_\Lambda b^\dagger(\mathbf{k}) U_\Lambda^\dagger &= \sqrt{\frac{\omega_{\Lambda\mathbf{k}}}{\omega_{\mathbf{k}}}} b^\dagger(\Lambda\mathbf{k}), \end{aligned}$$

where  $\Lambda\mathbf{k}$  denotes the spatial part of  $\Lambda k$ .

- (a) For a free scalar field theory, use the two oscillator transformation laws above to show that

$$U_\Lambda \phi(x) U_\Lambda^\dagger = \phi(\Lambda x).$$

- (b) In this and the remaining parts, consider a general interacting theory, for which the free-field mode expansion of  $\phi$  does not apply. If the system is translation and Lorentz invariant, the result of the preceding part and

$$U_y \phi(x) U_y^\dagger = \phi(x + y)$$

remain valid.<sup>58</sup> Use translation invariance to prove

$$G_+(x, x') = G_+(x - x').$$

The same conclusion holds for all the two-point functions defined above, but you need not prove each case separately.

- (c) Use Lorentz invariance and the preceding result to show that

$$G_+(x, x') = \theta(x^0 - x'^0) G((x - x')^2) + \theta(x'^0 - x^0) G^*((x - x')^2),$$

where  $G(\xi)$  is a function satisfying

$$G(\xi) = G^*(\xi), \quad \xi > 0.$$

- (d) Now take  $\phi$  to be real and show that

$$G_F(x, x') = G((x - x')^2),$$

where  $G$  is the same function as in the preceding part.

### S.6.2. Particle Production by an External Source

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Relativistic Quantum Field Theory I, Spring 2023, Assignment 4, Problem 2, pp. 3–6.)

This is the simplest example in which the  $S$ -matrix can be calculated exactly; the source sheet develops it from Peskin and Schroeder's example on p. 32 and Problem 4.1. Consider a free real scalar field coupled to a fixed external source  $J(x)$ , with Lagrangian density

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + J(x) \phi = \mathcal{L}_0 + J(x) \phi.$$

Assume

$$J(t, \mathbf{x}) \longrightarrow 0 \quad \text{as} \quad t \longrightarrow \pm\infty \quad \text{or} \quad |\mathbf{x}| \longrightarrow \infty,$$

and define

$$J(p) = \int d^4x e^{-ip \cdot x} J(x).$$

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<sup>58</sup>The latter relation can be proved by writing  $H$ ,  $P^i$ , and  $M^{\mu\nu}$  in terms of  $\phi$  and its conjugate momentum and using their canonical commutation relations.

Suppose that  $J(p)$  is analytic in the complex  $p^0$  plane. For reference, the Green functions used below are

$$\begin{aligned}\Delta(x, x') &= \langle \Omega | [\phi(x), \phi(x')] | \Omega \rangle, \\ G_R(x, x') &= \theta(x^0 - x'^0) \Delta(x, x'), \\ G_A(x, x') &= -\theta(x'^0 - x^0) \Delta(x, x'), \\ G_F(x, x') &= \langle \Omega | T \{ \phi(x) \phi(x') \} | \Omega \rangle.\end{aligned}$$

Because  $J(x)$  depends on time, the system does not have time-translation symmetry. In particular, the vacuum at past infinity is generally different from the vacuum at future infinity. Suppose the system starts in the past vacuum  $|0\rangle_{\text{in}}$ . In the Heisenberg picture it remains in this same state at all times. At future infinity it is therefore not the ground state, because  $|0\rangle_{\text{in}} \neq |0\rangle_{\text{out}}$ , and it contains particle excitations. In other words, turning on the source has produced particles. We shall find the relation between the past and future vacua and calculate the probability for producing particles. In the far past the field has the expansion

$$\phi(x) = \phi_{\text{in}}(x) \equiv \int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} \left[ a_{\text{in}}(\mathbf{k}) e^{ik \cdot x} + a_{\text{in}}^\dagger(\mathbf{k}) e^{-ik \cdot x} \right], \quad x^0 \longrightarrow -\infty,$$

where  $k^0 = \omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$  and

$$[a_{\text{in}}(\mathbf{k}), a_{\text{in}}^\dagger(\mathbf{k}')] = (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}').$$

Define

$$a_{\text{in}}(\mathbf{k})|0\rangle_{\text{in}} = 0, \quad {}_{\text{in}}\langle 0|0\rangle_{\text{in}} = 1,$$

and

$$|\mathbf{k}_1, \dots, \mathbf{k}_n\rangle_{\text{in}} = \sqrt{2\omega_{\mathbf{k}_1}} \cdots \sqrt{2\omega_{\mathbf{k}_n}} a_{\text{in}}^\dagger(\mathbf{k}_1) \cdots a_{\text{in}}^\dagger(\mathbf{k}_n) |0\rangle_{\text{in}}.$$

Thus

$${}_{\text{in}}\langle \mathbf{k} | \mathbf{k}' \rangle_{\text{in}} = 2\omega_{\mathbf{k}} (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}').$$

Similarly, in the far future,

$$\phi(x) = \phi_{\text{out}}(x) \equiv \int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} \left[ a_{\text{out}}(\mathbf{k}) e^{ik \cdot x} + a_{\text{out}}^\dagger(\mathbf{k}) e^{-ik \cdot x} \right], \quad x^0 \longrightarrow +\infty,$$

with

$$a_{\text{out}}(\mathbf{k})|0\rangle_{\text{out}} = 0, \quad {}_{\text{out}}\langle 0|0\rangle_{\text{out}} = 1,$$

and

$$|\mathbf{k}_1, \dots, \mathbf{k}_n\rangle_{\text{out}} = \sqrt{2\omega_{\mathbf{k}_1}} \cdots \sqrt{2\omega_{\mathbf{k}_n}} a_{\text{out}}^\dagger(\mathbf{k}_1) \cdots a_{\text{out}}^\dagger(\mathbf{k}_n) |0\rangle_{\text{out}}.$$

Because of the external source,  $a_{\text{in}}(\mathbf{k}) \neq a_{\text{out}}(\mathbf{k})$ . The full field for  $-\infty < x^0 < +\infty$  interpolates between the two asymptotic expansions. Take the initial state to be  $|0\rangle_{\text{in}}$ .

- (a) For general  $x^0$ , solve the classical field equation and show that the solution can be written

$$\phi(x) = \phi_0(x) + i \int d^4x' G(x - x') J(x'),$$

where

$$(-\square + m^2)\phi_0(x) = 0, \quad (-\square + m^2)G(x - x') = -i\delta^4(x - x').$$

Which of the retarded, advanced, and Feynman Green functions must be used?

- (b) Promote  $\phi$  to a quantum operator, treating the expression in part (a) as an operator equation. Show that  $\phi_0$  must be  $\phi_{\text{in}}$ .
- (c) Evaluate the solution at  $x^0 \rightarrow +\infty$  and find the relation between  $a_{\text{out}}(\mathbf{k})$  and  $a_{\text{in}}(\mathbf{k})$ .
- (d) Use the preceding relation to show that the expectation value  $\lambda$  of the total number of produced particles is

$$\lambda = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{|J(k)|^2}{2\omega_{\mathbf{k}}}, \quad k^0 = \omega_{\mathbf{k}}.$$

- (e) Show that

$$a_{\text{out}}(\mathbf{k}) = S^\dagger a_{\text{in}}(\mathbf{k}) S,$$

where the unitary operator  $S$  is

$$S = e^{iB}, \quad B = \int d^4x J(x) \phi_{\text{in}}(x).$$

- (f) Use the preceding result to show

$$S|0\rangle_{\text{out}} = |0\rangle_{\text{in}}, \quad S|\mathbf{k}_1, \dots, \mathbf{k}_n\rangle_{\text{out}} = |\mathbf{k}_1, \dots, \mathbf{k}_n\rangle_{\text{in}}.$$

These relations identify  $S$  as the scattering operator: for arbitrary free-theory states  $|\alpha\rangle$  and  $|\beta\rangle$ ,

$$S_{\beta\alpha} = {}_{\text{out}}\langle\beta||\alpha\rangle_{\text{in}} = {}_{\text{in}}\langle\beta|S|\alpha\rangle_{\text{in}}.$$

- (g) Show that  $S$  can be normal ordered as

$$S = e^{iB} = e^F e^G e^{-\lambda/2},$$

where

$$F = i \int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} J(k) a_{\text{in}}^\dagger(\mathbf{k}), \quad G = i \int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} J(-k) a_{\text{in}}(\mathbf{k}),$$

and  $\lambda$  is the quantity defined above.

- (h) Use the state relation and the normal-ordered form of  $S$  to find the vacuum-to-vacuum probability

$$P_0 = |{}_{\text{out}}\langle 0 || 0 \rangle_{\text{in}}|^2.$$

This is the probability that no particles are produced.

(i) Show that

$${}_{\text{out}}\langle \mathbf{k}_1, \dots, \mathbf{k}_n | 0 \rangle_{\text{in}} = i^n J(k_1) \cdots J(k_n) e^{-\lambda/2}.$$

(j) The probability  $dP$  of finding exactly  $n$  particles, with one particle in each momentum cell  $d^3\mathbf{k}_i$  around  $\mathbf{k}_i$ , is

$$dP = |{}_{\text{out}}\langle \mathbf{k}_1, \dots, \mathbf{k}_n | 0 \rangle_{\text{in}}|^2 \prod_{i=1}^n \left( \frac{d^3\mathbf{k}_i}{(2\pi)^3 2\omega_{\mathbf{k}_i}} \right).$$

Use this formula to show that the probability of producing exactly  $n$  particles is

$$P_n = e^{-\lambda} \frac{\lambda^n}{n!}.$$

Thus the multiplicity is Poisson distributed with mean  $\lambda$ . You are not being asked to derive the displayed differential-probability formula.

### Remarks on Problem 2.

1. Although the setup is simple, the fact that past and future vacua can be inequivalent in a time-dependent situation, with particle production as a result, is very general and is important in cosmology and black-hole physics. In particular, Hawking radiation is also a consequence of this phenomenon.

2. Only the on-shell component of  $J(k)$ , with  $k^0 = \omega_{\mathbf{k}}$ , contributes to particle production. This is the field-theory analogue of resonance in a forced harmonic oscillator.

3. If the source is turned on and off very slowly, the adiabatic theorem suggests that the system remains in its vacuum throughout. Indeed, in the adiabatic limit  $J(\omega, \mathbf{k})$  has support only near  $\omega = 0$ , so it has no massive on-shell component and no particles are produced.

### S.6.3. Counting a Four-Point Wick Contraction

(Inspired by MIT OpenCourseWare, 8.323 Recitation 6, Sections 1.1–1.2.)

In  $\phi^4$  theory, count the contractions that connect four specified external fields to one interaction vertex. Show how the count cancels the  $4!$  in the interaction and identify the resulting position-space vertex rule.

### S.6.4. Position-Space Feynman Rules in Cubic Scalar Theory

(Adapted from J. McGreevy, *Physics 215A, Fall 2023, Assignment 7, Problem 2.*)

Consider the real scalar theory

$$S[\phi] = \int d^Dx \left[ -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{g}{3!} \phi^3 \right].$$

(a) Briefly state its Feynman rules in position space, emphasizing how they differ from those of  $\phi^4$  theory.



- (b) Draw every diagram that corrects the normalized position-space two-point function at order  $g^2$ .
- (c) Find the symmetry factor of each diagram and verify the factors directly by counting Wick contractions.

The source writes the free action with the mostly-minus metric. It has been translated here to the mostly-plus convention of this volume; the interaction and all three requested tasks are unchanged.

### S.6.5. A Vacuum Bubble Cancels in the Ratio and A Vertex Delta Function from Fourier Phases

(Inspired by MIT OpenCourseWare, 8.323 Recitation 6, Sections 1.1–1.2 / MIT OpenCourseWare, 8.323 Recitation 6, Section 1.3.)

- (a) Expand the normalized two-point Gell-Mann–Low formula through second order in a generic interaction. Isolate a term consisting of the free two-point function times a vacuum bubble and demonstrate its cancellation against the denominator.
- (b) Fourier transform every line incident on a position-space vertex at  $x$ . Carry out the  $x$ -integral and derive the exact factor enforcing four-momentum conservation, including all powers of  $2\pi$ .

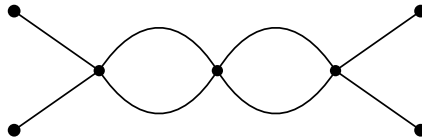
### S.6.6. A Two-Loop Four-Point Correlator

(Adapted from Daniel Harlow, Lecture Notes for Physics 8.323, Section 7.8, Homework Problem 4.)

Consider the real scalar theory

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$$

in four-dimensional Minkowski spacetime. The diagram below is a two-loop contribution to the Fourier transform of the connected Lorentzian time-ordered four-point function. Its black end-points are the four external positions and its three internal dots are quartic vertices.



Using the momentum-space Feynman rules, write the complete contribution of this diagram, including the overall momentum-conserving delta function, all four external propagators, all internal propagators, the vertex factors, and the symmetry factor. Assign a direction and momentum to every internal line. Carry out every momentum integral fixed by a vertex delta function, but leave the two independent loop integrations unevaluated.

### S.6.7. The Free Dirac Propagator, Its Residues, and Fermionic Signs

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 9, Problem 2.)

- (a) Starting from the free Dirac mode expansion and spin sums, show that the time-ordered contraction of  $\psi_\ell(x)$  with  $\bar{\psi}_m(y)$  is a first-order differential operator acting on the scalar Feynman propagator. Translate the result to the chapter's gamma-matrix convention.
- (b) Apply the free Dirac operator to the propagator obtained in S.6.13. Derive the spinor identity times  $\delta^4(x - y)$ , carefully retaining the contact term produced when the derivative acts on time ordering.
- (c) Find the momentum-space factors for contracting  $\partial_\mu \psi(x)$  with  $\bar{\psi}(y)$ , and  $\psi(x)$  with  $\partial_\mu \bar{\psi}(y)$ , when momentum flows from  $y$  to  $x$ . Explain the relative sign.
- (d) Factor the Dirac propagator denominator into its two energy poles. Evaluate the numerator at each pole and show that the residues reproduce the positive-energy  $u$ -spinor and negative-energy  $v$ -spinor completeness sums.
- (e) Derive the minus sign in the  $y^0 > x^0$  branch of  $T\{\psi(x)\bar{\psi}(y)\}$ . Show that omitting it prevents the two energy-pole residues from combining into a Lorentz-covariant propagator.
- (f) Follow the index contractions around a closed fermion loop with  $n$  bosonic insertions. Demonstrate why the matrices form a trace and why one additional minus sign remains relative to a bosonic loop.

### S.6.8. The Scalar Pole Prescription

(Inspired by David Tong, Lectures on Quantum Field Theory, Section 2.7.1.)

Locate the two scalar propagator poles in the complex energy plane under the chapter's  $-i0$  denominator convention. Explain how their displacement implements vacuum boundary conditions rather than retarded boundary conditions.

### S.6.9. Five-Particle Scattering in Cubic Scalar Theory

(Adapted from University of Cambridge, Part III, Quantum Field Theory, Paper 301, 2024, Question 2.)

Consider a real scalar field in four dimensions with the mostly-plus Lagrangian

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3,$$

where  $m$  and  $g$  are real and positive.

- (a) What is the mass dimension of  $g$ ? Briefly justify the answer.
- (b) State the Feynman rules for this theory. A brief but clear explanation is sufficient; a derivation of the rules is not required.

- (c) Consider two incoming particles with momenta  $p_1, p_2$  and three outgoing particles with momenta  $k_i$ ,  $i = 1, 2, 3$ . Starting from the expression for the scattering operator in terms of creation and annihilation operators, relate the corresponding matrix element

$${}_{\text{out}}\langle k_1, k_2, k_3 | S | p_1, p_2 \rangle_{\text{in}}$$

to a time-ordered correlation function. You may use either LSZ reduction or the Dyson formula.

- (d) Draw every connected tree diagram contributing to this process. If diagrams differ only by permutations of external particles, one representative may be drawn provided all allowed permutations are stated explicitly.
- (e) Use the Feynman rules to evaluate the contribution to the scattering matrix from every connected tree diagram in the preceding part. Again, a compact representative expression is acceptable only if all external permutations and their number are made explicit.

### S.6.10. Pair Creation Through Scalar Exchange

(Adapted from University of Cambridge, Part III Quantum Field Theory, Paper 301, 2023, Question 2.)

Consider two complex scalar fields  $\Psi_1$  and  $\Psi_2$ , of masses  $M_1$  and  $M_2$ , interacting with a real scalar field  $\phi$  of mass  $\mu$ . Let  $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}$ , where  $\mathcal{L}_0$  is the sum of the three free Lagrangians and

$$\mathcal{L}_{\text{int}} = -g\phi \sum_{i=1,2} \Psi_i^* \Psi_i.$$

Using Dyson's formula and Wick's theorem, calculate the tree-level scattering amplitude for the pair-creation process

$$\Psi_1(p) + \bar{\Psi}_1(q) \longrightarrow \Psi_2(p') + \bar{\Psi}_2(q'),$$

where every displayed four-momentum is on shell. Then work in the center-of-momentum frame,  $\mathbf{p} + \mathbf{q} = 0$ , and find the minimum value of  $|\mathbf{p}|$  for which the scattering amplitude is kinematically nonzero. Give the answer for both possible orderings of  $M_1$  and  $M_2$ .

### S.6.11. Exchange of Identical External Fermions

(Inspired by David Tong, Lectures on Quantum Field Theory, Section 5.7.)

Consider two identical fermions scattering through neutral-scalar exchange. Write both connected tree amplitudes and derive their relative minus sign from the ordering of external creation operators, not from the geometry of the drawings.

### S.6.12. Pairings from a Gaussian Generator

(Inspired by Daniel Harlow, Lecture Notes for Physics 8.323, Sections 7.2–7.3.)

Differentiate a multidimensional Gaussian generating function six times. Prove that the result is a sum over fifteen pairings and interpret every inverse quadratic kernel as a propagator line.

### S.6.13. Symmetry Factors from Diagram Automorphisms

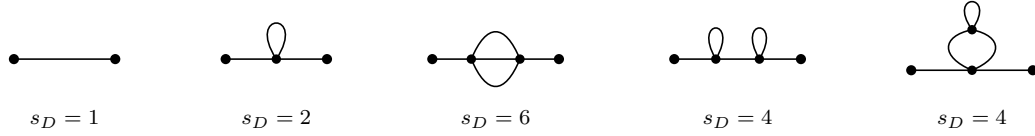
(Adapted from Daniel Harlow, Lecture Notes for Physics 8.323, Section 7.8, Homework Problem 3.)

Consider the real scalar theory

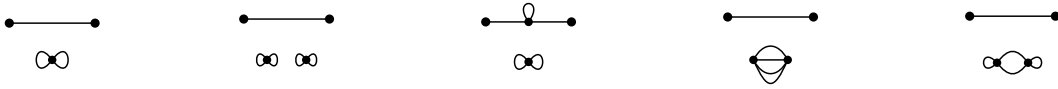
$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4.$$

Define the symmetry factor  $s_D$  of a diagram  $D$  as the number of its graph automorphisms, with every external point held fixed. A self-contracted propagator may be flipped, parallel propagators may be permuted, and interaction vertices may be permuted only when all incidences and all fixed external points are preserved.

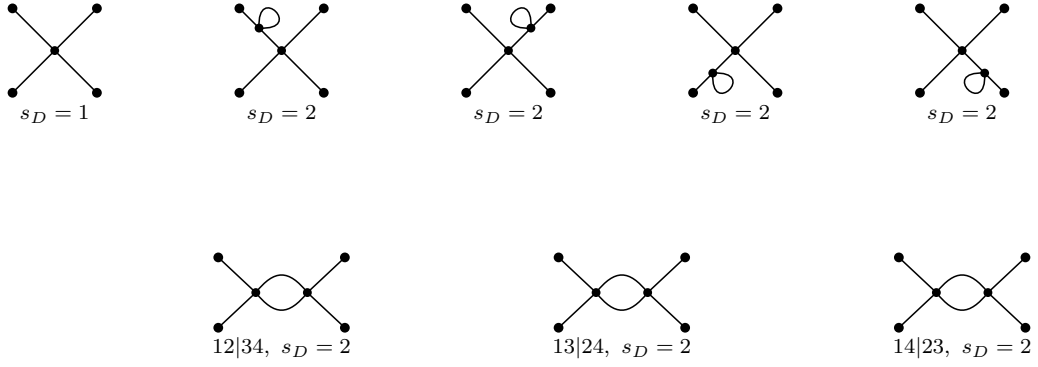
The connected part of source Figure 16 contains the following two-point diagrams. The factors quoted in its caption, in the displayed order, are  $s_D = 1, 2, 6, 4, 4$ . (The full numerator also contains products with vacuum bubbles; division by the vacuum functional removes those disconnected components.)



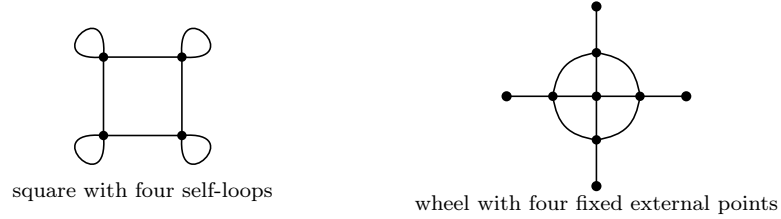
For completeness, the disconnected terms drawn in the same source Figure 16 are the free line times one vacuum figure-eight, the free line times two such vacuum components, the one-tadpole two-point graph times a vacuum figure-eight, and the free line times each of the two connected two-vertex vacuum topologies shown below. These are precisely the terms removed by the vacuum denominator.



Source Figure 18 gives all connected four-point diagrams through order  $\lambda^2$ : the one-vertex tree, four diagrams obtained by inserting a tadpole on one of its external legs, and the three pairings of the external points across a pair of vertices joined by two propagators. Its caption quotes  $s_D = 1$  for the tree and  $s_D = 2$  for each of the other seven diagrams. The complete topology data are displayed below; the labels on the last three diagrams record the three partitions of four fixed external points.



Finally, compute rather than assume the symmetry factors of the two diagrams from source Figure 20:



Verify every symmetry factor quoted for the Figure 16 and Figure 18 topologies, and determine both Figure 20 factors directly from their automorphism groups.

#### S.6.14. Connected Diagrams from a Logarithm

(Inspired by Daniel Harlow, *Lecture Notes for Physics 8.323, Section 7.4.*)

Let  $Z[J]$  be a perturbative vacuum functional and  $W[J] = \log Z[J]$ . Using the first four source derivatives, prove algebraically that derivatives of  $W$  generate connected correlation functions rather than all Wick partitions.

#### S.6.15. Propagator Corrections in a Solvable Theory

(Adapted from John McGreevy, *Physics 215A, Fall 2023, Assignment 7, Problem 4.*)

Consider a scalar field in  $D$  spacetime dimensions with action

$$S = S_0 + S_1, \quad S_0 = -\frac{1}{2} \int d^D x (\partial_\mu \phi \partial^\mu \phi + m_0^2 \phi^2), \quad S_1 = -\frac{1}{2} \int d^D x \delta m^2 \phi^2.$$

The mass term has been artificially split into two pieces. Perform perturbation theory in small  $\delta m^2$ , treating  $S_1$  as an interaction, and show that the resulting expansion reassembles the complete mass.

- (a) Write every Feynman rule for this perturbation theory.

- (b) Determine the one-particle-irreducible two-point function, defined by

$$-i\Sigma(p) \equiv \sum (\text{all 1PI diagrams with two external nubbins}).$$

- (c) Geometrically sum all propagator corrections and show that the result is the propagator one would have written without splitting  $m_0^2 + \delta m^2$ .

## Solutions to Supplementary Exercises

### S.6.1. Properties of Wightman and Feynman Functions

- (a) Use the free complex-scalar expansion supplied in the problem:

$$\phi(x) = \int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} \left[ a(\mathbf{k}) e^{ik \cdot x} + b^\dagger(\mathbf{k}) e^{-ik \cdot x} \right].$$

Apply  $U_\Lambda$  and use the supplied particle and antiparticle oscillator laws. In the annihilation term, for example, this gives

$$\int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} \sqrt{\frac{\omega_{\Lambda\mathbf{k}}}{\omega_{\mathbf{k}}}} a(\Lambda\mathbf{k}) e^{ik \cdot x}.$$

Set  $k' = \Lambda k$ . Invariance of the mass-shell measure,  $d^3\mathbf{k}/\omega_{\mathbf{k}} = d^3\mathbf{k}'/\omega_{\mathbf{k}'}$ , turns the measure and square-root factor into  $d^3\mathbf{k}'/\sqrt{2\omega_{\mathbf{k}'}}$ . Lorentz invariance of the inner product gives  $k \cdot x = k' \cdot (\Lambda x)$ . The antiparticle-creation term works identically, and therefore

$$U_\Lambda \phi(x) U_\Lambda^\dagger = \phi(\Lambda x).$$

- (b) Translation invariance of the vacuum means  $U_y|\Omega\rangle = |\Omega\rangle$ , after the vacuum energy has been subtracted. Insert  $U_y^\dagger U_y$  between the factors in the Wightman function and choose  $y = -x'$ :

$$\begin{aligned} G_+(x, x') &= \langle \Omega | U_{-x'}^\dagger (U_{-x'} \phi(x) U_{-x'}^\dagger) (U_{-x'} \phi^\dagger(x') U_{-x'}^\dagger) U_{-x'} | \Omega \rangle \\ &= \langle \Omega | \phi(x - x') \phi^\dagger(0) | \Omega \rangle \equiv G_+(x - x'). \end{aligned}$$

The argument uses only translation covariance of the interacting field, not a free-field expansion.

- (c) Put  $z = x - x'$ . Scalar covariance and vacuum invariance imply  $G_+(z) = G_+(\Lambda z)$ . In addition,

$$G_+(z)^* = \langle \Omega | \phi(0) \phi^\dagger(z) | \Omega \rangle = G_+(-z),$$

where the final equality is translation invariance. For timelike  $z$ , the proper orthochronous Lorentz group has separate future- and past-directed orbits. Define  $G(z^2)$  on the future-directed orbit; the identity above then fixes the past-directed value to be  $G^*(z^2)$ . Hence

$$G_+(x, x') = \theta(z^0) G(z^2) + \theta(-z^0) G^*(z^2).$$

For spacelike  $z$ , for which  $z^2 > 0$  in the mostly-plus convention, a proper Lorentz transformation can map  $z$  to  $-z$ . Lorentz invariance and the conjugation identity then give

$$G(z^2) = G^*(z^2), \quad z^2 > 0.$$

- (d) For a real scalar field,  $\phi^\dagger = \phi$ , so time ordering gives

$$G_F(x, x') = \theta(z^0)G_+(z) + \theta(-z^0)G_+(-z).$$

If  $z^0 > 0$ , the first term is  $G(z^2)$ . If  $z^0 < 0$ , then  $-z$  is future directed, so the second term is also  $G((-z)^2) = G(z^2)$ . Therefore

$$G_F(x, x') = G((x - x')^2),$$

with the same function  $G$  as in the preceding part.

### S.6.2. Particle Production by an External Source

- (a) Varying the action gives

$$(-\square + m^2)\phi(x) = J(x).$$

Let

$$\Delta(z) = \int \frac{d^3\mathbf{k}}{(2\pi)^3 2\omega_{\mathbf{k}}} \left( e^{ik \cdot z} - e^{-ik \cdot z} \right), \quad G_R(z) = \theta(z^0)\Delta(z).$$

The free equation annihilates  $\Delta$  away from  $z^0 = 0$ , while  $\Delta(0, \mathbf{z}) = 0$  and  $\partial_{z^0}\Delta(0, \mathbf{z}) = -i\delta^3(\mathbf{z})$ . It follows distributionally that

$$(-\square + m^2)G_R(z) = -i\delta^4(z).$$

Thus

$$\phi(x) = \phi_0(x) + i \int d^4x' G_R(x - x')J(x')$$

solves the sourced equation. The retarded Green function is required: it makes the response at  $x$  depend only on source values in its causal past. The advanced function would impose future data, while the Feynman function implements vacuum time ordering rather than a causal initial-value problem.

- (b) At times earlier than the support of  $J$ , the retarded integral vanishes. The full Heisenberg field must then approach the specified incoming free field, so

$$\phi_0(x) = \phi_{\text{in}}(x).$$

Because the source is a c-number, the added response is also a c-number and does not alter the canonical commutators.

- (c) At a time later than the support of the source, the step function in  $G_R$  is one throughout the integral. Using the mode representation of  $\Delta$  and the stated Fourier transform,

$$i \int d^4x' G_R(x - x')J(x') = i \int \frac{d^3\mathbf{k}}{(2\pi)^3 2\omega_{\mathbf{k}}} \left[ J(k)e^{ik \cdot x} - J(-k)e^{-ik \cdot x} \right].$$

Comparison with the future mode expansion gives the exact shifts

$$a_{\text{out}}(\mathbf{k}) = a_{\text{in}}(\mathbf{k}) + \frac{iJ(k)}{\sqrt{2\omega_{\mathbf{k}}}}$$

and

$$a_{\text{out}}^\dagger(\mathbf{k}) = a_{\text{in}}^\dagger(\mathbf{k}) - \frac{iJ(-k)}{\sqrt{2\omega_{\mathbf{k}}}}.$$

For a real source,  $J(-k) = J(k)^*$ , as required by Hermiticity of the field.

- (d) With the continuum normalization used in the prompt, the future number operator is

$$N_{\text{out}} = \int \frac{d^3\mathbf{k}}{(2\pi)^3} a_{\text{out}}^\dagger(\mathbf{k}) a_{\text{out}}(\mathbf{k}).$$

In the incoming vacuum, every term containing an incoming annihilator on the right or creator on the left vanishes. The remaining c-number term is

$$\lambda = {}_{\text{in}}\langle 0 | N_{\text{out}} | 0 \rangle_{\text{in}} = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{J(-k)J(k)}{2\omega_{\mathbf{k}}} = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{|J(k)|^2}{2\omega_{\mathbf{k}}}.$$

- (e) Insert the incoming expansion into  $B$ :

$$B = \int \frac{d^3\mathbf{k}}{(2\pi)^3 \sqrt{2\omega_{\mathbf{k}}}} \left[ J(-k) a_{\text{in}}(\mathbf{k}) + J(k) a_{\text{in}}^\dagger(\mathbf{k}) \right].$$

For real  $J$ , this operator is Hermitian, so  $S = e^{iB}$  is unitary. Since  $[a_{\text{in}}(\mathbf{k}), iB] = iJ(k)/\sqrt{2\omega_{\mathbf{k}}}$  is a c-number, all higher nested commutators vanish. Therefore

$$S^\dagger a_{\text{in}}(\mathbf{k}) S = e^{-iB} a_{\text{in}}(\mathbf{k}) e^{iB} = a_{\text{in}}(\mathbf{k}) + \frac{iJ(k)}{\sqrt{2\omega_{\mathbf{k}}}} = a_{\text{out}}(\mathbf{k}).$$

- (f) The preceding operator identity is equivalent to

$$a_{\text{in}}(\mathbf{k}) S = S a_{\text{out}}(\mathbf{k}).$$

Consequently  $a_{\text{in}}(\mathbf{k}) S |0\rangle_{\text{out}} = 0$  for every  $\mathbf{k}$ ; uniqueness of the normalized vacuum fixes the irrelevant common phase so that

$$S |0\rangle_{\text{out}} = |0\rangle_{\text{in}}.$$

Taking the adjoint of the oscillator relation also gives  $a_{\text{in}}^\dagger S = S a_{\text{out}}^\dagger$ . Acting on the normalized multiparticle definitions then yields

$$S |\mathbf{k}_1, \dots, \mathbf{k}_n\rangle_{\text{out}} = |\mathbf{k}_1, \dots, \mathbf{k}_n\rangle_{\text{in}}.$$

Taking the overlap with an incoming bra proves  ${}_{\text{out}}\langle \beta | | \alpha \rangle_{\text{in}} = {}_{\text{in}}\langle \beta | S | \alpha \rangle_{\text{in}}$ .



- (g) The expansion of  $B$  shows that  $iB = F + G$ , with  $F$  and  $G$  exactly as defined in the prompt. Their commutator is a c-number:

$$\begin{aligned} [F, G] &= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{J(k)J(-k)}{2\omega_{\mathbf{k}}} \\ &= \lambda. \end{aligned}$$

The Baker–Campbell–Hausdorff formula therefore terminates and gives

$$S = e^{F+G} = e^F e^G e^{-[F,G]/2} = e^F e^G e^{-\lambda/2}.$$

- (h) Using the scattering-operator relation from part (f),

$${}_{\text{out}}\langle 0|0\rangle_{\text{in}} = {}_{\text{in}}\langle 0|S|0\rangle_{\text{in}}.$$

In the normal-ordered form,  $G|0\rangle_{\text{in}} = 0$  and  ${}_{\text{in}}\langle 0|F = 0$ , so

$${}_{\text{out}}\langle 0|0\rangle_{\text{in}} = e^{-\lambda/2}, \quad \boxed{P_0 = e^{-\lambda}}.$$

- (i) The covariantly normalized bra contributes one factor  $\sqrt{2\omega_{\mathbf{k}_i}}a_{\text{in}}(\mathbf{k}_i)$  for each particle. Since

$$[a_{\text{in}}(\mathbf{k}), F] = \frac{iJ(k)}{\sqrt{2\omega_{\mathbf{k}}}},$$

commuting all annihilators through  $e^F$  cancels the square roots in the state normalization and gives

$$\boxed{{}_{\text{out}}\langle \mathbf{k}_1, \dots, \mathbf{k}_n | 0 \rangle_{\text{in}} = i^n J(k_1) \cdots J(k_n) e^{-\lambda/2}}.$$

- (j) Substitution of the amplitude into the supplied probability formula gives

$$dP = e^{-\lambda} \prod_{i=1}^n \left[ \frac{d^3\mathbf{k}_i}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k}_i}} |J(k_i)|^2 \right].$$

Integrating over all unordered  $n$ -particle configurations introduces the identical-particle factor  $1/n!$ . Each momentum integral equals  $\lambda$ , so

$$\boxed{P_n = e^{-\lambda} \frac{\lambda^n}{n!}}.$$

Indeed  $\sum_{n=0}^{\infty} P_n = 1$ , and differentiating the exponential generating function gives  $\sum_n n P_n = \lambda$ .

The formula for  $\lambda$  samples  $J(p)$  only at  $p^0 = \omega_{\mathbf{p}}$ , so only an on-shell source component can create particles. An adiabatically varying source has Fourier support concentrated near  $p^0 = 0$  and therefore produces no massive on-shell quanta. More generally, the calculation exhibits how a time-dependent background makes the natural past and future vacua inequivalent, the mechanism underlying many particle-production effects in cosmology and black-hole physics.

### S.6.3. Counting a Four-Point Wick Contraction

At first order the relevant term is

$$\frac{-i\lambda}{4!} \int d^4z \langle \Omega | T \{ \phi(x_1) \phi(x_2) \phi(x_3) \phi(x_4) \phi(z)^4 \} | \Omega \rangle.$$

For the connected tree contribution, the four labeled external fields must contract with four distinct fields at  $z$ . There are  $4! = 24$  bijections. They cancel the  $1/4!$ , leaving

$$(-i\lambda) \int d^4z \prod_{a=1}^4 D(x_a - z).$$

Thus a quartic position-space vertex contributes  $-i\lambda$ , with one propagator for each attached line; no factorial is appended to the diagrammatic rule.

### S.6.4. Position-Space Feynman Rules in Cubic Scalar Theory

(a) Write  $D_F(x - y) = \langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle$  for the free propagator. The position-space rules are:

- attach a factor  $D_F(x - y)$  to a line whose endpoints are  $x, y$ ;
- attach  $-ig \int d^Dz$  to each cubic interaction vertex;
- integrate every internal vertex position and divide by the graph's automorphism factor.

External insertions occur at their specified positions. Relative to  $\phi^4$  theory, every interaction vertex has three incident lines and the vertex factor is  $-ig$ , rather than four lines and  $-i\lambda$ . There is no order- $g$  two-point correction because the corresponding vacuum expectation value contains five fields.

(b) Vacuum normalization removes components touching neither external point. The three surviving order- $g^2$  topologies of the full normalized two-point function are

$$x \text{---} \bullet \text{---} \text{---} \bullet \text{---} y, \quad \begin{matrix} x \\ y \end{matrix} \text{---} \bullet \text{---} \bullet \text{---} \text{---} \bullet \text{---} y, \quad x \text{---} \bullet \text{---} \bullet \text{---} y \times \bullet \text{---} \bullet \text{---} y$$

They are, respectively, the two-line self-energy graph, the tadpole-chain graph (with both external insertions on one vertex), and the disconnected product of two one-point tadpoles. The last graph belongs to the full two-point function but not to its connected part.

(c) Before contractions the second-order coefficient is

$$\frac{(-ig)^2}{2!(3!)^2} \int d^Dz d^Dw = \frac{(-ig)^2}{72} \int d^Dz d^Dw.$$

For the self-energy graph there are

$$2 \cdot 3 \cdot 3 \cdot 2! = 36$$

contractions: choose which labeled vertex receives  $x$ , choose its field and the field receiving  $y$ , and pair the two remaining legs at each vertex. For the tadpole chain there are likewise

$$2 \cdot (3 \cdot 2) \cdot 3 = 36$$

contractions: choose the vertex receiving both labeled external fields, assign those fields to two of its legs, and choose the leg at the second vertex joined to the first. For the two one-point components there are

$$2 \cdot 3 \cdot 3 = 18$$

contractions. Thus the multiplicative graph factors are  $1/2, 1/2, 1/4$ , or equivalently the symmetry factors are  $2, 2, 4$ .

In formulas, with  $D_{ab} = D_F(a - b)$ , the complete non-vacuum part is

$$\begin{aligned} G^{(2)}(x, y) = (-ig)^2 \int d^D z d^D w \Big[ & \frac{1}{2} D_{xz} D_{zw}^2 D_{wy} \\ & + \frac{1}{2} D_{xz} D_{yz} D_{zw} D_{ww} \\ & + \frac{1}{4} D_{xz} D_{zz} D_{yw} D_{ww} \Big]. \end{aligned}$$

Diagrams in which  $x$  contracts directly with  $y$  and the two vertices form a vacuum graph cancel against the Gell-Mann–Low denominator. If the connected correlator is requested, the final  $1/4$  term is also omitted.

### S.6.5. A Vacuum Bubble Cancels in the Ratio and A Vertex Delta Function from Fourier Phases

(a) Let  $X = T\{\phi(x)\phi(y)\}$  and

$$N = \langle \Omega | T\{X e^{-i \int \mathcal{H}_I}\} | \Omega \rangle, \quad Z = \langle \Omega | T\{e^{-i \int \mathcal{H}_I}\} | \Omega \rangle.$$

Write  $Z = 1 + Z_1 + Z_2 + \dots$ . At second order, Wick's theorem contains a disconnected contribution  $N_2 \supset \langle \Omega | X | \Omega \rangle Z_2$ , and similarly at first order when allowed. Expanding the inverse,

$$Z^{-1} = 1 - Z_1 + (Z_1^2 - Z_2) + \dots,$$

shows that  $N/Z$  contains  $+\langle X || \Omega \rangle Z_2 - \langle X || \Omega \rangle Z_2 = 0$ . The cancellation is algebraic and applies separately to every vacuum component and its symmetry factor.

(b) Take all momenta as flowing into the vertex. A line of momentum  $q_a$  then supplies

$$D(x - x_a) = \int \frac{d^4 q_a}{(2\pi)^4} \tilde{D}(q_a) e^{iq_a \cdot (x - x_a)}.$$

The product of the  $x$ -dependent phases is  $\exp[ix \cdot \sum_a q_a]$ . Therefore

$$\int d^4x e^{ix \cdot \sum_a q_a} = (2\pi)^4 \delta^4\left(\sum_a q_a\right).$$

Multiplying by the interaction coefficient gives the vertex rule

$$-ig(2\pi)^4 \delta^4\left(\sum_{\text{in}} q - \sum_{\text{out}} q\right).$$

The powers of  $2\pi$  arise solely from the stated Fourier convention.

### S.6.6. A Two-Loop Four-Point Correlator

Take all four external momenta to point outward and put

$$p_1 + p_2 + p_3 + p_4 = 0, \quad P = p_1 + p_2 = -(p_3 + p_4).$$

Direct the two lines from the left vertex to the middle vertex and assign them momenta  $q$  and  $-P - q$ . Direct the two lines from the middle vertex to the right vertex and assign them  $k$  and  $-P - k$ . Momentum conservation then holds at each vertex. Of the twelve components initially present in the three vertex delta functions, eight fix the other internal momenta and four remain as the single overall delta function. Thus  $q$  and  $k$  are independent loop momenta.

For brevity define

$$D(\ell) = \ell^2 + m^2 - i0.$$

With the Fourier convention of Chapter 6, the requested contribution is

$$\begin{aligned} \tilde{G}_{c,D}^{(4)}(p_1, p_2, p_3, p_4) &= \frac{(-i\lambda)^3}{4} (2\pi)^4 \delta^4\left(\sum_{a=1}^4 p_a\right) \prod_{a=1}^4 \frac{-i}{D(p_a)} \\ &\times \int \frac{d^4q}{(2\pi)^4} \frac{d^4k}{(2\pi)^4} \frac{(-i)^4}{D(q)D(P+q)D(k)D(P+k)}. \end{aligned}$$

This retains the four external propagators because the source asks for the Fourier transform of the time-ordered correlator, not an amputated vertex. The factor  $1/4$  is the inverse symmetry factor: the two parallel lines on the left may be exchanged, and independently the two on the right may be exchanged. Hence  $s_D = 2!2! = 4$ . Finally  $I = 4$ ,  $V = 3$ , so  $L = I - V + 1 = 2$ , in agreement with the two remaining integrations.

### S.6.7. The Free Dirac Propagator, Its Residues, and Fermionic Signs

- (a) The two time branches of the mode expansion contain the completeness sums for  $u$  and  $v$ . In the chapter convention they combine into

$$\langle \Omega | T \{ \psi_\ell(x) \bar{\psi}_m(y) \} | \Omega \rangle = -i \Delta_{\ell m}(x - y),$$

where

$$\Delta_{\ell m}(z) = [(-\gamma^\mu \partial_\mu + M)\beta]_{\ell m} \Delta_F(z) = \int \frac{d^4 q}{(2\pi)^4} \frac{[(-i\gamma_\mu q^\mu + M)\beta]_{\ell m}}{q^2 + M^2 - i0} e^{iq \cdot z}.$$

Acting on the scalar propagator generates exactly the spinor numerator. The rightmost  $\beta = i\gamma^0$  reflects the definition  $\bar{\psi} = \psi^\dagger \beta$ .

- (b) Define  $D_F^D(z) = -i\Delta(z)$ . The numerator is the algebraic inverse of the free Dirac operator because

$$(i\gamma^\mu \partial_\mu - M)(-\gamma^\nu \partial_\nu + M) = -(\square - M^2)\mathbf{1},$$

using  $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ . Together with  $(\square - M^2)\Delta_F = -\delta^4$ , this yields the identity matrix times the delta source, with the fixed right factor  $\beta$  appropriate to the barred index. In the direct time-ordered derivation, the same delta function comes from differentiating the step functions and using  $\{\psi_\ell(t, \mathbf{x}), \psi_m^\dagger(t, \mathbf{y})\} = \delta_{\ell m} \delta^3(\mathbf{x} - \mathbf{y})$ .

- (c) For momentum  $q$  flowing from  $y$  to  $x$ , the propagator phase is  $e^{iq \cdot (x-y)}$ . Therefore

$$\partial_{x^\mu} e^{iq \cdot (x-y)} = iq_\mu e^{iq \cdot (x-y)}, \quad \partial_{y^\mu} e^{iq \cdot (x-y)} = -iq_\mu e^{iq \cdot (x-y)}.$$

Thus a derivative on  $\psi(x)$  multiplies the ordinary contraction by  $iq_\mu$ , whereas a derivative on  $\bar{\psi}(y)$  multiplies it by  $-iq_\mu$ . The relative sign is not a fermionic exchange sign; it is the elementary consequence of differentiating the two opposite arguments of a translation-invariant function.

- (d) With  $E_{\mathbf{q}} = \sqrt{\mathbf{q}^2 + M^2}$ ,

$$q^2 + M^2 - i0 = -(q^0 - E_{\mathbf{q}} + i0)(q^0 + E_{\mathbf{q}} - i0).$$

At  $q^0 = E_{\mathbf{q}}$ , the numerator  $[(-i\gamma_\mu q^\mu + M)\beta]$  equals the positive-energy spin sum, up to the normalization factor  $2E_{\mathbf{q}}$  fixed in Chapter 5. At  $q^0 = -E_{\mathbf{q}}$ , change  $\mathbf{q} \rightarrow -\mathbf{q}$ ; the corresponding residue is the negative-energy  $vv^\dagger$  sum. The opposite pole orientations then reproduce the two terms in the time-ordered mode expansion. This check fixes both the numerator and the  $i0$  assignment.

- (e) By definition,

$$T\{\psi(x)\bar{\psi}(y)\} = \theta(x^0 - y^0)\psi(x)\bar{\psi}(y) - \theta(y^0 - x^0)\bar{\psi}(y)\psi(x).$$

The minus sign is required because interchanging two fermionic operators is an odd permutation. Vacuum expectation values then give a positive-energy  $u$ -sum in the first branch and minus the negative-energy  $v$ -sum in the second. That relative sign is exactly what allows both residues to be written with the single covariant numerator  $-i\gamma_\mu q^\mu + M$ . With a plus sign, the coefficients of  $q^0$  and  $M$  would not transform as one Dirac matrix polynomial.

- (f) Writing spinor indices explicitly, a loop with insertions  $\Gamma_1, \dots, \Gamma_n$  contains

$$(\Gamma_1)_{a_1 a_2} D_{a_2 a_3} \cdots (\Gamma_n)_{a_n a_1} = \text{Tr}(\Gamma_1 D \cdots \Gamma_n D).$$

To obtain this cyclic contraction from the original operator string, one must move one fermionic operator around the complete loop so that every  $\psi$  is paired with the next  $\bar{\psi}$ . This is one additional odd permutation after the signs already represented by the propagators. Consequently every closed fermion loop carries a factor  $-1$ , independent of the number  $n$  of bosonic insertions.

### S.6.8. The Scalar Pole Prescription

The denominator factorizes as

$$q^2 + m^2 - i0 = -(q^0 - E_{\mathbf{q}} + i0)(q^0 + E_{\mathbf{q}} - i0).$$

Thus  $+E_{\mathbf{q}}$  is displaced below and  $-E_{\mathbf{q}}$  above the real axis. For positive time the Fourier contour selects the positive-energy pole; for negative time it selects the negative-energy pole. A retarded Green function would place both poles below the real axis, forcing the function to vanish for negative time. The Feynman placement instead propagates positive-frequency modes forward and negative-frequency modes backward, which implements vacuum time ordering rather than causal response to a classical source.

### S.6.9. Five-Particle Scattering in Cubic Scalar Theory

- (a) In four dimensions the action is dimensionless and  $[\mathcal{L}] = 4$ . The kinetic term gives  $[\phi] = 1$ , so

$$[g] + 3[\phi] = 4, \quad [g] = 1.$$

Thus the cubic coupling has dimensions of mass.

- (b) For every internal scalar line carrying momentum  $q$ , include

$$\frac{-i}{q^2 + m^2 - i0};$$

for every cubic vertex include  $-ig$  and enforce four-momentum conservation; integrate each undetermined internal momentum with  $d^4q/(2\pi)^4$ ; divide by the graph symmetry factor; and sum all topologically distinct connected graphs. For a correlation function one also includes an external propagator on every displayed field insertion. LSZ amputation removes those propagators for a scattering matrix element.

- (c) In terms of asymptotic creation and annihilation operators the connected matrix element is

$$S_{fi,c} = \langle \Omega | a_{\text{out}}(k_3) a_{\text{out}}(k_2) a_{\text{out}}(k_1) a_{\text{in}}^\dagger(p_1) a_{\text{in}}^\dagger(p_2) | \Omega \rangle_c.$$

Expressing each asymptotic oscillator as a large-time Klein–Gordon inner product with the Heisenberg field and integrating the time surfaces inward gives LSZ reduction. Let

$$G_c^{(5)}(x_1, x_2, x_3; y_1, y_2) = \langle \Omega | T \{ \phi(x_1) \phi(x_2) \phi(x_3) \phi(y_1) \phi(y_2) \} | \Omega \rangle_c.$$

With the chapter’s Fourier phases, and suppressing only the common external state-normalization factors, the result is

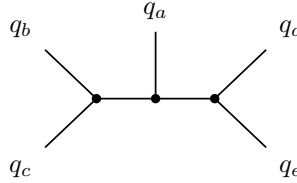
$$S_{fi,c} = Z^{-5/2} \prod_{r=1}^3 \left[ \int d^4 x_r e^{-ik_r \cdot x_r} (-i(\square_{x_r} - m^2)) \right] \\ \times \prod_{s=1}^2 \left[ \int d^4 y_s e^{+ip_s \cdot y_s} (-i(\square_{y_s} - m^2)) \right] G_c^{(5)}(x_1, x_2, x_3; y_1, y_2).$$

Indeed  $-i(\square - m^2)$  is the inverse of the position-space propagator whose momentum-space value is  $-i/(q^2 + m^2 - i0)$ . At tree level  $Z = 1$ . The same expression follows by expanding the Dyson exponential and contracting its fields with the five external oscillators.

(d) Put every external momentum in outgoing convention:

$$q_1 = -p_1, \quad q_2 = -p_2, \quad q_3 = k_1, \quad q_4 = k_2, \quad q_5 = k_3, \quad \sum_{a=1}^5 q_a = 0.$$

For a connected cubic tree with five external legs,  $3V = 2I + 5$  and  $I = V - 1$ , so  $V = 3$  and  $I = 2$ . Its unique topology is a chain of three vertices: one external momentum  $q_a$  attaches to the middle vertex, while two unordered pairs attach to the end vertices. A representative labelling  $a; bc|de$  is



The complete set of external assignments is

$$\mathcal{P} = \{1; 23|45, 1; 24|35, 1; 25|34, \\ 2; 13|45, 2; 14|35, 2; 15|34, \\ 3; 12|45, 3; 14|25, 3; 15|24, \\ 4; 12|35, 4; 13|25, 4; 15|23, \\ 5; 12|34, 5; 13|24, 5; 14|23\}.$$

Here  $a; bc|de$  means that  $q_a$  is attached to the middle vertex and the pairs  $(q_b, q_c)$ ,  $(q_d, q_e)$  to the two ends. There are  $5 \times 3 = 15$  labeled diagrams. This list is precisely the set of external permutations allowed by the one chain topology.

(e) Define

$$D_{ab} = (q_a + q_b)^2 + m^2 - i0.$$

The diagram  $a; bc|de$  has symmetry factor one once the five external momenta are fixed, and its reduced scattering-matrix contribution is

$$S_{fi,c}^{(a;bc|de)} = (2\pi)^4 \delta^4(p_1 + p_2 - k_1 - k_2 - k_3) \frac{-ig^3}{D_{bc}D_{de}}.$$

The numerator follows directly from  $(-ig)^3(-i)^2 = -ig^3$ . Consequently the complete tree answer is

$$S_{fi,c}^{\text{tree}} = (2\pi)^4 \delta^4(p_1 + p_2 - k_1 - k_2 - k_3) (-ig^3) \sum_{a;bc|de \in \mathcal{P}} \frac{1}{D_{bc}D_{de}},$$

again with the common external normalization factors understood. This compact sum evaluates every one of the fifteen diagrams and is manifestly symmetric under permutations of the three outgoing particles (indeed, under permutations of all five legs after crossing).

### S.6.10. Pair Creation Through Scalar Exchange

The first contribution connecting the two species is the cross term in the second-order Dyson expansion,

$$\frac{(-ig)^2}{2!} \int d^4x d^4y T\{\phi\Psi_1^*\Psi_1(x) \phi\Psi_2^*\Psi_2(y) + (x \leftrightarrow y)\}.$$

The two assignments of the species to  $x, y$  cancel the  $2!$ . Wick contracting the two real fields produces one internal  $\phi$  propagator; the charged fields at one vertex annihilate the incoming  $\Psi_1\bar{\Psi}_1$  pair and those at the other create the outgoing  $\Psi_2\bar{\Psi}_2$  pair. The position integrals impose four-momentum conservation and set

$$k = p + q = p' + q'.$$

There is no second tree topology: each vertex preserves the species and the corresponding complex-scalar charge.

With the convention

$${}_{\text{out}}\langle f || i \rangle_{\text{in}}|_c = i(2\pi)^4 \delta^4(p + q - p' - q') \mathcal{M}$$

and the mostly-plus propagator  $-i/(k^2 + \mu^2 - i0)$ , the diagram gives

$$i\mathcal{M} = (-ig)^2 \frac{-i}{k^2 + \mu^2 - i0},$$

$$\mathcal{M} = \boxed{\frac{g^2}{(p+q)^2 + \mu^2 - i0}} = \frac{g^2}{\mu^2 - s - i0} = -\frac{g^2}{s - \mu^2 + i0},$$



where  $s \equiv -(p + q)^2$ . The overall delta function and the standard external-state normalizations are displayed separately by the stated  $S$ -matrix convention.

In the center-of-momentum frame, put  $\mathbf{q} = -\mathbf{p}$ . The incoming energy is

$$E_{\text{in}} = 2\sqrt{M_1^2 + |\mathbf{p}|^2}.$$

The least possible final energy occurs when the outgoing particles are both at rest in their center-of-momentum frame and equals  $2M_2$ . Energy conservation therefore requires

$$2\sqrt{M_1^2 + |\mathbf{p}|^2} \geq 2M_2.$$

Consequently

$$|\mathbf{p}|_{\text{min}} = \begin{cases} \sqrt{M_2^2 - M_1^2}, & M_2 > M_1, \\ 0, & M_2 \leq M_1. \end{cases}$$

The exchanged-particle pole at  $s = \mu^2$  affects the value (and in an ideal stable theory the interpretation) of the amplitude, but not this kinematic pair-production threshold.

### S.6.11. Exchange of Identical External Fermions

Neutral-scalar exchange gives the direct chain

$$\mathcal{M}_d = g^2 \frac{\bar{u}(p'_1)\Gamma u(p_1) \bar{u}(p'_2)\Gamma u(p_2)}{(p_1 - p'_1)^2 + m^2 - i0}.$$

The crossed attachment is

$$\mathcal{M}_x = g^2 \frac{\bar{u}(p'_2)\Gamma u(p_1) \bar{u}(p'_1)\Gamma u(p_2)}{(p_1 - p'_2)^2 + m^2 - i0}.$$

To compare both contractions, the crossed final creation operators must be interchanged once to restore the reference order. Therefore  $\mathcal{M} = \mathcal{M}_d - \mathcal{M}_x$ . The minus sign enforces antisymmetry under  $p'_1 \leftrightarrow p'_2$ ; it is not inferred from whether the drawn lines cross on the page.

### S.6.12. Pairings from a Gaussian Generator

For

$$Z[B] = \exp\left(\frac{1}{2}B_i(A^{-1})_{ij}B_j\right),$$

each derivative must ultimately act on a distinct remaining  $B$  for the result to survive at  $B = 0$ . Thus the six indices are partitioned into three unordered pairs. Their number is

$$\frac{6!}{2^3 3!} = 15.$$

Consequently

$$\partial_{i_1} \cdots \partial_{i_6} Z[B]|_{B=0} = \sum_{15 \text{ pairings}} (A^{-1})_{i_a i_b} (A^{-1})_{i_c i_d} (A^{-1})_{i_e i_f}.$$

Each factor  $A^{-1}$  joins the two indices in one pair and is therefore represented by one propagator line.

### S.6.13. Symmetry Factors from Diagram Automorphisms

For the Figure 16 two-point topologies, external points are fixed throughout.

- The free propagator has no interaction vertex and only the identity automorphism, so  $s_D = 1$ .
- In the one-tadpole graph the two half-edges making the self-contracted loop may be exchanged. Nothing else moves, so  $s_D = 2$ .
- In the sunset graph the three propagators joining the two interaction vertices may be permuted arbitrarily. The external points prevent exchange of the vertices, hence  $s_D = 3! = 6$ .
- In the graph with two tadpoles in series, each self-loop may be flipped independently. Thus  $s_D = 2 \cdot 2 = 4$ .
- In the nested graph, exchange of the two propagators joining the lower and upper vertices gives one  $\mathbb{Z}_2$ , and flipping the self-loop at the upper vertex gives another. Hence  $s_D = 2 \cdot 2 = 4$ .

These are exactly the quoted 1, 2, 6, 4, 4.

For source Figure 18, the one-vertex tree has no nontrivial automorphism once the four external points are fixed, so its factor is one. Each of the four external-leg tadpole diagrams has precisely the flip of its self-loop and therefore has factor two. Each of the three two-vertex diagrams has precisely the exchange of its two parallel internal propagators and therefore also has factor two. This proves the complete caption statement, including all seven order- $\lambda^2$  diagrams.

For the left diagram of source Figure 20, the four interaction vertices form a square. The square has the dihedral automorphism group  $D_4$ , of order eight. Independently, the self-loop at each of the four vertices may be flipped. The automorphism group is consequently

$$D_4 \ltimes (\mathbb{Z}_2)^4, \quad s_D = 8 \cdot 2^4 = 128.$$

For the right diagram, each of the four outer interaction vertices is attached to a distinct fixed external point. Every automorphism must therefore fix those four vertices, and then it must also fix the central vertex and every edge. There are no parallel edges or self-loops to permute, so the group is trivial and

$$s_D = 1.$$

### S.6.14. Connected Diagrams from a Logarithm

At zero source, assume one-point functions vanish. Since  $W = \log Z$ ,

$$W_{12} = \frac{Z_{12}}{Z}, \quad W_{1234} = \frac{Z_{1234}}{Z} - \frac{Z_{12}Z_{34} + Z_{13}Z_{24} + Z_{14}Z_{23}}{Z^2}.$$

The three subtractions remove every partition of four insertions into two disconnected pairs. With nonzero one-point functions, further terms subtract the 1+3, 1+1+2, and

1+1+1+1 partitions by the same product rule. Thus derivatives of  $W$  are cumulants: precisely the parts that cannot be factored into correlators of proper subsets, which are the connected diagrams.

### S.6.15. Propagator Corrections in a Solvable Theory

- (a) The free propagator is

$$D_0(p) = \frac{-i}{p^2 + m_0^2 - i0}.$$

Expanding  $e^{iS_1}$ , a mass-insertion vertex with two scalar lines gives  $-i\delta m^2$  and conserves the momentum entering it. The factor 1/2 in  $S_1$  is canceled by the two attachments of its identical fields. There are no vertices with any other valence, no additional couplings, and no closed-loop integration unless a diagram contains an undetermined momentum. Each undetermined momentum is integrated with  $d^D p / (2\pi)^D$ , and ordinary graph symmetry factors apply.

- (b) The sole 1PI diagram with two external nubbins is one mass insertion:

$$-i\Sigma(p) = -i\delta m^2, \quad \Sigma(p) = \delta m^2.$$

Any diagram with two or more insertions contains a free propagator between successive insertions; cutting that line disconnects it. Such a graph is one-particle reducible and therefore is not another term in  $\Sigma$ . There are no loop corrections because the interaction is purely quadratic.

- (c) Put  $A = p^2 + m_0^2 - i0$  and  $V = -i\delta m^2$ . The complete two-point function is the geometric chain

$$\begin{aligned} D(p) &= D_0(p) + D_0(p)V D_0(p) \\ &\quad + D_0(p)V D_0(p)V D_0(p) + \cdots \\ &= \frac{-i}{A} \left[ 1 - \frac{\delta m^2}{A} + \left( \frac{\delta m^2}{A} \right)^2 - \cdots \right]. \end{aligned}$$

Summing gives

$$D(p) = \frac{-i}{A + \delta m^2} = \frac{-i}{p^2 + m_0^2 + \delta m^2 - i0}.$$

This is exactly the free propagator with the unsplit mass  $m^2 = m_0^2 + \delta m^2$ . Thus the perturbative organization has reassembled the original quadratic theory, including all signs and the full mass dependence.

### References

1. F. J. Dyson, *Phys. Rev.* **75**, 486, 1736 (1949).
2. The formal statement of this result is known as *Wick's theorem*; see G. C. Wick, *Phys. Rev.* **80**, 268 (1950).

3. I do not know who first proved this theorem. It was known in the early 1950s to several theorists, including M. Gell-Mann and F. E. Low.
4. J. Schwinger, *Phys. Rev.* **82**, 914 (1951).

## 7 The Canonical Formalism

Ever since the birth of quantum field theory in the papers of Born, Dirac, Fermi, Heisenberg, Jordan, and Pauli in the late 1920s, its development has been historically linked to the canonical formalism, so much so that it seems natural to begin any treatment of the subject today by postulating a Lagrangian and applying to it the rules of canonical quantization. This is the approach used in most books on quantum field theory. Yet historical precedent is not a very convincing reason for using this formalism. If we discovered a quantum field theory that led to a physically satisfactory  $S$ -matrix, would it bother us if it could not be derived by the canonical quantization of some Lagrangian?

To some extent this question is moot because, as we shall see in Section 7.1, all of the most familiar quantum field theories furnish canonical systems, and these can easily be put in a Lagrangian form. However, there is no proof that every conceivable quantum field theory can be formulated in this way. And even if it can, this does not in itself explain why we should *prefer* to use the Lagrangian formalism as a starting point in constructing various quantum field theories.

The point of the Lagrangian formalism is that it makes it easy to satisfy Lorentz invariance and other symmetries: a classical theory with a Lorentz-invariant Lagrangian density will when canonically quantized lead to a Lorentz-invariant quantum theory. That is, we shall see here that such a theory allows the construction of suitable quantum mechanical operators that satisfy the commutation relations of the Poincaré algebra, and therefore leads to a Lorentz-invariant  $S$ -matrix.

This is not so trivial. We saw in the previous chapter that in theories with derivative couplings or spins  $j \geq 1$ , it is not enough to take the interaction Hamiltonian as the integral over space of a scalar interaction density; we also need to add non-scalar terms to the interaction density to compensate for non-covariant terms in the propagators. The canonical formalism with a scalar Lagrangian density will automatically provide these extra terms. Later, when we come to non-Abelian gauge theories in Volume II, this extra convenience will become a necessity; it would be just about hopeless to try to guess at the form of the Hamiltonian in such theories without starting with a Lorentz-invariant and gauge-invariant Lagrangian density.

### 7.1 Canonical Variables

In this section we shall show that various quantum field theories that we have constructed so far satisfy the commutation rules and equations of motion of the Hamiltonian version of the canonical formalism. It is the Hamiltonian formalism that is needed to calculate the  $S$ -matrix (whether by operator or path-integral methods) but it is not always easy to choose Hamiltonians that yield a Lorentz-invariant  $S$ -matrix. In the balance of this chapter we shall take the Lagrangian version of the canonical formalism as our starting point, and use it to derive physically satisfactory Hamiltonians. The purpose of the present section is to identify the canonical fields and their conjugates in various field theories, to tell us how to separate the free-field terms in the Lagrangian, and incidentally to reassure us that the canonical formalism is indeed applicable to physically realistic theories.

We first show that the free fields constructed in Chapter 5 automatically provide a system of quantum operators  $q^n(\mathbf{x}, t)$  and canonical conjugates  $p_n(\mathbf{x}, t)$  that satisfy the familiar canonical commutation or anticommutation relations:

$$\begin{aligned} [q^n(\mathbf{x}, t), p_{\bar{n}}(\mathbf{y}, t)] &= i\delta^3(\mathbf{x} - \mathbf{y})\delta_{\bar{n}}^n, & \text{if either particle is a boson,} \\ \{q^n(\mathbf{x}, t), p_{\bar{n}}(\mathbf{y}, t)\} &= i\delta^3(\mathbf{x} - \mathbf{y})\delta_{\bar{n}}^n, & \text{if both particles are fermions.} \end{aligned} \quad (7.1.1)$$

$$\begin{aligned} [q^n(\mathbf{x}, t), q^{\bar{n}}(\mathbf{y}, t)] &= 0, & \text{if either particle is a boson,} \\ \{q^n(\mathbf{x}, t), q^{\bar{n}}(\mathbf{y}, t)\} &= 0, & \text{if both particles are fermions.} \end{aligned} \quad (7.1.2)$$

$$\begin{aligned} [p_n(\mathbf{x}, t), p_{\bar{n}}(\mathbf{y}, t)] &= 0, & \text{if either particle is a boson,} \\ \{p_n(\mathbf{x}, t), p_{\bar{n}}(\mathbf{y}, t)\} &= 0, & \text{if both particles are fermions.} \end{aligned} \quad (7.1.3)$$

In each equation, the commutator line applies if either of the particles created and destroyed by the two operators are bosons, and the anticommutator line applies if both particles are fermions. For instance, the real scalar field  $\phi(x)$  for a self-charge-conjugate particle of zero spin was found in Section 5.2 to obey the commutation relation

$$[\phi(x), \phi(y)] = \Delta(x - y),$$

where  $\Delta$  is the function

$$\Delta(x - y) \equiv \int \frac{d^3k}{2k^0(2\pi)^3} \left[ e^{ik \cdot (x - y)} - e^{-ik \cdot (x - y)} \right]$$

with  $k^0 = \sqrt{\mathbf{k}^2 + m^2}$ . We note that

$$\Delta(\mathbf{x}, 0) = 0, \quad \dot{\Delta}(\mathbf{x}, 0) = -i\delta^3(\mathbf{x}).$$

(A dot denotes the derivative with respect to the time  $x^0$ .) It is easy then to see that the field and its time-derivative  $\dot{\phi}$  obey the equal-time commutation relations:

$$[\phi(\mathbf{x}, t), \dot{\phi}(\mathbf{y}, t)] = i\delta^3(\mathbf{x} - \mathbf{y}), \quad (7.1.4)$$

$$[\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = 0, \quad (7.1.5)$$

$$[\dot{\phi}(\mathbf{x}, t), \dot{\phi}(\mathbf{y}, t)] = 0. \quad (7.1.6)$$

Therefore we may define canonical variables

$$q(\mathbf{x}, t) \equiv \phi(\mathbf{x}, t), \quad p(\mathbf{x}, t) \equiv \dot{\phi}(\mathbf{x}, t) \quad (7.1.7)$$

which satisfy the canonical commutation relations (7.1.1)–(7.1.3).

For the complex scalar field of a particle of spin zero with a distinct antiparticle, the commutation relations are

$$[\phi(x), \phi^\dagger(y)] = \Delta(x - y), \quad [\phi(x), \phi(y)] = 0.$$

We may therefore define the free-particle canonical variables as the complex operators

$$q(\mathbf{x}, t) \equiv \phi(\mathbf{x}, t), \quad (7.1.8)$$

$$p(\mathbf{x}, t) \equiv \dot{\phi}^\dagger(\mathbf{x}, t). \quad (7.1.9)$$

Equivalently, writing  $\phi \equiv (\phi_1 + i\phi_2)/\sqrt{2}$  with  $\phi_k$  Hermitian for  $k = 1, 2$ , we have canonical variables

$$q^k(\mathbf{x}, t) = \phi_k(\mathbf{x}, t), \quad (7.1.10)$$

$$p_k(\mathbf{x}, t) = \dot{\phi}_k(\mathbf{x}, t), \quad (7.1.11)$$

and these satisfy the commutation relations (7.1.1)–(7.1.3).

For the real vector field of a particle of spin one, the commutation relations are given by Section 5.3 as

$$[v^\mu(x), v^\nu(y)] = \left[ \eta^{\mu\nu} - \frac{\partial^\mu \partial^\nu}{m^2} \right] \Delta(x - y).$$

(We are using  $v^\mu$  rather than  $V^\mu$  for the vector field because we want to reserve upper case letters here for the fields in the Heisenberg picture.) Here the free-particle canonical variables may be taken as

$$q^i(\mathbf{x}, t) = v^i(\mathbf{x}, t), \quad (7.1.12)$$

$$p_i(\mathbf{x}, t) = \dot{v}^i(\mathbf{x}, t) + \frac{\partial v^0(\mathbf{x}, t)}{\partial x^i}, \quad (7.1.13)$$

with  $i = 1, 2, 3$ . The reader may check that (7.1.12) and (7.1.13) satisfy the commutation relations (7.1.1)–(7.1.3). The field equations (5.3.36) and (5.3.38) together with Eq. (7.1.13) allow us to express  $v^0$  in terms of the other variables as

$$v^0 = m^{-2} \nabla \cdot \mathbf{p}, \quad (7.1.14)$$

so  $v^0$  is not regarded as one of the  $q$ s. The extension of these results to complex vector fields may be handled just as for complex scalar fields.

For the Dirac field of a non-Majorana spin  $\frac{1}{2}$  particle, Section 5.6 shows that the anticommutator is

$$\{\psi_\ell(x), \psi_{\ell'}^\dagger(y)\} = [(-\gamma^\mu \partial_\mu + m)\beta]_{\ell\ell'} \Delta(x - y)$$

and

$$\{\psi_\ell(x), \psi_{\ell'}(y)\} = 0.$$

Here it would be inconsistent to take  $\psi_\ell$  and  $\psi_\ell^\dagger$  to be independent canonical variables, because their anticommutator does not vanish at equal times. It is conventional instead to define

$$q^\ell(x) \equiv \psi_\ell(x), \quad (7.1.15)$$

$$p_\ell(x) \equiv i\psi_\ell^\dagger(x). \quad (7.1.16)$$

It is easy then to see that (7.1.15) and (7.1.16) satisfy the canonical anticommutation relations (7.1.1)–(7.1.3).

For any system of operators that satisfy commutation or anticommutation relations like (7.1.1)–(7.1.3), we may define a quantum mechanical functional derivative: for an arbitrary bosonic functional  $F[q(t), p(t)]$  of  $q^n(\mathbf{x}, t)$  and  $p_n(\mathbf{x}, t)$  at a fixed time  $t$ , we define<sup>59</sup>

$$\frac{\delta F[q(t), p(t)]}{\delta q^n(\mathbf{x}, t)} = i[p_n(\mathbf{x}, t), F[q(t), p(t)]], \quad (7.1.17)$$

$$\frac{\delta F[q(t), p(t)]}{\delta p_n(\mathbf{x}, t)} = i[F[q(t), p(t)], q^n(\mathbf{x}, t)]. \quad (7.1.18)$$

This definition is motivated by the fact that if  $F[q(t), p(t)]$  is written with all  $qs$  to the left of all  $ps$ , then (7.1.17) and (7.1.18) are respectively just the left- and right-derivatives with respect to  $q^n$  and  $p_n$ . That is, for an arbitrary c-number<sup>60</sup> variation  $\delta q$  and  $\delta p$  of the  $qs$  and  $ps$ , we have

$$\delta F[q(t), p(t)] = \int d^3x \sum_n \left( \delta q^n(\mathbf{x}, t) \frac{\delta F[q(t), p(t)]}{\delta q^n(\mathbf{x}, t)} + \frac{\delta F[q(t), p(t)]}{\delta p_n(\mathbf{x}, t)} \delta p_n(\mathbf{x}, t) \right).$$

For more general functionals we need the definitions (7.1.17) and (7.1.18) to pin down various signs and equal-time commutators that may appear.

In particular,  $H_0$  is the generator of time-translations on free-particle states in the sense that:

$$q^n(\mathbf{x}, t) = \exp(iH_0 t) q^n(\mathbf{x}, 0) \exp(-iH_0 t), \quad (7.1.19)$$

$$p_n(\mathbf{x}, t) = \exp(iH_0 t) p_n(\mathbf{x}, 0) \exp(-iH_0 t), \quad (7.1.20)$$

so the free-particle operators have the time-dependence

$$\dot{q}^n(\mathbf{x}, t) = i[H_0, q^n(\mathbf{x}, t)] = \frac{\delta H_0}{\delta p_n(\mathbf{x}, t)}, \quad (7.1.21)$$

$$\dot{p}_n(\mathbf{x}, t) = -i[p_n(\mathbf{x}, t), H_0] = -\frac{\delta H_0}{\delta q^n(\mathbf{x}, t)}. \quad (7.1.22)$$

We recognize these as the familiar dynamical equations in the Hamiltonian formalism.

The free-particle Hamiltonian is given as always by

$$H_0 = \sum_{n,\sigma} \int d^3k a_{\mathbf{k},\sigma,n}^\dagger a_{\mathbf{k},\sigma,n} \sqrt{\mathbf{k}^2 + m_n^2}. \quad (7.1.23)$$

This  $H_0$  may be rewritten in terms of the  $qs$  and  $ps$  at time  $t$ . For instance, it is easy to see that for a real scalar field, Eq. (7.1.23) is equal up to a constant term to the functional

$$H_0 = \int d^3x \left[ \frac{1}{2} p^2 + \frac{1}{2} (\nabla q)^2 + \frac{1}{2} m^2 q^2 \right]. \quad (7.1.24)$$

<sup>59</sup>We are here using a notation that will be adopted henceforth; if  $f(x, y)$  is a function of two classes of variables collectively called  $x$  and  $y$ , then  $F[f(y)]$  indicates a functional that depends on the values of  $f(x, y)$  for all  $x$  at fixed  $y$ . By a bosonic functional we mean one in which each term contains only even numbers of fermionic fields.

<sup>60</sup>Where  $q^n$  and  $p_n$  are bosonic or fermionic,  $\delta q^n$  and  $\delta p_n$  are understood to commute or anticommute with all fermionic operators, respectively, and to commute with all bosonic operators.



To be more precise, using (7.1.7) and the Fourier representation of the scalar field  $\phi$ , we find that Eq. (7.1.24) becomes:

$$\begin{aligned} H_0 &= \frac{1}{2} \int d^3k k^0 \{a_{\mathbf{k}}, a_{\mathbf{k}}^\dagger\} \\ &= \int d^3k k^0 \left( a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2} \delta^3(\mathbf{k} - \mathbf{k}) \right). \end{aligned} \quad (7.1.25)$$

This is the same as Eq. (7.1.23), except for the infinite constant term. Such terms only affect the zero of energy, and have no physical significance in the absence of gravity.<sup>61</sup> Explicit forms for  $H_0$  as a functional of the  $q$  and  $p$  variables for other fields will be given in Section 7.5.

It is usual in textbooks on quantum field theory to derive Eq. (7.1.25) as a consequence of Eq. (7.1.24), which in turn is derived from a Lagrangian density. This seems to me backward, for Eq. (7.1.25) *must* hold; if some assumed free-particle Lagrangian did not give Eq. (7.1.25) up to a constant term, we would conclude that it was the wrong Lagrangian. Rather, we should ask what free-field Lagrangian gives Eq. (7.1.25) for spinless particles, or more generally, gives the free-particle Hamiltonian (7.1.23). This question may be answered by the well-known Legendre transformation from the Hamiltonian to the Lagrangian; the free-field Lagrangian is given by

$$L_0[q(t), \dot{q}(t)] \equiv \sum_n \int d^3x p_n(\mathbf{x}, t) \dot{q}^n(\mathbf{x}, t) - H_0, \quad (7.1.26)$$

it being understood that  $p_n$  is replaced everywhere by its expression in terms of  $q^n$  and  $\dot{q}^n$  (and, as we shall see, perhaps some auxiliary fields as well). For instance, from the Hamiltonian (7.1.24) and (7.1.7) we can derive the free-field Lagrangian for a scalar field:

$$\begin{aligned} L_0 &= \int d^3x \left[ p\dot{q} - \frac{1}{2}p^2 - \frac{1}{2}(\nabla q)^2 - \frac{1}{2}m^2q^2 \right] \\ &= \int d^3x \left[ -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 \right]. \end{aligned} \quad (7.1.27)$$

Whatever we suppose the complete Lagrangian of the scalar field may be, this is the term that must be separated out and treated as a term of zeroth order in perturbation theory. A similar exercise may be carried out for the other canonical systems described in this section, but from now on we shall content ourselves with guessing the form of the free-field Lagrangian and then confirming that it gives the correct free-particle Hamiltonian.

We have seen that various free-field theories can be formulated in canonical terms. It is then a short step to show that the same is true of the interacting fields. We can introduce canonical variables in what is called the ‘Heisenberg picture’, defined by

$$Q^n(\mathbf{x}, t) = \exp(iHt)q^n(\mathbf{x}, 0)\exp(-iHt), \quad (7.1.28)$$

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<sup>61</sup>However, changes in such terms due to changes in the boundary conditions for the fields, as for instance quantizing in the space between parallel plates rather than in infinite space, are physically significant, and have even been measured [1].

$$P_n(\mathbf{x}, t) = \exp(iHt)p_n(\mathbf{x}, 0)\exp(-iHt), \quad (7.1.29)$$

where  $H$  is the full Hamiltonian. Because this is a similarity transformation that commutes with  $H$ , the total Hamiltonian is the same functional of the Heisenberg picture operators as it was of the  $qs$  and  $ps$ :

$$H[Q, P] = e^{iHt}H[q, p]e^{-iHt} = H[q, p].$$

Also, because Eqs. (7.1.28)–(7.1.29) define a similarity transformation, the Heisenberg picture operators again satisfy the canonical commutation or anticommutation relations:

$$\begin{aligned} [Q^n(\mathbf{x}, t), P_{\bar{n}}(\mathbf{y}, t)] &= i\delta^3(\mathbf{x} - \mathbf{y})\delta_{\bar{n}}^n, & \text{if either particle is a boson,} \\ \{Q^n(\mathbf{x}, t), P_{\bar{n}}(\mathbf{y}, t)\} &= i\delta^3(\mathbf{x} - \mathbf{y})\delta_{\bar{n}}^n, & \text{if both particles are fermions.} \end{aligned} \quad (7.1.30)$$

$$\begin{aligned} [Q^n(\mathbf{x}, t), Q^{\bar{n}}(\mathbf{y}, t)] &= 0, & \text{if either particle is a boson,} \\ \{Q^n(\mathbf{x}, t), Q^{\bar{n}}(\mathbf{y}, t)\} &= 0, & \text{if both particles are fermions.} \end{aligned} \quad (7.1.31)$$

$$\begin{aligned} [P_n(\mathbf{x}, t), P_{\bar{n}}(\mathbf{y}, t)] &= 0, & \text{if either particle is a boson,} \\ \{P_n(\mathbf{x}, t), P_{\bar{n}}(\mathbf{y}, t)\} &= 0, & \text{if both particles are fermions.} \end{aligned} \quad (7.1.32)$$

However, they now have the time-dependence

$$\dot{Q}^n(\mathbf{x}, t) = i[H, Q^n(\mathbf{x}, t)] = \frac{\delta H}{\delta P_n(\mathbf{x}, t)}, \quad (7.1.33)$$

$$\dot{P}_n(\mathbf{x}, t) = -i[P_n(\mathbf{x}, t), H] = -\frac{\delta H}{\delta Q^n(\mathbf{x}, t)}. \quad (7.1.34)$$

For instance, we might take the Hamiltonian for a real scalar field as the free-particle term (7.1.24) plus the integral of a scalar interaction density  $\mathcal{H}_I$ , so that in terms of Heisenberg-picture variables

$$H = \int d^3x \left[ \frac{1}{2}P^2 + \frac{1}{2}(\nabla Q)^2 + \frac{1}{2}m^2Q^2 + \mathcal{H}_I(Q) \right]. \quad (7.1.35)$$

In this case the canonical conjugate to  $Q$  is given by the same formula as for free fields:

$$P = \dot{Q}. \quad (7.1.36)$$

However, as we shall see, the relation between the canonical conjugates  $P_n(x)$  and the field variables and their time-derivatives is in general not the same as for the free-particle operators, but must be inferred from Eqs. (7.1.33) and (7.1.34).

## 7.2 The Lagrangian Formalism

Having seen that various realistic theories may be cast in the canonical formalism, we must now face the question of how to choose the Hamiltonian. As we will see in the next section, the easiest way to enforce Lorentz invariance and other symmetries is to choose a suitable Lagrangian and use it to derive the Hamiltonian. There is not much loss of generality in this; given a realistic Hamiltonian, we can generally reconstruct a Lagrangian from which

it could be derived, by reversing the process that we are going to describe here of deriving Hamiltonians from Lagrangians. (The derivation of Eq. (7.1.26) gives one example of this reconstruction.) But although we can go from Hamiltonians to Lagrangians or Lagrangians to Hamiltonians, it is easier to explore physically satisfactory theories by listing possible Lagrangians, rather than Hamiltonians.

The Lagrangian is, in general, a functional<sup>62</sup>  $L[\Psi(t), \dot{\Psi}(t)]$  of a set of generic fields  $\Psi^\ell(\mathbf{x}, t)$  and their time-derivatives  $\dot{\Psi}^\ell(\mathbf{x}, t)$ . The conjugate fields  $\Pi_\ell(\mathbf{x}, t)$  are defined as the variational derivatives<sup>63</sup>

$$\Pi_\ell(\mathbf{x}, t) \equiv \frac{\delta L[\Psi(t), \dot{\Psi}(t)]}{\delta \dot{\Psi}^\ell(\mathbf{x}, t)}, \quad (7.2.1)$$

The equations of motion are

$$\dot{\Pi}_\ell(\mathbf{x}, t) = \frac{\delta L[\Psi(t), \dot{\Psi}(t)]}{\delta \Psi^\ell(\mathbf{x}, t)}. \quad (7.2.2)$$

These field equations can be usefully reformulated as a variational principle. We define a functional of  $\Psi^\ell(x)$  over all spacetime, known as the *action*

$$I[\Psi] \equiv \int_{-\infty}^{\infty} dt L[\Psi(t), \dot{\Psi}(t)]. \quad (7.2.3)$$

Under an arbitrary variation of  $\Psi(x)$ , the change in  $I[\Psi]$  is

$$\delta I[\Psi] = \int_{-\infty}^{\infty} dt \int d^3x \left[ \frac{\delta L}{\delta \Psi^\ell(x)} \delta \Psi^\ell(x) + \frac{\delta L}{\delta \dot{\Psi}^\ell(x)} \delta \dot{\Psi}^\ell(x) \right].$$

Assuming that  $\delta \Psi^\ell(x)$  vanishes for  $t \rightarrow \pm\infty$ , we may integrate by parts, and write

$$\delta I[\Psi] = \int d^4x \left[ \frac{\delta L}{\delta \Psi^\ell(x)} - \frac{d}{dt} \frac{\delta L}{\delta \dot{\Psi}^\ell(x)} \right] \delta \Psi^\ell(x). \quad (7.2.4)$$

We see that the action is stationary with respect to all variations  $\delta \Psi^\ell$  that vanish at  $t \rightarrow \pm\infty$  if and only if the fields satisfy the field equations (7.2.2).

Because the field equations are determined by the functional  $I[\Psi]$ , it is natural in trying to construct a Lorentz-invariant theory to make  $I[\Psi]$  a scalar functional. In particular, since  $I[\Psi]$  is a time-integral of  $L[\Psi(t), \dot{\Psi}(t)]$ , we guess that  $L$  should itself be a space-integral of an ordinary scalar function of  $\Psi(x)$  and  $\partial\Psi(x)/\partial x^\mu$ , known as the *Lagrangian density*  $\mathcal{L}$ :

$$L[\Psi(t), \dot{\Psi}(t)] = \int d^3x \mathcal{L}(\Psi(\mathbf{x}, t), \nabla\Psi(\mathbf{x}, t), \dot{\Psi}(\mathbf{x}, t)), \quad (7.2.5)$$

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<sup>62</sup>Recall that in the notation we use for functionals, a functional like  $L$  in which we display the variable  $t$  is understood to depend on the fields  $\Psi^\ell(\mathbf{x}, t)$  and  $\dot{\Psi}^\ell(\mathbf{x}, t)$ , with the undisplayed variables  $\ell$  and  $\mathbf{x}$  running over all their values at a fixed value of the displayed variable  $t$ . We use upper case  $\Psi$ s and  $\Pi$ s to indicate that these are interacting rather than free fields.

<sup>63</sup>Because the  $\Psi$ s and  $\dot{\Psi}$ s do not in general satisfy simple commutation or anticommutation relations, we cannot give a simple definition of the functional derivatives occurring here as we did for functional derivatives with respect to the  $Q$ s and  $P$ s in the previous section. Instead, we will simply specify that the variational derivatives are what they would be for c-number variables, with minus signs and equal-time commutators or anticommutators supplied as needed to make the formulas correct quantum-mechanically. As far as I know, no important issues hinge on the details here.

so that the action is

$$I[\Psi] = \int d^4x \mathcal{L}(\Psi(x), \partial\Psi(x)/\partial x^\mu). \quad (7.2.6)$$

All field theories used in current theories of elementary particles have Lagrangians of this form.

Varying  $\Psi^\ell(x)$  by an amount  $\delta\Psi^\ell(x)$ , and integrating by parts, we find a variation in  $L$ :

$$\begin{aligned} \delta L &= \int d^3x \left[ \frac{\partial \mathcal{L}}{\partial \Psi^\ell} \delta\Psi^\ell + \frac{\partial \mathcal{L}}{\partial(\nabla\Psi^\ell)} \nabla\delta\Psi^\ell + \frac{\partial \mathcal{L}}{\partial\dot{\Psi}^\ell} \delta\dot{\Psi}^\ell \right] \\ &= \int d^3x \left[ \left( \frac{\partial \mathcal{L}}{\partial \Psi^\ell} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial(\nabla\Psi^\ell)} \right) \delta\Psi^\ell + \frac{\partial \mathcal{L}}{\partial\dot{\Psi}^\ell} \delta\dot{\Psi}^\ell \right], \end{aligned}$$

so (with obvious arguments suppressed)

$$\frac{\delta L}{\delta\Psi^\ell} = \frac{\partial \mathcal{L}}{\partial \Psi^\ell} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial(\nabla\Psi^\ell)}, \quad (7.2.7)$$

$$\frac{\delta L}{\delta\dot{\Psi}^\ell} = \frac{\partial \mathcal{L}}{\partial\dot{\Psi}^\ell}. \quad (7.2.8)$$

The field equations (7.2.1) then read

$$\frac{\partial}{\partial x^\mu} \frac{\partial \mathcal{L}}{\partial(\partial\Psi^\ell/\partial x^\mu)} = \frac{\partial \mathcal{L}}{\partial \Psi^\ell}. \quad (7.2.9)$$

These are known as the *Euler–Lagrange equations*. As expected, if  $\mathcal{L}$  is a scalar then these equations are Lorentz-invariant.

In addition to being Lorentz-invariant, the action  $I$  is required to be *real*. This is because we want just as many field equations as there are fields. By breaking up any complex fields into their real and imaginary parts, we can always think of  $I$  as being a functional only of a number of *real* fields, say  $N$  of them. If  $I$  were complex, with independent real and imaginary parts, then the real and imaginary parts of the conditions that  $I$  be stationary (the Euler–Lagrange equations) would yield  $2N$  field equations for  $N$  fields, too many to be satisfied except in special cases. We will see in the next section that the reality of the action also ensures that the generators of various symmetry transformations are Hermitian operators.

Although the Lagrangian formalism makes it easy to construct theories that will satisfy Lorentz invariance and other symmetries, to calculate the  $S$ -matrix we need a formula for the interaction Hamiltonian. In general, the Hamiltonian is given by the *Legendre transformation*

$$H = \sum_\ell \int d^3x \Pi_\ell(\mathbf{x}, t) \dot{\Psi}^\ell(\mathbf{x}, t) - L[\Psi(t), \dot{\Psi}(t)]. \quad (7.2.10)$$

Although Eq. (7.2.1) does not in general allow  $\dot{\Psi}^\ell$  to be expressed uniquely in terms of  $\Psi^\ell$  and  $\Pi_\ell$ , it is easy to see that Eq. (7.2.10) has vanishing variational derivative with respect

to  $\dot{\Psi}^\ell$  for any  $\dot{\Psi}^\ell$  satisfying Eq. (7.2.1), so in general it is a functional only of  $\Psi^\ell$  and  $\Pi_\ell$ . Its variational derivatives with respect to these variables are

$$\begin{aligned}\left.\frac{\delta H}{\delta \Psi^\ell(\mathbf{x}, t)}\right|_\Pi &= \int d^3y \sum_{\ell'} \Pi_{\ell'}(\mathbf{y}, t) \left.\frac{\delta \dot{\Psi}^{\ell'}(\mathbf{y}, t)}{\delta \Psi^\ell(\mathbf{x}, t)}\right|_\Pi - \left.\frac{\delta L}{\delta \Psi^\ell(\mathbf{x}, t)}\right|_{\dot{\Psi}} \\ &\quad - \int d^3y \sum_{\ell'} \left.\frac{\delta L}{\delta \dot{\Psi}^{\ell'}(\mathbf{y}, t)}\right|_{\Psi} \left.\frac{\delta \dot{\Psi}^{\ell'}(\mathbf{y}, t)}{\delta \Psi^\ell(\mathbf{x}, t)}\right|_\Pi, \\ \left.\frac{\delta H}{\delta \Pi_\ell(\mathbf{x}, t)}\right|_\Psi &= \dot{\Psi}^\ell(\mathbf{x}, t) + \int d^3y \sum_{\ell'} \Pi_{\ell'}(\mathbf{y}, t) \left.\frac{\delta \dot{\Psi}^{\ell'}(\mathbf{y}, t)}{\delta \Pi_\ell(\mathbf{x}, t)}\right|_\Psi \\ &\quad - \int d^3y \sum_{\ell'} \left.\frac{\delta L}{\delta \dot{\Psi}^{\ell'}(\mathbf{y}, t)}\right|_\Psi \left.\frac{\delta \dot{\Psi}^{\ell'}(\mathbf{y}, t)}{\delta \Pi_\ell(\mathbf{x}, t)}\right|_\Psi,\end{aligned}$$

where subscripts denote the quantities held fixed in these variational derivatives. Using the defining equation (7.2.1) for  $\Pi_\ell$ , this simplifies to

$$\left.\frac{\delta H}{\delta \Psi^\ell(\mathbf{x}, t)}\right|_\Pi = - \left.\frac{\delta L}{\delta \Psi^\ell(\mathbf{x}, t)}\right|_{\dot{\Psi}}, \quad (7.2.11)$$

and

$$\left.\frac{\delta H}{\delta \Pi_\ell(\mathbf{x}, t)}\right|_\Psi = \dot{\Psi}^\ell(\mathbf{x}, t). \quad (7.2.12)$$

The equations of motion (7.2.2) are then equivalent to

$$\left.\frac{\delta H}{\delta \Psi^\ell(\mathbf{x}, t)}\right|_\Pi = -\dot{\Pi}_\ell(\mathbf{x}, t). \quad (7.2.13)$$

It is tempting now to identify the generic field variables  $\Psi^\ell$  and their conjugates  $\Pi_\ell$  with the canonical variables  $Q^n$  and  $P_n$  of the previous section, and impose on them the same canonical commutation relations (7.1.30)–(7.1.32), so that Eqs. (7.2.12) and (7.2.13) are the same as the Hamiltonian equations of motion (7.1.33) and (7.1.34). This is indeed the case for the simple example of the real scalar field  $\Phi$  with non-derivative coupling. Consider the Lagrangian density<sup>64</sup>

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - \frac{m^2}{2}\Phi^2 - \mathcal{H}_I(\Phi), \quad (7.2.14)$$

which can be obtained by adding a real function  $-\mathcal{H}_I(\Phi)$  of  $\Phi$  to the free-field Lagrangian density found in the previous section. The Euler–Lagrange equations here are

$$(\square - m^2)\Phi = \mathcal{H}'_I(\Phi). \quad (7.2.15)$$

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<sup>64</sup>We do not include a free constant factor in the term  $-\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi$ , because any such constant if positive can be absorbed into the normalization of  $\Phi$ . As we shall see, a negative constant here would lead to a Hamiltonian that is not bounded below. The constant  $m$  is known as the bare mass. The most general Lagrangian that satisfies the principle of renormalizability (discussed in Chapter 12) is of this form, with  $\mathcal{H}_I(\Phi)$  a quartic polynomial in  $\Phi$ .

From this Lagrangian density, we calculate a canonical conjugate to  $\Phi$ :

$$\Pi = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = \dot{\Phi}, \quad (7.2.16)$$

which is the same as Eq. (7.1.36) if we identify  $\Phi$  and  $\Pi$  with the canonical variables  $Q$  and  $P$ . The Hamiltonian is now given by Eq. (7.2.10) as<sup>65</sup>

$$\begin{aligned} H &= \int d^3x (\Pi \dot{\Phi} - \mathcal{L}) \\ &= \int d^3x \left[ \frac{1}{2} \Pi^2 + \frac{1}{2} (\nabla \Phi)^2 + \frac{1}{2} m^2 \Phi^2 + \mathcal{H}_I(\Phi) \right], \end{aligned} \quad (7.2.17)$$

which we recognize as the Hamiltonian (7.1.35). This little exercise should not be regarded as another derivation of this Hamiltonian, but rather as a validation of the Lagrangian (7.2.14) as a possible theory of scalar fields.

Matters are not always so simple. We have already seen in the previous section that there are field variables, such as the time component of a vector field or the Hermitian conjugate of a Dirac field, that are not canonical field variables  $Q^n$  and do not have canonical conjugates; yet Lorentz invariance dictates that these must appear in the Lagrangians for the vector and Dirac fields.

From the point of view of the Lagrangian formalism, the special character of field variables like the time component of a vector field or the Hermitian conjugate of a Dirac field arises from the fact that although they appear in the Lagrangian, their time-derivatives do not. We shall denote the field variables  $\Psi^\ell$  whose time-derivatives do not appear in the Lagrangian as  $C^r$ ; the remaining independent field variables are the canonical variables  $Q^n$ . The  $Q^n$  have canonical conjugates

$$P_n(\mathbf{x}, t) = \frac{\delta L[Q(t), \dot{Q}(t), C(t)]}{\delta \dot{Q}^n(\mathbf{x}, t)}, \quad (7.2.18)$$

and satisfy the commutation relations (7.1.30)–(7.1.32), but there are no canonical conjugates for the  $C^r$ . Because  $\delta L / \delta \dot{C}^r = 0$ , the Hamiltonian (7.2.10) is in general

$$H = \sum_n \int d^3x P_n \dot{Q}^n - L[Q(t), \dot{Q}(t), C(t)], \quad (7.2.19)$$

but this is not yet useful until we express the  $C^r$  and  $\dot{Q}^\ell$  in terms of the  $Q$ s and  $P$ s. The equations of motion of the  $C^r$  involve only fields and their first time-derivatives

$$0 = \frac{\delta L[Q(t), \dot{Q}(t), C(t)]}{\delta C^r(\mathbf{x}, t)}. \quad (7.2.20)$$

In the simple cases to be discussed in this chapter, these equations together with Eq. (7.2.18) can be solved to give the  $C^r$  and  $\dot{Q}^\ell$  in terms of the  $Q$ s and  $P$ s. Section 7.6 shows how in

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<sup>65</sup>In order for  $H$  to be interpreted as an energy, it should be bounded below. The positivity of the first two terms shows that we guessed correctly as to the sign in the first term in Eq. (7.2.14). The remaining condition is that  $\frac{1}{2} m^2 \Phi^2 + \mathcal{H}_I(\Phi)$  must be bounded below as a function of  $\Phi$ .

such cases one can avoid the task of actually solving for the  $C^r$  and  $\dot{Q}^\ell$ . In gauge theories like electrodynamics other methods must be used: either choosing a particular gauge, as in Chapter 8, or the more modern covariant methods to be discussed in Volume II.

Once we have derived a Hamiltonian as a functional of the Heisenberg picture  $Q$ s and  $P$ s, to use perturbation theory we must make a transition to the interaction picture. The Hamiltonian is time-independent, so it can be written in terms of the  $P_n$  and  $Q^n$  at  $t = 0$ , which are equal to the corresponding operators  $p_n$  and  $q^n$  in the interaction picture at  $t = 0$ . The Hamiltonian derived in this way may then be expressed in terms of the  $q$ s and  $p$ s of the interaction picture, and split into two parts, a suitable free-particle term  $H_0$  and an interaction  $H_{\text{int}}$ . Finally, the time-dependence equations (7.1.21) and (7.1.22) and the commutation or anticommutation relations (7.1.1)–(7.1.2) are used to express the  $q$ s and  $p$ s in  $H_I(t)$  as linear combinations of annihilation and creation operators.

We shall present a number of examples of this procedure in Section 7.5; for the moment we will give only one example of the simplest type, the scalar field with Hamiltonian (7.2.17). We split  $H$  into a free-particle term and an interaction

$$H = H_0 + H_{\text{int}}, \quad (7.2.21)$$

$$H_0 = \int d^3x \left[ \frac{1}{2} \Pi^2 + \frac{1}{2} (\nabla \Phi)^2 + \frac{1}{2} m^2 \Phi^2 \right], \quad (7.2.22)$$

$$H_{\text{int}} = \int d^3x \mathcal{H}_I(\Phi). \quad (7.2.23)$$

Here  $\Phi$  and  $\Pi$  are taken at the same time  $t$ , and  $H$  is independent of  $t$ , though  $H_0$  and  $H_{\text{int}}$  usually are not.

We now pass to the interaction representation. Taking  $t = 0$  in Eqs. (7.2.22) and (7.2.23), we can simply replace  $\Phi, \Pi$  with the interaction picture variables  $\phi, \pi$ , since they are defined by Eqs. (7.1.28) and (7.1.29) to be equal at that time. To calculate the interaction  $H_I(t)$  in the interaction picture, we apply the similarity transformation (3.5.5)

$$H_I(t) = \exp(iH_0 t) H_{\text{int}} \exp(-iH_0 t) = \int d^3x \mathcal{H}_I(\phi(\mathbf{x}, t)). \quad (7.2.24)$$

The same transformation applied to  $H_0$  leaves it constant:

$$\begin{aligned} H_0 &= \exp(iH_0 t) H_0 \exp(-iH_0 t) \\ &= \int d^3x \left[ \frac{1}{2} \pi^2(\mathbf{x}, t) + \frac{1}{2} (\nabla \phi(\mathbf{x}, t))^2 \right]. \end{aligned} \quad (7.2.25)$$

The relation between  $\pi$  and  $\dot{\phi}$  is dictated by Eq. (7.1.21)

$$\dot{\phi}(\mathbf{x}, t) = \frac{\delta H_0}{\delta \pi(\mathbf{x}, t)} = \pi(\mathbf{x}, t). \quad (7.2.26)$$

(This happens to be the same relation as in Eq. (7.2.16), but as we shall see this is not to be expected in general.) Also, the equation of motion for  $\phi$  is dictated by Eq. (7.1.22):

$$\ddot{\phi}(\mathbf{x}, t) = -\frac{\delta H_0}{\delta \phi(\mathbf{x}, t)} = +\nabla^2 \phi(\mathbf{x}, t) - m^2 \phi(\mathbf{x}, t), \quad (7.2.27)$$

which together with Eq. (7.2.26) yields the field equations

$$(\square - m^2)\phi(x) = 0. \quad (7.2.28)$$

The general real solution may be expressed as

$$\phi(x) = (2\pi)^{-3/2} \int d^3p (2p^0)^{-1/2} \left[ e^{ip \cdot x} a_{\mathbf{p}} + e^{-ip \cdot x} a_{\mathbf{p}}^\dagger \right] \quad (7.2.29)$$

with  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$  understood, and  $a_{\mathbf{p}}$  some as-yet-unknown operator function of  $\mathbf{p}$ . Eq. (7.2.26) then gives the canonical conjugate as

$$\pi(x) = -i(2\pi)^{-3/2} \int d^3p (p^0/2)^{1/2} \left[ e^{ip \cdot x} a_{\mathbf{p}} - e^{-ip \cdot x} a_{\mathbf{p}}^\dagger \right]. \quad (7.2.30)$$

In order to get the desired commutation relations,

$$[\phi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = i\delta^3(\mathbf{x} - \mathbf{y}), \quad (7.2.31)$$

$$[\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = 0, \quad (7.2.32)$$

$$[\pi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = 0, \quad (7.2.33)$$

we must take the  $a$ s to satisfy the familiar commutation relations

$$[a_{\mathbf{p}}, a_{\mathbf{p}'}^\dagger] = \delta^3(\mathbf{p} - \mathbf{p}'), \quad (7.2.34)$$

$$[a_{\mathbf{p}}, a_{\mathbf{p}'}] = 0. \quad (7.2.35)$$

Also, we have already shown in the previous section that using these expansions in Eq. (7.2.25) gives the usual formula (4.2.11) for the free-particle Hamiltonian, up to an inconsequential additive constant. As remarked before, these results should not be regarded so much as an alternative derivation of Eqs. (7.2.29), (7.2.34), and (7.2.35) (which were obtained in Chapter 5 on quite other grounds) but rather as a validation of the first two terms of Eq. (7.2.14) as the correct free-particle Lagrangian for a real scalar field. We can now proceed to use perturbation theory to calculate the  $S$ -matrix, taking (7.2.24) as  $H_I(t)$ , with the field  $\phi(x)$  given by Eq. (7.2.29).

The procedures illustrated here will be carried out for examples that are more complicated and more interesting in Section 7.5.

\* \* \*

In considering the various possible Lagrangian densities for physical theories it is common to apply integration by parts, treating as equivalent any Lagrangian densities that differ only by total derivatives  $\partial_\mu \mathcal{F}^\mu$ . It is obvious that such total derivative terms do not contribute to the action and hence do not affect the field equations. It is also obvious that a space-derivative term  $\nabla \cdot \mathcal{F}$  in the Lagrangian density does not contribute to the Lagrangian and hence does not affect the quantum theory defined by the Lagrangian.<sup>66</sup> What is less

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<sup>66</sup>This is under the usual assumption, that the fields vanish at infinity. These results do not necessary apply when we allow fields of different topology, as discussed in Volume II.



obvious and worth noting here is that a time-derivative  $\partial_0 \mathcal{F}^0$  in the Lagrangian density also does not affect the quantum structure of the theory. To see this, let's first consider the effect of adding a term to the Lagrangian of the more general form

$$\Delta L(t) = \int d^3x D_{n,\mathbf{x}}[Q(t)] \dot{Q}^n(\mathbf{x}, t), \quad (7.2.36)$$

where  $D$  is an arbitrary  $n$ - and  $\mathbf{x}$ -dependent functional of the values of  $Q$  at a given time. This changes the formula for the conjugate variables  $P(t)$  as functionals of  $Q(t)$  and  $\dot{Q}(t)$  by the amount

$$\Delta P_n(\mathbf{x}, t) = \frac{\delta \Delta L(t)}{\delta \dot{Q}^n(\mathbf{x}, t)} = D_{n,\mathbf{x}}[Q(t)]. \quad (7.2.37)$$

It follows that there is no change in the Hamiltonian as expressed as a functional of the  $Q(t)$  and  $\dot{Q}(t)$ :

$$\int d^3x \Delta P_n(\mathbf{x}, t) \dot{Q}^n(\mathbf{x}, t) - \Delta L(t) = 0. \quad (7.2.38)$$

Hence also there is no change in the Hamiltonian as expressed as a functional of the old canonical variables  $Q^n$  and  $P_n$ . However, the Hamiltonian is *not* the same functional of the *new* canonical variables  $Q^n$  and  $P_n + \Delta P_n$  as it was of the  $Q^n$  and  $P_n$ , and in a theory described by the new Lagrangian  $\mathcal{L} + \Delta \mathcal{L}$  it is the new canonical variables  $Q^n$  and  $P_n + \Delta P_n$  rather than the  $Q^n$  and  $P_n$  that would satisfy the canonical commutation relations. The commutators of the  $Q^n$  with each other and of the  $Q^n$  with the  $P_m$  are given by the usual canonical relations, but the commutators of the  $P_n$  with each other are now

$$\begin{aligned} [P_n(\mathbf{x}, t), P_m(\mathbf{y}, t)] &= [P_n(\mathbf{x}, t) + \Delta P_n(\mathbf{x}, t), P_m(\mathbf{y}, t) + \Delta P_m(\mathbf{y}, t)] \\ &\quad - [\Delta P_n(\mathbf{x}, t), P_m(\mathbf{y}, t) + \Delta P_m(\mathbf{y}, t)] \\ &\quad - [P_n(\mathbf{x}, t) + \Delta P_n(\mathbf{x}, t), \Delta P_m(\mathbf{y}, t)] \\ &\quad + [\Delta P_n(\mathbf{x}, t), \Delta P_m(\mathbf{y}, t)] \\ &= -i \frac{\delta D_{n,\mathbf{x}}[Q(t)]}{\delta Q^m(\mathbf{y}, t)} + i \frac{\delta D_{m,\mathbf{y}}[Q(t)]}{\delta Q^n(\mathbf{x}, t)}. \end{aligned} \quad (7.2.39)$$

In general this doesn't vanish, but if the added term in the Lagrangian is a total time-derivative

$$\Delta L = \frac{dG}{dt} = \int d^3x \frac{\delta G[Q(t)]}{\delta Q^n(\mathbf{x}, t)} \dot{Q}^n(\mathbf{x}, t), \quad (7.2.40)$$

then  $D$  in Eq. (7.2.36) is of the special form

$$D_{n,\mathbf{x}}[Q] = \frac{\delta G[Q(t)]}{\delta Q^n(\mathbf{x}, t)}. \quad (7.2.41)$$

In this case the commutator (7.2.39) vanishes, so the variables  $Q^n$  and  $P_n$  satisfy the usual commutation relations. We have seen that a change of the form (7.2.36) in the Lagrangian does not change the form of the Hamiltonian as a functional of the  $Q^n$  and  $P_n$ , and since, as we have now shown, the commutation relations of these variables are also unchanged, the addition to the Lagrangian of the term (7.2.36) has no effect on the quantum structure of the theory. Different Lagrangian densities obtained from each other by partial integration may therefore be regarded as equivalent in quantum as well as classical field theory.

### 7.3 Global Symmetries

We now come to the real point of the Lagrangian formalism, that it provides a natural framework for the quantum mechanical implementation of symmetry principles. This is because the dynamical equations in the Lagrangian formalism take the form of a variational principle, the principle of stationary action. Consider any infinitesimal transformation of the fields

$$\Psi^\ell(x) \longrightarrow \Psi^\ell(x) + i\epsilon \mathcal{F}^\ell(x) \quad (7.3.1)$$

that leaves the action (7.2.3) invariant:

$$0 = \delta I = i\epsilon \int d^4x \frac{\delta I[\Psi]}{\delta \Psi^\ell(x)} \mathcal{F}^\ell(x). \quad (7.3.2)$$

(With  $\epsilon$  a constant, such symmetries are known as *global symmetries*. In general,  $\mathcal{F}^\ell$  depends on the fields and their derivatives at  $x$ .) Of course Eq. (7.3.2) is automatically satisfied for *all* infinitesimal variations of the fields if the fields satisfy the dynamical equations; by an infinitesimal symmetry transformation we mean one that leaves the action invariant even when the dynamical equations are *not* satisfied. If we now consider the same transformation with  $\epsilon$  an arbitrary function of position in spacetime:

$$\Psi^\ell(x) \longrightarrow \Psi^\ell(x) + i\epsilon(x) \mathcal{F}^\ell(x), \quad (7.3.3)$$

then, in general, the variation of the action will not vanish, but it will have to be of the form

$$\delta I = - \int d^4x J^\mu(x) \frac{\partial \epsilon(x)}{\partial x^\mu} \quad (7.3.4)$$

in order that it should vanish when  $\epsilon(x)$  is constant. If we *now* take the fields in  $I[\Psi]$  to satisfy the field equations then  $I$  is stationary with respect to arbitrary field variations that vanish at large spacetime distances, including variations of the form (7.3.3), so in this case (7.3.4) should vanish. Integrating by parts, we see that  $J^\mu(x)$  must satisfy a conservation law:

$$0 = \frac{\partial J^\mu(x)}{\partial x^\mu}. \quad (7.3.5)$$

It follows immediately that

$$0 = \frac{dF}{dt}, \quad (7.3.6)$$

where

$$F \equiv \int d^3x J^0. \quad (7.3.7)$$

There is one such conserved current  $J^\mu$  and one constant of the motion  $F$  for each independent infinitesimal symmetry transformation. This represents a general feature of the canonical formalism, often referred to as Noether's theorem: *symmetries imply conservation laws*.

Many symmetry transformations leave the Lagrangian and not just the action invariant. This is the case, for instance, for translations and rotations in space and also isospin transformations and other internal symmetry transformations, though not for general Lorentz

transformations. When the Lagrangian is invariant we can go further, and write an explicit formula for the conserved quantities  $F$ . Consider a field variation (7.3.3) in which  $\epsilon(x)$  depends on  $t$  but not  $\mathbf{x}$ . In this case the variation in the action is

$$\delta I = i \int dt \int d^3x \left[ \frac{\delta L[\Psi(t), \dot{\Psi}(t)]}{\delta \Psi^\ell(\mathbf{x}, t)} \epsilon(t) \mathcal{F}^\ell(\mathbf{x}, t) + \frac{\delta L[\Psi(t), \dot{\Psi}(t)]}{\delta \dot{\Psi}^\ell(\mathbf{x}, t)} \frac{d}{dt} (\epsilon(t) \mathcal{F}^\ell(\mathbf{x}, t)) \right]. \quad (7.3.8)$$

The requirement that the Lagrangian be invariant under this transformation when  $\epsilon$  is a constant yields

$$0 = \int d^3x \left[ \frac{\delta L[\Psi(t), \dot{\Psi}(t)]}{\delta \Psi^\ell(\mathbf{x}, t)} \mathcal{F}^\ell(\mathbf{x}, t) + \frac{\delta L[\Psi(t), \dot{\Psi}(t)]}{\delta \dot{\Psi}^\ell(\mathbf{x}, t)} \frac{d}{dt} \mathcal{F}^\ell(\mathbf{x}, t) \right], \quad (7.3.9)$$

so for general fields (whether or not the field equations are satisfied) the variation in the action is

$$\delta I = i \int dt \int d^3x \frac{\delta L[\Psi(t), \dot{\Psi}(t)]}{\delta \dot{\Psi}^\ell(\mathbf{x}, t)} \dot{\epsilon}(t) \mathcal{F}^\ell(\mathbf{x}, t). \quad (7.3.10)$$

Comparing this with Eq. (7.3.4) gives

$$F = -i \int d^3x \frac{\delta L[\Psi(t), \dot{\Psi}(t)]}{\delta \dot{\Psi}^\ell(\mathbf{x}, t)} \mathcal{F}^\ell(\mathbf{x}, t). \quad (7.3.11)$$

Using the symmetry condition (7.3.9), the reader can easily check that this  $F$  is indeed time-independent for any fields that satisfy the dynamical equations (7.2.2).

Other symmetry transformations such as isospin rotations leave not only the action and the Lagrangian invariant but also the Lagrangian density. In such cases we can go even further, and write an explicit formula for the current  $J^\mu(x)$ . Writing the action as in Eq. (7.2.6) as the integral of the Lagrangian density, its variation under the transformation (7.3.3) with a general infinitesimal parameter  $\epsilon(x)$  is

$$\delta I[\Psi] = i \int d^4x \left[ \frac{\partial \mathcal{L}(\Psi(x), \partial_\mu \Psi(x))}{\partial \Psi^\ell(x)} \mathcal{F}^\ell(x) \epsilon(x) + \frac{\partial \mathcal{L}(\Psi(x), \partial_\mu \Psi(x))}{\partial (\partial_\mu \Psi^\ell(x))} \partial_\mu (\mathcal{F}^\ell(x) \epsilon(x)) \right]. \quad (7.3.12)$$

The invariance of the Lagrangian density when  $\epsilon$  is a constant requires that

$$0 = \frac{\partial \mathcal{L}(\Psi(x), \partial_\mu \Psi(x))}{\partial \Psi^\ell(x)} \mathcal{F}^\ell(x) + \frac{\partial \mathcal{L}(\Psi(x), \partial_\mu \Psi(x))}{\partial (\partial_\mu \Psi^\ell(x))} \partial_\mu \mathcal{F}^\ell(x), \quad (7.3.13)$$

so for arbitrary fields the variation of the action is

$$\delta I[\Psi] = i \int d^4x \frac{\partial \mathcal{L}(\Psi(x), \partial_\mu \Psi(x))}{\partial (\partial_\mu \Psi^\ell(x))} \partial_\mu \epsilon(x). \quad (7.3.14)$$

Comparison with Eq. (7.3.4) shows that

$$J^\mu = -i \frac{\partial \mathcal{L}}{\partial(\partial \Psi^\ell / \partial x^\mu)} \mathcal{F}^\ell. \quad (7.3.15)$$

Using the symmetry condition (7.3.13), it is easy to see directly that  $\partial_\mu J^\mu$  vanishes when the fields satisfy the Euler–Lagrange equations (7.2.9). Note also that the integral of the time component of the current (7.3.15) has the previously derived value (7.3.11).

So far everything we have said would apply to classical as well as quantum mechanical field theories. The quantum properties of the conserved quantities  $F$  are most easily seen for symmetries of the Lagrangian (not necessarily the Lagrangian density) that transform the canonical fields  $Q^n(\mathbf{x}, t)$  (that is, those of the  $\Psi^\ell$  whose time derivatives appear in the Lagrangian) into  $\mathbf{x}$ -dependent functionals of themselves at the same time. For such transformations, we have

$$\mathcal{F}^n(\mathbf{x}, t) = \mathcal{F}^n[Q(t); \mathbf{x}]. \quad (7.3.16)$$

As we shall see, infinitesimal spatial translations and rotations as well as all infinitesimal internal symmetry transformations are of the form (7.3.1), (7.3.16), with  $\mathcal{F}^n$  a linear functional of the  $Q^m$ , but we will not need to assume here that the symmetry is linear. For all such symmetries the operator  $F$  is not only conserved; it also acts in quantum mechanics as a *generator* of this symmetry.

To see this, note first that when  $\Psi^\ell$  is a canonical field  $Q^n$ , the functional derivative  $\delta L / \delta \dot{\Psi}^\ell$  is equal to the canonical conjugate  $P_n$ , while when  $\Psi^\ell$  is an auxiliary field  $C^r$ , this functional derivative vanishes; hence we may rewrite Eq. (7.3.11) in the form

$$F = -i \int d^3x P_n(\mathbf{x}, t) \mathcal{F}^n(\mathbf{x}, t) = -i \int d^3x \int d^3y P_n(\mathbf{x}, t) \mathcal{F}^n[Q(t); \mathbf{x}]. \quad (7.3.17)$$

To calculate the commutator (not anticommutator) of  $F$  with a canonical field  $Q^m(\mathbf{x}, t)$  at an arbitrary time  $t$ , we can invoke Eq. (7.3.6) to evaluate  $F$  as a functional of the  $Q$ s and  $P$ s at the time  $t$ , and then use the equal-time canonical commutation relations (7.1.30)–(7.1.32) to obtain<sup>67</sup>

$$[F, Q^m(\mathbf{x}, t)] = -\mathcal{F}^m(\mathbf{x}, t). \quad (7.3.18)$$

It is in this sense that  $F$  is the generator of the transformation with Eq. (7.3.16). Eq. (7.3.17) and the canonical commutation rules give also

$$[F, P_n(\mathbf{x}, t)] = \int d^3y P_m(\mathbf{y}, t) \frac{\delta \mathcal{F}^m[Q(t); \mathbf{y}]}{\delta Q^n(\mathbf{x}, t)}. \quad (7.3.19)$$

Where  $\mathcal{F}^m$  is linear, Eq. (7.3.19) tells us that  $P_n$  transforms contragrediently to  $Q^n$ .

As a first example, consider the symmetry transformation of spacetime translation:

$$\Psi^\ell(x) \longrightarrow \Psi^\ell(x + \epsilon) = \Psi^\ell(x) + \epsilon^\mu \partial_\mu \Psi^\ell(x). \quad (7.3.20)$$

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<sup>67</sup>We are here assuming that for  $Q^n$  bosonic or fermionic the variation  $\mathcal{F}^n$  is also respectively bosonic or fermionic, so that  $F$  is bosonic. The only exceptions are certain symmetries known as supersymmetries, for which  $F$  is fermionic and (7.3.18) is an anticommutator if  $Q^n$  is also fermionic.

This is of the form (7.3.1), with four independent parameters  $\epsilon^\mu$  and four corresponding transformation functions

$$\mathcal{F}_\mu^\ell = -i\partial_\mu \Psi^\ell. \quad (7.3.21)$$

In consequence we have four independent conserved currents, conventionally grouped together in the *energy-momentum tensor*  $T^\mu{}_\nu$ :

$$\partial_\mu T^\mu{}_\nu = 0 \quad (7.3.22)$$

from which we can derive time-independent quantities as the spatial integrals of the time components of the translation ‘currents’ (not to be confused with the canonical conjugate field variables  $P_n(\mathbf{x}, t)$ ):

$$P_\nu = \int d^3x T^0{}_\nu, \quad (7.3.23)$$

$$\frac{d}{dt} P_\nu = 0. \quad (7.3.24)$$

The Lagrangian is invariant under spatial translations, so in accordance with the above general results we can conclude that the spatial components of  $P_\nu$  take the form

$$\mathbf{P} = - \int d^3x P_n(\mathbf{x}, t) \nabla Q^n(\mathbf{x}, t). \quad (7.3.25)$$

Using the equal-time commutation relations (7.1.30)–(7.1.32), we also find the commutator of this operator with the canonical fields and conjugates:

$$[\mathbf{P}, Q^n(\mathbf{x}, t)] = i \nabla Q^n(\mathbf{x}, t), \quad (7.3.26)$$

$$[\mathbf{P}, P_n(\mathbf{x}, t)] = i \nabla P_n(\mathbf{x}, t). \quad (7.3.27)$$

It follows that for any function  $\mathcal{G}$  of  $Q$ s and  $P$ s that does not also depend explicitly on  $\mathbf{x}$ , we have

$$[\mathbf{P}, \mathcal{G}(\mathbf{x})] = i \nabla \mathcal{G}(\mathbf{x}). \quad (7.3.28)$$

These results show that the operator  $\mathbf{P}$  is indeed the generator of space translations.

In contrast, time-translations do not leave the Lagrangian  $L(t)$  invariant. However, we already know the generator of time-translations; it is the Hamiltonian  $P^0 \equiv H$ , which as we know satisfies the commutation relation

$$[H, \mathcal{G}(\mathbf{x}, t)] = -i \dot{\mathcal{G}}(\mathbf{x}, t) \quad (7.3.29)$$

for any function  $\mathcal{G}$  of Heisenberg picture operators.

If we further assume that the Lagrangian is the integral of a Lagrangian density, then we may also obtain an explicit formula for the energy-momentum tensor  $T^\mu{}_\nu$ . However, the Lagrangian density  $\mathcal{L}(x)$  is not invariant under spacetime translations, so we cannot use Eq. (7.3.15) here. Instead, note that the change in the action under a spacetime-dependent translation

$$\Psi^\ell(x) \longrightarrow \Psi^\ell(x + \epsilon(x)) = \Psi^\ell(x) + \epsilon^\mu(x) \partial_\mu \Psi^\ell(x) \quad (7.3.30)$$

is

$$\delta I[\Psi] = \int d^4x \left( \frac{\partial \mathcal{L}}{\partial \Psi^\ell} \epsilon^\mu \partial_\mu \Psi^\ell + \frac{\partial \mathcal{L}}{\partial (\partial_\nu \Psi^\ell)} \partial_\nu [\epsilon^\mu \partial_\mu \Psi^\ell] \right). \quad (7.3.31)$$

The Euler–Lagrange equations (7.2.9) show that the terms proportional to  $\epsilon$  add up to  $\epsilon^\mu \partial_\mu \mathcal{L}$ , so

$$\delta I[\Psi] = \int d^4x \left( \frac{\partial \mathcal{L}}{\partial x^\mu} \epsilon^\mu + \frac{\partial \mathcal{L}}{\partial (\partial_\nu \Psi^\ell)} \partial_\mu \Psi^\ell \partial_\nu \epsilon^\mu \right). \quad (7.3.32)$$

Integrating by parts, we see that this takes the form of Eq. (7.3.4)

$$\delta I = - \int d^4x T^\nu{}_\mu \partial_\nu \epsilon^\mu \quad (7.3.33)$$

with ‘currents’

$$T^\nu{}_\mu = \delta^\nu{}_\mu \mathcal{L} - \frac{\partial \mathcal{L}}{\partial (\partial_\nu \Psi^\ell)} \partial_\mu \Psi^\ell. \quad (7.3.34)$$

As a check, we may note that the spatial components of Eq. (7.3.23) are the same as our previous formula (7.3.25) for  $\mathbf{P}$ , while for  $\mu = 0$  Eq. (7.3.23) gives the usual formula for the Hamiltonian:

$$H \equiv -P_0 = \int d^3x \left[ \sum_n P_n \dot{Q}^n - \mathcal{L} \right]. \quad (7.3.35)$$

(A warning: the tensor  $T^{\mu\nu}$  obtained by raising the second index in Eq. (7.3.34) is not in general symmetric, and therefore cannot be used as the right-hand side of the field equations of general relativity. The correct energy–momentum tensor to use as the source of the gravitational field is the symmetric tensor  $\Theta^{\mu\nu}$  introduced in the next section.)

In many theories there are also one or more symmetry principles that state the invariance of the action, under a set of linear coordinate-independent transformations of the canonical fields

$$Q^n(x) \longrightarrow Q^n(x) + i\epsilon^a (t_a)^n{}_m Q^m(x) \quad (7.3.36)$$

together with a set of suitable transformations on any auxiliary fields  $C^r$ :

$$C^r(x) \longrightarrow C^r(x) + i\epsilon^a (\tau_a)^r{}_s C^s(x). \quad (7.3.37)$$

Here  $t_a$  and  $\tau_a$  are sets of Hermitian matrices furnishing some representations of the Lie algebra of the symmetry group, and we sum over repeated group indices  $a, b$ , etc. (For instance, in electrodynamics there is such a symmetry, for which the one matrix  $t^n{}_m$  is diagonal, with the charges carried by each field on the main diagonal.) From any such symmetry, we can infer the existence of another set of conserved currents  $J_a^\mu$ :

$$\partial_\mu J_a^\mu = 0, \quad (7.3.38)$$

whose time components are the densities of a set of time-independent operators

$$T_a \equiv \int d^3x J_a^0. \quad (7.3.39)$$

When the Lagrangian as well as the action is invariant under the transformation (7.3.36), Eq. (7.3.11) provides an explicit formula for the  $T_a$ :

$$T_a = -i \int d^3x P_n(\mathbf{x}, t) (t_a)^n{}_m Q^m(\mathbf{x}, t). \quad (7.3.40)$$

The equal-time commutation relations here give

$$[T_a, Q^n(x)] = -(t_a)^n{}_m Q^m(x), \quad (7.3.41)$$

$$[T_a, P_n(x)] = +(t_a)^m{}_n P_m(x). \quad (7.3.42)$$

(Where  $t_a$  is diagonal, this tells us that  $Q^n$  and  $P_n$  respectively lower and raise the value of  $T_a$  by an amount equal to the  $n$ th diagonal element of  $t_a$ .) Using these results, we can calculate the commutator of  $T_a$  with another generator  $T_b$ :

$$[T_a, T_b] = i \int d^3x \left[ -P_m(t_a)^m{}_n (t_b)^n{}_k Q^k + P_n(t_b)^n{}_k (t_a)^k{}_m Q^m \right]. \quad (7.3.43)$$

Thus, if the matrices  $t_a$  form a Lie algebra with structure constants  $f_{ab}{}^c$ ,

$$[t_a, t_b] = i f_{ab}{}^c t_c, \quad (7.3.44)$$

then so do the quantum operators  $T_a$ :

$$[T_a, T_b] = i f_{ab}{}^c T_c. \quad (7.3.45)$$

This confirms that the quantities (7.3.40) are correctly normalized to qualify as generators of the symmetry group.

Where the Lagrangian is the integral of a Lagrangian density which is invariant under (7.3.36) and (7.3.37) we can go further, and use Eq. (7.3.15) to provide an explicit formula for the currents associated with these global symmetries:

$$J_a^\mu = -i \frac{\partial \mathcal{L}}{\partial(\partial Q^n / \partial x^\mu)} (t_a)^n{}_m Q^m - i \frac{\partial \mathcal{L}}{\partial(\partial C^r / \partial x^\mu)} (\tau_a)^r{}_s C^s. \quad (7.3.46)$$

As an illustration, suppose we have two real scalar fields of equal mass, with Lagrangian density

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \Phi_1 \partial^\mu \Phi_1 - \frac{1}{2} m^2 \Phi_1^2 - \frac{1}{2} \partial_\mu \Phi_2 \partial^\mu \Phi_2 - \frac{1}{2} m^2 \Phi_2^2 - \mathcal{H}_I(\Phi_1^2 + \Phi_2^2). \quad (7.3.47)$$

This is invariant under a linear transformation like (7.3.36):

$$\delta \Phi_1 = -\epsilon \Phi_2, \quad \delta \Phi_2 = +\epsilon \Phi_1,$$

so there is a conserved current (7.3.46):

$$J^\mu = \Phi_2 \partial^\mu \Phi_1 - \Phi_1 \partial^\mu \Phi_2.$$

The explicit formula (7.3.46) for the current can be used to derive other useful commutation relations. In particular, since the Lagrangian density does not involve time-derivatives of the auxiliary fields, we have

$$J_a^0 = -iP_n(t_a)^n{}_m Q^m. \quad (7.3.48)$$

We can then derive the equal-time commutators of general fields not only with the symmetry generators  $T_a$ , but also with the densities  $J_a^0$ :

$$[J_a^0(\mathbf{x}, t), Q^n(\mathbf{y}, t)] = -\delta^3(\mathbf{x} - \mathbf{y})(t_a)^n{}_m Q^m(\mathbf{x}, t), \quad (7.3.49)$$

$$[J_a^0(\mathbf{x}, t), P_m(\mathbf{y}, t)] = \delta^3(\mathbf{x} - \mathbf{y})(t_a)^n{}_m P_n(\mathbf{x}, t). \quad (7.3.50)$$

If the auxiliary fields are constructed as local functions of the  $P$ s and  $Q$ s in such a way that they transform according to a representation of the symmetry algebra with generators  $\tau_a$ , then also

$$[J_a^0(\mathbf{x}, t), C^r(\mathbf{y}, t)] = -\delta^3(\mathbf{x} - \mathbf{y})(\tau_a)^r{}_s C^s(\mathbf{x}, t). \quad (7.3.51)$$

We often summarize Eqs. (7.3.49) and (7.3.51) in the single commutation relation

$$[J_a^0(\mathbf{x}, t), \Psi^\ell(\mathbf{y}, t)] = -\delta^3(\mathbf{x} - \mathbf{y})(t_a)^\ell{}_{\ell'} \Psi^{\ell'}(\mathbf{x}, t). \quad (7.3.52)$$

Commutation relations like (7.3.49)–(7.3.51) will be used in Chapter 10 to derive relations called Ward identities for matrix elements involving the current  $J^\mu$ .

## 7.4 Lorentz Invariance

We are now going to show that the Lorentz invariance of the Lagrangian density implies the Lorentz invariance of the  $S$ -matrix. Consider an infinitesimal Lorentz transformation

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad (7.4.1)$$

$$\omega_{\mu\nu} = -\omega_{\nu\mu}. \quad (7.4.2)$$

According to the analysis of the previous section, the invariance of the action under such transformations tells us immediately that there are a set of conserved ‘currents’  $\mathcal{M}^{\rho\mu\nu}$ :

$$\partial_\rho \mathcal{M}^{\rho\mu\nu} = 0, \quad (7.4.3)$$

$$\mathcal{M}^{\rho\mu\nu} = -\mathcal{M}^{\rho\nu\mu}, \quad (7.4.4)$$

one current for each independent component of  $\omega_{\mu\nu}$ . The integrals of the time-components of these ‘currents’ then provide us with a set of time-independent tensors:

$$J^{\mu\nu} \equiv \int d^3x \mathcal{M}^{0\mu\nu}, \quad (7.4.5)$$

$$\frac{d}{dt} J^{\mu\nu} = 0. \quad (7.4.6)$$

The  $J^{\mu\nu}$  will turn out to be the generators of the homogeneous Lorentz group.

We would like to have an explicit formula for the tensor  $\mathcal{M}^{\rho\mu\nu}$ , but Lorentz transformations act on the coordinates and hence cannot leave the Lagrangian density invariant, so we



cannot immediately use the results of the previous section. However, translation invariance allows us to formulate Lorentz invariance as a symmetry of the Lagrangian density under a set of transformations on the fields and field-derivatives alone. The fields undergo the matrix transformation

$$\delta\Psi^\ell = \frac{i}{2}\omega^{\mu\nu}(\mathcal{J}_{\mu\nu})^\ell{}_m\Psi^m, \quad (7.4.7)$$

where  $\mathcal{J}_{\mu\nu}$  are a set of matrices satisfying the algebra of the homogeneous Lorentz group

$$[\mathcal{J}_{\mu\nu}, \mathcal{J}_{\rho\sigma}] = i\mathcal{J}_{\rho\nu}\eta_{\mu\sigma} - i\mathcal{J}_{\sigma\nu}\eta_{\mu\rho} - i\mathcal{J}_{\mu\rho}\eta_{\nu\sigma} + i\mathcal{J}_{\mu\sigma}\eta_{\nu\rho}. \quad (7.4.8)$$

For example, for a scalar field  $\phi$  we have  $\delta\phi = 0$ , so  $\mathcal{J}_{\mu\nu} = 0$ , while for an irreducible field of type  $(A, B)$  we have

$$\mathcal{J}_{ij} = \epsilon_{ijk}(\mathcal{A}_k + \mathcal{B}_k), \quad \mathcal{J}_{k0} = -i(\mathcal{A}_k - \mathcal{B}_k),$$

where  $\mathcal{A}$  and  $\mathcal{B}$  are spin matrices for spin  $A$  and  $B$ , respectively. We specially note that for a covariant vector field,  $\delta V_\kappa = \omega_\kappa{}^\lambda V_\lambda$ , so here

$$(\mathcal{J}_{\mu\nu})_\kappa{}^\lambda = -i\eta_{\mu\kappa}\delta_\nu{}^\lambda + i\eta_{\nu\kappa}\delta_\mu{}^\lambda.$$

The derivative of a field that transforms as in Eq. (7.4.7) transforms like another such field, but with an extra vector index

$$\delta(\partial_\kappa\Psi^\ell) = \frac{i}{2}\omega^{\mu\nu}(\mathcal{J}_{\mu\nu})^\ell{}_m\partial_\kappa\Psi^m + \omega_\kappa{}^\lambda\partial_\lambda\Psi^\ell. \quad (7.4.9)$$

The Lagrangian density is assumed to be invariant under the combined transformations Eqs. (7.4.7) and (7.4.9), so

$$\begin{aligned} 0 &= \frac{\partial\mathcal{L}}{\partial\Psi^\ell} \frac{i}{2}\omega^{\mu\nu}(\mathcal{J}_{\mu\nu})^\ell{}_m\Psi^m + \frac{\partial\mathcal{L}}{\partial(\partial_\kappa\Psi^\ell)} \frac{i}{2}\omega^{\mu\nu}(\mathcal{J}_{\mu\nu})^\ell{}_m\partial_\kappa\Psi^m \\ &\quad + \frac{\partial\mathcal{L}}{\partial(\partial_\kappa\Psi^\ell)} \omega_\kappa{}^\lambda\partial_\lambda\Psi^\ell. \end{aligned}$$

Setting the coefficient of  $\omega^{\mu\nu}$  equal to zero gives

$$\begin{aligned} 0 &= \frac{i}{2} \frac{\partial\mathcal{L}}{\partial\Psi^\ell} (\mathcal{J}_{\mu\nu})^\ell{}_m\Psi^m + \frac{i}{2} \frac{\partial\mathcal{L}}{\partial(\partial_\kappa\Psi^\ell)} (\mathcal{J}_{\mu\nu})^\ell{}_m\partial_\kappa\Psi^m \\ &\quad + \frac{1}{2} \frac{\partial\mathcal{L}}{\partial(\partial_\kappa\Psi^\ell)} (\eta_{\mu\kappa}\partial_\nu - \eta_{\nu\kappa}\partial_\mu)\Psi^\ell. \end{aligned}$$

Using the Euler–Lagrange equations (7.2.9), and our formula (7.3.34) for the energy–momentum tensor  $T_{\mu\nu}$ , we may write this as

$$0 = \partial_\kappa \left[ \frac{i}{2} \frac{\partial\mathcal{L}}{\partial(\partial_\kappa\Psi^\ell)} (\mathcal{J}_{\mu\nu})^\ell{}_m\Psi^m \right] - \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}). \quad (7.4.10)$$

This immediately suggests the definition of a new energy–momentum tensor, known as the *Belinfante tensor* [2]:

$$\begin{aligned} \Theta^{\mu\nu} &= T^{\mu\nu} - \frac{i}{2}\partial_\kappa \left[ \frac{\partial\mathcal{L}}{\partial(\partial_\kappa\Psi^\ell)} (\mathcal{J}^{\mu\nu})^\ell{}_m\Psi^m - \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Psi^\ell)} (\mathcal{J}^{\kappa\nu})^\ell{}_m\Psi^m \right. \\ &\quad \left. - \frac{\partial\mathcal{L}}{\partial(\partial_\nu\Psi^\ell)} (\mathcal{J}^{\kappa\mu})^\ell{}_m\Psi^m \right]. \end{aligned} \quad (7.4.11)$$

The quantity in square brackets is manifestly antisymmetric in  $\mu$  and  $\kappa$ , so  $\Theta^{\mu\nu}$  satisfies the same conservation law as  $T^{\mu\nu}$ ,

$$\partial_\mu \Theta^{\mu\nu} = 0. \quad (7.4.12)$$

For the same reason, when we set  $\mu = 0$  in Eq. (7.4.11) the index  $\kappa$  runs over space components only, so the derivative term here drops out when we integrate over all space

$$\int \Theta^{0\nu} d^3x = \int T^{0\nu} d^3x = P^\nu, \quad (7.4.13)$$

where  $P^0 \equiv H$ . Thus  $\Theta^{\mu\nu}$  can be regarded as *the* energy-momentum tensor, just as well as  $T^{\mu\nu}$ . However, Eq. (7.4.10) tells us that, unlike  $T^{\mu\nu}$ , the Belinfante tensor  $\Theta^{\mu\nu}$  is not only conserved but also *symmetric*:

$$\Theta^{\mu\nu} = \Theta^{\nu\mu}. \quad (7.4.14)$$

It is  $\Theta^{\mu\nu}$  rather than  $T^{\mu\nu}$  that acts as a source of the gravitational field [3]. In consequence of the symmetry of  $\Theta^{\mu\nu}$ , we may construct one more conserved tensor density:

$$\mathcal{M}^{\lambda\mu\nu} \equiv x^\mu \Theta^{\lambda\nu} - x^\nu \Theta^{\lambda\mu}. \quad (7.4.15)$$

This is conserved, in the sense that

$$\partial_\lambda \mathcal{M}^{\lambda\mu\nu} = \Theta^{\mu\nu} - \Theta^{\nu\mu} = 0. \quad (7.4.16)$$

Thus Lorentz invariance allows us to define one more time-independent tensor

$$J^{\mu\nu} = \int \mathcal{M}^{0\mu\nu} d^3x = \int d^3x (x^\mu \Theta^{0\nu} - x^\nu \Theta^{0\mu}). \quad (7.4.17)$$

The rotation generator  $J_k = \epsilon_{ijk} J^{ij}/2$  is not only time-independent, but also has no explicit time-dependence, so it commutes with the Hamiltonian

$$[H, \mathbf{J}] = 0. \quad (7.4.18)$$

Also, applying Eq. (7.3.28) to the function  $\Theta^{0\nu}$ , we have

$$\begin{aligned} [P_j, J_i] &= \frac{1}{2} \epsilon_{k\ell i} [P_j, J^{k\ell}] \\ &= \frac{i}{2} \epsilon_{k\ell i} \int d^3x \left( x^k \frac{\partial}{\partial x^j} \Theta^{0\ell} - x^\ell \frac{\partial}{\partial x^j} \Theta^{0k} \right) \\ &= -i \epsilon_{ijk} \int d^3x \Theta^{0k} \end{aligned}$$

and therefore

$$[P_j, J_i] = -i \epsilon_{ijk} P_k. \quad (7.4.19)$$

On the other hand, the ‘boost’ generator  $K_k \equiv J^{k0}$ , though time-independent, does explicitly involve the time coordinate

$$K_k = \int d^3x (x^k \Theta^{00} - x^0 \Theta^{0k}),$$

or more explicitly

$$\mathbf{K} = -t\mathbf{P} + \int d^3x \mathbf{x} \Theta^{00}(\mathbf{x}, t). \quad (7.4.20)$$

Since this is a constant, we have  $0 = \dot{\mathbf{K}} = -\dot{\mathbf{P}} + i[H, \mathbf{K}]$ , and therefore

$$[H, \mathbf{K}] = -i\dot{\mathbf{P}}. \quad (7.4.21)$$

Also, applying Eq. (7.3.28) again gives

$$[P_j, K_k] = i \int d^3x x^k \frac{\partial}{\partial x^j} \Theta^{00} = -i\delta_{jk} \int d^3x \Theta^{00}$$

and therefore

$$[P_j, K_k] = -i\delta_{jk} H. \quad (7.4.22)$$

For any reasonable Lagrangian density, the operator (7.4.20) will be ‘smooth’ in the sense used in Section 3.3, i.e., the interaction terms in  $e^{iH_0 t} \int d^3x \mathbf{x} \Theta^{00}(\mathbf{x}, 0) e^{-iH_0 t}$  vanish<sup>68</sup> for  $t \rightarrow \pm\infty$ . (Note that the interaction terms in  $e^{iH_0 t} \int d^3x \Theta^{00}(\mathbf{x}, 0) e^{-iH_0 t}$  must vanish for  $t \rightarrow \pm\infty$  in order to allow the introduction of ‘in’ and ‘out’ states and the  $S$ -matrix.) With this smoothness assumption and the commutation relation (7.4.21) in hand, we can repeat the arguments of Section 3.3, and conclude that the  $S$ -matrix is Lorentz-invariant.

\* \* \*

The same arguments were also used in Section 3.3 to verify that the remaining commutation relations of the Lorentz group, those of the  $J^{\mu\nu}$  with each other, take the proper form. This can also be shown directly for the commutators of the rotation generators, which here take the form

$$J^{ij} \equiv \int d^3x \frac{\partial \mathcal{L}}{\partial \dot{\Psi}^\ell} \left( x^i \partial_j \Psi^\ell - x^j \partial_i \Psi^\ell - i(\mathcal{J}^{ij})^\ell{}_m \Psi^m \right). \quad (7.4.23)$$

Since the Lagrangian density does not depend on the time-derivatives of the auxiliary fields, and the rotation generators do not mix canonical and auxiliary fields, this can also be written as a sum over canonical fields alone:

$$J^{ij} = \int d^3x P_n \left( x^i \partial_j Q^n - x^j \partial_i Q^n - i(\mathcal{J}^{ij})^n{}_{n'} \Psi^{n'} \right). \quad (7.4.24)$$

It follows immediately then from the canonical commutation relations that

$$[J^{ij}, Q^n(x)] = -i(x_i \partial_j - x_j \partial_i) Q^n(x) - (\mathcal{J}^{ij})^n{}_{n'} Q^{n'}(x), \quad (7.4.25)$$

$$[J^{ij}, P_n(x)] = i(x_i \partial_j - x_j \partial_i) P_n(x) + (\mathcal{J}^{ij})^{n'}{}_n P_{n'}(x). \quad (7.4.26)$$

These results can be used to derive the usual commutation relations of the  $J^{ij}$  with each other and other generators.<sup>69</sup> If there are no auxiliary fields then the same arguments may

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<sup>68</sup>When we say that some interaction-picture operator vanishes for  $t \rightarrow \pm\infty$ , we mean that its matrix elements between states that are smooth superpositions of energy eigenstates vanish in this limit.

<sup>69</sup>Also, since  $J^{ij}$  commutes with  $H$  and  $P_n \dot{Q}^n$ , it commutes with  $L$ . The commutator of  $J^{ij}$  with the auxiliary fields must thus be consistent with the rotational invariance of  $L$ .

be applied to the ‘boost’ generators to complete the demonstration that the  $P^\mu$  and  $J^{\mu\nu}$  satisfy the commutation relations of the Poincaré group. However the ‘boost’ matrices  $\mathcal{J}^{i0}$  will in general mix canonical and auxiliary fields (such as the components  $V^i$  and  $V^0$  of a vector field), so the direct proof of the commutation relations of the  $J^{i0}$  with each other has to be given on a case by case basis. Fortunately, this is not needed for the proof of the Lorentz invariance of the  $S$ -matrix given in Section 3.3.

## 7.5 Transition to Interaction Picture: Examples

At the end of Section 7.2 we showed how to use the Lagrangian of a simple scalar field theory to derive the structure of the interaction and the free fields it contains in the interaction picture. We will now turn to somewhat more complicated and revealing examples.

### Scalar Field, Derivative Coupling

First let’s consider a neutral scalar field, but now with derivative coupling. We take the Lagrangian as

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2 - J^\mu\partial_\mu\Phi - \mathcal{H}_I(\Phi), \quad (7.5.1)$$

where  $J^\mu$  is either a c-number external current (unrelated to currents  $J^\mu$  introduced earlier), or a functional of various fields other than  $\Phi$  (in which case terms involving these other fields need to be added to (7.5.1)). The canonical conjugate to  $\Phi$  is now

$$\Pi = \frac{\partial\mathcal{L}}{\partial\dot{\Phi}} = \dot{\Phi} - J^0 \quad (7.5.2)$$

and the Hamiltonian is

$$\begin{aligned} H &= \int d^3x [\Pi\dot{\Phi} - \mathcal{L}] \\ &= \int d^3x \left[ \Pi(\Pi + J^0) + \frac{1}{2}(\nabla\Phi)^2 - \frac{1}{2}(\Pi + J^0)^2 + \frac{1}{2}m^2\Phi^2 \right. \\ &\quad \left. + \mathbf{J} \cdot \nabla\Phi + J^0(\Pi + J^0) + \mathcal{H}_I(\Phi) \right]. \end{aligned}$$

Collecting terms, we can write this as

$$H = H_0 + H_{\text{int}}, \quad (7.5.3)$$

$$H_0 = \int d^3x \left[ \frac{1}{2}\Pi^2 + \frac{1}{2}(\nabla\Phi)^2 + \frac{1}{2}m^2\Phi^2 \right], \quad (7.5.4)$$

$$H_{\text{int}} = \int d^3x \left[ \Pi J^0 + \mathbf{J} \cdot \nabla\Phi + \frac{1}{2}(J^0)^2 + \mathcal{H}_I(\Phi) \right]. \quad (7.5.5)$$

As explained in Section 7.2, we can pass to the interaction picture by simply replacing  $\Pi$  and  $\Phi$  with  $\pi$  and  $\phi$  (and likewise for any fields in the current  $J^\mu$ , though we will not bother to indicate this explicitly):

$$H_0 = \int d^3x \left[ \frac{1}{2}\pi^2(\mathbf{x}, t) + \frac{1}{2}(\nabla\phi(\mathbf{x}, t))^2 + \frac{1}{2}m^2\phi^2(\mathbf{x}, t) \right], \quad (7.5.6)$$

$$H_I(t) = \int d^3x \left[ \pi(\mathbf{x}, t) J^0(\mathbf{x}, t) + \mathbf{J}(\mathbf{x}, t) \cdot \nabla \phi(\mathbf{x}, t) + \frac{1}{2} [J^0(\mathbf{x}, t)]^2 + \mathcal{H}_I(\phi(\mathbf{x}, t)) \right]. \quad (7.5.7)$$

The free-particle Hamiltonian is just the same as Eq. (7.2.25), and leads as in Section 7.2 to Eqs. (7.2.26)–(7.2.35). Indeed, whatever the total Hamiltonian may be, we *must* take (7.5.6) as the part we split off and call the free-particle part, with the remainder called the interaction, because as we have seen it is this form of the free-particle Hamiltonian that leads to the correct expansion (7.2.29) of the scalar field in terms of creation and annihilation operators that satisfy the commutation relations (7.2.34), (7.2.35). The last step is to replace  $\pi$  in the interaction Hamiltonian with its value  $\dot{\phi}$  in the interaction picture (*not* its value  $\dot{\Phi} - J^0$  in the Heisenberg picture):

$$H_I(t) = \int d^3x \left[ J^\mu(\mathbf{x}, t) \partial_\mu \phi(\mathbf{x}, t) + \frac{1}{2} [J^0(\mathbf{x}, t)]^2 + \mathcal{H}_I(\phi(\mathbf{x}, t)) \right]. \quad (7.5.8)$$

The extra non-invariant term in Eq. (7.5.8) is just what we saw in Section 6.2 is needed to cancel a non-invariant term in the propagator of  $\partial_0 \phi$ .

### Vector Field, Spin One

Similar results are obtained in the canonical quantization of the vector field  $V_\mu$  for a particle of spin one. Let's here keep an open mind, and write the Lagrangian density in a fairly general form

$$\mathcal{L} = -\frac{1}{2} \alpha \partial_\mu V_\nu \partial^\mu V^\nu - \frac{1}{2} \beta \partial_\mu V_\nu \partial^\nu V^\mu - \frac{1}{2} m^2 V_\mu V^\mu - J_\mu V^\mu, \quad (7.5.9)$$

where  $\alpha$ ,  $\beta$ , and  $m^2$  are so far arbitrary constants, and  $J_\mu$  is either a c-number external current, or an operator depending on fields other than  $V^\mu$ , in which case additional terms involving these fields must be added to  $\mathcal{L}$ . The Euler–Lagrange field equations for  $V_\mu$  read

$$\alpha \square V^\nu + \beta \partial^\nu (\partial_\mu V^\mu) + m^2 V^\nu = -J^\nu.$$

Taking the divergence gives

$$(\alpha + \beta) \square \partial_\lambda V^\lambda + m^2 \partial_\lambda V^\lambda = -\partial_\lambda J^\lambda. \quad (7.5.10)$$

This is the equation for an ordinary scalar field with mass  $m^2/(\alpha + \beta)$  and source  $\partial_\lambda J^\lambda/(\alpha + \beta)$ . We want to describe a theory containing only particles of spin one, not spin zero, so to avoid the appearance of  $\partial_\lambda V^\lambda$  as an independently propagating scalar field, we take  $\alpha = -\beta$ , in which case  $\partial_\lambda V^\lambda$  can be expressed in terms of an external current or other fields, as  $-\partial_\lambda J^\lambda/m^2$ . The constant  $\alpha$  can be absorbed in the definition of  $V_\mu$ , so we can take  $\alpha = -\beta = 1$ , and therefore

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} m^2 V_\mu V^\mu - J_\mu V^\mu, \quad (7.5.11)$$

where

$$F_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu. \quad (7.5.12)$$

The derivative of the Lagrangian with respect to the time-derivative of the vector field is

$$\frac{\partial \mathcal{L}}{\partial \dot{V}^\mu} = -F^{0\mu}. \quad (7.5.13)$$

This is non-vanishing for  $\mu$  a spatial index  $i$ , so the  $V^i$  are canonical fields, with conjugates

$$\Pi^i = F^{i0} = \dot{V}^i + \partial_i V^0. \quad (7.5.14)$$

On the other hand  $F^{00} = 0$ , so  $\dot{V}^0$  does not appear in the Lagrangian, and  $V^0$  is therefore an auxiliary field. This causes no serious difficulty: the fact that  $\partial \mathcal{L} / \partial \dot{V}^0$  vanishes means that the field equation for  $V^0$  involves no second time-derivatives, and can therefore be used as a constraint that eliminates a field variable. Specifically, the Euler–Lagrange equation for  $\nu = 0$  is

$$\partial_i F^{i0} = m^2 V^0 + J^0 \quad (7.5.15)$$

or using Eq. (7.5.14)

$$V^0 = \frac{1}{m^2} (\nabla \cdot \mathbf{\Pi} - J^0). \quad (7.5.16)$$

Now let us calculate the Hamiltonian  $H = \int d^3x (\mathbf{\Pi} \cdot \dot{\mathbf{V}} - \mathcal{L})$  for this theory. Eq. (7.5.14) allows us to write  $\dot{\mathbf{V}}$  in terms of  $\mathbf{\Pi}$  and  $J^0$ :

$$\dot{\mathbf{V}} = -\nabla V^0 + \mathbf{\Pi} = \mathbf{\Pi} - \frac{1}{m^2} \nabla (\nabla \cdot \mathbf{\Pi} - J^0),$$

so

$$\begin{aligned} H = \int d^3x & \left[ \mathbf{\Pi}^2 + m^{-2} (\nabla \cdot \mathbf{\Pi}) (\nabla \cdot \mathbf{\Pi} - J^0) - \frac{1}{2} \mathbf{\Pi}^2 + \frac{1}{2} (\nabla \times \mathbf{V})^2 + \frac{1}{2} m^2 \mathbf{V}^2 \right. \\ & \left. - \frac{1}{2} m^{-2} (\nabla \cdot \mathbf{\Pi} - J^0)^2 + \mathbf{J} \cdot \mathbf{V} - m^{-2} J^0 (\nabla \cdot \mathbf{\Pi} - J^0) \right]. \end{aligned}$$

Again, we split this up into a free-particle term  $H_0$  and interaction  $H_{\text{int}}$ :

$$H = H_0 + H_{\text{int}}, \quad (7.5.17)$$

and pass to the interaction picture by replacing the Heisenberg-picture quantities  $\mathbf{V}$  and  $\mathbf{\Pi}$  with their interaction-picture counterparts  $\mathbf{v}$  and  $\boldsymbol{\pi}$  (and, though not shown explicitly, likewise for whatever fields and conjugates are present in  $J^\mu$ ):

$$H_0 = \int d^3x \left[ \frac{1}{2} \boldsymbol{\pi}^2 + \frac{1}{2m^2} (\nabla \cdot \boldsymbol{\pi})^2 + \frac{1}{2} (\nabla \times \mathbf{v})^2 + \frac{m^2}{2} \mathbf{v}^2 \right], \quad (7.5.18)$$

$$H_I(t) = \int d^3x \left[ \mathbf{J} \cdot \mathbf{v} - m^{-2} J^0 \nabla \cdot \boldsymbol{\pi} + \frac{1}{2m^2} (J^0)^2 \right]. \quad (7.5.19)$$

The relation between  $\boldsymbol{\pi}$  and  $\mathbf{v}$  is then

$$\dot{\mathbf{v}} = \frac{\delta H_0(\mathbf{v}, \boldsymbol{\pi})}{\delta \boldsymbol{\pi}} = \boldsymbol{\pi} - m^{-2} \nabla (\nabla \cdot \boldsymbol{\pi}) \quad (7.5.20)$$

and the ‘field equation’ is

$$\dot{\boldsymbol{\pi}} = -\frac{\delta H_0(\mathbf{v}, \boldsymbol{\pi})}{\delta \mathbf{v}} = +\nabla^2 \mathbf{v} - \nabla (\nabla \cdot \mathbf{v}) - m^2 \mathbf{v}. \quad (7.5.21)$$

Since  $V^0$  is not an independent field variable, it is not related by a similarity transformation to any interaction-picture object  $v^0$ . Instead, we can *invent* a quantity

$$v^0 \equiv m^{-2} \nabla \cdot \boldsymbol{\pi}. \quad (7.5.22)$$

Eq. (7.5.20) then allows us to write  $\boldsymbol{\pi}$  as

$$\boldsymbol{\pi} = \dot{\mathbf{v}} + \nabla v^0. \quad (7.5.23)$$

Inserting this in Eqs. (7.5.22) and (7.5.21) gives our field equations in the form

$$\begin{aligned} \nabla^2 v^0 + \nabla \cdot \dot{\mathbf{v}} - m^2 v^0 &= 0, \\ \nabla^2 \mathbf{v} - \nabla(\nabla \cdot \mathbf{v}) - \ddot{\mathbf{v}} - \nabla \dot{v}^0 - m^2 \mathbf{v} &= 0. \end{aligned}$$

These can be combined in the covariant form

$$\square v^\mu - \partial^\mu \partial_\nu v^\nu - m^2 v^\mu = 0. \quad (7.5.24)$$

Taking the divergence gives

$$\partial_\mu v^\mu = 0 \quad (7.5.25)$$

and hence

$$(\square - m^2)v^\mu = 0. \quad (7.5.26)$$

A real vector field satisfying Eqs. (7.5.25) and (7.5.26) can be expressed as a Fourier transform

$$\begin{aligned} v^\mu(x) = (2\pi)^{-3/2} \sum_{\sigma} \int d^3p (2p^0)^{-1/2} \Big\{ e^\mu(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma} e^{ip \cdot x} \\ + e^{\mu*}(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma}^\dagger e^{-ip \cdot x} \Big\}, \end{aligned} \quad (7.5.27)$$

where  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$ ; the  $e^\mu(\mathbf{p}, \sigma)$  for  $\sigma = +1, 0, -1$  are three independent vectors satisfying

$$p_\mu e^\mu(\mathbf{p}, \sigma) = 0 \quad (7.5.28)$$

and normalized so that

$$\sum_{\sigma} e^\mu(\mathbf{p}, \sigma) e^{\nu*}(\mathbf{p}, \sigma) = \eta^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}; \quad (7.5.29)$$

and the  $a_{\mathbf{p}, \sigma}$  are operator coefficients. It is straightforward using Eqs. (7.5.23), (7.5.27), and (7.5.29) to calculate that  $\mathbf{v}$  and  $\boldsymbol{\pi}$  satisfy the correct commutation relations

$$\begin{aligned} [v^i(\mathbf{x}, t), \pi^j(\mathbf{y}, t)] &= i\delta_{ij}\delta^3(\mathbf{x} - \mathbf{y}), \\ [v^i(\mathbf{x}, t), v^j(\mathbf{x}, t)] &= [\pi^i(\mathbf{x}, t), \pi^j(\mathbf{x}, t)] = 0, \end{aligned} \quad (7.5.30)$$

provided that  $a_{\mathbf{p}, \sigma}$  and  $a_{\mathbf{p}, \sigma}^\dagger$  satisfy the commutation relations

$$[a_{\mathbf{p}, \sigma}, a_{\mathbf{p}', \sigma'}^\dagger] = \delta^3(\mathbf{p}' - \mathbf{p})\delta_{\sigma'\sigma}, \quad (7.5.31)$$

$$[a_{\mathbf{p},\sigma}, a_{\mathbf{p}',\sigma'}] = 0. \quad (7.5.32)$$

We already know that the vector field for a spin one particle must take the form (7.5.27), so our derivation of these results serves to verify that Eq. (7.5.18) gives the correct free-particle Hamiltonian for a massive particle of spin one. It is easy to check also that Eq. (7.5.18) may be written (up to a constant term) in the standard form of a free-particle energy, as  $\sum_{\sigma} \int d^3p p^0 a_{\mathbf{p},\sigma}^{\dagger} a_{\mathbf{p},\sigma}$ . Finally, using Eq. (7.5.22) in Eq. (7.5.19) yields the interaction in the interaction picture

$$H_I(t) = \int d^3x \left[ J_{\mu} v^{\mu} + \frac{1}{2m^2} (J^0)^2 \right]. \quad (7.5.33)$$

The extra non-invariant term in Eq. (7.5.33) is just what we found in Chapter 6 is needed to cancel a non-invariant term in the propagator of the vector field.

### Dirac field, Spin One Half

For the Dirac field of a particle of spin 1/2, we tentatively take the Lagrangian as

$$\mathcal{L} = -\bar{\Psi}(\gamma^{\mu}\partial_{\mu} + m)\Psi - \mathcal{H}_I(\bar{\Psi}, \Psi) \quad (7.5.34)$$

with  $\mathcal{H}_I$  a real function of  $\bar{\Psi}$  and  $\Psi$ . This is not real, but the action is, because

$$\bar{\Psi}\gamma^{\mu}\partial_{\mu}\Psi - (\bar{\Psi}\gamma^{\mu}\partial_{\mu}\Psi)^{\dagger} = \bar{\Psi}\gamma^{\mu}\partial_{\mu}\Psi + (\partial_{\mu}\bar{\Psi})\gamma^{\mu}\Psi = \partial_{\mu}(\bar{\Psi}\gamma^{\mu}\Psi).$$

Hence the field equations obtained by requiring the action to be stationary with respect to  $\bar{\Psi}$  are the adjoints of those obtained by requiring the action to be stationary with respect to  $\Psi$ , as necessary if we are to avoid having too many field equations. The canonical conjugate to  $\Psi$  is

$$\Pi = \frac{\partial \mathcal{L}}{\partial \dot{\Psi}} = -\bar{\Psi}\gamma^0, \quad (7.5.35)$$

so we should not regard  $\bar{\Psi}$  as a field like  $\Psi$ , but rather as proportional to the canonical conjugate of  $\Psi$ . The Hamiltonian is

$$H = \int d^3x [\Pi \dot{\Psi} - \mathcal{L}] = \int d^3x [\Pi \gamma^0 (\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} + m) \Psi + \mathcal{H}_I].$$

We write this as

$$H = H_0 + H_{\text{int}}, \quad (7.5.36)$$

where

$$H_0 = \int d^3x \Pi \gamma^0 (\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} + m) \Psi, \quad (7.5.37)$$

$$H_{\text{int}} = \int d^3x \mathcal{H}_I(\bar{\Psi}, \Psi). \quad (7.5.38)$$

We now pass to the interaction picture. Since Eq. (7.5.35) does not involve the time, the similarity transformation (7.1.28), (7.1.29) yields immediately

$$\pi = -\bar{\psi}\gamma^0. \quad (7.5.39)$$



Likewise,  $H_0$  and  $H_I(t)$  can be calculated by replacing  $\Psi$  and  $\Pi$  with  $\psi$  and  $\pi$  in Eqs. (7.5.37) and (7.5.38). This gives the equation of motion

$$\dot{\psi} = \frac{\delta H_0}{\delta \pi} = \gamma^0(\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} + m)\psi \quad (7.5.40)$$

or more neatly

$$(\gamma^\mu \partial_\mu + m)\psi = 0. \quad (7.5.41)$$

(The other equation of motion,  $\dot{\pi} = -\delta H_0/\delta \psi$ , yields just the adjoint of this one.) Any field satisfying Eq. (7.5.41) can be written as a Fourier transform

$$\psi(x) = (2\pi)^{-3/2} \int d^3p \sum_{\sigma} \left\{ u(\mathbf{p}, \sigma) e^{ip \cdot x} a_{\mathbf{p}, \sigma} + v(\mathbf{p}, \sigma) e^{-ip \cdot x} b_{\mathbf{p}, \sigma}^\dagger \right\}, \quad (7.5.42)$$

where  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$ ;  $a_{\mathbf{p}, \sigma}$  and  $b_{\mathbf{p}, \sigma}^\dagger$  are operator coefficients; and  $u(\mathbf{p}, \pm \frac{1}{2})$  are the two independent solutions of

$$(i\gamma^\mu p_\mu + m)u(\mathbf{p}, \sigma) = 0 \quad (7.5.43)$$

and likewise

$$(-i\gamma^\mu p_\mu + m)v(\mathbf{p}, \sigma) = 0 \quad (7.5.44)$$

normalized so that<sup>70</sup>

$$\sum_{\sigma} u(\mathbf{p}, \sigma) \bar{u}(\mathbf{p}, \sigma) = \frac{-i\gamma^\mu p_\mu + m}{2p^0}, \quad (7.5.45)$$

$$\sum_{\sigma} v(\mathbf{p}, \sigma) \bar{v}(\mathbf{p}, \sigma) = -\frac{i\gamma^\mu p_\mu + m}{2p^0}. \quad (7.5.46)$$

In order to obtain the desired anticommutators

$$\begin{aligned} \{\psi_\alpha(\mathbf{x}, t), \bar{\psi}_\beta(\mathbf{y}, t)\} &= \{\psi_\alpha(\mathbf{x}, t), \pi_\gamma(\mathbf{y}, t)\} (\gamma^0)_{\gamma\beta} \\ &= i(\gamma^0)_{\alpha\beta} \delta^3(\mathbf{x} - \mathbf{y}), \end{aligned} \quad (7.5.47)$$

$$\{\psi_\alpha(\mathbf{x}, t), \psi_\beta(\mathbf{y}, t)\} = 0, \quad (7.5.48)$$

we must adopt the anticommutation relations

$$\{a_{\mathbf{p}, \sigma}, a_{\mathbf{p}', \sigma'}^\dagger\} = \{b_{\mathbf{p}, \sigma}, b_{\mathbf{p}', \sigma'}^\dagger\} = \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma' \sigma}, \quad (7.5.49)$$

$$\begin{aligned} \{a_{\mathbf{p}, \sigma}, a_{\mathbf{p}', \sigma'}\} &= \{b_{\mathbf{p}, \sigma}, b_{\mathbf{p}', \sigma'}\} = \\ \{a_{\mathbf{p}, \sigma}, b_{\mathbf{p}', \sigma'}\} &= \{a_{\mathbf{p}, \sigma}, b_{\mathbf{p}', \sigma'}^\dagger\} = 0, \end{aligned} \quad (7.5.50)$$

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<sup>70</sup>The matrix  $i\gamma^\mu p_\mu$  has eigenvalues  $\pm m$ , so  $\sum u \bar{u}$  and  $\sum v \bar{v}$  must be proportional to the projection matrices  $(-i\gamma^\mu p_\mu + m)/2m$  and  $(i\gamma^\mu p_\mu + m)/2m$ , respectively. The proportionality factor may be adjusted up to a sign by absorbing it in the definition of  $u$  and  $v$ . The overall sign is determined by positivity;  $\text{Tr} \sum u \bar{u} \beta = \sum u^\dagger u$  and  $\text{Tr} \sum v \bar{v} \beta = \sum v^\dagger v$  must be positive.

and their adjoints. These agree with the results obtained in Chapter 5, thus verifying that (7.5.37) is the correct free-particle Hamiltonian for spin 1/2. In terms of the  $a$ s and  $b$ s, this Hamiltonian is

$$H_0 = \sum_{\sigma} \int d^3p p^0 \left( a_{\mathbf{p},\sigma}^{\dagger} a_{\mathbf{p},\sigma} - b_{\mathbf{p},\sigma} b_{\mathbf{p},\sigma}^{\dagger} \right). \quad (7.5.51)$$

We can rewrite this as a more conventional free-particle Hamiltonian, plus another infinite c-number<sup>71</sup>

$$H_0 = \sum_{\sigma} \int d^3p p^0 \left[ a_{\mathbf{p},\sigma}^{\dagger} a_{\mathbf{p},\sigma} + b_{\mathbf{p},\sigma}^{\dagger} b_{\mathbf{p},\sigma} - \delta^3(\mathbf{p} - \mathbf{p}) \right]. \quad (7.5.52)$$

The c-number term in Eq. (7.5.52) is only important if we worry about gravitational phenomena; otherwise here, as for the scalar field, we can throw it away, since it only affects the zero of energy with respect to which all energies are measured. With this understanding,  $H_0$  is a positive operator, just as for bosons.

## 7.6 Constraints and Dirac Brackets

The chief obstacle to deriving the Hamiltonian from the Lagrangian is the occurrence of constraints. The standard analysis of this problem is that of Dirac [5], whose terminology we will follow here. Dirac's analysis is not really needed for the simple theories discussed in this chapter, where it is easy to identify the unconstrained canonical variables. We shall use the theory of a real massive vector field for illustration here, returning to Dirac's approach in the next chapter, where it will be actually useful.

*Primary constraints* are either imposed on the system (as when in the next chapter we choose a gauge for the electromagnetic field) or arise from the structure of the Lagrangian itself. For an example of the latter type, consider the Lagrangian (7.5.11) of a massive vector field  $V^{\mu}$  interacting with a current  $J_{\mu}$ :

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} m^2 V_{\mu} V^{\mu} - J_{\mu} V^{\mu} \quad (7.6.1)$$

where

$$F_{\mu\nu} = \partial_{\mu} V_{\nu} - \partial_{\nu} V_{\mu}. \quad (7.6.2)$$

Suppose we try to treat all four components of  $V^{\mu}$  on the same basis. We should then define the conjugates

$$\Pi_{\mu} \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 V^{\mu})} = -F^{0\mu}. \quad (7.6.3)$$

We immediately find the primary constraint:

$$\Pi_0 = 0. \quad (7.6.4)$$

More generally, we encounter primary constraints whenever the equations  $\Pi_{\ell} = \delta L / \delta \partial_0 \Psi^{\ell}$  cannot be solved to give all the  $\partial_0 \Psi^{\ell}$  (at least locally) in terms of  $\Pi_{\ell}$  and  $\Psi^{\ell}$ . This will be the case if and only if the matrix  $\delta^2 L / \delta(\partial_0 \Psi^{\ell}) \delta(\partial_0 \Psi^m)$  has vanishing determinant. Such Lagrangians are called *irregular*.

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<sup>71</sup>Note the negative sign of the c-number term. The conjectured symmetry known as supersymmetry [4] connects the numbers of boson and fermion fields, in such a way that the c-numbers in  $H_0$  all cancel.

Then there are *secondary constraints*, which arise from the requirement that the primary constraints be consistent with the equations of motion. For the massive vector field, this is just the Euler–Lagrange equation (7.5.16) for  $V^0$ :

$$\partial_i \Pi_i = m^2 V^0 + J^0. \quad (7.6.5)$$

Here we are finished, but in other theories we might encounter further constraints by requiring consistency of the secondary constraints with the field equations, and so on. The distinction between primary, secondary, etc. constraints is not important; we will treat them all together here.

There is another distinction between certain types of constraint that is more important. The constraints we have found for the massive vector field are of a type known as *second class*, for which there is a universal prescription for the commutation relations. To explain the distinction between first and second class constraints, and the prescription used to deal with second class constraints, it is useful first to recall the definition of the Poisson brackets of classical mechanics.

Consider any Lagrangian  $L(\Psi, \dot{\Psi})$  that depends on a set of variables  $\Psi^a(t)$  and their time-derivatives  $\dot{\Psi}^a(t)$ . (The Lagrangians of quantum field theory are a special case, with the index  $a$  running over all pairs of  $\ell$  and  $\mathbf{x}$ .) We can define canonical conjugates for *all* of these variables by

$$\Pi_a \equiv \frac{\partial L}{\partial \dot{\Psi}^a}. \quad (7.6.6)$$

The  $\Pi$ s and  $\Psi$ s will in general not be independent variables, but may instead be related by various constraint equations, both primary and secondary. The Poisson bracket is then defined by

$$[A, B]_P \equiv \frac{\partial A}{\partial \Psi^a} \frac{\partial B}{\partial \Pi_a} - \frac{\partial B}{\partial \Psi^a} \frac{\partial A}{\partial \Pi_a} \quad (7.6.7)$$

with the constraints ignored in calculating the derivatives with respect to  $\Psi^a$  and  $\Pi_a$ . In particular, we always have  $[\Psi^a, \Pi_b]_P = \delta_b^a$ . (Here and below all fields are taken at the same time, and time arguments are everywhere dropped.) These brackets have the same algebraic properties as commutators:

$$[A, B]_P = -[B, A]_P, \quad (7.6.8)$$

$$[A, BC]_P = [A, B]_P C + B[A, C]_P, \quad (7.6.9)$$

including the Jacobi identity

$$[A, [B, C]_P]_P + [B, [C, A]_P]_P + [C, [A, B]_P]_P = 0. \quad (7.6.10)$$

If we could adopt the usual commutation relations  $[\Psi^a, \Pi_b] = i\delta_b^a$ ,  $[\Psi^a, \Psi^b] = [\Pi_a, \Pi_b] = 0$ , then the commutator of any two functions of the  $\Psi$ s and  $\Pi$ s would be just  $[A, B] = i[A, B]_P$ . But the constraints do not always allow this.

The constraints may in general be expressed in the form  $\chi_N = 0$ , where the  $\chi_N$  are a set of functions of the  $\Psi$ s and  $\Pi$ s. Because we are including secondary constraints along

with the primary constraints, the set of all the constraints is necessarily consistent with the equations of motion  $\dot{A} = [A, H]_P$ , and therefore

$$[\chi_N, H]_P = 0 \quad (7.6.11)$$

when the constraint equations  $\chi_N = 0$  are satisfied.

We call a constraint *first class* if its Poisson bracket with all the other constraints vanishes when (*after* calculating the Poisson brackets) we impose the constraints. We shall see a simple example of such a constraint in the quantization of the electromagnetic field in the next chapter, where the first class constraint arises from a symmetry of the action, electromagnetic gauge invariance. In fact, the set of first class constraints  $\chi_N = 0$  is always associated with a group of symmetries, under which an arbitrary quantity  $A$  undergoes the infinitesimal transformation

$$\delta_N A \equiv \sum_N \epsilon_N [\chi_N, A]_P. \quad (7.6.12)$$

(In field theory these are local transformations, because the index  $N$  contains a spacetime coordinate.) Eq. (7.6.11) shows that this transformation leaves the Hamiltonian invariant, and for first class constraints it also respects all other constraints. Such first class constraints can be eliminated by a choice of gauge, or treated by gauge-invariant methods described in Volume II.

After all of the first class constraints have been eliminated, the remaining constraint equations  $\chi_N = 0$  are such that no linear combination  $\sum_N u_N [\chi_N, \chi_M]_P$  of the Poisson brackets of these constraints with each other vanishes. It follows that the matrix of the Poisson brackets of the remaining constraints is non-singular:

$$\text{Det } C \neq 0, \quad (7.6.13)$$

where

$$C_{NM} \equiv [\chi_N, \chi_M]_P. \quad (7.6.14)$$

Constraints of this sort are called *second class*. Note that there must always be an even number of second class constraints, because an antisymmetric matrix of odd dimensionality necessarily has vanishing determinant.

As we have seen, in the case of the massive real vector field the constraints are

$$\chi_{1\mathbf{x}} = \chi_{2\mathbf{x}} = 0, \quad (7.6.15)$$

where

$$\chi_{1\mathbf{x}} = \Pi_0(\mathbf{x}), \quad \chi_{2\mathbf{x}} = \partial_i \Pi_i(\mathbf{x}) - m^2 V^0(\mathbf{x}) - J^0(\mathbf{x}). \quad (7.6.16)$$

The Poisson bracket of these constraints is

$$C_{1\mathbf{x}, 2\mathbf{y}} = -C_{2\mathbf{y}, 1\mathbf{x}} = [\chi_{1\mathbf{x}}, \chi_{2\mathbf{y}}]_P = m^2 \delta^3(\mathbf{x} - \mathbf{y}) \quad (7.6.17)$$

and, of course,

$$C_{1\mathbf{x}, 1\mathbf{y}} = C_{2\mathbf{x}, 2\mathbf{y}} = 0. \quad (7.6.18)$$

This ‘matrix’ is obviously non-singular, so the constraints (7.6.15) are second class.

Dirac suggested that when all constraints are second class, the commutation relations will be given by

$$[A, B] = i[A, B]_D, \quad (7.6.19)$$

where  $[A, B]_D$  is a generalization of the Poisson bracket known as the *Dirac bracket*:

$$[A, B]_D \equiv [A, B]_P - [A, \chi_N]_P (C^{-1})^{NM} [\chi_M, B]_P. \quad (7.6.20)$$

(Here  $N$  and  $M$  are compound indices including the position in space, taking values like  $1, \mathbf{x}$  and  $2, \mathbf{x}$  in the vector field example.) He noted that the Dirac bracket like the Poisson bracket satisfies the same algebraic relations as the commutators

$$[A, B]_D = -[B, A]_D, \quad (7.6.21)$$

$$[A, BC]_D = [A, B]_D C + B[A, C]_D, \quad (7.6.22)$$

$$[A, [B, C]_D]_D + [B, [C, A]_D]_D + [C, [A, B]_D]_D = 0, \quad (7.6.23)$$

and also the relations

$$[\chi_N, B]_D = 0 \quad (7.6.24)$$

which make the commutation relations (7.6.19) consistent with the constraints  $\chi_N = 0$ . Also, the Dirac brackets are unchanged if we replace the  $\chi_N$  with any functions  $\chi'_N$  for which the equations  $\chi'_N = 0$  and  $\chi_N = 0$  define the same submanifold of phase space. But all these agreeable properties do not prove that the commutators are actually given by Eq. (7.6.19) in terms of the Dirac brackets.

This issue is illuminated if not settled by a powerful theorem proved by Maskawa and Nakajima [6]. They showed that for any set of canonical variables  $\Psi^a, \Pi_a$  governed by second class constraints, it is always possible by a canonical transformation<sup>72</sup> to construct two sets of variables  $Q^n, \mathcal{Q}^r$  and their respective conjugates  $P_n, \mathcal{P}_r$ , such that the constraints read  $\mathcal{Q}^r = \mathcal{P}_r = 0$ . Using these coordinates to calculate Poisson brackets, and redefining the constraint functions as  $\chi_{1r} = \mathcal{Q}^r, \chi_{2r} = \mathcal{P}_r$ , we have

$$C_{1r,2s} = [\mathcal{Q}^r, \mathcal{P}_s]_P = \delta_s^r,$$

$$C_{1r,1s} = [\mathcal{Q}^r, \mathcal{Q}^s]_P = 0, \quad C_{2r,2s} = [\mathcal{P}_r, \mathcal{P}_s]_P = 0,$$

and for any functions  $A, B$

$$[A, \chi_{1r}]_P = -\frac{\partial A}{\partial \mathcal{P}_r}, \quad [A, \chi_{2r}]_P = \frac{\partial A}{\partial \mathcal{Q}^r}.$$

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<sup>72</sup>Recall that by a canonical transformation, we mean a transformation from a set of phase space coordinates  $\Psi^a, \Pi_a$  to some other phase space coordinates  $\tilde{\Psi}^a, \tilde{\Pi}_a$ , such that  $[\tilde{\Psi}^a, \tilde{\Pi}_b]_P = \delta_b^a$  and  $[\tilde{\Psi}^a, \tilde{\Psi}^b]_P = [\tilde{\Pi}_a, \tilde{\Pi}_b]_P = 0$ , the Poisson brackets being calculated in terms of the  $\Psi^a$  and  $\Pi_a$ . It follows that the Poisson brackets for any functions  $A, B$  are the same whether calculated in terms of  $\Psi^a$  and  $\Pi_a$  or in terms of  $\tilde{\Psi}^a$  and  $\tilde{\Pi}_a$ . It also follows that if  $\Psi^a$  and  $\Pi_a$  satisfy the Hamiltonian equations of motion, then so do  $\tilde{\Psi}^a$  and  $\tilde{\Pi}_a$ , with the same Hamiltonian. The Lagrangian is changed by a canonical transformation, but only by a time-derivative, which does not affect the action.

This  $C$ -matrix has inverse  $C^{-1} = -C$ , so the Dirac brackets (7.6.20) are here

$$\begin{aligned}
[A, B]_D &= [A, B]_P + [A, \chi_{1r}]_P [\chi_{2r}, B]_P - [A, \chi_{2r}]_P [\chi_{1r}, B]_P \\
&= [A, B]_P - \frac{\partial A}{\partial Q^r} \frac{\partial B}{\partial \mathcal{P}_r} + \frac{\partial B}{\partial Q^r} \frac{\partial A}{\partial \mathcal{P}_r} \\
&= \frac{\partial A}{\partial Q^n} \frac{\partial B}{\partial P_n} - \frac{\partial B}{\partial Q^n} \frac{\partial A}{\partial P_n}.
\end{aligned} \tag{7.6.25}$$

In other words, *the Dirac bracket is equal to the Poisson bracket calculated in terms of the reduced set of unconstrained canonical variables  $Q^n$ ,  $P_n$* . If we assume that these unconstrained variables satisfy the canonical commutation relations, then the commutators of general operators  $A, B$  are given by Eq. (7.6.19) in terms of the Dirac brackets.<sup>73</sup>

We now return to the massive vector field, to see how it can be quantized using Dirac brackets. This is a case where it is easy to express the constrained variables  $V^0$  and  $\Pi_0$  in terms of the unconstrained ones<sup>74</sup>  $V_i$  and  $\Pi_i$ ; we have simply  $\Pi_0 = 0$ , and  $V^0$  is given by Eq. (7.6.5). From Eqs. (7.6.17) and (7.6.18), we see that  $C_{NM}$  here has the inverse

$$(C^{-1})^{1\mathbf{x}, 2\mathbf{y}} = -(C^{-1})^{2\mathbf{y}, 1\mathbf{x}} = -m^{-2} \delta^3(\mathbf{x} - \mathbf{y}), \tag{7.6.26}$$

$$(C^{-1})^{1\mathbf{x}, 1\mathbf{y}} = (C^{-1})^{2\mathbf{x}, 2\mathbf{y}} = 0. \tag{7.6.27}$$

Therefore the Dirac prescription (7.6.19), (7.6.20) yields the equal-time commutators

$$\begin{aligned}
[A, B] &= i[A, B]_P \\
&\quad + im^{-2} \int d^3z \left( [A, \Pi_0(\mathbf{z})]_P [\partial_i \Pi_i(\mathbf{z}) - m^2 V^0(\mathbf{z}) - J^0(\mathbf{z}), B]_P - A \leftrightarrow B \right).
\end{aligned} \tag{7.6.28}$$

By definition, we have

$$[V^\mu(\mathbf{x}), \Pi_\nu(\mathbf{y})]_P = \delta^3(\mathbf{x} - \mathbf{y}) \delta_\nu^\mu, \quad [V^\mu(\mathbf{x}), V^\nu(\mathbf{y})]_P = [\Pi_\mu(\mathbf{x}), \Pi_\nu(\mathbf{y})]_P = 0. \tag{7.6.29}$$

Hence

$$\begin{aligned}
[V^i(\mathbf{x}), V^j(\mathbf{y})] &= [V^0(\mathbf{x}), V^0(\mathbf{y})] = 0, \\
[V^i(\mathbf{x}), V^0(\mathbf{y})] &= -im^{-2} \partial_i \delta^3(\mathbf{x} - \mathbf{y}), \\
[V^i(\mathbf{x}), \Pi_j(\mathbf{y})] &= i\delta_j^i \delta^3(\mathbf{x} - \mathbf{y}), \\
[V^0(\mathbf{x}), \Pi_j(\mathbf{y})] &= [V^\mu(\mathbf{x}), \Pi_0(\mathbf{y})] = 0, \\
[\Pi^\mu(\mathbf{x}), \Pi^\nu(\mathbf{y})] &= 0.
\end{aligned} \tag{7.6.30}$$

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<sup>73</sup>It is still an open question whether we should adopt canonical commutation relations for the unconstrained variables  $Q^n$ ,  $P_n$  constructed by the Maskawa–Nakajima canonical transformation. Ultimately, the test of such canonical commutation relations is their consistency with the free-field commutation relations derived in Chapter 5, but to apply this test we need to know what the  $Q^n$  and  $P_n$  are. In the Appendix to this chapter we display two large classes of theories in which we can identify a set of unconstrained  $Q$ s and  $P$ s, such that the Dirac commutation relations (7.6.19) follow from the ordinary canonical commutation relations of the  $Q$ s and  $P$ s. We shall also show that in these cases, the Hamiltonian defined in terms of the unconstrained  $\Psi$ s and  $\Pi$ s may be written just as well in terms of the constrained variables.

<sup>74</sup>This is a special case of the theories discussed in Part A of the Appendix.

These are indeed just the commutation relations that we would find by assuming that the unconstrained variables satisfy the usual canonical commutation relations  $[V^i(\mathbf{x}), \Pi_j(\mathbf{y})] = i\delta_j^i \delta^3(\mathbf{x} - \mathbf{y})$ ,  $[V^i(\mathbf{x}), V^j(\mathbf{y})] = [\Pi_i(\mathbf{x}), \Pi_j(\mathbf{y})] = 0$ , and using the constraints to evaluate the commutators involving  $\Pi_0$  and  $V^0$ .

## 7.7 Field Redefinitions and Redundant Couplings<sup>75</sup>

Observables like masses and  $S$ -matrix elements are independent of some of the coupling parameters in any action, known as the *redundant parameters*. This is because changes in these parameters can be undone by simply redefining the field variables. A continuous redefinition of the fields, such as an infinitesimal local transformation  $\Psi^\ell(x) \rightarrow \Psi^\ell(x) + \epsilon F^\ell(\Psi(x), \partial_\mu \Psi(x), \dots)$ , clearly cannot affect any *observable* of the theory,<sup>76</sup> though, of course, it would change the values of matrix elements of the fields themselves.

How can we tell whether some variation in the parameters of a theory can be cancelled by a field redefinition? A continuous local field redefinition will produce a change in the action of the form

$$\delta I[\Psi] = \epsilon \sum_\ell \int d^4x \frac{\delta I[\Psi]}{\delta \Psi^\ell(x)} F^\ell(\Psi(x), \partial \Psi(x), \dots). \quad (7.7.1)$$

So any change  $\delta g_i$  in the coupling parameters  $g_i$ , for which the change in the action is of the form

$$\sum_i \frac{\partial I}{\partial g_i} \delta g_i = -\epsilon \sum_\ell \int d^4x \frac{\delta I[\Psi]}{\delta \Psi^\ell(x)} F^\ell(\Psi(x), \partial \Psi(x), \dots), \quad (7.7.2)$$

may be compensated by a field redefinition

$$\Psi^\ell(x) \rightarrow \Psi^\ell(x) + \epsilon F^\ell(\Psi(x), \partial_\mu \Psi(x), \dots),$$

and therefore can have no effect on any observables. In other words, *a coupling parameter is redundant if the change in the action when we vary this parameter vanishes when we use the field equations  $\delta I / \delta \Psi^\ell = 0$ .*

For example, suppose we write the Lagrangian density of a scalar field theory in the form

$$\mathcal{L} = -\frac{1}{2}Z(\partial^\mu \Phi \partial_\mu \Phi + m^2 \Phi^2) - \frac{1}{24}gZ^2 \Phi^4.$$

The constant  $Z$  is a redundant coupling, because

$$\frac{\partial I}{\partial Z} = \frac{1}{2} \int d^4x \Phi \left( \square \Phi - m^2 \Phi - \frac{1}{6}gZ \Phi^3 \right),$$

and this vanishes when we use the field equation

$$\square \Phi - m^2 \Phi = \frac{1}{6}gZ \Phi^3.$$

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<sup>75</sup>This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

<sup>76</sup>For instance, the theorem of Section 10.2 shows that as long as we multiply by the correct field renormalization constants,  $S$ -matrix elements can be obtained from the vacuum expectation value of a time-ordered product of *any* operators that have non-vanishing matrix elements between the vacuum and the one-particle states of the particles participating in the reaction.

On the other hand, neither the bare mass  $m$  nor the coupling  $g$  are redundant, and no function of  $m$  and  $g$  is redundant.

In this example, the field redefinition needed to compensate for a change of  $Z$  is a simple rescaling, in which  $F$  is proportional to  $\Phi$ . (For this reason  $Z$  is called a field renormalization constant.) This is the most general field transformation that leaves the general form of this action invariant. But for the more general actions considered in Sections 12.3 and 12.4, with arbitrary numbers of fields and derivatives, we would have to consider non-linear as well as linear field redefinitions, and an infinite subset of the parameters of the theory would be redundant.

## Appendix Dirac Brackets from Canonical Commutators

In this Appendix we shall show, in theories of two types, that the formula giving commutators as Dirac brackets times  $i$  follows from the usual canonical commutation relations for a reduced set of variables.

### A

Suppose (as in the case of a massive vector field  $V^\mu$ ) that the quantum variables  $\Psi^a$  and  $\Pi_a$  appearing in the Lagrangian  $L$  may be divided into two classes:<sup>77</sup> one set  $Q^n$  of independent canonical variables (like  $V^i(\mathbf{x})$ ) with independent canonical conjugates  $P_n = \partial L / \partial \dot{Q}^n$ ; and another set  $\mathcal{Q}^r(\mathbf{x})$  (like  $V^0$ ) whose time-derivatives do not appear in the Lagrangian. The primary constraints are the conditions  $\chi_{1r} = 0$ , where

$$\chi_{1r} = \mathcal{P}_r \tag{7.A.1}$$

are the variables conjugate to the  $\mathcal{Q}^r$ . The secondary constraints arise from the equations of motion  $0 = \partial L / \partial \mathcal{Q}^r$  for the  $\mathcal{Q}^r$ ; we suppose that these constraints can be ‘solved’—i.e., they may be written in the form  $\chi_{2r} = 0$ , with  $\chi_{2r}$  in the form

$$\chi_{2r} = \mathcal{Q}^r - f^r(Q, P). \tag{7.A.2}$$

(An example is provided by Eq. (7.6.5), which gives  $V^0$  in terms of the independent  $P$ s (here, the  $\Pi_i$ ) and  $Q$ s.) We assume that the independent  $Q$ s and  $P$ s satisfy the usual canonical commutation rules:

$$[Q^n, P_m] = i\delta_m^n, \quad [Q^n, Q^m] = [P_n, P_m] = 0. \tag{7.A.3}$$

The constraint  $\chi_{2r} = 0$  yields the commutators involving  $\mathcal{Q}^r$ :

$$[\mathcal{Q}^r, Q^n] = -i \frac{\partial f^r}{\partial P_n}, \quad [\mathcal{Q}^r, P_n] = i \frac{\partial f^r}{\partial Q^n}, \tag{7.A.4}$$

$$[\mathcal{Q}^r, \mathcal{Q}^s] = i\Gamma^{rs}, \tag{7.A.5}$$

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<sup>77</sup>We are again using a compact notation, in which labels like  $a$ ,  $n$ , and  $r$  include a space coordinate  $\mathbf{x}$  as well as discrete indices. Repeated labels are summed and integrated. All quantum variables are understood to be evaluated at the same time, with the common time argument dropped everywhere. The quantities  $\mathcal{Q}^r$  are the same as the  $C^r$  introduced in Section 7.2.



where  $\Gamma^{rs}$  is the Poisson bracket

$$\Gamma^{rs} \equiv [f^r, f^s]_{\text{P}}, \quad (7.A.6)$$

and, of course, all commutators involving  $\mathcal{P}_r$  vanish:

$$[\mathcal{P}_r, Q^n] = [\mathcal{P}_r, P_n] = [\mathcal{P}_r, \mathcal{Q}^s] = [\mathcal{P}_r, \mathcal{P}_s] = 0. \quad (7.A.7)$$

Now let us compare these commutators with the Dirac brackets. The Poisson brackets of the constraint functions are

$$C_{1r,1s} \equiv [\chi_{1r}, \chi_{1s}]_{\text{P}} = 0, \quad (7.A.8)$$

$$C_{1r,2s} = -C_{2s,1r} \equiv [\chi_{1r}, \chi_{2s}]_{\text{P}} = -\delta_r^s, \quad (7.A.9)$$

$$C_{2r,2s} \equiv [\chi_{2r}, \chi_{2s}]_{\text{P}} = [f^r(Q, P), f^s(Q, P)]_{\text{P}} = \Gamma^{rs}. \quad (7.A.10)$$

(In the example of the massive vector field  $\Gamma^{rs}$  vanishes, but the discussion here will apply also for non-vanishing  $\Gamma^{rs}$ .) It is easy to see that the  $C$ -matrix has the inverse

$$\begin{aligned} (C^{-1})^{1r,1s} &= \Gamma^{rs}, & (C^{-1})^{2r,2s} &= 0, \\ (C^{-1})^{1r,2s} &= -(C^{-1})^{2s,1r} = \delta_s^r. \end{aligned} \quad (7.A.11)$$

Also, the Poisson brackets of any function  $A$  with the constraint functions are

$$[A, \chi_{1r}]_{\text{P}} = \frac{\partial A}{\partial \mathcal{Q}^r}, \quad [A, \chi_{2r}]_{\text{P}} = -\frac{\partial A}{\partial \mathcal{P}_r} - [A, f^r(Q, P)]_{\text{P}}.$$

Hence the Dirac bracket is

$$\begin{aligned} [A, B]_{\text{D}} &= [A, B]_{\text{P}} - \frac{\partial A}{\partial \mathcal{Q}^r} \frac{\partial B}{\partial \mathcal{P}_r} + \frac{\partial B}{\partial \mathcal{Q}^r} \frac{\partial A}{\partial \mathcal{P}_r} \\ &\quad + \frac{\partial A}{\partial \mathcal{P}_r} \Gamma^{rs} \frac{\partial B}{\partial \mathcal{P}_s} - \frac{\partial A}{\partial \mathcal{P}_r} [B, f^r]_{\text{P}} + [A, f^r]_{\text{P}} \frac{\partial B}{\partial \mathcal{P}_r}. \end{aligned} \quad (7.A.12)$$

Now, if  $A$  and  $B$  are both functions only of the independent canonical variables  $Q^n$  and  $P_n$ , then  $\partial A / \partial \mathcal{Q}^r = \partial B / \partial \mathcal{Q}^r = 0$ , so the Dirac bracket is equal to the Poisson bracket. In particular,

$$[Q^n, P_m]_{\text{D}} = \delta_m^n, \quad [Q^n, Q^m]_{\text{D}} = [P_n, P_m]_{\text{D}} = 0. \quad (7.A.13)$$

Where  $A$  is  $\mathcal{Q}^r$  and  $B$  is a function of  $Q$ s and  $P$ s, it is only the fifth term on the right-hand side of Eq. (7.A.12) that contributes. In particular

$$[\mathcal{Q}^r, Q^n]_{\text{D}} = -\frac{\partial f^r}{\partial P_n}, \quad [\mathcal{Q}^r, P_n]_{\text{D}} = +\frac{\partial f^r}{\partial Q^n}. \quad (7.A.14)$$

Where both  $A$  and  $B$  are  $\mathcal{Q}$ s, we have only the fourth term

$$[\mathcal{Q}^r, \mathcal{Q}^s]_{\text{D}} = \Gamma^{rs}. \quad (7.A.15)$$

Finally, where  $A$  is  $\mathcal{P}_r$  and  $B$  is anything, we have only the first and third terms, which cancel:

$$[\mathcal{P}_r, B]_{\text{D}} = [\mathcal{P}_r, B]_{\text{P}} + \frac{\partial B}{\partial \mathcal{Q}^r} = 0. \quad (7.A.16)$$

Comparison of Eqs. (7.A.13)–(7.A.16) with Eqs. (7.A.3)–(7.A.7) shows that in all cases, the commutators are equal to the Dirac brackets times  $i$ . This is only to be expected, because as remarked in Section 7.6 all Dirac brackets involving the constraint functions vanish, so the Dirac brackets involving  $\mathcal{Q}^r$  and/or  $\mathcal{P}_s$  are given by using the constraint equations to express  $\mathcal{Q}^r$  and/or  $\mathcal{P}_s$  in terms of the independent  $Q$ s and  $P$ s.

## B

Next consider the case where the constraints take the form of conditions  $\chi_{1r}(\Psi) = 0$  on the  $\Psi^a$ , which can be solved by expressing them in terms of a smaller set of unconstrained variables  $Q^n$ , and an equal number of separate conditions  $\chi_{2r}(\Pi) = 0$  on the  $\Pi_a$ , which can be solved by expressing the  $\Pi_a$  in terms of a smaller set of unconstrained  $P_n$ . (We will see an example in the next chapter, where the constraints on the  $\Psi^a$  are gauge fixing conditions that are used to eliminate first class constraints, and the constraints on the  $\Pi_a$  are secondary constraints arising from the consistency of the first class constraints with the field equations.) We assume that the unconstrained variables satisfy the usual canonical commutation relations  $[Q^n, P_m] = i\delta_m^n$ ,  $[Q^n, Q^m] = [P_n, P_m] = 0$ . The constrained and unconstrained momenta are related by

$$P_n = \frac{\partial L}{\partial \dot{Q}^n} = \frac{\partial L}{\partial \dot{\Psi}^b} \frac{\partial \dot{\Psi}^b}{\partial \dot{Q}^n} = \Pi_b \frac{\partial \Psi^b}{\partial Q^n}. \quad (7.A.17)$$

It follows that

$$[\Psi^a, \Pi_b] \frac{\partial \Psi^b}{\partial Q^n} = [\Psi^a, P_n] = i \frac{\partial \Psi^a}{\partial Q^n}$$

or, in other words,

$$\{[\Psi^a, \Pi_b] - i\delta_b^a\} \frac{\partial \Psi^b}{\partial Q^n} = 0. \quad (7.A.18)$$

Now, the constraint  $\chi_{1r}(\Psi) = 0$  is satisfied for  $\Psi^a = \Psi^a(Q)$  for all  $Q$ , so

$$\frac{\partial \chi_{1r}}{\partial \Psi^b} \frac{\partial \Psi^b}{\partial Q^n} = 0. \quad (7.A.19)$$

Furthermore, the vectors  $(V_r)_b \equiv \partial \chi_{1r} / \partial \Psi^b$  form a complete set perpendicular to all the vectors  $(U_n)^b \equiv \partial \Psi^b / \partial Q^n$ , because if there were some other vector  $V_b$  with  $V_b(U_n)^b = 0$  for all  $n$ , then there would be additional constraints on the  $\Psi^a$ . Hence Eq. (7.A.18) implies that

$$[\Psi^a, \Pi_b] = i\delta_b^a + i c_r^a \frac{\partial \chi_{1r}}{\partial \Psi^b} \quad (7.A.20)$$

with some unknown coefficients  $c_r^a$ . To determine these coefficients, we make use of the other constraint, that  $\chi_{2r}(\Pi) = 0$ . It follows that

$$0 = [\Psi^a, \chi_{2r}(\Pi)] = [\Psi^a, \Pi_b] \frac{\partial \chi_{2r}(\Pi)}{\partial \Pi_b}.$$

Using Eq. (7.A.20), we have then

$$\frac{\partial \chi_{2r}(\Pi)}{\partial \Pi_a} = -c_s^a \frac{\partial \chi_{1s}(\Psi)}{\partial \Psi^b} \frac{\partial \chi_{2r}(\Pi)}{\partial \Pi_b}. \quad (7.A.21)$$

We recognize the factor multiplying  $c_s^a$  as the Poisson bracket

$$\frac{\partial \chi_{1s}(\Psi)}{\partial \Psi^b} \frac{\partial \chi_{2r}(\Pi)}{\partial \Pi_b} = [\chi_{1s}, \chi_{2r}]_P \equiv C_{1s,2r}.$$

Also, since  $\chi_{1s}$  depends only on the  $\Psi$  and  $\chi_{2r}$  depends only on the  $\Pi$ , these are the only non-vanishing Poisson brackets of constraints, so

$$C_{1r,1s} = C_{2r,2s} = 0.$$

Thus Eq. (7.A.21) may be written

$$\frac{\partial \chi_N}{\partial \Pi_a} = -c_s^a C_{1s,N} \quad (7.A.22)$$

with  $N$  running over all constraint functions. For second class constraints, this has the unique solution

$$c_s^a = -\frac{\partial \chi_N}{\partial \Pi_a} (C^{-1})^{N,1s} = -\frac{\partial \chi_{2r}}{\partial \Pi_a} (C^{-1})^{2r,1s}. \quad (7.A.23)$$

Using this in Eq. (7.A.20) shows that

$$[\Psi^a, \Pi_b] = i \left[ \delta_b^a - \frac{\partial \chi_{2r}}{\partial \Pi_a} (C^{-1})^{2r,1s} \frac{\partial \chi_{1s}}{\partial \Psi^b} \right]. \quad (7.A.24)$$

The Poisson brackets of  $\Psi^a$  and  $\Pi_b$  with the constraint functions are

$$\begin{aligned} [\Psi^a, \chi_{1r}]_P &= 0, & [\Psi^a, \chi_{2r}]_P &= \frac{\partial \chi_{2r}}{\partial \Pi_a}, \\ [\chi_{1r}, \Pi_b]_P &= \frac{\partial \chi_{1r}}{\partial \Psi^b}, & [\chi_{2r}, \Pi_b]_P &= 0, \end{aligned} \quad (7.A.25)$$

so the quantity in brackets on the right-hand side of Eq. (7.A.24) is the Dirac bracket

$$[\Psi^a, \Pi_b] = i[\Psi^a, \Pi_b]_D \quad (7.A.26)$$

as was to be shown. Also, we can easily see that because  $C^{-1}$  has no 11 or 22 components, the other Dirac brackets are

$$[\Psi^a, \Psi^b]_D = [\Pi_a, \Pi_b]_D = 0, \quad (7.A.27)$$

so trivially

$$[\Psi^a, \Psi^b] = i[\Psi^a, \Psi^b]_D, \quad [\Pi_a, \Pi_b] = i[\Pi_a, \Pi_b]_D. \quad (7.A.28)$$

\* \* \*

In addition to commutation rules, we also need an explicit formula for the Hamiltonian. The usual canonical formalism tells us to take

$$H = P_n \dot{Q}^n - L, \quad (7.A.29)$$

the sum running over independent canonical variables. In theories of both types considered in this Appendix, this Hamiltonian may be written in terms of the constrained variables as

$$H = \Pi_a \dot{\Psi}^a - L. \quad (7.A.30)$$

For theories of type A, this is trivial; the sum over  $a$  runs over values  $n$ , for which  $\Psi^n = Q^n$  and  $\Pi_n = P_n$  are the independent canonical variables, together with values  $r$ , for which  $\Pi_r = \mathcal{P}_r = 0$ . For theories of type B, we note that Eq. (7.A.17) gives

$$P_n \dot{Q}^n = \Pi_b \frac{\partial \Psi^b}{\partial Q^n} \dot{Q}^n = \Pi_b \dot{\Psi}^b$$

which again yields Eq. (7.A.30).

### Weinberg Exercises

**W.7.1.** Consider the theory of a set of real scalar fields  $\Phi^n$ , with Lagrangian density

$$\mathcal{L} = -\frac{1}{2} \sum_{mn} \partial_\mu \Phi^n \partial^\mu \Phi^m f_{mn}(\Phi),$$

where  $f_{mn}(\Phi)$  is an arbitrary real matrix function of the field. (This is called the non-linear  $\sigma$ -model.) Carry out the canonical quantization of this theory. Derive the interaction  $H_I[\phi(t), \dot{\phi}(t)]$  in the interaction picture.

**W.7.2.** Consider a theory of real scalar fields  $\Phi^n$  and Dirac fields  $\Psi^i$ , with Lagrangian density  $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1$ , where  $\mathcal{L}_0$  is the usual free-field Lagrangian density, and  $\mathcal{L}_1$  is an interaction term involving  $\Phi^n$  and  $\Psi^i$ , but not their derivatives. Derive an explicit expression for the symmetric energy-momentum tensor  $\Theta^{\mu\nu}$ .

**W.7.3.** In the theory described in Problem 2, suppose that the Lagrangian density is invariant under a global infinitesimal symmetry  $\delta\Phi^n = i\epsilon \sum_m t_m^n \Phi^m$ ,  $\delta\Psi^i = i\epsilon \sum_j \tau_j^i \Psi^j$ . Derive an explicit expression for the conserved current associated with this symmetry.

**W.7.4.** Consider the theory of a complex scalar field  $\Phi$  and a real vector field  $V^\mu$ , with Lagrangian density

$$\mathcal{L} = -(D_\mu \Phi)^\dagger D^\mu \Phi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} m^2 V_\mu V^\mu - \mathcal{H}(\Phi^\dagger \Phi),$$

where  $D_\mu \equiv \partial_\mu - igV_\mu$  and  $F_{\mu\nu} \equiv \partial_\mu V_\nu - \partial_\nu V_\mu$ , and  $\mathcal{H}$  is an arbitrary function. Carry out the canonical quantization of this theory. Derive the interaction in the interaction picture.

**W.7.5.** In the theory of Problem 4, derive expressions for the symmetric energy-momentum tensor  $\Theta^{\mu\nu}$  and for the conserved current associated with the symmetry under  $\delta\Phi = i\epsilon\Phi$ ,  $\delta V^\mu = 0$ .

**W.7.6.** Prove that the Dirac bracket satisfies the Jacobi identity (7.6.23).

**W.7.7.** Prove that the Dirac bracket is independent of the choice of constraint functions  $\chi_N$  used to describe a given submanifold of phase space.

## Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

**W.7.1.** Only the symmetric part of  $f_{mn}$  contributes, so we take  $f_{mn} = f_{nm}$ , with inverse  $f^{mn}$ . The conjugate fields are

$$\Pi_n = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}^n} = f_{nm}(\Phi) \dot{\Phi}^m, \quad \dot{\Phi}^n = f^{nm}(\Phi) \Pi_m.$$

Canonical quantization imposes, at equal times,

$$[\Phi^m(\mathbf{x}), \Pi_n(\mathbf{y})] = i \delta^m_n \delta^3(\mathbf{x} - \mathbf{y}), \quad [\Phi^m, \Phi^n] = [\Pi_m, \Pi_n] = 0.$$

The Legendre transform gives

$$H = \frac{1}{2} \int d^3x [\Pi_m f^{mn}(\Phi) \Pi_n + f_{mn}(\Phi) \nabla \Phi^m \cdot \nabla \Phi^n].$$

At the quantum level the first term is to be given a fixed Hermitian ordering; for example, one may use the displayed sandwich ordering after replacing  $f$  by a Hermitian multiplication operator.

Choose coordinates near  $\Phi = 0$  for which  $f_{mn}(0) = \delta_{mn}$ , and put  $h_{mn}(\phi) = f_{mn}(\phi) - \delta_{mn}$ . The free theory is a set of massless scalars with  $\pi_n = \dot{\phi}^n$ . At  $t = 0$ , and hence at every interaction-picture time, the interaction is

$$H_I(t) = \frac{1}{2} \int d^3x \left[ \dot{\phi}^m (f_{mn}^{-1}(\phi) - \delta_{mn}) \dot{\phi}^n + h_{mn}(\phi) \nabla \phi^m \cdot \nabla \phi^n \right].$$

Here  $f_{mn}^{-1}$  denotes the matrix inverse with its indices placed using  $\delta_{mn}$ . The occurrence of  $f^{-1}$ , rather than  $-h$ , is the characteristic result of doing the Legendre transform before replacing interacting fields by free ones.

**W.7.2.** It is convenient to replace the free Dirac density by the equivalent Hermitian density

$$\mathcal{L}_D = -\frac{1}{2} \sum_i \bar{\Psi}^i \gamma^\rho \overleftrightarrow{\partial}_\rho \Psi^i - \sum_i m_i \bar{\Psi}^i \Psi^i, \quad A \overleftrightarrow{\partial}_\rho B = A \partial_\rho B - (\partial_\rho A) B.$$

This differs from the usual first-order form only by a four-divergence. Applying the Belinfante construction, and noting that the nonderivative interaction contributes only through  $\eta^{\mu\nu} \mathcal{L}$ , gives

$$\Theta^{\mu\nu} = \sum_n \partial^\mu \Phi^n \partial^\nu \Phi^n - \frac{1}{4} \sum_i \bar{\Psi}^i \left( \gamma^\mu \overleftrightarrow{\partial}^\nu + \gamma^\nu \overleftrightarrow{\partial}^\mu \right) \Psi^i + \eta^{\mu\nu} \mathcal{L}. \quad (1)$$

The first line is already symmetric. The improvement supplied by the Dirac spin current converts the canonical fermion tensor into the second line, which is manifestly

symmetric. Direct differentiation of (1), followed by use of the scalar and adjoint Dirac equations, gives  $\partial_\mu \Theta^{\mu\nu} = 0$ . The terms obtained by differentiating  $\mathcal{L}_1(\Phi, \Psi, \bar{\Psi})$  cancel against the interaction terms in those equations. Surface terms do not change  $\int d^3x \Theta^{0\nu} = P^\nu$ .

**W.7.3.** For the real scalars,

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^n)} = -\partial^\mu \Phi^n.$$

For the Dirac fields it is simplest to use the unsymmetrized free Lagrangian; the answer obtained from the Hermitian form differs only by an identically conserved improvement. Equation (7.3.46) then gives

$$J^\mu = i \sum_{n,m} \partial^\mu \Phi^n t^n_m \Phi^m + i \sum_{i,j} \bar{\Psi}^i \gamma^\mu \tau^i_j \Psi^j.$$

Reality of the scalar transformation requires  $it$  to be a real antisymmetric matrix, while invariance of the Dirac inner product requires the corresponding flavor generator to be Hermitian. These properties also make  $J^\mu$  Hermitian. Differentiating the displayed current and inserting the field equations leaves precisely the variation of  $\mathcal{L}_1$  under the assumed symmetry; that variation vanishes, so  $\partial_\mu J^\mu = 0$ . Its charge agrees with Eq. (7.3.40) and therefore generates the stated transformations by equal-time commutators.

**W.7.4.** Write  $V_0 = -V^0$ , and use Euclidean notation for spatial indices. The nonzero canonical momenta are

$$\pi = (D_0 \Phi)^\dagger, \quad \pi^\dagger = D_0 \Phi, \quad \Pi_i = F_{0i} = \dot{V}_i - \partial_i V_0, \quad \Pi_0 = 0.$$

Thus  $\Pi_0 = 0$  is a primary constraint. The velocities are

$$\dot{\Phi} = \pi^\dagger + ig V_0 \Phi, \quad \dot{\Phi}^\dagger = \pi - ig V_0 \Phi^\dagger, \quad \dot{V}_i = \Pi_i + \partial_i V_0.$$

Define the canonical charge density

$$\rho \equiv ig(\pi \Phi - \pi^\dagger \Phi^\dagger).$$

After integrating  $\mathbf{\Pi} \cdot \nabla V_0$  by parts, the terms involving  $V_0$  are

$$V_0(\rho - \nabla \cdot \mathbf{\Pi}) - \frac{1}{2} m^2 V_0^2.$$

Its equation is the secondary constraint

$$V_0 = \frac{\rho - \nabla \cdot \mathbf{\Pi}}{m^2}.$$

Eliminating  $V_0$  gives the reduced Hamiltonian

$$H = \int d^3x \left[ \pi \pi^\dagger + (D_i \Phi)^\dagger D_i \Phi + \mathcal{H}(\Phi^\dagger \Phi) + \frac{1}{2} \mathbf{\Pi}^2 + \frac{1}{4} F_{ij} F_{ij} + \frac{1}{2} m^2 \mathbf{V}^2 + \frac{1}{2m^2} (\nabla \cdot \mathbf{\Pi} - \rho)^2 \right]. \quad (2)$$

The remaining independent fields have the usual equal-time canonical commutators. Equivalently, one may retain  $V^0, \Pi_0$  and use the Dirac brackets of the two second-class constraints.

To identify the interaction picture, write  $\mathcal{H}(z) = M^2 z + \mathcal{H}_I(z)$ . Let  $\phi, v^\mu, \pi_v$  be the corresponding free complex-scalar and Proca fields, with

$$\rho_I = ig(\dot{\phi}^\dagger \phi - \dot{\phi} \phi^\dagger), \quad \nabla \cdot \pi_v = -m^2 v_0.$$

Subtracting the free Hamiltonian from (2) yields

$$\boxed{H_I(t) = \int d^3x \left\{ \mathcal{H}_I(\phi^\dagger \phi) + ig v_\mu [\phi^\dagger \partial^\mu \phi - (\partial^\mu \phi^\dagger) \phi] + g^2 \mathbf{v}^2 \phi^\dagger \phi + \frac{\rho_I^2}{2m^2} \right\}. \quad (3)}$$

The last term is the local contact interaction forced by eliminating the auxiliary time component. It would be missed by setting  $H_I = - \int d^3x \mathcal{L}_I$ .

**W.7.5.** Varying the action with respect to the metric, or equivalently applying the Belinfante construction, gives

$$\boxed{\Theta^{\mu\nu} = (D^\mu \Phi)^\dagger D^\nu \Phi + (D^\nu \Phi)^\dagger D^\mu \Phi + F^{\mu\rho} F^\nu{}_\rho + m^2 V^\mu V^\nu + \eta^{\mu\nu} \mathcal{L}. \quad (4)}$$

It is symmetric by inspection. The vector equation,

$$\partial_\rho F^{\rho\mu} - m^2 V^\mu = ig \left[ \Phi^\dagger D^\mu \Phi - (D^\mu \Phi)^\dagger \Phi \right],$$

together with the scalar equation and its adjoint, verifies  $\partial_\mu \Theta^{\mu\nu} = 0$ . In particular,  $\Theta^{00}$  contains the positive electric, magnetic, scalar-gradient, and Proca mass terms when the constraints are imposed.

For the rigid phase symmetry, Noether's formula gives

$$\boxed{J^\mu = i \left[ (D^\mu \Phi)^\dagger \Phi - \Phi^\dagger D^\mu \Phi \right]. \quad (5)}$$

The vector is inert, so it supplies no additional term. The scalar equation

$$D_\mu D^\mu \Phi - \mathcal{H}'(\Phi^\dagger \Phi) \Phi = 0$$

and its adjoint make the divergence of (5) vanish. The source on the right side of the vector equation is  $-gJ^\mu$ , so its divergence also implies current conservation because the divergence of  $\partial_\rho F^{\rho\mu}$  vanishes identically and the Proca equation implies the corresponding subsidiary condition.

**W.7.6.** Write the Poisson bracket as  $\{, \}$ , put  $C_{MN} = \{\chi_M, \chi_N\}$ , and denote its inverse by  $C^{MN}$ . The two identities needed in expanding a nested Dirac bracket are

$$\{A, C^{MN}\} = -C^{MP}\{A, C_{PQ}\}C^{QN}, \quad (6)$$

and, by the Poisson Jacobi identity,

$$\{A, C_{PQ}\} = \{\{A, \chi_P\}, \chi_Q\} + \{\chi_P, \{A, \chi_Q\}\}. \quad (7)$$

Now substitute

$$\{A, B\}_D = \{A, B\} - \{A, \chi_M\}C^{MN}\{\chi_N, B\}$$

twice into  $\sum_{\text{cyc}}\{A, \{B, C\}_D\}_D$ . The terms with no  $C^{MN}$  vanish by the ordinary Jacobi identity. Use (6) on every term in which the outer bracket differentiates an inverse matrix, and then use (7). Terms in which the outer bracket hits  $A, B$ , or  $C$  cancel cyclically; the remaining terms pair with opposite signs after interchanging the dummy constraint indices and using  $C^{MN} = -C^{NM}$ . Hence

$$\sum_{\text{cyc}}\{A, \{B, C\}_D\}_D = 0.$$

This proof uses only the invertibility of the second-class constraint matrix. As a useful local check, in Maskawa–Nakajima coordinates the Dirac bracket is the ordinary Poisson bracket of the unconstrained pairs, Eq. (7.6.25), for which Jacobi is immediate.

**W.7.7.** Let  $\chi'_A$  be another regular set of functions defining the same constraint submanifold. The implicit-function theorem implies locally

$$\chi'_A = M_A{}^B(z)\chi_B + O(\chi^2),$$

where  $M$  is nonsingular on the submanifold. Therefore, with weak equality meaning equality after the constraints are imposed,

$$\{F, \chi'_A\} \simeq M_A{}^B\{F, \chi_B\}, \quad C'_{AB} \simeq M_A{}^C C_{CD} M_B{}^D.$$

The inverse matrix consequently obeys

$$(C'^{-1})^{AB} \simeq (M^{-1})_C{}^A C^{CD} (M^{-1})_D{}^B.$$

Substitution into the correction term gives

$$\begin{aligned} \{F, \chi'_A\}(C'^{-1})^{AB}\{\chi'_B, G\} \\ \simeq \{F, \chi_C\}C^{CD}\{\chi_D, G\}. \end{aligned}$$

Thus

$$\{F, G\}'_D \simeq \{F, G\}_D.$$

The qualifier is essential: a Dirac bracket is the Poisson structure induced on the constraint submanifold, and two extensions away from that submanifold may differ by terms proportional to the constraints. All brackets of physical observables, and hence all commutators obtained after imposing the constraints, are independent of the defining functions.



## Supplementary Exercises

### S.7.1. Symmetries, Currents, Charges, and Lorentz Boosts

(Adapted from *University of Cambridge, Part III Quantum Field Theory, Paper 301, 2023, Question 1.*)

Consider a relativistic theory of one real scalar field  $\phi(x)$ , with Lagrangian density  $\mathcal{L}$  and action

$$S = \int d^4x \mathcal{L}.$$

- (a) Define a continuous symmetry of  $S$ , a conserved current, and its conserved charge. Explain carefully how these three notions are related, including the assumptions under which the existence of any one implies either of the other two.
- (b) Define the energy–momentum tensor  $T^{\rho\sigma}$ . For the mostly-plus form of the source theory,

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi),$$

derive an explicit formula valid for arbitrary  $V$ . Explain why  $T^{\rho\sigma}$  supplies four conserved currents and find their charges.

- (c) Starting from invariance of the action under infinitesimal Lorentz transformations, show that

$$\partial_\mu (x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}) = 0.$$

- (d) Hence prove that the three-vector

$$V^i \equiv \frac{d}{dt} \int d^3x x^i T^{00}$$

is constant in time.

### S.7.2. The Maxwell Stress Tensor and Metric Response

(Adapted from *J. McGreevy, Physics 215A Quantum Field Theory, Fall 2023, Assignment 3, Problem 1; lightly edited there from Peskin and Schroeder, Problem 2.1.*)

Classical electromagnetism follows from

$$S[A] = \int d^4x \left( -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - j^\mu A_\mu \right), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

- (a) Treat the four components  $A_\mu(x)$  as the dynamical variables and derive the Euler–Lagrange equations

$$\frac{\delta S[A]}{\delta A_\mu(x)} = 0.$$

Write them in standard vector form using

$$E^i = -F^{0i}, \quad \varepsilon^{ijk} B^k = -F^{ij}.$$

In the source, the two resulting equations are called Gauss's and Faraday's laws; more precisely, variation of  $A_\mu$  gives Gauss's law and the Ampère–Maxwell law, while Faraday's law is the Bianchi identity.

- (b) Set  $j^\mu = 0$ . Construct the canonical energy–momentum tensor and show explicitly that it is neither symmetric nor gauge invariant. The source starts from its canonical convention

$$(T_{\text{src}})^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\rho)} \partial_\nu A_\rho - \mathcal{L} \delta^\mu{}_\nu.$$

Improve it according to

$$\tilde{T}_{\text{src}}^{\mu\nu} = T_{\text{src}}^{\mu\nu} + \partial_\lambda K_{\text{src}}^{\lambda\mu\nu}, \quad K_{\text{src}}^{\lambda\mu\nu} = F^{\mu\lambda} A^\nu.$$

This is the source's canonical-tensor convention. With the sign convention for  $T^\mu{}_\nu$  used in this volume, the translated improvement is

$$\tilde{T}^{\mu\nu} = T^{\mu\nu} - \partial_\lambda (F^{\mu\lambda} A^\nu).$$

Explain why the added term is identically conserved and does not change the global charges. Prove that  $\tilde{T}^{\mu\nu}$  is symmetric and gauge invariant, and recover

$$\mathcal{E} = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2), \quad \mathbf{S} = \mathbf{E} \times \mathbf{B}$$

as the electromagnetic energy and momentum densities.

- (c) As a bonus, couple the theory to a background metric:

$$S[A, g] = \int d^4x \sqrt{|g|} \left( -\frac{1}{4} F_{\mu\nu} F_{\rho\sigma} g^{\mu\rho} g^{\nu\sigma} + j^\mu A_\mu \right).$$

The displayed source uses the opposite sign for its source term here; the comparison below is made at  $j^\mu = 0$ , so this mismatch is irrelevant. The source's mostly-minus metric-response convention has a plus sign; in the volume's mostly-plus convention, compare the result of

$$T_{\mu\nu}^{(g)} = -\frac{2}{\sqrt{|g|}} \frac{\delta S[A, g]}{\delta g^{\mu\nu}} \Big|_{g_{\mu\nu} = \eta_{\mu\nu}}$$

with part (b), using

$$\frac{\delta g^{\mu\nu}(x)}{\delta g_{\rho\sigma}(y)} = -g^{\mu\rho} g^{\nu\sigma} \delta^D(x - y), \quad \frac{\delta \det g(x)}{\delta g_{\mu\nu}(y)} = \delta^D(x - y) \det g g^{\mu\nu}.$$

The source points to its accompanying notes for proofs of these two identities; they may be taken as supplied here. Explain why the metric-response tensor is automatically symmetric and gauge invariant.

### S.7.3. A Complex Scalar Field

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 1, Problem 3.)

Consider a complex scalar field  $\phi(x)$  with

$$S = \int d^4x \left[ -\partial_\mu \phi^\dagger \partial^\mu \phi - V(\phi^\dagger \phi) \right].$$

Treat  $\phi$  and  $\phi^\dagger$ , rather than their real and imaginary parts, as independent dynamical variables.

- (a) Check that the action is Lorentz invariant and derive the two Euler–Lagrange equations.
- (b) Find the momenta canonically conjugate to  $\phi$  and  $\phi^\dagger$ , and construct the Hamiltonian  $H$ .
- (c) The action is invariant under the rigid transformation

$$\phi \mapsto e^{i\alpha} \phi, \quad \phi^\dagger \mapsto e^{-i\alpha} \phi^\dagger,$$

where  $\alpha$  is constant. Use Noether’s theorem to find the corresponding conserved current  $j^\mu$  and charge  $Q$ .

- (d) Use the equations of motion from part (a) to verify directly that  $\partial_\mu j^\mu = 0$ .

### S.7.4. The Energy–Momentum Tensor of a Complex Scalar

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 1, Problem 4.)

Continue with the complex-scalar theory

$$S = \int d^4x \left[ -\partial_\mu \phi^\dagger \partial^\mu \phi - V(\phi^\dagger \phi) \right].$$

- (a) Under a spacetime translation

$$x^\mu \mapsto x'^\mu = x^\mu + a^\mu, \quad \phi'(x') = \phi(x),$$

show that the action is invariant under  $\phi(x) \mapsto \phi'(x)$ , and likewise for  $\phi^\dagger$ .

- (b) Write the infinitesimal transformations of  $\phi$  and  $\phi^\dagger$ , and use Noether’s theorem to derive the four conserved translation currents  $T^{\mu\nu}$ .
- (c) For a time translation, the conserved charge is

$$H = \int d^3x T^{00},$$

while a spatial translation gives

$$P^i = \int d^3x T^{0i}.$$

Write  $H$  and  $P^i$  explicitly and compare  $H$  with the Hamiltonian found in S.7.3(b).

(d) Use the field equations from S.7.3(a) to verify directly that  $\partial_\mu T^{\mu\nu} = 0$ .

### S.7.5. Non-Abelian Currents of Two Complex Scalars

(Adapted from J. McGreevy, *Physics 215A Quantum Field Theory, Fall 2023, Assignment 6, Problem 2.*)

Consider two complex scalar fields  $\Phi_\alpha$ ,  $\alpha = 1, 2$ , with the mostly-plus form of the source action

$$S[\Phi_\alpha] = \int d^d x dt \left[ -\frac{1}{2} \partial_\mu \Phi_\alpha^\dagger \partial^\mu \Phi_\alpha - V(\Phi_\alpha^\dagger \Phi_\alpha) \right].$$

Let  $\Pi_\alpha$  and  $\Pi_\alpha^\dagger$  be the canonical momenta, and consider the three objects written in the source as

$$Q^i \equiv \frac{1}{2} \int d^d x i \left( \Pi_\alpha^\dagger \sigma_{\alpha\beta}^i \Phi_\beta^\dagger \right) + \text{h.c.}, \quad i = 1, 2, 3,$$

where  $\sigma^i$  are the Pauli matrices.

- Determine the transformations generated by the three charges and prove that they are symmetries of  $S$ .
- Optionally verify directly that  $[Q^i, H] = 0$ .
- Evaluate  $[Q^i, Q^j]$ , show that the charges are non-Abelian, and identify the resulting familiar Lie algebra.
- Derive the Noether currents  $J_\mu^i$  for the transformations in part (a), and prove that

$$Q^i = \int d^d x J^{0i}$$

agrees with the supplied canonical expression. (The mostly-minus source writes the time component with a lower index.)

- Generalize the construction to  $N$  complex scalar fields.

### S.7.6. A Massive Vector Field: Positivity and the Longitudinal Mode

(Adapted from University of Cambridge, *Mathematical Tripos Part III, Quantum Field Theory, Paper 301, 2025, Question 2, printed p. 3.*)

Consider a real vector field  $V^\mu(x)$  in four-dimensional Minkowski space. In the mostly-plus convention of this volume, the signature-translated form of the source Lagrangian is

$$\mathcal{L} = a(\partial_\mu V_\nu)(\partial^\mu V^\nu) + b(\partial_\mu V_\nu)(\partial^\nu V^\mu) + c(\partial_\mu V^\mu)(\partial_\nu V^\nu) - m^2 V^\mu V_\mu,$$

where  $a, b, c$ , and  $m$  are real constants. (The source uses the mostly-minus metric, for which the last term is written  $+m^2 V^\mu V_\mu$ .)

- (i) Show that  $\mathcal{L}$  can be simplified, up to total derivatives, to

$$\mathcal{L} = \tilde{a} V_\nu \square V^\nu + \tilde{b} V^\nu \partial_\nu \partial_\mu V^\mu - m^2 V^\mu V_\mu, \quad (*)$$

with  $\tilde{a}, \tilde{b}$  real constants.

- (ii) Using (\*), derive the equations of motion for  $V^\mu$ .
- (iii) Using Noether's theorem, construct the energy-momentum tensor associated with this Lagrangian, and give an expression for the Hamiltonian.
- (iv) Consider the case  $\tilde{b} = 0$ . Show that, for any value of  $\tilde{a}$ , the energy of the system is not positive definite.
- (v) Assuming  $\tilde{a} > 0$ , find a relation between  $\tilde{b}$  and  $\tilde{a}$  such that the energy is positive definite on shell, up to total derivatives. Show that with this choice the vector-field equations imply  $\partial_\mu V^\mu = 0$ .
- (b) Another way to constrain  $\tilde{a}$  and  $\tilde{b}$  to their physical values is to decompose a general vector field as

$$V^\mu = A^\mu + \partial^\mu \pi, \quad \partial_\mu A^\mu = 0,$$

where  $\pi(x)$  is a scalar field.

- (i) Write (\*) in terms of  $A^\mu$  and  $\pi$ , and derive the equations of motion for both fields.
- (ii) Show that the propagator of  $\pi$ , up to a numerical factor, is

$$\langle 0|T\pi(x)\pi(y)|0\rangle = \int \frac{d^4k}{(2\pi)^4} \frac{i}{m^2} \left( \frac{1}{k^2} - \frac{\tilde{c}}{\tilde{c}k^2 + m^2} \right) e^{-ik \cdot (x-y)},$$

where  $\tilde{c}$  is a constant that you should specify. *Hint:* You may use without proof that the propagator is a Green's function. The plus sign in the second denominator is the mostly-plus translation of the source expression  $\tilde{c}k_{\text{src}}^2 - m^2$ .

Show that the term involving  $\tilde{c}$  vanishes when  $\tilde{a}$  and  $\tilde{b}$  obey the relation found in part (a)(v).

### S.7.7. Recovering Nonrelativistic Quantum Mechanics

(Adapted from J. McGreevy, *Physics 215A Quantum Field Theory, Fall 2023, Assignment 6, Problem 3*, printed pp. 1–2.)

Take the nonrelativistic limit of a complex scalar field by writing

$$\Phi(x) = \sqrt{2m} e^{-imt} \Psi(x), \quad |\dot{\Psi}| \ll m|\Psi|.$$

Antiparticles then decouple. At a fixed time, use the mode expansions

$$\Psi(\mathbf{x}) = \int_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{x}} a_{\mathbf{p}}, \quad \Psi^\dagger(\mathbf{x}) = \int_{\mathbf{p}} e^{-i\mathbf{p} \cdot \mathbf{x}} a_{\mathbf{p}}^\dagger, \quad \int_{\mathbf{p}} \equiv \int \frac{d^d \mathbf{p}}{(2\pi)^d},$$

with

$$[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = (2\pi)^d \delta^d(\mathbf{p} - \mathbf{q}), \quad H = \int_{\mathbf{p}} \frac{\mathbf{p}^2}{2m} a_{\mathbf{p}}^\dagger a_{\mathbf{p}}.$$

(a) Show that

$$P^i = \int_{\mathbf{p}} p^i a_{\mathbf{p}}^\dagger a_{\mathbf{p}}$$

generates spatial translations and commutes with the Hamiltonian.

(b) Define the second-quantized position operator and a localized one-particle state by

$$X^i = \int d^d \mathbf{x} \Psi^\dagger(\mathbf{x}) x^i \Psi(\mathbf{x}), \quad |\mathbf{x}\rangle = \Psi^\dagger(\mathbf{x}) |\Omega\rangle.$$

Prove that

$$X^i |\mathbf{x}\rangle = x^i |\mathbf{x}\rangle.$$

(c) For a general one-particle wave packet

$$|\psi\rangle = \int d^d \mathbf{x} \psi(\mathbf{x}) \Psi^\dagger(\mathbf{x}) |\Omega\rangle = \int d^d \mathbf{x} \psi(\mathbf{x}) |\mathbf{x}\rangle,$$

show that

$$X^i |\psi\rangle = \int d^d \mathbf{x} x^i \psi(\mathbf{x}) |\mathbf{x}\rangle,$$

and, with the surface term at spatial infinity discarded,

$$P^i |\psi\rangle = \int d^d \mathbf{x} \left( -i \frac{\partial}{\partial x^i} \psi(\mathbf{x}) \right) |\mathbf{x}\rangle.$$

Thus recover the usual position- and momentum-operator actions on a one-particle Schrödinger wavefunction.

### S.7.8. Field Redefinition and Vanishing Scalar Scattering

(Adapted from University of Cambridge, Mathematical Tripos Part III, Quantum Field Theory, Paper 301, 2025, Question 4, printed p. 6.)

Suppose that a massive scalar field in four dimensions has the mostly-plus Lagrangian

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi - \frac{m^2}{2} \Phi^2.$$

This is the signature-translated form of the source's mostly-minus free Lagrangian. Consider the field redefinition

$$\Phi(x) = \phi(x) + \lambda \phi(x)^2,$$

where  $\lambda$  is real. Scattering amplitudes for particles created by  $\Phi$  or  $\phi$  should be unchanged by this field redefinition. Check explicitly that physical observables do not change as follows.

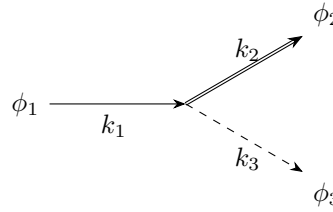
- (a) Work out the Lagrangian in terms of  $\phi$ . Identify its free and interaction terms.
- (b) For the Lagrangian in terms of  $\phi$ , write without derivation the momentum-space Feynman rules for the  $S$ -matrix.

- (c) For  $\phi\phi \rightarrow \phi\phi$ , draw the connected Feynman diagrams that contribute to the  $S$ -matrix at leading order in  $\lambda$ .
- (d) Using parts (b) and (c), evaluate the connected contribution to  $\phi\phi \rightarrow \phi\phi$  at leading order in  $\lambda$ . Comment on the answer.

*Hint for part (b).* If three scalar fields interact through

$$\mathcal{L}_{\text{int}} = \lambda \phi_1 (\partial_\mu \phi_2) (\partial^\mu \phi_3),$$

then, with  $k_1$  incoming and  $k_2, k_3$  outgoing, the derivative vertex and its rule are



$$= (i\lambda)(ik_2)_\mu (ik_3)^\mu (2\pi)^4 \delta^4(k_1 - k_2 - k_3).$$

Consider carefully how this rule applies when the three scalars are identical.

### S.7.9. Field Redefinitions, External Poles, and Order Bookkeeping

*(Inspired by Iain Stewart and MIT OpenCourseWare, 8.851 Lecture 3, Field Redefinitions.)*

- (a) Construct the inverse of  $\phi' = \phi + \epsilon\phi^2 + \epsilon^2 a\phi^3$  through order  $\epsilon^2$ , and identify the condition that makes this change of field variables locally admissible.
- (b) Suppose  $\phi' = \phi + \epsilon\phi^2$  has nonzero vacuum-to-one-particle overlap. Show from the one-particle pole of its two-point function why amputated scattering observables are unchanged after residue adjustment.
- (c) If a first-order field redefinition removes one dimension-six operator, show why its quadratic action shift produces dimension-eight interactions and why discarding them is consistent only at fixed order.

### S.7.10. Geometry, Hamiltonian, and Current of the $O(N)$ Sigma Model

*(Inspired by David Tong, Lectures on Statistical Field Theory, Section 4.3 / David Tong, Lectures on Statistical Field Theory, Sections 4.1 and 4.3.)*

- (a) Solve  $n^A n^A = 1$  by  $n^A = (\pi^a, \sqrt{1 - \pi^2})$  and derive the exact field-space metric multiplying  $\partial_\mu \pi^a \partial^\mu \pi^b$ .
- (b) Invert the sphere metric obtained in S.7.10(a), calculate the canonical momenta, and write the exact Hamiltonian density in terms of  $\pi$  and its conjugates.
- (c) For the constrained field  $n^A n^A = 1$ , derive the current associated with rotations in the  $AB$ -plane and verify its conservation using the constrained equation of motion.

### S.7.11. Quantization and Charge of a Complex Scalar Field

(Adapted from Daniel Harlow, Lecture Notes for Physics 8.323: Relativistic Quantum Field Theory I (2024), Section 3.8, Homework Problem 4, printed p. 44.)

In  $D$  spacetime dimensions, consider the free complex scalar theory

$$\mathcal{L} = -\partial_\mu \Phi^\dagger \partial^\mu \Phi - m^2 \Phi^\dagger \Phi, \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}.$$

The field, its canonical momentum, and the equal-time commutators are

$$\Phi(x) = \int \frac{d^{D-1}\mathbf{p}}{(2\pi)^{D-1}\sqrt{2E_{\mathbf{p}}}} \left( e^{ip \cdot x} a_{\mathbf{p}} + e^{-ip \cdot x} b_{\mathbf{p}}^\dagger \right), \quad \Pi(x) = \dot{\Phi}^\dagger(x),$$

$$[\Phi(t, \mathbf{x}), \dot{\Phi}^\dagger(t, \mathbf{y})] = [\Phi^\dagger(t, \mathbf{x}), \dot{\Phi}(t, \mathbf{y})] = i\delta^{D-1}(\mathbf{x} - \mathbf{y}),$$

with every other equal-time field commutator equal to zero. The global phase symmetry is  $\Phi'(x) = e^{i\theta} \Phi(x)$ . Use the volume's current convention

$$j^\mu = i \left[ \Phi^\dagger \partial^\mu \Phi - (\partial^\mu \Phi^\dagger) \Phi \right], \quad Q = \int d^{D-1}\mathbf{x} j^0.$$

- Starting from the displayed field expansion and canonical commutators, calculate all commutators among  $a_{\mathbf{p}}, b_{\mathbf{p}}, a_{\mathbf{p}}^\dagger$ , and  $b_{\mathbf{p}}^\dagger$ .
- Derive the Hamiltonian in terms of these creation and annihilation operators, retaining the vacuum constant before normal ordering.
- Evaluate the conserved charge  $Q$  in the same oscillator basis. Identify the charges of the one-particle states  $a_{\mathbf{p}}^\dagger|\Omega\rangle$  and  $b_{\mathbf{p}}^\dagger|\Omega\rangle$  in the stated current convention, and explain why the two species are particle and antiparticle.

### Solutions to Supplementary Exercises

#### S.7.1. Symmetries, Currents, Charges, and Lorentz Boosts

- A continuous symmetry is a differentiable family of field transformations

$$\phi(x) \mapsto \phi(x) + \epsilon \Delta\phi(x) + O(\epsilon^2)$$

under which the action is unchanged, possibly up to a boundary term. A conserved current is a local vector  $j^\mu$  satisfying  $\partial_\mu j^\mu = 0$  on the equations of motion. Its charge is

$$Q(t) = \int d^3x j^0(t, \mathbf{x}).$$

Current conservation implies

$$\dot{Q} = - \int d^3x \nabla \cdot \mathbf{j} = - \oint_{S_\infty} d\mathbf{S} \cdot \mathbf{j},$$

so  $Q$  is conserved when the flux at spatial infinity vanishes.



Noether's first theorem takes a differentiable action symmetry to an on-shell conserved current; it assumes the variational equations and keeps track of any boundary term in the action variation. A conserved local current gives a conserved charge only with suitable boundary conditions. Conversely, a charge gives a symmetry when it is a well-defined canonical generator, for example  $\delta\phi = i\epsilon[Q, \phi]$ , and conservation is equivalent to  $[Q, H] = 0$  when the charge has no explicit time dependence. An inverse Noether statement from a current to a local variational symmetry also requires a regular local action and is understood modulo identically conserved improvement currents. These qualifications are why none of the three implications is purely formal.

- (b) For translations, the scalar energy-momentum tensor in the volume's convention is

$$T^{\rho\sigma} = (\partial^\rho\phi)(\partial^\sigma\phi) + \eta^{\rho\sigma}\mathcal{L} = (\partial^\rho\phi)(\partial^\sigma\phi) - \eta^{\rho\sigma}\left[\frac{1}{2}\partial_\mu\phi\partial^\mu\phi + V(\phi)\right].$$

The field equation is  $\square\phi = V'(\phi)$ . Direct differentiation then gives

$$\partial_\rho T^{\rho\sigma} = (\square\phi - V')\partial^\sigma\phi = 0.$$

For each fixed  $\sigma$ ,  $T^{\rho\sigma}$  is a conserved current. The corresponding four charges are

$$P^\sigma = \int d^3x T^{0\sigma},$$

namely the energy and the three components of momentum, with the index-position signs fixed as in Section 7.3.

- (c) For a scalar,

$$\delta\phi = -\frac{1}{2}\omega_{\rho\sigma}(x^\rho\partial^\sigma - x^\sigma\partial^\rho)\phi.$$

Noether's theorem therefore gives the six Lorentz currents

$$M^{\mu\rho\sigma} = x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}.$$

Equivalently, differentiating and using symmetry of the scalar tensor,

$$\partial_\mu M^{\mu\rho\sigma} = T^{\rho\sigma} - T^{\sigma\rho} + x^\rho\partial_\mu T^{\mu\sigma} - x^\sigma\partial_\mu T^{\mu\rho} = 0.$$

- (d) Let

$$X^i(t) = \int d^3x x^i T^{00}.$$

Stress-tensor conservation and vanishing surface terms give

$$\dot{X}^i = - \int d^3x x^i \partial_j T^{j0} = \int d^3x T^{i0} = \int d^3x T^{0i} = P^i.$$

Thus the vector defined in the problem is  $V^i = \dot{X}^i = P^i$ . Since the translation charge is conserved,  $\dot{V}^i = \dot{P}^i = 0$ .

### S.7.2. The Maxwell Stress Tensor and Metric Response

(a) The variation of the field strength is

$$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu.$$

Using antisymmetry and integrating by parts,

$$\delta S = \int d^4x (\partial_\mu F^{\mu\nu} - j^\nu) \delta A_\nu,$$

so

$$\partial_\mu F^{\mu\nu} = j^\nu.$$

The  $\nu = 0$  equation is

$$\nabla \cdot \mathbf{E} = j^0,$$

and the spatial equations are

$$\nabla \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{j}.$$

The remaining two Maxwell equations follow instead from  $\partial_{[\lambda} F_{\mu\nu]} = 0$ :

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0.$$

This makes explicit the terminology correction recorded in the prompt.

(b) In the energy–momentum convention of this volume, the canonical tensor is

$$T_{\text{can}}^{\mu\nu} = F^{\mu\lambda} \partial^\nu A_\lambda - \frac{1}{4} \eta^{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}.$$

The explicit  $A_\lambda$  makes it gauge dependent, and  $T_{\text{can}}^{\mu\nu} \neq T_{\text{can}}^{\nu\mu}$ . In the source’s opposite canonical-tensor convention the stated improvement is  $+\partial_\lambda(F^{\mu\lambda}A^\nu)$ ; after translating to the volume’s convention it is

$$\tilde{T}^{\mu\nu} = T_{\text{can}}^{\mu\nu} - \partial_\lambda(F^{\mu\lambda}A^\nu).$$

For  $j^\mu = 0$ ,

$$\partial_\lambda(F^{\mu\lambda}A^\nu) = F^{\mu\lambda} \partial_\lambda A^\nu,$$

and hence

$$\tilde{T}^{\mu\nu} = F^{\mu\lambda} F^\nu{}_\lambda - \frac{1}{4} \eta^{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}.$$

This expression is symmetric and contains only  $F$ , so it is gauge invariant. Antisymmetry of the first two indices of the improvement superpotential makes its double divergence vanish identically; its spatial integral is a surface term, so the total four-momentum is unchanged.

Using  $F_{\rho\sigma} F^{\rho\sigma} = 2(\mathbf{B}^2 - \mathbf{E}^2)$ , one obtains

$$\tilde{T}^{00} = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2), \quad \tilde{T}^{0i} = (\mathbf{E} \times \mathbf{B})^i.$$

(c) At  $j^\mu = 0$ , vary

$$S[A, g] = -\frac{1}{4} \int d^4x \sqrt{|g|} F_{\mu\nu} F_{\rho\sigma} g^{\mu\rho} g^{\nu\sigma}.$$

The determinant variation in inverse-metric variables is

$$\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|} g_{\mu\nu} \delta g^{\mu\nu}.$$

The two inverse metrics in the Maxwell term give equal contributions. After symmetrizing the free metric indices,

$$\delta S = -\frac{1}{2} \int d^4x \sqrt{|g|} \left[ F_{\mu\rho} F_{\nu}{}^\rho - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma} \right] \delta g^{\mu\nu}.$$

Therefore the mostly-plus response definition in the prompt yields

$$T_{\mu\nu}^{(g)} = F_{\mu\rho} F_{\nu}{}^\rho - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}.$$

Raising both indices and taking  $g_{\mu\nu} = \eta_{\mu\nu}$  reproduces the improved tensor in part (b). Symmetry follows because the functional derivative is with respect to the symmetric metric, and gauge invariance follows because the metric-coupled action depends on  $A_\mu$  only through  $F_{\mu\nu}$  when  $j^\mu = 0$ .

### S.7.3. A Complex Scalar Field

(a) Under  $x \mapsto x' = \Lambda x$ , a scalar obeys  $\phi'(x') = \phi(x)$ , and similarly for  $\phi^\dagger$ . The measure  $d^4x$ , the contraction  $\partial_\mu \phi^\dagger \partial^\mu \phi$ , and  $\phi^\dagger \phi$  are Lorentz invariant, so  $S' = S$ . Independent variation of  $\phi$  and  $\phi^\dagger$  gives

$$\square\phi - V'(\phi^\dagger\phi)\phi = 0, \quad \square\phi^\dagger - V'(\phi^\dagger\phi)\phi^\dagger = 0.$$

(b) The momenta and Hamiltonian density are

$$\pi = \dot{\phi}^\dagger, \quad \pi^\dagger = \dot{\phi}, \quad \mathcal{H} = \pi\pi^\dagger + \nabla\phi^\dagger \cdot \nabla\phi + V(\phi^\dagger\phi).$$

Hamilton's equations reproduce the two equations in part 1.

(c) Under  $\delta\phi = i\epsilon\phi$  and  $\delta\phi^\dagger = -i\epsilon\phi^\dagger$ , the chapter's Noether normalization gives

$$j^\mu = i \left[ \phi^\dagger \partial^\mu \phi - (\partial^\mu \phi^\dagger) \phi \right].$$

The charge is therefore

$$Q = \int d^3x j^0 = i \int d^3x \left( \dot{\phi}^\dagger \phi - \phi^\dagger \dot{\phi} \right).$$

(d) Direct differentiation cancels the terms bilinear in first derivatives:

$$\partial_\mu j^\mu = i \left[ \phi^\dagger \square\phi - (\square\phi^\dagger) \phi \right].$$

Using the two equations from part (a), both terms become  $iV'(\phi^\dagger\phi)\phi^\dagger\phi$  with opposite signs. Thus  $\partial_\mu j^\mu = 0$ , as required.

#### S.7.4. The Energy–Momentum Tensor of a Complex Scalar

- (a) The scalar transformation law gives  $\phi'(x) = \phi(x - a)$  and  $\phi'^{\dagger}(x) = \phi^{\dagger}(x - a)$ . Hence

$$\begin{aligned} S' &= \int d^4x \left[ -\partial_{\mu}\phi^{\dagger}(x - a)\partial^{\mu}\phi(x - a) - V\left(\phi^{\dagger}(x - a)\phi(x - a)\right) \right] \\ &= S \end{aligned}$$

after the translation  $y = x - a$ , whose Jacobian is one.

- (b) Infinitesimally,

$$\delta\phi = -a^{\rho}\partial_{\rho}\phi, \quad \delta\phi^{\dagger} = -a^{\rho}\partial_{\rho}\phi^{\dagger}, \quad \delta\mathcal{L} = -a^{\rho}\partial_{\rho}\mathcal{L}.$$

Noether's theorem, with the coefficient of each independent  $a^{\rho}$  extracted, gives the symmetric tensor

$$T^{\mu\nu} = (\partial^{\mu}\phi^{\dagger})\partial^{\nu}\phi + (\partial^{\nu}\phi^{\dagger})\partial^{\mu}\phi + \eta^{\mu\nu}\mathcal{L}.$$

For every fixed  $\nu$ ,  $T^{\mu\nu}$  is the current associated with translation in that direction.

- (c) Because  $\partial^0 = -\partial_0$  and  $\eta^{00} = -1$ ,

$$T^{00} = \dot{\phi}^{\dagger}\dot{\phi} + \nabla\phi^{\dagger} \cdot \nabla\phi + V(\phi^{\dagger}\phi).$$

Thus

$$H = \int d^3x \left[ \dot{\phi}^{\dagger}\dot{\phi} + \nabla\phi^{\dagger} \cdot \nabla\phi + V(\phi^{\dagger}\phi) \right],$$

which agrees with the Legendre transform in S.7.3(b), since  $\pi = \dot{\phi}^{\dagger}$  and  $\pi^{\dagger} = \dot{\phi}$ . The spatial charges are

$$P^i = \int d^3x T^{0i} = - \int d^3x \left( \dot{\phi}^{\dagger}\partial_i\phi + \partial_i\phi^{\dagger}\dot{\phi} \right).$$

- (d) Differentiate the tensor found in part (b):

$$\begin{aligned} \partial_{\mu}T^{\mu\nu} &= (\Box\phi^{\dagger})\partial^{\nu}\phi + (\partial^{\nu}\phi^{\dagger})\Box\phi \\ &\quad - \partial^{\nu}\phi^{\dagger}V'(\phi^{\dagger}\phi)\phi - \phi^{\dagger}V'(\phi^{\dagger}\phi)\partial^{\nu}\phi \end{aligned}$$

after the second-derivative kinetic terms have canceled against  $\partial^{\nu}[-\partial_{\rho}\phi^{\dagger}\partial^{\rho}\phi]$ . Using  $\Box\phi = V'\phi$  and  $\Box\phi^{\dagger} = V'\phi^{\dagger}$ , with the signs inherited from the mostly-plus Lagrangian, cancels the remaining potential derivative terms. Hence  $\partial_{\mu}T^{\mu\nu} = 0$  for all four values of  $\nu$ .

#### S.7.5. Non-Abelian Currents of Two Complex Scalars

Write the fields as a column  $\Phi$ , set

$$t_i = -\frac{1}{2}\sigma_i^{\text{T}},$$

and impose

$$[\Phi_\alpha(\mathbf{x}), \Pi_\beta(\mathbf{y})] = i\delta_{\alpha\beta}\delta^d(\mathbf{x} - \mathbf{y}), \quad [\Phi_\alpha^\dagger(\mathbf{x}), \Pi_\beta^\dagger(\mathbf{y})] = i\delta_{\alpha\beta}\delta^d(\mathbf{x} - \mathbf{y}).$$

For the local mostly-plus action,

$$\Pi_\alpha = \frac{1}{2}\dot{\Phi}_\alpha^\dagger, \quad \Pi_\alpha^\dagger = \frac{1}{2}\dot{\Phi}_\alpha.$$

Expanding the source's “h.c.” term, reordering equal-time bosonic fields, and relabelling dummy indices gives exactly the canonical form

$$Q^i = i \int d^d x \left[ \Pi_\alpha(t_i)_{\alpha\beta} \Phi_\beta - \Phi_\alpha^\dagger(t_i)_{\alpha\beta} \Pi_\beta^\dagger \right].$$

The transpose simply realizes the conjugate doublet representation; for  $SU(2)$  it is equivalent to the usual Pauli-matrix basis.

(a) The canonical algebra gives

$$[Q^i, \Phi] = t_i \Phi, \quad [Q^i, \Phi^\dagger] = -\Phi^\dagger t_i.$$

Thus

$$\delta_i \Phi = i\epsilon_i t_i \Phi, \quad \delta_i \Phi^\dagger = -i\epsilon_i \Phi^\dagger t_i,$$

or, finitely,  $\Phi \mapsto U\Phi$  with  $U = \exp(i\epsilon_i t_i) \in SU(2)$ . Both  $\partial_\mu \Phi^\dagger \partial^\mu \Phi$  and  $\Phi^\dagger \Phi$  are invariant because  $U^\dagger U = \mathbf{1}$ . The complete action is therefore invariant.

(b) The Hamiltonian is

$$H = \int d^d x \left[ 2\Pi_\alpha \Pi_\alpha^\dagger + \frac{1}{2} \nabla \Phi_\alpha^\dagger \cdot \nabla \Phi_\alpha + V(\Phi^\dagger \Phi) \right].$$

Every flavor index is contracted with the identity. It follows either term by term from the canonical algebra or from finite  $SU(2)$  invariance that

$$[Q^i, H] = 0.$$

(c) Because

$$[t_i, t_j] = i\varepsilon_{ijk} t_k,$$

the commutator of two generated transformations on either canonical field is the transformation generated by  $i\varepsilon_{ijk} Q^k$ . Direct evaluation of the bilinear charges has no central remainder and gives

$$[Q^i, Q^j] = i\varepsilon_{ijk} Q^k.$$

This is the Lie algebra  $\mathfrak{su}(2)$ , the same angular-momentum algebra obeyed by the Pauli matrices.

- (d) Promote  $\epsilon_i$  temporarily to  $\epsilon_i(x)$ . The coefficient of  $\partial_\mu \epsilon_i$  in the action variation gives

$$J_i^\mu = \frac{i}{2} \left[ \Phi^\dagger t_i \partial^\mu \Phi - (\partial^\mu \Phi^\dagger) t_i \Phi \right].$$

The field equations make  $\partial_\mu J_i^\mu = 0$ ; the two potential terms cancel because  $V$  depends only on  $\Phi^\dagger \Phi$ . Using  $\partial^0 = -\partial_t$  and the canonical momenta,

$$\int d^d x J_i^0 = i \int d^d x \left[ \Pi_\alpha(t_i)_{\alpha\beta} \Phi_\beta - \Phi_\alpha^\dagger(t_i)_{\alpha\beta} \Pi_\beta^\dagger \right] = Q^i,$$

as required.

- (e) For  $N$  complex scalars, let  $t_A$  be Hermitian  $N \times N$  matrices satisfying

$$[t_A, t_B] = i f_{AB}{}^C t_C.$$

Then

$$\delta_A \Phi = i \epsilon_A t_A \Phi, \quad J_A^\mu = \frac{i}{2} \left[ \Phi^\dagger t_A \partial^\mu \Phi - (\partial^\mu \Phi^\dagger) t_A \Phi \right],$$

and  $[Q_A, Q_B] = i f_{AB}{}^C Q_C$ . All Hermitian generators give the full  $U(N)$  symmetry of the stated action; restricting to traceless generators gives its non-Abelian  $SU(N)$  subgroup.

### S.7.6. A Massive Vector Field: Positivity and the Longitudinal Mode

Put  $D \equiv \partial_\mu V^\mu$ .

- (a) (i) Integrating each quadratic derivative term once by parts gives

$$\begin{aligned} (\partial_\mu V_\nu)(\partial^\mu V^\nu) &\simeq -V_\nu \square V^\nu, \\ (\partial_\mu V_\nu)(\partial^\nu V^\mu) &\simeq -V^\nu \partial_\nu D, \\ D^2 &\simeq -V^\nu \partial_\nu D, \end{aligned}$$

where  $\simeq$  denotes equality up to a four-divergence. Therefore

$$\boxed{\tilde{a} = -a, \quad \tilde{b} = -(b+c)}.$$

- (ii) Varying the symmetric quadratic form  $(*)$ , integrating by parts, and dividing by the common factor of two gives

$$\boxed{\tilde{a} \square V_\nu + \tilde{b} \partial_\nu D - m^2 V_\nu = 0}.$$

Its divergence is

$$(\tilde{a} + \tilde{b}) \square D - m^2 D = 0.$$

- (iii) For the Noether calculation it is convenient to use the equivalent first-derivative representative

$$\mathcal{L} \simeq -\tilde{a}(\partial_\mu V_\rho)(\partial^\mu V^\rho) - \tilde{b} D^2 - m^2 V_\rho V^\rho.$$

The derivative of this density is

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu V^\rho)} = -2\tilde{a} \partial^\mu V_\rho - 2\tilde{b} D \delta^\mu_\rho.$$

Thus the canonical translation current in the convention of this volume is

$$T^\mu{}_\nu = \delta^\mu{}_\nu \mathcal{L} + 2\tilde{a} \partial^\mu V_\rho \partial_\nu V^\rho + 2\tilde{b} D \partial_\nu V^\mu, \quad \partial_\mu T^\mu{}_\nu = 0.$$

Writing  $V^\mu = (V^0, \mathbf{V})$ , the Hamiltonian  $H = - \int d^3x T^0_0$  is

$$H = \int d^3x \left\{ \tilde{a} [\dot{V}^i \dot{V}^i + (\partial_i V_j)(\partial_i V_j) - (\partial_i V^0)(\partial_i V^0)] - (\tilde{a} + \tilde{b})(\dot{V}^0)^2 + \tilde{b}(\partial_i V^i)^2 + m^2[V^i V^i - (V^0)^2] \right\}.$$

Surface terms can of course move derivatives among the displayed spatial terms without changing the charge.

- (iv) When  $\tilde{b} = 0$ , the time derivatives of  $V^0$  and  $V^i$  occur with opposite signs. If  $\tilde{a} > 0$ , a configuration with only rapidly varying  $V^0$  has arbitrarily negative energy; if  $\tilde{a} < 0$ , the same is true for a spatial component. At  $\tilde{a} = 0$ , the mass term  $m^2[\mathbf{V}^2 - (V^0)^2]$  is itself indefinite. Hence no value of  $\tilde{a}$  makes the energy positive definite.

- (v) The required relation is

$$\boxed{\tilde{b} = -\tilde{a}}.$$

Indeed, up to a total derivative,

$$\mathcal{L} = -\frac{\tilde{a}}{2} F_{\mu\nu} F^{\mu\nu} - m^2 V_\mu V^\mu, \quad F_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu.$$

Its momenta are

$$\Pi_i = 2\tilde{a} F_{0i}, \quad \Pi_0 = 0.$$

Before eliminating  $V^0$ , integration by parts in the Hamiltonian gives the  $V^0$ -dependent combination  $V^0 \partial_i \Pi_i - m^2 (V^0)^2$ . Its stationary value is

$$V^0 = \frac{\partial_i \Pi_i}{2m^2},$$

and substitution yields

$$H = \int d^3x \left[ \frac{\Pi_i \Pi_i}{4\tilde{a}} + \frac{\tilde{a}}{2} F_{ij} F_{ij} + m^2 V_i V_i + \frac{(\partial_i \Pi_i)^2}{4m^2} \right].$$

This is positive for  $\tilde{a} > 0$  and  $m^2 > 0$ . The divergence of the field equation simultaneously becomes  $-m^2 D = 0$ , and hence

$$\partial_\mu V^\mu = 0$$

for a massive field.

- (b) (i) Substitute  $V^\mu = A^\mu + \partial^\mu \pi$  into (\*). All mixed terms vanish after integration by parts because  $\partial_\mu A^\mu = 0$ . With

$$\tilde{c} \equiv \tilde{a} + \tilde{b},$$

the result may be written

$$\begin{aligned}\mathcal{L}_A &= \tilde{a} A_\nu \square A^\nu - m^2 A_\nu A^\nu, \\ \mathcal{L}_\pi &= \tilde{c} (\partial_\nu \pi) \square (\partial^\nu \pi) - m^2 \partial_\nu \pi \partial^\nu \pi \\ &\simeq -\tilde{c} (\square \pi)^2 + m^2 \pi \square \pi.\end{aligned}$$

The independent equations are therefore

$$(\tilde{a} \square - m^2) A_\mu = 0, \quad \square (\tilde{c} \square - m^2) \pi = 0.$$

- (ii) On  $e^{-ik \cdot x}$ , the mostly-plus wave operator acts as  $\square = -k^2$ . The scalar kinetic operator is consequently proportional to

$$k^2 (\tilde{c} k^2 + m^2).$$

Inverting it as a Green function and using partial fractions gives

$$\frac{i}{k^2 (\tilde{c} k^2 + m^2)} = \frac{i}{m^2} \left( \frac{1}{k^2} - \frac{\tilde{c}}{\tilde{c} k^2 + m^2} \right),$$

up to the common normalization and the usual pole prescriptions. This is the displayed propagator and specifies  $\tilde{c} = \tilde{a} + \tilde{b}$ . For the positive-energy choice  $\tilde{b} = -\tilde{a}$ , one has  $\tilde{c} = 0$ , so the second pole term vanishes exactly.

### S.7.7. Recovering Nonrelativistic Quantum Mechanics

- (a) The number-operator algebra gives

$$[P^i, a_{\mathbf{q}}] = -q^i a_{\mathbf{q}}, \quad [P^i, a_{\mathbf{q}}^\dagger] = q^i a_{\mathbf{q}}^\dagger.$$

Consequently

$$[P^i, \Psi(\mathbf{x})] = i \partial_i \Psi(\mathbf{x}), \quad [P^i, \Psi^\dagger(\mathbf{x})] = i \partial_i \Psi^\dagger(\mathbf{x}).$$

Thus  $e^{-i\epsilon \cdot \mathbf{P}} \Psi(\mathbf{x}) e^{i\epsilon \cdot \mathbf{P}} = \Psi(\mathbf{x} + \epsilon)$  to first order, so  $P^i$  generates translations. Since both  $H$  and  $P^i$  are diagonal integrals of mutually commuting number operators,

$$\boxed{[P^i, H] = 0.}$$

- (b) The field commutator is

$$[\Psi(\mathbf{y}), \Psi^\dagger(\mathbf{x})] = \delta^d(\mathbf{y} - \mathbf{x}).$$

Normal ordering makes  $X^i |\Omega\rangle = 0$ , and therefore

$$[X^i, \Psi^\dagger(\mathbf{x})] = \int d^d \mathbf{y} y^i \Psi^\dagger(\mathbf{y}) [\Psi(\mathbf{y}), \Psi^\dagger(\mathbf{x})] = x^i \Psi^\dagger(\mathbf{x}).$$

It follows immediately that

$$\boxed{X^i |\mathbf{x}\rangle = x^i |\mathbf{x}\rangle.}$$



(c) Linearity and the preceding commutator give

$$X^i|\psi\rangle = \int d^d\mathbf{x} x^i \psi(\mathbf{x})|\mathbf{x}\rangle.$$

For momentum,

$$\begin{aligned} P^i|\psi\rangle &= \int d^d\mathbf{x} \psi(\mathbf{x})[P^i, \Psi^\dagger(\mathbf{x})]|\Omega\rangle \\ &= i \int d^d\mathbf{x} \psi(\mathbf{x}) \partial_i |\mathbf{x}\rangle \\ &= -i \int d^d\mathbf{x} (\partial_i \psi(\mathbf{x})) |\mathbf{x}\rangle. \end{aligned}$$

The final equality is integration by parts; the assumed falloff removes the boundary term. Hence the one-particle sector carries exactly the Schrödinger representation  $X^i = x^i$  and  $P^i = -i\partial_i$ .

### S.7.8. Field Redefinition and Vanishing Scalar Scattering

(a) Since

$$\partial_\mu \Phi = (1 + 2\lambda\phi)\partial_\mu \phi,$$

direct substitution gives

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 \\ &\quad - 2\lambda\phi(\partial\phi)^2 - m^2\lambda\phi^3 \\ &\quad - 2\lambda^2\phi^2(\partial\phi)^2 - \frac{1}{2}m^2\lambda^2\phi^4. \end{aligned}$$

The first line is  $\mathcal{L}_0$ , the second is the cubic interaction  $\mathcal{L}_3$ , and the third is the quartic interaction  $\mathcal{L}_4$ . No higher powers occur because the field redefinition is quadratic.

(b) Take every momentum at a vertex to be incoming and suppress the common factor  $(2\pi)^4\delta^4(\sum p_i)$ . The rules are

$$\text{propagator:} \quad \frac{-i}{p^2 + m^2 - i0},$$

$$\text{cubic vertex:} \quad -2i\lambda \sum_{r=1}^3 (p_r^2 + m^2),$$

and

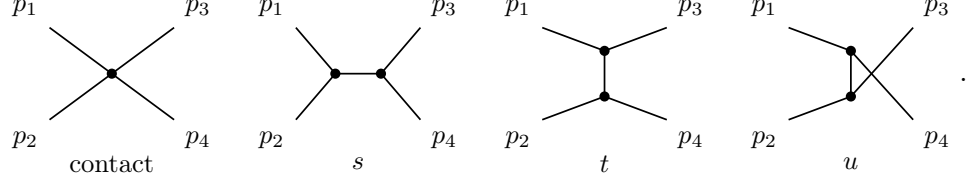
$$\text{quartic vertex:} \quad -4i\lambda^2 \left( \sum_{r=1}^4 p_r^2 + 3m^2 \right).$$

For example, the two differentiated slots in  $-2\lambda\phi(\partial\phi)^2$  can be assigned to the three identical legs in all ways. Their sum, together with the  $-m^2\lambda\phi^3$  term, is

$$4i\lambda \sum_{r<s} p_r \cdot p_s - 6i\lambda m^2 = -2i\lambda \sum_r (p_r^2 + m^2),$$

where momentum conservation was used. The same identical-field counting produces the displayed quartic rule.

- (c) Every leading connected graph is of order  $\lambda^2$ : there is one quartic contact graph and three graphs made from two cubic vertices, with the internal line in the  $s$ -,  $t$ -, or  $u$ -channel:



- (d) Let  $p_1, p_2, p_3, p_4$  denote all-incoming external momenta. Then

$$\sum_{i=1}^4 p_i = 0, \quad p_i^2 = -m^2.$$

For  $q_s = p_1 + p_2$ ,  $q_t = p_1 + p_3$ , and  $q_u = p_1 + p_4$ , these relations imply

$$q_s^2 + q_t^2 + q_u^2 = -4m^2.$$

At either end of an exchange line, the two external inverse propagators vanish on shell, so the cubic rule reduces to  $-2i\lambda(q_x^2 + m^2)$ . The contribution of one channel is therefore

$$[-2i\lambda(q_x^2 + m^2)]^2 \frac{-i}{q_x^2 + m^2 - i0} = 4i\lambda^2(q_x^2 + m^2).$$

Summing the three exchange graphs gives

$$\mathcal{A}_{\text{exch}} = 4i\lambda^2(q_s^2 + q_t^2 + q_u^2 + 3m^2) = -4i\lambda^2 m^2.$$

The on-shell contact rule is

$$\mathcal{A}_{\text{contact}} = -4i\lambda^2(-4m^2 + 3m^2) = 4i\lambda^2 m^2.$$

Consequently

$$\boxed{\mathcal{A}_{\text{connected}} = \mathcal{A}_{\text{contact}} + \mathcal{A}_{\text{exch}} = 0}.$$

The original  $\Phi$  theory is free, so this cancellation is exactly what invariance of the on-shell  $S$ -matrix under a local invertible field redefinition requires, even though the redefined Lagrangian contains nonzero off-shell vertices.

### S.7.9. Field Redefinitions, External Poles, and Order Bookkeeping

- (a) Seek

$$\phi = \phi' + \epsilon A \phi'^2 + \epsilon^2 B \phi'^3 + O(\epsilon^3).$$

Substitution into  $\phi' = \phi + \epsilon\phi^2 + \epsilon^2 a\phi^3$  gives

$$0 = \epsilon(A+1)\phi'^2 + \epsilon^2(B+2A+a)\phi'^3.$$

Thus  $A = -1$ ,  $B = 2 - a$ , and

$$\phi = \phi' - \epsilon\phi'^2 + \epsilon^2(2-a)\phi'^3 + O(\epsilon^3).$$

Local admissibility requires a nonzero functional Jacobian. Pointwise,  $\partial\phi'/\partial\phi = 1 + 2\epsilon\phi + 3a\epsilon^2\phi^2 \neq 0$  in the field region under consideration. Perturbatively this holds near the identity.

- (b) Near the physical one-particle pole, the Fourier transform of the new two-point function has the spectral form

$$\tilde{G}_{\phi'}(p) = \frac{Z'}{p^2 + m_{\text{phys}}^2 - i0} + \text{terms regular at the pole}.$$

The composite term  $\epsilon\phi^2$  changes the overlap  $\langle\Omega|\phi'(0)|p\rangle$ , and hence changes  $Z'$ , but it cannot move the pole because the mass is a property of the Hamiltonian spectrum. Amputation multiplies by the inverse pole and divides each external leg by  $\sqrt{Z'}$ . The overlap change then cancels, leaving the same on-shell scattering observable provided  $Z' \neq 0$ . Off-shell correlators need not agree.

- (c) For  $\phi \mapsto \phi + \epsilon F$ , Taylor expansion of the action gives

$$S[\phi + \epsilon F] = S[\phi] + \epsilon \int F \frac{\delta S}{\delta \phi} + \frac{\epsilon^2}{2} \int d^4x d^4y F(x) \frac{\delta^2 S}{\delta \phi(x) \delta \phi(y)} F(y) + \dots$$

The linear term can cancel a chosen dimension-six equation-of-motion operator. The quadratic term contains two powers of the suppressed coefficient and therefore contributes at the dimension-eight order in the same power counting. Omitting it is consistent in a calculation truncated at first order in  $\epsilon$ , but not in a second-order calculation, where it is required for equivalence.

### S.7.10. Geometry, Hamiltonian, and Current of the $O(N)$ Sigma Model

- (a) Write

$$n^A = (\pi^a, \sigma), \quad \sigma = \sqrt{1 - \pi^2}.$$

Then

$$\partial_\mu \sigma = -\frac{\pi_a \partial_\mu \pi^a}{\sqrt{1 - \pi^2}},$$

and hence

$$\partial_\mu n^A \partial^\mu n^A = g_{ab}(\pi) \partial_\mu \pi^a \partial^\mu \pi^b, \quad g_{ab} = \delta_{ab} + \frac{\pi_a \pi_b}{1 - \pi^2}.$$

The apparent interaction is therefore the metric induced on the unit sphere. The coordinate patch fails at  $\pi^2 = 1$ , where the eliminated component  $\sigma$  vanishes, even though the sphere itself is regular.

(b) The rank-one inverse is

$$g^{ab} = \delta^{ab} - \pi^a \pi^b, \quad g_{ac} g^{cb} = \delta_a^b.$$

For

$$\mathcal{L} = -\frac{1}{2e^2} g_{ab} \partial_\mu \pi^a \partial^\mu \pi^b,$$

the canonical momentum is

$$p_a = \frac{1}{e^2} g_{ab} \dot{\pi}^b.$$

Solving for the velocity and performing the Legendre transform gives

$$\mathcal{H} = \frac{e^2}{2} p_a g^{ab} p_b + \frac{1}{2e^2} g_{ab} \nabla \pi^a \cdot \nabla \pi^b.$$

The inverse target metric necessarily appears in the momentum term, just as in the general nonlinear sigma-model Hamiltonian.

(c) For a rotation in the  $AB$ -plane,  $\delta n^A = \epsilon n^B$  and  $\delta n^B = -\epsilon n^A$ . Noether's theorem gives, up to the orientation convention,

$$j_{AB}^\mu = \frac{1}{e^2} (n^A \partial^\mu n^B - n^B \partial^\mu n^A).$$

The constrained equation has the form

$$\square n^A = \lambda(x) n^A,$$

because the force normal to the sphere is supplied by a Lagrange multiplier. Therefore

$$\partial_\mu j_{AB}^\mu = \frac{1}{e^2} (n^A \square n^B - n^B \square n^A) = 0.$$

The unknown multiplier cancels, so conservation follows without solving for it.

### S.7.11. Quantization and Charge of a Complex Scalar Field

(a) At equal time, insert the mode expansion into  $[\Phi(\mathbf{x}), \dot{\Phi}^\dagger(\mathbf{y})]$ . The terms containing  $a$  and  $a^\dagger$  and those containing  $b^\dagger$  and  $b$  must reproduce the Fourier representation of the delta function. One finds two independent oscillator algebras,

$$[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = [b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = (2\pi)^{D-1} \delta^{D-1}(\mathbf{p} - \mathbf{q}),$$

with

$$[a_{\mathbf{p}}, a_{\mathbf{q}}] = [b_{\mathbf{p}}, b_{\mathbf{q}}] = [a_{\mathbf{p}}, b_{\mathbf{q}}] = [a_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = 0,$$

and the Hermitian conjugates of these relations. In particular the particle and antiparticle oscillators are unrelated, as required for a complex field.

(b) The Hamiltonian density is

$$\mathcal{H} = \dot{\Phi}^\dagger \dot{\Phi} + \nabla \Phi^\dagger \cdot \nabla \Phi + m^2 \Phi^\dagger \Phi.$$

The spatial integral kills the mixed terms; on the mass shell  $E_{\mathbf{p}}^2 = \mathbf{p}^2 + m^2$ . Hence

$$H = \int \frac{d^{D-1} \mathbf{p}}{(2\pi)^{D-1}} E_{\mathbf{p}} \left( a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + b_{\mathbf{p}} b_{\mathbf{p}}^\dagger \right).$$

Reordering the second term gives

$$H = \int \frac{d^{D-1} \mathbf{p}}{(2\pi)^{D-1}} E_{\mathbf{p}} \left( a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + b_{\mathbf{p}}^\dagger b_{\mathbf{p}} \right) + E_{\text{vac}},$$

where  $E_{\text{vac}} = \delta^{D-1}(0) \int d^{D-1} \mathbf{p} E_{\mathbf{p}}$  in the displayed continuum normalization. Normal ordering removes this state-independent constant.

(c) Since  $\partial^0 = -\partial_0$ , the supplied current gives

$$Q = i \int d^{D-1} \mathbf{x} \left( \dot{\Phi}^\dagger \Phi - \Phi^\dagger \dot{\Phi} \right).$$

Substitution of the modes yields, after normal ordering,

$$Q = \int \frac{d^{D-1} \mathbf{p}}{(2\pi)^{D-1}} \left( b_{\mathbf{p}}^\dagger b_{\mathbf{p}} - a_{\mathbf{p}}^\dagger a_{\mathbf{p}} \right).$$

Therefore

$$Q, a_{\mathbf{p}}^\dagger |\Omega\rangle = -a_{\mathbf{p}}^\dagger |\Omega\rangle, \quad Q, b_{\mathbf{p}}^\dagger |\Omega\rangle = +b_{\mathbf{p}}^\dagger |\Omega\rangle.$$

Thus the two equal-mass species have opposite conserved charges. Which one is called the particle is a convention tied to the sign chosen for the current; either way, one species is the antiparticle of the other.

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## 8 Electrodynamics

The original approach to quantum electrodynamics was to take for granted Maxwell's classical theory of electromagnetism, and quantize it. It will probably not surprise the reader that this book will follow a different path. We shall first infer the need for a principle of gauge invariance from the peculiar difficulties that arise in formulating a quantum theory of massless particles with spin, and then deduce the main features of electrodynamics from the gauge invariance principle. After that we shall follow a more conventional modern approach, in which one takes gauge invariance as the starting point and uses it to deduce the existence of a vector potential describing massless particles of unit spin.

It is too soon to tell which of these two alternatives corresponds to the logical order of nature. Most theorists have tended to take gauge invariance as a starting point, but in modern string theories [1] the argument runs the other way; one first notices a state of mass zero and unit spin among the normal modes of a string, and then from that deduces the gauge invariance of the effective field theory that describes such particles. At any rate, as we shall see, using either approach one is led to the quantized version of Maxwell's theory, still the paradigmatic example of a successful quantum field theory.

### 8.1 Gauge Invariance

Let's start by recalling the problems encountered in constructing covariant free fields for a massless particle of helicity  $\pm 1$ . We saw in Section 5.9 that there is no difficulty in constructing an antisymmetric tensor free field  $f_{\mu\nu}(x)$  for such particles. This field can be expressed in terms of the four-potential  $a_\mu(x)$ , given by Eq. (5.9.23), through the familiar relation

$$f_{\mu\nu}(x) = \partial_\mu a_\nu(x) - \partial_\nu a_\mu(x). \quad (8.1.1)$$

However, Eq. (5.9.23) shows that the  $a_\mu(x)$  transforms as a four-vector only up to a gauge transformation

$$U_0(\Lambda)a_\mu(x)U_0^{-1}(\Lambda) = \Lambda_\mu{}^\nu a_\nu(\Lambda x) + \partial_\mu \Omega(x, \Lambda). \quad (8.1.2)$$

There is, in fact, no way to construct a true four-vector as a linear combination of the creation and annihilation operators for helicity  $\pm 1$ . This is one way of understanding the presence of singularities at  $m = 0$  in the propagator of a massive vector field

$$\Delta_{\mu\nu}(x, y) = (2\pi)^{-4} \int d^4q e^{iq \cdot (x-y)} \frac{\eta_{\mu\nu} + q_\mu q_\nu / m^2}{q^2 + m^2 - i\epsilon},$$

which prevent us from dealing with massless particles of helicity  $\pm 1$  by simply passing to the limit  $m \rightarrow 0$  of the theory of a massive particle of spin one.

We could avoid these problems by demanding that all interactions involve only  $F_{\mu\nu}(x) \equiv \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)$  and its derivatives, not  $A_\mu(x)$ , but this is not the most general possibility, and not the one realized in nature. Instead of banishing  $A_\mu(x)$  from the

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<sup>78</sup>We now use  $A_\mu$  and  $F_{\mu\nu}$  for the electromagnetic potential vector and the field strength tensor because these are interacting fields.

action, we shall require instead that the part of the action  $I_M$  for matter and its interaction with radiation be invariant under the general gauge transformation

$$A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu \epsilon(x) \quad (8.1.3)$$

(at least when the matter fields satisfy the field equations) so that the extra term in Eq. (8.1.2) should have no effect. The change in the matter action under the transformations (8.1.3) may be written

$$\delta I_M = \int d^4x \frac{\delta I_M}{\delta A_\mu(x)} \partial_\mu \epsilon(x). \quad (8.1.4)$$

Hence the Lorentz invariance of  $I_M$  requires that

$$\partial_\mu \frac{\delta I_M}{\delta A_\mu(x)} = 0. \quad (8.1.5)$$

This is trivially true if  $I_M$  involves only  $F_{\mu\nu}(x)$  and its derivatives, along with matter fields. In this case

$$\frac{\delta I_M}{\delta A_\mu(x)} = 2\partial^\nu \frac{\delta I_M}{\delta F_{\mu\nu}(x)}.$$

But if  $I_M$  involves  $A_\mu(x)$  itself then Eq. (8.1.5) is a non-trivial constraint on the theory.

Now, what sort of theory will provide conserved currents to which we can couple the field  $A^\mu(x)$ ? We saw in Section 7.3 that infinitesimal internal symmetries of the action imply the existence of conserved currents. In particular, if the transformation<sup>79</sup>

$$\delta \Psi^\ell(x) = i\epsilon(x) q_\ell \Psi^\ell(x) \quad (8.1.6)$$

leaves the matter action invariant for a constant  $\epsilon$ , then for general infinitesimal functions  $\epsilon(x)$  the change in the matter action must take the form

$$\delta I_M = - \int d^4x J^\mu(x) \partial_\mu \epsilon(x). \quad (8.1.7)$$

When the matter fields satisfy their field equations, the matter action is stationary with respect to *any* variation of the  $\Psi_\ell$ , so in this case (8.1.7) must vanish, and hence

$$\partial_\mu J^\mu = 0. \quad (8.1.8)$$

In particular, we saw in Section 7.3 that if  $I_M$  is the integral of a function  $\mathcal{L}_M$  of  $\Psi^\ell$  and  $\partial_\mu \Psi^\ell$ , then the conserved current is given by<sup>80</sup>

$$J^\mu = -i \sum_\ell \frac{\partial \mathcal{L}_M}{\partial (\partial_\mu \Psi^\ell)} q_\ell \Psi^\ell,$$

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<sup>79</sup>Because the field transformation matrix is taken now to be diagonal it is not convenient here to use the summation convention for sums over field indices, so there is no sum over  $\ell$  in Eq. (8.1.6).

<sup>80</sup>Here  $\Psi^\ell$  is understood to run over all independent fields other than  $A^\mu$ . We use a capital psi to indicate that these are Heisenberg-picture fields, whose time-dependence includes the effects of interactions. Of course, this  $\Psi^\ell$  is not to be confused with a state-vector or wave function.



and this generates the transformations (8.1.6) in the sense that

$$[Q, \Psi^\ell(x)] = -q_\ell \Psi^\ell(x), \quad (8.1.9)$$

where  $Q$  is the time-independent charge operator

$$Q = \int d^3x J^0. \quad (8.1.10)$$

We can therefore construct a Lorentz-invariant theory by coupling the vector field  $A_\mu$  to the conserved current  $J^\mu$ , in the sense that  $\delta I_M / \delta A_\mu(x)$  is taken to be proportional to  $J^\mu(x)$ . Any constant of proportionality may be absorbed into the definition of the overall scale of the charges  $q_\ell$ , so we may simply set these quantities equal:

$$\frac{\delta I_M}{\delta A_\mu(x)} = J^\mu(x). \quad (8.1.11)$$

The conservation of electric charge only allows us to fix the values of all charges in terms of the value of any one of them, conventionally taken to be the electron charge, denoted  $-e$ . It is Eq. (8.1.11) that gives a definite meaning<sup>81</sup> to the value of  $e$ .

The requirement (8.1.11) may be restated as a principle of invariance [1a]: the matter action is invariant under the joint transformations

$$\delta A_\mu(x) = \partial_\mu \epsilon(x), \quad (8.1.12)$$

$$\delta \Psi_\ell(x) = i\epsilon(x)q_\ell \Psi_\ell(x). \quad (8.1.13)$$

A symmetry of this type with an arbitrary function  $\epsilon(x)$  is called a *local symmetry*, or a gauge invariance of the second kind. A symmetry under a transformation with  $\epsilon$  constant is called a *global symmetry*, or a gauge invariance of the first kind. Several exact local symmetries are now known, but the only purely global symmetries appear to be accidents enforced by other principles. (See Section 12.5.)

We have not yet said anything about the action for photons themselves. As a guess, we can take this to be the same as for massive vector fields, but with  $m = 0$ :

$$I_\gamma = -\frac{1}{4} \int d^4x F_{\mu\nu} F^{\mu\nu}. \quad (8.1.14)$$

This is the same as the action used in classical electrodynamics, but its real justification is that it is (up to a constant) the unique gauge-invariant functional that is quadratic in  $F_{\mu\nu}$ , without higher derivatives. Also, as we will see in the next section, it leads to a consistent quantum theory. If there are any terms in the action of  $A$  with higher derivatives and/or of higher order in  $F_{\mu\nu}$  they can be lumped into what we have called the matter action. Using Eqs. (8.1.11) and (8.1.14), the field equation for electromagnetism now reads

$$0 = \frac{\delta}{\delta A_\nu} [I_\gamma + I_M] = \partial_\mu F^{\mu\nu} + J^\nu. \quad (8.1.15)$$

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<sup>81</sup>Of course, Eq. (8.1.11) fixes the definition of  $e$  only after we have defined how we are normalizing  $A_\mu(x)$ . The question of electromagnetic field normalization is taken up in Section 10.4.

We recognize these as the usual inhomogeneous Maxwell equations, with current  $J^\nu$ . There are also other, homogeneous, Maxwell equations

$$0 = \partial_\mu F_{\nu\rho} + \partial_\rho F_{\mu\nu} + \partial_\nu F_{\rho\mu}, \quad (8.1.16)$$

which follow directly from the definition  $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$ .

In the above discussion, we have started with the existence of massless spin one particles, and have been led to infer the invariance of the matter action under a local gauge transformation (8.1.12), (8.1.13). As usually presented, the derivation runs in the opposite direction. That is, one starts with a global internal symmetry

$$\delta\Psi^\ell(x) = i\epsilon q_\ell \Psi^\ell(x) \quad (8.1.17)$$

and asks what must be done to promote this to a local symmetry

$$\delta\Psi^\ell(x) = i\epsilon(x)q_\ell \Psi^\ell(x). \quad (8.1.18)$$

If the Lagrangian density  $\mathcal{L}$  depended only on fields  $\Psi^\ell(x)$  and not on their derivatives then it would make no difference whether  $\epsilon$  is constant or not; invariance with  $\epsilon$  constant would imply invariance with  $\epsilon$  a function of spacetime position. But all realistic Lagrangians do involve field derivatives, and here we have the problem that derivatives of fields transform differently from fields themselves:

$$\delta\partial_\mu \Psi^\ell(x) = i\epsilon(x)q_\ell \partial_\mu \Psi^\ell(x) + iq_\ell \Psi^\ell(x) \partial_\mu \epsilon(x). \quad (8.1.19)$$

In order to cancel the second term here, we ‘invent’ a vector field  $A_\mu(x)$  with transformation rule

$$\delta A_\mu(x) = \partial_\mu \epsilon(x) \quad (8.1.20)$$

and require that the Lagrangian density depend on  $\partial_\mu \Psi^\ell$  and  $A_\mu$  only in the combination

$$D_\mu \Psi^\ell \equiv \partial_\mu \Psi^\ell - iq_\ell A_\mu \Psi^\ell, \quad (8.1.21)$$

which transforms just like  $\Psi^\ell$

$$\delta D_\mu \Psi^\ell(x) = i\epsilon(x)q_\ell D_\mu \Psi^\ell(x). \quad (8.1.22)$$

A matter Lagrangian density  $\mathcal{L}_M(\Psi, D\Psi)$  that is formed only out of  $\Psi^\ell$  and  $D_\mu \Psi^\ell$  will be invariant under the transformations (8.1.18), (8.1.20), with  $\epsilon(x)$  an arbitrary function, if it is invariant with  $\epsilon$  a constant. With the Lagrangian of this form, we have

$$\frac{\delta I_M}{\delta A_\mu} = \sum_\ell \frac{\partial \mathcal{L}_M}{\partial D_\mu \Psi^\ell} (-iq_\ell \Psi^\ell) = -i \sum_\ell \frac{\partial \mathcal{L}_M}{\partial \partial_\mu \Psi^\ell} q_\ell \Psi^\ell,$$

which is the same as Eq. (8.1.11). (We could also include  $F_{\mu\nu}$  and its derivatives in  $\mathcal{L}_M$ .) From this point of view, the masslessness of the particles described by  $A_\mu$  is a consequence of gauge invariance rather than an assumption: a term  $-\frac{1}{2}m^2 A_\mu A^\mu$  in the Lagrangian density would violate gauge invariance.

## 8.2 Constraints and Gauge Conditions

There are aspects of electrodynamics that stand in the way of quantizing the theory as we did for various theories of massive particles in the previous chapter. As usual, we may define the canonical conjugates to the electromagnetic vector potential by

$$\Pi^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)}. \quad (8.2.1)$$

Quantization by the usual rules would give

$$[A_\mu(\mathbf{x}, t), \Pi^\nu(\mathbf{y}, t)] = i\delta_\mu^\nu \delta^3(\mathbf{x} - \mathbf{y}).$$

But this is not possible here, because  $A_\mu$  and  $\Pi^\nu$  are subject to several constraints.

The first constraint arises from the fact that the Lagrangian density is independent<sup>82</sup> of the time-derivative of  $A_0$ , and therefore

$$\Pi^0(x) = 0. \quad (8.2.2)$$

This is called a *primary constraint*, because it follows directly from the structure of the Lagrangian. There is also a *secondary constraint* here, which follows from the field equation for the quantity fixed by the primary constraint:<sup>83</sup>

$$\partial_i \Pi^i = -\partial_i \frac{\partial \mathcal{L}}{\partial F_{i0}} = -\frac{\partial \mathcal{L}}{\partial A_0} \equiv -J^0, \quad (8.2.3)$$

the time-derivative term dropping out because  $F_{00} = 0$ . Even though the matter Lagrangian may generally depend on  $A^0$ , the charge density depends only on the canonical matter fields<sup>84</sup>  $Q^n$  and their canonical conjugates  $P_n$ :

$$J^0 = -i \sum_\ell \frac{\partial \mathcal{L}}{\partial(\partial_0 \Psi^\ell)} q_\ell \Psi^\ell = -i \sum_n P_n q_n Q^n. \quad (8.2.4)$$

Hence Eq. (8.2.3) is a functional relation among canonical variables. Both Eq. (8.2.2) and Eq. (8.2.3) are inconsistent with the usual assumptions that  $[A_\mu(\mathbf{x}, t), \Pi^\nu(\mathbf{y}, t)] = i\delta_\mu^\nu \delta^3(\mathbf{x} - \mathbf{y})$  and  $[Q^n(\mathbf{x}, t), \Pi^\nu(\mathbf{y}, t)] = [P_n(\mathbf{x}, t), \Pi^\nu(\mathbf{y}, t)] = 0$ .

We encountered a similar problem in the theory of the massive vector field. In that case we found two equivalent ways of dealing with it: either by the method of Dirac brackets or, more directly, by treating only  $A_i$  and  $\Pi^i$  as canonical variables, solving the analog of Eq. (8.2.3) to calculate  $A^0$  in terms of these variables. It is clear that here we cannot

<sup>82</sup>For  $\mathcal{L}_\gamma = -F_{\mu\nu}F^{\mu\nu}/4$ , we have  $\partial \mathcal{L}_\gamma / \partial(\partial_0 A_\mu) = -F^{0\mu}$ , which vanishes for  $\mu = 0$  because  $F^{\mu\nu}$  is antisymmetric. For matter Lagrangians  $\mathcal{L}_M$  that involve only  $\Psi^\ell$  and  $D_\mu \Psi^\ell$ , the prescription (8.1.21) tells us that  $\mathcal{L}_M$  does not depend on any derivatives of any  $A^\mu$ . Even if the matter Lagrangian depends also on  $F_{\mu\nu}$ ,  $\partial \mathcal{L}_M / \partial(\partial_\mu A_\nu)$  will be again antisymmetric in  $\mu$  and  $\nu$ , and therefore will vanish for  $\mu = \nu = 0$ .

<sup>83</sup>As usual,  $i, j$ , etc. run over the values 1, 2, 3.

<sup>84</sup>Due to exhaustion of alphabetic resources, I have had to adopt a notation here that is different from that of the previous chapter. The symbols  $Q^n$  and  $P_n$  are now reserved for the canonical matter fields and their canonical conjugates, respectively, while the canonical electromagnetic fields and canonical conjugates are  $A_i$  and  $\Pi_i$ .

use Dirac brackets; the constraint functions  $\chi$  here are  $\Pi^0$  and  $\partial_i \Pi_i - J^0$  (as compared with  $\partial_i \Pi_i - m^2 A^0 - J^0$ ) and these obviously have vanishing Poisson brackets. In Dirac's terminology, the constraints (8.2.2) and (8.2.3) are *first class*. Nor can we eliminate  $A^0$  as a dynamical variable by solving for it in terms of the other variables. Instead of giving  $A^0$  for all time, Eq. (8.2.3) is a mere initial condition; if Eq. (8.2.3) is satisfied at one time, then it is satisfied for all times, because (using the field equations for the other fields  $A^i$ ), we have

$$\begin{aligned}\partial_0 \left[ \partial_i \frac{\partial \mathcal{L}}{\partial F_{i0}} - J^0 \right] &= -\partial_i \partial_0 \frac{\partial \mathcal{L}}{\partial F_{0i}} - \partial_0 J^0 \\ &= +\partial_i \partial_j \frac{\partial \mathcal{L}}{\partial F_{ji}} - \partial_i J^i - \partial_0 J^0,\end{aligned}$$

and the current conservation condition then gives

$$\partial_0 \left[ \partial_i \frac{\partial \mathcal{L}}{\partial F_{i0}} - J^0 \right] = 0. \quad (8.2.5)$$

It should not be surprising that we still have four components of  $A^\mu$  with only three field equations, because this theory has a local gauge symmetry that makes it, in principle, impossible to infer the values of the fields at arbitrary times from their values and rates of change at any one time. Given any solution  $A_\mu(\mathbf{x}, t)$  of the field equations, we can always find another solution  $A_\mu(\mathbf{x}, t) + \partial_\mu \epsilon(\mathbf{x}, t)$  with the same value and time-derivative at  $t = 0$  (by choosing  $\epsilon$  so that its first and second derivatives vanish there) but which differs from  $A_\mu(\mathbf{x}, t)$  at later times.

Because of this partial arbitrariness of  $A_\mu(\mathbf{x}, t)$ , it is not possible to apply the canonical quantization procedure directly to  $A_\mu$  (or, as for finite mass, to  $\mathbf{A}$ ). Of the various approaches to this difficulty, two are particularly useful. One is the modern method of gauge-invariant quantization, to be discussed in Volume II. The other, which will be followed here, is to exploit the gauge invariance of the theory, to 'choose a gauge.' That is, we make a finite gauge transformation

$$A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu \lambda(x), \quad \Psi_\ell(x) \longrightarrow \exp(iq_\ell \lambda(x)) \Psi_\ell(x)$$

to impose a condition on  $A_\mu(x)$  that will allow us to apply the methods of canonical quantization. There are various gauges that have been found useful in various applications:<sup>85</sup>

$$\text{Lorentz (or Landau) gauge: } \partial_\mu A^\mu = 0,$$

$$\text{Coulomb gauge: } \nabla \cdot \mathbf{A} = 0,$$

$$\text{Temporal gauge: } A^0 = 0,$$

$$\text{Axial gauge: } A^3 = 0,$$

$$\text{Unitarity gauge: } \Phi \text{ real.}$$

---

<sup>85</sup>Here  $\Phi$  is any complex scalar field with  $q \neq 0$ ; this gauge condition is used when the gauge symmetry is spontaneously broken by a non-vanishing vacuum expectation value of  $\Phi$ .

The canonical quantization procedure works most easily in the axial or Coulomb gauge, but of course Coulomb gauge keeps manifest rotation invariance in a way that axial gauge does not, so we will adopt Coulomb gauge here [2].

To check that this is possible, note that if  $A^\mu$  does not satisfy the Coulomb gauge condition, then the gauge-transformed field  $A^\mu + \partial^\mu \lambda$  will, provided we choose  $\lambda$  so that  $\nabla^2 \lambda = -\nabla \cdot \mathbf{A}$ . From now on, we assume that this transformation has been made, so that

$$\nabla \cdot \mathbf{A} = 0. \quad (8.2.6)$$

It will be convenient henceforth to limit ourselves to theories in which the matter Lagrangian  $\mathcal{L}_M$  may depend on matter fields and their time-derivatives and also on  $A^\mu$  but not on derivatives of  $A^\mu$ . (The standard theories of the electrodynamics of scalar and Dirac fields have Lagrangians of this type.) Then the only term in the Lagrangian that depends on  $F_{\mu\nu}$  is the kinematic term  $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ , and the constraint equation (8.2.3) reads

$$-\partial_i F^{i0} = J^0. \quad (8.2.7)$$

Together with the Coulomb gauge condition (8.2.6), this yields

$$-\nabla^2 A^0 = J^0, \quad (8.2.8)$$

which can be solved to give

$$A^0(\mathbf{x}, t) = \int d^3y \frac{J^0(\mathbf{y}, t)}{4\pi|\mathbf{x} - \mathbf{y}|}. \quad (8.2.9)$$

The remaining degrees of freedom are  $A^i$ , with  $i = 1, 2, 3$ , subject to the gauge condition  $\nabla \cdot \mathbf{A} = 0$ .

As mentioned earlier, the charge density depends only on the canonical matter fields  $Q^n$  and their canonical conjugates  $P_n$ , so Eq. (8.2.9) represents an explicit solution for the auxiliary field  $A^0$ .

### 8.3 Quantization in Coulomb Gauge

There is still an impediment to the canonical quantization of electrodynamics in the Coulomb gauge. Even after we use Eq. (8.2.9) to eliminate  $A^0$  (and  $\Pi_0$ ) from the list of canonical variables, we cannot apply the usual canonical commutation relations to  $A^i$  and  $\Pi_i$ , because there are two remaining constraints on these variables.<sup>86</sup> One of them is the Coulomb gauge condition

$$\chi_{1\mathbf{x}} \equiv \partial_i A^i(\mathbf{x}) = 0. \quad (8.3.1)$$

The other is the secondary constraint Eq. (8.2.3), which requires that

$$\chi_{2\mathbf{x}} \equiv \partial_i \Pi^i(\mathbf{x}) + J^0(\mathbf{x}) = 0. \quad (8.3.2)$$

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<sup>86</sup>In this section  $i, j$ , etc. run over the values 1, 2, 3. We continue the practice of taking all operators at the same time, and omitting the time argument.

Neither constraint is consistent with the usual commutation relations  $[A^i(\mathbf{x}), \Pi_j(\mathbf{y})] = i\delta_j^i \delta^3(\mathbf{x} - \mathbf{y})$ , because operating on the right-hand side with either  $\partial/\partial x^i$  or  $\partial/\partial y^j$  does not give zero.

These constraints are of a type known as *second class*, for which there is a universal prescription for the commutation relations, discussed in Section 7.6. Note that the constraint functions have the Poisson brackets

$$\begin{aligned} C_{1\mathbf{x},2\mathbf{y}} &= -C_{2\mathbf{y},1\mathbf{x}} \equiv [\chi_{1\mathbf{x}}, \chi_{2\mathbf{y}}]_{\text{P}} = -\nabla^2 \delta^3(\mathbf{x} - \mathbf{y}), \\ C_{1\mathbf{x},1\mathbf{y}} &\equiv [\chi_{1\mathbf{x}}, \chi_{1\mathbf{y}}]_{\text{P}} = 0, \\ C_{2\mathbf{x},2\mathbf{y}} &\equiv [\chi_{2\mathbf{x}}, \chi_{2\mathbf{y}}]_{\text{P}} = 0, \end{aligned} \quad (8.3.3)$$

where here, for any functionals  $U$  and  $V$ ,

$$[U, V]_{\text{P}} \equiv \int d^3x \left[ \frac{\delta U}{\delta A^i(\mathbf{x})} \frac{\delta V}{\delta \Pi_i(\mathbf{x})} - \frac{\delta V}{\delta A^i(\mathbf{x})} \frac{\delta U}{\delta \Pi_i(\mathbf{x})} \right].$$

The ‘matrix’  $C_{N\mathbf{x},M\mathbf{y}}$  is non-singular, which identifies these as second class constraints. Also, the field variables  $A^i$  may be expressed in terms of independent canonical variables, which may, for instance, be taken as  $Q_{1\mathbf{x}} = A^1(\mathbf{x})$ ,  $Q_{2\mathbf{x}} = A^2(\mathbf{x})$ , with  $A^3$  given by the solution of Eq. (8.3.1):

$$A^3(\mathbf{x}) = - \int^{x^3} ds [\partial_1 A^1(x^1, x^2, s) + \partial_2 A^2(x^1, x^2, s)].$$

Using Eq. (8.3.2), the canonical conjugates  $\Pi_i$  to  $A^i$  may likewise be expressed in terms of the canonical conjugates  $P_{\mathbf{x}1}$  and  $P_{\mathbf{x}2}$ , to  $Q_{1\mathbf{x}}$  and  $Q_{2\mathbf{x}}$ . In such cases, Part B of the Appendix to the previous chapter tells us that if the independent variables  $Q_{1\mathbf{x}}$ ,  $Q_{2\mathbf{x}}$ ,  $P_{1\mathbf{x}}$ , and  $P_{2\mathbf{x}}$  satisfy the usual canonical commutation relations, then the commutators of the constrained variables and their canonical conjugates are given (aside from a factor  $i$ ) by the corresponding Dirac brackets (7.6.20). This prescription has the great advantage that we do not have to do use explicit expressions for the dependent variables in terms of the independent ones.

To calculate the Dirac brackets, we note that the matrix  $C$  has the inverse

$$\begin{aligned} (C^{-1})_{1\mathbf{x},2\mathbf{y}} &= -(C^{-1})_{2\mathbf{y},1\mathbf{x}} = - \int \frac{d^3k}{(2\pi)^3} \frac{e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}}{\mathbf{k}^2} = -\frac{1}{4\pi|\mathbf{x}-\mathbf{y}|}, \\ (C^{-1})_{1\mathbf{x},1\mathbf{y}} &= (C^{-1})_{2\mathbf{x},2\mathbf{y}} = 0. \end{aligned} \quad (8.3.4)$$

Also, the non-vanishing Poisson brackets of the  $A^i$  and  $\Pi_i$  with the constraint functions are

$$[A^i(\mathbf{x}), \chi_{2\mathbf{y}}]_{\text{P}} = -\frac{\partial}{\partial x^i} \delta^3(\mathbf{x} - \mathbf{y})$$

and

$$[\Pi_i(\mathbf{x}), \chi_{1\mathbf{y}}]_{\text{P}} = +\frac{\partial}{\partial x^i} \delta^3(\mathbf{x} - \mathbf{y}).$$

Hence according to Eqs. (7.6.19) and (7.6.20), the equal-time commutators are

$$\begin{aligned} [A^i(\mathbf{x}), \Pi_j(\mathbf{y})] &= i\delta_j^i \delta^3(\mathbf{x} - \mathbf{y}) + i \frac{\partial^2}{\partial x^i \partial x^j} \left( \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} \right), \\ [A^i(\mathbf{x}), A^j(\mathbf{y})] &= [\Pi_i(\mathbf{x}), \Pi_j(\mathbf{y})] = 0. \end{aligned} \quad (8.3.5)$$

Note that these are consistent with the Coulomb gauge conditions (8.3.1) and (8.3.2), as is guaranteed by the general properties of the Dirac bracket.

Now, what is  $\mathbf{\Pi}$  in electrodynamics? For the class of theories discussed in the previous section where only the kinematic term  $-\frac{1}{4} \int d^3x F_{\mu\nu} F^{\mu\nu}$  in the Lagrangian depends on  $\dot{\mathbf{A}}$ , varying the Lagrangian with respect to  $\dot{\mathbf{A}}$  without worrying about the constraint  $\nabla \cdot \dot{\mathbf{A}} = 0$  gives

$$\Pi_i = \frac{\delta L}{\delta \dot{A}^i(\mathbf{x})} = \dot{A}^i(\mathbf{x}) + \frac{\partial}{\partial x^i} A^0(\mathbf{x}). \quad (8.3.6)$$

But with  $\mathbf{A}$  constrained by the condition  $\nabla \cdot \mathbf{A} = 0$ , variational derivatives with respect to  $\dot{\mathbf{A}}$  are not really well defined. If the variation of  $L$  under a change  $\delta \dot{\mathbf{A}}$  in  $\dot{\mathbf{A}}$  is  $\delta L = \int d^3x \mathcal{P} \cdot \delta \dot{\mathbf{A}}$ , then since  $\nabla \cdot \delta \dot{\mathbf{A}} = 0$ , we also have  $\delta L = \int d^3x [\mathcal{P} + \nabla \mathcal{F}] \cdot \delta \dot{\mathbf{A}}$  for any scalar function  $\mathcal{F}(\mathbf{x})$ . Thus all we can conclude from inspection of the Lagrangian is that  $\mathbf{\Pi}$  equals  $\dot{\mathbf{A}}(\mathbf{x}) + \nabla A^0(\mathbf{x})$  plus the gradient of some scalar. This ambiguity is removed by condition (8.3.2), which requires that  $\nabla \cdot \mathbf{\Pi} = -J^0 = \nabla^2 A^0$ . Because  $\nabla \cdot \dot{\mathbf{A}} = 0$ , we conclude that Eq. (8.3.6) does indeed give the correct formula for  $\Pi^i$ .

Although the commutation relations (8.3.5) are reasonably simple, we must face the complication that  $\mathbf{\Pi}$  does not commute with matter fields and their canonical conjugates. If  $F$  is any functional of these matter degrees of freedom, then its Dirac bracket with  $\mathbf{A}$  vanishes, but its Dirac bracket with  $\mathbf{\Pi}$  is

$$\begin{aligned} [F, \mathbf{\Pi}(\mathbf{z})]_{\text{D}} &= - \int d^3x d^3y [F, \chi_{2\mathbf{x}}]_{\text{P}} \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} [\chi_{1\mathbf{y}}, \mathbf{\Pi}(\mathbf{z})]_{\text{P}} \\ &= - \int d^3x d^3y [F, J^0(\mathbf{x})]_{\text{P}} \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} \nabla \delta^3(\mathbf{y} - \mathbf{z}) \\ &= - \int d^3y [F, A^0(\mathbf{y})]_{\text{P}} \nabla \delta^3(\mathbf{y} - \mathbf{z}) \\ &= [F, \nabla A^0(\mathbf{z})]_{\text{P}} = [F, \nabla A^0(\mathbf{z})]_{\text{D}}. \end{aligned}$$

In order to facilitate the transition to the interaction picture, instead of expressing the Hamiltonian in terms of  $\mathbf{A}$  and  $\mathbf{\Pi}$ , we shall write it in terms of  $\mathbf{A}$  and  $\mathbf{\Pi}_{\perp}$ , where  $\mathbf{\Pi}_{\perp}$  is the solenoidal part of  $\mathbf{\Pi}$ :

$$\mathbf{\Pi}_{\perp} \equiv \mathbf{\Pi} - \nabla A^0 = \dot{\mathbf{A}}, \quad (8.3.7)$$

for which  $[F, \mathbf{\Pi}_{\perp}(\mathbf{z})]$  vanishes. By using the facts that  $\mathbf{\Pi}_{\perp}(\mathbf{x})$  commutes with  $\mathbf{\Pi}(\mathbf{y}) - \mathbf{\Pi}_{\perp}(\mathbf{y}) = \nabla A^0(\mathbf{y})$  and that  $\partial_i A^0(\mathbf{x})$  commutes with  $\partial_j A^0(\mathbf{y})$ , it is easy to see that  $\mathbf{\Pi}_{\perp}(\mathbf{x})$  satisfies the same commutation relations (8.3.5) as  $\mathbf{\Pi}(\mathbf{x})$ , and also the simple constraint

$$\nabla \cdot \mathbf{\Pi}_{\perp} = 0. \quad (8.3.8)$$

Now we need to construct a Hamiltonian. According to the general results of the Appendix to Chapter 7, we can apply the usual relation between the Hamiltonian and Lagrangian using the constrained variables  $\mathbf{A}$  and  $\mathbf{\Pi}_\perp$ , without first having explicitly to write the Hamiltonian in terms of the unconstrained  $Q$ s and  $P$ s. In electrodynamics, this gives

$$H = \int d^3x [\mathbf{\Pi}_\perp \cdot \dot{\mathbf{A}} + P_n \dot{Q}^n - \mathcal{L}], \quad (8.3.9)$$

where, as mentioned earlier,  $Q^n$  and  $P_n$  are to be understood as the matter canonical fields and their canonical conjugates. (We can use  $\mathbf{\Pi}_\perp$  in place of  $\mathbf{\Pi}$  in Eq. (8.3.9) because  $\nabla \cdot \dot{\mathbf{A}} = 0$ .)

To be specific, consider a theory with a Lagrangian density of the form

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + J_\mu A^\mu + \mathcal{L}_{\text{matter}}, \quad (8.3.10)$$

where  $J_\mu$  is a current that does not involve  $A^\mu$ , and  $\mathcal{L}_{\text{matter}}$  is the Lagrangian for whatever other fields do appear in  $J^\mu$ , aside from their electromagnetic interactions, which are given explicitly by the term  $J_\mu A^\mu$  in Eq. (8.3.10). (The electrodynamics of spin  $\frac{1}{2}$  particles has a Lagrangian of this form, but the electrodynamics of spinless particles is more complicated.) Replacing  $\dot{\mathbf{A}}$  everywhere with  $\mathbf{\Pi}_\perp$ , this gives a Hamiltonian (8.3.9) of the form

$$H = \int d^3x \left[ \mathbf{\Pi}_\perp^2 + \frac{1}{2}(\nabla \times \mathbf{A})^2 - \frac{1}{2}(\mathbf{\Pi}_\perp + \nabla A^0)^2 - \mathbf{J} \cdot \mathbf{A} + J^0 A^0 \right] + H_M,$$

where  $H_M$  is the Hamiltonian for matter fields, excluding their electromagnetic interactions

$$H_M \equiv \int d^3x (P_n \dot{Q}^n - \mathcal{L}_{\text{matter}}).$$

Using the solution (8.2.9) for  $A^0$ , this is

$$H = \int d^3x \left[ \frac{1}{2}\mathbf{\Pi}_\perp^2 + \frac{1}{2}(\nabla \times \mathbf{A})^2 - \mathbf{J} \cdot \mathbf{A} + \frac{1}{2}J^0 A^0 \right] + H_M. \quad (8.3.11)$$

The term  $\frac{1}{2}J^0 A^0$  may look peculiar, but this is nothing but the familiar Coulomb energy

$$\begin{aligned} V_{\text{Coul}} &= \frac{1}{2} \int d^3x J^0 A^0 \\ &= \frac{1}{2} \int d^3x \int d^3y \frac{J^0(\mathbf{x})J^0(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|}. \end{aligned} \quad (8.3.12)$$

The reader can verify, using the commutation relations (8.3.5), that the rate of change of any operator function  $F$  of  $\mathbf{A}$  and  $\mathbf{\Pi}$  is given by  $i\dot{F} = [F, H]$ , as it should be.



## 8.4 Electrodynamics in the Interaction Picture

We now break up the Hamiltonian (8.3.11) into a free-particle term  $H_0$  and an interaction  $H_{\text{int}}$

$$H = H_0 + H_{\text{int}}, \quad (8.4.1)$$

$$H_0 = \int d^3x \left[ \frac{1}{2} \mathbf{\Pi}_{\perp}^2 + \frac{1}{2} (\nabla \times \mathbf{A})^2 \right] + H_{\text{matter},0}, \quad (8.4.2)$$

$$H_{\text{int}} = - \int d^3x \mathbf{J} \cdot \mathbf{A} + V_{\text{Coul}} + H_{\text{int,matter}}, \quad (8.4.3)$$

where  $H_{\text{matter},0}$  and  $H_{\text{int,matter}}$  are the free-particle and interaction terms in  $H_{\text{matter}}$ , and  $V_{\text{Coul}}$  is the Coulomb interaction (8.3.12). The total Hamiltonian (8.4.1) is time-independent, so Eqs. (8.4.2) and (8.4.3) can be evaluated at any time we like (as long as both are evaluated at the same time), in particular at  $t = 0$ . As in Chapter 7, the transition to the interaction picture is made by applying the similarity transformation

$$\begin{aligned} H_I(t) &= \exp(iH_0t) H_{\text{int}}[\mathbf{A}, \mathbf{\Pi}_{\perp}, Q, P]_{t=0} \exp(-iH_0t) \\ &= H_{\text{int}}[\mathbf{a}(t), \boldsymbol{\pi}(t), q(t), p(t)], \end{aligned} \quad (8.4.4)$$

where  $P$  here denotes the canonical conjugates to the matter fields  $Q$ , and any operator  $O(\mathbf{x}, t)$  in the interaction picture is related to its value  $O(\mathbf{x}, 0)$  in the Heisenberg picture at  $t = 0$  by

$$O(\mathbf{x}, t) = \exp(iH_0t) O(\mathbf{x}, 0) \exp(-iH_0t), \quad (8.4.5)$$

so that

$$i\dot{O}(\mathbf{x}, t) = [O(\mathbf{x}, t), H_0]. \quad (8.4.6)$$

(We are dropping the subscript  $\perp$  on  $\boldsymbol{\pi}(\mathbf{x})$ .) Since Eq. (8.4.5) is a similarity transformation, the equal-time commutation relations are the same as in the Heisenberg picture:

$$[a^i(\mathbf{x}, t), \pi^j(\mathbf{y}, t)] = i \left[ \delta_{ij} \delta^3(\mathbf{x} - \mathbf{y}) + \frac{\partial^2}{\partial x^i \partial x^j} \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} \right], \quad (8.4.7)$$

$$[a^i(\mathbf{x}, t), a^j(\mathbf{y}, t)] = 0, \quad (8.4.8)$$

$$[\pi^i(\mathbf{x}, t), \pi^j(\mathbf{y}, t)] = 0, \quad (8.4.9)$$

and likewise for the matter fields and their conjugates. For the same reason, the constraints (8.2.6) and (8.3.8) still apply

$$\nabla \cdot \mathbf{a} = 0, \quad (8.4.10)$$

$$\nabla \cdot \boldsymbol{\pi} = 0. \quad (8.4.11)$$

To establish the relation between  $\boldsymbol{\pi}$  and  $\dot{\mathbf{a}}$ , we must use Eq. (8.4.6) to evaluate  $\dot{\mathbf{a}}$ :

$$\begin{aligned} i\dot{a}_i(\mathbf{x}, t) &= [a_i(\mathbf{x}, t), H_0] \\ &= i \int d^3y \left[ \delta_{ij} \delta^3(\mathbf{x} - \mathbf{y}) + \frac{\partial^2}{\partial x^i \partial x^j} \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} \right] \pi_j(\mathbf{y}, t). \end{aligned}$$

We can replace  $\partial/\partial x^j$  in the second term with  $-\partial/\partial y^j$ , integrate by parts, and use Eq. (8.4.11), yielding

$$\dot{\mathbf{a}} = \boldsymbol{\pi} \quad (8.4.12)$$

just as in the Heisenberg picture. The field equation is likewise determined by

$$\begin{aligned} i\dot{\pi}_i(\mathbf{x}, t) &= [\pi_i(\mathbf{x}, t), H_0] \\ &= -i \int d^3y \left[ \delta_{ij} \delta^3(\mathbf{x} - \mathbf{y}) + \frac{\partial^2}{\partial x^i \partial x^j} \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} \right] (\boldsymbol{\nabla} \times \boldsymbol{\nabla} \times \mathbf{a}(\mathbf{y}, t))_j, \end{aligned}$$

which (using Eqs. (8.4.10) and (8.4.12)) just yields the usual wave equation

$$\square \mathbf{a} = 0. \quad (8.4.13)$$

Since  $A^0$  is not an independent Heisenberg-picture field variable, but rather a functional (8.2.9) of the matter fields and their canonical conjugates that vanishes in the limit of zero charges, we do not introduce any corresponding operator  $a^0$  in the interaction picture, but rather take

$$a^0 = 0. \quad (8.4.14)$$

The most general real solution of Eqs. (8.4.10), (8.4.13), and (8.4.14) may be written

$$\begin{aligned} a^\mu(x) &= (2\pi)^{-3/2} \int \frac{d^3p}{\sqrt{2p^0}} \sum_{\sigma} \left[ e^{ip \cdot x} e^\mu(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma} \right. \\ &\quad \left. + e^{-ip \cdot x} e^{\mu*}(\mathbf{p}, \sigma) a_{\mathbf{p}, \sigma}^\dagger \right], \end{aligned} \quad (8.4.15)$$

where  $p^0 \equiv |\mathbf{p}|$ ;  $e^\mu(\mathbf{p}, \sigma)$  are any two independent ‘polarization vectors’ satisfying

$$\mathbf{p} \cdot \mathbf{e}(\mathbf{p}, \sigma) = 0, \quad (8.4.16)$$

$$e^0(\mathbf{p}, \sigma) = 0, \quad (8.4.17)$$

and  $a_{\mathbf{p}, \sigma}$  are a pair of operator coefficients, with  $\sigma$  a two-valued index. By adjusting the normalization of  $a_{\mathbf{p}, \sigma}$ , we can normalize the  $e^\mu(\mathbf{p}, \sigma)$  so that the completeness relation reads

$$\sum_{\sigma} e^i(\mathbf{p}, \sigma) e^j(\mathbf{p}, \sigma)^* = \delta_{ij} - \frac{p_i p_j}{|\mathbf{p}|^2}. \quad (8.4.18)$$

For instance, we could take the  $e(\mathbf{p}, \sigma)$  to be the same polarization vectors that we encountered in Section 5.9:

$$e^\mu(\mathbf{p}, \pm 1) = R(\hat{\mathbf{p}}) \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ \pm i/\sqrt{2} \\ 0 \end{bmatrix}, \quad (8.4.19)$$

where  $R(\hat{\mathbf{p}})$  is a standard rotation that carries the three-axis into the direction of  $\mathbf{p}$ . Using Eqs. (8.4.18) and (8.4.12), we can easily see that the commutation relations (8.4.7)–(8.4.9) are satisfied if (and in fact only if) the operator coefficients in Eq. (8.4.15) satisfy

$$[a_{\mathbf{p}, \sigma}, a_{\mathbf{p}', \sigma'}^\dagger] = \delta^3(\mathbf{p} - \mathbf{p}') \delta_{\sigma \sigma'}, \quad (8.4.20)$$

$$[a_{\mathbf{p}, \sigma}, a_{\mathbf{p}', \sigma'}] = 0. \quad (8.4.21)$$

As remarked before for massive particles, this result should be regarded not so much as an alternative derivation of Eqs. (8.4.20) and (8.4.21), but rather as a verification that Eq. (8.4.2) gives the correct Hamiltonian for free massless particles of helicity  $\pm 1$ . In the same spirit one can also use Eqs. (8.4.12) and (8.4.15) in Eq. (8.4.2) to calculate the free-photon Hamiltonian

$$\begin{aligned} H_0 &= \int d^3p \sum_{\sigma} \frac{1}{2} p^0 [a_{\mathbf{p},\sigma}, a_{\mathbf{p},\sigma}^{\dagger}]_+ \\ &= \int d^3p \sum_{\sigma} p^0 \left( a_{\mathbf{p},\sigma}^{\dagger} a_{\mathbf{p},\sigma} + \frac{1}{2} \delta^3(\mathbf{p} - \mathbf{p}) \right) \end{aligned} \quad (8.4.22)$$

which (aside from an inconsequential infinite  $c$ -number term) is just what we should expect.

Finally, we record that the interaction (8.4.4) in the interaction picture is

$$H_I(t) = - \int d^3x j_{\mu}(\mathbf{x}, t) a^{\mu}(\mathbf{x}, t) + V_{\text{Coul}}(t) + H_{I,\text{matter}}(t), \quad (8.4.23)$$

where in terms of the current  $J$  in the Heisenberg picture

$$j_{\mu}(\mathbf{x}, t) \equiv \exp(iH_0 t) J_{\mu}(\mathbf{x}, 0) \exp(-iH_0 t), \quad (8.4.24)$$

while  $V_{\text{Coul}}(t)$  is the Coulomb term

$$\begin{aligned} V_{\text{Coul}}(t) &= \exp(iH_0 t) V_{\text{Coul}} \exp(-iH_0 t) \\ &= \frac{1}{2} \int d^3x d^3y \frac{j^0(\mathbf{x}, t) j^0(\mathbf{y}, t)}{4\pi |\mathbf{x} - \mathbf{y}|} \end{aligned} \quad (8.4.25)$$

and  $H_{I,\text{matter}}(t)$  is the non-electromagnetic part of the matter field interaction in the interaction picture:

$$H_{I,\text{matter}}(t) = \exp(iH_0 t) H_{\text{int},\text{matter}} \exp(-iH_0 t). \quad (8.4.26)$$

We have written  $j_{\mu} a^{\mu}$  instead of  $\mathbf{j} \cdot \mathbf{a}$  in Eq. (8.4.23), but these are equal because  $a^0$  has been defined to have  $a^0 = 0$ .

## 8.5 The Photon Propagator

The general Feynman rules described in Chapter 6 dictate that an internal photon line in a Feynman diagram contributes a factor to the corresponding term in the  $S$ -matrix, given by the propagator:

$$-i\Delta_{\mu\nu}(x - y) \equiv \langle \Omega | T \{ a_{\mu}(x), a_{\nu}(y) \} | \Omega \rangle, \quad (8.5.1)$$

where  $T$  as usual denotes a time-ordered product. Inserting our formula (8.4.15) for the electromagnetic potential then yields

$$-i\Delta_{\mu\nu}(x - y) = \int \frac{d^3p}{(2\pi)^3 2|\mathbf{p}|} P_{\mu\nu}(\mathbf{p}) \left[ e^{ip \cdot (x-y)} \theta(x^0 - y^0) + e^{ip \cdot (y-x)} \theta(y^0 - x^0) \right], \quad (8.5.2)$$

where

$$P_{\mu\nu}(\mathbf{p}) \equiv \sum_{\sigma=\pm 1} e_{\mu}(\mathbf{p}, \sigma) e_{\nu}(\mathbf{p}, \sigma)^*, \quad (8.5.3)$$

and  $p^\mu$  in the exponentials is taken with  $p^0 = |\mathbf{p}|$ . We recall from Eqs. (8.4.18) and (8.4.17) that

$$\begin{aligned} P_{ij}(\mathbf{p}) &= \delta_{ij} - \frac{p^i p^j}{|\mathbf{p}|^2}, \\ P_{0i}(\mathbf{p}) &= P_{i0}(\mathbf{p}) = P_{00}(\mathbf{p}) = 0. \end{aligned} \quad (8.5.4)$$

As we saw in Chapter 6, the theta functions in Eq. (8.5.2) may be expressed as integrals over an independent time-component  $q^0$  of an off-shell four-momentum  $q^\mu$ , so that Eq. (8.5.2) may be rewritten

$$\Delta_{\mu\nu}(x-y) = (2\pi)^{-4} \int d^4 q \frac{P_{\mu\nu}(\mathbf{q})}{q^2 - i\epsilon} e^{iq \cdot (x-y)}. \quad (8.5.5)$$

Thus in using the Feynman rules in momentum space, the contribution of an internal photon line carrying four-momentum  $q$  that runs between vertices where the photon is created and destroyed by fields  $a^\mu$  and  $a^\nu$  is

$$\frac{-i}{(2\pi)^4} \frac{P_{\mu\nu}(\mathbf{q})}{q^2 - i\epsilon}. \quad (8.5.6)$$

It will be very useful (though apparently perverse) to rewrite Eq. (8.5.4) as

$$P_{\mu\nu}(\mathbf{q}) = \eta_{\mu\nu} + \frac{q^0 q_\mu n_\nu + q^0 q_\nu n_\mu - q_\mu q_\nu + q^2 n_\mu n_\nu}{|\mathbf{q}|^2}, \quad (8.5.7)$$

where  $n^\mu = (1, 0, 0, 0)$  is a fixed time-like vector,  $q^2$  as usual is  $|\mathbf{q}|^2 - (q^0)^2$ , but  $q^0$  is here entirely arbitrary. We shall choose  $q^0$  in Eq. (8.5.7) to be given by four-momentum conservation: it is the difference of the matter  $p^0$ s flowing in and out of the vertex where the photon line is created. The terms proportional to  $q_\mu$  and/or  $q_\nu$  then do not contribute to the  $S$ -matrix, because the factors  $q_\mu$  or  $q_\nu$  act like derivatives  $\partial_\mu$  and  $\partial_\nu$ , and the photon fields  $a_\mu$  and  $a_\nu$  are coupled to currents  $j^\mu$  and  $j^\nu$  that satisfy the conservation condition  $\partial_\mu j^\mu = 0$ .<sup>87</sup> The term proportional to  $n_\mu n_\nu$  contains a factor  $q^2$  that cancels the  $q^2$  in the denominator of the propagator, yielding a term that is the same as would be produced by a term in the action:

$$-i \frac{1}{2} \int d^4 x \int d^4 y [-ij^0(x)][-ij^0(y)] \frac{-i}{(2\pi)^4} \int \frac{d^4 q}{|\mathbf{q}|^2} e^{iq \cdot (x-y)}.$$

The integral over  $q^0$  here yields a delta function in time, so this is equivalent to a correction to the interaction Hamiltonian  $H_I(t)$ , of the form

$$-\frac{1}{2} \int d^3 x \int d^3 y \frac{j^0(\mathbf{x}, t) j^0(\mathbf{y}, t)}{4\pi |\mathbf{x} - \mathbf{y}|}.$$

This is just right to cancel the Coulomb interaction (8.4.25). Our result is that the photon propagator can be taken effectively as the covariant quantity

$$\Delta_{\mu\nu}^{\text{eff}}(x-y) = (2\pi)^{-4} \int d^4 q \frac{\eta_{\mu\nu}}{q^2 - i\epsilon} e^{iq \cdot (x-y)} \quad (8.5.8)$$

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<sup>87</sup>This argument as given here is little better than hand-waving. The result has been justified by a detailed analysis of Feynman diagrams [3], but the easiest way to treat this problem is by path-integral methods, as discussed in Section 9.6.

with the Coulomb interaction dropped from now on. We see that the apparent violation of Lorentz invariance in the instantaneous Coulomb interaction is cancelled by another apparent violation of Lorentz invariance, that as noted in Section 5.9 the fields  $a^\mu(x)$  are not four-vectors, and therefore have a non-covariant propagator. From a practical point of view, the important point is that in the momentum space Feynman rules, the contribution of an internal photon line is simply given by

$$\frac{-i}{(2\pi)^4} \frac{\eta_{\mu\nu}}{q^2 - i\epsilon} \quad (8.5.9)$$

and the Coulomb interaction is dropped.

## 8.6 Feynman Rules for Spinor Electrodynamics

We are now in a position to state the Feynman rules for calculating the  $S$ -matrix in quantum electrodynamics. For definiteness, we will consider the electrodynamics of a single species of spin  $\frac{1}{2}$  particles of charge  $q = -e$  and mass  $m$ . We will call these fermions electrons, but the same formalism applies to muons and other such particles. The simplest gauge- and Lorentz-invariant Lagrangian for this theory is<sup>88</sup>

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\Psi}(\gamma^\mu[\partial_\mu + ieA_\mu] + m)\Psi. \quad (8.6.1)$$

The electric current four-vector is then simply

$$J^\mu \equiv \frac{\partial \mathcal{L}}{\partial A_\mu} = -ie\bar{\Psi}\gamma^\mu\Psi. \quad (8.6.2)$$

The interaction (8.4.23) in the interaction picture is here

$$H_I(t) = +ie \int d^3x (\bar{\psi}(\mathbf{x}, t)\gamma^\mu\psi(\mathbf{x}, t))a_\mu(\mathbf{x}, t) + V_{\text{Coul}}(t). \quad (8.6.3)$$

(There is no  $H_{I,\text{matter}}$  here.) As we have seen, the Coulomb term  $V_{\text{Coul}}(t)$  just serves to cancel a part of the photon propagator that is non-covariant and local in time.

Following the general results of Section 6.3, we can state the momentum space Feynman rules for the connected part of the  $S$ -matrix in this theory as follows:

(i) Draw all Feynman diagrams with up to some given number of vertices. The diagrams consist of electron lines carrying arrows and photon lines without arrows, with the lines joined at vertices, at each of which there is one incoming and one outgoing electron line and one photon line. There is one external line coming into the diagram from below or going upwards out of the diagram for each particle in the initial or final states, respectively; electrons are represented by external lines carrying arrows pointing upwards into or out of the diagram, while positrons are represented by lines carrying arrows pointing downwards into or out of the diagram. There are also as many internal lines as are needed to give each

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<sup>88</sup>In Chapter 12 we will discuss reasons why more complicated terms are excluded from the Lagrangian density.

vertex the required number of attached lines. Each internal line is labelled with an off-mass-shell four-momentum flowing in a definite direction along the line (taken conventionally to flow along the direction of the arrow for electron lines). Each external line is labelled with the momentum and spin  $z$ -component or helicity of the electron or photon in the initial and final states.

(ii) Associate factors with the components of the diagram as follows:

#### *Vertices*

Label each vertex with a four-component Dirac index  $\alpha$  at the electron line with its arrow coming into the vertex, a Dirac index  $\beta$  at the electron line with its arrow going out of the vertex, and a spacetime index  $\mu$  at the photon line. For each such vertex, include a factor

$$(2\pi)^4 e(\gamma^\mu)_{\beta\alpha} \delta^4(k - k' + q), \quad (8.6.4)$$

where  $k$  and  $k'$  are the electron four-momenta entering and leaving the vertex, and  $q$  is the photon four-momentum entering the vertex (or minus the photon momentum leaving the vertex).

#### *External lines:*

Label each external line with the three-momentum  $\mathbf{p}$  and spin  $z$ -component or helicity  $\sigma$  of the particle in the initial or final state. For each line for an electron in the final state running out of a vertex carrying a Dirac label  $\beta$  on this line, include a factor <sup>89</sup>

$$\frac{\bar{u}_\beta(\mathbf{p}, \sigma)}{(2\pi)^{3/2}}. \quad (8.6.5)$$

For each line for a positron in the final state running into a vertex carrying a Dirac label  $\alpha$  on this line, include a factor

$$\frac{v_\alpha(\mathbf{p}, \sigma)}{(2\pi)^{3/2}}. \quad (8.6.6)$$

For each line for an electron in the initial state running into a vertex carrying a Dirac label  $\alpha$  on this line, include a factor

$$\frac{u_\alpha(\mathbf{p}, \sigma)}{(2\pi)^{3/2}}. \quad (8.6.7)$$

For each line for a positron in the initial state running out of a vertex carrying a Dirac label  $\beta$  on this line, include a factor

$$\frac{\bar{v}_\beta(\mathbf{p}, \sigma)}{(2\pi)^{3/2}}. \quad (8.6.8)$$

The  $u$ s and  $v$ s are the four-component spinors discussed in Section 5.5. For each line for a photon in the final state connected to a vertex carrying a spacetime label  $\mu$  on this line, include a factor

$$\frac{e_\mu^*(\mathbf{p}, \sigma)}{(2\pi)^{3/2} \sqrt{2p^0}}. \quad (8.6.9)$$

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<sup>89</sup>A matrix  $\beta$  has been extracted from the interaction in (8.6.4), so that  $\bar{u}$  and  $\bar{v}$  appear instead of  $u^\dagger$  and  $v^\dagger$ .

For each line for a photon in the initial state connected to a vertex carrying a spacetime label  $\mu$  on this line, include a factor

$$\frac{e_\mu(\mathbf{p}, \sigma)}{(2\pi)^{3/2} \sqrt{2p^0}}. \quad (8.6.10)$$

The  $e_\mu$  are the photon polarization four-vectors described in the previous section.

*Internal lines:*

For each internal electron line carrying a four-momentum  $k$  and running from a vertex carrying a Dirac label  $\beta$  to another vertex carrying a Dirac label  $\alpha$ , include a factor

$$\frac{-i}{(2\pi)^4} \frac{[-i\not{k} + m]_{\alpha\beta}}{k^2 + m^2 - i\epsilon}. \quad (8.6.11)$$

(We are here using the very convenient ‘Dirac slash’ notation; for any four-vector  $v^\mu$ ,  $\not{v}$  denotes  $\gamma_\mu v^\mu$ .) For each internal photon line carrying a four-momentum  $q$  that runs between two vertices carrying spacetime labels  $\mu$  and  $\nu$  include a factor

$$\frac{-i}{(2\pi)^4} \frac{\eta_{\mu\nu}}{q^2 - i\epsilon}. \quad (8.6.12)$$

(iii) Integrate the product of all these factors over the four-momenta carried by the internal lines, and sum over all Dirac and spacetime indices.

(iv) Add up the results obtained in this way from each Feynman diagram.

Additional combinatoric factors and fermionic signs may need to be included, as described in parts (v) and (vi) of Section 6.1.

The difficulty of evaluating Feynman diagrams increases rapidly with the number of internal lines and vertices, so it is important to have some idea of what numerical factors tend to suppress the contributions of the more complicated diagrams. We shall estimate these numerical factors including not only the factors of the electronic charge  $e$  associated with vertices, but also the factors of 2 and  $\pi$  from vertices, propagators, and momentum space integrals.

Consider a connected Feynman diagram with  $V$  vertices,  $I$  internal lines,  $E$  external lines, and  $L$  loops. These quantities are not independent, but are subject to relations already used in Section 6.3:

$$L = I - V + 1, \quad 2I + E = 3V.$$

There is a factor  $e(2\pi)^4$  from each vertex, a factor  $(2\pi)^{-4}$  from each internal line, and a four-dimensional momentum space integral for each loop. The volume element in four-dimensional Euclidean space in terms of a radius parameter  $\kappa$  is  $\pi^2 \kappa^2 d\kappa^2$ , so each loop contributes a factor  $\pi^2$ . Thus the diagram will contain a factor

$$(2\pi)^{4V} e^V (2\pi)^{-4I} \pi^{2L} = (2\pi)^4 e^{E-2} \left( \frac{e^2}{16\pi^2} \right)^L.$$

The number  $E$  of external lines is fixed for a given process, so we see that the expansion parameter that governs the suppression of Feynman graphs for each additional loop is

$$\frac{e^2}{16\pi^2} = \frac{\alpha}{4\pi} = 5.81 \cdot 10^{-4}.$$

Fortunately this is small enough that good accuracy can usually be obtained from Feynman diagrams with at most a few loops.

\* \* \*

We must say a little more about the spin states of photons and electrons in realistic experiments, where not every particle in the initial and final states has a definite known helicity or spin  $z$ -component. This consideration is especially important for photons, which in practice are often characterized by a state of transverse or elliptical polarization rather than helicity. As we saw in the previous section, for photons of helicity  $\pm 1$ , the polarization vectors are

$$e^\mu(\mathbf{p}, \pm 1) = R(\hat{\mathbf{p}}) \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ \pm i/\sqrt{2} \\ 0 \end{bmatrix},$$

where  $R(\hat{\mathbf{p}})$  is the standard rotation that takes the  $z$ -axis to the  $\mathbf{p}$  direction. These are not the only possible photon states; in general, a photon state can be a linear combination of helicity states  $|\mathbf{p}, \pm 1\rangle$

$$\alpha_+ |\mathbf{p}, +1\rangle + \alpha_- |\mathbf{p}, -1\rangle \quad (8.6.13)$$

which is properly normalized if

$$|\alpha_+|^2 + |\alpha_-|^2 = 1. \quad (8.6.14)$$

To calculate the  $S$ -matrix element for absorbing or emitting such a photon, we simply replace  $e_\mu(\mathbf{p}, \pm 1)$  in the Feynman rules with

$$e_\mu(\mathbf{p}) = \alpha_+ e_\mu(\mathbf{p}, +1) + \alpha_- e_\mu(\mathbf{p}, -1). \quad (8.6.15)$$

The polarization vectors for definite helicity satisfy the normalization condition

$$e_\mu^*(\mathbf{p}, \lambda') e^\mu(\mathbf{p}, \lambda) = \delta_{\lambda'\lambda} \quad (8.6.16)$$

and therefore in general

$$e_\mu^*(\mathbf{p}) e^\mu(\mathbf{p}) = 1. \quad (8.6.17)$$

The two extreme cases are *circular polarization*, for which  $\alpha_- = 0$  or  $\alpha_+ = 0$ , and *linear polarization*, for which  $|\alpha_+| = |\alpha_-| = 1/\sqrt{2}$ . For linear polarization, by an adjustment of the overall phase of the state (8.6.13), we can make  $\alpha_+$  and  $\alpha_-$  complex conjugates, so that they can be expressed as

$$\alpha_\pm = \exp(\mp i\phi)/\sqrt{2}. \quad (8.6.18)$$



Then in the Feynman rules we should use a polarization vector

$$e_\mu(\mathbf{p}) = R(\hat{\mathbf{p}}) \begin{bmatrix} 0 \\ \cos \phi \\ \sin \phi \\ 0 \end{bmatrix}. \quad (8.6.19)$$

That is,  $\phi$  is the azimuthal angle of the photon polarization in the plane perpendicular to  $\mathbf{p}$ . Note that the photon polarization vector here is real, which is only possible for linear polarization. In between the extremes of circular and linear polarization are the states of *elliptic polarization*, for which  $|\alpha_+|$  and  $|\alpha_-|$  are non-zero and unequal.

More generally, an initial photon may be prepared in a statistical mixture of spin states. In the most general case, an initial photon may have any number of possible polarization vectors  $e_\mu^{(r)}(\mathbf{p})$ , each with probability  $P_r$ . The rate for absorbing such a photon in a given process will then be of the form

$$\Gamma = \sum_r P_r |e_\mu^{(r)}(\mathbf{p}) M^\mu|^2 = M^{\mu*} M^\nu \rho_{\nu\mu}, \quad (8.6.20)$$

where  $\rho$  is the *density matrix*

$$\rho_{\nu\mu} \equiv \sum_r P_r e_\nu^{(r)}(\mathbf{p}) e_\mu^{(r)*}(\mathbf{p}). \quad (8.6.21)$$

Since  $\rho$  is obviously a Hermitian positive matrix of unit trace (because  $\sum_r P_r = 1$ ) with  $\rho_{\mu 0} = \rho_{0\mu} = 0$  and  $\rho_{\nu\mu} p^\nu = \rho_{\nu\mu} p^\mu = 0$ , it may be written as

$$\rho_{\nu\mu} = \sum_{s=1,2} \lambda_s e_\nu(\mathbf{p}; s) e_\mu^*(\mathbf{p}; s), \quad (8.6.22)$$

where  $e_\mu(\mathbf{p}; s)$  are the two orthonormal eigenvectors of  $\rho$  with

$$e_0(\mathbf{p}; s) = e_\mu(\mathbf{p}; s) p^\mu = 0 \quad (8.6.23)$$

and  $\lambda_s$  are the corresponding eigenvalues, with

$$\lambda_s \geq 0, \quad \sum_{s=1,2} \lambda_s = 1.$$

We may then write the rate for the photon absorption process as

$$\Gamma = \sum_{s=1,2} \lambda_s |e_\nu(\mathbf{p}; s) M^\nu|^2. \quad (8.6.24)$$

Thus any statistical mixture of initial photon states is always equivalent to having just two orthonormal polarizations  $e_\nu(\mathbf{p}; s)$  with probabilities  $\lambda_s$ .

In particular, if we know nothing whatever about the initial photon polarization, then the two probabilities  $\lambda_s$  for the polarization vectors  $e_\nu(\mathbf{p}; s)$  are equal, so that  $\lambda_1 = \lambda_2 = \frac{1}{2}$ ,

and the density matrix (and hence the absorption rate) is an average over initial polarizations

$$\rho_{ij} = \frac{1}{2} \sum_{s=1,2} e_i(\mathbf{p}; s) e_j^*(\mathbf{p}; s) = \frac{1}{2} (\delta_{ij} - \hat{p}_i \hat{p}_j). \quad (8.6.25)$$

Fortunately, this result does not depend on the particular pair of polarization vectors  $e_i(\mathbf{p}; s)$  over which we average; for unpolarized photons we can average the absorption rate over any pair of orthonormal polarization vectors. Similarly, if we make no attempt to measure the polarization of a photon in the final state, then the rate may be calculated by summing over any pair of orthonormal final photon polarization vectors.

The same remarks apply to electrons and positrons; if (as is usually the case) we make no attempt to prepare an electron or positron so that some spin states are more likely than others, then the rate is to be calculated by *averaging* over any two orthonormal initial spin states, such as those with spin  $z$ -component  $\sigma = \pm \frac{1}{2}$ ; if we make no attempt to measure a final electron's or positron's spin state, then we must *sum* the rate over any two orthonormal initial spin states, such as those with spin  $z$ -component  $\sigma = \pm \frac{1}{2}$ . Such sums may be performed using the relations (5.5.37) and (5.5.38):

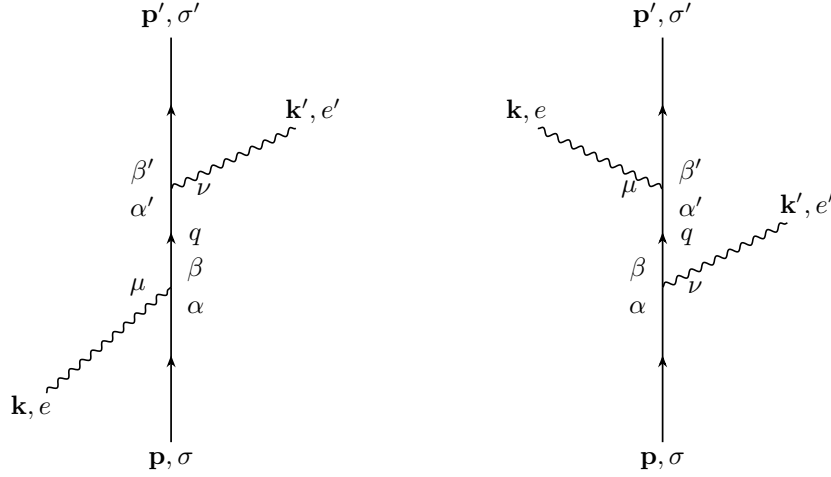
$$\sum_{\sigma} u_{\alpha}(\mathbf{p}, \sigma) \bar{u}_{\beta}(\mathbf{p}, \sigma) = \left( \frac{-i\not{p} + m}{2p^0} \right)_{\alpha\beta}, \quad (8.6.26)$$

$$\sum_{\sigma} v_{\alpha}(\mathbf{p}, \sigma) \bar{v}_{\beta}(\mathbf{p}, \sigma) = \left( \frac{-i\not{p} - m}{2p^0} \right)_{\alpha\beta}, \quad (8.6.27)$$

where  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$ . For instance, if the initial state contains an electron with momentum  $\mathbf{p}$  and spin  $z$ -component  $\sigma$ , and a positron with momentum  $\mathbf{p}'$  and spin  $z$ -component  $\sigma'$ , then the  $S$ -matrix element for the process will be of the form  $\bar{v}_{\alpha}(\mathbf{p}', \sigma') \mathcal{M}_{\alpha\beta} u_{\beta}(\mathbf{p}, \sigma)$ . Hence if neither electron nor positron spins are observed, the rate will be proportional to

$$\frac{1}{4} \sum_{\sigma', \sigma} |\bar{v}_{\alpha}(\mathbf{p}', \sigma') \mathcal{M}_{\alpha\beta} u_{\beta}(\mathbf{p}, \sigma)|^2 = \frac{1}{4} \text{Tr} \left\{ \beta \mathcal{M}^{\dagger} \left( \frac{-i\not{p}' - m}{2p'^0} \right) \mathcal{M} \left( \frac{-i\not{p} + m}{2p^0} \right) \right\}.$$

Techniques for the calculation of such traces are described in the Appendix to this chapter.



**Figure 8.1.** The two lowest-order Feynman diagrams for Compton scattering. Straight lines are electrons; wavy lines are photons.

## 8.7 Compton Scattering

As an example of the methods described in this chapter, we shall consider here the scattering of a photon by an electron (or other particle of spin  $\frac{1}{2}$  and charge  $-e$ ), to lowest order in  $e$ . We label the initial and final photon momenta and polarization vectors by  $k^\mu$ ,  $e^\mu$  and  $k'^\mu$ ,  $e'^\mu$ , where  $k^0 = |\mathbf{k}|$  and  $k'^0 = |\mathbf{k}'|$ . Also, the initial and final electron momenta and spin  $z$ -components are labelled  $p^\mu, \sigma$  and  $p'^\mu, \sigma'$ , where  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$  and  $p'^0 = \sqrt{\mathbf{p}'^2 + m^2}$ , with  $m$  the electron mass. The lowest order Feynman diagrams for this process are shown in Figure 8.1. Using the rules outlined in the previous section, the corresponding  $S$ -matrix element is

$$\begin{aligned}
& S(\mathbf{p}, \sigma + \mathbf{k}, e \rightarrow \mathbf{p}', \sigma' + \mathbf{k}', e') \\
&= \frac{\bar{u}(\mathbf{p}', \sigma')_{\beta'}}{(2\pi)^{3/2}} \frac{e'^*_{\nu}}{(2\pi)^{3/2} \sqrt{2k'^0}} \frac{u(\mathbf{p}, \sigma)_{\alpha}}{(2\pi)^{3/2}} \frac{e_{\mu}}{(2\pi)^{3/2} \sqrt{2k^0}} \int d^4q \left[ \frac{-i}{(2\pi)^4} \right] \left[ \frac{-i\not{q} + m}{q^2 + m^2 - i\epsilon} \right]_{\alpha'\beta} \\
&\cdot \left\{ [e(2\pi)^4 \gamma^{\nu}_{\beta'\alpha'} \delta^4(q - p' - k')] [e(2\pi)^4 \gamma^{\mu}_{\beta\alpha} \delta^4(q - p - k)] \right. \\
&\quad \left. + [e(2\pi)^4 \gamma^{\mu}_{\beta'\alpha'} \delta^4(q + k - p')] [e(2\pi)^4 \gamma^{\nu}_{\beta\alpha} \delta^4(q + k' - p)] \right\}. \tag{8.7.1}
\end{aligned}$$

Performing the (trivial)  $q$ -integral, collecting factors of  $i$  and  $2\pi$ , and rewriting the result in matrix notation, we have more simply

$$\begin{aligned}
S = \frac{-ie^2 \delta^4(p' + k' - p - k)}{(2\pi)^2 \sqrt{2k'^0} \sqrt{2k^0}} \bar{u}(\mathbf{p}', \sigma') & \left[ \not{\epsilon}'^* \frac{-i(\not{p} + \not{k}) + m}{(p + k)^2 + m^2} \not{\epsilon} \right. \\
& \left. + \not{\epsilon} \frac{-i(\not{p} - \not{k}') + m}{(p - k')^2 + m^2} \not{\epsilon}'^* \right] u(\mathbf{p}, \sigma). \tag{8.7.2}
\end{aligned}$$

(Here  $\not{\epsilon}^*$  means  $e_\mu^* \gamma^\mu$ , not  $(\not{\epsilon})^*$ . Also, we drop the  $-i\epsilon$ , because the denominators here do not vanish.) Because  $p^2 = -m^2$  and  $k^2 = k'^2 = 0$ , the denominators can be simplified

$$(p+k)^2 + m^2 = 2p \cdot k, \quad (8.7.3)$$

$$(p-k')^2 + m^2 = -2p \cdot k'. \quad (8.7.4)$$

Also, the 'Feynman amplitude'  $M$  is defined in general by Eq. (3.3.2), which (because some scattering is assumed to take place) here reads

$$S = -2\pi i \delta^4(p' + k' - p - k) M, \quad (8.7.5)$$

so

$$M = \frac{e^2}{4(2\pi)^3 \sqrt{k^0 k'^0}} \bar{u}(\mathbf{p}', \sigma') \left\{ \not{\epsilon}^{*\prime} [-i(\not{p} + \not{k}) + m] \not{\epsilon} / (p \cdot k) - \not{\epsilon} [-i(\not{p} - \not{k}') + m] \not{\epsilon}^{*\prime} / (p \cdot k') \right\} u(\mathbf{p}, \sigma). \quad (8.7.6)$$

The differential cross-section is given in terms of  $M$  by Eq. (3.4.15), which here reads

$$d\sigma = (2\pi)^4 u^{-1} |M|^2 \delta^4(p' + k' - p - k) d^3 p' d^3 k'. \quad (8.7.7)$$

Since one of the particles here is massless, Eq. (3.4.17) for the initial velocity gives

$$u = |p \cdot k| / p^0 k^0. \quad (8.7.8)$$

To go further, it will be convenient to adopt a specific coordinate frame. Since electrons in atoms move non-relativistically, the laboratory frame for high-energy (X ray or gamma ray) photon-electron scattering experiments is usually (though not always) one in which the initial electron can be taken to be at rest. We will adopt this frame here, so that

$$\mathbf{p} = 0, \quad p^0 = m. \quad (8.7.9)$$

The velocity (8.7.8) is then simply

$$u = 1. \quad (8.7.10)$$

To save writing, we denote the photon energies by

$$\omega = k^0 = |\mathbf{k}| = -p \cdot k / m, \quad (8.7.11)$$

$$\omega' = k'^0 = |\mathbf{k}'| = -p \cdot k' / m. \quad (8.7.12)$$

The three-momentum delta function in Eq. (8.7.7) just serves to eliminate the differential  $d^3 p'$ , setting  $\mathbf{p}' = \mathbf{k} - \mathbf{k}'$ . This leaves the remaining energy delta function

$$\delta(p'^0 + k'^0 - p^0 - k^0) = \delta\left(\sqrt{(\mathbf{k} - \mathbf{k}')^2 + m^2} + \omega' - m - \omega\right). \quad (8.7.13)$$

This fixes  $\omega'$  to satisfy

$$\sqrt{\omega^2 - 2\omega\omega' \cos \theta + \omega'^2 + m^2} = \omega + m - \omega',$$

where  $\theta$  is the angle between  $\mathbf{k}$  and  $\mathbf{k}'$ . Squaring both sides and cancelling  $\omega'^2$  terms gives<sup>90 91</sup>

$$\omega' = \omega \frac{m}{m + \omega(1 - \cos \theta)} \equiv \omega_c(\theta). \quad (8.7.14)$$

The energy delta function (8.7.13) can be written

$$\begin{aligned} \delta(p'^0 + k'^0 - p^0 - k^0) &= \frac{\delta(\omega' - \omega_c(\theta))}{|\partial[\sqrt{\omega^2 - 2\omega\omega' \cos \theta + \omega'^2 + m^2} + \omega']/\partial\omega'|} \\ &= \frac{\delta(\omega' - \omega_c(\theta))}{|(\omega' - \omega \cos \theta)/p'^0 + 1|} = \frac{p'^0 \omega'}{m\omega} \delta(\omega' - \omega_c(\theta)). \end{aligned} \quad (8.7.15)$$

Also, the differential  $d^3k'$  can be written

$$d^3k' = \omega'^2 d\omega' d\Omega, \quad (8.7.16)$$

where  $d\Omega$  is the solid angle into which the final photon is scattered. The final delta function in Eq. (8.7.15) just serves to eliminate the differential  $d\omega'$  in Eq. (8.7.16), leaving us with a differential cross-section

$$d\sigma = (2\pi)^4 |M|^2 \frac{p'^0 \omega'^3}{m\omega} d\Omega \quad (8.7.17)$$

with  $p'^0 = m + \omega - \omega'$ , and  $\omega'$  given by Eq. (8.7.14).

Usually we do not measure the spin  $z$ -component of the initial or final electron. In such cases, we must sum over  $\sigma'$  and average over  $\sigma$ , or in other words take half the sum over  $\sigma$  and  $\sigma'$ :

$$d\bar{\sigma}(\mathbf{p} + \mathbf{k}, e \rightarrow \mathbf{p}' + \mathbf{k}', e') \equiv \frac{1}{2} \sum_{\sigma, \sigma'} d\sigma(\mathbf{p}, \sigma + \mathbf{k}, e \rightarrow \mathbf{p}', \sigma' + \mathbf{k}', e'). \quad (8.7.18)$$

To calculate this, we use the standard formula

$$\sum_{\sigma} u_{\alpha}(\mathbf{p}, \sigma) \bar{u}_{\beta}(\mathbf{p}, \sigma) = \frac{(-i\not{p} + m)_{\alpha\beta}}{2p^0} \quad (8.7.19)$$

and likewise for the sum over  $\sigma'$ . It follows that for an arbitrary  $4 \times 4$  matrix  $A$

$$\begin{aligned} \sum_{\sigma, \sigma'} |\bar{u}(\mathbf{p}', \sigma') A u(\mathbf{p}, \sigma)|^2 &= \sum_{\sigma, \sigma'} (\bar{u}' A u) (\bar{u} \beta A^{\dagger} \beta u') \\ &= \text{Tr} \left\{ A \left( \frac{-i\not{p} + m}{2p^0} \right) \beta A^{\dagger} \beta \left( \frac{-i\not{p}' + m}{2p'^0} \right) \right\}. \end{aligned} \quad (8.7.20)$$

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<sup>90</sup>Equivalently, there is an increase in wavelength

$$\frac{1}{\omega'} - \frac{1}{\omega} = \frac{1 - \cos \theta}{m}.$$

<sup>91</sup>The verification of this formula in the scattering of X rays by electrons by A. H. Compton in 1922–3 played a key role in confirming Einstein's 1905 proposal of a quantum of light, which soon after Compton's experiments came to be known as the photon.

Recalling that  $\beta\gamma_\mu^\dagger\beta = -\gamma_\mu$ , Eq. (8.7.6) gives now

$$\sum_{\sigma,\sigma'} |M|^2 = \frac{e^4}{64(2\pi)^6\omega\omega'p^0p'^0} \text{Tr} \left[ \left\{ \not{\epsilon}'^* \frac{[-i(\not{p} + \not{k}) + m]}{p \cdot k} \not{\epsilon} - \not{\epsilon} \frac{[-i(\not{p} - \not{k}') + m]}{p \cdot k'} \not{\epsilon}'^* \right\} (-i\not{p} + m) \right. \\ \left. \cdot \left\{ \not{\epsilon}^* \frac{[-i(\not{p} + \not{k}) + m]}{p \cdot k} \not{\epsilon}' - \not{\epsilon}' \frac{[-i(\not{p} - \not{k}') + m]}{p \cdot k'} \not{\epsilon}^* \right\} (-i\not{p}' + m) \right]. \quad (8.7.21)$$

(Recall again that  $\not{\epsilon}^*$  means  $e_\mu^*\gamma^\mu$ , not  $(e_\mu\gamma^\mu)^*$ , and likewise for  $\not{\epsilon}'^*$ .) We work in a 'gauge' in which

$$e \cdot p = e^* \cdot p = e' \cdot p = e'^* \cdot p = 0, \quad (8.7.22)$$

such as for instance Coulomb gauge in the laboratory frame, where  $e^0 = e'^0 = 0$  and  $\mathbf{p} = 0$ . This implies that

$$[-i\not{p} + m]\not{\epsilon}[-i\not{p} + m] = \not{\epsilon}[i\not{p} + m][-i\not{p} + m] = \not{\epsilon}(p^2 + m^2) = \not{\epsilon}(p_\mu p^\mu + m^2) = 0$$

and likewise for  $\not{\epsilon}^*$ ,  $\not{\epsilon}'$ , and  $\not{\epsilon}'^*$ . Eq. (8.7.21) can therefore be written in the greatly simplified form

$$\sum_{\sigma,\sigma'} |M|^2 = \frac{-e^4}{64(2\pi)^6\omega\omega'p^0p'^0} \text{Tr} \left[ \left\{ \frac{\not{\epsilon}'^* \not{k} \not{\epsilon}}{p \cdot k} + \frac{\not{\epsilon} \not{k}' \not{\epsilon}'^*}{p \cdot k'} \right\} (-i\not{p} + m) \right. \\ \left. \cdot \left\{ \frac{\not{\epsilon}^* \not{k} \not{\epsilon}'}{p \cdot k} + \frac{\not{\epsilon}' \not{k}' \not{\epsilon}^*}{p \cdot k'} \right\} (-i\not{p}' + m) \right]. \quad (8.7.23)$$

The trace of any product of an odd number of gamma matrices vanishes, so this breaks up into terms of zeroth and second order in  $m$ :

$$\sum_{\sigma,\sigma'} |M|^2 = \frac{e^4}{64(2\pi)^6\omega\omega'p^0p'^0} \left( \frac{T_1}{(p \cdot k)^2} + \frac{T_2}{(p \cdot k)(p \cdot k')} + \frac{T_3}{(p \cdot k)(p \cdot k')} + \frac{T_4}{(p \cdot k')^2} \right. \\ \left. - \frac{m^2 t_1}{(p \cdot k)^2} - \frac{m^2 t_2}{(p \cdot k)(p \cdot k')} - \frac{m^2 t_3}{(p \cdot k)(p \cdot k')} - \frac{m^2 t_4}{(p \cdot k')^2} \right), \quad (8.7.24)$$

where

$$T_1 = \text{Tr}\{\not{\epsilon}'^* \not{k} \not{\epsilon} \not{p} \not{\epsilon}^* \not{k}' \not{p}'\}, \quad (8.7.25)$$

$$T_2 = \text{Tr}\{\not{\epsilon}'^* \not{k} \not{\epsilon} \not{p} \not{k}' \not{\epsilon}^* \not{p}'\}, \quad (8.7.26)$$

$$T_3 = \text{Tr}\{\not{\epsilon} \not{k}' \not{\epsilon}'^* \not{p} \not{\epsilon}^* \not{k} \not{p}'\}, \quad (8.7.27)$$

$$T_4 = \text{Tr}\{\not{\epsilon} \not{k}' \not{\epsilon}'^* \not{p} \not{k}' \not{\epsilon}^* \not{p}'\}, \quad (8.7.28)$$

$$t_1 = \text{Tr}\{\not{\epsilon}'^* \not{k} \not{\epsilon} \not{\epsilon}^* \not{k}'\}, \quad (8.7.29)$$

$$t_2 = \text{Tr}\{\not{\epsilon}'^* \not{k} \not{\epsilon} \not{k}' \not{\epsilon}^*\}, \quad (8.7.30)$$

$$t_3 = \text{Tr}\{\not{\epsilon} \not{k}' \not{\epsilon}'^* \not{\epsilon}^* \not{k}\}, \quad (8.7.31)$$

$$t_4 = \text{Tr}\{\not{\epsilon} \not{k}' \not{\epsilon}'^* \not{\epsilon}' \not{k}' \not{\epsilon}^*\}. \quad (8.7.32)$$

The Appendix to this chapter shows how to calculate any trace  $\text{Tr}\{\not{a}\not{b}\not{c}\not{d}\dots\}$  as a sum of products of scalar products of the four-vectors  $a, b, c, d, \dots$ . In general, traces of products of 6 or 8 gamma matrices like the  $t_k$  or  $T_k$  would be given by a sum of 15 or 105 terms, respectively, but fortunately here most scalar products vanish; in addition to Eq. (8.7.22), we also have  $k \cdot k = k' \cdot k' = 0$ . (Furthermore,  $e \cdot e^* = e' \cdot e'^* = 1$ .) To simplify the calculation further, let us specialize to the case of *linear* polarization, where  $e^\mu$  and  $e'^\mu$  are real. Dropping the asterisks in Eqs. (8.7.25)–(8.7.32), we have then

$$T_1 = \text{Tr}\{\not{e}'\not{k}\not{p}\not{e}\not{k}\not{e}'\not{p}'\}.$$

Since  $e^\mu p_\mu = 0$  and  $e^\mu e_\mu = 1$ , we have

$$\not{e}\not{p}\not{e} = -\not{p}\not{e}\not{e} = -\not{p},$$

so

$$T_1 = -\text{Tr}\{\not{e}'\not{k}\not{p}\not{k}\not{e}'\not{p}'\}.$$

Also,  $k^\mu k_\mu = 0$ , so

$$\not{k}\not{p}\not{k} = -\not{k}\not{k}\not{p} + 2\not{k}p \cdot k = 2\not{k}p \cdot k$$

and hence

$$T_1 = -2p \cdot k \text{Tr}\{\not{e}'\not{k}\not{e}'\not{p}'\}.$$

Using Eq. (8.A.6), this is

$$T_1 = -8p \cdot k[2e' \cdot k e' \cdot p' - k \cdot p'].$$

It is convenient to make the substitutions

$$\begin{aligned} e' \cdot p' &= e' \cdot [p + k - k'] = e' \cdot k, \\ k \cdot p' &= -\frac{1}{2}(p' - k)^2 - \frac{1}{2}m^2 = -\frac{1}{2}(p - k')^2 - \frac{1}{2}m^2 = p \cdot k', \end{aligned}$$

so

$$T_1 = -16p \cdot k(e' \cdot k)^2 + 8p \cdot k p \cdot k'. \quad (8.7.33)$$

A similar (though more lengthy) calculation gives

$$\begin{aligned} T_2 = T_3 &= -8(e \cdot k')^2(p \cdot k) + 16(e \cdot e')^2 p \cdot k' p \cdot k + 8(e \cdot e')^2 k \cdot k' m^2 \\ &\quad - 8(e \cdot e')m^2(k \cdot e')(k' \cdot e) + 8(e' \cdot k)^2 p \cdot k' - 4(k \cdot p)^2 \\ &\quad + 4(k \cdot k')(p \cdot p') - 4(k \cdot p')(p \cdot k'), \end{aligned} \quad (8.7.34)$$

$$T_4 = -16p \cdot k'(e \cdot k')^2 + 8(p \cdot k)(p \cdot k'), \quad (8.7.35)$$

$$t_1 = t_4 = 0, \quad (8.7.36)$$

$$t_2 = t_3 = -8(e \cdot e')(k \cdot e')(k' \cdot e) + 8(p \cdot k')(e \cdot e')^2 - 4(k \cdot k'). \quad (8.7.37)$$

Combining all these terms in Eq. (8.7.24) gives

$$\sum_{\sigma, \sigma'} |M|^2 = \frac{e^4}{64(2\pi)^6 \omega \omega' p^0 p'^0} \left[ \frac{8(k \cdot k')^2}{(k \cdot p)(k' \cdot p)} + 32(e \cdot e')^2 \right]. \quad (8.7.38)$$

All this applies in any Lorentz frame. In the laboratory frame, we have the special results

$$k \cdot k' = \omega\omega'(\cos\theta - 1) = m\omega\omega' \left( \frac{1}{\omega} - \frac{1}{\omega'} \right), \quad p \cdot k = -m\omega, \quad p \cdot k' = -m\omega'.$$

Combining Eq. (8.7.38) with Eq. (8.7.17), the laboratory frame cross-section is

$$\frac{1}{2} \sum_{\sigma, \sigma'} d\sigma(\mathbf{p}, \sigma + \mathbf{k}, e \rightarrow \mathbf{p}', \sigma' + \mathbf{k}', e') = \frac{e^4 \omega'^2 d\Omega}{64\pi^2 m^2 \omega^2} \left[ \frac{\omega}{\omega'} + \frac{\omega'}{\omega} - 2 + 4(e \cdot e')^2 \right]. \quad (8.7.39)$$

This is the celebrated formula derived (using old-fashioned perturbation theory) by O. Klein and Y. Nishina [4] in 1929.

As discussed in Section 8.6, if the incoming photon is (as usual) not prepared in a state with any particular polarization, then we must average over two orthonormal values of  $\mathbf{e}$ . This average gives

$$\frac{1}{2} \sum_{\mathbf{e}} e_i e_j = \frac{1}{2} (\delta_{ij} - \hat{k}_i \hat{k}_j)$$

and the differential cross-section is then

$$\frac{1}{4} \sum_{e, \sigma, \sigma'} d\sigma(\mathbf{p}, \sigma + \mathbf{k}, e \rightarrow \mathbf{p}', \sigma' + \mathbf{k}', e') = \frac{e^4 \omega'^2 d\Omega}{64\pi^2 m^2 \omega^2} \left[ \frac{\omega}{\omega'} + \frac{\omega'}{\omega} - 2(\hat{\mathbf{k}} \cdot \mathbf{e}')^2 \right]. \quad (8.7.40)$$

We see that the scattered photon is preferentially polarized in a direction perpendicular to the incident as well as the final photon direction, i.e., perpendicular to the plane in which the scattering takes place. This is a well-known result, responsible among other things for the polarization of light from eclipsing binary stars.<sup>92</sup>

To calculate the cross-section for experiments in which the final photon polarization is not measured, we must sum Eq. (8.7.40) over  $e'$ , using

$$\sum_{e'} e'_i e'_j = \delta_{ij} - \hat{k}'_i \hat{k}'_j.$$

This gives

$$\frac{1}{4} \sum_{e, e', \sigma, \sigma'} d\sigma(\mathbf{p}, \sigma + \mathbf{k}, e \rightarrow \mathbf{p}', \sigma' + \mathbf{k}', e') = \frac{e^4 \omega'^2 d\Omega}{32\pi^2 m^2 \omega^2} \left[ \frac{\omega}{\omega'} + \frac{\omega'}{\omega} - 1 + \cos^2 \theta \right], \quad (8.7.41)$$

where  $\theta$  is the angle between  $\hat{\mathbf{k}}$  and  $\hat{\mathbf{k}}'$ . In the non-relativistic case,  $\omega \ll m$ , Eq. (8.7.41) gives

$$\frac{1}{4} \sum_{e, e', \sigma, \sigma'} d\sigma = \frac{e^4 d\Omega}{32\pi^2 m^2} (1 + \cos^2 \theta). \quad (8.7.42)$$

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<sup>92</sup>The light from one of the stars is polarized when it is scattered by free electrons in the outer atmosphere of the other, cooler, star when both are along the same line of sight. This polarization is normally undetectable because it cancels when the astronomer adds up light from all parts of the star's disk. The polarization has been observed in eclipsing binary stars at times when the cooler star blocks the light from just one side of the hotter star.



The solid angle integral is

$$\int [1 + \cos^2 \theta] d\Omega = \int_0^{2\pi} d\phi \int_0^\pi [1 + \cos^2 \theta] \sin \theta d\theta = \frac{16\pi}{3},$$

giving a total cross-section for  $\omega \ll m$ :

$$\sigma_T = \frac{e^4}{6\pi m^2}. \quad (8.7.43)$$

This is often written  $\sigma_T = 8\pi r_0^2/3$ , where  $r_0 = e^2/4\pi m = 2.818 \cdot 10^{-13}$  cm is known as the *classical electron radius*. Expression (8.7.43) is called the *Thomson cross-section*, after J. J. Thomson, the discoverer of the electron. Eqs. (8.7.42) and (8.7.43) were originally derived using classical mechanics and electrodynamics, by calculating the reradiation of light by a non-relativistic point charge in a plane wave electromagnetic field.

## 8.8 Generalization: $p$ -Form Gauge Fields<sup>93</sup>

The antisymmetric field strength tensor  $F_{\mu\nu}$  of electromagnetism is a special case of a general class of tensors of special importance in physics and mathematics. A  $p$ -form is an antisymmetric covariant tensor of rank  $p$ . From a  $p$ -form  $t_{\mu_1\mu_2\cdots\mu_p}$  one may construct a  $(p+1)$ -form called the *exterior derivative*<sup>94</sup>  $dt$  by taking the derivative and then antisymmetrizing with respect to all indices:

$$\begin{aligned} (dt)_{\mu_1\mu_2\cdots\mu_{p+1}} &= \partial_{[\mu_1} t_{\mu_2\mu_3\cdots\mu_{p+1}]} \\ &\equiv \partial_{\mu_1} t_{\mu_2\mu_3\cdots\mu_{p+1}} - \partial_{\mu_2} t_{\mu_1\mu_3\cdots\mu_{p+1}} + \cdots + (-1)^p \partial_{\mu_{p+1}} t_{\mu_1\mu_2\cdots\mu_p}, \end{aligned} \quad (8.8.1)$$

with square brackets indicating antisymmetrization with respect to the indices within the brackets. Because derivatives commute, repeated exterior derivatives vanish

$$d(dt) = 0. \quad (8.8.2)$$

A  $p$ -form whose exterior derivative vanishes is called *closed*, while a  $p$ -form that is itself an exterior derivative is called *exact*. From Eq. (8.8.2) it follows that any exact  $p$ -form is closed; a famous theorem of Poincaré [6] states that in a simply connected region, any closed  $p$ -form is exact.<sup>95</sup> For instance, the homogeneous Maxwell equations (8.1.16) tell us that the electromagnetic field strength two-form  $F_{\mu\nu}$  is closed; Poincaré's theorem then shows that it is also exact, so that it can be written as an exterior derivative, i.e., as

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<sup>93</sup>This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

<sup>94</sup>Exterior derivatives and  $p$ -forms play a special role in general relativity, in part because the exterior derivative of a tensor transforms like a tensor even though it is calculated using ordinary rather than covariant derivatives [5].

<sup>95</sup>In multiply connected spaces closed forms are not necessarily exact; although it is possible to write a closed  $p$ -form as an exterior derivative locally, this cannot in general be done smoothly throughout the space. The set of closed  $p$ -forms, modulo exact  $p$ -forms, makes up what is called the  $p$ th de Rham cohomology group of the space. There is a deep relation between the de Rham cohomology groups of a space and its topology, which will be discussed further in Volume II.

$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ . Again using Eq. (8.8.2), we see that the two-form  $F_{\mu\nu}$  is invariant if  $A_\mu$  is changed by an exterior derivative, i.e., by a gauge transformation  $\delta A_\mu = \partial_\mu \Omega$ .

The formalism of  $p$ -forms and exterior derivatives makes it natural to consider the possibility of massless particles described by  $p$ -form field gauge fields<sup>96</sup>  $A_{\mu_1 \dots \mu_p}$ , with an invariance under gauge transformations

$$\delta A = d\Omega \quad (8.8.3)$$

or in more detail

$$\delta A_{\mu_1 \dots \mu_p} = \partial_{[\mu_1} \Omega_{\mu_2 \dots \mu_p]},$$

where  $\Omega_{\mu_1 \dots \mu_{p-1}}$  is an arbitrary  $(p-1)$ -form. From such a  $p$ -form gauge field, we can construct a gauge-invariant field strength tensor given by

$$F = dA \quad (8.8.4)$$

or in detail

$$F_{\mu_1 \dots \mu_{p+1}} = \partial_{[\mu_1} A_{\mu_2 \dots \mu_{p+1}]}. \quad (8.8.5)$$

(Alternatively, we can start with a  $(p+1)$ -form  $F$ , and from an assumed condition  $dF = 0$  infer the existence of a  $p$ -form  $A$  with  $F = dA$ .) By analogy with electrodynamics, we might expect the Lagrangian density for  $A$  to take the form

$$\mathcal{L} = -\frac{1}{2(p+1)} F_{\mu_1 \dots \mu_{p+1}} F^{\mu_1 \dots \mu_{p+1}} + J^{\mu_1 \dots \mu_p} A_{\mu_1 \dots \mu_p}, \quad (8.8.6)$$

where  $J$  is an antisymmetric tensor current (either a  $c$ -number, or a function of fields other than  $A$ ) that in order to make the action gauge-invariant must satisfy the conservation condition

$$\partial_{\mu_1} J^{\mu_1 \dots \mu_p} = 0. \quad (8.8.7)$$

The Euler–Lagrange equations are then

$$\partial_{\mu_1} F^{\mu_1 \dots \mu_{p+1}} = -J^{\mu_2 \dots \mu_{p+1}}. \quad (8.8.8)$$

These  $p$ -form gauge fields play an important role in theories with more than four spacetime dimensions. For instance, in the simplest string theories in 26 spacetime dimensions, there is a normal mode of the string represented at low energies by a two-form gauge field  $A_{\mu\nu}$ . But in four spacetime dimensions,  $p$ -forms offer no new possibilities.

To see this, note first that in  $D$  spacetime dimensions there are no antisymmetric tensors with more than  $D$  indices, so in general we must take  $p+1 \leq D$ . Like any other  $(p+1)$ -form with  $p+1 \leq D$ , the field strength  $F$  may be expressed in terms of a dual  $(D-p-1)$ -form  $\mathcal{F}$ , as

$$F^{\mu_1 \dots \mu_{p+1}} = \epsilon^{\mu_1 \dots \mu_D} \mathcal{F}_{\mu_{p+2} \dots \mu_D}. \quad (8.8.9)$$

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<sup>96</sup>We are speaking loosely in calling  $A_{\mu_1 \dots \mu_p}$  a  $p$ -form, because for  $F = dA$  to be a tensor it is only necessary for  $A$  to be a tensor up to a gauge transformation. In fact, we have already seen that in four spacetime dimensions, it is not possible to construct a four-vector field from the creation and annihilation operators of physical massless particles of helicity  $\pm 1$ , so we have to deal with an  $A^\mu(x)$  that according to Eq. (8.1.2) transforms as a four-vector only up to a gauge transformation.

Likewise, the  $p$ -form current  $J$  may be expressed in terms of a dual  $(D - p)$ -form current  $\mathcal{J}$ , as

$$J^{\mu_1 \cdots \mu_p} = \epsilon^{\mu_1 \cdots \mu_D} \mathcal{J}_{\mu_{p+1} \cdots \mu_D}. \quad (8.8.10)$$

The field equation (8.8.8) and conservation condition (8.8.7) then read simply

$$d\mathcal{F} = \mathcal{J}, \quad d\mathcal{J} = 0. \quad (8.8.11)$$

Because the dual current  $\mathcal{J}$  is closed, it may be written in terms of a  $(D - p - 1)$ -form  $\mathcal{S}$  as

$$\mathcal{J} = d\mathcal{S}. \quad (8.8.12)$$

Eqs. (8.8.11) and (8.8.12) tell us that the difference of  $\mathcal{F}$  and  $\mathcal{S}$  is closed, and therefore according to Poincaré's theorem, may be written as

$$\mathcal{F} = \mathcal{S} + d\phi, \quad (8.8.13)$$

with  $\phi$  a  $(D - p - 2)$ -form. There is an exception for the case  $p = D - 1$ , where  $\mathcal{F}$  and  $\mathcal{S}$  are zero-forms, i.e., scalars, and the condition  $d\mathcal{F} = d\mathcal{S}$  simply tells us that  $\mathcal{F}$  and  $\mathcal{S}$  differ only by a constant. In this case the gauge field describes no degrees of freedom at all. We may therefore concern ourselves only with the cases  $p \leq D - 2$ .

For  $p \leq D - 2$ , the homogeneous 'Maxwell' equations  $dF = 0$  read

$$\partial_{\mu_1} \mathcal{F}^{\mu_1 \cdots \mu_{D-p}} = 0, \quad (8.8.14)$$

which with Eq. (8.8.13) yields the field equation for  $\phi$ :

$$\partial_{\mu_1} (d\phi)^{\mu_1 \cdots \mu_{D-p}} = -(d\mathcal{S})^{\mu_2 \cdots \mu_{D-p}}. \quad (8.8.15)$$

This is invariant under a new set of gauge transformations  $\phi \rightarrow \phi + d\omega$ , except that where  $D - p - 2 = 0$  the gauge transformation that leaves  $F$  invariant is  $\phi \rightarrow \phi + c$ , where  $c$  is an arbitrary constant. *We see that in  $D$  spacetime dimensions, the theory of a  $p$ -form gauge field  $A$  with current  $J$  is equivalent to the theory of a  $(D - p - 2)$ -form gauge field  $\phi$ , with current  $-d\mathcal{S}$ .*

We can now understand why  $p$ -form gauge fields offer no new possibilities in four spacetime dimensions. As we have seen, we need only consider the cases  $p \leq D - 2$ , or  $p = 0$ , 1, or 2. A zero-form gauge field is a scalar  $S$ , for which Eq. (8.8.5) reads  $F_\mu = \partial_\mu S$ , and the field equations (8.8.8) read simply  $\square S = -J$ . The gauge invariance here is invariance under a shift  $S \rightarrow S + c$ , with  $c$  a constant. This is just the theory of a massless scalar field with only derivative interactions. A one-form gauge field is a four-vector  $A^\mu(x)$  coupled to a conserved four-vector current, just as in electrodynamics. Finally, according to the general result quoted above, a two-form gauge field in four spacetime dimensions is equivalent to a zero-form gauge field, which as we have seen is equivalent to a massless derivatively coupled scalar field.

## Appendix Traces

In calculating  $S$ -matrix elements and transition rates for processes involving particles of spin  $\frac{1}{2}$ , we often encounter traces of products of Dirac gamma matrices. It will therefore be useful to give formulas for these traces that can be used in all such calculations.

For products of even numbers of gamma matrices, the trace is given by

$$\text{Tr}\{\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_{2N}}\} = 4 \sum_{\text{pairings}} \delta_P \prod_{\text{pairs}} \eta_{\text{paired } \mu s}. \quad (8.A.1)$$

Here the sum is over all different ways of pairing the indices  $\mu_1, \dots, \mu_{2N}$ . A pairing can be regarded as a permutation of the integers  $1, 2, \dots, 2N$  into some order  $P1, P2, \dots, P(2N)$ , in which we pair  $\mu_{P1}$  with  $\mu_{P2}$ ,  $\mu_{P3}$  with  $\mu_{P4}$ , and so on. Permuting pairs or permuting  $\mu s$  within a pair yields the same pairing, so the number of different pairings is

$$\frac{(2N)!}{N!2^N} = (2N-1)(2N-3)\cdots 1 \equiv (2N-1)!! \quad (8.A.2)$$

We can avoid summing over equivalent pairings by requiring that

$$P1 < P2, \quad P3 < P4, \quad \dots, \quad P(2N-1) < P(2N) \quad (8.A.3)$$

and

$$P1 < P3 < P5 < \dots \quad (8.A.4)$$

With this convention, the factor  $\delta_P$  is  $+1$  or  $-1$  according to whether the pairing involves an even or odd permutation of indices. The product in Eq. (8.A.1) is over all  $N$  pairs, the  $n$ th pair contributing a factor  $\eta_{\mu_{P(2n-1)}\mu_{P(2n)}}$ . For instance (writing  $\mu, \nu, \rho, \sigma, \dots$  in place of  $\mu_1, \mu_2, \mu_3, \mu_4, \dots$ ), for  $N = 1, 2$ , and  $3$  we have<sup>97</sup>

$$\text{Tr}\{\gamma_\mu\gamma_\nu\} = 4\eta_{\mu\nu}, \quad (8.A.5)$$

$$\text{Tr}\{\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\} = 4[\eta_{\mu\nu}\eta_{\rho\sigma} - \eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho}], \quad (8.A.6)$$

$$\begin{aligned} \text{Tr}\{\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\gamma_\kappa\gamma_\eta\} = & 4[\eta_{\mu\nu}\eta_{\rho\sigma}\eta_{\kappa\eta} - \eta_{\mu\nu}\eta_{\rho\kappa}\eta_{\sigma\eta} + \eta_{\mu\nu}\eta_{\rho\eta}\eta_{\sigma\kappa} \\ & - \eta_{\mu\rho}\eta_{\nu\sigma}\eta_{\kappa\eta} + \eta_{\mu\rho}\eta_{\nu\kappa}\eta_{\sigma\eta} - \eta_{\mu\rho}\eta_{\nu\eta}\eta_{\sigma\kappa} + \eta_{\mu\sigma}\eta_{\nu\rho}\eta_{\kappa\eta} - \eta_{\mu\sigma}\eta_{\nu\kappa}\eta_{\rho\eta} \\ & + \eta_{\mu\sigma}\eta_{\nu\eta}\eta_{\rho\kappa} - \eta_{\mu\kappa}\eta_{\nu\rho}\eta_{\sigma\eta} + \eta_{\mu\kappa}\eta_{\nu\sigma}\eta_{\rho\eta} - \eta_{\mu\kappa}\eta_{\nu\eta}\eta_{\rho\sigma} + \eta_{\mu\eta}\eta_{\nu\rho}\eta_{\sigma\kappa} \\ & - \eta_{\mu\eta}\eta_{\nu\sigma}\eta_{\rho\kappa} + \eta_{\mu\eta}\eta_{\nu\kappa}\eta_{\rho\sigma}]. \end{aligned} \quad (8.A.7)$$

For an odd number of gamma matrices, the result is much simpler

$$\text{Tr}\{\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_{2N+1}}\} = 0. \quad (8.A.8)$$

The proof of Eq. (8.A.1) is by mathematical induction. First note that

$$\text{Tr}\{\gamma_\mu\gamma_\nu\} = -\text{Tr}\{\gamma_\nu\gamma_\mu\} + 2\text{Tr}\{\eta_{\mu\nu}\mathbf{1}\} = -\text{Tr}\{\gamma_\mu\gamma_\nu\} + 8\eta_{\mu\nu},$$

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<sup>97</sup>There are now computer programs [7] available for the calculation of traces of products of large numbers of Dirac matrices.

so  $\text{Tr}\{\gamma_\mu\gamma_\nu\} = 4\eta_{\mu\nu}$ , in agreement with Eq. (8.A.1). Next, suppose that Eq. (8.A.1) is true for  $N \leq M - 1$ . We then have

$$\begin{aligned}
\text{Tr}\{\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_{2M}}\} &= 2\eta_{\mu_1\mu_2}\text{Tr}\{\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\} - \text{Tr}\{\gamma_{\mu_2}\gamma_{\mu_1}\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\} \\
&= 2\eta_{\mu_1\mu_2}\text{Tr}\{\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\} - 2\eta_{\mu_1\mu_3}\text{Tr}\{\gamma_{\mu_2}\gamma_{\mu_4}\cdots\gamma_{\mu_{2M}}\} \\
&\quad + \text{Tr}\{\gamma_{\mu_2}\gamma_{\mu_3}\gamma_{\mu_1}\gamma_{\mu_4}\cdots\gamma_{\mu_{2M}}\} \\
&= 2\eta_{\mu_1\mu_2}\text{Tr}\{\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\} - 2\eta_{\mu_1\mu_3}\text{Tr}\{\gamma_{\mu_2}\gamma_{\mu_4}\cdots\gamma_{\mu_{2M}}\} \\
&\quad + 2\eta_{\mu_1\mu_4}\text{Tr}\{\gamma_{\mu_2}\gamma_{\mu_3}\gamma_{\mu_5}\cdots\gamma_{\mu_{2M}}\} - \cdots \\
&\quad + 2\eta_{\mu_1\mu_{2M}}\text{Tr}\{\gamma_{\mu_2}\cdots\gamma_{\mu_{2M-1}}\} - \text{Tr}\{\gamma_{\mu_2}\cdots\gamma_{\mu_{2M}}\gamma_{\mu_1}\}.
\end{aligned}$$

All commutators have zero trace, so the last term subtracted here is the same as the left-hand side, and so

$$\begin{aligned}
\text{Tr}\{\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_{2M}}\} &= \eta_{\mu_1\mu_2}\text{Tr}\{\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\} - \eta_{\mu_1\mu_3}\text{Tr}\{\gamma_{\mu_2}\gamma_{\mu_4}\cdots\gamma_{\mu_{2M}}\} \\
&\quad + \eta_{\mu_1\mu_4}\text{Tr}\{\gamma_{\mu_2}\gamma_{\mu_3}\gamma_{\mu_5}\cdots\gamma_{\mu_{2M}}\} - \cdots \\
&\quad + \eta_{\mu_1\mu_{2M}}\text{Tr}\{\gamma_{\mu_2}\cdots\gamma_{\mu_{2M-1}}\}.
\end{aligned} \tag{8.A.9}$$

If we assume that Eq. (8.A.1) correctly gives the trace of any product of  $2N - 2$  Dirac matrices, then Eq. (8.A.9) shows that Eq. (8.A.1) also correctly gives the trace of any product of  $2N$  Dirac matrices.

The easiest way to see that the trace of an odd number of Dirac matrices vanishes is to note that  $-\gamma_\mu$  is related to  $\gamma_\mu$  by a similarity transformation,  $-\gamma_\mu = \gamma_5\gamma_\mu(\gamma_5)^{-1}$ . Traces are unaffected by such similarity transformations, so the trace of an odd number of Dirac matrices is equal to minus itself, and hence vanishes.

One occasionally encounters another class of traces, of the form

$$\text{Tr}\{\gamma_5\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_n}\}.$$

This vanishes for odd  $n$  for the same reason as given above for traces without a  $\gamma_5$ . It also vanishes for  $n = 0$  and  $n = 2$ :

$$\text{Tr}\{\gamma_5\} = 0 \tag{8.A.10}$$

$$\text{Tr}\{\gamma_5\gamma_\mu\gamma_\nu\} = 0. \tag{8.A.11}$$

(To see this just recall that  $\gamma_5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$ , and note that there is no way of pairing the indices in  $\text{Tr}\{\gamma_0\gamma_1\gamma_2\gamma_3\}$  or in  $\text{Tr}\{\gamma_0\gamma_1\gamma_2\gamma_3\gamma_\mu\gamma_\nu\}$  so that the spacetime indices in each pair are equal.) For  $n = 4$  it is possible to pair the indices in  $\text{Tr}\{\gamma_0\gamma_1\gamma_2\gamma_3\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\}$  so that the spacetime indices in each pair are equal, but only if  $\mu, \nu, \rho, \sigma$  are some permutation of  $0, 1, 2, 3$ . Furthermore this trace must be odd under permutations of  $\mu, \nu, \rho, \sigma$  since gamma matrices with different indices anticommute. Thus the trace  $\text{Tr}\{\gamma_5\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\}$  must be proportional to the totally antisymmetric tensor  $\epsilon_{\mu\nu\rho\sigma}$ . The constant of proportionality may be worked out by letting  $\mu, \nu, \rho, \sigma$  take the values  $0, 1, 2, 3$ , and recalling that  $\epsilon_{0123} = -1$ . In this way we find

$$\text{Tr}\{\gamma_5\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\} = 4i\epsilon_{\mu\nu\rho\sigma}. \tag{8.A.12}$$

The trace of products of  $\gamma_5$  with six, eight, or more Dirac matrices may be calculated by the same methods used above to verify Eq. (8.A.1).

### Weinberg Exercises

- W.8.1.** Calculate the differential and total cross-sections for the process  $e^+e^- \rightarrow \mu^+\mu^-$  to lowest order in  $e$ . Assume that electron and muon spins are not observed. Use the simplest Lagrangian for the electrodynamics of electrons and muons.
- W.8.2.** Carry out the canonical quantization of the theory of a charged scalar field  $\Phi$  and its interaction with electromagnetism, with Lagrangian density:

$$\mathcal{L} = -(D_\mu \Phi)^\dagger (D^\mu \Phi) - m^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu},$$

where

$$D_\mu \Phi \equiv \partial_\mu \Phi - iq A_\mu \Phi, \quad F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Use Coulomb gauge. Express the Hamiltonian in terms of the fields  $\mathbf{A}$ ,  $\Phi$ , and  $\Phi^\dagger$  and their canonical conjugates. Evaluate the interaction  $H_I(t)$  in the interaction picture in terms of the interaction-picture fields and their derivatives.

- W.8.3.** Use the results of Problem 2 to calculate the differential and total cross-sections for photon scattering by a massive charged scalar particle to lowest order in  $e$ .
- W.8.4.** Write a gauge-invariant Lagrangian for a charged massive vector field interacting with the electromagnetic field.
- W.8.5.** Calculate the differential cross-section for electron-electron scattering to lowest order in  $e$ . Assume that final and initial spins are not measured.

### Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

- W.8.1.** Write the center-of-mass energy as  $s = 4\omega^2$ , and define

$$\beta_e = \sqrt{1 - \frac{m_e^2}{\omega^2}}, \quad \beta_\mu = \sqrt{1 - \frac{m_\mu^2}{\omega^2}}.$$

At tree level there is one annihilation graph. With the overall phase suppressed, its invariant amplitude is

$$\mathcal{M} = \frac{e^2}{s} \bar{v}(p_2) \gamma^\nu u(p_1) \bar{u}(k_1) \gamma_\nu v(k_2).$$

The completeness relations reduce the unobserved-spin sum to two traces. In the center-of-mass frame the result is

$$|\overline{\mathcal{M}}|^2 = e^4 \left[ 1 + \frac{m_e^2 + m_\mu^2}{\omega^2} + \beta_e^2 \beta_\mu^2 \cos^2 \theta \right].$$

The two-body phase-space formula therefore gives

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4s} \frac{\beta_\mu}{\beta_e} \left[ 1 + \frac{4(m_e^2 + m_\mu^2)}{s} + \left( 1 - \frac{4m_e^2}{s} \right) \left( 1 - \frac{4m_\mu^2}{s} \right) \cos^2 \theta \right], \quad \alpha = \frac{e^2}{4\pi}.}$$

This expression is zero below the muon threshold. Angular integration yields

$$\boxed{\sigma = \frac{4\pi\alpha^2}{3s} \frac{\beta_\mu}{\beta_e} \left[ 1 + \frac{2(m_e^2 + m_\mu^2)}{s} + \frac{4m_e^2 m_\mu^2}{s^2} \right].}$$

For  $m_e \rightarrow 0$ , this reduces to

$$\sigma = \frac{4\pi\alpha^2}{3s} \beta_\mu \left( 1 + \frac{2m_\mu^2}{s} \right).$$

**W.8.2.** Let

$$\pi \equiv \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = (D_0 \Phi)^\dagger, \quad \pi^\dagger \equiv \frac{\partial \mathcal{L}}{\partial \dot{\Phi}^\dagger} = D_0 \Phi.$$

The momentum conjugate to  $A^0$  vanishes. Coulomb gauge and Gauss's law form the electromagnetic constraints,

$$\nabla \cdot \mathbf{A} = 0, \quad -\nabla^2 A^0 = \rho, \quad \rho = iq(\Phi^\dagger \pi^\dagger - \pi \Phi).$$

Thus  $A^0$  is auxiliary:

$$A^0(\mathbf{x}) = \int d^3y \frac{\rho(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|}.$$

Writing  $\mathbf{\Pi}_\perp = \dot{\mathbf{A}}$ , the nonzero independent equal-time brackets are

$$[\Phi(\mathbf{x}), \pi(\mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}), \quad [A_i(\mathbf{x}), \Pi_{\perp j}(\mathbf{y})] = i\delta_{ij}^T(\mathbf{x} - \mathbf{y}),$$

where

$$\delta_{ij}^T(\mathbf{x} - \mathbf{y}) = \int \frac{d^3k}{(2\pi)^3} \left( \delta_{ij} - \frac{k_i k_j}{\mathbf{k}^2} \right) e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})}.$$

The constrained Legendre transform gives

$$H = \int d^3x \left[ \pi^\dagger \pi + (D_i \Phi)^\dagger (D_i \Phi) + m^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 + \frac{1}{2} \mathbf{\Pi}_\perp^2 + \frac{1}{2} \mathbf{B}^2 \right] + \frac{1}{2} \int d^3x d^3y \frac{\rho(\mathbf{x}) \rho(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|}.$$

This is expressed only in the requested canonical variables;  $A^0$  has been eliminated.

For the interaction picture, let  $\phi, \phi^\dagger, \mathbf{a}$  be free fields and set

$$\rho_I = iq(\phi^\dagger \dot{\phi} - \dot{\phi}^\dagger \phi).$$

Expanding the spatial covariant derivatives then gives

$$H_I(t) = \int d^3x \left\{ iq \mathbf{a} \cdot [\phi^\dagger \nabla \phi - (\nabla \phi^\dagger) \phi] + q^2 \mathbf{a}^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2 \right\} \\ + \frac{1}{2} \int d^3x d^3y \frac{\rho_I(\mathbf{x}, t) \rho_I(\mathbf{y}, t)}{4\pi |\mathbf{x} - \mathbf{y}|}.$$

The first line supplies the scalar–scalar–photon vertex, the seagull vertex, and the quartic matter vertex. The second line is the instantaneous Coulomb interaction. As in Section 8.5, its noncovariant contribution combines with transverse-photon exchange to reproduce the covariant photon propagator.

**W.8.3.** Let the initial scalar be at rest, let  $\omega$  and  $\omega'$  be the initial and final photon energies, and let  $\theta$  be their angle. Energy–momentum conservation gives

$$\omega' = \frac{\omega}{1 + (\omega/m)(1 - \cos \theta)}.$$

There are two scalar-exchange graphs and the  $q^2 \mathbf{a}^2 \phi^\dagger \phi$  seagull graph. In a gauge with both photon polarizations orthogonal to the initial scalar momentum, their gauge-invariant sum reduces to

$$\mathcal{M} = 2q^2 e'^* \cdot e.$$

The result in another gauge is identical because replacing either polarization by its photon momentum makes the full three-graph sum vanish. For unobserved photon polarizations,

$$\frac{1}{2} \sum_{e, e'} |e'^* \cdot e|^2 = \frac{1}{2} (1 + \cos^2 \theta).$$

The laboratory two-body phase space therefore gives

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2m^2} \left( \frac{\omega'}{\omega} \right)^2 (1 + \cos^2 \theta).}$$

To integrate, put  $x = \omega/m$  and  $\omega'/\omega = [1 + x(1 - \cos \theta)]^{-1}$ . Then

$$\int_{-1}^1 dc \frac{1 + c^2}{[1 + x(1 - c)]^2} = \frac{4(1 + x)^2}{x^2(1 + 2x)} - \frac{2(1 + x)}{x^3} \ln(1 + 2x).$$

Hence

$$\boxed{\sigma = \frac{\pi \alpha^2}{m^2} \left[ \frac{4(1 + x)^2}{x^2(1 + 2x)} - \frac{2(1 + x)}{x^3} \ln(1 + 2x) \right].}$$

The bracket tends to  $8/3$  as  $x \rightarrow 0$ , so the low-energy limit is the Thomson value  $8\pi\alpha^2/(3m^2)$ .

**W.8.4.** Let  $W_\mu$  be a complex vector field of charge  $q$ , transforming as

$$W_\mu \mapsto e^{iq\Lambda} W_\mu, \quad A_\mu \mapsto A_\mu + \partial_\mu \Lambda.$$



Then

$$D_\mu W_\nu = (\partial_\mu - iqA_\mu)W_\nu, \quad W_{\mu\nu} = D_\mu W_\nu - D_\nu W_\mu$$

both transform covariantly. One gauge-invariant answer is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}W_{\mu\nu}^\dagger W^{\mu\nu} - m^2 W_\mu^\dagger W^\mu.$$

Every Lorentz index is contracted between a field and its conjugate, so the local phase cancels. The mass term is therefore allowed even though a photon mass would not be.

Gauge invariance alone does not make this answer unique. For example the Hermitian, dimension-four term

$$i\kappa q F_{\mu\nu} W^{\mu\dagger} W^\nu$$

may be added with arbitrary real  $\kappa$ , as may gauge-invariant self-interactions. The displayed minimal Lagrangian has the Proca constraint and three physical polarizations. A charged vector arising as a gauge boson of a broken non-Abelian theory has the special magnetic coupling conventionally corresponding to  $\kappa = 1$ .

**W.8.5.** Label the process  $e^-(p_1)e^-(p_2) \rightarrow e^-(p_3)e^-(p_4)$ , and use

$$s = -(p_1 + p_2)^2, \quad t = -(p_1 - p_3)^2, \quad u = -(p_1 - p_4)^2, \quad s + t + u = 4m^2.$$

The two final electrons can be attached in two ways. Fermi statistics fixes their relative minus sign:

$$\mathcal{M} = e^2 \left[ \frac{\bar{u}(p_3)\gamma^\mu u(p_1) \bar{u}(p_4)\gamma_\mu u(p_2)}{t} - \frac{\bar{u}(p_4)\gamma^\mu u(p_1) \bar{u}(p_3)\gamma_\mu u(p_2)}{u} \right].$$

Spin completeness and the four-gamma trace give

$$|\overline{\mathcal{M}}|^2 = 2e^4 \left[ \frac{s^2 + u^2 + 8m^2(t - m^2)}{t^2} + \frac{s^2 + t^2 + 8m^2(u - m^2)}{u^2} + \frac{2(s^2 - 8m^2s + 12m^4)}{tu} \right].$$

Because the final particles are identical, using the full solid angle requires the factor  $1/2!$ :

$$\frac{d\sigma}{d\Omega} = \frac{|\overline{\mathcal{M}}|^2}{128\pi^2 s}.$$

Equivalently, one may omit the factor and retain only one of the two hemispheres. As a useful check, for  $s \gg m^2$  the full-solid-angle answer becomes

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2s} \frac{(3 + \cos^2 \theta)^2}{\sin^4 \theta}.$$

It is symmetric under  $\theta \leftrightarrow \pi - \theta$ , as required when the two outgoing electrons cannot be distinguished.

## Supplementary Exercises

### S.8.1. The Proca Field

(Adapted from University of Cambridge, Mathematical Tripos Part III, Quantum Field Theory, Paper 301, 2024, Question 1.)

In the volume's mostly-plus convention, the Proca Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \frac{1}{2}m^2 A^\mu A_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad m^2 > 0.$$

- (a) Verify that this theory is not invariant under the standard gauge transformation of  $A_\mu$ . Derive the equations of motion and show that they imply

$$(\square - m^2)A^\nu = 0, \quad \partial_\mu A^\mu = 0.$$

- (b) Evaluate the energy-momentum tensor of the Proca theory. Show that its on-shell energy density is positive definite, up to a total spatial derivative.
- (c) Find the plane-wave solutions of the equations from part (a), and show that the massive vector has three polarization states.
- (d) Construct the Proca propagator, giving an integral expression for

$$\Delta_{\mu\nu}(x-y) = \langle \Omega | T(A_\mu(x)A_\nu(y)) | \Omega \rangle.$$

*Hint:* Regard the propagator as a Green function. Write the equation of motion as  $\Pi_{\mu\nu}A^\nu = 0$ , then invert the operator  $\Pi_{\mu\nu}$ .

### S.8.2. Massless QED with a Pauli Interaction

(Adapted from University of Cambridge, Mathematical Tripos Part III, Quantum Field Theory, Paper 301, 2025, Question 1, printed p. 2.)

In four dimensions, let a massless Dirac spinor  $\psi$  couple to a Maxwell field  $A_\mu$ . The mostly-plus translation of the source Lagrangian is

$$\mathcal{L} = -\bar{\psi}\gamma^\mu D_\mu\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\lambda\bar{\psi}[\gamma^\mu, \gamma^\nu]F_{\mu\nu}\psi,$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad D_\mu = \partial_\mu + ieA_\mu,$$

and  $e$  and  $\lambda$  are real. The factors of  $i$  have been translated together with the gamma matrices so that  $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$  and  $\beta = i\gamma^0$ , as elsewhere in this volume.

- (a) Classify  $e$  and  $\lambda$  as relevant, marginal, or irrelevant couplings, and justify the classification by dimensional analysis.
- (b) State the Lorentz transformations of  $\psi(x)$ ,  $\bar{\psi}(x)$ , and  $A_\mu(x)$ . Explain why every term in  $\mathcal{L}$  is Lorentz invariant, and show that the Lagrangian is real up to a total derivative.

- (c) Derive the equations of motion of all the fields.
- (d) Show that the theory has both a local and a global  $U(1)$  symmetry. Derive the global Noether current  $j^\mu$  and charge  $Q$ , and verify directly from the equations of motion that  $Q$  is conserved. Compare  $j^\mu$  with the source in Maxwell's equation, identifying and interpreting every term by which they differ.
- (e) Consider the constant chiral transformation

$$\psi \mapsto e^{i\alpha\gamma_5}\psi, \quad \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3.$$

Under what condition is this a symmetry of the displayed massless theory? Does the same condition suffice if the fermion is massive? Can this symmetry be made local by any choices of  $e$  and  $\lambda$ ? Explain.

### S.8.3. Electron–Positron Annihilation to Muons

(Adapted from Bernhard Mistlberger and Kevin Zhou, *Stanford PHYS 330, Fall 2022, Problem Set 8, Problem 3.*)

One of the key successes of quantum electrodynamics is its description of particle creation and annihilation at relativistic energies. Consider

$$e^+(p_{e+}) + e^-(p_{e-}) \longrightarrow \mu^+(p_{\mu+}) + \mu^-(p_{\mu-}),$$

where the electron and muon are Dirac fields of charge magnitude  $e$ , with masses  $m_e$  and  $m_\mu$ . In the volume's mostly-plus convention define

$$\begin{aligned} s &= -(p_{e+} + p_{e-})^2, \\ t &= -(p_{e-} - p_{\mu-})^2, \\ u &= -(p_{e-} - p_{\mu+})^2. \end{aligned}$$

- (a) Find the scattering amplitude  $\mathcal{M}$  to leading nonvanishing order in  $e$ .
- (b) Let  $|\overline{\mathcal{M}}|^2$  denote the squared amplitude summed over final spins and averaged over initial spins. Compute it in terms of  $e, m_e, m_\mu, s, t, u$ , without neglecting the electron mass.
- (c) For the remainder of the problem work in the center-of-mass frame. Rewrite  $|\overline{\mathcal{M}}|^2$  in terms of  $e, m_e, m_\mu, s$  and the angle  $\theta$  between  $\mathbf{p}_{e-}$  and  $\mathbf{p}_{\mu-}$ .
- (d) Use

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} |\overline{\mathcal{M}}|^2$$

to compute the differential and total cross sections. The displayed two-body formula applies to distinct final particles; for  $n$  identical final particles the fully labelled phase-space integral must be divided by  $n!$ .

For orientation or to check the result, you may consult Peskin and Schroeder, Section 5.1. That discussion neglects the electron mass, whereas every part of this problem requires  $m_e$  to be retained.

### S.8.4. Gauge Symmetry, QED Rules, and Three Tree Processes

(Adapted from *University of Cambridge, Mathematical Tripos Part III, Quantum Field Theory, Paper 301, 2023, Question 4.*)

- (a) What is meant by a gauge symmetry? Write the Lagrangian density for quantum electrodynamics and demonstrate its gauge invariance explicitly. Use the volume's mostly-plus metric and gamma-matrix conventions.
- (b) Write down, without derivation, a complete set of momentum-space Feynman rules for calculating QED scattering amplitudes in a covariant gauge, such as Feynman gauge. State the arrow, external-state, and relative fermion-sign prescriptions as well as the vertex and propagators.
- (c) Draw every tree-level diagram contributing to each of the following two-particle processes:
  - i.  $e^-(p_1, s_1) + e^+(p_2, s_2) \rightarrow e^-(p_3, s_3) + e^+(p_4, s_4)$ ;
  - ii.  $e^-(p_1, s_1) + e^+(p_2, s_2) \rightarrow \gamma(k_1, \lambda_1) + \gamma(k_2, \lambda_2)$ ;
  - iii.  $\gamma(k_1, \lambda_1) + e^-(p_1, s_1) \rightarrow \gamma(k_2, \lambda_2) + e^-(p_2, s_2)$ .

For every initial and final particle, identify the momentum together with the spin component or photon helicity that characterizes its state, and label those quantum numbers on the diagrams.

- (d) Use the rules from part (b) to evaluate the complete tree-level scattering amplitude for each of the three processes. The two diagrams for a given process must be combined with their correct relative sign.

### S.8.5. Canonical Structure of Scalar Electrodynamics

(Adapted from *Daniel Harlow, Lecture Notes for Physics 8.324: Relativistic Quantum Field Theory II, Fall 2024, Section 9.8, Problem 5.*)

Scalar electrodynamics couples a complex scalar field  $\Phi$  to a Maxwell field  $A_\mu$ . In the volume's mostly-plus convention its Lagrangian density is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - (\partial_\mu\Phi^* + iqA_\mu\Phi^*)(\partial^\mu\Phi - iqA^\mu\Phi) - m^2|\Phi|^2.$$

- (a) Determine the current density that appears in Maxwell's equations.
- (b) Derive the equation of motion for  $\Phi$ .
- (c) Find the canonical conjugates of  $\Phi$  and  $\Phi^*$ , and express the full current density in terms of canonical coordinates and momenta.
- (d) Show directly that the action is invariant under the local transformation

$$A'_\mu = A_\mu + \partial_\mu\Omega, \quad \Phi' = e^{iq\Omega}\Phi.$$

### S.8.6. Magnetic Flux, Holonomy, and Dirac Quantization

(Adapted from Daniel Harlow, Lecture Notes for Physics 8.324: Relativistic Quantum Field Theory II (Fall 2024), Section 10.7, Problem 4, printed p. 121.)

One description of a magnetic monopole uses a gauge potential on a two-sphere surrounding the monopole. In polar coordinates  $\theta \in (0, \pi)$  and  $\varphi \in (0, 2\pi)$ , take the one-form

$$A = A_\varphi d\varphi, \quad A_\varphi = -\frac{n}{2e}(1 + \cos \theta).$$

Show that this potential produces magnetic flux through the sphere and that its integral is

$$\int_{S^2} \mathbf{B} \cdot d\mathbf{A} = \int_{S^2} F = \frac{2\pi n}{e}.$$

For a circle at fixed  $\theta$ , calculate the holonomy

$$h(\theta) = \int_0^{2\pi} d\varphi A_\varphi.$$

Does  $h$  vanish when  $\theta = \pi$ ? What happens when  $\theta = 0$ ? In a  $U(1)$  gauge theory, the Wilson line

$$W = e^{ieh}$$

must approach the identity when its circular contour shrinks to zero size. Apply this condition to the circle at  $\theta = 0$  and determine the allowed values of  $n$ .

Here  $A_\varphi$  is the  $\varphi$ -component of the one-form  $A$ , not the component of a spatial vector along the unit vector  $\hat{\varphi}$ . Thus vector-calculus curl formulas require a coordinate rescaling; it is simpler to integrate the two-form  $F = dA$  intrinsically on  $S^2$ .

### S.8.7. The Gupta–Bleuler Physical Space and Residual Gauge Modes

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 11, Problem 1.)

- (a) In covariantly gauge-fixed free electrodynamics, calculate an equal-time commutator of  $A_0$  with  $\partial_\mu A^\mu$ . Explain why its nonzero value forbids imposing the Lorenz condition as an operator identity.
- (b) Separate  $\partial_\mu A^\mu$  into annihilation and creation parts. Formulate the Gupta–Bleuler condition, express it in terms of time-like and longitudinal photon operators, and show why the vacuum satisfies it.
- (c) Form the sum and difference of the time-like and longitudinal photon annihilation operators. Compute all four mixed commutators and identify which creation operator generates the null excitations allowed by the physical-state condition.
- (d) Prove that every Gupta–Bleuler state is a transverse state plus a vector orthogonal to every physical state. Explain why quotienting by these null vectors leaves a positive-definite two-polarization Hilbert space.
- (e) Evaluate the one-particle wavefunction  $\langle \Omega | A_\mu(x) | \mathbf{k}, \text{null} \rangle$ . Show that it is a pure gradient  $\partial_\mu \lambda$ , with  $\square \lambda = 0$ , and interpret the result.

- (f) Replace the null photon of definite momentum by a normalizable wave packet. Construct the corresponding scalar gauge parameter explicitly and prove that the complete wavefunction remains a residual Lorenz-gauge transformation.

### S.8.8. Invariant Flux and Two-Body Phase Space

(Inspired by Hong Liu and MIT OpenCourseWare, 8.323 Assignment 12, Notes 3–5 / Hong Liu and MIT OpenCourseWare, 8.323 Assignment 12, Notes 2–6 / Hong Liu and MIT OpenCourseWare, 8.323 Assignment 12, Notes 2–3.)

- (a) Show that the flux denominator for two incident particles can be written as  $4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}$ . Reduce it in the center-of-mass frame and in the rest frame of the second particle.
- (b) Starting from the Lorentz-invariant final-state measure, perform the energy and radial momentum integrations for a general  $2 \rightarrow 2$  process. Derive the center-of-mass differential cross-section formula.
- (c) Explain why integrating labeled momenta over the full final-state phase space overcounts  $n$  identical particles by  $n!$ . Apply the argument specifically to the two electrons in Møller scattering.

### S.8.9. Relativistic Rutherford Scattering

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 12, Problem 3.)

- (a) Derive the first-Born scattering amplitude and differential cross section for a nonrelativistic particle in a static potential.
- (b) Fourier transform  $V(r) = Z\alpha/r$  and recover the Rutherford cross section.
- (c) Treat the same Coulomb field as an external field coupled to a Dirac electron and compute the spin-averaged matrix element.
- (d) Derive the relativistic Mott factor multiplying Rutherford and check its nonrelativistic limit.

### S.8.10. Electron–Muon Scattering Invariants and Heavy-Target Limit

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 12, Problem 4.)

- (a) Calculate the tree amplitude for  $e^- \mu^- \rightarrow e^- \mu^-$ . Sum and average over spins and express the squared amplitude entirely in terms of  $s, t, u, m_e, m_\mu$ .
- (b) Derive the center-of-mass differential cross section, including the unequal-mass flux and two-body phase space.
- (c) Convert the result to the muon rest frame and then take  $m_\mu \rightarrow \infty$  with the electron energy fixed. Recover the relativistic Rutherford formula and explain the disappearance of target recoil.

### S.8.11. Electric Charge from a Static Potential and Dirac Magnetic Moment

(Adapted from Bernhard Mistlberger and Kevin Zhou, Stanford PHYS 330 Problem Set 8, Problem 1.)

- (a) Evaluate the one-fermion QED matrix element in a weak static scalar potential. Show that the coefficient of  $A^0$  is the electric charge.
- (b) For a uniformly charged, uniformly massive classical sphere rotating slowly, compare magnetic moment with angular momentum and obtain  $g = 1$ .
- (c) Use the Gordon identity to reduce the spatial Dirac current in a static magnetic field and identify the spin magnetic moment.
- (d) Compare it with  $\boldsymbol{\mu} = gq\mathbf{S}/(2m)$  and derive the Dirac value  $g = 2$ .

### S.8.12. Nonminimal Electromagnetic Couplings

(Adapted from Bernhard Mistlberger and Kevin Zhou, Stanford PHYS 330, Fall 2022, Problem Set 8, Problem 4.)

Use the one-particle background-field formalism of the source's Problem 1. In the volume convention, the minimally coupled interaction Hamiltonian for a Dirac field of charge  $q$  is

$$H_I = -iq \int d^3x \bar{\Psi} \gamma^\mu \Psi A_\mu.$$

Take relativistically normalized states

$$|\mathbf{p}, s\rangle = \sqrt{2E_{\mathbf{p}}} a_s^\dagger(\mathbf{p}) |\Omega\rangle,$$

and assume that the electromagnetic field has negligible uncertainty, so that  $A_\mu$  may be replaced by a time-independent classical expectation value  $A_\mu^{\text{cl}}(\mathbf{x})$ . Then

$$\langle \mathbf{p}, s | H_I | \mathbf{p}', s' \rangle = -iq \int d^3x e^{-i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}} \bar{u}_s(p) \gamma^\mu u_{s'}(p') A_\mu^{\text{cl}}(\mathbf{x}).$$

This displayed matrix-element formula is to be used for  $|\mathbf{p}', s'\rangle \neq |\mathbf{p}, s\rangle$ , as in the source setup. For a nonrelativistic particle of charge  $q'$  and magnetic moment  $\boldsymbol{\mu} = gq'\mathbf{S}/(2m)$ ,

$$H_I^{\text{nr}} = q' A_0(\mathbf{x}) - \boldsymbol{\mu} \cdot \mathbf{B}(\mathbf{x}),$$

and matching relativistic to nonrelativistic normalization requires

$$\langle \mathbf{p}, s | H_I | \mathbf{p}', s' \rangle = 2m \langle \mathbf{p}, s | H_I^{\text{nr}} | \mathbf{p}', s' \rangle \quad (|\mathbf{p}|, |\mathbf{p}'| \ll m).$$

With  $\gamma^{\mu\nu} = \frac{1}{2}[\gamma^\mu, \gamma^\nu]$ , the Gordon identity in the volume convention is

$$-i\bar{u}(p')\gamma^\mu u(p) = \frac{1}{2m}\bar{u}(p')[(p' + p)^\mu + \gamma^{\mu\nu}(p' - p)_\nu]u(p).$$

Detailed calculations are not required below; qualitative answers with justification are sufficient.

- (a) The proton and neutron magnetic moments are not given by minimal Dirac coupling. Show that adding a term proportional to

$$i\bar{\Psi}[\gamma^\mu, \gamma^\nu]\Psi F_{\mu\nu}$$

shifts the magnetic dipole moment.

- (b) What is the physical effect of a term proportional to

$$\bar{\Psi}[\gamma^\mu, \gamma^\nu]\gamma_5\Psi F_{\mu\nu}?$$

These are the two dimension-five nonminimal couplings.

- (c) What is the physical effect of the dimension-six term

$$\bar{\Psi}\gamma^\mu\Psi\partial^\nu F_{\mu\nu}?$$

- (d) What is the physical effect of the dimension-six term

$$\bar{\Psi}\gamma^\mu\gamma_5\Psi\partial^\nu F_{\mu\nu}?$$

## Solutions to Supplementary Exercises

### S.8.1. The Proca Field

- (a) The Maxwell term is invariant under  $A_\mu \mapsto A_\mu + \partial_\mu \Lambda$ , but the mass term changes by

$$-\frac{1}{2}m^2[2A^\mu\partial_\mu\Lambda + \partial^\mu\Lambda\partial_\mu\Lambda],$$

which is not a total derivative for a general local  $\Lambda$ . Varying  $A_\nu$  gives

$$\partial_\mu F^{\mu\nu} - m^2 A^\nu = 0.$$

Taking a divergence and using the antisymmetry of  $F^{\mu\nu}$  yields

$$m^2\partial_\nu A^\nu = 0.$$

Since  $m^2 > 0$ , the field is transverse. Expanding the remaining equation then gives

$$\partial_\mu A^\mu = 0, \quad (\square - m^2)A^\nu = 0.$$

- (b) A metric variation gives the symmetric energy-momentum tensor

$$T_{\mu\nu} = F_{\mu\rho}F_{\nu}{}^\rho - \frac{1}{4}\eta_{\mu\nu}F_{\rho\sigma}F^{\rho\sigma} + m^2\left(A_\mu A_\nu - \frac{1}{2}\eta_{\mu\nu}A_\rho A^\rho\right).$$

The canonical Noether tensor differs from this expression, on the equations of motion, by the divergence of a local superpotential. Thus their integrated energies differ only by a spatial boundary term. With  $E_i = F_{0i}$  and  $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$ ,

$$T_{00} = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + \frac{m^2}{2}[(A^0)^2 + \mathbf{A}^2] \geq 0.$$

This is manifestly positive; the qualification about a total spatial derivative is needed only if one starts from the canonical tensor.



(c) For

$$A^\mu(x) = \epsilon^\mu(p) e^{-ip \cdot x},$$

the field equations become

$$p^2 = -m^2, \quad p_\mu \epsilon^\mu(p) = 0.$$

In the rest frame  $p^\mu = (m, \mathbf{0})$ , transversality sets  $\epsilon^0 = 0$ , leaving three independent spatial vectors. For general  $\mathbf{p}$ , choose two orthonormal vectors  $\mathbf{e}_1, \mathbf{e}_2 \perp \mathbf{p}$  and take

$$\epsilon_{1,2}^\mu = (0, \mathbf{e}_{1,2}), \quad \epsilon_L^\mu = \frac{1}{m} (|\mathbf{p}|, p^0 \hat{\mathbf{p}}).$$

All three have unit spacelike norm and are orthogonal to  $p^\mu$ . Their completeness relation is

$$\sum_{\lambda=1}^3 \epsilon_\mu(p, \lambda) \epsilon_\nu(p, \lambda)^* = \eta_{\mu\nu} + \frac{p_\mu p_\nu}{m^2}.$$

(d) Before imposing the constraint, the differential operator is

$$\Pi_{\mu\nu} = (\square - m^2) \eta_{\mu\nu} - \partial_\mu \partial_\nu.$$

In momentum space it becomes

$$\Pi_{\mu\nu}(p) = -(p^2 + m^2) \eta_{\mu\nu} + p_\mu p_\nu.$$

Its inverse is

$$[\Pi^{-1}(p)]_{\mu\nu} = -\frac{\eta_{\mu\nu} + p_\mu p_\nu / m^2}{p^2 + m^2},$$

as direct multiplication verifies. The Feynman prescription therefore gives

$$\Delta_{\mu\nu}(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{-i (\eta_{\mu\nu} + p_\mu p_\nu / m^2)}{p^2 + m^2 - i0} e^{ip \cdot (x-y)}.$$

The numerator is precisely the polarization sum found in part (c).

### S.8.2. Massless QED with a Pauli Interaction

For brevity define

$$C^{\mu\nu} = [\gamma^\mu, \gamma^\nu], \quad B^{\mu\nu} = \bar{\psi} C^{\mu\nu} \psi.$$

(a) In four dimensions,  $[psi] = 3/2$ ,  $[A_\mu] = 1$ , and  $[F_{\mu\nu}] = 2$ . Hence the minimal operator  $\bar{\psi} \gamma^\mu A_\mu \psi$  has dimension four, whereas  $B^{\mu\nu} F_{\mu\nu}$  has dimension five. Therefore

$$[e] = 0 \quad (\text{marginal}), \quad [\lambda] = -1 \quad (\text{irrelevant}).$$

This is the classical or engineering classification; quantum running can make the marginal coupling logarithmically scale dependent.

(b) For  $x' = \Lambda x$ ,

$$\psi'(x') = S(\Lambda)\psi(x), \quad \bar{\psi}'(x') = \bar{\psi}(x)S(\Lambda)^{-1}, \quad A'^{\mu}(x') = \Lambda^{\mu}_{\nu}A^{\nu}(x),$$

where  $S^{-1}\gamma^{\mu}S = \Lambda^{\mu}_{\nu}\gamma^{\nu}$ . Thus  $D_{\mu}\psi$  is a vector-spinor,  $F_{\mu\nu}$  is an antisymmetric tensor, and  $\bar{\psi}C^{\mu\nu}\psi$  is an antisymmetric tensor. Every index in the Lagrangian is contracted.

The volume gamma matrices obey

$$\beta^{-1}(\gamma^{\mu})^{\dagger}\beta = -\gamma^{\mu}, \quad \beta^{-1}(C^{\mu\nu})^{\dagger}\beta = -C^{\mu\nu}.$$

The Maxwell term is real,  $iB^{\mu\nu}F_{\mu\nu}$  is Hermitian, and the gauge interaction is Hermitian because  $\bar{\psi}\gamma^{\mu}\psi$  is anti-Hermitian. Finally,

$$(-\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi)^{\dagger} = -\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi + \partial_{\mu}(\bar{\psi}\gamma^{\mu}\psi).$$

Hence the action is real; the only mismatch in the density is the displayed total derivative.

(c) Independent variation of  $\bar{\psi}$  and  $\psi$  gives

$$\boxed{\gamma^{\mu}D_{\mu}\psi - i\lambda C^{\mu\nu}F_{\mu\nu}\psi = 0,}$$

$$\boxed{(D_{\mu}\bar{\psi})\gamma^{\mu} + i\lambda\bar{\psi}C^{\mu\nu}F_{\mu\nu} = 0,} \quad D_{\mu}\bar{\psi} = \partial_{\mu}\bar{\psi} - ieA_{\mu}\bar{\psi}.$$

Variation of the gauge field, including the derivative inside  $F_{\mu\nu}$ , yields

$$\boxed{\partial_{\mu}F^{\mu\nu} = ie\bar{\psi}\gamma^{\nu}\psi + 2i\lambda\partial_{\mu}(\bar{\psi}C^{\mu\nu}\psi).}$$

The factor two follows from the antisymmetry of  $C^{\mu\nu}$ .

(d) For an arbitrary real function  $\Lambda(x)$ ,

$$A'_{\mu} = A_{\mu} + \partial_{\mu}\Lambda, \quad \psi' = e^{-ie\Lambda}\psi, \quad \bar{\psi}' = \bar{\psi}e^{ie\Lambda}.$$

Then  $D'_{\mu}\psi' = e^{-ie\Lambda}D_{\mu}\psi$ , while  $F'_{\mu\nu} = F_{\mu\nu}$ ; all three terms are locally invariant. Taking  $\Lambda$  constant gives the global symmetry. With the unit-normalized infinitesimal transformation  $\delta\psi = -ie\psi$ , Noether's theorem gives

$$\boxed{j^{\mu} = i\bar{\psi}\gamma^{\mu}\psi, \quad quad Q = \int d^3\mathbf{x} j^0.}$$

Its divergence is

$$\partial_{\mu}j^{\mu} = i[(D_{\mu}\bar{\psi})\gamma^{\mu}\psi + \bar{\psi}\gamma^{\mu}D_{\mu}\psi] = 0,$$

because the two Pauli terms supplied by the conjugate field equations cancel. Thus  $\dot{Q} = 0$  when the spatial flux vanishes at infinity.

The current sourcing Maxwell's equation is

$$J_{\text{Max}}^{\nu} = ej^{\nu} + 2i\lambda\partial_{\mu}B^{\mu\nu}.$$

The second term is an identically conserved magnetization current:  $\partial_{\nu}\partial_{\mu}B^{\mu\nu} = 0$ . Its contribution to the integrated charge is a spatial boundary term, so for localized fields  $\int d^3\mathbf{x} J_{\text{Max}}^0 = eQ$ , even though the local source density is not simply  $ej^0$ .

- (e) Because  $\{\gamma_5, \gamma^\mu\} = 0$ , the massless kinetic and minimal coupling terms are invariant under a constant axial rotation; the adjoint transforms as  $\bar{\psi} \mapsto \bar{\psi} e^{i\alpha\gamma_5}$ . In contrast,  $[\gamma_5, C^{\mu\nu}] = 0$ , so

$$\bar{\psi} C^{\mu\nu} \psi \mapsto \bar{\psi} C^{\mu\nu} e^{2i\alpha\gamma_5} \psi.$$

For a continuous chiral symmetry one must therefore set

$$\boxed{\lambda = 0.}$$

If a mass term is present,  $\bar{\psi}\psi$  acquires the same  $e^{2i\alpha\gamma_5}$  factor, so one also needs  $m = 0$ .

For  $\alpha = \alpha(x)$ , the kinetic term produces

$$\delta\mathcal{L} = -i(\partial_\mu\alpha)\bar{\psi}\gamma^\mu\gamma_5\psi.$$

A variation of the existing vector gauge field couples instead to  $\bar{\psi}\gamma^\mu\psi$ ; adjusting  $e$  or  $\lambda$  cannot cancel the axial current. A new axial gauge field would be required, so no choice of the two displayed couplings makes this chiral symmetry local.

### S.8.3. Electron–Positron Annihilation to Muons

- (a) There is one tree diagram: the electron–positron pair annihilates into a virtual photon, which creates the muon pair. Suppressing a convention-dependent common phase, the amplitude is

$$\boxed{\mathcal{M} = \frac{e^2}{s} [\bar{v}_e(p_{e^+})\gamma^\mu u_e(p_{e^-})] [\bar{u}_\mu(p_{\mu^-})\gamma_\mu v_\mu(p_{\mu^+})].}$$

- (b) The standard spin sums, translated to the volume gamma convention, reduce the product of currents to two Dirac traces. Writing

$$A_t = m_e^2 + m_\mu^2 - t, \quad A_u = m_e^2 + m_\mu^2 - u,$$

the result is

$$\boxed{|\mathcal{M}|^2 = \frac{2e^4}{s^2} [A_t^2 + A_u^2 + 2s(m_e^2 + m_\mu^2)].}$$

Using the four-gamma trace identity and  $s + t + u = 2m_e^2 + 2m_\mu^2$  gives the boxed expression. It reduces to  $2e^4(t^2 + u^2)/s^2$  when both masses vanish.

- (c) Define

$$\beta_e = \sqrt{1 - \frac{4m_e^2}{s}}, \quad \beta_\mu = \sqrt{1 - \frac{4m_\mu^2}{s}}.$$

In the center-of-mass frame,

$$|\mathbf{p}_i| = \frac{\sqrt{s}}{2}\beta_e, \quad |\mathbf{p}_f| = \frac{\sqrt{s}}{2}\beta_\mu,$$

and

$$\begin{aligned} t &= m_e^2 + m_\mu^2 - \frac{s}{2} + \frac{s}{2} \beta_e \beta_\mu \cos \theta, \\ u &= m_e^2 + m_\mu^2 - \frac{s}{2} - \frac{s}{2} \beta_e \beta_\mu \cos \theta. \end{aligned}$$

Substitution gives

$$\boxed{|\mathcal{M}|^2 = e^4 \left[ 3 - \beta_e^2 - \beta_\mu^2 + \beta_e^2 \beta_\mu^2 \cos^2 \theta \right].}$$

(d) The supplied two-body formula now yields

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4s} \frac{\beta_\mu}{\beta_e} \left[ 3 - \beta_e^2 - \beta_\mu^2 + \beta_e^2 \beta_\mu^2 \cos^2 \theta \right]}, \quad \alpha = \frac{e^2}{4\pi}.$$

Using  $\int d\Omega \cos^2 \theta = 4\pi/3$ , the total cross section is

$$\boxed{\sigma(e^+ e^- \rightarrow \mu^+ \mu^-) = \frac{\pi \alpha^2}{s} \frac{\beta_\mu}{\beta_e} \left[ 3 - \beta_e^2 - \beta_\mu^2 + \frac{1}{3} \beta_e^2 \beta_\mu^2 \right].}$$

As a check, when  $m_e \rightarrow 0$ ,

$$\sigma = \frac{4\pi \alpha^2}{3s} \beta_\mu \left( 1 + \frac{2m_\mu^2}{s} \right),$$

the usual result. No identical-particle factor is present because  $\mu^+$  and  $\mu^-$  are distinct.

#### S.8.4. Gauge Symmetry, QED Rules, and Three Tree Processes

- (a) A gauge symmetry is a local redundancy in the field variables: functions related by the gauge transformation describe the same physical configuration. For an electron of charge  $-e$ , define

$$D_\mu = \partial_\mu + ieA_\mu, \quad \mathcal{L}_{\text{QED}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \bar{\Psi}(\gamma^\mu D_\mu + m)\Psi.$$

Under

$$A'_\mu = A_\mu + \partial_\mu \Lambda, \quad \Psi' = e^{-ie\Lambda} \Psi, \quad \bar{\Psi}' = \bar{\Psi} e^{ie\Lambda},$$

one has

$$D'_\mu \Psi' = e^{-ie\Lambda} D_\mu \Psi, \quad F'_{\mu\nu} = F_{\mu\nu}.$$

The phases cancel in the Dirac bilinear, and the Maxwell term is unchanged, which proves the local invariance.

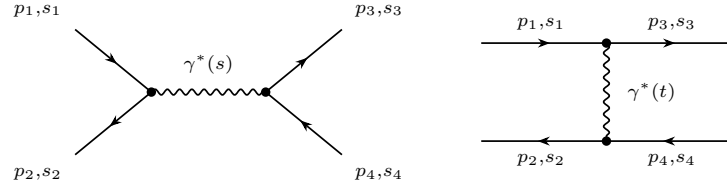
- (b) Suppressing the momentum-conserving delta functions and powers of  $2\pi$ , a consistent set of rules in the volume convention is as follows. Direct each fermion line along its arrow and use

$$\begin{aligned} \text{vertex: } & e\gamma^\mu, \\ \text{fermion line: } & \frac{-i[-i\not{q} + m]}{q^2 + m^2 - i0}, \\ \text{photon line: } & \frac{-i}{q^2 - i0} \left[ \eta_{\mu\nu} + (\xi - 1) \frac{q_\mu q_\nu}{q^2 - i0} \right]. \end{aligned}$$

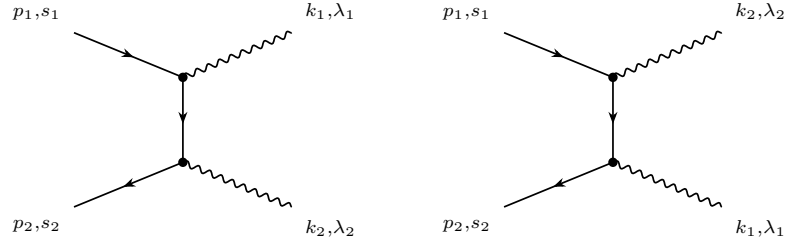
Feynman gauge is  $\xi = 1$ . Attach  $u, \bar{u}$  to incoming and outgoing electrons,  $\bar{v}, v$  to incoming and outgoing positrons, and  $\epsilon_\mu, \epsilon_\mu^*$  to incoming and outgoing photons, respectively. Integrate every undetermined internal momentum. Include a minus sign for each closed fermion loop and for every odd permutation needed to restore a fixed ordering of external fermionic operators. These rules, together with four-momentum conservation at each vertex, are complete for the requested tree graphs.

- (c) Each process has two tree topologies. The labels shown are also the complete one-particle quantum numbers required in the question.

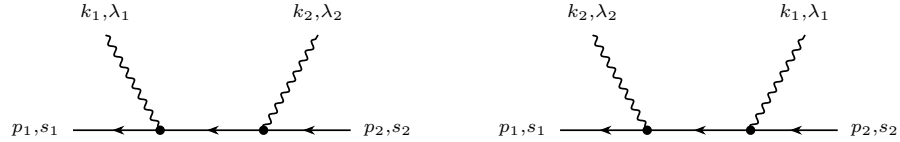
*Bhabha scattering: s-channel annihilation and t-channel exchange*



*Pair annihilation: the two orders of photon emission*



*Positron Compton scattering: the two orders along the positron line*



- (d) It is convenient to strip one common overall phase and define

$$G(q) \equiv \frac{-i\not{q} + m}{q^2 + m^2 - i0}, \quad \not{\epsilon} \equiv \gamma^\mu \epsilon_\mu.$$

For Bhabha scattering, a fixed ordering of the four external fermion operators gives

$$\mathcal{A}_{\text{Bh}} = e^2 \left[ \frac{[\bar{v}_2 \gamma^\mu u_1][\bar{u}_3 \gamma_\mu v_4]}{(p_1 + p_2)^2 - i0} - \frac{[\bar{u}_3 \gamma^\mu u_1][\bar{v}_2 \gamma_\mu v_4]}{(p_1 - p_3)^2 - i0} \right].$$

The relative minus sign is the fermion-interchange sign; a different common ordering changes only an overall sign.

For pair annihilation,  $p_1 + p_2 = k_1 + k_2$ , and the two possible photon orders give

$$\mathcal{A}_{\gamma\gamma} = e^2 \bar{v}_2 [\not{\epsilon}_2^* G(p_1 - k_1) \not{\epsilon}_1^* + \not{\epsilon}_1^* G(p_1 - k_2) \not{\epsilon}_2^*] u_1.$$

The expression is symmetric under interchange of the two final photons, as it must be.

Finally  $p_1 + k_1 = p_2 + k_2$  for positron Compton scattering. Following momentum along the fermion arrow gives

$$\mathcal{A}_{e^+\gamma} = e^2 \bar{v}_1 [\not{\epsilon}_1 G(-p_1 - k_1) \not{\epsilon}_2^* + \not{\epsilon}_2^* G(-p_1 + k_2) \not{\epsilon}_1] v_2.$$

In all three answers the omitted common phase depends only on the convention used to define  $S = \mathbf{1} + iT$ . Replacing either external photon polarization by its momentum makes the two terms cancel after the external Dirac equations are used, providing a check on the relative signs and momentum routing.

### S.8.5. Canonical Structure of Scalar Electrodynamics

Write

$$D_\mu \Phi = (\partial_\mu - iqA_\mu)\Phi, \quad (D_\mu \Phi)^* = (\partial_\mu + iqA_\mu)\Phi^*.$$

(a) Variation with respect to  $A_\mu$  gives

$$\partial_\nu F^{\nu\mu} = j^\mu, \quad j^\mu = iq [\Phi^* D^\mu \Phi - (D^\mu \Phi)^* \Phi].$$

Equivalently,

$$j^\mu = iq (\Phi^* \partial^\mu \Phi - \partial^\mu \Phi^* \Phi) + 2q^2 A^\mu |\Phi|^2.$$

Thus the term quadratic in  $q$ , which would be absent for a fixed background current, is required by the dynamical charged field.

(b) Varying  $\Phi^*$  and integrating the covariant derivative by parts gives

$$(D_\mu D^\mu - m^2)\Phi = 0.$$

The conjugate equation is  $(D_\mu^* D^{*\mu} - m^2)\Phi^* = 0$ . Multiplying these equations by  $\Phi^*$  and  $\Phi$ , subtracting, and using the neutrality of  $j^\mu$  also verifies  $\partial_\mu j^\mu = 0$ .

- (c) Because the mostly-plus scalar kinetic term is  $(D_0\Phi)^*D_0\Phi - (\mathbf{D}\Phi)^* \cdot \mathbf{D}\Phi$ , the canonical momenta are

$$\begin{aligned}\Pi_\Phi &= \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = (D_0\Phi)^* = \dot{\Phi}^* + iqA_0\Phi^*, \\ \Pi_{\Phi^*} &= \frac{\partial \mathcal{L}}{\partial \dot{\Phi}^*} = D_0\Phi = \dot{\Phi} - iqA_0\Phi.\end{aligned}$$

The charge density and spatial current are therefore

$$j^0 = iq(\Pi_\Phi\Phi - \Phi^*\Pi_{\Phi^*}),$$

$$\mathbf{j} = iq[\Phi^*\nabla\Phi - (\nabla\Phi^*)\Phi] + 2q^2\mathbf{A}|\Phi|^2.$$

These expressions contain only the canonical coordinates, their conjugate momenta,  $A_i$ , and spatial derivatives;  $A_0$  has disappeared from the charge density.

- (d) Under the stated transformation,

$$D'_\mu\Phi' = (\partial_\mu - iqA_\mu - iq\partial_\mu\Omega)e^{iq\Omega}\Phi = e^{iq\Omega}D_\mu\Phi.$$

Its conjugate transforms with the inverse phase, while

$$F'_{\mu\nu} = F_{\mu\nu} + \partial_\mu\partial_\nu\Omega - \partial_\nu\partial_\mu\Omega = F_{\mu\nu}.$$

Hence each of the Maxwell, scalar kinetic, and mass terms is separately invariant. The canonical variables transform as

$$\Phi' = e^{iq\Omega}\Phi, \quad \Pi'_\Phi = e^{-iq\Omega}\Pi_\Phi, \quad \Phi'^* = e^{-iq\Omega}\Phi^*, \quad \Pi'_{\Phi^*} = e^{iq\Omega}\Pi_{\Phi^*},$$

so the phase-space form of  $j^\mu$  is gauge invariant as well.

### S.8.6. Magnetic Flux, Holonomy, and Dirac Quantization

The field strength is the intrinsic two-form

$$F = dA = (\partial_\theta A_\varphi) d\theta \wedge d\varphi = \frac{n}{2e} \sin\theta d\theta \wedge d\varphi.$$

It is proportional to the area form on the unit sphere, so the radial magnetic flux density is constant. Its total flux is

$$\int_{S^2} F = \frac{n}{2e} \int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\varphi = \frac{2\pi n}{e}.$$

Along a latitude,  $A_\varphi$  is constant, and hence

$$h(\theta) = -\frac{\pi n}{e}(1 + \cos\theta).$$

At the south pole,  $\theta = \pi$ , this is zero. At the north pole,  $\theta = 0$ , the coordinate circle also shrinks to a point but

$$h(0) = -\frac{2\pi n}{e}.$$

The gauge potential is singular in this patch at that pole; the gauge-invariant Wilson line must nevertheless become the identity:

$$W(0) = e^{ieh(0)} = e^{-2\pi in} = 1.$$

Therefore

$$\boxed{n \in \mathbb{Z}.$$

Equivalently, the magnetic flux is quantized in units of  $2\pi/e$ . The apparent nonzero holonomy of a vanishing contour records the nontrivial transition function between gauge patches, not a physical singularity of the field strength.

### S.8.7. The Gupta–Bleuler Physical Space and Residual Gauge Modes

- (a) In Feynman gauge the four components behave as oscillators with an indefinite metric. The canonical equal-time algebra implies

$$[A_0(t, \mathbf{x}), \partial_\mu A^\mu(t, \mathbf{y})] = \pm i\delta^3(\mathbf{x} - \mathbf{y}),$$

where the displayed sign depends on whether the time index is raised before defining the canonical momentum. The crucial point is that the right-hand side is nonzero. If  $\partial_\mu A^\mu$  were the zero operator, its commutator with every operator would vanish, contradicting this canonical relation. Lorenz gauge must instead be imposed weakly as a condition on physical states.

- (b) Decompose the divergence into annihilation and creation pieces,

$$\partial_\mu A^\mu = (\partial \cdot A)_{\text{ann}} + (\partial \cdot A)_{\text{cre}}.$$

The Gupta–Bleuler condition is

$$(\partial \cdot A)_{\text{ann}}|\psi_{\text{phys}}\rangle = 0.$$

For each momentum, the divergence contains the combination

$$b_+(\mathbf{k}) = \frac{a_L(\mathbf{k}) + a_0(\mathbf{k})}{\sqrt{2}}$$

up to a harmless convention-dependent sign, so the condition is  $b_+(\mathbf{k})|\psi_{\text{phys}}\rangle = 0$ . It eliminates negative-norm excitations without requiring the creation part to annihilate the state. Because every annihilation operator kills the vacuum, the vacuum remains physical. Matrix elements of the full divergence between physical states then vanish.



- (c) Normalize the longitudinal and time-like modes by

$$[a_L, a_L^\dagger] = 1, \quad [a_0, a_0^\dagger] = -1,$$

and define  $b_\pm = (a_L \pm a_0)/\sqrt{2}$ . Direct expansion gives

$$[b_+, b_+^\dagger] = [b_-, b_-^\dagger] = 0, \quad [b_+, b_-^\dagger] = [b_-, b_+^\dagger] = 1,$$

with momentum delta functions understood. Since the physical condition is  $b_+|\psi\rangle = 0$ , a  $b_-^\dagger$  excitation is forbidden: commuting  $b_+$  through it leaves the state below. In contrast,  $b_+^\dagger$  excitations are allowed because their self-commutator is zero. They also have zero norm and are the null photons.

- (d) The oscillator algebra shows that a state satisfying  $b_+|\psi\rangle = 0$  can be built from transverse creation operators and  $b_+^\dagger$ , but not from  $b_-^\dagger$ . Move all  $b_+^\dagger$  factors to one side and write

$$|\psi\rangle = |\psi_T\rangle + |\chi\rangle,$$

where every term in  $|\chi\rangle$  contains at least one  $b_+^\dagger$ . Its inner product with any physical state vanishes: after Hermitian conjugation, a  $b_+$  can be commuted through all transverse and  $b_+^\dagger$  operators without producing a number, and then annihilates the vacuum. Thus the null vectors form a radical of the inner product. The quotient

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_{\text{small}}/\mathcal{N}$$

is represented uniquely by transverse states and has a positive-definite inner product with exactly two photon polarizations.

- (e) For a one-null-photon state, only the annihilation part of the field contributes to  $\langle\Omega|A_\mu(x)|\mathbf{k}, \text{null}\rangle$ . The sum of longitudinal and time-like polarization vectors is proportional to the null momentum:

$$\chi_\mu(x) = C k_\mu e^{ik \cdot x}.$$

Choose

$$\lambda(x) = \frac{C}{i} e^{ik \cdot x}.$$

Then  $\chi_\mu = \partial_\mu \lambda$ , while

$$\square \lambda = -k^2 \lambda = 0.$$

It is precisely a residual gauge transformation preserving  $\partial_\mu A^\mu = 0$ . The null state therefore changes a potential representative but produces no field strength and no physical photon.

- (f) Let

$$|f\rangle = \int d^3k f(\mathbf{k}) |\mathbf{k}, \text{null}\rangle,$$

where  $f$  is square integrable and sufficiently regular. Linearity of the field matrix element gives

$$\chi_\mu^{(f)}(x) = \int d^3k C(\mathbf{k}) f(\mathbf{k}) k_\mu e^{ik \cdot x}.$$

Define

$$\lambda_f(x) = \int d^3k \frac{C(\mathbf{k})f(\mathbf{k})}{i} e^{ik \cdot x}.$$

Every momentum in the integral is null, so  $\square \lambda_f = 0$ , and differentiating under the integral gives  $\chi_\mu^{(f)} = \partial_\mu \lambda_f$ . Superposing residual gauge transformations therefore never creates a physical transverse component.

### S.8.8. Invariant Flux and Two-Body Phase Space

- (a) For mostly-plus momenta  $p_i^2 = -m_i^2$ , the Lorentz scalar

$$\mathcal{F} = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}$$

is the invariant flux. In the center-of-mass frame,  $p_1 = (E_1, \mathbf{p})$  and  $p_2 = (E_2, -\mathbf{p})$ . Using  $E_i^2 = \mathbf{p}^2 + m_i^2$  gives

$$\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} = |\mathbf{p}|(E_1 + E_2) = |\mathbf{p}|\sqrt{s},$$

so  $\mathcal{F} = 4|\mathbf{p}|\sqrt{s}$ . In the rest frame of particle 2,  $p_2 = (m_2, \mathbf{0})$ , and the square root becomes  $m_2|\mathbf{p}_1|$ . Thus  $\mathcal{F} = 4m_2|\mathbf{p}_1|$ , the familiar target-density times incident-velocity normalization in invariant form.

- (b) The invariant measure is

$$d\Phi_2 = (2\pi)^4 \delta^4(P - k_1 - k_2) \frac{d^3k_1}{(2\pi)^3 2E'_1} \frac{d^3k_2}{(2\pi)^3 2E'_2}.$$

In the center-of-mass frame the spatial delta function sets  $\mathbf{k}_2 = -\mathbf{k}_1 \equiv -\mathbf{k}$ . The remaining delta function is  $\delta(\sqrt{s} - E'_1(k) - E'_2(k))$ . Since

$$\frac{d(E'_1 + E'_2)}{dk} = k \left( \frac{1}{E'_1} + \frac{1}{E'_2} \right) = \frac{k\sqrt{s}}{E'_1 E'_2},$$

the radial integral gives

$$d\Phi_2 = \frac{|\mathbf{k}|}{16\pi^2 \sqrt{s}} d\Omega.$$

Combining this with the flux  $4|\mathbf{p}|\sqrt{s}$  yields, for distinguishable final particles,

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{|\mathcal{M}|^2 |\mathbf{k}|}{64\pi^2 s |\mathbf{p}|}}.$$

- (c) The invariant phase-space integral labels the variables  $\mathbf{k}_1, \dots, \mathbf{k}_n$ . If the particles are identical, permuting these labels leaves the physical final state unchanged but generates  $n!$  points in the labeled integration domain. Division by  $n!$  removes the overcounting.

For Møller scattering, the point with one electron at angle  $\theta$  and the other at  $\pi - \theta$  is identical to the point obtained by exchanging the two momenta. One may therefore either integrate the labeled formula over one hemisphere without an extra factor, or integrate over the full solid angle and multiply by  $1/2!$ . Doing both would undercount, while doing neither would double the physical rate.

### S.8.9. Relativistic Rutherford Scattering

- (a) The first Born result is

$$f(\mathbf{q}) = -\frac{\mu_{\text{red}}}{2\pi} \tilde{V}(\mathbf{q}), \quad \frac{d\sigma}{d\Omega} = |f(\mathbf{q})|^2.$$

- (b) For  $V(r) = Z\alpha/r$ ,  $\tilde{V} = 4\pi Z\alpha/\mathbf{q}^2$  and  $\mathbf{q}^2 = 4p^2 \sin^2(\theta/2)$ , yielding

$$\frac{d\sigma_R}{d\Omega} = \frac{Z^2\alpha^2}{4\mu_{\text{red}}^2 v^4 \sin^4(\theta/2)}.$$

- (c) For a static source of charge  $Ze$ ,

$$A^0(\mathbf{q}) = \frac{Ze}{\mathbf{q}^2}, \quad \mathcal{M} = -e \bar{u}(p') \gamma^0 u(p) A^0(\mathbf{q}).$$

The spin average is evaluated with a trace:

$$\frac{1}{2} \sum |\bar{u}(p') \gamma^0 u(p)|^2 = \frac{1}{2} \text{tr}[(\not{p}' + m) \gamma^0 (\not{p} + m) \gamma^0].$$

- (d) Using  $|\mathbf{p}'| = |\mathbf{p}|$  and  $\mathbf{q}^2 = 4|\mathbf{p}|^2 \sin^2(\theta/2)$ , the external-field phase-space normalization gives the Mott formula

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{Z^2\alpha^2}{4|\mathbf{p}|^2 \beta^2 \sin^4(\theta/2)} \left[ 1 - \beta^2 \sin^2 \frac{\theta}{2} \right].}$$

The bracket is the relativistic spin correction. It tends to one for  $\beta \ll 1$ , recovering Rutherford scattering.

### S.8.10. Electron–Muon Scattering Invariants and Heavy-Target Limit

- (a) With  $p_1, p_2$  initial and  $k_1, k_2$  final electron and muon momenta, the one  $t$ -channel graph gives

$$\mathcal{M} = \frac{e^2}{t} \bar{u}(k_1) \gamma^\mu u(p_1) \bar{u}(k_2) \gamma_\mu u(p_2), \quad t = -(p_1 - k_1)^2.$$

Two independent four-gamma traces yield

$$\boxed{|\overline{\mathcal{M}}|^2 = \frac{2e^4}{t^2} [(s - m_e^2 - m_\mu^2)^2 + (u - m_e^2 - m_\mu^2)^2 + 2t(m_e^2 + m_\mu^2)].}$$

Here  $s + t + u = 2m_e^2 + 2m_\mu^2$ . Setting both masses to zero gives the standard check  $2e^4(s^2 + u^2)/t^2$ .

- (b) In the center-of-mass frame,

$$p_* = \frac{\lambda^{1/2}(s, m_e^2, m_\mu^2)}{2\sqrt{s}}, \quad \frac{d\sigma}{d\Omega} = \frac{|\overline{\mathcal{M}}|^2}{64\pi^2 s},$$

because elastic unequal-mass scattering has the same incoming and outgoing  $p_*$ .

- (c) In the muon rest frame, let the incident electron have energy  $E$ , momentum  $p$ , and speed  $\beta = p/E$ . For  $m_\mu \rightarrow \infty$ , the energy transfer to the target is  $O(1/m_\mu)$ , so

$$E' = E, \quad |\mathbf{p}'| = p, \quad t = -4p^2 \sin^2 \frac{\theta}{2}.$$

Substituting

$$s = m_\mu^2 + m_e^2 + 2m_\mu E + O(1), \quad u = m_\mu^2 + m_e^2 - 2m_\mu E - t + O(1)$$

into S.8.10(a) cancels the target-mass factors against the laboratory flux. The remaining expression is

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4p^2 \beta^2 \sin^4(\theta/2)} \left( 1 - \beta^2 \sin^2 \frac{\theta}{2} \right),$$

the  $Z = 1$  Mott–Rutherford result. Infinite target mass removes recoil but retains the electron’s relativistic spin factor.

### S.8.11. Electric Charge from a Static Potential and Dirac Magnetic Moment

- (a) For a weak classical field, the interaction matrix element is

$$\langle p, s | H_I | p', s' \rangle = q \int d^3x e^{-i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}} \bar{u}_s(p) \gamma^\mu u_{s'}(p') A_\mu(\mathbf{x}).$$

Set  $\mathbf{A} = 0$  and take  $|\mathbf{p}|, |\mathbf{p}'| \ll m$ . The temporal bilinear is

$$\bar{u}_s(p) \gamma^0 u_{s'}(p') = 2m \xi_s^\dagger \xi_{s'} + O(p^2/m).$$

After dividing by the  $2m$  that converts relativistic to nonrelativistic state normalization, the Hamiltonian is

$$H_{\text{NR}} = qA^0(\mathbf{x}).$$

Thus the charge measured by a static potential is exactly the coefficient  $q$  in the covariant derivative.

- (b) For a rigid sphere with identical normalized charge and mass profiles, the rotational integrals differ only by  $q/m$ , and the current carries one additional factor  $1/2$ :

$$\boldsymbol{\mu} = \frac{q}{2m} \mathbf{L}.$$

Hence a classical rotating distribution has  $g = 1$ .

- (c) The Gordon identity gives

$$\bar{u}(p') \gamma^\mu u(p) = \bar{u}(p') \left[ \frac{(p' + p)^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p' - p)_\nu}{2m} \right] u(p).$$

For a static magnetic field, the first term supplies orbital minimal coupling. Fourier transforming the second and using  $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}$  gives

$$H_{\text{spin}} = -\frac{q}{m} \mathbf{S} \cdot \mathbf{B}.$$

(d) Comparing with  $-\boldsymbol{\mu} \cdot \mathbf{B}$ , where  $\boldsymbol{\mu} = gq\mathbf{S}/(2m)$ , yields

$$\boxed{g = 2}$$

at tree level. The difference from the classical rigid-body value  $g = 1$  follows from the spinor current rather than a literal rotating charge distribution.

### S.8.12. Nonminimal Electromagnetic Couplings

For compact coefficients define the volume-convention Pauli tensor

$$\sigma^{\mu\nu} \equiv \frac{i}{2}[\gamma^\mu, \gamma^\nu].$$

Thus the operator in part (a) is proportional to  $\bar{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu}$ . Overall phases needed to make each Hamiltonian Hermitian are understood below; they are fixed by the volume's anti-Hermitian vector gamma matrices and do not change the nonrelativistic interpretation.

(a) Choose the conventional normalization

$$\Delta\mathcal{L} = -\frac{\kappa q}{4m} \bar{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu}.$$

For slowly moving positive-energy spinors, the spatial tensor bilinear reduces to the Pauli spin density, while  $F_{ij}$  is the magnetic field. The resulting Hamiltonian term is

$$\Delta H = -\frac{\kappa q}{m} \mathbf{S} \cdot \mathbf{B}.$$

Adding the minimal Dirac term gives

$$\boldsymbol{\mu} = \frac{q}{m}(1 + \kappa)\mathbf{S}, \quad g = 2(1 + \kappa).$$

Thus  $\kappa = (g - 2)/2$  is the anomalous magnetic moment in this normalization. Gauge invariance permits the operator because it contains the invariant field strength.

(b) With a real dipole coefficient  $d$ , a Hermitian convention is

$$\Delta\mathcal{L}_{\text{EDM}} = -\frac{id}{2} \bar{\psi}\sigma^{\mu\nu}\gamma_5\psi F_{\mu\nu}.$$

The dominant nonrelativistic bilinear converts the time-space components of  $F_{\mu\nu}$  into

$$\Delta H_{\text{EDM}} = -d \boldsymbol{\sigma} \cdot \mathbf{E} = -2d \mathbf{S} \cdot \mathbf{E},$$

up to whether  $d$  is defined for  $\boldsymbol{\sigma}$  or  $\mathbf{S}$ . Spin is an axial vector and  $\mathbf{E}$  a polar vector, so their dot product is odd under parity. Spin reverses under time reversal while  $\mathbf{E}$  does not, making it time-reversal odd as well. A permanent EDM therefore signals  $P$  and  $T$ , and under the usual assumptions  $CP$ , violation.

(c) Write the operator as

$$\Delta\mathcal{L} = \frac{c_V}{\Lambda^2} \bar{\psi} \gamma^\mu \psi \partial^\nu F_{\mu\nu}.$$

For a classical external source, Maxwell's equation gives  $\partial^\nu F_{\mu\nu} = j_\mu^{\text{ext}}$ , so

$$\Delta\mathcal{L} = \frac{c_V}{\Lambda^2} (\bar{\psi} \gamma^\mu \psi) j_\mu^{\text{ext}}.$$

It is a local current–current interaction, not a long-range dipole field. In momentum space the photon vertex is proportional to  $q^2 \gamma^\mu - q^\mu \not{q}$ . Against a conserved source the second term vanishes, leaving a contribution proportional to  $q^2$ . It therefore changes the slope of the electric form factor at  $q^2 = 0$ , often interpreted as a charge-radius term, while vanishing for an on-shell real photon.

(d) The axial analogue is

$$\Delta\mathcal{L}_A = \frac{a}{\Lambda^2} \bar{\psi} \gamma^\mu \gamma_5 \psi \partial^\nu F_{\mu\nu} = \frac{a}{\Lambda^2} (\bar{\psi} \gamma^\mu \gamma_5 \psi) j_\mu^{\text{ext}}.$$

For a slowly moving fermion, the spatial axial current is proportional to the spin, so the leading term couples  $\mathbf{S}$  to the external electric current rather than directly to  $\mathbf{E}$  or  $\mathbf{B}$ . The momentum-space vertex contains  $(q^2 \gamma^\mu - q^\mu \not{q}) \gamma_5$ , which vanishes for an on-shell transverse photon. It is therefore neither an electric nor a magnetic dipole and is observable only through sources or virtual photons. The anapole interaction is parity odd and time-reversal even; it is the standard electromagnetic moment associated with parity violation without a permanent EDM.

## References

1. See, e.g., M. B. Green, J. H. Schwarz, and E. Witten, *Superstring Theory* (Cambridge University Press, Cambridge, 1987): Section 2.2.
- 1a. V. Fock, *Z. f. Phys.* **39**, 226 (1927); H. Weyl, *Z. Phys.* **56**, 330 (1929). The term ‘gauge invariance’ derives from an analogy with earlier speculations about scale invariance by H. Weyl, in *Raum, Zeit, Materie*, 3rd ed. (Springer-Verlag, Berlin, 1920). Also see F. London, *Z. f. Phys.* **42**, 375 (1927). This history has been reviewed by C. N. Yang, talk at City College (unpublished).
2. The use of Coulomb gauge in electrodynamics was strongly advocated by Schwinger on pretty much the same grounds as here: that we ought not to introduce photons with helicities other than  $\pm 1$ . See J. Schwinger, *Phys. Rev.* **78**, 1439 (1948); **127**, 324 (1962); *Nuovo Cimento* **30**, 278 (1963).
3. R. P. Feynman, *Phys. Rev.* **101**, 769 (1949): Section 8.
4. O. Klein and Y. Nishina, *Z. f. Phys.* **52**, 853 (1929); Y. Nishina, *ibid.*, 869 (1929); also see I. Tamm, *Z. f. Phys.* **62**, 545 (1930).
5. See, e.g., S. Weinberg, *Gravitation and Cosmology* (Wiley, New York, 1972): Section 4.11.

6. For a readable general introduction to the geometry and topology of  $p$ -forms, see H. Flanders, *Differential Forms* (Academic Press, New York, 1963).
7. T. West, *Comput. Phys. Commun.* **77**, 286 (1993).

## 9 Path-Integral Methods

In Chapters 7 and 8 we applied the canonical quantization operator formalism to derive the Feynman rules for a variety of theories. In many cases, such as the scalar field with derivative coupling or the vector field with zero or non-zero mass, the procedure though straightforward was rather awkward. The interaction Hamiltonian turned out to contain a covariant term, equal to the negative of the interaction term in the Lagrangian, plus a non-covariant term, which served to cancel non-covariant terms in the propagator. In the case of electrodynamics this non-covariant term (the Coulomb energy) turned out to be not even spatially local, though it is local in time. Yet the final results are quite simple: the Feynman rules are just those we should obtain with covariant propagators, and using the negative of the interaction term in the Lagrangian to calculate vertex contributions. The awkwardness in obtaining these simple results, which was bad enough for the theories considered in Chapters 7 and 8, becomes unbearable for more complicated theories, like the non-Abelian gauge theories to be discussed in Volume II, and also general relativity. One would very much prefer a method of calculation that goes directly from the Lagrangian to the Feynman rules in their final, Lorentz-covariant form.

Fortunately, such a method does exist. It is provided by the path-integral approach to quantum mechanics. This was first presented in the context of non-relativistic quantum mechanics in Feynman's Princeton Ph. D. thesis [1], as a means of working directly with a Lagrangian rather than a Hamiltonian. In this respect, it was inspired by earlier work of Dirac [2]. The path-integral approach played a part (along with inspired guesswork) in Feynman's later derivation of his diagrammatic rules [3]. However, although Feynman diagrams became widely used in the 1950s, most physicists (including myself) tended to derive them using the operator methods of Schwinger and Tomonaga, which were shown by Dyson in 1949 to lead to the same diagrammatic rules that had been obtained by Feynman by his own methods.

The path-integral approach was revived in the late 1960s, when Faddeev and Popov [4] and De Witt [5] showed how to apply it to non-Abelian gauge theories and general relativity. For most theorists, the turning point came in 1971, when 't Hooft [6] used path-integral methods to derive the Feynman rules for spontaneously broken gauge theories (discussed in Volume II), including in particular the theory of weak and electromagnetic interactions, in a gauge that made the high energy behavior of these theories transparent. Soon after, as also discussed in Volume II, it was discovered that the path-integral method allows us to take account of contributions to the  $S$ -matrix that have an essential singularity at zero coupling constant and therefore cannot be discovered in any finite order of perturbation theory. Since then, the path-integral methods described here have become an indispensable part of the equipment of all physicists who make use of quantum field theory.

At this point the reader may be wondering why if the path-integral method is so convenient we bothered in Chapter 7 to introduce the canonical formalism. Indeed, Feynman seems at first to have thought of his path-integral approach as a substitute for the ordinary canonical formulation of quantum mechanics. There are two reasons for starting with the canonical formalism. The first is a point of principle: although the path-integral formalism



provides us with manifestly Lorentz-invariant diagrammatic rules, it does not make clear why the  $S$ -matrix calculated in this way is unitary. As far as I know, the only way to show that the path-integral formalism yields a unitary  $S$ -matrix is to use it to reconstruct the canonical formalism, in which unitarity is obvious. There is a kind of conservation of trouble here; we can use the canonical approach, in which unitarity is obvious and Lorentz invariance obscure, or the path-integral approach, which is manifestly Lorentz-invariant but far from manifestly unitary. Since the path-integral approach is here derived from the canonical approach, we know that the two approaches yield the same  $S$ -matrix, so that the  $S$ -matrix must indeed be both Lorentz-invariant and unitary.

The second reason for introducing the canonical formalism first is more practical: there are important theories in which the simplest version of the Feynman path-integral method, in which propagators and interaction vertices are taken directly from the Lagrangian, is simply wrong. One example is the non-linear  $\sigma$ -model, with Lagrangian density  $\mathcal{L} = -\frac{1}{2}g_{k\ell}(\phi)\partial_\mu\phi^k\partial^\mu\phi^\ell$ . In such theories, using the naive Feynman rules derived directly from the Lagrangian density would yield an  $S$ -matrix that is not only wrong but even non-unitary, and that also depends on the way in which we define the scalar field [7]. In this chapter we shall derive the path-integral formalism from the canonical formalism, and in this way we will see what additional sorts of vertices are needed to supplement the simplest version of the Feynman path-integral method.

## 9.1 The General Path-Integral Formula

We start with a general quantum mechanical system, with Hermitian operator ‘coordinates’  $Q_a$  and conjugate ‘momenta’  $P_b$ , satisfying the canonical commutation relations<sup>98</sup>:

$$[Q_a, P_b] = i\delta_{ab}, \quad (9.1.1)$$

$$[Q_a, Q_b] = [P_a, P_b] = 0. \quad (9.1.2)$$

(We shall restrict ourselves in this and the next three sections to bosonic operators, which satisfy commutation rather than anticommutation relations. Our results will be generalized to include fermionic operators in Section 9.5.) In a field theory, the index  $a$  consists of a position  $\mathbf{x}$  and a discrete Lorentz and species index  $m$ , and we conventionally write

$$Q_{\mathbf{x},m} \equiv Q_m(\mathbf{x}), \quad (9.1.3)$$

$$P_{\mathbf{x},m} \equiv P_m(\mathbf{x}). \quad (9.1.4)$$

Also, the Kronecker delta in Eq. (9.1.1) is interpreted in a field theory as

$$\delta_{\mathbf{x},m;\mathbf{y},n} \equiv \delta^3(\mathbf{x} - \mathbf{y})\delta_{mn}. \quad (9.1.5)$$

However, for the present it will be convenient to use the more compact notation of Eqs. (9.1.1) and (9.1.2). These are ‘Schrödinger-picture’ operators, taken at a fixed time (say,

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<sup>98</sup>We are tacitly assuming here that any first class constraints are eliminated by choosing a gauge, and any remaining second class constraints are ‘solved’ by writing the constrained degrees of freedom in terms of the unconstrained  $Q_a$  and  $P_a$ , as in Section 7.6. The direct application of path-integral methods to constrained systems is described by Faddeev [9].

$t = 0$ ). The time-dependent operators in the Heisenberg picture will be considered a little later.

Since the  $Q_a$  all commute, we can find a simultaneous eigenstate  $|q\rangle$ , with eigenvalues  $q_a$ :

$$Q_a|q\rangle = q_a|q\rangle. \quad (9.1.6)$$

(We are using lower case  $qs$  and  $ps$  here to denote eigenvalues rather than to denote operators in the interaction picture as in Chapter 7, but since we will not be using the interaction picture in this chapter no confusion should arise.) The eigenvectors can be taken to be orthonormal,

$$\langle q'|q\rangle = \prod_a \delta(q'_a - q_a) \equiv \delta(q' - q), \quad (9.1.7)$$

so that the completeness relation reads

$$\mathbf{1} = \int \prod_a dq_a |q\rangle \langle q|. \quad (9.1.8)$$

Similarly, we can find a complete orthonormal set of eigenstates of the  $P_a$ :

$$P_a|p\rangle = p_a|p\rangle, \quad (9.1.9)$$

$$\langle p'|p\rangle = \prod_a \delta(p'_a - p_a) \equiv \delta(p' - p), \quad (9.1.10)$$

$$\mathbf{1} = \int \prod_a dp_a |p\rangle \langle p|. \quad (9.1.11)$$

As usual, it follows from Eq. (9.1.1) that these two complete sets of eigenstates have the scalar product<sup>99</sup>

$$\langle q|p\rangle = \prod_a \frac{1}{\sqrt{2\pi}} \exp(iq_a p_a). \quad (9.1.12)$$

In the Heisenberg picture, the  $Q$  and  $P$  operators are given a time-dependence

$$Q_a(t) = \exp(iHt)Q_a \exp(-iHt), \quad (9.1.13)$$

$$P_a(t) = \exp(iHt)P_a \exp(-iHt), \quad (9.1.14)$$

where  $H$  is the total Hamiltonian. These have eigenstates  $|q; t\rangle$  and  $|p; t\rangle$

$$Q_a(t)|q; t\rangle = q_a|q; t\rangle, \quad (9.1.15)$$

$$P_a(t)|p; t\rangle = p_a|p; t\rangle, \quad (9.1.16)$$

given by

$$|q; t\rangle = \exp(iHt)|q\rangle, \quad (9.1.17)$$

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<sup>99</sup>The proof follows the same lines as in the quantum mechanics of point particles. From Eq. (9.1.1), we see that  $P_a$  acts as  $-i\partial/\partial q_a$  on wave functions in a  $q$ -basis. The right-hand side of Eq. (9.1.12) is then seen to be the wave function in this basis of an eigenstate of  $P$ . The factor  $\prod_a 1/\sqrt{2\pi}$  is fixed by the normalization requirement, Eq. (9.1.10).

$$|p; t\rangle = \exp(iHt)|p\rangle. \quad (9.1.18)$$

(Note that  $|q; t\rangle$  is the eigenstate of  $Q_a(t)$  with eigenvalue  $q_a$ , and *not* the result of letting the state  $|q\rangle$  evolve for a time  $t$ . This is why its time-dependence is given by a factor  $\exp(iHt)$  rather than  $\exp(-iHt)$ .) These states obviously satisfy the completeness and orthonormality conditions

$$\langle q'; t|q; t\rangle = \delta(q' - q), \quad (9.1.19)$$

$$\langle p'; t|p; t\rangle = \delta(p' - p), \quad (9.1.20)$$

$$\int \prod_a dq_a |q; t\rangle \langle q; t| = \mathbf{1}, \quad (9.1.21)$$

$$\int \prod_a dp_a |p; t\rangle \langle p; t| = \mathbf{1}, \quad (9.1.22)$$

and also

$$\langle q; t|p; t\rangle = \prod_a \frac{1}{\sqrt{2\pi}} \exp(iq_a p_a). \quad (9.1.23)$$

If, by measurements at time  $t$ , we find that our system is in a definite state  $|q; t\rangle$ , then the probability amplitude for measurements at time  $t'$  to give a state  $|q'; t'\rangle$  is the scalar product  $\langle q'; t'|q; t\rangle$ . Our central dynamical problem is to calculate this scalar product.

This is easy when  $t'$  and  $t$  are infinitesimally close, say  $t' = \tau + d\tau$  and  $t = \tau$ . Using Eq. (9.1.17), we have

$$\langle q'; \tau + d\tau|q; \tau\rangle = \langle q'; \tau|\exp(-iH d\tau)|q; \tau\rangle. \quad (9.1.24)$$

The Hamiltonian  $H$  is given as a function  $H(Q, P)$ , but since (9.1.13) and (9.1.14) are similarity transformations, and  $H$  commutes with itself, it can equally well be written as the *same* function of  $Q(t)$  and  $P(t)$

$$H \equiv H(Q, P) = e^{iHt} H(Q, P) e^{-iHt} = H(Q(t), P(t)). \quad (9.1.25)$$

This function can be written in various different forms, with different constant coefficients, by using the commutation relations (9.1.1) and (9.1.2) to move the  $Q$ s and  $P$ s past each other. It will be convenient to adopt a standard form, in which all  $Q$ s appear to the *left* of all  $P$ s. For instance, given a term in the Hamiltonian of form  $P_a Q_b P_c$ , we would rewrite it as  $P_a Q_b P_c = Q_b P_a P_c - i\delta_{ab} P_c$ . With this convention, the  $Q_a(t)$ s in the Hamiltonian in Eq. (9.1.24) may be replaced<sup>100</sup> with their eigenvalues  $q'_a$ . To deal with the  $P(t)$ , we use Eq. (9.1.23) to expand  $|q; \tau\rangle$  in  $P$ -eigenstates  $|p; \tau\rangle$ , and find

$$\begin{aligned} \langle q'; \tau + d\tau|q; \tau\rangle &= \int \prod_a dp_a \langle q'; \tau|\exp(-iH(Q(\tau), P(\tau))d\tau)|p; \tau\rangle \langle p; \tau|q; \tau\rangle \\ &= \int \prod_a \frac{dp_a}{2\pi} \exp\left[-iH(q', p)d\tau + i\sum_a (q'_a - q_a)p_a\right], \end{aligned} \quad (9.1.26)$$

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<sup>100</sup>This is only possible because with  $d\tau$  infinitesimal,  $\exp(-iH d\tau)$  is linear in  $H$ .

with each  $p_a$  integrated from  $-\infty$  to  $+\infty$ .

Now let's return to the more general case of a finite time-interval. To calculate  $\langle q'; t' | q; t \rangle$ , with  $t < t'$ , we break up the time-interval from  $t$  to  $t'$  into steps  $t, \tau_1, \tau_2, \dots, \tau_N, t'$ , with

$$\tau_{k+1} - \tau_k = d\tau = (t' - t)/(N + 1) \quad (9.1.27)$$

and sum over a complete set of states  $|q; \tau_k\rangle$  at each time  $\tau_k$ :

$$\langle q'; t' | q; t \rangle = \int dq_1 \cdots dq_N \langle q'; t' | q_N; \tau_N \rangle \langle q_N; \tau_N | q_{N-1}; \tau_{N-1} \rangle \cdots \langle q_1; \tau_1 | q; t \rangle. \quad (9.1.28)$$

Inserting Eq. (9.1.26), this becomes

$$\begin{aligned} \langle q'; t' | q; t \rangle = & \int \left[ \prod_{k=1}^N \prod_a dq_{k,a} \right] \left[ \prod_{k=0}^N \prod_a \frac{dp_{k,a}}{2\pi} \right] \\ & \cdot \exp \left[ i \sum_{k=1}^{N+1} \left\{ \sum_a (q_{k,a} - q_{k-1,a}) p_{k-1,a} - H(q_k, p_{k-1}) d\tau \right\} \right], \end{aligned} \quad (9.1.29)$$

where

$$q_0 \equiv q, \quad q_{N+1} \equiv q'. \quad (9.1.30)$$

Our result, Eq. (9.1.29), can be put in a much more elegant form. Define smooth interpolating functions,  $q(\tau)$  and  $p(\tau)$ , such that

$$q_a(\tau_k) = q_{k,a}, \quad p_a(\tau_k) = p_{k,a}. \quad (9.1.31)$$

In the limit  $d\tau \rightarrow 0$  (i.e.,  $N \rightarrow \infty$ ), the argument of the exponential in Eq. (9.1.29) becomes just an integral over  $\tau$

$$\begin{aligned} & \sum_{k=1}^{N+1} \left\{ \sum_a (q_{k,a} - q_{k-1,a}) p_{k-1,a} - H(q_k, p_{k-1}) d\tau \right\} \\ &= \sum_{k=1}^{N+1} \left\{ \sum_a \dot{q}_a(\tau_k) p_a(\tau_k) - H(q(\tau_k), p(\tau_k)) \right\} d\tau + O(d\tau^2) \\ &\longrightarrow \int_t^{t'} d\tau \left\{ \sum_a \dot{q}_a(\tau) p_a(\tau) - H(q(\tau), p(\tau)) \right\}. \end{aligned} \quad (9.1.32)$$

Further, we may define integrals *over the functions*  $q(\tau), p(\tau)$  by

$$\int \prod_{\tau,a} dq_a(\tau) \prod_{\tau,b} \frac{dp_b(\tau)}{2\pi} \cdots \equiv \lim_{d\tau \rightarrow 0} \int \prod_{k,a} dq_{k,a} \prod_{k,b} \frac{dp_{k,b}}{2\pi} \cdots. \quad (9.1.33)$$

Eq. (9.1.29) then becomes a constrained path integral

$$\begin{aligned} \langle q'; t' | q; t \rangle = & \int_{\substack{q_a(t)=q_a \\ q_a(t')=q'_a}} \prod_{\tau,a} dq_a(\tau) \prod_{\tau,b} \frac{dp_b(\tau)}{2\pi} \\ & \cdot \exp \left[ i \int_t^{t'} d\tau \left\{ \sum_a \dot{q}_a(\tau) p_a(\tau) - H(q(\tau), p(\tau)) \right\} \right]. \end{aligned} \quad (9.1.34)$$

This is called a *path integral*, because we integrate over all paths that take  $q(\tau)$  from  $q$  at  $\tau = t$  to  $q'$  at  $\tau = t'$ , as well as over all  $p(\tau)$ . The great advantage of writing matrix elements in this way is that, as shown in Section 9.3, the path integrals are easy to calculate when expanded in powers of the coupling constants in  $H$ .

The path-integral formalism allows us to calculate not only transition probability amplitudes like  $\langle q'; t' | q; t \rangle$ , but also the matrix elements between states  $\langle q'; t' |$  and  $| q; t \rangle$  of time-ordered products of general operators  $\mathcal{O}(P(t), Q(t))$ . It will be convenient to define these operators with (unlike  $H$ ) all  $P$ s moved to the *left* and all  $Q$ s to the *right*. Then by inserting any such operator  $\mathcal{O}[P(t), Q(t)]$  in Eq. (9.1.26), we have

$$\begin{aligned} & \langle q'; \tau + d\tau | \mathcal{O}(P(t), Q(t)) | q; \tau \rangle \\ &= \int \prod_a dp_a \langle q'; \tau | \exp(-iH(Q(\tau), P(\tau))d\tau) | p; \tau \rangle \langle p; \tau | \mathcal{O}(P(t), Q(t)) | q; \tau \rangle \\ &= \int \prod_a \frac{dp_a}{2\pi} \exp \left[ -iH(q', p)d\tau + i \sum_a (q'_a - q_a)p_a \right] \mathcal{O}(p, q). \end{aligned} \quad (9.1.35)$$

In order to calculate the matrix element of the operator product

$$\mathcal{O}_A(P(t_A), Q(t_A)) \mathcal{O}_B(P(t_B), Q(t_B)) \cdots$$

with  $t_A > t_B > \cdots$ , we can insert the  $\mathcal{O}$ -operators between the appropriate states in Eq. (9.1.28), and use Eq. (9.1.35). For instance, if the time  $t_A$  falls between  $\tau_k$  and  $\tau_{k+1}$ , then insert  $\mathcal{O}_A(P(t_A), Q(t_A))$  between  $\langle q_{k+1}; \tau_{k+1} |$  and  $| q_k; \tau_k \rangle$ . Note that in Eq. (9.1.28) each successive sum over states is at a later time, so this is only possible because of our assumption that  $t_A > t_B > \cdots$ . Following the same steps as before, we now find the general path-integral formula

$$\begin{aligned} & \langle q'; t' | \mathcal{O}_A(P(t_A), Q(t_A)) \mathcal{O}_B(P(t_B), Q(t_B)) \cdots | q; t \rangle \\ &= \int_{\substack{q_a(t)=q_a \\ q_a(t')=q'_a}} \prod_{\tau, a} dq_a(\tau) \prod_{\tau, b} \frac{dp_b(\tau)}{2\pi} \mathcal{O}_A(p(t_A), q(t_A)) \mathcal{O}_B(p(t_B), q(t_B)) \cdots \\ & \quad \cdot \exp \left[ i \int_t^{t'} d\tau \left\{ \sum_a \dot{q}_a(\tau) p_a(\tau) - H(q(\tau), p(\tau)) \right\} \right]. \end{aligned} \quad (9.1.36)$$

This result is only valid if the times are ordered, with

$$t' > t_A > t_B > \cdots > t. \quad (9.1.37)$$

However, nothing on the right-hand side of Eq. (9.1.36) refers to the order of time-arguments. Hence if we are presented with a path integral like the right-hand side of Eq. (9.1.36), with  $t_A, t_B, \dots$  in arbitrary order (all between  $t$  and  $t'$ , with  $t < t'$ ), then this path integral will equal a matrix element like the left-hand side of Eq. (9.1.36), but with the operators arranged in order (from left to right) of decreasing time. That is, for

$t_A, t_B, \dots$  in arbitrary order, we have

$$\begin{aligned}
& \langle q'; t' | T \{ \mathcal{O}_A(P(t_A), Q(t_A)), \mathcal{O}_B(P(t_B), Q(t_B)), \dots \} | q; t \rangle \\
&= \int_{\substack{q_a(t)=q_a \\ q_a(t')=q'_a}} \prod_{\tau, a} dq_a(\tau) \prod_{\tau, b} \frac{dp_b(\tau)}{2\pi} \mathcal{O}_A(p(t_A), q(t_A)) \mathcal{O}_B(p(t_B), q(t_B)) \cdots \\
&\quad \cdot \exp \left[ i \int_t^{t'} d\tau \left\{ \sum_a \dot{q}_a(\tau) p_a(\tau) - H(q(\tau), p(\tau)) \right\} \right], \tag{9.1.38}
\end{aligned}$$

where  $T$  denotes the usual time-ordered product.

It should perhaps be stressed that the c-number functions  $q_a(\tau)$ ,  $p_a(\tau)$  in Eq. (9.1.38) are mere variables of integration, and in particular are *not* constrained to obey the equations of motion of classical Hamiltonian dynamics

$$\dot{q}_a(\tau) - \frac{\partial H(q(\tau), p(\tau))}{\partial p_a(\tau)} = 0, \tag{9.1.39}$$

$$\dot{p}_a(\tau) + \frac{\partial H(q(\tau), p(\tau))}{\partial q_a(\tau)} = 0. \tag{9.1.40}$$

(For this reason, the Hamiltonian  $H(q(\tau), p(\tau))$  in Eq. (9.1.38) is *not* constant in  $\tau$ .) Nevertheless, there is a limited sense in which path integrals do respect these equations of motion. Suppose that one of the functions in Eq. (9.1.38), say  $\mathcal{O}_A(p(t_A), q(t_A))$ , happens to be the left-hand side of either Eq. (9.1.39) or Eq. (9.1.40). We note that (for  $t < t_A < t'$ )

$$\begin{aligned}
& \left( \dot{q}_a(t_A) - \frac{\partial H(q(t_A), p(t_A))}{\partial p_a(t_A)} \right) \exp(iI[q, p]) = -i \frac{\delta}{\delta p_a(t_A)} \exp(iI[q, p]), \\
& \left( \dot{p}_a(t_A) + \frac{\partial H(q(t_A), p(t_A))}{\partial q_a(t_A)} \right) \exp(iI[q, p]) = i \frac{\delta}{\delta q_a(t_A)} \exp(iI[q, p]),
\end{aligned}$$

where  $I$  is the argument of the exponential in Eq. (9.1.38):

$$I[q, p] \equiv \int_t^{t'} d\tau \left\{ \sum_a \dot{q}_a(\tau) p_a(\tau) - H(q(\tau), p(\tau)) \right\}.$$

As long as  $t_A$  does not approach  $t$  or  $t'$ , the integrations over  $q_a(t_A)$  and  $p_a(t_A)$  are unconstrained, and so with reasonable assumptions about the convergence of these integrals, the integral of such variational derivatives must vanish. Hence the path integral (9.1.38) vanishes if  $\mathcal{O}_A(p, q)$  is taken to be the left-hand side of either of the equations of motion (9.1.39) or (9.1.40).

This simple rule applies only if the integration variables  $q_a(t_A)$ ,  $p_a(t_A)$  are independent of any of the variables  $q_a(t_B)$ ,  $p_a(t_B)$ , etc. appearing in any of the other functions  $\mathcal{O}_B$ ,  $\mathcal{O}_C$ , etc. in Eq. (9.1.38), and hence only if we prohibit  $t_A$  from approaching  $t_B, t_C$ , etc. as well as  $t$  or  $t'$ . When  $t_A$  approaches, say,  $t_B$ , the path integral will be found to involve a non-zero term proportional to  $\delta(t_A - t_B)$  or its derivatives. These delta functions are the same as would be found in the operator formalism from time derivatives of the step functions implicit in the definition of the time-ordered product.

In evaluating the path integrals (9.1.34) and (9.1.38), we only need to know the classical Hamiltonian, the c-number function  $H(q, p)$ . If we were to define a theory by the path integrals, the question would naturally arise, which of many possible quantum mechanical Hamiltonians  $H(Q, P)$  (differing in the order of  $Q$ s and  $P$ s) governed the quantum theory that corresponds to these path integrals. Our derivation has provided an answer: the quantum Hamiltonian is to be taken with all  $Q$ s on the left, all  $P$ s on the right. But it would be a mistake to give this prescription too much significance. There are a great many ways of interpreting the measure  $\prod dq_a(\tau) \prod dp_b(\tau)$  appearing in path integrals like (9.1.34) or (9.1.38). Our prescription, of putting all  $Q$ s to the left of all  $P$ s, is appropriate only if the measure is interpreted according to Eqs. (9.1.31)–(9.1.33). Other measures would lead to other prescriptions for operator ordering. The question is not an urgent one, because different prescriptions for ordering operators in the Hamiltonian just correspond to different choices of the constants that appear as coefficients of the various terms in the Hamiltonian, and we generally formulate theories with these constants left as arbitrary parameters anyway.

It is difficult to use the general path integral in Eq. (9.1.38) for numerical calculations or as a source of rigorous theorems. For these purposes it is better to use the path-integral method to calculate amplitudes in Euclidean space, where  $t$  is replaced with an imaginary quantity  $-ix_4$ , and the argument of the exponential in Eq. (9.1.38) is a negative real quantity. In this way, instead of jagged paths producing rapid oscillations of the integrand from one path to another, all jagged paths are exponentially suppressed. Though we shall not go into it here, quantum field theory may be formulated from the beginning in terms of Feynman amplitudes in Euclidean spacetime [8a]. Under certain plausible assumptions, it is possible to reconstruct the Feynman amplitudes in Minkowskian spacetime from their Euclidean counterparts [8b]. But we may as well stick to the Minkowski space formulation of the path integral if we are only going to use it to calculate Feynman amplitudes in perturbation theory.

## 9.2 Transition to the $S$ -Matrix

As already mentioned, we can easily convert the general quantum mechanical results of Section 9.1 to a notation appropriate to quantum field theory, by letting the index  $a$  run over points  $\mathbf{x}$  in space and over a spin-and-species index  $m$ , and replacing  $Q_a(t)$  and  $P_a(t)$  with  $Q_m(\mathbf{x}, t)$  and  $P_m(\mathbf{x}, t)$ , respectively. Eq. (9.1.38) then reads<sup>101</sup>

$$\begin{aligned} & \langle q'; t' | T \{ \mathcal{O}_A[P(t_A), Q(t_A)], \mathcal{O}_B[P(t_B), Q(t_B)], \dots \} | q; t \rangle \\ &= \int_{\substack{q_m(\mathbf{x}, t) = q_m(\mathbf{x}) \\ q_m(\mathbf{x}, t') = q'_m(\mathbf{x})}} \prod_{\tau, \mathbf{x}, m} dq_m(\mathbf{x}, \tau) \prod_{\tau, \mathbf{x}, m} \frac{dp_m(\mathbf{x}, \tau)}{2\pi} \\ & \cdot \mathcal{O}_A[p(t_A), q(t_A)] \mathcal{O}_B[p(t_B), q(t_B)] \cdots \\ & \cdot \exp \left[ i \int_t^{t'} d\tau \left\{ \int d^3x \dot{q}_m(\mathbf{x}, \tau) p_m(\mathbf{x}, \tau) - H[q(\tau), p(\tau)] \right\} \right]. \end{aligned} \quad (9.2.1)$$

<sup>101</sup>We are now writing  $H$  and the  $\mathcal{O}$ s with square brackets, to remind us that  $H[q(t), p(t)]$  and  $\mathcal{O}[p(t), q(t)]$  are functionals of  $q_m(\mathbf{x}, t)$  and  $p_m(\mathbf{x}, t)$  at a fixed time  $t$ .

However, in field theory Eq. (9.2.1) is not exactly what we want. Experimentalists do not measure probability amplitudes for transitions between eigenstates  $\langle q'; t' |$  and  $| q; t \rangle$  of the quantum field  $Q$ , but rather  $S$ -matrix elements, the probability amplitudes for transitions between states that at  $t \rightarrow -\infty$  or  $t \rightarrow +\infty$  contain definite numbers of particles of various types. These are called ‘in’ and ‘out’ states,  $|\alpha\rangle_{\text{in}}$  and  $|\beta\rangle_{\text{out}}$ , where  $\alpha$  and  $\beta$  denote sets of particles characterized by the various particles’ momenta, spin  $z$ -component (or helicity), and species. To calculate a matrix element of a time-ordered product (perhaps empty) between such states, we need to multiply Eq. (9.2.1) by the ‘wave functions’  ${}_{\text{out}}\langle \beta | q'; t' \rangle$  and  $\langle q; t | \alpha \rangle_{\text{in}}$  at any fixed times  $t$  and  $t'$ , taken for convenience here to be  $-\infty$  and  $+\infty$ , respectively, and then perform an integral over the ‘arguments’  $q_m(\mathbf{x})$  and  $q'_m(\mathbf{x})$  of these wave functions. But instead of constraining the path integral over  $q_m(\mathbf{x}, \tau)$  by the conditions

$$q_m(\mathbf{x}, +\infty) = q'_m(\mathbf{x}), \quad q_m(\mathbf{x}, -\infty) = q_m(\mathbf{x}), \quad (9.2.2)$$

and then integrating over  $q'_m(\mathbf{x})$  and  $q_m(\mathbf{x})$ , we can just as well do an unconstrained integral over  $q_m(\mathbf{x}, \tau)$  (and also over  $p_m(\mathbf{x}, \tau)$ ), and set the arguments of the wave functions equal to the values given by Eq. (9.2.2):

$$\begin{aligned} & {}_{\text{out}}\langle \beta | T \{ \mathcal{O}_A[P(t_A), Q(t_A)], \mathcal{O}_B[P(t_B), Q(t_B)], \dots \} | \alpha \rangle_{\text{in}} \\ &= \int \prod_{\tau, \mathbf{x}, m} dq_m(\mathbf{x}, \tau) \prod_{\tau, \mathbf{x}, m} \frac{dp_m(\mathbf{x}, \tau)}{2\pi} \mathcal{O}_A[p(t_A), q(t_A)] \mathcal{O}_B[p(t_B), q(t_B)] \cdots \\ & \cdot \exp \left[ i \int_{-\infty}^{+\infty} d\tau \left\{ \int d^3x \dot{q}_m(\mathbf{x}, \tau) p_m(\mathbf{x}, \tau) - H[q(\tau), p(\tau)] \right\} \right] \\ & \cdot {}_{\text{out}}\langle \beta | q(+\infty); +\infty \rangle \langle q(-\infty); -\infty | \alpha \rangle_{\text{in}}. \end{aligned} \quad (9.2.3)$$

Incidentally, this result leads immediately<sup>102</sup> to Eq. (6.4.3), a theorem that we use repeatedly to relate sums of off-shell Feynman graphs to matrix elements of Heisenberg-picture operators between exact energy eigenstates.

It is necessary now to consider how to calculate the wave functions appearing as the final pair of factors in Eq. (9.2.3). Let’s first consider the simplest and most important case, the vacuum. (We saw in Section 6.4 that  $S$ -matrix elements may be easily calculated from the vacuum expectation values of time-ordered products.) We assume as usual that

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<sup>102</sup>It is only necessary to note that, for a Hamiltonian  $H[P(t), Q(t)] + \sum_A \int d^3x \epsilon_A(\mathbf{x}, t) \mathcal{O}_A(\mathbf{x}, t)$ , the  $S$ -matrix is given by Eq. (9.2.3) as

$$\begin{aligned} & {}_{\text{out}}\langle \beta | \alpha \rangle_{\text{in}, \epsilon} = \int \prod_{\tau, \mathbf{x}, m} dq_m(\mathbf{x}, \tau) \prod_{\tau, \mathbf{x}, m} \frac{dp_m(\mathbf{x}, \tau)}{2\pi} \\ & \cdot \exp \left[ i \int_{-\infty}^{+\infty} dt \left\{ \int d^3x \dot{q}_m(\mathbf{x}, t) p_m(\mathbf{x}, t) - H[q(t), p(t)] - \sum_A \int d^3x \epsilon_A(\mathbf{x}, t) \mathcal{O}_A(\mathbf{x}, t) \right\} \right] \\ & \cdot {}_{\text{out}}\langle \beta | q(+\infty); +\infty \rangle \langle q(-\infty); -\infty | \alpha \rangle_{\text{in}}. \end{aligned}$$

The left-hand side of Eq. (6.4.3) is the derivative of this expression with respect to  $\epsilon_A$ ,  $\epsilon_B$ , etc., at  $\epsilon = 0$ , which yields the right-hand side of Eq. (9.2.3), and using Eq. (9.2.3) again then immediately gives the right-hand side of Eq. (6.4.3).



for  $t \rightarrow \mp\infty$ , matrix elements may be calculated as if there were no interactions. The ‘in’ and ‘out’ vacua may thus be defined by the conditions

$$a_{\mathbf{p},\sigma,n,\text{in}}|\Omega\rangle_{\text{in}} = 0, \quad a_{\mathbf{p},\sigma,n,\text{out}}|\Omega\rangle_{\text{out}} = 0, \quad (9.2.4)$$

where  $a_{\mathbf{p},\sigma,n,\text{in}}$  and  $a_{\mathbf{p},\sigma,n,\text{out}}$  are the operators appearing in the coefficients of  $\exp(i\mathbf{p}\cdot\mathbf{x} - iEt)$  in the plane-wave expansion of the operator  $Q_m(\mathbf{x}, t)$  at  $t \rightarrow -\infty$  and  $t \rightarrow +\infty$ , respectively. For instance, for the real scalar field of a neutral spinless particle, we have in effect

$$\Phi(\mathbf{x}, t) \xrightarrow[t \rightarrow \mp\infty]{} (2\pi)^{-3/2} \int d^3p (2E)^{-1/2} \left[ a_{\text{out}}^{\text{in}}(\mathbf{p}) e^{ip\cdot x} + \text{H.c.} \right], \quad (9.2.5)$$

$$\Pi(\mathbf{x}, t) \xrightarrow[t \rightarrow \mp\infty]{} \dot{\Phi}(\mathbf{x}, t) \xrightarrow[t \rightarrow \mp\infty]{} -i(2\pi)^{-3/2} \int d^3p (E/2)^{1/2} \left[ a_{\text{out}}^{\text{in}}(\mathbf{p}) e^{ip\cdot x} - \text{H.c.} \right], \quad (9.2.6)$$

where  $p^0 \equiv E \equiv \sqrt{\mathbf{p}^2 + m^2}$ , and we here use the conventional  $\Phi$  and  $\Pi$  rather than  $Q$  and  $P$  for a scalar field, and drop the unnecessary labels  $m, \sigma, n$ . Inverting the Fourier transforms and taking a linear combination of the resulting expressions, we have

$$a_{\text{out}}^{\text{in}}(\mathbf{p}) = \lim_{t \rightarrow \mp\infty} \frac{e^{iEt}}{(2\pi)^{3/2}} \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} \left[ \sqrt{\frac{E}{2}} \Phi(\mathbf{x}, t) + i\sqrt{\frac{1}{2E}} \Pi(\mathbf{x}, t) \right]. \quad (9.2.7)$$

As mentioned in Section 9.1, the ‘momentum’  $\Pi(\mathbf{x}, t)$  acts on wave-functions in a  $\phi$ -basis as the variational derivative  $-i\delta/\delta\phi(\mathbf{x}, t)$ , so in this basis the conditions (9.2.4) read

$$0 = \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} \left[ \frac{\delta}{\delta\phi(\mathbf{x})} + E(\mathbf{p})\phi(\mathbf{x}) \right] \langle \phi(\mathbf{x}); \mp\infty | \Omega \rangle_{\text{out}}^{\text{in}}. \quad (9.2.8)$$

The analogous ordinary differential equation has a well-known Gaussian solution, so let’s try a Gaussian ansatz here:

$$\langle \phi(\mathbf{x}); \mp\infty | \Omega \rangle_{\text{out}}^{\text{in}} = \mathcal{N} \exp \left( -\frac{1}{2} \int d^3x d^3y \mathcal{E}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{x}) \phi(\mathbf{y}) \right), \quad (9.2.9)$$

with kernel  $\mathcal{E}$  and constant  $\mathcal{N}$  to be determined. Substituting this in Eq. (9.2.8), we see that the functional differential equation for the vacuum wave functional is satisfied if for all  $\phi$

$$0 = \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} \left[ \int d^3y \mathcal{E}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{y}) - E(\mathbf{p})\phi(\mathbf{x}) \right] \quad (9.2.10)$$

or, in other words, if

$$\int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} \mathcal{E}(\mathbf{x}, \mathbf{y}) = E(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{y}}. \quad (9.2.11)$$

The solution is easily found by inverting the Fourier transform

$$\mathcal{E}(\mathbf{x}, \mathbf{y}) = (2\pi)^{-3} \int d^3p e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} E(\mathbf{p}). \quad (9.2.12)$$

(Recall that  $E(\mathbf{p}) \equiv \sqrt{\mathbf{p}^2 + m^2}$ .) This is actually the most useful representation for the kernel  $\mathcal{E}$ , but we may note in passing that for  $\mathbf{x} \neq \mathbf{y}$ ,  $\mathcal{E}$  may also be written in terms of a Hankel function of negative order

$$\mathcal{E}(\mathbf{x}, \mathbf{y}) = \frac{m}{2\pi^2 r} \frac{d}{dr} \left( \frac{1}{r} K_{-1}(imr) \right), \quad (9.2.13)$$

where  $r \equiv |\mathbf{x} - \mathbf{y}|$ . The constant  $\mathcal{N}$  in Eq. (9.2.9) may be formally obtained from the normalization condition for the vacuum state, but we will not need this result.

According to Eq. (9.2.9), in calculating vacuum expectation values in the theory of a scalar field, the product of the last two factors in Eq. (9.2.3) is

$$\begin{aligned} & {}_{\text{out}}\langle\Omega|\phi(+\infty); +\infty\rangle\langle\phi(-\infty); -\infty|\Omega\rangle_{\text{in}} \\ &= |\mathcal{N}|^2 \exp\left(-\frac{1}{2} \int d^3x d^3y \mathcal{E}(\mathbf{x}, \mathbf{y}) [\phi(\mathbf{x}, +\infty)\phi(\mathbf{y}, +\infty) + \phi(\mathbf{x}, -\infty)\phi(\mathbf{y}, -\infty)]\right) \\ &= |\mathcal{N}|^2 \exp\left(-\frac{1}{2} \epsilon \int d^3x d^3y \int_{-\infty}^{+\infty} d\tau \mathcal{E}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{x}, \tau)\phi(\mathbf{y}, \tau) e^{-\epsilon|\tau|}\right), \end{aligned} \quad (9.2.14)$$

where  $\epsilon$  is a positive infinitesimal. To obtain the final expression, we have used the fact that for any reasonably smooth function  $f(\tau)$ ,

$$f(+\infty) + f(-\infty) = \lim_{\epsilon \rightarrow 0+} \epsilon \int_{-\infty}^{+\infty} d\tau f(\tau) e^{-\epsilon|\tau|}. \quad (9.2.15)$$

Inserting Eq. (9.2.14) in Eq. (9.2.3) now gives

$$\begin{aligned} & {}_{\text{out}}\langle\Omega|T\{\mathcal{O}_A[\Pi(t_A), \Phi(t_A)], \mathcal{O}_B[\Pi(t_B), \Phi(t_B)], \dots\}|\Omega\rangle_{\text{in}} \\ &= |\mathcal{N}|^2 \int \prod_{\tau, \mathbf{x}} d\phi(\mathbf{x}, \tau) \prod_{\tau, \mathbf{x}} \frac{d\pi(\mathbf{x}, \tau)}{2\pi} \mathcal{O}_A[\pi(t_A), \phi(t_A)] \mathcal{O}_B[\pi(t_B), \phi(t_B)] \cdots \\ &\quad \cdot \exp\left[i \int_{-\infty}^{+\infty} d\tau \left\{ \int d^3x \dot{\phi}(\mathbf{x}, \tau) \pi(\mathbf{x}, \tau) - H[\phi(\tau), \pi(\tau)] \right. \right. \\ &\quad \left. \left. + \frac{1}{2} i\epsilon \int d^3x d^3y \mathcal{E}(\mathbf{x}, \mathbf{y}) e^{-\epsilon|\tau|} \phi(\mathbf{x}, \tau) \phi(\mathbf{y}, \tau) \right\} \right]. \end{aligned} \quad (9.2.16)$$

We shall see in Section 9.4 that the whole effect of the last term in the argument of the exponential in Eq. (9.2.16) is to provide the  $-i\epsilon$  in the denominator of the scalar field propagator in momentum space,  $[p^2 + m^2 - i\epsilon]^{-1}$ . We will not go into the corresponding details for fields of general spin, but will simply state that in general

$$\begin{aligned} & {}_{\text{out}}\langle\Omega|T\{\mathcal{O}_A[P_A(t_A), Q(t_A)], \mathcal{O}_B[P_B(t_B), Q(t_B)], \dots\}|\Omega\rangle_{\text{in}} \\ &= |\mathcal{N}|^2 \int \left[ \prod_{\tau, \mathbf{x}, m} dq_m(\mathbf{x}, \tau) \right] \left[ \prod_{\tau, \mathbf{x}, m} \frac{dp_m(\mathbf{x}, \tau)}{2\pi} \right] \mathcal{O}_A[p(t_A), q(t_A)] \mathcal{O}_B[p(t_B), q(t_B)] \cdots \\ &\quad \cdot \exp\left[i \int_{-\infty}^{+\infty} d\tau \left\{ \int d^3x \sum_m \dot{q}_m(\mathbf{x}, \tau) p_m(\mathbf{x}, \tau) - H[q(\tau), p(\tau)] + i\epsilon \text{ terms} \right\} \right], \end{aligned} \quad (9.2.17)$$

where the ' $i\epsilon$  terms' just have the effect of putting the correct  $-i\epsilon$  in the denominators of all propagators.

This is a good place to mention that field-independent factors in Eq. (9.2.17), like the constant  $|\mathcal{N}|^2$ , are not important. This is because such factors contribute also to the matrix element  ${}_{\text{out}}\langle\Omega|\Omega\rangle_{\text{in}}$ . In calculating the *connected* part of vacuum expectation values of time-ordered products (or the  $S$ -matrix) we eliminate the contribution of disconnected

vacuum fluctuation subgraphs by dividing by  ${}_{\text{out}}\langle\Omega|\Omega\rangle_{\text{in}}$ , and any constant factors in the vacuum expectation values cancel in this ratio.

We could go on and calculate matrix elements between multi-particle states, by inserting the appropriate ‘wave functionals’ in Eq. (9.2.3). These can be calculated by applying the adjoints of annihilation operators such as (9.2.7) to the vacuum state; just as for the harmonic oscillator, these wave functionals turn out to be Hermite polynomials in the field times the vacuum Gaussian. We do not need to work all this out here, because as shown in Section 6.4, the vacuum expectation values (9.2.17) are all we need in order to be able to calculate  $S$ -matrix elements.

### 9.3 Lagrangian Version of the Path-Integral Formula

The integrand in the exponential in Eqs. (9.1.38) or (9.2.17) looks like the Lagrangian  $L$  associated with the Hamiltonian  $H$ . This appearance is somewhat misleading because here the ‘momenta’  $p_a(t)$  or  $p_n(\mathbf{x}, t)$  are independent variables, not yet related to  $q_a(t)$  or  $q_n(\mathbf{x}, t)$  or their derivatives. However, there is a large and important class of theories in which the integral over the ‘momenta’ can be done by just replacing them with the values dictated by the canonical formalism, in which case the integrand in the exponential in the path integrals really is the Lagrangian.

These are theories with a Hamiltonian that is quadratic in the ‘momenta’—in the language of field theory

$$\begin{aligned} H[Q, P] = & \frac{1}{2} \sum_{nm} \int d^3x d^3y A_{xn,ym}[Q] P_n(\mathbf{x}) P_m(\mathbf{y}) \\ & + \sum_n \int d^3x B_{xn}[Q] P_n(\mathbf{x}) + C[Q] \end{aligned} \quad (9.3.1)$$

with a ‘matrix’  $A$  that is real, symmetric, positive, and non-singular. The argument of the exponential in Eq. (9.2.17) is then quadratic in the  $ps$ :

$$\begin{aligned} & \int d\tau \left\{ \int d^3x \sum_n p_n(\mathbf{x}, \tau) \dot{q}_n(\mathbf{x}, \tau) - H[q(\tau), p(\tau)] \right\} \\ & = -\frac{1}{2} \sum_{nm} \int d^3x d^3y d\tau d\tau' \mathcal{A}_{xn\tau,ym\tau'}[q] p_n(\mathbf{x}, \tau) p_m(\mathbf{y}, \tau') \\ & \quad - \sum_n \int d^3x \int d\tau \mathcal{B}_{xn\tau}[q] p_n(\mathbf{x}, \tau) - \mathcal{C}[q], \end{aligned} \quad (9.3.2)$$

where

$$\mathcal{A}_{xn\tau,ym\tau'}[q] \equiv A_{xn,ym}[q(\tau)] \delta(\tau - \tau'), \quad (9.3.3)$$

$$\mathcal{B}_{xn\tau}[q] \equiv B_{xn}[q(\tau)] - \dot{q}_n(\mathbf{x}, \tau), \quad (9.3.4)$$

$$\mathcal{C}[q] \equiv \int d\tau C[q(\tau)]. \quad (9.3.5)$$

Now, in general the integral of the exponential of a quadratic expression like (9.3.2) will be proportional to the exponential evaluated at the stationary point of its argument. For

a finite number of real variables  $\xi_s$ , this formula reads

$$\begin{aligned} & \int_{-\infty}^{\infty} \left( \prod_s d\xi_s \right) \exp \left\{ -\frac{i}{2} \sum_{sr} \mathcal{A}_{sr} \xi_s \xi_r - i \sum_s \mathcal{B}_s \xi_s - i\mathcal{C} \right\} \\ &= \left( \text{Det} \left[ \frac{i\mathcal{A}}{2\pi} \right] \right)^{-1/2} \exp \left\{ -\frac{i}{2} \sum_{sr} \mathcal{A}_{sr} \bar{\xi}_s \bar{\xi}_r - i \sum_s \mathcal{B}_s \bar{\xi}_s - i\mathcal{C} \right\}, \end{aligned} \quad (9.3.6)$$

where  $\bar{\xi}$  is the stationary point

$$\bar{\xi}_s = - \sum_r (\mathcal{A}^{-1})_{sr} \mathcal{B}_r. \quad (9.3.7)$$

(For a proof of this formula, see the Appendix to this chapter.) Hence, as long as the  $\mathcal{O}_A, \mathcal{O}_B$ , etc. in Eq. (9.2.17) are independent of the  $ps$ , for such a Hamiltonian we can evaluate the path integral over the  $ps$  in Eq. (9.2.17) by setting these variables at the stationary point of the quadratic expression in the argument of the exponential. But the variational derivative of this quadratic is

$$\begin{aligned} & \frac{\delta}{\delta p_n(\mathbf{x}, \tau)} \int_{-\infty}^{\infty} d\tau \left\{ \int d^3x \sum_m \dot{q}_m(\mathbf{x}, \tau) p_m(\mathbf{x}, \tau) - H[q(\tau), p(\tau)] + i\epsilon \text{ terms} \right\} \\ &= \dot{q}_n(\mathbf{x}, \tau) - \frac{\delta}{\delta p_n(\mathbf{x}, \tau)} H[q(\tau), p(\tau)]. \end{aligned}$$

(The  $i\epsilon$  terms depend only on the  $qs$ .) Thus the stationary ‘point’  $\bar{p}_n(\mathbf{x}, t)$  where this vanishes is just the value of  $p_n(\mathbf{x}, t)$  dictated by the canonical formula

$$\dot{q}_n(\mathbf{x}, t) = \left[ \frac{\delta H[q(t), p(t)]}{\delta p_n(\mathbf{x}, t)} \right]_{p=\bar{p}}. \quad (9.3.8)$$

With  $p_n(\mathbf{x}, t)$  set equal to this value, the argument of the exponential in Eq. (9.2.17) is the ordinary Lagrangian

$$L[q(t), \dot{q}(t)] \equiv \int d^3x \left( \sum_n \dot{q}_n(\mathbf{x}, t) \bar{p}_n(\mathbf{x}, t) - H[q(t), \bar{p}(t)] \right) \quad (9.3.9)$$

and we can write Eq. (9.2.17) as

$$\begin{aligned} & {}_{\text{out}} \langle \Omega | T \{ \mathcal{O}_A[Q(t_A)], \mathcal{O}_B[Q(t_B)], \dots \} | \Omega \rangle_{\text{in}} \\ &= |\mathcal{N}|^2 \int \prod_{\tau, \mathbf{x}, n} dq_n(\mathbf{x}, \tau) (\text{Det}[2i\pi \mathcal{A}[q]])^{-1/2} \\ &\quad \cdot \mathcal{O}_A[q(t_A)] \mathcal{O}_B[q(t_B)] \cdots \\ &\quad \cdot \exp \left[ i \int_{-\infty}^{\infty} d\tau \{ L[q(\tau), \dot{q}(\tau)] + i\epsilon \text{ terms} \} \right]. \end{aligned} \quad (9.3.10)$$

(We have combined the  $1/2\pi$  factors in the integrals over the  $p_n$  with the determinant coming from Eq. (9.3.6).) This is the desired Lagrangian form of the path-integral formula.

In deriving Eq. (9.3.10), it was necessary to assume that the operators  $\mathcal{O}_A, \mathcal{O}_B, \dots$  were independent of the canonical ‘momenta’. This is not as restrictive as it may seem. For instance, in a scalar field theory for which the canonical conjugate to  $\Phi$  is  $\Pi = \dot{\Phi}$ , it is possible to calculate the matrix element of a time-ordered product of operators, one of which is  $\dot{\Phi}(t)$ , by taking the difference of matrix elements in which this operator is replaced with  $\Phi(t + d\tau)$  and  $\Phi(t)$ , and then dividing by  $d\tau$ , with  $d\tau \rightarrow 0$ . Equivalently, as long as  $t$  is not equal to any of the other time-arguments of the operator in Eq. (9.3.10), we can simply differentiate Eq. (9.3.10) with respect to  $t$ .

The one serious remaining complication in Eq. (9.3.10) is the determinant of  $\mathcal{A}[q]$ . If  $\mathcal{A}[q]$  is field-independent, then this is no problem; we have already noted that overall constants make no contribution to the connected parts of vacuum expectation values, in which we divide by a vacuum-vacuum amplitude proportional to the same constant factor. This is the case for instance for the theory of a set of scalar fields  $\Phi_n$  with non-derivative coupling to each other and/or derivative coupling to external currents  $J_n$ . The Lagrangian density here is

$$\mathcal{L} = - \sum_n \left[ \frac{1}{2} \partial_\mu \Phi_n \partial^\mu \Phi_n + J_n^\mu \partial_\mu \Phi_n \right] - V(\Phi).$$

An obvious extension of the results of Section 7.5 from one to several derivatively coupled scalars shows that this Lagrangian implies the Hamiltonian

$$H = \int d^3x \sum_n \left[ \frac{1}{2} \Pi_n^2 + \frac{1}{2} (\nabla \Phi_n)^2 + \mathbf{J}_n \cdot \nabla \Phi_n + J_n^0 \Pi_n + \frac{1}{2} (J_n^0)^2 \right] + \int d^3x V(\Phi).$$

(The  $\Phi_n$  are taken to be real scalars, but complex scalars can be accommodated by separating them into real and imaginary parts.) In general there is a non-trivial term that is linear in the  $\Pi_n$ , but the coefficient of the quadratic term is a constant, just the unit ‘matrix’:

$$\mathcal{A}_{xn, x'n'} = \delta^4(x - x') \delta_{nn'}.$$

The factor  $(\text{Det}[4i\pi\mathcal{A}[q]])^{-1/2}$  in Eq. (9.3.10) is here a field-independent constant, and therefore is without effect.

However, matters are not always so simple. As a second example, let us consider the so-called non-linear  $\sigma$ -model, with Lagrangian density

$$\mathcal{L} = -\frac{1}{2} \sum_{nm} \partial_\mu \Phi_n \partial^\mu \Phi_m [\delta_{nm} + U_{nm}(\Phi)] - V(\Phi).$$

A straightforward calculation gives the Hamiltonian as

$$H = \int d^3x \left[ \frac{1}{2} \Pi_n (\mathbf{1} + U(\Phi))_{nm}^{-1} \Pi_m + \frac{1}{2} \nabla \Phi_n \cdot \nabla \Phi_m (\mathbf{1} + U(\Phi))_{nm} + V(\Phi) \right].$$

Here  $\mathcal{A}$  is the field-dependent quantity

$$\mathcal{A}_{nx, my} = [\mathbf{1} + U(\Phi(x))]_{nm}^{-1} \delta^4(x - y).$$

In cases of this sort, the determinant may be reexpressed as a contribution to the effective Lagrangian, using the relation  $\text{Det } \mathcal{A} = \exp \text{Tr} \ln \mathcal{A}$ . By replacing the continuum of space-time positions with a discrete lattice of points surrounded by separate regions of very small spacetime volume  $\Omega$ , we may interpret the delta function in  $\mathcal{A}_{nx,my}$  as  $\delta^4(x-y) = \Omega^{-1} \delta_{x,y}$ , so that

$$(\ln \mathcal{A})_{nx,my} = \delta_{x,y} [\ln(\mathbf{1} + U(\Phi(x))) - \mathbf{1} \cdot \ln \Omega]_{nm}$$

with the logarithm of a matrix defined now by its power series expansion

$$\ln(\mathbf{1} + U) = U - \frac{U^2}{2} + \frac{U^3}{3} - \dots$$

To evaluate the trace, we note that  $\sum_x \dots = \Omega^{-1} \int d^4x \dots$ . The determinant factor here is then

$$\text{Det } \mathcal{A} \propto \exp \left[ -\Omega^{-1} \int d^4x \text{tr} \ln[\mathbf{1} + U(\Phi(x))] \right],$$

where ‘tr’ is to be understood as the trace in an ordinary matrix sense. The constant of proportionality (which arises from the  $-\ln \Omega$  term) is field-independent and therefore of no present interest. We can regard this determinant as providing a correction to the effective Lagrangian density

$$\Delta \mathcal{L} = -\frac{i}{2} \Omega^{-1} \text{tr} \ln[\mathbf{1} + U(\Phi(x))].$$

The factor  $\Omega^{-1}$  may be written as an ultraviolet divergent integral

$$\Omega^{-1} = \delta^4(x-x) = (2\pi)^{-4} \int d^4p \cdot 1.$$

We shall not show this here, but the extra terms in the Feynman diagrams for this theory contributed by  $\Delta \mathcal{L}$  could also have been derived in the canonical formalism by taking account of the equal-time-commutator terms in the propagator of time-derivatives of the scalar field [7]. Ignoring this correction would lead to a spurious dependence of the  $S$ -matrix on the way that the scalar field is defined, and would also be inconsistent with any symmetries of the Lagrangian under transformations of the scalar fields.

Even where the factor  $(\text{Det } \mathcal{A})^{-1/2}$  in the path-integral formula (9.3.10) is field-independent, the Lagrangian in this formula may not be the same as the one with which we started. As an example, let’s consider the theory of a set of real vector fields, with Lagrangian density

$$\begin{aligned} \mathcal{L} = - \sum_n \left[ \frac{1}{4} (\partial_\mu A_{n\nu} - \partial_\nu A_{n\mu}) (\partial^\mu A_n^\nu - \partial^\nu A_n^\mu) \right. \\ \left. + \frac{1}{2} m_n^2 A_{n\mu} A_n^\mu + J_n^\mu A_{n\mu} \right], \end{aligned}$$

where the currents  $J_n^\mu$  are either externally produced c-number quantities or depend on other fields (in which case terms describing these other fields are to be added to the Lagrangian).

By a simple extension of the results of Section 7.5, we see that the Hamiltonian is

$$H = \int d^3x \sum_n \left[ \frac{1}{2} \mathbf{\Pi}_n^2 + \frac{1}{2} (\nabla \times \mathbf{A}_n)^2 + \frac{1}{2} m_n^2 \mathbf{A}_n^2 + \frac{1}{2m_n^2} (\nabla \cdot \mathbf{\Pi}_n)^2 \right. \\ \left. + \mathbf{J}_n \cdot \mathbf{A}_n - \frac{1}{m_n^2} J_n^0 \nabla \cdot \mathbf{\Pi}_n + \frac{1}{2m_n^2} (J_n^0)^2 \right],$$

again with the understanding that other terms must be added involving any fields that appear in  $J_n^\mu$ . Here the coefficient of the quadratic term is somewhat more complicated than in our first example:

$$\mathcal{A}_{nix,mjy} = \delta_{nm} \left[ \delta_{ij} \delta^4(x-y) - \frac{1}{2m_n^2} \nabla_i \nabla_j \delta^4(x-y) \right],$$

but it is field-independent, so that the factor  $(\text{Det } \mathcal{A})^{-1/2}$  has no effect. On the other hand, the Lagrangian (9.3.9) is here not the one with which we started; it is expressed entirely in terms of  $\mathbf{A}$  and its spacetime derivatives, with no dependence on any time-component  $A^0$ . For this reason, the Lorentz invariance of Eq. (9.3.10) is far from obvious.

To remedy this, we may reintroduce the auxiliary field. Suppose we add to the Hamiltonian a term

$$\Delta H = -\frac{1}{2} \sum_n m_n^2 \int d^3x [A_n^0 - m_n^{-2} \nabla \cdot \mathbf{\Pi}_n + m_n^{-2} J_n^0]^2$$

and integrate over the  $A_n^0$  as well as the  $\mathbf{A}_n$  and  $\mathbf{\Pi}_n$ . This can only introduce a field-independent overall factor, since  $\Delta H$  is a quadratic in  $A^0$  (with a field-independent coefficient in the term of second order in  $A^0$ ) whose stationary value vanishes. However, suppose that we now integrate over the  $\mathbf{\Pi}_n$  *before* integrating over the  $A_n^0$ . The Hamiltonian in the path integral (9.2.17) is here replaced with

$$H + \Delta H = \int d^3x \sum_n \left[ \frac{1}{2} \mathbf{\Pi}_n^2 + \frac{1}{2} (\nabla \times \mathbf{A}_n)^2 + \frac{1}{2} m_n^2 \mathbf{A}_n^2 - \frac{1}{2} m_n^2 (A_n^0)^2 \right. \\ \left. + \mathbf{J}_n \cdot \mathbf{A}_n - J_n^0 A_n^0 + A_n^0 \nabla \cdot \mathbf{\Pi}_n \right].$$

This is still quadratic in  $\mathbf{\Pi}_n$ , with a field-independent (and somewhat simpler) coefficient of the quadratic term, so the integral over the  $\mathbf{\Pi}_n$ s can be done by just replacing  $\mathbf{\Pi}_n$  with its value at the stationary point of the functional  $\sum_n \int d^3x \mathbf{\Pi}_n \cdot \dot{\mathbf{A}}_n - H - \Delta H$ :

$$\mathbf{\Pi}_n = \dot{\mathbf{A}}_n + \nabla A_n^0.$$

With  $\mathbf{\Pi}_n$  eliminated in this way,  $\sum_n \int d^3x \mathbf{\Pi}_n \cdot \dot{\mathbf{A}}_n - H - \Delta H$  is just the Lorentz-invariant Lagrangian with which we started.

In order to take account of the possible need to introduce auxiliary fields like  $A_n^0$ , from now on we shall write the path-integral formula after elimination of the canonical conjugates

in terms of fields  $\psi_\ell$  that include both canonical fields  $q_n$  and auxiliary fields  $c_r$ :

$$\begin{aligned} & \text{out} \langle \Omega | T \{ \mathcal{O}_A[\Psi_A(t_A)], \mathcal{O}_B[\Psi_B(t_B)], \dots \} | \Omega \rangle_{\text{in}} \\ & \propto \int \prod_{\tau, \mathbf{x}, n} d\psi_n(\mathbf{x}, \tau) \mathcal{O}_A[\psi(t_A)] \mathcal{O}_B[\psi(t_B)] \cdots \\ & \quad \cdot \exp \left[ i \int_{-\infty}^{\infty} d\tau \left\{ L[\psi(\tau), \dot{\psi}(\tau)] + i\epsilon \text{ terms} \right\} \right], \end{aligned} \quad (9.3.11)$$

it now being understood that  $L$  includes any terms arising from a possible field-dependent factor  $(\text{Det } \mathcal{A})^{-1/2}$ .

#### 9.4 Path-Integral Derivation of Feynman Rules

We are now ready to use the path-integral formalism to derive the Feynman rules in a wide class of theories. We will concentrate here on the vacuum expectation values of time-ordered products of field operators (and their adjoints),

$$M_{\ell_A \ell_B \dots}(x_A x_B \dots) = \frac{\text{out} \langle \Omega | T \{ \Psi_{\ell_A}(x_A), \Psi_{\ell_B}(x_B), \dots \} | \Omega \rangle_{\text{in}}}{\text{out} \langle \Omega | \Omega \rangle_{\text{in}}}, \quad (9.4.1)$$

from which  $S$ -matrix elements may be obtained (as shown in Section 6.4) by stripping off the final propagators associated with each field, replacing them with the coefficient functions that multiply creation or annihilation operators in the corresponding free fields, and summing over the indices on these coefficient functions.

For the simpler theories whose Hamiltonian is quadratic in the  $\Pi$ s, Eq. (9.3.11) gives

$$M_{\ell_A \ell_B \dots}(x_A x_B \dots) = \frac{\int \left[ \prod_{x, \ell} d\psi_\ell(x) \right] \psi_{\ell_A}(x_A) \psi_{\ell_B}(x_B) \cdots e^{iI[\psi]}}{\int \left[ \prod_{x, \ell} d\psi_\ell(x) \right] e^{iI[\psi]}}, \quad (9.4.2)$$

where  $I[\psi]$  is the action

$$I[\psi] = \int_{-\infty}^{\infty} d\tau \left\{ L[\psi(\tau), \dot{\psi}(\tau)] + i\epsilon \text{ terms} \right\} \quad (9.4.3)$$

with  $L$  now including any terms that may arise from a field-dependent determinant in Eq. (9.3.10).

Let us now suppose that the Lagrangian is the integral of a Lagrangian density, consisting of a quadratic term  $\mathcal{L}_0$  which would be present in the absence of interactions, plus a Lagrangian interaction density  $\mathcal{L}_1$ :

$$\begin{aligned} L[\psi(t), \dot{\psi}(t)] &= \int d^3x \left[ \mathcal{L}_0(\psi(\mathbf{x}, t), \partial_\mu \psi(\mathbf{x}, t)) \right. \\ & \quad \left. + \mathcal{L}_1(\psi(\mathbf{x}, t), \partial_\mu \psi(\mathbf{x}, t)) \right]. \end{aligned} \quad (9.4.4)$$



That is, the action (9.4.3) is

$$I[\psi] = I_0[\psi] + I_1[\psi], \quad (9.4.5)$$

$$I_0[\psi] = \int d^4x \mathcal{L}_0(\psi(x), \partial_\mu \psi(x)) + i\epsilon \text{ terms}, \quad (9.4.6)$$

$$I_1[\psi] = \int d^4x \mathcal{L}_1(\psi(x), \partial_\mu \psi(x)). \quad (9.4.7)$$

Since  $\mathcal{L}_0$  and the ' $i\epsilon$  terms' are quadratic in the fields, we may always write  $I_0$  in the generalized quadratic form

$$I_0[\psi] = -\frac{1}{2} \int d^4x d^4x' \sum_{\ell, \ell'} \mathcal{D}_{\ell x, \ell' x'} \psi_\ell(x) \psi_{\ell'}(x'). \quad (9.4.8)$$

For instance, for a real scalar field of mass  $m$ , the unperturbed Lagrangian is

$$\mathcal{L}_0 = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \quad (9.4.9)$$

and the  $i\epsilon$  terms in  $I_0$  are given by Eq. (9.2.16), as

$$\frac{1}{2} i\epsilon \int dt \int d^3x d^3x' \mathcal{E}(\mathbf{x}, \mathbf{x}') \phi(\mathbf{x}, t) \phi(\mathbf{x}', t) \quad (9.4.10)$$

so here

$$\mathcal{D}_{x, x'} = \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x'_\mu} \delta^4(x - x') + m^2 \delta^4(x - x') - i\epsilon \mathcal{E}(\mathbf{x}, \mathbf{x}') \delta(t - t'). \quad (9.4.11)$$

(We are now dropping the factor  $e^{-\epsilon|t|}$  in the  $i\epsilon$  term, since it produces a correction of higher order in  $\epsilon$ .) To deal with interactions, we will expand the exponential in powers of  $I_1$ ,

$$\exp(iI[\psi]) = \exp(iI_0[\psi]) \sum_{N=0}^{\infty} \frac{i^N}{N!} (I_1[\psi])^N \quad (9.4.12)$$

and then expand  $I_1$  in powers of the fields. The general integrals that we encounter in the numerator and the denominator of Eq. (9.4.2) are of the form

$$\mathcal{I}_{\ell_1 \ell_2 \dots}(x_1 x_2 \dots) \equiv \int \left( \prod_{\ell, x} d\psi_\ell(x) \right) e^{iI_0[\psi]} \psi_{\ell_1}(x_1) \psi_{\ell_2}(x_2) \dots, \quad (9.4.13)$$

where the field factors  $\psi_{\ell_1}(x_1), \psi_{\ell_2}(x_2)$ , etc. arise from  $I_1[\psi]$  and/or from the field factors  $\psi_{\ell_A}(x_A)$  etc. originally present in the numerator of Eq. (9.4.2). With  $I_0[\psi]$  of the form (9.4.8), the integral (9.4.13) is of the same form as the integral evaluated in the Appendix to this chapter, with the discrete index  $s$  replaced with the pair of labels  $\ell, x$ . We can therefore use Eqs. (9.A.12) and (9.A.15), which give here

$$\mathcal{I}_{\ell_1 \ell_2 \dots}(x_1 x_2 \dots) = \left[ \text{Det} \left( \frac{i\mathcal{D}}{2\pi} \right) \right]^{-1/2} \sum_{\text{pairings of pairs of fields}} \prod_{\text{fields}} [-i\mathcal{D}^{-1}]_{\text{paired fields}}. \quad (9.4.14)$$

This just amounts to the coordinate-space Feynman rules for calculating the numerator of Eq. (9.4.2) in their covariant form: we expand in the interaction  $I_1$ , and then sum over the ways of pairing the fields in the  $I_1$ s with each other and with the fields  $\psi_{\ell_A}(x_A)$ , etc., with the contribution of each pairing being given by the spacetime integral of the product of the coefficients of the fields in  $I_1[\psi]$  and the product of the ‘propagators’  $-i\Delta$ , where

$$\Delta_{\ell_1\ell_2}(x_1, x_2) = (\mathcal{D}^{-1})_{\ell_1x_1, \ell_2x_2}. \quad (9.4.15)$$

(The factor  $[\text{Det}(i\mathcal{D}/2\pi)]^{-1/2}$  in Eq. (9.4.14) actually represents the contribution of graphs with unlimited numbers of single loops unattached to any other lines, but in any case this factor cancels in the ratio (9.4.2).)

It remains to calculate the propagators (9.4.15). We interpret Eq. (9.4.15) as an integral equation

$$\sum_{\ell_2} \int d^4x_2 \mathcal{D}_{\ell_1x_1, \ell_2x_2} \Delta_{\ell_2\ell_3}(x_2, x_3) = \delta^4(x_1 - x_3) \delta_{\ell_1\ell_3}. \quad (9.4.16)$$

In the absence of external fields, translation invariance will make  $\mathcal{D}$  necessarily a function only of  $x_1 - x_2$ , which can be written as a Fourier integral

$$\mathcal{D}_{\ell_1x_1, \ell_2x_2} = (2\pi)^{-4} \int d^4p e^{ip \cdot (x_1 - x_2)} \mathcal{D}_{\ell_1\ell_2}(p). \quad (9.4.17)$$

The solution of Eq. (9.4.16) is then

$$\Delta_{\ell_1\ell_2}(x_1, x_2) = (2\pi)^{-4} \int d^4p e^{ip \cdot (x_1 - x_2)} \mathcal{D}_{\ell_1\ell_2}^{-1}(p), \quad (9.4.18)$$

where  $\mathcal{D}^{-1}$  is the ordinary inverse of the matrix  $\mathcal{D}$ . As we will see, the  $i\epsilon$  terms have the effect of making the inverse well-defined for all real values of  $p$ . We have thus reduced the problem of calculating the propagator to that of taking the inverse of a finite matrix.

First consider a massive scalar field, for which the kernel  $\mathcal{D}$  takes the form (9.4.11). We can write this as a Fourier integral

$$\mathcal{D}_{x,y} = (2\pi)^{-4} \int d^4p e^{ip \cdot (x-y)} (p^2 + m^2 - i\epsilon E(p)),$$

so the propagator is

$$\Delta(x, y) = (2\pi)^{-4} \int d^4p e^{ip \cdot (x-y)} (p^2 + m^2 - i\epsilon E(p))^{-1}.$$

We recognize this as the same scalar propagator previously obtained by operator methods. (The difference between  $\epsilon$  and  $\epsilon E(p)$  is immaterial, since both are just positive infinitesimals.)

For a second example, consider a real massive vector field. The unperturbed Lagrangian is

$$\mathcal{L}_0 = -\frac{1}{4}(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) - \frac{1}{2}m^2 A_\mu A^\mu.$$

We can again write  $I_0[\psi]$  in the form (9.4.8), with kernel

$$\begin{aligned}\mathcal{D}_{\rho x, \sigma y} &= \left[ \eta_{\rho\sigma} \frac{\partial^2}{\partial x^\mu \partial y_\mu} - \frac{\partial^2}{\partial x^\sigma \partial y^\rho} + m^2 \eta_{\rho\sigma} \right] \delta^4(x - y) + i\epsilon \text{ terms} \\ &= (2\pi)^{-4} \int d^4 p e^{ip \cdot (x-y)} [\eta_{\rho\sigma} p^2 - p_\rho p_\sigma + m^2 \eta_{\rho\sigma} + i\epsilon \text{ terms}].\end{aligned}$$

We will not bother to show it here, but the ‘ $+i\epsilon$  terms’ here take the simple form  $-i\epsilon E(p)\eta_{\rho\sigma}$ . The vector field propagator is then given by simply inverting the  $4 \times 4$  matrix in the integrand

$$\Delta_{\rho\sigma}(x, y) = (2\pi)^{-4} \int \frac{d^4 p e^{ip \cdot (x-y)}}{p^2 + m^2 - i\epsilon E(p)} \left[ \eta_{\rho\sigma} + \frac{p_\rho p_\sigma}{m^2} \right].$$

(Terms proportional to  $\epsilon$  are dropped in the numerator. They are important in the denominator in defining how the integrand is to be treated near the mass shell,  $p^2 = -m^2$ .) This is the same as the propagator derived by operator methods, except that the non-covariant terms proportional to  $\delta(x^0 - y^0)$  are now absent. These non-covariant terms were previously needed to cancel non-covariant terms in the interaction Hamiltonian, but the vertex contributions in the Feynman rules are now obtained directly by inspection of the covariant Lagrangian, and no such cancellation is needed.

Theories with derivative coupling are equally simple. The factor arising from the pairing of a field derivative  $\partial_\mu \psi_\ell(x)$  with any other field  $\psi_m(y)$  (perhaps itself a field derivative) is

$$\begin{aligned}\overline{(\partial_\mu \psi_\ell(x) \psi_m(y))} &= \frac{\int \left[ \prod_{x, \ell} d\psi_\ell(x) \right] \partial_\mu \psi_\ell(x) \psi_m(y) e^{iI[\psi]}}{\int \left[ \prod_{x, \ell} d\psi_\ell(x) \right] e^{iI[\psi]}} \\ &= \frac{\partial}{\partial x^\mu} \overline{(\psi_\ell(x) \psi_m(y))}.\end{aligned}\tag{9.4.19}$$

Such propagators have no non-covariant pieces. For instance, for a real scalar field, the pairing of  $\partial_\mu \phi$  with  $\partial_\nu \phi$  gives a momentum space propagator  $k_\mu k_\nu / (k^2 + m^2 - i\epsilon)$ . Also, as we saw in the previous section, vertices in the theory of a scalar field with derivative couplings to other fields may be read off from the Lagrangian, and are separately covariant.

## 9.5 Path Integrals for Fermions

We now turn to the problem of extending the path-integral formalism to cover theories containing fermions as well as bosons. It would be easy to proceed in a purely formal way, by analogy with the bosonic case, with the justification that this gives the ‘right’ Feynman rules. Instead, we will here derive the path-integral formalism for fermions directly from the principles of quantum mechanics, as we did for bosons [9].

As before, we will start with a general quantum mechanical system, with ‘coordinates’  $Q_a$  and canonical conjugate ‘momenta’  $P_a$ , but now satisfying anticommutation rather than commutation relations:

$$\{Q_a, P_b\} = i\delta_{ab},\tag{9.5.1}$$

$$\{Q_a, Q_b\} = \{P_a, P_b\} = 0. \quad (9.5.2)$$

(These are Schrödinger-picture operators, or in other words Heisenberg-picture operators at time  $t = 0$ .) Later we will replace the discrete index  $a$  with a spatial position  $\mathbf{x}$  and a field index  $m$ .

We wish first to construct a complete basis for the states on which the  $Q$ s and  $P$ s act. Note that for any given  $a$ , we have

$$Q_a^2 = P_a^2 = 0. \quad (9.5.3)$$

It follows that there will always be a ‘ket’ state  $|\Omega\rangle$  annihilated by all  $Q_a$ :

$$Q_a|\Omega\rangle = 0, \quad (9.5.4)$$

and a ‘bra’ state  $\langle\Omega|$  annihilated (from the right) by all  $P_a$ :

$$\langle\Omega|P_a = 0. \quad (9.5.5)$$

For instance, we can take

$$|\Omega\rangle \propto \left( \prod_a Q_a \right) |f\rangle, \quad \langle\Omega| \propto \langle g| \left( \prod_a P_a \right),$$

where  $|f\rangle$  and  $\langle g|$  are any kets and bras for which these expressions do not vanish. (They cannot vanish for all  $|f\rangle$  and  $\langle g|$  unless the operators  $\prod_a Q_a$  and  $\prod_a P_a$  vanish, which we assume not to be the case.) These states satisfy Eqs. (9.5.4) and (9.5.5) by virtue of Eq. (9.5.3). They are not in general unique, because there may be other bosonic degrees of freedom that distinguish the various possible  $|\Omega\rangle$  and  $\langle\Omega|$ , but for simplicity we will limit ourselves here to the case where the only degrees of freedom are those described by the fermionic operators  $Q_a$  and  $P_a$ , and will assume that the states satisfying Eqs. (9.5.4) and (9.5.5) are unique up to constant factors, which we choose so that

$$\langle\Omega|\Omega\rangle = 1. \quad (9.5.6)$$

(Note that this normalization convention could not be imposed if we had defined  $\langle\Omega|$  as the left-eigenstate of the  $Q_a$  with eigenvalue zero, because in this case  $\langle\Omega|\{Q_a, P_b\}|\Omega\rangle$  would vanish, which with Eq. (9.5.1) would imply that  $\langle\Omega|\Omega\rangle = 0$ .)

As we saw in Section 7.5, in the Dirac theory  $Q_a$  is not Hermitian, but instead has an adjoint  $-iP_a$ , in which case  $\langle\Omega|$  can be regarded as simply the adjoint of  $|\Omega\rangle$ . However, there are fermionic operators (such as the ‘ghost’ fields to be introduced in Volume II) for which  $P_a$  is unrelated to the adjoint of  $Q_a$ . In what follows we will not need to assume anything about the adjoints of  $Q_a$  or  $P_a$ , or about any relation between  $|\Omega\rangle$  and  $\langle\Omega|$ .

A complete basis for the states of this system is provided by  $|\Omega\rangle$  and the states (antisymmetric in indices  $a, b, \dots$ )

$$|a, b, \dots\rangle \equiv P_a P_b \cdots |\Omega\rangle \quad (9.5.7)$$

with any number of *different*  $P$ s acting on  $|\Omega\rangle$ . That is, the result of acting on these states with any operator function of the  $P$ s and  $Q$ s can be written as a linear combination of the same set of states. In particular, if an index  $a$  is unequal to any of the indices appearing in  $|b, c, \dots\rangle$ , then

$$Q_a|b, c, \dots\rangle = 0, \quad (9.5.8)$$

$$P_a|b, c, \dots\rangle = |a, b, c, \dots\rangle. \quad (9.5.9)$$

On the other hand, if  $a$  is equal to one of the indices in the sequence,  $b, c, \dots$ , we can always rewrite the state (possibly changing its sign) so that  $a$  is the first of these indices, in which case we have

$$Q_a|a, b, c, \dots\rangle = i|b, c, \dots\rangle, \quad (9.5.10)$$

$$P_a|a, b, c, \dots\rangle = 0. \quad (9.5.11)$$

Similarly, we may define a complete dual basis, consisting of  $\langle\Omega|$  and the states (also antisymmetric in the indices)

$$\langle a, b, \dots| \equiv \langle\Omega| \cdots (-iQ_b)(-iQ_a). \quad (9.5.12)$$

Using Eqs. (9.5.4)–(9.5.6) and the anticommutation relation (9.5.1), we see that the scalar products of these states take the values

$$\begin{aligned} \langle c, d, \dots|a, b, \dots\rangle &= \langle\Omega| \cdots (-iQ_d)(-iQ_c)P_aP_b \cdots |\Omega\rangle \\ &= \begin{cases} 0 & \text{if } \{c, d, \dots\} \neq \{a, b, \dots\}, \\ 1 & \text{if } c = a, d = b, \text{ etc.} \end{cases} \end{aligned} \quad (9.5.13)$$

where  $\{\dots\}$  here denotes the set of indices within the brackets, irrespective of order.

In deriving the Feynman rules, we would like to be able to rewrite sums over intermediate states like (9.5.7) as integrals over eigenstates of the  $Q_a$  or the  $P_a$ . However, it is not possible for these operators to have eigenvalues (other than zero) in the usual sense. Suppose we try to find a state  $|q\rangle$  that satisfies (for all  $a$ )

$$Q_a|q\rangle = q_a|q\rangle. \quad (9.5.14)$$

From Eq. (9.5.2) we see that

$$q_a q_b + q_b q_a = 0, \quad (9.5.15)$$

which is impossible for ordinary numbers. However, nothing can stop us from introducing an algebra of ‘variables’ (known as *Grassmann variables*)  $q_a$ , which act like c-numbers as far as the physical Hilbert space is concerned, but which still satisfy the anticommutation relations (9.5.15). We will require further that

$$\{q_a, q'_b\} = \{q_a, Q_b\} = \{q_a, P_b\} = 0, \quad (9.5.16)$$

where  $q$  and  $q'$  denote any two ‘values’ of these variables. We can now construct eigenstates  $|q\rangle$  satisfying Eq. (9.5.14):

$$|q\rangle = \exp\left(-i \sum_a P_a q_a\right) |\Omega\rangle \quad (9.5.17)$$

with the exponential defined as usual by its power series expansion. (To verify Eq. (9.5.14), use the fact that all  $P_a q_a$  commute with one another and have zero square, so that

$$\begin{aligned} [Q_a - q_a]|q\rangle &= [Q_a - q_a] \exp(-iP_a q_a) \exp\left(-i \sum_{b \neq a} P_b q_b\right) |\Omega\rangle \\ &= [Q_a - q_a][1 - iP_a q_a] \exp\left(-i \sum_{b \neq a} P_b q_b\right) |\Omega\rangle \\ &= [-i\{Q_a, P_a\}q_a - q_a] \exp\left(-i \sum_{b \neq a} P_b q_b\right) |\Omega\rangle = 0 \end{aligned}$$

as required by Eq. (9.5.14). We can also define left-eigenstates  $\langle q|$  (*not* the adjoints of  $|q\rangle$ ), as

$$\begin{aligned} \langle q| &\equiv \langle \Omega| \left( \prod_a Q_a \right) \exp\left(-i \sum_a q_a P_a\right) \\ &= \langle \Omega| \left( \prod_a Q_a \right) \exp\left(+i \sum_a P_a q_a\right), \end{aligned} \quad (9.5.18)$$

where  $\prod_a$  is the product in whatever order we take as standard. By the same argument as for Eq. (9.5.14), we see that

$$\langle q|Q_a = \langle q|q_a. \quad (9.5.19)$$

These eigenstates have the scalar product

$$\begin{aligned} \langle q'|q\rangle &= \langle \Omega| \left( \prod_a Q_a \right) \exp\left(i \sum_b P_b (q'_b - q_b)\right) |\Omega\rangle \\ &= \langle \Omega| \left( \prod_a Q_a \right) \left( \prod_b [1 + iP_b (q'_b - q_b)] \right) |\Omega\rangle. \end{aligned}$$

Moving each  $Q_a$  to the right (starting with the rightmost) yields factors  $i^2(q'_a - q_a)$ , which we move to the right out of the scalar product, so

$$\langle q'|q\rangle = \prod_a (q_a - q'_a). \quad (9.5.20)$$

We shall see that Eq. (9.5.20) plays the role of a delta function in integrals over the  $q$ s.

In the same way, we can construct right- and left-eigenstates of the  $P_a$ :

$$P_a|p\rangle = p_a|p\rangle, \quad (9.5.21)$$

$$\langle p|P_a = \langle p|p_a, \quad (9.5.22)$$

where the  $p_a$  are like  $q_a$  anticommuting c-numbers (taken for convenience to anticommute with the  $q_a$  and all fermionic operators as well as each other), and

$$|p\rangle = \exp\left(-i \sum_a Q_a p_a\right) \left( \prod_b P_b \right) |\Omega\rangle, \quad (9.5.23)$$

$$\langle p| = \langle \Omega| \exp \left( -i \sum_a p_a Q_a \right) \quad (9.5.24)$$

with scalar product (now derived by moving the  $P$ s to the left)

$$\langle p'|p\rangle = \prod_a (p'_a - p_a). \quad (9.5.25)$$

The scalar products of these two sorts of eigenstate with each other are

$$\begin{aligned} \langle q|p\rangle &= \langle q| \exp \left( -i \sum_a Q_a p_a \right) \left( \prod_a P_a \right) |\Omega\rangle \\ &= \left( \prod_a \exp(-i q_a p_a) \right) \langle q| \left( \prod_a P_a \right) |\Omega\rangle \\ &= \left( \prod_a \exp(-i q_a p_a) \right) \langle \Omega| \left( \prod_a Q_a \right) \left( \prod_a P_a \right) |\Omega\rangle, \end{aligned}$$

and so

$$\langle q|p\rangle = \chi_N \exp \left( -i \sum_a q_a p_a \right) = \chi_N \exp \left( i \sum_a p_a q_a \right), \quad (9.5.26)$$

where  $\chi_N$  is a phase that depends only on the number  $N$  of  $Q_a$  operators:

$$\chi_N \equiv \langle \Omega| \left( \prod_a Q_a \right) \left( \prod_a P_a \right) |\Omega\rangle = i^N (-1)^{N(N-1)/2}.$$

Somewhat more simply, we also find

$$\langle p|q\rangle = \prod_a \exp(-i p_a q_a). \quad (9.5.27)$$

It is easy to see that the states  $|q\rangle$  are in a sense a complete set (and so also are the  $|p\rangle$ .) From the definitions (9.5.17), we see that the state  $|a, b, \dots\rangle$  in the general basis is (up to a phase) just the coefficient of the product  $q_a q_b \dots$  in an expansion of  $|q\rangle$  in a sum of products of  $q$ s. Therefore we can write any state  $|f\rangle$  in the form

$$|f\rangle = f_0 |q\rangle_0 + \sum_a f_a |q\rangle_a + \sum_{a \neq b} f_{ab} |q\rangle_{ab} + \dots,$$

where the  $f$ s are numerical coefficients, and a subscript  $a, b, \dots$  on  $|q\rangle$  denotes the coefficient of  $q_a q_b \dots$  in  $|q\rangle$ .

In summing over states, it will be very convenient to introduce a sort of integration over fermionic variables, known as *Berezin integration* [10], that is designed to pick out the coefficients of such products of anticommuting c-numbers. For any set of such variables  $\xi_n$  (either  $p$ s or  $q$ s or both together), the most general function  $f(\xi)$  (either a c-number or a state-vector like  $|q\rangle$ ) can be put in the form

$$f(\xi) = \left( \prod_n \xi_n \right) c + \text{terms with fewer } \xi \text{ factors} \quad (9.5.28)$$

and the integral over the  $\xi$ s is defined simply by

$$\int \left( \widetilde{\prod_n} d\xi_n \right) f(\xi) \equiv c \quad (9.5.29)$$

with the tilde in Eq. (9.5.29) indicating that we use the convenient convention that the differentials are written in an order *opposite* to that of the product of integration variables in Eq. (9.5.28). Since this product is antisymmetric under the interchange of any two  $\xi$ s, the integral is likewise antisymmetric under the interchange of any two  $d\xi$ s, so these ‘differentials’ effectively anticommute

$$d\xi_n d\xi_m + d\xi_m d\xi_n = 0. \quad (9.5.30)$$

Also, the coefficient  $c$  may itself depend on other unintegrated c-number variables that anticommute with the  $\xi$ s over which we integrate, in which case it is important to standardize the definition of  $c$  by moving all  $\xi$ s to the left of  $c$  before integrating over them, as we have done in Eq. (9.5.28).

For instance, the most general function of a pair of anticommuting c-numbers  $\xi_1$  and  $\xi_2$  takes the form

$$f(\xi_1, \xi_2) = \xi_1 \xi_2 c_{12} + \xi_1 c_1 + \xi_2 c_2 + d$$

because the squares and all higher powers of  $\xi_1$  and  $\xi_2$  vanish. This function has the integrals

$$\begin{aligned} \int d\xi_1 f(\xi_1, \xi_2) &= \xi_2 c_{12} + c_1, & \int d\xi_2 f(\xi_1, \xi_2) &= -\xi_1 c_{12} + c_2, \\ \int d\xi_2 d\xi_1 f(\xi_1, \xi_2) &= c_{12}. \end{aligned}$$

Note that the multiple integral is the same as a repeated integral:

$$\int d\xi_2 d\xi_1 f(\xi_1, \xi_2) = \int d\xi_2 \left[ \int d\xi_1 f(\xi_1, \xi_2) \right],$$

a result that can easily be extended to integrals over any number of fermionic variables. (It was in order to obtain this result without extra sign factors that we took the product of differentials in Eq. (9.5.29) to be in the opposite order to the product of variables in Eq. (9.5.28).) Indeed, we could have first defined the integral over a single anticommuting c-number  $\xi_1$ , and then defined multiple integrals in the usual way by iteration. The most general function of anticommuting c-numbers is linear in any one of them

$$f(\xi_1, \xi_2, \dots) = b(\xi_2, \dots) + \xi_1 c(\xi_2, \dots)$$

(because  $\xi_1^2 = 0$ ), and its integral over  $\xi_1$  is defined as

$$\int d\xi_1 f(\xi_1, \xi_2, \dots) = c(\xi_2, \dots).$$

Repeating this process leads to the same multiple integral as defined by Eqs. (9.5.28) and (9.5.29).



This definition of integration shares some other properties with multiple integrals (from  $-\infty$  to  $+\infty$ ) over ordinary real variables, but there are significant differences.

Obviously, Berezin integration is linear, in the sense that

$$\int \left( \widetilde{\prod}_n d\xi_n \right) [f(\xi) + g(\xi)] = \int \left( \widetilde{\prod}_n d\xi_n \right) f(\xi) + \int \left( \widetilde{\prod}_n d\xi_n \right) g(\xi) \quad (9.5.31)$$

and also

$$\int \left( \widetilde{\prod}_n d\xi_n \right) [f(\xi) a(\xi')] = \left[ \int \left( \widetilde{\prod}_n d\xi_n \right) f(\xi) \right] a(\xi'), \quad (9.5.32)$$

where  $a(\xi')$  is any function (including a constant) of any anticommuting c-numbers  $\xi'_m$  over which we are *not* integrating. However, linearity with respect to left-multiplication is not so obvious. If we are integrating over  $\nu$  variables, then since  $\xi'_m$  is assumed to anticommute with all  $\xi_n$ , we have

$$a((-1)^\nu \xi') \left( \prod_n \xi_n \right) = \left( \prod_n \xi_n \right) a(\xi')$$

and so

$$\int \left( \widetilde{\prod}_n d\xi_n \right) [a((-1)^\nu \xi') f(\xi)] = a(\xi') \int \left( \widetilde{\prod}_n d\xi_n \right) f(\xi). \quad (9.5.33)$$

It is therefore very convenient (though not strictly necessary) to take the differentials  $d\xi_n$  to anticommute with all anticommuting variables (including the  $\xi_n$ ):

$$(d\xi_n) \xi'_m + \xi'_m (d\xi_n) = 0 \quad (9.5.34)$$

in which case Eq. (9.5.33) reads more simply

$$\int a(\xi') \left( \widetilde{\prod}_n d\xi_n \right) f(\xi) = a(\xi') \int \left( \widetilde{\prod}_n d\xi_n \right) f(\xi). \quad (9.5.35)$$

Another similarity with ordinary integration is that, for an arbitrary anticommuting c-number  $\xi'$  independent of  $\xi$ ,

$$\int \left( \widetilde{\prod}_n d\xi_n \right) f(\xi + \xi') = \int \left( \widetilde{\prod}_n d\xi_n \right) f(\xi) \quad (9.5.36)$$

since shifting  $\xi$  by a constant only affects the terms in  $f$  with fewer than the total number of  $\xi$ -variables.

On the other hand, consider a change of variables

$$\xi_n \longrightarrow \xi'_n = \sum_m \mathcal{G}_{nm} \xi_m, \quad (9.5.37)$$

where  $\mathcal{G}$  is an arbitrary non-singular matrix of ordinary numbers. The product of the new variables is

$$\prod_n \xi'_n = \sum_{m_1 m_2 \dots} \left( \prod_n \mathcal{G}_{nm_n} \xi_{m_n} \right).$$

But  $\prod_n \xi_{m_n}$  here is just the same as the product (in the original order)  $\prod_n \xi_n$ , except for a sign  $\epsilon[m]$  which is  $+1$  or  $-1$  according to whether the permutation  $n \rightarrow m_n$  is an even or odd permutation of the original order:

$$\prod_n \xi'_n = \left[ \sum_{m_1 m_2 \dots} \left( \prod_n \mathcal{G}_{nm_n} \right) \epsilon[m] \right] \prod_n \xi_n = (\text{Det } \mathcal{G}) \prod_n \xi_n.$$

This applies whatever order we take for the  $\xi_n$ , as long as we take the  $\xi'_n$  in the same order. It follows that the coefficient of  $\prod_n \xi'_n$  in any function  $f(\xi)$  is just  $(\text{Det } \mathcal{G})^{-1}$  times the coefficient of  $\prod_n \xi_n$ , a statement we write as

$$\int \left( \widetilde{\prod}_n d\xi'_n \right) f = (\text{Det } \mathcal{G})^{-1} \int \left( \widetilde{\prod}_n d\xi_n \right) f. \quad (9.5.38)$$

This is the usual rule for changing variables of integration, except that  $(\text{Det } \mathcal{G})$  appears to the power  $-1$  instead of  $+1$ . We shall use Eq. (9.5.38) and the linearity properties (9.5.31), (9.5.32), and (9.5.35) later to evaluate the integrals encountered in deriving the Feynman rules for theories with fermions.

We can now use this definition of integration to write the completeness condition as a formula for an integral over eigenvalues. As already mentioned, any state  $|f\rangle$  can be expanded in a series of the states  $|\Omega\rangle$ ,  $|a\rangle$ ,  $|a, b\rangle$ , etc. and these states are (up to a phase) the coefficients of the products  $1$ ,  $q_a$ ,  $q_a q_b$ , etc. in the  $Q$ -eigenstate  $|q\rangle$ . According to the definition of integration here, we can pick out the coefficient of any product  $q_b q_c q_d \dots$  in the state  $|q\rangle$  by integrating the product of  $|q\rangle$  with all  $q_a$  with  $a$  *not* equal to  $b, c, d, \dots$ . Thus, by choosing a function  $f(q)$  as a suitable sum of such products of  $q$ s, we can write any state  $|f\rangle$  as an integral:

$$|f\rangle = \int \left( \widetilde{\prod}_a dq_a \right) |q\rangle f(q) = \int |q\rangle \left( \widetilde{\prod}_a dq_a \right) f(q). \quad (9.5.39)$$

(We can move  $|q\rangle$  to the left of the differentials without any sign changes because the exponential in Eq. (9.5.17) used to define  $|q\rangle$  involves only even numbers of fermionic quantities.) To find the function  $f(q)$  for a given state-vector  $|f\rangle$ , take the scalar product of Eq. (9.5.39) with some bra  $\langle q'|$  (with  $q'$  any fixed  $Q$ -eigenvalue). According to Eqs. (9.5.35) and (9.5.20), this is

$$\langle q'|f\rangle = \int \left( \prod_a (q_a - q'_a) \right) \left( \widetilde{\prod}_b dq_b \right) f(q).$$

Moving every factor  $(q_a - q'_a)$  to the right past every differential  $dq_b$  yields a sign factor  $(-)^{N^2} = (-)^N$ , where  $N$  is now the total number of  $q_a$  variables, so

$$\langle q'|f\rangle = (-)^N \int \left( \widetilde{\prod}_b dq_b \right) \left( \prod_a (q_a - q'_a) \right) f(q).$$

We can rewrite  $f(q)$  as  $f(q' + (q - q'))$  and expand in powers of  $q - q'$ . All terms beyond the lowest order vanish when multiplied with the product

$\prod_a(q_a - q'_a)$ , so

$$\left(\prod_a(q_a - q'_a)\right) f(q) = \left(\prod_a(q_a - q'_a)\right) f(q'), \quad (9.5.40)$$

which partly justifies our earlier remark that Eq. (9.5.20) plays the role of a delta function for integrals over the  $q$ s. Using Eq. (9.5.32), we now have

$$\langle q' | f \rangle = (-)^N \left[ \int \left( \widetilde{\prod}_b dq_b \right) \left( \prod_a(q_a - q'_a) \right) f(q') \right].$$

The term in the integrand proportional to  $\prod_a q_a$  has coefficient  $f(q')$ , so according to our definition of integration  $\langle q' | f \rangle = (-)^N f(q')$ . Inserting this back in Eq. (9.5.39) gives our completeness relation

$$|f\rangle = (-)^N \int |q\rangle \left( \widetilde{\prod}_b dq_b \right) \langle q | f \rangle,$$

or as an operator equation

$$\mathbf{1} = \int |q\rangle \left( \widetilde{\prod}_a(-dq_a) \right) \langle q|. \quad (9.5.41)$$

In exactly the same way, we can also show that

$$\mathbf{1} = \int |p\rangle \left( \widetilde{\prod}_a dp_a \right) \langle p|. \quad (9.5.42)$$

We are now in a position to calculate transition matrix elements. As before, we define time-dependent operators

$$Q_a(t) = \exp(iHt) Q_a \exp(-iHt) \quad (9.5.43)$$

$$P_a(t) = \exp(iHt) P_a \exp(-iHt) \quad (9.5.44)$$

and their right- and left-eigenstates

$$|q; t\rangle \equiv \exp(iHt) |q\rangle, \quad |p; t\rangle \equiv \exp(iHt) |p\rangle, \quad (9.5.45)$$

$$\langle q; t| \equiv \langle q| \exp(-iHt), \quad \langle p; t| \equiv \langle p| \exp(-iHt). \quad (9.5.46)$$

The scalar product between  $q$ -eigenstates defined at infinitesimally close times is then

$$\langle q'; \tau + d\tau | q; \tau \rangle = \langle q' | \exp(-iH d\tau) | q \rangle.$$

Now insert Eq. (9.5.42) to the left of the operator  $\exp(-iH d\tau)$ . It is convenient here to define the Hamiltonian operator  $H(P, Q)$  with all  $P$ s to the left of all  $Q$ s, so that (for  $d\tau$  infinitesimal)

$$\langle p | \exp[-iH(P, Q)d\tau] | q \rangle = \langle p | q \rangle \exp[-iH(p, q)d\tau].$$

(We could move the c-number  $H(p, q)$  to either side of the matrix element without any sign changes because each term in the Hamiltonian is assumed to contain an even number of fermionic operators.) This gives

$$\begin{aligned}\langle q'; \tau + d\tau | q; \tau \rangle &= \int \langle q' | p \rangle \left( \widetilde{\prod}_a dp_a \right) \langle p | \exp(-iH d\tau) | q \rangle \\ &= \int \langle q' | p \rangle \left( \widetilde{\prod}_a dp_a \right) \langle p | q \rangle \exp[-iH(p, q)d\tau].\end{aligned}$$

Using Eqs. (9.5.26) and (9.5.27), and noting that the products  $p_a q_a$  and  $p_a q'_a$  commute with all anticommuting c-numbers, we find

$$\langle q'; \tau + d\tau | q; \tau \rangle = \int \left( \widetilde{\prod}_a i dp_a \right) \exp \left[ i \sum_a p_a (q'_a - q_a) - iH(p, q)d\tau \right]. \quad (9.5.47)$$

The rest of the derivation follows the same lines as in Section 9.1. To calculate the matrix element  $\langle q'; t' | \mathcal{O}_A(P(t_A), Q(t_A)) \mathcal{O}_B(P(t_B), Q(t_B)) \cdots | q; t \rangle$  of a product of operators (with  $t' > t_A > t_B > \cdots > t$ ), divide the time-interval from  $t$  to  $t'$  into a large number of very close time steps; at each time step insert the completeness relation (9.5.41); use Eq. (9.5.47) to evaluate the resulting matrix elements (with  $\mathcal{O}_A, \mathcal{O}_B$ , etc. inserted where appropriate); move all differentials to the left (this introduces no sign changes, because at each step we have an equal number of  $dp$ s and  $dq$ s); and then introduce functions  $q_a(t)$  and  $p_a(t)$  that interpolate between the values of  $q_a$  and  $p_a$  at each step. We then find

$$\begin{aligned}&\langle q'; t' | T \{ \mathcal{O}_A(P(t_A), Q(t_A)), \mathcal{O}_B(P(t_B), Q(t_B)), \dots \} | q; t \rangle \\ &= (-i)^N \chi_N \int_{q_a(t)=q_a, q_a(t')=q'_a} \left( \widetilde{\prod}_{a\tau} dq_a(\tau) dp_a(\tau) \right) \\ &\quad \cdot \mathcal{O}_A[p(t_A), q(t_A)] \mathcal{O}_B[p(t_B), q(t_B)] \cdots \\ &\quad \cdot \exp \left[ i \int_t^{t'} d\tau \left\{ \sum_a p_a(\tau) \dot{q}_a(\tau) - H[p(\tau), q(\tau)] \right\} \right]. \quad (9.5.48)\end{aligned}$$

The symbol  $T$  here denotes the ordinary product if the times are in the order originally assumed,  $t_A > t_B > \cdots$ . However, the right-hand side is totally symmetric in the  $\mathcal{O}_A, \mathcal{O}_B, \dots$  (except for minus signs where anticommuting c-numbers are interchanged) so this formula holds for general times (between  $t$  and  $t'$ ), provided  $T$  is interpreted as the time-ordered product, with an overall minus sign if time-ordering the operators involves an odd number of permutations of fermionic operators.

Up to this point we have kept track of the overall phase factor  $(-i)^N \chi_N$ . But in fact these phases contribute only to the vacuum–vacuum transition amplitude, and hence will not be of importance to us.

The transition to quantum field theory follows along the same lines as described for bosonic fields in Section 9.2. The vacuum expectation value of a time-ordered product of

operators is given by a formula just like Eq. (9.2.17):

$$\begin{aligned}
& {}_{\text{out}}\langle\Omega|T\{\mathcal{O}_A[P(t_A), Q(t_A)], \\
& \quad \mathcal{O}_B[P(t_B), Q(t_B)], \dots\}|\Omega\rangle_{\text{in}} \\
& \propto \int \left[ \prod_{\tau, \mathbf{x}, m} dq_m(\mathbf{x}, \tau) \right] \left[ \prod_{\tau, \mathbf{x}, m} dp_m(\mathbf{x}, \tau) \right] \\
& \quad \cdot \mathcal{O}_A[p(t_A), q(t_A)] \mathcal{O}_B[p(t_B), q(t_B)] \cdots \\
& \quad \cdot \exp \left[ i \int_{-\infty}^{\infty} d\tau \left\{ \int d^3x \sum_m p_m(\mathbf{x}, \tau) \dot{q}_m(\mathbf{x}, \tau) \right. \right. \\
& \quad \quad \left. \left. - H[q(\tau), p(\tau)] + i\epsilon \text{ terms} \right\} \right], \tag{9.5.49}
\end{aligned}$$

where the proportionality constant is the same for all operators  $\mathcal{O}_A, \mathcal{O}_B$ , etc., and the ‘ $i\epsilon$  terms’ again arise from the wave function of the vacuum. As before, we have replaced each discrete index like  $a$  with a space position  $\mathbf{x}$  and a field index  $m$ . We are also dropping the tilde on the product of differentials, since it only affects the constant phase in the path integral.

A major difference between the fermionic and bosonic cases is that here we will not want to integrate out the  $ps$  before the  $qs$ . Indeed, in the standard model of electroweak interactions (and in other theories, such as the older Fermi theory of beta decay) the canonical conjugates  $p_m$  are auxiliary fields unrelated to the  $q_m$ , and the Lagrangian is linear in the  $\dot{q}_m$ , so that the quantity  $\int d^3x \sum_m p_m \dot{q}_m - H$  in Eq. (9.5.49) as it stands is the Lagrangian  $L$ . Each term in the Hamiltonian for a fermionic field that carries a non-vanishing quantum number (like the electron field in quantum electrodynamics) generally contains an equal number of  $ps$  (proportional to  $q_m^\dagger$ ) and  $qs$ . In particular, the free-particle term  $H_0$  in the Hamiltonian is bilinear in  $p$  and  $q$ , so that

$$\begin{aligned}
& \int_{-\infty}^{\infty} d\tau \left\{ \int d^3x \sum_m p_m(\mathbf{x}, \tau) \dot{q}_m(\mathbf{x}, \tau) - H_0[q(\tau), p(\tau)] + i\epsilon \text{ terms} \right\} \\
& = - \sum_{mn} \int d^4x d^4y \mathcal{D}_{m\mathbf{x}, n\mathbf{y}} p_m(x) q_n(y) \tag{9.5.50}
\end{aligned}$$

with  $\mathcal{D}$  some numerical ‘matrix’. The interaction Hamiltonian  $H_{\text{int}} = H - H_0$  is a sum of products of equal numbers of fermionic  $qs$  and  $ps$  (with coefficients that may depend on bosonic fields) so when we expand Eq. (9.5.49) in powers of the  $H_{\text{int}}$  we encounter a sum of fermionic integrals of the form

$$\begin{aligned}
& \mathcal{I}_{n_1 m_1 n_2 m_2 \dots n_N m_N}(x_1, y_1, x_2, y_2, \dots, x_N, y_N) \\
& \equiv \int \left[ \prod_{\tau, \mathbf{x}, m} dq_m(\mathbf{x}, \tau) \right] \left[ \prod_{\tau, \mathbf{x}, m} dp_m(\mathbf{x}, \tau) \right] q_{m_1}(x_1) p_{n_1}(y_1) q_{m_2}(x_2) p_{n_2}(y_2) \cdots \\
& \quad \cdot q_{m_N}(x_N) p_{n_N}(y_N) \\
& \quad \cdot \exp \left( -i \sum_{mn} \int d^4x d^4y \mathcal{D}_{m\mathbf{x}, n\mathbf{y}} p_m(x) q_n(y) \right), \tag{9.5.51}
\end{aligned}$$

one such term for each possible set of vertices in the Feynman diagram, with coefficients contributed by each vertex given by  $i$  times the coefficient of the product of fields in the corresponding term in the interaction.

To calculate this sort of integral, first consider a generating function for all these integrals:

$$\begin{aligned} \mathcal{J}(f, g) = & \int \left[ \prod_{\mathbf{x}, \tau, m} dq_m(\mathbf{x}, \tau) dp_m(\mathbf{x}, \tau) \right] \\ & \cdot \exp \left\{ \begin{aligned} & -i \sum_{mn} \int d^4x d^4y \mathcal{D}_{m\mathbf{x}, n\mathbf{y}} p_m(x) q_n(y) \\ & -i \sum_m \int d^4x p_m(x) f_m(x) \\ & -i \sum_n \int d^4y g_n(y) q_n(y) \end{aligned} \right\}, \end{aligned} \quad (9.5.52)$$

where  $f_m(x)$  and  $g_n(y)$  are arbitrary anticommuting c-number functions. We shift to new variables of integration

$$\begin{aligned} p'_m(x) &= p_m(x) + \sum_n \int d^4y g_n(y) (\mathcal{D}^{-1})_{n\mathbf{y}, m\mathbf{x}}, \\ q'_n(y) &= q_n(y) + \sum_m \int d^4x (\mathcal{D}^{-1})_{n\mathbf{y}, m\mathbf{x}} f_m(x). \end{aligned}$$

Using the translation invariance condition (9.5.36), we then find

$$\begin{aligned} \mathcal{J}(f, g) = & \exp \left( i \sum_{mn} \int d^4x d^4y (\mathcal{D}^{-1})_{n\mathbf{y}, m\mathbf{x}} g_n(y) f_m(x) \right) \\ & \cdot \int \left[ \prod_{\mathbf{x}, \tau, m} dq'_m(\mathbf{x}, \tau) dp'_m(\mathbf{x}, \tau) \right] \\ & \cdot \exp \left( -i \sum_{mn} \int d^4x d^4y \mathcal{D}_{m\mathbf{x}, n\mathbf{y}} p'_m(x) q'_n(y) \right). \end{aligned} \quad (9.5.53)$$

The integral is a constant (i.e., independent of the functions  $f$  and  $g$ ) which can be shown using Eq. (9.5.38) to be proportional to  $\text{Det } \mathcal{D}$ . Of more importance to us is the first factor. Expanding this factor in powers of  $gf$  and comparing with the direct expansion of Eq. (9.5.52), we see that

$$\begin{aligned} & \mathcal{I}_{n_1 m_1 n_2 m_2 \dots n_N m_N}(x_1, y_1, x_2, y_2, \dots, x_N, y_N) \\ & \propto \sum_{\text{pairings}} \delta_{\text{pairing}} \prod_{\text{pairs}} (-i \mathcal{D}^{-1})_{\text{paired } m\mathbf{x}, n\mathbf{y}}, \end{aligned} \quad (9.5.54)$$

with a proportionality constant that is independent of the  $x$ ,  $y$ ,  $m$ , or  $n$ , and also independent of the number of these variables. The sum is over all different ways of pairing  $ps$  with  $qs$ , not counting as different pairings that only differ in the order of the pairs. In other

words, we sum over the  $N!$  permutations either of the  $ps$  or the  $qs$ . The sign factor  $\delta_{\text{pairing}}$  is  $+1$  if this permutation is even;  $-1$  if it is odd.

This sign factor and sum over pairings are just the same as we encountered in our earlier derivation of the Feynman rules, with the sum over pairings corresponding to the sum over ways of connecting the lines associated with vertices in the Feynman diagrams, and the factors  $(\mathcal{D}^{-1})_{m\mathbf{x},n\mathbf{y}}$  playing the role of the propagator for the pairing of  $q_m(x)$  with  $p_n(y)$ . In the Dirac formalism for spin  $\frac{1}{2}$ , the free-particle action is

$$\begin{aligned} & \int_{-\infty}^{\infty} d\tau \left\{ \int d^3x \sum_m p_m(\mathbf{x}, \tau) \dot{q}_m(\mathbf{x}, \tau) - H_0[q(\tau), p(\tau)] \right\} \\ &= - \int d^4x \bar{\psi}(x) [\gamma^\mu \partial_\mu + m] \psi(x), \end{aligned} \quad (9.5.55)$$

where in the usual notation the canonical variables here are

$$q_m(x) = \psi_m(x), \quad p_m(x) = -[\bar{\psi}(x) \gamma^0]_m = i\psi_m^\dagger(x) \quad (9.5.56)$$

with  $m$  a four-valued Dirac index. Comparing this with Eq. (9.5.50), we find here

$$\begin{aligned} \mathcal{D}_{m\mathbf{x},n\mathbf{y}} &= \left\{ \gamma^0 \left[ \gamma^\mu \frac{\partial}{\partial x^\mu} + m - i\epsilon \right] \right\}_{mn} \delta^4(x - y) \\ &= \int \frac{d^4k}{(2\pi)^4} \{ \gamma^0 [i\gamma^\mu k_\mu + m - i\epsilon] \}_{mn} e^{ik \cdot (x-y)}. \end{aligned} \quad (9.5.57)$$

(Though we shall not work it out in detail, the  $i\epsilon$  term here arises in much the same way as for the scalar field in Section 9.2.) The propagator is then

$$(\mathcal{D}^{-1})_{m\mathbf{x},n\mathbf{y}} = \int \frac{d^4k}{(2\pi)^4} \{ [i\gamma^\mu k_\mu + m - i\epsilon]^{-1} [-\gamma^0] \}_{mn} e^{ik \cdot (x-y)}, \quad (9.5.58)$$

just as we found in the operator formalism. The extra factor  $-\gamma^0$  arises because this propagator is the vacuum expectation value of

$$T\{\psi_m(x), -[\bar{\psi}(y) \gamma^0]_n\}, \quad \text{not} \quad T\{\psi_m(x), \bar{\psi}_n(y)\}.$$

As one example of a problem that is easier to solve by path-integral than by operator methods, let us calculate the field dependence of the vacuum  $\rightarrow$  vacuum amplitude for a Dirac field that interacts only with an external field. Take the Lagrangian as

$$\mathcal{L} = -\bar{\psi}[\gamma^\mu \partial_\mu + m + \Gamma]\psi, \quad (9.5.59)$$

where  $\Gamma(x)$  is an  $x$ -dependent matrix representing the interaction of the fermion with the external field. According to Eq. (9.5.49), the vacuum persistence amplitude in the presence of this external field is

$$\begin{aligned} {}_{\text{out}}\langle \Omega | \Omega \rangle_{\text{in}, \Gamma} &\propto \int \left[ \prod_{\tau, \mathbf{x}, m} dq_m(\mathbf{x}, \tau) \right] \left[ \prod_{\tau, \mathbf{x}, m} dp_m(\mathbf{x}, \tau) \right] \\ &\cdot \exp \left\{ -i \int d^4x p^\top \gamma^0 [\gamma^\mu \partial_\mu + m + \Gamma - i\epsilon] q \right\} \end{aligned} \quad (9.5.60)$$

with a proportionality constant that is independent of  $\Gamma(x)$ . We write this as

$$\begin{aligned} \text{out} \langle \Omega | \Omega \rangle_{\text{in}, \Gamma} &\propto \int \left[ \prod_{\tau, \mathbf{x}, m} dq_m(\mathbf{x}, \tau) \right] \left[ \prod_{\tau, \mathbf{x}, m} dp_m(\mathbf{x}, \tau) \right] \\ &\cdot \exp \left\{ -i \sum_{mn} \int d^4x d^4y p_m(x) q_n(y) \mathcal{K}[\Gamma]_{m\mathbf{x}, n\mathbf{y}} \right\}, \end{aligned} \quad (9.5.61)$$

where

$$\mathcal{K}[\Gamma]_{m\mathbf{x}, n\mathbf{y}} = \{ \gamma^0 [\gamma^\mu \partial / \partial x^\mu + m + \Gamma(x) - i\epsilon] \}_{mn} \delta^4(x - y). \quad (9.5.62)$$

To evaluate this, we change the variables of integration  $q_m(x)$  to

$$q'_m(x) = \sum_n \int d^4y \mathcal{K}[\Gamma]_{m\mathbf{x}, n\mathbf{y}} q_n(y). \quad (9.5.63)$$

The remaining integral is now  $\Gamma$ -independent, so the whole dependence of the vacuum persistence amplitude is contained in the determinant arising according to Eq. (9.5.38) from the change of variables:

$$\text{out} \langle \Omega | \Omega \rangle_{\text{in}, \Gamma} \propto \text{Det } \mathcal{K}[\Gamma]. \quad (9.5.64)$$

To recover the results of perturbation theory, let us write

$$\mathcal{K}[\Gamma] \equiv \mathcal{D} + \mathcal{G}[\Gamma], \quad (9.5.65)$$

$$\mathcal{G}[\Gamma]_{m\mathbf{x}, n\mathbf{y}} = (\gamma^0 \Gamma(x))_{mn} \delta^4(x - y), \quad (9.5.66)$$

and expand in powers of  $\mathcal{G}[\Gamma]$ . Eq. (9.5.64) gives then

$$\begin{aligned} \text{out} \langle \Omega | \Omega \rangle_{\text{in}, \Gamma} &\propto \text{Det} (\mathcal{D} [\mathbf{1} + \mathcal{D}^{-1} \mathcal{G}[\Gamma]]) \\ &= [\text{Det } \mathcal{D}] \exp \left( \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \text{Tr}(\mathcal{D}^{-1} \mathcal{G}[\Gamma])^n \right). \end{aligned} \quad (9.5.67)$$

This is just what we should expect from the Feynman rules: the contributions from internal lines and vertices in this theory are  $-i\mathcal{D}^{-1}$  and  $-i\mathcal{G}[\Gamma]$ ; the trace of the product of  $n$  factors of  $-\mathcal{D}^{-1}\mathcal{G}[\Gamma]$  thus corresponds to a loop with  $n$  vertices connected by  $n$  internal lines;  $1/n$  is the usual combinatoric factor associated with such loops (see Section 6.1); the sign factor is  $(-1)^{n+1}$  rather than  $(-1)^n$  because an extra minus sign is associated with fermion loops; and the sum over  $n$  appears as the argument of an exponential because the vacuum persistence amplitude receives contributions from graphs with any number of disconnected loops. The  $\Gamma$ -independent factor  $\text{Det } \mathcal{D}$  is less easy to derive from the Feynman rules; it represents the contribution of any number of fermion loops that carry no vertices.

More to the point, a formula like Eq. (9.5.64) allows us to derive non-perturbative results by using topological theorems to derive information about the eigenvalues of kernels like  $\mathcal{K}[\Gamma]$ . This will be pursued further in Volume II.



## 9.6 Path-Integral Formulation of Quantum Electrodynamics

The path-integral approach to quantum field theory really comes into its own when applied to gauge theories of massless spin one particles, such as quantum electrodynamics. The derivation of the Feynman rules for quantum electrodynamics in the previous chapter involved a fair amount of hand-waving, in arguing that the terms in the photon propagator  $\Delta^{\mu\nu}(q)$  proportional to  $q^\mu$  or  $q^\nu$  could be dropped, and that the purely time-like terms would just cancel the Coulomb term in the Hamiltonian, so that the effective photon propagator could be taken as  $\eta^{\mu\nu}/q^2$ . To give a real justification of this result using the methods of Chapter 8 would involve us in a complicated analysis of Feynman diagrams. But as we shall now see, the path-integral approach yields the desired form of the photon propagator, without ever having to think about the details of Feynman diagrams.

In Chapter 8 we found that in Coulomb gauge, the Hamiltonian for the interaction of photons with charged particles takes the form

$$H[\mathbf{A}, \mathbf{\Pi}_\perp, \dots] = H_M + \int d^3x \left[ \frac{1}{2} \mathbf{\Pi}_\perp^2 + \frac{1}{2} (\nabla \times \mathbf{A})^2 - \mathbf{A} \cdot \mathbf{J} \right] + V_{\text{Coul}}. \quad (9.6.1)$$

Here  $\mathbf{A}$  is the vector potential, subject to the Coulomb gauge condition

$$\nabla \cdot \mathbf{A} = 0, \quad (9.6.2)$$

while  $\mathbf{\Pi}_\perp$  is the solenoidal part of its canonical conjugate, satisfying the same constraint

$$\nabla \cdot \mathbf{\Pi}_\perp = 0. \quad (9.6.3)$$

Also,  $H_M$  is the matter Hamiltonian and  $V_{\text{Coul}}$  is the Coulomb energy

$$V_{\text{Coul}}(t) = \frac{1}{2} \int d^3x d^3y \frac{J^0(\mathbf{x}, t) J^0(\mathbf{y}, t)}{4\pi |\mathbf{x} - \mathbf{y}|}. \quad (9.6.4)$$

Just as for any other Hamiltonian system, we can calculate vacuum expectation values of time-ordered products as path integrals<sup>103</sup>

$$\begin{aligned} \langle T\{\mathcal{O}_A \mathcal{O}_B \cdots\} \rangle_\Omega &= \int \left[ \prod_{x,i} da_i(x) \prod_{x,i} d\pi_i(x) \prod_{x,\ell} d\psi_\ell(x) \right] \\ &\quad \cdot \mathcal{O}_A \mathcal{O}_B \cdots \\ &\quad \cdot \exp \left\{ i \int d^4x \left[ \boldsymbol{\pi} \cdot \dot{\mathbf{a}} - \frac{1}{2} \boldsymbol{\pi}^2 - \frac{1}{2} (\nabla \times \mathbf{a})^2 \right. \right. \\ &\quad \left. \left. + \mathbf{a} \cdot \mathbf{j} + \mathcal{L}_M \right] - i \int dt V_{\text{Coul}} \right\} \\ &\quad \cdot \left[ \prod_x \delta(\nabla \cdot \mathbf{a}(x)) \right] \left[ \prod_x \delta(\nabla \cdot \boldsymbol{\pi}(x)) \right], \end{aligned} \quad (9.6.5)$$

<sup>103</sup>Note that  $\boldsymbol{\pi}(x)$  is the interpolating c-number field for the quantum operator  $\mathbf{\Pi}_\perp$ , whose commutation relations with each other and with  $\mathbf{A}$  are the same as those of  $\mathbf{\Pi}$ , but which unlike  $\mathbf{\Pi}$  commutes with all canonical matter variables.

where  $\psi_\ell(x)$  are generic matter fields. In writing Eq. (9.6.5) in terms of a matter Lagrangian density, we are assuming that  $H_M$  is local and either linear in the matter  $\pi$ s (as in spinor electrodynamics) or quadratic with field-independent coefficients (as in scalar electrodynamics). We have inserted delta functions<sup>104</sup> in Eq. (9.6.5) to enforce the constraints (9.6.2) and (9.6.3).

The argument of the exponential in Eq. (9.6.5) is evidently quadratic in the independent components of  $\boldsymbol{\pi}$  (say,  $\pi_1$  and  $\pi_2$ ), with field-independent coefficients in the term of second order in  $\boldsymbol{\pi}$ . Thus, according to Eq. (9.A.9), the integral over  $\boldsymbol{\pi}$  can be done (up to a constant factor) by setting  $\boldsymbol{\pi}$  equal to the stationary point of the argument of the exponential,  $\boldsymbol{\pi} = \dot{\mathbf{a}}$ :

$$\begin{aligned} \langle T\{\mathcal{O}_A \mathcal{O}_B \cdots\} \rangle_\Omega &= \int \left[ \prod_{x,i} da_i(x) \prod_{x,\ell} d\psi_\ell(x) \right] \\ &\cdot \mathcal{O}_A \mathcal{O}_B \cdots \\ &\cdot \exp \left\{ i \int d^4x \left[ \frac{1}{2} \dot{\mathbf{a}}^2 - \frac{1}{2} (\boldsymbol{\nabla} \times \mathbf{a})^2 \right. \right. \\ &\quad \left. \left. + \mathbf{a} \cdot \mathbf{j} + \mathcal{L}_M \right] - i \int dt V_{\text{Coul}} + i\epsilon \text{ terms} \right\} \\ &\cdot \left[ \prod_x \delta(\boldsymbol{\nabla} \cdot \mathbf{a}(x)) \right]. \end{aligned} \quad (9.6.6)$$

To bring out the essential covariance of this result, we use a trick. Introduce a new variable of integration  $a^0(x)$ , and replace the Coulomb term  $-\int dt V_{\text{Coul}}$  in the action with

$$\int d^4x \left[ -a^0(x) j^0(x) + \frac{1}{2} (\boldsymbol{\nabla} a^0(x))^2 \right]. \quad (9.6.7)$$

Since (9.6.7) is quadratic in  $a^0$ , the integral over  $a^0$  can be done (up to a constant factor) by setting  $a^0(x)$  equal to the stationary point of (9.6.7), i.e., to the solution of

$$-j^0(x) - \boldsymbol{\nabla}^2 a^0(x) = 0$$

or in other words, to

$$a^0(\mathbf{x}, t) = \int d^3y \frac{j^0(\mathbf{y}, t)}{4\pi|\mathbf{x} - \mathbf{y}|}. \quad (9.6.8)$$

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<sup>104</sup>This is not strictly accurate. If we take the canonical variables to be, say,  $a_1, a_2$  and  $\pi_1, \pi_2$ , with  $a_3$  and  $\pi_3$  regarded as functionals of these variables given by Eqs. (9.6.2) and (9.6.3), then we should insert the delta functions

$$\prod_x \delta(a_3(x) + \partial_3^{-1}[\partial_1 a_1(x) + \partial_2 a_2(x)]) \delta(\pi_3(x) + \partial_3^{-1}[\partial_1 \pi_1(x) + \partial_2 \pi_2(x)]).$$

However, this differs from the product of delta functions in Eq. (9.6.5) only by a factor  $\text{Det } \partial_3^2$ , which although infinite is field-independent and hence cancels in ratios like (9.4.1).

Using this in Eq. (9.6.7) just gives the Coulomb action  $-\int dt V_{\text{Coul}}$ . Hence we can rewrite the argument of the exponential in Eq. (9.6.6) as

$$\begin{aligned} & \frac{1}{2}\dot{\mathbf{a}}^2 - \frac{1}{2}(\nabla \times \mathbf{a})^2 + \mathbf{a} \cdot \mathbf{j} + \mathcal{L}_M - a^0 j^0 + \frac{1}{2}(\nabla a^0)^2 \\ &= -\frac{1}{4}f_{\mu\nu}f^{\mu\nu} + a_\mu j^\mu + \mathcal{L}_M + \text{total derivatives} \end{aligned}$$

with  $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$ , and integrate over  $a^0$  as well as over  $\mathbf{a}$  and matter fields. That is, the path integral (9.6.6) is now

$$\begin{aligned} \langle T\{\mathcal{O}_A \mathcal{O}_B \cdots\} \rangle_\Omega &\propto \int \left[ \prod_{x,\mu} da_\mu(x) \right] \left[ \prod_{x,\ell} d\psi_\ell(x) \right] \mathcal{O}_A \mathcal{O}_B \cdots \\ &\quad \cdot \exp(iI[a, \psi]) \prod_x \delta(\nabla \cdot \mathbf{a}(x)), \end{aligned} \quad (9.6.9)$$

where  $I$  is the original action

$$I[a, \psi] = \int d^4x \left[ -\frac{1}{4}f_{\mu\nu}f^{\mu\nu} + a_\mu j^\mu + \mathcal{L}_M \right] + i\epsilon \text{ terms.} \quad (9.6.10)$$

Now everything is manifestly Lorentz- and gauge-invariant, except for the final product of delta functions which enforce the Coulomb gauge condition.<sup>105</sup> To make further progress, we shall use a simple version of a trick [4, 5] that in Volume II will be used to treat the more difficult case of non-Abelian gauge theories. For simplicity, we shall deal here with the case where the operators  $\mathcal{O}_A[A, \Psi]$ ,  $\mathcal{O}_B[A, \Psi]$ ,  $\dots$  as well as the action  $I[a, \psi]$  and measure  $[\prod da][\prod d\psi]$  are gauge-invariant.

First, replace the field variables of integration  $a_\mu(x)$  and  $\psi_\ell(x)$  everywhere in Eq. (9.6.9) with the new variables

$$a_{\mu\Lambda}(x) \equiv a_\mu(x) + \partial_\mu \Lambda(x), \quad (9.6.11)$$

$$\psi_{\ell\Lambda}(x) \equiv \exp(iq_\ell \Lambda(x)) \psi_\ell(x) \quad (9.6.12)$$

with arbitrary finite  $\Lambda(x)$ . This step is a mathematical triviality, like changing an integral  $\int_{-\infty}^{\infty} f(x)dx$  to read  $\int_{-\infty}^{\infty} f(y)dy$ , and does not require use of the postulated gauge invariance of the theory. Next, use gauge invariance to replace  $a_{\mu\Lambda}(x)$  and  $\psi_{\ell\Lambda}(x)$  in the action, measure, and  $\mathcal{O}$ -functions with the original fields  $a_\mu(x)$  and  $\psi_\ell(x)$ , respectively. Eq. (9.6.9) then becomes

$$\begin{aligned} & \langle T\{\mathcal{O}_A[A, \Psi], \mathcal{O}_B[A, \Psi], \dots\} \rangle_\Omega \\ & \propto \int \left[ \prod_{x,\mu} da_\mu(x) \right] \left[ \prod_{x,\ell} d\psi_\ell(x) \right] \mathcal{O}_A[a, \psi] \mathcal{O}_B[a, \psi] \cdots \\ & \quad \cdot \exp(iI[a, \psi]) \prod_x \delta(\nabla \cdot \mathbf{a}(x) + \nabla^2 \Lambda(x)). \end{aligned} \quad (9.6.13)$$

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<sup>105</sup>Note that now  $a^0(x)$  is *not* equal to the value (9.6.8), but is an independent variable of integration. We will not integrate over  $a^0(x)$  first, which would lead back to Eq. (9.6.6), but instead will treat it in tandem with  $\mathbf{a}(x)$ .

Now, the function  $\Lambda(x)$  was chosen at random, so despite appearances the right-hand side of Eq. (9.6.13) cannot depend on this function. We shall exploit this fact to put the path integral in a much more convenient form. Multiply Eq. (9.6.13) by the functional

$$B[\Lambda, a] = \exp \left( -\frac{1}{2} i \alpha \int d^4 x [\partial_0 a^0 - \nabla^2 \Lambda]^2 \right) \quad (9.6.14)$$

(where  $\alpha$  is an arbitrary constant), and integrate over  $\Lambda(x)$ . By shifting the integration variable  $\Lambda(x)$ , and noting the actual  $\Lambda$ -independence of (9.6.13), we see that the effect is simply to multiply Eq. (9.6.13) with the field-independent constant

$$\int \left[ \prod_x d\Lambda(x) \right] \exp \left( -\frac{1}{2} i \alpha \int d^4 x (\nabla^2 \Lambda)^2 \right). \quad (9.6.15)$$

This factor cancels out in the connected part of the vacuum expectation value, and thus has no physical effect. But (9.6.13) is only  $\Lambda$ -independent *after* we integrate over  $a^\mu(x)$  and  $\psi(x)$ . We can just as well integrate over  $\Lambda(x)$  *before* we integrate over  $a^\mu(x)$  and  $\psi(x)$ , in which case the factor  $\prod_x \delta(\nabla \cdot \mathbf{a}(x) + \nabla^2 \Lambda)$  in Eq. (9.6.13) is replaced with

$$\begin{aligned} & \int \left[ \prod_x d\Lambda(x) \right] \exp \left( -\frac{1}{2} i \alpha \int d^4 x [\partial_0 a^0 - \nabla^2 \Lambda]^2 \right) \prod_x \delta(\nabla \cdot \mathbf{a}(x) + \nabla^2 \Lambda) \\ & \propto \exp \left( -\frac{1}{2} i \alpha \int d^4 x (\partial_\mu a^\mu)^2 \right), \end{aligned} \quad (9.6.16)$$

where “ $\propto$ ” again means proportional with a field-independent factor. Dropping constant factors, Eq. (9.6.9) now becomes

$$\langle T \{ \mathcal{O}_A \mathcal{O}_B \cdots \} \rangle_\Omega \propto \int \left[ \prod_{x,\mu} da_\mu(x) \right] \left[ \prod_{x,\ell} d\psi_\ell(x) \right] \mathcal{O}_A \mathcal{O}_B \cdots \exp(i I_{\text{eff}}[a, \psi]), \quad (9.6.17)$$

where

$$I_{\text{eff}}[a, \psi] = I[a, \psi] - \frac{1}{2} \alpha \int (\partial_\mu a^\mu)^2 d^4 x. \quad (9.6.18)$$

This is now manifestly Lorentz-invariant.

We consider the new term in (9.6.18) as a contribution to the unperturbed part of the action, whose photonic part now reads

$$\begin{aligned} I_0[a] &= \int d^4 x \left[ -\frac{1}{4} (\partial_\mu a_\nu - \partial_\nu a_\mu) (\partial^\mu a^\nu - \partial^\nu a^\mu) - \frac{1}{2} \alpha (\partial_\mu a^\mu)^2 + i\epsilon \text{ terms} \right] \\ &= -\frac{1}{2} \int d^4 x d^4 y a^\mu(x) a^\nu(y) \mathcal{D}_{\mu\nu, xy}, \end{aligned} \quad (9.6.19)$$

where

$$\begin{aligned} \mathcal{D}_{\mu\nu, xy} &= \left[ \eta_{\mu\nu} \frac{\partial^2}{\partial x^\rho \partial y^\rho} - (1 - \alpha) \frac{\partial^2}{\partial x^\mu \partial y^\nu} \right] \delta^4(x - y) + i\epsilon \text{ terms} \\ &= (2\pi)^{-4} \int d^4 q [\eta_{\mu\nu} q^2 - (1 - \alpha) q_\mu q_\nu - i\epsilon \eta_{\mu\nu}] e^{iq \cdot (x-y)}. \end{aligned} \quad (9.6.20)$$

The photon propagator is then found immediately by inverting the  $4 \times 4$  matrix in the integrand of Eq. (9.6.20)

$$\Delta_{\mu x, \nu y} = (2\pi)^{-4} \int d^4 q \left[ \frac{\eta^{\mu\nu}}{q^2 - i\epsilon} + \frac{1 - \alpha}{\alpha} \frac{q^\mu q^\nu}{(q^2 - i\epsilon)^2} \right] e^{iq \cdot (x-y)}. \quad (9.6.21)$$

We are free to choose  $\alpha$  as seems most convenient. Two common choices are  $\alpha = 1$ , which yields the propagator in *Feynman gauge*:

$$\Delta_{\mu x, \nu y}^{\text{Feynman}} = (2\pi)^{-4} \int d^4 q \left[ \frac{\eta^{\mu\nu}}{q^2 - i\epsilon} \right] e^{iq \cdot (x-y)}, \quad (9.6.22)$$

or  $\alpha = \infty$ , in which case the factor (9.6.14) acts as a delta function, and we obtain the propagator in *Landau gauge* (often also called *Lorentz gauge*):

$$\Delta_{\mu x, \nu y}^{\text{Landau}} = (2\pi)^{-4} \int d^4 q \left[ \frac{\eta^{\mu\nu}}{q^2 - i\epsilon} - \frac{q^\mu q^\nu}{(q^2 - i\epsilon)^2} \right] e^{iq \cdot (x-y)}. \quad (9.6.23)$$

Practical calculations are made far more convenient by working with such manifestly Lorentz-invariant interactions and propagators.

## 9.7 Varieties of Statistics<sup>106</sup>

We can now take up a question raised in Chapter 4: what are the possibilities for the change of state-vectors when we interchange identical particles?

For this purpose, we will consider the preparation of the initial or final states in a scattering process. Suppose that a set of indistinguishable particles in either of these states is brought to a particular configuration with momenta  $\mathbf{p}_1, \mathbf{p}_2$ , etc. from a standard configuration with momenta  $\mathbf{k}_1, \mathbf{k}_2$ , etc., by some sort of slowly varying external fields, keeping the particles far enough apart in the process to justify the use of non-relativistic quantum mechanics. (Spin indices are not shown explicitly here; they should be understood to accompany momentum labels.) To calculate the amplitude for this process we can use the path-integral method,<sup>107</sup> taking the  $qs$  and  $ps$  of Section 9.1 as particle positions and momenta, rather than fields and their canonical conjugates. These always satisfy canonical commutation rather than anticommutation relations, whether or not the particles are bosons or fermions or something else, so at this point we are not committing ourselves to any particular statistics. The path-integral formula (9.1.34) gives an amplitude  $\langle \mathbf{p}_1, \mathbf{p}_2, \dots | \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D$  as an integral over paths in which one particle is brought continuously from momentum  $\mathbf{k}_1$  to momentum  $\mathbf{p}_1$ , another identical particle is brought continuously from momentum  $\mathbf{k}_2$  to momentum  $\mathbf{p}_2$ , and so on. The subscript ‘D’ indicates

<sup>106</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

<sup>107</sup>Here I am following the discussion of Laidlaw and C. DeWitt [11], except that they apply the path-integral method to the whole scattering process, rather than just the preparation of initial or final states. In a relativistic theory the possibility of particle creation and annihilation makes it necessary to apply the path-integral method to fields rather than particle orbits. For us, this is not a problem, because we limit such calculations to sufficiently early or late times, when the particles participating in a scattering process are all far apart.

that this is the amplitude we would calculate for distinguishable particles. In particular, this amplitude is symmetric under permutations of the  $\mathbf{p}$ s *and* simultaneous permutations of the  $\mathbf{k}$ s, but has no particular symmetry under separate permutations of the  $\mathbf{p}$ s or  $\mathbf{k}$ s. But if the particles are really indistinguishable, then there are other paths that are topologically distinct, but that yield the same final configuration. For space dimensionality  $d \geq 3$ , the only such paths<sup>108</sup> are those that take  $\mathbf{k}_1, \mathbf{k}_2, \dots$  into some non-trivial permutation  $\mathcal{P}$  of  $\mathbf{p}_1, \mathbf{p}_2, \dots$ . Hence the true amplitude should be written

$$\langle \mathbf{p}_1, \mathbf{p}_2, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle = \sum_{\mathcal{P}} C_{\mathcal{P}} \langle \mathbf{p}_{\mathcal{P}1}, \mathbf{p}_{\mathcal{P}2}, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D, \quad (9.7.1)$$

the sum running over all  $N!$  permutations of the  $N$  indistinguishable particles in the state, and  $C_{\mathcal{P}}$  a set of complex constants. These amplitudes must satisfy a composition rule appropriate for indistinguishable particles:

$$\begin{aligned} & \langle \mathbf{p}_1, \mathbf{p}_2, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle \\ &= \frac{1}{N!} \int d^3 q_1 d^3 q_2 \cdots \langle \mathbf{p}_1, \mathbf{p}_2, \dots \mid \mathbf{q}_1, \mathbf{q}_2, \dots \rangle \langle \mathbf{q}_1, \mathbf{q}_2, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle. \end{aligned} \quad (9.7.2)$$

Using Eq. (9.7.1), this is the requirement

$$\begin{aligned} & \sum_{\mathcal{P}} C_{\mathcal{P}} \langle \mathbf{p}_{\mathcal{P}1}, \mathbf{p}_{\mathcal{P}2}, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D \\ &= \frac{1}{N!} \sum_{\mathcal{P}', \mathcal{P}''} C_{\mathcal{P}'} C_{\mathcal{P}''} \int d^3 q_1 d^3 q_2 \cdots \langle \mathbf{p}_{\mathcal{P}'1}, \mathbf{p}_{\mathcal{P}'2}, \dots \mid \mathbf{q}_1, \mathbf{q}_2, \dots \rangle_D \\ & \quad \cdot \langle \mathbf{q}_{\mathcal{P}''1}, \mathbf{q}_{\mathcal{P}''2}, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D. \end{aligned}$$

Applying a permutation  $\mathcal{P}''$  to both the initial and final states in the first amplitude on the right, this is

$$\begin{aligned} & \sum_{\mathcal{P}} C_{\mathcal{P}} \langle \mathbf{p}_{\mathcal{P}1}, \mathbf{p}_{\mathcal{P}2}, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D \\ &= \frac{1}{N!} \sum_{\mathcal{P}', \mathcal{P}''} C_{\mathcal{P}'} C_{\mathcal{P}''} \int d^3 q_1 d^3 q_2 \cdots \langle \mathbf{p}_{\mathcal{P}''\mathcal{P}'1}, \mathbf{p}_{\mathcal{P}''\mathcal{P}'2}, \dots \mid \mathbf{q}_{\mathcal{P}''1}, \mathbf{q}_{\mathcal{P}''2}, \dots \rangle_D \\ & \quad \cdot \langle \mathbf{q}_{\mathcal{P}''1}, \mathbf{q}_{\mathcal{P}''2}, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D. \end{aligned}$$

But the amplitudes  $\langle \mathbf{p}_1, \mathbf{p}_2, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D$  satisfy the composition rule for distinguishable particles

$$\begin{aligned} & \langle \mathbf{p}_1, \mathbf{p}_2, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D \\ &= \int d^3 q_1 d^3 q_2 \cdots \langle \mathbf{p}_1, \mathbf{p}_2, \dots \mid \mathbf{q}_1, \mathbf{q}_2, \dots \rangle_D \langle \mathbf{q}_1, \mathbf{q}_2, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D, \end{aligned} \quad (9.7.3)$$

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<sup>108</sup>This is expressed formally in the statement that the first homotopy group of configuration space in  $d \geq 3$  is the permutation group [12]. By ‘configuration space’ for  $N$  distinguishable particles is meant the space of  $N$   $d$ -vectors, excluding  $d$ -vectors that coincide with (or are within an arbitrary limiting distance of) each other, and identifying configurations that differ only by a permutation of the vectors.

so the composition rule for the physical amplitudes may be written

$$\begin{aligned} & \sum_{\mathcal{P}} C_{\mathcal{P}} \langle \mathbf{p}_{\mathcal{P}1}, \mathbf{p}_{\mathcal{P}2}, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D \\ &= \frac{1}{N!} \sum_{\mathcal{P}', \mathcal{P}''} C_{\mathcal{P}'} C_{\mathcal{P}''} \langle \mathbf{p}_{\mathcal{P}'\mathcal{P}''1}, \mathbf{p}_{\mathcal{P}'\mathcal{P}''2}, \dots \mid \mathbf{k}_1, \mathbf{k}_2, \dots \rangle_D, \end{aligned}$$

which will be satisfied if and only if

$$C_{\mathcal{P}'\mathcal{P}''} = C_{\mathcal{P}'} C_{\mathcal{P}''}. \quad (9.7.4)$$

That is, the coefficients  $C_{\mathcal{P}}$  must furnish a one-dimensional representation of the permutation group. But the permutation group has only two such representations: one is the identity, with  $C_{\mathcal{P}} = +1$  for all permutations, and the other is the alternating representation, with  $C_{\mathcal{P}} = +1$  or  $C_{\mathcal{P}} = -1$  according to whether  $\mathcal{P}$  is an even or odd permutation. These two possibilities correspond to Bose or Fermi statistics, respectively.<sup>109</sup>

The nice feature of this argument is that it makes it clear why the case of two space dimensions is an exception. In this case there is a much richer variety of topologically distinct paths.<sup>110</sup> For instance, a path in which one particle circles another a definite number of times cannot be deformed into a path where it does not. In consequence, in two space dimensions it is possible to have *anyons* [15], particles with more general permutation properties than just Fermi or Bose statistics [8a].

## Appendix Gaussian Multiple Integrals

We wish first to calculate the multiple integral, over a finite number of real variables  $\xi_r$ , of the exponential of a general quadratic function of  $\xi$ :

$$\mathcal{I} \equiv \int_{-\infty}^{\infty} \prod_r d\xi_r \exp\{-Q(\xi)\}, \quad (9.A.1)$$

$$Q(\xi) = \frac{1}{2} \sum_{rs} K_{rs} \xi_r \xi_s + \sum_r L_r \xi_r + M, \quad (9.A.2)$$

where  $K_{rs}$ ,  $L_r$ , and  $M$  are arbitrary constants, except that the matrix  $K$  is required to be symmetric and non-singular. For this purpose, we begin by considering the case where  $K_{rs}$ ,  $L_r$ , and  $M$  are all real, with  $K_{rs}$  also positive. The result in the general case can then be obtained by analytic continuation.

Any real symmetric matrix can be diagonalized by an orthogonal matrix. Therefore, there is a matrix  $\mathcal{S}$  with transpose  $\mathcal{S}^T = \mathcal{S}^{-1}$  such that

$$(\mathcal{S}^T K \mathcal{S})_{rs} = \delta_{rs} \kappa_r. \quad (9.A.3)$$

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<sup>109</sup>There has been much discussion in the literature of possibilities other than Bose or Fermi statistics, often under the label *parastatistics*. It has been shown [13] that parastatistics theories in  $d \geq 3$  space dimensions are equivalent to theories in which all particles are ordinary fermions or bosons, but carrying an extra quantum number, so that wave functions could have unusual properties under permutations of momenta and spins.

<sup>110</sup>This is expressed in the statement that the first homotopy group of configuration space in two space dimensions is not the permutation group, but a larger group known as the braid group [14].

Because  $K$  is assumed positive and non-singular, the eigenvalues  $\kappa_r$  are positive-definite. We can use the matrix  $\mathcal{S}$  to perform a change of variables:

$$\xi_r = \sum_s \mathcal{S}_{rs} \xi'_s. \quad (9.A.4)$$

The Jacobian  $|\text{Det } \mathcal{S}|$  of this transformation is unity, so the multiple integral (9.A.1) is now given by a product of ordinary integrals:

$$\begin{aligned} \mathcal{I} &= e^{-M} \prod_r \int_{-\infty}^{\infty} d\xi'_r \exp \left\{ -\frac{\kappa_r}{2} \xi'^2_r - (\mathcal{S}^\top L)_r \xi'_r \right\} \\ &= e^{-M} \prod_r \sqrt{\frac{2\pi}{\kappa_r}} \exp \left\{ \frac{1}{2\kappa_r} (\mathcal{S}^\top L)_r^2 \right\}. \end{aligned} \quad (9.A.5)$$

But the determinant and the reciprocal of Eq. (9.A.3) give

$$\text{Det } K = \prod_r \kappa_r, \quad K_{rs}^{-1} = \sum_t \mathcal{S}_{rt} \mathcal{S}_{st} \kappa_t^{-1},$$

so Eq. (9.A.5) may be written

$$\mathcal{I} = \left( \text{Det } \frac{K}{2\pi} \right)^{-1/2} \exp \left\{ \frac{1}{2} \sum_{rs} L_r L_s K_{rs}^{-1} - M \right\}. \quad (9.A.6)$$

Eq. (9.A.1) defines a function of  $K_{rs}$ ,  $L_r$ , and  $M$  that is analytic in  $K_{rs}$  in a finite region around the surface where  $K_{rs}$  is real and positive, where the integral converges, and for such  $K_{rs}$  is analytic everywhere in  $L_r$  and  $M$ . Since (9.A.6) equals (9.A.1) for  $K_{rs}$ ,  $L_r$ , and  $M$  all real, with  $K_{rs}$  also positive, Eq. (9.A.6) provides an analytic continuation of Eq. (9.A.1) to the whole complex plane, with a cut required by the square root. The sign of the square root is fixed by this analytic continuation. In field theory  $K_{rs}$  is actually imaginary, except for a small real part due to the ‘ $i\epsilon$  term’.

It is useful to express Eq. (9.A.6) in terms of the stationary point of the function (9.A.2):

$$\bar{\xi}_r = - \sum_s K_{rs}^{-1} L_s, \quad (9.A.7)$$

$$\frac{\partial Q(\xi)}{\partial \xi_r} = 0 \quad \text{at} \quad \xi = \bar{\xi}, \quad (9.A.8)$$

as

$$\mathcal{I} = \left( \text{Det } \frac{K}{2\pi} \right)^{-1/2} \exp \{ -Q(\bar{\xi}) \}. \quad (9.A.9)$$

This is the result to remember: *Gaussian integrals can be evaluated up to a determinant factor by setting the integration variable equal to the point where the argument of the exponential is stationary.*

We next wish to use this result to calculate the integrals

$$I_{r_1 r_2 \dots r_{2N}} \equiv \int \left( \prod_r d\xi_r \right) \xi_{r_1} \xi_{r_2} \dots \xi_{r_{2N}} \exp \left\{ -\frac{1}{2} \sum_{rs} K_{rs} \xi_r \xi_s \right\}. \quad (9.A.10)$$



(Integrals of this sort with an odd number of  $\xi$ -factors in the integrand obviously vanish.) From the power-series expression of  $\exp(-\sum_r L_r \xi_r)$  in Eq. (9.A.1), we have the sum rule

$$\begin{aligned}
& \sum_{N=0}^{\infty} \sum_{r_1 r_2 \dots r_{2N}} \frac{1}{(2N)!} I_{r_1 r_2 \dots r_{2N}} L_{r_1} L_{r_2} \dots L_{r_{2N}} \\
&= \int \left( \prod_r d\xi_r \right) \exp \left\{ -\sum_r L_r \xi_r - \frac{1}{2} \sum_{rs} K_{rs} \xi_r \xi_s \right\} \\
&= \left[ \text{Det} \left( \frac{K}{2\pi} \right) \right]^{-1/2} \exp \left\{ \frac{1}{2} \sum_{rs} L_r L_s K_{rs}^{-1} \right\} \\
&= \left[ \text{Det} \left( \frac{K}{2\pi} \right) \right]^{-1/2} \sum_{N=0}^{\infty} \frac{1}{N! 2^N} \left( \sum_{rs} L_r L_s K_{rs}^{-1} \right)^N. \tag{9.A.11}
\end{aligned}$$

Comparing the coefficients of  $L_{r_1} L_{r_2} \dots L_{r_{2N}}$  on both sides, we see that  $I_{r_1 r_2 \dots r_{2N}}$  must be proportional to a sum of products of elements of  $K^{-1}$ , which symmetry requires to take the form

$$I_{r_1 r_2 \dots r_{2N}} = c_N \sum_{\substack{\text{pairings} \\ \text{of } r_1 \dots r_{2N}}} \prod_{\text{pairs}} (K^{-1})_{\text{paired indices}}. \tag{9.A.12}$$

Here the sum is over all ways of pairing the indices  $r_1 \dots r_{2N}$ , with two pairings being considered the same if they differ only by the order of the pairs, or by the order of indices within a pair. To calculate the constant factor  $c_N$ , we note that the number  $\nu_N$  of terms in the sum over pairings in Eq. (9.A.12) is equal to the number  $(2N)!$  of permutations of the indices, divided by the number of ways  $N!$  of permuting index pairs and by the number  $2^N$  of permutations within index pairs

$$\nu_N = \frac{(2N)!}{N! 2^N}. \tag{9.A.13}$$

Therefore, Eq. (9.A.12) gives

$$\sum_{r_1 r_2 \dots r_{2N}} L_{r_1} L_{r_2} \dots L_{r_{2N}} I_{r_1 r_2 \dots r_{2N}} = \nu_N c_N \left( \sum_{rs} L_r L_s K_{rs}^{-1} \right)^N. \tag{9.A.14}$$

Comparing this with Eq. (9.A.11) shows that the factors  $(2N)!$  and  $N! 2^N$  are cancelled by  $\nu_N$ , leaving us with

$$c_N = \left[ \text{Det} \left( \frac{K}{2\pi} \right) \right]^{-1/2}. \tag{9.A.15}$$

For instance,

$$I_{r_1 r_2} = I_0 (K^{-1})_{r_1 r_2}, \tag{9.A.16}$$

$$\begin{aligned}
I_{r_1 r_2 r_3 r_4} = I_0 & \left[ (K^{-1})_{r_1 r_2} (K^{-1})_{r_3 r_4} + (K^{-1})_{r_1 r_3} (K^{-1})_{r_2 r_4} \right. \\
& \left. + (K^{-1})_{r_1 r_4} (K^{-1})_{r_2 r_3} \right], \tag{9.A.17}
\end{aligned}$$

and so on, where  $I_0$  is the integral with no indices

$$\begin{aligned} I_0 &= \int \left( \prod_r d\xi_r \right) \exp \left\{ -\frac{1}{2} \sum_{rs} K_{rs} \xi_r \xi_s \right\} \\ &= \left[ \text{Det} \left( \frac{K}{2\pi} \right) \right]^{-1/2}. \end{aligned} \quad (9.A.18)$$

### Weinberg Exercises

- W.9.1.** Consider a non-relativistic particle of mass  $m$ , moving along the  $x$ -axis in a potential  $V(x) = m\omega^2 x^2/2$ . Use path-integral methods to find the probability that if the particle is at  $x_1$  at time  $t_1$ , then it is between  $x$  and  $x + dx$  at time  $t$ .
- W.9.2.** Find the wave function in field space of a state consisting of a single spinless particle of mass  $m \neq 0$ . Use the result to derive the Feynman rules for emission or absorption of such a particle.
- W.9.3.** Find the wave function in field space of the vacuum in the theory of a neutral vector field of mass  $m \neq 0$ . Use the result to derive the form of the  $i\epsilon$  terms in the propagator of this field.
- W.9.4.** The Lagrangian density of the free spin 3/2 Rarita–Schwinger field  $\psi^\mu$  is

$$\mathcal{L} = -\bar{\psi}^\mu (\gamma^\nu \partial_\nu + m) \psi_\mu - \frac{1}{3} \bar{\psi}^\mu (\gamma_\mu \partial_\nu + \gamma_\nu \partial_\mu) \psi^\nu + \frac{1}{3} \bar{\psi}^\mu \gamma_\mu (\gamma^\sigma \partial_\sigma - m) \gamma^\nu \psi_\nu.$$

Use path-integral methods to find the propagator of this field.

### Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

- W.9.1.** Put  $T = t - t_1 > 0$ , and split a path into the classical path and a fluctuation with vanishing endpoints. The classical solution and its action are

$$\begin{aligned} x_{\text{cl}}(\tau) &= \frac{x_1 \sin \omega(t - \tau) + x \sin \omega(\tau - t_1)}{\sin \omega T}, \\ S_{\text{cl}} &= \frac{m\omega}{2 \sin \omega T} [(x^2 + x_1^2) \cos \omega T - 2xx_1]. \end{aligned}$$

Because the action is quadratic, the fluctuation integral is independent of the endpoints. Its determinant is fixed by the free-particle limit, giving

$$K(x, t; x_1, t_1) = \left( \frac{m\omega}{2\pi i \sin \omega T} \right)^{1/2} \exp(iS_{\text{cl}}).$$

The square root is defined by continuation from  $T = 0^+$ ; on crossing a zero of  $\sin \omega T$ , it acquires the usual Maslov phase. Away from those caustics,

$$dP = |K(x, t; x_1, t_1)|^2 dx = \frac{m\omega}{2\pi |\sin \omega T|} dx.$$

An exact position eigenstate is not normalizable, so this is a formal conditional transition density rather than a normalized probability distribution. A physical initial wave packet must be convolved with  $K$ ; then its probability integrates to one. At  $\sin \omega T = 0$ , the kernel is instead a phase times the appropriate delta distribution.

**W.9.2.** At an asymptotic time define

$$\tilde{\phi}(\mathbf{p}) = (2\pi)^{-3/2} \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} \phi(\mathbf{x}), \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}.$$

Equation (9.2.8) gives the normalized vacuum wave functional up to its field-independent constant:

$$\Psi_0[\phi] = \mathcal{N} \exp \left[ -\frac{1}{2} \int d^3p E_{\mathbf{p}} \tilde{\phi}(\mathbf{p}) \tilde{\phi}(-\mathbf{p}) \right].$$

In the field representation  $\Pi(\mathbf{x}) = -i\delta/\delta\phi(\mathbf{x})$ . Taking the adjoint of Eq. (9.2.7) at  $t = 0$ , and acting on  $\Psi_0$ , therefore gives

$$\Psi_{\mathbf{p}}[\phi] \equiv \langle \phi | \mathbf{p} \rangle = \sqrt{2E_{\mathbf{p}}} \tilde{\phi}(-\mathbf{p}) \Psi_0[\phi].$$

The result has the expected first Hermite polynomial multiplying the Gaussian. With  $\langle \mathbf{p} | \mathbf{q} \rangle = \delta^3(\mathbf{p} - \mathbf{q})$ , its norm follows from the Gaussian covariance  $\langle \tilde{\phi}(\mathbf{p}) \tilde{\phi}(\mathbf{q}) \rangle_0 = \delta^3(\mathbf{p} + \mathbf{q}) / (2E_{\mathbf{p}})$ .

For example, an incoming particle inserts  $\sqrt{2E_{\mathbf{p}}} \tilde{\phi}(-\mathbf{p})$  in the initial boundary functional, while an outgoing particle inserts its conjugate in the final boundary functional. Contracting that boundary field with a field at a vertex  $x$  supplies

| external state                        | factor at the attached vertex                            |
|---------------------------------------|----------------------------------------------------------|
| $ \mathbf{p}\rangle_{\text{in}}$      | $(2\pi)^{-3/2} (2E_{\mathbf{p}})^{-1/2} e^{ip \cdot x}$  |
| ${}_{\text{out}}\langle \mathbf{p}  $ | $(2\pi)^{-3/2} (2E_{\mathbf{p}})^{-1/2} e^{-ip \cdot x}$ |

with  $p^0 = E_{\mathbf{p}}$ . The conjugate factors describe absorption at the opposite boundary. These are precisely the external-line rules; all vacuum Gaussian factors cancel between the normalized numerator and denominator of the path integral.

**W.9.3.** Use the independent spatial components  $v_i$  and write

$$P_{ij}^T = \delta_{ij} - \frac{p_i p_j}{\mathbf{p}^2}, \quad P_{ij}^L = \frac{p_i p_j}{\mathbf{p}^2}, \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}.$$

In momentum space the free Proca Hamiltonian is a sum of oscillators whose momentum and coordinate quadratic forms are

$$A = P^T + \frac{E_{\mathbf{p}}^2}{m^2} P^L, \quad B = E_{\mathbf{p}}^2 P^T + m^2 P^L.$$

For a Gaussian  $\Psi_0[v] \propto \exp[-\frac{1}{2} \int v_i(-\mathbf{p}) \mathcal{E}_{ij}(\mathbf{p}) v_j(\mathbf{p}) d^3p]$ , the functional Schrödinger equation requires  $\mathcal{E}A\mathcal{E} = B$ . Since the projectors commute,

$$\mathcal{E}_{ij}(\mathbf{p}) = E_{\mathbf{p}} P_{ij}^T + \frac{m^2}{E_{\mathbf{p}}} P_{ij}^L = E_{\mathbf{p}} \left( \delta_{ij} - \frac{p_i p_j}{E_{\mathbf{p}}^2} \right).$$

Thus

$$\Psi_0[v] = \mathcal{N} \exp \left[ -\frac{1}{2} \int d^3p v_i(-\mathbf{p}) \mathcal{E}_{ij}(\mathbf{p}) v_j(\mathbf{p}) \right].$$

The product of the initial and final vacuum functionals adds the convergent boundary damping

$$-\epsilon \int dt d^3p v_i(-\mathbf{p}, t) \mathcal{E}_{ij}(\mathbf{p}) v_j(\mathbf{p}, t)$$

to  $iS$ , with an immaterial positive rescaling of  $\epsilon$ . After the auxiliary  $v^0$  is restored, inversion of the covariant quadratic form yields

$$\Delta^{\mu\nu}(p) = \frac{\eta^{\mu\nu} + p^\mu p^\nu / m^2}{p^2 + m^2 - i\epsilon}.$$

Terms of order  $\epsilon$  in the longitudinal numerator depend on how the damping is extended off shell and vanish as distributions when  $\epsilon \rightarrow 0^+$ . The important result fixed by the vacuum is the common Feynman displacement of the physical transverse and longitudinal poles.

**W.9.4.** Fourier transform with  $\partial_\mu \mapsto ip_\mu$ , and write the quadratic action as

$$I_0 = - \int \frac{d^4p}{(2\pi)^4} \bar{\psi}^\mu(-p) \mathcal{K}_{\mu\nu}(p) \psi^\nu(p),$$

where

$$\mathcal{K}_{\mu\nu} = (i\gamma \cdot p + m)\eta_{\mu\nu} + \frac{i}{3}(\gamma_\mu p_\nu + \gamma_\nu p_\mu) - \frac{1}{3}\gamma_\mu(i\gamma \cdot p - m)\gamma_\nu.$$

The Grassmann Gaussian integral says that the contraction is the inverse of this finite tensor–spinor matrix. Direct multiplication, using  $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ , gives the pole part

$$\Delta^{\mu\nu}(p) = \frac{-i\gamma \cdot p + m}{p^2 + m^2 - i\epsilon} \left[ \eta^{\mu\nu} - \frac{1}{3}\gamma^\mu \gamma^\nu + \frac{2p^\mu p^\nu}{3m^2} + \frac{i}{3m}(\gamma^\mu p^\nu - \gamma^\nu p^\mu) \right].$$

With the canonical Grassmann variables of Section 9.5, the corresponding  $\psi$ -canonical-momentum contraction has one additional right factor  $\beta = i\gamma^0$ ; the displayed expression is the conventional  $T\{\psi^\mu(x)\bar{\psi}^\nu(y)\}$  propagator.

On the mass shell the tensor in brackets projects onto the physical spin- $\frac{3}{2}$  sector: after multiplication by the Dirac numerator it is transverse and gamma-traceless,

$$p_\mu \Delta^{\mu\nu}|_{p^2=-m^2} = 0, \quad \gamma_\mu \Delta^{\mu\nu}|_{p^2=-m^2} = 0.$$

It therefore removes the spin- $\frac{1}{2}$  components of the vector spinor and leaves the four massive spin- $\frac{3}{2}$  polarizations. The vacuum boundary functional supplies the same  $-i\epsilon$  displacement as for the Dirac propagator.

## Supplementary Exercises

### S.9.1. One Phase-Space Time Slice and Integrating Out a Quadratic Momentum

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 5, Problem 1.)

- (a) For  $H = P^2/(2m) + V(Q)$ , insert a momentum resolution of the identity between two nearby position eigenstates. Derive the short-time phase-space kernel through first order in the time step, including its normalization and the endpoint at which  $V$  is evaluated.
- (b) Starting with the time-sliced phase-space path integral for  $H = P^2/(2m) + V(Q)$ , perform every momentum integral. Derive the configuration-space action and explain why the resulting determinant is field independent in this example.

### S.9.2. The Schrödinger Equation from One Step

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 5, Problem 2.)

Propagate a smooth wavefunction through one infinitesimal configuration-space time slice. Expand consistently in the short time and prove that the path-integral kernel generates the Schrödinger equation with the correct kinetic and potential terms. Track the Gaussian moments that survive through first order and state how the endpoint prescription for the potential enters the result.

### S.9.3. The Harmonic-Oscillator Propagator from the Path Integral

(Adapted from Daniel Harlow, Lecture Notes for Physics 8.323: Relativistic Quantum Field Theory I (2024), Section 5.8, Homework Problem 3, printed p. 68.)

Use the path integral to calculate the propagator

$$K_\omega(q', T; q, 0) = \langle q' | e^{-iHT} | q \rangle$$

of the simple harmonic oscillator

$$H = \frac{P^2}{2m} + \frac{k}{2}Q^2, \quad \omega = \sqrt{\frac{k}{m}}, \quad T > 0.$$

The endpoints are fixed:  $q(0) = q$  and  $q(T) = q'$ . For comparison, the preceding free-particle problem gives

$$K_0(q', T; q, 0) = \sqrt{\frac{m}{2\pi iT}} \exp \left[ \frac{im(q' - q)^2}{2T} \right],$$

where the square-root phase is selected by the short-time  $i0$  prescription.

Write each path as a classical solution plus a fluctuation,  $q(t) = q_{\text{cl}}(t) + \eta(t)$ , with  $\eta(0) = \eta(T) = 0$ . Expand  $\eta$  in its complete Fourier sine series and perform the Gaussian integrals over all mode coefficients. You may first keep only factors that depend on  $k$ ; fix every  $k$ -independent factor at the end by requiring the result to approach the displayed free-particle kernel as  $k \rightarrow 0$ . Determine both the full classical exponential and the determinant prefactor.

### S.9.4. The Cubic Anharmonic Oscillator through Order $\lambda^2$

(Adapted from University of Cambridge, Mathematical Tripos Part III, Advanced Quantum Field Theory, Paper 304 (2023), Question 1, p. 2, parts (a)–(c).)

Use the source convention

$$Z[J] = \int \mathcal{D}x \exp \left[ -iS[x] - i \int dt J(t)x(t) \right], \quad x(t) \longleftrightarrow i \frac{\delta}{\delta J(t)}.$$

The action for a harmonic oscillator of unit mass is

$$S[x] = \int dt \left( \frac{1}{2} \dot{x}^2(t) - \frac{1}{2} \omega^2 x^2(t) \right), \quad \omega > 0.$$

- (a) Show, by Fourier transform or otherwise, that the propagator is

$$D(t-t') = \frac{1}{2\omega} e^{i\omega|t-t'|}.$$

By introducing the source  $J(t)$ , show that the harmonic-oscillator generating functional can be written

$$Z_0[J] = Z_0[0] \exp \left[ -\frac{1}{2} \int dt \int dt' J(t) D(t-t') J(t') \right].$$

- (b) Now add the anharmonic term

$$S_\lambda[x] = -\frac{\lambda}{3!} \int dt x^3(t)$$

to the action. Write the interacting generating functional  $Z[J]$  as an infinite series involving  $Z_0[J]$  and its functional derivatives. Briefly comment on the convergence of this series.

- (c) Write down the Feynman rules for the anharmonic oscillator. Hence find an integral expression for the connected correlation function

$$\langle \Omega | T \{ x(t_1) x(t_2) \} | \Omega \rangle_{\text{conn}}$$

through order  $\lambda^2$ , including every connected contribution at that order. Without evaluating the integrals, do you expect this correlation function to diverge? Briefly explain your reasoning, including its ultraviolet behavior.

### S.9.5. Vacuum Survival under an External Force

(Adapted from Mark Srednicki, Quantum Field Theory, Chapter 7, Problem 7.4.)

A ground-state oscillator is acted on by a real force  $f(t)$  of compact support. Evaluate the Gaussian path integral and express its vacuum survival probability entirely in terms of the on-shell Fourier component of the force. Separate the phase from the modulus, identify the mean number of produced quanta, and verify the adiabatic no-production limit.

### S.9.6. The Free Scalar Source Functional and Wick Pairings by Differentiation

(Inspired by Mark Srednicki, *Quantum Field Theory*, Chapter 8, Sections 8.1–8.2 / Mark Srednicki, *Quantum Field Theory*, Chapter 8, Eqs. 8.14–8.17.)

- (a) For a real massive scalar coupled linearly to a source  $J$ , complete the functional square and derive  $Z_0[J]/Z_0[0]$ . State the momentum-space kernel and its  $i\epsilon$  prescription in the mostly-plus convention.
- (b) Differentiate the free scalar source functional four times and then set the source to zero. Derive the three pairings of the four-point function and generalize the counting to  $2n$  insertions.

### S.9.7. The Propagator Is a Green Function

(Adapted from John McGreevy, *Physics 215A QFT*, Fall 2023, Assignment 5, Problem 2, “The propagator is a Green’s function,” pp. 1–2, parts (a)–(d).)

Work in  $d + 1$  spacetime dimensions with the volume convention

$$\square_x \equiv \partial_{x^\mu} \partial_x^\mu = \nabla_{\mathbf{x}}^2 - \partial_{x^0}^2.$$

For reference, the free real scalar has the mode expansion

$$\phi_0(x) = \int \frac{d^d \mathbf{p}}{(2\pi)^d \sqrt{2E_{\mathbf{p}}}} \left[ a_{\mathbf{p}} e^{ip \cdot x} + a_{\mathbf{p}}^\dagger e^{-ip \cdot x} \right], \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \quad p^0 = E_{\mathbf{p}},$$

with

$$[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = (2\pi)^d \delta^d(\mathbf{p} - \mathbf{q}), \quad a_{\mathbf{p}} |\Omega\rangle = 0.$$

- (a) Consider the retarded propagator

$$D_R(x - y) \equiv \theta(x^0 - y^0) \langle \Omega | [\phi(x), \phi(y)] | \Omega \rangle.$$

Show that it is a Green function for the Klein–Gordon operator in the sense that

$$(\square_x - m^2) D_R(x - y) = a \delta^{d+1}(x - y),$$

and find  $a$ .

- (b) Add the external-source term

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \phi(x) j(x),$$

where  $j(x)$  is a fixed  $c$ -number function. Use the preceding result to generalize the free mode expansion to a solution of the sourced field equation with retarded boundary conditions.

(c) The time-ordered propagator is

$$\begin{aligned} D(x-y) &\equiv \langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle \\ &= \theta(x^0 - y^0) \langle \Omega | \phi(x) \phi(y) | \Omega \rangle + \theta(y^0 - x^0) \langle \Omega | \phi(y) \phi(x) | \Omega \rangle. \end{aligned}$$

Show that it too is a Green function for the Klein–Gordon operator, and determine its delta-function normalization.

*Hint.* Use the canonical equal-time commutation relations

$$[\phi(x^0, \mathbf{x}), \phi(x^0, \mathbf{y})] = 0, \quad [\partial_{x^0} \phi(x^0, \mathbf{x}), \phi(x^0, \mathbf{y})] = -i\delta^d(\mathbf{x} - \mathbf{y}).$$

Do not neglect  $\partial_t \theta(t) = \delta(t)$ : the time derivatives act on the time-ordering symbol.

(d) This is an example of a fact that is easier to understand from the path integral: explain why correlation functions solve the field equations only up to contact terms, and relate that statement to the delta function found in the preceding part.

### S.9.8. The Scalar Vacuum Wave Functional

(Adapted from Mark Srednicki, *Quantum Field Theory*, Chapter 8, Problem 8.8.)

Treat each spatial Fourier mode of a free real scalar as an oscillator. Derive the vacuum wave functional, its nonlocal position-space kernel, and the equal-time covariance of two field modes. Fix the normalization as a regulated product of oscillator normalizations and identify which part survives in normalized expectation values.

### S.9.9. Schwinger–Dyson Equations from Functional Integration by Parts

(Adapted from J. McGreevy, *Physics 215A Quantum Field Theory*, Fall 2023, Assignment 5, Problem 3, printed pp. 2–3.)

Consider the Lorentzian path integral

$$Z = \int \mathcal{D}\phi e^{iS[\phi]},$$

defined with the usual  $i0$  prescription. Invariance of the measure under a change of field variable implies

$$0 = \int \mathcal{D}\phi \frac{\delta}{\delta\phi(x)} (\text{anything}),$$

provided the differentiated expression has no boundary contribution at large field. To explain this rather than merely assume it, discretize spacetime. Then

$$\int \mathcal{D}\phi = \int \prod_z d\phi_z, \quad \phi_z \equiv \phi(z),$$

and the identity is just ordinary integration by parts in the variable  $\phi_x$ :

$$\int d\phi_x \frac{\partial}{\partial\phi_x} (\text{anything}) = 0.$$



These functional integration-by-parts identities are the Schwinger–Dyson equations. They are nonperturbative and implement the field equations inside correlation functions.

- (a) Evaluate

$$0 = \int \mathcal{D}\phi \frac{\delta}{\delta\phi(x)} \left( \phi(y) e^{iS[\phi]} \right)$$

and prove the normalized identity

$$\boxed{\frac{1}{Z} \int \mathcal{D}\phi e^{iS[\phi]} \frac{\delta S}{\delta\phi(x)} \phi(y) = i\delta^{d+1}(x-y).}$$

- (b) Apply this result to the free massive real scalar action in the volume convention,

$$S_0[\phi] = \int d^{d+1}x \left( -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right).$$

Show that the time-ordered two-point function obeys the classical field equation away from the other insertion and derive its contact term:

$$\boxed{(\Box_x - m^2) \langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = i\delta^{d+1}(x-y).}$$

- (c) Derive the corresponding Schwinger–Dyson equation for the time-ordered three-point function of the free field. Display every contact term and then simplify the result using the free-vacuum one-point function.

#### S.9.10. Low-Order Correlators in $\lambda\phi^4$

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 6, Problem 2.)

- (a) Enumerate every Wick topology contributing to the normalized two-point function through order  $\lambda^2$ . Give its position-space integral and symmetry factor, showing the cancellation of vacuum factors.
- (b) Do the same for the connected four-point function through order  $\lambda^2$ , including the three one-loop channels and the one-vertex tree term.

#### S.9.11. Vacuum Energy and Connected Vacuum Graphs

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 6, Problem 3.)

- (a) Show that translational invariance makes  $W_0 = -i \log Z[0]$  proportional to space-time volume and identify the coefficient with the vacuum-energy density.
- (b) Prove combinatorially that  $W_0$  is the sum of connected vacuum graphs, with one inverse symmetry factor for each topology.
- (c) In  $\lambda\phi^4/4!$  theory, calculate the vacuum-energy-density shift through order  $\lambda^2$ , displaying all connected integrals and their symmetry factors.

### S.9.12. Cancellation of Arbitrary Vacuum Bubbles

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 6, Problem 4.)

For a general polynomial scalar interaction, factor diagrams contributing to an unnormalized  $n$ -point function into vacuum and source-connected components. Prove to all perturbative orders that normalization by  $Z[0]$  cancels every vacuum component. State the argument as a generating functional identity and then interpret the same cancellation graph by graph, including symmetry factors.

### S.9.13. A Wave Functional as a Euclidean Boundary and A Parity-Twisted Euclidean Trace

(Inspired by David Skinner, Advanced Quantum Field Theory, Sections 1.2.3 and 3.1 / David Skinner, Advanced Quantum Field Theory, Example Sheet 1, Problem 6.)

- (a) Represent  $\langle \phi_f | e^{-TH} \psi_i \rangle$  as a Euclidean path integral with one fixed boundary and one boundary weighted by an arbitrary initial wave functional. Show how the ground-state functional is projected out as  $T \rightarrow \infty$ .
- (b) Find the Euclidean path-integral boundary condition that computes  $\text{Tr}(Pe^{-TH})$  for a particle on the line, where  $P$  is parity. Evaluate the trace for the harmonic oscillator and compare it with the energy-basis answer.

### S.9.14. Pfaffian Covariance and Canonical Form

(Adapted from David Skinner, Advanced Quantum Field Theory, Example Sheet 1, Problem 3.)

For  $2N$  real Grassmann variables and an invertible antisymmetric matrix  $A$ , derive the transformation of the Berezin measure under  $\theta = N\theta'$ . Use it to prove  $\text{Pf}(N^T A N) = \det(N) \text{Pf}(A)$ . Then reduce  $A$  by a change of basis to  $2 \times 2$  antisymmetric blocks and use that canonical form to prove  $\text{Pf}(A) = \pm \sqrt{\det A}$ . Fix the sign by continuity and check every formula for one Grassmann pair.

### S.9.15. An Exact Four-Grassmann Interaction

(Adapted from David Skinner, Advanced Quantum Field Theory, Example Sheet 1, Problem 4.)

Evaluate exactly the partition function of four real Grassmann variables with an antisymmetric quadratic kernel and a completely antisymmetric quartic coupling. Reproduce the same answer by enumerating the allowed vacuum contractions. Track the ordering convention for the Berezin measure, express the quadratic contribution as a Pfaffian, and check both the free and pure-quartic limits.

### S.9.16. Bosonic and Fermionic Determinant Powers and Fermionic Wick's Theorem from a Pfaffian

(Inspired by David Skinner, Advanced Quantum Field Theory, Sections 2.2 and 2.4 / David Skinner, Advanced Quantum Field Theory, Sections 2.4 and 3.2.)

- (a) Compare a finite-dimensional real bosonic Gaussian, a complex Grassmann Gaussian, and a real Grassmann Gaussian. Derive respectively the determinant powers  $-1/2$ ,  $+1$ , and the Pfaffian, including the assumptions on their kernels.
- (b) Introduce Grassmann sources for a real fermionic Gaussian. Complete the square, differentiate four times, and derive the signed sum over the three pairings. Explain how the result generalizes to  $2N$  insertions.

### S.9.17. Berezin Gaussians, Fermionic Sources, and Determinant Signs

(Adapted from Hong Liu and MIT OpenCourseWare, 8.323 Assignment 10, Problem 3.)

- (a) With a stated order for both variables and differentials, evaluate  $\int d\bar{\theta} d\theta e^{-\bar{\theta}a\theta}$  directly by expansion. Check translation invariance under independent Grassmann shifts.
- (b) For  $N$  independent pairs  $\theta_i, \bar{\theta}_i$ , diagonalize or triangularize the kernel and prove that their Gaussian integral equals  $\det A$ . Explain why the determinant is not inverted as it is for commuting variables.
- (c) Insert  $\theta_k \bar{\theta}_l$  into the complex Grassmann Gaussian. Derive the result by differentiating the determinant and identify carefully the ordering of the indices of  $A^{-1}$ .
- (d) Evaluate the normalized Gaussian expectation of  $\theta_i \bar{\theta}_j \theta_k \bar{\theta}_l$ . Derive the relative minus sign between its two pairings directly from Grassmann source differentiation.
- (e) Add independent Grassmann sources to the free Dirac action. Shift  $\psi, \bar{\psi}$ , derive the normalized generating functional, and recover the momentum-space Dirac propagator in the volume's gamma-matrix convention.
- (f) Integrate out a Dirac field coupled bilinearly to a background matrix  $\Gamma(x)$ . Expand the resulting determinant and show explicitly why each connected closed fermion loop carries one additional minus sign.

### S.9.18. The Complete Gaussian Generating Function

(Adapted from John McGreevy, Physics 215A, Fall 2023, Assignment 4, Problem 2.)

Let  $A$  be a real symmetric positive-definite  $N \times N$  matrix.

- (a) Derive the one-dimensional source integral

$$Z_a(j) = \int_{-\infty}^{\infty} dx e^{-ax^2/2+jx}$$

including its normalization. Establish the source-free Gaussian integral without assuming its value, and state the convergence condition.

- (b) Diagonalize  $A$  and evaluate

$$Z_A(J) = \int_{\mathbb{R}^N} d^N x e^{-x^\top A x/2 + J^\top x}$$

in terms of  $\det A$  and  $A^{-1}$ . Explain why an orthogonal change of basis introduces no additional Jacobian magnitude.

- (c) With normalized expectation values defined by  $Z_A(0)$ , obtain  $\langle x_i x_j \rangle$  both by differentiating  $Z_A$  and by differentiating  $\log Z_A$ . Identify what derivatives of the logarithm generate, and state the continuum functional version of the result.

### S.9.19. The Characteristic Function of a Gaussian Quantum Coordinate

(Adapted from John McGreevy, *Physics 215A, Fall 2023, Assignment 4, Problem 3.*)

Let  $q$  be a coordinate in the vacuum of a centered Gaussian quantum system, so its Hamiltonian is quadratic and  $\langle \Omega | q | \Omega \rangle = 0$ . Determine the full  $K$ -dependence of  $\langle \Omega | e^{iKq} | \Omega \rangle$  in terms of  $\langle \Omega | q^2 | \Omega \rangle$ . Derive the identity once from the Gaussian position-space measure and once from oscillator operator algebra. Then give the corrected formula when the Gaussian has nonzero mean  $\bar{q}$ .

## Solutions to Supplementary Exercises

### S.9.1. One Phase-Space Time Slice and Integrating Out a Quadratic Momentum

- (a) Let  $\delta t > 0$ ,  $q' = q(t + \delta t)$ , and  $q = q(t)$ . To the accuracy needed in a time-sliced product,

$$e^{-iH\delta t} = e^{-iP^2\delta t/(2m)} e^{-iV(Q)\delta t} + O(\delta t^2).$$

Inserting  $\mathbf{1} = \int dp |p\rangle \langle p|$ , with  $\langle q|p\rangle = (2\pi)^{-1/2} e^{ipq}$ , gives

$$\langle q' | e^{-iH\delta t} | q \rangle = \int \frac{dp}{2\pi} \exp \left\{ ip(q' - q) - i\delta t \left[ \frac{p^2}{2m} + V(q) \right] \right\} + O(\delta t^2).$$

The potential is evaluated at the right endpoint of the ket because of the displayed operator ordering. A midpoint prescription instead represents Weyl ordering; the distinction matters for nontrivial products of  $P$  and  $Q$ , but vanishes here in the continuum limit.

- (b) For each slice, complete the square:

$$p \frac{\Delta q}{\delta t} - \frac{p^2}{2m} = -\frac{1}{2m} \left( p - m \frac{\Delta q}{\delta t} \right)^2 + \frac{m}{2} \left( \frac{\Delta q}{\delta t} \right)^2.$$

Thus

$$\int \frac{dp}{2\pi} e^{i\delta t [p\Delta q/\delta t - p^2/(2m)]} = \left( \frac{m}{2\pi i\delta t} \right)^{1/2} e^{im(\Delta q)^2/(2\delta t)}.$$

Multiplying the slices and taking their continuum limit yields

$$K(q_f, t_f; q_i, t_i) = \int_{q_i}^{q_f} \mathcal{D}q \exp \left\{ i \int_{t_i}^{t_f} dt \left[ \frac{m}{2} \dot{q}^2 - V(q) \right] \right\}.$$

The coefficient of  $p^2$  is the constant  $1/m$ , so the product of Gaussian determinants depends only on the slicing and total time. A coordinate-dependent kinetic coefficient would instead leave a field-dependent measure factor.

### S.9.2. The Schrödinger Equation from One Step

Write  $y = x - \xi$  in

$$\psi(x, t + \delta t) = \left( \frac{m}{2\pi i \delta t} \right)^{1/2} \int d\xi e^{im\xi^2/(2\delta t)} e^{-i\delta t V(x-\xi)} \psi(x - \xi, t).$$

The oscillatory Gaussian localizes  $\xi = O(\sqrt{\delta t})$ . Expanding the last two factors and using

$$\langle \xi \rangle = 0, \quad \langle \xi^2 \rangle = \frac{i\delta t}{m},$$

gives

$$\psi(x, t + \delta t) = \psi + \frac{i\delta t}{2m} \partial_x^2 \psi - i\delta t V(x) \psi + O(\delta t^{3/2}).$$

After subtracting  $\psi(x, t)$ , dividing by  $\delta t$ , and multiplying by  $i$ , one obtains

$$i\partial_t \psi = \left[ -\frac{1}{2m} \partial_x^2 + V(x) \right] \psi.$$

The normalization of the short-time kernel is essential: it makes the zeroth Gaussian moment equal to one. Replacing  $V(x - \xi)$  by  $V(x)$  changes the integrand by a term linear in  $\xi$ , whose Gaussian average vanishes, and by terms of order  $\delta t^2$ . Thus either consistent endpoint convention gives the displayed equation at this order; differences first matter in the higher-order definition of the discretized path integral.

### S.9.3. The Harmonic-Oscillator Propagator from the Path Integral

The Lorentzian action is

$$S[q] = \frac{m}{2} \int_0^T dt (\dot{q}^2 - \omega^2 q^2).$$

The unique classical path away from the caustics  $\sin \omega T = 0$  is

$$q_{\text{cl}}(t) = \frac{q \sin[\omega(T - t)] + q' \sin(\omega t)}{\sin(\omega T)}.$$

Using the equation of motion to reduce the action to its endpoint term gives

$$S_{\text{cl}} = \frac{m\omega}{2 \sin(\omega T)} [(q'^2 + q^2) \cos(\omega T) - 2qq'].$$

Write  $q = q_{\text{cl}} + \eta$ . The cross term vanishes by the classical equation and the fixed endpoint conditions, leaving

$$S[q] = S_{\text{cl}} + \frac{m}{2} \int_0^T dt (\dot{\eta}^2 - \omega^2 \eta^2).$$

The complete Dirichlet expansion is

$$\eta(t) = \sum_{n=1}^{\infty} \eta_n \sin \frac{n\pi t}{T},$$

for which

$$S_{\text{fl}} = \frac{mT}{4} \sum_{n=1}^{\infty} \left[ \left( \frac{n\pi}{T} \right)^2 - \omega^2 \right] \eta_n^2.$$

Each mode is an ordinary Fresnel Gaussian. Dividing its determinant by the free-particle determinant removes every  $k$ -independent measure factor:

$$\frac{\mathcal{N}_{\omega}(T)}{\mathcal{N}_0(T)} = \prod_{n=1}^{\infty} \left[ 1 - \frac{\omega^2 T^2}{n^2 \pi^2} \right]^{-1/2}.$$

Euler's product

$$\frac{\sin z}{z} = \prod_{n=1}^{\infty} \left( 1 - \frac{z^2}{n^2 \pi^2} \right)$$

therefore gives

$$\mathcal{N}_{\omega}(T) = \sqrt{\frac{\omega T}{\sin(\omega T)}} \sqrt{\frac{m}{2\pi i T}} = \sqrt{\frac{m\omega}{2\pi i \sin(\omega T)}}.$$

The full answer is

$$K_{\omega}(q', T; q, 0) = \sqrt{\frac{m\omega}{2\pi i \sin(\omega T)}} \exp \left\{ \frac{im\omega}{2 \sin(\omega T)} [(q'^2 + q^2) \cos(\omega T) - 2qq'] \right\}.$$

The square root is continued from  $T = 0^+$  with the same  $i0$  prescription as the free kernel; crossing a zero of  $\sin \omega T$  produces the usual Maslov phase. Finally,  $\sin \omega T \sim \omega T$  and  $\cos \omega T \sim 1 - \omega^2 T^2/2$ , so the boxed result tends exactly to the supplied free-particle kernel when  $k = m\omega^2 \rightarrow 0$ .

#### S.9.4. The Cubic Anharmonic Oscillator through Order $\lambda^2$

(a) After integrating by parts, write

$$S_0[x] = -\frac{1}{2} \int dt x(t) K x(t), \quad K = \partial_t^2 + \omega^2.$$

The boundary value appropriate to the source convention in the problem is

$$D(\tau) = \lim_{\epsilon \rightarrow 0^+} \int \frac{dE}{2\pi} \frac{i e^{-iE\tau}}{\omega^2 - E^2 + i\epsilon}.$$

For  $\tau > 0$ , closing below selects the pole near  $E = -\omega$ ; for  $\tau < 0$ , closing above selects the pole near  $E = +\omega$ . The residues give

$$D(\tau) = \frac{1}{2\omega} e^{i\omega|\tau|}.$$

Equivalently, this follows from the homogeneous equation away from the origin and the jump

$$\dot{D}(0^+) - \dot{D}(0^-) = i, \quad KD(\tau) = i\delta(\tau).$$

Thus  $K^{-1} = -iD$ . The exponent in the free functional is

$$\frac{i}{2} xKx - iJx = \frac{i}{2} (x - K^{-1}J)K(x - K^{-1}J) - \frac{i}{2} JK^{-1}J,$$

where repeated time arguments are integrated. Translating the Gaussian measure leaves its source-free determinant unchanged, and therefore

$$Z_0[J] = Z_0[0] \exp \left[ -\frac{1}{2} \int dt dt' J(t)D(t-t')J(t') \right].$$

- (b) In the chosen convention, insertion of  $x(t)$  is  $i\delta/\delta J(t)$ . Since  $-iS_\lambda = +i\lambda \int x^3/3!$ , the factors of  $i$  combine to give

$$\begin{aligned} Z[J] &= \exp \left[ \frac{\lambda}{3!} \int dt \frac{\delta^3}{\delta J(t)^3} \right] Z_0[J] \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left( \frac{\lambda}{3!} \right)^n \prod_{r=1}^n \left[ \int dt_r \frac{\delta^3}{\delta J(t_r)^3} \right] Z_0[J]. \end{aligned}$$

This is a formal perturbation series, not a convergent definition of the real cubic oscillator. For either sign of real  $\lambda$ , the cubic potential is unbounded in one direction, so the real-contour functional integral itself requires a contour prescription. Moreover, the number of Wick contractions grows factorially. These facts are compatible with each fixed-order coefficient being finite.

- (c) In position space, every internal line contributes  $D(t-t')$ , every cubic vertex contributes  $i\lambda \int dt$ , and each graph is divided by its automorphism factor. Vacuum components cancel in normalized correlators, and only graphs connected to both external insertions enter the connected two-point function.

There is no order- $\lambda$  term because it would require pairing five oscillator coordinates. At order  $\lambda^2$ , there are two connected topologies: the two-line self-energy graph and the graph in which both external legs meet one vertex while the other vertex carries a tadpole. Each has symmetry factor  $1/2$ . Define

$$G_{\text{conn}}(t_1, t_2) = \langle \Omega | T \{ x(t_1) x(t_2) \} | \Omega \rangle_{\text{conn}}, \quad D_{a,b} = D(a-b).$$

Then

$$\begin{aligned} G_{\text{conn}}(t_1, t_2) &= D_{t_1, t_2} + \frac{(i\lambda)^2}{2} \int du dv D_{t_1, u} D_{u, v}^2 D_{v, t_2} \\ &\quad + \frac{(i\lambda)^2}{2} \int du dv D_{t_1, u} D_{t_2, u} D_{u, v} D_{v, v} + O(\lambda^4). \end{aligned}$$

The nominal order- $\lambda^3$  term also vanishes because an even correlator is even under the simultaneous replacement  $x \mapsto -x$ ,  $\lambda \mapsto -\lambda$ .

There is no ultraviolet divergence. In energy space an oscillator propagator falls as  $E^{-2}$ . The self-energy loop therefore behaves as  $\int dE E^{-4}$ , while the tadpole behaves as  $\int dE E^{-2}$ ; both converge in one integration dimension. The  $i\epsilon$  prescription is still needed to define the oscillatory large-time integrals. This fixed-order ultraviolet finiteness does not make the full perturbation series convergent.

### S.9.5. Vacuum Survival under an External Force

The forced action remains Gaussian. Completing its functional square gives

$$\frac{Z[f]}{Z[0]} = \exp \left[ -\frac{i}{2} \int dt dt' f(t) G_F(t-t') f(t') \right],$$

with the convention for  $G_F$  in S.9.7. Only the delta-function part of the pole prescription contributes to the modulus. For  $\tilde{f}(E) = \int dt e^{iEt} f(t)$  and real  $f$ ,

$$P_{0 \rightarrow 0} = \left| \frac{Z[f]}{Z[0]} \right|^2 = \exp \left[ -\frac{|\tilde{f}(\omega)|^2}{2\omega} \right].$$

The exponent is the mean occupation number of the coherent final state. It is dimensionless, nonnegative, and vanishes for an adiabatic force with no Fourier support at the oscillator frequency.

### S.9.6. The Free Scalar Source Functional and Wick Pairings by Differentiation

- (a) After integrating by parts, the free action is quadratic. In momentum space it reads

$$S_0 + S_J = -\frac{1}{2} \int \frac{d^4 p}{(2\pi)^4} \tilde{\phi}(-p) (p^2 + m^2 - i0) \tilde{\phi}(p) + \int \frac{d^4 p}{(2\pi)^4} \tilde{J}(-p) \tilde{\phi}(p).$$

Shift  $\tilde{\phi}(p) \mapsto \tilde{\phi}(p) - (p^2 + m^2 - i0)^{-1} \tilde{J}(p)$ . Translation invariance of the measure then yields

$$\frac{Z_0[J]}{Z_0[0]} = \exp \left[ \frac{i}{2} \int \frac{d^4 p}{(2\pi)^4} \frac{\tilde{J}(-p) \tilde{J}(p)}{p^2 + m^2 - i0} \right].$$

The positive infinitesimal is fixed by the vacuum boundary functionals, not by the algebraic completion of the square.

- (b) Let

$$G_F(x-y) \equiv \frac{1}{i^2 Z_0[0]} \frac{\delta^2 Z_0[J]}{\delta J(x) \delta J(y)} \Big|_{J=0}.$$

Because the exponent of  $Z_0[J]$  is quadratic, four derivatives can be nonzero at  $J=0$  only if they differentiate it in pairs. Therefore

$$\langle T \phi_1 \phi_2 \phi_3 \phi_4 \rangle_0 = G_{F,12} G_{F,34} + G_{F,13} G_{F,24} + G_{F,14} G_{F,23}.$$



For  $2n$  fields, every complete pairing occurs once. The number of terms is

$$(2n-1)!! = \frac{(2n)!}{2^n n!}.$$

An odd number of derivatives leaves one source unpaired and hence gives zero at  $J = 0$ .

### S.9.7. The Propagator Is a Green Function

(a) Set  $\tau = x^0 - y^0$  and

$$C(x, y) = \langle \Omega | [\phi(x), \phi(y)] | \Omega \rangle.$$

Both fields obey the homogeneous equation, so  $(\square_x - m^2)C = 0$ . The only possible terms in  $(\square_x - m^2)[\theta(\tau)C]$  therefore come from time derivatives of the step function. At equal time,

$$C(x, y)|_{\tau=0} = 0, \quad \partial_{x^0} C(x, y)|_{\tau=0} = -i\delta^d(\mathbf{x} - \mathbf{y}).$$

The first derivative of  $\theta C$  has no contact term. Differentiating once more and remembering that  $\square$  contains  $-\partial_{x^0}^2$  gives

$$\begin{aligned} (\square_x - m^2)D_R(x - y) &= -\delta(\tau) \partial_{x^0} C(x, y)|_{\tau=0} \\ &= i\delta^{d+1}(x - y). \end{aligned}$$

Hence

$$\boxed{a = i.}$$

(b) Varying the sourced action gives

$$(\square - m^2)\phi(x) = -j(x).$$

Since the preceding part established  $(\square - m^2)D_R = i\delta^{d+1}$ , the solution with no response before the source acts is

$$\boxed{\phi(x) = \phi_0(x) + i \int d^{d+1}y D_R(x - y)j(y).}$$

Indeed, applying the wave operator to the second term gives  $i^2 j = -j$ , and the step function in  $D_R$  enforces the retarded boundary condition.

To display the sourced modes explicitly, define

$$\tilde{j}(t, \mathbf{p}) = \int d^d \mathbf{y} e^{-i\mathbf{p} \cdot \mathbf{y}} j(t, \mathbf{y}).$$

The free expansion gives

$$D_R(t, \mathbf{x}) = -i\theta(t) \int \frac{d^d \mathbf{p}}{(2\pi)^d} \frac{\sin(E_{\mathbf{p}} t)}{E_{\mathbf{p}}} e^{i\mathbf{p} \cdot \mathbf{x}}.$$

Therefore the complete retarded solution is

$$\phi(x) = \phi_0(x) + \int \frac{d^d \mathbf{p}}{(2\pi)^d} e^{i\mathbf{p} \cdot \mathbf{x}} \int_{-\infty}^{x^0} dy^0 \frac{\sin[E_{\mathbf{p}}(x^0 - y^0)]}{E_{\mathbf{p}}} \tilde{j}(y^0, \mathbf{p}).$$

For real  $j$ , the response is Hermitian as required.

(c) Write  $D = \theta(\tau)W_> + \theta(-\tau)W_<$ , where

$$W_>(x, y) = \langle \Omega | \phi(x) \phi(y) | \Omega \rangle, \quad W_<(x, y) = \langle \Omega | \phi(y) \phi(x) | \Omega \rangle.$$

Away from  $\tau = 0$ , the wave operator annihilates both terms. The coefficient generated by the first time derivative is  $W_> - W_< = \langle \Omega | [\phi(x), \phi(y)] | \Omega \rangle$ , which vanishes at equal time. The second derivative leaves

$$\delta(\tau) \langle \Omega | [\partial_{x^0} \phi(x), \phi(y)] | \Omega \rangle = -i\delta(\tau) \delta^d(\mathbf{x} - \mathbf{y}).$$

The minus sign in  $\square = \nabla^2 - \partial_{x^0}^2$  reverses this sign. Thus

$$\boxed{(\square_x - m^2)D(x - y) = i\delta^{d+1}(x - y).}$$

The retarded and time-ordered functions have different boundary conditions, but they invert the same differential operator with the same normalization.

(d) The path-integral statement follows from functional integration by parts. For the free action and the usual vacuum boundary prescription,

$$\begin{aligned} 0 &= \int \mathcal{D}\phi \frac{\delta}{\delta\phi(x)} \left[ \phi(y) e^{iS_0[\phi]} \right] \\ &= Z[0] \delta^{d+1}(x - y) + iZ[0] \langle \Omega | T \{ \phi(y) (\square_x - m^2) \phi(x) \} | \Omega \rangle. \end{aligned}$$

Solving this identity reproduces the contact term in the preceding part. More generally,

$$\begin{aligned} &\langle \Omega | T \left\{ (\square_x - m^2) \phi(x) \prod_{r=1}^n \phi(x_r) \right\} | \Omega \rangle \\ &= i \sum_{r=1}^n \delta^{d+1}(x - x_r) \langle \Omega | T \left\{ \prod_{\substack{s=1 \\ s \neq r}}^n \phi(x_s) \right\} | \Omega \rangle. \end{aligned}$$

With a source,  $(\square - m^2)\phi$  on the left is replaced by  $(\square - m^2)\phi + j$ . Thus the field equation holds inside a correlator except where the varied field meets another insertion. Those coincident points are precisely the contact terms.

### S.9.8. The Scalar Vacuum Wave Functional

For each  $\mathbf{p}$ , the scalar Hamiltonian is an oscillator of frequency  $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$ . The product of their ground states is

$$\Psi_0[\phi] = \mathcal{N} \exp \left[ -\frac{1}{2} \int d^3p E_{\mathbf{p}} \tilde{\phi}(\mathbf{p}) \tilde{\phi}(-\mathbf{p}) \right].$$

In position space the kernel is

$$\mathcal{E}(\mathbf{x} - \mathbf{y}) = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})},$$

so it is nonlocal even though the Hamiltonian density is local. Gaussian integration gives the equal-time covariance

$$\langle \tilde{\phi}(\mathbf{p}) \tilde{\phi}(\mathbf{q}) \rangle_0 = \frac{\delta^3(\mathbf{p} + \mathbf{q})}{2E_{\mathbf{p}}}.$$

This inverse-frequency factor agrees with the canonical mode expansion. With a momentum cutoff and finite spatial volume,  $\mathcal{N}$  is the product  $\prod_{\mathbf{p}} (E_{\mathbf{p}}/\pi)^{1/4}$ , with the usual adjustment for the reality relation between  $\mathbf{p}$  and  $-\mathbf{p}$ . This regulator-dependent constant cancels between normalized matrix elements, whereas the kernel  $E_{\mathbf{p}}$  and its inverse covariance remain.

### S.9.9. Schwinger–Dyson Equations from Functional Integration by Parts

(a) The functional product rule gives

$$0 = \int \mathcal{D}\phi \left[ \delta^{d+1}(x - y) + i\phi(y) \frac{\delta S}{\delta \phi(x)} \right] e^{iS[\phi]}.$$

The  $i0$  prescription makes the boundary term in field space vanish. After division by  $Z$ , solving for the second term gives

$$\left\langle \frac{\delta S}{\delta \phi(x)} \phi(y) \right\rangle = i\delta^{d+1}(x - y).$$

The brackets denote the normalized, time-ordered path-integral expectation value. No expansion in the couplings was used.

(b) Varying the free action and integrating the kinetic term by parts gives

$$\frac{\delta S_0}{\delta \phi(x)} = (\square_x - m^2)\phi(x).$$

Substitution in the first result yields

$$(\square_x - m^2)\langle \Omega | T\{\phi(x)\phi(y)\} | \Omega \rangle = i\delta^{d+1}(x - y).$$

Thus the classical equation holds whenever  $x \neq y$ ; the failure at coincidence is the contact term generated when the functional derivative acts on the insertion.

(c) Use  $\phi(y)\phi(z)e^{iS_0}$  as the differentiated expression. The two possible insertion derivatives give

$$\begin{aligned} & (\square_x - m^2)\langle \Omega | T\{\phi(x)\phi(y)\phi(z)\} | \Omega \rangle \\ &= i\delta^{d+1}(x - y)\langle \Omega | \phi(z) | \Omega \rangle + i\delta^{d+1}(x - z)\langle \Omega | \phi(y) | \Omega \rangle. \end{aligned}$$

The free vacuum is invariant under  $\phi \mapsto -\phi$ , so both one-point functions vanish. Hence the three-point function and the right-hand side are zero, but the unsimplified equation displays the two contact terms and generalizes directly to theories with nonzero one-point functions.

### S.9.10. Low-Order Correlators in $\lambda\phi^4$

(a) At first order,

$$\langle T\phi(x)\phi(y) \rangle = \frac{\langle T\phi(x)\phi(y)[1 - i\lambda \int d^4z \phi(z)^4/4!] \rangle_0}{\langle T[1 - i\lambda \int d^4z \phi(z)^4/4!] \rangle_0}.$$

For the connected tadpole, choose the vertex field joined to  $x$  in 4 ways and that joined to  $y$  in 3 ways; the remaining two pair. Thus  $12/4! = 1/2$ , and

$$G_F(x-y) - \frac{i\lambda}{2} \int d^4z G_F(x-z)G_F(0)G_F(z-y) + O(\lambda^2).$$

The other numerator contraction pairs  $x$  with  $y$  and contracts the four vertex fields among themselves in three ways. It is exactly the free two-point function times the first-order denominator and therefore cancels.

At second order the three non-vacuum topologies are

$$\begin{aligned} & \frac{(-i\lambda)^2}{6} \int dz dw G(x-z)G(z-w)^3G(w-y), \\ & \frac{(-i\lambda)^2}{4} \int dz dw G(x-z)G(z-w)G(w-y)G(0)^2, \\ & \frac{(-i\lambda)^2}{4} \int dz dw G(x-z)G(y-z)G(z-w)^2G(0), \end{aligned}$$

with distinct external attachments included. The coefficients follow by pairing the half-edges at the two labeled vertices; every omitted pairing has a vacuum component and cancels against  $Z[0]$ .

(b) The first connected contribution is

$$-\frac{i\lambda}{4!} \int d^4z \langle T\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\phi(z)^4 \rangle_0.$$

Connectivity at tree level requires each external field to contract with a different field at  $z$ . There are  $4!$  bijections, canceling the factor in the interaction. Consequently

$$G_{\text{conn}}^{(4)}(x_1, \dots, x_4) = -i\lambda \int d^4z \prod_{a=1}^4 G_F(x_a - z) + O(\lambda^2).$$

Contractions that pair external fields among themselves are disconnected; pure vacuum factors cancel against  $Z[0]$ .

At order  $\lambda^2$ , the three partitions of the external labels give the  $s, t, u$  fish graphs, each with factor  $1/2$ :

$$\frac{(-i\lambda)^2}{2} \int dz dw G(x_a - z)G(x_b - z)G(z-w)^2G(x_c - w)G(x_d - w).$$

The other connected class consists of four external-leg tadpole dressings: three external legs meet one vertex, one line joins the second, and that vertex carries the remaining external leg and a tadpole. Its factor is  $1/2$ . The vertex-degree constraints show that these classes exhaust order  $\lambda^2$ .

### S.9.11. Vacuum Energy and Connected Vacuum Graphs

- (a) Translation invariance leaves one unconstrained vertex integral in each vacuum graph. Consequently

$$Z[0] = e^{iW_0}, \quad W_0 = -\mathcal{V}_4 \mathcal{E}_{\text{vac}}, \quad \mathcal{V}_4 = \int d^4x.$$

- (b) Let  $C_r$  denote the complete weight of one connected vacuum-graph type  $r$ , including its coupling, integrals, and inverse symmetry factor. A vacuum diagram containing  $n_r$  indistinguishable copies of each type has weight  $\prod_r C_r^{n_r}/n_r!$ . Summing over all multiplicities gives

$$Z[0] = \prod_r \sum_{n_r=0}^{\infty} \frac{C_r^{n_r}}{n_r!} = \prod_r e^{C_r} = \exp\left(\sum_r C_r\right).$$

Therefore

$$\log Z[0] = \sum_r C_r.$$

The factorials are precisely those obtained by permuting identical connected components. Thus the logarithm retains each connected vacuum graph once, with the symmetry factor already contained in  $C_r$ .

- (c) For  $\lambda\phi^4/4!$ , the one-vertex figure-eight has factor  $1/8$ . At second order, the connected four-line basketball has factor  $1/48$ , and the topology with two joining lines and one tadpole at each vertex has factor  $1/16$ . Thus

$$\begin{aligned} iW_0 &= \frac{-i\lambda}{8} \int dx G(0)^2 \\ &+ (-i\lambda)^2 \int dx dy \left[ \frac{G(x-y)^4}{48} + \frac{G(x-y)^2 G(0)^2}{16} \right] + O(\lambda^3). \end{aligned}$$

Dividing by  $\mathcal{V}_4$  and using part 1 yields the vacuum-energy shift, with the overall sign fixed by the propagator convention.

### S.9.12. Cancellation of Arbitrary Vacuum Bubbles

Every diagram contributing to the unnormalized insertion functional can be decomposed uniquely into a part  $D$  containing all external insertions and a multiset  $V$  of vacuum components. The sum over  $V$  is independent of the external points and equals the vacuum functional:

$$\tilde{G}_n(x_1, \dots, x_n) = Z[0] G_n^{\text{no vac}}(x_1, \dots, x_n).$$

Dividing by  $Z[0]$  therefore gives

$$G_n = \frac{\tilde{G}_n}{Z[0]} = G_n^{\text{no vac.}}$$

This argument uses only component factorization and the exponential counting of indistinguishable vacuum graphs, not the form or order of the polynomial interaction. It consequently holds to every perturbative order.

### S.9.13. A Wave Functional as a Euclidean Boundary and A Parity-Twisted Euclidean Trace

(a) Insert a field basis at the initial boundary:

$$\langle \phi_f | e^{-TH} \psi_i \rangle = \int \mathcal{D}\phi_i K_E[\phi_f, T; \phi_i, 0] \Psi_i[\phi_i] = \int_{\phi(T)=\phi_f} \mathcal{D}\phi e^{-S_E[\phi]} \Psi_i[\phi(0)].$$

The spectral expansion is

$$\sum_n e^{-E_n T} \Psi_n[\phi_f] \langle n | \psi_i \rangle.$$

If the initial state has nonzero vacuum overlap, then as  $T \rightarrow \infty$

$$e^{E_0 T} \langle \phi_f | e^{-TH} \psi_i \rangle \longrightarrow \Psi_0[\phi_f] \langle \Omega | \psi_i \rangle.$$

Thus a semi-infinite Euclidean path integral prepares the ground-state wave functional, while changing its far-boundary weight changes only the overlap constant.

(b) Since  $P|x\rangle = |-x\rangle$ ,

$$\text{Tr}(Pe^{-TH}) = \int dx \langle x | Pe^{-TH} x \rangle = \int dx \langle -x | e^{-TH} x \rangle.$$

The corresponding Euclidean paths obey the twisted boundary condition

$$x(T) = -x(0).$$

For the oscillator, parity of the  $n$ th energy eigenstate is  $(-1)^n$ , so

$$\text{Tr}(Pe^{-TH}) = \sum_{n=0}^{\infty} (-1)^n e^{-\omega T(n+1/2)} = \frac{1}{2 \cosh(\omega T/2)}.$$

Integrating the quoted oscillator heat kernel with endpoints  $x$  and  $-x$  gives the same Gaussian result, verifying the twisted path-integral boundary condition.

### S.9.14. Pfaffian Covariance and Canonical Form

For  $\theta = N\theta'$ , the top Grassmann monomial transforms as  $\theta_1 \cdots \theta_{2N} = \det(N)\theta'_1 \cdots \theta'_{2N}$ . Berezin integration must therefore transform inversely:

$$d^{2N}\theta = \det(N)^{-1} d^{2N}\theta'.$$

Now evaluate

$$\int d^{2N} \theta e^{\frac{1}{2} \theta^\top A \theta} = \text{Pf}(A)$$

after the change of variables. The exponent contains  $N^\top A N$ , while the measure contributes the inverse determinant, so

$$\text{Pf}(A) = \det(N)^{-1} \text{Pf}(N^\top A N).$$

Rearrangement proves

$$\boxed{\text{Pf}(N^\top A N) = \det(N) \text{Pf}(A)}.$$

For one pair,  $A = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}$ , so  $\text{Pf}(A) = a$ . The formula then follows immediately for a diagonal rescaling and fixes the sign on the component continuously connected to the identity; polynomiality extends it to every invertible  $N$ .

There is a congruence transformation for which

$$N^\top A N = \bigoplus_{r=1}^N \begin{pmatrix} 0 & a_r \\ -a_r & 0 \end{pmatrix}.$$

The Grassmann Gaussian factorizes, so this canonical matrix has Pfaffian  $\prod_r a_r$  and determinant  $\prod_r a_r^2$ . Using congruence covariance on both quantities proves

$$\text{Pf}(A)^2 = \det A, \quad \text{Pf}(A) = \pm \sqrt{\det A}.$$

Continuity from an oriented canonical basis fixes the sign.

#### S.9.15. An Exact Four-Grassmann Interaction

Choose  $d^4 \theta = d\theta_4 d\theta_3 d\theta_2 d\theta_1$ ,  $\int d^4 \theta \theta_1 \theta_2 \theta_3 \theta_4 = 1$ , and write

$$S = \frac{1}{2} A_{ab} \theta_a \theta_b + \lambda \theta_1 \theta_2 \theta_3 \theta_4.$$

Only the top monomial contributes. Since the quadratic term squared supplies  $\text{Pf}(A) \theta_1 \theta_2 \theta_3 \theta_4$ ,

$$Z = \int d^4 \theta e^{-S} = \text{Pf}(A) - \lambda.$$

No higher term exists because every Grassmann variable squares to zero. In diagram language,  $\text{Pf}(A)$  is the sum over the three signed Gaussian pairings, while the single quartic vacuum vertex has weight  $-\lambda$ . There are no multi-vertex graphs, so the diagrammatic enumeration is already exact. Thus  $Z \rightarrow \text{Pf}(A)$  when  $\lambda \rightarrow 0$ , while for  $A = 0$  only the saturated quartic monomial survives and  $Z = -\lambda$ , confirming both limiting cases with the chosen measure orientation.

#### S.9.16. Bosonic and Fermionic Determinant Powers and Fermionic Wick's Theorem from a Pfaffian

- (a) For a positive real symmetric  $K$ ,

$$\int d^N x e^{-\frac{1}{2}x^\top K x} = (2\pi)^{N/2}(\det K)^{-1/2}.$$

For independent complex Grassmann pairs and an arbitrary invertible  $A$ ,

$$\int d^N \bar{\theta} d^N \theta e^{-\bar{\theta} A \theta} = \det A.$$

Finally, for  $2N$  real Grassmann variables and antisymmetric  $B$ ,

$$\int d^{2N} \theta e^{\frac{1}{2}\theta^\top B \theta} = \text{Pf}(B), \quad \text{Pf}(B)^2 = \det B.$$

Commuting Gaussians integrate inverse widths, whereas Berezin integration extracts the highest polynomial coefficient; this reverses the determinant power. Antisymmetry of  $B$  is required because the symmetric part multiplies an antisymmetric variable product and vanishes.

- (b) For antisymmetric invertible  $A$ , introduce real Grassmann sources:

$$Z[\eta] = \int d^{2N} \theta e^{\frac{1}{2}\theta^\top A \theta + \eta^\top \theta} = \text{Pf}(A) e^{\frac{1}{2}\eta^\top A^{-1} \eta},$$

with the source and differential orders held fixed. Four left derivatives give

$$\langle \theta_a \theta_b \theta_c \theta_d \rangle = (A^{-1})_{ab}(A^{-1})_{cd} - (A^{-1})_{ac}(A^{-1})_{bd} + (A^{-1})_{ad}(A^{-1})_{bc}.$$

The middle pairing is odd because forming it requires one interchange of Grassmann objects. With  $2N$  insertions, differentiation produces the Pfaffian of the restricted covariance matrix, equivalently the signed sum over all complete pairings.

### S.9.17. Berezin Gaussians, Fermionic Sources, and Determinant Signs

- (a) Fix

$$\int d\bar{\theta} d\theta \theta \bar{\theta} = 1.$$

Because  $(\bar{\theta}\theta)^2 = 0$ ,

$$e^{-\bar{\theta}a\theta} = 1 - \bar{\theta}a\theta = 1 + a\theta\bar{\theta},$$

and hence

$$\int d\bar{\theta} d\theta e^{-\bar{\theta}a\theta} = a.$$

Under  $\theta \rightarrow \theta + \xi$  and  $\bar{\theta} \rightarrow \bar{\theta} + \bar{\xi}$ , all new terms have degree lower than two in the integration variables or differ by a Berezin total derivative. Their integrals vanish. Thus the integral is translation invariant even though Berezin integration is algebraic differentiation rather than an ordinary limiting sum.



- (b) Change variables so that  $A$  is triangular; Berezin measures transform with the inverse Jacobian, so the result is basis independent. For a diagonal kernel,

$$\prod_{i=1}^N \int d\bar{\theta}_i d\theta_i e^{-a_i \bar{\theta}_i \theta_i} = \prod_i a_i = \det A.$$

Therefore, generally,

$$\int \prod_i (d\bar{\theta}_i d\theta_i) e^{-\bar{\theta}_i A_{ij} \theta_j} = \det A.$$

The exponential terminates, and the integral selects the coefficient containing every  $\theta_i$  and  $\bar{\theta}_i$ . That coefficient is the antisymmetrized product defining  $\det A$ , rather than its reciprocal.

- (c) Differentiate the identity  $Z(A) = \det A$ :

$$\frac{\partial Z}{\partial A_{lk}} = \det A (A^{-1})_{kl}.$$

On the integral side,

$$\frac{\partial Z}{\partial A_{lk}} = \int d\bar{\theta} d\theta (-\bar{\theta}_l \theta_k) e^{-\bar{\theta} A \theta} = \int d\bar{\theta} d\theta \theta_k \bar{\theta}_l e^{-\bar{\theta} A \theta}.$$

Consequently

$$\int d\bar{\theta} d\theta \theta_k \bar{\theta}_l e^{-\bar{\theta} A \theta} = \det A (A^{-1})_{kl}.$$

The transpose-looking order is fixed by differentiating  $A_{lk}$ , not by a convention imposed afterward.

- (d) Introduce sources and complete the square:

$$\frac{Z[\bar{\eta}, \eta]}{Z[0]} = \exp(\bar{\eta} A^{-1} \eta).$$

Taking ordered source derivatives corresponding to  $\theta_i \bar{\theta}_j \theta_k \bar{\theta}_l$  yields

$$\langle \theta_i \bar{\theta}_j \theta_k \bar{\theta}_l \rangle = (A^{-1})_{ij} (A^{-1})_{kl} - (A^{-1})_{il} (A^{-1})_{kj}.$$

The second contraction crosses the middle pair. Moving the associated Grassmann derivatives next to each other requires one odd interchange, which supplies the minus sign. This is the finite-dimensional origin of the sign rule for interchanging fermion lines.

- (e) Let

$$\mathcal{D}(p) = i\gamma^\mu p_\mu + m - i0$$

and use the source convention

$$-i\bar{\psi} \mathcal{D} \psi - i\bar{\psi} \eta - i\bar{\eta} \psi$$

in the exponent. The shifts  $\psi \rightarrow \psi + \mathcal{D}^{-1}\eta$  and  $\bar{\psi} \rightarrow \bar{\psi} + \bar{\eta}\mathcal{D}^{-1}$  give

$$\frac{Z[\bar{\eta}, \eta]}{Z[0]} = \exp \left( i \int \frac{d^4 p}{(2\pi)^4} \bar{\eta}(-p) \mathcal{D}^{-1}(p) \eta(p) \right).$$

Since

$$\mathcal{D}^{-1}(p) = \frac{-i\gamma \cdot p + m}{p^2 + m^2 - i0},$$

ordered source differentiation reproduces

$$\langle T\psi(p)\bar{\psi}(-p) \rangle = -i \frac{-i\gamma \cdot p + m}{p^2 + m^2 - i0}$$

for the displayed exponent convention. The pole is fixed by the fermionic vacuum wave functional just as in the scalar case.

(f) For a background bilinear, the fermionic integral is

$$Z[\Gamma] \propto \det(\mathcal{D} + \Gamma) = \det \mathcal{D} \exp [\text{Tr} \log(1 + \mathcal{D}^{-1}\Gamma)].$$

Expanding,

$$\text{Tr} \log(1 + \mathcal{D}^{-1}\Gamma) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \text{Tr}(\mathcal{D}^{-1}\Gamma)^n.$$

The trace closes the chain of  $n$  propagators and vertices into one loop;  $1/n$  removes cyclic choices of its starting vertex. Relative to the bosonic expansion, the alternating determinant series differs by one overall minus sign for every connected closed chain. Equivalently, closing the chain requires an odd cyclic reordering of Grassmann fields. Thus each closed fermion loop carries the standard extra factor  $-1$ .

### S.9.18. The Complete Gaussian Generating Function

(a) First set  $j = 0$  and write  $I(a) = \int_{\mathbb{R}} dx e^{-ax^2/2}$ . For  $a > 0$ ,

$$I(a)^2 = \int_{\mathbb{R}^2} dx dy e^{-a(x^2+y^2)/2} = 2\pi \int_0^{\infty} r dr e^{-ar^2/2} = \frac{2\pi}{a}.$$

Positivity fixes  $I(a) = \sqrt{2\pi/a}$ . Completing the square,

$$-\frac{a}{2}x^2 + jx = -\frac{a}{2} \left( x - \frac{j}{a} \right)^2 + \frac{j^2}{2a},$$

and translation of the integration variable gives

$$\boxed{Z_a(j) = \sqrt{\frac{2\pi}{a}} e^{j^2/(2a)}}.$$

For a real contour, absolute convergence requires  $a > 0$ .

- (b) Choose an orthogonal matrix  $O$  with  $O^T A O = \text{diag}(a_1, \dots, a_N)$ , where every  $a_i > 0$ . With  $x = O y$  and  $J' = O^T J$ , the absolute Jacobian is  $|\det O| = 1$ . The integral factors into the result of part (a):

$$\begin{aligned} Z_A(J) &= \prod_{i=1}^N \sqrt{\frac{2\pi}{a_i}} \exp\left(\frac{J_i'^2}{2a_i}\right) \\ &= \frac{(2\pi)^{N/2}}{\sqrt{\det A}} \exp\left(\frac{1}{2} J'^T A^{-1} J'\right). \end{aligned}$$

- (c) Normalized moments follow from source differentiation:

$$\langle x_i x_j \rangle = \frac{1}{Z_A(0)} \frac{\partial^2 Z_A}{\partial J_i \partial J_j} \Big|_{J=0} = (A^{-1})_{ij}.$$

Because the mean vanishes, the second derivative of  $\log Z_A$  gives the same answer. In general, derivatives of  $\log Z$  generate connected cumulants, while derivatives of  $Z/Z[0]$  generate full moments. Replacing indices by space-time points and  $A$  by a quadratic kernel gives

$$\frac{Z[J]}{Z[0]} = \exp\left(\frac{1}{2} \int d^d x d^d y J(x) A^{-1}(x, y) J(y)\right),$$

with the appropriate factors of  $i$  after continuation to the real-time convention.

### S.9.19. The Characteristic Function of a Gaussian Quantum Coordinate

In the position representation, the normalized vacuum probability density is a centered Gaussian,

$$|\Psi_0(q)|^2 = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-q^2/(2\sigma^2)}, \quad \sigma^2 = \langle \Omega | q^2 | \Omega \rangle.$$

The source formula of S.9.18 with  $j = iK$  gives

$$\langle \Omega | e^{iKq} | \Omega \rangle = e^{-K^2\sigma^2/2}.$$

For an operator derivation, write  $q = c(a + a^\dagger)$ , where  $[a, a^\dagger] = 1$  and  $c^2 = \sigma^2$ . Since  $[iKca^\dagger, iKca] = K^2c^2$  is central, the Baker–Campbell–Hausdorff formula yields

$$e^{iKq} = e^{-K^2c^2/2} e^{iKca^\dagger} e^{iKca}.$$

The rightmost exponential acts trivially on  $|\Omega\rangle$ , and every nonconstant term in the left exponential is killed by  $\langle \Omega |$ . This reproduces the Gaussian answer and fixes the coefficient without choosing a position basis.

If the mean is  $\bar{q}$ , write  $q = \bar{q} + \delta q$  with  $\langle \delta q \rangle = 0$ . The characteristic function becomes

$$\langle e^{iKq} \rangle = e^{iK\bar{q}} \exp\left[-\frac{K^2}{2} \langle (q - \bar{q})^2 \rangle\right].$$

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## 10 Non-Perturbative Methods

We are now going to begin our study of higher-order contributions to physical processes, corresponding to Feynman diagrams involving one or more loops. It will be very useful in this work to have available a method of deriving results valid to all orders in perturbation theory (and in some cases beyond perturbation theory). In this chapter we will exploit the field equations and commutation relations of the interacting fields in the Heisenberg picture for this purpose. The essential bridge between the Heisenberg picture and the Feynman diagrams of perturbation theory is provided by the theorem proved in Section 6.4: the sum of all diagrams for a process  $\alpha \rightarrow \beta$  with extra vertices inserted corresponding to operators  $O_a(x)$ ,  $O_b(y)$ , etc. is given by the matrix element of the time-ordered product of the corresponding Heisenberg-picture operators

$${}_{\text{out}}\langle\beta|T\{-iO_a(x), -iO_b(y), \dots\}|\alpha\rangle_{\text{in}}.$$

As a special case, where the operators  $O_a(x)$ ,  $O_b(x)$ , etc. are elementary particle fields, this matrix element equals the sum of all Feynman diagrams with incoming lines on the mass shell corresponding to the state  $\alpha$ , outgoing lines on the mass shell corresponding to the state  $\beta$ , and lines off the mass shell (including propagators) corresponding to the operators  $O_a(x)$ ,  $O_b(x)$ , etc. After exploring some of the non-perturbative results that can be obtained in this way we will be in a good position to take up the perturbative calculation of radiative corrections.

### 10.1 Symmetries

One obvious but important use of the theorem quoted above is to extend the application of symmetry principles from  $S$ -matrix elements, where all external lines have four-momenta on the mass shell, to parts of Feynman diagrams, with some or all external lines off the mass shell.

For instance, consider the symmetry of spacetime translational invariance. This symmetry has as a consequence the existence of a Hermitian four-vector operator  $P^\mu$ , with the property that, for any local function  $O(x)$  of field operators and their canonical conjugates,

$$[P_\mu, O(x)] = i \frac{\partial}{\partial x^\mu} O(x). \quad (10.1.1)$$

(See Eqs. (7.3.28) and (7.3.29).) Also, the states  $\alpha$  and  $\beta$  are usually chosen to be eigenstates of the four-momentum:

$$P^\mu |\alpha\rangle_{\text{in}} = p_\alpha^\mu |\alpha\rangle_{\text{in}}, \quad P^\mu |\beta\rangle_{\text{out}} = p_\beta^\mu |\beta\rangle_{\text{out}}. \quad (10.1.2)$$

It follows that for any set of local functions  $O_a(x)$ ,  $O_b(x)$ , etc. of fields and/or field derivatives

$$\begin{aligned} & (p_{\beta\mu} - p_{\alpha\mu})_{\text{out}} \langle\beta|T\{O_a(x_1), O_b(x_2), \dots\}|\alpha\rangle_{\text{in}} \\ &= {}_{\text{out}}\langle\beta|[P_\mu, T\{O_a(x_1), O_b(x_2), \dots\}]|\alpha\rangle_{\text{in}} \\ &= i \left( \frac{\partial}{\partial x_1^\mu} + \frac{\partial}{\partial x_2^\mu} + \dots \right) {}_{\text{out}}\langle\beta|T\{O_a(x_1), O_b(x_2), \dots\}|\alpha\rangle_{\text{in}}. \end{aligned} \quad (10.1.3)$$

This has the solution

$$\begin{aligned} & {}_{\text{out}}\langle\beta|T\{O_a(x_1), O_b(x_2), \dots\}|\alpha\rangle_{\text{in}} \\ & = \exp(i(p_\beta - p_\alpha) \cdot x) F_{ab\dots}(x_1 - x_2, \dots), \end{aligned} \quad (10.1.4)$$

where  $x$  is any sort of average spacetime coordinate

$$x^\mu = c_1 x_1^\mu + c_2 x_2^\mu + \dots, \quad c_1 + c_2 + \dots = 1 \quad (10.1.5)$$

and  $F$  depends only on differences among the  $x$ s. (In particular, a vacuum expectation value can depend only on the coordinate differences.) We can Fourier transform Eq. (10.1.4) by integrating separately over  $x^\mu$  and the coordinate differences, with the result that

$$\begin{aligned} & \int d^4x_1 d^4x_2 \dots {}_{\text{out}}\langle\beta|T\{O_a(x_1), O_b(x_2), \dots\}|\alpha\rangle_{\text{in}} \\ & \cdot \exp(-ik_1 \cdot x_1 - ik_2 \cdot x_2 - \dots) \propto \delta^4(p_\beta - p_\alpha - k_1 - k_2 - \dots). \end{aligned} \quad (10.1.6)$$

We saw in Section 6.4 that the matrix element of the time-ordered product is given by applying the usual coordinate-space Feynman rules to the sum of all graphs with incoming particles corresponding to particles in  $\alpha$ , outgoing particles in  $\beta$ , and external lines that simply terminate in vertices at  $x_1, x_2, \dots$ . The Fourier transform (10.1.6) is correspondingly given by applying the momentum-space Feynman rules to the same sum of Feynman diagrams, with off-shell external lines carrying four-momenta  $k_1, k_2, \dots$  into the diagrams. Eq. (10.1.6) is then just the statement that this sum of Feynman graphs conserves four-momentum. The result is obvious in perturbation theory, because four-momentum is conserved at every vertex, so it is not surprising to see the same result emerging without having to rely on perturbation theory.

With somewhat more effort, one can use the Lorentz transformation properties of the Heisenberg-picture fields and the ‘in’ and ‘out’ states to show that the sum of all graphs with a given set of on- and off-shell lines satisfies the same Lorentz transformation conditions as the lowest-order terms.

Similar arguments apply to the conservation of internal quantum numbers, like electric charge. As shown in Section 7.3, a field or other operator  $O_a(x)$  that destroys a charge  $q_a$  (or creates a charge  $-q_a$ ) will satisfy

$$[Q, O_a(x)] = -q_a O_a(x)$$

in the Heisenberg and interaction pictures alike. Also, if the free-particle states  $\alpha$  and  $\beta$  have charges  $q_\alpha$  and  $q_\beta$ , then so do the corresponding ‘in’ and ‘out’ states. We then have

$$\begin{aligned} & (q_\beta - q_\alpha) {}_{\text{out}}\langle\beta|T\{O_a(x), O_b(y), \dots\}|\alpha\rangle_{\text{in}} \\ & = {}_{\text{out}}\langle\beta|[Q, T\{O_a(x), O_b(y), \dots\}]|\alpha\rangle_{\text{in}} \\ & = -(q_a + q_b + \dots) {}_{\text{out}}\langle\beta|T\{O_a(x), O_b(y), \dots\}|\alpha\rangle_{\text{in}}. \end{aligned}$$

Thus the amplitude  ${}_{\text{out}}\langle\beta|T\{O_a(x), O_b(y), \dots\}|\alpha\rangle_{\text{in}}$  vanishes unless charge is conserved

$$q_\beta = q_\alpha - q_a - q_b - \dots \quad (10.1.7)$$

A somewhat less trivial example is provided by the symmetry of charge-conjugation invariance. As we saw in Chapter 5, there is an operator  $C$  that interchanges electron and positron operators

$$\begin{aligned} C a_{\mathbf{p},\sigma,e^-} C^{-1} &= \xi^* a_{\mathbf{p},\sigma,e^+}, \\ C a_{\mathbf{p},\sigma,e^+} C^{-1} &= \xi a_{\mathbf{p},\sigma,e^-}, \end{aligned}$$

with  $\xi$  a phase factor. For the free electron field  $\psi(x)$ , this gives

$$C\psi(x)C^{-1} = \xi^* \beta C\psi(x)^*,$$

where  $\beta C$  is a  $4 \times 4$  matrix, which (for the Dirac matrix representation we have been using, with  $\gamma_5$  diagonal) takes the form

$$\beta C = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

Applied to the free-particle electric current in spinor electrodynamics, this gives

$$C(\bar{\psi}\gamma^\mu\psi)C^{-1} = -\bar{\psi}C\gamma^{\mu T}C^{-1}\psi = -\bar{\psi}\gamma^\mu\psi.$$

If  $C$  is to be conserved in electrodynamics, it must then also be defined to anticommute with the free photon field

$$C(a^\mu)C^{-1} = -a^\mu.$$

In theories like electrodynamics for which  $C$  commutes with the interaction as well as  $H_0$ , it also commutes with the similarity transformation  $\Omega(t)$  between the Heisenberg and interaction pictures, and so it anticommutes with the electric current of the interacting fields

$$C(\bar{\Psi}\gamma^\mu\Psi)C^{-1} = -\bar{\Psi}\gamma^\mu\Psi \quad (10.1.8)$$

and the electromagnetic field in the Heisenberg-picture

$$C(A^\mu)C^{-1} = -A^\mu. \quad (10.1.9)$$

It follows then that the vacuum expectation value of the time-ordered product of any odd number of electromagnetic currents and/or fields vanishes. Therefore the sum of all Feynman graphs with an odd number of external photon lines (off or on the photon mass shell) and no other external lines vanishes.

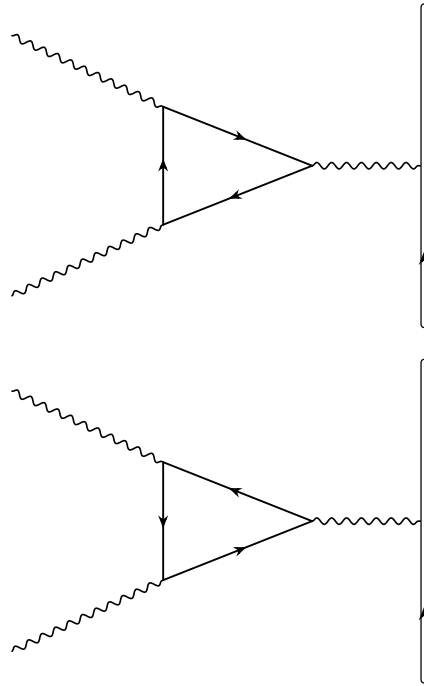
This result is known as Furry's theorem [1]. It can be proved perturbatively by noting that a graph consisting of electron loops  $\ell$ , to each of which are attached  $n_\ell$  photon lines, must have numbers  $I$  and  $E$  of internal and external photon lines related by an analog of Eq. (6.3.11):

$$2I + E = \sum_{\ell} n_{\ell}.$$

Hence if  $E$  is odd at least one of the loops must have attached an odd number of photon lines. For any such loop there is a cancellation between the two diagrams in which the electron arrows circulate around the loop in opposite directions. Hence Furry's theorem is a somewhat less trivial consequence of a symmetry principle than translation or Lorentz invariance; it is not true of individual diagrams, but rather of certain sums of diagrams. Figure 10.1 illustrates the application of Furry's theorem that was historically most important, its use to show that the scattering of a photon by an external electromagnetic field receives no contributions of first order (or any odd order) in the external field.

## 10.2 Polology

One of the most important uses of the non-perturbative methods described in this chapter is to clarify the pole structure of Feynman amplitudes as



**Figure 10.1.** The lowest-order diagrams for the scattering of a photon by an electromagnetic field. Here straight lines represent virtual electrons; wavy lines represent real and virtual photons; and the double line represents a heavy particle like an atomic nucleus that serves as a source of an electromagnetic field. The contributions of these two diagrams cancel, as required by charge-conjugation invariance.

functions of the momenta carried by external lines. Often the  $S$ -matrix for a physical process can be well approximated by the contribution of a single pole. Also, an understanding of this pole structure will help us later in dealing with radiative corrections to particle propagators.



Consider the momentum-space amplitude

$$\begin{aligned} \int d^4x_1 \cdots d^4x_n e^{-iq_1 \cdot x_1} \cdots e^{-iq_n \cdot x_n} \langle T\{A_1(x_1) \cdots A_n(x_n)\} \rangle_0 \\ \equiv G(q_1 \cdots q_n). \end{aligned} \quad (10.2.1)$$

The  $A$ s are Heisenberg-picture operators of arbitrary Lorentz type, and  $\langle \cdots \rangle_0$  denotes the expectation value in the true vacuum  $|\Omega\rangle_{\text{in}} = |\Omega\rangle_{\text{out}} \equiv |\Omega\rangle$ . As discussed in Section 6.4, if  $A_1, \dots, A_n$  are ordinary fields appearing in the Lagrangian, then (10.2.1) is a sum of the terms calculated using the ordinary Feynman rules, for all graphs with external lines corresponding to the fields  $A_1, \dots, A_n$ , carrying off-shell four-momenta  $q_1 \cdots q_n$  into the graph. However, we will not be limited to this case; the  $A_i$  may be arbitrary local functions of fields and field derivatives.

We are interested in poles of  $G$  at certain values of the invariant squares of the total four-momenta carried by various subsets of the external lines. To be definite, let's consider  $G$  as a function of  $q^2$ , where

$$q \equiv q_1 + \cdots + q_r = -q_{r+1} - \cdots - q_n \quad (10.2.2)$$

with  $1 \leq r \leq n-1$ . We will show that  $G$  has a pole at  $q^2 = -m^2$ , where  $m$  is the mass of any one-particle state that has non-vanishing matrix elements with the states  $A_1^\dagger \cdots A_r^\dagger |\Omega\rangle$  and  $A_{r+1} \cdots A_n |\Omega\rangle$ , and that the residue at this pole is given by

$$\begin{aligned} G \longrightarrow \frac{-2i\sqrt{\mathbf{q}^2 + m^2}}{q^2 + m^2 - i\epsilon} (2\pi)^4 \delta^4(q_1 + \cdots + q_n) \\ \cdot \sum_{\sigma} M_{0;\mathbf{q},\sigma}(q_2 \cdots q_r) M_{\mathbf{q},\sigma;0}(q_{r+2} \cdots q_n), \end{aligned} \quad (10.2.3)$$

where the  $M$ s are defined by<sup>111</sup>

$$\begin{aligned} \int d^4x_1 \cdots d^4x_r e^{-iq_1 \cdot x_1} \cdots e^{-iq_r \cdot x_r} \langle \Omega | T\{A_1(x_1) \cdots A_r(x_r)\} | \mathbf{p}, \sigma \rangle \\ = (2\pi)^4 \delta^4(q_1 + \cdots + q_r - p) M_{0;\mathbf{p},\sigma}(q_2 \cdots q_r), \end{aligned} \quad (10.2.4)$$

$$\begin{aligned} \int d^4x_{r+1} \cdots d^4x_n e^{-iq_{r+1} \cdot x_{r+1}} \cdots e^{-iq_n \cdot x_n} \langle \mathbf{p}, \sigma | T\{A_{r+1}(x_{r+1}) \cdots A_n(x_n)\} | \Omega \rangle \\ = (2\pi)^4 \delta^4(q_{r+1} + \cdots + q_n + p) M_{\mathbf{p},\sigma;0}(q_{r+2} \cdots q_n) \end{aligned} \quad (10.2.5)$$

(with  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$ ), and the sum is over all spin (or other) states of the particle of mass  $m$ .

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<sup>111</sup>Recall that in the absence of time-varying external fields, there is no distinction between ‘in’ and ‘out’ one-particle states, so that  $|\mathbf{p}, \sigma\rangle_{\text{in}} = |\mathbf{p}, \sigma\rangle_{\text{out}} \equiv |\mathbf{p}, \sigma\rangle$ .

Before proceeding to the proof, it will help to clarify the significance of (10.2.3) if we write it in the somewhat long-winded form

$$\begin{aligned}
G(q_1 \cdots q_n) \longrightarrow & \int d^4 k \left[ (2\pi)^4 \delta^4(q_1 + \cdots + q_r - k) \right. \\
& \cdot (2\pi)^{3/2} (2\sqrt{\mathbf{k}^2 + m^2})^{1/2} M_{0;\mathbf{k},\sigma}(q_2 \cdots q_r) \left. \right] \\
& \cdot \left[ \frac{-i}{(2\pi)^4} \frac{1}{k^2 + m^2 - i\epsilon} \right] \\
& \cdot \left[ (2\pi)^4 \delta^4(k + q_{r+1} + \cdots + q_n) \right. \\
& \cdot (2\pi)^{3/2} (2\sqrt{\mathbf{k}^2 + m^2})^{1/2} M_{\mathbf{k},\sigma;0}(q_{r+2} \cdots q_n) \left. \right]. \tag{10.2.6}
\end{aligned}$$

This is just what we should expect from a Feynman diagram with a single internal line for a particle of mass  $m$  connecting the first  $r$  and the last  $n - r$  external lines.<sup>112</sup> However, it is not necessary that the particle of mass  $m$  correspond to a field that appears in the Lagrangian of the theory. Eqs. (10.2.3) and (10.2.6) apply even if this particle is a bound state of the so-called elementary particles whose fields do appear in the Lagrangian. In this case, the pole arises not from single Feynman diagrams, like Figure 10.2, but rather from infinite sums of diagrams, such as the one shown in Figure 10.3. This is the first place where the methods of this chapter take us beyond results that could be derived as properties of each order of perturbation theory.

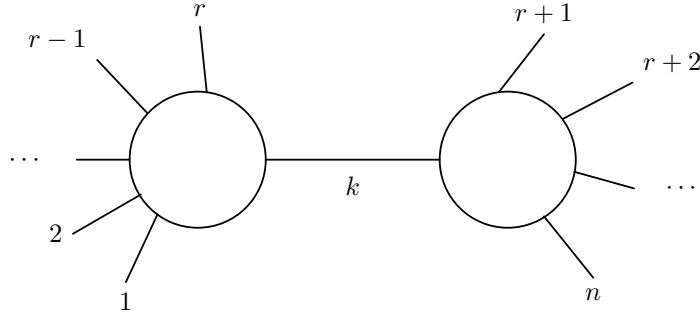
Now to the proof. Among the  $n!$  possible orderings of the times  $x_1^0 \cdots x_n^0$  in Eq. (10.2.1), there are  $n!/r!(n-r)!$  for which the first  $r$  of the  $x_i^0$  are all larger than the last  $n-r$ . Isolating the contribution of this part of the volume of integration in Eq. (10.2.1), we have

$$\begin{aligned}
G(q_1 \cdots q_n) = & \int d^4 x_1 \cdots d^4 x_n e^{-iq_1 \cdot x_1} \cdots e^{-iq_n \cdot x_n} \\
& \cdot \theta(\min[x_1^0 \cdots x_r^0] - \max[x_{r+1}^0 \cdots x_n^0]) \\
& \cdot \langle \Omega | T\{A_1(x_1) \cdots A_r(x_r)\} T\{A_{r+1}(x_{r+1}) \cdots A_n(x_n)\} | \Omega \rangle \\
& + \text{OT}, \tag{10.2.7}
\end{aligned}$$

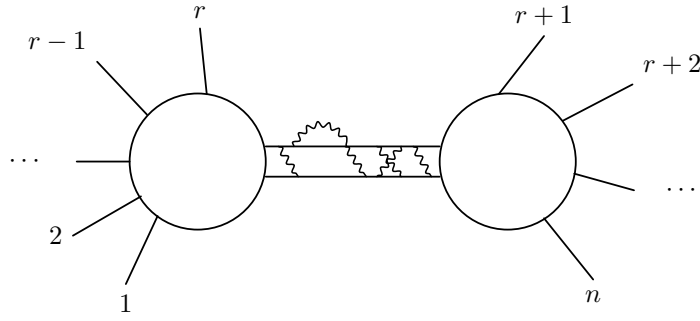
where ‘OT’ denotes the other terms arising from different time-orderings. We can evaluate the matrix element here by inserting a complete set of intermediate states between time-ordered products. Among these may

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<sup>112</sup>See Figure 10.2. The factors  $(2\pi)^{3/2} [2(\mathbf{k}^2 + m^2)]^{1/2}$  just serve to remove kinematic factors associated with the mass  $m$  external line in  $M_{0;\mathbf{k},\sigma}$  and  $M_{\mathbf{k},\sigma;0}$ . Also, the sum over  $\sigma$  of the product of coefficient-function factors from these two matrix elements yields the numerator of the propagator associated with the internal line in Figure 10.2.



**Figure 10.2.** A Feynman diagram with the pole structure (10.2.6). Here the line carrying a momentum  $k$  represents an elementary particle, one whose field appears in the Lagrangian.



**Figure 10.3.** A Feynman diagram of the class whose sum has the pole structure (10.2.6). Here the pole is due to a composite particle, a bound state of two elementary particles. The elementary particles are represented by straight lines, and interact by the exchange of particles represented by wavy lines.

be the single-particle state  $|\mathbf{p}, \sigma\rangle$  of a definite species of mass  $m$ . Further isolating the contribution of these one-particle intermediate states, we have

$$\begin{aligned}
G(q_1 \cdots q_n) = & \int d^4x_1 \cdots d^4x_n e^{-iq_1 \cdot x_1} \cdots e^{-iq_n \cdot x_n} \\
& \cdot \theta(\min[x_1^0 \cdots x_r^0] - \max[x_{r+1}^0 \cdots x_n^0]) \sum_{\sigma} \int d^3p \\
& \cdot \langle \Omega | T\{A_1(x_1) \cdots A_r(x_r)\} | \mathbf{p}, \sigma \rangle \langle \mathbf{p}, \sigma | T\{A_{r+1}(x_{r+1}) \cdots A_n(x_n)\} | \Omega \rangle \\
& + \text{OT}, \tag{10.2.8}
\end{aligned}$$

where ‘OT’ now denotes other terms, here arising not only from other time-orderings, but also from other intermediate states. It will be convenient to shift variables of integration, so that

$$\begin{aligned}
x_i &= x_1 + y_i, & i &= 2, 3, \dots, r, \\
x_i &= x_{r+1} + y_i, & i &= r+2, \dots, n,
\end{aligned}$$

and use the results of the previous section to write

$$\begin{aligned} & \langle \Omega | T \{ A_1(x_1) \cdots A_r(x_r) \} | \mathbf{p}, \sigma \rangle \\ &= e^{ip \cdot x_1} \langle \Omega | T \{ A_1(0) A_2(y_2) \cdots A_r(y_r) \} | \mathbf{p}, \sigma \rangle, \end{aligned} \quad (10.2.9)$$

$$\begin{aligned} & \langle \mathbf{p}, \sigma | T \{ A_{r+1}(x_{r+1}) \cdots A_n(x_n) \} | \Omega \rangle \\ &= e^{-ip \cdot x_{r+1}} \langle \mathbf{p}, \sigma | T \{ A_{r+1}(0) \cdots A_n(y_n) \} | \Omega \rangle. \end{aligned} \quad (10.2.10)$$

Also, the argument of the theta function becomes

$$\begin{aligned} & \min[x_1^0 \cdots x_r^0] - \max[x_{r+1}^0 \cdots x_n^0] \\ &= x_1^0 - x_{r+1}^0 + \min[0, y_2^0 \cdots y_r^0] - \max[0, y_{r+1}^0 \cdots y_n^0]. \end{aligned}$$

We also insert the Fourier representation (6.2.15) of the step function

$$\theta(\tau) = -\frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{d\omega e^{-i\omega\tau}}{\omega + i\epsilon}.$$

The integrals over  $x_1$  and  $x_{r+1}$  now just yield delta functions:

$$\begin{aligned} G(q_1 \cdots q_n) &= \int d^4 y_2 \cdots d^4 y_r d^4 y_{r+2} \cdots d^4 y_n \\ &\cdot e^{-iq_2 \cdot y_2} \cdots e^{-iq_r \cdot y_r} e^{-iq_{r+2} \cdot y_{r+2}} \cdots e^{-iq_n \cdot y_n} \\ &\cdot \left( -\frac{1}{2\pi i} \right) \int_{-\infty}^{\infty} \frac{d\omega}{\omega + i\epsilon} \exp(-i\omega [\min[0, y_2^0 \cdots y_r^0] - \max[0, y_{r+1}^0 \cdots y_n^0]]) \\ &\cdot \sum_{\sigma} \int d^3 p \langle \Omega | T \{ A_1(0) \cdots A_r(y_r) \} | \mathbf{p}, \sigma \rangle \\ &\cdot \langle \mathbf{p}, \sigma | T \{ A_{r+1}(0) \cdots A_n(y_n) \} | \Omega \rangle \\ &\cdot (2\pi)^4 \delta^3(\mathbf{p} - \mathbf{q}_1 - \cdots - \mathbf{q}_r) \delta\left(\sqrt{\mathbf{p}^2 + m^2} + \omega - q_1^0 - \cdots - q_r^0\right) \\ &\cdot (2\pi)^4 \delta^3(\mathbf{q}_{r+1} + \cdots + \mathbf{q}_n + \mathbf{p}) \delta\left(q_{r+1}^0 + \cdots + q_n^0 + \sqrt{\mathbf{p}^2 + m^2} + \omega\right) \\ &+ \text{OT}. \end{aligned} \quad (10.2.11)$$

We are interested here only in the pole that arises from the vanishing of the denominator  $\omega + i\epsilon$ , so for our present purposes we can set the factor  $\exp(-i\omega[\min - \max])$  equal to unity. The integrals over both  $\mathbf{p}$  and  $\omega$  are now trivial, and yield the pole

$$\begin{aligned} G(q_1 \cdots q_n) &\longrightarrow i(2\pi)^7 \delta^4(q_1 + \cdots + q_n) \left[ q^0 - \sqrt{\mathbf{q}^2 + m^2} + i\epsilon \right]^{-1} \\ &\cdot \sum_{\sigma} M_{0;-\mathbf{q},\sigma}(q_2 \cdots q_n) M_{\mathbf{q},\sigma;0}(q_{r+2} \cdots q_n) + \cdots \end{aligned} \quad (10.2.12)$$

where now

$$q \equiv q_1 + \cdots + q_r = -q_{r+1} - \cdots - q_n,$$

$$M_{0;\mathbf{q},\sigma}(q_2 \cdots q_r) \equiv \int d^4 y_2 \cdots d^4 y_r e^{-iq_2 \cdot y_2} \cdots e^{-iq_r \cdot y_r} \cdot \langle \Omega | T \{ A_1(0) A_2(y_2) \cdots A_n(y_r) \} | \mathbf{q}, \sigma \rangle, \quad (10.2.13)$$

$$M_{\mathbf{q},\sigma;0}(q_{r+2} \cdots q_n) \equiv \int d^4 y_{r+2} \cdots d^4 y_n e^{iq_{r+2} \cdot y_{r+2}} \cdots e^{-iq_n \cdot y_n} \cdot \langle \mathbf{p}, \sigma | T \{ A_{r+1}(0) A_{r+2}(y_{r+2}) \cdots A_n(y_n) \} | \Omega \rangle, \quad (10.2.14)$$

and the final ‘ $\cdots$ ’ in Eq. (10.2.12) denotes terms that do not exhibit this particular pole. (The ‘other terms’ arising from other single-particle states produce poles in  $q$  at different positions, while those arising from multi-particle states produce branch points in  $q$ , and those arising from other time-orderings produce poles and branch cuts in other variables.) Using Eqs. (10.2.9) and (10.2.10), it is easy to see that these  $M$ s are the same as defined by Eqs. (10.2.4) and (10.2.5). Also, near the pole we can write

$$\frac{1}{q^0 - \sqrt{\mathbf{q}^2 + m^2} + i\epsilon} = \frac{-q^0 - \sqrt{\mathbf{q}^2 + m^2} + i\epsilon}{-(q^0)^2 + (\sqrt{\mathbf{q}^2 + m^2} - i\epsilon)^2} \longrightarrow \frac{-2\sqrt{\mathbf{q}^2 + m^2}}{q^2 + m^2 - i\epsilon}.$$

(We again redefine  $\epsilon$  by a positive factor  $2\sqrt{\mathbf{q}^2 + m^2}$ , which is permissible since  $\epsilon$  stands for any positive infinitesimal.) Eq. (10.2.12) is thus the same as the desired result (10.2.3).

This result has a classic application to the theory of nuclear forces. Let  $\Phi_a(x)$  be any real field or combination of fields (for instance, proportional to a quark–antiquark bilinear  $\bar{q}\gamma_5\tau_a q$ ) that has a non-vanishing matrix element between a one-pion state of isospin  $a$  and the vacuum, normalized so that

$$\langle \Omega | \Phi_a(0) | \pi_b, \mathbf{p} \rangle = (2\pi)^{-3/2} (2p^0)^{-1/2} \delta_{ab}. \quad (10.2.15)$$

The matrix element of  $\Phi_a$  between one-nucleon states with four-momenta  $p, p'$  then has a pole at  $(p - p')^2 \rightarrow -m_\pi^2$ , which isospin and Lorentz invariance (including space inversion invariance) dictate must take the form<sup>113</sup>

$$\langle N', \sigma', \mathbf{p}' | \Phi_a(0) | N, \sigma, \mathbf{p} \rangle \longrightarrow i(2\pi)^{-3} G_\pi \frac{\bar{u}' \gamma_5 \tau_a u}{(p - p')^2 + m_\pi^2}, \quad (10.2.16)$$

where  $u$  and  $u'$  are the initial and final nucleon spinor coefficient functions, including the nucleon wave functions in isospin space, and  $\tau_a$  with  $a = 1, 2, 3$  are the  $2 \times 2$  Pauli isospin matrices. The constant  $G_\pi$  is known as the *pion–nucleon coupling constant*. This pole is not

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<sup>113</sup>Lorentz and isospin invariance requires this matrix to take the form  $(\bar{u}' \Gamma \tau_a u)$ , where  $\Gamma$  is a  $4 \times 4$  matrix for which the bilinear  $(\bar{\psi} \Gamma \psi)$  transforms as a pseudoscalar. Like any  $4 \times 4$  matrix,  $\Gamma$  can be expanded as a sum of terms proportional to the Dirac matrices  $\mathbf{1}$ ,  $\gamma_\mu$ ,  $[\gamma_\mu, \gamma_\nu]$ ,  $\gamma_5 \gamma_\mu$ , and  $\gamma_5$ . The coefficients must be respectively pseudoscalar, pseudovector, pseudotensor, vector, and scalar. Out of the two momenta  $p$  and  $p'$  it is possible to construct no pseudoscalars or pseudovectors; just one pseudotensor, proportional to  $\epsilon^{\mu\nu\rho\sigma} p_\rho p'_\sigma$ ; two independent vectors, proportional to  $p_\mu$  or  $p'_\mu$ ; and a scalar proportional to unity, in each case with a proportionality factor depending on the only independent scalar variable,  $(p - p')^2$ . By using the momentum-space Dirac equations for  $u$  and  $u'$ , it is easy to see that the tensor and pseudovector matrices in  $\Gamma$  give contributions proportional to  $\gamma_5$ .

actually in the physical region for the matrix element (10.2.16), for which  $(p - p')^2 \geq 0$ , but it can be reached by analytic extension of this matrix element, for instance by considering the off-shell matrix element

$$\int d^4x d^4x' e^{-ip \cdot x} e^{ip' \cdot x'} \langle \Omega | T \{ \Phi_a(0) \bar{N}(x) N'(x') \} | \Omega \rangle,$$

where  $N$  and  $N'$  are appropriate components of a field operator or product of field operators with non-vanishing matrix elements between one-nucleon states and the vacuum. The theorem proved above in this section shows then that exchange of a pion in the scattering of two nucleons with initial four-momenta  $p_1, p_2$ , and final four-momenta  $p'_1, p'_2$  yields a pole at  $(p_1 - p'_1)^2 = (p_2 - p'_2)^2 \rightarrow -m_\pi^2$ .

$$S_{N'_1 N'_2, N_1 N_2} \longrightarrow -i(2\pi)^4 \delta^4(p'_1 + p'_2 - p_1 - p_2) \frac{G_\pi^2}{(p_1 - p'_1)^2 + m_\pi^2} \cdot (2\pi)^{-3} [\bar{u}_1 \gamma_5 \tau_a u_1] \cdot (2\pi)^{-3} [\bar{u}_2 \gamma_5 \tau_a u_2]. \quad (10.2.17)$$

(The easiest way to get the phases and numerical factors right in such formulas is to use Feynman diagrams; our theorem just says that the pole structure is the same as would be found in a field theory in which the Lagrangian involved an elementary pion field.) Again, this pion pole is not actually in the physical region for scattering of nucleons on the mass shell, for which  $(p_1 - p'_1)^2 \geq 0$ , but it can be reached by analytic extension of the  $S$ -matrix element, for instance by considering the off-shell matrix

element

$$\int d^4x_1 d^4x_2 d^4x'_1 d^4x'_2 e^{-ip_1 \cdot x_1} e^{-ip_2 \cdot x_2} e^{ip'_1 \cdot x'_1} e^{ip'_2 \cdot x'_2} \cdot \langle \Omega | T \{ \bar{N}_1(x_1), \bar{N}_2(x_2), N'_1(x'_1), N'_2(x'_2) \} | \Omega \rangle.$$

Although this pole is not in the physical region for nucleon–nucleon scattering, the pion mass is small enough so that the pole is quite near the physical region, and under some circumstances may dominate the scattering amplitude, as for instance for large  $\ell$  in the partial wave expansion.

Interpreted in coordinate space, a pole like this at  $(p_1 - p'_1)^2 = (p_2 - p'_2)^2 \rightarrow -m_\pi^2$  implies a force of range  $1/m_\pi$ . For instance, in Yukawa's original theory [2] of nuclear force the exchange of mesons (then assumed scalar rather than pseudoscalar) produced a local potential of the form  $\exp(-m_\pi r)/4\pi r$ , which in the first Born approximation yields an  $S$ -matrix for non-relativistic nucleon scattering proportional to the Fourier transform:

$$\begin{aligned} & \int d^3\mathbf{x}_1 d^3\mathbf{x}_2 d^3\mathbf{x}'_1 d^3\mathbf{x}'_2 e^{-i\mathbf{x}_1 \cdot \mathbf{p}_1} e^{-i\mathbf{x}_2 \cdot \mathbf{p}_2} e^{i\mathbf{x}'_1 \cdot \mathbf{p}'_1} e^{i\mathbf{x}'_2 \cdot \mathbf{p}'_2} \\ & \cdot \frac{\exp(-m_\pi |\mathbf{x}_1 - \mathbf{x}_2|)}{4\pi |\mathbf{x}_1 - \mathbf{x}_2|} \delta^3(\mathbf{x}_1 - \mathbf{x}'_1) \delta^3(\mathbf{x}_2 - \mathbf{x}'_2) \\ & = -(2\pi)^3 \delta^3(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 - \mathbf{p}'_2) \frac{1}{(\mathbf{p}_1 - \mathbf{p}'_1)^2 + m_\pi^2}. \end{aligned}$$

The factor  $1/[(\mathbf{p}_1 - \mathbf{p}'_1)^2 + m_\pi^2]$  is just the non-relativistic limit of the propagator  $1/[(p_1 - p'_1)^2 + m_\pi^2]$  in (10.2.17). (In (10.2.17) the energy transfer  $p_1^0 - p'_1{}^0$  for  $|\mathbf{p}_1| \ll m_N$  and

$|\mathbf{p}'_1| \ll m_N$  equals  $[\mathbf{p}_1^2 - \mathbf{p}'_1{}^2]/2m_N$ , which is negligible compared with the magnitude  $|\mathbf{p}_1 - \mathbf{p}'_1|$  of the momentum transfer.) When Yukawa's theory was first proposed, it was generally supposed that this sort of momentum-dependence arises from the appearance of a meson field in the theory. It was not until the 1950s that it became generally understood that the existence of a pole at  $(p_1 - p'_1)^2 \rightarrow -m_\pi^2$  follows from the existence of a pion *particle* and has nothing to do with whether this is an elementary particle with its own field in the Lagrangian.

### 10.3 Field and Mass Renormalization

We will now use a special case of the result of the previous section to clarify the treatment of radiative corrections in the internal and external line of general processes.

The special case that concerns us here is the one in which the four-momentum of a single external line approaches the mass shell. (In the notation of the previous section, this corresponds to taking  $r = 1$ .) We will consider a function

$$G_\ell(q_1 q_2 \cdots) = \int d^4 x_1 d^4 x_2 \cdots e^{-iq_1 \cdot x_1} e^{-iq_2 \cdot x_2} \cdots \cdot \langle \Omega | T \{ \mathcal{O}_\ell(x_1), A_2(x_2), \dots \} | \Omega \rangle, \quad (10.3.1)$$

where  $\mathcal{O}_\ell(x)$  is a Heisenberg-picture operator, with the Lorentz transformation properties of some sort of free field  $\psi_\ell$  belonging to an irreducible representation of the homogeneous Lorentz group (or the Lorentz group including space inversion for theories that conserve parity), as labelled by the subscript  $\ell$ , and  $A_2, A_3$ , etc. are arbitrary Heisenberg-picture operators. Suppose there is a one-particle state  $|q_1, \sigma\rangle$  that has non-vanishing matrix elements with the states  $\mathcal{O}_\ell^\dagger |\Omega\rangle$  and with  $A_2 A_3 \cdots |\Omega\rangle$ . Then according to the theorem proved in the previous section,  $G_\ell$  has a pole at  $q_1^2 = -m^2$ , with

$$G_\ell(q_1 q_2 \cdots) \rightarrow \frac{-2i\sqrt{\mathbf{q}_1^2 + m^2}}{q_1^2 + m^2 - i\epsilon} (2\pi)^3 \sum_\sigma \langle \Omega | \mathcal{O}_\ell(0) | q_1, \sigma \rangle \cdot \int d^4 x_2 \cdots e^{-iq_2 \cdot x_2} \cdots \langle q_1, \sigma | T \{ A_2(x_2) \cdots \} | \Omega \rangle. \quad (10.3.2)$$

We use Lorentz invariance to write

$$\langle \Omega | \mathcal{O}_\ell(0) | q_1, \sigma \rangle = (2\pi)^{-3/2} N u_\ell(\mathbf{q}_1, \sigma), \quad (10.3.3)$$

where  $u_\ell(\mathbf{q}, \sigma)$  is (aside from the factor  $(2\pi)^{-3/2}$ ) the coefficient function<sup>114</sup> appearing in the free field  $\psi_\ell$  with the same Lorentz transformation properties as  $\mathcal{O}_\ell$ , and  $N$  is a constant. (It was in order to obtain Eq. (10.3.3) with a single free constant  $N$  that we had to assume that  $\mathcal{O}_\ell$  transforms irreducibly.) We also define a ‘truncated’ matrix element  $M_{\ell'}$  by

$$\begin{aligned} & \int d^4 x_2 \cdots e^{-iq_2 \cdot x_2} \cdots \langle q_1, \sigma | T \{ A_2(x_2) \cdots \} | \Omega \rangle \\ & \equiv N^{-1} (2\pi)^{-3/2} \sum_{\ell'} u_{\ell'}^*(\mathbf{q}_1, \sigma) M_{\ell'}(q_2 \cdots). \end{aligned} \quad (10.3.4)$$

<sup>114</sup>For instance, for a conventionally normalized free scalar field,  $u_\ell(\mathbf{q}_1, \sigma) = [2\sqrt{\mathbf{q}_1^2 + m^2}]^{-1/2}$ .

Eq. (10.3.2) then reads, for  $q_1^2 \rightarrow -m^2$ ,

$$G_\ell \rightarrow \frac{-2i\sqrt{\mathbf{q}_1^2 + m^2}}{q_1^2 + m^2 - i\epsilon} \sum_{\sigma, \ell'} u_\ell(\mathbf{q}_1, \sigma) u_{\ell'}^*(\mathbf{q}_1, \sigma) M_{\ell'}. \quad (10.3.5)$$

According to Eqs. (6.2.2) and (6.2.18), the quantity multiplying  $M_{\ell'}$  in (10.3.5) is the momentum space matrix propagator  $-i\Delta_{\ell\ell'}(q_1)$  for the free field with the Lorentz transformation properties of  $\mathcal{O}_\ell$  (or at least its limiting behavior for  $q_1^2 \rightarrow -m^2$ ), so (10.3.5) allows us to identify  $M_{\ell'}$  as the sum of all graphs with external lines carrying momenta  $q_1, q_2, \dots$  corresponding to the operators  $\mathcal{O}_\ell, A_2, \dots$ , but with the final propagator for the  $\mathcal{O}_\ell$  line stripped away. Eq. (10.3.4) is then just the usual prescription for how to calculate the matrix element for emission of a particle from the sum of Feynman diagrams: strip away the particle propagator, and contract with the usual external line factor  $(2\pi)^{-3/2} u_{\ell'}^*$ . The only discrepancy with the usual Feynman rules is the factor  $N$ .

The above theorem is a famous result due to Lehmann, Symanzik, and Zimmerman [3], known as the *reduction formula*, which we have proved here by a somewhat different method that has allowed us easily to generalize this result to the case of arbitrary spin. One important aspect of this result is that it applies to any sort of operator;  $\mathcal{O}_\ell$  need not be some field that actually appears in the Lagrangian, and the particle it creates may be a bound state composed of those particles whose fields do occur in the Lagrangian. It provides an important lesson even where  $\mathcal{O}_\ell$  is some field  $\Psi_\ell$  in the Lagrangian: if we are to use the usual Feynman rules to calculate  $S$ -matrix elements, then we should first redefine the normalization of the fields by a factor  $1/N$ , so that (with apologies for the multiple use of the symbol  $\Psi$ ):

$$\langle \Omega | \Psi_\ell(0) | q, \sigma \rangle = (2\pi)^{-3/2} u_\ell(\mathbf{q}, \sigma). \quad (10.3.6)$$

A field normalized as in Eq. (10.3.6) is called a *renormalized field*.

The field renormalization constant  $N$  shows up in another place. Suppose that there is just one of the operators  $A_2, A_3, \dots$  in Eq. (10.3.1), and take it to be the adjoint of a member of the same field multiplet as  $\mathcal{O}_\ell$ . Then Eq. (10.3.2) reads

$$\begin{aligned} & \int d^4x_1 \int d^4x_2 e^{-iq_1 \cdot x_1} e^{-iq_2 \cdot x_2} \langle \Omega | T \{ \mathcal{O}_\ell(x_1) \mathcal{O}_{\ell'}^\dagger(x_2) \} | \Omega \rangle \\ & \xrightarrow{q_1^2 \rightarrow -m^2} \frac{-2i\sqrt{\mathbf{q}_1^2 + m^2} (2\pi)^3}{q_1^2 + m^2 - i\epsilon} \sum_{\sigma} \langle \Omega | \mathcal{O}_\ell(0) | q_1, \sigma \rangle \\ & \quad \cdot \int d^4x_2 e^{-iq_2 \cdot x_2} e^{-iq_1 \cdot x_2} \langle q_1, \sigma | \mathcal{O}_{\ell'}^\dagger(0) | \Omega \rangle \\ & = \frac{-2i|N|^2\sqrt{\mathbf{q}_1^2 + m^2}}{q_1^2 + m^2 - i\epsilon} \sum_{\sigma} u_\ell(\mathbf{q}_1, \sigma) u_{\ell'}^*(\mathbf{q}_1, \sigma) (2\pi)^4 \delta^4(q_1 + q_2). \end{aligned}$$

This is just the usual behavior of a propagator (the sum of all graphs with two external lines) near its pole, except for the factor  $|N|^2$ . According to Eq. (10.3.6), this factor is absent in the propagator of the renormalized field  $\Psi_\ell$ . Thus *a renormalized field is one whose propagator has the same behavior near its pole as for a free field, and the renormalized mass is defined by the position of the pole.*



To see how this works in practice, consider the theory of a real self-interacting scalar field  $\Phi_B$ , the subscript B being added here to remind us that so far this is a ‘bare’ (i.e., unrenormalized) field. The Lagrangian density is taken as usual as

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\Phi_B\partial^\mu\Phi_B - \frac{1}{2}m_B^2\Phi_B^2 - V_B(\Phi_B). \quad (10.3.7)$$

In general there would be no reason to expect that the field  $\Phi_B$  would satisfy condition (10.3.6), nor that the pole in  $q^2$  would be at  $-m_B^2$ , so let us introduce a renormalized field and mass

$$\Phi \equiv Z^{-1/2}\Phi_B, \quad (10.3.8)$$

$$m^2 \equiv m_B^2 + \delta m^2, \quad (10.3.9)$$

with  $Z$  to be chosen so that  $\Phi$  does satisfy Eq. (10.3.6), and  $\delta m^2$  chosen so that the pole of the propagator is at  $q^2 = -m^2$ . (The use of the symbol  $Z$  in this context has become conventional; there is a different  $Z$  for each field in the Lagrangian.) The Lagrangian density (10.3.7) may then be rewritten

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1, \quad (10.3.10)$$

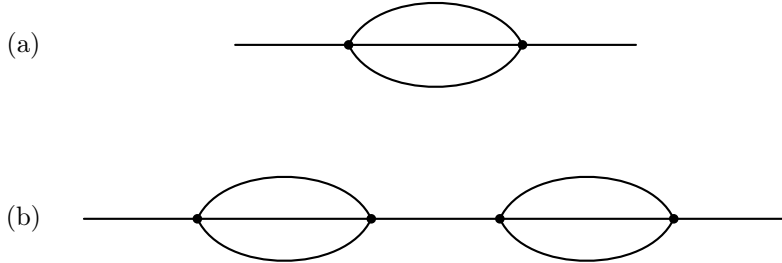
$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2, \quad (10.3.11)$$

$$\mathcal{L}_1 = -\frac{1}{2}(Z-1)[\partial_\mu\Phi\partial^\mu\Phi + m^2\Phi^2] + \frac{1}{2}Z\delta m^2\Phi^2 - V(\Phi), \quad (10.3.12)$$

where

$$V(\Phi) \equiv V_B(\sqrt{Z}\Phi).$$

In calculating the corrections to the complete momentum space propagator of the renormalized scalar field, conventionally called  $\Delta'(q)$ , it is convenient to consider separately the *one-particle-irreducible* graphs: those connected graphs (excluding a graph consisting of a single scalar line) that cannot be disconnected by cutting through any one internal scalar line. An example is shown in Figure 10.4. It is conventional to write the sum of all such graphs, with the two external line propagator factors  $-i(2\pi)^{-4}(q^2 + m^2 - i\epsilon)^{-1}$  omitted, as  $i(2\pi)^4\Pi^*(q^2)$ , with the asterisk to remind us that these are one-particle-irreducible graphs. Then the corrections to the complete propagator are given by a sum of chains of one, two, or more of these one-particle-irreducible subgraphs connected with the usual uncorrected



**Figure 10.4.** Diagrams that (a) are, or (b) are not, one-particle-irreducible. These diagrams are drawn for a theory with some sort of quadrilinear interaction, like the theory of a scalar field  $\phi$  with interaction proportional to  $\phi^4$ .

propagator factors:

$$\begin{aligned} \frac{-i}{(2\pi)^4} \Delta'(q) &= \frac{-i}{(2\pi)^4} \frac{1}{q^2 + m^2 - i\epsilon} \\ &+ \left[ \frac{-i}{(2\pi)^4} \frac{1}{q^2 + m^2 - i\epsilon} \right] [i(2\pi)^4 \Pi^*(q^2)] \left[ \frac{-i}{(2\pi)^4} \frac{1}{q^2 + m^2 - i\epsilon} \right] \\ &+ \left[ \frac{-i}{(2\pi)^4} \frac{1}{q^2 + m^2 - i\epsilon} \right] [i(2\pi)^4 \Pi^*(q^2)] \left[ \frac{-i}{(2\pi)^4} \frac{1}{q^2 + m^2 - i\epsilon} \right] \\ &\cdot [i(2\pi)^4 \Pi^*(q^2)] \left[ \frac{-i}{(2\pi)^4} \frac{1}{q^2 + m^2 - i\epsilon} \right] + \dots \end{aligned} \quad (10.3.13)$$

or more simply

$$\begin{aligned} \Delta'(q) &= [q^2 + m^2 - i\epsilon]^{-1} + [q^2 + m^2 - i\epsilon]^{-1} \Pi^*(q^2) [q^2 + m^2 - i\epsilon]^{-1} \\ &+ [q^2 + m^2 - i\epsilon]^{-1} \Pi^*(q^2) [q^2 + m^2 - i\epsilon]^{-1} \Pi^*(q^2) [q^2 + m^2 - i\epsilon]^{-1} + \dots \end{aligned} \quad (10.3.14)$$

Summing the geometric series, this gives

$$\Delta'(q) = [q^2 + m^2 - \Pi^*(q^2) - i\epsilon]^{-1}. \quad (10.3.15)$$

In calculating  $\Pi^*$ , we encounter a tree graph arising from a single insertion of vertices corresponding to the terms in Eq. (10.3.12) proportional to  $\partial_\mu \Phi \partial^\mu \Phi$  and  $\Phi^2$ , plus a term  $\Pi_{\text{LOOP}}^*$  arising from loop graphs like that in Figure 10.4(a):

$$\Pi^*(q^2) = -(Z - 1)[q^2 + m^2] + Z\delta m^2 + \Pi_{\text{LOOP}}^*(q^2). \quad (10.3.16)$$

The condition that  $m^2$  is the true mass of the particle is that the pole of the propagator should be at  $q^2 = -m^2$ , so that

$$\Pi^*(-m^2) = 0. \quad (10.3.17)$$

Also, the condition that the pole of the propagator at  $q^2 = -m^2$  should have a unit residue (like the uncorrected propagator) is that

$$\left[ \frac{d}{dq^2} \Pi^*(q^2) \right]_{q^2=-m^2} = 0. \quad (10.3.18)$$

These conditions allow us to evaluate  $Z$  and  $\delta m^2$ :

$$Z\delta m^2 = -\Pi_{\text{LOOP}}^*(0), \quad (10.3.19)$$

$$Z = 1 + \left[ \frac{d}{dq^2} \Pi_{\text{LOOP}}^*(q^2) \right]_{q^2=-m^2}. \quad (10.3.20)$$

This incidentally shows that  $Z\delta m^2$  and  $Z - 1$  are given by a series of terms containing one or more coupling constant factors, justifying the treatment of the first two terms in Eq. (10.3.12) as part of the interaction  $\mathcal{L}_1$ .

In actual calculations it is simplest just to say that from the loop terms  $\Pi_{\text{LOOP}}^*(q^2)$  we must subtract a first-order polynomial in  $q^2$  with coefficients chosen so that the difference satisfies Eqs. (10.3.17) and (10.3.18). As we shall see, this subtraction procedure incidentally

cancels the infinities that arise from the momentum space integrals in  $\Pi_{\text{LOOP}}^*$ . However, as this discussion should make clear, *the renormalization of masses and fields has nothing directly to do with the presence of infinities, and would be necessary even in a theory in which all momentum space integrals were convergent.*

An important consequence of the conditions (10.3.17) and (10.3.18) is that it is not necessary to include radiative corrections in external lines on the mass shell. That is,

$$\left[ \begin{aligned} &\Pi^*(q^2)[q^2 + m^2 - i\epsilon]^{-1} \\ &+ \Pi^*(q^2)[q^2 + m^2 - i\epsilon]^{-1}\Pi^*(q^2)[q^2 + m^2 - i\epsilon]^{-1} + \dots \end{aligned} \right]_{q^2 \rightarrow -m^2} = 0. \quad (10.3.21)$$

Similar remarks apply to particles of arbitrary spin. For instance, for the ‘bare’ Dirac field the Lagrangian is

$$\mathcal{L} = -\bar{\Psi}_B[\not{\partial} + m_B]\Psi_B - V_B(\bar{\Psi}_B\Psi_B). \quad (10.3.22)$$

We introduce renormalized fields and masses

$$\Psi \equiv Z_2^{-1/2}\Psi_B, \quad (10.3.23)$$

$$m \equiv m_B + \delta m. \quad (10.3.24)$$

(The subscript 2 on  $Z_2$  is conventionally used to distinguish the renormalization constant of a fermion field.) The Lagrangian density is then rewritten

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1, \quad (10.3.25)$$

$$\mathcal{L}_0 = -\bar{\Psi}[\not{\partial} + m]\Psi, \quad (10.3.26)$$

$$\mathcal{L}_1 = -(Z_2 - 1)[\bar{\Psi}(\not{\partial} + m)\Psi] + Z_2\delta m\bar{\Psi}\Psi - V_B(Z_2\bar{\Psi}\Psi). \quad (10.3.27)$$

Let  $i(2\pi)^4\Sigma^*(\not{k})$  be the sum of all connected graphs, with one fermion line coming in with four-momentum  $k$  and one going out with the same four-momentum, that cannot be disconnected by cutting through any single internal fermion line, and with external line propagator factors  $-i(2\pi)^{-4}$  and  $[i\not{k} + m - i\epsilon]^{-1}$  omitted. (Lorentz invariance is being used to justify writing  $\Sigma^*$  as an ordinary function of the Lorentz scalar matrix  $\not{k} = k_\mu\gamma^\mu$ .) Then the complete fermion propagator is

$$\begin{aligned} S'(k) &= [i\not{k} + m - i\epsilon]^{-1} + [i\not{k} + m - i\epsilon]^{-1}\Sigma^*(\not{k})[i\not{k} + m - i\epsilon]^{-1} \\ &\quad + [i\not{k} + m - i\epsilon]^{-1}\Sigma^*(\not{k})[i\not{k} + m - i\epsilon]^{-1}\Sigma^*(\not{k})[i\not{k} + m - i\epsilon]^{-1} + \dots \\ &= [i\not{k} + m - \Sigma^*(\not{k}) - i\epsilon]^{-1}. \end{aligned} \quad (10.3.28)$$

In calculating  $\Sigma^*(\not{k})$  we take into account the tree graphs from the terms in Eq. (10.3.27) proportional to  $\bar{\Psi}\not{\partial}\Psi$  and  $\bar{\Psi}\Psi$  as well as loop contributions:

$$\Sigma^*(\not{k}) = -(Z_2 - 1)[i\not{k} + m] + Z_2\delta m + \Sigma_{\text{LOOP}}^*(\not{k}). \quad (10.3.29)$$

The condition that the complete propagator has a pole at  $k^2 = -m^2$  with the same residue as the uncorrected propagator is then that

$$\Sigma^*(im) = 0, \quad (10.3.30)$$

$$\left. \frac{\partial \Sigma^*(\not{k})}{\partial \not{k}} \right|_{\not{k}=im} = 0, \quad (10.3.31)$$

and hence

$$Z_2 \delta m = -\Sigma_{\text{LOOP}}^*(im), \quad (10.3.32)$$

$$Z_2 = 1 - i \left. \frac{\partial \Sigma_{\text{LOOP}}^*(\not{k})}{\partial \not{k}} \right|_{\not{k}=im}. \quad (10.3.33)$$

Just as for scalars, the vanishing of  $[i\not{k} + m]^{-1} \Sigma^*(\not{k})$  in the limit  $\not{k} \rightarrow im$  tells us that radiative corrections may be ignored in external fermion lines. Corresponding results for the photon propagator will be derived in Section 10.5.

#### 10.4 Renormalized Charge and Ward Identities

The use of the commutation and conservation relations of Heisenberg-picture operators allows us to make a connection between the charges (or other similar quantities) in the Lagrangian density and the properties of physical states. Recall that the invariance of the Lagrangian density with respect to global gauge transformations  $\Psi_\ell \rightarrow \exp(iq_\ell \alpha) \Psi_\ell$  (with  $\alpha$  an arbitrary constant phase) implies the existence of a current

$$J^\mu = -i \sum_\ell \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Psi_\ell)} q_\ell \Psi_\ell, \quad (10.4.1)$$

satisfying the conservation condition

$$\partial_\mu J^\mu = 0. \quad (10.4.2)$$

This implies that the space-integral of the time component of  $J^\mu$  is time-independent:

$$i \frac{d}{dt} Q = [Q, H] = 0, \quad (10.4.3)$$

where

$$Q \equiv \int d^3x J^0. \quad (10.4.4)$$

(There is a very important possible exception here, that the integral (10.4.4) may not exist if there are long-range forces due to massless scalars in the system. We will return to this point when we consider broken symmetries in Volume II.) Also, since it is a space-integral,  $Q$  is manifestly translation-invariant

$$[\mathbf{P}, Q] = 0 \quad (10.4.5)$$

and since  $J^\mu$  is a four-vector,  $Q$  is invariant with respect to homogeneous Lorentz transformations

$$[J^{\mu\nu}, Q] = 0. \quad (10.4.6)$$

It follows that  $Q$  acting on the true vacuum  $|\Omega\rangle$  must be another Lorentz-invariant state of zero energy and momentum, and hence (assuming no vacuum degeneracy) must be

proportional to  $|\Omega\rangle$  itself. But the proportionality constant must vanish, because Lorentz invariance requires that  $\langle\Omega|J_\mu|\Omega\rangle$  vanish. Hence

$$Q|\Omega\rangle = 0. \quad (10.4.7)$$

Also,  $Q$  acting on any one-particle state  $|\mathbf{p}, \sigma, n\rangle$  must be another state with the same energy, momentum, and Lorentz transformation properties, and thus (assuming no degeneracy of one-particle states) must be proportional to the same one-particle state

$$Q|\mathbf{p}, \sigma, n\rangle = q_{(n)}|\mathbf{p}, \sigma, n\rangle. \quad (10.4.8)$$

The Lorentz invariance of  $Q$  ensures that the eigenvalue  $q_{(n)}$  is independent of  $\mathbf{p}$  and  $\sigma$ , depending only on the species of the particle. This eigenvalue is what is known as the electric charge (or whatever other quantum number of which  $J^\mu$  may be the current) of the one-particle state. To relate this to the  $q_\ell$  parameters in the Lagrangian, we note that the canonical commutation relations give

$$[J^0(\mathbf{x}, t), \Psi_\ell(\mathbf{y}, t)] = -q_\ell \Psi_\ell(\mathbf{y}, t) \delta^3(\mathbf{x} - \mathbf{y}), \quad (10.4.9)$$

or integrating over  $\mathbf{x}$ :

$$[Q, \Psi_\ell(y)] = -q_\ell \Psi_\ell(y). \quad (10.4.10)$$

The same is true of any local function  $F(y)$  of the fields and field derivatives and their adjoints, containing definite numbers of each:

$$[Q, F(y)] = -q_F F(y), \quad (10.4.11)$$

where  $q_F$  is the sum of the  $q_\ell$  for all fields and field derivatives in  $F(y)$ , minus the sum of the  $q_\ell$  for all field adjoints and their derivatives. Taking the matrix element of this equation between a one-particle state and the vacuum, and using Eqs. (10.4.7) and (10.4.8), we have

$$\langle\Omega|F(y)|\mathbf{p}, \sigma, n\rangle(q_F - q_{(n)}) = 0. \quad (10.4.12)$$

Hence we must have

$$q_{(n)} = q_F \quad (10.4.13)$$

as long as

$$\langle\Omega|F(y)|\mathbf{p}, \sigma, n\rangle \neq 0. \quad (10.4.14)$$

As we saw in the previous section, Eq. (10.4.14) is the condition that assures that momentum space Greens functions involving  $F$  have poles corresponding to the one-particle state  $|\mathbf{p}, \sigma, n\rangle$ . For a one-particle state corresponding to one of the fields in the Lagrangian we could take  $F = \Psi_\ell$ , in which case  $q_F = q_\ell$ , but our results here apply to general one-particle states, whether or not their fields appear in the Lagrangian.

This almost, but not quite, tells us that despite all the possible high-order graphs that affect the emission and absorption of photons by charged particles, the physical electric charge is just equal to a parameter  $q_\ell$  appearing in the Lagrangian (or to a sum of such parameters, like  $q_F$ .) The qualification that has to be added here is that the requirement,

that the Lagrangian be invariant under the transformations  $\Psi_\ell \rightarrow \exp(iq_\ell \alpha) \Psi_\ell$ , does nothing to fix the over-all scale of the quantities  $q_\ell$ . The physical electric charges are those that determine the response of matter fields to a given *renormalized* electromagnetic field  $A^\mu$ . That is, the scale of the  $q_\ell$  is fixed by requiring that the renormalized electromagnetic field appears in the matter Lagrangian  $\mathcal{L}_M$  in the linear combinations  $[\partial_\mu - iq_\ell A_\mu] \Psi_\ell$ , so that the current  $J^\mu$  is

$$J^\mu = \frac{\delta \mathcal{L}_M}{\delta A_\mu}. \quad (10.4.15)$$

But  $A^\mu$  and  $q_\ell$  are not the same as the ‘bare electromagnetic field’  $A_B^\mu$  and ‘bare charges’  $q_{B\ell}$  that appear in the Lagrangian when we write it in its simplest form

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4}(\partial_\mu A_{B\nu} - \partial_\nu A_{B\mu})(\partial^\mu A_B^\nu - \partial^\nu A_B^\mu) \\ & + \mathcal{L}_M(\Psi_\ell, [\partial_\mu - iq_{B\ell} A_{B\mu}] \Psi_\ell). \end{aligned} \quad (10.4.16)$$

The renormalized electromagnetic field (defined to have a complete propagator whose pole at  $p^2 = 0$  has unit residue) is conventionally written in terms of  $A_B^\mu$  as

$$A^\mu = Z_3^{-1/2} A_B^\mu, \quad (10.4.17)$$

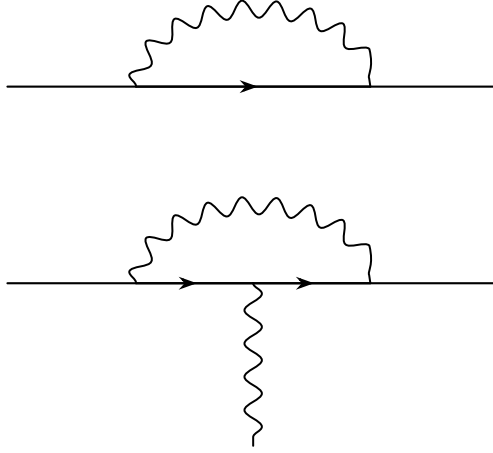
so in order for the charge  $q_\ell$  to characterize the response of the charged particles to a given renormalized electromagnetic field, we should define the renormalized charges by

$$q_\ell = \sqrt{Z_3} q_{B\ell}. \quad (10.4.18)$$

We see that the physical electric charge  $q$  of any particle is just proportional to a parameter  $q_B$  related to those appearing in the Lagrangian, with a proportionality constant  $Z_3^{-1/2}$  that is the same for all particles. This helps us to understand how a particle like the proton, that is surrounded by a cloud of virtual mesons and other strongly interacting particles, can have the same charge as the positron, whose interactions are all much weaker. It is only necessary to assume that for some reason the charges  $q_{B\ell}$  in the Lagrangian are equal and opposite for the electron and for those particles (two  $u$  quarks and one  $d$  quark) that make up the proton; the effect of higher-order corrections then appears solely in the common factor  $Z_3^{-1/2}$ .

In order for charge renormalization to arise only from radiative corrections to the photon propagator, there must be cancellations among the great variety of other radiative corrections to the propagators and electromagnetic vertices of the charged particles. We can see a little more deeply into the nature of these cancellations by making use of the celebrated relations between these charged particle propagators and vertices known as the *Ward identities*.

For instance, consider the Greens function for an electric current  $J^\mu(x)$  together with a Heisenberg-picture Dirac field  $\Psi_n(y)$  of charge  $q$  and its covariant adjoint  $\bar{\Psi}_m(z)$ . We define the electromagnetic vertex function  $\Gamma^\mu$



**Figure 10.5.** Diagrams for the first corrections to the electron propagator and vertex function in quantum electrodynamics. Here straight lines are electrons; wavy lines are photons.

of the charged particle by

$$\begin{aligned} & \int d^4x d^4y d^4z e^{-ip \cdot x} e^{-ik \cdot y} e^{+i\ell \cdot z} \langle \Omega | T \{ J^\mu(x) \Psi_n(y) \bar{\Psi}_m(z) \} | \Omega \rangle \\ & \equiv -iq S'_{nn'}(k) \Gamma_{n'm'}^\mu(k, \ell) S'_{m'm}(\ell) \delta^4(p + k - \ell), \end{aligned} \quad (10.4.19)$$

where

$$-i S'_{nm}(k) \delta^4(k - \ell) \equiv \int d^4y d^4z \langle \Omega | T \{ \Psi_n(y) \bar{\Psi}_m(z) \} | \Omega \rangle e^{-ik \cdot y} e^{+i\ell \cdot z}. \quad (10.4.20)$$

According to the theorem of Section 6.4, Eq. (10.4.20) gives the sum of all Feynman graphs with one incoming and one outgoing fermion line, i.e., the complete Dirac propagator. Also, Eq. (10.4.19) gives the sum of all such graphs with an extra photon line attached, so  $\Gamma^\mu$  is the sum of “vertex” graphs with one incoming Dirac line, one outgoing Dirac line, and one photon line, but with the complete Dirac external line propagators and the bare photon external line propagator stripped away. To make the normalization of  $S'$  and  $\Gamma^\mu$  perfectly clear, we mention that in the limit of no interactions, these functions take the values

$$S'(k) \rightarrow [i\gamma_\nu k^\nu + m - i\epsilon]^{-1}, \quad \Gamma^\mu(k, \ell) \rightarrow \gamma^\mu.$$

The one-loop diagrams that provide corrections to these limiting values are shown in Figure 10.5.

We can derive a relation between  $\Gamma^\mu$  and  $S'$  by use of the identity

$$\begin{aligned} \frac{\partial}{\partial x^\mu} T \{ J^\mu(x) \Psi_n(y) \bar{\Psi}_m(z) \} &= T \{ \partial_\mu J^\mu(x) \Psi_n(y) \bar{\Psi}_m(z) \} \\ &+ \delta(x^0 - y^0) T \{ [J^0(x), \Psi_n(y)] \bar{\Psi}_m(z) \} \\ &+ \delta(x^0 - z^0) T \{ \Psi_n(y) [J^0(x), \bar{\Psi}_m(z)] \}. \end{aligned} \quad (10.4.21)$$

where the delta functions arise from time-derivatives of step functions. The conservation condition (10.4.2) tells us that the first term vanishes, while the second and third terms can be calculated using the commutation relations (10.4.9), which here give

$$[J^0(\mathbf{x}, t), \Psi_n(\mathbf{y}, t)] = -q\Psi_n(\mathbf{y}, t)\delta^3(\mathbf{x} - \mathbf{y}) \quad (10.4.22)$$

and its adjoint

$$[J^0(\mathbf{x}, t), \bar{\Psi}_n(\mathbf{y}, t)] = q\bar{\Psi}_n(\mathbf{y}, t)\delta^3(\mathbf{x} - \mathbf{y}). \quad (10.4.23)$$

Eq. (10.4.21) then reads

$$\begin{aligned} \frac{\partial}{\partial x^\mu} T\{J^\mu(x)\Psi_n(y)\bar{\Psi}_m(z)\} = & -q\delta^4(x-y)T\{\Psi_n(y)\bar{\Psi}_m(z)\} \\ & + q\delta^4(x-z)T\{\Psi_n(y)\bar{\Psi}_m(z)\}. \end{aligned} \quad (10.4.24)$$

Inserting this in the Fourier transform (10.4.19) gives

$$(\ell - k)_\mu S'(k)\Gamma^\mu(k, \ell)S'(\ell) = iS'(\ell) - iS'(k)$$

or in other words

$$(\ell - k)_\mu \Gamma^\mu(k, \ell) = iS'^{-1}(k) - iS'^{-1}(\ell). \quad (10.4.25)$$

This is known as the *generalized Ward identity*, first derived (by these methods) by Takahashi [4]. The original Ward identity, derived earlier by Ward [5] from a study of perturbation theory, can be obtained from Eq. (10.4.25) by letting  $\ell$  approach  $k$ . In this limit, Eq. (10.4.25) gives

$$\Gamma^\mu(k, k) = -i\frac{\partial}{\partial k_\mu} S'^{-1}(k). \quad (10.4.26)$$

The fermion propagator is related to the self-energy insertion  $\Sigma^*(k)$  by Eq. (10.3.28)

$$S'^{-1}(k) = i\not{k} + m - \Sigma^*(k),$$

so Eq. (10.4.26) may be written

$$\Gamma^\mu(k, k) = \gamma^\mu + i\frac{\partial}{\partial k_\mu} \Sigma^*(k). \quad (10.4.27)$$

For a renormalized Dirac field, Eqs. (10.3.31) and (10.4.27) tell us that on the mass shell

$$\bar{u}_{k'}\Gamma^\mu(k, k)u_k = \bar{u}_{k'}\gamma^\mu u_k, \quad (10.4.28)$$

where  $[i\gamma_\mu k^\mu + m]u_k = [i\gamma_\mu k'^\mu + m]u_{k'} = 0$ . Thus the renormalization of the fermion field ensures that the radiative corrections to the vertex function  $\Gamma_\mu$  cancel when a fermion on the mass shell interacts with an electromagnetic field with zero momentum transfer, as is the case when we set out to measure the fermion's electric charge. If we had not used a renormalized fermion field then the corrections to the vertex function would have just cancelled the corrections due to radiative corrections to the external fermion lines, leaving the electric charge again unchanged.



## 10.5 Gauge Invariance

The conservation of electric charge may be used to prove a useful result for the quantities

$$M_{\beta\alpha}^{\mu\mu'\cdots}(q, q', \dots) \equiv \int d^4x \int d^4x' \dots e^{-iq \cdot x} e^{-iq' \cdot x'} \dots \cdot {}_{\text{out}}\langle\beta|T\{J^\mu(x), J^{\mu'}(x'), \dots\}|\alpha\rangle_{\text{in}}. \quad (10.5.1)$$

In theories like spinor electrodynamics in which the electromagnetic interaction is linear in the field  $A^\mu$ , this is the matrix element for emission (and/or absorption) of on- or off-shell photons having four-momenta  $q, q'$ , etc. (and/or  $-q, -q'$ , etc.), with external line photon coefficient functions or propagators omitted, in an arbitrary transition  $\alpha \rightarrow \beta$ . Our result is that Eq. (10.5.1) vanishes when contracted with any one of the photon four-momenta:

$$q_\mu M_{\beta\alpha}^{\mu\mu'\cdots}(q, q', \dots) = q'_{\mu'} M_{\beta\alpha}^{\mu\mu'\cdots}(q, q', \dots) = \dots = 0. \quad (10.5.2)$$

Since  $M$  is defined symmetrically with respect to the photon lines, it will be sufficient to show the vanishing of the first of these quantities.

For this purpose, note that by an integration by parts

$$q_\mu M_{\beta\alpha}^{\mu\mu'\cdots}(q, q', \dots) = -i \int d^4x \int d^4x' \dots e^{-iq \cdot x} e^{-iq' \cdot x'} \dots \cdot {}_{\text{out}}\langle\beta|\frac{\partial}{\partial x^\mu} T\{J^\mu(x), J^{\mu'}(x'), \dots\}|\alpha\rangle_{\text{in}}. \quad (10.5.3)$$

The electric current  $J^\mu(x)$  is conserved, but this does not immediately imply that Eq. (10.5.3) vanishes, because we still have to take account of the  $x^0$ -dependence contained in the theta functions that appear in the definition of the time-ordered product. For instance, for just two currents

$$T\{J^\mu(x)J^\nu(y)\} = \theta(x^0 - y^0)J^\mu(x)J^\nu(y) + \theta(y^0 - x^0)J^\nu(y)J^\mu(x)$$

so, taking account of the conservation of  $J^\mu(x)$ :

$$\begin{aligned} \frac{\partial}{\partial x^\mu} T\{J^\mu(x)J^\nu(y)\} &= \delta(x^0 - y^0)J^0(x)J^\nu(y) - \delta(y^0 - x^0)J^\nu(y)J^0(x) \\ &= \delta(x^0 - y^0)[J^0(x), J^\nu(y)]. \end{aligned} \quad (10.5.4)$$

With more than two currents, we get an equal-time commutator like this (inside the time-ordered product) for each current aside from  $J^\mu(x)$  itself. To evaluate this commutator, we recall that (as shown in the previous section) for any product  $F$  of field operators and their adjoints and/or derivatives

$$[J^0(\mathbf{x}, t), F(\mathbf{y}, t)] = -q_F F(\mathbf{x}, t)\delta^3(\mathbf{x} - \mathbf{y}),$$

where  $q_F$  is the sum of the  $q_\ell$ s for the fields and field derivatives in  $F$ , minus the sum of the  $q_\ell$ s for the field adjoints and their derivatives. For the electric current,  $q_J$  is zero;  $J^\nu(y)$  is itself an electrically neutral operator. It follows that

$$[J^0(\mathbf{x}, t), J^\nu(\mathbf{y}, t)] = 0 \quad (10.5.5)$$

and therefore Eq. (10.5.4) vanishes, so that Eq. (10.5.3) gives

$$q_\mu M_{\beta\alpha}^{\mu\mu'\cdots}(q, q', \dots) = 0 \quad (10.5.6)$$

as was to be proved.

There is an important qualification here. In deriving Eq. (10.5.5) we should take into account the fact that a product of fields at the same spacetime point  $y$  like the current operator  $J^\nu(y)$  can only be properly defined through some regularization procedure that deals with the infinities in such products. In many cases it turns out that there are non-vanishing contributions to the commutator of  $J^0(\mathbf{x}, t)$  with the regulated current  $J^\nu(\mathbf{y}, t)$ , known as Schwinger terms [6]. Where the current includes terms arising from a charged scalar field  $\Phi$ , there are additional regulator-independent Schwinger terms involving  $\Phi^\dagger\Phi$ . However, all these Schwinger terms are cancelled in multi-photon amplitudes by the contribution of additional interactions that are quadratic in the electromagnetic field, either arising from the regulator procedure (if gauge-invariant) or, as for charged scalars, directly from terms in the Lagrangian. We will be dealing mostly with charged spinor fields, and will use a regularization procedure (dimensional regularization) that does not lead to Schwinger terms, so in what follows we will ignore this issue and continue to use the naive commutation relation (10.5.5).

The same argument yields a result like Eq. (10.5.2) even if other particles besides photons are off the mass shell, provided that all charged particles are taken on the mass shell, i.e., kept in the states  $|\beta\rangle_{\text{out}}$  and  $|\alpha\rangle_{\text{in}}$ . Otherwise, the left-hand side of Eq. (10.5.2) receives contributions from non-vanishing equal-time commutators, such as those we encountered in the derivation of the Ward identity in the previous section.

One consequence of Eq. (10.5.2) is that  $S$ -matrix elements are unaffected if we change any photon propagator  $\Delta_{\mu\nu}(q)$  by

$$\Delta_{\mu\nu}(q) \longrightarrow \Delta_{\mu\nu}(q) + \alpha_\mu q_\nu + q_\mu \beta_\nu \quad (10.5.7)$$

or if we change any photon polarization vector by

$$e_\rho(\mathbf{k}, \lambda) \longrightarrow e_\rho(\mathbf{k}, \lambda) + ck_\rho, \quad (10.5.8)$$

where  $k^0 \equiv |\mathbf{k}|$ , and  $\alpha_\mu$ ,  $\beta_\nu$ , and  $c$  are entirely arbitrary (not necessarily constants, and not necessarily the same for all propagators or polarization vectors.) This is (somewhat loosely) called the gauge invariance of the  $S$ -matrix.

To prove this result it is only necessary to display the explicit dependence of the  $S$ -matrix on photon polarization vectors and propagators

$$\begin{aligned} S_{\beta\alpha} \propto & \int d^4q_1 d^4q_2 \cdots \Delta_{\mu_1\nu_1}(q_1) \Delta_{\mu_2\nu_2}(q_2) \cdots \\ & \cdot e_{\rho_1}^*(\mathbf{k}'_1, \lambda'_1) e_{\rho_2}^*(\mathbf{k}'_2, \lambda'_2) \cdots e_{\sigma_1}(\mathbf{k}_1, \lambda_1) e_{\sigma_2}(\mathbf{k}_2, \lambda_2) \cdots \\ & \cdot M_{ba}^{\mu_1\mu_2\cdots\nu_1\nu_2\cdots\rho_1\rho_2\cdots\sigma_1\sigma_2\cdots}(-q_1, -q_2, \dots, q_1, q_2, \dots, -k'_1, -k'_2, \dots, k_1, k_2, \dots). \end{aligned} \quad (10.5.9)$$

where  $M^{\mu\sigma\cdots}$  is the matrix element (10.5.1) calculated in the absence of electromagnetic interactions.<sup>115</sup> The invariance of Eq. (10.5.9) under the ‘gauge transformations’ (10.5.7) and (10.5.8) follows immediately from the conservation conditions (10.5.2). (In Section 9.6 we used the path-integral formalism to prove a special case of this theorem, that vacuum expectation values of time-ordered products of gauge-invariant operators are independent of the constant  $\alpha$  in the propagator (9.6.21).) This result is not as elementary as it looks, as it applies not to individual diagrams, but only to sums of diagrams in which the current vertices are inserted in all possible places in the diagrams.

There is a particularly important application of Eq. (10.5.2) to the calculation of the photon propagator. The complete photon propagator, conventionally called  $\Delta'_{\mu\nu}(q)$ , takes the form

$$\Delta'_{\mu\nu}(q) = \Delta_{\mu\nu}(q) + \Delta_{\mu\rho}(q)M^{\rho\sigma}(q)\Delta_{\sigma\nu}(q), \quad (10.5.10)$$

where  $M^{\rho\sigma}$  is proportional to the matrix element (10.5.1) with two currents and  $\alpha$  and  $\beta$  both the vacuum state, and  $\Delta_{\mu\nu}$  is the bare photon propagator, written here in a general Lorentz-invariant gauge as

$$\Delta_{\mu\nu}(q) = \frac{\eta_{\mu\nu} - \xi(q^2)q_\mu q_\nu / q^2}{q^2 - i\epsilon}. \quad (10.5.11)$$

From Eq. (10.5.2) we have here  $q^\mu M_{\mu\nu}(q) = 0$ , so that

$$q^\mu \Delta'_{\mu\nu}(q) = q^\mu \Delta_{\mu\nu}(q) = \frac{q_\nu [1 - \xi(q^2)]}{q^2 - i\epsilon}. \quad (10.5.12)$$

On the other hand, just as we did for scalar and spinor fields in Section 10.3, we may express the complete photon propagator in terms of a sum  $\Pi^*(q)$  of graphs with two external photon lines that (unlike  $M$ ) are one-photon-irreducible:

$$\begin{aligned} \Delta'(q) &= \Delta(q) + \Delta(q)\Pi^*(q)\Delta(q) + \Delta(q)\Pi^*(q)\Delta(q)\Pi^*(q)\Delta(q) + \cdots \\ &= [\Delta(q)^{-1} - \Pi^*(q)]^{-1}. \end{aligned} \quad (10.5.13)$$

or in other words

$$\Delta'_{\mu\nu}(q) = \Delta_{\mu\nu}(q) + \Delta_{\mu\rho}(q)\Pi^{*\rho\sigma}(q)\Delta'_{\sigma\nu}(q). \quad (10.5.14)$$

Then in order to satisfy Eq. (10.5.12), we must have

$$q_\rho \Pi^{*\rho\sigma}(q) = 0. \quad (10.5.15)$$

This together with Lorentz invariance tells us that  $\Pi^*(q)$  must take the form

$$\Pi^{*\rho\sigma}(q) = (q^2 \eta^{\rho\sigma} - q^\rho q^\sigma) \pi(q^2). \quad (10.5.16)$$

Then Eq. (10.5.13) yields a complete propagator of the form

$$\Delta'_{\mu\nu}(q) = \frac{\eta_{\mu\nu} - \bar{\xi}(q^2)q_\mu q_\nu / q^2}{[q^2 - i\epsilon][1 - \pi(q^2)]}, \quad (10.5.17)$$

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<sup>115</sup>The states  $a$  and  $b$  are the same as  $\alpha$  and  $\beta$ , but with photons deleted. Note that the arguments of  $M$  are all taken to be *incoming* four-momenta, which is why we have to insert various signs for some of the arguments of  $M$  in Eq. (10.5.9).

where

$$\bar{\xi}(q^2) = \xi(q^2)[1 - \pi(q^2)] + \pi(q^2). \quad (10.5.18)$$

Now, because  $\Pi_{\mu\nu}^*(q)$  receives contributions only from one-photon-irreducible graphs, it is expected not to have any pole at  $q^2 = 0$ . (There is an important exception in the case of broken gauge symmetry, to be discussed in Volume II.) In particular, the absence of poles at  $q^2 = 0$  in the  $q_\mu q_\nu$  term in  $\Pi_{\mu\nu}^*(q)$  tells us that the function  $\pi(q^2)$  in Eq. (10.5.16) also has no such pole, and so the pole in the complete propagator (10.5.17) is

still at  $q^2 = 0$ , indicating that *radiative corrections do not give the photon a mass*.

For a renormalized electromagnetic field, radiative corrections should also not alter the gauge-invariant part of the residue of the photon pole in Eq. (10.5.17), so

$$\pi(0) = 0. \quad (10.5.19)$$

This condition leads to a determination of the electromagnetic field renormalization constant  $Z_3$ . Recall that when expressed in terms of the renormalized field (10.4.17), the electrodynamic Lagrangian takes the form

$$\mathcal{L} = -\frac{1}{4}Z_3(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) + \mathcal{L}_M(\Psi_\ell, [\partial_\mu - iZ_3 q_\ell A_\mu]\Psi_\ell).$$

The function  $\pi(q^2)$  in the one-photon-irreducible amplitude is then

$$\pi(q^2) = 1 - Z_3 + \pi_{\text{LOOP}}(q^2), \quad (10.5.20)$$

where  $\pi_{\text{LOOP}}$  is the contribution of loop diagrams. It follows that

$$Z_3 = 1 + \pi_{\text{LOOP}}(0). \quad (10.5.21)$$

In practice, we just calculate the loop contributions and subtract a constant in order to make  $\pi(0)$  vanish.

Incidentally, Eq. (10.5.18) shows that for  $q^2 \neq 0$  the gauge term in the photon propagator is altered by radiative corrections. The one exception is the case of Landau gauge, for which  $\bar{\xi} = \xi = 1$  for all  $q^2$ .

## 10.6 Electromagnetic Form Factors and Magnetic Moment

Suppose that we want to calculate the scattering of a particle by an external electromagnetic field (or by the electromagnetic field of another particle), to first order in this electromagnetic field, but to all orders in all other interactions (including electromagnetic) of our particle. For this purpose, we need to know the sum of the contributions of all Feynman diagrams with one incoming and one outgoing particle line, both on the mass shell, plus a photon line, which may be on or off the mass shell. According to the theorem of Section 6.4, this sum is given by the one-particle matrix element of the electromagnetic current  $J^\mu(x)$ . Let us see what governs the general form of this matrix element.

According to spacetime translation invariance, the one-particle matrix element of the electromagnetic current takes the form

$$\langle p', \sigma' | J^\mu(x) | p, \sigma \rangle = \exp(i(p - p') \cdot x) \langle p', \sigma' | J^\mu(0) | p, \sigma \rangle. \quad (10.6.1)$$

The current conservation condition  $\partial_\mu J^\mu = 0$  then requires

$$(p' - p)_\mu \langle p', \sigma' | J^\mu(0) | p, \sigma \rangle = 0. \quad (10.6.2)$$

Also, setting  $\mu = 0$  and integrating over all  $\mathbf{x}$  gives

$$\langle p', \sigma' | Q | p, \sigma \rangle = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \langle p', \sigma' | J^0(0) | p, \sigma \rangle.$$

Using Eq. (10.4.8), this gives

$$\langle p', \sigma' | J^0(0) | p, \sigma \rangle = (2\pi)^{-3} q \delta_{\sigma'\sigma}, \quad (10.6.3)$$

where  $q$  is the particle charge.

We also have at our disposal the constraints on the current matrix elements imposed by Lorentz invariance. To explore these, we will limit ourselves to the simplest cases: spin zero and spin  $\frac{1}{2}$ . The analysis presented here provides an example of techniques that are useful for other currents, such as those of the semi-leptonic weak interactions.

### *Spin Zero*

For spin zero, Lorentz invariance requires the one-particle matrix element of the current to take the general form

$$\langle p' | J^\mu(0) | p \rangle = q(2\pi)^{-3} (2p'^0)^{-1/2} (2p^0)^{-1/2} \mathcal{J}^\mu(p', p), \quad (10.6.4)$$

where  $p^0$  and  $p'^0$  are the mass-shell energies ( $p^0 = \sqrt{\mathbf{p}^2 + m^2}$ ), and  $\mathcal{J}^\mu(p', p)$  is a four-vector function of the two four-vectors  $p'^\mu$  and  $p^\mu$ . (We have extracted a factor of the charge  $q$  of the particle from  $\mathcal{J}$  for future convenience.) Obviously, the most general such four-vector function takes the form of a linear combination of  $p'^\mu$  and  $p^\mu$ , or equivalently of  $p'^\mu + p^\mu$  and  $p'^\mu - p^\mu$ , with scalar coefficients. But the scalars  $p^2$  and  $p'^2$  are fixed at the values  $p'^2 = p^2 = -m^2$ , so the scalar variables that can be formed from  $p^\mu$  and  $p'^\mu$  can be taken as functions only of  $p \cdot p'$ , or equivalently of

$$k^2 \equiv (p - p')^2 = -2m^2 - 2p \cdot p'. \quad (10.6.5)$$

Thus the function  $\mathcal{J}^\mu(p', p)$  must take the form

$$\mathcal{J}^\mu(p', p) = (p' + p)^\mu F(k^2) + i(p' - p)^\mu H(k^2). \quad (10.6.6)$$

The fact that  $J^\mu$  is Hermitian implies that  $\mathcal{J}^\mu(p', p)^* = \mathcal{J}^\mu(p, p')$ , so that both  $F(k^2)$  and  $H(k^2)$  are real.

Now  $(p' - p) \cdot (p' + p)$  vanishes, while  $(p' - p)^2 = k^2$  is not generally zero, so the condition of current conservation is simply

$$H(k^2) = 0. \quad (10.6.7)$$

Also, setting  $\mathbf{p}' = \mathbf{p}$  and  $\mu = 0$  in Eq. (10.6.4), and comparing with Eq. (10.6.3), we find that

$$F(0) = 1. \quad (10.6.8)$$

The function  $F(k^2)$  is called the *electromagnetic form factor* of the particle.

$$Spin \frac{1}{2}$$

For spin  $\frac{1}{2}$ , Lorentz invariance requires the one-particle matrix element of the current to take the general form

$$\langle p', \sigma' | J^\mu(0) | p, \sigma \rangle = iq(2\pi)^{-3} \bar{u}(p', \sigma') \Gamma^\mu(p', p) u(p, \sigma) \quad (10.6.9)$$

where  $\Gamma^\mu$  is a four-vector  $4 \times 4$  matrix function of  $p^\nu$ ,  $p'^\nu$ , and  $\gamma^\nu$ , and  $u$  is the usual Dirac coefficient function. We have extracted a factor  $iq$  to make the normalization of  $\Gamma^\mu$  the same as in the previous section.

Just as for any  $4 \times 4$  matrix, we may expand  $\Gamma^\mu$  in the 16 covariant matrices  $\mathbf{1}$ ,  $\gamma_\rho$ ,  $[\gamma_\rho, \gamma_\sigma]$ ,  $\gamma_5 \gamma_\rho$ , and  $\gamma_5$ . The most general four-vector  $\Gamma^\mu$  can therefore be written as a linear combination of

$$\begin{aligned} \mathbf{1} : & \quad p^\mu, \quad p'^\mu, \\ \gamma_\rho : & \quad \gamma^\mu, \quad p^\mu \not{p}, \quad p'^\mu \not{p}, \quad p^\mu \not{p}', \quad p'^\mu \not{p}', \\ [\gamma_\rho, \gamma_\sigma] : & \quad [\gamma^\mu, \not{p}], \quad [\gamma^\mu, \not{p}'], \quad [\not{p}, \not{p}'] p^\mu, \quad [\not{p}, \not{p}'] p'^\mu, \\ \gamma_5 \gamma_\rho : & \quad \gamma_5 \gamma_\rho \epsilon^{\rho\mu\nu\sigma} p_\nu p'_\sigma, \\ \gamma_5 : & \quad \text{NONE} \end{aligned}$$

with the coefficient of each term a function of the only scalar variable in the problem, the quantity (10.6.5). This can be greatly simplified by using the Dirac equations satisfied by  $u$  and  $\bar{u}$ :

$$\bar{u}(p', \sigma') (i \not{p}' + m) = 0, \quad (i \not{p} + m) u(p, \sigma) = 0.$$

In consequence, we can drop<sup>116</sup> all but the first three entries:  $p^\mu$ ,  $p'^\mu$ , and  $\gamma^\mu$ . We conclude that, on the fermion mass shell,  $\Gamma^\mu$  may be expressed as a linear combination of  $\gamma^\mu$ ,  $p^\mu$ , and  $p'^\mu$ , which we choose to write as

$$\begin{aligned} \bar{u}(p', \sigma') \Gamma^\mu(p', p) u(p, \sigma) = \bar{u}(p', \sigma') \left[ \gamma^\mu F(k^2) - \frac{i}{2m} (p + p')^\mu G(k^2) \right. \\ \left. + \frac{(p - p')^\mu}{2m} H(k^2) \right] u(p, \sigma). \end{aligned} \quad (10.6.10)$$

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<sup>116</sup>This is obvious for the terms  $p^\mu \not{p}$ ,  $p'^\mu \not{p}$ ,  $p^\mu \not{p}'$ , and  $p'^\mu \not{p}'$ , which may be replaced respectively with  $imp^\mu$ ,  $imp'^\mu$ ,  $imp^\mu$ , and  $imp'^\mu$ , which are the same as terms already on our list. Also, we can write

$$[\gamma^\mu, \not{p}] = 2\gamma^\mu \not{p} - \{\gamma^\mu, \not{p}\} = 2\gamma^\mu \not{p} - 2p^\mu,$$

which may be replaced with  $2im\gamma^\mu - 2p^\mu$ , a linear combination of terms already on our list. The same applies to  $[\gamma^\mu, \not{p}']$ . Also,

$$[\not{p}, \not{p}'] = -2\not{p}' \not{p} + \{\not{p}, \not{p}'\} = -2\not{p}' \not{p} + 2p \cdot p',$$

which may be replaced with  $2m^2 + 2p \cdot p' = -k^2$ . Hence the terms  $[\not{p}, \not{p}'] p^\mu$  and  $[\not{p}, \not{p}'] p'^\mu$  give nothing new. Finally, to deal with the last term we may use the relation

$$\gamma_5 \gamma_\rho \epsilon^{\rho\mu\nu\sigma} = \frac{i}{6} (\gamma^\mu \gamma^\nu \gamma^\sigma + \gamma^\sigma \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\sigma \gamma^\mu - \gamma^\mu \gamma^\sigma \gamma^\nu - \gamma^\nu \gamma^\mu \gamma^\sigma - \gamma^\sigma \gamma^\nu \gamma^\mu).$$

Contracting this with  $p_\nu$  and  $p'_\sigma$  and then moving all  $\not{p}$  factors to the right and  $\not{p}'$  factors to the left again gives a linear combination of  $p^\mu$ ,  $p'^\mu$ , and  $\gamma^\mu$ .

The hermiticity of  $J^\mu(0)$  implies that

$$\beta \Gamma^{\mu\dagger}(p', p) \beta = -\Gamma^\mu(p, p'), \quad (10.6.11)$$

so that  $F(k^2)$ ,  $G(k^2)$ , and  $H(k^2)$  must all be *real* functions of  $k^2$ .

The conservation condition (10.6.2) is automatically satisfied by the first two terms in Eq. (10.6.10), because

$$(p' - p)_\mu \gamma^\mu = -i[(i\not{p}' + m) - (i\not{p} + m)]$$

and

$$(p' - p) \cdot (p' + p) = p'^2 - p^2.$$

On the other hand,  $(p' - p)^2$  does not in general vanish, so current conservation requires the third term to vanish

$$H(k^2) = 0. \quad (10.6.12)$$

Also, letting  $\mathbf{p}' \rightarrow \mathbf{p}$  in Eqs. (10.6.9) and (10.6.10), we find

$$\langle p, \sigma' | J^\mu(0) | p, \sigma \rangle = iq(2\pi)^{-3} \bar{u}(p, \sigma') \left[ \gamma^\mu F(0) - \frac{i}{m} p^\mu G(0) \right] u(p, \sigma).$$

Using the identity  $\{\gamma^\mu, i\not{p} + m\} = 2m\gamma^\mu + 2ip^\mu$ , we also have

$$\bar{u}(p, \sigma') \gamma^\mu u(p, \sigma) = -\frac{ip^\mu}{m} \bar{u}(p, \sigma') u(p, \sigma).$$

Recall also that

$$\bar{u}(p, \sigma') u(p, \sigma) = \delta_{\sigma'\sigma} m/p^0$$

and therefore

$$\langle p, \sigma' | J^\mu(0) | p, \sigma \rangle = q(2\pi)^{-3} (p^\mu/p^0) \delta_{\sigma'\sigma} [F(0) + G(0)]. \quad (10.6.13)$$

Comparing this with Eq. (10.6.3) yields the normalization condition

$$F(0) + G(0) = 1. \quad (10.6.14)$$

It may be useful to note that the electromagnetic vertex matrix  $\Gamma^\mu$  is commonly written in terms of two other matrices, as

$$\begin{aligned} \bar{u}(p', \sigma') \Gamma^\mu(p', p) u(p, \sigma) &= \bar{u}(p', \sigma') \left[ \gamma^\mu F_1(k^2) \right. \\ &\quad \left. + \frac{i}{2} [\gamma^\mu, \gamma^\nu] (p' - p)_\nu F_2(k^2) \right] u(p, \sigma). \end{aligned} \quad (10.6.15)$$

As already mentioned, we may rewrite the matrix appearing in the second term in terms of those used in defining  $F(k^2)$  and  $G(k^2)$ :

$$\begin{aligned} \bar{u}(p', \sigma') \frac{i}{2} [\gamma^\mu, \gamma^\nu] (p' - p)_\nu u(p, \sigma) &= \bar{u}(p', \sigma') \left[ -i\not{p}' \gamma^\mu + \frac{i}{2} \{\gamma^\mu, \not{p}'\} - i\gamma^\mu \not{p} + \frac{i}{2} \{\gamma^\mu, \not{p}\} \right] u(p, \sigma) \\ &= \bar{u}(p', \sigma') [i(p^\mu + p'^\mu) + 2m\gamma^\mu] u(p, \sigma). \end{aligned} \quad (10.6.16)$$

Comparing Eq. (10.6.15) with Eq. (10.6.10), we find

$$F(k^2) = F_1(k^2) + 2mF_2(k^2) \quad (10.6.17)$$

$$G(k^2) = -2mF_2(k^2). \quad (10.6.18)$$

The normalization condition (10.6.14) now reads

$$F_1(0) = 1.$$

In order to evaluate the magnetic moment of our particle in terms of its form factors, let us consider the spatial part of the vertex function in the case of small momenta,  $|\mathbf{p}|, |\mathbf{p}'| \ll m$ . For this purpose, it is useful to use Eq. (10.6.16) to rewrite Eq. (10.6.10) (with  $H = 0$ ) in a third form:

$$\begin{aligned} \bar{u}(p', \sigma') \Gamma^\mu(p', p) u(p, \sigma) = & -\frac{i}{2m} \bar{u}(p', \sigma') \left[ (p + p')^\mu \{F(k^2) + G(k^2)\} \right. \\ & \left. - \frac{1}{2} [\gamma^\mu, \gamma^\nu] (p' - p)_\nu F(k^2) \right] u(p, \sigma). \end{aligned} \quad (10.6.19)$$

For zero momenta the matrix elements of the commutators of Dirac matrices are given by Eqs. (5.4.19) and (5.4.20) as

$$\bar{u}(0, \sigma') [\gamma^i, \gamma^j] u(0, \sigma) = 4i\epsilon_{ijk} (J_k^{(1/2)})_{\sigma'\sigma}, \quad \bar{u}(0, \sigma') [\gamma^i, \gamma^0] u(0, \sigma) = 0,$$

where  $\mathbf{J}^{(1/2)} = \frac{1}{2}\boldsymbol{\sigma}$  is the angular momentum matrix for spin  $\frac{1}{2}$ . Hence to first order in the small momenta,

$$\bar{u}(p', \sigma') \Gamma^i(p', p) u(p, \sigma) \longrightarrow \frac{1}{m} [(\mathbf{p} - \mathbf{p}') \cdot \mathbf{J}^{(1/2)}]_{i\sigma'\sigma} F(0). \quad (10.6.20)$$

In a very weak time-independent external vector potential  $\mathbf{A}(\mathbf{x})$  the matrix element of the interaction Hamiltonian  $H' = -\int d^3x \mathbf{J}(\mathbf{x}) \cdot \mathbf{A}(\mathbf{x})$  between one-particle states of small momentum is therefore

$$\begin{aligned} \langle p', \sigma' | H' | p, \sigma \rangle = & -\frac{iqF(0)}{m(2\pi)^3} \int d^3x e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}} \mathbf{A}(\mathbf{x}) \cdot [(\mathbf{p} - \mathbf{p}') \cdot \mathbf{J}^{(1/2)}]_{\sigma'\sigma} \\ = & -\frac{qF(0)}{m(2\pi)^3} \int d^3x e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}} (\mathbf{J}^{(1/2)})_{\sigma'\sigma} \cdot \mathbf{B}(\mathbf{x}), \end{aligned} \quad (10.6.21)$$

where  $\mathbf{B} = \nabla \times \mathbf{A}$  is the magnetic field. Hence in the limit of a slowly varying weak magnetic field, the matrix element of the interaction Hamiltonian is

$$\langle p', \sigma' | H' | p, \sigma \rangle = -\frac{qF(0)}{m} (\mathbf{J}^{(1/2)})_{\sigma'\sigma} \cdot \mathbf{B} \delta^3(\mathbf{p} - \mathbf{p}'). \quad (10.6.22)$$

The magnetic moment  $\mu$  for an arbitrary particle of general spin  $j$  is defined by the statement that the matrix element of the interaction of the particle with a weak static slowly varying magnetic field is

$$\langle p', \sigma' | H' | p, \sigma \rangle = -\frac{\mu}{j} (\mathbf{J}^{(j)})_{\sigma'\sigma} \cdot \mathbf{B} \delta^3(\mathbf{p} - \mathbf{p}'). \quad (10.6.23)$$



Hence Eq. (10.6.22) gives the magnetic moment of a particle of charge  $q$ , mass  $m$ , and spin  $\frac{1}{2}$  as:

$$\mu = \frac{qF(0)}{2m}. \quad (10.6.24)$$

This contains as a special case the celebrated Dirac result [7]  $\mu = q/2m$  for a spin  $\frac{1}{2}$  particle without radiative corrections.

We mention without proof that the form factors  $F(k^2)$  and  $G(k^2)$  of the proton may be measured for  $k^2 > 0$  by comparison of experimental data for electron–proton scattering with the Rosenbluth formula [8] for the laboratory frame differential cross-section:

$$\begin{aligned} \frac{d\sigma}{d\Omega} = & \frac{e^4}{4(4\pi)^2 E_0^2} \frac{\cos^2(\theta/2)}{\sin^4(\theta/2)} \left[ 1 + \frac{2E_0}{m} \sin^2(\theta/2) \right]^{-1} \\ & \cdot \left\{ (F(k^2) + G(k^2))^2 + \frac{k^2}{4m^2} (2F^2(k^2) \tan^2(\theta/2) + G^2(k^2)) \right\}, \end{aligned}$$

where  $E_0$  is the energy of the incident electron (taken here with  $E_0 \gg m_e$ );  $\theta$  is the scattering angle; and

$$k^2 = \frac{4E_0^2 \sin^2(\theta/2)}{1 + (2E_0/m) \sin^2(\theta/2)}.$$

## 10.7 The Källén–Lehmann Representation<sup>117</sup>

We saw in Section 10.2 that the presence of one-particle intermediate states leads to poles in Fourier transforms of matrix elements of time-ordered products, like (10.2.1). Multi-particle intermediate states lead to more complicated singularities, which are difficult to describe in general. But in the special case of a vacuum expectation value involving just two operators, we have a convenient representation that explicitly displays the analytic structure of the Fourier transform. In particular, this representation may be used for propagators, where the two operators are the fields of elementary particles. When combined with the positivity requirements of quantum mechanics, this representation yields interesting bounds on the asymptotic behavior of propagators and the magnitude of renormalization constants.

Consider a complex scalar Heisenberg-picture operator  $\Phi(x)$ , which may or may not be an elementary particle field. The vacuum expectation value of a product  $\Phi(x)\Phi^\dagger(y)$  may be expressed as

$$\langle \Omega | \Phi(x) \Phi^\dagger(y) | \Omega \rangle = \sum_n \langle \Omega | \Phi(x) | n \rangle \langle n | \Phi^\dagger(y) | \Omega \rangle, \quad (10.7.1)$$

where the sum runs over any complete set of states. (Here the sum over  $n$  includes integrals over continuous labels as well as sums over discrete labels.) Choosing these states as eigenstates of the momentum four-vector  $P^\mu$ , translational invariance tells us that

$$\begin{aligned} \langle \Omega | \Phi(x) | n \rangle &= \exp(ip_n \cdot x) \langle \Omega | \Phi(0) | n \rangle, \\ \langle n | \Phi^\dagger(y) | \Omega \rangle &= \exp(-ip_n \cdot y) \langle n | \Phi^\dagger(0) | \Omega \rangle \end{aligned} \quad (10.7.2)$$

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<sup>117</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

and so

$$\langle \Omega | \Phi(x) \Phi^\dagger(y) | \Omega \rangle = \sum_n \exp(ip_n \cdot (x - y)) |\langle \Omega | \Phi(0) | n \rangle|^2. \quad (10.7.3)$$

It is convenient to rewrite this in terms of a *spectral function*. Note that the sum  $\sum_n \delta^4(p - p_n) |\langle \Omega | \Phi(0) | n \rangle|^2$  is a scalar function of the four-vector  $p^\mu$ , and therefore may depend only on  $p^2$  and (for  $p^2 \leq 0$ ) on the step function  $\theta(p^0)$ . In fact, the intermediate states in Eq. (10.7.3) all have  $p^2 \leq 0$  and  $p^0 > 0$ , so this sum takes the form

$$\sum_n \delta^4(p - p_n) |\langle \Omega | \Phi(0) | n \rangle|^2 = (2\pi)^{-3} \theta(p^0) \rho(-p^2) \quad (10.7.4)$$

with  $\rho(-p^2) = 0$  for  $p^2 > 0$ . (The factor  $(2\pi)^{-3}$  is extracted from  $\rho$  for future convenience.) The spectral function  $\rho(-p^2)$  is clearly real and positive. With this definition, we can rewrite Eq. (10.7.3) as

$$\begin{aligned} \langle \Omega | \Phi(x) \Phi^\dagger(y) | \Omega \rangle &= (2\pi)^{-3} \int d^4p \\ &\quad \cdot \exp[ip \cdot (x - y)] \theta(p^0) \rho(-p^2) \\ &= (2\pi)^{-3} \int d^4p \int_0^\infty d\mu^2 \\ &\quad \cdot \exp[ip \cdot (x - y)] \theta(p^0) \rho(\mu^2) \delta(p^2 + \mu^2). \end{aligned} \quad (10.7.5)$$

Interchanging the order of integration over  $p^\mu$  and  $\mu^2$ , this may be expressed as

$$\langle \Omega | \Phi(x) \Phi^\dagger(y) | \Omega \rangle = \int_0^\infty d\mu^2 \rho(\mu^2) \Delta_+(x - y; \mu^2), \quad (10.7.6)$$

where  $\Delta_+$  is the familiar function

$$\Delta_+(x - y; \mu^2) \equiv (2\pi)^{-3} \int d^4p \exp[ip \cdot (x - y)] \theta(p^0) \delta(p^2 + \mu^2). \quad (10.7.7)$$

In just the same way, we can show that

$$\langle \Omega | \Phi^\dagger(y) \Phi(x) | \Omega \rangle = \int_0^\infty d\mu^2 \bar{\rho}(\mu^2) \Delta_+(y - x; \mu^2) \quad (10.7.8)$$

with a second spectral function  $\bar{\rho}(\mu^2)$  defined by

$$\sum_n \delta^4(p + p_n) \left| \langle n | \Phi^\dagger(0) | \Omega \rangle \right|^2 = (2\pi)^{-3} \theta(p^0) \bar{\rho}(-p^2). \quad (10.7.9)$$

We now make use of the causality requirement, that the commutator  $[\Phi(x), \Phi^\dagger(y)]$  must vanish for space-like separations  $x - y$ . The vacuum expectation value of the commutator is

$$\begin{aligned} \langle \Omega | [\Phi(x), \Phi^\dagger(y)] | \Omega \rangle &= \int_0^\infty d\mu^2 (\rho(\mu^2) \Delta_+(x - y; \mu^2) \\ &\quad - \bar{\rho}(\mu^2) \Delta_+(y - x; \mu^2)). \end{aligned} \quad (10.7.10)$$

As noted in Section 5.2, for  $x - y$  space-like the function  $\Delta_+(x - y)$  does not vanish, but it does become *even*. In order for (10.7.10) to vanish for arbitrary space-like separations, it is thus necessary that

$$\rho(\mu^2) = \bar{\rho}(\mu^2). \quad (10.7.11)$$

This is a special case of the CPT theorem, proved here without the use of perturbation theory; for whatever states with  $p^2 = -\mu^2$  have the quantum numbers of the operator  $\Phi$ , there must be corresponding states with  $p^2 = -\mu^2$  that have the quantum numbers of the operator  $\Phi^\dagger$ .

Using Eq. (10.7.11), the vacuum expectation of the time-ordered product is

$$\langle \Omega | T \{ \Phi(x) \Phi^\dagger(y) \} | \Omega \rangle = -i \int_0^\infty d\mu^2 \rho(\mu^2) \Delta_F(x - y; \mu^2), \quad (10.7.12)$$

where  $\Delta_F(x - y; \mu^2)$  is the Feynman propagator for a spinless particle of mass  $\mu$ :

$$\begin{aligned} -i\Delta_F(x - y; \mu^2) &\equiv \theta(x^0 - y^0) \Delta_+(x - y; \mu^2) \\ &\quad - \theta(y^0 - x^0) \Delta_+(y - x; \mu^2). \end{aligned} \quad (10.7.13)$$

Borrowing the notation introduced in Section 10.3 for complete propagators, we introduce the momentum space function

$$-i\Delta'(p) \equiv \int d^4x \exp[-ip \cdot (x - y)] \langle \Omega | T \{ \Phi(x) \Phi^\dagger(y) \} | \Omega \rangle. \quad (10.7.14)$$

Recall that

$$\int d^4x \exp[-ip \cdot (x - y)] \Delta_F(x - y; \mu^2) = \frac{1}{p^2 + \mu^2 - i\epsilon}. \quad (10.7.15)$$

This yields our spectral representation [9]:

$$\Delta'(p) = \int_0^\infty \rho(\mu^2) \frac{d\mu^2}{p^2 + \mu^2 - i\epsilon}. \quad (10.7.16)$$

One immediate consequence of this result and the positivity of  $\rho(\mu^2)$  is that  $\Delta'(p)$  cannot vanish for  $|p^2| \rightarrow \infty$  faster<sup>118</sup> than the bare propagator  $1/(p^2 + m^2 - i\epsilon)$ . From time to time the suggestion is made to include higher derivative terms in the unperturbed Lagrangian, which would make the propagator vanish faster than  $1/p^2$  for  $|p^2| \rightarrow \infty$ , but the spectral representation shows that this would necessarily entail a departure from the positivity postulates of quantum mechanics.

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<sup>118</sup>In fact, it is not even certain that  $\Delta'(p)$  vanishes for  $|p^2| \rightarrow \infty$  at all, even though this would seem to follow from the spectral representation. The problem arises from the interchange of the integrals over  $p^\mu$  and  $\mu^2$ . What is certain is that  $\Delta'(p)$  is an analytic function of  $-p^2$  with a discontinuity across the positive real axis  $-p^2 = \mu^2$  given by  $\pi\rho(\mu^2)$ , as can be shown by the methods of the next section. From this, it follows that  $\Delta'(p)$  is given by a dispersion relation with spectral function  $\rho(\mu^2)$  and possible subtractions:

$$\Delta'(p) = P(p^2) + (-p^2 + \mu_0^2)^n \int_0^\infty \frac{\rho(\mu^2)}{(\mu^2 + \mu_0^2)^n} \frac{d\mu^2}{p^2 + \mu^2 - i\epsilon},$$

where  $n$  is a positive integer,  $\mu_0^2$  is an arbitrary positive constant, and  $P(p^2)$  is a  $\mu_0^2$ -dependent polynomial in  $p^2$  of order  $n - 1$  that is absent for  $n = 0$ .

We can use the spectral representation together with equal-time commutation relations to derive an interesting sum rule for the spectral function. If  $\Phi(x)$  is a conventionally normalized (not renormalized) canonical field operator, then

$$\left[ \frac{\partial \Phi(\mathbf{x}, t)}{\partial t}, \Phi^\dagger(\mathbf{y}, t) \right] = -i\delta^3(\mathbf{x} - \mathbf{y}). \quad (10.7.17)$$

We note that

$$\frac{\partial}{\partial x^0} \Delta_+(x - y) \Big|_{x^0=y^0} = -i\delta^3(\mathbf{x} - \mathbf{y})$$

so the spectral representation (10.7.10) and the commutation relations (10.7.17) together tell us that

$$\int_0^\infty \rho(\mu^2) d\mu^2 = 1. \quad (10.7.18)$$

This implies that for  $|p^2| \rightarrow \infty$ , the momentum space propagator (10.7.16) of the unrenormalized fields has the free-field asymptotic behavior

$$\Delta'(p) \longrightarrow \frac{1}{p^2}.$$

This result is only meaningful within a suitable scheme for regulating ultraviolet divergences; in perturbation theory the unrenormalized fields have infinite matrix elements, and their propagator is ill-defined.

Now consider the possibility that there is a one-particle state  $|k\rangle$  of mass  $m$  with a non-vanishing matrix element with the state  $\langle \Omega | \Phi(0)$ . Lorentz invariance requires this matrix element to take the form

$$\langle \Omega | \Phi(0) | k \rangle = (2\pi)^{-3/2} \left( 2\sqrt{\mathbf{k}^2 + m^2} \right)^{-1/2} N, \quad (10.7.19)$$

where  $N$  is a constant. According to the general results of Section 10.3, the propagator  $\Delta'(p)$  of the unrenormalized fields should have a pole at  $p^2 \rightarrow -m^2$  with residue  $Z \equiv |N|^2 > 0$ . That is,

$$\rho(\mu^2) = Z\delta(\mu^2 - m^2) + \sigma(\mu^2), \quad (10.7.20)$$

where  $\sigma(\mu^2) \geq 0$  is the contribution of multi-particle states. Together with Eq. (10.7.18), this has the consequence that

$$1 = Z + \int_0^\infty \sigma(\mu^2) d\mu^2 \quad (10.7.21)$$

and so

$$Z \leq 1 \quad (10.7.22)$$

with the equality reached only for a free particle, for which  $\langle \Omega | \Phi(x)$  has no matrix elements with multi-particle states.

Because  $Z$  is positive, Eq. (10.7.21) can also be regarded as providing an upper bound on the coupling of the field  $\Phi$  to multi-particle states:

$$\int_0^\infty \sigma(\mu^2) d\mu^2 \leq 1 \quad (10.7.23)$$

with the equality reached for  $Z = 0$ . The limit  $Z = 0$  has an interesting interpretation as a condition for a particle to be composite rather than elementary [10]. In this context, a ‘composite’ particle may be understood to be one whose field does not appear in the Lagrangian. Consider such a particle, say a neutral particle of spin zero, and suppose that its quantum numbers allow it to be destroyed by an operator  $F(\Psi)$  constructed out of other fields. We can freely introduce a field  $\Phi$  for this particle by adding a term to the Lagrangian density of the form<sup>119</sup>  $\Delta\mathcal{L} = (\Phi - F(\Psi))^2$ , because the path integral over  $\Phi$  can be done by setting it equal to the stationary point  $\Phi = F(\Psi)$ , at which  $\Delta\mathcal{L} = 0$ . But suppose instead we write  $\Delta\mathcal{L} = \Delta\mathcal{L}_0 + \Delta\mathcal{L}_1$ , where  $\Delta\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2$  is the usual free-field Lagrangian, and treat  $\Delta\mathcal{L}_1 = \Delta\mathcal{L} - \Delta\mathcal{L}_0$  as an interaction. A term  $\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi$  in the interaction is nothing new. We encountered such a term in Eq. (10.3.12), multiplied by a factor  $(1 - Z)$ ; the only new thing is that here  $Z = 0$ . Instead of adjusting  $Z$  to satisfy the field renormalization condition  $\Pi^{*'}(0) = 0$ , here we must regard this as a condition on the

coupling constants of the composite particle. Unfortunately, it has not been possible to implement this procedure in quantum field theories, because as we have seen  $Z = 0$  means that the particle couples as strongly as possible to its constituents, and this rules out the use of perturbation theory. The condition  $Z = 0$  does prove useful in non-relativistic quantum mechanics; for instance, it fixes the coupling of the deuteron to the neutron and proton [12].

Although the spectral representation has been derived here only for a spinless field, it is easy to generalize these results to other fields. Indeed, in the next chapter we shall show that to order  $e^2$ , the  $Z$ -factor for the electromagnetic field (conventionally called  $Z_3$ ) is given by

$$Z_3 = 1 - \frac{e^2}{12\pi^2} \ln\left(\frac{\Lambda^2}{m_e^2}\right)$$

(where  $\Lambda \gg m_e$  is an ultraviolet cutoff), in agreement with the bound (10.7.22).

## 10.8 Dispersion Relations<sup>120</sup>

The failure of early attempts to apply perturbative quantum field theory to the strong and weak nuclear forces had led theorists by the late 1950s to attempt the use of the analyticity and unitarity of scattering amplitudes as a way of deriving general non-perturbative results that would not depend on any particular field theory. This started with a revival of interest in dispersion relations. In its original form [13], a dispersion relation was a formula giving the real part of the index of refraction in terms of an integral over its imaginary part. It was derived from an analyticity property of the index of refraction as a function of frequency, which followed from the condition that electromagnetic signals in a medium cannot travel faster than light in a vacuum. By expressing the index of refraction in terms of the forward photon scattering amplitude, the dispersion relation could be rewritten as a formula for the real part of the forward scattering amplitude as an integral of its imaginary

<sup>119</sup>This is known in condensed matter physics as a ‘Hubbard–Stratonovich transformation’ [11]. It will be used to introduce fields for pairs of electrons in our discussion of superconductivity in Volume II.

<sup>120</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

part, and hence via unitarity in terms of the total cross-section. One of the exciting things about this relation was that it provided an alternative to conventional perturbation theory; given the scattering amplitude to order  $e^2$ , one could calculate the cross-section and the imaginary part of the scattering amplitude to order  $e^4$ , and then use the dispersion relation to calculate the real part of the forward scattering amplitude to this order, without having ever to calculate a loop graph.

The modern approach to dispersion relations began in 1954 with the work of Gell-Mann, Goldberger, and Thirring [14]. Instead of considering the propagation of light in a medium, they derived the analyticity of the scattering amplitude directly from the condition of microscopic causality, which states that commutators of field operators vanish when the points at which the operators are evaluated are separated by a space-like interval. This approach allowed Goldberger [15] soon thereafter to derive a very useful dispersion relation for the forward pion-nucleon scattering amplitude.

To see how to use the principle of microscopic causality, consider the forward scattering in the laboratory frame of a massless boson of any spin on an arbitrary target  $\alpha$  of mass  $m_\alpha > 0$  and  $\mathbf{p}_\alpha = 0$ . (This has important applications to the scattering not only of photons but also pions in the limit  $m_\pi = 0$ , to be discussed in Volume II.) By a repeated use of Eq. (10.3.4) or the Lehmann-Symanzik-Zimmermann theorem [3], the  $S$ -matrix element here is

$$S = \frac{1}{(2\pi)^3 \sqrt{4\omega\omega'} |N|^2} \lim_{k^2 \rightarrow 0} \lim_{k'^2 \rightarrow 0} \cdot \int d^4x \int d^4y e^{-ik' \cdot y} e^{ik \cdot x} (i\Box_y)(i\Box_x) \langle \alpha | T \{ A^\dagger(y), A(x) \} | \alpha \rangle. \quad (10.8.1)$$

Here  $k$  and  $k'$  are the initial and final boson four-momenta, with  $\omega = k^0$ ,  $\omega' = k'^0$ ;  $A(x)$  is any Heisenberg-picture operator with a non-vanishing matrix element  $\langle \Omega | A(x) | k \rangle = (2\pi)^{-3/2} (2\omega)^{-1/2} N e^{ik \cdot x}$  between the one-boson state  $|k\rangle$  and the vacuum; and  $N$  is the constant in this matrix element. In photon scattering  $A(x)$  would be one of the transverse components of the electromagnetic field, while for massless pion scattering it would be a pseudoscalar function of hadron fields. The differential operators  $-i\Box_x$  and  $-i\Box_y$  are inserted to supply factors of  $ik'^2$  and  $ik^2$  that are needed to cancel the external line boson propagators. Letting these operators act on  $A^\dagger(y)$  and  $A(x)$ , we have

$$S = \frac{-1}{(2\pi)^3 \sqrt{4\omega\omega'} |N|^2} \lim_{k^2 \rightarrow 0} \lim_{k'^2 \rightarrow 0} \cdot \int d^4x \int d^4y e^{-ik' \cdot y} e^{ik \cdot x} \langle \alpha | T \{ J^\dagger(y), J(x) \} | \alpha \rangle + \text{ETC}, \quad (10.8.2)$$

where  $J(x) \equiv \Box_x A(x)$ , and ‘ETC’ denotes the Fourier transform of equal time commutator terms arising from the derivative acting on the step functions in the time-ordered product. The commutators of operators like  $A(x)$  and  $A^\dagger(y)$  (or their derivatives) vanish for  $x^0 = y^0$  unless  $\mathbf{x} = \mathbf{y}$ , so the ‘ETC’ term is the Fourier transform of a differential operator acting on  $\delta^4(x - y)$ , and is hence a polynomial function of the boson four-momenta. We are concerned here with the analytic properties of the  $S$ -matrix element, so the details of this polynomial will be irrelevant.

Using translation invariance, Eq. (10.8.2) gives the  $S$ -matrix element as  $S = -2\pi i \delta^4(k' - k) M(\omega)$ , where

$$M(\omega) = \frac{-i}{2\omega|N|^2} F(\omega), \quad (10.8.3)$$

$$F(\omega) \equiv \int d^4x e^{i\omega\ell \cdot x} \langle \alpha | T \{ J^\dagger(0), J(x) \} | \alpha \rangle + \text{ETC}, \quad (10.8.4)$$

it now being understood that  $k^\mu = \omega \ell^\mu$ , where  $\ell$  is a fixed four-vector with  $\ell^\mu \ell_\mu = 0$  and  $\ell^0 = 1$ .

The time-ordered product can be rewritten in terms of commutators in two different ways:

$$\begin{aligned} T \{ J^\dagger(0), J(x) \} &= \theta(-x^0) [J^\dagger(0), J(x)] + J(x) J^\dagger(0) \\ &= -\theta(x^0) [J^\dagger(0), J(x)] + J^\dagger(0) J(x). \end{aligned} \quad (10.8.5)$$

Correspondingly, we can write

$$F(\omega) = F_A(\omega) + F_+(\omega) = F_R(\omega) + F_-(\omega), \quad (10.8.6)$$

where

$$F_A(\omega) \equiv \int d^4x \theta(-x^0) \langle \alpha | [J^\dagger(0), J(x)] | \alpha \rangle e^{i\omega\ell \cdot x} + \text{ETC}, \quad (10.8.7)$$

$$F_R(\omega) \equiv - \int d^4x \theta(x^0) \langle \alpha | [J^\dagger(0), J(x)] | \alpha \rangle e^{i\omega\ell \cdot x} + \text{ETC}, \quad (10.8.8)$$

$$F_+(\omega) \equiv \int d^4x \langle \alpha | J(x) J^\dagger(0) | \alpha \rangle e^{i\omega\ell \cdot x}, \quad (10.8.9)$$

$$F_-(\omega) \equiv \int d^4x \langle \alpha | J^\dagger(0) J(x) | \alpha \rangle e^{i\omega\ell \cdot x}. \quad (10.8.10)$$

Microscopic causality tells us that the integrands in (10.8.7) and (10.8.8) vanish unless  $x^\mu$  is within the light cone, and the step functions then require that  $x^\mu$  is in the backward light cone in (10.8.7), so that  $x \cdot \ell > 0$ , and in the forward light cone in Eq. (10.8.8), so that  $x \cdot \ell < 0$ . We conclude that  $F_A(\omega)$  is analytic for  $\text{Im } \omega > 0$  and  $F_R(\omega)$  is analytic for  $\text{Im } \omega < 0$ , because in both cases the factor  $e^{i\omega\ell \cdot x}$  provides a cutoff for the integral over  $x^\mu$ . (Recall that the ‘ETC’ term is a polynomial, and hence analytic at all finite points.) We may then define a function

$$\mathcal{F}(\omega) \equiv \begin{cases} F_A(\omega), & \text{Im } \omega > 0, \\ F_R(\omega), & \text{Im } \omega < 0, \end{cases} \quad (10.8.11)$$

which is analytic in the whole complex  $\omega$  plane, except for a cut on the real axis.

We can now derive the dispersion relation. According to Eq. (10.8.6), the discontinuity of  $\mathcal{F}(\omega)$  across the cut at any real  $E$  is

$$\mathcal{F}(E + i\epsilon) - \mathcal{F}(E - i\epsilon) = F_A(E) - F_R(E) = F_-(E) - F_+(E). \quad (10.8.12)$$

If  $\mathcal{F}(\omega)/\omega^n$  vanishes as  $|\omega| \rightarrow \infty$  in the upper or lower half-plane, then by dividing by any polynomial  $P(\omega)$  of order  $n$  we obtain a function that vanishes for  $|\omega| \rightarrow \infty$  and is analytic

except for the cut on the real axis and poles at the zeroes  $\omega_\nu$  of  $P(\omega)$ . (Where  $\mathcal{F}(\omega)$  itself vanishes as  $|\omega| \rightarrow \infty$ , we can take  $P(\omega) = 1$ .) According to the method of residues, we then have

$$\frac{\mathcal{F}(\omega)}{P(\omega)} + \sum_\nu \frac{\mathcal{F}(\omega_\nu)}{(\omega_\nu - \omega)P'(\omega_\nu)} = \frac{1}{2\pi i} \oint_C \frac{\mathcal{F}(z) dz}{(z - \omega)P(z)}, \quad (10.8.13)$$

where  $\omega$  is any point off the real axis, and  $C$  is a contour consisting of two segments: one running just above the real axis from  $-\infty + i\epsilon$  to  $+\infty + i\epsilon$  and then around a large semi-circle back to  $-\infty + i\epsilon$ , and the other just below the real axis from  $+\infty - i\epsilon$  to  $-\infty - i\epsilon$  and then around a large semi-circle back to  $+\infty - i\epsilon$ . Because the function  $\mathcal{F}(z)/P(z)$  vanishes for  $|z| \rightarrow \infty$ , we can neglect the contribution from the large semi-circles. Using Eq. (10.8.12), Eq. (10.8.13) becomes

$$\mathcal{F}(\omega) = Q(\omega) + \frac{P(\omega)}{2\pi i} \int_{-\infty}^{+\infty} \frac{F_-(E) - F_+(E)}{(E - \omega)P(E)} dE, \quad (10.8.14)$$

where  $Q(\omega)$  is the  $(n - 1)$ th-order polynomial

$$Q(\omega) = -P(\omega) \sum_\nu \frac{\mathcal{F}(\omega_\nu)}{(\omega_\nu - \omega)P'(\omega_\nu)}.$$

A dispersion relation of this form, with  $P(\omega)$  and  $Q(\omega)$  of order  $n$  and  $n - 1$  respectively, is said to have  $n$  *subtractions*. If we can take  $P = 1$  then  $Q = 0$ , and the dispersion relation is said to be unsubtracted.

If we now let  $\omega$  approach the real axis from above, Eq. (10.8.14) gives

$$F_A(\omega) = Q(\omega) + \frac{P(\omega)}{2\pi i} \int_{-\infty}^{+\infty} \frac{F_-(E) - F_+(E)}{(E - \omega - i\epsilon)P(E)} dE. \quad (10.8.15)$$

Recalling Eqs. (10.8.6) and (3.1.25), this is

$$\begin{aligned} F(\omega) &= Q(\omega) + \frac{1}{2}F_-(\omega) + \frac{1}{2}F_+(\omega) \\ &\quad + \frac{P(\omega)}{2\pi i} \int_{-\infty}^{+\infty} \frac{F_-(E) - F_+(E)}{(E - \omega)P(E)} dE \end{aligned} \quad (10.8.16)$$

with  $1/(E - \omega)$  now interpreted as the principal value function  $\mathcal{P}/(E - \omega)$ .

This result is useful because the functions  $F_\pm(E)$  may be expressed in terms of measurable cross-sections. Summing over a complete set of multi-particle intermediate states  $\beta$  in Eqs. (10.8.9) and (10.8.10) (including integrations over the momenta of the particles in  $\beta$ ) and using translation invariance again, we have

$$F_+(E) = (2\pi)^4 \sum_\beta \left| {}_{\text{out}}\langle \beta | J(0)^\dagger | \alpha \rangle \right|^2 \delta^4(-p_\alpha + E\ell + p_\beta), \quad (10.8.17)$$

$$F_-(E) = (2\pi)^4 \sum_\beta | {}_{\text{out}}\langle \beta | J(0) | \alpha \rangle |^2 \delta^4(p_\alpha + E\ell - p_\beta). \quad (10.8.18)$$



But the matrix elements for the absorption of the massless scalar boson  $B$  in  $B + \alpha \rightarrow \beta$  or its antiparticle  $B^c$  in  $B^c + \alpha \rightarrow \beta$  are

$$-2i\pi M_{B^c+\alpha\rightarrow\beta} = \frac{(2\pi)^4}{(2\pi)^{3/2}\sqrt{2E_{B^c}}N} \text{out} \langle \beta | J^\dagger(0) | \alpha \rangle, \quad (10.8.19)$$

$$-2i\pi M_{B+\alpha\rightarrow\beta} = \frac{(2\pi)^4}{(2\pi)^{3/2}\sqrt{2E_B}N} \text{out} \langle \beta | J(0) | \alpha \rangle. \quad (10.8.20)$$

Comparing with Eq. (3.4.15), we see that  $F_\pm(E)$  may be expressed in terms of total cross-sections<sup>121</sup> at energies  $\mp E$ :

$$F_+(E) = \theta(-E) \frac{2|E||N|^2}{(2\pi)^3} \sigma_{\alpha+B^c}(|E|), \quad (10.8.21)$$

$$F_-(E) = \theta(E) \frac{2E|N|^2}{(2\pi)^3} \sigma_{\alpha+B}(E). \quad (10.8.22)$$

The scattering amplitude (10.8.3) is now, for real  $\omega > 0$ ,

$$M(\omega) = \frac{-iQ(\omega)}{2\omega|N|^2} - \frac{i}{2(2\pi)^3} \sigma_{\alpha+B}(\omega) - \frac{P(\omega)}{\omega(2\pi)^4} \int_0^\infty \left[ \frac{\sigma_{\alpha+B}(E)}{(E-\omega)P(E)} + \frac{\sigma_{\alpha+B^c}(E)}{(E+\omega)P(-E)} \right] E dE. \quad (10.8.23)$$

It is more usual to express this dispersion relation in terms of the amplitude  $f(\omega)$  for forward scattering in the laboratory frame, defined so that the laboratory frame differential cross-section in the forward direction is  $|f(\omega)|^2$ . This amplitude is given in terms of  $M(\omega)$  by  $f(\omega) = -4\pi^2\omega M(\omega) = 2\pi^2 i F(\omega)/|N|^2$ , so Eq. (10.8.23) now reads

$$f(\omega) = R(\omega) + \frac{i\omega}{4\pi} \sigma_{\alpha+B}(\omega) + \frac{P(\omega)}{4\pi^2} \int_0^\infty \left[ \frac{\sigma_{\alpha+B}(E)}{(E-\omega)P(E)} + \frac{\sigma_{\alpha+B^c}(E)}{(E+\omega)P(-E)} \right] E dE,$$

where  $R(\omega) \equiv 2i\pi^2 Q(\omega)/|N|^2$ . The optical theorem (3.6.4) tells us that the second term on the right-hand side equals  $i \text{Im} f(\omega)$ , so this can just as well be written in the more conventional form

$$\text{Re} f(\omega) = R(\omega) + \frac{P(\omega)}{4\pi^2} \int_0^\infty \left[ \frac{\sigma_{\alpha+B}(E)}{(E-\omega)P(E)} + \frac{\sigma_{\alpha+B^c}(E)}{(E+\omega)P(-E)} \right] E dE. \quad (10.8.24)$$

In particular, we see that  $R(\omega)$  is real if we choose  $P(\omega)$  real.

The forward scattering amplitude also satisfies an important symmetry condition. By changing the integration variable in Eqs. (10.8.7) and (10.8.8) from  $x$  to  $-x$  and then using the translation-invariance property

$$\langle \alpha | [J^\dagger(0), J(-x)] | \alpha \rangle = \langle \alpha | [J^\dagger(x), J(0)] | \alpha \rangle$$

<sup>121</sup>In some cases where selection rules allow the transition  $\alpha \rightarrow \alpha + B$  and  $\alpha \rightarrow \alpha + B^c$ , the functions  $F_\pm(E)$  also contain terms proportional to  $\delta(E)$  arising from the contribution of the one-particle state  $\alpha$  in the sum over intermediate states  $\beta$ . This does not occur for transversely polarized photons, or for pseudoscalar pions in the limit  $m_\pi \rightarrow 0$ .

we see that for  $\text{Im } \omega \leq 0$ ,  $F_A(-\omega)$  is the same as  $F_R(\omega)$ , except for an interchange of  $J$  with  $J^\dagger$ . That is,

$$F_A(-\omega) = F_R^c(\omega) \quad \text{for} \quad \text{Im } \omega \leq 0,$$

where a superscript  $c$  indicates that the amplitude refers to the scattering of the antiparticle  $B^c$  on  $\alpha$ . (We leave it to the reader to show that this relation is not upset by the equal-time commutator terms in Eqs. (10.8.7) and (10.8.8).) In the same way, we find

$$F_R(-\omega) = F_A^c(\omega) \quad \text{for} \quad \text{Im } \omega \geq 0,$$

and for real  $\omega$

$$F_\pm(-\omega) = F_\mp^c(\omega).$$

Using these relations in (10.8.6), and recalling that  $f(\omega)$  is proportional to  $F(\omega)$ , we find the *crossing symmetry* relation, that for real  $\omega$

$$f(-\omega) = f^c(\omega). \quad (10.8.25)$$

We are free to take  $P(\omega)$  as any polynomial of sufficiently high order, but  $R(\omega)$  then depends not only on  $P(\omega)$  but also on the values of  $\mathcal{F}(\omega)$  at the zeroes of  $P(\omega)$ . For  $P(\omega)$  real and of  $n$ th order, the only free parameters in Eq. (10.8.16) are the  $n$  real coefficients in the real  $(n-1)$ th order polynomial  $R(\omega)$ . Hence Eq. (10.8.16) contains just  $n$  unknown real independent constants, the coefficients in the polynomial  $R(\omega)$  for a given  $P(\omega)$ . We therefore wish to take the order  $n$  of the otherwise arbitrary polynomial  $P(\omega)$  to be as small as possible.

We might try taking  $P(\omega) = 1$ , but this doesn't work. The analysis of Section 3.7 suggests that the forward scattering amplitude should grow like  $\omega$  or perhaps as fast as  $\omega \ln^2 \omega$ . In this case for  $f(\omega)/P(\omega)$  to vanish as  $\omega \rightarrow 0$ , it is sufficient to take  $P(\omega)$  as a second order polynomial, so that  $R(\omega)$  is linear in  $\omega$ . Choosing  $P(E) = E^2$  for convenience, Eq. (10.8.24) then becomes

$$\text{Re } f(\omega) = a + b\omega + \frac{\omega^2}{4\pi^2} \int_0^\infty \left[ \frac{\sigma_{\alpha+B}(E)}{E-\omega} + \frac{\sigma_{\alpha+B^c}(E)}{E+\omega} \right] \frac{dE}{E}, \quad (10.8.26)$$

with  $a$  and  $b$  unknown real constants. The crossing symmetry condition (10.8.25) tells us that the corresponding constants in the dispersion relation for the antiparticle scattering amplitude  $f^c(\omega)$  are

$$a^c = a, \quad b^c = -b. \quad (10.8.27)$$

If we assume for instance that the cross-sections  $\sigma_{\alpha+B}(E)$  and  $\sigma_{\alpha+B^c}(E)$  behave for  $E \rightarrow \infty$  as different constants times  $(\ln E)^r$ , then (10.8.26) would give

$$\text{Re } f(\omega) \sim [\sigma_{\alpha+B}(\omega) - \sigma_{\alpha+B^c}(\omega)] \ln \omega \sim \omega (\ln \omega)^{r+1} \quad (10.8.28)$$

so the real part of the scattering amplitude would grow faster than the imaginary part by a factor  $\ln \omega$ . This is implausible; we saw in Section 3.7 that the real part of the forward scattering amplitude is expected to become much smaller than the imaginary part

for  $\omega \rightarrow \infty$ , as confirmed by experiment. We conclude that if  $\sigma_{\alpha+B}(E)$  and  $\sigma_{\alpha+B^c}(E)$  do behave for  $E \rightarrow \infty$  as constants times  $(\ln E)^r$  then the constants must be the same. Because we are concerned here with the high-energy limit, this result does not depend on the assumption that  $B$  is a massless boson, so in the same sense, *the ratio of the cross-sections of any particle and its antiparticle on a fixed target should approach unity at high energy*. This result is a somewhat generalized version of what is known as Pomeranchuk's theorem [16]. (Pomeranchuk considered only the case  $r = 0$ , while Section 3.7 and the observed behavior of cross-sections both suggest that  $r = 2$  is more likely.)

Although Pomeranchuk took his estimates of the asymptotic behavior of scattering amplitudes from arguments like those of Section 3.7, today high energy behavior is usually inferred from Regge pole theory [17]. It would take us too far from our subject to go into details about this; suffice it to say that for hadronic processes the asymptotic behavior of  $f(\omega)$  as  $\omega$  goes to infinity is a sum over terms proportional to  $\omega^{\alpha_n(0)}$ , where  $\alpha_n(t)$  are a set of 'Regge trajectories', each representing the exchange of an infinite family of different one-hadron states in the collision process. The leading trajectory (actually, a complex of many trajectories) in

hadron scattering is the 'Pomeron,' for which  $\alpha(0)$  is close to unity. It is this trajectory that gives cross-sections that are approximately constant for  $E \rightarrow \infty$ . According to Pomeranchuk's theorem, the Pomeron couples equally to any hadron and its antiparticle. We can estimate  $\alpha_n(0)$  for the lower Regge trajectories from the spectrum of hadronic states. A necessary though not sufficient condition [18] for a mesonic resonance of spin  $j$  to occur at a mass  $m$  is that  $m^2$  equals the value of  $t$  where one of trajectories  $\alpha_n(t)$  equals  $j$ . Apart from the Pomeron, the leading trajectory in pion-nucleon scattering is that on which we find the  $j = 1$   $\rho$  meson at  $m = 770$  MeV, the  $j = 3$   $g$  meson at  $m = 1690$  MeV, and a  $j = 5$  meson at  $m = 2350$  MeV. Extrapolating these values of  $\alpha(t)$  down to  $t = 0$ , we can estimate that this trajectory has  $\alpha(0) \approx 0.5$ . This trajectory couples with opposite sign to  $\pi^+$  and  $\pi^-$ , so for pion-nucleon scattering we expect  $f(\omega) - f^c(\omega)$  to behave roughly like  $\sqrt{\omega}$ .

For photon scattering there is no distinction between  $B$  and  $B^c$ , so here Eq. (10.8.27) gives  $b = 0$ , and Eq. (10.8.26) reads

$$f(\omega) = a + \frac{\omega^2}{2\pi^2} \int_0^\infty \frac{\sigma(E)}{E^2 - \omega^2} dE. \quad (10.8.29)$$

This is essentially the original Kramers-Kronig [13] relation. As we shall see in Section 13.5, for a target of charge  $e$  and mass  $m$  the constant  $a$  has the known value  $\text{Re } f(0) = -e^2/m$ .

### Weinberg Exercises

- W.10.1.** Consider a neutral vector field  $v_\mu(x)$ . What conditions must be imposed on the sum  $\Pi_{\mu\nu}^*(k)$  of one-particle-irreducible graphs with two external vector field lines in order that the field should be properly renormalized and describe a particle of renormalized mass  $m$ ? How do we split the free-field and interacting terms in the Lagrangian to achieve this?

- W.10.2.** Derive the generalized Ward identity that governs the electromagnetic vertex function of a charged scalar field.
- W.10.3.** What is the most general form of the matrix element  $\langle p_2, \sigma_2 | J^\mu(x) | p_1, \sigma_1 \rangle$  of the electromagnetic current  $J^\mu(x)$  between two spin  $\frac{1}{2}$  one-particle states of *different* masses  $m_1$  and  $m_2$  and equal parity? What if the parities were opposite? (Assume parity conservation throughout.)
- W.10.4.** Derive the spectral (Källén–Lehmann) representation for the vacuum expectation value  $\langle \Omega | T \{ J^\mu(x) J^\nu(y) \} | \Omega \rangle$ , where  $J^\mu(x)$  is a complex conserved current.
- W.10.5.** Derive the spectral (Källén–Lehmann) representation for the vacuum expectation value  $\langle \Omega | T \{ \psi_n(x) \bar{\psi}_m(y) \} | \Omega \rangle$ , where  $\psi(x)$  is a Dirac field.
- W.10.6.** Without using any assumptions about the asymptotic behavior of the scattering amplitude or cross-sections, show that it is impossible for forward photon scattering amplitudes to satisfy unsubtracted dispersion relations.
- W.10.7.** Derive the spectral (Källén–Lehmann) representation for a complex scalar field by using the methods of dispersion theory.
- W.10.8.** Use dispersion theory and the results of Section 8.7 to calculate the amplitude for forward photon–electron scattering in the electron rest frame to order  $e^4$ .

### Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

- W.10.1.** Introduce the transverse and longitudinal projectors

$$P_{\mu\nu}^T(k) = \eta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}, \quad P_{\mu\nu}^L(k) = \frac{k_\mu k_\nu}{k^2}.$$

The inverse free Proca propagator is

$$K_{\mu\nu}^{(0)} = (k^2 + m^2)P_{\mu\nu}^T + m^2 P_{\mu\nu}^L.$$

Lorentz invariance permits the one-particle-irreducible two-point function to be written

$$\Pi_{\mu\nu}^*(k) = \Pi_T(k^2)P_{\mu\nu}^T + \Pi_L(k^2)P_{\mu\nu}^L.$$

Consequently the complete propagator has a transverse denominator  $k^2 + m^2 - \Pi_T(k^2)$  and a longitudinal denominator  $m^2 - \Pi_L(k^2)$ . A spin-one particle of physical mass  $m$  and unit pole residue therefore requires

$$\boxed{\Pi_T(-m^2) = 0, \quad \Pi_L'(-m^2) = 0.}$$

The longitudinal denominator must be nonzero at  $k^2 = -m^2$ ; otherwise an additional scalar state would occur at the same mass. More generally, any zero of  $m^2 - \Pi_L(k^2)$

represents an extra spin-zero excitation and must be absent if the field is to describe only the stated particle.

Starting from

$$\mathcal{L}_B = -\frac{1}{4}v_{B\mu\nu}v_B^{\mu\nu} - \frac{1}{2}m_B^2 v_{B\mu}v_B^\mu - V_B(v_B), \quad v_{B\mu\nu} = \partial_\mu v_{B\nu} - \partial_\nu v_{B\mu},$$

put

$$v_{B\mu} = \sqrt{Z} v_\mu, \quad m^2 = m_B^2 + \delta m^2.$$

Then one convenient split is

$$\mathcal{L}_0 = -\frac{1}{4}v_{\mu\nu}v^{\mu\nu} - \frac{1}{2}m^2 v_\mu v^\mu$$

and

$$\mathcal{L}_1 = -\frac{1}{4}(Z-1)v_{\mu\nu}v^{\mu\nu} - \frac{1}{2}[(Z-1)m^2 - Z\delta m^2]v_\mu v^\mu - V_B(\sqrt{Z}v).$$

The two local counterterms are chosen so that the transverse loop self-energy obeys the two boxed on-shell conditions.

**W.10.2.** Let  $\Phi$  be a scalar of charge  $q$ , and define its complete propagator and amputated current vertex by

$$\begin{aligned} -i\Delta'(k)\delta^4(k-\ell) &= \int d^4y d^4z e^{-ik\cdot y + i\ell\cdot z} \langle \Omega | T \{ \Phi(y) \Phi^\dagger(z) \} | \Omega \rangle, \\ \int d^4x d^4y d^4z e^{-ip\cdot x - ik\cdot y + i\ell\cdot z} \langle \Omega | T \{ J^\mu(x) \Phi(y) \Phi^\dagger(z) \} | \Omega \rangle \\ &= -iq \Delta'(k) \Gamma^\mu(k, \ell) \Delta'(\ell) \delta^4(p+k-\ell). \end{aligned}$$

Differentiating the time-ordered product gives contact terms. Current conservation and the equal-time charge commutators

$$\begin{aligned} [J^0(\mathbf{x}, t), \Phi(\mathbf{y}, t)] &= -q\Phi(\mathbf{y}, t)\delta^3(\mathbf{x}-\mathbf{y}), \\ [J^0(\mathbf{x}, t), \Phi^\dagger(\mathbf{y}, t)] &= q\Phi^\dagger(\mathbf{y}, t)\delta^3(\mathbf{x}-\mathbf{y}) \end{aligned}$$

give

$$\partial_\mu^x T \{ J^\mu(x) \Phi(y) \Phi^\dagger(z) \} = -q\delta^4(x-y) T \{ \Phi(y) \Phi^\dagger(z) \} + q\delta^4(x-z) T \{ \Phi(y) \Phi^\dagger(z) \}.$$

Fourier transformation and amputation therefore yield the generalized scalar Ward identity

$$(\ell-k)_\mu \Gamma^\mu(k, \ell) = i\Delta'^{-1}(k) - i\Delta'^{-1}(\ell).$$

At zero momentum transfer this becomes the ordinary Ward identity

$$\Gamma^\mu(k, k) = -i \frac{\partial}{\partial k_\mu} \Delta'^{-1}(k).$$

For a free scalar  $\Delta'^{-1}(k) = k^2 + m^2$ , so  $\Gamma^\mu(k, \ell) = -i(k+\ell)^\mu$ , which verifies both formulas and fixes the convention-dependent factors of  $i$ .

**W.10.3.** Write  $q = p_2 - p_1$ ,  $P = p_2 + p_1$ , and suppress the state-normalization factor. For states of equal intrinsic parity, parity and Lorentz covariance allow

$$\langle p_2, \sigma_2 | J^\mu(0) | p_1, \sigma_1 \rangle = \bar{u}_2 [\gamma^\mu F(q^2) + P^\mu G(q^2) + q^\mu H(q^2)] u_1.$$

The on-shell equations imply

$$\bar{u}_2 \not{q} u_1 = i(m_2 - m_1) \bar{u}_2 u_1, \quad q \cdot P = -(m_2^2 - m_1^2).$$

Thus current conservation imposes one relation,

$$i(m_2 - m_1)F - (m_2^2 - m_1^2)G + q^2 H = 0,$$

leaving two independent form factors. A manifestly transverse basis is

$$\bar{u}_2 \left[ \left( \gamma^\mu - \frac{q^\mu \not{q}}{q^2} \right) F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{m_1 + m_2} F_2(q^2) \right] u_1,$$

where  $\sigma^{\mu\nu} \equiv \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ ; harmless factors of  $i$  may instead be absorbed in the definitions of  $F_1, F_2$ . For unequal masses, regularity at  $q^2 = 0$  requires  $F_1(0) = 0$ . This is also the statement that the conserved charge cannot connect distinct mass eigenstates.

If the two states have opposite intrinsic parity, the allowed matrix element is obtained by inserting  $\gamma_5$ :

$$\bar{u}_2 \left[ \left( \gamma^\mu - \frac{q^\mu \not{q}}{q^2} \right) \tilde{F}_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{m_1 + m_2} \tilde{F}_2(q^2) \right] \gamma_5 u_1.$$

Equivalently one may start with  $(\gamma^\mu \tilde{F} + P^\mu \tilde{G} + q^\mu \tilde{H})\gamma_5$ . In that basis the conservation relation contains  $m_2 + m_1$ , because  $\bar{u}_2 \not{q} \gamma_5 u_1 = i(m_2 + m_1) \bar{u}_2 \gamma_5 u_1$ . Again  $\tilde{F}_1(0) = 0$  for a nonsingular transition matrix element.

**W.10.4.** Define

$$G^{\mu\nu}(q) = \int d^4x e^{-iq \cdot x} \langle \Omega | T \{ J^\mu(x) J^\nu(0) \} | \Omega \rangle.$$

Inserting a complete set of momentum eigenstates into the two Wightman functions shows that the nonlocal part is supported at  $p^0 > 0$ ,  $p^2 = -\mu^2$ . Lorentz covariance and  $p_\mu \langle \Omega | J^\mu(0) | n, p \rangle = 0$  force its tensor spectral density to be transverse:

$$\rho^{\mu\nu}(p) = (p^\mu p^\nu - p^2 \eta^{\mu\nu}) \rho(\mu^2).$$

Time ordering then gives, up to local contact terms,

$$G^{\mu\nu}(q) = (q^2 \eta^{\mu\nu} - q^\mu q^\nu) \left[ P(q^2) + \int_0^\infty \frac{\rho(\mu^2) d\mu^2}{q^2 + \mu^2 - i\epsilon} \right].$$

Here  $P$  is a subtraction polynomial; changing it corresponds to changing local terms in the product of the two currents. Such terms are unavoidable in general because differentiating a time-ordered product can produce equal-time commutators.

For a genuinely complex current, the spectral function in the displayed  $JJ$  correlator need not be nonnegative: it is a sum of products  $\langle \Omega | J^\mu | n \rangle \langle n | J^\nu | \Omega \rangle$ , not absolute squares. The mixed correlator  $\langle \Omega | T \{ J^\mu(x) J^{\nu\dagger}(0) \} | \Omega \rangle$  has the same form with a non-negative transverse spectral measure. If a stable spin-one state of mass  $M$  couples to the current, its contribution is a positive pole term  $Z\delta(\mu^2 - M^2)$  in the mixed spectral function. Scalar intermediate states are excluded by current conservation except for possible massless or contact contributions.

**W.10.5.** Insert a complete set between  $\psi(x)$  and  $\bar{\psi}(y)$ , use translation invariance, and organize the resulting spinor matrix by Lorentz covariance. If parity is conserved, only a scalar and a vector Dirac structure occur. The exact momentum-space propagator therefore has the form

$$S'(p) = \int_0^\infty d\mu^2 \frac{-i\not{p} \rho_1(\mu^2) + \rho_2(\mu^2)}{p^2 + \mu^2 - i\epsilon}.$$

It is often useful to resolve the spectral measure into intermediate states of the two rest-frame parities:

$$\rho_1(\mu^2) = \rho_+(\mu^2) + \rho_-(\mu^2), \quad \rho_2(\mu^2) = \mu[\rho_+(\mu^2) - \rho_-(\mu^2)].$$

Hilbert-space positivity gives

$$\rho_\pm(\mu^2) \geq 0, \quad \boxed{\rho_1(\mu^2) \geq 0, \quad |\rho_2(\mu^2)| \leq \mu \rho_1(\mu^2)}.$$

These inequalities follow equally by evaluating the spectral spinor matrix in the rest frame and requiring its two energy-projector eigenvalues to be nonnegative.

If  $\psi$  interpolates a stable fermion of mass  $m$ , then

$$\rho_1(\mu^2) = Z\delta(\mu^2 - m^2) + \rho_1^c(\mu^2), \quad \rho_2(\mu^2) = Zm\delta(\mu^2 - m^2) + \rho_2^c(\mu^2),$$

and hence

$$S'(p) = Z \frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} + \text{continuum}.$$

For a canonically normalized bare field, the equal-time anticommutator gives the sum rule  $\int_0^\infty \rho_1(\mu^2) d\mu^2 = 1$ , subject to the usual ultraviolet regulator qualification.

**W.10.6.** Assume, for contradiction, that the forward amplitude for a real photon on a stable target obeys an unsubtracted dispersion relation. A photon is its own antiparticle, so crossing makes the spin-averaged forward amplitude even in the laboratory energy. The optical theorem and the unsubtracted relation would then give at zero energy

$$\text{Re } f(0) = \frac{1}{2\pi^2} \int_0^\infty \sigma_{\gamma\alpha}(E) dE.$$

The integrand is a total cross-section and is nonnegative. It is not identically zero for a target that can absorb or scatter photons, so the right-hand side is positive.

The low-energy theorem gives a contradictory result. For a target of mass  $m$  and electric charge  $Qe$ , the forward Thomson amplitude is

$$f(0) = -\frac{\alpha Q^2}{m} \leq 0.$$

For a neutral target the leading constant is zero, which is still inconsistent with a strictly positive integral whenever the target has a nonzero photoabsorption cross-section. Thus an unsubtracted relation cannot be valid.

No assumption about the large-energy behavior of either  $f$  or  $\sigma$  entered the contradiction. The conclusion is instead that the contour at infinity cannot be discarded without additional high-energy information. At least one subtraction constant—and in the general forward relation two subtractions—must retain precisely the low-energy information that the positive absorptive integral cannot reproduce.

**W.10.7.** Regard the complete scalar propagator as a function of  $z = -p^2$ . Microcausality makes it analytic in the complex  $z$ -plane away from the positive real axis, while unitarity gives the discontinuity

$$\text{Disc } \Delta'(z) \equiv \Delta'(z + i0) - \Delta'(z - i0) = 2\pi i \rho(z), \quad \rho(z) \geq 0.$$

Apply Cauchy's theorem to

$$\frac{\Delta'(z')}{(z' - z)(z' + \mu_0^2)^n},$$

where  $n$  is large enough that the circle at infinity vanishes. Collapsing the contour onto the cut yields

$$\Delta'(p) = P_{n-1}(p^2) + (-p^2 + \mu_0^2)^n \int_0^\infty \frac{\rho(\mu^2) d\mu^2}{(\mu^2 + \mu_0^2)^n (p^2 + \mu^2 - i\epsilon)}.$$

The polynomial  $P_{n-1}$  contains the  $n$  subtraction constants and represents local contact terms.

For a canonically normalized scalar field, the equal-time commutator fixes the leading large- $|p^2|$  behavior and gives  $\int \rho(\mu^2) d\mu^2 = 1$ . The unsubtracted form is then

$$\Delta'(p) = \int_0^\infty \frac{\rho(\mu^2) d\mu^2}{p^2 + \mu^2 - i\epsilon}.$$

A stable particle contributes

$$\rho(\mu^2) = Z\delta(\mu^2 - m^2) + \sigma(\mu^2), \quad \sigma \geq 0,$$

so the isolated pole has residue  $Z > 0$ , while the continuum begins at the lightest multiparticle threshold. This is the Källén–Lehmann representation obtained directly from dispersion theory.



**W.10.8.** Let  $f(\omega)$  be the spin-averaged forward amplitude in the electron rest frame, normalized so that the forward differential cross-section is  $|f|^2$ . The renormalized low-energy theorem fixes

$$f(0) = -\frac{\alpha}{m};$$

there is no independent order- $e^4$  correction to this subtraction constant. Crossing makes the spin-independent amplitude even in  $\omega$ . The twice-subtracted dispersion relation and optical theorem therefore give

$$f(\omega) = -\frac{\alpha}{m} + \frac{\omega^2}{2\pi^2} \mathcal{P} \int_0^\infty \frac{\sigma_{\text{KN}}(E)}{E^2 - \omega^2} dE + \frac{i\omega}{4\pi} \sigma_{\text{KN}}(\omega) + O(e^6).$$

Everything after the Thomson term is of order  $e^4$ . The required absorptive input is the tree-level total Klein–Nishina cross-section. With  $x = E/m$ ,

$$\sigma_{\text{KN}}(E) = \frac{2\pi\alpha^2}{m^2} \left[ \frac{1+x}{x^3} \left\{ \frac{2x(1+x)}{1+2x} - \ln(1+2x) \right\} + \frac{\ln(1+2x)}{2x} - \frac{1+3x}{(1+2x)^2} \right].$$

This follows by integrating the differential result of Section 8.7. The imaginary part in the boxed expression is exactly  $\text{Im } f = \omega \sigma_{\text{KN}}/(4\pi)$ , so it satisfies the optical theorem. The real part is a single convergent principal-value integral and therefore completes the order- $e^4$  calculation without a direct loop integration.

There is an infrared subtlety at  $\omega = 0$ . Because the elastic electron–photon cut begins there and  $\sigma_{\text{KN}}(E) = \sigma_T[1 - 2E/m + O(E^2/m^2)]$ , the dispersive integral may not be expanded term by term in  $\omega^2/E^2$ . Its low-energy boundary value instead contains

$$\text{Im } f^{(4)}(\omega) = \frac{\sigma_T}{4\pi} \omega + O(\omega^2),$$

$$\text{Re } f^{(4)}(\omega) = -\frac{\sigma_T}{\pi^2 m} \omega^2 \ln \frac{m}{|\omega|} + C\omega^2 + O(\omega^4 \ln |\omega|),$$

where the finite constant  $C$  is fixed by the full Klein–Nishina integral. The logarithm is the expected nonanalyticity from a massless threshold.

## Supplementary Exercises

### S.10.1. Positive Spectral Measures, Pole Weights, and Sum Rules

(Inspired by Mark Srednicki, *Quantum Field Theory, Chapter 13* / Mark Srednicki, *Quantum Field Theory, Chapters 13 and 15*.)

- (a) For a Hermitian scalar field with vanishing vacuum expectation value, insert exact energy–momentum eigenstates into its Wightman two-point function. Derive a Lorentz-invariant spectral density, prove that it is nonnegative, and identify the part produced by a stable one-particle state.

- (b) Suppose a scalar field interpolates a stable particle of mass  $m$ , while the lightest multiparticle state with the same quantum numbers has invariant mass  $M_{\text{th}}$ . Determine the support of the pole and continuum pieces of its spectral density and the resulting analytic structure of the exact propagator.
- (c) Starting from the Källén–Lehmann representation of a canonically normalized scalar field, take the equal-time derivative of its commutator. Derive the normalization sum rule for the spectral density and use it to bound the isolated pole residue.

### S.10.2. The Thermal Lehmann Representation

(Adapted from John McGreevy, Physics 215B Quantum Field Theory, Winter 2025, Assignment 7, Problem 3, p. 1.)

In the vacuum, a scalar operator has the time-ordered two-point function

$$-iD(x) = \langle 0 | T \{ O(x) O^\dagger(0) \} | 0 \rangle.$$

Derive the corresponding spectral representation when the system is in thermal equilibrium at nonzero temperature  $T = \beta^{-1}$ , starting from

$$\begin{aligned} -iD_\beta(x) &\equiv \text{tr} \left( \frac{e^{-\beta H}}{Z_\beta} T \{ O(x) O^\dagger(0) \} \right) \\ &= \frac{1}{Z_\beta} \sum_n e^{-\beta E_n} \langle n | T \{ O(x) O^\dagger(0) \} | n \rangle, \quad Z_\beta \equiv \text{tr} e^{-\beta H}. \end{aligned}$$

Make the thermal spectral weight and its detailed-balance relation explicit, and verify that the limit  $\beta \rightarrow \infty$  reproduces the vacuum representation. You may take  $O = O^\dagger$ .

### S.10.3. The Ward–Takahashi Identity in Euclidean QED

(Adapted from University of Cambridge, Mathematical Tripos Part III, Advanced Quantum Field Theory, Paper 304, 2022, Question 3, p. 4.)

For this problem only, work in Euclidean spacetime, with indices contracted using  $\delta_{\mu\nu}$  and Euclidean gamma matrices satisfying  $\{\gamma_E^\mu, \gamma_E^\nu\} = 2\delta^{\mu\nu}$ . The QED Lagrangian is

$$\mathcal{L}_E = \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} (\not{D}_E + m) \psi, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad \not{D}_E = \gamma_E^\mu (\partial_\mu - ieA_\mu).$$

- (a) Show that the Lagrangian is invariant under the global  $U(1)$  transformations

$$\psi(x) \mapsto e^{i\alpha} \psi(x), \quad \bar{\psi}(x) \mapsto \bar{\psi}(x) e^{-i\alpha}, \quad A_\mu(x) \mapsto A_\mu(x).$$

Use this invariance to show that the current

$$j^\mu(x) = \bar{\psi}(x) \gamma_E^\mu \psi(x)$$

is conserved.

- (b) Derive the Schwinger–Dyson equation

$$\partial_\mu \langle j^\mu(x) \psi(x_1) \bar{\psi}(x_2) \rangle = - \left[ \delta^{(4)}(x - x_1) - \delta^{(4)}(x - x_2) \right] \langle \psi(x_1) \bar{\psi}(x_2) \rangle.$$

- (c) Use the preceding result to derive the Ward–Takahashi identity involving the exact vertex function  $V_3^\mu$  and exact fermion propagator  $G(\not{p})$ :

$$iq_\mu V_3^\mu(q, p_1, p_2) = ie \left[ G^{-1}(\not{p}_1) - G^{-1}(\not{p}_1 - \not{q}) \right], \quad p_2 = p_1 - q.$$

Here  $\not{p} = \gamma_E^\mu p_\mu$ , and the charge  $e$  is included in the normalization of  $V_3^\mu$ .

- (d) Explain what the Ward–Takahashi identity implies for renormalization of terms in the Lagrangian, for example in the on-shell renormalization scheme. Comment on what, if any, conclusions continue to hold nonperturbatively.

#### S.10.4. Charge Conjugation and Odd-Photon Selection Rules

(Adapted from *University of Cambridge, Mathematical Tripos Part III, Quantum Field Theory, Paper 301, 2024, Question 3, p. 4.*)

- (a) Scalar QED consists of a complex scalar  $\phi$  of mass  $M$  and an electromagnetic field  $A_\mu$ , coupled only by minimal coupling with coupling  $e$ . Write its Lagrangian  $\mathcal{L}_{\text{SQED}}$ . Likewise write  $\mathcal{L}_{\text{QED}}$  for a Dirac field  $\psi$  of mass  $m$  minimally coupled to  $A_\mu$ . Give the local  $U(1)$  gauge transformations of both theories and derive, in each case, the Noether current  $j^\mu$  for the global symmetry obtained by taking the gauge parameter constant.
- (b) Let the linear, unitary charge-conjugation operator  $\hat{C}$  act on the scalar field by

$$\hat{C}^{-1} \phi(x) \hat{C} = \phi^\dagger(x), \quad \hat{C}^{-1} \phi^\dagger(x) \hat{C} = \phi(x), \quad \hat{C}^\dagger \hat{C} = \mathbf{1}.$$

Determine the necessary transformation of  $A_\mu$ , and show that  $\mathcal{L}_{\text{SQED}}$  is invariant. Operator-ordering issues such as normal ordering may be ignored.

- (c) For the Dirac field, take

$$\hat{C}^{-1} \psi(x) \hat{C} = \mathbf{C} \bar{\psi}(x)^\top, \quad \hat{C}^{-1} \bar{\psi}(x) \hat{C} = \psi(x)^\top \mathbf{C},$$

where the  $4 \times 4$  matrix  $\mathbf{C}$  obeys

$$\mathbf{C}^\top = \mathbf{C}^{-1} = -\mathbf{C}, \quad \mathbf{C}^{-1}(\gamma^\mu)^\top \mathbf{C} = -\gamma^\mu.$$

Using the transformation of  $A_\mu$  found in the preceding part, prove that  $\mathcal{L}_{\text{QED}}$  is invariant. You may use

$$\bar{\psi} M \psi = -\psi^\top M^\top \bar{\psi}^\top$$

for any spinor matrix  $M$ , and again may ignore normal-ordering issues.

- (d) Assume that the vacuum  $|\Omega\rangle$  of either theory is invariant under charge conjugation. Without using Feynman diagrams, prove to all orders in  $e$  that, for odd  $n$ ,

$$\langle\Omega|T\{A^{\mu_1}(x_1)\cdots A^{\mu_n}(x_n)\}|\Omega\rangle=0, \quad \langle\Omega|T\{j^{\mu_1}(x_1)\cdots j^{\mu_n}(x_n)\}|\Omega\rangle=0.$$

What do these identities imply physically for photon scattering in scalar and spinor QED?

### S.10.5. Scalar and Pseudoscalar Decays from Schwinger–Dyson and LSZ

(Adapted from *University of Cambridge*, Mathematical Tripos Part III, Quantum Field Theory, Paper 301, 2025, Question 3, pp. 4–5.)

Consider in four dimensions a massive real scalar field  $\phi(x)$  and a massive Dirac field  $\psi(x)$ . In the volume’s mostly-plus conventions, the free Lagrangian is

$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}M^2\phi^2 - \bar{\psi}(\gamma^\mu\partial_\mu + m)\psi,$$

where  $M$  and  $m$  are the scalar and fermion masses.

- (a) Consider first  $\mathcal{L}_0 + \mathcal{L}_1$ , with

$$\mathcal{L}_1 = -\lambda\phi\bar{\psi}\psi.$$

- i. Write down the classical equations of motion. State, without derivation, the three Schwinger–Dyson equations for this theory.
- ii. Following the LSZ approach, derive the  $S$ -matrix element for the decay  $\phi \rightarrow \psi\bar{\psi}$  by defining the asymptotic states and identifying the correlation function that enters the matrix element.
- iii. To leading order in  $\lambda$ , evaluate the connected contribution to the correlation function appearing in  $\phi \rightarrow \psi\bar{\psi}$  by using the Schwinger–Dyson equations. Hence evaluate the  $S$ -matrix element to leading order and identify the corresponding invariant amplitude  $\mathcal{A}_{s,s'}$ .
- iv. Compute the spin-summed/averaged squared matrix element

$$\mathcal{P} \equiv \frac{1}{4} \sum_{s,s'=1}^2 |\mathcal{A}_{s,s'}|^2,$$

where  $s$  and  $s'$  are the spins of the final particles.

- (b) Now consider instead  $\mathcal{L}_0 + \mathcal{L}_2$ , with

$$\mathcal{L}_2 = -ig\phi\bar{\psi}\gamma_5\psi, \quad \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3.$$

What is the  $S$ -matrix element for  $\phi \rightarrow \psi\bar{\psi}$ ? You need not repeat the derivation: state the appropriate modification of the answer in part (a)(iii). Evaluate the spin-summed/averaged squared matrix element in this case. Could an experiment distinguish the effects of  $\mathcal{L}_1$  and  $\mathcal{L}_2$ ?

You may use without proof the following free-field mode expansions and spinor identities. With  $p^0 > 0$ , define

$$\tilde{\omega}_{\mathbf{p}}^2 = M^2 + \mathbf{p}^2, \quad \omega_{\mathbf{p}}^2 = m^2 + \mathbf{p}^2, \quad \hat{p} \equiv -i\gamma^\mu p_\mu.$$

The last definition translates the source's mostly-minus slash to the volume convention. The mode expansions are

$$\begin{aligned} \phi(x) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\tilde{\omega}_{\mathbf{p}}}} \left[ a_{\mathbf{p}} e^{ip \cdot x} + a_{\mathbf{p}}^\dagger e^{-ip \cdot x} \right], \\ \psi(x) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \sum_{r=1}^2 \left[ b_{\mathbf{p}}^r u^r(\mathbf{p}) e^{ip \cdot x} + c_{\mathbf{p}}^{r\dagger} v^r(\mathbf{p}) e^{-ip \cdot x} \right]. \end{aligned}$$

For on-shell  $p$ ,

$$\begin{aligned} (\hat{p} - m)u^s(\mathbf{p}) &= 0, & (\hat{p} + m)v^s(\mathbf{p}) &= 0, \\ \sum_{s=1}^2 u^s(\mathbf{p}) \bar{u}^s(\mathbf{p}) &= \hat{p} + m, & \sum_{s=1}^2 v^s(\mathbf{p}) \bar{v}^s(\mathbf{p}) &= \hat{p} - m, \end{aligned}$$

and

$$\begin{aligned} u^{s'\dagger}(\mathbf{p})u^s(\mathbf{p}) &= v^{s'\dagger}(\mathbf{p})v^s(\mathbf{p}) = 2\omega_{\mathbf{p}}\delta^{ss'}, \\ \bar{u}^{s'}(\mathbf{p})u^s(\mathbf{p}) &= -\bar{v}^{s'}(\mathbf{p})v^s(\mathbf{p}) = 2m\delta^{ss'}, \\ u^{s\dagger}(\mathbf{p})v^{s'}(-\mathbf{p}) &= v^{s\dagger}(\mathbf{p})u^{s'}(-\mathbf{p}) = 0. \end{aligned}$$

#### S.10.6. Two Form Factors for an Elastic Dirac Current and Charge and Magnetic Moment at Zero Transfer

(Inspired by David Tong, Lectures on Quantum Field Theory, Sections 4.4 and 6.5.)

- (a) Use Lorentz covariance, parity, the on-shell Dirac equations, and current conservation to reduce an elastic spin- $\frac{1}{2}$  electromagnetic current matrix element to two independent form factors. Give both a Dirac–Pauli basis and a basis using  $p + p'$ .
- (b) Take the nonrelativistic, small-momentum-transfer limit of the Dirac–Pauli current vertex. Identify the electric charge and magnetic moment in terms of  $F_1(0)$  and  $F_2(0)$ , and state the value for a minimally coupled point Dirac particle at tree level.

#### S.10.7. The Ordered Time Integrals in the General LSZ Pole

(Adapted from Daniel Harlow, Lecture Notes for Physics 8.323: Relativistic Quantum Field Theory I (2024), Section 9.5, Homework Problem 4, printed p. 120, with Eqs. (9.44)–(9.53), printed pp. 115–116.)

Let  $O_1^{b_1}, \dots, O_M^{b_M}$  and  $O_1'^{a_1}, \dots, O_N'^{a_N}$  be Heisenberg operators in a quantum field theory with a scattering description, transforming in irreducible Lorentz representations. With  $O(k) = \int d^d x e^{-ik \cdot x} O(x)$ , introduce the regulated Fourier transform

$$\begin{aligned} G_\epsilon &= \langle \Omega | T \{ O_N'^{a_N}(k'_N) \cdots O_1'^{a_1}(k'_1) O_1^{b_1 \dagger}(k_1) \cdots O_M^{b_M \dagger}(k_M) \} | \Omega \rangle_\epsilon \\ &:= \int d^d x_1 \cdots d^d x_{M+N} e^{-i(k_1 \cdot x_1 + \cdots + k_M \cdot x_M + k'_1 \cdot x_{M+1} + \cdots + k'_N \cdot x_{M+N})} \\ &\quad \times \langle \Omega | T \{ O_N'^{a_N}(x_{M+N}) \cdots O_1'^{a_1}(x_{M+1}) O_1^{b_1 \dagger}(x_1) \cdots O_M^{b_M \dagger}(x_M) \} | \Omega \rangle \\ &\quad \times e^{-\epsilon(t_{\max} - t_{\min})}. \end{aligned} \quad (9.44)$$

Here  $t_{\min}$  and  $t_{\max}$  are the least and greatest of the times  $t_1, \dots, t_{M+N}$ . The incoming and outgoing on-shell limits are

$$k_i^0 \longrightarrow -\omega_{n_i, \mathbf{k}_i}, \quad k_j'^0 \longrightarrow \omega_{n'_j, \mathbf{k}'_j}. \quad (9.45)$$

The pole of interest comes from the ordered region

$$t_1 \leq t_2 \leq \cdots \leq t_{M+N}. \quad (9.46)$$

Insert complete sets of scattering states, writing the  $r$ -particle states as  $|\alpha_r; \text{in}\rangle$  and  $|\beta_r; \text{out}\rangle$ . The contribution from this ordered region is

$$\begin{aligned} G_\epsilon &\supset \int_{t_1 \leq \cdots \leq t_{M+N}} d^d x_1 \cdots d^d x_{M+N} \int d\alpha_1 \cdots d\alpha_M d\beta_1 \cdots d\beta_N \\ &\quad \times e^{-i(k_1 \cdot x_1 + \cdots + k_M \cdot x_M + k'_1 \cdot x_{M+1} + \cdots + k'_N \cdot x_{M+N}) - \epsilon(t_{M+N} - t_1)} \\ &\quad \times \langle \Omega | O_N'^{a_N}(x_{M+N}) | \beta_1; \text{out} \rangle \cdots \langle \beta_{N-1}; \text{out} | O_1'^{a_1}(x_{M+1}) | \beta_N; \text{out} \rangle \\ &\quad \times \langle \beta_N; \text{out} | \alpha_M; \text{in} \rangle \langle \alpha_M; \text{in} | O_1^{b_1 \dagger}(x_1) | \alpha_{M-1}; \text{in} \rangle \cdots \langle \alpha_1; \text{in} | O_M^{b_M \dagger}(x_M) | \Omega \rangle. \end{aligned} \quad (9.48)$$

Use translation invariance in the form

$$\langle \gamma | O(x) | \delta \rangle = e^{-i(p_\gamma - p_\delta) \cdot x} \langle \gamma | O(0) | \delta \rangle. \quad (9.49)$$

After extracting the matrix elements at the origin, the spacetime dependence is therefore

$$\begin{aligned} G_\epsilon &\supset \int d\alpha_1 \cdots d\alpha_M d\beta_1 \cdots d\beta_N \langle \beta_N; \text{out} | \alpha_M; \text{in} \rangle \int_{t_1 \leq \cdots \leq t_{M+N}} d^d x_1 \cdots d^d x_{M+N} \\ &\quad \times \exp \left\{ -i \left[ (k_1 + p_{\alpha_M} - p_{\alpha_{M-1}}) \cdot x_1 + \cdots + (k_M + p_{\alpha_1}) \cdot x_M \right. \right. \\ &\quad \left. \left. + (k'_1 - p_{\beta_N} + p_{\beta_{N-1}}) \cdot x_{M+1} + \cdots + (k'_N - p_{\beta_1}) \cdot x_{M+N} \right] \right. \\ &\quad \left. - \epsilon(t_{M+N} - t_1) \right\} \\ &\quad \times \langle \Omega | O_N'^{a_N}(0) | \beta_1; \text{out} \rangle \cdots \langle \beta_{N-1}; \text{out} | O_1'^{a_1}(0) | \beta_N; \text{out} \rangle \\ &\quad \times \langle \alpha_M; \text{in} | O_1^{b_1 \dagger}(0) | \alpha_{M-1}; \text{in} \rangle \cdots \langle \alpha_1; \text{in} | O_M^{b_M \dagger}(0) | \Omega \rangle. \end{aligned} \quad (9.50)$$

The spatial integrations give delta functions. The remaining nested time integral is

$$\begin{aligned}
\mathcal{T} := & \int_{-\infty}^{\infty} dt_1 e^{i(k_1^0 + p_{\alpha_M}^0 - p_{\alpha_{M-1}}^0 - i\epsilon)t_1} \int_{t_1}^{\infty} dt_2 e^{i(k_2^0 + p_{\alpha_{M-1}}^0 - p_{\alpha_{M-2}}^0)t_2} \\
& \times \cdots \int_{t_{M-1}}^{\infty} dt_M e^{i(k_M^0 + p_{\alpha_1}^0)t_M} \\
& \times \int_{t_M}^{\infty} dt_{M+1} e^{i(k_1'^0 - p_{\beta_N}^0 + p_{\beta_{N-1}}^0)t_{M+1}} \cdots \\
& \times \int_{t_{M+N-2}}^{\infty} dt_{M+N-1} e^{i(k_{N-1}'^0 - p_{\beta_2}^0 + p_{\beta_1}^0)t_{M+N-1}} \\
& \times \int_{t_{M+N-1}}^{\infty} dt_{M+N} e^{i(k_N'^0 - p_{\beta_1}^0 + i\epsilon)t_{M+N}}.
\end{aligned} \tag{9.51}$$

Define

$$k_{\text{tot}} = k_1 + \cdots + k_M, \quad k'_{\text{tot}} = k'_1 + \cdots + k'_N. \tag{9.53}$$

Evaluate the integrations in  $\mathcal{T}$ , going from right to left, and show that

$$\begin{aligned}
\mathcal{T} = & 2\pi\delta(k_{\text{tot}}^0 + k'_{\text{tot}}{}^0 + p_{\alpha_M}^0 - p_{\beta_N}^0) \\
& \times \frac{-i}{k_1^0 + p_{\alpha_M}^0 - p_{\alpha_{M-1}}^0 - i\epsilon} \frac{-i}{k_1^0 + k_2^0 + p_{\alpha_M}^0 - p_{\alpha_{M-2}}^0 - i\epsilon} \\
& \times \cdots \frac{-i}{k_{\text{tot}}^0 - k_M^0 + p_{\alpha_M}^0 - p_{\alpha_1}^0 - i\epsilon} \\
& \times \frac{i}{k_{\text{tot}}^0 - p_{\beta_N}^0 + i\epsilon} \cdots \frac{i}{k_{N-1}'^0 + k_N'^0 - p_{\beta_2}^0 + i\epsilon} \frac{i}{k_N'^0 - p_{\beta_1}^0 + i\epsilon}.
\end{aligned} \tag{9.52}$$

Be prepared to use the final delta function to rewrite the poles involving the incoming energies  $k_i^0$ .

### S.10.8. The Scalar Charge-Radius Expansion and Transverse Decomposition of Vacuum Polarization

(Inspired by David Tong, Lectures on Quantum Field Theory, Sections 2.2 and 6.1 / David Tong, Lectures on Quantum Field Theory, Sections 6.1–6.4.)

- Derive the most general conserved electromagnetic current between equal-mass spin-zero states. Normalize its form factor at zero momentum transfer and show how the coefficient of its small-spacelike-momentum expansion defines the mean-square charge radius.
- Starting with the most general Lorentz-covariant rank-two function of one momentum, impose the electromagnetic Ward identity. Derive the transverse form of the vacuum-polarization tensor and show from Dyson resummation why the photon remains massless when the renormalized scalar function is finite at  $q^2 = 0$ .

### S.10.9. Subtracted Dispersion Relations and Positivity

(Inspired by Bernd A. Kniehl, Dispersion Relations in Loop Calculations, Section 2 / Bernd A. Kniehl, Dispersion Relations in Loop Calculations, Sections 2–3.)

- (a) Let  $F(z)$  be analytic away from a cut  $z \geq s_0$ , real below the cut, and no worse than a constant at infinity. Derive a once-subtracted dispersion relation at a spacelike point  $z_0 < s_0$  in terms of  $\text{Im } F(s + i0)$ .
- (b) Assume  $F(z) = O(|z|^N \log |z|)$  on a large circle. Determine a sufficient number of subtractions for its dispersion integral. Show explicitly how changing a subtraction point changes only a polynomial and not the discontinuity.
- (c) Suppose a once-subtracted amplitude has a nonnegative absorptive part on its physical cut. Differentiate its dispersive representation at a real point below threshold and prove positivity statements for all derivatives of order two and higher. State precisely why the first derivative may require separate care.

#### S.10.10. Muon Vacuum Polarization from the Optical Theorem

(Adapted from John McGreevy, Physics 215B Quantum Field Theory, Winter 2025, Assignment 7, Problem 4, p. 2.)

For general initial and final states  $a$  and  $b$ , the source sheet states the optical theorem as

$$-i[\mathcal{M}(a \rightarrow b) - \mathcal{M}(b \rightarrow a)] = \sum_f \int d\Phi_f \mathcal{M}(b \rightarrow f)^* \mathcal{M}(a \rightarrow f).$$

*Editorial source note.* The sheet omits complex conjugation from the second amplitude on the left. The identity following from  $S^\dagger S = \mathbf{1}$  contains  $\mathcal{M}(b \rightarrow a)^*$  there. Keep track of this printed omission explicitly and state which form is used in the calculation.

- (a) In QED with electrons of charge  $Q_e$  and muons of charge  $Q_\mu$ , consider forward  $e^-e^+ \rightarrow e^-e^+$  scattering. Isolate the contribution to its imaginary part proportional to  $Q_e^2 Q_\mu^2$ . Draw the relevant Feynman diagram and compute the absorptive part  $\text{Im } \Pi_\mu$  of the muon vacuum polarization as a function of the physical timelike invariant

$$s \equiv -(k_1 + k_2)^2 > 0,$$

where  $k_1, k_2$  are the incoming electron and positron momenta. (The minus sign is the translation of the source's mostly-minus definition to this volume's metric.) Results from the tree calculation of  $e^+e^- \rightarrow \mu^+\mu^-$  may be reused. Check that the imaginary part is independent of the ultraviolet cutoff.

- (b) Use the optical theorem and

$$\sigma(e^+e^- \rightarrow \mu^+\mu^-) \geq 0$$

to show that every Feynman diagram containing a closed fermion loop must carry a minus sign. With that sign included, verify the optical theorem for the contribution calculated in the preceding part.



### S.10.11. The Complete Compton Trace

(Adapted from Daniel Harlow, Lecture Notes for Physics 8.324: Relativistic Quantum Field Theory II, Fall 2024, Section 11.6, Problem 3, p. 133, with Eqs. (11.38), (11.45)–(11.47), pp. 129–131.)

Consider tree-level Compton scattering  $e^-(k_1) + \gamma(k_2) \rightarrow e^-(k'_1) + \gamma(k'_2)$ . The direct and crossed electron-exchange diagrams are both included, with

$$k_1 + k_2 = k'_1 + k'_2, \quad k_1^2 = k_1'^2 = -m^2, \quad k_2^2 = k_2'^2 = 0.$$

For their gauge-invariant sum, current conservation permits the physical photon polarization sums to be replaced by

$$\sum_{\lambda} \epsilon_{\mu}(k, \lambda) \epsilon_{\nu}(k, \lambda)^* \longrightarrow \eta_{\mu\nu},$$

and the electron spin sums are  $\sum_{\sigma} u(k, \sigma) \bar{u}(k, \sigma) = \not{k} + im$ . Consequently Harlow's Eq. (11.46), translated only by replacing the charge  $q$  by this volume's  $e$ , is

$$\begin{aligned} \sum_{\sigma, \sigma'} |\widetilde{\mathcal{M}}_c|^2 &= \frac{e^4}{4} \eta_{\mu\alpha} \eta_{\nu\beta} \text{tr} \left[ (\not{k}'_1 + im) \left( \frac{\gamma^{\nu} \not{k}_2 \gamma^{\mu} + 2k_1^{\mu} \gamma^{\nu}}{k_1 \cdot k_2} + \frac{\gamma^{\mu} \not{k}'_2 \gamma^{\nu} - 2k_1^{\nu} \gamma^{\mu}}{k_1 \cdot k'_2} \right) \right. \\ &\quad \left. \times (\not{k}_1 + im) \left( \frac{\gamma^{\alpha} \not{k}_2 \gamma^{\beta} + 2k_1^{\alpha} \gamma^{\beta}}{k_1 \cdot k_2} + \frac{\gamma^{\beta} \not{k}'_2 \gamma^{\alpha} - 2k_1^{\beta} \gamma^{\alpha}}{k_1 \cdot k'_2} \right) \right]. \end{aligned}$$

Evaluate this trace and prove

$$\begin{aligned} \frac{1}{4} \sum_{\sigma, \sigma'} |\widetilde{\mathcal{M}}_c|^2 &= 2e^4 \left[ \frac{k_1 \cdot k'_2}{k_1 \cdot k_2} + \frac{k_1 \cdot k_2}{k_1 \cdot k'_2} \right. \\ &\quad \left. + 2m^2 \left( \frac{1}{k_1 \cdot k'_2} - \frac{1}{k_1 \cdot k_2} \right) + m^4 \left( \frac{1}{k_1 \cdot k'_2} - \frac{1}{k_1 \cdot k_2} \right)^2 \right]. \end{aligned}$$

You should first use the  $d$ -dimensional contraction identities

$$\begin{aligned} \gamma^{\mu} \gamma_{\mu} &= d, & \gamma^{\mu} \gamma^{\alpha} \gamma_{\mu} &= -(d-2) \gamma^{\alpha}, \\ \gamma^{\mu} \gamma^{\alpha} \gamma^{\beta} \gamma_{\mu} &= (d-4) \gamma^{\alpha} \gamma^{\beta} + 4 \eta^{\alpha\beta}, \\ \gamma^{\mu} \gamma^{\alpha} \gamma^{\beta} \gamma^{\gamma} \gamma_{\mu} &= -2 \gamma^{\gamma} \gamma^{\beta} \gamma^{\alpha} - (d-4) \gamma^{\alpha} \gamma^{\beta} \gamma^{\gamma}, \end{aligned}$$

so that no trace of more than four gamma matrices need be evaluated.

### S.10.12. The Photon Propagator Contour and Scalar QED Rules

(Adapted from University of Cambridge, Mathematical Tripos Part III, Quantum Field Theory, Paper 301, 2021, Question 4, printed p. 4, parts (a)–(b).)

For this exercise use the source metric  $\eta_s = \text{diag}(+1, -1, -1, -1)$ .

- (a) Starting from an appropriate mode expansion of the Heisenberg-picture field operator  $\hat{A}_\mu(x)$ , which may be stated without proof, derive the integral representation of the photon propagator in Feynman gauge,

$$\langle 0|T\{\hat{A}^\mu(x)\hat{A}^\nu(y)\}|0\rangle = \int_{C_F} \frac{d^4k}{(2\pi)^4} \frac{-i\eta_s^{\mu\nu}}{k^2} e^{-ik\cdot(x-y)},$$

and specify the contour  $C_F$ , with justification.

- (b) A complex scalar field  $\Phi(x)$  of mass  $M$  has Lagrangian density

$$\mathcal{L} = \partial_\mu \Phi^* \partial^\mu \Phi - M^2 \Phi^* \Phi.$$

Identify a global  $U(1)$  symmetry and explain what it means to gauge this symmetry. Define a covariant derivative and write a gauge-invariant Lagrangian for  $\Phi$  coupled to an electromagnetic four-potential  $A_\mu$  with coupling  $e$ . Identify every interaction term and derive the momentum-space Feynman rules for all internal lines and vertices that can appear. External-line factors need not be discussed.

### S.10.13. Callan–Symanzik Evolution and Asymptotic Freedom

(Adapted from *University of Cambridge, Mathematical Tripos Part III, Advanced Quantum Field Theory, Paper 304, 2021, Question 3, printed p. 4.*)

Consider a theory of a single scalar field with one dimensionless coupling  $g$  and no mass parameters. Let

$$G^{(n)}(x_1, \dots, x_n) = \langle \phi(x_1) \cdots \phi(x_n) \rangle_{\text{conn}}$$

be the renormalized connected  $n$ -point function.

- (a) Explain why this correlation function depends on an additional scale  $\mu$ , define  $\beta(g)$  and  $\gamma(g)$ , and derive

$$\left( \mu \frac{\partial}{\partial \mu} + \beta(g) \frac{\partial}{\partial g} + n\gamma(g) \right) G^{(n)}(x_1, \dots, x_n) = 0. \quad (*)$$

- (b) What is the significance of the beta function and of a fixed point  $g_*$ ? What does the sign of  $\beta(g)$  say about the behavior of the coupling as  $\mu$  is increased from a reference scale  $\mu_0$ ?
- (c) Take as given that the renormalized propagator obeys

$$\int d^4x e^{ip\cdot x} G^{(2)}(x, 0) = \frac{1}{p^2} C\left(\frac{p^2}{\mu^2}, g(\mu)\right).$$

For  $n = 2$ , solve (\*) in the form

$$C\left(\frac{p^2}{\mu^2}, g(\mu)\right) = f(\mu) C\left(\frac{p^2}{\mu_0^2}, g(\mu_0)\right)$$

for arbitrary  $\mu_0$  and unspecified running coupling  $g(\mu)$ . Show that  $f(\mu) = e^{h(\mu)}$ , where

$$h(\mu) \propto \int_{g(\mu_0)}^{g(\mu)} dg \frac{\gamma(g)}{\beta(g)}.$$

*Hint.* Use  $t = \log(\mu/\mu_0)$ .

(d) Now assume

$$\beta(g) = -bg^3, \quad b > 0.$$

Find every fixed point and describe its nature. Express  $g(\mu)$  in terms of  $g(\mu_0)$  and  $\mu/\mu_0$ . Let  $\Lambda$  be the scale at which this expression makes  $g(\mu)$  diverge; rewrite  $g(\mu)$  in terms of  $\mu$  and  $\Lambda$ .

(e) Assuming further that  $\gamma(g) = cg^2$ , evaluate

$$\frac{C(p^2/\mu^2, g(\mu))}{C(1, g(p))}.$$

## Solutions to Supplementary Exercises

### S.10.1. Positive Spectral Measures, Pole Weights, and Sum Rules

(a) Translation invariance gives

$$\langle \Omega | \phi(x) | n, p \rangle = e^{ip \cdot x} \langle \Omega | \phi(0) | n, p \rangle.$$

Inserting the identity and collecting states with the same invariant mass therefore yields

$$\langle \Omega | \phi(x) \phi(0) | \Omega \rangle = \int_0^\infty d\mu^2 \rho(\mu^2) \Delta_+(x; \mu^2),$$

where, schematically,

$$\rho(\mu^2) = \sum_n (2\pi)^3 \delta(\mu^2 - \mu_n^2) |\langle \Omega | \phi(0) | n \rangle|^2.$$

The omitted factors are just those fixed by the chosen invariant normalization. Every summand is an absolute square, so  $\rho(\mu^2) \geq 0$ . If a stable particle of mass  $m$  has  $\langle \Omega | \phi(0) | p \rangle \neq 0$ , its orbit under translations produces

$$\rho(\mu^2) \supset Z \delta(\mu^2 - m^2), \quad Z > 0.$$

All states with continuously variable relative momenta contribute a nonnegative continuum rather than an isolated delta function.

(b) The one-particle contribution and the continuum separate as

$$\rho(\mu^2) = Z \delta(\mu^2 - m^2) + \theta(\mu^2 - M_{\text{th}}^2) \rho_c(\mu^2), \quad \rho_c \geq 0.$$

Thus

$$D(p) = \frac{Z}{p^2 + m^2 - i0} + \int_{M_{\text{th}}^2}^\infty \frac{\rho_c(\mu^2) d\mu^2}{p^2 + \mu^2 - i0}.$$

There is an isolated simple pole at  $p^2 = -m^2$ , with positive residue  $Z$ . The continuum produces a branch point at  $p^2 = -M_{\text{th}}^2$  and a cut extending along the timelike axis. The assumption of stability means  $m < M_{\text{th}}$ ; otherwise the pole lies on the cut and generally moves away from the physical sheet, describing a resonance rather than an asymptotic particle.

(c) The spectral commutator is

$$\langle \Omega | [\phi(x), \phi^\dagger(y)] | \Omega \rangle = \int_0^\infty d\mu^2 \rho(\mu^2) [\Delta_+(x-y; \mu^2) - \Delta_+(y-x; \mu^2)].$$

Differentiate with respect to  $x^0$  and set  $x^0 = y^0$ . For every  $\mu$ , the bracket on the right gives  $-i\delta^3(\mathbf{x} - \mathbf{y})$ . Comparing with the canonical commutator gives

$$\boxed{\int_0^\infty \rho(\mu^2) d\mu^2 = 1.}$$

Writing  $\rho = Z\delta(\mu^2 - m^2) + \rho_c$  and using  $\rho_c \geq 0$  gives

$$1 = Z + \int \rho_c, \quad 0 \leq Z \leq 1.$$

The statement concerns a regulated, canonically normalized bare field. For a renormalized field the right-hand side is rescaled with the field normalization.

### S.10.2. The Thermal Lehmann Representation

Choose simultaneous energy-momentum eigenstates,

$$H|n\rangle = E_n|n\rangle, \quad \mathbf{P}|n\rangle = \mathbf{p}_n|n\rangle,$$

and define

$$O_{nm} = \langle n | O(0) | m \rangle, \quad \omega_{mn} = E_m - E_n, \quad \mathbf{q}_{mn} = \mathbf{p}_m - \mathbf{p}_n.$$

For Hermitian  $O$ , inserting a complete set on each side of the time ordering and interchanging  $n$  and  $m$  in the negative-time term gives

$$-iD_\beta(t, \mathbf{x}) = \frac{1}{Z_\beta} \sum_{n,m} |O_{nm}|^2 e^{-i\omega_{mn}t + i\mathbf{q}_{mn} \cdot \mathbf{x}} [\theta(t)e^{-\beta E_n} + \theta(-t)e^{-\beta E_m}].$$

This is already the thermal Lehmann representation in position space. With the Fourier convention  $F(\omega, \mathbf{p}) = \int d^4x e^{i\omega t - i\mathbf{p} \cdot \mathbf{x}} F(x)$ , it becomes

$$\begin{aligned} -iD_\beta(\omega, \mathbf{p}) &= \frac{i}{Z_\beta} \sum_{n,m} (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}_{mn}) |O_{nm}|^2 \\ &\times \left[ \frac{e^{-\beta E_n}}{\omega - \omega_{mn} + i0} - \frac{e^{-\beta E_m}}{\omega - \omega_{mn} - i0} \right]. \end{aligned}$$

It is useful to separate the two thermal Wightman functions. Their common positive-frequency weight and the commutator spectral function are

$$G_\beta^>(\omega, \mathbf{p}) \equiv \frac{(2\pi)^4}{Z_\beta} \sum_{n,m} e^{-\beta E_n} |O_{nm}|^2 \delta(\omega - \omega_{mn}) \delta^3(\mathbf{p} - \mathbf{q}_{mn}),$$

$$\rho_\beta(\omega, \mathbf{p}) \equiv \frac{(2\pi)^4}{Z_\beta} \sum_{n,m} (e^{-\beta E_n} - e^{-\beta E_m}) |O_{nm}|^2 \delta(\omega - \omega_{mn}) \delta^3(\mathbf{p} - \mathbf{q}_{mn}).$$

Since  $E_m = E_n + \omega$  on the support of the delta function,

$$G_\beta^<(\omega, \mathbf{p}) = e^{-\beta\omega} G_\beta^>(\omega, \mathbf{p}), \quad \rho_\beta = (1 - e^{-\beta\omega}) G_\beta^>.$$

This is detailed balance, or the frequency-space KMS relation. In terms of  $n_B(\omega) = (e^{\beta\omega} - 1)^{-1}$ , the time-ordered answer may also be written

$$-iD_\beta(\omega, \mathbf{p}) = i \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \rho_\beta(\omega', \mathbf{p}) \left[ \frac{1 + n_B(\omega')}{\omega - \omega' + i0} - \frac{n_B(\omega')}{\omega - \omega' - i0} \right].$$

The heat bath selects a rest frame, so the thermal density depends separately on  $\omega$  and  $\mathbf{p}$ , rather than only on a Lorentz invariant mass.

Assume a unique vacuum with  $E_0 = 0$ . As  $\beta \rightarrow \infty$ , only  $n = 0$  survives in the positive-time term and only  $m = 0$  in the negative-time term. Thus

$$-iD(\omega, \mathbf{p}) = i \sum_m (2\pi)^3 \left[ \frac{|O_{0m}|^2 \delta^3(\mathbf{p} - \mathbf{p}_m)}{\omega - E_m + i0} - \frac{|O_{m0}|^2 \delta^3(\mathbf{p} + \mathbf{p}_m)}{\omega + E_m - i0} \right],$$

which is the usual vacuum Lehmann representation; Lorentz invariance then reorganizes the intermediate states into the standard invariant-mass spectral density.

### S.10.3. The Ward–Takahashi Identity in Euclidean QED

- (a) Under a constant phase,  $D_\mu \psi$  transforms in the same way as  $\psi$ , while  $F_{\mu\nu}$  is unchanged. Every term in  $\mathcal{L}_E$  is therefore invariant. To extract the current, promote the phase to  $\alpha(x)$ . To first order,

$$\delta\psi = i\alpha\psi, \quad \delta\bar{\psi} = -i\alpha\bar{\psi}, \quad \delta\mathcal{L}_E = i(\partial_\mu \alpha) \bar{\psi} \gamma_E^\mu \psi.$$

Thus the Noether current is  $j^\mu = \bar{\psi} \gamma_E^\mu \psi$ . Equivalently, the Euclidean equations of motion are

$$(\not{D}_E + m)\psi = 0, \quad \bar{\psi}(-\overleftarrow{\not{D}}_E + m) = 0,$$

where  $\bar{\psi} \overleftarrow{D}_\mu = \partial_\mu \bar{\psi} + ie A_\mu \bar{\psi}$ . They give

$$\partial_\mu j^\mu = (\partial_\mu \bar{\psi}) \gamma_E^\mu \psi + \bar{\psi} \gamma_E^\mu \partial_\mu \psi = 0.$$

- (b) Use the Euclidean functional integral with weight  $e^{-S_E}$ , and make the position-dependent change of integration variables used in the preceding part. For  $\mathcal{O} = \psi(x_1)\bar{\psi}(x_2)$ ,

$$\delta\mathcal{O} = i[\alpha(x_1) - \alpha(x_2)]\mathcal{O}, \quad \delta S_E = -i \int d^4x \alpha(x) \partial_\mu j^\mu(x).$$

Invariance of both the measure and the integral gives  $\langle \delta\mathcal{O} - \mathcal{O} \delta S_E \rangle = 0$ . Because  $\alpha(x)$  is arbitrary, the coefficient of it is

$$\partial_\mu \langle j^\mu(x) \psi(x_1) \bar{\psi}(x_2) \rangle = - \left[ \delta^{(4)}(x - x_1) - \delta^{(4)}(x - x_2) \right] \langle \psi(x_1) \bar{\psi}(x_2) \rangle.$$

The two contact terms have opposite signs because  $\psi$  and  $\bar{\psi}$  carry opposite  $U(1)$  charges.

- (c) Let  $p_2 = p_1 - q$ , and normalize the amputated vertex so that it includes the charge  $e$ . Fourier transforming the connected current insertion and factoring its two exact external propagators gives

$$iq_\mu G(\not{p}_1) V_3^\mu(q, p_1, p_2) G(\not{p}_2) = ie \left[ G(\not{p}_2) - G(\not{p}_1) \right].$$

Multiplication by the inverse propagators on the two sides yields the exact identity

$$iq_\mu V_3^\mu(q, p_1, p_2) = ie \left[ G^{-1}(\not{p}_1) - G^{-1}(\not{p}_1 - \not{q}) \right].$$

At zero momentum transfer this becomes

$$V_3^\mu(0, p, p) = e \frac{\partial G^{-1}(\not{p})}{\partial p_\mu}.$$

This derivative form is a useful check on every factor of  $i$  and on the momentum routing.

- (d) Write the renormalized fermion kinetic and vertex terms schematically as

$$Z_2 \bar{\psi} \not{\partial}_E \psi - ie Z_1 \bar{\psi} \gamma_E^\mu A_\mu \psi.$$

The derivative identity says that the ultraviolet coefficient of the zero-momentum vertex is the same as that of the inverse fermion propagator. Hence

$$Z_1 = Z_2.$$

In an on-shell scheme the residue counterterm fixed at the fermion pole therefore fixes the vertex counterterm in the Thomson limit. With  $A_{0\mu} = Z_3^{1/2} A_\mu$ , this also gives

$$e_0 = Z_3^{-1/2} e$$

in the usual notation: charge renormalization is determined by photon field-strength renormalization, not by an independent fermion–photon vertex counterterm. The same gauge symmetry makes the vacuum polarization transverse and forbids a photon mass counterterm in the unbroken theory.

Nothing in the change-of-variables derivation used a loop expansion. Consequently the identity between the exact propagator and exact vertex, current conservation, and transversality hold nonperturbatively whenever the regulator and functional measure preserve the vector  $U(1)$  symmetry. The vector current of QED has no anomaly. A symmetry-breaking regulator would require symmetry-restoring counterterms, while a genuinely broken phase would change the assumptions rather than invalidate the identity.

#### S.10.4. Charge Conjugation and Odd-Photon Selection Rules

(a) Set

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad D_\mu = \partial_\mu + ieA_\mu.$$

In the volume’s mostly-plus convention the two Lagrangians are

$$\mathcal{L}_{\text{SQED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - (D_\mu\phi)^\dagger D^\mu\phi - M^2\phi^\dagger\phi,$$

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}(\gamma^\mu D_\mu + m)\psi.$$

They are invariant under

$$A_\mu \mapsto A_\mu + \partial_\mu\Lambda, \quad \phi \mapsto e^{-ie\Lambda}\phi, \quad \psi \mapsto e^{-ie\Lambda}\psi,$$

because  $D_\mu\phi$  and  $D_\mu\psi$  acquire the same phase as their fields and  $F_{\mu\nu}$  is unchanged.

For a constant phase  $\varepsilon = e\Lambda$ , so that  $\delta\phi = -i\varepsilon\phi$  and  $\delta\psi = -i\varepsilon\psi$ , the unit-normalized Noether currents are

$$J_\phi^\mu = -i[\phi^\dagger D^\mu\phi - (D^\mu\phi)^\dagger\phi], \quad J_\psi^\mu = i\bar{\psi}\gamma^\mu\psi.$$

Multiplication by the coupling gives the corresponding gauge-source normalization. The Euler–Lagrange equations give  $\partial_\mu J_\phi^\mu = \partial_\mu J_\psi^\mu = 0$ .

(b) Charge conjugation must send

$$\widehat{C}^{-1}A_\mu\widehat{C} = -A_\mu.$$

Indeed, it then maps

$$D_\mu\phi \mapsto (\partial_\mu - ieA_\mu)\phi^\dagger = (D_\mu\phi)^\dagger, \quad F_{\mu\nu} \mapsto -F_{\mu\nu}.$$

Both the Maxwell term and every scalar term are therefore invariant. The same calculation gives  $\widehat{C}^{-1}J_\phi^\mu\widehat{C} = -J_\phi^\mu$ .

- (c) The supplied matrix identities imply  $\mathbf{C}^2 = -\mathbf{1}$  and  $\mathbf{C}(\gamma^\mu)^\top \mathbf{C} = \gamma^\mu$ . Therefore

$$\begin{aligned}\widehat{C}^{-1}(\bar{\psi}\psi)\widehat{C} &= \psi^\top \mathbf{C}^2 \bar{\psi}^\top = -\psi^\top \bar{\psi}^\top = \bar{\psi}\psi, \\ \widehat{C}^{-1}(\bar{\psi}\gamma^\mu\psi)\widehat{C} &= \psi^\top \mathbf{C}\gamma^\mu \mathbf{C}\bar{\psi}^\top = -\bar{\psi}(\mathbf{C}\gamma^\mu \mathbf{C})^\top \psi = -\bar{\psi}\gamma^\mu\psi.\end{aligned}$$

For the kinetic term the same Grassmann rearrangement gives

$$\bar{\psi}^C \gamma^\mu \partial_\mu \psi^C = -(\partial_\mu \bar{\psi}) \gamma^\mu \psi = \bar{\psi} \gamma^\mu \partial_\mu \psi - \partial_\mu (\bar{\psi} \gamma^\mu \psi).$$

It is invariant in the action. The interaction is invariant because both  $A_\mu$  and  $\bar{\psi}\gamma^\mu\psi$  are odd, while the mass and Maxwell terms are even. Hence  $\mathcal{L}_{\text{QED}}$  is charge-conjugation invariant up to the displayed total derivative, and  $\widehat{C}^{-1}J_\psi^\mu\widehat{C} = -J_\psi^\mu$ .

- (d) Let  $X_i$  be either  $A^{\mu_i}(x_i)$  or  $J^{\mu_i}(x_i)$ . Since the vacuum and the time-ordering operation are invariant,

$$\begin{aligned}\langle \Omega | T\{X_1 \cdots X_n\} | \Omega \rangle &= \langle \Omega | \widehat{C}^{-1} T\{X_1 \cdots X_n\} \widehat{C} | \Omega \rangle \\ &= (-1)^n \langle \Omega | T\{X_1 \cdots X_n\} | \Omega \rangle.\end{aligned}$$

For odd  $n$  the correlator equals its negative and must vanish. This is an exact operator argument, so it is valid to all orders rather than merely diagram by diagram. LSZ reduction then says that any amplitude having an odd number of external photons and no charged external particles vanishes: three-photon scattering or splitting is forbidden, while even-photon light-by-light scattering is not. Perturbatively this is Furry's theorem, the cancellation between the two orientations of each charged loop with an odd number of photon attachments.

### S.10.5. Scalar and Pseudoscalar Decays from Schwinger–Dyson and LSZ

- (a) i. Varying  $\phi$ ,  $\bar{\psi}$ , and  $\psi$ , respectively, gives

$$\begin{aligned}(\square - M^2)\phi &= \lambda \bar{\psi}\psi, \\ (\gamma^\mu \partial_\mu + m + \lambda\phi)\psi &= 0, \\ \bar{\psi}(\overleftarrow{\partial}_\mu \gamma^\mu - m - \lambda\phi) &= 0.\end{aligned}$$

The relative signs in the scalar equation follow from  $\mathcal{L}_\phi = -\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}M^2\phi^2$ . For completeness, the three Schwinger–Dyson identities can be stated in a form that covers any time-ordered insertion  $\mathcal{X}$ . With left Grassmann differentiation in the second line and right differentiation in the third,

$$\begin{aligned}\langle T\{[(\square - M^2)\phi - \lambda\bar{\psi}\psi](x)\mathcal{X}\} \rangle &= i \left\langle T \frac{\delta \mathcal{X}}{\delta \phi(x)} \right\rangle, \\ \langle T\{[(\gamma^\mu \partial_\mu + m + \lambda\phi)\psi](x)\mathcal{X}\} \rangle &= -i \left\langle T \frac{\delta \mathcal{X}}{\delta \bar{\psi}(x)} \right\rangle, \\ \langle T\{\mathcal{X}[\bar{\psi}(\overleftarrow{\partial}_\mu \gamma^\mu - m - \lambda\phi)](x)\} \rangle &= i \left\langle T \mathcal{X} \frac{\overleftarrow{\delta}}{\delta \psi(x)} \right\rangle.\end{aligned}$$



These equations include all contact terms: differentiating a field inside  $\mathcal{X}$  produces its delta function. For example, the free limits give  $(\square - M^2)D_F = i\delta^{(4)}$  and  $(\gamma^\mu \partial_\mu + m)S_F = -i\delta^{(4)}$ , consistent with the volume's propagator phases.

- ii. Use covariantly normalized one-particle states. In terms of the noncovariantly normalized operators in the supplied mode expansions,

$$\begin{aligned} |\phi(k)\rangle_{\text{in}} &= \sqrt{2\tilde{\omega}_{\mathbf{k}}} a_{\text{in},\mathbf{k}}^\dagger |\Omega\rangle, \\ |\psi(p, s), \bar{\psi}(q, s')\rangle_{\text{out}} &= \sqrt{2\omega_{\mathbf{p}} 2\omega_{\mathbf{q}}} b_{\text{out},\mathbf{p}}^{s\dagger} c_{\text{out},\mathbf{q}}^{s'\dagger} |\Omega\rangle. \end{aligned}$$

The connected correlation function having all three required one-particle poles is

$$G_{\alpha\beta}(x, y, z) = [\langle \Omega | T \{ \psi_\alpha(x) \bar{\psi}_\beta(y) \phi(z) \} | \Omega \rangle]_c.$$

Define its Fourier transform with the external-state phases by

$$\tilde{G}_{\alpha\beta}(p, q; k) = \int d^4x d^4y d^4z e^{-ip \cdot x - iq \cdot y + ik \cdot z} G_{\alpha\beta}(x, y, z).$$

Near the three mass shells it factors as

$$\begin{aligned} \tilde{G}(p, q; k) &= (2\pi)^4 \delta^{(4)}(k - p - q) \\ &\quad \times S_F(p) \Gamma(p, q; k) S_F(-q) D_F(k) + R(p, q; k). \end{aligned}$$

Here  $R$  denotes terms missing at least one of the three external one-particle poles. LSZ multiplies by the three inverse propagators, takes  $p^2 = q^2 = -m^2$ ,  $k^2 = -M^2$ , and contracts the fermion residues with  $\bar{u}^s(\mathbf{p})$  and  $v^{s'}(\mathbf{q})$ . Thus, defining the amputated invariant vertex by  $\Gamma = -i\mathcal{V}$ ,

$$\begin{aligned} \text{out} \langle \psi(p, s), \bar{\psi}(q, s') | S | \phi(k) \rangle_{\text{in}} &= i(2\pi)^4 \delta^{(4)}(k - p - q) \mathcal{A}_{s,s'}, \\ \mathcal{A}_{s,s'} &= \bar{u}^s(\mathbf{p}) \mathcal{V}(p, q; k) v^{s'}(\mathbf{q}). \end{aligned}$$

The mode expansions supplied in the prompt fix both the pole residues and the square-root normalizations in this formula.

- iii. Apply the scalar Schwinger–Dyson equation to  $G$ . Its contact term vanishes because the other insertion contains no scalar field, and its right-hand side contains  $\langle T \{ \bar{\psi} \psi \psi \bar{\psi} \} \rangle$ . At lowest order Wick contraction gives two free fermion propagators. Applying the two fermion Schwinger–Dyson equations then exposes their delta-function sources. Equivalently, before amputation the result is

$$G_c(x, y, z) = -i\lambda \int d^4w D_F(z - w) S_F(x - w) S_F(w - y) + O(\lambda^3),$$

with the displayed order of spinor indices. Fourier transformation gives

$$\tilde{G}_c(p, q; k) = (2\pi)^4 \delta^{(4)}(k - p - q) S_F(p) (-i\lambda) S_F(-q) D_F(k) + O(\lambda^3).$$

The three LSZ amputations therefore leave

$$\mathcal{V} = -\lambda \mathbf{1}, \quad \mathcal{A}_{s,s'} = -\lambda \bar{u}^s(\mathbf{p}) v^{s'}(\mathbf{q}),$$

and

$$\begin{aligned} \text{out} \langle \psi(p, s), \bar{\psi}(q, s') | S | \phi(k) \rangle_{\text{in}} = & -i\lambda(2\pi)^4 \delta^{(4)}(k - p - q) \\ & \times \bar{u}^s(\mathbf{p}) v^{s'}(\mathbf{q}) + O(\lambda^3). \end{aligned}$$

iv. The supplied completeness relations give

$$\begin{aligned} \sum_{s,s'} |\mathcal{A}_{s,s'}|^2 &= \lambda^2 \text{tr}[(\hat{p} + m)(\hat{q} - m)] \\ &= -4\lambda^2(p \cdot q + m^2). \end{aligned}$$

Momentum conservation and the mostly-plus mass shells imply

$$-M^2 = k^2 = (p + q)^2 = -2m^2 + 2p \cdot q, \quad p \cdot q = m^2 - \frac{M^2}{2}.$$

Consequently

$$\sum_{s,s'} |\mathcal{A}_{s,s'}|^2 = 2\lambda^2(M^2 - 4m^2), \quad \mathcal{P}_S = \frac{\lambda^2}{2}(M^2 - 4m^2).$$

The expression is nonnegative precisely when the decay is kinematically allowed,  $M \geq 2m$ .

(b) The pseudoscalar interaction replaces the scalar vertex  $-i\lambda \mathbf{1}$  by  $g\gamma_5$ . Hence

$$\mathcal{A}_{s,s'}^{(5)} = -ig \bar{u}^s(\mathbf{p}) \gamma_5 v^{s'}(\mathbf{q}),$$

and

$$\text{out} \langle \psi(p, s), \bar{\psi}(q, s') | S | \phi(k) \rangle_{\text{in}} = g(2\pi)^4 \delta^{(4)}(k - p - q) \bar{u}^s(\mathbf{p}) \gamma_5 v^{s'}(\mathbf{q}) + O(g^3).$$

Since  $(\bar{u}\gamma_5 v)^* = -\bar{v}\gamma_5 u$ , the spin sum is

$$\begin{aligned} \sum_{s,s'} |\mathcal{A}_{s,s'}^{(5)}|^2 &= -g^2 \text{tr}[(\hat{p} + m)\gamma_5(\hat{q} - m)\gamma_5] \\ &= 2g^2 M^2, \end{aligned}$$

where  $\{\gamma_5, \hat{p}\} = 0$  was used. Therefore

$$\mathcal{P}_P = \frac{g^2 M^2}{2}.$$

The distinction is physical. Writing  $\beta = \sqrt{1 - 4m^2/M^2}$ , the actual decay widths, which use the full final-spin sum  $4\mathcal{P}$ , are

$$\Gamma_S = \frac{\lambda^2 M}{8\pi} \beta^3, \quad \Gamma_P = \frac{g^2 M}{8\pi} \beta.$$

Thus scalar and pseudoscalar couplings have different threshold behavior and different final-state spin/parity correlations. A single width could be imitated by choosing different numerical couplings, but threshold or angular/spin information distinguishes the two interactions.

#### S.10.6. Two Form Factors for an Elastic Dirac Current and Charge and Magnetic Moment at Zero Transfer

- (a) For  $q = p' - p$ , parity permits only vector bilinears. The Dirac equations and the Gordon identity reduce all candidates to

$$\Gamma^\mu(q) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m} F_2(q^2).$$

Both terms are conserved between equal-mass on-shell spinors. Using

$$\bar{u}' i\sigma^{\mu\nu} q_\nu u = \bar{u}' (2m\gamma^\mu - (p' + p)^\mu) u$$

in the conventional gamma-matrix normalization gives the equivalent basis

$$\bar{u}' \Gamma^\mu u = \bar{u}' \left[ \gamma^\mu (F_1 + F_2) - \frac{(p' + p)^\mu}{2m} F_2 \right] u.$$

Factors of  $i$  move between the Gordon identity and the coefficient functions in the volume's mostly-plus convention, but the two independent scalar functions and all observables are unchanged.

- (b) In the nonrelativistic expansion, the temporal current is

$$J^0 = e F_1(0) \chi'^\dagger \chi + O(|\mathbf{q}|/m),$$

so charge normalization gives  $F_1(0) = Q$  when the overall coupling is  $e$ . The spatial current contains

$$\mathbf{J}_{\text{spin}} = \frac{e}{2m} [F_1(0) + F_2(0)] \chi'^\dagger i\mathbf{q} \times \boldsymbol{\sigma} \chi.$$

Hence

$$\boldsymbol{\mu} = \frac{e}{2m} [F_1(0) + F_2(0)] \boldsymbol{\sigma}.$$

For a minimally coupled point Dirac particle at tree level,  $F_1(0) = Q$  and  $F_2(0) = 0$ ; for  $Q = 1$ , this is the Dirac value  $g = 2$ . Radiative corrections appear as the anomalous Pauli form factor  $F_2(0)$ .

#### S.10.7. Why a Transition Charge Must Vanish and Opposite-Parity Electromagnetic Transitions

- (a) A useful conserved basis for unequal masses is

$$\bar{u}_2 \left[ \left( \gamma^\mu - \frac{q^\mu \not{q}}{q^2} \right) F_T(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{m_1 + m_2} G_T(q^2) \right] u_1.$$

Contracting with  $q_\mu$  annihilates both structures identically. The first, however, contains  $q^\mu/q^2$ . Since

$$\bar{u}_2 \not{q} u_1 = i(m_2 - m_1) \bar{u}_2 u_1$$

does not vanish for  $m_2 \neq m_1$ , absence of a kinematic pole at  $q^2 = 0$  requires

$$\boxed{F_T(0) = 0.}$$

Equivalently, the zero-momentum charge operator commutes with the Hamiltonian and cannot connect energy eigenstates of different mass.

- (b) A vector current between opposite-parity states must be represented by an axial-looking spinor bilinear:

$$\bar{u}_2 \left[ \gamma^\mu \tilde{F} + P^\mu \tilde{G} + q^\mu \tilde{H} \right] \gamma_5 u_1.$$

Anticommuting  $\gamma_5$  through the initial Dirac equation gives

$$\bar{u}_2 \not{q} \gamma_5 u_1 = i(m_2 + m_1) \bar{u}_2 \gamma_5 u_1.$$

Therefore current conservation reads

$$i(m_2 + m_1) \tilde{F} - (m_2^2 - m_1^2) \tilde{G} + q^2 \tilde{H} = 0.$$

The sum appears because  $\gamma_5$  reverses the sign of the initial  $\not{p}_1$ . A manifestly transverse basis is obtained by multiplying the two structures of the preceding part by  $\gamma_5$ , and its transition-charge coefficient again vanishes at  $q^2 = 0$ .

### S.10.8. The Scalar Charge-Radius Expansion and Transverse Decomposition of Vacuum Polarization

- (a) For equal scalar masses, covariance gives

$$\langle p' | J^\mu(0) | p \rangle = \mathcal{N} [(p' + p)^\mu F(q^2) + q^\mu H(q^2)].$$

Because  $q \cdot (p' + p) = p'^2 - p^2 = 0$ , conservation requires  $q^2 H = 0$ , and regularity gives  $H = 0$ . Charge normalization fixes

$$F(0) = 1.$$

In the Breit frame,  $q^0 = 0$  and the Fourier transform of the static charge distribution is  $F(\mathbf{q}^2)$ . Its small- $\mathbf{q}$  expansion is

$$\boxed{F(q^2) = 1 - \frac{1}{6} \langle r^2 \rangle q^2 + O(q^4), \quad \langle r^2 \rangle = -6F'(0),}$$

where  $q^2 = \mathbf{q}^2 > 0$  in the volume's mostly-plus convention.

- (b) The most general symmetric rank-two function is

$$\Pi^{\mu\nu}(q) = A(q^2)\eta^{\mu\nu} + B(q^2)q^\mu q^\nu.$$

The Ward identity gives  $A + q^2 B = 0$ , hence

$$\boxed{\Pi^{\mu\nu}(q) = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\pi(q^2).}$$

Dyson summation changes the transverse photon denominator from  $q^2$  to  $q^2[1 - \pi(q^2)]$ , up to the convention used for the sign of  $\Pi$ . The explicit factor  $q^2$  remains. If the renormalized  $\pi(q^2)$  is finite at the origin and its residue is normalized by  $\pi(0) = 0$ , the propagator retains a pole at  $q^2 = 0$  with the chosen unit transverse residue: radiative corrections do not generate a photon mass.

### S.10.9. Subtracted Dispersion Relations and Positivity

- (a) Apply Cauchy's theorem to

$$\frac{F(\zeta)}{(\zeta - z)(\zeta - z_0)}$$

and collapse the contour around the cut. Real analyticity gives  $\text{Disc } F(s) = 2i \text{Im } F(s + i0)$ . The result is

$$\boxed{F(z) = F(z_0) + \frac{z - z_0}{\pi} \int_{s_0}^{\infty} \frac{\text{Im } F(s + i0) ds}{(s - z_0)(s - z - i0)}.$$

One subtraction is sufficient because the extra denominator makes the large-circle contribution vanish when  $F$  grows no faster than a constant. The formula reproduces  $F(z_0)$  identically and has the required discontinuity on the cut.

- (b) After  $n$  subtractions, the contour integrand behaves on a large circle as  $F(\zeta)/\zeta^{n+1}$ . Its arc integral therefore scales as  $R^{N-n} \log R$ . A sufficient condition for it to vanish is

$$\boxed{n > N.}$$

Thus  $n = N + 1$  subtractions suffice for the stated bound. The general answer is

$$F(z) = P_{n-1}(z) + \frac{(z - z_0)^n}{\pi} \int_{s_0}^{\infty} \frac{\text{Im } F(s) ds}{(s - z_0)^n (s - z - i0)}.$$

Changing  $z_0$  rearranges the first  $n$  Taylor data into a different polynomial  $P_{n-1}$ . The integral's discontinuity remains  $2i \text{Im } F$ , so two subtraction prescriptions differ only by an entire polynomial of degree at most  $n - 1$ .

- (c) For one subtraction at  $z_0 < s_0$ ,

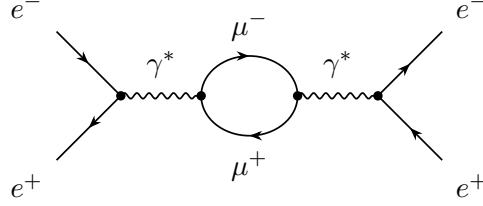
$$F^{(n)}(z_0) = \frac{n!}{\pi} \int_{s_0}^{\infty} \frac{\text{Im } F(s) ds}{(s - z_0)^{n+1}}, \quad n \geq 1.$$

Every denominator is positive, so a nonnegative absorptive part makes all these derivatives nonnegative. In particular the asserted inequalities hold for  $n \geq 2$ .

There is a qualification for the first derivative. If the actual high-energy behavior requires two or more subtractions, a free linear polynomial is present; then  $F'(z_0)$  contains an undetermined subtraction constant, while derivatives above the polynomial's degree remain dispersive and positive. If a genuinely once-subtracted relation is justified, the displayed formula also proves  $F'(z_0) \geq 0$ .

#### S.10.10. Muon Vacuum Polarization from the Optical Theorem

- (a) The required graph is the forward  $s$ -channel photon exchange with a muon loop inserted into the photon line:



Write the transverse muon contribution as

$$\Pi_\mu^{\alpha\beta}(q) = (q^\alpha q^\beta - q^2 \eta^{\alpha\beta}) \Pi_\mu(s), \quad s = -q^2.$$

Cutting the two muon propagators is equivalent to inserting the on-shell  $\mu^+\mu^-$  states in the current two-point function. With

$$\beta_\mu = \sqrt{1 - \frac{4m_\mu^2}{s}}, \quad x_\pm = \frac{1}{2}(1 \pm \beta_\mu),$$

the scalar absorptive part is

$$\begin{aligned} \text{Im } \Pi_\mu(s + i0) &= \frac{Q_\mu^2}{2\pi} \int_{x_-}^{x_+} dx x(1-x) \\ &= \boxed{\frac{Q_\mu^2}{12\pi} \left(1 + \frac{2m_\mu^2}{s}\right) \sqrt{1 - \frac{4m_\mu^2}{s}} \theta(s - 4m_\mu^2)}. \end{aligned}$$

Indeed, a regulator-dependent representation has the form

$$\Pi_\mu(s) = C_{\text{UV}} - \frac{Q_\mu^2}{2\pi^2} \int_0^1 dx x(1-x) \log[m_\mu^2 - sx(1-x) - i0],$$

up to a real convention-dependent additive constant. The cutoff occurs only in the real local term  $C_{\text{UV}}$ ; the logarithm acquires an imaginary part only for  $x_- < x < x_+$ . The boxed answer is consequently cutoff independent.

For negligible electron mass, the same cut gives

$$\sigma(e^+e^- \rightarrow \mu^+\mu^-) = \frac{Q_e^2}{s} \text{Im } \Pi_\mu(s + i0) = \frac{Q_e^2 Q_\mu^2}{12\pi s} \left(1 + \frac{2m_\mu^2}{s}\right) \beta_\mu,$$

which identifies precisely the  $Q_e^2 Q_\mu^2$  contribution to the forward amplitude.

(b) Write  $S = \mathbf{1} + iT$ . Unitarity gives

$$-i(T - T^\dagger) = T^\dagger T,$$

and hence the correctly conjugated matrix-element identity is

$$-i[\mathcal{M}(a \rightarrow b) - \mathcal{M}(b \rightarrow a)^*] = \sum_f \int d\Phi_f \mathcal{M}(b \rightarrow f)^* \mathcal{M}(a \rightarrow f).$$

Thus the missing conjugation in the source display is a typographical omission, not a changed unitarity convention. Setting  $b = a$  gives

$$2 \operatorname{Im} \mathcal{M}(a \rightarrow a) = \sum_f \int d\Phi_f |\mathcal{M}(a \rightarrow f)|^2 \geq 0.$$

For the  $\mu^+ \mu^-$  intermediate state, division by the incident flux reproduces

$$\frac{Q_e^2}{s} \operatorname{Im} \Pi_\mu(s) = \sigma(e^+ e^- \rightarrow \mu^+ \mu^-) \geq 0,$$

which is the optical theorem for this cut.

In a Wick expansion, closing a chain of fermion contractions into a cycle requires one additional odd permutation of Grassmann fields. Each independent closed fermion loop therefore contributes  $(-1)$ . Directly in the present example, omitting that sign reverses the cut of the loop and would give  $\operatorname{Im} \Pi_\mu = -Q_\mu^2 \beta_\mu (1 + 2m_\mu^2/s)/(12\pi)$ , making the cross-section negative. Including the closed-loop minus sign gives the positive absorptive part above and exactly restores unitarity.

### S.10.11. The Complete Compton Trace

Put

$$a = k_1 \cdot k_2, \quad b = k_1 \cdot k'_2.$$

Momentum conservation and the mass-shell conditions imply

$$k_2 \cdot k'_2 = a - b, \quad k'_1 \cdot k_2 = b, \quad k'_1 \cdot k'_2 = a, \quad k'_1 \cdot k_1 = -m^2 + a - b.$$

These relations eliminate every scalar product generated by the trace in favor of  $a, b, m^2$ .

Denote the two numerator structures in the first parenthesis of the prompt by  $N_s^{\mu\nu}$  and  $N_u^{\mu\nu}$ , and those in the second by  $\bar{N}_s^{\alpha\beta}$  and  $\bar{N}_u^{\alpha\beta}$ , before their denominators  $a$  and  $b$  are included. Apply the supplied contractions with  $d = 4$ , and then use

$$\operatorname{tr} \mathbf{1} = 4, \quad \operatorname{tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu},$$

and

$$\operatorname{tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(\eta^{\mu\nu} \eta^{\rho\sigma} - \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}).$$

All traces with an odd number of gamma matrices vanish. The four pieces of the contracted trace reduce to

| numerator pair | $\eta_{\mu\alpha}\eta_{\nu\beta} \text{tr}[(\not{k}'_1 + im)N^{\mu\nu}(\not{k}_1 + im)\bar{N}^{\alpha\beta}]$ |
|----------------|---------------------------------------------------------------------------------------------------------------|
| $s-s$          | $32(ab - m^2a + m^4)$                                                                                         |
| $u-u$          | $32(ab + m^2b + m^4)$                                                                                         |
| $s-u$          | $16m^2(a - b) - 32m^4$                                                                                        |
| $u-s$          | $16m^2(a - b) - 32m^4$                                                                                        |

The denominators are respectively  $a^2, b^2, ab, ab$ . Their sum is therefore

$$\mathcal{T} = 32 \left[ \frac{b}{a} + \frac{a}{b} + 2m^2 \left( \frac{1}{b} - \frac{1}{a} \right) + m^4 \left( \frac{1}{b} - \frac{1}{a} \right)^2 \right].$$

Finally Eq. (11.46) contributes the prefactor  $e^4/4$ , and the requested unpolarized average supplies another factor  $1/4$ . Hence

$$\boxed{\frac{1}{4} \sum_{\sigma, \sigma'} |\widetilde{\mathcal{M}}_c|^2 = 2e^4 \left[ \frac{b}{a} + \frac{a}{b} + 2m^2 \left( \frac{1}{b} - \frac{1}{a} \right) + m^4 \left( \frac{1}{b} - \frac{1}{a} \right)^2 \right]},$$

which is precisely the stated result after restoring  $a = k_1 \cdot k_2$  and  $b = k_1 \cdot k'_2$ .

#### S.10.12. Integrating the Klein–Nishina Distribution

With  $x = \omega/m$  and  $\omega'/\omega = [1 + x(1 - \cos\theta)]^{-1}$ , angular integration gives

$$\sigma_{\text{KN}}(x) = \frac{2\pi\alpha^2}{m^2} \left[ \frac{1+x}{x^3} \left( \frac{2x(1+x)}{1+2x} - \ln(1+2x) \right) + \frac{\ln(1+2x)}{2x} - \frac{1+3x}{(1+2x)^2} \right].$$

Series expansion at small  $x$  gives

$$\sigma_{\text{KN}} = \frac{8\pi\alpha^2}{3m^2} [1 - 2x + O(x^2)].$$

At high energy,

$$\sigma_{\text{KN}} = \frac{\pi\alpha^2}{m^2x} \left[ \ln(2x) + \frac{1}{2} + O\left(\frac{\ln x}{x}\right) \right].$$

Thus the result approaches the Thomson cross-section in the infrared and falls as  $(\ln x)/x$  in the ultraviolet.

#### S.10.13. Forward Compton Crossing and Dispersion

(a) For a null momentum  $k$  and reference vector  $n$ , the physical sum can be written

$$P_{\mu\nu}(k, n) = \eta_{\mu\nu} - \frac{k_\mu n_\nu + n_\mu k_\nu}{k \cdot n} + \frac{n^2 k_\mu k_\nu}{(k \cdot n)^2},$$



up to the overall metric-sign convention. If  $\mathcal{M} = \epsilon_\mu J^\mu$  and  $k \cdot J = 0$ , every  $n$ -dependent term contains a factor  $k \cdot J$  or its conjugate. Hence  $P_{\mu\nu}$  may be replaced by  $\eta_{\mu\nu}$ .

A single Compton ordering is not conserved:

$$k_\nu \bar{u}' \gamma^\mu S_F(p+k) \gamma^\nu u = \bar{u}' \gamma^\mu u \neq 0.$$

Using the covariant replacement on that diagram alone would retain unphysical polarizations. Only the direct-plus-crossed sum satisfies the Ward identity and permits the replacement.

(b) In the target rest frame one may write

$$T(\omega) = \boldsymbol{\epsilon}'^* \cdot \boldsymbol{\epsilon} f(\omega) + i \boldsymbol{\sigma} \cdot (\boldsymbol{\epsilon}'^* \times \boldsymbol{\epsilon}) g(\omega).$$

Interchanging the incoming and outgoing photon reverses  $\omega$  and interchanges the polarization vectors. The scalar product is unchanged, whereas the cross product changes sign. Thus

$$\boxed{f(-\omega) = f(\omega), \quad g(-\omega) = -g(\omega).}$$

The subtraction polynomial for  $f$  may contain only even powers (a constant at the minimal twice-subtracted level), while that for  $g$  may contain only odd powers (beginning with a term proportional to  $\omega$ ). Crossing therefore removes half of the otherwise allowed subtraction constants.

(c) The optical theorem supplies

$$\text{Im } f^{(4)}(\omega) = \frac{\omega}{4\pi} \sigma_{\text{KN}}(\omega).$$

Crossing combines the positive- and negative-energy cuts, and the renormalized low-energy theorem fixes the even subtraction constant. Consequently

$$\boxed{f(\omega) = -\frac{\alpha}{m} + \frac{\omega^2}{2\pi^2} \mathcal{P} \int_0^\infty \frac{\sigma_{\text{KN}}(E)}{E^2 - \omega^2} dE + \frac{i\omega}{4\pi} \sigma_{\text{KN}}(\omega) + O(e^6).}$$

At small  $\omega$ ,

$$\text{Im } f^{(4)}(\omega) = \frac{\sigma_T}{4\pi} \omega + O(\omega^2), \quad \sigma_T = \frac{8\pi\alpha^2}{3m^2}.$$

One must not replace the dispersive denominator by a power series in  $\omega^2/E^2$ : the cut begins at  $E = 0$  and  $\int_0^\infty \sigma_{\text{KN}}(E) dE/E^2$  diverges. Using  $\sigma_{\text{KN}}(E) = \sigma_T[1 - 2E/m + O(E^2/m^2)]$  in the low-energy part of the principal-value integral instead gives

$$\text{Re } f^{(4)}(\omega) = -\frac{\sigma_T}{\pi^2 m} \omega^2 \ln \frac{m}{|\omega|} + C\omega^2 + O(\omega^4 \ln |\omega|).$$

The finite  $C$  is determined by the complete integral. The  $\omega^2 \ln |\omega|$  term is the nonanalytic imprint of the massless electron–photon threshold.

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## 11 One-Loop Radiative Corrections in Quantum Electrodynamics

In this chapter we shall proceed to carry out some of the classic one-loop calculations in the theory of charged leptons—massive spin  $\frac{1}{2}$  particles that interact only with the electromagnetic field. There are three known species or “flavors” of leptons: the electron and muon, and the heavier, more recently discovered tauon. For definiteness we shall refer to the charged particles in our calculations here as “electrons,” though most of our calculations will apply equally to muons and tauons. After some generalities in Section 11.1, we will move on to the calculation of the vacuum polarization in Section 11.2, the anomalous magnetic moment of the electron in Section 11.3, and the electron self-energy in Section 11.4. Along the way, we will introduce a number of the mathematical techniques that prove useful in such calculations, including the use of Feynman parameters, Wick rotation, and both the dimensional regularization of ’t Hooft and Veltman and the older regularization method of Pauli and Villars. Although we shall encounter infinities, it will be seen that the final results are finite if expressed in terms of the renormalized charge and mass. In the next chapter we shall extend what we have learned here about renormalization to general theories in arbitrary orders of perturbation theory.

### 11.1 Counterterms

The Lagrangian density for electrons and photons is taken in the form<sup>122</sup>

$$\mathcal{L} = -\frac{1}{4}F_B^{\mu\nu}F_{B\mu\nu} - \bar{\psi}_B[\gamma_\mu(\partial^\mu + ie_B A_B^\mu) + m_B]\psi_B, \quad (11.1.1)$$

where  $F_B^{\mu\nu} \equiv \partial^\mu A_B^\nu - \partial^\nu A_B^\mu$ ;  $A_B^\mu$  and  $\psi_B$  are the bare (i.e., unrenormalized) fields of the photon and electron, and  $-e_B$  and  $m_B$  are the bare charge and mass of the electron. As described in the previous chapter, we introduce renormalized fields and charge and mass:

$$\psi \equiv Z_2^{-1/2}\psi_B, \quad (11.1.2)$$

$$A^\mu \equiv Z_3^{-1/2}A_B^\mu, \quad (11.1.3)$$

$$e \equiv Z_3^{+1/2}e_B, \quad (11.1.4)$$

$$m \equiv m_B + \delta m, \quad (11.1.5)$$

with the constants  $Z_2$ ,  $Z_3$ , and  $\delta m$  adjusted so that the propagators of the renormalized fields have poles in the same position and with the same residues as the propagators of the free fields in the absence of interactions. The Lagrangian may then be written in terms of renormalized quantities, as

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_2, \quad (11.1.6)$$

where

$$\mathcal{L}_0 = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \bar{\psi}[\gamma_\mu\partial^\mu + m]\psi, \quad (11.1.7)$$

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<sup>122</sup>In this chapter we will not be making transformations between Heisenberg- and interaction-picture operators, so we shall return to a conventional notation, in which an upper case  $A$  and a lower case  $\psi$  are used to denote the photon and charged particle fields, respectively.

$$\mathcal{L}_1 = -ieA_\mu \bar{\psi} \gamma^\mu \psi, \quad (11.1.8)$$

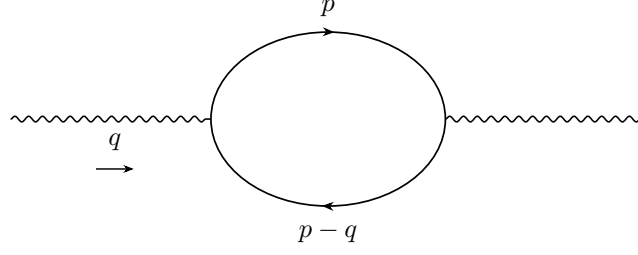
and  $\mathcal{L}_2$  is a sum of “counterterms”

$$\begin{aligned} \mathcal{L}_2 = & -\frac{1}{4}(Z_3 - 1)F^{\mu\nu}F_{\mu\nu} - (Z_2 - 1)\bar{\psi}[\gamma_\mu \partial^\mu + m]\psi \\ & + Z_2 \delta m \bar{\psi} \psi - ie(Z_2 - 1)A_\mu \bar{\psi} \gamma^\mu \psi. \end{aligned} \quad (11.1.9)$$

It will turn out that all of the terms in  $\mathcal{L}_2$  are of second order and higher order in  $e$ , and that these terms just suffice to cancel the ultraviolet divergences that arise from loop graphs.

## 11.2 Vacuum Polarization

We now begin our first calculation of a radiative correction involving loop graphs, the so-called vacuum polarization effect, consisting of the corrections to the propagator associated with an internal photon line. Vacuum polarization produces measurable shifts in the energy levels of hydrogen, and makes an important correction to the energies of muons bound in atomic orbits around heavy nuclei. Also, as we shall see in Volume II, the calculation of the vacuum polarization provides a key element in the calculation of the high energy behavior of electrodynamics and other gauge theories.



**Figure 11.1.** The one-loop diagram for the vacuum polarization in quantum electrodynamics. Here wavy lines represent photons; lines carrying arrows represent electrons.

As in Section 10.5, we define  $i(2\pi)^4 \Pi^{*\mu\nu}(q)$  as the sum of all connected graphs with two external photon lines with polarization indices  $\mu$  and  $\nu$  and carrying four-momentum  $q$  into and out of the diagram, not including photon propagators for the two external lines, and with the asterisk indicating that we exclude diagrams that can be disconnected by cutting through some internal photon line. The complete photon propagator  $\Delta'^{\mu\nu}(q)$  is given by Eq. (10.5.13):

$$\Delta' = \Delta [1 - \Pi^* \Delta]^{-1}, \quad (11.2.1)$$

where  $\Delta^{\mu\nu}(q)$  is the photon propagator without radiative corrections. Our task here is to calculate the leading contributions to  $\Pi^{*\mu\nu}(q)$ .

In lowest order there is a one-loop contribution to  $\Pi^*$ , corresponding to the diagram in Figure 11.1:

$$i(2\pi)^4 \Pi_{1 \text{ loop}}^{*\rho\sigma}(q) = - \int d^4p \operatorname{Tr} \left\{ \left[ \frac{-i}{(2\pi)^4} \frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} \right] [(2\pi)^4 e\gamma^\rho] \right. \\ \left. \cdot \left[ \frac{-i}{(2\pi)^4} \frac{-i(\not{p} - \not{q}) + m}{(p - q)^2 + m^2 - i\epsilon} \right] [(2\pi)^4 e\gamma^\sigma] \right\}. \quad (11.2.2)$$

with the first minus sign on the right required by the presence of a fermion loop. More simply, this is

$$\Pi_{1 \text{ loop}}^{*\rho\sigma}(q) = \frac{-ie^2}{(2\pi)^4} \int d^4p \frac{\operatorname{Tr}\{[-i\not{p} + m]\gamma^\rho[-i(\not{p} - \not{q}) + m]\gamma^\sigma\}}{(p^2 + m^2 - i\epsilon)((p - q)^2 + m^2 - i\epsilon)}. \quad (11.2.3)$$

The first step in doing this integral is to use a trick introduced by Feynman [1]. We use the elementary formula

$$\frac{1}{AB} = \int_0^1 \frac{dx}{[(1 - x)A + xB]^2} \quad (11.2.4)$$

to write the product of scalar propagators in Eq. (11.2.3) as

$$\begin{aligned}
& \frac{1}{(p^2 + m^2 - i\epsilon)((p - q)^2 + m^2 - i\epsilon)} \\
&= \int_0^1 [(p^2 + m^2 - i\epsilon)(1 - x) + ((p - q)^2 + m^2 - i\epsilon)x]^{-2} dx \\
&= \int_0^1 [p^2 + m^2 - i\epsilon - 2p \cdot qx + q^2x]^{-2} dx \\
&= \int_0^1 [(p - qx)^2 + m^2 - i\epsilon + q^2x(1 - x)]^{-2} dx.
\end{aligned}$$

(This is a special case of a class of integrals given in the Appendix to this chapter.) We can now shift the variable of integration in momentum space<sup>123</sup>

$$p \longrightarrow p + qx,$$

so that Eq. (11.2.3) becomes

$$\begin{aligned}
\Pi_{1\text{ loop}}^{*\rho\sigma}(q) &= \frac{-ie^2}{(2\pi)^4} \int_0^1 dx \int d^4p [p^2 + m^2 - i\epsilon + q^2x(1 - x)]^{-2} \\
&\quad \cdot \text{Tr}\{[-i(\not{p} + \not{q}x) + m]\gamma^\rho[-i(\not{p} - \not{q}(1 - x)) + m]\gamma^\sigma\}.
\end{aligned} \tag{11.2.5}$$

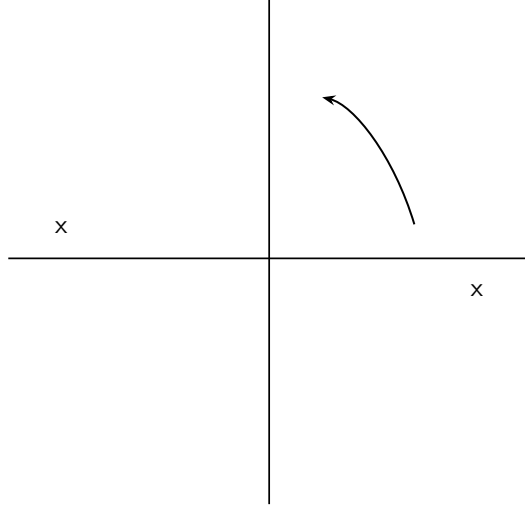
Using the results of the Appendix to Chapter 8, the trace here can easily be calculated as

$$\begin{aligned}
& \text{Tr}\{[-i(\not{p} + \not{q}x) + m]\gamma^\rho[-i(\not{p} - \not{q}(1 - x)) + m]\gamma^\sigma\} \\
&= 4[-(p + qx)^\rho(p - q(1 - x))^\sigma + (p + qx) \cdot (p - q(1 - x))\eta^{\rho\sigma} \\
&\quad - (p + qx)^\sigma(p - q(1 - x))^\rho + m^2\eta^{\rho\sigma}].
\end{aligned} \tag{11.2.6}$$

Our next step is called a Wick rotation [2]. As long as  $-q^2 < 4m^2$ , the quantity  $m^2 + q^2x(1 - x)$  is positive for all  $x$  between 0 and 1, so the poles in the integrand of Eq. (11.2.5) are at  $p^0 = \pm\sqrt{\mathbf{p}^2 + m^2 + q^2x(1 - x)} - i\epsilon$ , i.e., just above the negative real axis and just below the positive real axis. (See Figure 11.2.) We can rotate the contour of integrations of  $p^0$  counterclockwise without crossing either of these poles, so that instead of integrating  $p^0$  on the real axis from  $-\infty$  to  $+\infty$ , we integrate it on the imaginary axis from  $-i\infty$  to  $+i\infty$ . That is, we can write  $p^0 = ip^4$ , with  $p^4$  integrated over real values from  $-\infty$  to  $+\infty$ . (If an  $i\epsilon$  instead of  $-i\epsilon$  had appeared in the denominator of the propagator, then we would have been setting  $p^0 = -ip^4$ , with  $p^4$  again integrated over real values from  $-\infty$  to  $+\infty$ . The effect would be a change of sign of  $\Pi_{1\text{ loop}}^{*\rho\sigma}(q)$ .)

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<sup>123</sup>Strictly speaking, this step is only valid in convergent integrals. In principle, in order to justify the shift of variables, we should introduce some regulator scheme to make all integrals converge, such as the dimensional regularization scheme discussed below.



**Figure 11.2.** Wick rotation of the  $p^0$  contour of integration. Small x's mark the poles in the  $p^0$  complex plane; the arrow indicates the direction of rotation of the contour of integration, from the real to the imaginary  $p^0$ -axes.

Equation (11.2.5) now becomes

$$\begin{aligned} \Pi_{1\text{ loop}}^{*\rho\sigma}(q) = & \frac{4e^2}{(2\pi)^4} \int_0^1 dx \int (d^4p)_E [p^2 + m^2 + q^2 x(1-x)]^{-2} \\ & \cdot [-(p+qx)^\rho (p-q(1-x))^\sigma + (p+qx) \cdot (p-q(1-x)) \eta^{\rho\sigma} \\ & - (p+qx)^\sigma (p-q(1-x))^\rho + m^2 \eta^{\rho\sigma}], \end{aligned} \quad (11.2.7)$$

where

$$(d^4p)_E = dp^1 dp^2 dp^3 dp^4$$

and all scalar products are evaluated using the Euclidean norm

$$a \cdot b = a^1 b^1 + a^2 b^2 + a^3 b^3 + a^4 b^4$$

with the understanding that  $q^4 \equiv -iq^0$ . Also,  $\eta^{\rho\sigma}$  can be taken as either the Kronecker delta, with the indices running over 1, 2, 3, 4, or as the usual Minkowski tensor, with the indices running over 0, 1, 2, 3.

The integral (11.2.7) is badly divergent. Eventually all infinities will cancel, but to see this it is necessary at intermediate stages of the calculation to use some sort of regularization technique that makes the integrals finite. It would not do simply to cut off the integrals at some maximum momentum  $\Lambda$ , integrating only over  $p^\mu$  with  $p^2 < \Lambda^2$ , because this would amount to introducing a step function  $\theta(\Lambda^2 - p^2)$  into the electron propagator, and the Ward identity (10.4.25) shows that in order to maintain gauge invariance, any modification of the electron propagator must be accompanied with a modification of the electron-photon vertex. In fact, with an ordinary cutoff  $\Lambda$ , radiative corrections would induce a photon mass, a clear violation of the requirements of gauge invariance.



Experience has shown that the most convenient method for regulating divergent integrals without impairing gauge invariance is the dimensional regularization technique introduced by 't Hooft and Veltman [3] in 1972, based on a continuation from four to an arbitrary number  $d$  of spacetime dimensions. This amounts to carrying out angular averages in integrals like (11.2.7) by dropping all terms that are odd in  $p$ , and replacing the terms that have even numbers of  $p$ -factors with<sup>124</sup>

$$p^\mu p^\nu \longrightarrow p^2 \eta^{\mu\nu} / d, \quad (11.2.8)$$

$$p^\mu p^\nu p^\rho p^\sigma \longrightarrow (p^2)^2 [\eta^{\mu\nu} \eta^{\rho\sigma} + \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}] / d(d+2). \quad (11.2.9)$$

Also, after writing the integrand in this way as a function only of  $p^2$ , the volume element  $d^4 p_E$  is to be replaced with  $\Omega_d \kappa^{d-1} d\kappa$ , where  $\kappa \equiv \sqrt{p^2}$ , and  $\Omega_d$  is the area of a unit sphere in  $d$  dimensions

$$\Omega_d = 2\pi^{d/2} / \Gamma(d/2). \quad (11.2.10)$$

The integral (11.2.7) now converges for complex spacetime dimensionality  $d$ . We can continue the integral through complex  $d$ -values to  $d = 4$ , the infinities then reappearing as factors  $(d-4)^{-1}$ .

For the integral (11.2.7), dimensional regularization gives

$$\begin{aligned} \Pi_{1 \text{ loop}}^{*\rho\sigma}(q) = & \frac{4e^2 \Omega_d}{(2\pi)^4} \int_0^1 dx \int_0^\infty \kappa^{d-1} d\kappa [\kappa^2 + m^2 + q^2 x(1-x)]^{-2} \\ & \cdot \left[ -\frac{2\kappa^2}{d} \eta^{\rho\sigma} + 2q^\rho q^\sigma x(1-x) + (\kappa^2 - q^2 x(1-x)) \eta^{\rho\sigma} + m^2 \eta^{\rho\sigma} \right]. \end{aligned}$$

The integrals over  $\kappa$  can be carried out for any complex  $d$  (or for any real  $d$ , aside from the even integers). We use the well-known formulas (given in greater generality in the Appendix to this chapter):

$$\int_0^\infty \kappa^{d-1} [\kappa^2 + v^2]^{-2} d\kappa = \frac{1}{2} (v^2)^{d/2-2} \Gamma(d/2) \Gamma(2-d/2), \quad (11.2.11)$$

$$\int_0^\infty \kappa^{d+1} [\kappa^2 + v^2]^{-2} d\kappa = \frac{1}{2} (v^2)^{d/2-1} \Gamma(1+d/2) \Gamma(1-d/2). \quad (11.2.12)$$

and find

$$\begin{aligned} \Pi_{1 \text{ loop}}^{*\rho\sigma}(q) = & \frac{2e^2 \Omega_d}{(2\pi)^4} \int_0^1 dx \left[ \left(1 - \frac{2}{d}\right) \eta^{\rho\sigma} (m^2 + q^2 x(1-x))^{d/2-1} \Gamma(1+d/2) \Gamma(1-d/2) \right. \\ & + (2q^\rho q^\sigma x(1-x) - q^2 \eta^{\rho\sigma} x(1-x) + m^2 \eta^{\rho\sigma}) (m^2 + q^2 x(1-x))^{d/2-2} \\ & \left. \cdot \Gamma(d/2) \Gamma(2-d/2) \right]. \end{aligned}$$

The two terms in the integrand can be combined, using

$$(1 - 2/d) \Gamma(1+d/2) \Gamma(1-d/2) = -\Gamma(d/2) \Gamma(2-d/2).$$

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<sup>124</sup>These expressions may most easily be derived by noting that their form is dictated by Lorentz invariance and the symmetry among the indices  $\mu, \nu, \rho$ , etc., while the factors may be found by requiring that both sides give the same result when contracted with  $\eta$ 's.

We find

$$\begin{aligned} \Pi_{1\text{ loop}}^{*\rho\sigma}(q) &= \frac{4e^2\Omega_d}{(2\pi)^4} \Gamma(d/2) \Gamma(2-d/2) (q^\rho q^\sigma - q^2 \eta^{\rho\sigma}) \\ &\cdot \int_0^1 dx x(1-x) (m^2 + q^2 x(1-x))^{d/2-2}. \end{aligned} \quad (11.2.13)$$

We note the very important result that this contribution to  $\Pi^{*\rho\sigma}$  satisfies the relation

$$q_\rho \Pi_{1\text{ loop}}^{*\rho\sigma}(q) = 0 \quad (11.2.14)$$

that was derived in Section 10.5 on the basis of the conservation and neutrality of the electric current. It was precisely to achieve this result that we adopted the dimensional regularization scheme. The reason that dimensional regularization gives this result is that the conservation of current does not depend on the dimensionality of spacetime.

The gamma function  $\Gamma(2-d/2)$  in Eq. (11.2.13) blows up for  $d \rightarrow 4$ . Fortunately, as we saw in Section 11.1, there is another term that must be added to  $\Pi^{*\rho\sigma}(q)$ , arising from the term  $-\frac{1}{4}(Z_3-1)F_{\mu\nu}F^{\mu\nu}$  in the interaction Lagrangian. This term has a structure like Eq. (11.2.13)

$$\Pi_{\mathcal{L}_2}^{*\rho\sigma}(q) = -(Z_3-1)(q^2 \eta^{\rho\sigma} - q^\rho q^\sigma), \quad (11.2.15)$$

so to order  $e^2$ , the complete  $\Pi^*$  has the form

$$\Pi^{*\rho\sigma}(q) = (q^2 \eta^{\rho\sigma} - q^\rho q^\sigma) \pi(q^2), \quad (11.2.16)$$

with

$$\pi(q^2) = -\frac{4e^2\Omega_d}{(2\pi)^4} \Gamma(d/2) \Gamma(2-d/2) \int_0^1 dx x(1-x) (m^2 + q^2 x(1-x))^{d/2-2} - (Z_3-1). \quad (11.2.17)$$

As we saw in Section 10.5, the definition of the renormalized electromagnetic field requires that  $\pi(0) = 0$  (in order that the residue of the pole in the complete photon propagator at  $q^2 = 0$  should be the same as for the bare propagator, aside from gauge-dependent terms). Therefore, to order  $e^2$ ,

$$Z_3 = 1 - \frac{4e^2\Omega_d}{(2\pi)^4} \Gamma(d/2) \Gamma(2-d/2) (m^2)^{d/2-2} \int_0^1 x(1-x) dx, \quad (11.2.18)$$

so that, to order  $e^2$ ,

$$\begin{aligned} \pi(q^2) &= -\frac{4e^2\Omega_d}{(2\pi)^4} \Gamma(d/2) \Gamma(2-d/2) \int_0^1 dx x(1-x) \\ &\cdot \left[ (m^2 + q^2 x(1-x))^{d/2-2} - (m^2)^{d/2-2} \right]. \end{aligned} \quad (11.2.19)$$

Now we can remove the regularization, allowing  $d$  to approach its physical value  $d = 4$ . As mentioned before, there is an infinity in the one-loop contribution, arising from the limiting behavior of the Gamma function

$$\Gamma(2-d/2) \longrightarrow \frac{1}{2-d/2} - \gamma,$$

where  $\gamma$  is the Euler constant,  $\gamma = 0.5772157$ . The infinite part of  $Z_3 - 1$  is given by using  $1/(2 - d/2)$  for  $\Gamma(2 - d/2)$ , and replacing  $d$  everywhere else by 4:

$$(Z_3 - 1)_\infty = -\frac{4e^2 \cdot 2\pi^2}{6(2\pi)^4} \frac{1}{2 - d/2} = \frac{e^2}{6\pi^2} \frac{1}{d - 4}. \quad (11.2.20)$$

We shall see in Volume II that this result may be used to derive the leading term in the renormalization group equation for the electric charge.

The poles at  $d = 4$  obviously cancel in  $\pi(q^2)$ , because for  $d = 4$  both  $(m^2 + q^2x(1 - x))^{d/2-2}$  and  $(m^2)^{d/2-2}$  have the same limit, unity. For the same reason, the term  $-\gamma$  in  $\Gamma(2 - d/2)$  cancels in the total  $\pi(q^2)$ , though it does make a finite contribution to  $Z_3 - 1$ . There are other finite contributions to  $Z_3 - 1$ , that arise from the product of the pole in  $\Gamma(2 - d/2)$  with the linear terms in the expansion of  $\Omega_d\Gamma(d/2)$  around  $d = 4$ , but these also cancel in the total  $\pi(q^2)$ . Indeed, in carrying out our dimensional regularization, we might have replaced  $(2\pi)^{-4}$  with  $(2\pi)^{-d}$ , and the factor  $\text{Tr } \mathbf{1} = 4$  might have been replaced with the dimensionality  $2^{d/2}$  of gamma matrices in arbitrary even spacetime dimensionalities  $d$ , and these too would have contributed to the finite part of  $Z_3 - 1$ , but not of  $\pi(q^2)$ . Moreover,  $e^2$  cannot be supposed to be  $d$ -independent, because as shown by inspection of Eq. (11.2.13), it has the  $d$ -dependent dimensionality  $[\text{mass}]^{4-d}$ . If we take  $e^2 \propto \mu^{4-d}$ , where  $\mu$  is some quantity with the units of mass, then there are additional finite terms in  $Z_3 - 1$ , arising from the product of the pole in  $\Gamma(2 - d/2)$  with the term  $(4 - d)\ln \mu$  in the expansion of  $\mu^{4-d}$  in powers of  $4 - d$ , but again, these cancel between  $Z_3 - 1$  and the one-loop contributions to  $\pi(q^2)$ .

The only terms that do contribute to  $\pi(q^2)$  in the limit  $d \rightarrow 4$  are those arising from the product of the pole in  $\Gamma(2 - d/2)$  with the linear terms in the expansion of  $(m^2 + q^2x(1 - x))^{d/2-2}$  and  $(m^2)^{d/2-2}$  in powers of  $d - 4$ :

$$(m^2 + q^2x(1 - x))^{d/2-2} - (m^2)^{d/2-2} \longrightarrow \left(\frac{d}{2} - 2\right) \ln \left(1 + \frac{q^2x(1 - x)}{m^2}\right). \quad (11.2.21)$$

This gives at last

$$\pi(q^2) = \frac{e^2}{2\pi^2} \int_0^1 x(1 - x) \ln \left(1 + \frac{q^2x(1 - x)}{m^2}\right) dx. \quad (11.2.22)$$

The physical significance of the vacuum polarization can be explored by considering its effect on the scattering of two charged particles of spin  $\frac{1}{2}$ . The Feynman diagrams of Figure 11.3 make contributions to the scattering  $S$ -matrix element of the form

$$\begin{aligned} S_a(1, 2 \rightarrow 1', 2') &= (2\pi)^{-12/2} \delta^4(p_{1'} + p_{2'} - p_1 - p_2) [e_1(2\pi)^4 \bar{u}_{1'} \gamma^\mu u_1] \\ &\quad \cdot \left[ -i(2\pi)^{-4} \frac{1}{q^2} \right] [e_2(2\pi)^4 \bar{u}_{2'} \gamma_\mu u_2], \\ S_b(1, 2 \rightarrow 1', 2') &= (2\pi)^{-12/2} \delta^4(p_{1'} + p_{2'} - p_1 - p_2) [e_1(2\pi)^4 \bar{u}_{1'} \gamma^\mu u_1] \\ &\quad \cdot \left[ -i(2\pi)^{-4} \frac{1}{q^2} \right]^2 [i(2\pi)^4 (q^2 \eta_{\mu\nu} - q_\mu q_\nu) \pi(q^2)] [e_2(2\pi)^4 \bar{u}_{2'} \gamma^\nu u_2], \end{aligned}$$

where  $e_1$  and  $e_2$  are the charges of the two particles being scattered;  $\pi(q^2)$  is calculated using for  $e$  in Eq. (11.2.22) the magnitude of the charge of the particle circulating in the loop in Figure 11.3; and  $q^\mu$  is the momentum transfer  $q = p_1 - p_{1'} = p_{2'} - p_2$ . Using the conservation property  $q_\mu \bar{u}_{1'} \gamma^\mu u_1 = 0$  the two diagrams together yield an  $S$ -matrix element:

$$S_{a+b}(1, 2 \rightarrow 1', 2') = \frac{-ie_1 e_2}{4\pi^2 q^2} [1 + \pi(q^2)] \delta^4(p_{1'} + p_{2'} - p_1 - p_2) \cdot [\bar{u}_{1'} \gamma^\mu u_1] [\bar{u}_{2'} \gamma_\mu u_2]. \quad (11.2.23)$$

In the non-relativistic limit,  $\bar{u}_{1'} \gamma^0 u_1 \simeq -i\delta_{\sigma_{1'}, \sigma_1}$  while  $\bar{u}_{1'} \gamma^i u_1 \simeq 0$ , and likewise for particle 2. Also, in this limit  $q^0$  is negligible compared with  $|\mathbf{q}|$ . Equation (11.2.23) in this limit becomes

$$S_{a+b}(1, 2 \rightarrow 1', 2') = \frac{-ie_1 e_2}{4\pi^2 \mathbf{q}^2} [1 + \pi(\mathbf{q}^2)] \delta^4(p_{1'} + p_{2'} - p_1 - p_2) \delta_{\sigma_{1'}, \sigma_1} \delta_{\sigma_{2'}, \sigma_2}. \quad (11.2.24)$$

This may be compared with the  $S$ -matrix in the Born approximation due to a local spin-independent central potential  $V(r)$ :

$$S_{\text{Born}}(1, 2 \rightarrow 1', 2') = -2\pi i \delta(E_{1'} + E_{2'} - E_1 - E_2) T_{\text{Born}}(1, 2 \rightarrow 1', 2'). \quad (11.2.25)$$

$$T_{\text{Born}}(1, 2 \rightarrow 1', 2') = \delta_{\sigma_{1'}, \sigma_1} \delta_{\sigma_{2'}, \sigma_2} \int d^3 x_1 \int d^3 x_2 V(|\mathbf{x}_1 - \mathbf{x}_2|) \cdot (2\pi)^{-12/2} e^{-i\mathbf{p}_{1'} \cdot \mathbf{x}_1} e^{-i\mathbf{p}_{2'} \cdot \mathbf{x}_2} e^{i\mathbf{p}_1 \cdot \mathbf{x}_1} e^{i\mathbf{p}_2 \cdot \mathbf{x}_2}. \quad (11.2.26)$$

Setting  $\mathbf{x}_1 = \mathbf{x}_2 + \mathbf{r}$ , this gives

$$S_{\text{Born}} = \frac{-i}{4\pi^2} \delta^4(p_{1'} + p_{2'} - p_1 - p_2) \delta_{\sigma_{1'}, \sigma_1} \delta_{\sigma_{2'}, \sigma_2} \int d^3 r V(r) e^{-i\mathbf{q} \cdot \mathbf{r}}. \quad (11.2.27)$$

Comparing this with Eq. (11.2.23) shows that in the non-relativistic limit the diagrams of Figure 11.3 yield the same  $S$ -matrix element as a potential  $V(r)$  such that

$$\int d^3 r V(r) e^{-i\mathbf{q} \cdot \mathbf{r}} = e_1 e_2 \frac{1 + \pi(\mathbf{q}^2)}{\mathbf{q}^2}$$

or, inverting the Fourier transform,

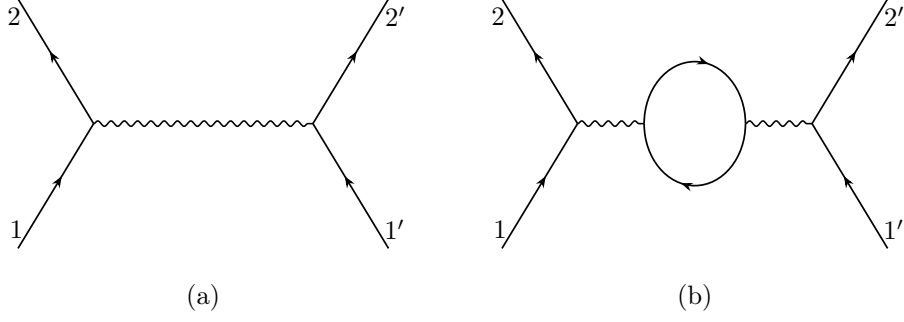
$$V(r) = \frac{e_1 e_2}{(2\pi)^3} \int d^3 q e^{i\mathbf{q} \cdot \mathbf{r}} \left[ \frac{1 + \pi(\mathbf{q}^2)}{\mathbf{q}^2} \right]. \quad (11.2.28)$$

Equation (11.2.28) is to first order in the radiative correction the same potential energy that would be produced by the electrostatic interaction of two extended charge distributions  $e_1 \eta(\mathbf{x})$  and  $e_2 \eta(\mathbf{y})$  at a distance  $\mathbf{r}$ :

$$V(|\mathbf{r}|) = e_1 e_2 \int d^3 x \int d^3 y \frac{\eta(\mathbf{x}) \eta(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y} + \mathbf{r}|}, \quad (11.2.29)$$

where

$$\eta(\mathbf{r}) = \delta^3(\mathbf{r}) + \frac{1}{2(2\pi)^3} \int d^3 q \pi(\mathbf{q}^2) e^{i\mathbf{q} \cdot \mathbf{r}}. \quad (11.2.30)$$



**Figure 11.3.** Two diagrams for the scattering of charged particles. Here lines carrying arrows are charged particles; wavy lines are photons. Diagram (b) represents the lowest-order vacuum polarization correction to the tree approximation graph (a).

Note that

$$\int d^3r \eta(\mathbf{r}) = 1 + \frac{1}{2}\pi(0) = 1, \quad (11.2.31)$$

so the total charges of particles 1 and 2, as determined from the long-range part of the Coulomb potential, are the same constants  $e_1$  and  $e_2$  that govern the interactions of the renormalized electromagnetic field.

For  $|\mathbf{r}| \neq 0$  the integral (11.2.30) can be carried out by a straightforward contour integration:

$$\eta(\mathbf{r}) = -\frac{e^2}{8\pi^3 r^3} \int_0^1 x(1-x) dx \left[ 1 + \frac{mr}{\sqrt{x(1-x)}} \right] \exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right).$$

This expression is negative everywhere. However, we have seen that the integral of  $\eta(\mathbf{r})$  over all  $\mathbf{r}$  equals +1. Therefore,  $\eta(\mathbf{r})$  must contain a term  $(1+L)\delta^3(\mathbf{r})$  that is singular at  $\mathbf{r} = 0$ , with  $L$  chosen to satisfy Eq. (11.2.31):

$$L = \frac{e^2}{8\pi^3} \int \frac{d^3r}{r^3} \int_0^1 x(1-x) dx \left[ 1 + \frac{mr}{\sqrt{x(1-x)}} \right] \exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right). \quad (11.2.32)$$

The complete expression for the charge distribution function is then

$$\begin{aligned} \eta(\mathbf{r}) = & (1+L)\delta^3(\mathbf{r}) - \frac{e^2}{8\pi^3 r^3} \int_0^1 x(1-x) dx \\ & \cdot \left[ 1 + \frac{mr}{\sqrt{x(1-x)}} \right] \exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right). \end{aligned} \quad (11.2.33)$$

The physical interpretation of this result is that a bare point charge attracts particles of charge of opposite sign out of the vacuum, repelling their antiparticles to infinity, so that the bare charge is partially shielded, yielding a renormalized charge smaller by a factor  $1/(1+L)$ . As a check, we may note that if we cut off the divergent integral (11.2.32) by taking the integral to extend only over  $r \geq a$ , we find that the part that is divergent for  $a \rightarrow 0$  is

$$L_\infty = \frac{e^2}{12\pi^2} \ln a^{-1}. \quad (11.2.34)$$

Hence if we identify the momentum space cutoff  $\Lambda$  with  $a^{-1}$ , the divergent part of  $L$  is related to the divergent part of  $Z_3 - 1$  by

$$(Z_3 - 1)_\infty = -2L_\infty, \quad (11.2.35)$$

because to order  $e^2$  the renormalized charge (10.4.18) is given by

$$e_r = Z_3^{1/2} e_{Br} \simeq \left(1 + \frac{1}{2}(Z_3 - 1)\right) e_{Br} \simeq (1 + L)^{-1} e_{Br}. \quad (11.2.36)$$

Equation (11.2.35) is confirmed below.

Vacuum polarization has a measurable effect on muonic atomic energy levels. As we shall see in Chapter 14, the effect of Feynman graph (b) in Figure 11.3 is to shift the energy of an atomic state with wave function  $\psi(\mathbf{r})$  by

$$\Delta E = \int d^3r \Delta V(\mathbf{r}) |\psi(\mathbf{r})|^2, \quad (11.2.37)$$

where  $\Delta V(\mathbf{r})$  is the perturbation in the potential (11.2.28):

$$\Delta V(\mathbf{r}) = \frac{e_1 e_2}{(2\pi)^3} \int d^3q e^{i\mathbf{q}\cdot\mathbf{r}} \left[ \frac{\pi(\mathbf{q}^2)}{\mathbf{q}^2} \right]. \quad (11.2.38)$$

This perturbation falls off exponentially for  $r \gg m^{-1}$ . On the other hand, the wave function of electrons in ordinary atoms will generally be confined within a much larger radius  $a \gg m^{-1}$ ; for instance, for hydrogenic orbits of electrons around a nucleus of charge  $Ze$  we have  $a = 137/Zm$  (where here  $m = m_e$ ). The energy shift will then depend only on the behavior of the wave function for  $r \ll a$ . For orbital angular momentum  $\ell$ , the wave function behaves like  $r^\ell$  for  $r \ll a$ , so Eq. (11.2.37) gives  $\Delta E$  proportional to a factor  $(ma)^{-2\ell-3}$ . The effect of vacuum polarization is therefore very much larger for  $\ell = 0$  than for higher orbital angular momenta. For  $\ell = 0$  the wave function is approximately equal to the constant  $\psi(0)$  for  $r$  less than or of the order of  $m^{-1}$ , so Eq. (11.2.37) becomes

$$\Delta E = |\psi(0)|^2 \int d^3r \Delta V(\mathbf{r}). \quad (11.2.39)$$

Using Eqs. (11.2.38) and (11.2.22), the integral of the shift in the potential (for  $e_1 e_2 = -Ze^2$ ) is

$$\int d^3r \Delta V(\mathbf{r}) = -Ze^2 \pi'(0) = -\frac{4Z\alpha^2}{15m^2}. \quad (11.2.40)$$

Also, in states of hydrogenic atoms with  $\ell = 0$  and principal quantum number  $n$  the wave function at the origin is

$$\psi(0) = \frac{2}{\sqrt{4\pi}} \left( \frac{Z\alpha m}{n} \right)^{3/2}, \quad (11.2.41)$$

so the energy shift (11.2.39) is

$$\Delta E = -\frac{4Z^4 \alpha^5 m}{15\pi n^3}. \quad (11.2.42)$$

For instance, in the  $2s$  state of hydrogen this energy shift is  $-1.122 \cdot 10^{-7}$  eV, corresponding to a frequency shift  $\Delta E/2\pi\hbar$  of  $-37.13$  MHz. This is sometimes called the Uehling effect [4]. As discussed in Chapter 1, such tiny energy shifts became measurable because in the absence of various radiative corrections the pure Dirac theory would predict exact degeneracy of the  $2s$  and  $2p$  states of hydrogen. As we shall see in Chapter 14, most of the  $+1058$  MHz “Lamb shift” between the  $2s$  and the  $2p$  states comes from other radiative corrections, but the agreement between theory and experiment is good enough to verify the presence of the  $-37.13$  MHz shift due to vacuum polarization.

Although vacuum polarization contributes only a small part of the radiative corrections in ordinary atoms, it dominates the radiative corrections in muonic atoms, in which a muon takes the place of the orbiting electron. This is because most radiative corrections give energy shifts in muonic atoms that on dimensional grounds are proportional to  $m_\mu$ , while the integrated vacuum polarization energy  $\int d^3r \Delta V$  due to an electron loop is still proportional to  $m_e^{-2}$  as in Eq. (11.2.40), giving an energy shift proportional to  $m_\mu^3 m_e^{-2} = (210)^2 m_\mu$ . However, in this case the muonic atomic radius is not much larger than the electron Compton wavelength, so the approximate result (11.2.39) only gives the order of magnitude of the energy shift due to vacuum polarization.

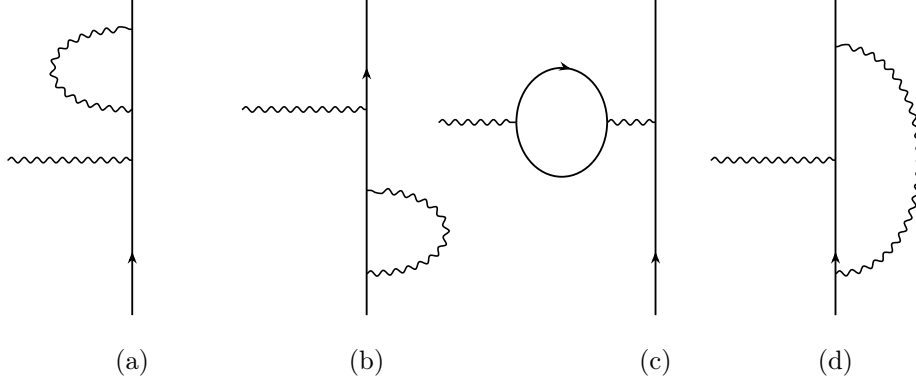
\* \* \*

For the purposes of comparison with later calculations, note that if we had cut off the integral (11.2.7) at  $\kappa = \Lambda$ , then in place of Eq. (11.2.20) we would have encountered an integral of the form

$$(Z_3 - 1)_\infty = -\frac{e^2}{6\pi^2} \int_\mu^\Lambda \kappa^{d-5} d\kappa = \frac{e^2}{6\pi^2} \frac{\mu^{d-4} - \Lambda^{d-4}}{d-4},$$

where  $\mu$  is an infrared effective cutoff of the order of the mass of the charged particle circulating in the loop of Figure 11.1. (The easiest way to find the constant factor here is to require that the limit of this expression for  $d < 3$  and  $\Lambda \rightarrow \infty$  matches Eq. (11.2.20).) With such an ultraviolet cutoff in place, we can pass to the limit  $d \rightarrow 4$ , and obtain

$$(Z_3 - 1)_\infty = -\frac{e^2}{6\pi^2} \ln(\Lambda/\mu). \quad (11.2.43)$$



**Figure 11.4.** One-loop diagrams for the photon-lepton vertex function  $\Gamma^{\mu*}$ . Here wavy lines represent photons; other lines represent electrons or muons. Diagrams (a) and (b) are cancelled by lepton field renormalization terms; diagram (c) arises from the vacuum polarization calculated in Section 11.2; and (d) is the term calculated in Section 11.3.

### 11.3 Anomalous Magnetic Moments and Charge Radii

For our next example, we shall calculate the shift in the magnetic moment and the charge radius of an electron or muon due to lowest-order radiative corrections. The one-loop graphs and renormalization corrections for the photon-lepton vertex are shown in Figure 11.4. Of these graphs, those involving insertions in incoming or outgoing lepton lines vanish because the lepton is on the mass shell, as discussed in Section 10.3. The graph involving an insertion in the external photon line is the vacuum polarization effect, discussed in the previous section. This leaves one one-loop graph (the last in Figure 11.4) that needs to be calculated here:

$$\Gamma_{1\text{ loop}}^{\mu}(p', p) = \int d^4k [e\gamma^{\rho}(2\pi)^4] \left[ \frac{-i}{(2\pi)^4} \frac{-i(\not{p}' - \not{k}) + m}{(p' - k)^2 + m^2 - i\epsilon} \right] [\gamma^{\mu}] \cdot \left[ \frac{-i}{(2\pi)^4} \frac{-i(\not{p} - \not{k}) + m}{(p - k)^2 + m^2 - i\epsilon} \right] [e\gamma_{\rho}(2\pi)^4] \left[ \frac{-i}{(2\pi)^4} \frac{1}{k^2 - i\epsilon} \right], \quad (11.3.1)$$

where  $p'$  and  $p$  are the final and initial lepton four-momenta, respectively. (The contribution of the vertex connecting the external photon line and the internal lepton line is taken as  $\gamma^{\mu}$ , because a factor  $e(2\pi)^4$  was extracted in defining  $\Gamma^{\mu}$ .)

This integral has an obvious ultraviolet divergence, roughly like  $\int d^4k/(k^2)^2$ . Unlike the case of the vacuum polarization, here we do not need a fancy regularization procedure like dimensional regularization to maintain the structure required by gauge invariance, because the photon is a neutral particle and so the integral may be rendered finite by suitable modifications of the photon propagator (for instance by including a factor  $M^2/(k^2 + M^2)$  with a large cutoff mass  $M$ ), without having to introduce modifications elsewhere to maintain gauge invariance. In any case, as we shall see, the anomalous magnetic moment and charge radii can be calculated without encountering any ultraviolet divergences at all. In what follows we shall leave the integrals for the vertex function in their infinite form,



with it being understood that if necessary any divergent integrals can be expressed in terms of a cutoff mass  $M$ .

We start by combining denominators, using a repeated version of the Feynman trick described in the Appendix to this chapter

$$\frac{1}{ABC} = 2 \int_0^1 dx \int_0^x dy [Ay + B(x-y) + C(1-x)]^{-3}. \quad (11.3.2)$$

Applied to the denominators in Eq. (11.3.1), this gives

$$\begin{aligned} & \frac{1}{(p' - k)^2 + m^2 - i\epsilon} \frac{1}{(p - k)^2 + m^2 - i\epsilon} \frac{1}{k^2 - i\epsilon} \\ &= 2 \int_0^1 dx \int_0^x dy \left[ ((p' - k)^2 + m^2 - i\epsilon)y + ((p - k)^2 + m^2 - i\epsilon)(x - y) \right. \\ & \quad \left. + (k^2 - i\epsilon)(1 - x) \right]^{-3} \\ &= 2 \int_0^1 dx \int_0^x dy \left[ (k - p'y - p(x - y))^2 + m^2 x^2 + q^2 y(x - y) - i\epsilon \right]^{-3}, \end{aligned} \quad (11.3.3)$$

where  $q \equiv p - p'$  is the momentum transferred to the photon. Shifting the variable of integration

$$k \longrightarrow k + p'y + p(x - y)$$

the integral (11.3.1) becomes

$$\begin{aligned} \Gamma_{1 \text{ loop}}^\mu(p', p) &= \frac{2ie^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int \frac{d^4 k}{[k^2 + m^2 x^2 + q^2 y(x - y) - i\epsilon]^3} \\ & \quad \cdot \gamma^\rho [-i(\not{p}'(1 - y) - \not{k} - \not{p}(x - y)) + m] \gamma^\mu \\ & \quad \cdot [-i(\not{p}(1 - x + y) - \not{k} - \not{p}'y) + m] \gamma_\rho. \end{aligned} \quad (11.3.4)$$

Our next step is a Wick rotation. As explained in the previous section, the  $-i\epsilon$  in the denominator dictates that when we rotate the  $k^0$  contour of integration to the imaginary axis we must rotate counterclockwise, so that the integral over  $k^0$  from  $-\infty$  to  $+\infty$  is replaced with an integral over imaginary values from  $-i\infty$  to  $+i\infty$ , or equivalently over real values of  $k^4 \equiv -ik^0$  from  $-\infty$  to  $+\infty$ . We also exploit the rotational symmetry of the denominator in Eq. (11.3.4); we drop terms in the denominator of odd order in  $k$ , replace  $k^\mu k^\nu$  with  $\eta^{\mu\nu} \kappa^2/4$ , and replace the volume element  $d^4 k = i dk^1 dk^2 dk^3 dk^4$  with  $2i\pi^2 \kappa^3 d\kappa$ , where  $\kappa$  is the Euclidean length of the four-vector  $k$ . Putting this all together, Eq. (11.3.4) now becomes

$$\begin{aligned} \Gamma_{1 \text{ loop}}^\mu(p', p) &= \frac{-4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \left\{ -\frac{\kappa^2}{4} \gamma^\rho \gamma^\sigma \gamma^\mu \gamma_\sigma \gamma_\rho \right. \\ & \quad \left. + \gamma^\rho [-i(\not{p}'(1 - y) - \not{p}(x - y)) + m] \gamma^\mu [-i(\not{p}(1 - x + y) - \not{p}'y) + m] \gamma_\rho \right\} \\ & \quad \cdot [\kappa^2 + m^2 x^2 + q^2 y(x - y)]^{-3}. \end{aligned} \quad (11.3.5)$$

We are interested here only in the matrix element  $\bar{u}'\Gamma^\mu u$  of the vertex function between Dirac spinors that satisfy the relations

$$\bar{u}'[i\not{p}' + m] = 0, \quad [i\not{p} + m]u = 0.$$

We can therefore simplify this expression by using the anticommutation relations of the Dirac matrices to move all factors  $\not{p}'$  to the right and all factors  $\not{p}$  to the left, replacing them when they arrive on the right or left with  $im$ . After a straightforward but tedious calculation, Eq. (11.3.5) then becomes

$$\begin{aligned} \bar{u}'\Gamma_{\text{one loop}}^\mu(p', p)u &= \frac{-4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \bar{u}' \left\{ \gamma^\mu [-\kappa^2 + 2m^2(x^2 - 4x + 2) \right. \\ &\quad \left. + 2q^2(y(x - y) + 1 - x)] \right. \\ &\quad \left. + 4im\not{p}'^\mu(y - x + xy) + 4im\not{p}^\mu(x^2 - xy - y) \right\} u \\ &\quad \cdot [\kappa^2 + m^2x^2 + q^2y(x - y)]^{-3}. \end{aligned} \quad (11.3.6)$$

We next exploit the symmetry of the final factor under the reflection  $y \rightarrow x - y$ . Under this reflection, the functions  $y - x + xy$  and  $x^2 - xy - y$  that multiply  $\not{p}'^\mu$  and  $\not{p}^\mu$  are interchanged, so both may be replaced with their average:

$$\frac{1}{2}(y - x + xy) + \frac{1}{2}(x^2 - xy - y) = -\frac{1}{2}x(1 - x).$$

This gives finally

$$\begin{aligned} \bar{u}'\Gamma_{\text{one loop}}^\mu(p', p)u &= \frac{-4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \\ &\quad \cdot \bar{u}' \left\{ \gamma^\mu [-\kappa^2 + 2m^2(x^2 - 4x + 2) + 2q^2(y(x - y) + 1 - x)] \right. \\ &\quad \left. - 2im(\not{p}'^\mu + \not{p}^\mu)x(1 - x) \right\} u [\kappa^2 + m^2x^2 + q^2y(x - y)]^{-3}. \end{aligned} \quad (11.3.7)$$

Note that  $p^\mu$  and  $p'^\mu$  now enter only in the combination  $p^\mu + p'^\mu$ , as required by current conservation.

There are other diagrams that need to be taken into account. Of course, there is the zeroth-order term  $\gamma^\mu$  in  $\Gamma^\mu$ . The term proportional to  $Z_2 - 1$  in the correction term (11.1.9) yields a term in  $\Gamma^\mu$

$$\Gamma_{\mathcal{L}_2}^\mu = (Z_2 - 1)\gamma^\mu. \quad (11.3.8)$$

Also, the effect of insertions of corrections to the external photon propagator is a term:

$$\Gamma_{\text{vac pol}}^\mu(p', p) = \frac{1}{(p' - p)^2 - i\epsilon} \Pi^{\mu\nu}(p' - p)\gamma_\nu. \quad (11.3.9)$$

The form of each of these terms is in agreement with the general result (10.6.10) (with  $H(q^2) = 0$ )

$$\bar{u}'\Gamma^\mu(p', p)u = \bar{u}' \left[ \gamma^\mu F(q^2) - \frac{i}{2m}(p + p')^\mu G(q^2) \right] u. \quad (11.3.10)$$

To order  $e^2$ , the form factors are

$$F(q^2) = Z_2 + \pi(q^2) + \frac{4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \cdot \frac{\kappa^2 - 2m^2(x^2 - 4x + 2) - 2q^2(y(x - y) + 1 - x)}{[\kappa^2 + m^2 x^2 + q^2 y(x - y)]^3}, \quad (11.3.11)$$

$$G(q^2) = \frac{-4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \frac{4m^2 x(1 - x) \kappa^3 d\kappa}{[\kappa^2 + m^2 x^2 + q^2 y(x - y)]^3}, \quad (11.3.12)$$

where  $\pi(q^2)$  is the vacuum polarization function (11.2.22).

The integral for the form factor  $G(q^2)$  is finite as it stands:

$$G(q^2) = -\frac{e^2 m^2}{4\pi^2} \int_0^1 dx \int_0^x dy \frac{x(1 - x)}{m^2 x^2 + q^2 y(x - y)}. \quad (11.3.13)$$

This makes it easy to calculate the anomalous magnetic moment. As noted in Section 10.6, it is only the  $\gamma^\mu$  term that contributes to the magnetic moment, so the effect of radiative corrections is to multiply the Dirac value  $e/2m$  of the magnetic moment by a factor  $F(0)$ . But the definition of  $e$  as the true lepton charge requires that

$$F(0) + G(0) = 1, \quad (11.3.14)$$

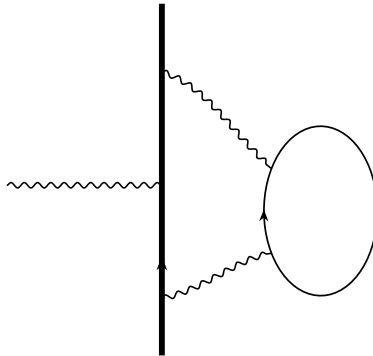
so the magnetic moment may be expressed as

$$\mu = \frac{e}{2m} (1 - G(0)). \quad (11.3.15)$$

From Eq. (11.3.13), we find

$$-G(0) = \frac{e^2}{8\pi^2} = 0.001161. \quad (11.3.16)$$

This is the famous  $\alpha/2\pi$  correction first calculated by Schwinger [5].



**Figure 11.5.** A two-loop diagram for the muon magnetic moment. Here the heavy straight line represents a muon; the light wavy lines are photons; and the other light lines are electrons. This diagram makes a relatively large contribution to the fourth-order muon gyromagnetic ratio, proportional to  $\ln(m_\mu/m_e)$ .

Of course, this is only the first term in the radiative corrections to the magnetic moment. Even in just the next order, fourth order in  $e$ , there are so many terms that the calculations become quite complicated. However, because of the large muon–electron mass ratio, there is one fourth-order term in the magnetic moment of the muon that is somewhat larger than any of the others. It arises from the insertion of an electron loop in the virtual photon line of the second-order diagram, as shown in Figure 11.5. The effect of this electron loop is to change the photon propagator  $1/k^2$  in Eq. (11.3.1) to  $(1 + \pi_e(k^2))/k^2$ , where  $\pi_e(k^2)$  is given by Eq. (11.2.22), but with the mass  $m$  taken as the electron mass:

$$\pi_e(k^2) = \frac{e^2}{2\pi^2} \int_0^1 x(1-x) \ln \left( 1 + \frac{k^2 x(1-x)}{m_e^2} \right) dx.$$

Inspection of Eq. (11.3.12) shows that in calculating the muon magnetic moment the effective cutoff on the virtual photon momentum  $k$  is  $m_\mu$ . The ratio  $m_\mu/m_e$  is so large that for  $k^2$  of order  $m_\mu^2$  we may approximate

$$\pi_e(k^2) \simeq \frac{e^2}{2\pi^2} \int_0^1 x(1-x) \ln(m_\mu^2/m_e^2) = \frac{e^2}{12\pi^2} \ln(m_\mu^2/m_e^2), \quad (11.3.17)$$

with the neglected terms having coefficients of order unity in place of  $\ln(m_\mu^2/m_e^2)$ . Since this is a constant, the change in  $-G(0)$  produced by adding an electron loop in the virtual photon line is simply given by multiplying our previous result (11.3.16) for  $-G(0)$  by Eq. (11.3.17), so that now

$$\mu_\mu = \frac{e}{2m_\mu} \left( \frac{e^2}{8\pi^2} + \frac{e^4}{96\pi^4} \left[ \ln \frac{m_\mu^2}{m_e^2} + O(1) \right] \right). \quad (11.3.18)$$

(As we shall see in Volume II, this argument is a primitive version of the method of the renormalization group.) The result (11.3.18) may be compared with the full fourth-order result [6]

$$\mu_\mu = \frac{e}{2m_\mu} \left\{ \frac{e^2}{8\pi^2} + \frac{e^4}{96\pi^4} \left[ \ln \frac{m_\mu^2}{m_e^2} - \frac{25}{6} + \frac{197}{24} + \frac{\pi^2}{2} + \frac{9\zeta(3)}{2} - 3\pi^2 \ln 2 + O\left(\left(\frac{m_e}{m_\mu}\right)^2\right) \right] \right\}. \quad (11.3.19)$$

It turns out that the  $O(1)$  terms multiplying  $e^4/96\pi^4$  add up to  $-6.137$ , which is not very much smaller than  $\ln(m_\mu^2/m_e^2) = 10.663$ , so the approximation (11.3.18) gives the fourth-order terms only to a factor of order 2. The correct fourth-order result (11.3.19) gives  $\mu_\mu = 1.00116546e/2m_\mu$ , in comparison with the second-order result  $\mu_\mu = 1.001161e/2m_\mu$  and the current experimental value [7],  $\mu_\mu = 1.001165923(8)e/2m_\mu$ .

Now let us turn to the other form factor. The integral in Eq. (11.3.11) for  $F(q^2)$  has an ultraviolet divergence. However, in order to satisfy the charge-non-renormalization condition (11.3.14), it is necessary that  $Z_2$  take the value

$$Z_2 = 1 + \frac{e^2}{8\pi^2} - \frac{4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \frac{\kappa^2 - 2m^2(x^2 - 4x + 2)}{[\kappa^2 + m^2 x^2]^3}. \quad (11.3.20)$$

(Recall that  $\pi(0) = 0$ .) This is itself ultraviolet divergent, with an infinite part

$$(Z_2 - 1)_\infty = -\frac{e^2}{8\pi^2} \int_0^\infty \frac{d\kappa}{\kappa}. \quad (11.3.21)$$

Inserting Eq. (11.3.20) back into Eq. (11.3.11) gives

$$\begin{aligned} F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \\ \cdot \left\{ \frac{\kappa^2 - 2m^2(x^2 - 4x + 2) - 2q^2[y(x - y) + 1 - x]}{[\kappa^2 + m^2x^2 + q^2y(x - y)]^3} \right. \\ \left. - \frac{\kappa^2 - 2m^2(x^2 - 4x + 2)}{[\kappa^2 + m^2x^2]^3} \right\}. \end{aligned} \quad (11.3.22)$$

The integral over  $\kappa$  is now convergent:

$$\begin{aligned} F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \left\{ \frac{-m^2[x^2 - 4x + 2] - q^2[y(x - y) + 1 - x]}{m^2x^2 + q^2y(x - y)} \right. \\ \left. + \frac{x^2 - 4x + 2}{x^2} - \ln \left[ \frac{m^2x^2 + q^2y(x - y)}{m^2x^2} \right] \right\}. \end{aligned} \quad (11.3.23)$$

However, we see that the integral over  $x$  and  $y$  now diverges logarithmically at  $x = 0$  and  $y = 0$ , because there are two powers of  $x$  and/or  $y$  in the denominators, and just two differentials  $dx dy$  in the numerator. This divergence can be traced to the vanishing of the denominator  $[\kappa^2 + m^2x^2 + q^2y(x - y)]^3$  in Eq. (11.3.11) at  $x = 0$ ,  $y = 0$ , and  $\kappa = 0$ . Because this infinity comes from the region of small rather than large  $\kappa$ , it is termed an *infrared divergence* rather than an ultraviolet divergence.

We shall give a comprehensive treatment of the infrared divergences in Chapter 13. It will be shown there that infrared divergences in the cross-section for processes like electron-electron scattering, such as those that are introduced by the infrared divergence in the electron form factor  $F(q^2)$ , are cancelled when we include the emission of low-energy photons as well as elastic scattering. Also, as we shall see in Chapter 14, when we calculate radiative corrections to atomic energy levels the infrared divergence in  $F(q^2)$  is cut off because the bound electron is not exactly on the free-particle mass shell. For the present we shall continue our calculation by simply introducing a fictitious photon mass  $\mu$  to cut off the infrared divergence in  $F(q^2)$ , leaving it for Chapter 14 to see how to use this result.

With a photon mass  $\mu$ , the denominator  $k^2 - i\epsilon$  in Eq. (11.3.1) would be replaced with  $k^2 + \mu^2 - i\epsilon$ . The effect would then be to add a term  $\mu^2(1 - x)$  to the cubed quantity in the denominators of Eqs. (11.3.3)–(11.3.7), (11.3.11), (11.3.20), and (11.3.22). Eq. (11.3.23) then is replaced with

$$\begin{aligned} F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \left\{ \frac{-m^2[x^2 - 4x + 2] - q^2[y(x - y) + 1 - x]}{m^2x^2 + q^2y(x - y) + \mu^2(1 - x)} \right. \\ \left. + \frac{m^2[x^2 - 4x + 2]}{m^2x^2 + \mu^2(1 - x)} - \ln \left[ \frac{m^2x^2 + q^2y(x - y) + \mu^2(1 - x)}{m^2x^2 + \mu^2(1 - x)} \right] \right\}. \end{aligned} \quad (11.3.24)$$

This integral is now completely convergent. It can be expressed in terms of Spence functions, but the result is not particularly illuminating. For our purposes in Chapter 14, it will be sufficient to calculate the behavior of  $F(q^2)$  for small  $q^2$ . We already know from the Ward identity that  $F(0) = 1 - G(0) = 1 + e^2/8\pi^2$ , so let us consider the first derivative  $F'(q^2)$  at  $q^2 = 0$ . According to Eq. (11.3.24), this is

$$F'(0) = \pi'(0) + \frac{2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \left\{ \frac{2y(x-y) + 1-x}{m^2 x^2 + \mu^2(1-x)} + \frac{m^2[x^2 - 4x + 2]y(x-y)}{[m^2 x^2 + \mu^2(1-x)]^2} \right\}. \quad (11.3.25)$$

The vacuum polarization contribution is given by Eq. (11.2.22) as

$$\pi'(0) = \frac{e^2}{60\pi^2 m^2}. \quad (11.3.26)$$

Dropping all terms proportional to powers of  $\mu/m$  in Eq. (11.3.25), we then have<sup>125</sup>

$$F'(0) = \frac{e^2}{24\pi^2 m^2} \left[ \ln \left( \frac{\mu^2}{m^2} \right) + \frac{2}{5} + \frac{1}{4} \right] \quad (11.3.27)$$

with the term  $2/5$  the contribution of vacuum polarization. On the other hand, Eq. (11.3.13) shows that  $G(q^2)$  has a finite derivative at  $q^2 = 0$ ,

$$G'(0) = \frac{e^2}{48\pi^2 m^2}. \quad (11.3.28)$$

These results are most conveniently expressed in terms of the charge form factor  $F_1(q^2)$ , defined by the alternative representation (10.6.15) of the vertex function

$$\begin{aligned} \bar{u}(p', \sigma') \Gamma^\mu(p', p) u(p, \sigma) \\ = \bar{u}(p', \sigma') \left[ \gamma^\mu F_1(q^2) + \frac{1}{2} i [\gamma^\mu, \gamma^\nu] (p' - p)_\nu F_2(q^2) \right] u(p, \sigma). \end{aligned} \quad (11.3.29)$$

According to Eqs. (10.6.17) and (10.6.18),

$$F_1(q^2) = F(q^2) + G(q^2). \quad (11.3.30)$$

For  $|q^2| \ll m^2$ , this form factor is approximately

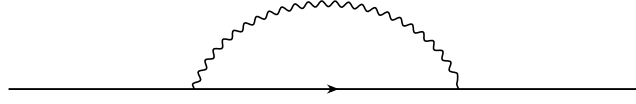
$$F_1(q^2) \simeq 1 + \frac{e^2}{24\pi^2} \frac{q^2}{m^2} \left[ \ln \left( \frac{\mu^2}{m^2} \right) + \frac{2}{5} + \frac{3}{4} \right]. \quad (11.3.31)$$

This may be expressed in terms of a *charge radius*  $a$ , defined by the limiting behavior of the charge form factor for  $q^2 \rightarrow 0$ :

$$F_1(q^2) \rightarrow 1 - q^2 a^2 / 6. \quad (11.3.32)$$

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<sup>125</sup>The  $y$ -integral is trivial. The  $x$ -integral is most easily calculated in the limit  $\mu \ll m$  by dividing the range of integration into two parts, one from 0 to  $s$ , where  $\mu/m \ll s \ll 1$ , and the second from  $s$  to 1.



**Figure 11.6.** The one-loop diagram for the electron self-energy function. As usual, the straight line represents an electron; the wavy line is a photon.

(This definition is motivated by the fact that the average of  $\exp(i\mathbf{q} \cdot \mathbf{x})$  over a spherical shell of radius  $a$  goes as  $1 - \mathbf{q}^2 a^2/6$  for  $\mathbf{q}^2 a^2 \ll 1$ .) We see that the charge radius of the electron is given by

$$a^2 = -\frac{e^2}{4\pi^2 m^2} \left[ \ln \left( \frac{\mu^2}{m^2} \right) + \frac{2}{5} + \frac{3}{4} \right]. \quad (11.3.33)$$

We will see in Chapter 14 that for electrons in atoms the role of the photon mass is played by an effective infrared cutoff that is much less than  $m$ , so the logarithm here is large and negative, yielding a positive value for  $a^2$ .

#### 11.4 Electron Self-Energy

We conclude this chapter with a calculation of the electron self-energy function. This by itself does not have any direct experimental implications, but some of the results here will be useful in Chapter 14 and Volume II.

As in Section 10.3, we define  $i(2\pi)^2[\Sigma^*(p)]_{\beta,\alpha}$  as the sum of all graphs with one incoming and one outgoing electron line carrying momenta  $p$  and Dirac indices  $\alpha$  and  $\beta$  respectively, with the asterisk indicating that we exclude diagrams that can be disconnected by cutting through some internal electron line, and with propagators omitted for the two external lines. The complete electron propagator is then given by the sum

$$[-i(2\pi)^{-4}S'(p)] = [-i(2\pi)^{-4}S(p)] + [-i(2\pi)^{-4}S(p)][i(2\pi)^4\Sigma^*(p)][-i(2\pi)^{-4}S(p)] + \cdots, \quad (11.4.1)$$

where

$$S(p) = \frac{-i\not{p} + m_e}{p^2 + m_e^2 - i\epsilon}. \quad (11.4.2)$$

The sum is trivial, and gives

$$S'(p) = [i\not{p} + m_e - \Sigma^*(p) - i\epsilon]^{-1}. \quad (11.4.3)$$

In lowest order there is a one-loop contribution to  $\Sigma^*$ , given by Figure 11.6:

$$\begin{aligned} i(2\pi)^4\Sigma_{1\text{ loop}}^*(p) &= \int d^4k \left[ \frac{-i}{(2\pi)^4} \frac{\eta_{\rho\sigma}}{k^2 - i\epsilon} \right] \cdot [(2\pi)^4 e\gamma^\rho] \\ &\quad \cdot \left[ \frac{-i}{(2\pi)^4} \frac{-i\not{p} + i\not{k} + m_e}{(p-k)^2 + m_e^2 - i\epsilon} \right] [(2\pi)^4 e\gamma^\sigma] \end{aligned}$$

or more simply

$$\Sigma_{1\text{ loop}}^*(p) = \frac{ie^2}{(2\pi)^4} \int d^4k \left[ \frac{1}{k^2 - i\epsilon} \right] \left[ \frac{\gamma^\rho(-i\not{p} + i\not{k} + m_e)\gamma_\rho}{(p-k)^2 + m_e^2 - i\epsilon} \right]. \quad (11.4.4)$$

(This is in Feynman gauge; amplitudes with charged particles off the mass shell are not gauge-invariant.) For use in our calculation of the Lamb shift, it will be convenient to use a method of regularization introduced by Pauli and Villars [8]. We replace the photon propagator  $(k^2 - i\epsilon)^{-1}$  with

$$\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon},$$

so that the electron self-energy function is

$$\Sigma_{1 \text{ loop}}^*(p) = \frac{ie^2}{(2\pi)^4} \int d^4k \left[ \frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right] \cdot \left[ \frac{\gamma^\rho(-i\not{p} + i\not{k} + m_e)\gamma_\rho}{(p-k)^2 + m_e^2 - i\epsilon} \right]. \quad (11.4.5)$$

Later we can drop the regulator by letting the regulator mass  $\mu$  go to infinity. In Chapter 14 we will also be interested in the case where  $\mu \ll m_e$ .

We again use the Feynman trick to combine denominators, and recall that  $\gamma^\rho \gamma^\kappa \gamma_\rho = -2\gamma^\kappa$  and  $\gamma^\rho \gamma_\rho = 4$ . This gives

$$\begin{aligned} \Sigma_{1 \text{ loop}}^*(p) &= \frac{ie^2}{(2\pi)^4} \int d^4k [2i(\not{p} - \not{k}) + 4m_e] \\ &\quad \cdot \int_0^1 dx \left[ \frac{1}{\{(k - px)^2 + p^2x(1-x) + m_e^2x - i\epsilon\}^2} \right. \\ &\quad \left. - \frac{1}{\{(k - px)^2 + p^2x(1-x) + m_e^2x + \mu^2(1-x) - i\epsilon\}^2} \right]. \end{aligned} \quad (11.4.6)$$

Shifting the variable of integration  $k \rightarrow k + px$  and rotating the contour of integration gives

$$\begin{aligned} \Sigma_{1 \text{ loop}}^*(p) &= \frac{-2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx [2i(1-x)\not{p} + 4m_e] \int_0^\infty d\kappa \kappa^3 \\ &\quad \cdot \left[ \frac{1}{\{\kappa^2 + p^2x(1-x) + m_e^2x\}^2} - \frac{1}{\{\kappa^2 + p^2x(1-x) + m_e^2x + \mu^2(1-x)\}^2} \right]. \end{aligned} \quad (11.4.7)$$

The  $\kappa$ -integral is trivial:

$$\Sigma_{1 \text{ loop}}^*(p) = \frac{-\pi^2 e^2}{(2\pi)^4} \int_0^1 dx [2i(1-x)\not{p} + 4m_e] \ln \left( \frac{p^2x(1-x) + m_e^2x + \mu^2(1-x)}{p^2x(1-x) + m_e^2x} \right). \quad (11.4.8)$$

The interaction (11.1.9) also contributes a renormalization counterterm  $-(Z_2 - 1)(i\not{p} + m_e) + Z_2 \delta m_e$  in  $\Sigma^*(p)$ , with  $Z_2$  and  $\delta m_e$  determined by the condition that the complete propagator  $S'(p)$  regarded as a function of  $i\not{p}$  should have a pole at  $i\not{p} = -m_e$  with residue unity. (As we shall see in the next chapter, this makes  $\Sigma^*$  finite as  $\mu \rightarrow \infty$  to all orders in  $e$ .) In lowest order, this gives

$$\begin{aligned} \delta m_e &= - \Sigma_{1 \text{ loop}}^*|_{i\not{p} = -m_e} \\ &= \frac{2m_e \pi^2 e^2}{(2\pi)^4} \int_0^1 dx [1+x] \ln \left( \frac{m_e^2 x^2 + \mu^2(1-x)}{m_e^2 x^2} \right), \end{aligned} \quad (11.4.9)$$



$$\begin{aligned}
Z_2 - 1 &= -i \left. \frac{\partial \Sigma_{1 \text{ loop}}^*}{\partial \not{p}} \right|_{i\not{p} = -m_e} \\
&= \frac{-2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \left\{ (1-x) \ln \left( \frac{m_e^2 x^2 + \mu^2(1-x)}{m_e^2 x^2} \right) - \frac{2\mu^2(1-x)^2(1+x)}{x[m_e^2 x^2 + \mu^2(1-x)]} \right\}.
\end{aligned} \tag{11.4.10}$$

(To this order, we do not distinguish between  $\delta m_e$  and  $Z_2 \delta m_e$ .) Dropping terms that vanish for  $\mu^2 \rightarrow \infty$ , Eqs. (11.4.8)–(11.4.10) yield

$$\Sigma_{1 \text{ loop}}^*(p) = \frac{-\pi^2 e^2}{(2\pi)^4} \int_0^1 dx [2i(1-x)\not{p} + 4m_e] \ln \left( \frac{\mu^2(1-x)}{p^2 x(1-x) + m_e^2 x} \right), \tag{11.4.11}$$

$$\delta m_e = \frac{2m_e \pi^2 e^2}{(2\pi)^4} \int_0^1 dx [1+x] \ln \left( \frac{\mu^2(1-x)}{m_e^2 x^2} \right), \tag{11.4.12}$$

$$Z_2 - 1 = \frac{-2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \left\{ (1-x) \ln \left( \frac{\mu^2(1-x)}{m_e^2 x^2} \right) - \frac{2(1-x^2)}{x} \right\}. \tag{11.4.13}$$

Inspection then shows that in the complete self-energy function the  $\ln \mu^2$  terms cancel, leaving us with

$$\begin{aligned}
\Sigma_{\text{order } e^2}^*(p) &= \Sigma_{1 \text{ loop}}^*(p) - (Z_2 - 1)(i\not{p} + m_e) + Z_2 \delta m_e \\
&= \frac{2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \left\{ [i(1-x)\not{p} + 2m_e] \ln \left( \frac{m_e^2(1-x)}{p^2 x(1-x) + m_e^2 x} \right) \right. \\
&\quad \left. - m_e[1+x] \ln \left( \frac{1-x}{x^2} \right) \right. \\
&\quad \left. - (i\not{p} + m_e) \left[ (1-x) \ln \left( \frac{1-x}{x^2} \right) - \frac{2(1-x^2)}{x} \right] \right\}.
\end{aligned} \tag{11.4.14}$$

There is still a divergence from the behavior of the last term as  $x \rightarrow 0$ , which can be traced to the singular behavior of the integral over the photon momentum  $k$  in Eq. (11.4.5) at  $k^2 = 0$ , when we take  $p^2$  at the point  $p^2 = -m_e^2$  where we evaluated  $Z_2 - 1$ . Such infrared divergences will be discussed in detail in Chapter 13. For the present, the point that concerns us is that the ultraviolet divergence has cancelled.

\* \* \*

The result (11.4.9) for  $\delta m_e$  is of some interest in itself. Note that  $\delta m_e/m_e > 0$ , as we would expect for the electromagnetic self-energy due to the interaction of a charge with its own field. But unlike the classical estimates of electromagnetic self-energy by Poincaré, Abraham, and others [9], Eq. (11.4.9) is only logarithmically divergent in the limit  $\mu \rightarrow \infty$  where the cutoff is removed. In this limit:

$$\delta m_e \rightarrow \frac{6m_e \pi^2 e^2}{(2\pi)^4} \ln \left( \frac{\mu}{m_e} \right). \tag{11.4.15}$$

In our calculation of the Lamb shift in Section 14.3 we will be interested in the opposite limit,  $\mu \ll m_e$ . Here Eq. (11.4.9) gives

$$\delta m_e \rightarrow \frac{e^2 \mu}{8\pi} \left[ 1 - \frac{\mu}{2\pi m_e} + \cdots \right]. \tag{11.4.16}$$

## Appendix Assorted Integrals

In order to combine the denominators of  $N$  propagators, we need to replace a product like  $D_1^{-1}D_2^{-1}\cdots D_N^{-1}$  with an integral of a function that involves a linear combination of  $D_1, D_2, \dots, D_N$ . For this purpose it is often convenient to make use of the formula

$$\frac{1}{D_1 D_2 \cdots D_N} = (N-1)! \int_0^1 dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{N-2}} dx_{N-1} \cdot [D_1 x_{N-1} + D_2 (x_{N-2} - x_{N-1}) + \cdots + D_N (1 - x_1)]^{-N}. \quad (11.A.1)$$

In this chapter we have used special cases of this formula for  $N = 2$  and  $N = 3$ .

After combining denominators, shifting the four-momentum variable of integration, Wick rotating, and using four-dimensional rotational invariance, we commonly encounter integrals of the form

$$\int d^4 k \frac{(k^2)^n}{(k^2 + \nu^2)^m}$$

with  $(k^2 + \nu^2)^m$  coming from the combined propagator denominators, and  $(k^2)^n$  coming from the propagator numerators and vertex momentum factors. This is divergent for  $2n + 4 \geq 2m$ , but the integral can be given a finite value by analytically continuing the spacetime dimensionality from 4 to a complex value  $d$ . To evaluate the resulting integral, we use the well-known formula

$$\int_0^\infty d\kappa \frac{\kappa^{\ell-1}}{(\kappa^2 + \nu^2)^m} = \nu^{\ell-2m} \frac{\Gamma(\ell/2) \Gamma(m - \ell/2)}{2\Gamma(m)}, \quad (11.A.2)$$

where  $\ell = d + 2n$ . We used this formula in the special cases  $n = 0, m = 2$  and  $n = 1, m = 2$  in Section 11.2.

Ultraviolet divergences manifest themselves in Eq. (11.A.2) as poles in the factor  $\Gamma(m - \ell/2) = \Gamma(m - n - d/2)$  as  $d \rightarrow 4$  with fixed integer  $n$ . For  $2 + n = m$ , this factor goes as

$$\Gamma\left(\frac{4-d}{2}\right) \rightarrow \frac{2}{d-4} + \gamma, \quad (11.A.3)$$

where  $\gamma = 0.5772157 \cdots$  is the Euler constant. The limiting behavior for  $2 + n > m$  can be obtained from (11.A.3) and the recursion relation for Gamma functions.

## Weinberg Exercises

- W.11.1.** Calculate the contributions to the vacuum polarization function  $\pi(q^2)$  and to  $Z_3$  of one-loop graphs involving a charged spinless particle of mass  $m_s$ . What effect does this have on the energy shift of the  $2s$  state of hydrogen, if  $m_s \gg Z\alpha m_e$ ?
- W.11.2.** Suppose that a neutral scalar field  $\phi$  of mass  $m_\phi$  has an interaction  $g\phi\bar{\psi}\psi$  with the electron field. To one-loop order, what effect does this have on the magnetic moment of the electron? On  $Z_2$ ?
- W.11.3.** Consider a neutral scalar field  $\phi$  with mass  $m_\phi$  and self-interaction  $g\phi^3/6$ . To one-loop order, calculate the  $S$ -matrix element for scalar-scalar scattering.
- W.11.4.** To one-loop order, calculate the effect of the neutral scalar field of Problem 2 on the mass shift  $\delta m_e$  of the electron.

## Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

- W.11.1.** Both scalar-QED graphs must be retained: the loop with two three-point vertices and the seagull graph. Their non-transverse pieces cancel. With the normalization

$$\Pi_s^{\mu\nu}(q) = (q^2 \eta^{\mu\nu} - q^\mu q^\nu) \pi_s(q^2), \quad \pi_s(0) = 0,$$

the result for one charged complex scalar is

$$\pi_s(q^2) = \frac{e^2}{16\pi^2} \int_0^1 dx (1-2x)^2 \ln \left( 1 + \frac{q^2 x(1-x)}{m_s^2} \right).$$

For comparison, the coefficient of the large- $q^2$  logarithm is  $e^2/(48\pi^2)$ , one quarter of the coefficient from a Dirac lepton. Dimensional regularization gives the corresponding divergent part

$$(Z_3 - 1)_{\text{div}} = \frac{e^2}{24\pi^2} \frac{1}{d-4}.$$

As in the fermion calculation, finite terms in  $Z_3$  depend on the definition of the  $d$ -dimensional scale, while  $\pi_s(q^2)$  above does not.

For  $q^2 \ll m_s^2$ ,

$$\pi_s(q^2) = \frac{e^2 q^2}{16\pi^2 m_s^2} \int_0^1 dx (1-2x)^2 x(1-x) + O(q^4) = \frac{e^2 q^2}{480\pi^2 m_s^2} + O(q^4).$$

The local potential shift therefore satisfies

$$\int d^3r \Delta V(\mathbf{r}) = -Ze^2 \pi'_s(0) = -\frac{Z\alpha^2}{30m_s^2}.$$

Using  $|\psi_{n0}(0)|^2 = (Z\alpha m_e)^3/(\pi n^3)$ , we obtain

$$\Delta E_{ns}^{(s)} = -\frac{Z^4 \alpha^5 m_e^3}{30\pi n^3 m_s^2}, \quad \Delta E_{2s}^{(s)} = -\frac{\alpha^5 m_e^3}{240\pi m_s^2} \quad (Z=1).$$

The assumed hierarchy  $m_s \gg Z\alpha m_e$  is precisely what permits the replacement of the short-range polarization potential by this contact term.

- W.11.2.** Let  $r = m_\phi/m_e$ . The one-loop vertex graph with a scalar and two electron propagators is ultraviolet finite in its Pauli form factor. After combining denominators and using the on-shell Dirac equations, it gives

$$\delta a_e \equiv \frac{g_e - 2}{2} = \frac{g^2}{8\pi^2} \int_0^1 dz \frac{(1-z)^2(1+z)}{(1-z)^2 + zr^2}.$$

Thus the scalar correction to the magnetic moment is  $\delta\mu_e = (e/2m_e)\delta a_e$ . It is positive. Useful checks are

$$\delta a_e \longrightarrow \frac{3g^2}{16\pi^2} \quad (r \rightarrow 0), \quad \delta a_e = \frac{g^2}{8\pi^2} \frac{m_e^2}{m_\phi^2} \left[ \ln \frac{m_\phi^2}{m_e^2} - \frac{7}{6} + O\left(\frac{m_e^2}{m_\phi^2} \ln r\right) \right] \quad (r \gg 1).$$

For completeness, the wave-function effect can be displayed without hiding the subtraction convention. Put

$$C_d = \frac{2}{4-d} - \gamma + \ln 4\pi, \quad \Delta(x, p^2) = xm_\phi^2 + (1-x)m_e^2 + x(1-x)p^2.$$

With the self-energy convention

$$S'^{-1}(p) = i\not{p} + m_e - \Sigma_\phi(p),$$

$$\Sigma_\phi(p) = -\frac{g^2}{16\pi^2} \int_0^1 dx [-ix\not{p} + m_e] \left[ C_d - \ln \frac{\Delta(x, p^2)}{\mu^2} \right],$$

terms vanishing as  $d \rightarrow 4$  being omitted, the on-shell residue condition is

$$Z_2 - 1 = -i \left. \frac{\partial \Sigma_\phi}{\partial \not{p}} \right|_{i\not{p} = -m_e}.$$

Writing  $\Delta_{\text{os}}(x) = xm_\phi^2 + (1-x)^2 m_e^2$ , this becomes

$$\boxed{Z_2 - 1 = \frac{g^2}{16\pi^2} \int_0^1 dx \left[ x \left( C_d - \ln \frac{\Delta_{\text{os}}(x)}{\mu^2} \right) + \frac{2m_e^2 x(1-x)(1+x)}{\Delta_{\text{os}}(x)} \right].}$$

Unlike the Pauli form factor,  $Z_2$  contains an ultraviolet pole and is scheme dependent.

**W.11.3.** It is clearest to state the answer in terms of renormalized one-particle irreducible functions. For a four-momentum  $P$ , define

$$D(P) = P^2 + m_\phi^2 - i\epsilon$$

and define the renormalized self-energy  $\Pi_R$  by the on-shell conditions

$$\Pi_R(-m_\phi^2) = 0, \quad \Pi'_R(-m_\phi^2) = 0.$$

The one-loop building blocks are

$$i\Pi_{\text{loop}}(P) = \frac{1}{2}(-ig)^2 \int \frac{d^d k}{(2\pi)^d} \frac{i}{D(k)} \frac{i}{D(P-k)},$$

$$i\Gamma_\Delta(P_1, P_2, P_3) = (-ig)^3 \int \frac{d^4 k}{(2\pi)^4} \frac{i}{D(k)} \frac{i}{D(k+P_1)} \frac{i}{D(k-P_2)}.$$

The triangle is ultraviolet finite in four dimensions. Its finite renormalized version  $\Gamma_R$  includes the chosen finite definition of  $g$ . The box for a cyclic ordering  $(P_1, P_2, P_3, P_4)$  is

$$iB(P_1, P_2, P_3, P_4) = (-ig)^4 \int \frac{d^4 k}{(2\pi)^4} \prod_{j=0}^3 \frac{i}{D(k+K_j)},$$

where  $K_0 = 0$ ,  $K_1 = P_1$ ,  $K_2 = P_1 + P_2$ , and  $K_3 = P_1 + P_2 + P_3$ . It too is ultraviolet finite.

Let all external momenta be incoming and put

$$P_s = p_1 + p_2, \quad P_t = p_1 + p_3, \quad P_u = p_1 + p_4.$$

Through order  $g^4$ , the amputated on-shell amplitude is

$$\begin{aligned} i\mathcal{M} = & \sum_{X=s,t,u} \left[ (-ig)^2 \frac{i}{D(P_X)} + (-ig)^2 \frac{i}{D(P_X)} i\Pi_R(P_X) \frac{i}{D(P_X)} \right] \\ & + \sum_{X=s,t,u} \left[ 2(-ig) \frac{i}{D(P_X)} i\Gamma_R(P_X) \right] + iB(s,t) + iB(s,u) + iB(t,u) + O(g^6). \end{aligned}$$

Here  $\Gamma_R(P_X)$  denotes the triangle correction at either end of the exchange line; the factor two counts the two ends. The three  $B$ 's are the inequivalent box orderings. External self-energy insertions are absent after LSZ because the pole residue is unity. The only ultraviolet divergence at this order lies in the two-point subgraph and is removed by the mass and field-strength counterterms used in  $\Pi_R$ . This expression is the complete one-loop  $S$ -matrix element, with the overall momentum-conserving delta function supplied by the usual  ${}_{\text{out}}\langle p_3, p_4 | S | p_1, p_2 \rangle_{\text{in}}$  normalization.

**W.11.4.** Use the same self-energy convention as in Problem 2. The mass counterterm is fixed by the pole condition,

$$\delta m_e^{(\phi)} = - \Sigma_\phi(p)|_{i\not{p}=-m_e}.$$

On the mass shell,

$$\Delta_{\text{os}}(x) = x m_\phi^2 + (1-x)^2 m_e^2, \quad [-ix\not{p} + m_e]_{i\not{p}=-m_e} = m_e(1+x),$$

and hence

$$\boxed{\delta m_e^{(\phi)} = \frac{g^2 m_e}{16\pi^2} \int_0^1 dx (1+x) \left[ C_d - \ln \frac{x m_\phi^2 + (1-x)^2 m_e^2}{\mu^2} \right].}$$

This is the bare on-shell counterterm in dimensional regularization. In the  $\overline{\text{MS}}$  scheme the bracket  $C_d$  is subtracted; in an on-shell scheme the full expression is used so that the dressed propagator has its pole at the measured mass. The ultraviolet pole has coefficient

$$\delta m_e^{(\phi)} \Big|_{\text{div}} = \frac{3g^2 m_e}{32\pi^2} C_d,$$

which has the required mass dimension and vanishes with  $m_e$ , as demanded by the chiral symmetry restored when the electron mass is zero.

## Supplementary Exercises

### S.11.1. A Census of One-Loop QED Graphs

(Adapted from Bernhard Mistlberger and Kevin Zhou, *Stanford PHYS 330, Problem Set 9, Problem 1.*)

- (a) For  $e^-e^+ \rightarrow \mu^-\mu^+$ , classify every connected diagram of order  $e^4$ , including external-leg, vacuum-polarization, vertex, box, crossed-box, and counterterm graphs.
- (b) For each class, write the corresponding dimensionally regulated loop integral with an explicit momentum routing, numerator, denominator, and overall sign. State how on-shell renormalization reorganizes the external-leg and subdivergent pieces.

### S.11.2. Feynman Parameters

(Adapted from Bernhard Mistlberger and Kevin Zhou, *Stanford PHYS 330: QFT I, Problem Set 9 (2022), Problem 2, “Feynman parameters,” pp. 1–2.*)

Loop integrals take a few generic forms. For simplicity, consider a scalar theory, so that the loop-integrand numerator contains no nontrivial factors. In the volume’s mostly-plus convention, define the tadpole, bubble, and triangle families by

$$I_T(a; m^2) = \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 + m^2 - i0)^a}, \quad (3)$$

$$I_B(a, b; p^2) = \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 - i0)^a ((k+p)^2 - i0)^b}, \quad (4)$$

$$I_\Delta(a, b, c; p_1, p_2) = \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 - i0)^a ((k+p_1)^2 - i0)^b ((k-p_2)^2 - i0)^c}. \quad (5)$$

They are named after the Feynman diagrams in which they appear. The source suppresses the displayed  $i0$  terms, but they are essential in the last part and have therefore been restored here. Regard  $a, b, c$  as general real exponents (although in applications they are usually positive integers), and regard  $d$  as a real parameter. We use dimensional regularization with  $d = 4 - 2\epsilon$  when taking the final check.

- (a) The Gamma function is defined by

$$\Gamma(z) = \int_0^\infty dx x^{z-1} e^{-x}, \quad (6)$$

and for a positive integer  $n$  it satisfies  $\Gamma(n) = (n-1)!$ . Show, for real  $\nu$  in the convergence domain and then by analytic continuation, that

$$\frac{1}{A^\nu} = \frac{1}{\Gamma(\nu)} \int_0^\infty dx x^{\nu-1} e^{-xA}. \quad (7)$$

- (b) Show that for general  $\nu_i$ ,

$$\frac{1}{\prod_{i=1}^n A_i^{\nu_i}} = \frac{\Gamma(\sum_{i=1}^n \nu_i)}{\prod_{i=1}^n \Gamma(\nu_i)} \int_0^1 \left( \prod_{i=1}^n dx_i \right) \delta\left(1 - \sum_{i=1}^n x_i\right) \frac{\prod_{i=1}^n x_i^{\nu_i-1}}{(\sum_{i=1}^n x_i A_i)^{\sum_{i=1}^n \nu_i}}. \quad (8)$$

Here the  $x_i$  are Feynman parameters.

*Hint.* Do not base the answer on the Peskin–Schroeder derivation, which works only for integer  $\nu_i$ . Instead start from the result of the preceding part.

- (c) Apply (8) to  $I_B(a, b; p^2)$  to obtain an integrand whose denominator is a single quantity raised to the  $a + b$  power.
- (d) Complete the evaluation of  $I_B(a, b; p^2)$  by Wick rotating to Euclidean signature and performing all remaining integrals. The Euclidean measure may be written  $d^d k_E = |k_E|^{d-1} d|k_E| d\Omega_d$ , where the area of the unit sphere in  $d$  dimensions is

$$\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \quad (9)$$

You may also use

$$\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} = \int_0^1 dx x^{a-1} (1-x)^{b-1}. \quad (10)$$

As a check, after translating the source’s mostly-minus invariant to the volume convention and setting  $d = 4 - 2\varepsilon$ , the answer must obey

$$I_B(1, 1; p^2) = \left( \frac{p^2 - i0}{4\pi} \right)^{-\varepsilon} \frac{i \Gamma(1-\varepsilon)^2 \Gamma(1+\varepsilon)}{(4\pi)^2 (1-2\varepsilon) \varepsilon \Gamma(1-2\varepsilon)}. \quad (11)$$

### S.11.3. Rank-Four Rotational Averaging and Tensor Reduction of a Bubble

(Adapted from Bernhard Mistlberger and Kevin Zhou, *Stanford PHYS 330, Problem Set 9, Problem 3.*)

- (a) For a rotationally invariant function  $f(k^2)$  in  $d$  Euclidean dimensions, reduce the integrals with numerators  $k^\mu k^\nu k^\rho$  and  $k^\mu k^\nu k^\rho k^\sigma$  to scalar integrals. Determine every coefficient by contraction.
- (b) Reduce the massless rank-one and rank-two bubble integrals to the scalar bubble  $I_B(a, b; p^2)$  and tadpoles. In dimensional regularization, use the vanishing of scaleless tadpoles to obtain the particularly simple result for  $a = b = 1$ .

### S.11.4. Self-Energy Corrections in QED

(Adapted from Bernhard Mistlberger and Kevin Zhou, *Stanford PHYS 330: QFT I, Problem Set 9 (2022), Problem 4, “Self-energy corrections in QED,” pp. 3–4.*)

Loop corrections to two-point correlation functions play a key role in renormalization and are often called self-energy corrections. The one-loop electron and photon self-energies in QED are the following labelled graphs:

The top diagram shows the electron self-energy  $\Sigma(p^2)$ . It consists of a horizontal fermion line with momentum  $p$  entering from the left and  $p$  exiting to the right. A wavy photon line forms a loop on top of the fermion line, with momentum  $k$  flowing clockwise. The internal fermion line has momentum  $k+p$ . The bottom diagram shows the photon vacuum polarization  $\Pi^{\mu\nu}(p^2)$ . It consists of a horizontal wavy photon line with momentum  $p$  entering from the left and  $p$  exiting to the right. A closed fermion loop is attached to the photon line. The left vertex of the loop has index  $\mu$  and the right vertex has index  $\nu$ . The top arc of the loop has momentum  $k+p$  and the bottom arc has momentum  $k$ .

$$\Sigma(p^2) = \text{diagram} \quad (20)$$

$$\Pi^{\mu\nu}(p^2) = \text{diagram}$$

Throughout this problem work in Feynman gauge and massless QED,  $m_e = 0$ , using dimensional regularization with  $d = 4 - 2\varepsilon$ . In the volume convention an internal photon carries the scalar denominator  $(k^2 - i0)^{-1}$ , a massless internal fermion carries the numerator  $-i\not{k}$  over  $q^2 - i0$ , a vertex contributes  $e\gamma^\mu$ , and a closed fermion loop contributes an additional minus sign.

For self-containment, the complete scalar and tensor results needed from Problems 2 and 3 of the source are supplied here. With  $I_B \equiv I_B(1, 1; p^2)$ ,

$$I_B = \left( \frac{p^2 - i0}{4\pi} \right)^{-\varepsilon} \frac{i \Gamma(1 - \varepsilon)^2 \Gamma(1 + \varepsilon)}{(4\pi)^2 (1 - 2\varepsilon) \varepsilon \Gamma(1 - 2\varepsilon)},$$

$$\int \frac{d^d k}{(2\pi)^d} \frac{k^\mu}{(k^2 - i0)((k+p)^2 - i0)} = -\frac{p^\mu}{2} I_B,$$

$$\int \frac{d^d k}{(2\pi)^d} \frac{k^\mu k^\nu}{(k^2 - i0)((k+p)^2 - i0)} = \frac{dp^\mu p^\nu - p^2 \eta^{\mu\nu}}{4(d-1)} I_B.$$

The last two formulas use the dimensional-regularization rule that scaleless tadpoles vanish.

- Using the Feynman rules, write down  $\Sigma(p^2)$  and  $\Pi^{\mu\nu}(p^2)$ . Because these are corrections to the electron and photon propagators, do *not* include external-leg factors such as  $u_s(p)$  or  $\epsilon_\mu$  to form a scalar matrix element. Instead,  $\Sigma$  must remain a  $4 \times 4$  spinor matrix and  $\Pi^{\mu\nu}$  a rank-two Lorentz tensor.
- Evaluate the loop integrals, expressing the final results in terms of  $p^2$  and  $\varepsilon$ .  
*Hint.* Once the gamma matrices have been handled, every integral that remains is one of the supplied scalar or tensor integrals.
- The Euler–Mascheroni constant is defined by

$$\gamma_E = - \int_0^\infty dx e^{-x} \log x \simeq 0.577. \quad (21)$$

Show that

$$\Gamma(1 - \varepsilon) = 1 + \gamma_E \varepsilon + O(\varepsilon^2). \quad (22)$$



- (d) Using (22), expand the results from the second part, dropping all terms that vanish as  $\varepsilon \rightarrow 0$ .

### S.11.5. Euclidean Master Integrals and Shifted Tensor Integrals

(Adapted from Heidelberg University, Quantum Field Theory 1, Sheet 11, Problem 1.)

- (a) Evaluate the  $d$ -dimensional Gaussian both as a product of one-dimensional integrals and in radial coordinates. Derive  $\Omega_d = 2\pi^{d/2}/\Gamma(d/2)$  and state its analytic continuation in  $d$ .
- (b) Evaluate

$$J_n(m^2) = \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + m^2)^n}$$

for complex  $d$  and  $n$ . State its convergence domain before analytic continuation and locate its ultraviolet poles.

- (c) Use a justified momentum shift to evaluate

$$\int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + m^2 + 2q \cdot p)^n}.$$

Specify the condition on  $m^2 - p^2$  in the Euclidean domain and then give the analytically continued result.

- (d) Evaluate the shifted master integrals of Exercise S.11.9 with numerators  $q^\mu$  and  $q^\mu q^\nu$ . Express the answers in terms of  $p^\mu$ ,  $\delta^{\mu\nu}$ , and Gamma functions, and check them by differentiating the scalar master integral.

### S.11.6. The Spinor Vacuum-Polarization Numerator and Subtracted Vacuum Polarization at Small Momentum

(Adapted from Heidelberg University, Quantum Field Theory 1, Sheet 11, Problem 2.)

- (a) Starting from the closed-fermion-loop rules, take the Dirac trace in the one-loop photon self-energy, introduce one Feynman parameter, and shift the loop momentum. Show before integration how the terms organize into a transverse tensor plus a rotational average.
- (b) Renormalize the spinor-QED vacuum polarization by imposing  $\pi(0) = 0$ . Derive the finite parameter integral for  $\pi(q^2)$  and its first two terms for  $|q^2| \ll m^2$ .

### S.11.7. QED Counterterms and On-Shell Renormalization

(Inspired by Mark Srednicki, Quantum Field Theory, Chapter 62, Eqs. (62.1)–(62.6) / Mark Srednicki, Quantum Field Theory, Chapter 62, Eqs. (62.7)–(62.25) / Mark Srednicki, Quantum Field Theory, Chapter 62, Eqs. (62.26)–(62.39).)

- (a) Use Lorentz invariance, gauge invariance, charge conjugation, parity, and power counting to list all local QED counterterms of mass dimension at most four. Explain why no photon mass, photon tadpole, or Pauli term belongs to the renormalizable counterterm Lagrangian.
- (b) Dyson-sum a transverse photon self-energy and show how the residue at  $q^2 = 0$  fixes the on-shell condition  $\pi(0) = 0$ . Relate this condition to the definition of the measured electric charge.
- (c) Write the most general parity-even fermion self-energy as  $\Sigma(p) = A(p^2)i\not{p} + B(p^2)m$ . Derive the two on-shell conditions that fix the mass and field-strength counterterms, and show that the renormalized self-energy has a double zero in the inverse propagator expansion about the physical pole.

#### S.11.8. One-Loop QED Renormalization Constants

(Adapted from Mark Srednicki, *Quantum Field Theory*, Chapter 62, Exercise 2; with the Ward-identity discussion in Chapter 68.)

Work in  $d = 4 - \epsilon$  and  $R_\xi$  gauge.

- (a) Compute the coefficients of  $e^2/\epsilon$  in  $Z_1, Z_2, Z_3$ , and  $Z_m$ .
- (b) Verify directly that the vertex and electron wave-function poles are equal, and derive the same equality from the zero-momentum Ward identity.
- (c) Specialize to Lorenz gauge  $\xi = 0$ , showing that  $Z_1 = Z_2 = 1 + O(e^4)$ , and explain why charge renormalization is then controlled only by  $Z_3$ .

#### S.11.9. Why Four-Photon Scattering Is Ultraviolet Finite

(Adapted from Mark Srednicki, *Quantum Field Theory*, Chapter 62, Exercise 3.)

Each one-loop box ordering with four external photons is superficially logarithmically divergent. Use gauge invariance and the absence of an allowed dimension-four four-photon counterterm to show that the complete sum is ultraviolet finite. Also explain why the three-photon amplitude vanishes.

#### S.11.10. From the Pauli Form Factor to $g - 2$

(Inspired by Mark Srednicki, *Quantum Field Theory*, Chapters 63–64.)

Decompose the on-shell QED vertex into Dirac and Pauli form factors. Couple it to a slowly varying magnetic field and prove that  $(g_e - 2)/2 = F_2(0)$ . Identify which normalization fixes  $F_1(0)$ .

#### S.11.11. Running Charge, Screening, and Heavy-Threshold Decoupling

(Inspired by David Skinner, *Advanced Quantum Field Theory*, Section 6.3.3 / David Skinner, *Advanced Quantum Field Theory*, Section 6.3.4 / David Skinner, *Advanced Quantum Field Theory*, Sections 6.3.3–6.3.4.)

- (a) Starting from the pole part of the photon field counterterm in minimal subtraction, derive the one-loop beta function for one Dirac fermion,  $\beta(e) = e^3/(12\pi^2)$ . Keep the  $d$ -dimensional classical scaling term until the pole cancellation is explicit.
- (b) Integrate the one-loop QED beta function between two scales. Show that the effective charge grows at short distance and explain the sign using the polarization cloud around a test charge.
- (c) Use the on-shell-subtracted vacuum-polarization integral to show that a charged fermion of mass  $M$  contributes only  $O(q^2/M^2)$  to processes with  $|q^2| \ll M^2$ . Reconcile this decoupling with the mass-independent minimal-subtraction beta function.

#### S.11.12. The Euler–Heisenberg Effective Interaction

*(Inspired by David Skinner, Advanced Quantum Field Theory, Section 7.2 | David Skinner, Advanced Quantum Field Theory, Sections 7.1–7.2.)*

- (a) At energies well below the electron mass, classify the independent Lorentz- and parity-even operators built from four electromagnetic field strengths and no derivatives. Show that only two invariants are required and give them both covariantly and in terms of  $\mathbf{E}, \mathbf{B}$ .
- (b) For a constant electromagnetic background, the renormalized one-loop spinor determinant may be written in Schwinger proper time. Expand it through fourth order in the fields and recover the Euler–Heisenberg interaction, including its numerical coefficients.
- (c) Use the Euler–Heisenberg interaction to determine the energy and fine-structure-constant dependence of the photon–photon scattering amplitude and total cross section for  $\omega \ll m_e$ . Explain why no lower power of  $\omega/m_e$  is possible.

#### S.11.13. Pauli Operators, Nonrelativistic Reduction, and Simplex Parameters

*(Inspired by Hong Liu and MIT OpenCourseWare, 8.324 Lecture 15, Eqs. (1)–(4) | Hong Liu and MIT OpenCourseWare, 8.324 Lecture 15, Eqs. (5)–(11) | Hong Liu and MIT OpenCourseWare, 8.324 Lecture 15, Eqs. (12)–(17).)*

- (a) Show that the zero-momentum Pauli form factor is reproduced by a local operator  $\bar{\psi}\sigma^{\mu\nu}F_{\mu\nu}\psi$ . Match its coefficient to the on-shell vertex and verify its mass dimension and discrete symmetries.
- (b) Split the Dirac spinor into large and small two-component fields in a slowly varying electromagnetic background. Eliminate the small field through order  $1/m$ , and derive the magnetic Hamiltonian including  $F_2(0)$ .
- (c) Apply the three-denominator Feynman identity to the one-loop QED vertex. For on-shell external electrons, perform the loop-momentum shift and derive the common denominator in terms of three simplex parameters and  $q^2$ .

### S.11.14. The One-Loop Pauli Form Factor

(Inspired by Hong Liu and MIT OpenCourseWare, 8.324 Lecture 15, Eqs. (18)–(22) | Hong Liu and MIT OpenCourseWare, 8.324 Lecture 15, Eqs. (22)–(24) | Hong Liu and MIT OpenCourseWare, 8.324 Lecture 15, Eqs. (24)–(25).)

- (a) Use symmetric integration, gamma-matrix identities, and the external Dirac equations to isolate the part of the one-loop vertex numerator proportional to  $p^\mu + p'^\mu$ . Explain why this is the part needed for  $F_2$ .
- (b) Wick rotate and evaluate the convergent loop integral

$$\int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 + \Delta)^3}.$$

Use it to reduce the one-loop Pauli form factor to a finite simplex integral, retaining a nonzero momentum transfer  $q$ .

- (c) Take  $q^2 \rightarrow 0$  in the simplex representation of the Pauli form factor and perform every remaining parameter integral. Derive  $F_2(0) = \alpha/(2\pi)$  and  $g_e = 2 + \alpha/\pi$ .

### Solutions to Supplementary Exercises

#### S.11.1. A Census of One-Loop QED Graphs

- (a) The tree graph contains one virtual photon and two fermion–photon vertices. At order  $e^4$ , the connected loop topologies are:
  - i. a photon vacuum-polarization insertion (with every charged flavor that can circulate in the loop);
  - ii. a one-loop vertex correction at the electron or muon annihilation vertex;
  - iii. self-energy insertions on each of the four external fermion legs; and
  - iv. the box and crossed-box two-photon-exchange graphs.

There are also counterterm graphs for the photon propagator, both vertices, and the fermion two-point functions. In an on-shell scheme the external-leg self-energies combine with their counterterms and LSZ factors to leave unit pole residue. The vacuum-polarization and vertex subgraphs are replaced by their subtracted, finite versions, while the box graphs are ultraviolet finite after their orderings are summed. Thus a convenient organization is

$$\mathcal{M}_{\text{NLO}} = \mathcal{M}_{\text{vac.pol.}}^R + \mathcal{M}_{\text{vertex,e}}^R + \mathcal{M}_{\text{vertex,\mu}}^R + \mathcal{M}_{\text{box}} + \mathcal{M}_{\text{crossed}}.$$

This grouping is gauge invariant only after all members required at this order are included.

- (b) For  $q = p_1 + p_2$ , representative regulated subgraph integrals are

$$i\Pi^{\mu\nu}(q) = -(-ie)^2 \mu^{2\varepsilon} \int_k \frac{\text{tr}[\gamma^\mu(\not{k} + m_f)\gamma^\nu(\not{k} + \not{q} + m_f)]}{(k^2 - m_f^2 + i0)((k+q)^2 - m_f^2 + i0)},$$

$$\Lambda_f^\mu = -e^2 \mu^{2\varepsilon} \int_k \frac{\gamma^\alpha (\not{k} + \not{p}' + m_f) \gamma^\mu (\not{k} + \not{p} + m_f) \gamma_\alpha}{(k^2 + i0)((k + p')^2 - m_f^2 + i0)((k + p)^2 - m_f^2 + i0)},$$

and

$$\Sigma_f(\not{p}) = -ie^2 \mu^{2\varepsilon} \int_k \frac{\gamma^\alpha (\not{p} - \not{k} + m_f) \gamma_\alpha}{(k^2 + i0)((p - k)^2 - m_f^2 + i0)}.$$

The box has two photon and two fermion denominators; exchanging the photon-attachment order gives the crossed box. Inserting these tensors between external currents supplies every class in part 1. Local  $\delta Z_1, \delta Z_2, \delta Z_3, \delta m$  insertions subtract the subdivergences, while the on-shell LSZ condition removes explicit external self-energy factors.

### S.11.2. Feynman Parameters

- (a) For  $\Re \nu > 0$ , rescale the defining Gamma-function integral by  $y = Ax$ , initially taking  $\Re A > 0$ :

$$\Gamma(\nu) = \int_0^\infty dy y^{\nu-1} e^{-y} = A^\nu \int_0^\infty dx x^{\nu-1} e^{-Ax}.$$

Dividing by  $A^\nu \Gamma(\nu)$  proves (7). Both sides are analytic in their common convergence domain, and the formula elsewhere is defined by analytic continuation with the branch of  $A^{-\nu}$  fixed by the prescription in the propagator denominator.

- (b) Apply (7) independently to all  $n$  factors:

$$\frac{1}{\prod_i A_i^{\nu_i}} = \frac{1}{\prod_i \Gamma(\nu_i)} \int_0^\infty \prod_i (dt_i t_i^{\nu_i-1}) \exp \left[ - \sum_i t_i A_i \right].$$

Introduce an overall scale  $T = \sum_i t_i$  and simplex coordinates  $x_i = t_i/T$ . The measure is

$$\prod_i dt_i = T^{n-1} dT \left( \prod_i dx_i \right) \delta \left( 1 - \sum_i x_i \right),$$

so, with  $\nu = \sum_i \nu_i$ ,

$$\begin{aligned} \frac{1}{\prod_i A_i^{\nu_i}} &= \frac{1}{\prod_i \Gamma(\nu_i)} \int_0^1 \left( \prod_i dx_i \right) \delta \left( 1 - \sum_i x_i \right) \prod_i x_i^{\nu_i-1} \\ &\quad \times \int_0^\infty dT T^{\nu-1} e^{-T \sum_i x_i A_i}. \end{aligned}$$

The last integral is  $\Gamma(\nu)(\sum_i x_i A_i)^{-\nu}$ , giving precisely (8). No integer-exponent step has been used. As a check, when every  $\nu_i = 1$ , the prefactor becomes  $\Gamma(n) = (n-1)!$ .

- (c) For the bubble choose  $A = k^2 - i0$  and  $B = (k + p)^2 - i0$ . Equation (8) for two factors gives

$$I_B(a, b; p^2) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 dx x^{a-1} (1-x)^{b-1} \times \int \frac{d^d k}{(2\pi)^d} \frac{1}{[x(k^2 - i0) + (1-x)((k+p)^2 - i0)]^{a+b}}.$$

Since

$$xk^2 + (1-x)(k+p)^2 = [k + (1-x)p]^2 + x(1-x)p^2,$$

the shift  $\ell = k + (1-x)p$  yields the requested single denominator:

$$I_B(a, b; p^2) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 dx x^{a-1} (1-x)^{b-1} \times \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{[\ell^2 + x(1-x)p^2 - i0]^{a+b}}.$$

Dimensional regularization justifies the translation of loop momentum by analytic continuation from a convergence domain.

- (d) Write  $N = a+b$  and  $\Delta = x(1-x)p^2$ . Wick rotation in the mostly-plus convention gives

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 + \Delta - i0)^N} = i \frac{\Omega_d}{(2\pi)^d} \int_0^\infty dr \frac{r^{d-1}}{(r^2 + \Delta - i0)^N}.$$

With  $u = r^2/(\Delta - i0)$ , the radial integral is

$$\frac{1}{2}(\Delta - i0)^{d/2-N} \frac{\Gamma(d/2)\Gamma(N-d/2)}{\Gamma(N)}.$$

Using (9), the loop integral therefore becomes

$$\frac{i}{(4\pi)^{d/2}} \frac{\Gamma(N-d/2)}{\Gamma(N)} (\Delta - i0)^{d/2-N}.$$

Substitution in the preceding part leaves a beta integral,

$$\int_0^1 dx x^{d/2-b-1} (1-x)^{d/2-a-1} = \frac{\Gamma(d/2-b)\Gamma(d/2-a)}{\Gamma(d-a-b)}.$$

Consequently

$$I_B(a, b; p^2) = \frac{i}{(4\pi)^{d/2}} (p^2 - i0)^{d/2-a-b} \frac{\Gamma(a+b-d/2)\Gamma(d/2-a)\Gamma(d/2-b)}{\Gamma(a)\Gamma(b)\Gamma(d-a-b)}.$$

For  $a = b = 1$  and  $d = 4 - 2\varepsilon$ , use

$$\Gamma(\varepsilon) = \frac{\Gamma(1+\varepsilon)}{\varepsilon}, \quad \Gamma(2-2\varepsilon) = (1-2\varepsilon)\Gamma(1-2\varepsilon),$$

and  $(4\pi)^{-d/2}(p^2 - i0)^{-\varepsilon} = (4\pi)^{-2}[(p^2 - i0)/(4\pi)]^{-\varepsilon}$ . The result is exactly the required check (11):

$$I_B(1, 1; p^2) = \left( \frac{p^2 - i0}{4\pi} \right)^{-\varepsilon} \frac{i \Gamma(1 - \varepsilon)^2 \Gamma(1 + \varepsilon)}{(4\pi)^2 (1 - 2\varepsilon) \varepsilon \Gamma(1 - 2\varepsilon)}.$$

The Gamma function  $\Gamma(\varepsilon)$  displays the logarithmic ultraviolet pole.

### S.11.3. Rank-Four Rotational Averaging and Tensor Reduction of a Bubble

(a) An odd tensor integral vanishes because the measure and  $f(k^2)$  are even:

$$\int d^d k k^\mu k^\nu k^\rho f(k^2) = 0.$$

Rotational invariance and index symmetry imply

$$\int d^d k k^\mu k^\nu k^\rho k^\sigma f(k^2) = A(\delta^{\mu\nu} \delta^{\rho\sigma} + \delta^{\mu\rho} \delta^{\nu\sigma} + \delta^{\mu\sigma} \delta^{\nu\rho}).$$

Contracting  $\mu$  with  $\nu$  and  $\rho$  with  $\sigma$  gives

$$\int d^d k (k^2)^2 f(k^2) = A(d^2 + 2d),$$

so

$$\boxed{\int d^d k k^\mu k^\nu k^\rho k^\sigma f(k^2) = \frac{\delta^{\mu\nu} \delta^{\rho\sigma} + \delta^{\mu\rho} \delta^{\nu\sigma} + \delta^{\mu\sigma} \delta^{\nu\rho}}{d(d+2)} \int d^d k (k^2)^2 f(k^2).}$$

The analogous rank-two coefficient is  $\delta^{\mu\nu}/d$ .

(b) Write  $I = I_B(1, 1; p^2)$ . The change of variables  $k \mapsto -k - p$  immediately gives

$$I^\mu \equiv \int \frac{d^d k}{(2\pi)^d} \frac{k^\mu}{k^2 (k+p)^2} = -\frac{p^\mu}{2} I.$$

For  $I^{\mu\nu} = A\eta^{\mu\nu} + Bp^\mu p^\nu$ , contraction with the metric leaves a scaleless tadpole:

$$dA + p^2 B = \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k+p)^2} = 0.$$

Also  $2p \cdot k = (k+p)^2 - k^2 - p^2$ . Squaring this relation inside the integral shows, after scaleless terms are dropped, that

$$p_\mu p_\nu I^{\mu\nu} = \frac{p^4}{4} I.$$

Solving the two linear equations gives

$$\boxed{I^{\mu\nu} = \frac{I}{4(d-1)} (dp^\mu p^\nu - p^2 \eta^{\mu\nu}).}$$

Its trace vanishes, consistently with dimensional regularization of the scaleless tadpole.

### S.11.4. Self-Energy Corrections in QED

- (a) Use the momentum routing printed on the two diagrams. The volume's Feynman-gauge rules give

$$\Sigma(p^2) = ie^2 \int \frac{d^d k}{(2\pi)^d} \frac{\gamma_\rho [-i(\not{k} + \not{p})] \gamma^\rho}{(k^2 - i0)((k+p)^2 - i0)}$$

and

$$\Pi^{\mu\nu}(p^2) = -ie^2 \int \frac{d^d k}{(2\pi)^d} \frac{\text{tr}[(-i\not{k})\gamma^\mu(-i(\not{k} + \not{p}))\gamma^\nu]}{(k^2 - i0)((k+p)^2 - i0)}.$$

The overall minus sign in the second expression is the closed-fermion-loop sign. No external spinor or polarization occurs, so the first expression is a spinor matrix and the second carries its two free Lorentz indices.

- (b) In  $d$  dimensions,

$$\gamma_\rho \gamma_\alpha \gamma^\rho = (2-d)\gamma_\alpha.$$

Writing  $I_B = I_B(1, 1; p^2)$ , the supplied rank-one reduction gives

$$\begin{aligned} \Sigma(p^2) &= e^2(2-d) \int \frac{d^d k}{(2\pi)^d} \frac{\not{k} + \not{p}}{(k^2 - i0)((k+p)^2 - i0)} \\ &= \frac{e^2(2-d)}{2} \not{p} I_B = -e^2(1-\varepsilon) \not{p} I_B. \end{aligned}$$

Substituting the supplied exact bubble yields

$$\Sigma(p^2) = -\frac{ie^2}{(4\pi)^2} \not{p} \left( \frac{p^2 - i0}{4\pi} \right)^{-\varepsilon} \frac{(1-\varepsilon)\Gamma(1-\varepsilon)^2\Gamma(1+\varepsilon)}{(1-2\varepsilon)\varepsilon\Gamma(1-2\varepsilon)}.$$

For the photon, first take the trace:

$$\text{tr}[\not{k}\gamma^\mu(\not{k} + \not{p})\gamma^\nu] = 4[k^\mu(k+p)^\nu + k^\nu(k+p)^\mu - \eta^{\mu\nu}k \cdot (k+p)].$$

The two factors  $-i$  in the fermion numerators turn the prefactor into  $+ie^2$ . Let

$$\begin{aligned} I^\mu &= \int \frac{d^d k}{(2\pi)^d} \frac{k^\mu}{(k^2 - i0)((k+p)^2 - i0)}, \\ I^{\mu\nu} &= \int \frac{d^d k}{(2\pi)^d} \frac{k^\mu k^\nu}{(k^2 - i0)((k+p)^2 - i0)}. \end{aligned}$$

Scaleless tadpoles give  $I^\rho_\rho = 0$ , while the supplied formulas give  $I^\mu = -p^\mu I_B/2$  and

$$I^{\mu\nu} = \frac{dp^\mu p^\nu - p^2 \eta^{\mu\nu}}{4(d-1)} I_B.$$

It follows that

$$\begin{aligned} \Pi^{\mu\nu}(p^2) &= 4ie^2 [2I^{\mu\nu} + p^\mu I^\nu + p^\nu I^\mu - \eta^{\mu\nu}(I^\rho_\rho + p \cdot I)] \\ &= ie^2 \frac{2(d-2)}{d-1} (p^2 \eta^{\mu\nu} - p^\mu p^\nu) I_B. \end{aligned}$$



With  $d = 4 - 2\varepsilon$ , this is

$$\Pi^{\mu\nu}(p^2) = -\frac{e^2}{(4\pi)^2}(p^2\eta^{\mu\nu} - p^\mu p^\nu) \left(\frac{p^2 - i0}{4\pi}\right)^{-\varepsilon} \frac{4(1-\varepsilon)\Gamma(1-\varepsilon)^2\Gamma(1+\varepsilon)}{(3-2\varepsilon)(1-2\varepsilon)\varepsilon\Gamma(1-2\varepsilon)}.$$

The exact tensor is transverse, which checks both the tensor reduction and the relative numerator terms.

(c) From the defining integral,

$$\begin{aligned}\Gamma(1-\varepsilon) &= \int_0^\infty dx x^{-\varepsilon} e^{-x} \\ &= \int_0^\infty dx e^{-x} [1 - \varepsilon \log x + O(\varepsilon^2)] \\ &= 1 + \gamma_E \varepsilon + O(\varepsilon^2),\end{aligned}$$

where the last equality is exactly definition (21). This proves (22).

(d) Equation (22), with  $\varepsilon$  replaced by  $-\varepsilon$  or  $2\varepsilon$ , gives

$$\Gamma(1+\varepsilon) = 1 - \gamma_E \varepsilon + O(\varepsilon^2), \quad \Gamma(1-2\varepsilon) = 1 + 2\gamma_E \varepsilon + O(\varepsilon^2).$$

Thus

$$\frac{\Gamma(1-\varepsilon)^2\Gamma(1+\varepsilon)}{\Gamma(1-2\varepsilon)} = 1 - \gamma_E \varepsilon + O(\varepsilon^2).$$

Set

$$L_p \equiv \log\left(\frac{p^2 - i0}{4\pi}\right), \quad \left(\frac{p^2 - i0}{4\pi}\right)^{-\varepsilon} = 1 - \varepsilon L_p + O(\varepsilon^2).$$

The rational factors expand as

$$\frac{1-\varepsilon}{1-2\varepsilon} = 1 + \varepsilon + O(\varepsilon^2), \quad \frac{4(1-\varepsilon)}{(3-2\varepsilon)(1-2\varepsilon)} = \frac{4}{3} \left(1 + \frac{5}{3}\varepsilon\right) + O(\varepsilon^2).$$

Therefore, after dropping terms that vanish with  $\varepsilon$ ,

$$\Sigma(p^2) = -\frac{ie^2}{(4\pi)^2} \not{p} \left[ \frac{1}{\varepsilon} + 1 - \gamma_E - L_p \right] + O(\varepsilon).$$

and

$$\Pi^{\mu\nu}(p^2) = -\frac{e^2}{12\pi^2}(p^2\eta^{\mu\nu} - p^\mu p^\nu) \left[ \frac{1}{\varepsilon} + \frac{5}{3} - \gamma_E - L_p \right] + O(\varepsilon).$$

If one inserts the conventional dimensional-regularization scale through  $e^2 \mapsto e^2 \mu^{2\varepsilon}$ , replace  $L_p$  by  $\log[(p^2 - i0)/(4\pi\mu^2)]$ . This restores a dimensionless logarithm without changing the source's requested Laurent coefficients.

### S.11.5. Euclidean Master Integrals and Shifted Tensor Integrals

- (a) Cartesian factorization gives

$$\int_{\mathbb{R}^d} d^d x e^{-x^2} = \left( \int_{-\infty}^{\infty} e^{-u^2} du \right)^d = \pi^{d/2}.$$

In radial coordinates the same integral is

$$\Omega_d \int_0^{\infty} dr r^{d-1} e^{-r^2} = \frac{\Omega_d}{2} \Gamma(d/2).$$

Equating them gives

$$\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}.$$

The right side is meromorphic in complex  $d$  and defines the continuation used by dimensional regularization. For positive integer  $d$ , it agrees with the geometric area; for general  $d$ , only the analytic formula is needed.

- (b) Using radial coordinates and  $u = q^2/m^2$ ,

$$\begin{aligned} J_n(m^2) &= \frac{\Omega_d}{(2\pi)^d} \int_0^{\infty} dq \frac{q^{d-1}}{(q^2 + m^2)^n} \\ &= \frac{\Omega_d}{2(2\pi)^d} (m^2)^{d/2-n} \int_0^{\infty} du \frac{u^{d/2-1}}{(1+u)^n}. \end{aligned}$$

The last integral is  $B(d/2, n - d/2)$ , so

$$J_n(m^2) = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(n - d/2)}{\Gamma(n)} (m^2)^{d/2-n}.$$

Before continuation it converges for  $\Re d > 0$  and  $\Re n > \Re d/2$ . Ultraviolet poles occur when  $n - d/2 = 0, -1, -2, \dots$ , exactly the poles of the Gamma function.

- (c) Complete the square:

$$q^2 + 2q \cdot p + m^2 = (q + p)^2 + M^2, \quad M^2 = m^2 - p^2.$$

For  $M^2 > 0$  in the Euclidean convergence domain, translation invariance of the regulated integral justifies  $\ell = q + p$ . Exercise S.11.8 then gives

$$\int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + m^2 + 2q \cdot p)^n} = \frac{\Gamma(n - d/2)}{(4\pi)^{d/2} \Gamma(n)} (m^2 - p^2)^{d/2-n}.$$

The expression elsewhere is defined by analytic continuation with the appropriate pole prescription. A cutoff that obstructed the momentum shift would not preserve this simple identity.

- (d) After  $\ell = q + p$ , odd powers of  $\ell$  integrate to zero. Denoting the scalar answer of S.11.9 by  $J_n(M^2)$ ,  $M^2 = m^2 - p^2$ , we find

$$\int \frac{d^d q}{(2\pi)^d} \frac{q^\mu}{(q^2 + m^2 + 2q \cdot p)^n} = -p^\mu J_n(M^2).$$

For rank two,

$$\int \frac{d^d q}{(2\pi)^d} \frac{q^\mu q^\nu}{(q^2 + m^2 + 2q \cdot p)^n} = p^\mu p^\nu J_n(M^2) + \frac{\delta^{\mu\nu}}{d} [J_{n-1}(M^2) - M^2 J_n(M^2)].$$

Equivalently, the second line is

$$\frac{\delta^{\mu\nu}}{2(4\pi)^{d/2}} \frac{\Gamma(n-1-d/2)}{\Gamma(n)} (M^2)^{d/2+1-n}.$$

Differentiating the scalar integral with respect to  $p_\mu$  produces the same vector and tensor relations, providing an independent check.

### S.11.6. The Spinor Vacuum-Polarization Numerator and Subtracted Vacuum Polarization at Small Momentum

(a) The closed loop gives

$$\Pi^{\mu\nu}(p) = -e^2 \int \frac{d^d k}{(2\pi)^d} \frac{\text{Tr}[\gamma^\mu(-i\not{k} + m)\gamma^\nu(-i(\not{k} + \not{p}) + m)]}{(k^2 + m^2)((k+p)^2 + m^2)}.$$

The trace identity reduces the numerator to a sum of  $k^\mu k^\nu$ ,  $k^{(\mu} p^{\nu)}$ , and metric terms. Introducing  $x$ , shifting  $\ell = k + xp$ , and dropping odd powers gives

$$\Delta_x = m^2 + x(1-x)p^2,$$

$$\Pi^{\mu\nu}(p) = -4e^2 \int_0^1 dx \int \frac{d^d \ell}{(2\pi)^d} \frac{N^{\mu\nu}(\ell, p; x)}{(\ell^2 + \Delta_x)^2},$$

$$N^{\mu\nu}(\ell, p; x) = 2\ell^\mu \ell^\nu - \eta^{\mu\nu} \ell^2 - 2x(1-x)p^\mu p^\nu + \eta^{\mu\nu} \Delta_x.$$

up to the common Wick-rotation factor fixed by convention. Now use  $\ell^\mu \ell^\nu \rightarrow \eta^{\mu\nu} \ell^2/d$ . The scalar master-integral recursion cancels the remaining non-transverse metric term, leaving

$$\Pi^{\mu\nu}(p) = (p^2 \eta^{\mu\nu} - p^\mu p^\nu) \Pi(p^2).$$

The cancellation would be spoiled by an uncorrelated hard momentum cutoff.

(b) The on-shell photon normalization subtracts the unrenormalized scalar coefficient at  $q^2 = 0$ . The pole and all scheme-dependent constants then cancel, leaving

$$\pi(q^2) = \frac{e^2}{2\pi^2} \int_0^1 dx x(1-x) \ln \left( 1 + \frac{q^2 x(1-x)}{m^2} \right).$$

For small  $q^2$ , expand the logarithm. The needed beta integrals are

$$\int_0^1 x^2(1-x)^2 dx = \frac{1}{30}, \quad \int_0^1 x^3(1-x)^3 dx = \frac{1}{140}.$$

Therefore

$$\pi(q^2) = \frac{e^2}{60\pi^2} \frac{q^2}{m^2} - \frac{e^2}{560\pi^2} \frac{q^4}{m^4} + O(q^6/m^6).$$

In particular  $\pi(0) = 0$ , and a heavy charged particle decouples from low-momentum propagation as  $q^2/m^2$ .

### S.11.7. QED Counterterms and On-Shell Renormalization

- (a) The fields have dimensions  $[A_\mu] = 1$ ,  $[\psi] = 3/2$ , and  $[\partial_\mu] = 1$ . The local, Lorentz- and gauge-invariant,  $C$ - and  $P$ -even operators of dimension at most four that are already present in QED are

$$-\frac{\delta_3}{4} F_{\mu\nu} F^{\mu\nu}, \quad -\delta_2 \bar{\psi} \gamma^\mu \partial_\mu \psi, \quad \delta_m m \bar{\psi} \psi, \quad -i\delta_1 e A_\mu \bar{\psi} \gamma^\mu \psi.$$

Gauge fixing renormalizes with the photon field and introduces no new physical coupling. A photon mass  $A_\mu A^\mu$  violates gauge invariance. A photon tadpole is odd under charge conjugation. The Pauli operator  $\bar{\psi} \sigma^{\mu\nu} F_{\mu\nu} \psi$  has dimension five, so its finite coefficient can be predicted in renormalizable QED but it cannot absorb a dimension-four ultraviolet divergence. This census explains why the counterterms in the chapter suffice.

- (b) Current conservation makes the self-energy transverse:

$$\Pi^{\mu\nu}(q) = (q^2 \eta^{\mu\nu} - q^\mu q^\nu) \pi(q^2).$$

On the transverse subspace, Dyson summation is scalar and gives

$$D'_T(q) = \frac{D_T(q)}{1 - \pi(q^2)} \propto \frac{1}{q^2[1 - \pi(q^2)]}.$$

Hence the massless pole has residue multiplied by  $[1 - \pi(0)]^{-1}$ . The on-shell field normalization demands unit residue, so

$$\pi(0) = 0.$$

At very small momentum transfer, the Coulomb amplitude is then  $-e^2/q^2$  with  $e$  equal to the measured long-distance charge. A different finite subtraction would merely define a different parameter  $e(\mu)$ ; converting it to the Thomson-limit charge restores the same observable amplitude.

- (c) Let  $z = i\not{p}$ . Parity permits  $\Sigma(p) = A(p^2)z + B(p^2)m$ , and the full inverse propagator is

$$S_R^{-1}(p) = z + m - \Sigma_R(z).$$

A pole at the measured mass requires

$$\Sigma_R(-m) = 0,$$

while unit residue requires

$$\left. \frac{\partial \Sigma_R}{\partial z} \right|_{z=-m} = 0.$$

The first condition fixes the mass counterterm and the second fixes the field-strength counterterm. Taylor expansion then gives

$$\Sigma_R(z) = O((z+m)^2), \quad S_R^{-1}(p) = (z+m)[1 + O(z+m)].$$

Thus  $S_R(p)$  has a simple pole with residue one. The phrase “double zero” applies to the subtracted self-energy, not to the full inverse propagator.

### S.11.8. One-Loop QED Renormalization Constants

- (a) In the  $d = 4 - \epsilon$  convention, direct extraction of the pole parts gives

$$\boxed{Z_1 = Z_2 = 1 - \frac{\xi e^2}{8\pi^2\epsilon}, \quad Z_3 = 1 - \frac{e^2}{6\pi^2\epsilon}, \quad Z_m = 1 - \frac{e^2}{2\pi^2\epsilon}} + O(e^4).$$

The longitudinal photon propagator produces the  $\xi$ -dependent kinetic and vertex poles; the mass and transverse photon poles are gauge independent.

- (b) The Ward–Takahashi identity for the proper vertex is

$$q_\mu \Gamma^\mu(p+q, p) = S^{-1}(p+q) - S^{-1}(p).$$

Taking  $q \rightarrow 0$  gives

$$\Gamma^\mu(p, p) = \frac{\partial S^{-1}(p)}{\partial p_\mu}.$$

The local ultraviolet divergence on the left is proportional to  $\delta_1 \gamma^\mu$ ; the divergence on the right is the derivative of the kinetic counterterm and is proportional to  $\delta_2 \gamma^\mu$ . Therefore

$$\boxed{Z_1 = Z_2}$$

to one loop and, with a gauge-preserving regulator, to all orders. Since  $e_0 = \mu^\epsilon Z_1 Z_2^{-1} Z_3^{-1/2} e$ , the equality leaves

$$e_0 = \mu^\epsilon Z_3^{-1/2} e.$$

Thus charge renormalization is determined entirely by photon wave-function renormalization.

- (c) In Lorenz gauge  $\xi = 0$ , the first result reduces to

$$Z_1 = Z_2 = 1 + O(e^4).$$

Moreover  $Z_1 Z_2^{-1} = 1$ , so the bare-charge relation depends only on  $Z_3^{-1/2}$ .

### S.11.9. Why Four-Photon Scattering Is Ultraviolet Finite

Power counting assigns logarithmic superficial degree to a fermion loop with four external photons. Any remaining local divergence would therefore have to be cancelled by a gauge-invariant operator of dimension four containing four photon fields. No such operator exists: gauge invariance requires photons to occur through  $F_{\mu\nu}$ , and an  $F^4$  operator has dimension eight. Consequently the logarithmic pieces of the six oriented box orderings cancel, and the complete amplitude is ultraviolet finite.

For an odd number of photons, reversing the fermion-loop orientation gives a factor  $(-1)^n$  under charge conjugation. At  $n = 3$  the two orientations cancel. This is Furry's theorem. The first nonzero low-energy local interaction is therefore  $F^4/m^4$ , consistent with both claims.

### S.11.10. From the Pauli Form Factor to $g - 2$

For on-shell spinors, Lorentz covariance and current conservation give

$$\bar{u}(p')\Gamma^\mu u(p) = e \bar{u}(p') \left[ \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m} F_2(q^2) \right] u(p), \quad q = p' - p.$$

The Thomson-limit definition of charge fixes  $F_1(0) = 1$ . In a slowly varying magnetic field, the nonrelativistic Hamiltonian obtained from this vertex contains

$$H_{\text{mag}} = -\frac{e}{m}[F_1(0) + F_2(0)]\mathbf{S} \cdot \mathbf{B}.$$

Comparing with  $H_{\text{mag}} = -g_e(e/2m)\mathbf{S} \cdot \mathbf{B}$  yields

$$g_e = 2[1 + F_2(0)], \quad \boxed{\frac{g_e - 2}{2} = F_2(0)}.$$

The Dirac form factor normalizes the charge; the Pauli form factor measures the anomalous magnetic moment.

### S.11.11. Running Charge, Screening, and Heavy-Threshold Decoupling

(a) The Ward identity gives

$$e_0 = \mu^\varepsilon Z_3^{-1/2} e, \quad d = 4 - 2\varepsilon.$$

For one Dirac fermion, minimal subtraction yields

$$Z_3 = 1 - \frac{e^2}{12\pi^2\varepsilon} + O(e^4).$$

Differentiate  $\ln e_0$  with respect to  $\ln \mu$  at fixed  $e_0$ :

$$0 = \varepsilon + \frac{\beta(e)}{e} - \frac{1}{2}\beta(e)\frac{\partial \ln Z_3}{\partial e}.$$

Solving perturbatively, while retaining the classical term until it multiplies the pole, gives

$$\boxed{\beta(e) = -\varepsilon e + \frac{e^3}{12\pi^2} + O(e^5)}.$$

At  $d = 4$ ,  $\varepsilon = 0$ , so the quantum term is positive.

(b) At one loop,

$$\frac{de}{d \ln \mu} = \frac{e^3}{12\pi^2}.$$

Therefore

$$\frac{d}{d \ln \mu} \frac{1}{e^2} = -\frac{1}{6\pi^2},$$

and integration gives

$$\boxed{\frac{1}{e^2(\mu_2)} = \frac{1}{e^2(\mu_1)} - \frac{1}{6\pi^2} \ln \frac{\mu_2}{\mu_1}.}$$

Thus  $e(\mu)$  grows as the distance scale  $1/\mu$  decreases. In position space, virtual electron–positron pairs align so that charge of opposite sign lies closer to a test charge. The cloud screens the source at long distance, making the infrared charge smaller than the short-distance charge. The positive beta function is the momentum-space expression of this screening.

(c) For a fermion of mass  $M$ , on-shell subtraction gives

$$\pi_R(q^2) = \frac{e^2}{2\pi^2} \int_0^1 dx x(1-x) \ln \left( 1 + \frac{q^2 x(1-x)}{M^2} \right).$$

At low momentum,

$$\boxed{\pi_R(q^2) = \frac{e^2}{60\pi^2} \frac{q^2}{M^2} + O(q^4/M^4).}$$

The heavy fermion therefore changes low-energy amplitudes only through higher-dimension operators suppressed by  $M^{-2}$ .

Minimal subtraction removes only poles and is insensitive to masses, so its beta function formally includes the heavy fermion even for  $\mu \ll M$ . The effective theory resolves this by matching at  $\mu \simeq M$ : a finite threshold correction relates the couplings above and below  $M$ , after which the heavy field is omitted from the low-energy beta function.

### S.11.12. The Euler–Heisenberg Effective Interaction

(a) In four dimensions, the two independent parity-even scalars quartic in an abelian field strength may be chosen as

$$\mathcal{O}_1 = (F_{\mu\nu} F^{\mu\nu})^2, \quad \mathcal{O}_2 = (F_{\mu\nu} \tilde{F}^{\mu\nu})^2,$$

where  $\tilde{F}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$ . Other contractions of four antisymmetric tensors reduce to these in four dimensions.

With the mostly-plus convention,

$$F_{\mu\nu} F^{\mu\nu} = 2(\mathbf{B}^2 - \mathbf{E}^2), \quad F_{\mu\nu} \tilde{F}^{\mu\nu} = -4\mathbf{E} \cdot \mathbf{B}$$

up to the sign fixed by the displayed dual convention; its square is unambiguous. Thus the two physical invariants are

$$(\mathbf{E}^2 - \mathbf{B}^2)^2, \quad (\mathbf{E} \cdot \mathbf{B})^2.$$

- (b) For constant fields, introduce invariants  $a, b$  such that the eigenvalues of  $F^\mu{}_\nu$  are encoded by an electric-like and magnetic-like pair. After subtracting the zero-field vacuum term and the quadratic charge renormalization term, the spinor determinant has the schematic form

$$\mathcal{L}_1 = -\frac{1}{8\pi^2} \int_0^\infty \frac{ds}{s^3} e^{-m_e^2 s} \left[ (es)^2 ab \coth(eas) \cot(ebs) - 1 - \frac{e^2 s^2}{3} (a^2 - b^2) \right].$$

Using  $\coth z = z^{-1} + z/3 - z^3/45 + \dots$  and  $\cot z = z^{-1} - z/3 - z^3/45 + \dots$ , the first surviving term is quartic. Since  $\int_0^\infty ds s e^{-m_e^2 s} = m_e^{-4}$ , one obtains

$$\mathcal{L}_{\text{EH}} = \frac{2\alpha^2}{45m_e^4} [(\mathbf{E}^2 - \mathbf{B}^2)^2 + 7(\mathbf{E} \cdot \mathbf{B})^2].$$

Equivalently,

$$\mathcal{L}_{\text{EH}} = \frac{\alpha^2}{90m_e^4} \left[ (F_{\mu\nu} F^{\mu\nu})^2 + \frac{7}{4} (F_{\mu\nu} \tilde{F}^{\mu\nu})^2 \right].$$

- (c) Each external photon enters  $\mathcal{L}_{\text{EH}}$  through one field strength, so it contributes one power of its momentum. The coefficient of the four-photon vertex is  $O(\alpha^2/m_e^4)$ . Consequently

$$\mathcal{M}(\gamma\gamma \rightarrow \gamma\gamma) = O\left(\alpha^2 \frac{\omega^4}{m_e^4}\right).$$

For relativistic  $2 \rightarrow 2$  scattering, the cross section scales as  $|\mathcal{M}|^2/s$ , with  $s = O(\omega^2)$ . Hence

$$\sigma = O\left(\alpha^4 \frac{\omega^6}{m_e^8}\right).$$

No lower power is allowed: gauge invariance requires every external photon to appear through  $F_{\mu\nu}$ , while locality below threshold makes the amplitude analytic in momenta. The next corrections contain additional derivatives and are further suppressed by  $\omega^2/m_e^2$ .

### S.11.13. Pauli Operators, Nonrelativistic Reduction, and Simplex Parameters

- (a) Define  $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ . The on-shell vertex at small momentum transfer contains

$$e\Gamma^\mu = e\gamma^\mu + e \frac{i\sigma^{\mu\nu} q_\nu}{2m} F_2(0) + O(q^2).$$

The local interaction

$$\Delta\mathcal{L} = \frac{eF_2(0)}{4m} \bar{\psi} \sigma^{\mu\nu} F_{\mu\nu} \psi$$



produces precisely this vertex: differentiating  $F_{\mu\nu}$  with respect to  $A_\mu$  supplies two equal antisymmetric contributions and one factor of  $iq_\nu$ . The operator has dimension five, so its coefficient has dimension  $-1$ . It is gauge invariant and  $C$ -,  $P$ -, and  $T$ -even for a magnetic dipole; replacing it by the dual field strength would instead describe an electric dipole and violate  $P$  and  $T$ .

- (b) Write the Dirac spinor after removing its rest-energy phase as  $\psi = e^{-imt}(\Phi, X)^\top$ , and put  $\boldsymbol{\pi} = \mathbf{p} - e\mathbf{A}$ . To leading order the lower component obeys

$$X = \frac{\boldsymbol{\sigma} \cdot \boldsymbol{\pi}}{2m} \Phi + O(m^{-2}).$$

Substitution into the upper-component equation uses

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2 = \boldsymbol{\pi}^2 + e\boldsymbol{\sigma} \cdot \mathbf{B}$$

with the sign adjusted if  $e$  denotes the positive magnitude of the electron charge. Including the Pauli operator gives

$$i\partial_t \Phi = \left[ \frac{\boldsymbol{\pi}^2}{2m} + eA^0 + \frac{e}{2m}[1 + F_2(0)]\boldsymbol{\sigma} \cdot \mathbf{B} \right] \Phi.$$

Thus, in terms of  $\mathbf{S} = \boldsymbol{\sigma}/2$ ,

$$H_{\text{mag}} = -g_e \frac{e}{2m} \mathbf{S} \cdot \mathbf{B}, \quad \boxed{g_e = 2[1 + F_2(0)]},$$

where the overall sign follows the particle's signed charge.

- (c) For denominators  $A_1, A_2, A_3$ ,

$$\frac{1}{A_1 A_2 A_3} = 2 \int_0^1 \prod_{i=1}^3 dx_i \frac{\delta(1 - x_1 - x_2 - x_3)}{(x_1 A_1 + x_2 A_2 + x_3 A_3)^3}.$$

Take  $A_1 = (k + p')^2 + m^2 - i0$ ,  $A_2 = (k + p)^2 + m^2 - i0$ , and  $A_3 = k^2 - i0$ , with  $p^2 = p'^2 = -m^2$  and  $q = p' - p$ . Shifting  $\ell = k + x_1 p' + x_2 p$  gives

$$\boxed{x_1 A_1 + x_2 A_2 + x_3 A_3 = \ell^2 + m^2(x_1 + x_2)^2 + q^2 x_1 x_2 - i0.}$$

Since  $x_1 + x_2 = 1 - x_3$ , this is  $\ell^2 + m^2(1 - x_3)^2 + q^2 x_1 x_2 - i0$ . The common form makes the finite Pauli projection manifest.

#### S.11.14. The One-Loop Pauli Form Factor

- (a) After the shift in S.11.13(c), terms odd in  $\ell$  vanish. Symmetric integration replaces  $\ell^\alpha \ell^\beta$  by  $\eta^{\alpha\beta} \ell^2/d$ . Gamma identities reduce the even  $\ell^2$  term to a multiple of  $\gamma^\mu$ , so it contributes only to the Dirac form factor and its ultraviolet subtraction.

For the remaining terms, move  $\not{p}$  and  $\not{p}'$  onto the external spinors and use  $(i\not{p} + m)u(p) = 0$  and  $\bar{u}(p')(i\not{p}' + m) = 0$ . The part independent of  $\ell$  that is not proportional to  $\gamma^\mu$  reduces to

$$N_{\text{Pauli}}^\mu = 2im(x_1 + x_2)x_3(p^\mu + p'^\mu).$$

The Gordon identity rewrites  $p^\mu + p'^\mu$  between on-shell spinors as a combination of  $\gamma^\mu$  and  $i\sigma^{\mu\nu}q_\nu$ . After the  $F_1(0)$  piece is fixed by charge renormalization, the remainder is exactly the Pauli form factor.

(b) Radial integration in four Euclidean dimensions gives

$$\begin{aligned} \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 + \Delta)^3} &= \frac{2\pi^2}{(2\pi)^4} \int_0^\infty dk \frac{k^3}{(k^2 + \Delta)^3} \\ &= \boxed{\frac{1}{32\pi^2\Delta}}. \end{aligned}$$

Combining this with the numerator from S.11.14(a) yields the finite simplex representation

$$F_2(q^2) = \frac{e^2}{4\pi^2} \int_0^1 \prod_{i=1}^3 dx_i \delta(1 - \sum_i x_i) \frac{m^2 x_3 (1-x_3)}{m^2 (1-x_3)^2 + q^2 x_1 x_2}.$$

No ultraviolet regulator remains: the Pauli projection has lowered the superficial degree sufficiently for convergence.

(c) At  $q^2 = 0$ , the integrand in S.11.14(b) becomes

$$\frac{x_3}{1 - x_3}.$$

For fixed  $x_3$ , the line segment  $x_1 + x_2 = 1 - x_3$  has parameter measure  $1 - x_3$ . Therefore

$$\begin{aligned} F_2(0) &= \frac{e^2}{4\pi^2} \int_0^1 dx_3 (1 - x_3) \frac{x_3}{1 - x_3} \\ &= \frac{e^2}{4\pi^2} \int_0^1 x_3 dx_3 = \frac{e^2}{8\pi^2} = \boxed{\frac{\alpha}{2\pi}}. \end{aligned}$$

Using  $g_e = 2[1 + F_2(0)]$  then gives

$$g_e = 2 + \frac{\alpha}{\pi} + O(\alpha^2), \quad \frac{g_e - 2}{2} = \frac{\alpha}{2\pi} + O(\alpha^2).$$

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## 12 General Renormalization Theory

We saw in the previous chapter that calculations in quantum electrodynamics involving one-loop graphs yield divergent integrals over momentum space, but that these infinities cancel when we express all parameters of the theory in terms of “renormalized” quantities, such as the masses and charges that are actually measured. In 1949 Dyson [1] sketched a proof that this cancellation would take place to all orders in quantum electrodynamics. It was immediately apparent (and will be shown here in Sections 12.1 and 12.2) that Dyson’s arguments apply to a larger class of theories with finite numbers of relatively simple interactions, the so-called renormalizable theories, of which quantum electrodynamics is just one simple example.

For some years it was widely thought that any sensible physical theory would have to take the form of a renormalizable quantum field theory. The requirement of renormalizability played a crucial role in the development of the modern “standard model” of weak, electromagnetic, and strong interactions. But as we shall see here, the cancellation of ultraviolet divergences does not really depend on renormalizability; as long as we include every one of the infinite number of interactions allowed by symmetries, the so-called non-renormalizable theories are actually just as renormalizable as renormalizable theories.

It is generally believed today that the realistic theories that we use to describe physics at accessible energies are what are known as “effective field theories.” As discussed in Section 12.3, these are low-energy approximations to a more fundamental theory that may not be a field theory at all. Any effective field theory necessarily includes an infinite number of non-renormalizable interactions. Nevertheless, as discussed in Sections 12.3 and 12.4, we expect that at sufficiently low energy all the non-renormalizable interactions in such effective field theories are highly suppressed. Renormalizable theories like quantum electrodynamics and the standard model thus retain their special status in physics, though for reasons that are somewhat different from those that originally motivated the assumption of renormalizability in these theories.

### 12.1 Degrees of Divergence

Let us consider a very general sort of theory, containing interactions of varying types labelled  $i$ . Each interaction may be characterized by the number  $n_{if}$  of fields of each type  $f$ , and by the number  $d_i$  of derivatives acting on these fields.

We will start by calculating the “superficial degree of divergence”  $D$  of an arbitrary connected one-particle irreducible Feynman diagram in such a theory. This is the number of factors of momentum in the numerator minus the number in the denominator of the integrand, plus four for every independent four-momentum over which we integrate. The superficial divergence is the actual degree of divergence of the integration over the region of momentum space in which the momenta of all internal lines go to infinity together. That is, if  $D > 0$ , then the part of the amplitude where all internal momenta go to infinity with a common factor  $\kappa \rightarrow \infty$  will diverge like

$$\int^\infty \kappa^{D-1} d\kappa. \quad (12.1.1)$$

In the same sense, an integral with degree of divergence  $D = 0$  is logarithmically divergent, and an integral with  $D < 0$  is convergent, at least as far as this region of momentum space is concerned. We will come back later to the problem posed by subintegrations that behave worse than the integral over this region.

To calculate  $D$ , we will need to know the following about the diagram:

$$\begin{aligned} I_f &\equiv \text{number of internal lines of field type } f, \\ E_f &\equiv \text{number of external lines of field type } f, \\ N_i &\equiv \text{number of vertices of interaction type } i. \end{aligned}$$

We will write the asymptotic behavior of the propagator  $\Delta_f(k)$  of a field of type  $f$  in the form

$$\Delta_f(k) \sim k^{-2+2s_f}. \quad (12.1.2)$$

Looking back at Chapter 6, we see that  $s_f = 0$  for scalar fields,  $s_f = \frac{1}{2}$  for Dirac fields, and  $s_f = 1$  for massive vector fields. More generally, it can be shown that for massive fields of Lorentz transformation type  $(A, B)$ , we have  $s_f = A + B$ . Speaking loosely, we may call  $s_f$  the “spin.” However, dropping terms that because of gauge invariance have no effect, the effective photon propagator  $\eta_{\mu\nu}/k^2$  has  $s_f = 0$ . A similar result applies to a massive vector field coupled to a conserved current, provided the current does not depend on the vector field. It can also be shown that, in the same sense, the graviton field  $g_{\mu\nu}$  has a propagator also with  $s_f = 0$ .

According to Eq. (12.1.2), the propagators make a total contribution to  $D$  equal to

$$\sum_f I_f (2s_f - 2). \quad (12.1.3)$$

Also, the derivatives in each interaction of type  $i$  introduce  $d_i$  momentum factors into the integrand, yielding a total contribution to  $D$  equal to

$$\sum_i N_i d_i. \quad (12.1.4)$$

Finally, we need the total number of independent momentum variables of integration. Each internal line can be labelled with a four-momentum, but these are not all independent; the delta function associated with each vertex imposes a linear relation among these internal momenta, except that one delta function only serves to enforce conservation of the external momenta. Thus, the momentum space integration volume elements contribute to  $D$  a term

$$4 \left[ \sum_f I_f - \left( \sum_i N_i - 1 \right) \right], \quad (12.1.5)$$

which, of course, is just four times the number of independent loops in the diagram. Adding the contributions (12.1.3), (12.1.4), and (12.1.5), we find

$$D = \sum_f I_f (2s_f + 2) + \sum_i N_i (d_i - 4) + 4. \quad (12.1.6)$$

Eq. (12.1.6) is not very convenient as it stands, because it gives a value for  $D$  that seems to depend on the internal details of the Feynman diagram. Fortunately, it can be simplified by using the topological identities

$$2I_f + E_f = \sum_i N_i n_{if}. \quad (12.1.7)$$

(Each internal line contributes two of the lines attached to vertices, while each external line contributes only one.) Using Eq. (12.1.7) to eliminate  $I_f$ , we see that Eq. (12.1.6) becomes

$$D = 4 - \sum_f E_f(s_f + 1) - \sum_i N_i \Delta_i, \quad (12.1.8)$$

where  $\Delta_i$  is a parameter characterizing interactions of type  $i$ :

$$\Delta_i \equiv 4 - d_i - \sum_f n_{if}(s_f + 1). \quad (12.1.9)$$

This result could have been obtained by simple dimensional analysis, without considering the structure of Feynman diagrams. The propagator of a field is a four-dimensional Fourier transform of the vacuum expectation value of a time-ordered product of a pair of free fields, so a conventionally normalized field  $f$  whose dimensionality<sup>126</sup> in powers of momentum is  $\mathcal{D}_f$  will have a propagator of dimensionality  $-4 + 2\mathcal{D}_f$ . Hence if the propagator behaves like  $k^{-2+2s_f}$  when  $k$  is much larger than the mass, then the field must have a dimensionality with  $-4 + 2\mathcal{D}_f = -2 + 2s_f$ , or  $\mathcal{D}_f = 1 + s_f$ . An interaction  $i$  with  $n_{if}$  such fields and  $d_i$  derivatives will then have dimensionality  $d_i + \sum_f n_{if}(1 + s_f)$ . But the action must be dimensionless, so each term in the Lagrangian density must have dimensionality  $+4$  to cancel the dimensionality  $-4$  of  $d^4x$ . Hence the interaction must have a coupling constant of dimensionality  $4 - d_i - \sum_f n_{if}(1 + s_f)$ , which is just the parameter  $\Delta_i$ . The momentum space amplitude corresponding to a connected Feynman graph with  $E_f$  external lines of type  $f$  is the Fourier transform over  $4\sum_f E_f$  coordinates of a vacuum expectation value of the time-ordered product of fields with a total dimensionality  $\sum_f E_f(1 + s_f)$ , so it has dimensionality  $\sum_f E_f(-3 + s_f)$ . Of this dimensionality,  $-4$  comes from a momentum space delta function, and  $\sum_f E_f(-2 + 2s_f)$  is the dimensionality of the propagators for the external lines, so the momentum space integral itself together with all coupling constant factors has dimensionality

$$\sum_f E_f(-3 + s_f) - (-4) - \sum_f E_f(-2 + 2s_f) = 4 - \sum_f E_f(s_f + 1).$$

The coupling constants for a given Feynman graph have total dimensionality  $\sum_i N_i \Delta_i$ , leaving the momentum space integral with dimensionality  $4 - \sum_f E_f(s_f + 1) - \sum_i N_i \Delta_i$ . As long as we are interested in the region of integration where all momenta go to infinity

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<sup>126</sup>In this chapter, “dimensionality” will always refer to the dimensionality in powers of mass or momentum, in units with  $\hbar = c = 1$ . We are using fields that are conventionally normalized, in the sense that the term in the free-field Lagrangian with the largest number of derivatives (which determines the asymptotic behavior of the propagator) has a dimensionless coefficient.

**Table 12.1.** Terms in the Lagrangian density for quantum electrodynamics. Here  $d_i$ ,  $n_{i\gamma}$ , and  $n_{ie}$  are the numbers of derivatives, photon fields, and electron fields in the interaction, and  $\Delta_i$  is the dimensionality of the corresponding coefficient. (Recall that  $s_\gamma = 0$ ,  $s_e = \frac{1}{2}$ .)

| Interaction                                 | $d_i$ | $n_{i\gamma}$ | $n_{ie}$ | $\Delta_i$      |
|---------------------------------------------|-------|---------------|----------|-----------------|
| $-ie\bar{\psi}\not{A}\psi$                  | 0     | 1             | 2        | $4 - 1 - 3 = 0$ |
| $-\frac{1}{4}(Z_3 - 1)F_{\mu\nu}F^{\mu\nu}$ | 2     | 2             | 0        | $4 - 2 - 2 = 0$ |
| $-(Z_2 - 1)\bar{\psi}\not{\partial}\psi$    | 1     | 0             | 2        | $4 - 1 - 3 = 0$ |
| $[-(Z_2 - 1)m + Z_2\delta m]\bar{\psi}\psi$ | 0     | 0             | 2        | $4 - 3 = 1$     |

together, the degree of divergence of the momentum space integral is its dimensionality, thus justifying Eq. (12.1.8).

If all interactions have  $\Delta_i \geq 0$ , then Eq. (12.1.8) provides an upper bound on  $D$  that depends only on the numbers of external lines of each type, i.e., on the physical process whose amplitude is being calculated:

$$D \leq 4 - \sum_f E_f(s_f + 1). \quad (12.1.10)$$

For example, in the simple version of quantum electrodynamics studied in the previous chapter, the Lagrangian included terms of the types shown in Table 12.1. All interactions here have  $\Delta_i \geq 0$ , and hence a Feynman diagram with  $E_\gamma$  external photon lines and  $E_e$  external Dirac lines will

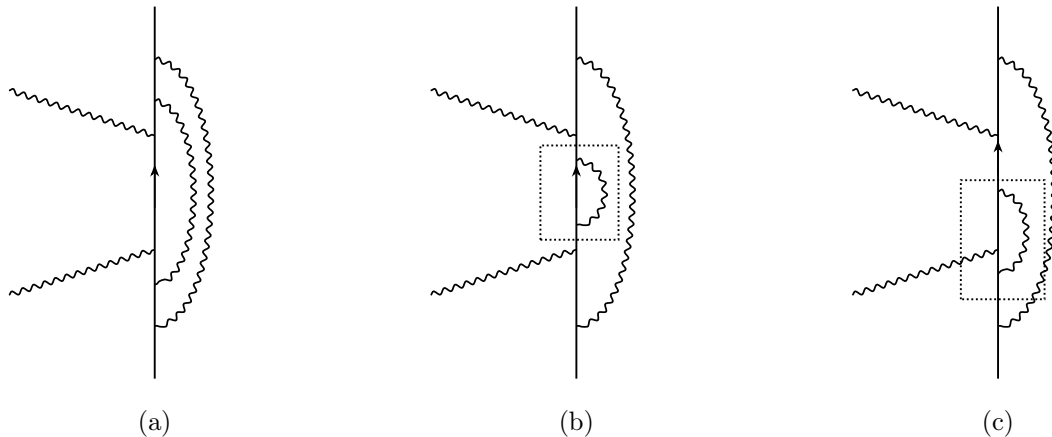
have superficial degree of divergence bounded by Eq. (12.1.10):

$$D \leq 4 - \frac{3}{2}E_e - E_\gamma. \quad (12.1.11)$$

Only a finite number of sets of external lines can yield superficially divergent integrals; these will be enumerated in Section 12.2. We are going to show that the limited number of divergences that appear in theories with  $\Delta_i \geq 0$  for all interactions are automatically removed by a redefinition of a finite number of physical constants and a renormalization of fields. For this reason, such theories are called renormalizable. In Section 12.3 we will catalog all the renormalizable theories, and discuss the significance of renormalizability as a criterion for physical theories.

The term “renormalizable” is also applied to individual interactions. Renormalizable interactions are those with  $\Delta_i \geq 0$ , whose coupling constants have positive or zero dimensionality. Sometimes one distinguishes between interactions with  $\Delta_i = 0$ , called simply renormalizable, and those with  $\Delta_i > 0$ , called superrenormalizable. Since adding additional fields or derivatives always lowers  $\Delta_i$ , there can only be a finite number of renormalizable interactions involving fields of any given types. We have seen that all the interactions in the simplest version of quantum electrodynamics are renormalizable, with the  $\bar{\psi}\psi$  terms superrenormalizable.

On the other hand, if any interaction has  $\Delta_i < 0$ , the degree of divergence (12.1.8) becomes larger and larger the more such vertices we include. No matter how large we take



**Figure 12.1.** Some two-loop graphs for Compton scattering. Here straight lines are electrons; wavy lines are photons. The momentum space integral for diagram (a) is convergent, while for (b) and (c) it is divergent, due to the subintegration associated with the subgraphs surrounded by dotted lines.

the various  $E_f$ , eventually with enough vertices of type  $i$  for which  $\Delta_i < 0$ , Eq. (12.1.8) will become positive (or zero), and the integral will diverge. Such interactions, whose couplings have negative-definite dimensionality, are called non-renormalizable;<sup>127</sup> theories with any non-renormalizable interactions are also known as non-renormalizable. But this does not mean that such theories are hopeless; we shall see that these divergences may also be absorbed into a redefinition of the parameters of the theory, but here we need an infinite number of couplings.

It should be kept in mind that we have here calculated the degree of divergence of Feynman diagrams arising only from regions of momentum space in which all internal four-momenta go to infinity together. Divergences can also arise from regions in which only the four-momenta of lines belonging to some subgraph go to infinity. For instance, in quantum electrodynamics Eq. (12.1.11) gives  $D \leq -1$  for Compton scattering (where  $E_e = 2$ ,  $E_\gamma = 2$ ), and indeed graphs like Figure 12.1(a) are convergent, but a graph like Figure 12.1(b) or 12.1(c) is logarithmically divergent, because these graphs contain subgraphs (indicated by dotted boxes) with  $D \geq 0$ . We can think of the divergence of these graphs as being due to an anomalously bad asymptotic behavior that occurs when the eight components of the two independent internal four-momenta of these graphs go to infinity on a particular four-dimensional subspace, namely, that subspace in which the only four-momentum actually going to infinity is the one circulating in the loops that are inserted in the internal lines or at an electron-photon vertex.

It has been shown [2] that the requirement for the actual convergence of

the amplitude corresponding to any graph is that power-counting should give  $D < 0$  not only for the complete multiple integral for the whole amplitude, but also for any

<sup>127</sup>In perturbative statistical mechanics, non-renormalizable interactions are called irrelevant, because they become less important in the limit of low energies. Renormalizable and super-renormalizable interactions are called marginal and relevant, respectively.



subintegration defined by holding any one or more linear combinations of the loop momenta fixed. (The graphs shown in Figures 12.1(b) and 12.1(c) fail this test because  $D \geq 0$  for the subintegrations in which only the momenta for the loops within the dotted squares are integrated.) We will not repeat the rather long proof here, because it is well treated in earlier books [3], and in any case the method of proof has little to do with how we actually do calculations. The next section will describe how this requirement is fulfilled.

## 12.2 Cancellation of Divergences

Consider a Feynman diagram, or part of a Feynman diagram, with positive superficial degree of divergence,  $D \geq 0$ . The part of the momentum space integral where all internal momenta go to infinity together will then diverge, like  $\int^\infty k^{D-1} dk$ . If we differentiate  $D+1$  times with respect to any external momentum, we lower the net number of momentum factors in the integrand by  $D+1$ ,<sup>128</sup> and hence render this part of the momentum space integral convergent. There may still be divergences arising from subgraphs, like those in Figures 12.1(b) and 12.1(c); for the moment we will ignore this possibility, returning to it later in this section. Since differentiation  $D+1$  times renders the integral finite, it follows that the contribution of such a graph or subgraph can be written as a polynomial of order  $D$  in external momenta, with divergent coefficients, plus a finite remainder.

To see how this works without irrelevant complications, consider the logarithmically divergent one-dimensional integral

$$\mathcal{J}(q) = \int_0^\infty \frac{dk}{k+q}$$

with  $D = 1 - 1 = 0$ . Differentiating once gives

$$\mathcal{J}'(q) = - \int_0^\infty \frac{dk}{(k+q)^2} = -\frac{1}{q},$$

so

$$\mathcal{J}(q) = -\ln q + c.$$

The constant  $c$  is obviously divergent, but the rest of the integral is perfectly finite. In exactly the same way, we can evaluate the  $D = 1$  integral

$$\int_0^\infty \frac{k dk}{k+q} = a + bq + q \ln q$$

with divergent constants  $a$  and  $b$ .

Now, a polynomial term in external momenta is just what would be produced by adding suitable terms to the Lagrangian: if a graph with  $E_f$  external lines of type  $f$  has degree of divergence  $D \geq 0$ , then the ultraviolet divergent polynomial is the same as would be produced by adding various interactions  $i$  with  $n_{if} = E_f$  fields of type  $f$  and  $d_i \leq D$  derivatives.

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<sup>128</sup>For instance, if an internal scalar field line carries a momentum  $k+p$ , where  $p$  is a linear combination of external four-momenta and  $k$  is a four-momentum variable of integration, then the derivative of the propagator  $[(k+p)^2 + m^2]^{-1}$  with respect to  $p^\mu$  gives  $-2(k_\mu + p_\mu)[(k+p)^2 + m^2]^{-2}$ , which goes as  $k^{-3}$  rather than  $k^{-2}$  for  $k \rightarrow \infty$ .

If there already are such interactions in the Lagrangian, then the ultraviolet divergences simply add corrections to the coupling constants of these interactions. Hence these infinities can be cancelled by including suitable infinite terms in these coupling constants. All that we ever measure is the sum of the bare coupling constant and the corresponding coefficient from one of the divergent polynomials, so if we demand that the sum equals the (presumably finite) measured value, then the bare coupling must automatically contain an infinity that cancels the infinity from the divergent integral over internal momenta. (One qualification: where the divergence occurs in a graph or subgraph with just two external lines, which appears as a radiative correction to a particle propagator, we must demand not that some effective coupling constant equals its measured value, but rather that the complete propagator has a pole at the same position and with the same residue as for free particles.) In this way, all infinities are absorbed into a redefinition of couplings constants, masses, and fields.

For this renormalization program to work, it is essential that the Lagrangian include *all* interactions that correspond to the ultraviolet divergent parts of Feynman amplitudes. (There are exceptions to this rule in supersymmetric theories [4].) The interactions in the Lagrangian are, of course, limited by various symmetry principles, such as Lorentz invariance, gauge invariance, etc., but these constrain the ultraviolet divergences in the same way. (It takes some work to prove that non-Abelian gauge symmetries constrain infinities in the same way that they constrain interactions. This will be shown in Volume II.) In the general case, there are no other limitations on the ultraviolet divergences, so *the Lagrangian must include every possible term consistent with symmetry principles*.

But there is an important class of theories with only a finite number of interactions, where the renormalization program also works. These are the so-called renormalizable theories, whose interactions all have  $\Delta_i \geq 0$ . Equation (12.1.8) then gives

$$D \leq 4 - \sum_f E_f(s_f + 1),$$

so divergent polynomials arise in only a limited number of Feynman graphs or subgraphs: those with few enough external lines so that  $D \geq 0$ . The contribution of such divergent polynomials is just the same as would be produced by replacing the divergent graph or subgraph with a single vertex arising from a term in the Lagrangian with  $E_f$  fields of type  $f$  and  $0, 1, \dots, D$  derivatives. But, comparing with Eq. (12.1.9), we see that *these are precisely the same as the interactions that satisfy the renormalizability requirement  $\Delta_i \geq 0$ , or in other words,*

$$0 \leq d_i \leq 4 - \sum_f n_{if}(s_f + 1).$$

In order for all infinities to cancel in a renormalizable theory, it is usually necessary that *all* renormalizable interactions that are allowed by symmetries must actually appear in the Lagrangian.<sup>129</sup> For instance, if there is a scalar (or pseudoscalar) field  $\phi$  and fermion field  $\psi$  with interactions  $\bar{\psi}\psi\phi$  (or  $\bar{\psi}\gamma_5\psi\phi$ ) then we cannot exclude an interaction  $\phi^4$ ; otherwise

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<sup>129</sup>In addition, interactions and mass terms that are not allowed by global symmetries may appear in the Lagrangian, as long as they are superrenormalizable, that is, with  $\Delta_i > 0$ . This is because the presence

there would be no counterterm to cancel the logarithmic divergence arising from fermion loops with four attached scalar or pseudoscalar lines.

Let's see in more detail how the cancellation of infinities works in the simplest version of quantum electrodynamics. Equation (12.1.11) shows that the only graphs or subgraphs that could possibly yield divergent integrals are the following:

$$E_e = 2, E_\gamma = 1$$

This is the electron-photon vertex  $\Gamma_\mu^{*(\ell)}(p', p)$ . (The superscript  $\ell$  indicates that this includes only contributions from graphs with loops.) It has  $D = 0$ , so its divergent part is momentum-independent. Lorentz invariance then only allows this divergent constant to be proportional to  $\gamma_\mu$ , so

$$\Gamma_\mu^{*(\ell)} = L\gamma_\mu + \Gamma_\mu^{(f)}, \quad (12.2.1)$$

with  $L$  a logarithmically divergent constant, and  $\Gamma_\mu^{(f)}$  finite. This does not uniquely define the constant  $L$ , since we can always move a finite term  $\delta L\gamma_\mu$  from  $\Gamma_\mu^{(f)}$  to  $L\gamma_\mu$ . To complete the definition, we may note that as shown in Section 9.7, the mass-shell matrix element of  $\Gamma_\mu^*(p, p)$  and hence of  $\Gamma_\mu^{(f)}(p, p)$  between mass-shell Dirac spinors is proportional to the same matrix element of  $\gamma^\mu$ , so we may define  $L$  by the prescription that

$$\bar{u}(p, \sigma') \Gamma_\mu^{(f)}(p, p) u(p, \sigma) = 0 \quad (12.2.2)$$

for  $p^2 + m_e^2 = 0$ .

$$E_e = 2, E_\gamma = 0$$

This is the electron self-energy insertion  $\Sigma^*(p)$ . It has  $D = 1$ , so its divergent part is linear in the momentum  $p^\mu$  carried by the incoming and outgoing fermion. Lorentz invariance (including parity conservation) will only allow it to be a function of  $\not{p}$ , so we may write the loop contribution as

$$\Sigma^{*(\ell)}(p) = A - (i\not{p} + m)B + \Sigma^{(f)}(\not{p}), \quad (12.2.3)$$

where  $A$  and  $B$  are divergent constants, and  $\Sigma^{(f)}$  is finite. Again, this does not uniquely define the constants  $A$  and  $B$ , because we can always shift  $\Sigma^{(f)}$  by a finite first-order polynomial in  $\not{p}$ . We will define  $A$  and  $B$  by the prescription that

$$\Sigma^{(f)} = \frac{\partial \Sigma^{(f)}}{\partial \not{p}} = 0 \quad \text{for} \quad i\not{p} = -m. \quad (12.2.4)$$

Actually,  $B$  is not a new divergent constant. As long as we use a regularization procedure that respects current conservation,  $\Gamma_\mu$  and  $\Sigma$  will be related by the Ward identity (10.4.27)

$$\Gamma^\mu(p, p) = \gamma^\mu + i \frac{\partial}{\partial p_\mu} \Sigma(p)$$

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of a superrenormalizable coupling lowers the degree of divergence, so that the symmetry breaking does not affect those divergences that are cancelled by the strictly renormalizable couplings with  $\Delta_i = 0$ . Note that it is the *bare* strictly renormalizable couplings that would exhibit the symmetry; renormalized couplings that are defined in terms of mass-shell matrix elements generally show the effect of symmetry breaking.

and therefore

$$L\gamma_\mu + \Gamma_\mu^{(f)}(p, p) = B\gamma_\mu + i\frac{\partial\Sigma^{(f)}(p)}{\partial p^\mu}. \quad (12.2.5)$$

Taking the matrix element of this equation between  $\bar{u}(p, \sigma')$  and  $u(p, \sigma)$  and using Eqs. (12.2.2) and (12.2.4), we find

$$L = B. \quad (12.2.6)$$

$E_\gamma = 2, E_e = 0$

This is the photon self-energy insertion  $\Pi_{\mu\nu}^*(q)$ . It has  $D = 2$ , so its divergent part is a second-order polynomial in  $q$ . Lorentz invariance only allows  $\Pi_{\mu\nu}^*$  to take the form of a linear combination of  $\eta_{\mu\nu}$  and  $q_\mu q_\nu$  with coefficients depending only on  $q^2$ , so the loop contributions take the form

$$\Pi_{\mu\nu}^{*(\ell)}(q) = C_1\eta_{\mu\nu} + C_2\eta_{\mu\nu}q^2 + C_3q_\mu q_\nu + \text{finite terms},$$

where  $C_1, C_2$ , and  $C_3$  are divergent constants. As long as we use a regularization technique that respects current conservation, we must have

$$q^\mu \Pi_{\mu\nu}^{*(\ell)}(q) = 0.$$

The same must then be true for the divergent terms, so  $C_1q_\nu + (C_2 + C_3)q^2q_\nu$  must be finite for all  $q$ . It follows that  $C_1$  and  $C_2 + C_3$  must be finite, and can therefore be lumped into the finite part of  $\Pi_{\mu\nu}^{*(\ell)}(q)$ . Thus

$$\Pi_{\mu\nu}^{*(\ell)}(q) = (\eta_{\mu\nu}q^2 - q_\mu q_\nu)(C + \pi(q^2)), \quad (12.2.7)$$

where  $\pi(q^2)$  is finite and  $C$  is the sole remaining divergence in  $\Pi_{\mu\nu}$ . To pin down the definition of  $C$ , we may move any finite constant  $\pi(0)$  into  $C$ , so that

$$\pi(0) = 0. \quad (12.2.8)$$

$E_\gamma = 4, E_e = 0$

This is the amplitude  $M_{\mu\nu\rho\sigma}$  for scattering of light by light. It has  $D = 0$ , so using Lorentz invariance and Bose statistics, it may be written (there is no non-loop contribution)

$$M_{\mu\nu\rho\sigma} = K(\eta_{\mu\nu}\eta_{\rho\sigma} + \eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho}) + \text{finite terms}$$

with  $K$  a potentially divergent constant. However, current conservation gives

$$q^\mu M_{\mu\nu\rho\sigma} = 0$$

and so  $K(q_\nu\eta_{\rho\sigma} + q_\rho\eta_{\nu\sigma} + q_\sigma\eta_{\nu\rho})$  is finite. In order for this to be true for  $q \neq 0$ ,  $K$  must itself be finite. This is a nice example of the role of symmetry principles in the renormalization program: if  $K$  had turned out to be infinite it could not be removed by renormalization of the coupling constant for an interaction  $(A_\mu A^\mu)^2$ , because no such interaction is allowed by gauge invariance, but  $K$  is finite because of current conservation conditions that are imposed by gauge invariance.

$E_\gamma = 1, E_e = 0$  **and**  $E_e = 1, E_\gamma = 0, 1, 2$

These have  $D = 3$  and  $D = \frac{5}{2}, \frac{3}{2}$ , and  $\frac{1}{2}$ , respectively, but Lorentz invariance makes all such graphs vanish.

$E_\gamma = 3, E_e = 0$

This has  $D = 1$ , but vanishes because of charge-conjugation invariance.

The reader will perhaps have noticed that the independent divergent constants  $A, B, C$  are in one-to-one correspondence with the independent parameters  $Z_2, Z_3$ , and  $\delta m$  in the counterterm part (11.1.9) of the Lagrangian for quantum electrodynamics. These counterterms make a direct contribution  $Z_2\delta m - (Z_2 - 1)(i\not{p} + m)$  to  $\Sigma^*(p)$ . The requirement that the position and residue of the one-particle pole be the same as in the free-field propagator means we must choose  $Z_2$  and  $\delta m$  so that the total  $\Sigma^*(p)$  satisfies Eq. (12.2.4), i.e.,

$$Z_2\delta m = -A, \quad (12.2.9)$$

$$Z_2 - 1 = -B, \quad (12.2.10)$$

so that the complete electron self-energy insertion is just the finite function  $\Sigma^{(f)}(p)$ :

$$\Sigma(p) = \Sigma^{(f)}(p). \quad (12.2.11)$$

Also,  $\mathcal{L}_2$  makes a direct contribution to  $\Gamma_\mu$  equal to  $(Z_2 - 1)\gamma_\mu$ . Using Eq. (12.2.6), we see that the full vertex is

$$\Gamma_\mu = \gamma_\mu + (Z_2 - 1)\gamma_\mu + \Gamma_\mu^{*(\ell)} = \gamma_\mu + \Gamma_\mu^{(f)}. \quad (12.2.12)$$

This is not only finite, but satisfies the condition

$$\bar{u}(p, \sigma')\Gamma_\mu(p, p)u(p, \sigma) = \bar{u}(p, \sigma')\gamma_\mu u(p, \sigma), \quad (12.2.13)$$

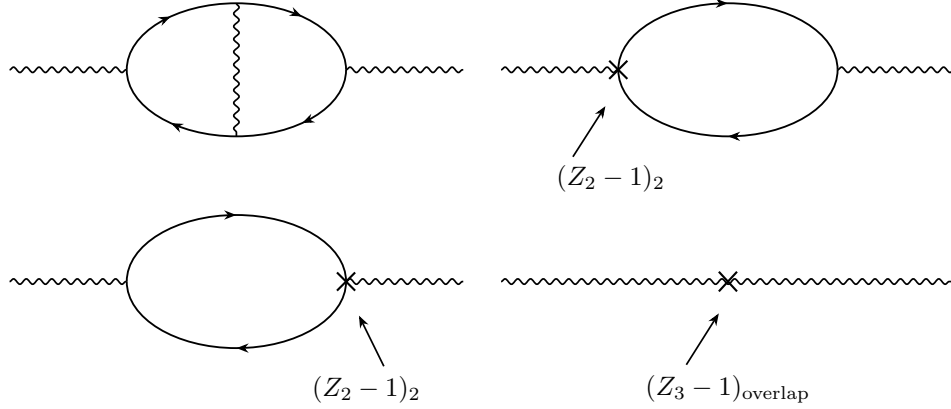
as can also be seen from Eqs. (10.6.13) and (10.6.14). Finally,  $\mathcal{L}_2$  makes a contribution  $-(Z_3 - 1)(q^2\eta_{\mu\nu} - q_\mu q_\nu)$  to  $\Pi_{\mu\nu}^*(q)$ . In order that the photon propagator should have a pole with the same residue as for free fields we need the coefficient of  $q^2\eta_{\mu\nu} - q_\mu q_\nu$  in the total  $\Pi_{\mu\nu}(q)$  to vanish, so

$$Z_3 = 1 + C \quad (12.2.14)$$

and the photon propagator is then finite:

$$\Pi_{\mu\nu}(q) = (\eta_{\mu\nu}q^2 - q_\mu q_\nu)\pi(q^2). \quad (12.2.15)$$

So far, we have only checked that the divergences, arising from the region of momentum space in which all internal momenta are large (and with generic ratios), are polynomials in external momenta that are cancelled by suitable counterterms. Such graphs are called *superficially convergent*. Before we conclude that all ultraviolet divergences actually are removed by renormalization, we need to consider the ultraviolet divergences arising in higher-order graphs when some subset of the momentum space integration variables rather than all of them go to infinity. For instance, in quantum electrodynamics the superficial divergences in subintegrations come from subgraphs that are either photon self-energy parts  $\Pi^*$ , or electron self-energy parts  $\Sigma^*$ , or electron–electron–photon vertices  $\Gamma^\mu$ .



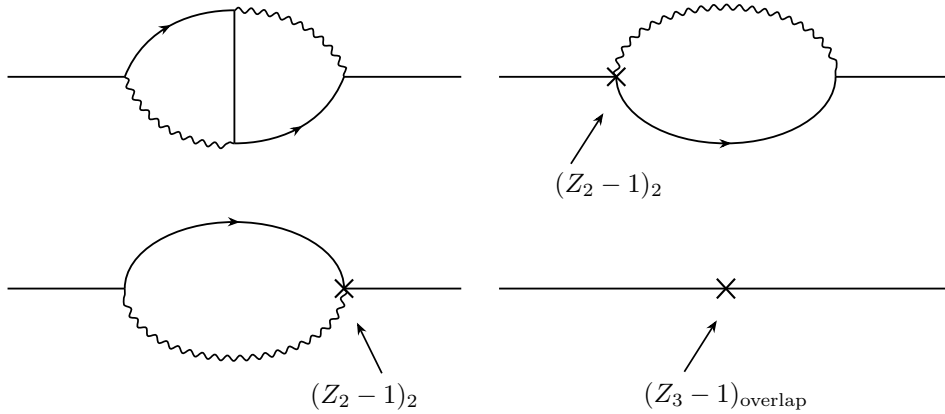
**Figure 12.2.** Some fourth-order graphs for the photon self-energy in quantum electrodynamics that involve overlapping divergences. Lines carrying arrows are electrons; wavy lines photons. The crosses mark the contribution of counterterms.

The problem with such divergences is that they cannot be removed by differentiation with respect to external momenta; we are left with terms where the derivatives act only on internal lines in the parts of the graph which are *not* in the divergent subgraphs, and therefore do not reduce the degree of divergence of these subgraphs. As mentioned in the previous section, a graph or sum of graphs is actually convergent only if it and all its subintegrations are superficially convergent, in the sense of counting powers of momentum. But wherever such a divergent subgraph appears, it always comes accompanied with an infinite counterterm. In electrodynamics, these are the terms in Eq. (11.1.9): a term  $-(Z_3 - 1)(q^2 \eta_{\mu\nu} - q_\mu q_\nu)$  for each  $\Pi_{\mu\nu}^*(q)$ ; a term  $Z_2 \delta m - (Z_2 - 1)(i\not{p} + m)$  for each  $\Sigma^*(p)$ ; and a term  $(Z_2 - 1)\gamma^\mu$  for each  $\Gamma^\mu$ . Just as for the graph as a whole, these counterterms cancel the infinities from the divergent subgraphs.

Unfortunately, there is a flaw in this simple argument—the possibility of overlapping divergences. That is, it is possible that two divergent subgraphs may share an internal line, so that we cannot regard them as independent divergent integrals. In quantum electrodynamics this happens only when two electron–electron–photon vertices overlap inside a photon or an electron self-energy insertion,<sup>130</sup> as shown in Figures 12.2 and 12.3.

A complete treatment of renormalization that takes account of overlapping divergences should include a prescription for eliminating superficial ultraviolet divergences, not only in the overall integration but in all subintegrations as well, together with a proof that this prescription is (at least formally) implemented by renormalization of masses, fields, and coupling constants. The theorem of Ref. [2] then ensures that all Green’s functions of renormalized fields are finite when expressed in terms of renormalized masses and couplings.

<sup>130</sup>The sharing of a line in two self-energy insertions or in a self-energy insertion and a vertex part would not leave enough external lines to attach such a subgraph to the rest of the diagram. Historically, the Ward identity (10.4.26) was used to by-pass the problem of overlapping divergences in the electron self-energy, by expressing the electron self-energy in terms of the vertex function, where overlapping divergences do not occur. This approach will not be followed here, as it is unnecessary, and in any case does not solve the problem for the self-energy of the photon or other neutral particles.



**Figure 12.3.** Some fourth-order graphs for the electron self-energy in quantum electrodynamics that involve overlapping divergences. Wavy lines are photons, other lines are electrons. The crosses mark the contribution of counterterms.

The first proof that the renormalization of fields, couplings, and masses renders the whole integration and all its subintegrations superficially divergent was offered by Salam [5]. A more specific prescription for eliminating ultraviolet divergences was given by Bogoliubov and Parasiuk [6], and corrected by Hepp [7], and shown by them to be equivalent to a renormalization of fields, masses, and coupling constants. Finally, Zimmermann [8] proved that this prescription does eliminate all superficial divergences in the whole integration and all its subintegrations, and used the theorem of Ref. [2] to conclude from this that the renormalized Feynman momentum-space integrals are convergent.

Briefly, the ‘BPHZ’ prescription for eliminating superficial divergences requires that we consider all possible ways (called ‘forests’) of surrounding a whole graph and/or its subgraphs with boxes that may be nested within each other but do not overlap. (An example is given below.) For each forest we define a subtraction term by replacing the integrand for any subgraph of superficial divergence  $D$  within a box (starting with the innermost boxes and working outwards) with the first  $D + 1$  terms of its Taylor series expansion in the momenta flowing into or out of that box.<sup>131</sup> The subtracted Feynman amplitude is given by the original graph, minus all these subtraction terms, including the subtraction term for a forest consisting of a single box surrounding the whole graph.

It is fairly easy to see that the subtracted Feynman amplitude that is calculated in this way is the same as would be obtained by replacing all fields, coupling constants, and masses in the original Lagrangian by their renormalized counterparts. The difference between this procedure and the sort of renormalization we carried out in Chapter 11 is that the renormalized fields, coupling constants, and masses are defined in terms of amplitudes at an unconventional renormalization point, where all four-momenta vanish. (In this respect, the one-dimensional divergent integrals discussed at the start of this section provide an elemen-

<sup>131</sup>As described here, this prescription applies to unrenormalizable as well as renormalizable theories. In renormalizable theories it implies that there is no subtraction unless the box contains one of the limited number of graphs corresponding to renormalizable terms in the Lagrangian.

tary example of the BPHZ method of separating divergent terms.) But there is nothing special about this renormalization point; once a Feynman amplitude is made convergent by expressing it in terms of these unconventional renormalized quantities, it can be rewritten in terms of conventionally renormalized fields, couplings, and masses without introducing new infinities.

It is not necessary to use the BPHZ subtraction prescription in practice. Replacing fields, masses, and couplings with their renormalized counterparts (defined using any convenient renormalization points) automatically provides counterterms that cancel all infinities. Instead of proving that the BPHZ subtraction prescription really does make all integrals converge, we shall just look at one example that shows how renormalization works, even in the presence of overlapping divergences.

Consider the fourth-order contribution to the photon self-energy insertion  $\Pi_{\mu\nu}^*(q)$  shown in Figure 12.2. (The forests here consist of the whole integral over  $p$  and  $p'$ , the subintegration over  $p$  alone, and the subintegration over  $p'$  alone.) Including the corresponding counterterms for the vertex parts and photon field renormalization, this has the value

$$\begin{aligned} [\Pi_{\mu\nu}^*(q)]_{\text{overlap}} = & -\frac{e^4}{(2\pi)^8} \int d^4p \int d^4p' \frac{1}{(p-p')^2 - i\epsilon} \\ & \cdot \text{Tr}\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\} \\ & - 2(Z_2 - 1)_2 \frac{ie^2}{(2\pi)^4} \int d^4p \text{Tr}\{\gamma_\nu S(p+q)\gamma_\mu S(p)\} \\ & - (Z_3 - 1)_{\text{overlap}}(q^2\eta_{\mu\nu} - q_\mu q_\nu), \end{aligned} \quad (12.2.16)$$

where  $S(p) \equiv [-i\not{p} + m]/[p^2 + m^2 - i\epsilon]$ ;  $(Z_2 - 1)_2$  is the term in  $Z_2 - 1$  of second order in  $e$ ; and  $(Z_3 - 1)_{\text{overlap}}$  is a logarithmically divergent constant of fourth order in  $e$  that cancels the terms in  $[\Pi_{\mu\nu}^*(q)]_{\text{overlap}}$  of second order in  $q^\mu$ . The factor 2 in the second term arises because there is a renormalization counterterm  $Z_2 - 1$  for each of the two vertices in the second-order photon self-energy. Note, however, that the first term here can be thought of *either* as the insertion of a vertex correction given by the  $p'$ -integral into a photon self-energy given by the  $p$ -integral, or as the insertion of a vertex correction given by the  $p$ -integral into a photon self-energy given by the  $p'$ -integral, but *not* as the insertion of two independent vertex corrections, because there is only one photon propagator.

To see how to handle the infinities in Eq. (12.2.16), note that

$$[(Z_2 - 1)_2 + R_2]\gamma_\mu = \frac{ie^2}{(2\pi)^4} \int \frac{d^4p'}{p'^2 - i\epsilon} \gamma_\rho S(p')\gamma_\mu S(p')\gamma^\rho, \quad (12.2.17)$$

where  $R_2$  is a finite remainder. (Lorentz invariance tells us that the integral on the right is proportional to  $\gamma_\mu$ . The difference between this integral and  $(Z_2 - 1)_2\gamma_\mu$  equals the complete renormalized electron–electron–photon vertex to second order in  $e$  at zero electron and photon momenta, and is therefore finite.) This allows us to rewrite Eq. (12.2.16) in



the form

$$\begin{aligned}
[\Pi_{\mu\nu}^*(q)]_{\text{overlap}} = & -\frac{e^4}{(2\pi)^8} \int d^4p \int d^4p' \left[ \frac{1}{(p-p')^2 - i\epsilon} \right. \\
& \cdot \text{Tr}\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\} \\
& - \frac{1}{p'^2 - i\epsilon} \text{Tr}\{S(p')\gamma_\nu S(p')\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\} \\
& \left. - \frac{1}{p^2 - i\epsilon} \text{Tr}\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p)\gamma_\mu S(p)\gamma_\rho\} \right] \\
& - 2R_2 \frac{ie^2}{(2\pi)^4} \int d^4p \text{Tr}\{\gamma_\nu S(p+q)\gamma_\mu S(p)\} \\
& - (Z_3 - 1)_{\text{overlap}}(q^2\eta_{\mu\nu} - q_\mu q_\nu). \tag{12.2.18}
\end{aligned}$$

First consider integration over  $p'$  alone. Each of the first two terms is logarithmically divergent, but their difference is finite. The third term is also logarithmically divergent (with a gauge-invariant regulator), but the divergence in this term (unlike the first two terms) takes the form of a second-order polynomial in  $q$ , with the remainder finite. This remaining divergence is cancelled by the term  $-(Z_3 - 1)(q^2\eta^{\mu\nu} - q^\mu q^\nu)$  that cancels all second-order terms in the expansion of  $\Pi_{\mu\nu}^*(q)$ . So the  $p'$ -subintegration gives a finite result. The symmetry of Eq. (12.2.16) shows that in exactly the same way the  $p$  subintegration also gives a finite result. Generic subintegrations over  $p$  and  $p'$  with  $ap + bp'$  held fixed (where  $a$  and  $b$  are arbitrary non-zero constants) are manifestly convergent, and the integration over  $p$  and  $p'$  together is made finite by the counterterm  $-(Z_3 - 1)(q^2\eta^{\mu\nu} - q^\mu q^\nu)$ . Thus Eq. (12.2.18) and any of its subintegrations satisfy the power-counting requirements for convergence, and therefore according to the theorem [2] quoted in the previous section, the whole expression actually converges.

\* \* \*

In electrodynamics there is a natural definition of renormalized couplings as well as renormalized masses and fields. This is not always the case. For example, consider the theory of a single real scalar field  $\phi(x)$  with Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{1}{24}g\phi^4. \tag{12.2.19}$$

To one-loop order, the S-matrix for scalar-scalar scattering is given by the Feynman rules as

$$S(q_1q_2 \rightarrow q'_1q'_2) = \frac{-i(2\pi)^4\delta^4(p'_1 + p'_2 - p_1 - p_2)}{(2\pi)^6(16E'_1E'_2E_1E_2)^{1/2}} F(q_1q_2 \rightarrow q'_1q'_2), \tag{12.2.20}$$

where

$$\begin{aligned}
-i(2\pi)^4 F(q_1 q_2 \rightarrow q'_1 q'_2) = & -i(2\pi)^4 g + \frac{1}{2} [-i(2\pi)^4 g]^2 \left[ \frac{-i}{(2\pi)^4} \right]^2 \\
& \cdot \int d^4 k \left[ \frac{1}{[(q_1 + k)^2 + m^2 - i\epsilon][(q_2 - k)^2 + m^2 - i\epsilon]} \right. \\
& \left. + (q_2 \rightarrow -q'_1) + (q_2 \rightarrow -q'_2) \right], \tag{12.2.21}
\end{aligned}$$

and  $q_1, q_2$  and  $q'_1, q'_2$  are the incoming and outgoing four-momenta. Combining denominators and rotating the  $k^0$ -integration contour as usual, this is

$$\begin{aligned}
F = g - \frac{g^2}{16\pi^2} \int_0^\infty k^3 dk \int_0^1 dx \Big\{ [k^2 + m^2 - sx(1-x)]^{-2} \\
+ [k^2 + m^2 - tx(1-x)]^{-2} + [k^2 + m^2 - ux(1-x)]^{-2} \Big\}, \tag{12.2.22}
\end{aligned}$$

where  $s, t$ , and  $u$  are the Mandelstam variables

$$s = -(p_1 + p_2)^2, \quad t = -(p_1 - p'_1)^2, \quad u = -(p_1 - p'_2)^2, \tag{12.2.23}$$

related by  $s + t + u = 4m^2$ ; also,  $x$  is the Feynman parameter introduced in combining denominators. With an ultraviolet cutoff at  $k = \Lambda$ , this gives the result (for  $\Lambda \gg m$ )

$$\begin{aligned}
F = g - \frac{g^2}{32\pi^2} \int_0^1 dx \Big\{ \ln \left( \frac{\Lambda^2}{m^2 - sx(1-x)} \right) + \ln \left( \frac{\Lambda^2}{m^2 - tx(1-x)} \right) \\
+ \ln \left( \frac{\Lambda^2}{m^2 - ux(1-x)} \right) - 3 \Big\}. \tag{12.2.24}
\end{aligned}$$

We can define the renormalized coupling  $g_R$  as the value of  $F$  at any point  $s, t, u$  we like, provided we stay in the region where  $F$  is real. For instance, suppose that in order to maintain the symmetry among the scalars, we choose to renormalize at the off-mass-shell point <sup>132</sup>  $p_1^2 = p_2^2 = p_1'^2 = p_2'^2 = \mu^2$ ,  $s = t = u = -4\mu^2/3$ . Defining the renormalized coupling  $g_R$  as the value of  $F$  at this point, we have

$$g = g_R + \frac{3g_R^2}{32\pi^2} \left[ \ln \left( \frac{\Lambda^2}{\mu^2} \right) - 1 - \int_0^1 dx \ln \left( \frac{4x(1-x)}{3} + \frac{m^2}{\mu^2} \right) \right] + \dots \tag{12.2.25}$$

The cutoff dependence then cancels in Eq. (12.2.24) to order  $g_R^2$ , leaving a finite formula for  $F$  in terms of  $g_R$ :

$$\begin{aligned}
F = g_R - \frac{g_R^2}{32\pi^2} \int_0^1 dx \Big\{ \ln \left( \frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - sx(1-x)} \right) + \ln \left( \frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - tx(1-x)} \right) \\
+ \ln \left( \frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - ux(1-x)} \right) \Big\} + \dots \tag{12.2.26}
\end{aligned}$$

<sup>132</sup>Going back over the derivation of Eq. (12.2.25), one may check that in this derivation we have not used the conditions  $p_1^2 = p_2^2 = p_1'^2 = p_2'^2 = -m^2$ , so Eq. (12.2.24) is valid whatever we take for the external line masses.

Here  $\mu^2$  may be taken to be any real quantity greater than  $-3m^2$ , in which range  $g_R$  is real. The explicit  $\mu$ -dependence in Eq. (12.2.26) is, of course, cancelled by the  $\mu$ -dependence of the renormalized coupling. This freedom to change the renormalization prescription (which of course exists also in electrodynamics and other realistic theories) will be of great importance to us when we come to the renormalization group method in Volume II.

### 12.3 Is Renormalizability Necessary?

In the previous section we found a special class of theories having only a finite number of terms in the Lagrangian, to which the renormalization program is nevertheless applicable. These are theories in which all interactions satisfy the renormalizability condition

$$\Delta_i \equiv 4 - d_i - \sum_f n_{if}(s_f + 1) \geq 0,$$

where  $d_i$  and  $n_{if}$  are the numbers of derivatives and fields of type  $f$  in interactions of type  $i$ , and  $s_f$  is (with some qualifications) the spin of fields of type  $f$ . For renormalization to work in such theories, it is also usually necessary that all renormalizable interactions that are allowed by symmetry principles should actually appear in the Lagrangian.

It is important that there are only a limited number of such interaction types.  $\Delta_i$  becomes negative if we have too many fields or derivatives, or fields of too high spin. Barring special cancellations, there are no renormalizable interactions at all involving fields with  $s_f \geq 1$ , because the only possible term in the Lagrangian with  $\Delta_i \geq 0$  that involves such a field along with two or more other fields would involve a single  $s_f = 1$  field along with two scalars and no derivatives, which would not be Lorentz-invariant. We shall see in Volume II that general, massless, spin one gauge fields in a suitable gauge effectively have  $s_f = 0$ , like the photon. Also, in Volume II we shall see that even massive gauge fields may effectively have  $s_f = 0$ , depending on where their mass comes from. Leaving aside these special cases, Table 12.2 gives a list of *all* renormalizable terms in the Lagrangian density that are allowed by Lorentz invariance and gauge invariance involving scalars ( $s = 0$ ), photons ( $s = 0$ ), and spin  $\frac{1}{2}$  fermions ( $s = \frac{1}{2}$ ).

We see that the requirement of renormalizability puts severe restrictions on the variety of physical theories that we may consider. Such restrictions provide a valuable key to the structure of physical theories. For instance, Lorentz and gauge invariance by themselves would allow the introduction of a ‘Pauli’ term proportional to  $\bar{\psi}[\gamma_\mu, \gamma_\nu]\psi F^{\mu\nu}$  in the Lagrangian of quantum electrodynamics, which would make the magnetic moment of the electron an adjustable parameter, but we exclude such terms because they are not renormalizable. The successful predictions of quantum electrodynamics, such as the calculation of the magnetic moment of the electron outlined in Section 11.3, may be regarded as validations of the principle of renormalizability. The same applies to the standard model of weak, electromagnetic, and strong interactions, to be discussed in Volume II; there are any number of terms that might be added to this theory, such as four-fermion interactions among quarks and leptons, that would invalidate all the predictions of the standard model, and are excluded only because they are non-renormalizable.

**Table 12.2.** Allowed renormalizable terms in a Lagrangian density involving scalars  $\phi$ , Dirac fields  $\psi$ , and photon fields  $A^\mu$ . Here  $n_{if}$  and  $d_i$  are the number of fields of type  $f$  and the number of derivatives in an interaction of type  $i$ , and  $\Delta_i$  is the dimensionality of the associated coefficient.

|         | $n_{if}$ |                    | $d_i$ | $\Delta_i$ | $\mathcal{H}_i$                           |
|---------|----------|--------------------|-------|------------|-------------------------------------------|
| Scalars | Photons  | Spin $\frac{1}{2}$ |       |            |                                           |
| 1       | 0        | 0                  | 0     | 3          | $\phi$                                    |
| 2       | 0        | 0                  | 0     | 2          | $\phi^2$                                  |
| 2       | 0        | 0                  | 2     | 0          | $\partial_\mu \phi \partial^\mu \phi$     |
| 3       | 0        | 0                  | 0     | 1          | $\phi^3$                                  |
| 4       | 0        | 0                  | 0     | 0          | $\phi^4$                                  |
| 2       | 1        | 0                  | 1     | 0          | $\phi \partial_\mu \phi A^\mu$            |
| 2       | 2        | 0                  | 0     | 0          | $\phi^2 A_\mu A^\mu$                      |
| 1       | 0        | 2                  | 0     | 0          | $\phi \bar{\psi} \psi$                    |
| 0       | 2        | 0                  | 2     | 0          | $F_{\mu\nu} F^{\mu\nu}$                   |
| 0       | 0        | 2                  | 0     | 1          | $\bar{\psi} \psi$                         |
| 0       | 0        | 2                  | 1     | 0          | $\bar{\psi} \gamma^\mu \partial_\mu \psi$ |
| 0       | 1        | 2                  | 0     | 0          | $\bar{\psi} \gamma^\mu A_\mu \psi$        |

Must we believe that the Lagrangian is restricted to contain only renormalizable interactions? As we saw in the previous section, if we include in the Lagrangian *all* of the infinite number of interactions allowed by symmetries, then there will be a counterterm available to cancel every ultraviolet divergence. In this sense, as said earlier, non-renormalizable theories are just as renormalizable as renormalizable theories, as long as we include all possible terms in the Lagrangian.

In recent years it has become increasingly apparent that renormalizability is not a fundamental physical requirement, and that in fact any realistic quantum field theory will contain non-renormalizable as well as renormalizable terms. This change in point of view can be traced in part to the continued failure to find a renormalizable theory of gravitation. In the general class of metric theories of gravitation governed by Einstein’s principle of equivalence there are no renormalizable interactions at all—generally covariant interactions must be constructed from the curvature tensor and its generally covariant derivatives, and hence, even in a ‘gauge’ where the graviton propagator goes as  $k^{-2}$ , these interactions involve too many derivatives of the metric for renormalizability. In particular, we can easily see that general relativity is non-renormalizable from the fact that its coupling constant  $8\pi G_N = (2.43 \cdot 10^{18} \text{ GeV})^{-2}$  has negative dimensionality. Even if nothing else did, the cancellation of divergences due to virtual gravitons would require that the Lagrangian contain all interactions allowed by symmetries—not only interactions involving gravitons, but involving any particles.

But if renormalizability is not a fundamental physical principle, then how do we explain the success of renormalizable theories like quantum electrodynamics and the standard

model? The answer can be seen by simple dimensional analysis. We have already noted that the coupling constant of an interaction of type  $i$  has dimensionality

$$[g_i] \sim [\text{mass}]^{\Delta_i}, \quad (12.3.1)$$

where  $\Delta_i$  is the index (12.1.9). Non-renormalizable interactions are just those whose coupling constants have the dimensionality of *negative* powers of mass. Now, it is not unreasonable to guess from (12.3.1) that the coupling constants not only have dimensionalities governed by  $\Delta_i$ , but are roughly of order

$$g_i \approx M^{\Delta_i}, \quad (12.3.2)$$

where  $M$  is some common mass. (This is found to be actually the case in the effective field theories discussed below and in more detail in Volume II.) In calculating physical processes at a characteristic momentum scale  $k \ll M$ , the inclusion of a non-renormalizable interaction of type  $i$  with  $\Delta_i < 0$  will introduce a factor  $g_i \approx M^{\Delta_i}$ , which on dimensional grounds must be accompanied by a factor  $k^{-\Delta_i}$ , and so the effect of such an interaction is suppressed<sup>133</sup> for  $k \ll M$  by a factor  $(k/M)^{-\Delta_i} \ll 1$ . (This argument will be made more carefully using the method of the renormalization group in Volume II.) The success of the renormalizable theories of electroweak and strong energies shows only that  $M$  is very much larger than the energy scale at which these theories have been tested.

For instance, the leading non-renormalizable corrections to the conventional electrodynamics of electrons or muons would be those interactions of dimension 5, which are suppressed by only one factor of  $1/M$ . There is just one such interaction allowed by Lorentz, gauge, and CP invariance, a Pauli term of order  $(ie/2M)\bar{\psi}[\gamma_\mu, \gamma_\nu]\psi F^{\mu\nu}$ . According to Eqs. (10.6.24), (10.6.17), and (10.6.19), such a term would contribute an amount of order  $4e/M$  to the magnetic moment of the electron or muon. The calculated value of the magnetic moment of the electron agrees with experiment to within terms of order  $10^{-10}e/2m_e$ , so  $M$  must be greater than about  $8 \cdot 10^{10}m_e = 4 \cdot 10^7 \text{ GeV}$ .

This limit may be weakened if other symmetries restrict the form of the non-renormalizable interactions. For instance, the conventional Lagrangian of quantum electrodynamics is invariant under a chiral transformation  $\psi \rightarrow \gamma_5 \psi$ , except for a change of sign of the fermion mass term  $-m\bar{\psi}\psi$ . If we assume that the full Lagrangian is invariant under a formal symmetry  $\psi \rightarrow \gamma_5 \psi$ ,  $m \rightarrow -m$ , then a Pauli term in the Lagrangian would have to appear with an extra factor  $m/M$ , so that its contribution to the magnetic moment would be only of order  $4em/M^2$ . Because of the extra factor of  $m$ , here it is the muon rather than

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<sup>133</sup>It is essential at this point to assume that the ultraviolet divergences have been removed by renormalization, so that there are no factors of an ultraviolet cutoff  $\Lambda$  to mess up our dimensional analysis. Otherwise, dimensional analysis tells us that for  $\Lambda \rightarrow \infty$ , each additional non-renormalizable coupling constant factor  $g_i$  with  $\Delta_i < 0$  would be accompanied with a growing factor  $\Lambda^{-\Delta_i}$ . This dimensional argument led Heisenberg [9] very early to classify interactions according to the dimensionality of their coupling constants, and to suggest [10] that new effects might arise at energies of order  $g_i^{1/\Delta_i}$ , as for instance at the energy  $G_F^{-1/2} \approx 300 \text{ GeV}$ , where  $G_F$  is the four-fermion coupling constant of the Fermi beta decay theory. After the development of renormalization theory it was noted by Sakata *et al.* [11] that the non-renormalizable theories are those whose coupling constants have negative dimensionality.

the electron that provides the most useful limit on  $M$ . The calculated value of the magnetic moment of the muon agrees with experiment to within terms of order  $10^{-8}e/2m_e$ , so  $M$  must be greater than about  $\sqrt{8 \cdot 10^8} m_\mu = 3 \cdot 10^3$  GeV. In any case, if  $M$  is anywhere near as large as  $10^{18}$  GeV, then we are certainly justified in neglecting any non-renormalizable interactions that might appear in quantum electrodynamics.

These considerations help us to cope with some of the puzzles associated with higher-derivative terms in the Lagrangian. For instance, in the general theory of a real scalar field  $\phi$ , we would expect to find terms in the Lagrangian density of the form  $\phi \square^n \phi$ . Any one such term would make a direct contribution to the scalar self-energy function  $\Pi^*(q^2)$  proportional to  $(q^2)^n$ . If we were to include this contribution to all orders, but ignore all other effects of non-renormalizable interactions, then the propagator  $\Delta'(q^2) = 1/(q^2 + m^2 - \Pi^*(q^2))$  would not have the simple pole in  $q^2$  at negative  $q^2$  expected from the general arguments of Section 10.7, but  $n$  such poles (some of which may coincide), generally at complex values of  $q^2$ . But if the non-renormalizable term  $\phi \square^n \phi$  has a coefficient of order  $M^{-2(n-1)}$ , where  $M \gg m$ , then the extra poles are at  $q^2$  of order  $M^2$ , where it is illegitimate to ignore the infinite number of other non-renormalizable interactions that must also appear in the Lagrangian. Thus the appearance of higher-derivative terms in a general non-renormalizable Lagrangian is not in conflict with the general principles underlying quantum field theory that were used in Section 10.7. But by the same token, we also cannot use higher-derivative terms to avoid ultraviolet divergences altogether, as has been repeatedly proposed. A term  $M^{-2(n-1)} \phi \square^n \phi$  in the Lagrangian density provides a cutoff at momenta  $q^2 \approx M^2$ , but at these momenta we cannot ignore all the other non-renormalizable interactions that must be present.

Although highly suppressed, non-renormalizable interactions may be detectable if they have effects that would otherwise be forbidden. For instance, we will see in Section 12.5 that the symmetries of charge-conjugation and space-inversion invariance are an automatic consequence of the structure of the electromagnetic interactions that is imposed by gauge invariance, Lorentz invariance, and renormalizability, but we can easily imagine non-renormalizable terms that would violate these symmetries, such as an electron electric dipole moment term  $\bar{\psi} \gamma_5 [\gamma_\mu, \gamma_\nu] \psi F^{\mu\nu}$ , or the Fermi interaction  $\bar{\psi} \gamma_5 \gamma_\mu \psi \bar{\psi} \gamma^\mu \psi$ . It is widely believed today that the conservation of baryon and lepton number is violated by very small effects of highly suppressed non-renormalizable interactions. Another example of a detectable non-renormalizable interaction is provided by gravitation. As mentioned before, gravitons have no renormalizable interactions at all. But, of course, we detect gravitation, because it has the special property that the gravitational fields of all the particles in a macroscopic body add up coherently.

Although non-renormalizable theories involve an infinite number of free parameters, they retain considerable predictive power: [12] they allow us to calculate the non-analytic parts of Feynman amplitudes, like the  $\ln q$  and  $q \ln q$  terms in the one-dimensional examples at the beginning of the previous section. Such calculations just reproduce the results required by the axiom of  $S$ -matrix theory, that the  $S$ -matrix has only those singularities required by unitarity.

Paradoxically, it is just in the case where symmetry principles forbid renormalizable interactions that non-renormalizable quantum field theories prove the most useful. In such

cases we can derive a useful perturbation theory by expanding in powers of  $k/M$ . This has been worked out in detail for the theory of low-energy pions [12, 13], to be discussed in detail in Volume II, and the theory of low-energy gravitons [14]. For a simpler example, consider the theory of a real scalar field, satisfying the principle of invariance under the field translation

$$\phi(x) \longrightarrow \phi(x) + \epsilon$$

with  $\epsilon$  an arbitrary constant. This symmetry forbids any renormalizable interactions or scalar mass, but it allows an infinite number of non-renormalizable derivative interactions

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{g}{4}(\partial_\mu\phi\partial^\mu\phi)^2 - \dots;$$

where  $g \approx M^{-4}$ , and ‘ $\dots$ ’ denotes terms with more derivatives or fields. (For simplicity, it is assumed here that the theory also has a symmetry under the reflection  $\phi \rightarrow -\phi$ .) According to the above dimensional analysis, the graph for a general reaction in which all energies and momenta are of order  $k \ll M$  is suppressed by a factor  $(k/M)^\nu$ , where

$$\nu = -\sum_i V_i \Delta_i = \sum_i V_i (d_i + n_i - 4),$$

with  $n_i$  and  $d_i$  the numbers of scalar fields and derivatives in an interaction of type  $i$ , and  $V_i$  the number of vertices for these interactions in our graph. For  $k \ll M$ , the dominant contributions to any process are those with the smallest value of  $\nu$ . The formula for  $\nu$  can be put in a more useful form by using the familiar topological identities for a connected graph:

$$\sum_i V_i = I - L + 1, \quad \sum_i V_i n_i = 2I + E,$$

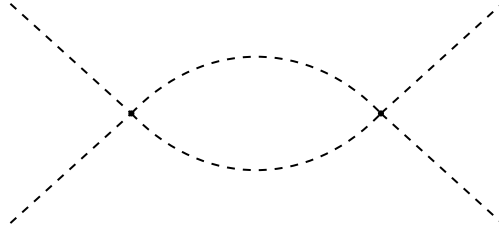
where  $I$ ,  $E$ , and  $L$  are the numbers of internal lines, external lines, and loops in our graph. Combining these relations gives

$$\nu = 2E - 4 + 4L + \sum_i V_i (d_i - n_i),$$

Now, the field translation symmetry requires that every field must be accompanied with at least one derivative, so the quantity  $d_i - n_i$  as well as  $L$  is non-negative for all interactions. Thus for a given process (that is, a fixed number  $E$  of external lines) the dominant terms will be those constructed solely from *tree* graphs (i.e.,  $L = 0$ ), and interactions with the minimum number  $d_i = n_i$  of derivatives. That is, in leading order we can take the Lagrangian density to depend only on *first* derivatives of the field. Higher-order corrections may involve loops and/or interactions with more derivatives on some fields. But to any given order  $\nu$  in  $k/M$ , we need only consider a finite number of graphs, those with  $L \leq (4 - 2E + \nu)/4$ , and only a finite number of interaction types.

For instance, scalar-scalar scattering is given in leading order by the one-vertex tree graph calculated using the interaction  $-g(\partial_\mu\phi\partial^\mu\phi)^2$  in first order. According to our formula for  $\nu$ , the leading correction, suppressed at low energy by a factor  $(k/M)^2$ , arises from another single-vertex tree graph, produced by an interaction with two additional derivatives

of the form<sup>134</sup>  $\partial_\mu \partial_\nu \phi \partial^\mu \partial^\nu \phi \partial_\rho \phi \partial^\rho \phi$ . The next corrections, suppressed at low energy by two further factors of  $k/M$ , arise both from the one-loop diagram of Figure 12.4 (including permutations of external lines), calculated using only the interaction  $-g(\partial_\mu \phi \partial^\mu \phi)^2$ , and also from tree graphs with a single vertex arising from a quartic interaction with eight derivatives, whose couplings contain infinite parts that cancel the ultraviolet divergence from the loop graph.<sup>135</sup> The loop graph also yields finite terms in the scattering amplitude proportional to terms like  $s^4 \ln s + t^4 \ln t + u^4 \ln u$ ,  $s^2 t^2 \ln u + t^2 u^2 \ln s + u^2 s^2 \ln t$ , etc., with calculable coefficients proportional to  $g^2$ . These finite terms simply represent the correction to the lowest-order scattering amplitude needed to ensure the unitarity of the  $S$ -matrix, but perturbative quantum field theory is by far the easiest way of calculating them.



**Figure 12.4.** One-loop diagram for scalar-scalar scattering in the theory with derivative quadri-linear interactions.

Although non-renormalizable theories can provide useful expansions in powers of energy, they inevitably lose all predictive power at energies of the order of the common mass scale  $M$  that characterizes the various couplings. If we were to take these expansions literally, the results for  $S$ -matrix elements would violate unitarity bounds for  $E \gg M$ . There seem to be just two possibilities about what happens at such energies. One is that the growing strength of the effects of the non-renormalizable couplings somehow saturates, avoiding any conflict with unitarity [15]. The other is that new physics of some sort enters at the scale  $M$ . In this case, the non-renormalizable theories that describe nature at energies  $E \ll M$  are just *effective field theories* rather than truly fundamental theories.

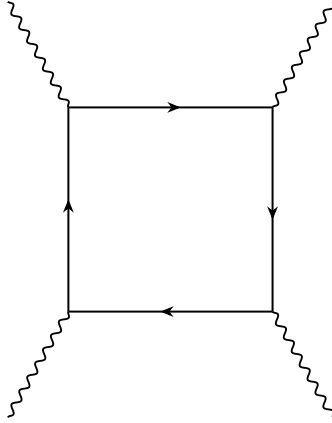
Probably the earliest example of an effective field theory was derived in the 1930s by Euler *et al.* [16], as a theory of low-energy photon-photon interactions. (See Section 1.3.) In effect, they calculated the contribution to photon-photon scattering of Feynman diagrams such as Figure 12.5, and found that at energies much less than  $m_e$  the scattering of light by light was the same as would be calculated with an effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{2\alpha^2}{45m_e^4} [(\mathbf{E}^2 - \mathbf{B}^2)^2 + 7(\mathbf{E} \cdot \mathbf{B})^2] \\ + \text{higher orders in } \frac{eE}{m_e^2} \text{ \& } \frac{eB}{m_e^2}$$

<sup>134</sup>In accordance with the remarks of Section 7.7, we are excluding interactions involving  $\square\phi$ , because the field equation for  $\phi$  can be used to express such interactions in terms of the others.

<sup>135</sup>These are the only ultraviolet divergences encountered in one-loop graphs if we use dimensional regularization. For other methods of regularization there are also quartic and quadratic divergences, which are cancelled by counterterms in the four-scalar interactions with four or six derivatives.





**Figure 12.5.** Diagram for photon–photon scattering, whose effect at low energy can be calculated from the effective Lagrangian of Euler *et al.* [16]. Straight lines are electrons; wavy lines are photons.

Euler *et al.* used this effective Lagrangian only in the tree approximation, to calculate the leading terms in photon interaction matrix elements. It was not until much later that such Lagrangians, though non-renormalizable, were used beyond the tree approximation [12, 17].

In modern jargon, we say that in deriving this Lagrangian the electron is ‘integrated out’, because in the one-loop approximation we have

$$\exp\left(i\int\mathcal{L}_{\text{eff}}(\mathbf{E},\mathbf{B})d^4x\right)=\int\left[\prod_x d\psi_e(x)\right]\exp\left(i\int\mathcal{L}_{\text{QED}}(\psi_e,A)d^4x\right).$$

A more general procedure is simply to write down the most general non-renormalizable effective Lagrangian, use it to calculate various amplitudes as an expansion in energies and momenta, and then choose the constants in the effective Lagrangian by matching the results it gives for these amplitudes to those derived from the underlying theory.

We will encounter effective field theories again, especially in considering broken symmetries in Volume II. As we shall see, effective field theories are useful even where they cannot be derived from an underlying theory, either because the theory is unknown, or because its interactions are too strong to allow the use of perturbation theory. Indeed, even if we knew nothing about the properties of charged particles, the scattering of photons at sufficiently low energy would have to be described by an effective Lagrangian consisting of the terms  $(\mathbf{E}^2 - \mathbf{B}^2)^2$  and  $(\mathbf{E} \cdot \mathbf{B})^2$ , because these are the unique quartic Lorentz- and gauge-invariant terms with no derivatives acting on  $\mathbf{E}$  and  $\mathbf{B}$ . Terms with such derivatives would be suppressed at low photon energies  $E$  by additional factors of  $E/M$ , where  $M$  is some typical mass of the charged particles that are being integrated out. We can go further: we shall see that effective field theories are useful even where the light particles they describe are not present in the underlying theory at all, but composites of the heavy particles that are integrated out. The underlying theory might not even be a field theory at all—the problem of incorporating gravitation has led many theorists to believe that, in

fact, it is a string theory. But wherever an effective field theory comes from, it is inevitably a non-renormalizable theory.

## 12.4 The Floating Cutoff<sup>136</sup>

Before closing this chapter, it is worth commenting on the relation between conventional renormalization theory and an approach pioneered by Wilson [18]. In Wilson’s method one imposes a “floating” finite ultraviolet cutoff (either sharp or smooth) at momenta with components of order  $\Lambda$ , and instead of taking  $\Lambda \rightarrow \infty$ , one requires that the bare constants of the theory (those appearing in the Lagrangian) depend on  $\Lambda$  in such a way that all observable quantities are  $\Lambda$ -independent.

It is convenient to work with dimensionless parameters. If a bare coupling or mass parameter  $g_i(\Lambda)$  has dimensionality  $[\text{mass}]^{\Delta_i}$ , we define the corresponding dimensionless parameter  $\mathcal{G}_i$  by

$$\mathcal{G}_i(\Lambda) \equiv \Lambda^{-\Delta_i} g_i(\Lambda). \quad (12.4.1)$$

Ordinary dimensional analysis tells us that the value of  $\mathcal{G}_i$  at one value  $\Lambda'$  of the cutoff can be expressed as a function of the values of the  $\mathcal{G}_j$  at another value  $\Lambda$  of the cutoff, and the dimensionless ratio  $\Lambda'/\Lambda$ :

$$\mathcal{G}_i(\Lambda') = F_i(\mathcal{G}(\Lambda), \Lambda'/\Lambda). \quad (12.4.2)$$

No dimensional parameters other than  $\Lambda'$  and  $\Lambda$  can appear in  $F$ , because no ultraviolet or infrared divergences can enter here; the difference between the constants at  $\Lambda$  and at  $\Lambda'$  arises from diagrams whose internal lines are restricted to have momenta between  $\Lambda$  and  $\Lambda'$ . Differentiating Eq. (12.4.2) with respect to  $\Lambda'$  and then setting  $\Lambda'$  equal to  $\Lambda$  yields a differential equation for  $\mathcal{G}_i$ :

$$\Lambda \frac{d}{d\Lambda} \mathcal{G}_i(\Lambda) = \beta_i(\mathcal{G}(\Lambda)), \quad (12.4.3)$$

where  $\beta_i(\mathcal{G}) \equiv [\partial F_i(\mathcal{G}, z)/\partial z]_{z=1}$ . The functions  $\beta_i(\mathcal{G})$  may be calculated for small couplings in perturbation theory. This is Wilson’s version of the “renormalization group” equation, which will be discussed in somewhat different terms in Volume II.

The Lagrangian for any finite value of the cutoff defines an effective field theory, in which instead of (or in addition to) integrating out “heavy” particles like the electron in the work of Euler *et al.*, one integrates out *all* particles with momenta greater than  $\Lambda$ . Even if one starts with a theory with a finite number of coupling parameters  $\mathcal{G}_i^0$  at some cutoff  $\Lambda_0$ , at any other value of the cutoff the differential equation (12.4.3) will generally yield non-zero values for all couplings allowed by symmetry principles.<sup>137</sup>

We now distinguish between the renormalizable and unrenormalizable couplings, labelled  $\mathcal{G}_a$  and  $\mathcal{G}_n$ , respectively, with  $a$  running over the finite number  $N$  of couplings (including masses) for which  $\Delta_a \geq 0$ , and  $n$  running over the infinite number of couplings with dimensionalities  $\Delta_n < 0$ . We want to show that if the couplings  $\mathcal{G}_a(\Lambda_0)$  and  $\mathcal{G}_n(\Lambda_0)$  at some initial cutoff value  $\Lambda_0$  lie on a generic  $N$ -dimensional initial surface  $\mathcal{S}_0$ , then (with some

<sup>136</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

<sup>137</sup>The only known exceptions to this rule are in theories based on supersymmetry [4].

qualifications) for  $\Lambda \ll \Lambda_0$  they will approach a fixed surface  $\mathcal{S}$  that is independent of both  $\Lambda_0$  and the initial surface.<sup>138</sup> This fixed surface is stable, in the sense that from any point on the surface, the trajectory generated by Eq. (12.4.3) stays on the surface. Such a stable surface defines a finite-parameter set of theories whose physical content is cutoff-independent, which as argued in the previous section, is the essential property of renormalizable theories. Furthermore, this construction shows that a generic theory defined with cutoff  $\Lambda_0$  will look for  $\Lambda \ll \Lambda_0$  like a renormalizable theory.<sup>139</sup>

To prove these results, consider any small perturbation  $\delta\mathcal{G}_i(\Lambda)$  in the values of the  $\mathcal{G}_i(\Lambda)$  satisfying Eq. (12.4.3). It will satisfy the differential equation

$$\Lambda \frac{d}{d\Lambda} \delta\mathcal{G}_i(\Lambda) = \sum_j M_{ij}(\mathcal{G}(\Lambda)) \delta\mathcal{G}_j(\Lambda), \quad (12.4.4)$$

where

$$M_{ij}(\mathcal{G}) \equiv \frac{\partial}{\partial\mathcal{G}_j} \beta_i(\mathcal{G}). \quad (12.4.5)$$

This equation couples the renormalizable and unrenormalizable couplings, making it difficult to see the difference in their behavior. To decouple them, we introduce the linear combinations

$$\xi_n \equiv \delta\mathcal{G}_n - \sum_{ab} \frac{\partial\mathcal{G}_n}{\partial\mathcal{G}_a^0} \left( \frac{\partial\mathcal{G}}{\partial\mathcal{G}^0} \right)_{ab}^{-1} \delta\mathcal{G}_b, \quad (12.4.6)$$

where  $\mathcal{G}_a^0$  are the values of the renormalizable couplings at cutoff  $\Lambda_0$ , which we shall use as coordinates for the initial surface, and  $\mathcal{G}_n$  are the values of the non-renormalizable couplings at cutoff  $\Lambda$  derived from the differential equation (12.4.3), with initial value for  $\Lambda_0$  at the point on the initial surface with coordinates  $\mathcal{G}_a^0$ . To calculate the derivative of  $\xi_n$  with respect to  $\Lambda$ , we note that the derivatives  $\partial\mathcal{G}_i/\partial\mathcal{G}_a^0$  satisfy the same differential equation (12.4.4) as the  $\delta\mathcal{G}_i$ . It is an elementary exercise then to show that

$$\Lambda \frac{d}{d\Lambda} \xi_n = \sum_m N_{nm} \xi_m, \quad (12.4.7)$$

where

$$N_{nm} \equiv M_{nm} - \sum_{ab} \frac{\partial\mathcal{G}_n}{\partial\mathcal{G}_a^0} \left( \frac{\partial\mathcal{G}}{\partial\mathcal{G}^0} \right)_{ab}^{-1} M_{bm}. \quad (12.4.8)$$

Now we must estimate the elements of  $N_{nm}$ . For a free-field theory no cutoff is needed, so for very small coupling all bare parameters  $g_i(\Lambda)$  become  $\Lambda$ -independent. Hence for small couplings the dimensionless parameters  $\mathcal{G}_i$  simply scale as  $\Lambda^{-\Delta_i}$ , and the matrix  $M_{ij}$  is given by

$$M_{ij} \approx -\Delta_i \delta_{ij}. \quad (12.4.9)$$

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<sup>138</sup>This theorem is due to Polchinski [19]. What follows here is a shortened and less rigorous version. (In Polchinski's proof, the initial surface is taken to be that with all non-renormalizable couplings vanishing. As we shall see here, the couplings approach the same fixed surface for generic initial surfaces.)

<sup>139</sup>Of course some theories have symmetries and a field content that do not allow any renormalizable interactions. This is the case for theories containing only fermion fields, or only the gravitational field. Such theories for  $\Lambda \ll \Lambda_0$  look like free-field theories.

It follows that the matrix  $N_{nm}$  is given approximately by  $-\Delta_n \delta_{nm}$ . The defining characteristic of the non-renormalizable couplings is that  $\Delta_n < 0$ , so Eq. (12.4.7) tells us that, at least for couplings in some finite range, where  $N_{nm}$  is positive-definite, the  $\xi_n$  decay for  $\Lambda \ll \Lambda_0$  like positive powers of  $\Lambda/\Lambda_0$ . In this limit, then, the perturbations are related by

$$\delta \mathcal{G}_n = \sum_{ab} \frac{\partial \mathcal{G}_n}{\partial \mathcal{G}_a^0} \left( \frac{\partial \mathcal{G}}{\partial \mathcal{G}^0} \right)_{ab}^{-1} \delta \mathcal{G}_b. \quad (12.4.10)$$

In particular, if we make a small change in the initial surface  $\mathcal{S}_0$  and/or the starting point on that surface and/or the initial cutoff  $\Lambda_0$ , such that the perturbations  $\delta \mathcal{G}_a$  in the renormalizable couplings vanish at some cutoff  $\Lambda \ll \Lambda_0$ , then the perturbations  $\delta \mathcal{G}_n$  in all the other couplings at cutoff  $\Lambda$  also vanish. Thus the non-renormalizable couplings  $\mathcal{G}_n(\Lambda)$  for  $\Lambda \ll \Lambda_0$  can depend only on the renormalizable couplings  $\mathcal{G}_a(\Lambda)$ , not separately on the initial surface or the starting point on that surface or the initial cutoff  $\Lambda_0$ . At cutoff  $\Lambda \ll \Lambda_0$  all the couplings therefore approach an  $N$ -dimensional surface  $\mathcal{S}$ , with coordinates  $\mathcal{G}_a(\Lambda)$ , which is independent of both the initial surface and of  $\Lambda_0$ . Note that the non-renormalizable couplings  $\mathcal{G}_n$  are not generally small on  $\mathcal{S}$ ; the important point is that they become functions of the renormalizable couplings. Changes in  $\Lambda$  with  $\Lambda$  remaining much less than  $\Lambda_0$  will change the couplings, but the couplings will remain close to  $\mathcal{S}$  (at least as long as the couplings do not become so large that  $N_{nm}$  is no longer a positive-definite matrix). Hence  $\mathcal{S}$  is a stable surface, as was to be proved.

We have seen that all physical quantities may be expressed in terms of  $\Lambda$  and the  $\mathcal{G}_n(\Lambda)$ , and are  $\Lambda$ -independent. This is true in particular of the  $N$  conventional renormalized couplings and masses, like  $e$  and  $m_e$  in quantum electrodynamics. But we can then invert this relation, and express the  $\mathcal{G}_n(\Lambda)$  in terms of the conventional parameters and  $\Lambda$ . In this way we can justify the usual renormalization program; all physical quantities are expressed in a cutoff-independent way in terms of the conventional renormalized couplings and masses.

The Wilson approach has some advantages in practice. One does not have to worry about subintegrations and overlapping divergences; the momentum cutoff applies to all internal lines. Also, some of the non-renormalization theorems of supersymmetry theory, which tell us that certain couplings are not affected by radiative corrections, work only for the cutoff-dependent bare couplings [20].

On the other hand, there are disadvantages to the Wilson approach. One must give up the special simplicities of working with renormalizable theories like quantum electrodynamics; once one starts integrating out particles with momenta above some scale  $\Lambda$ , the resulting effective field theory will contain all Lorentz- and gauge-invariant interactions, with  $\Lambda$ -dependent couplings. (Nevertheless, in physical processes at energies  $E \ll \Lambda$ , the dominant couplings will still be the renormalizable ones.) Also, the cutoff generally destroys *manifest* gauge invariance, and either manifest Lorentz invariance or unitarity. None of this is a problem in condensed matter physics, the original context of Wilson's approach, because no one would expect a realistic condensed matter theory to be strictly renormalizable, and there are no fundamental physical principles that are necessarily violated by a cutoff. In fact, in crystals there is a cutoff on phonon momenta, provided by the inverse lattice spacing.

At bottom, the difference between the conventional and the Wilson approach is one of mathematical convenience rather than of physical interpretation. Indeed, conventional renormalization already provides a sort of adjustable cutoff; when we express our answer in terms of coupling constants that are defined as the values of physical amplitudes at some momenta of order  $\mu$  (as for the scalar field theory discussed in the previous section), the cancellations that make integrals converge begin to operate at virtual momenta of order  $\mu$ . Conversely, the  $\Lambda$ -dependent coupling constants of the Wilson approach must ultimately be expressed in terms of observable masses and charges, and when this is done the results are, of course, the same as those obtained by conventional means.

## 12.5 Accidental Symmetries<sup>140</sup>

In Section 12.3 we saw that there are good reasons to adopt renormalizable field theories as approximate descriptions of nature at sufficiently low energy. It often happens that condition of renormalizability is so stringent that the effective Lagrangian automatically obeys one or more symmetries, which are not symmetries of the underlying theory, and may therefore be violated by the suppressed non-renormalizable terms in the effective Lagrangian. Indeed, most of the experimentally discovered symmetries of elementary particle physics are “accidental symmetries” of this sort.

A classic example is provided by the inversions and flavor conservation in the electrodynamics of charged leptons. The most general renormalizable and gauge- and Lorentz-invariant Lagrangian density for photons and fields  $\psi_i$  of spin  $\frac{1}{2}$  and charge  $-e$  takes the form

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4}Z_3F_{\mu\nu}F^{\mu\nu} \\ & -\sum_{ij}Z_{Lij}\bar{\psi}_{Li}[\not{\partial}+ie\not{A}]\psi_{Lj}-\sum_{ij}Z_{Rij}\bar{\psi}_{Ri}[\not{\partial}+ie\not{A}]\psi_{Rj} \\ & -\sum_{ij}M_{ij}\bar{\psi}_{Li}\psi_{Rj}-\sum_{ij}M_{ij}^\dagger\bar{\psi}_{Ri}\psi_{Lj},\end{aligned}\tag{12.5.1}$$

where  $i, j$  are summed over the three lepton flavors ( $e$ ,  $\mu$ , and  $\tau$ ),  $\psi_{iL}$  and  $\psi_{iR}$  are the left- and right-handed parts of the field  $\psi_i$ , defined by

$$\psi_{Li} = \frac{1}{2}(\mathbf{1} + \gamma_5)\psi_i, \quad \psi_{Ri} = \frac{1}{2}(\mathbf{1} - \gamma_5)\psi_i,\tag{12.5.2}$$

and  $Z_L$ ,  $Z_R$ , and  $M$  are numerical matrices. We are not assuming anything about lepton flavor conservation, so the matrices  $Z_{Lij}$ ,  $Z_{Rij}$  and  $M_{ij}$  need not be diagonal. Also we are not assuming anything about invariance under P, C, or T invariance, so there is no necessary relation between  $Z_L$  and  $Z_R$ , or between  $M$  and  $M^\dagger$ . The only constraints on these matrices come from the reality of the Lagrangian density, which requires that  $Z_{Lij}$  and  $Z_{Rij}$  are Hermitian, and from the canonical anticommutation relations, which require that  $Z_{Lij}$  and  $Z_{Rij}$  are positive-definite.

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<sup>140</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

Now suppose we replace the lepton fields  $\psi_L, \psi_R$  with new fields  $\psi'_L, \psi'_R$  defined by

$$\psi_L = S_L \psi'_L, \quad \psi_R = S_R \psi'_R, \quad (12.5.3)$$

where  $S_{L,R}$  are non-singular matrices that can be chosen as we like. The Lagrangian density when expressed in terms of these new fields then takes the same form as in Eq. (12.5.1), but with new matrices

$$Z'_L = S_L^\dagger Z_L S_L, \quad Z'_R = S_R^\dagger Z_R S_R, \quad M' = S_L^\dagger M S_R. \quad (12.5.4)$$

We can choose  $S_L$  and  $S_R$  so that  $Z'_L = Z'_R = \mathbf{1}$ . (Take  $S_{L,R} = U_{L,R} D_{L,R}$ , where  $U_{L,R}$  are the unitary matrices that diagonalize the positive-definite Hermitian matrices  $Z_{L,R}$ , and the  $D_{L,R}$  are the diagonal matrices whose elements are the inverse square roots of the eigenvalues of  $Z_{L,R}$ .)

Now make another transformation, to lepton fields  $\psi''_L$  defined by

$$\psi'_L = S'_L \psi''_L, \quad \psi'_R = S'_R \psi''_R. \quad (12.5.5)$$

The Lagrangian density again takes the same form when expressed in terms of these new fields, with new matrices

$$Z''_L = S'^{\dagger}_L S'_L, \quad Z''_R = S'^{\dagger}_R S'_R, \quad M'' = S'^{\dagger}_L M' S'_R. \quad (12.5.6)$$

This time we take  $S'_{L,R}$  unitary, so that again  $Z''_L = Z''_R = \mathbf{1}$ . We choose these unitary matrices so that  $M''$  is real and diagonal. (By the polar decomposition principle,  $M'$  like any square matrix may be put in the form  $M' = V H$ , where  $V$  is unitary and  $H$  is Hermitian. Take  $S'_L = S'^{\dagger}_R V^\dagger$  and choose  $S'_R$  as the unitary matrix that diagonalizes  $H$ .) Dropping primes, the Lagrangian density now takes the form

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4} Z_3 F_{\mu\nu} F^{\mu\nu} - \sum_i \bar{\psi}_{Li} [\not{\partial} + ie \not{A}] \psi_{Li} - \sum_i \bar{\psi}_{Ri} [\not{\partial} + ie \not{A}] \psi_{Ri} \\ & - \sum_i m_i \bar{\psi}_{Li} \psi_{Ri} - \sum_i m_i \bar{\psi}_{Ri} \psi_{Li}, \end{aligned} \quad (12.5.7)$$

where  $m_i$  are real numbers, the eigenvalues of the Hermitian matrix  $H$ . Finally, this can be put in the more familiar form

$$\mathcal{L} = -\frac{1}{4} Z_3 F_{\mu\nu} F^{\mu\nu} - \sum_i \bar{\psi}_i [\not{\partial} + ie \not{A}] \psi_i - \sum_i m_i \bar{\psi}_i \psi_i. \quad (12.5.8)$$

With the Lagrangian taking this form, it is now apparent that any renormalizable Lagrangian for lepton electrodynamics automatically conserves P, C, and T, as well as the numbers of leptons (minus the numbers of antileptons) of each flavor: electron, muon, and tauon.<sup>141</sup> In particular, despite the appearance of Eq. (12.5.1), this theory does not allow

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<sup>141</sup>This was first shown by Feinberg, Kabir, and myself [21]. Feinberg [22] had earlier noted that weak interaction effects in a theory with only one neutrino species would give rise to an observable rate for the process  $\mu \rightarrow e + \gamma$ , a difficulty that was only resolved by the discovery of a second neutrino species.

such processes as  $\mu \rightarrow e + \gamma$ . The reader may perhaps worry whether it is correct to identify the lepton fields as the  $\psi_i$  (previously called  $\psi_i''$ ) appearing in Eq. (12.5.8), which obviously conserves lepton flavor, rather than the  $\psi_i$  appearing in Eq. (12.5.1), which seems to allow processes like  $\mu \rightarrow e + \gamma$ .

Such worries may be put aside; as stressed in Section 10.3, there is no one field that can be identified as *the* field of the electron or muon. In fact, although Eq. (12.5.1) yields a non-vanishing matrix element for radiative decay of lepton 1 into lepton 2 *off* the lepton mass shell, by taking the lepton momenta on the mass shell we find a vanishing  $S$ -matrix for all such processes even when calculated using Eq. (12.5.1).

It was essential in deriving these results that the same electric charge appeared in Eq. (12.5.1) for both the left- and right-handed parts of the lepton fields or, in other words, that both left- and right-handed parts of the lepton fields transform in the same way under electromagnetic gauge transformations. As we shall see in Volume II, for similar reasons the modern renormalizable theory of strong interactions known as quantum chromodynamics automatically conserves C, and (aside from certain non-perturbative effects) P and T, as well as the numbers of quarks (minus the numbers of antiquarks) of each quark flavor. We shall also see in Volume II that the simplest version of the renormalizable standard model of weak and electromagnetic interactions automatically conserves lepton flavor (though not C and P) for reasons similar to those described here for electrodynamics. It remains an open possibility that non-renormalizable interactions arising from higher mass scales may violate any of these conservation laws.

### Weinberg Exercises

- W.12.1.** List all the renormalizable (or superrenormalizable) Lorentz-invariant terms in the Lagrangian of a single scalar field for space-time dimensionalities 2, 3, and 6.
- W.12.2.** Show how the overlapping divergence in the electron self-energy is cancelled in quantum electrodynamics.
- W.12.3.** Consider the theory of a scalar field  $\phi$  and spinor field  $\psi$ , with interaction Hamiltonian  $g\phi\bar{\psi}\psi$ . Write the one-loop part of the scalar self-energy function  $\Pi^*(q)$  as a divergent polynomial in  $p^\mu$ , plus an explicit convergent integral.
- W.12.4.** Suppose that the quantum electrodynamics of electrons and photons is actually an effective field theory, derived by integrating out unknown particles of mass  $M \gg m_e$ . Assume gauge invariance and Lorentz invariance, but not invariance under C, P, or T. What are the non-renormalizable terms in the Lagrangian of leading order in  $1/M$ ? Of next to leading order?

### Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

- W.12.1.** In  $D$  space-time dimensions a canonically normalized real scalar has

$$[\phi] = \frac{D-2}{2}, \quad [\partial_\mu] = 1.$$

An operator is power-counting renormalizable when its dimension is at most  $D$ ; it is superrenormalizable when the inequality is strict.

In  $D = 3$ ,  $[\phi] = 1/2$ . Apart from a constant and the kinetic term, the complete polynomial list is

$$\phi, \quad \phi^2, \quad \phi^3, \quad \phi^4, \quad \phi^5, \quad \phi^6.$$

The first five monomials are superrenormalizable and  $\phi^6$  is marginal. No additional derivative interaction is allowed: after integration by parts the first candidate is  $\phi^n(\partial\phi)^2$ , of dimension  $3 + n/2$ , and only  $n = 0$  gives the kinetic term.

In  $D = 6$ ,  $[\phi] = 2$ , so the list is

$$\phi, \quad \phi^2, \quad \phi^3,$$

together with the kinetic term and a constant. The cubic interaction is marginal; the lower powers are superrenormalizable.

The two-dimensional case is exceptional because  $[\phi] = 0$ . Every polynomial potential is superrenormalizable:

$$V(\phi) = \sum_{n \geq 0} g_n \phi^n, \quad [g_n] = 2.$$

At the level of local power counting one may also write

$$\frac{1}{2} Z(\phi) (\partial_\mu \phi) (\partial^\mu \phi),$$

with an arbitrary polynomial  $Z$ ; all of its terms are marginal. For one scalar and nonsingular  $Z$ , however, the field redefinition  $d\chi/d\phi = \sqrt{Z(\phi)}$  restores a canonical kinetic term. Thus an operator basis modulo field redefinitions consists simply of the kinetic term and the arbitrary potential. Terms containing more than two derivatives are nonrenormalizable in  $D = 2$ .

**W.12.2.** Let  $G$  be the two-loop electron self-energy graph in which the internal photon joins two points of the electron line. There are two logarithmically divergent one-loop vertex subgraphs,  $\gamma_1$  and  $\gamma_2$ , according to which external electron segment is included. They share an internal electron line and therefore overlap. In particular, a BPHZ forest may contain  $\gamma_1$  or  $\gamma_2$ , but not both.

For a vertex subgraph let  $t_\gamma$  denote its degree-zero Taylor operation in the momenta entering the subgraph. If no other subdivergence is present, the subdivergence-subtracted integrand is

$$\bar{R}I_G = I_G - t_{\gamma_1} I_G - t_{\gamma_2} I_G.$$

Each subtraction replaces the corresponding one-loop vertex by its local divergent part,

$$t_{\gamma_i} \Gamma_{1 \text{ loop}}^\mu = L_{\text{div}} \gamma^\mu.$$



This is exactly the graph produced by inserting the one-loop QED vertex counterterm at that location. The Ward identity gives  $L_{\text{div}} = (Z_2 - 1)_{\text{div}}$ , so the same coefficient is already fixed by electron field renormalization.

Consider a subintegration in which the loop momentum of  $\gamma_1$  becomes large while the other independent momentum is held fixed. The difference  $(1 - t_{\gamma_1})I_G$  falls by one additional inverse power and is convergent. The analogous statement holds for  $\gamma_2$ . There is no double subtraction  $t_{\gamma_1}t_{\gamma_2}I_G$ , precisely because the two subgraphs overlap rather than form a forest. Generic linear combinations of the two loop momenta are already superficially convergent.

After these two local vertex subtractions, only the overall electron self-energy divergence remains. Its degree is one, so Lorentz invariance restricts its Taylor polynomial to

$$t_G \bar{R}I_G = A_{\text{div}} + B_{\text{div}} i\not{p}.$$

The mass and field-strength counterterms cancel these two terms. Equivalently, the full forest formula is

$$RI_G = (1 - t_G)(I_G - t_{\gamma_1}I_G - t_{\gamma_2}I_G),$$

with each Taylor term represented by an ordinary QED counterterm graph. Thus every ultraviolet subintegration, including the two overlapping regions, and the overall integration are finite.

**W.12.3.** It is useful to make both the polynomial and the convergence of the remainder explicit. Work in Euclidean momentum and define  $\Pi_E(q)$  as the additive correction to the inverse scalar propagator. The closed fermion loop gives

$$\Pi_E^{(1)}(q) = 4g^2 \int_0^1 dx \int \frac{d^4\ell}{(2\pi)^4} \frac{m_\psi^2 - \ell^2 + x(1-x)q^2}{[\ell^2 + m_\psi^2 + x(1-x)q^2]^2},$$

where the shift  $\ell = k + xq$  has been made. A regulator is understood in the two terms that follow.

Put  $a = x(1-x)$ ,  $u = \ell^2 + m_\psi^2$ , and

$$F(\ell, q^2; x) = \frac{m_\psi^2 - \ell^2 + aq^2}{(u + aq^2)^2}.$$

Taylor's formula through first order gives

$$\Pi_E^{(1)}(q) = A + Bq^2 + \Pi_{\text{conv}}(q^2),$$

where

$$A = 4g^2 \int_0^1 dx \int \frac{d^4\ell}{(2\pi)^4} \frac{m_\psi^2 - \ell^2}{u^2},$$

$$B = 4g^2 \int_0^1 dx \int \frac{d^4\ell}{(2\pi)^4} \frac{a(3\ell^2 - m_\psi^2)}{u^3},$$

and

$$\Pi_{\text{conv}}(q^2) = 4g^2 \int_0^1 dx \int \frac{d^4\ell}{(2\pi)^4} \left[ F(\ell, q^2; x) - \frac{m_\psi^2 - \ell^2}{u^2} - q^2 \frac{a(3\ell^2 - m_\psi^2)}{u^3} \right].$$

The first line in the bracket has had its constant and  $q^2$  Taylor coefficients removed. Consequently the bracket is  $O(q^4/\ell^6)$  for large  $\ell$ , and the four-dimensional integral is absolutely ultraviolet convergent.

The divergent constants  $A$  and  $B$  are local: they multiply  $\phi^2$  and  $(\partial\phi)^2$ , respectively. A scalar mass counterterm and a scalar field-strength counterterm therefore remove them. Analytic continuation  $q_E^2 = -q^2 - i0$  gives the requested Minkowski self-energy and its physical cut.

**W.12.4.** Choose an operator basis modulo integrations by parts and the leading QED equations of motion. The first corrections have dimension five. For one Dirac field there are two independent Hermitian dipoles,

$$\mathcal{L}^{(5)} = \frac{c_M}{M} \bar{\psi} \sigma^{\mu\nu} \psi F_{\mu\nu} + \frac{c_E}{M} i \bar{\psi} \sigma^{\mu\nu} \gamma_5 \psi F_{\mu\nu}.$$

The first is the magnetic Pauli operator and the second is the electric dipole operator. The latter would be forbidden by P or T, but those symmetries were not assumed. Operators such as  $\bar{\psi} D^2 \psi$  reduce, by

$$(i\not{D})^2 = -D^2 - \frac{e}{2} \sigma^{\mu\nu} F_{\mu\nu},$$

to these dipoles and lower-dimensional terms.

At dimension six an economical on-shell basis is most transparent in chiral notation:

$$\mathcal{L}^{(6)} = \frac{1}{M^2} [c_{LL} (\bar{\psi}_L \gamma_\mu \psi_L) (\bar{\psi}_L \gamma^\mu \psi_L) + c_{RR} (\bar{\psi}_R \gamma_\mu \psi_R) (\bar{\psi}_R \gamma^\mu \psi_R) + c_{LR} (\bar{\psi}_L \gamma_\mu \psi_L) (\bar{\psi}_R \gamma^\mu \psi_R)]$$

Fierz identities reduce all other four-fermion contractions for a single species to these three. Unequal  $c_{LL}$  and  $c_{RR}$  and the equivalent vector-axial form make clear that discrete symmetries need not be present.

An off-shell basis may instead retain the two anapole/charge-radius operators

$$(\bar{\psi} \gamma_\mu \psi) \partial_\nu F^{\nu\mu}, \quad (\bar{\psi} \gamma_\mu \gamma_5 \psi) \partial_\nu F^{\nu\mu},$$

and the photon operator  $(\partial_\mu F^{\mu\nu})(\partial^\rho F_{\rho\nu})$ . The Maxwell equation  $\partial_\nu F^{\nu\mu} = e \bar{\psi} \gamma^\mu \psi$  converts them into the four-fermion basis above, so they must not be counted twice. A cubic Abelian field-strength scalar vanishes in four dimensions, while the remaining two-fermion derivative terms reduce by integration by parts and the Dirac equation. Hence the two dipoles are the complete leading  $1/M$  set, and the three current-current structures give a complete next-to-leading  $1/M^2$  on-shell set.

## Supplementary Exercises

### S.12.1. Canonical Dimensions in Arbitrary Dimension and A $Z_2$ -Even Scalar Operator Census

(Inspired by Iain W. Stewart, MIT 8.851 Effective Field Theory, Lectures 1–2 / Iain W. Stewart, MIT 8.851 Effective Field Theory, Lecture 2.)

- (a) In  $D$  space-time dimensions, determine the canonical dimensions of a real scalar, a Dirac spinor, an Abelian gauge potential, and their free kinetic operators. For a local operator containing  $n_\phi$  scalars,  $n_\psi$  spinors,  $n_F$  field strengths, and  $r$  derivatives, derive the mass dimension of its Wilson coefficient and classify it as relevant, marginal, or irrelevant.
- (b) For a real scalar in four dimensions with  $\phi \mapsto -\phi$ , list a nonredundant Lorentz-invariant operator basis through dimension six. Work modulo total derivatives and the leading equation of motion. Explain why  $\phi^3 \square \phi$ ,  $\phi^2 (\partial \phi)^2$ , and  $\phi^6$  do not all have to be retained as independent operators.

### S.12.2. The Jacobian of a Local Field Coordinate

(Inspired by Iain W. Stewart, MIT 8.851 Effective Field Theory, Lecture 3.)

In the scalar path integral make the local change of variable  $\phi(x) = \chi(x) + a\chi^3(x)/M^2$ . Compute its functional Jacobian. Show why the Jacobian is unity in dimensional regularization, describe the local terms it produces with a hard cutoff, and track how a linear source  $J\phi$  changes. Reconcile the resulting change of off-shell Green functions with invariance of on-shell amplitudes.

### S.12.3. Tree-Level Matching of a Heavy Scalar and Matching Is Not Merely Dropping the Heavy Field

(Inspired by Iain W. Stewart, MIT 8.851 Effective Field Theory, Lectures 1 and 4.)

- (a) Let a heavy real field  $H$  have Euclidean quadratic operator  $M^2 - \square$  and interaction  $\kappa H \phi^2/2$ . Integrate out  $H$  classically, obtain the exact nonlocal light-field action, and expand it through  $M^{-4}$ . Verify the coefficients by expanding the full-theory four- $\phi$  exchange amplitude at small external momentum.
- (b) For the heavy-scalar model of Exercise S.12.4, compare three low-energy descriptions: setting  $H = 0$ , retaining only the induced  $\phi^4$  operator, and retaining the first derivative correction. Determine the relative error in the four-point amplitude at momentum scale  $E \ll M$  in each case and state the energy at which the local expansion fails.

### S.12.4. Chiral Spurions and Dipole Suppression

(Inspired by Iain W. Stewart, MIT 8.851 Effective Field Theory, Lectures 1–2.)

Massless QED is invariant under independent phase rotations of  $\psi_L$  and  $\psi_R$ . Treat the fermion mass as a spurion and show that a dipole operator connecting  $\psi_L$  and  $\psi_R$  must carry one chiral-symmetry-breaking insertion. Compare the natural estimates  $c/M$  and  $cm/M^2$  for its coefficient and the corresponding contributions to a magnetic moment.

### S.12.5. Massless $\phi^4$ Theory at a Symmetric Subtraction Point

(Adapted from *University of Cambridge, Mathematical Tripos Part III, 2023, Paper 304: Advanced Quantum Field Theory, Question 2.*)

In the volume's mostly-plus convention, consider the massless real scalar theory in four-dimensional Minkowski space with Lagrangian

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{\lambda}{4!}\phi^4.$$

- (a) Without derivation, write down the momentum-space Feynman rules for the theory, including counterterms. Draw the diagrams that contribute to the renormalized 1PI four-point function  $\Gamma_4$  through order  $\lambda^2$ .
- (b) The function  $\Gamma_4$  may be written in terms of the Mandelstam variables  $s, t, u$ , defined as in Eq. (12.2.23). Define the effective coupling  $\lambda_{\text{eff}}$  at an energy scale  $M$  by

$$-i\lambda_{\text{eff}} = \Gamma_4 \quad \text{when} \quad s = t = u = -M^2.$$

Using dimensional regularization in  $d = 4 - \epsilon$ , evaluate the relevant integrals and show that

$$-i\lambda_{\text{eff}} = -i\lambda - \frac{i(-i\lambda)^2}{32\pi^2} \int_0^1 dx \left( \frac{6}{\epsilon} - 3\gamma + \dots \right) - i\delta\lambda + \dots.$$

Here  $\gamma$  is the Euler–Mascheroni constant, and the ellipses inside the parentheses contain the finite scale- and Feynman-parameter-dependent terms. Find a suitable choice of  $\delta\lambda$  such that  $\lambda_{\text{eff}} = \lambda$  at the renormalization scale  $M$ .

- (c) State, without derivation, the Callan–Symanzik equation for this theory. Assuming that there is no wavefunction renormalization at this order in  $\lambda$ , show that

$$\beta(\lambda) = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3).$$

You may use, for integers  $a, b$ ,

$$\int_{\mathbb{R}^d} \frac{d^d\ell}{(2\pi)^d} \frac{(\ell^2)^a}{(\ell^2 + \Delta)^b} = \frac{\Gamma(b-a-d/2)\Gamma(a+d/2)}{(4\pi)^{d/2}\Gamma(b)\Gamma(d/2)} \Delta^{-(b-a-d/2)},$$

as well as

$$\Gamma(z) = \frac{1}{z} - \gamma + O(z) \quad (z \rightarrow 0).$$

### S.12.6. Wilsonian Shell Integration in $\Phi^3$ Theory

(Adapted from University of Cambridge, Mathematical Tripos Part III, 2024, Paper 304: Advanced Quantum Field Theory, Question 3.)

Consider a scalar field theory in  $d$ -dimensional Euclidean space with action

$$S[\Phi] = \int d^d x \left( \frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi + \frac{g}{3!} \Phi^3 \right).$$

The quantum field  $\Phi(x)$  is defined with a momentum cutoff  $\Lambda$ .

- (a) Decompose its Fourier transform into low- and high-momentum modes at an arbitrary scale  $b\Lambda < \Lambda$ , where  $0 < b < 1$ , by

$$\tilde{\Phi}(p) = \begin{cases} \tilde{\phi}(p), & p^2 < b^2 \Lambda^2, \\ \tilde{\chi}(p), & b^2 \Lambda^2 < p^2 < \Lambda^2. \end{cases}$$

Show that

$$\Phi(x) = \phi(x) + \chi(x),$$

where  $\tilde{\phi}(p)$  and  $\tilde{\chi}(p)$  are the Fourier transforms of  $\phi(x)$  and  $\chi(x)$ , respectively.

- (b) Define the effective action by

$$e^{-W[\phi]} = \int \mathcal{D}\chi e^{-S[\Phi]}.$$

Show that it may be written as

$$e^{-W[\phi]} = e^{-S[\phi]} \left[ \exp \left( -S_{\text{int}} \left[ \phi, -\frac{\delta}{\delta J} \right] \right) Z_\chi[J] \right]_{J=0},$$

where  $Z_\chi[J]$  is the generating functional of the free massless high-momentum field  $\chi$ . Find  $S_{\text{int}}$ , define the source sign in  $Z_\chi[J]$  so that the displayed identity holds, and briefly explain how to evaluate the action perturbatively.

- (c) Assume that, to the order of interest, the effective action has the form

$$W[\phi] = \int d^d x \left( \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + \frac{g_{\text{eff}}}{3!} \phi^3 + \dots \right),$$

where the omitted effective-Lagrangian terms may be neglected. Draw all Feynman diagrams needed to determine  $g_{\text{eff}}$  through order  $g^3$ . Find  $g_{\text{eff}}$  to this order and show that for  $d = 6$

$$g_{\text{eff}} = g + C g^3 \ln b + \dots,$$

specifying  $C$ . You may neglect wavefunction rescaling.

You may use

$$\Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}, \quad \Gamma(n) = (n-1)! \quad (n \in \mathbb{Z}_+),$$

where  $\Omega_{d-1}$  is the area of the unit  $(d-1)$ -sphere.

### S.12.7. The One-Loop Yukawa Beta Function

(Adapted from Daniel Harlow, *Lecture Notes for Physics 8.324: Relativistic Quantum Field Theory II*, Section 12.8, Problem 2, with the Yukawa Lagrangian from Eq. (4.34).)

Consider the Yukawa theory used in the source problem: a real scalar  $\phi$  and a Dirac field  $\psi$  with the mostly-plus Lagrangian

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{m_\phi^2}{2}\phi^2 - i(\bar{\psi}\not{\partial}\psi + m_\psi\bar{\psi}\psi) - ig\phi\bar{\psi}\psi.$$

Using dimensional regularization, compute the complete one-loop renormalization of the Yukawa coupling  $g$ , including the field-strength renormalizations required when the interaction is expressed in canonically normalized fields, and extract the beta function. You may take the masses to zero after choosing nonexceptional external momenta, because the ultraviolet pole and beta function are mass-independent. State whether additional scalar interactions needed to close the full theory under renormalization affect this one-loop result.

### S.12.8. Euler–Heisenberg, Rayleigh Scattering, and Fermi Theory

(Inspired by Aneesh V. Manohar, *Introduction to Effective Field Theories*, Chapter 3 / Aneesh V. Manohar, *Introduction to Effective Field Theories*, Chapter 3, Section 7 / Aneesh V. Manohar, *Introduction to Effective Field Theories*, Chapter 3, Sections 8–9.)

- (a) Below the electron mass, use Lorentz and gauge invariance to show that the first local photon self-interactions occur at dimension eight. Construct the two parity-even invariants, estimate their one-loop coefficients, and derive the low-energy scaling of the photon–photon amplitude and total cross-section.
- (b) Represent a neutral spinless atom by a nonrelativistic field  $h$  and write its leading rotationally invariant couplings to  $\mathbf{E}^2$  and  $\mathbf{B}^2$ . If the atomic size is  $a_0$ , use dimensional analysis to obtain the frequency dependence of the elastic photon-scattering amplitude and cross-section. Explain why no frequency-independent term is allowed.
- (c) Expand the propagator of a vector boson of mass  $M$  exchanged between two approximately conserved currents. Match the first two local operators, determine their dimensions, and show that the ratio of the derivative correction to the leading Fermi interaction is  $O(p^2/M^2)$ . Distinguish the Wilson coefficient scale from the actual radius of convergence.

### S.12.9. Loop Power Counting and Expansion by Regions

(Inspired by Aneesh V. Manohar, *Introduction to Effective Field Theories*, Chapters 3–4 / Aneesh V. Manohar, *Introduction to Effective Field Theories*, Chapter 4, Section 4 / Aneesh V. Manohar, *Introduction to Effective Field Theories*, Chapter 4.)

- (a) Consider a graph containing vertices with operator dimensions  $d_i$  and multiplicities  $V_i$ . In dimensional regularization, prove that loops cannot generate positive powers of the heavy scale that compensate the factors  $M^{4-d_i}$ . Derive a formula for the net inverse power of  $M$  and explain why counterterms at a fixed EFT order form a finite set.
- (b) For  $m \ll M$ , study

$$I(m, M) = \mu^{2\varepsilon} \int \frac{d^{4-2\varepsilon} k_E}{(2\pi)^{4-2\varepsilon}} \frac{1}{(k_E^2 + m^2)(k_E^2 + M^2)}.$$

Evaluate it exactly and expand in  $m^2/M^2$ . Then expand only the heavy propagator before integrating and show which nonanalytic light-mass terms the EFT loops reproduce. Identify the analytic difference as matching data.

- (c) Prove by a scaling argument that a dimensionally regulated integral with no mass or external-momentum scale vanishes. Separate a logarithmic example formally into ultraviolet and infrared poles, explain why its total is zero, and show how the same zero permits matching calculations to discard common infrared contributions without losing ultraviolet information.

#### S.12.10. Heavy-Vector Matching and Infrared Cancellation

*(Inspired by Witold Skiba, TASI Lectures on Effective Field Theory and Precision Electroweak Measurements, Sections 2–3 / Witold Skiba, TASI Lectures on Effective Field Theory and Precision Electroweak Measurements, Sections 3–4.)*

- (a) A massive vector  $X_\mu$  with mass  $M$  couples as  $g_X X_\mu (J_1^\mu + J_2^\mu)$  to conserved light currents. Integrate it out at tree level and determine the coefficients of the three induced current–current operators. Show how the relative sign can be checked from the low-momentum full-theory exchange amplitude.
- (b) Compare an operator generated by single heavy-particle exchange with one first generated by a heavy one-loop graph. Derive their natural coefficient estimates, including couplings and the  $16\pi^2$  loop factor. Determine when a loop-generated dimension-six effect can dominate a tree-generated dimension-eight effect at energy  $E$ .
- (c) Suppose a one-loop full-theory amplitude contains both  $\ln(M^2/\mu^2)$  and  $\ln(m^2/\mu^2)$ , with  $m$  a light mass. Explain, using equality of low-energy amplitudes, why the same  $\ln(m^2/\mu^2)$  must occur in the EFT loop. Show algebraically that the Wilson coefficient obtained by subtraction is infrared finite and depends only on short-distance data.

#### S.12.11. Dipoles, Anapoles, and Threshold Decoupling

*(Inspired by Witold Skiba, TASI Lectures on Effective Field Theory and Precision Electroweak Measurements, Sections 2–3 / Witold Skiba, TASI Lectures on Effective Field Theory and Precision Electroweak Measurements, Sections 3–4.)*

- (a) Without imposing C, P, or T, classify all gauge- and Lorentz-invariant dimension-five operators made from one Dirac field and the electromagnetic field. Reduce derivative bilinears with the equations of motion, impose Hermiticity, and identify separately the magnetic and electric dipoles.
- (b) At dimension six consider

$$(\bar{\psi}\gamma_{\mu}\psi)\partial_{\nu}F^{\nu\mu}, \quad (\bar{\psi}\gamma_{\mu}\gamma_5\psi)\partial_{\nu}F^{\nu\mu}, \quad (\partial_{\mu}F^{\mu\nu})^2.$$

Use the leading Maxwell equation to trade these operators for current–current terms. Explain why either off-shell basis is valid but why combining both lists overcounts physical parameters.

- (c) A heavy charged Dirac fermion contributes to the photon vacuum polarization. Expand its renormalized one-loop result for  $|q^2| \ll M^2$ , identify the local  $F_{\mu\nu}\Box F^{\mu\nu}$  correction, and explain why a mass-independent subtraction scheme still requires an explicit threshold matching of the electric charge.

### S.12.12. Asymptotic Regions, the Hierarchy Problem, and Shift Symmetry

(Inspired by Timothy Cohen, *As Scales Become Separated*, Sections 2–4 / Timothy Cohen, *As Scales Become Separated*, Sections 5–6 / Timothy Cohen, *As Scales Become Separated*, Sections 2 and 6.)

- (a) For  $0 < m \ll M$ , evaluate

$$J(m, M) = \int_0^{\infty} \frac{dk}{(k+m)(k+M)}.$$

Recover its small- $m/M$  expansion by adding a soft-region expansion  $k \sim m$  and a hard-region expansion  $k \sim M$ . Introduce an intermediate separator, show that it cancels, and identify which region contains the nonanalytic  $\ln m$ .

- (b) Let light and heavy real scalars interact through  $-\lambda_{\ell h}\phi^2 H^2/4$ , with  $M \gg m$ . Compute the heavy one-loop contribution to the renormalized light mass in  $\overline{\text{MS}}$ , keeping its  $M^2$  dependence. Match at  $\mu = M$ , and quantify the tuning required to maintain  $m^2 \ll M^2$  when no symmetry forbids the correction.
- (c) Assume a real scalar has an exact shift symmetry  $\phi \mapsto \phi + \epsilon$ . Classify its lowest local interactions, prove that every external scalar enters an amplitude with at least one power of its momentum, and explain why a mass counterterm is forbidden. Then introduce a small spurion that breaks the symmetry and determine how radiative mass corrections must depend on it.

### S.12.13. Symmetry Is Attractive: Two-Scalar Renormalization

(Adapted from John McGreevy, *Physics 215B Quantum Field Theory*, Winter 2025, Assignment 6, Problem 3.)

In  $D = 3 + 1$ , consider two real scalar fields of equal mass with the mostly-plus interaction

$$\mathcal{L}_{\text{int}} = -\frac{g}{4!}(\phi_1^4 + \phi_2^4) - \frac{2\lambda}{4!}\phi_1^2\phi_2^2.$$



- (a) Show that the model has an  $O(2)$  symmetry when  $\lambda = g$ .
- (b) For general  $g, \lambda$ , determine whether counterterms of the form  $\phi_1\phi_2$  or  $\phi_1\Box\phi_2$  are required. If not, explain which symmetry forbids them.
- (c) Renormalize the theory through one loop using, for example, a Euclidean momentum cutoff or Pauli–Villars regularization. Add all necessary counterterms and impose renormalization conditions on the propagators in the case where both scalars are massless and on the independent  $2 \rightarrow 2$  amplitudes at nonexceptional kinematic values  $s_0, t_0, u_0$ . You may reuse the ordinary  $\phi^4$  bubble integral.
- (d) In the low-energy limit  $s_0, t_0, u_0 \ll \Lambda^2$ , where  $\Lambda$  is the cutoff, tune both propagator poles to  $p^2 = 0$ . Show that for positive couplings with  $\lambda/g < 3$  the low-energy coupling ratio flows toward the  $O(2)$ -symmetric value when it begins nearby. Equivalently, analyze the beta function of  $r = \lambda/g$  and relate the flow to the source’s fixed-physical-coupling, large-cutoff formulation.

#### S.12.14. Scale Symmetry, the Dilatation Current, and Its Generator

(Adapted from John McGreevy, *Physics 215A, Fall 2023, Assignment 4, Problem 1.*)

In  $D > 2$  space-time dimensions consider a real scalar with action

$$S = \int d^D x \left[ \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - g \phi^p \right].$$

- (a) Under the active transformation  $\phi_\lambda(x) = \lambda^\Delta \phi(\lambda x)$ , determine  $\Delta$  and  $p$  for which the classical action is invariant. Relate  $\Delta$  to the engineering dimension of the field.
- (b) Find the infinitesimal field transformation. Construct an improved, conserved stress tensor whose on-shell trace vanishes, and use it to obtain the conserved dilatation current and charge.
- (c) Using the equal-time canonical commutator, verify that the charge generates the infinitesimal scale transformation of  $\phi$ . Finally, state how the dilatation current of a general first-derivative Lagrangian differs from  $x_\nu T^{\mu\nu}$  when a nontrivial virial current is present.

### Solutions to Supplementary Exercises

#### S.12.1. Canonical Dimensions in Arbitrary Dimension and A $Z_2$ -Even Scalar Operator Census

- (a) The action is dimensionless, so every Lagrangian density has dimension  $D$ . The scalar and gauge kinetic terms and the Dirac kinetic term give

$$2 + 2[\phi] = D, \quad 2 + 2[A] = D, \quad 1 + 2[\psi] = D.$$

Consequently

$$[\phi] = [A_\mu] = \frac{D-2}{2}, \quad [\psi] = \frac{D-1}{2}, \quad [F_{\mu\nu}] = \frac{D}{2}.$$

The operator in the question has

$$d_{\mathcal{O}} = r + \frac{D-2}{2}n_{\phi} + \frac{D-1}{2}n_{\psi} + \frac{D}{2}n_F, \quad [C_{\mathcal{O}}] = D - d_{\mathcal{O}}.$$

Under a dilation toward low energy its dimensionless coupling is multiplied by  $b^{D-d_{\mathcal{O}}}$ . Thus  $d_{\mathcal{O}} < D$  is relevant,  $d_{\mathcal{O}} = D$  is marginal, and  $d_{\mathcal{O}} > D$  is irrelevant at the Gaussian fixed point. Quantum anomalous dimensions can refine this classification, but do not alter the canonical result.

(b) Before eliminating redundancies, convenient representatives are

|     |                                                                 |
|-----|-----------------------------------------------------------------|
| $d$ | operators                                                       |
| 0   | 1                                                               |
| 2   | $\phi^2$                                                        |
| 4   | $(\partial\phi)^2, \phi^4$                                      |
| 6   | $\phi^6, \phi^2(\partial\phi)^2, \phi^3\Box\phi, (\Box\phi)^2.$ |

Integration by parts gives

$$\phi^3\Box\phi = -3\phi^2(\partial\phi)^2 + \partial_{\mu}(\phi^3\partial^{\mu}\phi).$$

The leading equation of motion has the form  $\Box\phi = m^2\phi + \lambda\phi^3/6$ , with signs adjusted to the chosen metric. It converts  $\phi^3\Box\phi$  into a linear combination of  $m^2\phi^4$  and  $\lambda\phi^6$ , and converts  $(\Box\phi)^2$  similarly. A minimal basis through dimension six is therefore

$$\boxed{1, \quad \phi^2, \quad (\partial\phi)^2, \quad \phi^4, \quad \phi^6.}$$

Keeping  $\phi^2(\partial\phi)^2$  instead of  $\phi^6$  is an equally valid basis choice; physical amplitudes cannot depend on that choice.

### S.12.2. The Jacobian of a Local Field Coordinate

The functional derivative is diagonal in position space:

$$\frac{\delta\phi(x)}{\delta\chi(y)} = \left(1 + \frac{3a}{M^2}\chi^2(x)\right)\delta^{(d)}(x-y).$$

Hence

$$\ln \mathcal{J} = \text{Tr} \ln \left(1 + \frac{3a\chi^2}{M^2}\right) = \delta^{(d)}(0) \int d^d x \ln \left(1 + \frac{3a\chi^2}{M^2}\right).$$

Dimensional regularization sets  $\delta^{(d)}(0) = \int d^d k / (2\pi)^d = 0$ , so  $\mathcal{J} = 1$ . With a hard cutoff,  $\delta^{(d)}(0) \sim \Lambda^d$ ; expanding the logarithm produces only local terms  $\chi^{2n}$ , which are absorbed into the Wilson coefficients already allowed in the action.

The source term becomes

$$\int d^d x J\phi = \int d^d x \left(J\chi + \frac{a}{M^2}J\chi^3\right).$$

Thus derivatives with respect to  $J$  insert a composite nonlinear operator, and off-shell correlators of the named field do change. The path integral itself is merely rewritten, however. Transforming the action, source, and local Jacobian consistently leaves pole locations and LSZ-amputated amplitudes invariant. This is the equivalence theorem; it does not assert equality of arbitrary off-shell Green functions.

### S.12.3. Tree-Level Matching of a Heavy Scalar and Matching Is Not Merely Dropping the Heavy Field

(a) Put  $K = M^2 - \square$ . Completing the square gives

$$\frac{1}{2}HKH + \frac{\kappa}{2}H\phi^2 = \frac{1}{2}\left(H + \frac{\kappa}{2}K^{-1}\phi^2\right)K\left(H + \frac{\kappa}{2}K^{-1}\phi^2\right) - \frac{\kappa^2}{8}\phi^2K^{-1}\phi^2.$$

Thus  $H_{\text{cl}} = -(\kappa/2)K^{-1}\phi^2$  and

$$\Delta\mathcal{L}_{\text{eff}} = -\frac{\kappa^2}{8}\phi^2\frac{1}{M^2 - \square}\phi^2 = -\frac{\kappa^2}{8M^2}\phi^4 - \frac{\kappa^2}{8M^4}\phi^2\square\phi^2 + O(M^{-6}).$$

After integration by parts, the second operator is  $+\kappa^2[\partial_\mu(\phi^2)]^2/(8M^4)$ . The full four-point amplitude is the sum of heavy exchange in the three channels. Expanding each Euclidean denominator,

$$\frac{1}{M^2 + p^2} = \frac{1}{M^2} - \frac{p^2}{M^4} + \dots,$$

reproduces precisely the contact vertex and its derivative correction.

(b) At low momentum a single heavy propagator contributes schematically

$$\mathcal{A}_{\text{full}} = \frac{\kappa^2}{M^2 + p^2} = \frac{\kappa^2}{M^2} \left(1 - \frac{p^2}{M^2} + \frac{p^4}{M^4} - \dots\right),$$

with a sum over channels understood. Setting  $H = 0$  and adding nothing misses the leading amplitude, so its relative error is  $O(1)$ . The induced  $\phi^4$  operator reproduces the first term and has relative error  $O(E^2/M^2)$ . Including the dimension-six derivative correction leaves relative error  $O(E^4/M^4)$ .

The expansion is a Taylor series about zero momentum. It fails when a channel invariant approaches the nearest heavy pole or threshold,  $|p^2| \sim M^2$ . Adding more local terms cannot extend the radius of convergence through that physical singularity; the heavy degree of freedom must then be restored.

### S.12.4. Chiral Spurions and Dipole Suppression

The dipole contains opposite chiralities:

$$\bar{\psi}\sigma^{\mu\nu}\psi = \bar{\psi}_L\sigma^{\mu\nu}\psi_R + \bar{\psi}_R\sigma^{\mu\nu}\psi_L.$$

Under  $U(1)_L \times U(1)_R$ , the first term has phase  $e^{i(\alpha_R - \alpha_L)}$ . The mass spurion in  $m\bar{\psi}_L\psi_R + \text{H.c.}$  has the compensating transformation. If it is the only source of chiral breaking, the Wilson coefficient must therefore be proportional to  $m$ :

$$\frac{c}{M} \longrightarrow \frac{cm}{M^2}.$$

A generic dimension-five dipole gives a dimensionless anomalous moment of order  $cm/M$ , whereas the chirally protected estimate is

$$\delta a = O\left(c \frac{m^2}{M^2}\right), \quad \delta\mu = O\left(\frac{ecm}{M^2}\right).$$

Setting  $m = 0$  restores the symmetry and forces the correction to vanish, which makes the small coefficient technically natural.

### S.12.5. Massless $\phi^4$ Theory at a Symmetric Subtraction Point

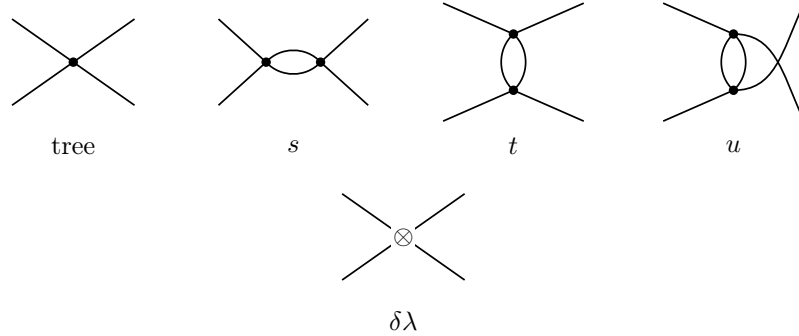
(a) A convenient counterterm organization in the volume's convention is

$$\mathcal{L} = -\frac{1}{2}(1 + \delta Z)\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}\delta m^2\phi^2 - \frac{\lambda + \delta\lambda}{4!}\phi^4.$$

The scalar propagator, ordinary quartic vertex, two-point counterterm, and four-point counterterm are respectively

$$\frac{-i}{p^2 - i0}, \quad -i\lambda, \quad -i(\delta Z p^2 + \delta m^2), \quad -i\delta\lambda.$$

Through order  $\lambda^2$ , the 1PI four-point diagrams are the tree vertex, the one-loop bubble in each of the  $s, t, u$  channels (each with symmetry factor  $1/2$ ), and the four-point counterterm:



The massless tadpole is scaleless in dimensional regularization, and the one-loop two-point function has no momentum-dependent pole, so  $\delta m^2 = 0$  and  $\delta Z = 0$  at the order used here.

(b) After Wick rotation, Feynman parametrization, and a shift of loop momentum, the bubble at a Euclidean momentum  $Q$  is

$$\begin{aligned} B(Q^2) &\equiv \mu^\epsilon \int \frac{d^{4-\epsilon}\ell_E}{(2\pi)^{4-\epsilon}} \frac{1}{\ell_E^2(\ell_E + Q)^2} \\ &= \frac{1}{16\pi^2} \int_0^1 dx \left[ \frac{2}{\epsilon} - \gamma + \ln(4\pi) + \ln \frac{\mu^2}{x(1-x)Q^2} \right] + O(\epsilon). \end{aligned}$$

Thus

$$\Gamma_4 = -i\lambda + \frac{i\lambda^2}{2} [B(-s) + B(-t) + B(-u)] - i\delta\lambda + O(\lambda^3),$$

where the arguments indicate the positive Euclidean invariants reached by continuation. At  $s = t = u = -M^2$ ,

$$\begin{aligned} -i\lambda_{\text{eff}} = & -i\lambda + \frac{i\lambda^2}{32\pi^2} \int_0^1 dx \left[ \frac{6}{\epsilon} - 3\gamma + 3\ln(4\pi) + 3\ln \frac{\mu^2}{x(1-x)M^2} \right] \\ & - i\delta\lambda + O(\lambda^3), \end{aligned}$$

which is the displayed form in the question after using  $-i(-i\lambda)^2 = +i\lambda^2$ .

Choose  $\mu = M$ . Since  $\int_0^1 dx [-\ln(x(1-x))] = 2$ , a momentum-subtraction counterterm that enforces  $\lambda_{\text{eff}}(M) = \lambda$  is

$$\boxed{\delta\lambda_{\text{MOM}} = \frac{3\lambda^2}{32\pi^2} \left( \frac{2}{\epsilon} - \gamma + \ln(4\pi) + 2 \right)}.$$

The  $\overline{\text{MS}}$  choice instead removes only

$$\delta\lambda_{\overline{\text{MS}}} = \frac{3\lambda^2}{16\pi^2} \frac{1}{\bar{\epsilon}}, \quad \frac{1}{\bar{\epsilon}} = \frac{1}{\epsilon} - \frac{\gamma}{2} + \frac{1}{2} \ln(4\pi),$$

and therefore differs by a finite redefinition of the coupling.

(c) For the renormalized 1PI  $n$ -point function the Callan–Symanzik equation is

$$\left[ \mu \frac{\partial}{\partial \mu} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma_\phi(\lambda) \right] \Gamma^{(n)} = 0.$$

Here  $\gamma_\phi = O(\lambda^2)$ , so it does not enter the one-loop four-point calculation. The pole part may be encoded as

$$\lambda_0 = \mu^\epsilon \left[ \lambda + \frac{3\lambda^2}{16\pi^2\epsilon} + O(\lambda^3) \right].$$

The bare coupling is independent of  $\mu$ ; differentiating gives

$$\beta(\lambda) = -\epsilon\lambda + \frac{3\lambda^2}{16\pi^2} + O(\lambda^3).$$

Taking  $\epsilon \rightarrow 0$  therefore yields

$$\boxed{\beta(\lambda) = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3)}.$$

### S.12.6. Wilsonian Shell Integration in $\Phi^3$ Theory

(a) Use the Fourier convention

$$\Phi(x) = \int_{p^2 < \Lambda^2} \frac{d^d p}{(2\pi)^d} e^{ip \cdot x} \tilde{\Phi}(p).$$

The two characteristic functions in the definition of  $\tilde{\phi}$  and  $\tilde{\chi}$  partition the cutoff ball (up to its measure-zero boundary), so

$$\tilde{\Phi}(p) = \tilde{\phi}(p) + \tilde{\chi}(p).$$

Fourier inversion immediately gives  $\Phi(x) = \phi(x) + \chi(x)$ . Their momentum supports are disjoint, which also makes the kinetic cross term vanish:  $\int d^d x \partial\phi \cdot \partial\chi = 0$ .

(b) Expanding the cubic action gives

$$S[\phi + \chi] = S[\phi] + S_{\chi,0} + S_{\text{int}}[\phi, \chi],$$

where

$$S_{\chi,0} = \frac{1}{2} \int d^d x \partial_\mu \chi \partial^\mu \chi$$

and

$$S_{\text{int}}[\phi, \chi] = g \int d^d x \left( \frac{1}{2} \phi^2 \chi + \frac{1}{2} \phi \chi^2 + \frac{1}{3!} \chi^3 \right).$$

To reproduce the functional-derivative sign printed in the question, define

$$Z_\chi[J] = \mathcal{N} \int \mathcal{D}\chi \exp \left[ -S_{\chi,0} - \int d^d x J\chi \right], \quad Z_\chi[0] = 1.$$

Then each insertion of  $\chi(x)$  is generated by  $-\delta/\delta J(x)$ , and direct expansion of the exponential proves

$$e^{-W[\phi]} = e^{-S[\phi]} \left[ \exp \left( -S_{\text{int}} \left[ \phi, -\frac{\delta}{\delta J} \right] \right) Z_\chi[J] \right]_{J=0}.$$

Perturbation theory is obtained either by expanding this operator in powers of  $g$  and performing Gaussian derivatives, or equivalently by using the connected-cumulant expansion

$$W[\phi] = S[\phi] + \langle S_{\text{int}} \rangle_{>} - \frac{1}{2} \langle S_{\text{int}}^2 \rangle_{>,c} + \frac{1}{3!} \langle S_{\text{int}}^3 \rangle_{>,c} + \cdots,$$

where every internal propagator is restricted to  $b\Lambda < |p| < \Lambda$ .

(c) Neglecting wavefunction rescaling, the diagrams relevant to the local three-low-field vertex through order  $g^3$  are the original tree-level cubic vertex and the one-loop triangle made from three  $g\phi\chi^2/2$  vertices. A vertex containing only one high-mode line cannot contribute at zero external momentum: momentum conservation would force that line outside the high shell. Disconnected vacuum factors cancel from  $W = -\ln Z$ .

The connected Wick contraction is

$$\langle \chi^2(x) \chi^2(y) \chi^2(z) \rangle_{>,c} = 8G_{>}(x-y)G_{>}(y-z)G_{>}(z-x).$$

Its coefficient in the third cumulant is

$$\frac{1}{3!} \left(\frac{g}{2}\right)^3 8 = \frac{g^3}{3!}.$$

For external momenta small compared with the shell momenta, the triangle is local, and hence

$$\frac{g_{\text{eff}}}{3!} = \frac{g}{3!} + \frac{g^3}{3!} I_d + O(g^5), \quad I_d = \int_{b\Lambda < |k| < \Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2)^3}.$$

For  $d \neq 6$ ,

$$I_d = \frac{\Omega_{d-1}}{(2\pi)^d} \frac{\Lambda^{d-6} - (b\Lambda)^{d-6}}{d-6}.$$

At  $d = 6$ ,  $\Omega_5 = \pi^3$ , so

$$I_6 = \frac{\pi^3}{(2\pi)^6} \ln \frac{1}{b} = \frac{1}{64\pi^3} \ln \frac{1}{b}.$$

Therefore

$$g_{\text{eff}} = g + \frac{g^3}{64\pi^3} \ln \frac{1}{b} + O(g^5) = g - \frac{g^3}{64\pi^3} \ln b + O(g^5), \quad C = -\frac{1}{64\pi^3}.$$

### S.12.7. The One-Loop Yukawa Beta Function

Work in  $d = 4 - \epsilon$  and introduce

$$\phi_0 = Z_\phi^{1/2} \phi, \quad \psi_0 = Z_\psi^{1/2} \psi, \quad g_0 = \mu^{\epsilon/2} Z_g g.$$

It is useful to write the renormalized interaction and its vertex counterterm as

$$\mathcal{L}_{\text{Yuk}} = -i\mu^{\epsilon/2} g Z_1 \phi \bar{\psi} \psi, \quad Z_1 = Z_g Z_\psi Z_\phi^{1/2}.$$

With Harlow's phases, also used in the prompt, the tree-level Yukawa vertex is  $+g$ .

There are three ultraviolet computations: the fermion self-energy with one internal fermion and one scalar, the scalar self-energy from a closed fermion loop, and the proper Yukawa vertex triangle. At nonexceptional momenta their poles can all be extracted from

$$\mu^\epsilon \int \frac{d^{4-\epsilon} \ell_E}{(2\pi)^{4-\epsilon}} \frac{1}{(\ell_E^2 + \Delta)^2} = \frac{1}{16\pi^2} \left( \frac{2}{\epsilon} + O(1) \right).$$

For the fermion self-energy, Feynman parametrization leaves the pole proportional to

$$2 \int_0^1 dx (1-x) i \not{p} = i \not{p}.$$

For the scalar self-energy, the fermion trace and the two terms obtained after the momentum shift give four times the corresponding pole in the coefficient of  $p^2$ . The required minimal-subtraction field factors are therefore

$$Z_\psi = 1 - \frac{g^2}{16\pi^2\epsilon} + O(g^4), \quad Z_\phi = 1 - \frac{4g^2}{16\pi^2\epsilon} + O(g^4).$$

These values give  $\gamma_\psi = g^2/(32\pi^2)$  and  $\gamma_\phi = g^2/(8\pi^2)$ , a useful normalization check. The divergent part of the one-loop proper vertex, relative to the tree-level rule, is

$$\Gamma_{\text{loop}}^{(3)}|_{\text{UV}} = -g \frac{2g^2}{16\pi^2\epsilon}.$$

It is canceled by

$$Z_1 = 1 + \frac{2g^2}{16\pi^2\epsilon} + O(g^4).$$

Consequently the coupling renormalization includes both the proper vertex and the three external-field square roots:

$$\begin{aligned} Z_g &= Z_1 Z_\psi^{-1} Z_\phi^{-1/2} \\ &= 1 + \frac{5g^2}{16\pi^2\epsilon} + O(g^4), \end{aligned}$$

and hence

$$g_0 = \mu^{\epsilon/2} g \left[ 1 + \frac{5g^2}{16\pi^2\epsilon} + O(g^4) \right].$$

Holding  $g_0$  fixed while differentiating with respect to  $\mu$  gives

$$\beta_g = -\frac{\epsilon}{2}g + \frac{5g^3}{16\pi^2} + O(g^5).$$

Thus in four dimensions

$$\beta_g = \frac{5g^3}{16\pi^2} + O(g^5).$$

The Lagrangian displayed in the source is not, by itself, the most general renormalizable scalar–fermion theory. Unless an additional exact symmetry is imposed, fermion loops require scalar-potential counterterms, including the allowed linear, cubic, and quartic terms. Their couplings do not enter the one-loop pole of the Yukawa vertex or the one-loop wavefunction factors above, so they do not change this one-loop beta function. They must still be included when renormalizing the full theory beyond the restricted question.

### S.12.8. Euler–Heisenberg, Rayleigh Scattering, and Fermi Theory

- (a) Gauge invariance requires photons to occur through  $F_{\mu\nu}$ . There is no nonzero Abelian scalar cubic in  $F$ , so the first self-interactions contain four field strengths:

$$\mathcal{O}_1 = (F_{\mu\nu}F^{\mu\nu})^2, \quad \mathcal{O}_2 = (F_{\mu\nu}\tilde{F}^{\mu\nu})^2.$$

Both have dimension eight. Integrating out an electron gives

$$\Delta\mathcal{L} = \frac{\alpha^2}{m_e^4}(c_1\mathcal{O}_1 + c_2\mathcal{O}_2) + \cdots,$$



where  $c_i$  are numerical one-loop constants. Each external field strength supplies one power of photon energy, so

$$\mathcal{A}_{\gamma\gamma\rightarrow\gamma\gamma} = O\left(\alpha^2 \frac{\omega^4}{m_e^4}\right).$$

Two-body phase space contributes  $1/\omega^2$ , yielding

$$\sigma = O\left(\alpha^4 \frac{\omega^6}{m_e^8}\right).$$

The energy power is fixed without evaluating the electron loop.

- (b) For a slowly moving neutral spinless atom, the leading local interactions are its electric and magnetic polarizabilities,

$$\Delta\mathcal{L} = h^\dagger h \left( c_E a_0^3 \mathbf{E}^2 + c_B a_0^3 \mathbf{B}^2 \right).$$

The coefficient has dimension  $-3$ , and the atomic size supplies that scale. Each field in a radiation wave is proportional to  $\omega$ , so

$$\mathcal{A} = O(a_0^3 \omega^2).$$

The nonrelativistic scattering normalization then gives

$$\sigma = O(a_0^6 \omega^4).$$

A frequency-independent coupling would require a term linear in the potential or a net electric charge. Neutrality removes the charge monopole, while gauge invariance forbids a local interaction built directly from  $A_\mu$ . This is why the first response is a polarizability and why shorter-wavelength light is scattered more strongly.

- (c) For conserved currents the longitudinal propagator does not contribute. The relevant factor expands as

$$\frac{g^2}{M^2 - q^2} = \frac{g^2}{M^2} \left( 1 + \frac{q^2}{M^2} + O(q^4/M^4) \right),$$

with signs changed in the Euclidean continuation. Matching gives

$$\Delta\mathcal{L}_{\text{eff}} = -\frac{g^2}{2M^2} J_\mu J^\mu - \frac{g^2}{2M^4} J_\mu \square J^\mu + \dots$$

The current has dimension three, so these are dimension-six and dimension-eight operators. Their matrix elements differ by  $O(p^2/M^2)$ .

The coefficient  $g^2/M^2$  may be conventionally written as a different scale, such as  $G_F \sim 1/v^2$ . The derivative series nevertheless knows about the propagator singularity and loses convergence at  $p^2 \sim M^2$ , not necessarily at  $p^2 \sim v^2$ .

### S.12.9. Loop Power Counting and Expansion by Regions

(a) Write each interaction as

$$\frac{c_i}{M^{d_i-4}} \mathcal{O}_{d_i}.$$

A graph with  $V_i$  insertions contains the fixed factor

$$M^{-P}, \quad P = \sum_i V_i (d_i - 4).$$

After the heavy fields have been removed, loop denominators contain only light masses and external momenta. Dimensional regularization introduces the book-keeping scale  $\mu$ , but no physical positive power of  $M$ . Consequently a loop cannot lower  $P$ . Its ultraviolet pole is a polynomial in light momenta and masses multiplying the same  $M^{-P}$ ; that polynomial is canceled by an operator at the corresponding EFT order.

For fixed  $P$ , there are only finitely many nonnegative multiplicities  $V_i$  and finitely many local operators of bounded dimension, once field content and symmetries are fixed. This establishes order-by-order counterterm closure even though the complete effective Lagrangian contains infinitely many terms.

(b) Partial fractions reduce the integral to tadpoles:

$$I = \frac{A(m^2) - A(M^2)}{M^2 - m^2}, \quad A_R(a) = \frac{a}{16\pi^2} \left( \ln \frac{a}{\mu^2} - 1 \right)$$

in the stated  $\overline{\text{MS}}$  convention. Therefore

$$I_R = \frac{1}{16\pi^2} \frac{m^2(\ln(m^2/\mu^2) - 1) - M^2(\ln(M^2/\mu^2) - 1)}{M^2 - m^2}.$$

With  $r = m^2/M^2$ ,

$$I_R = \frac{1}{16\pi^2} \left[ 1 - \ln \frac{M^2}{\mu^2} + r \ln \frac{m^2}{M^2} + O(r^2 \ln r) \right].$$

Expanding the heavy propagator gives

$$\frac{1}{k^2 + M^2} = \frac{1}{M^2} \sum_{n \geq 0} \left( -\frac{k^2}{M^2} \right)^n.$$

The resulting light tadpoles generate every  $r^n \ln m^2$  term. They cannot generate the analytic constants and  $\ln M^2$  pieces; those are the Wilson coefficients obtained by matching the hard region.

(c) For

$$I_\alpha = \mu^{2\varepsilon} \int d^d k (k^2)^{-\alpha},$$

the change  $k \mapsto \lambda k$  gives  $I_\alpha = \lambda^{d-2\alpha} I_\alpha$ . With no scale that could compensate this factor, analytic continuation sets  $I_\alpha = 0$ . In the logarithmic case the same statement may be displayed as

$$\int \frac{d^d k}{(k^2)^2} \propto \frac{1}{\varepsilon_{\text{UV}}} - \frac{1}{\varepsilon_{\text{IR}}} = 0.$$

The equality does not say that ultraviolet and infrared behavior are separately absent; it says they cancel in the scaleless regulated integral.

In matching, the full theory and EFT have identical soft behavior. If external momenta and light masses are set to zero, common soft graphs often become scaleless and vanish. The remaining full-theory hard part is then the Wilson coefficient. Keeping track of the UV versus IR interpretation of poles ensures that the EFT counterterm receives the correct ultraviolet information.

### S.12.10. Heavy-Vector Matching and Infrared Cancellation

(a) At leading order in derivatives, write

$$\mathcal{L}_X = \frac{1}{2} M^2 X_\mu X^\mu + g_X X_\mu (J_1^\mu + J_2^\mu).$$

The field equation gives

$$X_\mu = -\frac{g_X}{M^2} (J_{1\mu} + J_{2\mu}) + O(M^{-4}),$$

and substitution yields

$$\Delta\mathcal{L}_{\text{eff}} = -\frac{g_X^2}{2M^2} J_1^2 - \frac{g_X^2}{M^2} J_1 \cdot J_2 - \frac{g_X^2}{2M^2} J_2^2 + O(M^{-4}).$$

The cross coefficient is twice either diagonal coefficient because it comes from expanding  $(J_1 + J_2)^2$ . Expanding the full exchange kernel  $g_X^2/(q^2 - M^2)$  at small  $q^2$  fixes the same relative sign. An overall sign changes if the quadratic-term convention changes, but the exchange amplitude and matched contact term always agree.

(b) Single heavy exchange naturally gives

$$C_6^{\text{tree}} \sim \frac{g_1 g_2}{M^2}, \quad C_8^{\text{tree}} \sim \frac{g_T^2}{M^4}.$$

If symmetry or topology forbids tree generation, a one-loop coefficient is instead

$$C_6^{\text{loop}} \sim \frac{g_L^n}{16\pi^2 M^2}.$$

Their contributions to a dimensionless amplitude scale as

$$\mathcal{A}_6^{\text{loop}} \sim \frac{g_L^n}{16\pi^2} \frac{E^2}{M^2}, \quad \mathcal{A}_8^{\text{tree}} \sim g_T^2 \frac{E^4}{M^4}.$$

Thus the loop-generated dimension-six term dominates when

$$\boxed{\frac{E}{M} < \left( \frac{g_L^n}{16\pi^2 g_T^2} \right)^{1/2}}.$$

Canonical dimension alone therefore does not give a complete numerical ordering; coupling and loop counting must accompany it.

- (c) Write the low-energy expansion of the renormalized full amplitude as

$$\mathcal{A}_{\text{full}} = H(M, \mu) + S(m, \mu),$$

where  $S$  contains the light logarithm. The EFT has

$$\mathcal{A}_{\text{EFT}} = C(\mu) + S(m, \mu),$$

because it contains the same light particles, interactions, and infrared regulator. Equality of amplitudes gives

$$\boxed{C(\mu) = \mathcal{A}_{\text{full}} - \mathcal{A}_{\text{EFT}}^{\text{loop}} = H(M, \mu)}.$$

For example,

$$[a \ln(M^2/\mu^2) + b \ln(m^2/\mu^2) + c] - [b \ln(m^2/\mu^2) + c_{\text{soft}}] = a \ln(M^2/\mu^2) + c - c_{\text{soft}}.$$

The light logarithm and any infrared pole cancel. A failure of this cancellation signals that the EFT omitted a light mode or used inconsistent infrared conventions, not that a Wilson coefficient can depend on long-distance physics.

### S.12.11. Dipoles, Anapoles, and Threshold Decoupling

- (a) A gauge-invariant fermion bilinear has dimension three and  $F_{\mu\nu}$  has dimension two. Lorentz indices require the antisymmetric tensor bilinear. Hermiticity leaves

$$\boxed{\mathcal{O}_M = \bar{\psi} \sigma^{\mu\nu} \psi F_{\mu\nu}, \quad \mathcal{O}_E = i \bar{\psi} \sigma^{\mu\nu} \gamma_5 \psi F_{\mu\nu}}.$$

They are the magnetic and electric dipoles. The factor  $i$  in  $\mathcal{O}_E$  makes its coefficient real in a Hermitian Lagrangian under the stated gamma-matrix convention. Candidate derivative bilinears such as  $\bar{\psi} D^2 \psi$  reduce by

$$(i \not{D})^2 = -D^2 - \frac{e}{2} \sigma^{\mu\nu} F_{\mu\nu}$$

and the Dirac equation to a dipole plus lower-dimensional mass terms. Total derivatives add nothing. No pure-photon dimension-five scalar exists. Both dipoles are therefore required when discrete symmetries are not imposed; P or T would eliminate the electric one.

(b) The leading Maxwell equation is

$$\partial_\nu F^{\nu\mu} = eJ_V^\mu, \quad J_V^\mu = \bar{\psi}\gamma^\mu\psi.$$

Consequently, modulo higher-order corrections,

$$\begin{aligned} J_{V\mu}\partial_\nu F^{\nu\mu} &\doteq eJ_V^2, \\ J_{A\mu}\partial_\nu F^{\nu\mu} &\doteq eJ_A \cdot J_V, \\ (\partial_\nu F^{\nu\mu})^2 &\doteq e^2 J_V^2, \end{aligned}$$

where  $\doteq$  denotes equality in the on-shell operator quotient. Equivalently, local redefinitions of  $A_\mu$  move coefficients between the derivative-photon and four-fermion operators.

An off-shell Green-function basis may keep the three operators on the left; then their Wilson coefficients specify contact terms in that convention. An on-shell basis may keep the currents on the right. Both give identical  $S$ -matrix elements, but retaining both related sets introduces spurious parameters and a singular anomalous-dimension matrix.

(c) With the subtraction  $\pi(0) = 0$ , a heavy Dirac fermion gives

$$\pi(q^2) = \frac{e^2}{2\pi^2} \int_0^1 dx x(1-x) \ln \left[ 1 + \frac{x(1-x)q^2}{M^2} \right].$$

For  $|q^2| \ll M^2$ ,

$$\pi(q^2) = \frac{e^2}{60\pi^2} \frac{q^2}{M^2} + O(q^4/M^4).$$

This is reproduced by a local  $F_{\mu\nu}\Box F^{\mu\nu}$  operator (with coefficient  $e^2/(240\pi^2 M^2)$ , up to the metric and integration- by-parts sign convention).

In a mass-dependent charge definition the constant heavy contribution is automatically subtracted at low momentum. In  $\overline{\text{MS}}$ , the beta function does not know that  $\mu < M$ ; the heavy fermion would continue to run the charge. One must match the full and light-flavor theories near  $\mu = M$ , absorb the finite threshold term into the low-energy charge, and then run with only light fields.

### S.12.12. Asymptotic Regions, the Hierarchy Problem, and Shift Symmetry

(a) Partial fractions give the exact result

$$J(m, M) = \frac{\ln(M/m)}{M - m}.$$

Choose  $m \ll \nu \ll M$ . At leading order, the soft part is

$$J_s = \int_0^\nu \frac{dk}{M(k+m)} = \frac{1}{M} \ln \frac{\nu+m}{m} = \frac{1}{M} \ln \frac{\nu}{m} + O(m/\nu),$$

while the hard part is

$$J_h = \int_{\nu}^{\infty} \frac{dk}{k(k+M)} = \frac{1}{M} \ln \frac{\nu+M}{\nu} = \frac{1}{M} \ln \frac{M}{\nu} + O(\nu/M).$$

Their sum is  $M^{-1} \ln(M/m)$ ; the separator cancels. Expanding the integrands to further orders reconstructs

$$J = \frac{1}{M} \left( 1 + \frac{m}{M} + \dots \right) \ln \frac{M}{m}.$$

The nonanalytic  $\ln m$  originates in the soft region, while the hard region supplies the matching dependence on  $M$ .

- (b) The heavy tadpole has symmetry factor  $1/2$ . In  $\overline{\text{MS}}$ , with the interaction normalization in the question,

$$\boxed{\delta m^2(\mu) = \frac{\lambda_{\ell h} M^2}{32\pi^2} \left( \ln \frac{M^2}{\mu^2} - 1 \right)}$$

up to the overall self-energy sign convention. At the natural matching point  $\mu = M$ , its magnitude is

$$|\delta m^2(M)| = \frac{|\lambda_{\ell h}| M^2}{32\pi^2}.$$

The low-energy mass is the sum of the matched light parameter and this threshold correction. Maintaining  $m_{\text{phys}}^2 \ll M^2$  requires a fractional cancellation of order

$$\Delta^{-1} \sim \frac{32\pi^2 m_{\text{phys}}^2}{|\lambda_{\ell h}| M^2}.$$

This is a finite threshold sensitivity, not an artifact of a momentum cutoff. A symmetry relating the light scalar to protected degrees of freedom could force the  $M^2$  term to cancel; absent such a symmetry it is generic.

- (c) An invariant local operator cannot contain an undifferentiated  $\phi$ . The action begins with

$$-\frac{1}{2}(\partial\phi)^2 + \frac{c_4}{M^4}(\partial\phi)^4 + \dots,$$

where higher terms contain derivatives on every field, modulo integrations by parts. The corresponding Ward identity implies an Adler zero: taking one external scalar momentum  $q$  soft makes the amplitude  $O(q)$ . A potential and in particular  $m^2\phi^2$  violate the symmetry, so symmetric regularization cannot generate a mass counterterm.

If a spurion  $\epsilon$  is assigned a transformation that makes  $\epsilon V(\phi)$  invariant, every symmetry-breaking counterterm must carry at least one  $\epsilon$ . Thus

$$\delta m^2 = \epsilon M^2 \cdot \{\text{couplings and loop factors}\},$$

or a higher power if the spurion charges require it. The limit  $\epsilon \rightarrow 0$  restores the exact symmetry, making a small scalar mass radiatively stable.

### S.12.13. Symmetry Is Attractive: Two-Scalar Renormalization

It is convenient to define

$$h \equiv \frac{\lambda}{3}.$$

Then the completely symmetric quartic-coupling tensor has nonzero components

$$\lambda_{1111} = \lambda_{2222} = g, \quad \lambda_{1122} = \lambda_{1212} = \cdots = h,$$

because the six permutations of 1122 turn  $-h\phi_1^2\phi_2^2/4$  into the cross-interaction printed in the question.

(a) At  $\lambda = g$ ,

$$\mathcal{L}_{\text{int}} = -\frac{g}{4!} (\phi_1^4 + 2\phi_1^2\phi_2^2 + \phi_2^4) = -\frac{g}{4!} (\phi_1^2 + \phi_2^2)^2.$$

This depends only on the length of the vector  $(\phi_1, \phi_2)$ , and is therefore  $O(2)$ -invariant.

(b) For arbitrary  $g, \lambda$ , the action is invariant under the two independent reflections

$$\phi_1 \mapsto -\phi_1, \quad \phi_2 \mapsto -\phi_2,$$

as well as under  $\phi_1 \leftrightarrow \phi_2$ . Both  $\phi_1\phi_2$  and  $\phi_1\Box\phi_2$  are odd under either one of the individual reflections. A regulator that respects these symmetries cannot generate either counterterm. The exchange symmetry also keeps the two diagonal masses and wavefunction factors equal.

(c) The symmetry-allowed counterterm Lagrangian may be written

$$\begin{aligned} \mathcal{L}_{\text{ct}} = & -\frac{\delta Z}{2} \sum_{a=1}^2 \partial_\mu \phi_a \partial^\mu \phi_a - \frac{\delta m^2}{2} \sum_{a=1}^2 \phi_a^2 \\ & - \frac{\delta g}{4!} (\phi_1^4 + \phi_2^4) - \frac{2\delta\lambda}{4!} \phi_1^2 \phi_2^2. \end{aligned}$$

There is no momentum-dependent one-loop self-energy in a quartic scalar theory, so  $\delta Z = 0$  at this order. With a Euclidean cutoff, the two equal zero-momentum self-energies are

$$\Sigma_{11}(0) = \Sigma_{22}(0) = \frac{g+h}{2} I_\Lambda, \quad I_\Lambda = \int_{|k|<\Lambda} \frac{d^4k}{(2\pi)^4} \frac{1}{k^2} = \frac{\Lambda^2}{16\pi^2}.$$

The massless pole condition fixes  $\delta m^2 = -\Sigma_{aa}(0)$  when the inverse propagator is written as  $p^2 + \delta m^2 + \Sigma_{aa}(p^2)$ . No off-diagonal self-energy occurs.

For the four-point functions, define the dimensionally regulated massless bubble  $B(q^2)$  so that

$$B(q^2)|_{\text{UV}} = \frac{2}{16\pi^2\epsilon}.$$

The two independent 1PI amplitudes are

$$\begin{aligned}\Gamma_{1111} &= -ig + \frac{i}{2}(g^2 + h^2)[B(s) + B(t) + B(u)] - i\delta g, \\ \Gamma_{1122} &= -ih + \frac{i}{2}[2ghB(s) + 2h^2B(t) + 2h^2B(u)] - i\delta h,\end{aligned}$$

where  $\delta h = \delta\lambda/3$ , and the assignment of  $s, t, u$  in the second line follows the displayed external ordering. These formulas also give an explicit diagram census: three bubbles for four identical external fields; for 1122, the  $s$ -channel has either internal species and the crossed channels contain the mixed vertex twice.

At a chosen nonexceptional subtraction point, momentum subtraction is implemented by

$$\begin{aligned}\delta g_{\text{MOM}} &= \frac{1}{2}(g^2 + h^2)[B(s_0) + B(t_0) + B(u_0)], \\ \delta h_{\text{MOM}} &= \frac{1}{2}[2ghB(s_0) + 2h^2B(t_0) + 2h^2B(u_0)].\end{aligned}$$

These choices impose  $\Gamma_{1111}(s_0, t_0, u_0) = -ig$  and  $\Gamma_{1122}(s_0, t_0, u_0) = -ih$ . Their pole parts, equivalently the minimal-subtraction counterterms, are

$$\boxed{\delta g = \frac{3(g^2 + h^2)}{16\pi^2\epsilon}, \quad \delta h = \frac{2gh + 4h^2}{16\pi^2\epsilon}.}$$

Thus every one-loop divergence is canceled by a term already present in the symmetry-closed Lagrangian.

(d) The general one-loop quartic-tensor formula

$$16\pi^2\beta_{ijkl} = \lambda_{ijmn}\lambda_{mnkl} + \lambda_{ikmn}\lambda_{mnjl} + \lambda_{ilmn}\lambda_{mnjk}$$

gives, in four dimensions,

$$\boxed{\beta_g = \frac{3g^2 + \lambda^2/3}{16\pi^2}, \quad \beta_\lambda = \frac{2g\lambda + 4\lambda^2/3}{16\pi^2}.}$$

The equality  $\lambda = g$  is preserved: both beta functions become  $10g^2/(48\pi^2)$  there, as required by  $O(2)$ .

For  $r = \lambda/g$ ,

$$\boxed{\mu \frac{dr}{d\mu} = \frac{gr}{48\pi^2}(1-r)(r-3).}$$

Let  $\ell = \ln(\mu_0/\mu)$  increase toward the infrared. Then

$$\frac{dr}{d\ell} = \frac{gr}{48\pi^2}(r-1)(r-3).$$

For positive couplings,  $0 < r < 1$  gives  $dr/d\ell > 0$ , while  $1 < r < 3$  gives  $dr/d\ell < 0$ . Hence every trajectory in  $0 < r < 3$ , apart from the decoupled endpoint  $r = 0$ , is attracted to

$$\boxed{r = 1}$$



at low energy. The second fixed ratio  $r = 3$  is the boundary of this basin.

This also states precisely how to read the source's large-cutoff formulation: integrating from a regulator scale  $\Lambda$  down to fixed physical momenta produces an RG time  $\ln(\Lambda/\mu)$ , so increasing the scale separation drives the low-energy ratio toward one. With the standard convention  $\beta_r = \mu, dr/d\mu$ , the symmetric direction is infrared-attractive, not ultraviolet-attractive; solving the matching relation backward for bare parameters reverses the direction. Finally, positive four-dimensional scalar theory has a Landau pole, so a literal interacting  $\Lambda \rightarrow \infty$  limit lies outside one-loop perturbation theory. The controlled claim is the logarithmic low-energy attraction demonstrated above.

#### S.12.14. Scale Symmetry, the Dilatation Current, and Its Generator

- (a) Set  $y = \lambda x$ . The kinetic term transforms with the overall factor  $\lambda^{2\Delta+2-D}$ , so invariance requires

$$\Delta = \frac{D-2}{2}.$$

The interaction transforms with  $\lambda^{p\Delta-D}$ , and hence

$$p = \frac{D}{\Delta} = \frac{2D}{D-2}.$$

The same  $\Delta$  follows from requiring the kinetic action to be dimensionless:  $[d^D x] = -D$  and  $[\partial_\mu] = 1$  give  $[\phi] = (D-2)/2$ .

- (b) For  $\lambda = 1 + \varepsilon$ ,

$$\delta\phi(x) = \varepsilon (x^\nu \partial_\nu + \Delta) \phi(x).$$

With the critical power found above, use the improved tensor

$$\begin{aligned} T_{\text{imp}}^{\mu\nu} &= \partial^\mu \phi \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L} \\ &+ \xi (\eta^{\mu\nu} \square - \partial^\mu \partial^\nu) \phi^2, \quad \xi = \frac{D-2}{4(D-1)}. \end{aligned}$$

The improvement is identically conserved. Combining its trace with the field equation gives

$$T_{\text{imp}\mu}^\mu = 0 \quad \text{on shell.}$$

Therefore

$$j_D^\mu = x_\nu T_{\text{imp}}^{\mu\nu}, \quad \partial_\mu j_D^\mu = T_{\text{imp}\mu}^\mu = 0,$$

and the time-independent charge is  $\mathcal{D} = \int d^{D-1}x j_D^0$ .

- (c) At equal time, use  $[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] = i\delta^{D-1}(\mathbf{x} - \mathbf{y})$ . Writing the charge in canonical variables and integrating spatial derivatives by parts gives

$$[\mathcal{D}, \phi(x)] = -i (x^\nu \partial_\nu + \Delta) \phi(x)$$

in the convention  $U(\varepsilon) = e^{-i\varepsilon\mathcal{D}}$ . Consequently

$$U(\varepsilon)\phi(x)U(\varepsilon)^{-1} = \phi(x) + \varepsilon(x \cdot \partial + \Delta)\phi(x) + O(\varepsilon^2),$$

which is the required infinitesimal dilation.

For a general first-derivative theory the canonical Noether current has the form

$$j_D^\mu = x_\nu T^{\mu\nu} - V^\mu, \quad T^\mu{}_\mu = \partial_\mu V^\mu$$

on shell. If the virial current  $V^\mu$  is itself a divergence that can be removed by an improvement, one may choose a traceless stress tensor and write simply  $j_D^\mu = x_\nu T_{\text{imp}}^{\mu\nu}$ ; otherwise the virial term is essential.

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## 13 Infrared Effects

In the study of radiative corrections a special role is played by those corrections due to ‘soft’ photons: photons whose energy and momentum are much less than the masses and energies characteristic of the process in question. Not only are these corrections often so large that they must be summed to all orders of perturbation theory; they are so simple that this summation is not difficult. The contribution of photons of infinitely long wavelength takes the form of divergent integrals, but as we shall see these ‘infrared divergences’ all cancel [1].

In most of this chapter we will deal with photons interacting with charged particles of arbitrary type and spin, including particles like atomic nuclei that have strong as well as electromagnetic interactions. But it is not difficult to adapt the calculations presented here to the infrared effects of other massless particles, such as the gluons of quantum chromodynamics. In Section 13.4 we shall explicitly consider very general theories of massless particles, and will show the cancellation of infrared divergences on general grounds.

After these generalities, we shall return to photons, and take up two topics of practical importance: the scattering of soft photons by charged particles with arbitrary non-electromagnetic interactions and arbitrary spin, and the treatment of heavy charged particles like atomic nuclei as a source of an external electromagnetic field.

### 13.1 Soft Photon Amplitudes

In this section we shall derive a universal formula that gives the amplitude for emission of any number of very-low-energy photons in a process  $\alpha \rightarrow \beta$  involving any number of higher-energy charged particles of any types.

Let us start with the amplitude for emission of just one soft photon. If we attach the soft photon line with outgoing momentum  $q$  and polarization index  $\mu$  to an outgoing charged-particle line that leaves some connected Feynman diagram for the process  $\alpha \rightarrow \beta$ , as in Figure 13.1(a), then we must multiply the  $S$ -matrix element for  $\alpha \rightarrow \beta$  with an additional charged-particle propagator carrying the momentum  $p + q$  that the charged particle had before emitting the photon, together with the contribution of the new charged-particle-photon vertex. For charged particles of spin zero, mass  $m$ , and charge  $+e$ , these factors are

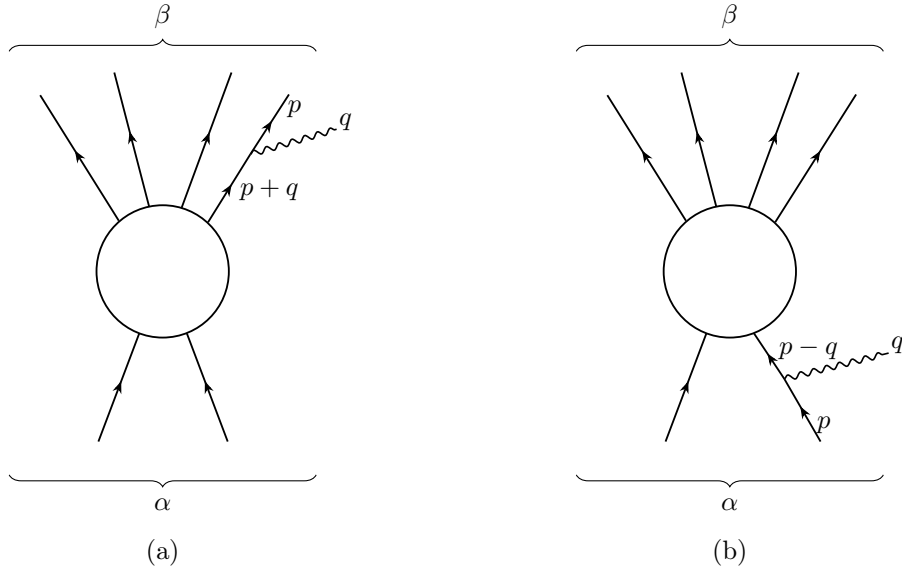
$$\left[ i(2\pi)^4 e(2p^\mu + q^\mu) \right] \left[ \frac{-i}{(2\pi)^4} \frac{1}{(p+q)^2 + m^2 - i\epsilon} \right],$$

which in the limit  $q \rightarrow 0$  becomes

$$\frac{ep^\mu}{p \cdot q - i\epsilon}. \quad (13.1.1)$$

(We are freely redefining the scale of the positive infinitesimal  $\epsilon$ , being careful only to keep track of its sign.) This result is actually true for charged particles of any spin. For instance, for a particle of spin  $\frac{1}{2}$  and charge  $+e$ , we must replace the coefficient function  $\bar{u}(p, \sigma)$  for the outgoing charged particle with

$$\bar{u}(p, \sigma) \left[ -i(2\pi)^4 e \gamma^\mu \right] \left[ \frac{-i}{(2\pi)^4} \frac{-i(\not{p} + \not{q}) + m}{(p+q)^2 + m^2 - i\epsilon} \right].$$



**Figure 13.1.** Dominant graphs for the emission of soft photons in an arbitrary process  $\alpha \rightarrow \beta$ . Straight lines are particles in the states  $\alpha$  and  $\beta$  (including possible hard photons); wavy lines are soft photons.

In the limit  $q \rightarrow 0$  the numerator of the propagator is given by a sum of dyads:

$$-i\not{p} + m = 2p^0 \sum_{\sigma'} u(p, \sigma') \bar{u}(p, \sigma'),$$

so we have a sum of equal-momentum matrix elements of  $\gamma^\mu$ , given by

$$\bar{u}(p, \sigma) \gamma^\mu u(p, \sigma') = -i\delta_{\sigma, \sigma'} p^\mu / p^0,$$

and again the effect again is to multiply the matrix element for the process  $\alpha \rightarrow \beta$  by the factor (13.1.1). More generally, for any spin in the limit  $q \rightarrow 0$  the four-momentum  $p + q$  of the new internal charged particle line approaches the mass shell, so the numerator of the propagator approaches a sum of dyads of coefficient functions which convert the new vertex matrix into a factor proportional to  $p^\mu$  and a unit matrix in helicity indices, leading again to the factor (13.1.1). Furthermore, as we saw in Chapter 10, higher-order corrections do not affect either the residue of the mass-shell poles in the propagators or the matrix element of the electric current between states of the same particle at equal momentum, so (13.1.1) gives the correct factor associated with the emission of a soft photon from an outgoing charged-particle line to all orders of perturbation theory.

The same reasoning applies to a photon emitted from an incoming charged-particle line of the process  $\alpha \rightarrow \beta$ , except that after the incoming particle emits a photon of four-momentum  $q$  the charged-particle line has

four-momentum  $p - q$ , so in place of (13.1.1) we find a factor

$$\frac{ep^\mu}{-p \cdot q - i\epsilon}. \quad (13.1.2)$$

The photon can also, of course, be emitted from an internal line of the process  $\alpha \rightarrow \beta$ , but in this case there is no factor that goes as  $(p \cdot q)^{-1}$  for  $q \rightarrow 0$ . The amplitude  $M_{\beta\alpha}^\mu(q)$  (the  $S$ -matrix without the energy-momentum conservation delta function) for emitting a single soft photon with four-momentum  $q$  and polarization index  $\mu$  in the process  $\alpha \rightarrow \beta$  is therefore given in the limit  $q \rightarrow 0$  by multiplying the matrix element  $M_{\beta\alpha}$  for  $\alpha \rightarrow \beta$  with a sum of terms like (13.1.1) and (13.1.2), one for each outgoing or incoming charged particle:

$$M_{\beta\alpha}^\mu(q) \longrightarrow M_{\beta\alpha} \sum_n \frac{\eta_n e_n p_n^\mu}{p_n \cdot q - i\eta_n \epsilon}, \quad (13.1.3)$$

where  $p_n$  and  $+e_n$  are the four-momentum and charge of the  $n$ th particle in the initial and final states, and  $\eta_n$  is a sign factor with the value  $+1$  for particles in the final state  $\beta$  and  $-1$  for particles in the initial state  $\alpha$ .

Before going on to consider the emission of more than one soft photon, it is worth mentioning an important feature [2] of the formula (13.1.3). To calculate the amplitude for emission of a photon of definite helicity, we must contract this expression with the corresponding photon polarization vector  $e_\mu(q, \pm)$ . But as we saw in Section 5.9,  $e^\mu(q, \pm)$  is not a four-vector; under a Lorentz transformation  $\Lambda^\mu{}_\nu$ , the polarization vector is transformed into  $\Lambda^\mu{}_\nu e^\nu(q, \pm)$  plus a term proportional to  $q^\mu$ . In order for this last term not to spoil Lorentz invariance, it is therefore necessary that  $M_{\beta\alpha}^\mu(q)$  should vanish when contracted with  $q_\mu$ . But for  $q \rightarrow 0$ , (13.1.3) gives

$$q_\mu M_{\beta\alpha}^\mu(q) \longrightarrow M_{\beta\alpha} \sum_n \eta_n e_n. \quad (13.1.4)$$

The coefficient of  $M_{\beta\alpha}$  on the right-hand side is just the total charge in the final state minus the total charge in the initial state, so the condition that it vanishes is just the condition that charge is conserved. Thus without any independent assumptions about gauge invariance, we see that *for particles of spin one and mass zero, Lorentz invariance requires the conservation of whatever coupling constant like electric charge governs the interaction of these particles at low energies.*

Incidentally, the amplitude for emitting a soft graviton of four-momentum  $q$  and tensor indices  $\mu, \nu$  in a process  $\alpha \rightarrow \beta$  is given by a formula [3] analogous to (13.1.3):

$$M_{\beta\alpha}^{\mu\nu}(q) \longrightarrow M_{\beta\alpha} \sum_n \frac{\eta_n f_n p_n^\mu p_n^\nu}{p_n \cdot q - i\eta_n \epsilon}, \quad (13.1.5)$$

where  $f_n$  is the coupling constant of the soft graviton to particles of type  $n$ . Lorentz invariance here requires that this vanish when contracted with  $q_\mu$ . But

$$q_\mu M_{\beta\alpha}^{\mu\nu}(q) \longrightarrow M_{\beta\alpha} \sum_n \eta_n f_n p_n^\nu, \quad (13.1.6)$$

so the sum  $\sum_n f_n p_n^\nu$  is conserved. However, the only linear combination of the four-momenta that can be conserved without forbidding all non-trivial scattering processes is the total four-momentum, so in order for (13.1.6) to vanish, the  $f_n$  must all be equal. (The common value of all  $f_n$  may be identified as  $\sqrt{8\pi G_N}$ , where  $G_N$  is Newton's constant of gravitation.) Thus

Lorentz invariance requires the result that low-energy massless particles of spin two couple in the same way to all forms of energy and momentum. This goes a long way toward showing that Einstein's principle of equivalence is a necessary consequence of Lorentz invariance as applied to massless particles of spin two. Likewise, the amplitude for emitting a soft massless particle of four-momentum  $q$  and spin  $j \geq 3$  in a process  $\alpha \rightarrow \beta$  is of the form

$$M_{\beta\alpha}^{\mu\nu\rho\cdots}(q) \longrightarrow M_{\beta\alpha} \sum_n \frac{\eta_n g_n p_n^\mu p_n^\nu p_n^\rho \cdots}{p_n \cdot q - i\eta_n \epsilon}.$$

Lorentz invariance here requires that the sum  $\sum_n g_n p_n^\nu p_n^\rho \cdots$  must be conserved. But no such quantity can be conserved without prohibiting all non-trivial scattering processes, so the  $g_n$  must all vanish. Massless particles of spin  $j \geq 3$  may exist, but they cannot have couplings that survive in the limit of low energy, and in particular they cannot mediate inverse square law forces.

Now let us consider the emission of two soft photons. The contribution to the matrix element from a graph in which the two photons are emitted from different external lines of the process  $\alpha \rightarrow \beta$  is given by multiplying the matrix element for  $\alpha \rightarrow \beta$  by a product of factors like (13.1.1) or (13.1.2). Perhaps surprisingly, the same is true even if the two photons are emitted from the *same* external line. For example, if photon 1 is emitted from an external line of charge  $+e$  and energy-momentum four-vector  $p$  after photon 2 we get a factor

$$\left[ \frac{\eta e p^{\mu_1}}{p \cdot q_1 - i\eta \epsilon} \right] \left[ \frac{\eta e p^{\mu_2}}{p \cdot (q_2 + q_1) - i\eta \epsilon} \right],$$

while if photon 2 is emitted after photon 1 the factor is

$$\left[ \frac{\eta e p^{\mu_2}}{p \cdot q_2 - i\eta \epsilon} \right] \left[ \frac{\eta e p^{\mu_1}}{p \cdot (q_1 + q_2) - i\eta \epsilon} \right].$$

(See Figure 13.2. Again,  $\eta$  is  $+1$  or  $-1$  according to whether the charged-particle line is outgoing or incoming.) These two factors add up to

$$\left[ \frac{\eta e p^{\mu_1}}{p \cdot q_1 - i\eta \epsilon} \right] \left[ \frac{\eta e p^{\mu_2}}{p \cdot q_2 - i\eta \epsilon} \right],$$

which is just a product of the same factors encountered in the emission of a single photon.

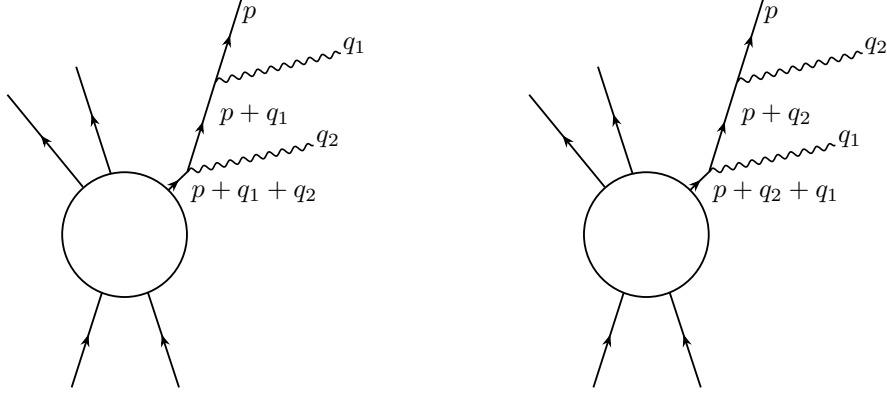
More generally, in emitting an arbitrary number of photons from a single external line we encounter a sum of the form<sup>142</sup>

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<sup>142</sup>This identity may be proved by mathematical induction. We have already seen that it is true for two photons. Suppose that it is true for  $N - 1$  photons. For  $N$  photons we may then write the sum over permutations as a sum over the choice of the first photon to be emitted together with a sum over permutations of the remaining photons:

$$\begin{aligned} & [p \cdot q_1 - i\eta \epsilon]^{-1} [p \cdot (q_1 + q_2) - i\eta \epsilon]^{-1} \cdots [p \cdot (q_1 + q_2 + \cdots + q_N) - i\eta \epsilon]^{-1} + \text{permutations} \\ &= \sum_{r=1}^N \left[ p \cdot \left( \sum_{s=1}^N q_s \right) - i\eta \epsilon \right]^{-1} \prod_{s \neq r} [p \cdot q_s - i\eta \epsilon]^{-1} \\ &= \sum_{r=1}^N \left[ p \cdot \left( \sum_{s=1}^N q_s \right) - i\eta \epsilon \right]^{-1} [p \cdot q_r - i\eta \epsilon] \prod_{s=1}^N [p \cdot q_s - i\eta \epsilon]^{-1} = \prod_{s=1}^N [p \cdot q_s - i\eta \epsilon]^{-1}, \end{aligned}$$

as was to be proved.



**Figure 13.2.** Graphs for the emission of two soft photons from the same outgoing charged particle. Straight lines are hard particles; wavy lines are soft photons.

$$\begin{aligned}
& [p \cdot q_1 - i\eta\epsilon]^{-1} [p \cdot (q_1 + q_2) - i\eta\epsilon]^{-1} [p \cdot (q_1 + q_2 + q_3) - i\eta\epsilon]^{-1} \dots \\
& \quad + \text{permutations} \\
& = [p \cdot q_1 - i\eta\epsilon]^{-1} [p \cdot q_2 - i\eta\epsilon]^{-1} [p \cdot q_3 - i\eta\epsilon]^{-1} \dots .
\end{aligned} \tag{13.1.7}$$

It follows then that the amplitude  $M_{\beta\alpha}^{\mu_1 \dots \mu_N}(q_1 \dots q_N)$  for emitting  $N$  very soft photons with polarization indices  $\mu_1, \dots, \mu_N$  and four-momenta  $q_1, \dots, q_N$  in the process  $\alpha \rightarrow \beta$  is given in the limit  $q \rightarrow 0$  by multiplying the matrix element  $M_{\beta\alpha}$  for  $\alpha \rightarrow \beta$  by a product of factors like that in (13.1.3), one for each photon:

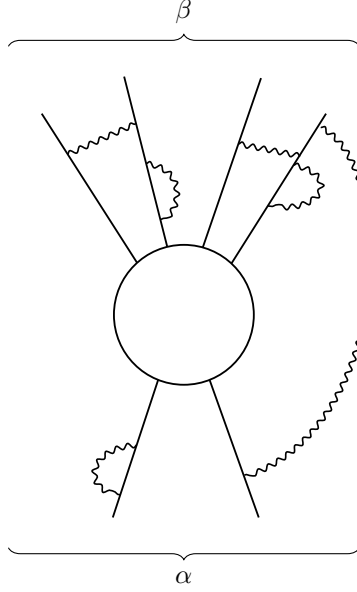
$$M_{\beta\alpha}^{\mu_1 \dots \mu_N}(q_1 \dots q_N) \longrightarrow M_{\beta\alpha} \prod_{r=1}^N \left( \sum_n \frac{\eta_n e_n p_n^{\mu_r}}{p_n \cdot q_r - i\eta_n \epsilon} \right). \tag{13.1.8}$$

### 13.2 Virtual Soft Photons

We shall now use the results of the previous section to calculate the effect to all orders of radiative corrections involving virtual soft photons exchanged among the charged particle lines of a process  $\alpha \rightarrow \beta$ , as in Figure 13.3. By a ‘soft’ photon we mean one that carries momentum less than  $\Lambda$ , where  $\Lambda$  is some convenient dividing point chosen low enough to justify the approximations made in the previous section. We shall find that these soft photons introduce infrared divergences, so as a stop-gap

we will have to introduce also a lower bound  $\lambda$  on the photon momenta. It is important to recognize the difference between these two limits on the soft photon momenta. The upper cutoff  $\Lambda$  just serves to define what we mean by ‘soft’ photons; the  $\Lambda$ -dependence of the soft-photon radiative corrections is cancelled by the  $\Lambda$ -dependence of the rest of the amplitude, which is calculated including only virtual photons with momenta *greater than*  $\Lambda$ . On the other hand, the lower cutoff  $\lambda$  must eventually be removed by taking  $\lambda \rightarrow 0$ ; as we shall see, the infrared divergences in this limit will be cancelled by the effects of real soft photon emission.





**Figure 13.3.** A typical dominant graph for the radiative corrections due to virtual soft photons to the  $S$ -matrix for the process  $\alpha \rightarrow \beta$ . Straight lines are particles in the states  $\alpha$  and  $\beta$  (including possible hard photons); wavy lines are soft photons.

For each virtual soft photon we must supply a propagator factor

$$\frac{-i}{(2\pi)^4} \frac{\eta_{\mu\nu}}{q^2 - i\epsilon}, \quad (13.2.1)$$

then multiply the amplitude (13.1.8) with the product of these propagators, contract photon polarization indices, and integrate over photon four-momenta. In addition for  $N$  virtual photons we must divide by a factor  $2^N N!$ , because the sum over all places to which we may attach the two ends of the soft photon lines includes spurious sums over the  $N!$  permutations of the photon lines and over interchanges of the two ends of these lines. The effect of radiative corrections involving  $N$  soft photons is then to multiply the matrix element  $M_{\beta\alpha}$  for the process without such radiative corrections by a factor

$$\frac{1}{N!2^N} \left[ \frac{1}{(2\pi)^4} \sum_{nm} e_n e_m \eta_n \eta_m J_{nm} \right]^N, \quad (13.2.2)$$

where

$$J_{nm} \equiv -i(p_n \cdot p_m) \int_{\lambda \leq |\mathbf{q}| \leq \Lambda} \frac{d^4 q}{[q^2 - i\epsilon][p_n \cdot q - i\eta_n \epsilon][p_m \cdot q - i\eta_m \epsilon]}. \quad (13.2.3)$$

Note that we have changed the sign of  $p_m \cdot q$  in the denominator in (13.2.3), because if we define  $q$  as the momentum emitted by line  $n$  then  $-q$  is the momentum emitted by line  $m$ .

Summing over  $N$ , we conclude that the matrix element for a process including radiative corrections due to any number of soft photons with momenta  $|\mathbf{q}| \geq \lambda$  is given by

$$M_{\beta\alpha}^\lambda = M_{\beta\alpha}^\Lambda \exp \left[ \frac{1}{2(2\pi)^4} \sum_{nm} e_n e_m \eta_n \eta_m J_{nm} \right], \quad (13.2.4)$$

where  $M_{\beta\alpha}^\Lambda$  is the amplitude including virtual photons only with momenta greater than  $\Lambda$ .

The integral over  $q^0$  in (13.2.3) may be done by the method of residues. The integrand is analytic in  $q^0$  except for four poles, at

$$\begin{aligned} q^0 &= |\mathbf{q}| - i\epsilon, & q^0 &= -|\mathbf{q}| + i\epsilon, \\ q^0 &= \mathbf{v}_n \cdot \mathbf{q} - i\eta_n\epsilon, & q^0 &= \mathbf{v}_m \cdot \mathbf{q} + i\eta_m\epsilon, \end{aligned}$$

where  $\mathbf{v}_n \equiv \mathbf{p}_n/p_n^0$ , and likewise for  $\mathbf{v}_m$ . If particle  $n$  is outgoing and particle  $m$  is incoming, then  $\eta_n = +1$ ,  $\eta_m = -1$ , so by closing the  $q^0$  contour in the upper half-plane we avoid the contributions from the poles at  $q^0 = \mathbf{v}_n \cdot \mathbf{q} - i\eta_n\epsilon$  or  $q^0 = \mathbf{v}_m \cdot \mathbf{q} + i\eta_m\epsilon$ . Similarly, if  $n$  is incoming and  $m$  is outgoing we can avoid these two poles by closing the contour in the lower half-plane. In these two cases it is only one of the poles at  $q^0 = \pm(|\mathbf{q}| - i\epsilon)$  that contributes, and we find a purely real integral:

$$J_{nm} = -\pi(p_n \cdot p_m) \int_{\lambda \leq |\mathbf{q}| \leq \Lambda} \frac{d^3q}{|\mathbf{q}|^3 (E_n - \hat{\mathbf{q}} \cdot \mathbf{p}_n)(E_m - \hat{\mathbf{q}} \cdot \mathbf{p}_m)} \quad (\text{for } \eta_n = -\eta_m = \pm 1). \quad (13.2.5)$$

On the other hand, if particles  $n$  and  $m$  are both outgoing or both incoming, then the poles at  $\mathbf{v}_n \cdot \mathbf{q} - i\eta_n\epsilon$  and  $\mathbf{v}_m \cdot \mathbf{q} + i\eta_m\epsilon$  lie on opposite sides of the real  $q^0$ -axis, and we cannot avoid a contribution from one of them whichever way we close the contour:

$$\begin{aligned} J_{nm} = & -\pi(p_n \cdot p_m) \int_{\lambda \leq |\mathbf{q}| \leq \Lambda} \frac{d^3q}{|\mathbf{q}|^3 (E_n - \hat{\mathbf{q}} \cdot \mathbf{p}_n)(E_m - \hat{\mathbf{q}} \cdot \mathbf{p}_m)} \\ & - \frac{4i\pi^3}{\beta_{nm}} \ln\left(\frac{\Lambda}{\lambda}\right) \quad (\text{for } \eta_n = \eta_m = \pm 1), \end{aligned} \quad (13.2.6)$$

where  $\beta_{nm}$  is the relative velocity of particles  $n$  and  $m$  in the rest frame of either:

$$\beta_{nm} \equiv \sqrt{1 - \frac{m_n^2 m_m^2}{(p_n \cdot p_m)^2}}. \quad (13.2.7)$$

The imaginary term in Eq. (13.2.6) leads to an infrared-divergent phase factor [4] in Eq. (13.2.4), which drops out when we take the absolute value of the matrix element to calculate the rate for the process  $\alpha \rightarrow \beta$ . (This infinite phase factor is the relativistic counterpart of the well-known feature of non-relativistic Coulomb scattering, that the outgoing wave part of the Schrödinger wave function has a dependence on the radial coordinate  $r$  of the form  $\exp(ipr - i\nu \ln r)/r$  instead of  $\exp(ipr)/r$ , where  $\nu$  is the product of charges divided by the relative velocity [5].) The reaction rate is affected by the real part of  $J_{nm}$ , which for all  $\eta_n$  and  $\eta_m$  takes the value

$$\text{Re } J_{nm} = -\pi(p_n \cdot p_m) \int_{\lambda \leq |\mathbf{q}| \leq \Lambda} \frac{d^3q}{|\mathbf{q}|^3 (E_n - \hat{\mathbf{q}} \cdot \mathbf{p}_n)(E_m - \hat{\mathbf{q}} \cdot \mathbf{p}_m)}. \quad (13.2.8)$$

An elementary calculation gives

$$\text{Re } J_{nm} = \frac{2\pi^2}{\beta_{nm}} \ln\left(\frac{1 + \beta_{nm}}{1 - \beta_{nm}}\right) \ln\left(\frac{\Lambda}{\lambda}\right). \quad (13.2.9)$$

Using this in the absolute value squared of Eq. (13.2.4) gives the effect of soft virtual photons on the rates for the process  $\Gamma_{\beta\alpha}$ , as

$$\Gamma_{\beta\alpha}^\lambda = \left(\frac{\lambda}{\Lambda}\right)^{A(\alpha \rightarrow \beta)} \Gamma_{\beta\alpha}^\Lambda, \quad (13.2.10)$$

where  $\Gamma_{\beta\alpha}^\lambda$  and  $\Gamma_{\beta\alpha}^\Lambda$  are the rates for the process  $\alpha \rightarrow \beta$  including radiative corrections of soft photons only with momenta greater than  $\lambda$  or  $\Lambda$ , respectively, and  $A$  is the exponent

$$A(\alpha \rightarrow \beta) = -\frac{1}{8\pi^2} \sum_{nm} \frac{e_n e_m \eta_n \eta_m}{\beta_{nm}} \ln \left( \frac{1 + \beta_{nm}}{1 - \beta_{nm}} \right). \quad (13.2.11)$$

Note that this makes sense only because the correction factor  $(\lambda/\Lambda)^A$  has turned out to be the ratio of a function of  $\lambda$  to the same function of  $\Lambda$ , since the two rates in Eq. (13.2.10) can only depend on  $\lambda$  and  $\Lambda$ , respectively.

The exponent  $A$  is always positive. For instance, in the scattering of a single charged particle by a neutral particle or an external potential, we must add terms in Eq. (13.2.11) where both  $n$  and  $m$  are the initial or the final charged particle (in which case  $\eta_n \eta_m = +1$  and  $\beta_{nm} = 0$ ), or  $n$  is the initial or final charged particle and  $m$  is the other one (in which case  $\eta_n \eta_m = -1$  and  $\beta_{nm} = \beta$ , where  $1 > \beta > 0$ .) This gives

$$A = -\frac{e^2}{8\pi^2} \left[ 4 - \frac{2}{\beta} \ln \left( \frac{1 + \beta}{1 - \beta} \right) \right],$$

which is positive for all  $1 > \beta > 0$ . Because  $A$  is positive the effect of the infrared divergences introduced by soft virtual photons when summed to all orders is to make the rate for any given charged particle process  $\alpha \rightarrow \beta$  *vanish* in the limit  $\lambda \rightarrow 0$ .

\* \* \*

Before we go on to consider how soft real photon emission cancels these infrared divergences, we should pause to note a technicality in the above calculation which as far as I know has always been ignored in the literature. In calculating these radiative corrections we have included diagrams in which the virtual photon is absorbed and emitted at the *same* external charged-particle line, as well as those in which it is emitted and absorbed at different lines. But as we learned in Chapter 10, in calculating the  $S$ -matrix we are not supposed to include radiative corrections arising from insertions of self-energy subgraphs in external lines. This might suggest that we should drop the terms  $n = m$  in Eq. (13.2.11), but then the cancellation of infrared divergences we will find in the next section would not be complete.

The resolution of this problem can be found in the observation that soft virtual photons produce infrared divergences not only directly, but also through their effect on the renormalization constants  $Z_n$  of the charged particle fields. (The renormalization constant  $Z_n$  is the one usually called  $Z_2$  in theories like quantum electrodynamics with a single charged field of spin  $\frac{1}{2}$ .) It is the counterterms proportional to  $Z_n - 1$  that cancel the effect of radiative corrections in external lines. To be specific, the renormalized field of a charged

particle of type  $n$  is a factor  $Z_n^{-1/2}$  times the unrenormalized field, so when we calculate the  $S$ -matrix using renormalized fields (corresponding to the omission of radiative corrections in external lines) we are introducing an extra factor  $\prod_n Z_n^{-1/2}$ , the product running over all charged particles in the initial and final states. (Of course there are also factors  $Z_n^{-1/2}$  for neutral particles, but these are not infrared-divergent.) In a slightly different notation, this factor is

$$\prod_f Z_f^{-E_f/2},$$

where  $Z_f$  is the field renormalization constant for fields of type  $f$ ,  $E_f$  is the number of external lines of type  $f$ , and the product now runs over *all* charged field types. However, these field renormalization constants also appear in the interiors of diagrams; expressing an interaction of type  $i$  that involves  $N_{if}$  charged-particle fields of type  $f$  in terms of renormalized fields introduces an infrared-divergent factor

$$\prod_f (Z_f)^{N_{if}/2}.$$

(For instance, the counterterm  $-ie(Z_2 - 1)A_\mu \bar{\psi} \gamma^\mu \psi$  in Eq. (11.1.9) together with the ordinary electromagnetic interaction  $-ie\bar{\psi} \gamma^\mu \psi$  yields a total interaction  $-iZ_2 e \bar{\psi} \gamma^\mu \psi$ . It was the infrared divergence in this  $Z_2$  factor that was responsible for the infrared divergence arising from the second term in the brackets in Eq. (11.3.23), and the last term in Eq. (11.4.14).) There is also an infrared divergence in the propagators of the renormalized fields; the propagator of a renormalized charged field of type  $f$  when expressed in terms of the propagator of the unrenormalized field introduces a factor  $Z_f^{-1}$ . Putting this all together, the total number of factors of  $Z_f$  for each charged field type  $f$ , introduced by the counterterms to interactions and to radiative corrections on internal as well as external lines, is

$$\frac{1}{2} \sum_i V_i N_{if} - I_f - \frac{1}{2} E_f,$$

where  $I_f$  and  $E_f$  are the numbers of internal and external lines of type  $f$ , and  $V_i$  is the number of vertices for interactions of type  $i$ . We have already noted in Section 6.3 that this quantity vanishes for each  $f$ . Thus the counterterms which cancelled the radiative corrections on external lines are themselves cancelled by the  $Z_f$  factors arising from internal lines and vertices. Eq. (13.2.11) is therefore correct as it stands, including the terms with  $n = m$ .

### 13.3 Real Soft Photons; Cancellation of Divergences

The resolution of the infrared divergence problem encountered in the previous section is found in the observation that it is not really possible to measure the rate  $\Gamma_{\beta\alpha}$  for a reaction  $\alpha \rightarrow \beta$  involving definite numbers of photons and charged particles, because photons of very low energy can always escape undetected. What can be measured is the rate  $\Gamma_{\beta\alpha}(E, E_T)$  for such a reaction to take place with no unobserved photon having an energy greater than some small quantity  $E$ , and with not more than some small total energy  $E_T$  going into any

number of unobserved photons. (Of course,  $E \leq E_T$ . In an experiment without soft photon detectors, one can rely on measurements of the energies of the ‘hard’ particles in  $\alpha$  and  $\beta$  to put a limit  $E_T$  on the total energy going into soft photons, and in this case we just set  $E = E_T$ .) We now turn to a calculation of this rate.

The  $S$ -matrix for emitting  $N$  real soft photons in a process  $\alpha \rightarrow \beta$  is obtained by contracting each of the  $N$  photon polarization indices  $\mu_1, \mu_2, \dots$  on the amplitude (13.1.8) with the appropriate coefficient function

$$\frac{\epsilon_\mu^*(\mathbf{q}, h)}{(2\pi)^{3/2} \sqrt{2|\mathbf{q}|}},$$

where  $\mathbf{q}$  is the photon momentum,  $h = \pm 1$  is its helicity, and  $\epsilon^\mu$  is the corresponding photon polarization ‘vector’.<sup>143</sup> This gives a photon emission matrix element (the  $S$ -matrix element with delta function omitted) as

$$M_{\beta\alpha}^\lambda(\mathbf{q}_1, h_1, \mathbf{q}_2, h_2, \dots) = M_{\beta\alpha}^\lambda \prod_{r=1}^N (2\pi)^{-3/2} (2|\mathbf{q}_r|)^{-1/2} \sum_n \frac{\eta_n e_n [p_n \cdot \epsilon^*(\mathbf{q}_r, h_r)]}{p_n \cdot q_r}. \quad (13.3.1)$$

(The superscript  $\lambda$  is to remind us that these amplitudes are to be calculated with an infrared cutoff  $\lambda$  on the momenta of virtual photons. Eventually we shall take  $\lambda \rightarrow 0$ . The presence of soft virtual photons does not interfere with the result (13.3.1) because of the factorization discussed in Section 13.1.) The differential rate for emitting  $N$  soft photons into a volume  $\prod_r d^3 q_r$  of momentum space is given by squaring this matrix element, summing over helicities, and multiplying with  $\prod_r d^3 q_r$ . We recall from Eq. (8.5.7) that for  $q^2 = 0$ , the helicity sums take the form

$$\sum_{h=\pm 1} \epsilon^\mu(\mathbf{q}, h) \epsilon^{\nu*}(\mathbf{q}, h) = \eta^{\mu\nu} + q^\mu c^\nu + q^\nu c^\mu, \quad (13.3.2)$$

where  $\mathbf{c} \equiv -\mathbf{q}/2|\mathbf{q}|^2$  and  $c^0 \equiv 1/2|\mathbf{q}|$ . The charge conservation condition  $\sum_n \eta_n e_n = 0$  allows us to drop the terms in (13.3.2) involving  $q_\mu$  or  $q_\nu$ , yielding a differential rate<sup>144</sup>

$$d\Gamma_{\beta\alpha}^\lambda(\mathbf{q}_1, \dots, \mathbf{q}_N) = \Gamma_{\beta\alpha}^\lambda \prod_{r=1}^N \frac{d^3 q_r}{(2\pi)^3 (2|\mathbf{q}_r|)} \sum_{nm} \frac{\eta_n \eta_m e_n e_m (p_n \cdot p_m)}{(p_n \cdot q_r)(p_m \cdot q_r)}. \quad (13.3.3)$$

To calculate the differential rate for the emission of  $N$  soft photons with definite energies  $\omega_r = |\mathbf{q}_r|$  we must integrate (13.3.3) over the directions of the photon momenta  $\mathbf{q}_r$ . These integrals are the same as those we encountered in the integrals (13.2.8),

$$-\pi(p_n \cdot p_m) \int \frac{d^2 \hat{\mathbf{q}}}{(E_n - \hat{\mathbf{q}} \cdot \mathbf{p}_n)(E_m - \hat{\mathbf{q}} \cdot \mathbf{p}_m)} = \frac{2\pi^2}{\beta_{nm}} \ln \left( \frac{1 + \beta_{nm}}{1 - \beta_{nm}} \right). \quad (13.3.4)$$

<sup>143</sup>We are using  $\epsilon^\mu$  instead of  $e^\mu$  for photon polarization vectors, to avoid confusion with our use here of  $e_n$  for electric charges.

<sup>144</sup>The result for  $|\mathbf{q}| d\Gamma_{\beta\alpha}^\lambda(\mathbf{q})/\Gamma_{\beta\alpha}^\lambda$  in the case  $N = 1$  corresponds to the distribution of energy emitted classically by a discontinuously changing current density four-vector  $J^\mu(x) = \sum_n \eta_n e_n \delta^3(\mathbf{x} - \mathbf{v}_n t) p_n^\mu/E_n$ , with the sum here running only over particles in the initial state for  $t < 0$  and in the final state for  $t > 0$ .

Integrating (13.3.3) over photon directions thus gives the differential rate for photons of energy  $\omega_1, \dots, \omega_N$ :

$$d\Gamma_{\beta\alpha}^\lambda(\omega_1, \dots, \omega_N) = \Gamma_{\beta\alpha}^\lambda A(\alpha \rightarrow \beta)^N \frac{d\omega_1}{\omega_1} \dots \frac{d\omega_N}{\omega_N}, \quad (13.3.5)$$

where  $A(\alpha \rightarrow \beta)$  is the same constant encountered in the previous section:

$$A(\alpha \rightarrow \beta) = -\frac{1}{8\pi^2} \sum_{nm} \frac{e_n e_m \eta_n \eta_m}{\beta_{nm}} \ln \left( \frac{1 + \beta_{nm}}{1 - \beta_{nm}} \right).$$

We see from (13.3.5) that an unrestricted integral over the energies of the emitted photons would introduce another infrared divergence. However, unitarity demands that if we use an infrared cutoff for the momenta of the virtual photons (as implied by the superscript  $\lambda$ ) then we must use the same infrared cutoff for the real photons. To calculate the rate  $\Gamma_{\beta\alpha}^\lambda(E, E_T)$  for the reaction  $\alpha \rightarrow \beta$  with not more than energy  $E$  going into any one unobserved photon and not more than energy  $E_T$  going into any number of unobserved photons (with  $E$  and  $E_T$  chosen small enough to justify the approximations used in deriving (13.3.1)), we must integrate (13.3.5) over all photon energies, subject to the limits  $E \geq \omega_r \geq \lambda$  and  $\sum_r \omega_r \leq E_T$ , then divide by  $N!$  because this integral includes configurations that differ only by permutations of the  $N$  soft photons, and finally sum over  $N$ . This gives

$$\Gamma_{\beta\alpha}^\lambda(E, E_T) = \Gamma_{\beta\alpha}^\lambda \sum_{N=0}^{\infty} \frac{A(\alpha \rightarrow \beta)^N}{N!} \int_{\substack{E \geq \omega_r \geq \lambda \\ \sum_r \omega_r \leq E_T}} \prod_{r=1}^N \frac{d\omega_r}{\omega_r}. \quad (13.3.6)$$

This integral would factor into the product of  $N$  integrals over the individual  $\omega_r$  were it not for the restriction  $\sum_r \omega_r \leq E_T$ . This restriction may be implemented by including as a factor in the integrand a step function

$$\theta \left( E_T - \sum_r \omega_r \right) = \frac{1}{\pi} \int_{-\infty}^{\infty} du \frac{\sin E_T u}{u} \exp \left( iu \sum_r \omega_r \right). \quad (13.3.7)$$

Eq. (13.3.6) then becomes

$$\Gamma_{\beta\alpha}^\lambda(E, E_T) = \frac{1}{\pi} \int_{-\infty}^{\infty} du \frac{\sin E_T u}{u} \exp \left( A(\alpha \rightarrow \beta) \int_{\lambda}^E \frac{d\omega}{\omega} e^{i\omega u} \right) \Gamma_{\beta\alpha}^\lambda. \quad (13.3.8)$$

The integral in the exponent can be done in the limit  $\lambda \ll E_T$  by writing it as the sum of the integral of  $(e^{i\omega u} - 1)/\omega$ , in which we can set  $\lambda = 0$ , and the integral of  $1/\omega$ , which is trivial. Rescaling the  $u$  and  $\omega$  variables, this gives for  $\lambda \ll E$ :

$$\Gamma_{\beta\alpha}^\lambda(E, E_T) \rightarrow \mathcal{F}(E/E_T; A(\alpha \rightarrow \beta)) \left( \frac{E}{\lambda} \right)^{A(\alpha \rightarrow \beta)} \Gamma_{\beta\alpha}^\lambda, \quad (13.3.9)$$

where

$$\begin{aligned} \mathcal{F}(x; A) &\equiv \frac{1}{\pi} \int_{-\infty}^{\infty} du \frac{\sin u}{u} \exp \left( A \int_0^x \frac{d\omega}{\omega} (e^{i\omega u} - 1) \right) \\ &= 1 - \frac{A^2 \theta(x - \frac{1}{2})}{2} \int_{1-x}^x \frac{d\omega}{\omega} \ln \left( \frac{x}{1-\omega} \right) + \dots \end{aligned} \quad (13.3.10)$$

For  $E$  and  $E_T$  of the same order and  $A \ll 1$  the factor  $\mathcal{F}(E/E_T; A)$  in (13.3.9) is close to unity; for instance,

$$\mathcal{F}(1; A) \simeq 1 - \frac{1}{12}\pi^2 A^2 + \dots$$

Because  $A(\alpha \rightarrow \beta) > 0$ , the factor  $(E/\lambda)^{A(\alpha \rightarrow \beta)}$  in (13.3.9) becomes infinite in the limit  $\lambda \rightarrow 0$ . However, Eq. (13.2.10) shows that the rate  $\Gamma_{\beta\alpha}^\lambda$  vanishes in this limit:

$$\Gamma_{\beta\alpha}^\lambda = \left(\frac{\lambda}{\Lambda}\right)^{A(\alpha \rightarrow \beta)} \Gamma_{\beta\alpha}^\Lambda.$$

Using this in (13.3.9) shows that the infrared cutoff  $\lambda$  drops out in the limit  $\lambda \ll E$ :

$$\Gamma_{\beta\alpha}^\lambda(E, E_T) \rightarrow \mathcal{F}(E/E_T; A(\alpha \rightarrow \beta)) \left(\frac{E}{\Lambda}\right)^{A(\alpha \rightarrow \beta)} \Gamma_{\beta\alpha}^\Lambda. \quad (13.3.11)$$

We remind the reader that the energy  $\Lambda$  is just a convenient dividing point between ‘soft’ photons which are taken into account explicitly in (13.3.11) and ‘hard’ photons whose effects are buried in  $\Gamma_{\beta\alpha}^\Lambda$ . The right-hand side of (13.3.11) is independent of  $\Lambda$  because  $\Gamma_{\beta\alpha}^\Lambda \propto \Lambda^A$ . However, in theories with a small coupling constant like quantum electrodynamics it is frequently a good strategy to take  $\Lambda$  to be sufficiently small compared with the typical energies  $W$  involved in the collision so that the approximations made here apply for photon energies less than  $\Lambda$ , but large enough so that  $A(\alpha \rightarrow \beta) \ln(W/\Lambda) \ll 1$ . Then it may be a good approximation to calculate  $\Gamma_{\beta\alpha}^\Lambda$  in lowest-order perturbation theory, with the dominant radiative corrections for  $E \ll \Lambda$  given by the factor  $(E/\Lambda)^A$  in (13.3.11).

\* \* \*

The same cancellation of infrared divergences occurs for soft gravitons [3]. The rate for any process  $\alpha \rightarrow \beta$ , with not more than an energy  $E$  going into soft gravitons, turns out to be proportional to  $E^B$ , where

$$B = \frac{G}{2\pi} \sum_{nm} \eta_n \eta_m m_n m_m \frac{1 + \beta_{nm}^2}{\beta_{nm} \sqrt{1 - \beta_{nm}^2}} \ln \left( \frac{1 + \beta_{nm}}{1 - \beta_{nm}} \right). \quad (13.3.12)$$

### 13.4 General Infrared Divergences

The infrared divergence due to soft photons that we have been considering up to now in this chapter is just one example of a variety of infrared divergences that are encountered in various physical theories. Another example is provided by quantum electrodynamics with massless charged particles. Here even after the cancellation of infrared divergences due to soft photons, we find a logarithmic divergence in the exponent  $A$  in (13.3.11). According to Eqs. (13.2.11) and (13.2.7), for a process in which all charged particles are electrons, in the limit  $m_e \rightarrow 0$  the exponent goes as

$$A \rightarrow -\frac{1}{4\pi^2} \sum_n e_n^2 - \frac{1}{4\pi^2} \sum_{n \neq m} e_n e_m \eta_n \eta_m \ln \left( \frac{2|p_n \cdot p_m|}{m_e^2} \right) \rightarrow -\frac{\ln m_e}{2\pi^2} \sum_n e_n^2.$$

(In the last step we have used the charge conservation condition  $\sum_n e_n \eta_n = 0$ .) The infrared divergence in this formula arises from soft photons that are emitted in a direction *parallel* to the momentum of one of the ‘hard’ electrons in the initial or final state, but it occurs also even if the photon like the electron is not soft, because the propagator denominator  $(p_n \pm q)^2$  vanishes for  $p_n^2 = q^2 = 0$  if  $\mathbf{p}_n$  is parallel to  $\mathbf{q}$ . To be a little more specific, for  $p_n^2 = q^2 = 0$  the integral of this factor<sup>145</sup> over photon directions takes the form

$$\int d^2\hat{\mathbf{q}} (p \pm q)^{-2} = \frac{2\pi}{\mathbf{p}^2 \mathbf{q}^2} \int_0^\pi \frac{\sin \theta d\theta}{1 - \cos \theta},$$

where  $\theta$  is the angle between the momenta of the photon and the charged particle. This integral diverges logarithmically at  $\theta = 0$ .

Of course, in the real world there are no massless electrically charged particles, but in reactions in which the typical value  $E^2$  of the scalar products  $|p_n \cdot p_m|$  is much larger than  $m_e^2$ , it is of interest to identify the places where large  $\ln(m_e/E)$  factors appear. The dominant radiative correction in this case is often given by the term  $-\ln(m_e/E) \sum_n e_n^2/2\pi^2$  in  $A$ . More importantly, in quantum chromodynamics there are massless particles, the gluons, that carry a conserved quantum number known as color that is analogous to electric charge, so that infrared divergences arise from the emission of parallel hard gluons from hard gluons or other hard colored particles in the initial or final states.

These infrared divergences are not, in general, eliminated by summing over suitable sets of final states. However, Lee and Nauenberg [6] have pointed out that the infrared divergences can be made to cancel if we not only sum over suitable final states, but also assume a certain probabilistic distribution of *initial* states. What follows is a modified version of their argument, which will immediately make clear why in the case of electrodynamics with massive electrons it was sufficient to sum over final states.

For these purposes it is convenient to return to ‘old-fashioned’ perturbation theory, in which the  $S$ -matrix is given by Eq. (3.2.7) and Eq. (3.5.3) as

$$S_{ba} = \delta(b - a) - 2i\pi\delta(E_a - E_b)T_{ba}, \quad (13.4.1)$$

where

$$T_{ba} = (H_{\text{int}})_{ba} + \sum_{\nu=1}^{\infty} \int dc_1 \cdots dc_\nu \frac{(H_{\text{int}})_{bc_\nu} (H_{\text{int}})_{c_\nu c_{\nu-1}} \cdots (H_{\text{int}})_{c_1 a}}{(E_a - E_{c_1} + i\epsilon) \cdots (E_a - E_{c_\nu} + i\epsilon)}. \quad (13.4.2)$$

(The integrals over  $c_1 \cdots c_\nu$  should be understood to include sums over the spins and types of particles in these states as well as integrals over the three-momenta of these particles.) Infrared divergences arise from (and only from) the vanishing of one or more of the energy denominators in this expression.

However, not all vanishing energy denominators give rise to infrared divergences. A general intermediate state  $c$  may have  $E_c = E_a$ , but usually this is just one point in the

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<sup>145</sup>This factor is not squared, because the divergence occurs only in the interference between this term in the  $S$ -matrix element and terms in which the photon is emitted from some other charged particle line  $m \neq n$ . For  $m = n$  the integral (13.2.8) is proportional to  $m_n^2$ .



interior of the range of integration, and the integral over this range is rendered convergent by the prescription implied by the  $i\epsilon$  in the denominator. In order for an intermediate state  $c$  to produce an infrared divergence, it is necessary that the energy  $E_c = E_a$  be reached at the *endpoint* of the range of integration. This happens for instance if the first intermediate state  $c_1$  in (13.4.2) consists of the particles in the initial state  $a$ , with any of the massless particles in this state replaced with *jets*, consisting of any number of nearly parallel massless particles with a total momentum equal to that of the particle the jet replaces. In this case, the endpoint at which  $E_{c_1} = E_a$  is the point in momentum space at which all of the massless particles in each jet are parallel. More generally, we can have any number of the massless particles in  $a$  replaced with jets of nearly parallel massless particles, plus any number of additional soft massless particles. The set of all such states will be called  $D(a)$ . (To be precise, we need to introduce a small angle  $\Theta$  as well as a small energy  $\Lambda$  to define what we mean by ‘nearly parallel’ and ‘soft’. We will not bother to show the dependence of the set  $D(a)$  on  $\Theta$  and  $\Lambda$ .) The states in  $D(a)$  are ‘dangerous’, in the sense that the vanishing of the energy denominator  $E_a - E_{c_1}$  at the endpoint can introduce an infrared divergence; the endpoint at which  $E_{c_1} = E_a$  is the point at which all massless particles in each jet are parallel, and all soft massless particles have zero energy.

Furthermore, if  $c_1, \dots, c_n$  are each in the set  $D(a)$ , then an intermediate state  $c_{n+1}$  in  $D(a)$  is also dangerous in the same sense. On the other hand, if some intermediate state  $c_m$  is *not* in  $D(a)$ , then a later state  $c_k$  with  $k > m$  would not be dangerous even if it belonged to the set  $D(a)$ , because the configuration of hard particles or jets with three-momenta equal to those of particles in the state  $a$  would be just an ordinary point inside the range of integration. In exactly the same way, we may define a set of states  $D(b)$  in which one or more of the massless particles in the state  $b$  are replaced with jets of nearly parallel massless particles, each having the same total three-momentum as the particle it replaces, and we add any number of soft massless particles. An intermediate state  $c_m$  is dangerous if it belongs to the set  $D(b)$  and if the later states  $c_k$  with  $k > m$  all belong to  $D(b)$ .

To isolate the effects of these dangerous states we rewrite (13.4.2) in the form

$$T_{ba} = (H_{\text{int}})_{ba} + \sum_{\nu=1}^{\infty} \left( H_{\text{int}} \left[ \frac{\mathcal{P}_a + \mathcal{P}_b + \mathcal{P}_{\not a,b}}{E_a - H_0 + i\epsilon} H_{\text{int}} \right]^\nu \right)_{ba}, \quad (13.4.3)$$

where  $\mathcal{P}_a$ ,  $\mathcal{P}_b$ , and  $\mathcal{P}_{\not a,b}$  are the projection operators respectively on  $D(a)$ ,  $D(b)$ , and on all other states. (It is assumed here that none of the charged particles in  $b$  have momenta close to that of some charged particle in  $a$ , so that the sets  $D(a)$  and  $D(b)$  do not overlap.) Now, for  $\Lambda \rightarrow 0$  and  $\Theta \rightarrow 0$ , the dangerous intermediate states occupy so little phase space that they may be neglected wherever they do not lead to infrared divergences. The power series (13.4.3) therefore becomes

$$T_{ba} = \sum_{r=0}^{\infty} \sum_{s=0}^{\infty} \sum_{\nu=0}^{\infty} \left( \left[ H_{\text{int}} \frac{\mathcal{P}_b}{E_a - H_0 + i\epsilon} \right]^r H_{\text{int}} \left[ \frac{\mathcal{P}_{\not a,b}}{E_a - H_0 + i\epsilon} H_{\text{int}} \right]^\nu \cdot \left[ \frac{\mathcal{P}_a}{E_a - H_0 + i\epsilon} H_{\text{int}} \right]^s \right)_{ba}. \quad (13.4.4)$$

This would be exact if all of the projection operators  $\mathcal{P}_{\not\in a,b}$  between the leftmost and the rightmost were replaced with  $\mathcal{P}_a + \mathcal{P}_b + \mathcal{P}_{\not\in a,b}$ , and if  $\mathcal{P}_b$  and  $\mathcal{P}_a$  on the left and right were replaced with  $\mathcal{P}_b + \mathcal{P}_a$ , but as remarked above this would have a negligible effect on the final result when  $\Lambda$  and  $\Theta$  are sufficiently small.

Eq. (13.4.4) may be written in a more compact form:

$$T_{ba} = \left\{ (\Omega_b^-)^\dagger T_S \Omega_a^+ \right\}_{ba}, \quad (13.4.5)$$

where, for future use, we define  $\Omega_\alpha^+$  and  $\Omega_\beta^-$  for general states  $\alpha$  and  $\beta$  as:

$$(\Omega_\alpha^+)_{ca} \equiv \sum_{r=0}^{\infty} \left( \left[ \frac{\mathcal{P}_\alpha}{E_\alpha - H_0 + i\epsilon} H_{\text{int}} \right]^r \right)_{ca}, \quad (13.4.6)$$

$$(\Omega_\beta^-)_{db} \equiv \sum_{r=0}^{\infty} \left( \left[ \frac{\mathcal{P}_\beta}{E_\beta - H_0 - i\epsilon} H_{\text{int}} \right]^r \right)_{db}, \quad (13.4.7)$$

and  $T_S$  is the ‘safe’ operator<sup>146</sup>

$$(T_S)_{dc} \equiv \sum_{\nu=0}^{\infty} \left( H_{\text{int}} \left[ \frac{\mathcal{P}_{\not\in c,d}}{E_c - H_0 + i\epsilon} H_{\text{int}} \right]^\nu \right)_{dc}. \quad (13.4.8)$$

All of the infrared divergences have now been isolated in the two operator factors  $\Omega_\beta^-$  and  $\Omega_\alpha^+$ .

To eliminate these infrared divergences it is now only necessary to note that if it were *not* for the projection operators on the dangerous states, the operators  $\Omega_\beta^-$  and  $\Omega_\alpha^+$  would be just the unitary operators that according to Eq. (3.1.16) convert free-particle states into ‘out’ or ‘in’ states, respectively. These operators are therefore *unitary if confined to the subspaces  $D(\beta)$  and  $D(\alpha)$  of states that would be dangerous for some given final state  $\beta$  and some given initial state  $\alpha$* . That is, for general  $\alpha$  and  $\beta$

$$\Omega_\beta^- \mathcal{P}_\beta (\Omega_\beta^-)^\dagger = \mathcal{P}_\beta, \quad (13.4.9)$$

$$\Omega_\alpha^+ \mathcal{P}_\alpha (\Omega_\alpha^+)^\dagger = \mathcal{P}_\alpha. \quad (13.4.10)$$

*The transition rate is therefore free of infrared divergences if summed over the subspaces of states that would be dangerous for any given final and initial states  $\beta$  and  $\alpha$ :*

$$\begin{aligned} \sum_{a \in D(\alpha)} \sum_{b \in D(\beta)} |T_{ba}|^2 &= \text{Tr} \left\{ \Omega_\beta^- \mathcal{P}_\beta (\Omega_\beta^-)^\dagger T_S \Omega_\alpha^+ \mathcal{P}_\alpha (\Omega_\alpha^+)^\dagger T_S^\dagger \right\} \\ &= \text{Tr} \left\{ \mathcal{P}_\beta T_S \mathcal{P}_\alpha T_S^\dagger \right\} = \sum_{a \in D(\alpha)} \sum_{b \in D(\beta)} |(T_S)_{ba}|^2. \end{aligned} \quad (13.4.11)$$

In order to be satisfied that this really does solve the general problem of infrared divergences, it is necessary to argue that it is only sums like that in (13.4.11) that are

<sup>146</sup>In  $(\Omega_b^-)_{db}$  we are using the fact that  $T_{ba}$  is calculated with  $E_b = E_a$ , and in  $(T_S)_{dc}$  we are using the fact that the projection operators  $\mathcal{P}_a$  make  $(\Omega_a^+)_{ca}$  vanish unless  $E_c$  is very close to  $E_a$ . Also, the factors  $(\Omega_b^-)^\dagger$  and  $\Omega_a^+$  in (13.4.5) make  $\mathcal{P}_{\not\in c,d} = \mathcal{P}_{\not\in a,b}$ .

experimentally measurable. It is plausible that we should have to sum over dangerous final states in order to have a measurable transition rate, since it is not possible experimentally to distinguish an outgoing charged (or colored) massless particle from a jet of massless particles with nearly parallel momenta and the same total energy [7], together with an arbitrary number of very soft quanta, all with the same total charge (or color). The sum over initial states is more problematic. Presumably one may argue that truly massless particles are always produced as jets accompanied by an ensemble of soft quanta that is uniform within some volume of momentum space. However, to the best of my knowledge no one has given a complete demonstration that the sums of transition rates that are free of infrared divergences are the only ones that are experimentally measurable.

This problem does not arise in quantum electrodynamics (with massive charged particles), where as we have seen it is only necessary to sum over final states in order to eliminate infrared divergences. The reason for this difference can be traced to the fact that in electrodynamics the states  $a, b, c, \dots$  are direct products of states (labelled with Greek letters) with fixed numbers of charged particles and hard photons, times states containing only soft photons having energy less than some small quantity  $\Lambda$ . Then for a reaction in which some set of soft photons  $f$  is produced in a reaction  $\alpha \rightarrow \beta$  among charged particles and hard photons, (13.4.5) simplifies to

$$T_{\beta f, \alpha} = \langle f | (\Omega^-(\beta))^\dagger \Omega^+(\alpha) | \Omega \rangle (T_S)_{\beta \alpha}, \quad (13.4.12)$$

where  $|\Omega\rangle$  denotes the soft photon vacuum, and  $\Omega^\pm$  are calculated as before, but in the reduced Hilbert space consisting only of soft photons, and with the interactions of these photons taken as the interaction Hamiltonian with all charged particles in the fixed states indicated by the arguments  $\beta$  or  $\alpha$ . Just as before, these operators are unitary in the ‘dangerous’ Hilbert space  $\mathcal{D}$  of soft photons, so<sup>147</sup>

$$\begin{aligned} \sum_{f \in \mathcal{D}} |T_{\beta f, \alpha}|^2 &= |(T_S)_{\beta \alpha}|^2 \langle \Omega | (\Omega^+(\alpha))^\dagger \Omega^-(\beta) (\Omega^-(\beta))^\dagger \Omega^+(\alpha) | \Omega \rangle \\ &= |(T_S)_{\beta \alpha}|^2 \langle \Omega | (\Omega^+(\alpha))^\dagger \Omega^+(\alpha) | \Omega \rangle = |(T_S)_{\beta \alpha}|^2 \end{aligned} \quad (13.4.13)$$

without having to sum over initial states.

### 13.5 Soft Photon Scattering<sup>148</sup>

In our treatment in this chapter of soft photon interactions, we have up to now considered only processes in which the soft photons are emitted or absorbed in a process  $\alpha \rightarrow \beta$  which was going on anyway. It is also possible to make useful general statements about processes in which the process  $\alpha \rightarrow \beta$  is trivial, and the soft photons play an essential part in producing an interesting reaction. We will consider here the simplest and most important example of

<sup>147</sup>The reason that we are now not encountering any factor  $(E/\Lambda)^A$  like that in (13.3.11) is that we are here identifying the maximum energy  $E$  of the real soft photon states over which we are summing with the maximum energy  $\Lambda$  of the ‘dangerous’ soft photon states over which we sum in calculating  $\Omega^\pm$ .

<sup>148</sup>This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

this sort, the scattering of a soft photon from a massive particle of arbitrary type and spin, where  $\alpha$  and  $\beta$  are single-particle states. The complication here is that the leading term in the soft photon scattering amplitude does not come from the pole terms, but from non-pole terms that are related to the pole terms by the condition of current conservation.

The  $S$ -matrix for photon scattering may be put in the form

$$S(q, \lambda; p, \sigma \rightarrow q', \lambda'; p', \sigma') = i(2\pi)^4 \delta^4(q + p - q' - p') \cdot \frac{\epsilon_\nu^*(q', \lambda') \epsilon_\mu(q, \lambda) M_{\sigma', \sigma}^{\nu\mu}(q; p', p)}{(2\pi)^6 \sqrt{4q^0 q'^0}}. \quad (13.5.1)$$

where  $q$  and  $q'$  are the initial and final photon four-momenta,  $p$  and  $p'$  are the initial and final target four-momenta,  $\lambda$  and  $\lambda'$  are the initial and final photon helicities,  $\epsilon_\nu(q', \lambda')$  and  $\epsilon_\mu(q, \lambda)$  are the corresponding photon polarization vectors, and  $\sigma$  and  $\sigma'$  are the initial and final target spin  $z$ -components. According to the theorem of Section 6.4, the amplitude  $M^{\nu\mu}$  may be expressed as

$$(2\pi)^{-3} M_{\sigma', \sigma}^{\nu\mu}(q; p', p) = \int d^4x e^{iq \cdot x} \langle p', \sigma' | T \{ J^\nu(0), J^\mu(x) \} | p, \sigma \rangle + \dots. \quad (13.5.2)$$

where  $J^\mu(x)$  is the electromagnetic current, and the dots indicate possible ‘seagull’ terms such as those in the theory of charged scalar fields in which the two photons interact at a single vertex instead of with separate currents. We now repeat the standard poleology arguments described in Chapter 10 and already used in Section 13.1. Inserting a complete set of intermediate states between the current operators in Eq. (13.5.2), integrating over  $x$  and isolating the one-particle intermediate states gives

$$M^{\nu\mu}(q; p', p) = \frac{G^\nu(p', p + q) G^\mu(p + q, p)}{E(p + q) - E(p) - q^0 - i\epsilon} + \frac{G^\mu(p', p' - q) G^\nu(p' - q, p)}{E(p' - q) - E(p') + q^0 - i\epsilon} + N^{\nu\mu}(q; p', p), \quad (13.5.3)$$

where  $G^\mu$  is the one-particle matrix element of the current

$$(2\pi)^{-3} G_{\sigma', \sigma}^\mu(p', p) \equiv \langle p', \sigma' | J^\mu(0) | p, \sigma \rangle \quad (13.5.4)$$

and  $N^{\nu\mu}$  represents the contribution of states other than the one-particle state itself, plus any seagull terms. (Eq. (13.5.3) is to be understood in the sense of matrix multiplication, with spin indices not shown explicitly.) About  $N^{\nu\mu}$  we know very little, except that it does not have the singularity at  $q^\mu \rightarrow 0$  exhibited by the first two terms, and therefore may be expanded in powers of  $q^\mu$ .

We now use the current conservation (or gauge invariance) conditions:

$$q_\mu M^{\nu\mu}(q; p', p) = 0, \quad (13.5.5)$$

$$\mathbf{q} \cdot \mathbf{G}(p + q, p) = [E(p + q) - E(p)] G^0(p + q, p), \quad (13.5.6)$$

$$\mathbf{q} \cdot \mathbf{G}(p', p' - q) = [E(p') - E(p' - q)] G^0(p', p' - q). \quad (13.5.7)$$

Applied to Eq. (13.5.3), these conditions yield the condition we need on  $N^{\nu\mu}$ :

$$q_\mu N^{\nu\mu}(q; p', p) = -G^\nu(p', p+q)G^0(p+q, p) + G^0(p', p'-q)G^\nu(p'-q, p). \quad (13.5.8)$$

We also note that  $M^{\nu\mu}$  satisfies the ‘crossing symmetry’ condition

$$M^{\nu\mu}(q; p', p) = M^{\mu\nu}(p' - p - q; p', p) \quad (13.5.9)$$

and since the pole terms in Eq. (13.5.3) evidently satisfy this condition, so also does  $N^{\nu\mu}$ :

$$N^{\nu\mu}(q; p', p) = N^{\mu\nu}(p' - p - q; p', p). \quad (13.5.10)$$

We will use these conditions to determine the first terms in the expansion of  $N^{\nu\mu}$  in powers of momenta.

First we need to say something about the expansion of the one-particle current matrix elements  $G^\mu(p', p)$  in powers of the momenta  $p'$  and  $p$ . Space inversion invariance (to the extent that it is applicable) tells us that the expansion of  $G^0$  and  $G^i$  (with  $i = 1, 2, 3$ ) contain terms respectively of only even and odd order in the momenta. According to Eq. (10.6.3), the term in  $G_{\sigma', \sigma}^0$  of zeroth order in momenta is  $e\delta_{\sigma', \sigma}$ , where  $e$  is the particle charge. The current conservation condition then tells us that to second order in momenta,

$$(\mathbf{p}' - \mathbf{p}) \cdot \mathbf{G}_{\sigma', \sigma}(\mathbf{p}', \mathbf{p}) = \left( \frac{\mathbf{p}'^2}{2m} - \frac{\mathbf{p}^2}{2m} \right) e\delta_{\sigma', \sigma}.$$

The terms in  $\mathbf{G}$  of first order in momenta are thus given by  $e(\mathbf{p}' + \mathbf{p})\delta_{\sigma', \sigma}/2m$ , plus a possible first-order term orthogonal to  $\mathbf{p}' - \mathbf{p}$ , which rotational invariance tells us must be proportional to  $(\mathbf{p}' - \mathbf{p}) \times \mathbf{J}_{\sigma', \sigma}$ , where  $\mathbf{J}$  is the familiar spin matrix of the charged particle. Summarizing these results, we have the expansions

$$G^0(\mathbf{p}', \mathbf{p}) = e\mathbf{1} + \text{quadratic}, \quad (13.5.11)$$

$$\mathbf{G}(\mathbf{p}', \mathbf{p}) = \frac{e\mathbf{1}}{2m}(\mathbf{p}' + \mathbf{p}) + \frac{i\mu}{j}\mathbf{J} \times (\mathbf{p}' - \mathbf{p}) + \text{cubic}, \quad (13.5.12)$$

where ‘ $\mathbf{1}$ ’ is the unit spin matrix, and ‘quadratic’ and ‘cubic’ refer to the order of the neglected terms in powers of the small momenta  $\mathbf{p}$  and  $\mathbf{p}'$ . The coefficient  $\mu/j$  in Eq. (13.5.12) is real because the current is Hermitian. With the coefficient written in this way (with  $j$  the spin of the charged particle),  $\mu$  is the quantity known as the magnetic moment of the particle.

Now let us return to  $N^{\nu\mu}$ , and consider the expansion of Eq. (13.5.8) in powers of the small momenta  $q^\mu$ ,  $\mathbf{p}$  and  $\mathbf{p}'$ . Taking  $\nu = 0$  in Eq. (13.5.8) shows that  $q_\mu N^{0\mu}$  is at least quadratic in these small quantities. There is no constant vector orthogonal to  $q^\mu$ , so  $N^{0\mu}$  must be at least of first order in small momenta. The crossing symmetry condition (13.5.10) then tells us that  $N^{\mu 0}$  must also be at least of first order in small momenta. Taking  $\mu = i$  in Eq. (13.5.8) and using Eq. (13.3.12) then tells us that

$$q_k N^{ik} = -\frac{e^2 q^i}{m} + \text{quadratic}$$

and hence

$$N^{ik} = -\frac{e^2}{m}\delta_{ik} + \text{linear}. \quad (13.5.13)$$

Since  $G^i$  is at least of first order in the small momenta, so are the pole terms in Eq. (13.5.3) for  $M^{ik}$ , leaving us in zero order with only the *non-pole* term  $N^{ik}$ :

$$M^{ik}(0; 0, 0) = N^{ik}(0; 0, 0) = -\frac{e^2}{m}\delta_{ik}. \quad (13.5.14)$$

From this we can calculate the soft photon scattering cross-section. But there is no need for this calculation; now that we know that the photon scattering amplitude in the limit of zero momentum depends only on the target particle mass and charge, and is of second order in the charge, we can immediately use the results of any second-order calculation of the photon scattering cross-section for target particles of any given spin, such as our result (8.7.42) for the differential photon scattering cross-section in quantum electrodynamics:

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{32\pi^2 m^2} (1 + \cos^2 \theta). \quad (13.5.15)$$

We now see that this is a universal formula, valid in the low-energy limit for target particles of mass  $m$  and charge  $e$  and of arbitrary type and spin, even if these particles are composite and strongly interacting, like atomic nuclei. Gell-Mann and Goldberger and Low [8] have shown that these results may be extended to give the next-to-leading term in the soft photon scattering amplitude in terms of the target particle's mass, charge, and magnetic moment.

### 13.6 The External Field Approximation<sup>149</sup>

It is intuitively obvious that a heavy charged particle like the nucleus of an atom acts approximately like the source of a classical external field. In this section we will see how to justify this approximation, and will gain some idea of its limitations.

Consider a Feynman diagram or a part of a Feynman diagram in which a heavy charged particle passing through the diagram from the initial to the final state emits  $N$  off-shell photons with four-momenta  $q_1, q_2, \dots, q_N$  and polarization indices  $\mu_1, \mu_2, \dots, \mu_N$ . The sum of all such graphs or subgraphs (not including the  $N$  photon propagators) yields an amplitude

$$\begin{aligned} & \int d^4x_1 d^4x_2 \dots d^4x_N e^{-iq_1 \cdot x_1} e^{-iq_2 \cdot x_2} \dots e^{-iq_N \cdot x_N} \\ & \cdot \langle p', \sigma' | T \{ J^{\mu_1}(x_1), J^{\mu_2}(x_2), \dots, J^{\mu_N}(x_N) \} | p, \sigma \rangle \\ & \equiv \mathcal{G}_{\sigma', \sigma}^{\mu_1 \mu_2 \dots \mu_N}(q_1, q_2, \dots, q_N; p). \end{aligned} \quad (13.6.1)$$

with the matrix element calculated including all interactions in which the heavy particle may participate, including strong nuclear forces. This amplitude has a multiple pole at  $q_1, q_2, \dots, q_N \rightarrow 0$ , arising from terms in the matrix elements of the product of currents in which the intermediate states consist of just the same heavy particle as in the initial and final states. This multiple pole dominates (13.6.1) when all components of  $q_1, q_2, \dots, q_N$

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<sup>149</sup>This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

are small compared with all energies and momenta associated with the dynamics of the (perhaps composite) heavy particle. In this case the methods of Section 10.2 give<sup>150</sup>

$$\begin{aligned} \mathcal{G}_{\sigma',\sigma}^{\mu_1\mu_2\cdots\mu_N}(q_1, q_2, \dots, q_N; p) &\rightarrow \frac{(-i)^{N-1}}{2p^0(2\pi)^3} (2\pi)^4 \\ &\cdot \delta^4(p' + q_1 + q_2 + \cdots + q_N - p) \sum_{\sigma_1, \sigma_2, \dots, \sigma_{N-1}} \\ &\cdot \frac{\mathcal{G}_{\sigma',\sigma_1}^{\mu_1}(p) \mathcal{G}_{\sigma_1,\sigma_2}^{\mu_2}(p) \cdots \mathcal{G}_{\sigma_{N-1},\sigma}^{\mu_N}(p)}{[2p \cdot q_1 - i\epsilon][2p \cdot (q_1 + q_2) - i\epsilon] \cdots [2p \cdot (q_1 + \cdots + q_{N-1}) - i\epsilon]} \\ &+ \text{permutations.} \end{aligned} \quad (13.6.2)$$

where

$$\frac{\mathcal{G}_{\sigma',\sigma}^{\mu}(p)}{2p^0(2\pi)^3} \equiv \langle p, \sigma' | J^{\mu}(0) | p, \sigma \rangle \quad (13.6.3)$$

and ‘+permutations’ indicates that we are to sum over all permutations of the  $N$  photons. For applications to atomic systems it is important to recognize that (13.6.1) applies for particles of arbitrary spin that have strong as well as electromagnetic interactions, like atomic nuclei.

We also note that for particles of arbitrary spin and charge  $Ze$ , the matrix elements of the electric current between states of equal four-momenta are<sup>151</sup>

$$\langle p, \sigma' | J^{\mu}(0) | p, \sigma \rangle = \frac{Ze p^{\mu} \delta_{\sigma',\sigma}}{p^0(2\pi)^3}, \quad (13.6.4)$$

so that

$$\mathcal{G}_{\sigma',\sigma}^{\mu}(p) = 2Ze p^{\mu} \delta_{\sigma',\sigma}. \quad (13.6.5)$$

The important thing about Eq. (13.6.5) is that these matrices all commute, so their product can be factored out of the sum over permutations:

$$\begin{aligned} \mathcal{G}_{\sigma',\sigma}^{\mu_1\mu_2\cdots\mu_N}(q_1, q_2, \dots, q_N; p) &\rightarrow \frac{(-i)^{N-1} (Ze)^N p^{\mu_1} p^{\mu_2} \cdots p^{\mu_N}}{p^0(2\pi)^3} (2\pi)^4 \\ &\cdot \delta^4(p' + q_1 + q_2 + \cdots + q_N - p) \delta_{\sigma',\sigma} \\ &\cdot \left[ \frac{1}{[p \cdot q_1 - i\epsilon][p \cdot (q_1 + q_2) - i\epsilon] \cdots [p \cdot (q_1 + \cdots + q_{N-1}) - i\epsilon]} \right. \\ &\quad \left. + \text{permutations} \right]. \end{aligned} \quad (13.6.6)$$

<sup>150</sup>In perturbation theory, the denominators come from the denominators of the propagators:

$$(p' + q_1 + \cdots + q_r)^2 + m^2 - i\epsilon \rightarrow 2p' \cdot (q_1 + \cdots + q_r) - i\epsilon \rightarrow 2p \cdot (q_1 + \cdots + q_r) - i\epsilon,$$

while the numerators of the propagators provide factors  $\sum_{\sigma} u u^{\dagger}$  that together with the photon emission vertex matrices yield the matrix elements (13.6.3). The matrix  $\mathcal{G}^{\mu}$  differs from the matrix  $G^{\mu}$  of the previous section by a factor  $2p^0$ .

<sup>151</sup>This is most easily proved by first noting that in the Lorentz frame in which the particle is at rest, rotational invariance requires that the matrix elements of the current have vanishing space components and a time component proportional to  $\delta_{\sigma',\sigma}$ , with no other dependence on  $\sigma$  or  $\sigma'$ . The constant of proportionality is supplied by Eq. (10.6.3), and a Lorentz transformation then gives Eq. (13.6.4).

To leading order in the  $q$ 's the delta function here may be written

$$\delta^4(p' + q_1 + \cdots + q_N - p) = p^0 \delta^3(\mathbf{p}' + \mathbf{q}_1 + \cdots + \mathbf{q}_N - \mathbf{p}) \cdot \delta(p \cdot (q_1 + \cdots + q_N)). \quad (13.6.7)$$

Fortunately, it turns out that the result of summing over permutations here is much simpler than the individual terms. For  $p \cdot (q_1 + \cdots + q_N) = 0$ , we have

$$\left[ \frac{1}{[p \cdot q_1 - i\epsilon][p \cdot (q_1 + q_2) - i\epsilon] \cdots [p \cdot (q_1 + \cdots + q_{N-1}) - i\epsilon]} + \text{permutations} \right] = (2i\pi)^{N-1} \delta(p \cdot q_1) \delta(p \cdot q_2) \cdots \delta(p \cdot q_{N-1}). \quad (13.6.8)$$

For instance, for  $N = 2$  this reads:

$$\frac{1}{[p \cdot q_1 - i\epsilon]} + \frac{1}{[p \cdot q_2 - i\epsilon]} = \frac{1}{[p \cdot q_1 - i\epsilon]} + \frac{1}{[-p \cdot q_1 - i\epsilon]} = 2i\pi \delta(p \cdot q_1).$$

The general result (13.6.8) can be obtained most easily as the Fourier transform of the identity

$$\theta(\tau_1 - \tau_2) \theta(\tau_2 - \tau_3) \cdots \theta(\tau_{N-1} - \tau_N) + \text{permutations} = 1.$$

Inserting Eq. (13.6.8) in Eq. (13.6.6) gives our final result for the amplitude (13.6.1):

$$\begin{aligned} \mathcal{G}_{\sigma', \sigma}^{\mu_1 \mu_2 \cdots \mu_N}(q_1, q_2, \dots, q_N; p) &\rightarrow (Ze)^N (2\pi)^N \delta_{\sigma', \sigma} p^{\mu_1} p^{\mu_2} \cdots p^{\mu_N} \\ &\cdot \delta^3(\mathbf{p}' + \mathbf{q}_1 + \mathbf{q}_2 + \cdots + \mathbf{q}_N - \mathbf{p}) \\ &\cdot \delta(p \cdot q_1) \delta(p \cdot q_2) \cdots \delta(p \cdot q_N). \end{aligned} \quad (13.6.9)$$

This result applies to relativistic as well as slowly moving heavy particles, and can be used to derive the ‘Weizsäcker–Williams’ approximation [9] for charged-particle scattering. In the special case of a non-relativistic heavy charged particle, with  $|\mathbf{p}| \ll p^0$ , Eq. (13.6.9) further simplifies to

$$\begin{aligned} \mathcal{G}_{\sigma', \sigma}^{\mu_1 \mu_2 \cdots \mu_N}(q_1, q_2, \dots, q_N; p) &\rightarrow (Ze)^N (2\pi)^N n^{\mu_1} n^{\mu_2} \cdots n^{\mu_N} \\ &\cdot \delta^3(\mathbf{p}' + \mathbf{q}_1 + \mathbf{q}_2 + \cdots + \mathbf{q}_N - \mathbf{p}) \\ &\cdot \delta(q_1^0) \delta(q_2^0) \cdots \delta(q_N^0) \delta_{\sigma', \sigma}, \end{aligned} \quad (13.6.10)$$

where  $n$  is a unit time-like vector

$$n^0 = 1, \quad \mathbf{n} = 0.$$

Now suppose that a single heavy non-relativistic particle of charge  $Ze$  with normalized momentum space wave function  $\chi_\sigma(\mathbf{p})$  appears in both the initial and final states. Using the Fourier representation of the delta function in Eq. (13.6.9), the matrix element of  $\mathcal{G}$  in this state is

$$\begin{aligned} &\int d^3p d^3p' \chi_{\sigma'}^*(\mathbf{p}') \chi_\sigma(\mathbf{p}) \\ &\cdot \mathcal{G}_{\sigma', \sigma}^{\mu_1 \mu_2 \cdots \mu_N}(q_1, q_2, \dots, q_N; p) \\ &\rightarrow \int d^3X \sum_\sigma |\psi_\sigma(\mathbf{X})|^2 \prod_{r=1}^N 2\pi Ze n^{\mu_r} \delta(q_r^0) e^{-i\mathbf{q}_r \cdot \mathbf{X}}. \end{aligned} \quad (13.6.11)$$



where  $\psi_\sigma(\mathbf{X})$  is the coordinate space wave function:

$$\psi_\sigma(\mathbf{X}) = (2\pi)^{-3/2} \int d^3p \chi_\sigma(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{X}}. \quad (13.6.12)$$

Because of the factorization in Eq. (13.6.11), the effect of including a heavy charged particle in this state is then the same as that of adding any numbers of a new kind of vertex in the momentum space Feynman rules, in which light Dirac particles of charge  $-e$  such as electrons interact with an external field, with each such vertex contributing to the overall amplitude a factor<sup>152</sup> (now including the photon propagator and electron–photon vertex)

$$i \int d^4q \left[ \frac{-i}{(2\pi)^4} \frac{1}{q^2 - i\epsilon} \right] [2\pi Z e n_\mu \delta(q^0) e^{-i\mathbf{q}\cdot\mathbf{X}}] [(2\pi)^4 e \gamma^\mu \delta^4(k - k' - q)]. \quad (13.6.13)$$

where  $k$  and  $k'$  are the initial and final electron four-momenta. The complete scattering amplitude must then be averaged over the heavy particle position  $\mathbf{X}$ , with weight function  $\sum_\sigma |\psi_\sigma(\mathbf{X})|^2$ . The factor (13.6.13) is the same as would be produced by a new term in the interaction Lagrangian

$$\mathcal{L}_{\text{ext}}(x) = \mathcal{A}_\mu(x) J_e^\mu(x), \quad (13.6.14)$$

where  $J_e^\mu \equiv -ie\bar{\Psi}\gamma^\mu\Psi$  is the electric current of the electrons, and  $\mathcal{A}^\mu$  is an external vector potential

$$\mathcal{A}^\mu(x) = \frac{1}{(2\pi)^4} \int d^4q e^{iq\cdot x} \left[ \frac{2\pi Z e n^\mu \delta(q^0) e^{-i\mathbf{q}\cdot\mathbf{X}}}{q^2 - i\epsilon} \right]. \quad (13.6.15)$$

This, of course, is just the usual Coulomb potential:

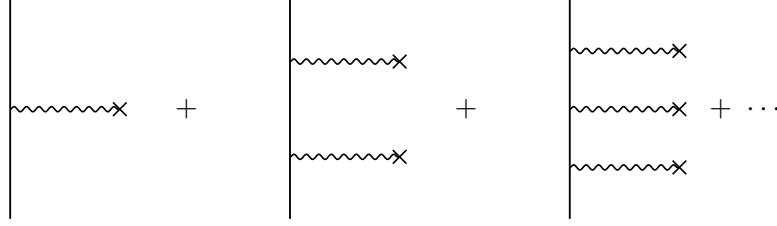
$$\mathcal{A}^0(x) = \frac{Ze}{4\pi|\mathbf{x} - \mathbf{X}|}, \quad \mathcal{A}(x) = 0. \quad (13.6.16)$$

If there is more than one heavy charged particle (as in a molecule) we must express  $\mathcal{A}^\mu(x)$  as a sum of terms like (13.6.16), each with its own charge  $Ze$  and position  $\mathbf{X}$ .

It is useful to keep in mind what diagrams we are summing in using the external field approximation. Consider the interaction of a single electron (relativistic or non-relativistic) with a single heavy charged particle such as a proton or deuteron. If we ignore all other interactions, then the Feynman diagrams for the scattering of the electron due to its interaction with the external field are just those with any number of insertions of the electron–external field vertex (13.6.14) in the electron line. (See Figure 13.4.) But as shown by the sum over permutations in Eq. (13.6.2), these diagrams in the external field approximation come from diagrams in the underlying theory in which the photons attached to the electron line are attached to the heavy charged particle line in all possible orders. (See Figure 13.5.) The ‘uncrossed ladder’ diagrams (labelled L) of Figure 13.5 do not dominate this sum unless the electron as well as the heavy charged particle is non-relativistic. (These diagrams include contributions from terms in old-fashioned perturbation theory whose intermediate states contain the same particles as the initial and final states, leading to small energy denominators when the electron and heavy charged particle are both non-relativistic, while all

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<sup>152</sup>The first factor here is the usual factor of  $i$  accompanying the constants in the interaction Lagrangian of the heavy charged particle in the Feynman rules.



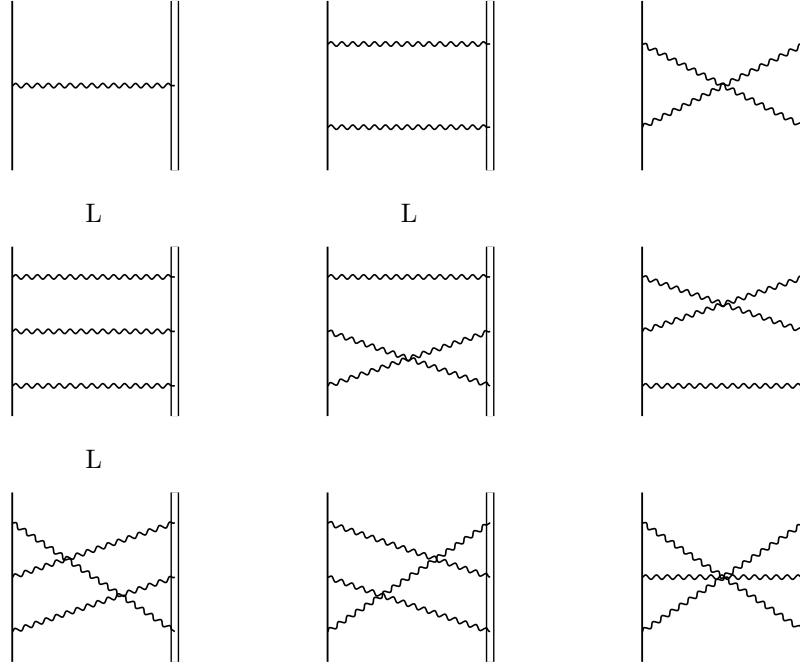
**Figure 13.4.** Diagrams for the scattering of an electron by an external electromagnetic field. Here straight lines represent the electron; wavy lines ending in crosses represent its interaction with an external field.

other diagrams of Figure 13.5 correspond to intermediate states with either extra photons, electron–positron pairs, or heavy particle–antiparticle pairs, which are suppressed by large energy denominators.) The uncrossed ladders can be summed by solving an integral equation, known as the Bethe–Salpeter equation [10], but there is no rationale for selecting out this subset of diagrams unless both particles are non-relativistic, in which case the Bethe–Salpeter equation just reduces to the ordinary non-relativistic Schrödinger equation, plus relativistic corrections associated with the spin–orbit coupling that can be treated as small perturbations. It must be said that the theory of relativistic effects and radiative corrections in bound states is not yet in entirely satisfactory shape.

In the derivation of the external field (13.6.16) we evaluated the interaction of the heavy charged particle with the electromagnetic field only to leading order in the photon momentum. There are corrections of higher order in the photon momentum arising from the heavy particle’s magnetic dipole moment, electric quadrupole moment, etc. Also, of course, there are radiative corrections arising from Feynman diagrams beyond those of Figure 13.4, such as diagrams in which photons are emitted and absorbed from the electron line or electron loops are inserted into photon lines. We shall see in the next chapter that in bound states the diagrams of Figure 13.4 must be included to all orders, but all other corrections of higher order in photon momenta or  $e$  may be included as perturbations to these diagrams.

### Weinberg Exercises

- W.13.1.** Consider the process  $e^+ + e^- \rightarrow \pi^+ + \pi^-$  in the center-of-mass frame at an energy of 1 GeV and scattering angle of  $90^\circ$ . Suppose that by measuring the energies of the final pions we determine that an energy of not more than  $E_T \ll 1$  GeV is emitted in soft photons. How does the reaction rate depend on  $E_T$ ?
- W.13.2.** Consider a massless spinless particle described by a scalar field  $\phi$ , whose interaction Lagrangian density is of the form  $\phi(x)J(x)$ , where  $J(x)$  involves only massive particle fields. Derive a formula for the rate of emission of arbitrary numbers of soft scalar particles in a process  $\alpha \rightarrow \beta$ , with these soft scalars having a total energy less than some small quantity  $E_T$ . Include radiative corrections due to soft scalars with energy less than some small quantity  $\Lambda$ .



**Figure 13.5.** Diagrams for the scattering of an electron by a heavy, charged target particle, which in the limit of large target mass yield the same result as the diagrams of Figure 13.4. Here the single straight line is the electron; the double straight line is the heavy target particle; and wavy lines are virtual photons. Diagrams marked ‘L’ are called uncrossed ladder graphs; they dominate the sum when the electron as well as the target particle is non-relativistic.

**W.13.3.** Derive a formula for the next term beyond Eq. (13.5.14) for low-energy photon scattering on an arbitrary target.

**W.13.4.** Suppose that a spin one particle of very small mass  $m$  is described by a vector field  $V^\mu(x)$ , which couples only to a much heavier fermion described by a Dirac field  $\psi(x)$ , with an interaction Lagrangian density of the form  $gV^\mu\bar{\psi}\gamma_\mu\psi$ . Suppose that the heavy fermion normally decays into other particles that have no interaction with  $V^\mu$ , releasing an energy  $W$  much greater than  $m$ . Consider such a decay process, but where an additional  $V^\mu$  particle is emitted along with the other decay products, with the vector particle energy less than an upper limit  $E$ , in the range  $W \gg E \gg m$ . How does the rate of this process depend on  $E$  and  $m$ ? Ignore radiative corrections.

**W.13.5.** Prove that Eq. (13.6.8) holds when  $p \cdot (q_1 + \cdots + q_N) = 0$ .

### Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg’s original text.*

**W.13.1.** Write

$$f(\beta) = \frac{1}{\beta} \ln \frac{1+\beta}{1-\beta}.$$

In the center-of-mass frame the four signed charges  $c_n = \eta_n e_n / e$ , in the order  $e^+, e^-, \pi^+, \pi^-$ , are  $(-1, +1, +1, -1)$ . At  $90^\circ$  every electron–pion relative speed is the same, so the four mixed terms in the double sum in Eq. (13.2.8) cancel. The two opposite-particle pairs remain, including both orderings, and the diagonal terms give  $f(0) = 2$ . Hence

$$A = \frac{\alpha}{\pi} \{f(\beta_e) + f(\beta_\pi) - 4\}, \quad \beta_a = \frac{2v_a}{1 + v_a^2}, \quad v_a = \sqrt{1 - \frac{4m_a^2}{s}}, \quad s = (1 \text{ GeV})^2.$$

Numerically  $f(\beta_e) \simeq 30.31$ ,  $f(\beta_\pi) \simeq 7.80$ , and  $A \simeq 7.9 \times 10^{-2}$ .

There is no separate single-photon resolution in the measurement stated in the question, so set  $E = E_T$  in Eq. (13.3.17). The inclusive rate is

$$\Gamma(E_T) = F(1; A) \left( \frac{E_T}{\Lambda} \right)^A \Gamma^\Lambda \propto E_T^{0.079}.$$

The unphysical separator  $\Lambda$  is cancelled by the hard rate  $\Gamma^\Lambda$ . Also  $F(1; A) = 1 - \pi^2 A^2 / 12 + O(A^3)$ , so it does not alter the requested power of  $E_T$ .

**W.13.2.** For each massive external particle define the on-shell soft residue  $g_n$  of  $J$ . With  $\eta_n = +1$  for an outgoing leg and  $-1$  for an incoming leg, emission of one scalar of momentum  $q = \omega(1, \hat{\mathbf{q}})$  has the universal limit

$$\mathcal{M}_{\beta\alpha}(q) = \left[ \sum_n \frac{\eta_n g_n}{p_n \cdot q - i\eta_n 0} \right] \mathcal{M}_{\beta\alpha} + O(\omega^0).$$

The leading term for  $N$  scalars is the product of  $N$  such factors. Real-emission phase space and the virtual soft graphs therefore exponentiate exactly as in the photon calculation, with

$$A_\phi = \frac{1}{2(2\pi)^3} \int d\Omega_{\hat{\mathbf{q}}} \left| \sum_n \frac{\eta_n g_n}{p_n^0 - \mathbf{p}_n \cdot \hat{\mathbf{q}}} \right|^2 \geq 0.$$

Equivalently,

$$A_\phi = -\frac{1}{8\pi^2} \sum_{n,m} \frac{\eta_n \eta_m g_n g_m}{(p_n \cdot p_m) \beta_{nm}} \ln \frac{1 + \beta_{nm}}{1 - \beta_{nm}},$$

with the coincident-velocity value understood by continuity.

After summing every multiplicity whose total scalar energy is below  $E_T$ , and including virtual scalars below  $\Lambda$ , the regulator-independent answer is

$$\Gamma_\phi(E_T) = F(1; A_\phi) \left( \frac{E_T}{\Lambda} \right)^{A_\phi} \Gamma_\phi^\Lambda.$$

Here  $\Gamma_\phi^\Lambda$  contains the hard radiative corrections. A temporary scalar mass or lower energy cutoff cancels between its real and virtual appearances.

**W.13.3.** Let  $\omega$  collectively denote the incident and outgoing photon energies. The low-energy theorem is most compactly stated by introducing the target's on-shell electromagnetic vertex through first order in the momentum transfer,

$$\langle p', \sigma' | J^0(0) | p, \sigma \rangle = e \delta_{\sigma' \sigma} + O(k^2),$$

$$\langle p', \sigma' | \mathbf{J}(0) | p, \sigma \rangle = \left[ \frac{e(\mathbf{p}' + \mathbf{p})}{2m} + \frac{i\mu}{j} \mathbf{J} \times (\mathbf{p}' - \mathbf{p}) \right]_{\sigma' \sigma} + O(k^2).$$

Form the two elastic pole graphs with this vertex, one with intermediate momentum  $p + q$  and the crossed one with momentum  $p - q'$ , and add the contact term required by the two Ward identities. Denote the resulting gauge-invariant tensor by  $B^{\mu\nu}(e, m, \mu, j; q', q)$ . This gives the following explicit prescription for the term asked for:

$$\mathcal{M} = -\frac{e^2}{m} \epsilon'^* \cdot \epsilon \omega \frac{\partial}{\partial \omega} \left[ \epsilon'^*_{\mu} B^{\mu\nu}(e, m, \mu, j; \omega q', \omega q) \epsilon_{\nu} \right] \Big|_{\omega=0} + O(\omega^2).$$

This is often called the Born form of the subleading Low–Gell-Mann–Goldberger theorem. It is a calculational formula: the displayed current fixes every pole residue, while the Ward identities fix the seagull. Thus the entire  $O(\omega)$  coefficient depends only on  $e, m, \mu$  and the spin matrices; charge radii, polarizabilities, and all other target structure first enter at  $O(\omega^2)$ . For a spinless target the magnetic term is absent.

**W.13.4.** Take the decaying fermion at rest,  $p^{\mu} = (M, \mathbf{0})$ , and let  $q^{\mu} = (\omega, \mathbf{q})$ . The soft attachment to the incoming fermion gives

$$\mathcal{M}_V \simeq g \frac{p \cdot \epsilon(q)}{p \cdot q} \mathcal{M}_0.$$

For the two transverse polarizations  $p \cdot \epsilon_T = 0$  at this order. For the longitudinal polarization,

$$\epsilon_L^{\mu} = \frac{1}{m} (|\mathbf{q}|, \omega \hat{\mathbf{q}}), \quad \left| \frac{p \cdot \epsilon_L}{p \cdot q} \right|^2 = \frac{\mathbf{q}^2}{m^2 \omega^2}.$$

Neglecting recoil corrections of relative order  $E/W$ , factorization of the remaining decay phase space therefore gives

$$\frac{\Gamma_V(E)}{\Gamma_0} = \frac{g^2}{4\pi^2 m^2} \int_0^{\sqrt{E^2 - m^2}} dq \frac{q^4}{(q^2 + m^2)^{3/2}}.$$

In the range specified in the question this is

$$\Gamma_V(E) = \Gamma_0 \frac{g^2}{8\pi^2} \frac{E^2}{m^2} \left[ 1 + O\left( \frac{m^2}{E^2} \ln \frac{E}{m}, \frac{E}{W} \right) \right].$$

The power enhancement, rather than a soft logarithm, comes from emission of the longitudinal mode by a current that is not conserved by the decay interaction.

**W.13.5.** Put  $x_r = p \cdot q_r$ , so that  $\sum_r x_r = 0$ , and use

$$\frac{1}{x - i0} = i \int_0^\infty dt e^{-ixt}.$$

For a fixed permutation the  $N - 1$  denominators in Eq. (13.6.8) become integrals over the successive positive time separations between  $N$  ordered insertions. Summing over all permutations partitions the whole space of relative insertion times: away from coincident times, exactly one ordering applies. The common translation drops out because  $\sum_r x_r = 0$ . Fourier transformation over the remaining  $N - 1$  unrestricted relative times then gives

$$(2\pi i)^{N-1} \delta(x_1) \cdots \delta(x_{N-1}).$$

Consequently, as a distribution on the hyperplane  $\sum_r x_r = 0$ ,

$$\sum_{\pi \in S_N} \prod_{s=1}^{N-1} \frac{1}{x_{\pi(1)} + \cdots + x_{\pi(s)} - i0} = (2\pi i)^{N-1} \prod_{r=1}^{N-1} \delta(x_r),$$

which is Eq. (13.6.8). Coincident-time boundaries have measure zero and are already assigned correctly by the common  $i0$  prescription.

### Supplementary Exercises

#### S.13.1. Soft Power Counting in the QED Vertex and Why the Inclusive Rate Is Observable

(Inspired by Hong Liu and MIT OpenCourseWare, 8.324 Lecture 16.)

- In the one-loop on-shell QED vertex, let the photon momentum  $k$  become soft while every external invariant remains fixed. Power count the measure, the photon propagator, and the two neighboring electron propagators. Show that the soft region is logarithmically divergent in four dimensions.
- Explain why a detector with resolution  $E$  cannot distinguish a hard charged final state from that state accompanied by photons of energy below  $E$ . Write the order- $e^2$  inclusive probability and identify the real-virtual cancellation that makes it finite.

#### S.13.2. The Eikonal Factor from an External Fermion

(Inspired by Hong Liu and MIT OpenCourseWare, 8.324 Lecture 16.)

Attach a photon of momentum  $q$  to an incoming or outgoing on-shell fermion. Use the Dirac equation to derive the leading factor  $\eta e p \cdot \epsilon / (p \cdot q - i\eta 0)$ , including the sign  $\eta$ .

#### S.13.3. Gauge Terms in the Polarization Sum

(Inspired by Hong Liu and MIT OpenCourseWare, 8.324 Lecture 16.)

Show that replacing the physical photon polarization sum by a covariant sum changes a soft-emission probability only by terms proportional to  $\sum_n \eta_n e_n$ . Explain precisely why charge conservation removes those terms.

#### S.13.4. Soft Photons

(Adapted from John McGreevy, *Physics 215B QFT, Winter 2025, Assignment 5, Problem 4.*)

Let an electron be deflected from on-shell momentum  $p$  to  $p'$ , with  $p^2 = p'^2 = -m_e^2$ , and write  $q = p' - p$ . Regulate the infrared region by giving the photon a small mass  $m_\gamma$ . In the mostly-plus convention of this volume a spacelike transfer has  $Q^2 \equiv q^2 > 0$ , and write  $\overline{d^4k} \equiv d^4k/(2\pi)^4$ .

Check that the contribution from a single virtual photon (Eq. (2.60) of McGreevy's lecture notes, with  $n = 2$ ) is

$$\begin{aligned} \frac{e^2}{2} \int \overline{d^4k} \frac{-i\eta_{\rho\sigma}}{k^2 + m_\gamma^2 - i0} \left( \frac{p'}{p' \cdot k} - \frac{p}{p \cdot k} \right)^\rho \left( \frac{p'}{-p' \cdot k} - \frac{p}{-p \cdot k} \right)^\sigma \\ = -\frac{\alpha}{2\pi} f_{\text{IR}}(Q^2) \ln\left(\frac{Q^2}{m_\gamma^2}\right) + \text{finite}, \end{aligned}$$

where

$$f_{\text{IR}}(Q^2) = \int_0^1 dx \frac{m_e^2 + Q^2/2}{m_e^2 + x(1-x)Q^2} - 1.$$

Hints: Wick rotate. Scale out the overall magnitude of  $k$ , writing  $k^\mu = k \hat{k}^\mu$ . Use a Feynman parameter to combine  $(p \cdot \hat{k})(p' \cdot \hat{k})$ . Observe that this same integral appears in the cross section involving the emission of one real soft photon.

*Editorial note.* The source prints one dummy integration variable in the measure and another in the integrand; the display above consistently calls that variable  $k$ . The signs of  $Q^2$  and  $m_\gamma^2$  have also been translated from the source's mostly-minus convention.

#### S.13.5. The Infrared Vertex Function

(Inspired by Nicolai Lang, *University of Stuttgart, QFT Problem Set 10.*)

- (a) Regulate the on-shell QED vertex by a photon mass  $\lambda$ . After Feynman parametrization, identify the parameter-space endpoint that produces the infrared divergence of the Dirac form factor  $F_1(q^2)$ , and extract the coefficient of  $\ln \lambda$  without evaluating finite terms.
- (b) For the three-denominator vertex simplex, set  $x = uv$ ,  $y = u(1-v)$ ,  $z = 1-u$ . Compute the Jacobian and show how the soft corner becomes  $u \rightarrow 1$ . Use the new variables to separate the logarithmically singular radial integral from a finite angular integral.

- (c) For spacelike transfer write the coefficient of the photon-mass logarithm as an integral over one Feynman parameter. Show that it is nonnegative, vanishes at zero transfer after charge normalization, and grows logarithmically at large spacelike momentum.

### S.13.6. Bremsstrahlung and Regulator Cancellation

(Inspired by Nicolai Lang, University of Stuttgart, QFT Problem Set 10.)

- (a) For an electron deflected from momentum  $p$  to  $p'$ , derive the soft-radiation factor

$$e^2 \int_{\lambda < \omega < E} \frac{d^3 q}{(2\pi)^3 2\omega} \left( \frac{p'}{p' \cdot q} - \frac{p}{p \cdot q} \right)^2.$$

Perform the energy integral and leave the finite angular coefficient in Lorentz-invariant form.

- (b) Combine the infrared part of the on-shell vertex correction with the bremsstrahlung probability of Exercise S.13.10. Show explicitly that the coefficient of  $\ln \lambda$  cancels and state which logarithm of the experimental resolution survives.
- (c) Compare dimensional regularization, a Pauli–Villars subtraction, and a small photon mass in the electron self-energy. Identify which endpoint each regulator controls. Explain why an ultraviolet counterterm cannot remove a dependence on experimental soft resolution.

### S.13.7. Equivalent-Photon Approximation

(Adapted from John McGreevy, *Physics 215B QFT, Winter 2025, Assignment 6, Problem 6.*)

Consider a process in which very-high-energy electrons scatter from a target. At leading order in  $\alpha$ , the electron line is connected to the rest of the diagram by a single photon propagator. If the initial and final electron energies are  $E$  and  $E'$ , the exchanged photon carries momentum  $q = p - p'$  with

$$q^2 = 2EE'(1 - \cos \theta)$$

when the electron mass  $m \ll E$  is neglected. Here and below the signs have been translated to the mostly-plus convention, in which this spacelike  $q$  has  $q^2 > 0$ . In the forward limit  $\theta \rightarrow 0$ , the photon approaches its mass shell. To what extent may it be treated as a real photon?

- (a) The matrix element for the scattering process can be written as

$$\mathcal{M} = (-ie) \bar{u}(p') \gamma^\mu u(p) \frac{-i\eta_{\mu\nu}}{q^2 - i0} \widehat{\mathcal{M}}^\nu(q),$$



where  $\widehat{\mathcal{M}}^\nu$  represents the coupling of the virtual photon to the target. Let  $q = (q^0, \mathbf{q})$  and define  $\tilde{q} = (q^0, -\mathbf{q})$ . Parametrize the electron current as

$$\bar{u}(p')\gamma^\mu u(p) = Aq^\mu + B\tilde{q}^\mu + C\epsilon_1^\mu + D\epsilon_2^\mu,$$

where  $\epsilon_\alpha$  are unit spacelike vectors transverse to  $\mathbf{q}$ , with  $\epsilon_\alpha^0 = 0$ . Show, by contracting with  $q$ , that  $B$  is at most of order  $\theta^2$ , and hence may be ignored at leading order about forward scattering. Why is the coefficient  $A$  irrelevant?

- (b) Work in the frame  $p = (E, 0, 0, E)$ , and compute

$$\bar{u}(p')\gamma \cdot \epsilon_\alpha u(p)$$

explicitly for massless electrons in states of definite helicity. The vectors  $\epsilon_{\alpha=\parallel, \perp}$  are respectively parallel and perpendicular to the scattering plane. Keep terms through order  $\theta$ . Note that the small  $\hat{\mathbf{z}}$  component of  $\epsilon_\parallel$  matters. For definiteness one may take  $p' = (E', E' \sin \theta, 0, E' \cos \theta)$ .

- (c) Write the electron–target cross section in terms of  $|\widehat{\mathcal{M}}^\mu|^2$  and the target phase-space integral. This expression must be integrated over the final electron momentum  $\mathbf{p}'$ ; the integral over  $p'^3$  is an integral over the electron’s energy loss. Show that the integral over  $\mathbf{p}'_\perp$  diverges logarithmically as  $\mathbf{p}'_\perp$ , or equivalently  $\theta$ , tends to zero.
- (d) The electron mass regulates the divergence that was exposed after setting  $m = 0$ . Reintroduce it in the mostly-plus form of the source’s supplied relation,

$$q^2 = 2(EE' - pp' \cos \theta) - 2m^2, \quad p = |\mathbf{p}|, \quad p' = |\mathbf{p}'|,$$

and show that the forward divergence is replaced by a factor  $\ln(s/m^2)$ .

- (e) Assemble all factors. Assuming that the target cross sections are independent of photon polarization, show that the leading part of the electron–target cross section is obtained by treating the electron as a source of real photons with energy distribution

$$N_\gamma(x) dx = \frac{dx}{x} \frac{\alpha}{2\pi} [1 + (1-x)^2] \ln\left(\frac{s}{m^2}\right), \quad x \equiv \frac{E_\gamma}{E}.$$

This is the Weizsäcker–Williams equivalent-photon approximation, a precursor of the theory of jets and partons in QCD.

### S.13.8. Collinear Logarithms and KLN Safety

(Inspired by Mark Srednicki, *Quantum Field Theory*, Chapter 26.)

- (a) Give the scalar a small mass  $m$  as a collinear regulator. Evaluate the singular part of the unresolved splitting integral and show that it has the form  $C \ln(s_{\text{res}}/m^2)$ . Express  $C$  as an integral over the momentum fraction.

- (b) Show schematically, with the same normalization on both sides of the cut, that the collinear logarithm from a virtual external-leg correction cancels the logarithm from unresolved real splitting. State the completeness sum that is essential to the cancellation.
- (c) Define a measurement function  $F_n(p_1, \dots, p_n)$ . Derive the condition on  $F_{n+1}$  when two momenta become collinear that makes an inclusive rate collinear safe. Test the condition for particle multiplicity and for total energy flow.

### S.13.9. Gauge-Invariant Charged Dressings and Branch-Point Propagators

(Inspired by Daniel Harlow, *Lecture Notes for Physics 8.324.*)

- (a) Let

$$\Psi_D(x) = \exp \left[ -ie \int d^3y \frac{\partial_i A_i(x^0, \mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|} \right] \psi(x).$$

Using  $\nabla^2(4\pi|\mathbf{x} - \mathbf{y}|)^{-1} = -\delta^3(\mathbf{x} - \mathbf{y})$ , prove that  $\Psi_D$  is gauge invariant.

- (b) Use the canonical commutator between the electric field and vector potential to compute the electric field created by  $\Psi_D^\dagger(x)$ . Show that the dressing supplies the Coulomb field required by Gauss's law.
- (c) Assume the charged two-point function in time behaves at long times as  $e^{-imt}t^{-\gamma}$ , with  $\gamma > 0$ . Fourier transform its singular part and show that the isolated pole is replaced by a branch behavior  $(p^0 - m + i0)^{\gamma-1}$ . Relate this to the absence of a bare charged one-particle state.

### S.13.10. Soft Photon Clouds, Dressed States, and Inclusive Rates

(Inspired by Daniel Harlow, *Lecture Notes for Physics 8.324.*)

- (a) For a coherent dressing with mode amplitude  $\alpha_\lambda(\mathbf{q}) \propto p \cdot \epsilon_\lambda / (p \cdot q)$ , compute the photon number and energy between  $\lambda$  and  $E$ . Show that the number is logarithmic while the energy remains finite as  $\lambda \rightarrow 0$ .
- (b) Compare two cures for the vanishing bare charged-particle  $S$ -matrix: coherent asymptotic dressings and a sum over unresolved photons. Show at leading eikonal order that both use the same classical current and explain which experimental question each construction answers.
- (c) Replace both internal electron propagators in the one-loop vertex by their eikonal limits. Derive the scalar soft integral

$$-ie^2 p \cdot p' \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 + \lambda^2 - i0)(p \cdot k - i0)(p' \cdot k + i0)}$$

and explain why its logarithm exponentiates.

### S.13.11. The Leading Soft-Photon Theorem and Crossing

(Inspired by Andrew Strominger, *Lectures on the Infrared Structure of Gravity and Gauge Theory*.)

- (a) Starting from external-line attachments, prove the leading soft theorem

$$\mathcal{M}_{n+1}(q, \epsilon) = \left[ \sum_k \eta_k e_k \frac{p_k \cdot \epsilon}{p_k \cdot q} \right] \mathcal{M}_n + O(q^0).$$

State clearly how incoming and outgoing legs are represented.

- (b) Replace  $\epsilon^\mu$  by  $q^\mu$  in the leading soft factor. Prove that invariance under this replacement is equivalent to  $\sum_k \eta_k e_k = 0$ , and interpret the result as the global Ward identity.
- (c) Cross an outgoing charged particle to an incoming antiparticle. Track the momentum, charge, and  $\eta$  signs and prove that its contribution to the soft factor is unchanged under the all-outgoing convention.

### S.13.12. Soft Gravitons?

(Adapted from John McGreevy, *Physics 215B QFT, Winter 2025, Assignment 6, Problem 5*.)

Photons are massless, so measured cross sections include soft photons that the detector does not resolve; omitting them gives infrared-divergent nonsense. Gravitons are also massless. Must they be treated in the same way? The following connected parts sketch how to answer this question.

- (a) Consider the schematic gauge-fixed action

$$S_0[h_{\mu\nu}] = \int d^4x \frac{1}{2} h_{\mu\nu} \square h^{\mu\nu}.$$

This is a kinetic term for too many polarizations of a symmetric tensor  $h_{\mu\nu} = h_{\nu\mu}$ , regarded as a small fluctuation of the metric,  $g_{\mu\nu} \simeq \eta_{\mu\nu} + h_{\mu\nu}$ . As with the photon, rely on the coupling to matter to prevent the production of unphysical polarizations. Write the propagator for  $h$ , continuing to raise and lower indices with  $\eta_{\mu\nu}$ .<sup>153</sup>

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<sup>153</sup>The source gives two warnings. First, the Einstein–Hilbert term is schematically  $\int d^4x (8\pi G_N)^{-1} \sqrt{|g|} R = \int d^4x (8\pi G_N)^{-1} (\partial h)^2 + \dots$  it contains a factor of  $G_N^{-1}$ . Since  $R$  has dimensions of inverse length squared and  $g$  is dimensionless,  $G_N$  has dimensions of length squared, with  $8\pi G_N = M_{\text{Pl}}^{-2}$ . A factor of  $\sqrt{G_N}$  has been absorbed into  $h$  so that its kinetic term has unit coefficient. Second,  $(\partial h)^2$  suppresses several index contractions, and de Donder gauge fixing is needed to obtain the simple expression above. The field denoted by  $h_{\mu\nu}$  here is actually the trace-reversed field  $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$ , with  $h = \eta^{\mu\nu} h_{\mu\nu}$ ; the source deliberately drops the bar. Details, which are not needed here, are in [Chapter 10 of McGreevy’s GR notes](#).

- (b) Couple the graviton to the electron field through

$$S_G = \int d^4x G h^{\mu\nu} T_{\mu\nu}, \quad T_{\mu\nu} \equiv \bar{\psi}(\gamma_\mu \partial_\nu + \gamma_\nu \partial_\mu)\psi.$$

What is the engineering dimension of  $G$ ? What is the new Feynman rule?

- (c) Draw the tree-level diagrams for production of gravitational radiation when an electron is accelerated by absorbing a photon,  $e\gamma \rightarrow eh$ . Evaluate them if desired, and estimate or calculate the cross section. Hint: use dimensional analysis.
- (d) Study the one-loop diagram in which a graviton corrects the QED vertex. Is it infrared divergent? If not, why not?
- (e) If the preceding part is troublesome, replace the graviton by a massless scalar  $\pi(x)$ . Compare derivative coupling  $\lambda \partial_\mu \pi \bar{\psi} \gamma^\mu \psi$  with direct Yukawa coupling  $y \pi \bar{\psi} \psi$ . Warning: although this example resembles the graviton case, its conclusion is different.
- (f) Much of the coupling of gravity to matter is fixed by coordinate invariance, which plays a role analogous to QED gauge invariance. On the small fluctuation the infinitesimal transformation is

$$h_{\mu\nu}(x) \longrightarrow h_{\mu\nu}(x) + \partial_\mu \lambda_\nu(x) + \partial_\nu \lambda_\mu(x).$$

What condition does invariance impose on the  $T_{\mu\nu}$  appearing in the coupling above?

### Solutions to Supplementary Exercises

#### S.13.1. Soft Power Counting in the QED Vertex and Why the Inclusive Rate Is Observable

- (a) In the soft region  $k^\mu \sim \rho$ , the loop measure is  $d^4k \sim \rho^4$ , the photon propagator is  $k^{-2} \sim \rho^{-2}$ , and

$$(p+k)^2 + m^2 = 2p \cdot k + O(k^2) \sim \rho$$

for each neighboring on-shell electron line. Their product therefore scales as  $\rho^4 \rho^{-2} \rho^{-1} \rho^{-1} = \rho^0$ . Equivalently the radial integral is  $d\rho/\rho$ . The soft part is logarithmic; neither the numerator nor generic hard external invariants change this conclusion.

- (b) A finite-time, finite-size detector projects onto an energy bin, not onto an exact photon-number eigenstate. Through order  $e^2$  the probability assigned to the bin is

$$P_{\text{inc}}(E) = P_0 + 2\Re(\mathcal{M}_0^* \mathcal{M}_{\text{virt}}) + \int_{\omega < E} d\Phi_\gamma |\mathcal{M}_\gamma|^2.$$

The virtual interference contains  $-A \ln(\Lambda/\lambda) P_0$ , while the real term contains  $+A \ln(E/\lambda) P_0$ . Their  $\ln \lambda$  terms cancel, leaving  $A \ln(E/\Lambda) P_0$ . An exclusive zero-photon probability is thus regulator dependent, whereas the detector-defined inclusive probability is finite.

### S.13.2. The Eikonal Factor from an External Fermion

For an outgoing fermion the relevant numerator is

$$\bar{u}(p)\gamma^\mu[-i(\not{p} + \not{q}) + m] = -2ip^\mu\bar{u}(p) + O(q),$$

where the on-shell Dirac equation was used. Its denominator is  $(p+q)^2 + m^2 - i0 = 2p \cdot q - i0 + O(q^2)$ . Contracting with the photon polarization leaves  $e p \cdot \epsilon / (p \cdot q - i0)$ . Reversing the external attachment for an incoming line reverses both the momentum flow and the pole prescription. Together,

$$\mathcal{M}_\gamma = \eta e \frac{p \cdot \epsilon}{p \cdot q - i\eta 0} \mathcal{M}_0 + O(q^0),$$

with  $\eta = +1$  outgoing and  $-1$  incoming.

### S.13.3. Gauge Terms in the Polarization Sum

A general covariant polarization sum differs from the physical one by  $q_\mu f_\nu + f_\mu q_\nu$ . Contracting  $q_\mu$  into the eikonal current

$$J_{\text{soft}}^\mu = \sum_n \eta_n e_n \frac{p_n^\mu}{p_n \cdot q}$$

gives  $q \cdot J_{\text{soft}} = \sum_n \eta_n e_n$ . Charge conservation makes this zero, so every gauge-dependent term drops out. The physical probability may consequently be evaluated with a convenient covariant sum, provided the complete set of charged external legs is retained.

### S.13.4. Soft Photons

Put

$$J^\mu(k) = \frac{p'^\mu}{p' \cdot k} - \frac{p^\mu}{p \cdot k}.$$

The second eikonal bracket in the problem is  $-J^\mu(k)$ . After the Wick rotation, write  $k_E = \kappa \hat{k}$ . The photon propagator and the two eikonal denominators separate the singular integral into a radial factor  $d\kappa/\kappa$  and a finite angular factor. For the mixed term use

$$\frac{1}{(p \cdot \hat{k})(p' \cdot \hat{k})} = \int_0^1 dx \frac{1}{[(xp + (1-x)p') \cdot \hat{k}]^2}.$$

The interpolating momentum obeys

$$[xp + (1-x)p']^2 = -m_e^2 - x(1-x)Q^2, \quad p \cdot p' = -m_e^2 - \frac{Q^2}{2}.$$

Consequently the angular integral of the mixed term, minus the two self-terms, is proportional to

$$\int_0^1 dx \left[ \frac{m_e^2 + Q^2/2}{m_e^2 + x(1-x)Q^2} - 1 \right] = f_{\text{IR}}(Q^2).$$

All dependence on the photon mass is now in the radial integral. Its singular part is

$$\int_0^{O(Q)} \frac{\kappa d\kappa}{\kappa^2 + m_\gamma^2} = \frac{1}{2} \ln\left(\frac{Q^2}{m_\gamma^2}\right) + O(1).$$

Keeping the measure and the prefactor  $e^2/2$ , the angular normalization then gives

$$I_{\text{virt}}^{\text{IR}} = -\frac{\alpha}{2\pi} f_{\text{IR}}(Q^2) \ln\left(\frac{Q^2}{m_\gamma^2}\right),$$

with everything independent of  $m_\gamma$  included in “finite.” As a check,  $f_{\text{IR}}(0) = 0$ , because an undeflected charge does not radiate.

Cutting the virtual photon replaces its propagator by the positive-energy on-shell delta function. The remaining eikonal current is the same  $J^\mu(k)$ , so precisely the same angular integral  $f_{\text{IR}}(Q^2)$  multiplies the real-emission logarithm. The opposite real–virtual signs are the origin of the inclusive cancellation.

### S.13.5. The Infrared Vertex Function

- (a) After combining the photon and two electron denominators, the singular part has the form

$$F_1(q^2)|_{\text{IR}} = -\frac{\alpha}{2\pi} \int_0^1 dv c(v, q^2) \int_0^1 \frac{du}{(1-u)m^2 + u\lambda^2 + \dots}.$$

Only the corner in which the photon parameter tends to one, equivalently the two electron parameters vanish together, can make all three denominators soft. Holding the ratio  $v$  fixed, the radial integral is  $-\ln \lambda^2$  plus finite terms. Therefore

$$F_1(q^2)|_{\text{IR}} = \frac{\alpha}{2\pi} \ln \frac{\lambda^2}{m^2} \int_0^1 dv c(v, q^2),$$

which extracts the full infrared coefficient without computing the regular region.

- (b) The inverse map is  $u = x + y$ ,  $v = x/(x + y)$ , and

$$\left| \frac{\partial(x, y)}{\partial(u, v)} \right| = u.$$

Thus  $dx dy dz \delta(1 - x - y - z) = u du dv$ . If  $z$  is assigned to the photon line, its soft corner is  $z \rightarrow 1$ , or  $u \rightarrow 0$ ; with the opposite conventional assignment it is  $u \rightarrow 1$ . In either labeling set  $r$  equal to the sum of the two small electron parameters. The singular piece separates as

$$\int_0^1 dv C(v) \int_0^1 \frac{r dr}{r^2 m^2 + \lambda^2},$$

and the last integral is logarithmic. The  $v$  integral is finite and supplies the kinematic coefficient.

- (c) For spacelike  $q^2 = Q^2 > 0$ , an economical representation is

$$c(Q^2) = \int_0^1 dv \left[ \frac{m^2 + \frac{1}{2}Q^2}{m^2 + Q^2 v(1-v)} - 1 \right].$$

After symmetrizing  $v \leftrightarrow 1-v$ , the integrand can be rewritten as a positive multiple of  $Q^2(1-2v)^2/[m^2 + Q^2 v(1-v)]$ . Hence  $c(Q^2) \geq 0$ , and  $c(0) = 0$ , as required by the charge subtraction  $F_1(0) = 1$ . For  $Q^2 \gg m^2$ , the endpoint regions  $v \sim m^2/Q^2$  and  $1-v \sim m^2/Q^2$  yield  $c(Q^2) = \ln(Q^2/m^2) + O(1)$ .

### S.13.6. Bremsstrahlung and Regulator Cancellation

- (a) The incoming and outgoing attachments give the current displayed in the problem. Put  $q = \omega \hat{q}$ . Since each denominator is linear in  $\omega$ , the phase-space measure separates:

$$P_{\text{soft}} = \frac{e^2}{2(2\pi)^3} \ln \frac{E}{\lambda} \int d\Omega \left( \frac{p'}{p' \cdot \hat{q}} - \frac{p}{p \cdot \hat{q}} \right)^2.$$

Writing  $\beta$  for the relative speed of  $p$  and  $p'$ , the angular integral gives

$$P_{\text{soft}} = \frac{\alpha}{\pi} \left[ \frac{1}{\beta} \ln \frac{1+\beta}{1-\beta} - 2 \right] \ln \frac{E}{\lambda}.$$

The bracket is Lorentz invariant and nonnegative.

- (b) Define

$$C(\beta) = \frac{\alpha}{\pi} \left[ \frac{1}{\beta} \ln \frac{1+\beta}{1-\beta} - 2 \right].$$

The virtual interference is  $-C(\beta) \ln(\Lambda/\lambda) P_0$ , whereas unresolved real emission is  $+C(\beta) \ln(E/\lambda) P_0$ . Their sum is

$$\Delta P_{\text{inc}} = C(\beta) \ln \frac{E}{\Lambda} P_0 + \text{finite}.$$

The fictitious photon mass cancels. The surviving logarithm involves the detector resolution  $E$  and the hard matching scale  $\Lambda$ , as a physical inclusive observable must.

- (c) Large loop momentum produces the ultraviolet endpoint. Dimensional continuation or a Pauli–Villars field suppresses that endpoint and fixes local counterterms. The infrared endpoint occurs when the photon momentum vanishes while adjacent charged propagators go on shell; a photon mass  $\lambda$  or  $d > 4$  regulates it. Dimensional regularization can label both poles, but their momentum regions and signs remain distinct. A local UV counterterm cannot contain a detector threshold  $E$ , so it cannot remove  $\ln E$ . Only the real–virtual inclusive sum replaces the unphysical infrared regulator by that physical resolution.

### S.13.7. Equivalent-Photon Approximation

- (a) Take  $q = p - p'$ . For a massless electron the Dirac equations give the electron Ward identity

$$q_\mu \bar{u}(p') \gamma^\mu u(p) = \bar{u}(p') (\not{p} - \not{p}') u(p) = 0.$$

The two transverse polarization vectors have zero contraction with  $q$ , so the parametrization in the problem implies

$$0 = Aq^2 + Bq \cdot \tilde{q}, \quad B = -A \frac{q^2}{q \cdot \tilde{q}}.$$

Now  $q^2 = 2EE'(1 - \cos \theta) = O(EE'\theta^2)$ , whereas  $q \cdot \tilde{q} = -(q^0)^2 - |\mathbf{q}|^2$  is generically of hard order. Thus  $B/A = O(\theta^2)$ . Finally the target Ward identity  $q_\nu \widehat{\mathcal{M}}^\nu = 0$  removes the entire term proportional to  $A$ .

- (b) One representation compatible with the volume conventions is obtained by taking  $\gamma^\mu = i\gamma_W^\mu$ , where  $\gamma_W^\mu$  are the usual Weyl matrices for the opposite metric signature. A convenient phase choice gives

$$\begin{aligned} u_+(p) &= \sqrt{2E}(0, 0, 1, 0)^\top, & u_-(p) &= \sqrt{2E}(0, 1, 0, 0)^\top, \\ u_+(p') &= \sqrt{2E'}(0, 0, c, s)^\top, & u_-(p') &= \sqrt{2E'}(-s, c, 0, 0)^\top, \end{aligned}$$

where  $c = \cos(\theta/2)$  and  $s = \sin(\theta/2)$ . With  $q = p - p'$ , choose

$$\epsilon_\perp = (0, 0, 1, 0), \quad \epsilon_\parallel = \frac{(0, E - E' \cos \theta, 0, E' \sin \theta)}{\sqrt{E^2 + E'^2 - 2EE' \cos \theta}}.$$

Both are unit spacelike vectors orthogonal to  $\mathbf{q}$ . For  $E > E'$ ,

$$\epsilon_\parallel = \left(0, 1, 0, \frac{E'}{E - E'}\theta\right) + O(\theta^2).$$

Using  $\bar{u} = u^\dagger \beta$ , direct multiplication yields, for helicity  $h = \pm 1$ ,

$$\begin{aligned} \bar{u}_h(p') \gamma \cdot \epsilon_\perp u_h(p) &= h\sqrt{EE'}\theta + O(\theta^2), \\ \bar{u}_h(p') \gamma \cdot \epsilon_\parallel u_h(p) &= -i\sqrt{EE'} \frac{E + E'}{E - E'}\theta + O(\theta^2), \\ \bar{u}_h(p') \gamma \cdot \epsilon_\alpha u_{-h}(p) &= 0. \end{aligned}$$

Individual phases change if the helicity spinors are rephased, but the squared magnitudes and the essential conclusion do not: every nonzero transverse matrix element is  $O(\theta)$ . The longitudinal component of  $\epsilon_\parallel$  is required to obtain the factor  $(E + E')/(E - E')$ .

- (c) Define the inclusive target tensor

$$W^{\mu\nu}(q) = \sum_X \int d\Phi_X (2\pi)^4 \delta^4(P + q - p_X) \widehat{\mathcal{M}}_X^\mu(q) \widehat{\mathcal{M}}_X^\nu(q)^*.$$

Apart from the standard incident flux  $\mathcal{F}_{eT}$ , the inclusive cross section is

$$d\sigma_{eT} = \frac{1}{\mathcal{F}_{eT}} \frac{d^3p'}{(2\pi)^3 2E'} \frac{e^2}{(q^2)^2} j_\mu j_\nu^* W^{\mu\nu}(q), \quad j^\mu = \bar{u}(p') \gamma^\mu u(p).$$



Only the transverse projection contributes at leading order. Part (b) gives  $j_\perp j_\perp^* = O(EE'\theta^2)$ , while  $(q^2)^2 = O(E^2 E'^2 \theta^4)$ . At fixed energy loss,  $d^2 \mathbf{p}'_\perp \simeq E'^2 \theta d\theta d\phi$ . Hence the singular part is

$$\int \frac{\theta d\theta \theta^2}{\theta^4} = \int \frac{d\theta}{\theta},$$

which is logarithmic.

- (d) Let  $x = (E - E')/E$ , so  $E' = (1 - x)E$  to leading power. Expanding  $p = \sqrt{E^2 - m^2}$ ,  $p' = \sqrt{E'^2 - m^2}$ , and the supplied exact kinematic relation at small angle gives

$$q^2 = EE'\theta^2 + m^2 \frac{(E - E')^2}{EE'} + O(\theta^4, m^2 \theta^2) = \frac{\mathbf{p}'_\perp^2 + x^2 m^2}{1 - x} + \dots$$

Thus  $q^2$  cannot fall below  $q_{\min}^2 = m^2 x^2 / (1 - x)$ . With a hard upper limit of order  $s$ ,

$$\int_{q_{\min}^2}^{q_{\max}^2} \frac{dq^2}{q^2} = \ln\left(\frac{s}{m^2}\right) + O(1),$$

where the omitted terms depend on  $x$  but are not enhanced by the large collinear logarithm.

- (e) Projecting the spin-averaged electron tensor onto the two transverse polarizations, or equivalently squaring the bilinears in part (b), gives the universal collinear splitting probability

$$dN_\gamma = \frac{\alpha}{2\pi} \frac{1 + (1 - x)^2}{x} dx \frac{dq^2}{q^2} + \text{terms without a collinear logarithm.}$$

In the nearly on-shell region the target tensor is the same one that enters the real-photon target cross section. If that cross section is independent of polarization, the electron–target result therefore factorizes as

$$d\sigma_{eT} = N_\gamma(x) dx d\sigma_{\gamma T}(xE) + \text{subleading terms.}$$

Using the integral from part (d) gives

$$N_\gamma(x) dx = \frac{dx}{x} \frac{\alpha}{2\pi} [1 + (1 - x)^2] \ln\left(\frac{s}{m^2}\right),$$

which is the Weiszäcker–Williams spectrum requested.

### S.13.8. Collinear Logarithms and KLN Safety

- (a) A small mass changes the singular denominator to  $k_\perp^2 + m^2 h(z)$ , with a positive  $h(z)$  of order unity away from the endpoints. The singular integral is

$$C \int_0^{z(1-z)s_{\text{res}}} \frac{d\kappa^2}{\kappa^2 + m^2 h(z)} = C \left[ \ln \frac{s_{\text{res}}}{m^2} + \ln \frac{z(1-z)}{h(z)} + O(m^2/s_{\text{res}}) \right].$$

Thus the logarithmic coefficient is

$$C = \mathcal{N} g^2 \int_0^1 dz z(1-z),$$

where  $\mathcal{N}$  is the fixed phase-space normalization. Regulator details affect only the finite  $z$ -integral.

- (b) Cutting the external self-energy graph gives exactly the unresolved one-to-two phase space with the same squared splitting amplitude. Unitarity therefore fixes

$$2\Re \delta Z_{\text{coll}} = - \int_{\text{degenerate}} d\Phi_2 |\mathcal{S}_{1 \rightarrow 2}|^2.$$

Multiplying the left side by the hard one-particle rate and adding the right side as a real final state cancels  $\ln(s_{\text{hard}}/m^2)$  against  $\ln(s_{\text{res}}/m^2)$ . The completeness sum must include every one- and two-particle state identified by the measurement; omitting the degenerate state destroys the cancellation.

- (c) Infrared safety requires

$$F_{n+1}(\dots, zP, (1-z)P, \dots) \longrightarrow F_n(\dots, P, \dots)$$

in every collinear limit. Then real and virtual singularities multiply the same measurement weight and cancel. Particle multiplicity fails: it changes from  $n$  to  $n+1$ . Total energy flow into a fixed angular cell passes because  $zP^0 + (1-z)P^0 = P^0$ . The same criterion is the measurement-function form of the KLN degeneracy sum.

### S.13.9. Gauge-Invariant Charged Dressings and Branch-Point Propagators

- (a) Under  $A_i \rightarrow A_i + \partial_i \alpha$ , the exponent changes by

$$-ie \int d^3y \frac{\nabla^2 \alpha(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|} = +ie \alpha(\mathbf{x}),$$

after integrating by parts and taking  $\alpha$  to vanish at spatial infinity. The exponential thus contributes  $e^{+ie\alpha(x)}$ , while  $\psi(x) \rightarrow e^{-ie\alpha(x)}\psi(x)$  in the stated convention. The phases cancel, so  $\Psi_D$  is gauge invariant.

- (b) Using  $[E_i(\mathbf{z}), A_j(\mathbf{y})] = -i\delta_{ij}\delta^3(\mathbf{z} - \mathbf{y})$ , the Baker–Campbell–Hausdorff series terminates after the first commutator. Acting on the vacuum,

$$E_i(\mathbf{z})\Psi_D^\dagger(\mathbf{x})|0\rangle = \frac{e(z-x)_i}{4\pi|\mathbf{z} - \mathbf{x}|^3}\Psi_D^\dagger(\mathbf{x})|0\rangle + \text{vacuum fluctuations}.$$

Its divergence is  $e\delta^3(\mathbf{z} - \mathbf{x})$ , so the dressing supplies exactly the Coulomb flux demanded by Gauss’s law.

- (c) Near threshold set  $\Delta = p^0 - m$ . The long-time contribution is

$$\int_0^\infty dt e^{i(\Delta+i0)t} t^{-\gamma} = e^{i\pi(1-\gamma)/2} \Gamma(1-\gamma) (\Delta+i0)^{\gamma-1}.$$

For  $\gamma = 0$  this reduces to a simple pole. Any positive noninteger  $\gamma$  instead gives a branch point and a continuum spectral density starting at  $m$ . The charged excitation is inseparable from arbitrarily soft photons, so a bare charged Fock state is not an isolated eigenstate.

### S.13.10. Soft Photon Clouds, Dressed States, and Inclusive Rates

- (a) For  $q = \omega \hat{q}$ , the coherent occupation is

$$N = \sum_\lambda \int \frac{d^3 q}{(2\pi)^3 2\omega} |\alpha_\lambda(\mathbf{q})|^2 = A \int_\lambda^E \frac{d\omega}{\omega} = A \ln \frac{E}{\lambda},$$

where the finite angular integral is denoted by  $A$ . Its energy is

$$E_{\text{cloud}} = A \int_\lambda^E d\omega = A(E - \lambda).$$

Thus the dressing leaves every finite-energy measurement well defined while containing infinitely many zero-energy quanta.

- (b) Both constructions use

$$J_{\text{as}}^\mu(q) = \sum_n \eta_n e_n \frac{p_n^\mu}{p_n \cdot q}.$$

Exponentiating  $J_{\text{as}}$  as a displacement operator builds the coherent photon cloud into the incoming and outgoing states; exponentiating its squared norm in a real-emission sum traces instead over unresolved photon states. The first defines an infrared-finite amplitude between specified dressed charges. The second predicts the probability assigned by a finite-resolution detector. Their identical eikonal current explains the same cancellation of the regulator, though the observables are not identical.

- (c) For  $k \rightarrow 0$ ,

$$\frac{-i(\not{p} + \not{k}) + m}{(p+k)^2 + m^2 - i0} \longrightarrow \frac{\not{p} - im}{2p \cdot k - i0},$$

and similarly on the other external line. The Dirac equations reduce the numerator to  $p \cdot p'$ , yielding precisely the integral in the problem. Its radial part is logarithmic. Multiple soft virtual photons attach to the same classical external-line current; disconnected attachments carry the factor  $1/N!$ , so their sum is the exponential of this one-loop soft integral.

### S.13.11. The Leading Soft-Photon Theorem and Crossing

- (a) Only attachments to external charged lines possess a propagator that goes on shell as  $q \rightarrow 0$ . Expanding that propagator and using the appropriate external wave equation gives

$$\mathcal{M}_{n+1} = \left[ \sum_k \eta_k e_k \frac{p_k \cdot \epsilon}{p_k \cdot q - i\eta_k 0} \right] \mathcal{M}_n + O(q^0).$$

Here  $p_k$  is a positive-energy physical momentum and  $\eta_k = +1$  for an outgoing state and  $-1$  for an incoming state. Internal attachments remain off shell and therefore contribute only to the  $O(q^0)$  remainder.

- (b) The replacement  $\epsilon^\mu \rightarrow \epsilon^\mu + cq^\mu$  changes the soft factor by

$$c \sum_k \eta_k e_k \frac{p_k \cdot q}{p_k \cdot q} = c \sum_k \eta_k e_k.$$

It vanishes for arbitrary  $c$  exactly when total outgoing charge equals total incoming charge. This is the global  $U(1)$  Ward identity, now visible as the gauge invariance of the leading soft theorem.

- (c) An outgoing particle contributes  $+e p \cdot \epsilon / (p \cdot q)$ . Crossing sends its all-outgoing momentum  $p \rightarrow -p$ , replaces it by an incoming antiparticle of charge  $-e$ , and changes  $\eta : +1 \rightarrow -1$ . The product becomes

$$(-1)(-e) \frac{(-p) \cdot \epsilon}{(-p) \cdot q} = e \frac{p \cdot \epsilon}{p \cdot q}.$$

Thus the soft factor is crossing covariant; apparent sign changes cancel when momentum, charge, and in/out assignment are transformed together.

### S.13.12. Soft Gravitons?

- (a) For a symmetric tensor the identity operator is

$$I_{\mu\nu,\rho\sigma} = \frac{1}{2} (\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho}).$$

The inverse of the schematic quadratic kernel in the problem is therefore

$$D_{\mu\nu,\rho\sigma}(k) = \frac{-i}{k^2 - i0} I_{\mu\nu,\rho\sigma}.$$

This is the propagator for the trace-reversed, gauge-fixed variable used in the statement. If one rewrites it for the unreversed metric perturbation, the familiar numerator is  $I_{\mu\nu,\rho\sigma} - \frac{1}{2}\eta_{\mu\nu}\eta_{\rho\sigma}$ , up to gauge-dependent momentum terms. Conserved sources remove those momentum terms.

- (b) In four dimensions the canonical fields have  $[h] = 1$ ,  $[\psi] = 3/2$ , and hence  $[T_{\mu\nu}] = 4$ . Since the Lagrangian density has dimension four,

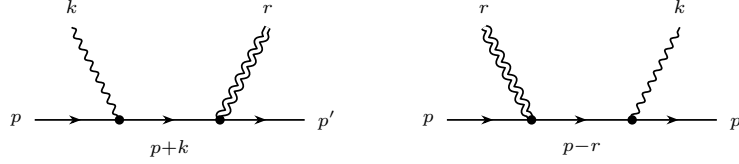
$$[G] = -1,$$

in agreement with  $G \sim \sqrt{G_N}$ . If momentum  $p$  flows into the vertex along the differentiated  $\psi$  line, the interaction as written gives, up to the overall Fourier-transform phase convention,

$$\mathcal{V}_{\mu\nu}(p) = G(\gamma_\mu p_\nu + \gamma_\nu p_\mu).$$

For the Hermitian, symmetrized stress tensor the corresponding rule has  $p$  replaced by one half the sum of the incoming and outgoing fermion momenta. Either form displays the crucial derivative momentum and the dimensionful coupling.

- (c) Let the incoming photon and outgoing graviton momenta be  $k$  and  $r$ , so  $p + k = p' + r$ . There are two Compton-like orderings on the electron line: absorb  $k$  before emitting  $r$ , or emit  $r$  before absorbing  $k$ . The graviton is the doubled wavy line:



With  $\mathcal{V}_{\mu\nu}$  denoting the preceding graviton vertex, their sum is schematically

$$\begin{aligned} \mathcal{M}_{e\gamma \rightarrow eh} = e\bar{u}(p') & [\epsilon_h^{*\mu\nu} \mathcal{V}_{\mu\nu} S_F(p+k) \gamma \cdot \epsilon_\gamma \\ & + \gamma \cdot \epsilon_\gamma S_F(p-r) \epsilon_h^{*\mu\nu} \mathcal{V}_{\mu\nu}] u(p). \end{aligned}$$

At energy  $E \gg m$ , dimensional analysis gives  $\mathcal{M} \sim eGE$ . The two-body cross section contains a factor  $1/E^2$ , and therefore

$$\sigma(e\gamma \rightarrow eh) \sim e^2 G^2 \sim \alpha G^2$$

up to numerical constants and angular dependence. This has the correct dimension of area and is energy independent at leading high-energy power. Gauge-dependent terms in the graviton polarization sum vanish against the conserved stress tensor.

- (d) It *is* infrared divergent. The derivatives in the matter stress tensor act on the hard electron fields, so a soft graviton attachment supplies hard momenta rather than powers of the soft loop momentum  $k$ . The eikonal region of the vertex correction therefore contains

$$G^2 \int \frac{d^4 k}{(2\pi)^4} \frac{\mathcal{N}(p, p')}{(k^2 - i0)(p \cdot k - i0)(p' \cdot k + i0)},$$

where  $\mathcal{N}(p, p')$  is nonzero as  $k \rightarrow 0$ . Radially this behaves as  $k^3 dk / (k^2 k k) = dk/k$ . Equivalently the leading soft-graviton factor is proportional to  $Gp^\mu p^\nu / (p \cdot k)$ . Thus virtual graviton corrections have a logarithmic infrared divergence, and a physical rate must include unresolved real soft gravitons. The conditional wording in the source prompt should not be read as asserting the opposite conclusion.

- (e) For derivative scalar coupling each soft-scalar vertex contains a factor of its momentum  $k_\mu$ . Two such vertices change the preceding soft power count by  $k^2$ , making the origin integrable. More strongly, integration by parts gives

$$\lambda \partial_\mu \pi \bar{\psi} \gamma^\mu \psi = -\lambda \pi \partial_\mu (\bar{\psi} \gamma^\mu \psi) + \text{a total derivative},$$

which vanishes for the conserved vector current, subject to the usual boundary conditions. Direct Yukawa coupling has no soft momentum in its vertex, so its two eikonal denominators remain and the virtual correction is logarithmically infrared divergent for a massive fermion. This is why the derivative-scalar analogy does not reproduce the graviton answer: the graviton derivative acts on a hard matter field, not on the soft graviton.

- (f) Using the symmetry of  $T^{\mu\nu}$  and discarding a boundary term,

$$\begin{aligned} \delta S_G &= G \int d^4x (\partial_\mu \lambda_\nu + \partial_\nu \lambda_\mu) T^{\mu\nu} \\ &= -2G \int d^4x \lambda_\nu \partial_\mu T^{\mu\nu}. \end{aligned}$$

Invariance for arbitrary  $\lambda_\nu(x)$  therefore requires

$$\boxed{\partial_\mu T^{\mu\nu} = 0}.$$

Thus the object coupled to the graviton must be a conserved stress tensor.

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## 14 Bound States in External Fields

In our calculations of radiative corrections in Chapter 11 we went just one step beyond the lowest order in perturbation theory. However, there is a very important class of problems where even the simplest calculation requires that from the beginning we consider classes of Feynman diagrams of arbitrarily high order in coupling constants like  $e$ . These problems are those involving bound states—in electrodynamics, either ordinary atoms and molecules, or such exotic atoms as positronium or muonium.

It is easy to see that such problems necessarily involve a breakdown of ordinary perturbation theory. Consider for instance the amplitude for electron–proton scattering as a function of the center-of-mass energy  $E$ . As shown in Section 10.3, the existence of a bound state like the ground state of hydrogen implies the existence of a pole in this amplitude at  $E = m_p + m_e - 13.6$  eV. However, no single term in the perturbation series for electron–proton scattering has such a pole. The pole therefore can only arise from a divergence of the sum over all diagrams at center-of-mass energies near  $m_p + m_e$ .

The reason for this divergence of the perturbation series is also easy to see, especially if for the moment we consider the time-ordered diagrams of old-fashioned perturbation theory instead of Feynman diagrams. Suppose that in the center-of-mass system the electron and proton both have momenta of magnitude  $q \ll m_e$ , and consider an intermediate state in which the electron and proton momenta are different but also of order  $q$ . The energy denominator factor contributed by this state will be of order  $[q^2/m_e]^{-1}$ . Each such state will also contribute a matrix element of the Coulomb interaction of order  $e^2/q^2$  (the Fourier transform of  $e^2/r$ ), and the corresponding momentum space integration will contribute a factor of order  $q^3$ . Putting this together, we see that each additional Coulomb interaction contributes an overall factor of order

$$[q^2/m_e]^{-1}[e^2/q^2][q^3] = e^2 m_e / q.$$

Thus the perturbation theory should break down when  $q$  is less than or of the order of  $e^2 m_e$ , or in other words when the kinetic and potential energies, which are of order  $q^2/m_e$ , are no larger than about  $e^4 m_e$ , which, of course, is of the order of the binding energy of hydrogen.

Our problem here is to learn how to use perturbation theory to evaluate radiative corrections in bound-state problems, summing to all orders those diagrams that need to be summed to all orders, and keeping only a finite number of those that don't.

### 14.1 The Dirac Equation

We shall limit ourselves in this chapter to problems in which bound states arise because of the Coulomb interaction of electrons (or muons) with heavy charged particles such as atomic nuclei. As shown in Section 13.6, this interaction may be taken into account by adding to the interaction Lagrangian a term<sup>154</sup> representing the effects of a c-number external vector

<sup>154</sup>In this chapter we return to the use of an upper case  $\Psi$  to denote the electron field in the Heisenberg picture, reserving a lower case  $\psi$  for the Dirac field with time-dependence governed solely by the c-number external field  $\mathcal{A}^\mu(x)$ .



potential  $\mathcal{A}^\mu(x)$ :

$$\begin{aligned}\mathcal{L}_{\mathcal{A}} = & -ie\bar{\Psi}\gamma^\mu\Psi\mathcal{A}_\mu - \frac{1}{4}(Z_3 - 1)(\partial^\mu\mathcal{A}^\nu - \partial^\nu\mathcal{A}^\mu)(\partial_\mu A_\nu - \partial_\nu A_\mu) \\ & - ie(Z_2 - 1)\mathcal{A}_\mu\bar{\Psi}\gamma^\mu\Psi,\end{aligned}\tag{14.1.1}$$

which is obtained by replacing the quantum vector potential  $A^\mu$  with  $A^\mu + \mathcal{A}^\mu$  in the interaction part of Eq. (11.1.6). For instance, for a single heavy particle of charge  $Ze$  at the origin,

$$\mathcal{A}^0(x) = \frac{Ze}{4\pi|\mathbf{x}|}, \quad \mathbf{A}(x) = 0.\tag{14.1.2}$$

It is the interaction (14.1.1) that must be taken into account to all orders. In this section we will consider the theory with only this interaction, leaving radiative corrections to subsequent sections.

Physicists learn in kindergarten to approach this sort of problem by solving the wave equation of Dirac in the presence of the external field. It might seem unnecessary to derive this equation here, but as emphasized in Chapter 1, Dirac's original motivation for this equation as a sort of relativistic Schrödinger equation does not stand up to inspection. Also, in the course of our derivation we will discover the normalization conditions that have to be imposed on the solutions of the Dirac equation, which seemed somewhat *ad hoc* in Dirac's approach. The solutions of the Dirac equation discussed here will be important ingredients in our treatment of radiative corrections in the next section.

We will work here in a version of the Heisenberg picture, in which the time-dependence of operators is determined by a Hamiltonian including the external field interaction (14.1.1), but no other interactions. The electron field  $\psi(x)$  in this picture satisfies the field equation

$$\left[\gamma^\lambda \frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda \mathcal{A}_\lambda(x)\right]\psi(x) = 0.\tag{14.1.3}$$

This is not the Dirac equation in the original sense of Dirac [1], because  $\psi(x)$  here is not a c-number wave function but a quantum operator. The c-number Dirac wave functions are defined by

$$u_N(x) \equiv \langle \Omega | \psi(x) | N \rangle,\tag{14.1.4}$$

$$v_N(x) \equiv \langle N | \psi(x) | \Omega \rangle,\tag{14.1.5}$$

where the  $|N\rangle$  are a complete orthonormal set of state-vectors, with  $|\Omega\rangle$  the vacuum. It follows immediately from Eq. (14.1.3) that these functions satisfy the homogeneous Dirac equation

$$\begin{aligned}\left[\gamma^\lambda \frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda \mathcal{A}_\lambda(x)\right]u_N(x) &= \left[\gamma^\lambda \frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda \mathcal{A}_\lambda(x)\right]v_N(x) \\ &= 0.\end{aligned}\tag{14.1.6}$$

We can also derive a normalization condition from the equal-time anticommutation relations for the Dirac field. These are unaffected by the interaction (14.1.1), and therefore take the same form as for the free fields:

$$\{\psi(\mathbf{x}, t), \bar{\psi}(\mathbf{y}, t)\} = i\gamma^0\delta^3(\mathbf{x} - \mathbf{y}).\tag{14.1.7}$$

Taking the vacuum expectation value and inserting a sum over the states  $|N\rangle$ , we find

$$\sum_N u_N(\mathbf{x}, t) u_N^\dagger(\mathbf{y}, t) + \sum_N v_N(\mathbf{x}, t) v_N^\dagger(\mathbf{y}, t) = \delta^3(\mathbf{x} - \mathbf{y}), \quad (14.1.8)$$

it being understood that the sum over  $N$  includes an integral over continuum states as well as a sum over any discrete bound states.

We are chiefly interested in the case of a time-independent external field, like (14.1.2). In such cases, the states  $|N\rangle$  may be taken as eigenstates of the Hamiltonian (including the interaction (14.1.1)) with energies  $E_N$ . Time-translation invariance then tells us that the  $u_N(x)$  and  $v_N(x)$  have the time-dependence:

$$u_N(\mathbf{x}, t) = e^{-iE_N t} u_N(\mathbf{x}), \quad v_N(\mathbf{x}, t) = e^{+iE_N t} v_N(\mathbf{x}). \quad (14.1.9)$$

The homogeneous Dirac equations (14.1.6) then become

$$i\gamma^0 [\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} + m + ie\gamma^\lambda \mathcal{A}_\lambda(\mathbf{x})] u_N(\mathbf{x}) = E_N u_N(\mathbf{x}), \quad (14.1.10)$$

$$i\gamma^0 [\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} + m + ie\gamma^\lambda \mathcal{A}_\lambda(\mathbf{x})] v_N(\mathbf{x}) = -E_N v_N(\mathbf{x}). \quad (14.1.11)$$

The minus sign on the right-hand side of Eq. (14.1.11) shows that the  $v_N$  are the famous ‘negative-energy’ solutions of Dirac. As shown by Eq. (14.1.8), these negative-energy solutions are needed to make up a complete set of wave functions. Of course, for moderate external fields there are no negative-energy states in the theory, so all  $E_N$  are positive, but there is still an important difference between the states with non-vanishing  $u_N$  or  $v_N$ : the definitions (14.1.4) and (14.1.5) show that a state can have  $u_N \neq 0$  or  $v_N \neq 0$  only if it has charge  $-e$  or  $+e$ , respectively. It is in this sense that negative-energy solutions of the Dirac equation have something to do with the existence of antiparticles. However, this argument has nothing to do with the details of the Dirac equation, or even with the spin of the electron.

From the Dirac wave equations (14.1.10) and (14.1.11) we can easily see that wave functions of different energy are orthogonal. That is,

$$(E_M - E_N^*)(u_N^\dagger u_M) = \boldsymbol{\nabla} \cdot (u_N^\dagger i\gamma^0 \boldsymbol{\gamma} u_M),$$

so if  $|\mathbf{x}|^2(u_N^\dagger i\gamma^0 \boldsymbol{\gamma} u_M)$  remains bounded as  $|\mathbf{x}| \rightarrow 0$  and  $|\mathbf{x}| \rightarrow \infty$ , then

$$\int d^3x (u_N^\dagger(\mathbf{x}) u_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq E_M^*. \quad (14.1.12)$$

With similar boundary conditions for the  $v_N$ , we find in the same way that

$$\int d^3x (v_N^\dagger(\mathbf{x}) v_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq E_M^*, \quad (14.1.13)$$

$$\int d^3x (u_N^\dagger(\mathbf{x}) v_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq -E_M^*. \quad (14.1.14)$$

Taking  $N = M$ , Eqs. (14.1.12) and (14.1.13) tell us that the energies are all real. Dropping the complex conjugation of  $E_M$  in Eqs. (14.1.12)–(14.1.14), we see that  $u$ s of different

energy are orthogonal, *vs* of different energy are orthogonal, and (as long as the potential is not strong enough to produce negative energy states) all *us* are orthogonal to all *vs*. By a suitable choice of the discrete quantum numbers that characterize the states along with the energy, we can then always arrange that

$$\int d^3x (u_N^\dagger(\mathbf{x})u_M(\mathbf{x})) = 0 \quad \text{if } N \neq M, \quad (14.1.15)$$

$$\int d^3x (v_N^\dagger(\mathbf{x})v_M(\mathbf{x})) = 0 \quad \text{if } N \neq M, \quad (14.1.16)$$

$$\int d^3x (u_N^\dagger(\mathbf{x})v_M(\mathbf{x})) = 0. \quad (14.1.17)$$

Multiplying Eq. (14.1.8) on the right with  $u_M(\mathbf{y})$  or  $v_M(\mathbf{y})$ , we find then that these wave functions must satisfy the normalization conditions

$$\int d^3y (u_N^\dagger(\mathbf{y})u_M(\mathbf{y})) = \int d^3y (v_N^\dagger(\mathbf{y})v_M(\mathbf{y})) = \delta_{NM}, \quad (14.1.18)$$

where  $\delta_{NM}$  is a product of Kronecker deltas and momentum space delta functions, with normalization adapted to that used in defining  $\sum_N$ , in such a way that  $\sum_N \delta_{NM} = 1$ . These normalization conditions have nothing directly to do with any probabilistic interpretation of the Dirac wave functions, but arise instead from the anticommutation relations (14.1.7) for the fields.

Let us now specialize to the case of a pure electrostatic external field with  $\mathbf{A} = 0$ . In our standard representation of the Dirac matrices, we have

$$\gamma = i \begin{pmatrix} 0 & -\boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad i\gamma^0 = \beta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix},$$

where  $\boldsymbol{\sigma}$  is the usual three-vector of  $2 \times 2$  Pauli matrices, and ‘ $\mathbf{1}$ ’ and ‘ $0$ ’ here are the  $2 \times 2$  unit and zero matrices. We introduce two-component wave functions  $f_N$  and  $g_N$  by setting

$$u_N = \frac{1}{\sqrt{2}} \begin{pmatrix} f_N + ig_N \\ f_N - ig_N \end{pmatrix}. \quad (14.1.19)$$

The energy eigenvalue condition (14.1.10) then takes the form:

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})f_N = (E_N + e\mathcal{A}^0 + m)g_N, \quad (14.1.20)$$

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})g_N = -(E_N + e\mathcal{A}^0 - m)f_N. \quad (14.1.21)$$

In the non-relativistic case where  $e\mathcal{A}^0 r \approx Z\alpha \ll 1$  the binding energy  $m - E_N$  is of order  $Z^2\alpha^2 m$ , while the gradient operator is of the order of  $Z\alpha m$ , so  $g_N$  is smaller than  $f_N$  by a factor of order  $Z\alpha$ . (To find the positron wave functions  $v_N$  we replace  $E_N$  everywhere by  $-E_N$ , so in this case  $f_N$  is smaller than  $g_N$  by the same factor.) We shall return to this non-relativistic case at the end of this section.

Physical states may be classified as even or odd under space inversion:

$$P|N\rangle = \eta_N|N\rangle, \quad (14.1.22)$$

where  $\eta_N$  is a sign factor,  $\pm 1$ . Recall that with the intrinsic parity of the electron defined to be  $+1$ , the Dirac field has the space-inversion property

$$P\psi(\mathbf{x}, t)P^{-1} = \beta\psi(-\mathbf{x}, t),$$

so Eqs. (14.1.4) and (14.1.22) show that the Dirac wave functions satisfy the parity condition

$$u_N(\mathbf{x}) = \eta_N \beta u_N(-\mathbf{x}), \quad (14.1.23)$$

or equivalently

$$f_N(\mathbf{x}) = \eta_N f_N(-\mathbf{x}), \quad g_N(\mathbf{x}) = -\eta_N g_N(-\mathbf{x}). \quad (14.1.24)$$

Note that the parity of the state is the same as the parity of  $f_N(\mathbf{x})$ , not  $g_N(\mathbf{x})$ .

Where the potential  $\mathcal{A}^0$  is rotationally invariant, the solutions of the wave equations here may be classified according to their total angular momentum  $j$  and parity  $\eta$ . For a given  $j$ , the components  $f$  and  $g$  may be expanded in spherical harmonics with orbital angular momentum  $\ell = j + \frac{1}{2}$  and  $\ell = j - \frac{1}{2}$ , but for a definite parity  $\eta = (-1)^{j \mp \frac{1}{2}}$ , Eqs. (14.1.24) show that we can only have  $\ell = j \mp \frac{1}{2}$  in  $f$  and  $\ell = j \pm \frac{1}{2}$  in  $g$ . The usual rules of angular-momentum addition then show that for a state of total angular momentum  $j$ , total angular-momentum  $z$ -component  $\mu$ , and parity  $(-1)^{j \mp \frac{1}{2}}$ , the ‘large’ two-component wave function  $f$  has the form

$$f(\mathbf{x}) = \begin{pmatrix} C_{j \mp \frac{1}{2}, \frac{1}{2}}(j \mu; \mu - \frac{1}{2}, \frac{1}{2}) Y_{j \mp \frac{1}{2}}^{\mu - \frac{1}{2}}(\hat{\mathbf{x}}) \\ C_{j \mp \frac{1}{2}, \frac{1}{2}}(j \mu; \mu + \frac{1}{2}, -\frac{1}{2}) Y_{j \mp \frac{1}{2}}^{\mu + \frac{1}{2}}(\hat{\mathbf{x}}) \end{pmatrix} F(|\mathbf{x}|), \quad (14.1.25)$$

where  $C$  and  $Y$  are the usual Clebsch–Gordan coefficients and spherical harmonics [2]. Also, given any wave function of definite total angular momentum and parity we can construct another wave function with the same  $j$  and  $\mu$  but opposite parity by applying the operator  $\boldsymbol{\sigma} \cdot \hat{\mathbf{x}}$ , so the ‘small’ components may be put in the form

$$g(\mathbf{x}) = (\boldsymbol{\sigma} \cdot \hat{\mathbf{x}}) \begin{pmatrix} C_{j \pm \frac{1}{2}, \frac{1}{2}}(j \mu; \mu - \frac{1}{2}, \frac{1}{2}) Y_{j \pm \frac{1}{2}}^{\mu - \frac{1}{2}}(\hat{\mathbf{x}}) \\ C_{j \pm \frac{1}{2}, \frac{1}{2}}(j \mu; \mu + \frac{1}{2}, -\frac{1}{2}) Y_{j \pm \frac{1}{2}}^{\mu + \frac{1}{2}}(\hat{\mathbf{x}}) \end{pmatrix} G(|\mathbf{x}|). \quad (14.1.26)$$

It is conventional to define the orbital angular momentum  $\ell$  of the state as the orbital angular momentum of the ‘large’ components  $f(\mathbf{x})$ ,

$$\ell = j \mp \frac{1}{2}, \quad (14.1.27)$$

so that the parity is always  $(-1)^\ell$ .

Inserting Eqs. (14.1.25) and (14.1.26) in Eqs. (14.1.20) and (14.1.21) yields the coupled differential equations

$$\frac{dG}{dr} + \frac{k+1}{r}G + (E + e\mathcal{A}^0 - m)F = 0, \quad (14.1.28)$$

$$\frac{dF}{dr} - \frac{k-1}{r}F - (E + e\mathcal{A}^0 + m)G = 0, \quad (14.1.29)$$

where for parity  $\eta = (-1)^{j \mp \frac{1}{2}}$ ,

$$k = \pm \left( j + \frac{1}{2} \right). \quad (14.1.30)$$

Let us now concentrate on the simple Coulomb field (14.1.2), for which  $e\mathcal{A}^0 = Z\alpha/r$ . The treatment of the Dirac equation in this case [3] is familiar, so it will be summarized briefly here just for completeness. It is easy to see that the solutions near the origin go as  $r^{s-1}$ , with  $s^2 = k^2 - Z^2\alpha^2$ . (Note that  $k^2 \geq 1$ , so the exponent  $s$  is real for  $Z\alpha \leq 1$ .) We must reject the solutions with  $s < 0$  as being inconsistent with the normalization condition (14.1.18). The condition that the wave functions do not blow up for  $r \rightarrow \infty$  then fixes the allowed values of the energy eigenvalues:

$$E_{n,j} = m \left[ 1 + \frac{Z^2\alpha^2}{\left[ n - j - \frac{1}{2} + \sqrt{(j + \frac{1}{2})^2 - Z^2\alpha^2} \right]^2} \right]^{-1/2}, \quad (14.1.31)$$

where  $n$  is a ‘principal quantum number’ with

$$j + \frac{1}{2} \leq n. \quad (14.1.32)$$

It is noteworthy that these energies do not depend on the parity or  $\ell$ , but only on  $n$  and  $j$ . For each  $n$  and  $j$  there are two solutions, corresponding to the two signs of  $k$  or the two possible parities, except that for  $n = j + \frac{1}{2}$  we only have  $k > 0$  and parity  $(-1)^{j - \frac{1}{2}}$ , so that  $\ell = j - \frac{1}{2}$ . With Eq. (14.1.32), this is the same as the familiar non-relativistic restriction that  $\ell \leq n - 1$ .

For light atoms with  $Z\alpha \ll 1$ , Eq. (14.1.31) yields the power series

$$E = m \left[ 1 - \frac{Z^2\alpha^2}{2n^2} + \frac{Z^4\alpha^4}{n^4} \left( \frac{3}{8} - \frac{n}{2j+1} \right) + \cdots \right]. \quad (14.1.33)$$

The first two terms, of course, just represent the rest energy and the binding energy as given by the non-relativistic Schrödinger equation. The leading term that depends on  $j$  as well as  $n$  is the third term, the first relativistic correction. For  $n = 1$  there is only one value of the total angular momentum,  $j = \frac{1}{2}$ , and since here  $n = j + \frac{1}{2}$  there is also only one parity,  $(-1)^{j - \frac{1}{2}} = +1$ , corresponding to  $\ell = 0$ . It is therefore difficult to see the effects of the relativistic corrections in Eq. (14.1.33) in the  $n = 1$  states of hydrogenic atoms, though as we shall see in Section 14.3, this has recently become possible. On the other hand, for  $n = 2$  we have a  $j = \frac{1}{2}$  state with both parities (i.e.,  $2s_{1/2}$  and  $2p_{1/2}$ ) as well as a  $2p_{3/2}$  state with  $j = \frac{3}{2}$  and negative parity. Eq. (14.1.33) gives the splitting between the  $p$  states in hydrogen as

$$E(2p_{3/2}) - E(2p_{1/2}) = \frac{\alpha^4 m_e}{32} = 4.5283 \cdot 10^{-5} \text{ eV}. \quad (14.1.34)$$

Such relativistic line splitting is known as the *fine structure* of the atomic state. From the beginning it was known that this prediction is in good agreement with the observed fine structure. On the other hand, the Dirac equation does not yield any energy difference in the

$2s_{1/2}$  and  $2p_{1/2}$  states, so this is a good place to look for the effects of further corrections, to be considered in Section 14.3.

Before closing this section we shall consider the approximate forms for the wave functions and matrix elements in the non-relativistic case for a general electrostatic potential  $\mathcal{A}^0$ . (For a Coulomb potential, this is the limit  $Z\alpha \ll 1$ .) Since here  $E_N + m \simeq 2m \gg |e\mathcal{A}^0|$ , the ‘small’ components of the electron wave function are given approximately in terms of the large components by

$$g_N \simeq (\boldsymbol{\sigma} \cdot \boldsymbol{\nabla}) f_N / 2m_e. \quad (14.1.35)$$

Eq. (14.1.21) then becomes just the non-relativistic Schrödinger equation

$$\left[ -\frac{\nabla^2}{2m_e} - e\mathcal{A}^0 \right] f_N \simeq (E_N - m) f_N. \quad (14.1.36)$$

Since there is no longer any coupling between spin and orbital degrees of freedom in the equation for  $f_N$ , we may find a complete set of solutions of this equation in the form

$$f_N = \chi_N \psi_N(\mathbf{x}),$$

where  $\chi_N$  is a two-component constant spinor, and  $\psi_N(\mathbf{x})$  is an ordinary one-component solution of the Schrödinger equation. However, we often work with states that have definite values of the total angular momentum  $j$ , for which  $f_N$  is (for non-vanishing orbital angular momentum) a sum of such terms.

In the non-relativistic approximation, the four-component Dirac wave function takes the form

$$u_N \simeq \frac{1}{\sqrt{2}} \begin{bmatrix} (1 + i\boldsymbol{\sigma} \cdot \boldsymbol{\nabla}/2m) f_N \\ (1 - i\boldsymbol{\sigma} \cdot \boldsymbol{\nabla}/2m) f_N \end{bmatrix} \quad (14.1.37)$$

and Eq. (14.1.18) gives the normalization condition

$$\int d^3x (f_N^\dagger f_M) \simeq \delta_{NM} - \frac{1}{4} (v^2)_{NM}, \quad (14.1.38)$$

where

$$(v^2)_{NM} \equiv -\frac{1}{m_e^2} \int d^3x f_N^\dagger(\mathbf{x}) \nabla^2 f_M(\mathbf{x}).$$

In relating matrix elements in an external field to free-particle matrix elements, it is useful to note that the momentum space wave function in an energy eigenstate  $N$  may be written

$$u_N(\mathbf{p}) \equiv (2\pi)^{-3/2} \int d^3p e^{-i\mathbf{p} \cdot \mathbf{x}} u_N(\mathbf{x}) \simeq \sum_{\sigma} u(\mathbf{p}, \sigma) [f_N(\mathbf{p})]_{\sigma}, \quad (14.1.39)$$

where  $u(\mathbf{p}, \sigma)$  is the free-particle Dirac spinor

$$u(\mathbf{p}, \sigma) \simeq \frac{1}{\sqrt{2}} \begin{bmatrix} (1 - \mathbf{p} \cdot \boldsymbol{\sigma}/2m_e) \chi_{\sigma} \\ (1 + \mathbf{p} \cdot \boldsymbol{\sigma}/2m_e) \chi_{\sigma} \end{bmatrix}, \quad \chi_{+\frac{1}{2}} \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_{-\frac{1}{2}} \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and  $f_N(\mathbf{p})$  is the Fourier transform of the two-component Schrödinger wave function

$$f_N(\mathbf{p}) \equiv (2\pi)^{-3/2} \int d^3p e^{-i\mathbf{p} \cdot \mathbf{x}} f_N(\mathbf{x}).$$

In closing, as an aid in calculating the effects of various perturbations, we note that the leading terms in the electron matrix elements of the sixteen independent  $4 \times 4$  matrices are

$$(\bar{u}_M u_N) \simeq (f_M^\dagger f_N) - \frac{1}{4m_e^2} (\nabla f_M^\dagger \cdot \boldsymbol{\sigma} \boldsymbol{\sigma} \cdot \nabla f_N), \quad (14.1.40)$$

$$i(\bar{u}_M \gamma^0 u_N) \simeq (f_M^\dagger f_N) + \frac{1}{4m_e^2} (\nabla f_M^\dagger \cdot \boldsymbol{\sigma} \boldsymbol{\sigma} \cdot \nabla f_N), \quad (14.1.41)$$

$$(\bar{u}_M \gamma^i u_N) \simeq \frac{1}{2m_e} \left[ (\nabla f_M^\dagger \cdot \boldsymbol{\sigma} \sigma_i f_N) - (f_M^\dagger \sigma_i \boldsymbol{\sigma} \cdot \nabla f_N) \right], \quad (14.1.42)$$

$$(\bar{u}_M [\gamma^0, \gamma^i] u_N) \simeq \frac{i}{m_e} \left[ (\nabla f_M^\dagger \cdot \boldsymbol{\sigma} \sigma_i f_N) + (f_M^\dagger \sigma_i \boldsymbol{\sigma} \cdot \nabla f_N) \right], \quad (14.1.43)$$

$$(\bar{u}_M [\gamma^i, \gamma^j] u_N) \simeq 2i\epsilon_{ijk} (f_M^\dagger \sigma_k f_N), \quad (14.1.44)$$

$$(\bar{u}_M \gamma_5 \gamma^i u_N) \simeq -i(f_M^\dagger \sigma_i f_N), \quad (14.1.45)$$

$$(\bar{u}_M \gamma_5 \gamma^0 u_N) \simeq \frac{1}{2m_e} \left[ (\nabla f_M^\dagger \cdot \boldsymbol{\sigma} f_N) - (f_M^\dagger \boldsymbol{\sigma} \cdot \nabla f_N) \right], \quad (14.1.46)$$

$$(\bar{u}_M \gamma_5 u_N) \simeq \frac{i}{2m_e} \left[ (\nabla f_M^\dagger \cdot \boldsymbol{\sigma} f_N) + (f_M^\dagger \boldsymbol{\sigma} \cdot \nabla f_N) \right]. \quad (14.1.47)$$

## 14.2 Radiative Corrections in External Fields

We now consider radiative corrections to the results of the previous section, due to the interaction of electrons with the quantum electromagnetic field as well as the external field of the heavy charged particles. These radiative corrections can be calculated using Feynman diagrams of the usual sort, with the whole effect of the external field being to modify the propagator of the electron field in the presence of the external electromagnetic field (and to supply the external-field-dependent renormalization counterterms shown in Eq. (14.1.1)). To be specific, the effect of inserting any number of vertices corresponding to the first term of the interaction (14.1.1) in an internal electron line of any graph is to replace the bare coordinate space propagator  $-iS(x-y)$  with a corrected propagator

$$\begin{aligned} -iS_A(x, y) \equiv & -iS(x-y) + (-i)^2 \int d^4 z_1 S(x-z_1) e\gamma^\mu \mathcal{A}_\mu(z_1) S(z_1-y) \\ & + (-i)^3 \int d^4 z_1 \int d^4 z_2 S(x-z_1) e\gamma^\mu \mathcal{A}_\mu(z_1) S(z_1-z_2) \\ & \cdot e\gamma^\nu \mathcal{A}_\nu(z_2) S(z_2-y) + \cdots, \end{aligned} \quad (14.2.1)$$

where as usual

$$S(x-y) \equiv \frac{1}{(2\pi)^4} \int d^4 p \frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} e^{ip \cdot (x-y)}.$$

(We must write  $S_A$  as a function of  $x$  and  $y$  rather than of  $x-y$ , because the external field invalidates translation invariance.) The theorem proved in Section 6.4 tells us that Eq. (14.2.1) is the same as

$$-iS_A(x, y) = \mathcal{A} \langle \Omega | T \{ \psi(x), \bar{\psi}(y) \} | \Omega \rangle_{\mathcal{A}}. \quad (14.2.2)$$

with the subscript  $\mathcal{A}$  on the right indicating that the vacuum state  $|\Omega\rangle_{\mathcal{A}}$  and electron field  $\psi(x)$  are to be defined in a Heisenberg picture in which the only interaction taken into account is the interaction (14.1.1) with the external field. Inserting a complete set of intermediate states  $|N\rangle_{\mathcal{A}}$  in Eq. (14.2.2) yields an expression for the propagator in terms of the Dirac wave functions  $u_N$  and  $v_N$  introduced in the previous section

$$-iS_{\mathcal{A}}(x, y) = \theta(x^0 - y^0) \sum_N u_N(x) \bar{u}_N(y) - \theta(y^0 - x^0) \sum_M v_M(x) \bar{v}_M(y). \quad (14.2.3)$$

It is also possible to obtain the propagator (14.2.2) as the solution of the *inhomogeneous Dirac equation*:

$$\left[ \gamma^\mu \frac{\partial}{\partial x^\mu} + m + ie\gamma^\mu \mathcal{A}_\mu(x) \right] S_{\mathcal{A}}(x, y) = \delta^4(x - y), \quad (14.2.4)$$

which follows from the field equations (14.1.3) and anticommutation relations (14.1.7), or formally from the perturbation series (14.2.1). Also Eq. (14.2.3) tells us that the propagator satisfies boundary conditions; its Fourier decomposition contains only ‘positive frequency terms’ proportional to  $\exp\{-iE(x^0 - y^0)\}$  with  $E > 0$  for  $x^0 - y^0 \rightarrow \infty$ , and only ‘negative frequency terms’ proportional to  $\exp\{+iE(x^0 - y^0)\}$  with  $E > 0$  for  $x^0 - y^0 \rightarrow -\infty$ . The inhomogeneous Dirac equation with these boundary conditions may be used to obtain a numerical solution [4] for this propagator even in cases where the external field is too strong to allow the use of the perturbation series (14.2.1). Once the propagator  $S_{\mathcal{A}}(x, y)$  has been calculated, the amplitudes for scattering in an external field can be calculated using ordinary Feynman diagrams, but with  $S_{\mathcal{A}}(x, y)$  in place of  $S(x - y)$  (and with  $\mathcal{A}$ -dependent renormalization counterterms inserted where appropriate).

Now let us see how to use the perturbation series with this corrected propagator to calculate the shifts of bound-state energy levels. Consider the full electron propagator  $S'_{\mathcal{A}}(x, y)$ , involving interactions of the electron with the quantum electromagnetic field as well as the external field:

$$-iS'_{\mathcal{A}}(x, y) = {}_{\mathcal{A}}\langle \Omega | T \{ \Psi(x), \bar{\Psi}(y) \} | \Omega \rangle_{\mathcal{A}} \quad (14.2.5)$$

with  $\Psi(x)$  the electron field in a Heisenberg picture including all interactions, and  $|\Omega\rangle$  the vacuum eigenstate of the full Hamiltonian. For a time-independent external potential we can find a complete orthonormal set  $|N\rangle$  of eigenstates of the full Hamiltonian with energies  $E'_N$ . Inserting a sum over these states in the operator product in Eq. (14.2.5), we find

$$\begin{aligned} -iS'_{\mathcal{A}}(x, y) &= \theta(x^0 - y^0) e^{-iE'_N(x^0 - y^0)} \sum_N U_N(\mathbf{x}) \bar{U}_N(\mathbf{y}) \\ &\quad - \theta(y^0 - x^0) e^{-iE'_N(y^0 - x^0)} \sum_N V_N(\mathbf{x}) \bar{V}_N(\mathbf{y}), \end{aligned} \quad (14.2.6)$$

where

$$\langle \Omega | \Psi(\mathbf{x}, t) | N \rangle \equiv e^{-iE'_N t} U_N(\mathbf{x}), \quad (14.2.7)$$

$$\langle N | \bar{\Psi}(\mathbf{x}, t) | \Omega \rangle \equiv e^{+iE'_N t} \bar{V}_N(\mathbf{x}). \quad (14.2.8)$$



(The sum includes an integral over continuum states as well as a sum over discrete bound states. As before,  $U_N$  and  $V_N$  are non-zero only if the state  $|N\rangle$  has charge  $-e$  or  $+e$ , respectively.) We can redefine the propagator as a function of energy rather than time

$$S'_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) \equiv \int_{-\infty}^{\infty} dx^0 e^{iE(x^0 - y^0)} S'_{\mathcal{A}}(x, y). \quad (14.2.9)$$

(Time-translation invariance dictates that  $S'_{\mathcal{A}}(x, y)$  is a function of  $x^0 - y^0$  but not of  $x^0$  and  $y^0$  separately.) From Eq. (14.2.6) we see that

$$S'_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) = \sum_N \frac{U_N(\mathbf{x}) \bar{U}_N(\mathbf{y})}{E'_N - E - i\epsilon} - \sum_N \frac{V_N(\mathbf{x}) \bar{V}_N(\mathbf{y})}{E'_N + E - i\epsilon}. \quad (14.2.10)$$

In particular,  $S'_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E)$  has a pole at any electron bound-state energy, and also at the negative of any positron bound-state energy. (Of course, positrons do not have bound states in the Coulomb field of an ordinary positively charged nucleus.)

Let us now consider the lowest-order radiative corrections to the complete propagator. The Feynman rules here give the complete propagator to this order as  $S'_{\mathcal{A}} = S_{\mathcal{A}} + \delta S_{\mathcal{A}}$ , with a correction term

$$\delta S_{\mathcal{A}}(x, y) = \int d^4 z \int d^4 w S_{\mathcal{A}}(x, z) \Sigma_{\mathcal{A}}^*(z, w) S_{\mathcal{A}}(w, y), \quad (14.2.11)$$

where  $i\Sigma_{\mathcal{A}}^*$  is the sum of all one-loop diagrams with one incoming and one outgoing electron line (excluding final electron propagators) calculated using  $S_{\mathcal{A}}(x, y)$  in place of  $S(x - y)$  for internal electron lines, plus second-order renormalization counterterms. Using energy variables in place of time variables, this is

$$\delta S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) = \int d^3 z \int d^3 w S_{\mathcal{A}}(\mathbf{x}, \mathbf{z}; E) \Sigma_{\mathcal{A}}^*(\mathbf{z}, \mathbf{w}; E) S_{\mathcal{A}}(\mathbf{w}, \mathbf{y}; E), \quad (14.2.12)$$

where

$$\Sigma_{\mathcal{A}}^*(\mathbf{z}, \mathbf{w}; E) \equiv \int dz^0 e^{iE(z^0 - w^0)} \Sigma_{\mathcal{A}}^*(z, w). \quad (14.2.13)$$

The effect of these radiative corrections is to change the wave functions to  $U_N = u_N + \delta u_N$  and  $V_N = v_N + \delta v_N$  and the bound-state energies to  $E'_N = E_N + \delta E_N$ , so that the complete propagator is

$$\begin{aligned} S'_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) &\simeq S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) \\ &+ \sum_N \frac{\delta u_N(\mathbf{x}) \bar{u}_N(\mathbf{y}) + u_N(\mathbf{x}) \delta \bar{u}_N(\mathbf{y})}{E_N - E} \\ &- \sum_N \frac{\delta v_N(\mathbf{x}) \bar{v}_N(\mathbf{y}) + v_N(\mathbf{x}) \delta \bar{v}_N(\mathbf{y})}{E_N + E} \\ &- \sum_N \frac{u_N(\mathbf{x}) \bar{u}_N(\mathbf{y}) \delta E_N}{(E_N - E)^2} + \sum_N \frac{v_N(\mathbf{x}) \bar{v}_N(\mathbf{y}) \delta E_N}{(E_N + E)^2}. \end{aligned} \quad (14.2.14)$$

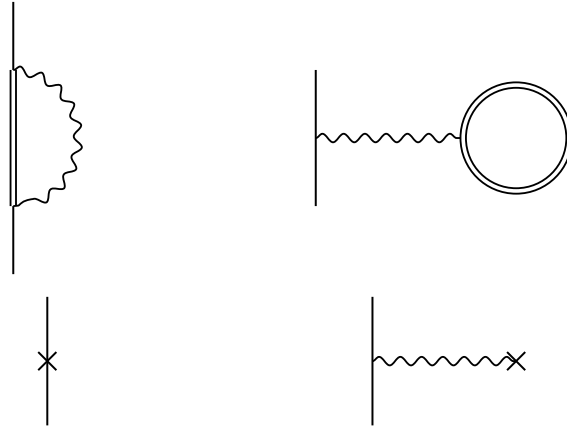
(We are dropping the  $i\epsilon$  terms because we are now not taking  $E$  to lie in the continuum of scattering states.) We see that the shift  $\delta E_N$  of an electron bound-state energy is given by

the coefficient of  $-u_N(\mathbf{x})\bar{u}_N(\mathbf{y})/(E_N - E)^2$  in the complete propagator. To calculate this, we note that Eq. (14.2.3) gives

$$S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) = \sum_N \frac{u_N(\mathbf{x})\bar{u}_N(\mathbf{y})}{E_N - E - i\epsilon} - \sum_N \frac{v_N(\mathbf{x})\bar{v}_N(\mathbf{y})}{E_N + E - i\epsilon}. \quad (14.2.15)$$

Inserting this in Eq. (14.2.12) gives

$$\begin{aligned} \delta S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) &= \sum_{N,M} \frac{u_N(\mathbf{x})\bar{u}_M(\mathbf{y})}{(E_N - E)(E_M - E)} \\ &\quad \cdot \int d^3z \int d^3w \bar{u}_N(\mathbf{z})\Sigma_{\mathcal{A}}^*(\mathbf{z}, \mathbf{w}; E)u_M(\mathbf{w}) + \dots \end{aligned} \quad (14.2.16)$$



**Figure 14.1.** Lowest-order Feynman diagrams for the electron self-energy function  $\Sigma_{\mathcal{A}}^*(x, y)$  in the presence of an external field. Here double straight lines represent electron propagators  $S_{\mathcal{A}}$  that include effects of the external field; single straight lines are the incoming and outgoing electron lines; wavy lines are virtual photons; the cross represents renormalization counterterms.

where the dots denote additional terms involving at least one negative-energy pole. Comparing the coefficients of  $(E_N - E)^{-2}$  here and in Eq. (14.2.14), we find

$$\delta E_N = - \int d^3x \int d^3y \bar{u}_N(\mathbf{x})\Sigma_{\mathcal{A}}^*(\mathbf{x}, \mathbf{y}; E_N)u_N(\mathbf{y}). \quad (14.2.17)$$

The  $u_N$  are solutions of the homogeneous Dirac equation satisfying the normalization condition (14.1.8), so this is very much like ordinary first-order perturbation theory, but with  $-\Sigma^*$  in place of a perturbation to the Hamiltonian.

Generally  $\delta E_N$  turns out to be complex. This is simply a consequence of the instability of atomic energy levels to radiative decay to lower levels; we saw in Chapter 3 that an unstable state of energy  $E$  and decay rate  $\Gamma$  produces poles in various amplitudes at the complex energy  $E - i\Gamma/2$ . The imaginary part of Eq. (14.2.17) therefore equals  $-\Gamma/2$ , while its real part gives the energy shift.

The Feynman diagrams for  $\Sigma^*$  are shown in Figure 14.1. (Note that there are new photon tadpole diagrams here because the external field breaks the Lorentz invariance and

charge-conjugation invariance that forbids such diagrams in the ordinary Feynman rules.) Application of the position space Feynman rules to these diagrams gives

$$\begin{aligned}
i\Sigma_{\mathcal{A}}^*(x, y) = & [e\gamma^\mu][-iS_{\mathcal{A}}(x, y)][e\gamma_\mu][-iD(x - y)] \\
& - [e\gamma^\mu\delta^4(x - y)] \int d^4z [-iD(x - z)] \text{Tr}\{[-iS_{\mathcal{A}}(z, z)][e\gamma_\mu]\} \\
& - i(Z_2 - 1)(\gamma^\mu\partial_\mu + m)\delta^4(x - y) + i\delta m\delta^4(x - y) \\
& + e\gamma^\mu(Z_2 - 1)\delta^4(x - y)\mathcal{A}_\mu(x) \\
& + i(Z_3 - 1)[e\gamma_\mu]\delta^4(x - y) \int d^4z [-iD(x - z)]\partial_\nu(\partial^\nu\mathcal{A}^\mu(z) - \partial^\mu\mathcal{A}^\nu(z)).
\end{aligned} \tag{14.2.18}$$

with the renormalization constants  $(Z_2 - 1)$ ,  $(Z_3 - 1)$ , and  $\delta m$  calculated to second order in  $e$ . (The minus sign in the second term is the usual one that accompanies closed fermion loops.)

For strong external fields with  $Z\alpha$  of order unity it is necessary to calculate the configuration space electron propagator  $S_{\mathcal{A}}$  and the integrals in Eqs. (14.2.17) and (14.2.18) numerically [4]. However, for weak fields we can use the first few terms of the series (14.2.1) in Eq. (14.2.18) and calculate these integrals in closed form. For this purpose, it is more convenient to work in momentum space, defining:

$$S_{\mathcal{A}}(x, y) = (2\pi)^{-4} \int d^4p' d^4p e^{ip'\cdot x} e^{-ip\cdot y} S_{\mathcal{A}}(p', p), \tag{14.2.19}$$

$$\Sigma_{\mathcal{A}}^*(x, y) = (2\pi)^{-4} \int d^4p' d^4p e^{ip'\cdot x} e^{-ip\cdot y} \Sigma_{\mathcal{A}}^*(p', p), \tag{14.2.20}$$

$$u_N(\mathbf{x}) = (2\pi)^{-3/2} \int d^3p e^{i\mathbf{p}\cdot\mathbf{x}} u_N(\mathbf{p}), \tag{14.2.21}$$

$$\mathcal{A}^\mu(x) = \int d^4q e^{iq\cdot x} \mathcal{A}^\mu(q). \tag{14.2.22}$$

(We are here committing the impropriety of using the same symbol for a function and its Fourier transform, leaving it to the displayed arguments of the functions to indicate which is which.) Then Eqs. (14.2.1) and (14.2.18) become

$$S_{\mathcal{A}}(p', p) = \frac{-i\not{p}' + m}{p'^2 + m^2 - i\epsilon} - ie \frac{-i\not{p}' + m}{p'^2 + m^2 - i\epsilon} \mathcal{A}(p' - p) \frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} + \dots \tag{14.2.23}$$

and

$$\begin{aligned}
\Sigma_{\mathcal{A}}^*(p', p) = & \frac{ie^2}{(2\pi)^4} \int \frac{d^4k}{k^2 - i\epsilon} \gamma^\mu S_{\mathcal{A}}(p' - k, p - k) \gamma_\mu \\
& + [-(Z_2 - 1)(i\not{p} + m) + Z_2\delta m] \delta^4(p' - p) - ie(Z_2 - 1) \mathcal{A}(p' - p) \\
& - \frac{ie^2\gamma^\mu}{(2\pi)^4[(p - p')^2 - i\epsilon]} \int d^4q \text{Tr}\{S_{\mathcal{A}}(q, q + p' - p) \gamma_\mu\} \\
& + \frac{ie(Z_3 - 1)}{(p - p')^2 - i\epsilon} [(p - p')^2 \mathcal{A}(p' - p) - (\not{p} - \not{p}')(p - p') \cdot \mathcal{A}(p' - p)].
\end{aligned} \tag{14.2.24}$$

Because the external field is time-independent,  $S_{\mathcal{A}}(x, y)$  and  $\Sigma_{\mathcal{A}}^*(x, y)$  can depend on  $x^0$  and  $y^0$  only through the difference  $x^0 - y^0$ , so  $S_{\mathcal{A}}(p', p)$  and  $\Sigma_{\mathcal{A}}^*(p', p)$  as well as  $\mathcal{A}^\mu(p' - p)$  must be proportional to  $\delta(p'^0 - p^0)$ :

$$\mathcal{A}^\mu(p' - p) = \delta(p'^0 - p^0) \mathcal{A}^\mu(\mathbf{p}' - \mathbf{p}), \quad (14.2.25)$$

$$S_{\mathcal{A}}(p', p) = \delta(p'^0 - p^0) S_{\mathcal{A}}(\mathbf{p}', \mathbf{p}; p^0), \quad (14.2.26)$$

$$\Sigma_{\mathcal{A}}^*(p', p) = \delta(p'^0 - p^0) \Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; p^0). \quad (14.2.27)$$

The energy shift is then given by Eqs. (14.2.17) and (14.2.13) as

$$\delta E_N = - \int d^3 p' \int d^3 p \bar{u}_N(\mathbf{p}') \Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; E_N) u_N(\mathbf{p}), \quad (14.2.28)$$

with  $\Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; E_N)$  given by Eqs. (14.2.23), (14.2.24), and (14.2.27). This is the master formula we shall use in the next section to calculate energy shifts in weak external fields.

### 14.3 The Lamb Shift in Light Atoms

Let us now consider radiative corrections to the energy levels of a non-relativistic electron in a general electrostatic field, such as an electron in the Coulomb field of a light nucleus with  $Z\alpha \ll 1$ . It is natural in this limit to treat the Coulomb field as a weak perturbation, but we will see that this would lead to an infrared divergence, related to that found in Section 11.3. The infrared divergence is really fictitious, because the four-momenta  $\mathbf{p}, E_N$  and  $\mathbf{p}', E_N$  are not on the electron mass shell, but it does force us to proceed with some care.

Usually this problem is dealt with by dividing the integral over virtual photon energies into a low-energy range, within which we can treat the electrons non-relativistically but must include effects to all orders in the external field, and a high-energy range, within which we have to include relativistic effects but may include only effects of lowest order in the external field. Instead, we shall here introduce a fictitious photon mass  $\mu$ , chosen to be much larger than typical electron kinetic energies, but much less than typical electron momenta. For a Coulomb field, this amounts to the requirement that

$$(Z\alpha)^2 m_e \ll \mu \ll Z\alpha m_e. \quad (14.3.1)$$

We write the photon propagator in the first two terms of Eq. (14.2.24) (including the formulas for the counterterms  $Z_2 - 1$  and  $Z_2 \delta m$ ) in the form

$$\frac{1}{k^2 - i\epsilon} = \left[ \frac{1}{k^2 + \mu^2 - i\epsilon} \right] + \left[ \frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right]. \quad (14.3.2)$$

The energy shift is correspondingly a sum of two terms, a ‘high-energy’ and a ‘low-energy’ term: The high-energy term is calculated by using the first term in the photon propagator (14.3.2) in the first three terms in Eq. (14.2.24), and adding the result to the last two terms (the vacuum polarization terms) in Eq. (14.2.24) which are not infrared divergent anyway; the low-energy term is calculated using the second term in Eq. (14.3.2) in the first three

terms of Eq. (14.2.24). One advantage of this procedure is that we shall be able to use the results of the relativistic calculations of Sections 11.3 and 11.4 directly, without the rather tricky conversion from a photon mass to an infrared energy cutoff. Of course, in the end we shall have to check that the dependence on the photon mass  $\mu$  in the high-energy and low-energy contributions to the energy shift cancel, leaving the total energy shift  $\mu$ -independent.

### A High-Energy Term

Because  $\mu$  is taken much larger than the atomic binding energies we can here keep only terms of lowest order in the external field. The one-loop radiative correction to atomic energy levels in a general time-independent external vector potential  $\mathcal{A}^\mu(\mathbf{x})$  is given in momentum space by Eq. (14.2.28), with the self-energy insertion  $\Sigma_{\mathcal{A}}(p', p)$  given by Eqs. (14.2.24) and (14.2.23). The terms of zeroth order in the external field simply cancel: the  $\delta m$  term cancels the first term with  $\mathcal{A} = 0$ ; the  $Z_3 - 1$  term cancels the third term with  $\mathcal{A} = 0$ ; and the  $Z_2 - 1$  terms vanish because  $u(p)$  satisfies the Dirac equation. The term in  $\Sigma_{\mathcal{A}}(p', p)$  that is of first order in  $\mathcal{A}^\mu$  may be put in the form

$$\Sigma_{\mathcal{A}1}(p', p) = -ie\mathcal{A}_\mu(p' - p)\Gamma_1^\mu(p', p) \quad (14.3.3)$$

with

$$\begin{aligned} \Gamma_1^\mu(p', p) = & \frac{ie^2}{(2\pi)^4} \int \frac{d^4k}{k^2 + \mu^2 - i\epsilon} \left[ \frac{-i(\not{p}' - \not{k}) + m_e}{(p' - k)^2 + m_e^2 - i\epsilon} \right] \gamma^\mu \left[ \frac{-i(\not{p} - \not{k}) + m_e}{(p - k)^2 + m_e^2 - i\epsilon} \right] \\ & + (Z_2 - 1)\gamma^\mu \\ & - \frac{ie^2\gamma_\nu}{(2\pi)^4[(p - p')^2 - i\epsilon]} \int d^4l \operatorname{Tr} \left\{ \left[ \frac{-i\not{l} + m_e}{l^2 + m_e^2} \right] \gamma^\mu \left[ \frac{-i\not{l} - i\not{p}' + i\not{p} + m_e}{(l + p' - p)^2 + m_e^2} \right] \gamma^\nu \right\} \\ & - \frac{Z_3 - 1}{(p' - p)^2 - i\epsilon} \\ & \cdot [(p - p')^2 \eta_{\mu\nu} - (p' - p)_\mu(p' - p)_\nu] \gamma^\nu. \end{aligned} \quad (14.3.4)$$

Comparison of the first two terms with Eqs. (11.3.1) and (11.3.8) and the second two terms with Eqs. (11.3.9), (11.2.3), and (11.2.15) reveals that  $\Gamma_1^\mu(p', p)$  is the complete one-loop vertex function, including vacuum polarization and all counterterms, whose mass-shell matrix elements we have already calculated in Section 11.3. Using Eqs. (14.2.26) and (14.2.25), this contribution to the energy shift (14.2.28) is given by

$$[\delta E_N]_{\text{high energy}} = ie \int d^3p' \int d^3p (\bar{u}_N(\mathbf{p}')\Gamma_1^\mu(\mathbf{p}', E_N, \mathbf{p}, E_N)u_N(\mathbf{p}))\mathcal{A}_\mu(\mathbf{p}' - \mathbf{p}). \quad (14.3.5)$$

(This could have been guessed at, by simply replacing the  $\gamma^\mu$  in the interaction of the electron with the external field with  $\Gamma_1^\mu$ .) As discussed in Section 14.1, because  $Z\alpha \ll 1$  we can approximate the Dirac wave function  $u_N$  in Eq. (14.3.5) as

$$[u_N(\mathbf{p})]_\alpha = \sum_\sigma u_\alpha(\mathbf{p}, \sigma)[f_N(\mathbf{p})]_\sigma, \quad (14.3.6)$$

where  $f_N$  is the non-relativistic two-component wave function of an electron in the external Coulomb field, and  $u(\mathbf{p}, \sigma)$  is the four-component normalized solution of the momentum space Dirac equation

$$[i\gamma_\mu p^\mu + m_e]u(\mathbf{p}, \sigma) = 0 \quad (14.3.7)$$

for spin  $z$ -component  $\sigma$ . Since  $u_N(\mathbf{p})$  approximately satisfies the free-particle Dirac equation, Eq. (10.6.15) gives the general form of the matrix element of  $\Gamma_1^\mu$  as

$$\bar{u}_N(\mathbf{p}')[\gamma^\mu + \Gamma_1^\mu(p', p)]u_N(\mathbf{p}) = \bar{u}_N(\mathbf{p}') \left[ \gamma^\mu F_1(q^2) + \frac{i}{2}[\gamma^\mu, \gamma^\nu]q_\nu F_2(q^2) \right] u_N(\mathbf{p}), \quad (14.3.8)$$

where  $q \equiv p' - p$ . The wave functions  $u_N(\mathbf{p})$  fall off very rapidly for  $|\mathbf{p}| \gg Z\alpha m_e$ , so we only need  $F_1(q^2)$  and  $F_2(q^2)$  in the limit  $|q^2| \ll m_e^2$ . In this limit, Eqs. (11.3.31), (10.6.18), and (11.3.16) give

$$F_1(q^2) \simeq 1 + \frac{e^2}{24\pi^2} \left( \frac{q^2}{m_e^2} \right) \left[ \ln \left( \frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right], \quad (14.3.9)$$

$$F_2(q^2) \simeq \frac{e^2}{16m_e\pi^2}. \quad (14.3.10)$$

Let us first consider the contribution of the  $F_1$  term in Eq. (14.3.8), which makes by far the largest contribution to the energy shift, and raises the most interesting problems in its calculation. For a pure electrostatic field with  $\mathcal{A} = 0$ , Eqs. (14.3.5), (14.3.8), and (14.3.9) give

$$[\delta E_N]_{F_1} = -\frac{e^2}{24\pi^2 m_e^2} \left[ \ln \left( \frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right] \int d^3 p' \int d^3 p \bar{u}_N(\mathbf{p}') \cdot (-ie\mathcal{A}^0(\mathbf{p}' - \mathbf{p}))\gamma^0(\mathbf{p}' - \mathbf{p})^2 u_N(\mathbf{p}). \quad (14.3.11)$$

To calculate this contribution, we may use the leading term in the non-relativistic matrix element (14.1.41), and find

$$[\delta E_N]_{F_1} = -\frac{e^2}{24\pi^2 m_e^2} \left[ \ln \left( \frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right] \int d^3 p' \int d^3 p f_N^\dagger(\mathbf{p}') e\mathcal{A}^0(\mathbf{p}' - \mathbf{p}) [\mathbf{p}' - \mathbf{p}]^2 f_N(\mathbf{p}) \quad (14.3.12)$$

or in position space

$$[\delta E_N]_{F_1} = \frac{e^2}{24\pi^2 m_e^2} \left[ \ln \left( \frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right] \int d^3 x f_N^\dagger(\mathbf{x}) [e\nabla^2 \mathcal{A}^0(\mathbf{x})] f_N(\mathbf{x}). \quad (14.3.13)$$

In particular, for the Coulomb potential (14.1.2), we have  $e\nabla^2 \mathcal{A}^0(\mathbf{x}) = -Ze^2\delta^3(\mathbf{x})$ , and the label  $N$  consists of a principal quantum number  $n$  and angular-momentum quantum numbers  $j, m, \ell$ , while (11.2.41) gives  $[f_{njm\ell}(0)]_\sigma = 2(Z\alpha m_e/n)^{3/2} \delta_{\ell,0} \delta_{\sigma,m} / \sqrt{4\pi}$ . The energy shift (14.3.11) is then

$$[\delta E_{nj\ell}]_{F_1} = -\frac{2Z^4 \alpha^5 m_e}{3\pi n^3} \left[ \ln \left( \frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right] \delta_{\ell,0}. \quad (14.3.14)$$

(The lack of dependence of  $\delta E$  on the total angular momentum  $z$ -component  $m$  is guaranteed by rotational invariance.) The term  $2/5$  in the brackets in Eqs. (14.3.12) and (14.3.13) arises from vacuum polarization, and yields just the energy shift calculated somewhat heuristically in Section 11.2.

Before going on to calculate the magnetic and low-energy contributions to the energy shift, it is worth noting that the result we have obtained so far yields a fair order-of-magnitude estimate of the Lamb shift without further work. We can anticipate that the low-energy terms will contain a term proportional to  $\ln(\mu/B)$ , with a coefficient such as to cancel the  $\mu$ -dependence in Eq. (14.3.12). The constant  $B$  here is an energy which must be included to make the argument of the logarithm dimensionless; since it is the binding of the electron in the atom that will eventually provide our infrared cutoff, we may guess that  $B$  is a typical atomic binding energy, of order  $B \simeq (Z\alpha)^2 m_e$ . The total energy shift in a state  $N$  with principal quantum number  $n$  and orbital angular momentum  $\ell$  is therefore of the form

$$\delta E_N = -\frac{2Z^4\alpha^5 m_e}{3\pi n^3} [\ln(Z^4\alpha^4)\delta_{\ell,0} + O(1)]. \quad (14.3.15)$$

For the  $2s$  state of hydrogen, the logarithmic term alone gives

$$\delta E_{2s} = -\frac{\alpha^5 m_e}{12\pi} \ln(\alpha^4) = 5.5 \cdot 10^{-6} \text{ eV} = 1300 \text{ MHz} \cdot 2\pi\hbar.$$

As we shall see, the ‘ $O(1)$ ’ terms in Eq. (14.3.15) will lower the total energy shift by about 25%.

Now let us consider the contribution of the  $F_2$  term in the matrix element of  $\Gamma_1^\mu$ , which as we saw in Section 10.6 may be interpreted as a radiative correction to the magnetic moment of the electron. Using Eqs. (14.3.10), (14.3.8), and (14.3.6) in Eq. (14.3.5), we find that this term gives an energy shift

$$[\delta E_N]_{F_2} = -\frac{e^2}{32\pi^2 m_e} \int d^3 p' \int d^3 p (\bar{u}_N(\mathbf{p}') [\gamma^\mu, \gamma^\nu] u_N(\mathbf{p})) e \mathcal{A}_\mu(\mathbf{p}' - \mathbf{p}) (p' - p)_\nu. \quad (14.3.16)$$

or in position space

$$[\delta E_N]_{F_2} = \frac{ie^2}{64\pi^2 m_e} \int d^3 x (\bar{u}_N(\mathbf{x}) [\gamma^\mu, \gamma^\nu] u_N(\mathbf{x})) e \mathcal{F}_{\mu\nu}(\mathbf{x}), \quad (14.3.17)$$

where

$$\mathcal{F}_{\mu\nu}(\mathbf{x}) = \partial_\mu \mathcal{A}_\nu(\mathbf{x}) - \partial_\nu \mathcal{A}_\mu(\mathbf{x}). \quad (14.3.18)$$

For a pure electrostatic field with  $\mathcal{A} = \mathbf{0}$ , this is

$$[\delta E_N]_{F_2} = -\frac{ie^2}{32\pi^2 m_e} \int d^3 x (\bar{u}_N(\mathbf{x}) [\gamma^0, \gamma] u_N(\mathbf{x})) \cdot \nabla [e \mathcal{A}^0(\mathbf{x})]. \quad (14.3.19)$$

In the non-relativistic limit  $Z\alpha \ll 1$ , we can use the approximate result (14.1.43), which here reads

$$\begin{aligned} (\bar{u}_N [\gamma^0, \gamma] u_N) &\simeq \frac{i}{m_e} \left[ (\nabla f_N^\dagger) \cdot \boldsymbol{\sigma} \boldsymbol{\sigma} f_N + f_N^\dagger \boldsymbol{\sigma} \boldsymbol{\sigma} \cdot \nabla f_N \right] \\ &= \frac{i}{m_e} \left[ \nabla (f_N^\dagger f_N) - i(\nabla f_N^\dagger \cdot \boldsymbol{\sigma}) f_N - i f_N^\dagger (\boldsymbol{\sigma} \cdot \nabla f_N) \right]. \end{aligned} \quad (14.3.20)$$

Using this in Eq. (14.3.8) and integrating by parts, this part of the energy shift is

$$[\delta E_N]_{F_2} = -\frac{e^2}{32\pi^2 m_e^2} \int d^3x \left[ -|f_N(\mathbf{x})|^2 \nabla^2 (e\mathcal{A}^0(\mathbf{x})) + 2if_N^\dagger(\mathbf{x})\boldsymbol{\sigma} \cdot (\nabla(e\mathcal{A}^0(\mathbf{x})) \cdot \nabla f_N(\mathbf{x})) \right]. \quad (14.3.21)$$

Combining Eqs. (14.3.12) and (14.3.21) gives the total high-energy contribution to the energy shift in an arbitrary electrostatic potential  $\mathcal{A}^0$ :

$$[\delta E_N]_{\text{high energy}} = \frac{e^2}{24\pi^2 m_e^2} \left[ \ln\left(\frac{\mu^2}{m_e^2}\right) + \frac{2}{5} \right] \int d^3x f_N^\dagger(\mathbf{x}) [e\nabla^2 \mathcal{A}^0(\mathbf{x})] f_N(\mathbf{x}) + \frac{ie^2}{16\pi^2 m_e^2} \int d^3x f_N^\dagger(\mathbf{x}) \boldsymbol{\sigma} \cdot (\nabla[e\mathcal{A}^0(\mathbf{x})] \cdot \nabla f_N(\mathbf{x})). \quad (14.3.22)$$

## B Low-Energy Term

The low-energy contribution to the energy shift is obtained from the first three terms in Eq. (14.2.24), making the replacement in the photon propagator

$$\frac{1}{k^2 - i\epsilon} \longrightarrow \frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon}. \quad (14.3.23)$$

This substitution will eventually serve to cut off the integral over components of the photon four-momentum  $k$  at values of order  $\mu$ , but it is not possible to see this until we carefully take mass renormalization into account, so we shall defer making any non-relativistic approximations until then. Also, we are now including photon momenta as small or smaller than the binding energies of the atomic states, so we must treat the electrostatic forces responsible for this binding to all orders.

Instead of working with the momentum space formula (14.2.24), it will be convenient to return to the configuration space formula (14.2.18). This gives the low-energy contribution to the electron self-energy function as

$$[\Sigma_{\mathcal{A}}^*(x, y)]_{\text{low energy}} = ie^2 \gamma^\mu S_{\mathcal{A}}(x, y) \gamma_\mu D(x - y; \mu) + \delta m_e(\mu) \delta^4(x - y) - i(Z_2(\mu) - 1) \{ \gamma^\mu [\partial_\mu + ie\mathcal{A}_\mu] + m_e \} \delta^4(x - y), \quad (14.3.24)$$

where  $D(x - y; \mu)$  is the modified photon propagator

$$D(x - y; \mu) = \frac{1}{(2\pi)^4} \int d^4k e^{-ik \cdot (x-y)} \left[ \frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right], \quad (14.3.25)$$

and the counterterms  $Z_2(\mu) - 1$  and  $\delta m_e(\mu)$  are calculated using this modified propagator. Converting from time to energy variables, the low-energy contribution to the function (14.2.13) is then

$$[\Sigma_{\mathcal{A}}^*(\mathbf{x}, \mathbf{y}; E)]_{\text{low energy}} = \frac{ie^2}{(2\pi)^4} \int d^4k \gamma^\mu \cdot S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E - k^0) \gamma_\mu \left[ \frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right] e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} - (Z_2(\mu) - 1) (\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} + i\gamma^0 E + ie\gamma^\nu \mathcal{A}_\nu + m_e) \delta^3(\mathbf{x} - \mathbf{y}) + \delta m_e(\mu) \delta^3(\mathbf{x} - \mathbf{y}). \quad (14.3.26)$$



The low-energy contribution to the energy shift is then given by Eq. (14.2.17) as

$$\begin{aligned}
[\delta E_N]_{\text{low energy}} &= - \int d^3x \int d^3y \bar{u}_N(\mathbf{x}) [\Sigma_{\mathcal{A}}^*(\mathbf{x}, \mathbf{y}; E_N)]_{\text{low energy}} u_N(\mathbf{y}) \\
&= - \frac{ie^2}{(2\pi)^4} \int d^4k \int d^3x \int d^3y \bar{u}_N(\mathbf{x}) \gamma^\mu S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E_N - k^0) \gamma_\mu u_N(\mathbf{y}) \\
&\quad \cdot \left[ \frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right] e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} \\
&\quad - \delta m_e(\mu) \int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x}).
\end{aligned} \tag{14.3.27}$$

Note that the terms proportional to  $Z_2(\mu) - 1$  have dropped out because the Dirac wave function  $u_N(\mathbf{x})$  satisfies the Dirac equation Eq. (14.1.10). For the electron propagator in the presence of the Coulomb field, we use Eq. (14.2.15):

$$S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) = \sum_M \frac{u_M(\mathbf{x}) \bar{u}_M(\mathbf{y})}{E_M - E - i\epsilon} - \sum_M \frac{v_M(\mathbf{x}) \bar{v}_M(\mathbf{y})}{E_M + E - i\epsilon},$$

with the sums in the first and second terms running over all one-electron and one-positron states, respectively. The  $k^0$  integrals can be done most easily by closing the contour of integration with a large semi-circle in the lower half-plane in the first term and in the upper half-plane in the second term:

$$\begin{aligned}
&\int dk^0 \left( \frac{1}{k^2 + \mu^2 - i\epsilon} \right) \left( \frac{1}{E_M \mp E_N \pm k^0 - i\epsilon} \right) \\
&= \frac{i\pi}{\sqrt{\mathbf{k}^2 + \mu^2}} \left( \frac{1}{E_M \mp E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon} \right),
\end{aligned}$$

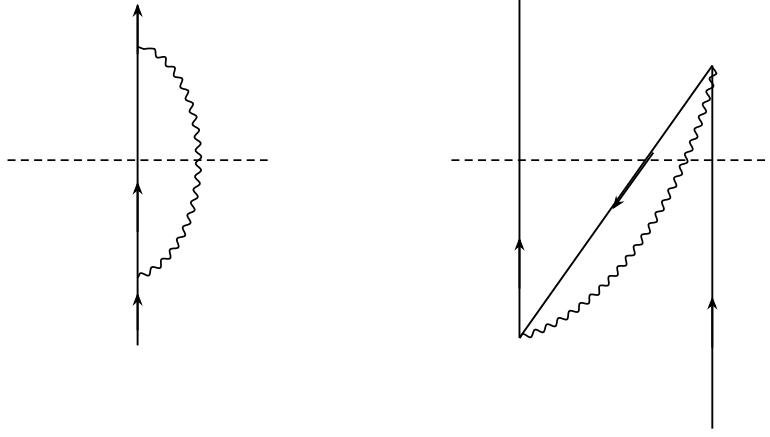
and likewise if  $\mu$  is replaced with zero. The energy shift (14.3.27) now becomes

$$\begin{aligned}
[\delta E_N]_{\text{low energy}} &= - \frac{e^2}{2(2\pi)^3} \int d^3k \sum_M \left[ \Gamma_{MN}^\rho(\mathbf{k})^* \Gamma_{\rho MN}(\mathbf{k}) \left( \frac{1}{|\mathbf{k}|(E_M - E_N + |\mathbf{k}| - i\epsilon)} \right. \right. \\
&\quad \left. \left. - \frac{1}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right) \right. \\
&\quad \left. - \tilde{\Gamma}_{MN}^\rho(\mathbf{k})^* \tilde{\Gamma}_{\rho MN}(\mathbf{k}) \left( \frac{1}{|\mathbf{k}|(E_M + E_N + |\mathbf{k}| - i\epsilon)} \right. \right. \\
&\quad \left. \left. - \frac{1}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M + E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right) \right] \\
&\quad - \delta m_e(\mu) \int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x}),
\end{aligned} \tag{14.3.28}$$

where

$$\Gamma_{MN}^\rho(\mathbf{k}) \equiv \int d^3y e^{-i\mathbf{k} \cdot \mathbf{y}} \bar{u}_M(\mathbf{y}) \gamma^\rho u_N(\mathbf{y}), \tag{14.3.29}$$

$$\tilde{\Gamma}_{MN}^\rho(\mathbf{k}) \equiv \int d^3y e^{-i\mathbf{k} \cdot \mathbf{y}} \bar{v}_M(\mathbf{y}) \gamma^\rho u_N(\mathbf{y}). \tag{14.3.30}$$



**Figure 14.2.** Old-fashioned perturbation theory diagrams for the low-energy part of the electron energy shift. Here solid lines are electrons; wavy lines are photons; the dashed line cuts through particle lines corresponding to the intermediate states appearing in the first two terms in (14.3.28)

(Of course, the ‘sum’ over  $M$  in Eq. (14.3.28) receives contributions only from electron states in the first term and positron states in the second term.) Eq. (14.3.28) could have been derived more directly from old-fashioned perturbation theory; the energy denominators  $E_M - E_N + \omega$  and  $E_M + E_N + \omega$  are the result of subtracting the energy  $E_N$  of the initial state from the energy of an intermediate state consisting of either an electron of energy  $E_M$  and a photon of energy  $\omega$ , or else a positron of energy  $E_M$ , a photon of energy  $\omega$ , and both the final and the initial electron. (See Figure 14.2.)

Before making any approximations to Eq. (14.3.28), it will be convenient to express the time components of the matrix elements  $\Gamma_{MN}^\rho$  and  $\tilde{\Gamma}_{MN}^\rho$  in terms of the corresponding space components, using relations<sup>155</sup> derived from the conservation of the electric current:

$$k_i \Gamma_{MN}^i(\mathbf{k}) = (E_N - E_M) \Gamma_{MN}^0(\mathbf{k}), \quad (14.3.31)$$

$$k_i \tilde{\Gamma}_{MN}^i(\mathbf{k}) = (E_N + E_M) \tilde{\Gamma}_{MN}^0(\mathbf{k}). \quad (14.3.32)$$

Furthermore, by using the completeness relation (14.1.8) it is straightforward to show that

$$\sum_M \left[ |\Gamma_{MN}^0(\mathbf{k})|^2 + |\tilde{\Gamma}_{MN}^0(\mathbf{k})|^2 \right] = 1 \quad (14.3.33)$$

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<sup>155</sup>To derive Eq. (14.3.31), note that

$$\begin{aligned} k_i \Gamma_{MN}^i(\mathbf{k}) &= -i \int d^3x e^{-i\mathbf{k}\cdot\mathbf{x}} \nabla \cdot (\bar{u}_M(\mathbf{x}) \boldsymbol{\gamma} u_N(\mathbf{x})) \\ &= i \int d^3x e^{-i\mathbf{k}\cdot\mathbf{x}} \partial_0 \left[ \bar{u}_M(\mathbf{x}) \gamma^0 u_N(\mathbf{x}) e^{-i(E_N - E_M)x^0} \right]_{x^0=0} = (E_N - E_M) \Gamma_{MN}^0(\mathbf{k}). \end{aligned}$$

Eq. (14.3.32) is derived in the same way.

and

$$\begin{aligned}
\sum_M \left[ |\Gamma_{MN}^0(\mathbf{k})|^2 (E_M - E_N) - |\tilde{\Gamma}_{MN}^0(\mathbf{k})|^2 (E_M + E_N) \right] &= \sum_M \left[ -\Gamma_{MN}^{0*}(\mathbf{k}) \mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k}) \right. \\
&\quad \left. - \tilde{\Gamma}_{MN}^{0*}(\mathbf{k}) \mathbf{k} \cdot \tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k}) \right] \\
&= -i\mathbf{k} \cdot \int d^3x \bar{u}_N(\mathbf{x}) \boldsymbol{\gamma} u_N(\mathbf{x}) = 0,
\end{aligned} \tag{14.3.34}$$

the last step following from the parity condition (14.1.23). In this way, Eq. (14.3.28) can be rewritten

$$\begin{aligned}
[\delta E_N]_{\text{low energy}} &= -\frac{e^2}{2(2\pi)^3} \int d^3k \sum_M \left[ \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / \mathbf{k}^2}{|\mathbf{k}|(E_M - E_N + |\mathbf{k}| - i\epsilon)} \right. \\
&\quad \left. - \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / (\mathbf{k}^2 + \mu^2)}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right] \\
&\quad + \frac{e^2}{2(2\pi)^3} \int d^3k \sum_M \left[ \frac{|\tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k})|^2 / \mathbf{k}^2}{|\mathbf{k}|(E_M + E_N + |\mathbf{k}|)} \right. \\
&\quad \left. - \frac{|\tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k})|^2 / (\mathbf{k}^2 + \mu^2)}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M + E_N + \sqrt{\mathbf{k}^2 + \mu^2})} \right] \\
&\quad - \frac{e^2}{(2\pi)^3} \int d^3k \sum_M |\tilde{\Gamma}_{MN}^0(\mathbf{k})|^2 \left( \frac{1}{\mathbf{k}^2} - \frac{1}{\mathbf{k}^2 + \mu^2} \right) \\
&\quad + \frac{1}{2} \alpha \mu - \delta m_e(\mu) \int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x}).
\end{aligned} \tag{14.3.35}$$

In the next-to-last term, we have used the elementary integral

$$\int d^3k \left( \frac{1}{\mathbf{k}^2} - \frac{1}{\mathbf{k}^2 + \mu^2} \right) = 2\mu\pi^2.$$

So far, this has been an exact rewriting of Eq. (14.3.28). We must now invoke several approximations. First, consider mass renormalization. We have already calculated  $\delta m_e(\mu)$  to order  $\alpha$  in Section 11.4; it is

$$\delta m_e(\mu) = \frac{2m_e\pi^2 e^2}{(2\pi)^4} \int_0^1 dx (1+x) \ln \left( \frac{m_e^2 x^2 + \mu^2(1-x)}{m_e^2 x^2} \right). \tag{14.3.36}$$

Although in Section 11.4 we regarded  $\mu$  as a regulator mass, to be taken much larger than  $m_e$ , we can just as well use Eq. (14.3.36) to provide a value for  $\delta m_e(\mu)$  in the case that interests us here,  $\mu \ll m_e$ . In this limit Eq. (14.3.36) gives

$$\delta m_e(\mu) \longrightarrow \frac{\alpha\mu}{2} \left[ 1 - \frac{3\mu}{2\pi m_e} + \dots \right]. \tag{14.3.37}$$

We also recall that for  $Z\alpha \ll 1$ ,  $u_N(\mathbf{x})$  is given by Eq. (14.1.37) as

$$u_N(\mathbf{x}) = \frac{1}{\sqrt{2}} \left[ \frac{(1 - \boldsymbol{\sigma} \cdot \mathbf{v}/2 + \dots) f_N(\mathbf{x})}{(1 + \boldsymbol{\sigma} \cdot \mathbf{v}/2 + \dots) f_N(\mathbf{x})} \right], \tag{14.3.38}$$

where dots indicate terms of higher order in  $Z\alpha$ ;  $\mathbf{v}$  is the non-relativistic velocity operator  $-i\nabla/m_e$ ; and  $f_N(\mathbf{x})$  is a two-component spinor solution of the non-relativistic Schrödinger equation, normalized according to Eq. (14.1.38) so that

$$\int |f_N(\mathbf{x})|^2 = 1 - \frac{1}{4}(\mathbf{v}^2)_{NN} + \dots \quad (14.3.39)$$

This gives the coefficient of  $\delta m_e(\mu)$  in Eq. (14.3.35) as

$$\int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x}) = 1 - \frac{1}{2}(\mathbf{v}^2)_{NN} + \dots \quad (14.3.40)$$

We note immediately that the leading term in  $-\delta m_e(\mu) \int d^3x \bar{u}_N u_N$  cancels the term  $\alpha\mu/2$  in Eq. (14.3.35). Indeed, we could have anticipated this cancellation, because the term  $\alpha\mu/2$  in Eq. (14.3.35) survives in the limit  $Z\alpha \rightarrow 0$ , and the definition of  $m_e(\mu)$  as the renormalized electron mass implies that there must be no energy shift in this limit. By the same argument we can anticipate that the term of order  $\alpha\mu^2/m_e$  in  $\delta m_e(\mu)$  (which is larger than of order  $\alpha(Z\alpha)^4 m_e$ , and therefore cannot simply be neglected) cancels the second and third terms in Eq. (14.3.35), which are also of this order.<sup>156</sup> On the other hand, the product of the  $\alpha\mu^2/m_e$  term in  $\delta m_e$  with the second term in the matrix element (14.3.40) is of order  $(Z\alpha)^2 \alpha\mu^2/m_e \ll \alpha(Z\alpha)^4 m_e$ , and may therefore be neglected. To order  $\alpha(Z\alpha)^4 m_e$ , the only remaining effect of mass renormalization is to leave us with the product of the leading term in  $\delta m_e(\mu)$  with the term of order  $(Z\alpha)^2$  in  $\int d^3x \bar{u}_N u_N$ :

$$-\left[\frac{e^2\mu}{8\pi}\right] \left[-\frac{1}{2}(\mathbf{v}^2)_{NN}\right] = \frac{e^2\mu}{16\pi}(\mathbf{v}^2)_{NN}.$$

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<sup>156</sup>The cancellation can be shown as follows. We anticipate that the second and third terms in Eq. (14.3.35) are small enough so that they can be evaluated using the extreme nonrelativistic approximation  $\beta u_N(\mathbf{x}) = u_N(\mathbf{x})$  for the Dirac equation satisfied by  $u_N(\mathbf{x})$ . On the other hand, although the Coulomb force may be neglected in the positron wave functions  $v_M(\mathbf{x})$ , the sum over  $M$  in the third term receives important contributions from relativistic positrons, so we use the approximation  $v_{\mathbf{p},\sigma}(\mathbf{x}) = (2\pi)^{-3/2} v(\mathbf{p},\sigma) e^{i\mathbf{p}\cdot\mathbf{x}}$ , where  $v(\mathbf{p},\sigma)$  is the positron spinor introduced in Section 5.5, normalized so that  $\bar{v}(\mathbf{p},\sigma') v(\mathbf{p},\sigma) = \delta_{\sigma'\sigma}$ . Thus the sums over  $M$  in the second and third terms of Eq. (14.3.35) are approximately given by

$$\begin{aligned} \frac{1}{2} \left[ \sum_M \tilde{\Gamma}_{MN}^{i*}(\mathbf{k}) \tilde{\Gamma}_{MN}^j(\mathbf{k}) + (i \leftrightarrow j) \right] &\simeq \delta_{ij} \left( \frac{\sqrt{\mathbf{k}^2 + m_e^2} + m_e}{2\sqrt{\mathbf{k}^2 + m_e^2}} \right), \\ \sum_M |\tilde{\Gamma}_{MN}^0(\mathbf{k})|^2 &\simeq \left( \frac{\sqrt{\mathbf{k}^2 + m_e^2} - m_e}{2\sqrt{\mathbf{k}^2 + m_e^2}} \right). \end{aligned}$$

To leading order in  $\mu/m_e$ , the second and third terms in Eq. (14.3.35) are then, respectively,

$$\begin{aligned} -\frac{e^2}{4m_e(2\pi)^3} \int d^3k \left[ \frac{2}{k} - \frac{3 - k^2/(k^2 + \mu^2)}{\sqrt{k^2 + \mu^2}} \right] \left( \frac{\sqrt{k^2 + m_e^2} + m_e}{2\sqrt{k^2 + m_e^2}} \right) &\simeq \frac{\alpha\mu^2}{4\pi m_e}, \\ -\frac{e^2}{2(2\pi)^3} \int d^3k \left( \frac{1}{k^2} - \frac{1}{k^2 + \mu^2} \right) \left( \frac{\sqrt{k^2 + m_e^2} - m_e}{2\sqrt{k^2 + m_e^2}} \right) &\simeq -\frac{\alpha\mu^2}{\pi m_e}. \end{aligned}$$

(Eq. (14.3.32) rules out the possibility that a relativistic correction to the latter expression might not be suppressed by the factor  $k^2/m_e^2$  that appears in this expression for  $|\mathbf{k}|^2 \ll m_e^2$ .) These two terms are cancelled by the term  $+3\mu^2/4\pi m_e$  in  $-\delta m_e(\mu) \int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x})$ . Finally, we note that relativistic corrections to the above estimates for the sums over positron states would involve additional factors of  $v^2/c^2 \simeq (Z\alpha)^2$ , yielding contributions of order  $\alpha(Z\alpha)^2 \mu^2/m_e \ll \alpha(Z\alpha)^4 m_e$ , which justifies the non-relativistic approximations used here.

(This is the effect of mass renormalization on the electron kinetic energy, mentioned in Section 1.3.) It turns out that this is just the negative of what the first term in Eq. (14.3.35) would be if we neglected the difference between energy levels. To see this, note that the integral in this term is effectively cut off at  $|\mathbf{k}| \sim \mu \ll Z\alpha m_e$ , so we can evaluate the matrix element  $\mathbf{\Gamma}_{MN}(\mathbf{k})$  in the limit  $\mathbf{k} \rightarrow 0$ . To lowest order in  $Z\alpha$ , Eq. (14.1.42) gives

$$\mathbf{\Gamma}_{MN}(0) = (\mathbf{v})_{MN} \quad (14.3.41)$$

and using the completeness of the solutions  $f_N$  of the non-relativistic Schrödinger equation, we then have

$$\sum_M \Gamma_{MN}^{i*}(\mathbf{k}) \Gamma_{MN}^j(\mathbf{k}) \simeq (v^i v^j)_{NN}, \quad (14.3.42)$$

so to this order

$$\begin{aligned} & -\frac{e^2}{2(2\pi)^3} \int d^3k \sum_M \left[ \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / \mathbf{k}^2}{\mathbf{k}^2} - \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / (\mathbf{k}^2 + \mu^2)}{\mathbf{k}^2 + \mu^2} \right] \\ & \simeq -\frac{e^2}{2(2\pi)^3} (\mathbf{v}^2)_{NN} \int d^3k \left[ \frac{2}{3\mathbf{k}^2} - \frac{1 - \mathbf{k}^2/3(\mathbf{k}^2 + \mu^2)}{\mathbf{k}^2 + \mu^2} \right] \\ & = -\frac{e^2 \mu}{16\pi} (\mathbf{v}^2)_{NN}. \end{aligned}$$

Thus after mass renormalization we are left with just the first term in Eq. (14.3.35), less the same with energy differences  $E_N - E_M$  dropped:

$$\begin{aligned} [\delta E_N]_{\text{low energy}} &= \frac{e^2}{2(2\pi)^3} \int d^3k \sum_M (E_M - E_N) \left[ \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / \mathbf{k}^2}{\mathbf{k}^2 (E_M - E_N + |\mathbf{k}| - i\epsilon)} \right. \\ & \quad \left. - \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / (\mathbf{k}^2 + \mu^2)}{(\mathbf{k}^2 + \mu^2) (E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right]. \quad (14.3.43) \end{aligned}$$

Again using Eq. (14.3.41), this is

$$\begin{aligned} [\delta E_N]_{\text{low energy}} &= \frac{e^2}{2(2\pi)^3} \sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 \int d^3k \left[ \frac{2}{3\mathbf{k}^2 (E_M - E_N + |\mathbf{k}| - i\epsilon)} \right. \\ & \quad \left. - \frac{1 - \mathbf{k}^2/3(\mathbf{k}^2 + \mu^2)}{(\mathbf{k}^2 + \mu^2) (E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right]. \quad (14.3.44) \end{aligned}$$

Even though typical values of the electron momentum are much larger than typical atomic energy differences, this is not true of typical values of  $|\mathbf{k}|$  in this integral, because the integral would be infrared divergent if we did not keep the  $E_M - E_N$  terms in the denominators. The integral in Eq. (14.3.44) may be evaluated in the limit  $\mu \gg |E_M - E_N| \sim (Z\alpha)^2 m_e$  by dividing the range of integration of  $|\mathbf{k}|$  into two segments, from zero to  $\lambda$  and from  $\lambda$  to

infinity, with  $\lambda$  chosen so that  $|E_M - E_N| \ll \lambda \ll \mu$  but otherwise arbitrary. In this way we find

$$\int_0^\infty k^2 dk \left[ \frac{2}{3k^2(E_M - E_N + k) - i\epsilon} - \frac{1 - k^2/3(k^2 + \mu^2)}{(k^2 + \mu^2)(E_M - E_N + \sqrt{k^2 + \mu^2} - i\epsilon)} \right] \\ \simeq \frac{2}{3} \left[ \ln \left( \frac{\mu}{2|E_M - E_N|} \right) + \frac{5}{6} + i\pi\theta(E_N - E_M) \right].$$

The imaginary term here reflects the possibility of decay of the atom in state  $N$  to states  $M$  of lower energy. This term contributes to the decay rate, given by the imaginary part of the energy shift. We are interested here in the real part of the energy shift, and so will drop this imaginary term in what follows. Eq. (14.3.44) now gives

$$[\delta E_N]_{\text{low energy}} = \frac{e^2}{6\pi^2} \sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 \left[ \ln \left( \frac{\mu}{2|E_N - E_M|} \right) + \frac{5}{6} \right]. \quad (14.3.45)$$

### C Total Energy Shift

We need to make a connection between the sum in Eq. (14.3.45) and the matrix element in the high-energy term (14.3.22). For this purpose, let's first see what value the sum in Eq. (14.3.45) would have if we could ignore the logarithm. We note that  $(E_M - E_N)v_{NM} = [v, H]_{NM}$ , so

$$\sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 = \frac{1}{2} \sum_M ([v^i, H]_{NM} v_{MN}^i + v_{NM}^i [H, v^i]_{MN}) \\ = -\frac{1}{2m_e^2} ([p^i, [p^i, H]])_{NN}.$$

The only term in the non-relativistic Hamiltonian  $H$  that does not commute with the momentum operator  $\mathbf{p}$  is the potential term  $-e\mathcal{A}^0(\mathbf{x})$ , so this gives

$$\sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 = -\frac{e}{2m_e^2} (\nabla^2 \mathcal{A}^0(\mathbf{x}))_{NN}. \quad (14.3.46)$$

Inspection of Eqs. (14.3.45) and (14.3.22) now shows that the term proportional to  $\ln \mu$  in the high-energy term is cancelled by the term proportional to  $\ln \mu$  in the low-energy term:

$$\delta E_N = [\delta E_N]_{\text{high energy}} + [\delta E_N]_{\text{low energy}} \\ = \frac{e^2}{6\pi^2} \sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 \left[ \ln \left( \frac{m_e}{2|E_N - E_M|} \right) + \frac{5}{6} - \frac{1}{5} \right] \\ - \frac{e^2}{16\pi^2 m_e^2} (\boldsymbol{\sigma} \cdot (\nabla(e\mathcal{A}^0(\mathbf{x})) \cdot \mathbf{p}))_{NN}. \quad (14.3.47)$$

So far, this has been for a general electrostatic field  $\mathcal{A}^0(\mathbf{x})$ . Let us now specialize to a pure Coulomb field, with

$$\mathcal{A}^0(\mathbf{x}) = Ze/|\mathbf{x}|. \quad (14.3.48)$$

In this case, Eq. (14.3.46) reads

$$\sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 = \frac{Ze^2}{2m_e^2} (\delta^3(\mathbf{x}))_{NN} = \frac{Ze^2}{2m_e^2} (f_N^\dagger(0) f_N(0)). \quad (14.3.49)$$

This is non-vanishing only for  $\ell = 0$ . Also, the matrix element in the last term in Eq. (14.3.47) has the value

$$(\boldsymbol{\sigma} \cdot (\nabla(e\mathcal{A}^0(\mathbf{x})) \cdot \mathbf{p}))_{NN} = -Ze^2 \left( \frac{1}{r^3} \boldsymbol{\sigma} \cdot \mathbf{L} \right)_{NN}, \quad (14.3.50)$$

which is non-vanishing only for  $\ell \neq 0$ . It is therefore useful at this point to divide our consideration between the two cases,  $\ell = 0$  and  $\ell \neq 0$ .

**i.**  $\ell = 0$  It is convenient here to define a mean excitation energy  $\Delta E_N$ :

$$\begin{aligned} \sum_M |\mathbf{v}_{MN}|^2 (E_M - E_N) \ln |E_N - E_M| &= \ln \Delta E_N \sum_M |\mathbf{v}_{MN}|^2 (E_M - E_N) \\ &= \frac{Ze^2}{2m_e^2} \ln \Delta E_N (f_N^\dagger(0) f_N(0)). \end{aligned} \quad (14.3.51)$$

For  $s$ -wave hydrogenic states, the label  $N$  consists of a principal quantum number  $n$  and spin  $z$ -component  $m$ , and  $[f_{nm}(0)]_\sigma = 2(Z\alpha m_e/n)^{3/2} \delta_{\sigma,m}/\sqrt{4\pi}$ , so

$$(f_N^\dagger(0) f_N(0)) = \frac{1}{\pi} \left( \frac{Z\alpha m_e}{n} \right)^3. \quad (14.3.52)$$

Using Eqs. (14.3.51) and (14.3.52) in Eq. (14.3.47) gives the energy shift in these states as

$$[\delta E]_{n,\ell=0} = \frac{4\alpha(Z\alpha)^4 m_e}{3\pi n^3} \left[ \ln \left( \frac{m_e}{2\Delta E_{n,\ell=0}} \right) + \frac{19}{30} \right]. \quad (14.3.53)$$

**ii.**  $\ell \neq 0$  For non-vanishing orbital angular momentum the sum (14.3.49) vanishes, so the definition (14.3.51) is inappropriate. Instead, it is conventional here to define a mean excitation energy  $\Delta E_N$  by

$$\sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 \ln |E_N - E_M| = \frac{2(Z\alpha)^4 m_e}{n^3} \ln \left( \frac{2\Delta E_N}{Z^2 \alpha^2 m_e} \right). \quad (14.3.54)$$

(Because Eq. (14.3.49) vanishes, it makes no difference what units are used to measure  $E_N - E_M$  in Eq. (14.3.54).) Also, in a state of total angular momentum  $j$  and orbital angular momentum  $\ell$ , the scalar product  $\boldsymbol{\sigma} \cdot \mathbf{L}$  has the familiar value  $j(j+1) - \ell(\ell+1) - 3/4$ , and for principal quantum number  $n$  the operator  $1/r^3$  has the expectation value

$$\int d^3r |f_N|^2 / r^3 = \frac{2Z^3 \alpha^3 m_e^3}{n^3 \ell(\ell+1)(2\ell+1)}. \quad (14.3.55)$$

Putting this all together in Eq. (14.3.47), we have for  $\ell \neq 0$ :

$$\begin{aligned} [\delta E]_{nj\ell} &= - \frac{4\alpha(Z\alpha)^4 m_e}{3\pi n^3} \ln \left( \frac{2\Delta E_{nj\ell}}{Z^2 \alpha^2 m_e} \right) \\ &\quad + \frac{\alpha(Z\alpha)^4 m_e}{2\pi n^3} \left[ \frac{j(j+1) - \ell(\ell+1) - \frac{3}{4}}{\ell(\ell+1)(2\ell+1)} \right]. \end{aligned} \quad (14.3.56)$$

It only remains to use these results to give numerical value for the energy shifts. The mean excitation energies here must be calculated numerically; using non-relativistic hydrogen wave functions, they have the values [5]:

$$\Delta E_{1s} = 19.769266917(6) \text{ Ry},$$

$$\Delta E_{2s} = 16.63934203(1) \text{ Ry},$$

$$\Delta E_{2p} = 0.9704293186(3) \text{ Ry},$$

where  $1 \text{ Ry} \equiv m_e \alpha^2 / 2 = 13.6057 \text{ eV}$ . Eq. (14.3.53) then gives

$$\begin{aligned} [\delta E]_{1s} &= \frac{4\alpha^5 m_e}{3\pi} \left[ \ln \left( \frac{m_e}{2\Delta E_{1s}} \right) + \frac{19}{30} \right] = 3.3612 \cdot 10^{-5} \text{ eV} \\ &= 2\pi\hbar \cdot 8127.4 \text{ MHz}, \end{aligned} \quad (14.3.57)$$

$$\begin{aligned} [\delta E]_{2s} &= \frac{\alpha^5 m_e}{6\pi} \left[ \ln \left( \frac{m_e}{2\Delta E_{2s}} \right) + \frac{19}{30} \right] = 4.2982 \cdot 10^{-5} \text{ eV} \\ &= 2\pi\hbar \cdot 1039.31 \text{ MHz}, \end{aligned} \quad (14.3.58)$$

while Eq. (14.3.56) gives

$$\begin{aligned} [\delta E]_{2p_{1/2}} &= \frac{\alpha^5 m_e}{6\pi} \left[ \ln \left( \frac{\alpha^2 m_e}{2\Delta E_{2p}} \right) - \frac{1}{8} \right] = -5.3267 \cdot 10^{-8} \text{ eV} \\ &= 2\pi\hbar \cdot (-12.88) \text{ MHz}. \end{aligned} \quad (14.3.59)$$

The classic Lamb shift is the energy difference between the  $2s$  and  $2p_{1/2}$  states of the hydrogen atom, states that would be degenerate in the absence of radiative corrections. Our calculation has given

$$[\delta E]_{2s} - [\delta E]_{2p_{1/2}} = 4.35152 \cdot 10^{-5} \text{ eV} = 2\pi\hbar \cdot 1052.19 \text{ MHz}.$$

This is numerically close (though analytically not identical) to the old result of Kroll and Lamb [6] and French and Weisskopf [7], which they obtained using the techniques of old-fashioned perturbation theory. Earlier in this section we made a crude estimate of 1300 MHz by considering only the high-energy contribution to the  $2s$  energy shift, with an infrared cutoff guessed to be of order  $\alpha^2 m_e = 2 \text{ Ry}$ . We can now see that this was an overestimate, arising mostly from the fact that the true value of the effective infrared cutoff  $\Delta E_{2s} = 16.64 \text{ Ry}$  is considerably larger than we had guessed. On the other hand, as described in Section 1.3, in 1947 Hans Bethe [8] was able to make a rather good estimate of the Lamb shift, 1040 MHz, by considering only the low-energy contribution to the  $2s$  energy shift, with an ultraviolet cutoff guessed to be  $m_e$ . (Bethe made the first estimate of the excitation energy,  $\Delta E_{2s} \simeq 17.8 \text{ Ry}$ .)

The calculation of the Lamb shift described here has been improved by the inclusion of higher-order radiative corrections and nuclear size and recoil effects. At present the greatest uncertainty is due to a doubt about the correct value of the rms charge radius  $r_p$  of the



proton. For  $r_p = 0.862 \cdot 10^{-13}$  cm or  $r_p = 0.805 \cdot 10^{-13}$  cm, one calculation [9] gives a Lamb shift of either 1057.87 MHz or 1057.85 MHz, while another [10] gives either 1057.883 MHz or 1057.865 MHz. Given the uncertainty in proton radius, the agreement is excellent with the present experimental value [11], 1057.845(9) MHz. The accuracy of this experimental value is limited chiefly by the  $\sim 100$  MHz natural linewidth of the  $2p$  state in hydrogen, so further improvements here will be very difficult.

In the last few years there has been an important improvement in measurements of the energy shift of the  $1s$  ground state itself, by direct comparison of the frequency of the  $1s$ – $2s$  resonance with four times the frequencies of the  $2s$ – $4s$  and  $2s$ – $4d$  two-photon resonances. These  $s$  and  $d$  states are much narrower than the  $2p$  state, so these frequency differences can be measured more accurately than the classic Lamb shift. For a brief while it seemed that there was a discrepancy here between theory and experiment. Calculations [12,13] showed that for a proton radius  $r_p = 0.862(11) \cdot 10^{-13}$  cm or  $0.805(11) \cdot 10^{-13}$  cm, the inclusion of proton size and other corrections increases the theoretical  $1s$  energy shift from the above result of 8127.4 MHz to 8173.12(6) MHz or 8172.94(9) MHz, respectively. This result for a proton radius  $r_p = 0.862(11) \cdot 10^{-13}$  cm, which is believed to be more reliable, was not quite in agreement with the measured value [13] of 8172.86(5) MHz. But a later calculation [14] that adopts this proton radius and includes terms of order  $\alpha^2(Z\alpha)^5$  yields energy shifts for the  $1s$ ,  $2s$ , and  $4s$  states in agreement with experiment. So apparently quantum electrodynamics wins again.

### Weinberg Exercises

- W.14.1.** Consider a charged scalar particle of mass  $m \neq 0$ , described by a field  $\phi(x)$  whose only interaction is with an external time-independent electromagnetic field  $\mathcal{A}^\mu(x)$ . Let  $|N\rangle$  be a complete set of normalized one-boson or one-antiboson states with energies  $E_N$ , and define

$$u_N(\mathbf{x})e^{-iE_N t} = \langle \Omega | \phi(\mathbf{x}, t) | N \rangle, \quad v_N(\mathbf{x})e^{iE_N t} = \langle N | \phi(\mathbf{x}, t) | \Omega \rangle,$$

where  $|\Omega\rangle$  is the vacuum. Show that the  $u_N$  and  $v_N$  together form a complete set, and give formulas for the coefficients of  $u_N$  and  $v_N$  in the expansion of a general function  $f(\mathbf{x})$ .

- W.14.2.** Suppose we include radiative corrections in the theory of Problem 1. Let  $i\Pi^*(x, y)$  be the sum of all diagrams with one incoming and one outgoing charged scalar line (excluding final scalar propagators) to order  $\alpha$ . Derive a formula for the shift in the energies  $E_N$  of the one-boson states due to these radiative corrections, in terms of  $u_N(\mathbf{x})$  and  $\Pi^*(x, y)$ .
- W.14.3.** Use the results of Section 14.3 to calculate the radiative decay rate of the  $2p$  states of hydrogen.
- W.14.4.** Suppose that the electron has an interaction with a light scalar field  $\phi$ , of the form  $g\phi\bar{\psi}_e\psi_e$ . Suppose that the scalar mass  $m_\phi$  is in the range  $(Z\alpha)^2 m_e \ll m_\phi \ll Z\alpha m_e$ .

Calculate the change in the energy of the  $1s$  state of hydrogenic atoms due to this interaction.

**W.14.5.** Carry out the calculation of Problem 4 for  $m_\phi = 0$ .

### Solutions to Weinberg Exercises

*These editorial solutions were added by Codex and are not part of Weinberg's original text.*

**W.14.1.** Choose the gauge in which the static external field is time independent and write the charged Klein–Gordon field as

$$\phi(\mathbf{x}, t) = \sum_N \left[ a_N u_N(\mathbf{x}) e^{-iE_N t} + b_N^\dagger v_N(\mathbf{x}) e^{iE_N t} \right].$$

(A sum over  $N$  includes continuum integration.) For a purely electrostatic field define

$$\epsilon_N^{(+)}(\mathbf{x}) = E_N + e\mathcal{A}^0(\mathbf{x}), \quad \epsilon_N^{(-)}(\mathbf{x}) = E_N - e\mathcal{A}^0(\mathbf{x}).$$

The equal-time commutators  $[\phi(\mathbf{x}), \phi^\dagger(\mathbf{y})] = 0$  and  $[\phi(\mathbf{x}), \pi(\mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y})$  give the two completeness identities

$$\begin{aligned} \sum_N \{u_N(\mathbf{x}) u_N^*(\mathbf{y}) - v_N(\mathbf{x}) v_N^*(\mathbf{y})\} &= 0, \\ \sum_N \{u_N(\mathbf{x}) \epsilon_N^{(+)}(\mathbf{y}) u_N^*(\mathbf{y}) + v_N(\mathbf{x}) \epsilon_N^{(-)}(\mathbf{y}) v_N^*(\mathbf{y})\} &= \delta^3(\mathbf{x} - \mathbf{y}). \end{aligned}$$

Consequently every square-integrable test function has the expansion

$$f(\mathbf{x}) = \sum_N [c_N u_N(\mathbf{x}) + d_N v_N(\mathbf{x})]$$

with

$$\begin{aligned} c_N &= \int d^3y u_N^*(\mathbf{y}) \epsilon_N^{(+)}(\mathbf{y}) f(\mathbf{y}), \\ d_N &= \int d^3y v_N^*(\mathbf{y}) \epsilon_N^{(-)}(\mathbf{y}) f(\mathbf{y}). \end{aligned}$$

These are the charged Klein–Gordon inner-product projections. If a static vector potential is also present, the same formulas hold because only the covariant time derivative enters the equal-time product.

**W.14.2.** Time-translation invariance permits the energy transform

$$\Pi_{\mathcal{A}}^*(\mathbf{x}, \mathbf{y}; E) = \int d\tau e^{iE\tau} \Pi_{\mathcal{A}}^*(x^0, \mathbf{x}; x^0 - \tau, \mathbf{y}).$$

Near an unperturbed one-particle pole, insert the mode expansion from Problem 1 on both sides of the Dyson equation. A single proper self-energy insertion produces a double pole. Comparing its coefficient with

$$\frac{1}{E - E_N - \delta E_N} = \frac{1}{E - E_N} + \frac{\delta E_N}{(E - E_N)^2} + \cdots$$

gives

$$\delta E_N = - \int d^3x d^3y u_N^*(\mathbf{x}) \Pi_{\mathcal{A}}^*(\mathbf{x}, \mathbf{y}; E_N) u_N(\mathbf{y}).$$

The Klein–Gordon normalization used in Problem 1 already contains the pole residue, so no additional  $1/(2E_N)$  occurs. In a degenerate subspace the right-hand side is a matrix that must be diagonalized. Its real part shifts the level, while an unstable level has  $\Gamma_N = -2 \operatorname{Im} \delta E_N$ .

**W.14.3.** The imaginary part of Eq. (14.3.45) gives the electric-dipole rate

$$\Gamma_N = \frac{4\alpha}{3} \sum_{E_M < E_N} (E_N - E_M) |\mathbf{v}_{MN}|^2 = \frac{4\alpha}{3} \sum_{E_M < E_N} (E_N - E_M)^3 |\mathbf{x}_{MN}|^2.$$

For a  $2p$  state the only lower E1 level is  $1s$ , and

$$\omega_{21} = \frac{3}{8} \alpha^2 m_e, \quad \int_0^\infty dr r^3 R_{10}(r) R_{21}(r) = \frac{256}{81\sqrt{6}} a_0.$$

The angular integral gives, for each magnetic sublevel,

$$\sum_i |\langle 1s | x^i | 2p, m \rangle|^2 = \frac{1}{3} \left( \frac{256 a_0}{81\sqrt{6}} \right)^2 = \frac{32768}{59049} a_0^2.$$

It follows that

$$\Gamma_{2p \rightarrow 1s} = \frac{2^8}{3^8} \alpha^5 m_e.$$

Numerically this is  $6.27 \times 10^8 \text{ s}^{-1}$ , or  $\tau_{2p} \simeq 1.60 \text{ ns}$ . Fine-structure and recoil corrections are beyond the requested order.

**W.14.4.** Let  $H$  be the Schrödinger–Coulomb Hamiltonian and  $\Delta_{MN} = E_M - E_N$ . After the free on-shell mass and residue subtractions, old-fashioned perturbation theory gives the state-dependent part of the scalar self-energy as

$$-\frac{g^2}{2} \int \frac{d^3k}{(2\pi)^3 \omega_k} \left\langle N \left| e^{i\mathbf{k} \cdot \mathbf{x}} \frac{1}{H - E_N + \omega_k} e^{-i\mathbf{k} \cdot \mathbf{x}} \right| N \right\rangle_{\text{sub}}, \quad \omega_k = \sqrt{\mathbf{k}^2 + m_\phi^2}.$$

In the stated range  $\Delta_{MN} \ll m_\phi \ll Z\alpha m_e$ , expand the resolvent in  $H - E_N$  and use isotropy of the  $1s$  state. The constant and linear terms are precisely the on-shell mass and residue subtractions. The first surviving term is

$$\delta E_N = -\frac{g^2 \langle \mathbf{p}^2 \rangle_N}{6m_e^2} \int \frac{d^3k}{(2\pi)^3} \frac{\mathbf{k}^2}{(\mathbf{k}^2 + m_\phi^2)^2}.$$

Dimensional regularization isolates the nonanalytic, scheme-independent term,

$$\int \frac{d^3k}{(2\pi)^3} \frac{\mathbf{k}^2}{(\mathbf{k}^2 + m_\phi^2)^2} = -\frac{3m_\phi}{8\pi}.$$

Since  $\langle \mathbf{p}^2 \rangle_{1s} = (Z\alpha m_e)^2$ ,

$$\delta E_{1s} = \frac{g^2}{16\pi} (Z\alpha)^2 m_\phi + O\left(g^2 (Z\alpha)^4 m_e, g^2 (Z\alpha)^2 \frac{m_\phi^2}{Z\alpha m_e}\right).$$

The positive sign means that the level is made less deeply bound. Terms analytic in  $m_\phi^2$  belong to the shorter-distance matching indicated in the remainder.

**W.14.5.** The expansion used in Problem 4 is not uniform when  $m_\phi = 0$ . After the on-shell mass and residue pieces have been removed, the dipole term  $e^{i\mathbf{k}\cdot\mathbf{x}} = 1 + i\mathbf{k}\cdot\mathbf{x} + \dots$  leaves

$$\delta E_N^{\text{soft}} = \frac{g^2}{12\pi^2} \sum_M \Delta_{MN}^3 |\mathbf{x}_{MN}|^2 \ln \frac{\Lambda}{|\Delta_{MN}|} + \text{nonlogarithmic matching terms.}$$

The artificial scale  $\Lambda$  cancels against the hard part. For the ground state define the scalar Bethe mean excitation energy by

$$\ln \Delta E_{1s}^{(\phi)} = \frac{\sum_M \Delta_{M1} |\mathbf{v}_{M1}|^2 \ln \Delta_{M1}}{\sum_M \Delta_{M1} |\mathbf{v}_{M1}|^2}.$$

The double-commutator sum rule gives

$$\sum_M \Delta_{M1} |\mathbf{v}_{M1}|^2 = \frac{Ze^2}{2m_e^2} |\psi_{1s}(0)|^2 = 2(Z\alpha)^4 m_e.$$

Thus, to the leading logarithmic accuracy fixed without a full hard one-loop matching calculation,

$$\delta E_{1s}(m_\phi = 0) = \frac{g^2}{6\pi^2} (Z\alpha)^4 m_e \left[ \ln \frac{m_e}{\Delta E_{1s}^{(\phi)}} + O(1) \right].$$

Because  $\Delta E_{1s}^{(\phi)}$  is of atomic order, the leading dependence may be displayed more simply as

$$\delta E_{1s}(0) = \frac{g^2}{6\pi^2} (Z\alpha)^4 m_e \left[ \ln \frac{1}{(Z\alpha)^2} + O(1) \right].$$

This also explains why simply setting  $m_\phi = 0$  in the leading answer to Problem 4 is invalid: the linear term disappears and atomic excitation energies themselves regulate the infrared region.

## Supplementary Exercises

### S.14.1. From the Dirac Equation to the Pauli Hamiltonian

(Inspired by Michael Dine, *Physics 217 Homework 1 Solutions*.)

- (a) For a charge  $q$  in a static electromagnetic background, write a positive energy Dirac spinor as  $e^{-i(m+\varepsilon)t}(\varphi, \chi)^\top$ . Derive the coupled equations for  $\varphi$  and  $\chi$ , solve for  $\chi$  through relative order  $1/m$ , and obtain the leading nonrelativistic Hamiltonian for  $\varphi$ .

- (b) Use  $(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2 = \pi^2 - q\boldsymbol{\sigma} \cdot \mathbf{B}$ , where  $\boldsymbol{\pi} = \mathbf{p} - q\mathbf{A}$ , to reduce the result of Exercise S.14.1. Identify the orbital and spin couplings to a weak magnetic field and check gauge covariance.
- (c) Compare the spin term in the Pauli Hamiltonian with  $-\boldsymbol{\mu} \cdot \mathbf{B}$ , taking  $\mathbf{S} = \boldsymbol{\sigma}/2$ . Determine the tree-level gyromagnetic ratio and explain which coefficient is changed by the Pauli form factor  $F_2(0)$ .

### S.14.2. Darwin, Spin–Orbit, and Coulombic Power Counting

*(Inspired by Michael Dine, Physics 217 Homework 1 Solutions.)*

- (a) Continue the Foldy–Wouthuysen reduction in a static electric field through order  $m^{-2}$ . Isolate the spin-independent term proportional to  $\boldsymbol{\nabla} \cdot \mathbf{E}$ , and evaluate its first-order energy shift for a Coulomb field in a hydrogenic  $n\ell m$  state.
- (b) Extract the spin–orbit operator at order  $m^{-2}$  for a central electrostatic potential  $V(r) = qA^0(r)$ . Express it in terms of  $\mathbf{L} \cdot \mathbf{S}$ , and show explicitly where the factor associated with Thomas precession appears.
- (c) For a hydrogenic state with principal quantum number of order unity, assign orders in  $Z\alpha$  to  $r^{-1}$ ,  $\mathbf{p}$ , the binding energy,  $\mathbf{E}$ , the Darwin operator, and the spin–orbit operator. Use these assignments to predict the order of the fine-structure energy shift.

### S.14.3. Feynman–Hellmann Relations for Hydrogenic Matrix Elements

*(Inspired by MIT OpenCourseWare 8.06, Problem Set 2.)*

- (a) Let  $H(\lambda)$  have a normalized, nondegenerate eigenstate  $|n(\lambda)\rangle$ . Derive the Feynman–Hellmann theorem by differentiating the eigenvalue equation. State what must be done inside a degenerate eigenspace.
- (b) Apply Exercise S.14.7 to  $H(\kappa) = \mathbf{p}^2/(2m) - \kappa/r$ . Starting only from the exact Coulomb energy  $E_n = -m\kappa^2/(2n^2)$ , find  $\langle n\ell m | r^{-1} | n\ell m \rangle$  and check the virial theorem.
- (c) Add  $\lambda/r^2$  to the Coulomb Hamiltonian and absorb it into an effective angular momentum. Differentiate the resulting energy at  $\lambda = 0$  to obtain  $\langle r^{-2} \rangle_{n\ell}$ , including its dependence on  $\ell$ .

### S.14.4. The Stark Effect and Polarizability

*(Inspired by MIT OpenCourseWare 8.06, Problem Set 2.)*

- (a) Place hydrogen in a uniform static electric field  $\boldsymbol{\mathcal{E}} = \mathcal{E}\hat{\mathbf{z}}$ , so that  $H' = e\mathcal{E}z$ . Prove that the  $1s$  energy has no term linear in  $\mathcal{E}$ , using both parity and the angular integral.

- (b) Write the second-order  $1s$  Stark shift as  $-\alpha_E \mathcal{E}^2/2$ . Use completeness and the fact that every dipole-accessible excitation energy is at least  $E_2 - E_1$  to derive an upper bound on  $\alpha_E$ . Evaluate the needed expectation  $\langle 1s|z^2|1s\rangle$ .
- (c) Ignoring fine structure and the Lamb shift, diagonalize  $e\mathcal{E}z$  within the four orbital states of the  $n = 2$  hydrogen manifold. Identify the states that mix, compute the nonzero matrix element, and give all first-order shifts.

#### S.14.5. Einstein Coefficients and Electric-Dipole Radiation

(Inspired by David Tong, *Topics in Quantum Mechanics*, Chapter 4.)

- (a) For two atomic levels in a thermal photon bath, include absorption, stimulated emission, and spontaneous emission. Combine detailed balance with the Planck distribution to derive the relation between the Einstein  $A$  and  $B$  coefficients, including level degeneracies.
- (b) Starting from  $H_I = -q \mathbf{x} \cdot \mathbf{E}(0)$  in the dipole approximation, apply Fermi's golden rule, sum the two photon polarizations, and integrate over directions. Derive the total E1 rate in natural units.
- (c) Treat the three Cartesian components of  $\mathbf{x}$  as a parity-odd rank-one tensor. Use parity and angular-momentum addition to derive the E1 selection rule for orbital angular momentum and explain why  $\ell = 0 \rightarrow \ell' = 0$  is forbidden.

#### S.14.6. Polarization Rules, the $2p$ Lifetime, and Stark Quenching

(Inspired by David Tong, *Topics in Quantum Mechanics*, Chapter 4.)

- (a) Rewrite  $\mathbf{x} \cdot \boldsymbol{\epsilon}$  in spherical components  $x_q$  with  $q = 0, \pm 1$ . Use the Wigner-Eckart theorem to obtain the magnetic-quantum-number rule and associate  $\pi$  and  $\sigma^\pm$  polarizations with the allowed changes.
- (b) Evaluate the radial and angular matrix elements for  $2p \rightarrow 1s + \gamma$ . Insert them in the E1 formula, simplify the answer to a rational multiple of  $\alpha^5 m_e$ , and estimate the lifetime.
- (c) A weak dc electric field mixes  $2s$  with  $2p, m = 0$ . Including their physical separation  $\Delta_{LS} = E_{2s} - E_{2p}$ , find the admixture coefficient and the induced one-photon width of  $2s$ . State the perturbative condition on the field.

#### S.14.7. Vacuum-Polarization Contribution to the Lamb Shift

(Adapted from Charles B. Thorn, *Quantum Field Theory Problem Set 13*, Problem 48.)

After charge renormalization, the one-loop relation between the electric displacement and electric field, through first order in  $e^2$ , is

$$\tilde{\mathbf{E}}(k) = \left[ 1 + \frac{e^2}{2\pi^2} \int_0^1 dx x(1-x) \ln \left( 1 + x(1-x) \frac{\mathbf{k}^2}{m_e^2} \right) \right] \tilde{\mathbf{D}}(k).$$

Consequently the potential energy of the electron in hydrogen is modified from its lowest-order value  $-e^2/(4\pi r)$ . Calculate this modification in lowest-order nonrelativistic perturbation theory.

- (a) Explain why the momentum transfers obey  $\mathbf{k}^2/m_e^2 \ll 1$  and  $(k^0)^2 \ll \mathbf{k}^2$ .
- (b) Use those estimates to expand the logarithm and evaluate the parameter integral. Show that the additional potential has the form

$$\delta V(\mathbf{r}) = -V_1 \delta^3(\mathbf{r})$$

and determine  $V_1$ .

- (c) Explain why this perturbation affects only  $s$ -levels ( $\ell = 0$ ).
- (d) Calculate the energy shift of the hydrogen  $2s$  state in electron-volts. Natural units have been used above; restore  $\hbar$  and  $c$  at a convenient late stage to check the dimensions.

### S.14.8. The Magnetic Moment of a Dirac Fermion

(Adapted from J. McGreevy, *Physics 215A Quantum Field Theory, Fall 2023, Assignment 10, Problem 5.*)

Consider the interaction Hamiltonian density

$$\mathcal{H}_I = -iq \bar{\Psi} \gamma^\mu \Psi A_\mu,$$

where the vector potential  $A_\mu$  is a fixed classical electromagnetic background. The factor  $-i$  translates the source interaction to the volume's gamma-matrix convention. Define one-particle states by

$$\langle 0 | \Psi(x) | \mathbf{p}, s \rangle = e^{-ip \cdot x} u_s(p).$$

The goal is to show that, in the rest frame, the one-particle Hamiltonian contains a magnetic-dipole term proportional to  $\mathbf{S} \cdot \mathbf{B}$ , where  $\mathbf{S} = \boldsymbol{\sigma}/2$ .

- (a) The charge  $q$  is real. What condition on  $A_\mu$  is required for

$$H_I = \int d^3x \mathcal{H}_I$$

to be Hermitian?

- (b) As a bonus, determine how  $A_\mu$  must transform under parity  $P$  and charge conjugation  $C$  for  $H_I$  to be invariant. How do  $\mathbf{E}$  and  $\mathbf{B}$  transform? Explain why these symmetries allow a magnetic dipole moment but forbid an electric dipole moment.
- (c) Let

$$\gamma^{\mu\nu} \equiv \frac{1}{2} [\gamma^\mu, \gamma^\nu].$$

In the nonrelativistic limit, show that

$$\bar{u}(p')\gamma^{\mu\nu}u(p)F_{\mu\nu} = a\xi'^{\dagger}\boldsymbol{\sigma}\cdot\mathbf{B}\xi$$

and determine  $a$ . In a Weyl basis compatible with the volume conventions, use

$$u(p) \longrightarrow \sqrt{m_{\Psi}}\begin{pmatrix} \xi \\ \xi \end{pmatrix}, \quad u(p') \longrightarrow \sqrt{m_{\Psi}}\begin{pmatrix} \xi' \\ \xi' \end{pmatrix}.$$

- (d) Take  $A_{\mu}$  to describe a magnetic field  $\mathbf{B}$  that is uniform in space and time. Show in the nonrelativistic limit that

$$\langle \mathbf{p}', s' | H_I | \mathbf{p}, s \rangle = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') h(\xi', \xi, \mathbf{B}) + \text{terms independent of spin.}$$

Find  $h$ , using the Gordon identity if helpful. Convert to one-particle states with nonrelativistic normalization and interpret the result as a gyromagnetic ratio  $g_{\text{mag}} = 2$ .

- (e) Optionally determine how the answer changes after adding

$$\Delta\mathcal{H} = \frac{c}{M}\bar{\Psi}F_{\mu\nu}[\gamma^{\mu}, \gamma^{\nu}]\Psi.$$

Include the condition on  $c$  required for this Hamiltonian term to be Hermitian in the stated gamma-matrix convention.

#### S.14.9. Bubble-Chain Bound States in $2+1$ Dimensions

(Adapted from J. McGreevy, *Physics 215B Quantum Field Theory*, Winter 2025, Assignment 8, Problem 1.)

Besides a one-particle delta function at  $s = m^2$  and a continuum above  $s = (2m)^2$ , the spectral density of an operator that creates a massive particle can contain bound-state delta functions with  $m^2 < s < (2m)^2$ . To see how such a pole can arise, work in  $D = 2+1$  dimensions with the mostly-plus form of the source action

$$S[\phi] = \int d^3x \left[ -\frac{1}{2}\partial_{\mu}\phi\partial^{\mu}\phi - \frac{1}{2}m^2\phi^2 - \frac{g}{4!}\phi^4 \right], \quad m, g \in \mathbb{R}.$$

Consider both signs of  $g$ , setting aside vacuum stability; as the source notes, a small stabilizing  $\phi^6$  interaction may be imagined without affecting this calculation.

- (a) For elastic  $\phi\phi \rightarrow \phi\phi$  scattering, let  $s = E_T^2$ , where  $E_T$  is the total center-of-mass energy. Compute  $\mathcal{M}(s)$  in the bubble-chain approximation defined by

$$\text{Contact} + \text{One-bubble} + \text{Two-bubble} + \dots$$

These are respectively the contact term, the one-bubble term, the two-bubble term, and their continuation. Do not justify the approximation or the convergence of the series. The loop may be left as a Feynman-parameter integral.



- (b) Verify the optical theorem explicitly within this approximation. In the center-of-mass frame in three spacetime dimensions,

$$\frac{d\sigma}{d\theta} = \frac{|\mathcal{M}|^2}{32\pi p E_T^2},$$

where  $p$  is the magnitude of either spatial momentum.

- (c) Look for a real pole at  $s = M_B^2$  below threshold,

$$0 < M_B^2 < 4m^2.$$

Determine the signs and range of  $g$  for which the bubble-chain approximation predicts such a bound state. An analytic formula for  $M_B$  is not required.

#### S.14.10. Nonrelativistic Forces from QED and Pseudoscalar Exchange

(Adapted from J. McGreevy, *Physics 215A Quantum Field Theory, Fall 2023, Assignment 10, Problem 6.*)

- (a) Starting from QED, derive the repulsive  $1/r^2$  force between two nonrelativistic electrons.
- (b) Consider a massive Dirac field  $\Psi$  and a massive pseudoscalar  $\phi$ , interacting through

$$V_5 = g_5 \bar{\Psi} \gamma_5 \Psi \phi.$$

- i. Assign parity transformations to both fields and prove that the interaction is parity invariant.
- ii. List all propagator and vertex rules needed at leading order.
- iii. Draw and evaluate every leading-order diagram for  $\Psi\Psi \rightarrow \Psi\Psi$ , including the exchange required for identical fermions.
- iv. Take  $m_\Psi \gg |\mathbf{p}|$  and derive the effective nonrelativistic interaction Hamiltonian. If the naive leading term vanishes, continue to the first nonzero term, as emphasized in the source.

#### Solutions to Supplementary Exercises

##### S.14.1. From the Dirac Equation to the Pauli Hamiltonian

- (a) With  $\boldsymbol{\pi} = \mathbf{p} - q\mathbf{A}$ , the two-component equations are

$$(\varepsilon - qA^0)\varphi = \boldsymbol{\sigma} \cdot \boldsymbol{\pi} \chi, \quad (2m + \varepsilon - qA^0)\chi = \boldsymbol{\sigma} \cdot \boldsymbol{\pi} \varphi.$$

Thus

$$\chi = \frac{\boldsymbol{\sigma} \cdot \boldsymbol{\pi}}{2m} \varphi + O(m^{-2}),$$

and the upper component obeys

$$i\partial_t \varphi = \left[ m + qA^0 + \frac{(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2}{2m} \right] \varphi + O(m^{-2}).$$

The rest energy may be dropped. The small component is suppressed by the particle velocity, as required for a consistent nonrelativistic expansion.

- (b) The covariant momenta satisfy  $[\pi_i, \pi_j] = iq\epsilon_{ijk}B_k$ . Hence

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2 = \boldsymbol{\pi}^2 - q\boldsymbol{\sigma} \cdot \mathbf{B}$$

and

$$H_P = qA^0 + \frac{\boldsymbol{\pi}^2}{2m} - \frac{q}{2m}\boldsymbol{\sigma} \cdot \mathbf{B}.$$

For a spatially constant field, the term linear in  $\mathbf{A}$  inside  $\boldsymbol{\pi}^2$  is the orbital magnetic coupling, while the last term is the spin coupling. Both  $\boldsymbol{\pi}$  and the wave function transform covariantly, so the result does not depend on the gauge used for  $\mathbf{A}$ .

- (c) Writing the spin term as

$$-\frac{q}{m}\mathbf{S} \cdot \mathbf{B} = -\boldsymbol{\mu} \cdot \mathbf{B}, \quad \boldsymbol{\mu} = g_{\text{mag}}\frac{q}{2m}\mathbf{S},$$

gives

$$g_{\text{mag}} = 2$$

at tree level. For an on-shell electromagnetic vertex written with Dirac and Pauli form factors, charge normalization gives  $F_1(0) = 1$ , whereas

$$g_{\text{mag}} = 2[F_1(0) + 2mF_2(0)]$$

in the dimensionful convention of Chapter 14. Thus the radiative anomalous moment is carried by  $F_2(0)$ , not by a change of the electric charge.

### S.14.2. Darwin, Spin–Orbit, and Coulombic Power Counting

- (a) The spin-independent order- $m^{-2}$  term in the Foldy–Wouthuysen Hamiltonian is

$$H_D = -\frac{q}{8m^2}\boldsymbol{\nabla} \cdot \mathbf{E}.$$

For an electron in the field of a point nucleus,  $q\boldsymbol{\nabla} \cdot \mathbf{E} = -4\pi Z\alpha\delta^3(\mathbf{r})$ . Therefore

$$H_D = \frac{\pi Z\alpha}{2m^2}\delta^3(\mathbf{r}), \quad \delta E_D = \frac{\pi Z\alpha}{2m^2}|\psi_{n\ell m}(0)|^2.$$

Using  $|\psi_{n\ell m}(0)|^2 = (Z\alpha m)^3\delta_{\ell 0}/(\pi n^3)$ , one obtains

$$\delta E_D = \frac{(Z\alpha)^4 m}{2n^3}\delta_{\ell 0}.$$

The contact form makes the restriction to  $s$  waves manifest.

- (b) The electric spin term is

$$H_{\text{SO}} = -\frac{q}{4m^2}\boldsymbol{\sigma} \cdot (\mathbf{E} \times \mathbf{p}).$$

Since  $V = qA^0$ ,  $\mathbf{E} = -\nabla A^0$ , and  $\nabla V = (dV/dr)\hat{\mathbf{r}}$ ,

$$H_{\text{SO}} = \frac{1}{2m^2r} \frac{dV}{dr} \mathbf{L} \cdot \mathbf{S}.$$

A naive rest-frame magnetic-field argument would give twice this coefficient. The factor  $1/2$  is the Thomas factor; in the Dirac reduction it is already included by the consistent elimination of the small component.

(c) The virial theorem gives

$$|\mathbf{p}| \sim Z\alpha m, \quad r^{-1} \sim Z\alpha m, \quad E_{\text{bind}} \sim (Z\alpha)^2 m.$$

For a Coulomb field,  $q|\mathbf{E}| \sim Z\alpha/r^2$ , so

$$q|\mathbf{E}| \sim (Z\alpha)^3 m^2, \quad \frac{q \nabla \cdot \mathbf{E}}{m^2} \sim \frac{q \mathbf{E} \cdot \mathbf{p}}{m^2} \sim (Z\alpha)^4 m.$$

Both Darwin and spin-orbit effects therefore begin at  $m(Z\alpha)^4$ , two powers of  $Z\alpha$  below the leading binding energy. The same counting applies to the relativistic kinetic correction  $-\mathbf{p}^4/(8m^3)$ .

### S.14.3. Feynman–Hellmann Relations for Hydrogenic Matrix Elements

(a) Differentiate  $H(\lambda)|n\rangle = E_n|n\rangle$ , multiply on the left by  $\langle n|$ , and use normalization. The terms containing  $\partial_\lambda|n\rangle$  cancel, leaving

$$\frac{dE_n}{d\lambda} = \left\langle n \left| \frac{\partial H}{\partial \lambda} \right| n \right\rangle.$$

For a degenerate level, one must first diagonalize  $P(\partial_\lambda H)P$  in the degenerate subspace  $P$ . Its eigenvalues are the possible first derivatives of the split levels.

(b) Here  $\partial H/\partial \kappa = -1/r$ , while

$$\frac{\partial E_n}{\partial \kappa} = -\frac{m\kappa}{n^2}.$$

Feynman–Hellmann therefore yields

$$\left\langle \frac{1}{r} \right\rangle_{n\ell} = \frac{m\kappa}{n^2}.$$

It follows that  $\langle V \rangle = -m\kappa^2/n^2 = 2E_n$  and  $\langle T \rangle = -E_n$ , which is exactly the Coulomb virial theorem.

(c) The centrifugal coefficient becomes

$$\frac{\ell(\ell+1)}{2m} + \lambda = \frac{\ell_\lambda(\ell_\lambda+1)}{2m}, \quad \left. \frac{d\ell_\lambda}{d\lambda} \right|_0 = \frac{m}{\ell+1/2}.$$

With  $n = n_r + \ell + 1$ ,

$$E(\lambda) = -\frac{m\kappa^2}{2[n_r + \ell_\lambda + 1]^2}.$$

Differentiating at zero auxiliary coupling gives

$$\left\langle \frac{1}{r^2} \right\rangle_{n\ell} = \frac{m^2 \kappa^2}{n^3(\ell + 1/2)}.$$

The stronger  $\ell$  dependence reflects the centrifugal suppression near the origin.

#### S.14.4. The Stark Effect and Polarizability

(a) The first-order shift is

$$\delta E_{1s}^{(1)} = e\mathcal{E}\langle 1s|z|1s\rangle.$$

The  $1s$  wave function is even under parity whereas  $z$  is odd, so the integrand is odd and the matrix element vanishes. Directly,  $\int d\Omega \cos\theta = 0$ . Hence

$$\delta E_{1s}^{(1)} = 0.$$

The leading Stark effect is quadratic because the perturbation admixes odd  $p$  states into the even ground state.

(b) Second-order perturbation theory gives

$$\alpha_E = 2e^2 \sum_{n \neq 1} \frac{|\langle n|z|1s\rangle|^2}{E_n - E_1}.$$

All nonzero dipole matrix elements have  $E_n - E_1 \geq E_2 - E_1 = 3m\alpha^2/8$ , including the continuum. Completeness and  $\langle z \rangle = 0$  therefore imply

$$\alpha_E \leq \frac{2e^2 \langle 1s|z^2|1s\rangle}{E_2 - E_1}.$$

For  $\psi_{1s} \propto e^{-r/a_0}$ ,  $\langle r^2 \rangle = 3a_0^2$ , so  $\langle z^2 \rangle = a_0^2$ . Thus

$$\alpha_E \leq \frac{16e^2 a_0^2}{3m\alpha^2} = \frac{64\pi}{3} a_0^3$$

in Heaviside–Lorentz units.

(c) Parity makes every diagonal element vanish. The  $m = \pm 1$  states cannot couple to  $2s$ , and the only nonzero orbital matrix element is

$$\langle 2s|z|2p, m=0\rangle = -3a_0$$

up to a phase convention. The  $2s, 2p_0$  block is therefore

$$\begin{pmatrix} 0 & -3ea_0\mathcal{E} \\ -3ea_0\mathcal{E} & 0 \end{pmatrix}.$$

Its eigenvectors are the equal even–odd combinations, with shifts

$$\boxed{\delta E^{(1)} = \pm 3ea_0\mathcal{E},}$$

while  $2p, m = \pm 1$  remain unshifted at first order.

#### S.14.5. Einstein Coefficients and Electric-Dipole Radiation

- (a) For lower and upper populations  $N_1, N_2$ , equilibrium requires

$$N_1 B_{12} u(\omega) = N_2 [B_{21} u(\omega) + A_{21}].$$

Using  $N_2/N_1 = (g_2/g_1)e^{-\beta\omega}$  and

$$u(\omega) = \frac{\omega^3}{\pi^2(e^{\beta\omega} - 1)}$$

forces the coefficients of the constant and Boltzmann-dependent pieces to agree:

$$\boxed{g_1 B_{12} = g_2 B_{21}, \quad \frac{A_{21}}{B_{21}} = \frac{\omega^3}{\pi^2}.}$$

Spontaneous emission is therefore required for thermal equilibrium with the full Planck spectrum.

- (b) For a photon of momentum  $\mathbf{k}$  and polarization  $\lambda$ , the matrix element is proportional to  $q\sqrt{\omega/2}\epsilon_\lambda^* \cdot \mathbf{x}_{fi}$ . Golden-rule phase space fixes  $\omega = E_i - E_f$ . Using

$$\sum_\lambda \epsilon_\lambda^i \epsilon_\lambda^{j*} = \delta^{ij} - \hat{k}^i \hat{k}^j, \quad \int d\Omega (\delta^{ij} - \hat{k}^i \hat{k}^j) = \frac{8\pi}{3} \delta^{ij},$$

gives

$$\boxed{\Gamma_{i \rightarrow f}^{\text{E1}} = \frac{q^2}{3\pi} \omega^3 |\mathbf{x}_{fi}|^2 = \frac{4\alpha}{3} \omega^3 |\mathbf{x}_{fi}|^2}$$

for  $|q| = e$ .

- (c) Because  $\mathbf{x}$  has odd parity, a nonzero matrix element requires  $(-1)^{\ell'} = -(-1)^\ell$ ; hence  $\ell + \ell'$  is odd. A rank-one operator also satisfies the triangle rule  $|\ell - 1| \leq \ell' \leq \ell + 1$ . Combining the two conditions gives

$$\boxed{\Delta\ell = \pm 1.}$$

For  $\ell = \ell' = 0$ , the triangle condition itself reads  $|0 - 1| \leq 0$ , which is false; equivalently, the relevant rank-one Clebsch–Gordan coefficient vanishes.

#### S.14.6. Polarization Rules, the $2p$ Lifetime, and Stark Quenching

- (a) The spherical components  $x_q$  form a rank-one tensor, so

$$\langle \ell' m' | x_q | \ell m \rangle \propto C_{\ell m, 1 q}^{\ell' m'}.$$

The Clebsch–Gordan coefficient vanishes unless  $m' = m + q$ . Therefore

$$\Delta m = 0, \pm 1.$$

The  $q = 0$  component corresponds to linear, or  $\pi$ , polarization along the quantization axis. The  $q = \pm 1$  components correspond to the two circular  $\sigma^\pm$  polarizations, with the sign fixed by the spherical basis convention.

- (b) The energy gap and radial integral are

$$\omega = \frac{3}{8} \alpha^2 m_e, \quad \int_0^\infty dr r^3 R_{10} R_{21} = \frac{256 a_0}{81 \sqrt{6}}.$$

For any initial magnetic sublevel, the angular sum gives one third of the squared radial integral. Inserting this in the spontaneous-emission formula from S.14.5(b) yields

$$\Gamma_{2p \rightarrow 1s} = \frac{2^8}{3^8} \alpha^5 m_e = 6.27 \times 10^8 \text{ s}^{-1}, \quad \tau_{2p} = 1.60 \text{ ns}.$$

Rotational invariance ensures the same total width for all three  $2p$  magnetic substates.

- (c) To first order in the dc field,

$$|\tilde{2s}\rangle = |2s\rangle + c |2p, 0\rangle, \quad c = \frac{\langle 2p, 0 | e \mathcal{E} z | 2s \rangle}{\Delta_{\text{LS}}} = -\frac{3ea_0 \mathcal{E}}{\Delta_{\text{LS}}}.$$

The  $2p$  component decays by its allowed E1 transition, so

$$\Gamma_{\tilde{2s}} = \left( \frac{3ea_0 \mathcal{E}}{\Delta_{\text{LS}}} \right)^2 \Gamma_{2p}.$$

This perturbative expression requires  $3ea_0 |\mathcal{E}| \ll |\Delta_{\text{LS}}|$ . It explains why even a modest laboratory electric field can quench the otherwise long-lived  $2s$  level.

#### S.14.7. Vacuum-Polarization Contribution to the Lamb Shift

- (a) The Coulomb virial estimate gives

$$|\mathbf{p}| \sim m_e \alpha, \quad r \sim (m_e \alpha)^{-1}, \quad |E_{\text{bind}}| \sim m_e \alpha^2.$$

The electron wave functions therefore restrict a typical spatial momentum transfer to  $|\mathbf{k}| \sim m_e \alpha$ , while an energy transfer is only  $|k^0| \sim m_e \alpha^2$ . Consequently

$$\frac{\mathbf{k}^2}{m_e^2} = O(\alpha^2) \ll 1, \quad \frac{(k^0)^2}{\mathbf{k}^2} = O(\alpha^2) \ll 1.$$

(b) Expanding the supplied logarithm gives

$$\begin{aligned}\tilde{\mathbf{E}}(k) &= \left[ 1 + \frac{e^2}{2\pi^2} \frac{\mathbf{k}^2}{m_e^2} \int_0^1 dx x^2 (1-x)^2 + O(\mathbf{k}^4/m_e^4) \right] \tilde{\mathbf{D}}(k) \\ &= \left[ 1 + \frac{\alpha}{15\pi} \frac{\mathbf{k}^2}{m_e^2} + O(\mathbf{k}^4/m_e^4) \right] \tilde{\mathbf{D}}(k),\end{aligned}$$

where  $\int_0^1 x^2(1-x)^2 dx = 1/30$  and  $e^2 = 4\pi\alpha$ . Applied to the proton Coulomb field, the momentum-independent correction to the electron's potential energy is

$$\delta\tilde{V}(\mathbf{k}) = -\frac{4\alpha^2}{15m_e^2}.$$

Thus

$$\delta V(\mathbf{r}) = -V_1 \delta^3(\mathbf{r}), \quad V_1 = \frac{4\alpha^2}{15m_e^2}.$$

(c) First-order perturbation theory gives

$$\delta E_{n\ell m} = -V_1 |\psi_{n\ell m}(0)|^2.$$

A regular Coulomb radial wave function behaves as  $R_{n\ell}(r) \propto r^\ell$  near the origin, so  $\psi_{n\ell m}(0) = 0$  unless  $\ell = 0$ . Only  $s$ -levels shift.

(d) For the hydrogen  $2s$  state,

$$R_{20}(r) = \frac{1}{\sqrt{2a_0^3}} \left( 1 - \frac{r}{2a_0} \right) e^{-r/(2a_0)}, \quad a_0 = \frac{1}{m_e \alpha},$$

and hence

$$|\psi_{2s}(0)|^2 = \frac{1}{8\pi a_0^3}.$$

The shift is

$$\delta E_{2s} = -\frac{V_1}{8\pi a_0^3} = -\frac{m_e \alpha^5}{30\pi} \simeq -1.1 \times 10^{-7} \text{ eV}.$$

Equivalently,  $\delta E/h \simeq -27$  MHz. The observed  $2s - 2p$  Lamb splitting is about  $4 \times 10^{-6}$  eV, is opposite in sign, and is dominated by the electron self-energy rather than this smaller vacuum-polarization contribution.

### S.14.8. The Magnetic Moment of a Dirac Fermion

(a) With  $\beta = i\gamma^0$ , the volume current

$$j^\mu = -i\bar{\Psi}\gamma^\mu\Psi$$

is Hermitian. Since  $q$  is real,  $H_I$  is Hermitian precisely when the classical four-potential is real:

$$A_\mu(x)^* = A_\mu(x).$$

(b) A polar four-vector transforms under parity as

$$A^0(t, \mathbf{x}) \mapsto A^0(t, -\mathbf{x}), \quad \mathbf{A}(t, \mathbf{x}) \mapsto -\mathbf{A}(t, -\mathbf{x}),$$

so

$$\mathbf{E} \xrightarrow{P} -\mathbf{E}, \quad \mathbf{B} \xrightarrow{P} +\mathbf{B}.$$

The Dirac vector current transforms in the same way, making  $j^\mu A_\mu$  parity even. Under charge conjugation the vector current is odd, so invariance requires

$$A_\mu \xrightarrow{C} -A_\mu, \quad \mathbf{E} \xrightarrow{C} -\mathbf{E}, \quad \mathbf{B} \xrightarrow{C} -\mathbf{B}.$$

Spin is an axial vector. Therefore  $\mathbf{S} \cdot \mathbf{B}$  is parity even, whereas  $\mathbf{S} \cdot \mathbf{E}$  is parity odd. A magnetic dipole coupling is allowed, but an electric dipole coupling would violate parity (and, in the relativistic theory, the corresponding  $CP$  symmetry).

(c) In the Weyl basis used in the prompt, the spatial commutators are

$$\gamma^{ij} = i\varepsilon^{ijk} \begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}, \quad F_{ij} = -\varepsilon_{ij\ell} B^\ell.$$

The terms containing  $F_{0i}$  are velocity suppressed at rest. Consequently

$$\gamma^{\mu\nu} F_{\mu\nu} \longrightarrow -2i \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{B} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \mathbf{B} \end{pmatrix}.$$

Using the two rest spinors and  $\bar{u} = u^\dagger \beta$  gives

$$\boxed{\bar{u}(p') \gamma^{\mu\nu} u(p) F_{\mu\nu} = -4im_\Psi \xi'^\dagger \boldsymbol{\sigma} \cdot \mathbf{B} \xi, \quad a = -4im_\Psi.}$$

The imaginary coefficient is a consequence of the volume's anti-Hermitian vector gamma convention; the physical current contains the compensating factor  $-i$ .

(d) The Gordon identity in the same convention may be written

$$-i\bar{u}(p') \gamma^\mu u(p) = \frac{1}{2m_\Psi} \bar{u}(p') [(p' + p)^\mu + \gamma^{\mu\nu} (p' - p)_\nu] u(p).$$

The first term gives the spin-independent convection current. The second term, integrated against a uniform background field, gives

$$\langle \mathbf{p}', s' | H_I | \mathbf{p}, s \rangle = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \left[ -q \xi'^\dagger \boldsymbol{\sigma} \cdot \mathbf{B} \xi + \text{spin-independent terms} \right]$$

for relativistically normalized one-particle states. Dividing by  $\sqrt{2E_{\mathbf{p}'} 2E_{\mathbf{p}}} \rightarrow 2m_\Psi$  gives the nonrelativistic kernel

$$\boxed{h_{\text{spin}} = -\frac{q}{2m_\Psi} \xi'^\dagger \boldsymbol{\sigma} \cdot \mathbf{B} \xi.}$$

Thus

$$H_{\text{spin}} = -\frac{q}{m_\Psi} \mathbf{S} \cdot \mathbf{B} = -g_{\text{mag}} \frac{q}{2m_\Psi} \mathbf{S} \cdot \mathbf{B}, \quad \boxed{g_{\text{mag}} = 2.}$$



- (e) The bilinear  $\bar{\Psi} F_{\mu\nu} [\gamma^\mu, \gamma^\nu] \Psi$  is anti-Hermitian in the stated convention. Hence

$$\boxed{c^* = -c.}$$

Write  $c = ic_H$  with  $c_H$  real. Since  $[\gamma^\mu, \gamma^\nu] = 2\gamma^{\mu\nu}$ , part (c) gives, after conversion to nonrelativistic normalization,

$$\Delta h_{\text{spin}} = \frac{4c_H}{M} \xi'^\dagger \boldsymbol{\sigma} \cdot \mathbf{B} \xi.$$

The magnetic coefficient becomes

$$-\frac{q}{2m_\Psi} + \frac{4c_H}{M} = -\frac{g_{\text{mag}} q}{4m_\Psi},$$

or

$$\boxed{g_{\text{mag}} = 2 - \frac{16c_H m_\Psi}{qM}.}$$

An opposite convention for the sign of the added Pauli operator correspondingly reverses the sign of the shift; its invariant content is a correction of order  $m_\Psi/M$ .

#### S.14.9. Bubble-Chain Bound States in 2 + 1 Dimensions

- (a) Include the factor 1/2 that exchanges the two identical lines in each bubble and define

$$B(s) = \frac{1}{16\pi} \int_0^1 \frac{dx}{\sqrt{m^2 - x(1-x)s - i0}}.$$

This follows by combining the two propagators with a Feynman parameter, shifting the loop momentum, and using the elementary three-dimensional Euclidean integral. Below threshold it can also be written

$$B(s) = \frac{1}{8\pi\sqrt{s}} \operatorname{artanh}\left(\frac{\sqrt{s}}{2m}\right), \quad 0 < s < 4m^2.$$

With the tree amplitude normalized as  $-g$ , the displayed chain is

$$\mathcal{M}(s) = -g + g^2 B(s) - g^3 B(s)^2 + \dots = \boxed{-\frac{g}{1 + gB(s)}}.$$

No assertion about convergence of the geometric series is needed; the last expression defines the bubble-chain resummation.

- (b) For  $s > 4m^2$ , continuation through the two-particle cut gives

$$\operatorname{Im} B(s) = \frac{1}{16\sqrt{s}}.$$

Consequently

$$\operatorname{Im} \mathcal{M}(s) = \frac{g^2 \operatorname{Im} B(s)}{|1 + gB(s)|^2} = \frac{|\mathcal{M}(s)|^2}{16E_T}.$$

This is exactly the two-body optical theorem. To see it from the cross section supplied in the problem, avoid double counting identical final scalars by integrating over an angular range of length  $\pi$  (equivalently integrate over  $2\pi$  and divide by 2!):

$$\sigma_{\text{el}} = \frac{|\mathcal{M}|^2}{32pE_T^2}.$$

Multiplication by the center-of-mass flux  $4pE_T$  gives

$$4pE_T \sigma_{\text{el}} = \frac{|\mathcal{M}|^2}{8E_T} = 2 \operatorname{Im} \mathcal{M},$$

which verifies unitarity within the approximation.

- (c) Below threshold  $B(s)$  is real, positive, and monotone, with

$$B(0) = \frac{1}{16\pi m}, \quad \lim_{s \rightarrow 4m^2-} B(s) = +\infty.$$

A pole requires

$$1 + gB(M_B^2) = 0.$$

It is therefore possible only for attractive coupling  $g < 0$ . The value  $-1/g$  lies strictly inside the range of  $B(s)$  for

$$\boxed{-16\pi m < g < 0.}$$

At  $g = -16\pi m$  the pole has reached  $M_B^2 = 0$ , and for more negative  $g$  there is no pole in the requested interval  $0 < M_B^2 < 4m^2$ .

#### S.14.10. Nonrelativistic Forces from QED and Pseudoscalar Exchange

- (a) For momentum transfer  $q^\mu \simeq (0, \mathbf{q})$ , the temporal Dirac current dominates:

$$\bar{u}(\mathbf{p}') \gamma^0 u(\mathbf{p}) \simeq 2m_\Psi.$$

The direct one-photon graph therefore gives

$$\mathcal{M}_{\text{NR}} \simeq \frac{4m_\Psi^2 e^2}{\mathbf{q}^2}.$$

Removing the two relativistic external-state normalization factors leaves the Born kernel

$$\tilde{V}(\mathbf{q}) = \frac{e^2}{\mathbf{q}^2}.$$

Fourier transformation yields

$$V(r) = \frac{e^2}{4\pi r} = \frac{\alpha}{r}, \quad \mathbf{F} = -\nabla V = \frac{\alpha}{r^2} \hat{\mathbf{r}}.$$

The force is outward and hence repulsive. The identical-fermion exchange graph antisymmetrizes the two-particle state but does not alter this long-range direct force between separated wave packets.

(b) i. Let

$$\mathbf{P} \Psi(t, \mathbf{x}) \mathbf{P}^{-1} = \eta_P \beta \Psi(t, -\mathbf{x}), \quad \mathbf{P} \phi(t, \mathbf{x}) \mathbf{P}^{-1} = -\phi(t, -\mathbf{x}),$$

where  $\beta = i\gamma^0$  in the volume's convention and  $|\eta_P| = 1$ . Since  $\beta^{-1}\gamma_5\beta = -\gamma_5$ ,  $\bar{\Psi}\gamma_5\Psi$  is a pseudoscalar. Its product with the pseudoscalar field  $\phi$  is parity even.

ii. Taking  $\mathcal{L}_{\text{int}} = -g_5 \bar{\Psi}\gamma_5\Psi\phi$ , a consistent mostly-plus set of rules is

$$S_F(k) = \frac{-i\not{k} + m_\Psi}{k^2 + m_\Psi^2 - i0}, \quad \Delta_F(k) = \frac{-i}{k^2 + m_\phi^2 - i0}, \quad \Gamma_5 = -ig_5\gamma_5,$$

together with momentum conservation at the vertex and a minus sign for exchanging identical external fermions. An overall phase of  $\Gamma_5$  changes if the Hermitian interaction is written with an explicit  $i$ ; none of the following force laws changes.

iii. For  $\Psi(p_1)\Psi(p_2) \rightarrow \Psi(p_3)\Psi(p_4)$ , the two leading graphs are the direct and exchange scalar-exchange diagrams:

$$i\mathcal{M}_t = [\bar{u}(p_3)\Gamma_5 u(p_1)] \Delta_F(p_1 - p_3) [\bar{u}(p_4)\Gamma_5 u(p_2)],$$

$$i\mathcal{M}_u = -[\bar{u}(p_4)\Gamma_5 u(p_1)] \Delta_F(p_1 - p_4) [\bar{u}(p_3)\Gamma_5 u(p_2)].$$

The relative minus sign is fixed by Fermi statistics. These two expressions are the complete leading-order relativistic answer.

iv. At zero three-momentum,  $\bar{u}\gamma_5 u = 0$ . Keeping the first nonzero term gives, with  $\mathbf{q} = \mathbf{p}' - \mathbf{p}$ ,

$$\bar{u}(\mathbf{p}')\gamma_5 u(\mathbf{p}) = -\xi'^\dagger \boldsymbol{\sigma} \cdot \mathbf{q} \xi + O(|\mathbf{p}|^3/m_\Psi^2).$$

After converting relativistic external states to nonrelativistic normalization, the direct interaction is

$$\boxed{\tilde{V}_5(\mathbf{q}) = -\frac{g_5^2}{4m_\Psi^2} \frac{(\boldsymbol{\sigma}_1 \cdot \mathbf{q})(\boldsymbol{\sigma}_2 \cdot \mathbf{q})}{\mathbf{q}^2 + m_\phi^2}}.$$

Thus the naive momentum-independent force vanishes; the first force is spin dependent and suppressed by  $1/m_\Psi^2$ .

For completeness, let

$$Y(r) = \frac{e^{-m_\phi r}}{4\pi r}, \quad S_{12} = 3(\boldsymbol{\sigma}_1 \cdot \hat{\mathbf{r}})(\boldsymbol{\sigma}_2 \cdot \hat{\mathbf{r}}) - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2.$$

Fourier transformation gives the effective coordinate-space Hamiltonian

$$V_5(\mathbf{r}) = \frac{g_5^2}{12m_\Psi^2} \left\{ (\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) [m_\phi^2 Y(r) - \delta^3(\mathbf{r})] \right. \\ \left. + S_{12} \left( m_\phi^2 + \frac{3m_\phi}{r} + \frac{3}{r^2} \right) Y(r) \right\}.$$

For identical particles this kernel acts on an antisymmetric two-fermion wave function; that antisymmetrization is the nonrelativistic counterpart of the  $u$ -channel graph.

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